

FUNDAMENTALS OF MATHEMATICAL LOGIC

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PETER G. HINMAN

Fundamentals of Mathematical Logic



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Fundamentals of Mathematical Logic

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Preface

In nearly forty years of teaching mathematical logic to both undergraduate and graduate students, I have tried a wide variety of pedagogical approaches in the attempt to enable students with varying levels of interest and ability to come to a deep understanding of this beautiful subject. This book, written over nearly twenty of those years, is my best attempt to set down what I think has been most successful. All parts have been tried out at least once in the classroom and heavily revised in light of the experience.

In some ways it is a relatively standard development of the central results of 20th-century logic as they look from the beginning of the 21st: presentations of propositional, first-order and more general logics are followed by the Gödel incompleteness results and excursions into Set Theory, Model Theory and Recursion (Computability) Theory. However, along the way I have incorporated some non-standard expository devices that I have found useful and will comment on below. The material is roughly what I would hope that a very broadly (in logic) educated graduate student would have mastered by the time she begins more specialized research. A notable lacuna is Proof Theory, which fails to appear largely due to the incompetence of the author in this area.

The principal intended audience is the mathematics or theoretical computer science graduate student, although students from other disciplines, especially philosophy, are welcome. [Chapters 1–2](#), [Sections 3.1–3.5](#) and [Section 4.1](#) provide a thorough one-semester introduction to the subject suitable

for the future specialist in any area. The remaining material can serve as the core content for three additional courses for either the future logician or the more interested amateur: Model Theory (Sections 3.6–3.7 and Chapter 7), Set Theory (Chapter 6) and Recursion (Computability) Theory (Sections 4.2–4.6, Chapter 5 and Chapter 8). The over 400 exercises, many with substantial hints, comprise a significant extension. No previous experience with logic as a subject is presumed, only the maturity and capacity for abstraction expected of a first- or second-year graduate student.

One of the central expositional tenets of the book is the introduction of difficult ideas in the simplest possible context, even if they are of little intrinsic interest there. For example, many of the basic results for theories rely mainly on propositional ideas so can be developed in the vastly simpler context of propositional logic before they show up in earnest for first-order and stronger logics. In Sections 1.4–1.6 ideas of axiomatizability, decidability and constructions of models are explored for propositional theories. Proofs of the Propositional Compactness Theorem via ultraproducts and Boolean algebra contain many of the key ideas at the heart of the first-order version. The relationship between effective axiomatizability of a theory and the decidability or effective enumerability of the theorems of the theory is exactly the same in the propositional case.

Here and up to Section 4.2, all notions of effectiveness are the informal variety relying on intuitions of computability rather than any formal definition. This is a concerted effort to develop these intuitions — which in my experience are not native to most students — before introducing the rather abstract formalism of recursiveness and recursive enumerability. After this, it is routine (although not always simple) to convert the earlier intuitive arguments to precise mathematical ones. The capstone of this development is a “proof” in Section 4.1 of the First Incompleteness Theorem on the basis of the Definability and Representability *Hypotheses*, which are eventually replaced by the corresponding Theorems by the end of Chapter 4. This has also turned out to be a nice way to give students a sense of these grand ideas in a course with not nearly enough time for a full presentation.

Most texts derive the Compactness Theorem for first-order logic from the Completeness Theorem, but the direct proof of compactness is considerably easier, since the necessary properties of finite logical consequence are a lot easier than the corresponding properties of formal provability. Furthermore, very little time/space is lost since the construction of the Henkin model is exactly the same once the lemmas are in place. To boot, after proving compactness, it is natural to give the very nice proof of the Enumerability Theorem using the semantic version of Herbrand’s Theorem.

One other noteworthy instance of this principle is the introduction of priority arguments in the context of many-one reducibility in Section 8.1

even before Turing reducibility has been defined. This has eased the path into the tangle of priority for many students.

Chapter 6 (Set Theory) has several interesting features. One is the consistent use of classes. Since all of the axioms except Extensionality and Foundation are of the form (1) “a certain class operation is a set operation” — that is, maps sets to sets — for each of the instances where one wants to verify that something is a class model — the intuitive universe of sets V , the constructible universe L , a forcing extension $m[G]$ — one can develop general tools for showing (1) and then work out the details for each instance. The proofs aren’t really different from the usual ones, but they exhibit more of the underlying unity. One of my main goals in this chapter was to try to help the reader keep straight the three levels of set theory that come into play: Informal Set Theory (what all mathematicians use everyday), Axiomatic Set Theory (a precise version of the informal theory sufficient for most purposes) and Formalized Axiomatic Set Theory (the embedding of the axiomatic theory into set theory). This third version is just one instance of the mantra “all mathematics can be formulated in terms of sets and the results derived from the axioms”, and I devote two sections of **Chapter 6** to explaining this, particularly for logic itself. These distinctions are essential for a deep understanding of constructibility and forcing. A nice fringe benefit is (my version of) Tomas Jech’s beautiful and really complete (as opposed to almost all published proofs for arithmetic) proof of the Second Incompleteness Theorem for set theory.

For those (probably many) students who will cover only a portion (or even none at all) of this material in formal courses and therefore lack the guidance of an experienced instructor, here are a few further comments on the structure of the book. Of course, less-experienced instructors may also find these guidelines useful. Every reader new to logic should begin with **Section 1.1** and **Sections 1.3–1.5**. **Section 1.2** is technical background for all of the many inductive and recursive definitions used throughout the subject, but it is probably unwise for the novice to slog through all of this very abstract material before seeing more of the examples in **Chapters 2–3**, so this section is best considered a reference on first reading. **Sections 1.6–1.7** are preparation for the corresponding **Section 3.3**, and can be omitted by the reader trying to plot the shortest path to some of the material of **Chapters 4–8**. **Sections 2.1–2.3** introduce the basic structure of first-order languages and structures and should be covered thoroughly. **Sections 2.4–2.5** are in the nature of examples which illustrate the fundamental concepts of first-order logic, and the reader may pick and choose among them, although **Section 2.5** is important background for the Gödel Incompleteness Theorems of **Chapters 4–5**. **Section 2.6** is, like **Section 1.2**, a collection of facts and methods around a theme which is best treated as a reference section to be covered as needed.

Sections 3.1–3.2 are part of the heart of the subject and therefore central to any program of study. Section 3.3 may be omitted on first reading as may be parts of Sections 3.4–3.5, but to understand the power of Compactness, the reader should study at least the Upward Löwenheim-Skolem and Łoś-Tarski theorems. Sections 3.6–3.7 deal with extensions of first-order logic and are important for the future reader of Chapter 7, but in the interest of expediency may be entirely omitted by the reader more interested in other chapters.

Every student of logic should come to understand at some level the Gödel Incompleteness Theorems. Section 4.1 is designed to make these accessible with minimal preparation by avoiding all the technical machinery of recursive functions in favor of intuitive computability. The rest of Chapter 4 is devoted to filling in these details for the First Incompleteness Theorem, and Sections 5.1–5.3 do the same for the Second Incompleteness Theorem, but much of Section 5.2 is not used until Chapter 8 and may be postponed. Section 4.4 presents alternative formal characterizations of the class of recursive functions in the context of a discussion of the claim (Church’s Thesis) that these are exactly the intuitively computable functions. Section 5.5 uses the idea of interpreting one theory in another (explained in Section 2.6) to carry over results on undecidability and incompleteness from arithmetic to other familiar theories.

Each of Chapters 6–8 is intended as an introduction to the area of its title. In practice they have served as the core material for independent courses, but as introductions they necessarily omit much that specialists would consider central to the area and invite supplemental material for a complete course. They are almost completely independent of each other.

As is discussed at its beginning, Chapter 6 (Set Theory) is the longest because of the complicated and central role that sets play in mathematics. The student who has some familiarity with axiomatic set theory can skim Sections 6.1–6.4 to absorb the notation and basic framework. Sections 6.5–6.7 are largely devoted to the consistency of the Axiom of Choice (AC) and the Generalized Continuum Hypothesis (GCH) via Gödel’s constructible universe; the program is laid out in general terms at the beginning of Section 6.5. Sections 6.8–6.9 develop enough of Cohen’s forcing method to establish the full independence of the GCH. The student interested in the foundations of mathematics will want to pay particular attention to Sections 6.2 and 6.6 where the idea is explored that most notions of mathematics can be formulated in terms of sets and most mathematical theories can be viewed as part of set theory. Sections 6.10–6.11 consider some possible extensions of set theory.

Chapter 7 (Model Theory) is a quite focussed development of the material necessary for one proof of Morley’s Theorem, although numerous other topics

show up along the way. The reader who skipped earlier material on Boolean algebras (Sections 1.7 and 3.3) or infinitary languages (Section 3.7) will need to cover these first. Model theorists will complain that an introduction to model theory in the 21st century should include more material on stability, Morley rank, etc., but I felt that the book was already getting large and that several other books (for example, [Bue] or [Mar]) provide good sources for this material. The final Section 7.6 is somewhat independent of much of the rest of the chapter and shows that first-order logic is really quite unique among potentially competing logics.

Chapter 8 similarly takes the tack of developing one theme — priority arguments leading to the density of the r.e. Turing degrees — rather than giving a comprehensive introduction to the area. The prerequisites are Section 1.5, Sections 4.2–4.3 and Section 5.2, but actually very little of the other material of Chapters 1–7.

General mathematical notation used in the book is fairly standard and I mention here only a few items which might give trouble to the reader. The double implication $\implies$ and bi-implication $\iff$ symbols are always used in the meta-language in contrast to $\rightarrow$ and $\leftrightarrow$ used as symbols in formal languages. In the first part of the book the symbols $\wedge$ and $\vee$ are also reserved for use in formal languages, with words “and” and “or” being used in the meta-language. At some later points this convention is relaxed. The assignment operator $:=$ is generally (but probably not quite always) used to designate a definition of the object on the left side; occasionally $\hat{=}$ is used when it is more convenient to put the *definiendum* to the right. Similarly I use $:\iff$ and $\iff:$ for definitions of predicates or assertions.

The set-theoretic difference of two sets is denoted $A \sim B$ rather than the other common notations $A \setminus B$ or $A - B$. The symbol ω always denotes the set $\{0, 1, 2, \dots\}$ of natural numbers. For $k \in \omega$, ordered k -tuples are written $(x_0, \dots, x_{k-1})$, sometimes without the parentheses. $X \times Y$ is the usual Cartesian product consisting of all pairs (x, y) with $x \in X$ and $y \in Y$; X^k is the set of all k -tuples with all $x_i \in X$, and $X^{<\omega}$ is the union of X^k for all $k \in \omega$ — that is, the set of finite sequences from X . The domain and image (range) of a function F are denoted $\text{Dom}(F)$ and $\text{Im}(F)$ or sometimes simply $\text{Dom}F$ and $\text{Im}F$. $F : X \rightarrow Y$ means that $\text{Dom}(F) = X$ and $\text{Im}(F) \subseteq Y$. If $Z \subseteq X$, then $F \upharpoonright Z$ is the restriction of the function to arguments in Z .

Letters κ , λ and μ are used for infinite cardinal numbers. $\text{Card}(X)$ denotes the cardinality or power of a set X . $\wp(A)$ is the powerset (set of all subsets) of A , and $\wp_{<\omega(A)}$ and $\wp_{<\kappa}(A)$ denote, respectively, the set of finite subsets of A and the set of subsets of A of cardinality less than κ . In the early parts of the book, an attempt is made to limit where possible the set-theoretic prerequisites. This leads to a focus on countable languages where cardinality considerations and use of the Axiom of Choice play a smaller role.

Eventually, however, tools such as cardinal arithmetic and Zorn's Lemma are indispensable and are used without explanation. The reader for whom this poses a problem can refer to the development of [Sections 6.2–6.4](#) for help. The end of a proof is denoted by $\square$; the same symbol following the statement of a corollary or proposition indicates that the proof is straightforward and is omitted.

The list of References and the Symbol and Subject indices are fairly standard. I have included one further index of Item References that is designed to answer questions of the sort: where in the sequel is this notion or result referred to or used. For example, the entry “4.1.21 376, 377, 394, 396, 403, 736” means that the Diagonal Lemma 4.1.21 is referred to explicitly by name and number on the pages indicated. Of course, for frequently used items, there may be other mentions just by name that will not be in the index.

As noted, I have benefitted from the comments and reactions of a generation of students, far too many to name. More recently I have had valuable feedback from Andreas Blass, Doug Cenzer, Beth Chen, Eric Hall, Rob Houck, Bart Kastermans, Mike Katelman, Guha Krishnamurthi, Geoff LaForte, Jessica Metcalf-Burton, Rory Molinary, Igor Pavlovsky and Ellen Veomett. Those who know the book will appreciate the debt that I, like others of my and later generations, owe to Joe Shoenfield; his text reprinted as [Sh01b] set a standard for the field. Other books that were influential in my understanding of various parts of the material are those of Barwise-Feferman [Ba-Fe], Chang-Keisler [Ch-Ke], Drake [Dra], Jech [Jec], Kunen [Kun], Marker [Mar], Monk [Mon], Moschovakis [Mo80], Odifreddi [Od89] and [Od 99], Rothmaler [Rot] and Soare [Soa].

June, 2005
Ann Arbor, Michigan

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Introduction

From most perspectives, mathematics is a very precise discipline. Mathematicians (ideally) define their concepts carefully, state their assumptions unambiguously and in full, and give detailed tightly reasoned proofs of their conclusions. Still, there are two aspects of this process that are frequently left unspecified: the language to be used in stating the assumptions and conclusions and the methods to be admitted in the proofs. At first glance these seem unimportant omissions — there is a high level of agreement among practicing mathematicians as to what constitutes a sensible mathematical statement and what is a correct proof. One of the primary goals of Mathematical Logic is to exploit the advantages to be gained by a precise specification of both the language in which mathematics is expressed and the circumstances under which one statement is a consequence of (or follows from) a collection of assumptions.

Let us consider first the question of language. Mathematics is typically written (we are not concerned here with spoken language) in what might be called “Mathematical English” (or German or Russian). This consists of a relatively small part of standard English (omitting all those expressions concerned with parts of human experience irrelevant to mathematics) enriched with a large number of special technical terms and symbols. More accurately, in any given context a dialect of Mathematical English containing the appropriate terms and symbols will be used. We shall call such a

language *informal* in contrast to the *formal languages* that we shall develop in the course of this book.

Every expression of a dialect of Mathematical English consists of a string or finite sequence of unit inscriptions: lower and upper-case letters, numbers, punctuation marks and special symbols. Some of these such as

$\int_b^a f(x) dx$ is the signed area enclosed by the curves

$$y = f(x), x = a, x = b, \text{ and the } x\text{-axis}$$

are universally recognized as meaningful expressions, while others are either clearly recognizable as meaningless, e.g., “zx A#% 0”, or need further explanation before one can decide, e.g., “let $A|*|B$ be a unilateral aardvarkian”.

Formal languages will also have as their expressions strings of symbols drawn from a specified list, but beyond this similarity there are very significant differences. First, for each formal language will be given a precise criterion which can be applied to a string of symbols to determine whether or not it is a meaningful expression of the language. This criterion will be purely *syntactic* — that is, concerned only with the arrangement of the symbols, not with any meaning attached to them. Related to this, the symbols will fall into two classes: *logical symbols*, which have a fixed meaning, and *non-logical symbols*, which are subject to multiple interpretations. Finally, the logical symbols and their meanings are chosen to make evident the logical structure of each expression.

It will take some development of the subject before we can fully appreciate the benefits of these differences, but let us consider here an example to illustrate the notion of logical structure. Anyone who has studied linear algebra has encountered some version of the following statement, which might occur as either a definition or as a theorem:

- (1) A set B of elements of a vector space V is a *basis* if and only if every $v \in V$ is expressible as a linear combination of elements of B in one and only one way.

Presumably the reader of (1) has a relatively clear understanding of the technical terms used here: set, element, vector space and linear combination. The difficulty that students typically have in understanding (1) is due largely to its somewhat involved logical structure, which is concealed by the way it is stated.

As a first step in analyzing this structure, we note that (1) is of the form

$$\phi \leftrightarrow \psi,$$

where ϕ stands for ‘ B is a basis of V ’, ψ stands for ‘every $v \in V$ is expressible as a linear combination of elements of B in one and only one way’ and $\leftrightarrow$ stands for ‘if and only if’. Each of ϕ and ψ has something to do with linear algebra whereas $\leftrightarrow$ expresses a purely logical relationship. ϕ is the notion being defined or characterized and requires no further analysis. ψ can be further broken down as follows:

$$\forall v [v \in V \rightarrow \exists l (l \in \text{LC}(B) \wedge v = l \wedge \forall l' [l' \in \text{LC}(B) \wedge v = l' \rightarrow l \text{ is identical with } l'])].$$

Here $\text{LC}(B)$ denotes the class of linear combinations of elements of B ; $\forall$, $\exists$, $\wedge$, and $\rightarrow$ stand for ‘for all’, ‘there exists’, ‘and’, and ‘implies’, respectively. Notice that this structure corresponds exactly to the procedure for verifying that some given B is a basis: for a given (arbitrary) $v \in V$, we must first show that v is a linear combination of elements of B and then that this representation is unique in the sense that any other (putatively different) linear combination equal to v is really the same one.

Finally, we may further expand the statement by noting that $l \in \text{LC}(B)$ is an abbreviation for

$$\exists n \exists b_0 \dots \exists b_n \exists r_0 \dots \exists r_n (l = r_0 b_0 + \dots + r_n b_n),$$

where, of course, we understand by convention that n is a non-negative integer, each $b_i \in B$, and each r_i is an element of the underlying field. In its full implementation, an expression in a formal language for (1) would need to make explicit all of these (and other) conventions. Formal expressions thus tend to be rather long and involved; this is an inevitable consequence of building into them all of the assumptions that a human reader makes in interpreting a statement of Mathematical English and one should think of them as “expressions of record”, just as legal documents codify the law and computer programs provide complete descriptions of algorithms in forms somewhat obscure to the casual reader.

Let us turn now to the notion of consequence. Most of mathematics is, or at least could in principle be, carried on axiomatically. At the beginning we assume a number of statements called **axioms** and then by a process of deduction or proof we derive other statements, which we call **theorems** (also lemmas, propositions etc.) In some cases the underlying hypothesis is that the axioms are true about one mathematical object (for example, the real line) and hence so are the theorems. In other cases, the axioms are true about many objects (for example, the axioms of group theory) and the theorems are then also true about all of these objects. Once we have made precise the language in which the axioms and theorems are to be stated, it is possible to define precisely which statements of this language are theorems

from a given set of axioms. In fact, there are two natural definitions which turn out to be equivalent.

The first involves a formal version of the sort of proof that every mathematician writes: a **formal proof** is a finite sequence of statements (of the appropriate formal language) such that each statement in the sequence is either an axiom or follows from earlier statements via one of a (short) list of simple **rules of inference** — that is, atomic proof-steps. Such a proof is a formal proof of the last statement in the sequence. We shall write $\Gamma \vdash \phi$ to mean that there exists a proof of this sort with axioms taken from the set Γ and ϕ as the last statement in the proof.

The second notion of consequence derives from the principle discussed above that all of the non-logical symbols of a formal language are considered susceptible of multiple interpretations. Hence a set Γ of axioms will in general have many possible **models** — that is, interpretations of the symbols which make all of the members of Γ true. Then we write $\Gamma \models \phi$ to mean that ϕ is true in every model of Γ . It is obvious that the rules of inference should be chosen so that proof preserves truth — that is, so that

$$\Gamma \vdash \phi \quad \text{implies} \quad \Gamma \models \phi.$$

One of the main results of elementary logic is the **Completeness Theorem** which says that they may be chosen so that in fact

$$\Gamma \vdash \phi \quad \text{if and only if} \quad \Gamma \models \phi.$$

To make some of these ideas more concrete, let us consider some particular formal languages and some of their interpretations. The logical symbols of each language are those we have mentioned above together with $\vee$ (or), $\neg$ (negation, not), the equality sign $\doteq$ and an infinite list of **variables** $x, y, \dots$. Here and on many future occasions we shall use a common (informal) mathematical sign with a dot over it to represent a formal symbol in a formal language which denotes the informal object (**dot convention**). Let L_{Gp} have in addition to the logical symbols the symbols $\ast$, $^{-1}$, and e . $\ast$ is to be a **binary operation** or **function symbol**, so that it always occurs in a context $t \ast u$. $^{-1}$ is a **unary operation symbol** which we use in a context of the form t^{-1} . e is a **constant symbol**. The intention is that L_{Gp} is a language for talking about groups, which we think of as structures consisting of a set G together with the (binary) group operation $\times$, the (unary) inverse operation $^{-1}$, and the identity element e — formally, we write $\mathcal{G} = (G, \ast, ^{-1}, e)$. Statements of the language are formed pretty much as one would expect, except that things are written somewhat more explicitly than is common in informal mathematics — for example

$$(2) \quad \forall x \forall y \forall z [(x \ast y) \ast z \doteq x \ast (y \ast z)]$$

$$(3) \quad \forall x [x \dot{*} \dot{e} \dot{=} x] \quad \text{and} \quad \forall x [\dot{e} \dot{*} x \dot{=} x]$$

$$(4) \quad \forall x [x \dot{*} x^{\dot{-}1} \dot{=} \dot{e}] \quad \text{and} \quad \forall x [x^{\dot{-}1} \dot{*} x \dot{=} \dot{e}].$$

Clearly, (2) expresses the associative law, (3) the fact that $\dot{e}$ is to be the identity and (4) the fact that $^{\dot{-}1}$ is the inverse operation. In fact, these three statements together are exactly the **axioms of group theory** and a structure $\mathcal{G}$ satisfies the set $\Gamma_{\text{Gp}} = \{(2), (3), (4)\}$ just in case it is a group. Hence for any other statement ϕ of this language, $\Gamma_{\text{Gp}} \models \phi$ just in case ϕ is also true in all groups — for example, this holds if ϕ is the statement

$$\forall x \forall y \left[(x \dot{*} y)^{\dot{-}1} \dot{=} y^{\dot{-}1} \dot{*} x^{\dot{-}1} \right],$$

but not if ϕ is $\forall x \forall y [x \dot{*} y \dot{=} y \dot{*} x]$, since this statement is true only in Abelian groups and not all groups are Abelian.

On the other hand, by the Completeness Theorem, it is also true that $\Gamma_{\text{Gp}} \models \phi$ just in case there is a formal proof of ϕ from Γ , so we could (and often would) establish that

$$\Gamma_{\text{Gp}} \models \forall x \forall y \left[(x \dot{*} y)^{\dot{-}1} \dot{=} y^{\dot{-}1} \dot{*} x^{\dot{-}1} \right]$$

by producing such a proof. However, to verify that

$$\Gamma_{\text{Gp}} \not\models \forall x \forall y [x \dot{*} y \dot{=} y \dot{*} x],$$

the approach using models is crucial, since otherwise we would have somehow to show that no sequence of statements could constitute a formal proof of $\forall x \forall y [x \dot{*} y \dot{=} y \dot{*} x]$ from Γ_{Gp} .

As a slightly more complicated example, consider the language L_{Ω} . As non-logical symbols we take the symbols $\dot{+}$, $\dot{\times}$, $\dot{<}$, $\dot{S}$ and $\dot{0}$. Here we have two binary operation symbols, one unary operation symbol, one constant symbol and one **relation symbol** $\dot{<}$. In this case there is an intended or primary interpretation (the **standard model**) in which the variables range over the set ω of non-negative integers (**natural numbers**), $\dot{+}$ and $\dot{\times}$ are interpreted as the usual binary operations $+$ and $\times$ of addition and multiplication of natural numbers, $\dot{S}$ is the successor function, $\dot{0}$ is 0 and $\dot{<}$ is interpreted as the usual ordering. We write this interpretation as $\Omega := (\omega, +, \times, <, \text{Sc}, 0)$.

Some sample statements of L_{Ω} are

$$(5) \quad \forall x \forall y \forall z [x \dot{<} y \wedge \neg z \dot{=} \dot{0} \rightarrow x \dot{\times} z \dot{<} y \dot{\times} z]$$

$$(6) \quad \forall x \forall y [x \dot{<} y \leftrightarrow \exists z (\neg z \dot{=} \dot{0} \wedge x \dot{+} z \dot{=} y)]$$

$$(7) \quad \forall x \exists y [x \dot{<} y \wedge \text{'y is prime'} \wedge \text{'(y \dot{+} \dot{S}\dot{S}\dot{0}) is prime'}]$$

where ‘ y is prime’ is an abbreviation for the expression

$$\forall z \forall w [z \dot{\times} w \doteq y \rightarrow z \doteq \dot{S}0 \vee w \doteq \dot{S}0].$$

Clearly under the intended interpretation, statement (5) expresses the monotonicity of multiplication, (6) a simple relationship between the ordering and addition and (7) the **twin-prime conjecture**: there exist an infinite number of pairs p and $p + 2$ of successive odd prime integers.

Any statement of the language L_Ω makes an assertion about Ω which may be either true or false. (5) and (6) above are true, but the current state of mathematical knowledge does not include whether (7) is true or false. If we write down any set Γ of statements all of which are true about Ω and $\Gamma \models \phi$ as described above, then also ϕ is true about Ω .

To understand the meaning of $\Gamma \models \phi$, we need to consider also **non-standard interpretations**. These are **structures** of the form

$$\mathfrak{A} = (A, F, G, R, r, s),$$

where A is any non-empty set, F and G are binary functions on A — that is, $F, G : A \times A \rightarrow A$ — R is a binary relation on A , and r and s are elements of A . Clearly any statement of the language can be interpreted as making a (true or false) assertion about any such structure — for example, (5) asserts that for any elements a, b and c of A , if $R(a, b)$ and $c \neq r$, then $R(G(a, c), G(b, c))$. Now $\Gamma \models \phi$ means that for any such $\mathfrak{A}$, whenever all of the statements in Γ are true about $\mathfrak{A}$, then also ϕ is true about $\mathfrak{A}$.

In most (non-logical) mathematical contexts we are concerned only with *positive* results of the form: ϕ follows from Γ . As we have seen, the methods described above allow us also to establish *negative* results of the form: ϕ does *not* follow from Γ . Of course, by this we do not mean simply that we are currently unable to prove ϕ from Γ but rather that there can be no such proof; by the Completeness Theorem this is equivalent to $\Gamma \not\models \phi$ — that is, there exists some structure $\mathfrak{A}$ such that all the statements in Γ are true about $\mathfrak{A}$, but ϕ is not. In particular, if ϕ is some contradictory statement, say $\exists x (\neg x \doteq x)$, it is natural to call Γ **consistent** if ϕ does not follow from Γ . Then Γ is consistent just in case Γ has at least one model, since ϕ will be false there.

Another question which is of interest in many contexts is the following: given a set Γ of axioms, is it the case that for every statement ϕ (of the relevant formal language), one of ϕ and $\neg \phi$ follows from Γ — that is, on the basis of Γ either the truth or falsity of ϕ can be established. Such a Γ is called **complete**; intuitively it is strong enough to answer all questions which can be expressed in the language. Note that to prove that some set Γ is complete it is essential to have a precisely defined set of statements.

On the other hand, to demonstrate that Γ is *incomplete*, we need to find a particular statement ϕ and show as above that neither ϕ nor $\neg \phi$ follows from Γ . Note that this use of the word “complete” is different from and only weakly related to its use in the term “completeness theorem” above.

Mathematics is often concerned with theories. Informally a theory involves a collection of results and notions around a common theme. A major concern of Mathematical Logic is the analysis of mathematical theories, and for this purpose we require a more precise specification. A theory will be determined by a particular language and a set T of statements of this language which is closed under logical consequence: if $T \models \phi$, then already $\phi \in T$. Then a theory is consistent iff it contains no contradictory statement and complete iff for every statement ψ of the language, either $\psi \in T$ or $\neg \psi \in T$.

Theories arise in two quite different ways: if Γ is a set of statements, then $\text{Th}(\Gamma) = \{ \psi : \Gamma \models \psi \}$ is the theory generated by the members of Γ considered as axioms, the **logical consequences** of Γ ; T is called an **axiomatic theory**; if $\mathfrak{A}$ is a structure, then $\text{Th}(\mathfrak{A}) = \{ \psi : \psi \text{ is true about } \mathfrak{A} \}$ is the **theory of the structure**. Although the theory of a structure is automatically both consistent and complete, an axiomatic theory may be either, both, or neither. A question that has played a major role in mathematics is to find for a given theory $T = \text{Th}(\mathfrak{A})$ a ‘nice’ set Γ of axioms such that also $T = \text{Th}(\Gamma)$. Of course, with no restriction on the sort of axioms allowed there is the trivial solution $\Gamma = T$, but this is intuitively unsatisfying and technically uninteresting in most cases.

One interpretation for ‘nice’ is ‘finite’, but this turns out to be too restrictive. A more liberal interpretation is ‘decidable’. A set Γ of statements of a language L is called **decidable** iff there exists an algorithm which applied to any statement ϕ of L will decide correctly whether or not ϕ belongs to Γ . In practical terms it is entirely reasonable to insist that a set of working axioms be decidable, since otherwise it is hard to see how one could devise proofs from the axioms. Furthermore, we shall see that all traditional axiom systems are decidable. Thus the axiomatization question becomes: for which structures $\mathfrak{A}$ does there exist a decidable set Γ such that $\text{Th}(\Gamma) = \text{Th}(\mathfrak{A})$? Surprisingly, the key tool for the solution is again the Completeness Theorem, from which it follows that from the existence of such a Γ it follows that the theory $\text{Th}(\mathfrak{A})$ itself is decidable. This is quite a different property.

In practical axiomatic mathematics we try to determine whether a given statement ϕ belongs to an axiomatic theory $T = \text{Th}(\Gamma)$ by trying to find a proof of ϕ from the axioms Γ . If $\phi \in T$ and we are clever, we may be able to verify this. If, however, $\phi \notin T$, no amount of struggling to find a proof will discover this. If T is complete, then we have the alternative strategy of looking also for a proof of $\neg \phi$ with the assurance that at least one of ϕ and $\neg \phi$ has a proof from Γ . Even for incomplete T , this is a

common practical strategy, although one which is not guaranteed to succeed. However, if T is decidable, then there is no need for all of our cleverness and erudition; we may simply apply the decision algorithm for T to ϕ and get a definitive answer to the question whether or not $\phi \in T$. This makes T in some sense trivial, and if central theories of mathematics were decidable, it would threaten to replace the creative mathematician by a computer. It is a deep result of [Chapter 4](#) that this is not the case for many theories, in particular the theory $\text{Th}(\Omega)$ of the standard model for arithmetic discussed above. Hence there can be no decidable set Γ of axioms such that $\text{Th}(\Gamma) = \text{Th}(\Omega)$ and questions such as the twin-prime conjecture still remain to be solved by other as yet unknown means.

In the more than two thousand year history of mathematics, the perspective we have sketched, and therefore the field of mathematical logic, is relatively new. Of course, the idea of setting down axioms and deriving theorems from them via proofs viewed as finite sequences of statements goes back at least to Euclid in the fourth century BCE. At least two central aspects of the modern view of logic can be traced to inadequacies of Euclid's system.

First, he viewed the axioms of plane geometry as basic true statements about the world (the *standard model*) and the process of proof a way of discovering other truths about the world. The *primitive terms* of the system, such things as 'point', 'line' and 'plane' were intended to refer to our usual notion of these objects. The idea of interpreting them in some other non-standard way did not arise until the 19th century in connection with the geometries of Lobachevski, Bolyai and others. Whereas Euclid, and generations of mathematicians after him, expected that the so-called parallel postulate could be derived from the other axioms, it was eventually seen that alternative interpretations of the primitive terms lead to non-standard models in which the parallel postulate is false but the other axioms remain true. It follows that there can be no derivation of the parallel postulate.

This method led fairly directly to the complete separation of syntax (the language) and semantics (the interpretations) described above: the primitive terms (non-logical symbols) are free to be given any coherent interpretation (a unary function symbol as a unary function etc.) in any universe. Then a statement follows from the axioms only if it is true in all of these interpretations under which the axioms are true.

A second problem with Euclid's system from a modern perspective was his failure to codify the rules of proof sufficiently to ensure that no assumptions apart from the stated axioms were used. In fact, there were such assumptions, and it was not until the end of the 19th century that these were identified and incorporated into a more complete axiom system by Pasch and Hilbert. This experience demonstrated the need for the formality and rigor of the proof system described above and developed in [Section 3.2](#).

At the same time that these developments were taking place, interest in formal axiomatization arose for a different reason. Mathematics had for centuries been regarded as the most secure of the sciences; new discoveries might render theories invalid in subjects such as astronomy, chemistry, biology or physics, but in mathematics a theorem once proved was forever true. Although as we have noted the basic assumptions behind such proofs were simply the fabric of mathematical practice and had never been made explicit, they had always worked flawlessly and thus never been questioned. However, in the early part of the 20th century the discovery of several apparent contradictions and paradoxes shattered the confidence of mathematicians that their subject was on a secure foundation.

A complete exposition of this crisis would be beyond the scope of this introduction, but we mention three examples. The first is actually an ancient paradox of ordinary language known as the *liar paradox*, which in its simplest form simply asks whether the sentence “This sentence is false” is true or false. Equivalently, if a person says “I am lying”, is he telling the truth or not? Clearly either answer leads to a contradiction.

The second is due to Bertrand Russell in the context of Georg Cantor’s new (in 1902) theory of abstract sets. One of Cantor’s seemingly obvious principles was that any collection of objects could be gathered together to be the elements of a set. Hence in particular there exists a set X such that

$$X = \{ y : y \text{ is a set and } y \notin y \}.$$

Now clearly $X \in X \iff X \notin X$, a contradiction.

The third example of Jules Richard (1905) concerns functions from natural numbers to natural numbers. Consider all of the infinitely many English expressions which define such functions and list them in order of length, with expressions of the same length in alphabetical order; this leads to an infinite list $F_0, F_1, \dots, F_n, \dots$ of functions. Now define a function G by $G(n) := F_n(n) + 1$. This symbolic definition can be accurately described by an English expression, so the function G must occur somewhere in our list — that is, for some natural number p , G is F_p . But $G(p) = F_p(p) + 1 \neq F_p(p)$, a contradiction. These matters can be pursued in the excellent source book [vHe] or in [Kl52; Chapter III].

There were many responses to this crisis by both philosophers and mathematicians. Some of them proposed greatly limiting the scope of mathematics in the hope of avoiding areas which lead to contradiction. To others, particularly David Hilbert, this cure seemed worse than the disease and over the course of the next twenty or so years he formulated another approach. In outline, although he never expressed it in just this way, Hilbert’s idea was that we could secure the foundations of each area of mathematics by carrying out the following steps, which have come to be called *Hilbert’s Program*:

- (A) make precise the fundamental notions of the area and set up a formal language which provides the means to express these notions;
- (B) codify the rules of logic;
- (C) find a reasonable set of axioms which express the basic truths about the fundamental notions;
- (D) show that the set of theorems provable from these axioms using these rules either (a) coincides with the set of true statements about the notions (assuming that there is a secure definition of truth) or (b) could be taken as the *definition* of this set of true statements (thus avoiding the potentially problematic notion of truth altogether);
- (E) prove in a completely secure way that no contradiction could be derived from the axioms.

Steps (A)–(C) are a continuation of the work described above — in particular Hilbert’s own work on the axiomatization of geometry. In (D), he expresses the view that any well-posed mathematical problem should be solvable; he intends that in case (b) that the set of theorems provable from the axioms should be complete in the sense mentioned above that for every formal statement ϕ , either ϕ or $\neg \phi$ is a theorem.

The most problematic step is (E). If we are to believe that our system is free from contradiction, then the proof that this is so must not itself be subject to doubt. For Hilbert the advantage of formulating a mathematical theory in this formal axiomatic way was this: no matter how complex the subject matter of the theory, the process of proving theorems in a formal system is just the manipulation of finite sequences of symbols according to simple finite rules. Therefore, we can hope to prove directly that no such finite sequence can be a proof of a contradiction.

The results of Gödel in 1931 showed surprisingly but conclusively that this program cannot be carried out. The First Incompleteness Theorem shows that for any part of mathematics at least as complex as arithmetic, there can be no reasonable complete system of axioms. The Second Incompleteness Theorem shows that even for an incomplete system, if it incorporates even a modest amount of mathematics, then its consistency cannot be proved in the system itself. Hence, if the system includes all of what we regard as finitary mathematics, there can be no finitary proof of its consistency, and if the system includes all of classical mathematics, then there can be no mathematical proof of its consistency. These results will be developed in [Chapters 4–6](#).

The part of Mathematical Logic we have been describing is often called *metamathematics*; although it is a part of mathematics it also has the whole of mathematics as its subject matter. Much of [Chapters 1–5](#) are concerned with topics roughly of this sort. From its original concern with these questions,

however, the subject has developed over the last century into an independent branch of mathematics with four principal sub-areas: Set Theory, Model Theory, Recursion (or Computability) Theory and Proof Theory. [Chapters 6–8](#) provide an introduction to the first three of these.



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Chapter 1

Propositional Logic and Other Fundamentals

In this chapter we shall introduce the syntax and semantics of Propositional Logic. From the point of view of its ability to express mathematical ideas, the propositional language has very limited power. The basic building blocks are not arbitrary mathematical objects and relationships among them as we shall want in our more expressive languages, but only unanalyzed mathematical assertions (propositions or sentences). The only relationships which can be expressed are those corresponding to the English words “and”, “or”, “implies” etc. Nevertheless, the structure of the propositional language has much in common with the much more useful first-order languages, and its study serves as a good training ground for the basic principles of syntax and semantics. Indeed we shall use various aspects of this structure to motivate a number of basic tools, notably induction, recursion and decidability, which will play important roles throughout our study.

1.1 The propositional language

Specifying the syntax of any language L follows a common pattern. First a set of symbols is given, and we define an L -expression to be any finite sequence of these symbols. Then we specify one or more sets of L -expressions which we regard as meaningful. The meaningful expressions are generally

described as those constructed by following certain rules or algorithms, and the set of them is characterized as the smallest set of expressions which is closed under these formation rules. Technically this takes the form of an inductive definition as described below.

For the remainder of this section, we fix the letter L to denote the **propositional language** about to be described. The set Symb_L of L -symbols consists of the denumerable set $\{p_n : n \in \omega\}$ of **sentence symbols** and the five **connectives**: $\neg$ (**not**), $\wedge$ (**and**), $\vee$ (**or**), $\rightarrow$ (**implies**), and $\leftrightarrow$ (**iff**), where the names in parentheses suggest the meanings we shall later attach to these symbols. The set of all finite sequences of these symbols is the set of L -expressions denoted by Expr_L . We use Greek letters $\phi, \psi, \dots$ to denote L -expressions. The basic operation on expressions is **concatenation** denoted by juxtaposition: for example, if ϕ denotes the expression $s_0 \dots s_{k-1}$ (or more pedantically $(s_0, \dots, s_{k-1})$) containing k symbols (ϕ is of **length** k), and ψ denotes $t_0 \dots t_{l-1}$, then $\wedge \phi \psi$ denotes the expression $\wedge s_0 \dots s_{k-1} t_0 \dots t_{l-1}$ of length $k + l + 1$. Two expressions are considered equal if and only if they are identical symbol by symbol. A symbol which is part of an expression is said to **occur** in the expression.

Definition 1.1.1 An L -atomic sentence is an L -expression consisting of a single sentence symbol.

Note that technically an atomic sentence is a different mathematical object from the single sentence symbol which occurs in it; we shall generally identify these two objects.

We want next to characterize the set of L -expressions to which we shall attach meaning, the L -sentences. Each L -atomic sentence is an L -sentence and any two L -sentences can be combined with a connective to form a new L -sentence. For technical reasons to be made clear shortly, we shall write $\wedge \phi \psi$ instead of the more natural $\phi \wedge \psi$ to represent the sentence ' ϕ and ψ ', and similarly for the other connectives. We need to make mathematically precise the intuitive idea that the L -sentences are *exactly* the expressions which are generated from the atomic sentences by these connective operations, so that, for example, $\vee \neg p_2 \wedge p_4 \neg p_1$ will be counted as an L -sentence, but $\neg p_2 p_1 \vee p_3$ will not. There are several ways to do this, which we shall explore in the next section, but for now we take the simplest of them as our definition.

Definition 1.1.2

- (i) Sent_0 := the set of L -atomic sentences;
- (ii) for each $n \in \omega$,

$$\begin{aligned} \text{Sent}_{n+1} := & \text{Sent}_n \cup \{ \neg \phi : \phi \in \text{Sent}_n \} \\ & \cup \{ \bullet \phi \psi : \phi, \psi \in \text{Sent}_n, \bullet \text{ is one of } \vee, \wedge, \rightarrow, \leftrightarrow \}; \end{aligned}$$

(iii) $\text{Sent}_L := \bigcup_{n \in \omega} \text{Sent}_n$.

A member of Sent_L is called an *L-sentence* or simply a sentence. If $\phi \in \text{Sent}_n$ but $\phi \notin \text{Sent}_i$ for any $i < n$, we say that ϕ is a sentence of *level* n .

Lemma 1.1.3 *Sent_L is the smallest set of expressions which includes Sent_0 and is **closed** under the connectives $\neg$, $\vee$, $\wedge$, $\rightarrow$ and $\leftrightarrow$. That is,*

- (i) $\text{Sent}_0 \subseteq \text{Sent}_L$;
- (ii) if $\phi, \psi \in \text{Sent}_L$, then also $\neg\phi$, $\vee\phi\psi$, $\wedge\phi\psi$, $\rightarrow\phi\psi$ and $\leftrightarrow\phi\psi$ belong to Sent_L ;
- (iii) if Γ is any other set of expressions with these two properties, then $\text{Sent}_L \subseteq \Gamma$.

Proof. Clearly $\text{Sent}_0 \subseteq \text{Sent}_L$. If $\phi, \psi \in \text{Sent}_L$, then for some m and n , $\phi \in \text{Sent}_m$ and $\psi \in \text{Sent}_n$, so for $p = \max(m, n)$, both $\phi, \psi \in \text{Sent}_p$. Then $\neg\phi$ and all $\bullet\phi\psi$ belong to $\text{Sent}_{p+1} \subseteq \text{Sent}_L$. Finally, if Γ is any other set with these properties, then it is straightforward to prove by induction that for all n , $\text{Sent}_n \subseteq \Gamma$ and hence $\text{Sent}_L \subseteq \Gamma$. $\square$

Part (iii) of this lemma provides an efficient way of proving assertions about sentences:

Proposition 1.1.4 (*proof by sentence induction*) *For any property $\mathcal{P}$ which applies to L -expressions, if*

- (i) (**basis**) *every L -atomic sentence has the property $\mathcal{P}$; and*
- (ii) (**induction steps**) *for any L -sentences ϕ and ψ , if both ϕ and ψ have the property $\mathcal{P}$, then so do $\neg\phi$, $\vee\phi\psi$, $\wedge\phi\psi$, $\rightarrow\phi\psi$ and $\leftrightarrow\phi\psi$;*

then all L -sentences have the property $\mathcal{P}$.

Proof. Let Γ denote the set of all expressions which have the property $\mathcal{P}$. The hypotheses (i) and (ii) correspond exactly to (i) and (ii) of the lemma, and we conclude that $\text{Sent}_L \subseteq \Gamma$, which is exactly the desired conclusion. $\square$

We shall make frequent use of this principle and variations on it throughout the book. Note that it is a natural generalization of proof by ordinary mathematical induction: the L -atomic sentences play the role of 0 and the mappings $\phi \mapsto \neg\phi$ and $\phi, \psi \mapsto \bullet\phi\psi$ play the role of the successor function. In this connection, the hypothesis of (ii) that both ϕ and ψ have the property $\mathcal{P}$ is usually referred to as the **induction hypothesis**. We shall explore further this connection in the next section.

The first and most basic property of L -sentences is that they are *uniquely readable* — that is, for each L -sentence there is a unique way to break it up into its component parts. For example the sentence $\vee \neg p_2 \wedge p_4 \neg p_1$ mentioned above can only be described as $\vee \phi\psi \in \text{Sent}_3$, with $\phi = \neg p_2 \in$

Sent_1 and $\psi = \wedge p_4 \neg p_1 \in \text{Sent}_2$, since no other analysis is consistent with the definition. The precise statement is

Proposition 1.1.5 (*unique readability for propositional sentences*) For any L -sentence θ , exactly one of the following holds:

- (i) θ is atomic;
- (ii) for some L -sentence ϕ , θ is $\neg \phi$, the **negation** of ϕ ;
- (iii) for some L -sentences ϕ and ψ , θ is $\vee \phi \psi$, the **disjunction** of ϕ and ψ ;
- (iv) for some L -sentences ϕ and ψ , θ is $\wedge \phi \psi$, the **conjunction** of ϕ and ψ ;
- (v) for some L -sentences ϕ and ψ , θ is $\rightarrow \phi \psi$, the **implication** of ϕ and ψ ;
- (vi) for some L -sentences ϕ and ψ , θ is $\leftrightarrow \phi \psi$, the **bi-implication** of ϕ and ψ .

Furthermore, in (ii)–(vi), ϕ and ψ are uniquely determined.

Proof. There are three assertions to verify:

- (1) for any L -sentence θ , at least one of (i)–(vi) holds;
- (2) for any L -sentence θ , at most one of (i)–(vi) holds;
- (3) for any L -sentence θ , in (ii)–(vi), ϕ and ψ are unique.

Claim (2) is immediate, since the required first symbol of the expression is different in each case. (1) may be proved by sentence induction as follows. Let $\mathcal{P}$ be the property of an expression θ that at least one of conditions (i)–(vi) holds; it suffices to verify the two conditions of the proposition. Certainly the basis holds, since every L -atomic sentence satisfies condition (i). For the induction steps, assume as induction hypothesis that ϕ and ψ are L -sentences which have the property $\mathcal{P}$. Then $\neg \phi$ satisfies condition (ii), $\vee \phi \psi$ satisfies condition (iii) etc. Thus by sentence induction every L -sentence has the property $\mathcal{P}$ as desired.

To prove (3) it is useful to introduce a technical lemma:

- (4) For any finite sequences $\phi_0, \dots, \phi_k$ and $\psi_0, \dots, \psi_l$ of sentences, if the expressions $\phi_0 \dots \phi_k$ and $\psi_0 \dots \psi_l$ are identical, then $k = l$ and for all $i \leq k$, $\phi_i = \psi_i$.

The proof of (4) goes by induction on the length of $\phi_0 \dots \phi_k$ (the total number of symbols, generally larger than k); assume as induction hypothesis that (4) holds for all shorter instances. By (1) and (2), we may assume that ϕ_0 falls under exactly one of cases (i)–(vi). If ϕ_0 is atomic, then the first symbol of ψ_0 is a sentence symbol, so also ψ_0 is atomic and $\phi_0 = \psi_0$. Then $\phi_1 \dots \phi_k = \psi_1 \dots \psi_l$ and are shorter, so the induction hypothesis implies that $k = l$ and for $1 \leq i \leq k$, $\phi_i = \psi_i$.

If $\phi_0 = \neg\theta$, then the first symbol of ψ_0 is $\neg$, so ψ_0 is of the form $\neg\chi$ and the expressions $\theta\phi_1 \dots \phi_k$ and $\chi\psi_1 \dots \psi_l$ are identical and shorter, so again the desired conclusion follows from the induction hypothesis. Finally, if ϕ_0 is $\bullet\theta\theta'$, then ψ_0 is $\bullet\chi\chi'$ so we have $\theta\theta'\phi_1 \dots \phi_k = \chi\chi'\psi_1 \dots \psi_l$ and the desired conclusion again follows from the induction hypothesis. Note in these two cases that it is essential that the induction is on the *length* of the expression, not on the number of sentences concatenated to form it.

Now, what is claimed in (3) is that the only way that $\bullet\phi\psi$ and $\bullet\phi'\psi'$ can be identical is for ϕ and ϕ' to be identical and ψ and ψ' to be identical. This is immediate from the lemma with $k = l = 2$. $\square$

So far we have been concerned only with the **syntax** of L : properties of L -expressions which depend only on how the symbols are arranged. We begin next to explore the **semantics**: properties which involve an interpretation or assignment of meaning. For this simple language, an interpretation just specifies which of the atomic sentences are true and which are false.

Definition 1.1.6

- (i) An **atomic truth assignment** is any function $V_0 : \text{Sent}_0 \rightarrow \{\text{T}, \text{F}\}$;
- (ii) a **truth assignment** is any function $V : \text{Sent}_L \rightarrow \{\text{T}, \text{F}\}$ such that for all sentences ϕ and ψ ,

$$\begin{aligned}
 \text{(a)} \quad V(\neg \phi) &= \begin{cases} \text{T}, & \text{if } V(\phi) = \text{F}; \\ \text{F}, & \text{otherwise;} \end{cases} \\
 \text{(b)} \quad V(\vee \phi\psi) &= \begin{cases} \text{T}, & \text{if } V(\phi) = \text{T} \text{ or } V(\psi) = \text{T}; \\ \text{F}, & \text{otherwise;} \end{cases} \\
 \text{(c)} \quad V(\wedge \phi\psi) &= \begin{cases} \text{T}, & \text{if } V(\phi) = \text{T} \text{ and } V(\psi) = \text{T}; \\ \text{F}, & \text{otherwise;} \end{cases} \\
 \text{(d)} \quad V(\rightarrow \phi\psi) &= \begin{cases} \text{T}, & \text{if } V(\phi) = \text{F} \text{ or } V(\psi) = \text{T}; \\ \text{F}, & \text{otherwise;} \end{cases} \\
 \text{(e)} \quad V(\leftrightarrow \phi\psi) &= \begin{cases} \text{T}, & \text{if } V(\phi) = V(\psi); \\ \text{F}, & \text{otherwise.} \end{cases}
 \end{aligned}$$

We refer to clauses (a)–(e) by saying that V **respects connectives**. Note that this definition does not make it clear that any truth assignments exist; we prove first

Theorem 1.1.7 *Every atomic truth assignment has a unique extension to a truth assignment.*

Proof. Fix an atomic truth assignment V_0 . We shall define recursively a sequence of functions $V_n : \mathbf{Sent}_n \rightarrow \{\mathbf{T}, \mathbf{F}\}$ for $n \in \omega$ such that for each n , V_{n+1} is an extension of V_n and respects connectives. Indeed, given such a V_n , we simply use the clauses (a)–(e) to define the values of V_{n+1} on the members of $\mathbf{Sent}_{n+1} \sim \mathbf{Sent}_n$. Thus for $\phi, \psi \in \mathbf{Sent}_n$, set

$$\begin{aligned} V_{n+1}(\neg \phi) &= \begin{cases} \mathbf{T}, & \text{if } V_n(\phi) = \mathbf{F}; \\ \mathbf{F}, & \text{otherwise;} \end{cases} \\ V_{n+1}(\phi \wedge \psi) &= \begin{cases} \mathbf{T}, & \text{if } V_n(\phi) = \mathbf{T} \text{ and } V_n(\psi) = \mathbf{T}; \\ \mathbf{F}, & \text{otherwise;} \end{cases} \end{aligned}$$

and similarly for the other connectives. We rely here on unique readability to ensure that exactly one of these conditions applies to each member of $\mathbf{Sent}_{n+1} \sim \mathbf{Sent}_n$. Now we may define $V : \mathbf{Sent}_L \rightarrow \{\mathbf{T}, \mathbf{F}\}$ by: $V(\theta) = V_n(\theta)$ for some (any) n such that $\theta \in \mathbf{Sent}_n$. Clearly V extends V_0 and respects connectives because all of the V_n do.

Finally, to prove the uniqueness of the extension, suppose that V and W are two truth assignments which extend a single atomic truth assignment V_0 . We prove by sentence induction that all L -sentences ϕ have the property: $V(\phi) = W(\phi)$. Every L -atomic sentence has this property since for atomic ϕ , $V(\phi) = V_0(\phi) = W(\phi)$. Assume as induction hypothesis that ϕ and ψ are L -sentences such that $V(\phi) = W(\phi)$ and $V(\psi) = W(\psi)$. Then, for example,

$$\begin{aligned} V(\phi \vee \psi) = \mathbf{T} &\iff V(\phi) = \mathbf{T} \text{ or } V(\psi) = \mathbf{T} \\ &\iff W(\phi) = \mathbf{T} \text{ or } W(\psi) = \mathbf{T} \\ &\iff W(\phi \vee \psi) = \mathbf{T}, \end{aligned}$$

and hence also

$$\begin{aligned} V(\phi \vee \psi) = \mathbf{F} &\iff V(\phi \vee \psi) \neq \mathbf{T} \\ &\iff W(\phi \vee \psi) \neq \mathbf{T} \iff W(\phi \vee \psi) = \mathbf{F}. \end{aligned}$$

Since these are the only two values, in all cases $V(\phi \vee \psi) = W(\phi \vee \psi)$. The proofs for the other connectives are analogous. $\square$

The definition of truth assignment is *effective* in the sense that there is an algorithm for computing the value of $V(\theta)$ from the values $V(p_n)$ for all of the sentence symbols p_n which occur in θ . For example, if θ is $\neg p_2 \wedge p_4 \neg p_1$ and $V(p_1) = \mathbf{T}$, $V(p_2) = \mathbf{F}$, and $V(p_4) = \mathbf{T}$, we may apply successively the clauses of the definition to compute $V(\neg p_2) = \mathbf{T}$, $V(\neg p_1) = \mathbf{F}$, $V(\neg p_2 \wedge p_4 \neg p_1) = \mathbf{F}$, and finally $V(\theta) = \mathbf{T}$. Of course, there are only finitely many (8, in this case) distinct ways to assign $\mathbf{T}$ or $\mathbf{F}$ to

the finitely many sentence symbols which occur in θ ; the standard way of organizing the computation of $V(\theta)$ for all of them is called a **truth table**:

p_1	p_2	p_4	$\neg p_2$	$\neg p_1$	$\wedge p_4 \neg p_1$	$\vee \neg p_2 \wedge p_4 \neg p_1$
T	T	T	F	F	F	F
T	T	F	F	F	F	F
T	F	T	T	F	F	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	T	F	F	T	F	F
F	F	T	T	T	T	T
F	F	F	T	T	F	T

Remark 1.1.8 The official form we have adopted for L -sentences with connectives preceding their arguments (e.g., $\wedge \phi\psi$) is called **prefix** or **Polish notation**. It is convenient for proving unique readability and other technical matters, but it is clearly not well designed for *human* readability. Beginning with [Section 1.3](#) we shall write sentences with the more intuitive **infix notation**; that is we shall write the example in the truth table above $(\neg p_2) \vee (p_4 \wedge \neg p_1)$. Note that we need to introduce parentheses to make clear the structure of the sentence by specifying explicitly the **scope** of some of the connectives: the scope of the first $\neg$ is limited to p_2 while the scope of the $\vee$ is extended to all of $(p_4 \wedge \neg p_1)$. Formally, we regard such expressions as abbreviations (even if they are longer!) of the official prefix expressions; in particular, parentheses are *not* symbols of the language. For the record, all of the advantages and disadvantages of prefix notation are shared by **postfix notation**, in which a conjunction would be written $\phi\psi \wedge$.

Choosing to set things up this way is a matter of taste; it is perfectly possible to make the infix notation official and include parentheses among the symbols.

Theorem 1.1.7 is an instance of the following general

Sentence Recursion Theorem 1.1.9 *For any set Z , any function $F_0 : \text{Sent}_0 \rightarrow Z$ and any functions $G_{\neg} : Z \rightarrow Z$ and $G_{\bullet} : Z \times Z \rightarrow Z$, for $\bullet$ any of $\vee, \wedge, \rightarrow, \leftrightarrow$, there exists a unique function $F : \text{Sent}_L \rightarrow Z$ such that F extends F_0 and for all L -sentences ϕ and ψ ,*

$$F(\neg \phi) = G_{\neg}(F(\phi)) \quad \text{and} \quad F(\bullet \phi \psi) = G_{\bullet}(F(\phi), F(\psi)). \quad \square$$

This and other similar theorems are proved in the following section.

Collateral Reading. Many mathematical logic texts do not treat propositional logic as a separate topic but only as a part of first-order logic — for example, [Eb-Fl-Th] and [Sh01b]. Here we have chosen to spend considerable space developing in the simpler context of propositional logic notions and techniques which are largely common to the two areas. Other texts which do this are [Hod], [Men] and particularly [En01].

Exercises

Exercise 1.1.10 Give a precise definition of the set of sentences using infix notation and prove unique readability for your definition.

Exercise 1.1.11 For any expression $\phi = s_0 \dots s_k$, a *proper initial segment* of ϕ is any sequence of symbols $s_0 \dots s_l$ with $l < k$. Prove that no proper initial segment of a sentence is a sentence, and show that this can be used as an alternative to (4) as a technical lemma for the proof of Proposition 1.1.5.

Exercise 1.1.12 Give a careful proof of Proposition 1.1.9, and show how Theorem 1.1.7 is an application of it.

Exercise 1.1.13 For each of the following expressions, decide whether or not it is a sentence, and if it is, write it in infix notation with parentheses:

- (i) $\neg \vee \neg p_1 p_2$;
- (ii) $\wedge p_1 p_2 \neg p_3$;
- (iii) $\rightarrow \wedge p_1 p_2 \vee \neg p_3 \leftrightarrow p_8 p_3$.

1.2 Induction and recursion

Constructions analogous to those of Sent_L and V above will come up so often that it is worthwhile developing some abstract machinery for facilitating their introduction and use, and to do so we make a somewhat lengthy detour away from propositional logic. The reader may want to scan this section quickly at first and return to it for reference as the need arises.

Induction

Mathematical induction is a standard method for proving assertions of the form: every natural number has the property $\mathcal{P}$ — in symbols, $(\forall n \in \omega) \mathcal{P}(n)$. The proof typically consists of the *basis*, the verification that $\mathcal{P}(0)$ holds, and the induction step: from the *induction hypothesis* $\mathcal{P}(n)$, for any fixed n , derive $\mathcal{P}(n+1)$. We may paraphrase this argument as follows. Let $Y_{\mathcal{P}} = \{n \in \omega : \mathcal{P}(n)\}$. The basis consists in showing that $0 \in Y_{\mathcal{P}}$ and

the induction step consists in showing that $Y_{\mathcal{P}}$ is **closed under the successor function** — that is, that $n \in Y_{\mathcal{P}} \implies n + 1 \in Y_{\mathcal{P}}$. The conclusion is that $\omega \subseteq Y_{\mathcal{P}}$, or equivalently that $\omega = Y_{\mathcal{P}}$. The validity of this argument derives simply from the fact that ω is the *smallest* set Y with these two properties: $0 \in Y$ and Y is closed under the successor function — that is, ω is a subset of any such Y .

The key idea here is that the set Sent_L and many other sets of interest below have a similar character; each is (by definition) the smallest set containing certain given objects (“generalized zeroes”) and closed under certain functions (“generalized successor functions”). For each such set there is a corresponding principle of proof by induction which provides an efficient means for establishing that all members of the set enjoy a property $\mathcal{P}$. We formulate this abstractly as follows.

Definition 1.2.1 An **induction system** is a triple $\mathcal{X} = (X, X_0, \mathcal{H})$ such that X is a non-empty set, $X_0 \subseteq X$ and $\mathcal{H}$ is a set of **finitary functions** on X — that is, for each $H \in \mathcal{H}$, $H : X^{k_H} \rightarrow X$, for some $k_H \in \omega$. With an induction system we associate the sequence of sets defined recursively by

$$X_{n+1} := X_n \cup \{ H(x_0, \dots, x_{k_H-1}) : H \in \mathcal{H} \text{ and } x_0, \dots, x_{k_H-1} \in X_n \},$$

and the set $\overline{X} := \bigcup_{n \in \omega} X_n$. $\overline{X}$ is called the **inductive closure** of X_0 under $\mathcal{H}$, and the whole process is called an **inductive definition** of the set $\overline{X}$. We call $x = H(x_0, \dots, x_{k_H-1})$ a **successor** of $x_0, \dots, x_{k_H-1}$ and each x_i a **predecessor** of x . When the additional precision is necessary, we write $\mathcal{X}_n$ in place of X_n and $\overline{\mathcal{X}}$ in place of $\overline{X}$.

Let us first check that our two main examples fit into this framework. For the set ω of natural numbers, we may use the induction system $\mathcal{X}_{\omega} = (\mathbb{R}, \{0\}, \{\text{Sc}\})$, where $\mathbb{R}$ is the set of real numbers and Sc is the function such that for any real number r , $\text{Sc}(r) = r + 1$. Then $X_n = \{0, 1, \dots, n\}$ and $\overline{X} = \omega$. Note that the choice of $X = \mathbb{R}$ is mainly arbitrary; we could as well have chosen for X the sets of rational numbers, complex numbers, countable ordinal numbers or any superset of ω endowed with a successor function which extends the successor on ω .

For the corresponding characterization of Sent_L , we may take the induction system

$$\mathcal{X}_{\text{Sent}} := (\text{Expr}_L, \text{Sent}_0, \mathcal{H}_{\text{Sent}}) \quad \text{and} \quad \mathcal{H}_{\text{Sent}} := \{ H_{\neg}, H_{\vee}, H_{\wedge}, H_{\rightarrow}, H_{\leftrightarrow} \},$$

where for any expressions ϕ and ψ , $H_{\neg}(\phi) = \neg \phi$, $H_{\wedge}(\phi, \psi) = \phi \wedge \psi$ etc. Then $X_n = \text{Sent}_n$ and $\overline{X} = \text{Sent}_L$.

Definition 1.2.2 For any induction system $\mathcal{X} = (X, X_0, \mathcal{H})$ and any $Y \subseteq X$, Y is **$\mathcal{X}$ -closed** iff $X_0 \subseteq Y$ and for all $H \in \mathcal{H}$ and all $x_0, \dots, x_{k_H-1} \in Y$, also $H(x_0, \dots, x_{k_H-1}) \in Y$.

Proposition 1.2.3 For any induction system $\mathcal{X} = (X, X_0, \mathcal{H})$, $\overline{X}$ is the **smallest $\mathcal{X}$ -closed set** — that is,

- (i) $\overline{X}$ is $\mathcal{X}$ -closed;
- (ii) for any $\mathcal{X}$ -closed set Y , $\overline{X} \subseteq Y$.

Proof. For (i), clearly $X_0 \subseteq \overline{X}$. If $H \in \mathcal{H}$ and all $x_0, \dots, x_{k_H-1} \in \overline{X}$, then for each $i < k_H$, there is an n_i such that $x_i \in X_{n_i}$, so if n is the maximum of the n_i , then all $x_0, \dots, x_{k_H-1} \in X_n$, so $H(x_0, \dots, x_{k_H-1}) \in X_{n+1} \subseteq \overline{X}$. For (ii), suppose that Y is $\mathcal{X}$ -closed. Then $X_0 \subseteq Y$ and a simple proof by induction establishes that for all n , $X_n \subseteq Y$ — given $X_n \subseteq Y$, each element of X_{n+1} is either in X_n , so in $\overline{X}$ or of the form $H(x_0, \dots, x_{k_H-1})$ for $x_0, \dots, x_{k_H-1} \in X_n$. By the induction hypothesis, also $x_0, \dots, x_{k_H-1} \in Y$, so by the assumption the Y is $\mathcal{X}$ -closed, also $H(x_0, \dots, x_{k_H-1}) \in Y$. Hence $X_{n+1} \subseteq Y$. It follows immediately that $\overline{X} \subseteq Y$. $\square$

As a restatement of the proposition, we have

Corollary 1.2.4 For any induction system $\mathcal{X} = (X, X_0, \mathcal{H})$,

$$\overline{X} = \bigcap \{Y \subseteq X : Y \text{ is } \mathcal{X}\text{-closed}\}. \quad \square$$

Theorem 1.2.5 (proof by generalized induction) For any induction system $\mathcal{X} = (X, X_0, \mathcal{H})$ and any property $\mathcal{P}$ which applies to elements of X , if

$$Y_{\mathcal{P}} = \{x \in X : x \text{ has the property } \mathcal{P}\}$$

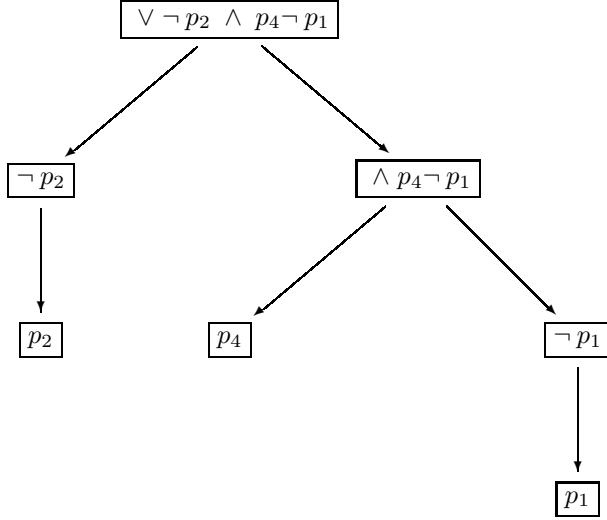
is $\mathcal{X}$ -closed, then every $x \in \overline{X}$ has the property $\mathcal{P}$.

Proof. Immediate; if $Y_{\mathcal{P}}$ is $\mathcal{X}$ -closed, then $\overline{X} \subseteq Y_{\mathcal{P}}$, which is just another way of expressing that every $x \in \overline{X}$ has the property $\mathcal{P}$. $\square$

In applying this principle we use the terminology inherited from the case $\mathcal{X}_\omega$. Given a property $\mathcal{P}$, we refer to checking that every member of X_0 has the property $\mathcal{P}$ (i.e. $X_0 \subseteq Y_{\mathcal{P}}$) as the **basis** of the induction, and describe as the **induction step** the verification that for each $H \in \mathcal{H}$ and $x_0, \dots, x_{k_H-1}$ from X , if we assume as **induction hypothesis** that $\mathcal{P}(x_0), \dots, \mathcal{P}(x_{k_H-1})$ hold, then we may conclude that also $\mathcal{P}(H(x_0, \dots, x_{k_H-1}))$ holds. The whole is called a proof by **$\mathcal{X}$ -induction** or (slightly ambiguously) proof by **$\overline{X}$ -induction** and is generally more efficient and elegant than a proof which refers explicitly to the sets X_n .

It is often useful to think of an inductively defined set $\overline{X}$ as being generated in time, with the elements of X_n appearing at moment n . If we focus on some particular $x \in \overline{X}$, we may restrict attention to the part of this process which is required to generate x . This may be represented either in the form

of a tree or a linear sequence, either of which naturally serves as a witness or proof that x really does belong to $\overline{X}$. For example, for the L -sentence $\vee \neg p_2 \wedge p_4 \neg p_1$ we have the tree



or the sequence $(p_2, \neg p_2, p_1, p_4, \neg p_1, \wedge p_4 \neg p_1, \vee \neg p_2 \wedge p_4 \neg p_1)$. Neither of these is completely unique since terms may be rearranged, but unique readability guarantees that the same terms must occur in any such rearrangement. The tree form is often best for visualization, but the linear form is easier to deal with formally.

Definition 1.2.6 For any induction system $\mathcal{X} = (X, X_0, \mathcal{H})$, an $\mathcal{X}$ -**derivation** is a non-empty finite sequence $(x_0, \dots, x_n)$ of elements of X such that for all $i \leq n$, either

- (i) $x_i \in X_0$ or
- (ii) for some $H \in \mathcal{H}$ and some $j_0, \dots, j_{k_H-1} < i$, $x_i = H(x_{j_0}, \dots, x_{j_{k_H-1}})$.

An $\mathcal{X}$ -derivation $(x_0, \dots, x_n)$ is called an $\mathcal{X}$ -**derivation of** its last term x_n .

Proposition 1.2.7 For any induction system $\mathcal{X} = (X, X_0, \mathcal{H})$ and any $z \in X$,

$$z \in \overline{X} \iff \text{there exists an } \mathcal{X}\text{-derivation of } z.$$

Proof. Left as an exercise. $\square$

We conclude this discussion with two useful general properties of inductive definitions.

Proposition 1.2.8 *For any induction system $\mathcal{X} = (X, X_0, \mathcal{H})$, if X_0 and $\mathcal{H}$ are countable, then so is the inductive closure $\overline{\mathcal{X}}$.*

Proof. We prove by induction that for all n , X_n is countable. Since $\overline{\mathcal{X}} = \bigcup_{n \in \omega} X_n$, it follows that $\overline{\mathcal{X}}$ as a countable union of countable sets is countable. X_0 is countable by hypothesis. Assume as induction hypothesis that X_n is countable. For each $H \in \mathcal{H}$ let

$$X_{n,H} = \{ H(x_0, \dots, x_{k_H-1}) : x_0, \dots, x_{k_H-1} \in X_n \},$$

so that $X_{n+1} = X_n \cup \bigcup_{H \in \mathcal{H}} X_{n,H}$. Since X_n is countable, so is $(X_n)^{k_H}$ — that is, there are countably many sequences $x_0, \dots, x_{k_H-1}$ with all $x_i \in X_n$. Hence each $X_{n,H}$ is countable. Finally, since $\mathcal{H}$ is countable by assumption, X_{n+1} is a countable union of countable sets so is countable. $\square$

Corollary 1.2.9 *Sent_L is denumerable.*

Proof. This is immediate from the preceding proposition, but also follows from the fact that Sent_L is a subset of Expr_L , which is denumerable. $\square$

Note that the hypothesis of the proposition does not restrict the size of X . For the purposes of the next proposition, we call $\mathcal{Y} = (X, Y_0, \mathcal{I})$ a **(finite) subsystem** of $\mathcal{X} = (X, X_0, \mathcal{H})$ iff Y_0 is a (finite) subset of X_0 and $\mathcal{I}$ is a (finite) subset of $\mathcal{H}$.

Lemma 1.2.10 *For any induction systems, $\mathcal{X}$ and $\mathcal{Y}$, if $\mathcal{Y}$ is a subsystem of $\mathcal{X}$, then $\overline{\mathcal{Y}} \subseteq \overline{\mathcal{X}}$.*

Proof. Under the hypothesis it is immediate that $\overline{\mathcal{X}}$ is $\mathcal{Y}$ -closed. $\square$

The next result may be summarized by saying that the closure of an induction system is the union of its finitely-generated parts or that the system has finite character.

Proposition 1.2.11 *For any induction system $\mathcal{X} = (X, X_0, \mathcal{H})$,*

$$\overline{\mathcal{X}} = \bigcup \{ \overline{\mathcal{Y}} : \mathcal{Y} \text{ is a finite subsystem of } \mathcal{X} \}.$$

Proof. The inclusion $\supseteq$ is immediate from the lemma; for the converse we need to show that the set on the right is $\mathcal{X}$ -closed. For any $x \in X_0$, x belongs to the closure of the finite subsystem $(X, \{x\}, \emptyset)$. Suppose that $H \in \mathcal{H}$ and $x_0, \dots, x_{k_H-1}$ belong to the right-hand side, say x_i is in the closure of the finite subsystem $(X, Y_i, \mathcal{I}_i)$. Then $H(x_0, \dots, x_{k_H-1})$ is in the closure of the finite subsystem

$$\left(X, \bigcup_{i < k_H} Y_i, \bigcup_{i < k_H} \mathcal{I}_i \cup \{H\} \right). \quad \square$$

Applied to the induction system $\mathcal{X}_{\text{Sent}}$, this means simply that a particular sentence ϕ belongs to the closure of any finite subsystem $\mathcal{Y}$ such that Y_0 contains the finitely many sentence symbols which occur in ϕ .

Recursion

We want next to analyze more closely the method used in the proof of Theorem 1.1.7 and its more general version Proposition 1.1.9. The underlying idea here is a general notion of *recursion* as a method for defining functions. A simple and familiar example of a recursive definition is that of the usual factorial function:

$$0! = 1; \quad \text{and for all } n \in \omega, \quad (n+1)! = n! \times (n+1).$$

The general format for such a definition is: given a set Z , an element $z_0 \in Z$ and a function $G : Z \times \omega \rightarrow Z$, we specify $F : \omega \rightarrow Z$ by

$$F(0) = z_0; \quad \text{and for all } n \in \omega, \quad F(n+1) = G(F(n), n).$$

Thus in the example, $Z = \omega$ and $G(m, n) = m \times (n+1)$. Note that Z may be any set; it is only the structure of the domain ω that matters.

The important point to notice is that although we treat this (correctly) as a definition, we need to explain away the apparent circularity of the second clause; F is on the face of it being defined in terms of itself. *A priori*, the condition expressed in the displayed line might be satisfied by no F , by exactly one F , or by many F ; for it to have the force of a definition, it must be satisfied by exactly one F . Of course it is clear that the two equations determine successively the values $F(0) = z_0$, $F(1) = G(z_0, 0)$, $F(2) = G(F(1), 1)$ and so on and thus indeed are satisfied by exactly one F . The precise statement of this is

Theorem 1.2.12 (definition by recursion) *For any set Z , any $z_0 \in Z$ and any function $G : Z \times \omega \rightarrow Z$, there exists a unique function $F : \omega \rightarrow Z$ such that $F(0) = z_0$ and for all $n \in \omega$, $F(n+1) = G(F(n), n)$.*

Proof. Given Z , z_0 and G as in the hypothesis, let $F(n) := z$ iff

$$\text{there exist } z_1, \dots, z_n = z \text{ such that for all } i < n, \quad z_{i+1} = G(z_i, i).$$

It is now easy to prove by induction that for all n the right side is satisfied by a unique z , that F is the desired function and that any other function satisfying the condition in fact coincides with F . $\square$

We have, in fact, already used this principle twice: in the definition of the sequence $n \mapsto \text{Sent}_n$ and the sequence $n \mapsto V_n$ in the proof of Theorem

1.1.7. For example, in the definition of Sent_n , the set Z is the set of all sets of L -expressions, and G is the function which describes Sent_{n+1} in terms of Sent_n .

The idea for the general version is suggested by the terminology introduced above. For an induction system $\mathcal{X} = (X, X_0, \mathcal{H})$ and a set Z , we would like to specify a unique function $F : \overline{X} \rightarrow Z$ by giving explicitly the values of $F(x)$ for $x \in X_0$ (a ‘generalized zero’) and providing ‘transition functions’ G_H such that for each $H \in \mathcal{H}$ and $x_0, \dots, x_{k_H-1} \in \overline{X}$,

$$F(H(x_0, \dots, x_{k_H-1})) = G_H(F(x_0), \dots, F(x_{k_H-1}), x_0, \dots, x_{k_H-1}).$$

In other words, G_H computes the value of F at a successor point $x = H(x_0, \dots, x_{k_H-1})$ from the values of F at the predecessors x_i of x . Here there is a potential problem: a given x might be a successor in more than one way — that is, we might have

$$H(x_0, \dots, x_{k_H-1}) = x = H'(x'_0, \dots, x'_{k_{H'}-1})$$

with either H distinct from H' or some x_i distinct from x'_i or both. Then if also

$$\begin{aligned} G_H(F(x_0), \dots, F(x_{k_H-1}), x_0, \dots, x_{k_H-1}) \\ \neq G_{H'}(F(x'_0), \dots, F(x'_{k_{H'}-1}), x'_0, \dots, x'_{k_{H'}-1}), \end{aligned}$$

it would be impossible to assign an appropriate value to $F(x)$. Of course, this doesn’t arise in the case of $\mathcal{X}_\omega$ since there is only one successor function and it is injective; we need to make a similar restriction in the general case.

Definition 1.2.13 The induction system $\mathcal{X} = (X, X_0, \mathcal{H})$ has **unique readability** iff the images (ranges) of $H \in \mathcal{H}$ are pairwise disjoint and disjoint from X_0 , and every $H \in \mathcal{H}$ is injective on $\overline{X}$.

As the name suggests, this property is just what we established in Theorem 1.1.5; part (2) of that proof says exactly that the images of H_- , $H_\vee$, $\dots$ are pairwise disjoint and disjoint from Sent_0 , and (3) asserts that on Sent_L these functions are injective. Thus $\mathcal{X}_{\text{Sent}}$ has unique readability; of course, $\mathcal{X}_\omega$ does also.

Theorem 1.2.14 (definition by generalized recursion) Let $\mathcal{X} = (X, X_0, \mathcal{H})$ be an induction system which has unique readability. For any set Z , any function $F_0 : X_0 \rightarrow Z$ and any family $\langle G_H : H \in \mathcal{H} \rangle$ of functions such that for all $H \in \mathcal{H}$, $G_H : Z^{k_H} \times X^{k_H} \rightarrow Z$, there exists a unique function $F : \overline{X} \rightarrow Z$ such that F extends F_0 and for each $H \in \mathcal{H}$ and $x_0, \dots, x_{k_H-1} \in \overline{X}$,

$$F(H(x_0, \dots, x_{k_H-1})) = G_H(F(x_0), \dots, F(x_{k_H-1}), x_0, \dots, x_{k_H-1}).$$

Proof. We proceed as in the proof of Theorem 1.1.7, which is a special case, and define by (ordinary) recursion a sequence of functions $F_n : X_n \rightarrow Z$. F_0 is given and we define F_{n+1} from F_n as follows. For each $x \in X_{n+1}$, either (i) $x \in X_n$, or (ii) for some $H \in \mathcal{H}$ and $x_0, \dots, x_{k_H-1} \in X_n$, $x = H(x_0, \dots, x_{k_H-1})$. In case (i), we set $F_{n+1}(x) = F_n(x)$ to ensure that F_{n+1} is an extension of F_n . In case (ii) unique readability ensures that H and $x_0, \dots, x_{k_H-1}$ are uniquely determined, and we set

$$F_{n+1}(x) = G_H(F_n(x_0), \dots, F_n(x_{k_H-1}), x_0, \dots, x_{k_H-1}).$$

Now define $F(x) = F_n(x)$ for some (any) n such that $x \in X_n$. It is easy to prove that F satisfies the required equations.

To establish that there is a unique such F , suppose that two functions F and F' both satisfy the equations. Let $\mathcal{P}(x)$ denote the property $F(x) = F'(x)$; we prove by $\mathcal{X}$ -induction that $\mathcal{P}(x)$ holds for all $x \in \overline{X}$ and hence that $F = F'$. For the basis of the induction consider $x \in X_0$; since both F and F' extend F_0 , we have $F(x) = F_0(x) = F'(x)$ so $\mathcal{P}(x)$ holds. For the induction step, fix $H \in \mathcal{H}$ and $x_0, \dots, x_{k_H-1}$ and assume as induction hypothesis that $\mathcal{P}(x_0), \dots, \mathcal{P}(x_{k_H-1})$ hold. Then by our assumption

$$\begin{aligned} F(H(x_0, \dots, x_{k_H-1})) &= G_H(F(x_0), \dots, F(x_{k_H-1}), x_0, \dots, x_{k_H-1}) \\ &= G_H(F'(x_0), \dots, F'(x_{k_H-1}), x_0, \dots, x_{k_H-1}) \\ &= F'(H(x_0, \dots, x_{k_H-1})), \end{aligned}$$

and we conclude that also $\mathcal{P}(H(x_0, \dots, x_{k_H-1}))$ as required. $\square$

For a particular $\mathcal{X}$, we call an application of this principle a **definition by $\mathcal{X}$ -recursion** or **definition by $\overline{X}$ -recursion**.

Examples 1.2.15

- (i) Using this principle, the proof of Theorem 1.1.7 is now simply the following: given any atomic truth assignment $V_0 : \text{Sent}_0 \rightarrow \{\text{T}, \text{F}\}$, its unique extension to a truth assignment is the unique extension $V : \text{Sent}_L \rightarrow \{\text{T}, \text{F}\}$ which respects connectives.
- (ii) The set of sentence symbols which occur in a sentence ϕ is given by the unique function $\text{At} : \text{Sent}_L \rightarrow \wp(\{p_n : n \in \omega\})$ which satisfies the conditions

- (a) $\text{At}(p_n) = \{p_n\}$;
- (b) $\text{At}(\neg \psi) = \text{At}(\psi)$;
- (c) $\text{At}(\bullet \psi \theta) = \text{At}(\psi) \cup \text{At}(\theta)$;

where $\bullet$ represents any one of the binary connectives.

- (iii) A **subsentence** of a sentence ϕ is a sentence χ which occurs as part of ϕ — that is, ϕ is $s_0 \dots s_{k-1} \chi t_0 \dots t_{l-1}$ for some symbols s_i and t_j . Another natural way of characterizing this notion is by the following recursive definition: $\text{SubSent} : \text{Sent}_L \rightarrow \wp(\text{Sent}_L)$ is the unique function such that
- (a) $\text{SubSent}(p_n) = \{p_n\}$;
 - (b) $\text{SubSent}(\neg \psi) = \text{SubSent}(\psi) \cup \{\neg \psi\}$;
 - (c) $\text{SubSent}(\bullet \psi \theta) = \text{SubSent}(\psi) \cup \text{SubSent}(\theta) \cup \{\bullet \psi \theta\}$.

The equivalence of these characterizations is left as an exercise.

- (iv) Examples (ii) and (iii) are instances of the following general example. Given $\mathcal{X} = (X, X_0, \mathcal{H})$ and $Y \subseteq \overline{X}$ with unique readability, there is a unique function $\text{Supp}_Y : \overline{X} \rightarrow \wp(Y)$ such that for $x \in X_0$,

$$\text{Supp}_Y(x) = \begin{cases} \{x\}, & \text{if } x \in Y; \\ \emptyset, & \text{otherwise;} \end{cases}$$

and for all $H \in \mathcal{H}$, $x_0, \dots, x_{k_H-1} \in \overline{X}$ and $x = H(x_0, \dots, x_{k_H-1})$,

$$\text{Supp}_Y(x) = \text{Supp}_Y(x_0) \cup \dots \cup \text{Supp}_Y(x_{k_H-1}) \cup \begin{cases} \{x\}, & \text{if } x \in Y; \\ \emptyset, & \text{otherwise.} \end{cases}$$

$\text{Supp}_Y(x)$ is called the *Y-support* of x and is intuitively the set of elements of Y which are “used” in the generation of x .

In applying the principle of definition by recursion we often need a form which allows the introduction of **parameters**. For example, the ordinary function $+$ on ω is the unique function such that for all (values of a parameter) n ,

$$+(0, n) = n; \quad \text{and for all } m \in \omega, \quad +(\text{Sc}(m), n) = \text{Sc}(+(m, n)),$$

where Sc is the successor function: $\text{Sc}(m) = m + 1$. We give the general form for a single parameter; the version for several parameters is an obvious modification. Note that like the image set Z , the parameter set U need not be related to the induction system in any special way.

Theorem 1.2.16 *Let $\mathcal{X} = (X, X_0, \mathcal{H})$ be an induction system which has unique readability. For any sets Z and U , any function $F_0 : X_0 \times U \rightarrow Z$ and any family $\langle G_H : H \in \mathcal{H} \rangle$ of functions such that for all $H \in \mathcal{H}$, $G_H : Z^{k_H} \times X^{k_H} \times U \rightarrow Z$, there exists a unique function $F : \overline{X} \times U \rightarrow Z$ such that F extends F_0 and for each $H \in \mathcal{H}$, $u \in U$ and $x_0, \dots, x_{k_H-1} \in \overline{X}$,*

$$F(H(x_0, \dots, x_{k_H-1}), u) = G_H(F(x_0, u), \dots, F(x_{k_H-1}, u), x_0, \dots, x_{k_H-1}, u).$$

Proof. One proof is a simple modification of the proof of the preceding theorem; we present here a different proof which suggests a still more general version of the theorem (see the following remark). Given the data of the theorem, let (as usual)

$${}^U Z = \{ f : \text{Dom}(f) = U \text{ and } \text{Im}(f) \subseteq Z \},$$

and define $F_0^* : X_0 \rightarrow {}^U Z$ by

$$F_0^*(x)(u) = F_0(x, u)$$

and $G_H^* : ({}^U Z)^{k_H} \times X^{k_H} \rightarrow {}^U Z$ by

$$\begin{aligned} G_H^*(f_0, \dots, f_{k_H-1}, x_0, \dots, x_{k_H-1})(u) \\ = G_H(f_0(u), \dots, f_{k_H-1}(u), x_0, \dots, x_{k_H-1}, u). \end{aligned}$$

Thus by Theorem 1.2.14 there exists a unique function $F^* : \overline{X} \rightarrow {}^U Z$ which extends F_0^* and such that for each $H \in \mathcal{H}$ and $x_0, \dots, x_{k_H-1} \in \overline{X}$,

$$F^*(H(x_0, \dots, x_{k_H-1})) = G_H^*(F^*(x_0), \dots, F^*(x_{k_H-1}), x_0, \dots, x_{k_H-1}),$$

and hence for each $u \in U$,

$$\begin{aligned} F^*(H(x_0, \dots, x_{k_H-1}))(u) \\ = G_H^*(F^*(x_0), \dots, F^*(x_{k_H-1}), x_0, \dots, x_{k_H-1})(u) \\ = G_H(F^*(x_0)(u), \dots, F^*(x_{k_H-1})(u), x_0, \dots, x_{k_H-1}, u). \end{aligned}$$

Thus the desired function F is obtained by setting $F(x, u) = F^*(x)(u)$. $\square$

Remark 1.2.17 The technique of the preceding proof may also be used in situations where it is desired to specify the value of $F(H(x_0, \dots, x_{k_H-1}), u)$ in terms of values $F(x_i, v)$ for different values v of the parameter. In such a case we will specify the functions G_H^* directly; in particular, the value of $G_H^*(f_0, \dots, f_{k_H-1}, x_0, \dots, x_{k_H-1})(u)$ may depend on various values $f_i(v)$ rather than only on $f_i(u)$. Then taking F^* and F as above will give the desired result. As a simple, although artificial example, consider the equations

$$F(0, n) = n + 1; \quad F(\text{Sc}(m), n) = F(m, F(m, n)).$$

Intuitively it is clear that these determine a unique function and indeed the function $F(m, n) = n + 2^m$. This may be established by this method by taking

$$G^*(f, m)(n) = f(f(n)),$$

F^* to be the unique function $\omega \rightarrow {}^\omega\omega$ such that

$$F^*(0)(n) = n + 1 \text{ and for all } m \in \omega, F^*(\text{Sc}(m)) = G^*(F^*(m), m),$$

and setting $F(m, n) = F^*(m)(n)$. A more interesting example is given in the exercises.

Collateral Reading. Relatively few logic texts treat induction and recursion in as much detail as is done here, and some which do use the term “inductive” for both sorts of definition. The closest version in the literature is [En01]. On the other hand, as will become clear throughout this book, these topics are central to almost all parts of logic and have been thoroughly studied to the point that a whole book [Mo74] is devoted to them. A (slightly) more accessible introduction to the general theory is Peter Aczel’s article [Ba77; Section C7]. We also return to the general development of inductive definability in [Section 6.2](#).

Exercises

Exercise 1.2.18 An alternative way to develop the theory of induction systems is to take

$$\overline{X} := \bigcap \{ Y \subseteq X : Y \text{ is } \mathcal{X}\text{-closed} \}$$

as the definition of $\overline{X}$. This gives by definition that $\overline{X} \subseteq Y$ for any $\mathcal{X}$ -closed set Y , but still requires a proof that $\overline{X}$ itself is $\mathcal{X}$ -closed. Give this proof.

Exercise 1.2.19 Prove Proposition 1.2.16 by the first method suggested there.

Exercise 1.2.20 Give a proof of the principle of definition by generalized recursion which does not depend on the special case of ordinary recursion over ω .

Hint: Regard F_0 as a set of ordered pairs, a subset of $\overline{X} \times Z$. Describe an induction system $\mathcal{Y} = (\overline{X} \times Z, F_0, \mathcal{I})$ whose inductive closure is the desired F . Note that part of the proof that F is the desired function is to show that it is a function at all — that is, that for all $x \in \overline{X}$, there is exactly one $z \in Z$ such that $(x, z) \in F$.

Exercise 1.2.21 Prove the characterization of $\overline{X}$ in terms of $\mathcal{X}$ -derivations (Proposition 1.2.7).

Exercise 1.2.22 The notion of an induction system is a special case of a much more general notion which we will study further in [Section 6.2](#). A pair $(X, \mathcal{I})$ such that $\mathcal{I} : \wp(X) \rightarrow \wp(X)$ and for $Y \subseteq Z \subseteq X$, $\mathcal{I}(Y) \subseteq \mathcal{I}(Z)$ is called a **monotone induction system**. Note that any induction system $(X, X_0, \mathcal{H})$ is also a monotone induction system with

$$\mathcal{I}(Y) = X_0 \cup \{H(x_0, \dots, x_{k_H-1}) : H \in \mathcal{H} \text{ and } x_0, \dots, x_{k_H-1} \in Y\}.$$

Then Y is $\mathcal{X}$ -closed iff Y is $\mathcal{I}$ -closed — that is, $\mathcal{I}(Y) \subseteq Y$. Let $(X, \mathcal{I})$ be any monotone induction system.

- (i) Show that there exists a unique smallest $\mathcal{I}$ -closed set $\overline{X}$.
- (ii) Let $X_0 = \mathcal{I}(\emptyset)$ and for all n , $X_{n+1} = \mathcal{I}(X_n)$. Prove that for all $n \in \omega$, $X_n \subseteq X_{n+1} \subseteq \overline{X}$.
- (iii) Give an example to show that $\bigcup_{n \in \omega} X_n$ may *not* be $\mathcal{I}$ -closed.
- (iv) Find a further hypothesis on $(X, \mathcal{I})$ which implies that $\bigcup_{n \in \omega} X_n$ is $\mathcal{I}$ -closed and hence equal to $\overline{X}$ (other than the trivial one $\mathcal{I}(\emptyset) = \emptyset$!).

Exercise 1.2.23 Apply the method of Remark 1.2.17 to prove that there exists a unique function $A : \omega \times \omega \rightarrow \omega$ such that for all p and n , $A(0, n) = n + 1$,

$$A(p + 1, 0) = A(p, 1) \quad \text{and} \quad A(p + 1, n + 1) = A(p, A(p + 1, n)).$$

A is called the **Ackermann function** and will play a role in [Section 4.4](#). In understanding A it is helpful to write out formulas for $A^*(0), \dots, A^*(3)$.

Exercise 1.2.24 Prove the equivalence of the two characterizations of the notion of *subsentence* given in Example 1.2.15(iii).

Hint: Extend the result of Exercise 1.1.11 to show that no proper initial segment of a sentence is a terminal segment of a sentence.

Exercise 1.2.25 Since an L -sentence ϕ is a finite sequence of L -symbols, it has a well-defined **length** $\text{lg}(\phi)$, the number of symbols occurring in ϕ . Give an alternative definition of this function by Sent_L -recursion, and prove the equivalence of the two definitions.

1.3 Propositional semantics

We return now to propositional logic. From now on we write $\phi \bullet \psi$ instead of $\bullet \phi \psi$ using parentheses where necessary. It will be useful to introduce a notation for iterated conjunction and disjunction. We write $\phi_0 \wedge \cdots \wedge \phi_n$ to denote simply ϕ_0 if $n = 0$ and otherwise to denote $(\phi_0 \wedge \cdots \wedge \phi_{n-1}) \wedge \phi_n$. We may regard this as a recursive specification of an n -fold conjunction operator or (more simply) in terms of the prefix notation, $\phi_0 \wedge \cdots \wedge \phi_n$ is $\wedge \cdots \wedge \phi_0 \dots \phi_n$, where the initial string of $\wedge$ has n entries. $\phi_0 \vee \cdots \vee \phi_n$ is defined similarly. It follows easily that for any truth assignment V , $V(\phi_0 \wedge \cdots \wedge \phi_n) = \mathbf{T}$ iff $V(\phi_i) = \mathbf{T}$ for *all* $i \leq n$, and $V(\phi_0 \vee \cdots \vee \phi_n) = \mathbf{T}$ iff $V(\phi_i) = \mathbf{T}$ for *some* $i \leq n$. From now on in this chapter, the letters $\phi, \psi, \dots$ always denote L -sentences.

A first result we shall need is the precise expression of the “obvious” fact that a value $V(\phi)$ depends only on the values $V(p_n)$ for those p_n which actually occur in ϕ , that is, $p_n \in \text{At}(\phi)$ in the terminology of Example 1.2.15(ii). Let us temporarily write $V =_\phi W$ to mean that $V(p_n) = W(p_n)$ for all p_n which occur in ϕ .

Proposition 1.3.1 *For all sentences ϕ and all truth assignments V and W , if $V =_\phi W$, then $V(\phi) = W(\phi)$.*

Proof. We proceed by Sent_L -induction. If ϕ is atomic, the claim is obvious. Suppose as induction hypothesis that it is true for ϕ and that $V =_{\neg \phi} W$. Then clearly also $V =_\phi W$ (since $\text{At}(\neg \phi) = \text{At}(\phi)$), so by the induction hypothesis, $V(\phi) = W(\phi)$, and thus also $V(\neg \phi) = W(\neg \phi)$, since each of these is the opposite truth value. The cases corresponding to the binary connectives are similar. $\square$

Formally, this proposition is the basis for the construction of truth tables; there is no need to take account of the values of a truth assignment on atomic sentences which do not occur in the sentence. We next introduce an alternative notation for asserting the truth value of a sentence which betrays our preference for truth over falsity.

Definition 1.3.2 For any sentences ϕ and ψ ,

- (i) ϕ is a **tautology** — in symbols, $\models \phi$ — iff for all truth assignments V , $V(\phi) = \mathbf{T}$; ϕ is **tautologically false** iff $\models \neg \phi$;
- (ii) ϕ and ψ are **tautologically equivalent** — in symbols, $\phi \models \psi$ — iff for all truth assignments V , $V(\phi) = V(\psi)$;
- (iii) ψ is a **tautological consequence** of ϕ — in symbols $\phi \models \psi$ — iff for all truth assignments V , if $V(\phi) = \mathbf{T}$, then $V(\psi) = \mathbf{T}$.

Examples 1.3.3

- (i) $p_0 \vee \neg p_0$, $p_1 \leftrightarrow p_1$ and $p_1 \wedge p_2 \rightarrow p_2$ are tautologies.
- (ii) $p_0 \wedge p_1$ is tautologically equivalent to $p_1 \wedge p_0$.
- (iii) $p_0 \wedge (p_1 \vee p_2)$ is taut. equivalent to $(p_0 \wedge p_1) \vee (p_0 \wedge p_2)$.
- (iv) p_2 is a tautological consequence of $p_1 \wedge p_2$ and of $p_1 \wedge (p_1 \rightarrow p_2)$.

These notions are closely related:

Proposition 1.3.4 For any ϕ and ψ ,

- (i) $\phi \models \psi$ iff $\models \phi \leftrightarrow \psi$;
- (ii) $\phi \models \psi$ iff $\models \phi \rightarrow \psi$.

Proof. Immediate from the definitions. $\square$

Tautologies and tautological equivalences and consequences correspond to valid patterns of reasoning or inference. For example, that from a statement of the form $__$ and $\dots$ we may validly conclude $__$ is reflected by the tautology $\models \phi \wedge \psi \rightarrow \phi$ or equivalently by the tautological consequence $\phi \wedge \psi \models \phi$. To take a slightly less trivial case, the principle of proof by cases corresponds to the fact that for any L -sentences ϕ , ψ and χ ,

$$(\phi \rightarrow \chi) \wedge (\psi \rightarrow \chi) \models (\phi \vee \psi) \rightarrow \chi.$$

Of course, not every valid pattern of inference is represented by a tautology. For example, from a statement $\neg x$, involving a variable x , and the statement $x = y$, we may validly conclude $\neg y$ but not tautologically; this is a first-order inference which will be considered further in [Chapter 2](#).

We list on the following page some of the more important instances of tautologies and tautological consequences. The names we have given them reflect either historical usage or their roles in informal mathematical proofs. Each may be checked by a straightforward application of the definitions. For example, we verify proof by cases. Suppose that V is a truth assignment such that $V(\phi \rightarrow \chi) = \mathbf{T}$ and $V(\psi \rightarrow \chi) = \mathbf{T}$. If $V(\chi) = \mathbf{T}$, then also $V((\phi \vee \psi) \rightarrow \chi) = \mathbf{T}$. Otherwise, both $V(\phi) = \mathbf{F}$ and $V(\psi) = \mathbf{F}$, so $V(\phi \vee \psi) = \mathbf{F}$ and again $V((\phi \vee \psi) \rightarrow \chi) = \mathbf{T}$.

It will be convenient to have available a canonical tautology, say $p_0 \vee \neg p_0$, which we denote by $\mathbf{T}$ and call *truth* and a tautological falsehood, say $p_0 \wedge \neg p_0$, which we denote by $\mathbf{F}$ and call *falsity*. For every truth assignment V , $V(\mathbf{T}) = \mathbf{T}$ and $V(\mathbf{F}) = \mathbf{F}$.

The main results of this section concern the expressive power of various combinations of connectives. For example, the fact that

$$\phi \wedge \psi \models \neg((\neg \phi) \vee (\neg \psi))$$

Tautologies, tautological consequences, and tautological equivalences

$\models \phi \vee \neg \phi$	<i>excluded middle</i>
$\models \neg(\phi \wedge \neg \phi)$	<i>contradiction</i>
$\phi \models \neg \neg \phi$	<i>double negation</i>
$(\phi \rightarrow \psi) \wedge (\psi \rightarrow \chi) \models \phi \rightarrow \chi$	<i>transitivity of implication</i>
$\phi \wedge (\phi \rightarrow \psi) \models \psi$	<i>modus ponens</i>
$\neg \psi \wedge (\phi \rightarrow \psi) \models \neg \phi$	<i>modus tollendo tollens</i>
$(\phi \rightarrow \psi) \wedge (\phi \rightarrow \chi) \models \phi \rightarrow (\psi \wedge \chi)$	<i>proof by parts</i>
$(\phi \rightarrow \chi) \wedge (\psi \rightarrow \chi) \models (\phi \vee \psi) \rightarrow \chi$	<i>proof by cases</i>
$\phi \rightarrow (\psi \rightarrow \chi) \models (\phi \wedge \psi) \rightarrow \chi$	<i>exportation</i>
$\phi \rightarrow \psi \models (\neg \phi) \vee \psi \models \neg(\phi \wedge (\neg \psi))$	<i>conditional equivalents</i>
$\phi \wedge \psi \models \neg(\phi \rightarrow (\neg \psi))$	
$\phi \vee \psi \models (\neg \phi) \rightarrow \psi$	
$\phi \vee \psi \models \neg((\neg \phi) \wedge (\neg \psi))$	<i>De Morgan's laws</i>
$\phi \wedge \psi \models \neg((\neg \phi) \vee (\neg \psi))$	
$\phi \vee \psi, \neg \phi \models \psi$	<i>disjunctive syllogism</i>
$\psi \rightarrow \chi \models \phi \wedge \psi \rightarrow \chi$	<i>strengthening the antecedent</i>
$\phi \rightarrow \psi \models \phi \rightarrow (\psi \vee \chi)$	<i>weakening the consequent</i>
$\phi \wedge \psi \models \phi$	<i>simplification</i>
$\phi \models \phi \vee \psi$	<i>expansion</i>
$\phi \vee \phi \models \phi \models \phi \wedge \phi$	<i>idempotency</i>
$\phi \wedge \psi \models \psi \wedge \phi \quad \phi \vee \psi \models \psi \vee \phi$	<i>commutative law</i>
$\phi \vee (\phi \wedge \psi) \models \phi \quad \phi \wedge (\phi \vee \psi) \models \phi$	<i>absorption</i>
$\phi \models \phi \wedge \mathbf{T} \models \phi \vee \mathbf{F}$	
$\phi \vee (\psi \vee \chi) \models (\phi \vee \psi) \vee \chi$	<i>associative law</i>
$\phi \wedge (\psi \wedge \chi) \models (\phi \wedge \psi) \wedge \chi$	
$\phi \wedge (\psi \vee \chi) \models (\phi \wedge \psi) \vee (\phi \wedge \chi)$	<i>distributive law</i>
$\phi \vee (\psi \wedge \chi) \models (\phi \vee \psi) \wedge (\phi \vee \chi)$	
$\neg \phi \rightarrow \mathbf{F} \models \phi \models \mathbf{T} \rightarrow \phi$	<i>proof by contradiction</i>
$\neg \phi \models \phi \rightarrow \mathbf{F} \models \mathbf{T} \rightarrow \neg \phi$	
$\phi \rightarrow \psi \models (\neg \psi) \rightarrow (\neg \phi)$	<i>contraposition</i>
$\neg \phi \models \phi \rightarrow \psi$	<i>denial of the antecedent</i>
$\psi \models \phi \rightarrow \psi$	<i>affirmation of the consequent</i>
$\phi \vee \psi, \neg \phi \vee \chi \models \psi \vee \chi$	<i>cut</i>

suggests that from a formal perspective the connective $\wedge$ is superfluous; it can always be replaced by a combination of $\neg$ and $\vee$. The next proposition makes this precise.

Definition 1.3.5 $\text{Sent}_{\neg, \vee}$ is the smallest set of expressions which includes the atomic sentences and is closed under $\neg$ and $\vee$; similarly for other lists of connectives.

In terms of the abstract framework of [Section 1.2](#), $\text{Sent}_{\neg, \vee}$ is the inductive closure of the set of atomic sentences under the set $\{H_{\neg}, H_{\vee}\}$. Clearly $\text{Sent}_{\neg, \vee} \subseteq \text{Sent}_L$, and the inclusion is proper. However, $\text{Sent}_{\neg, \vee}$ includes all sentences ‘up to tautological equivalence’ in the following sense.

Proposition 1.3.6 *For every sentence ϕ , there is a sentence $\phi^* \in \text{Sent}_{\neg, \vee}$ such that $\phi \models \phi^*$.*

Proof. We define by Sent_L -recursion a function $^* : \text{Sent}_L \rightarrow \text{Sent}_{\neg, \vee}$ and then prove by Sent_L -induction that for all ϕ , $\phi \models \phi^*$. For atomic ϕ , $\phi^* = \phi$ and we require

$$\begin{aligned} (\neg \phi)^* &= \neg \phi^*; \\ (\phi \vee \psi)^* &= \phi^* \vee \psi^*; \\ (\phi \wedge \psi)^* &= \neg(\neg \phi^* \vee \neg \psi^*); \\ (\phi \rightarrow \psi)^* &= \neg \phi^* \vee \psi^*; \\ (\phi \leftrightarrow \psi)^* &= \neg(\neg \phi^* \vee \neg \psi^*) \vee \neg(\phi^* \vee \psi^*). \end{aligned}$$

The basis of the induction is trivial. The induction step for $\wedge$, for example, requires verifying that

$$\text{if } \phi \models \phi^* \text{ and } \psi \models \psi^*, \text{ then } \phi \wedge \psi \models \neg(\neg \phi^* \vee \neg \psi^*),$$

which is straightforward. $\square$

Similar proofs can be carried out for other reduced sets of connectives and are left to the exercises.

These results say that in a semantic sense the original list of five connectives is longer than it needs to be; we want next to verify that it is long enough — that is, we have not inadvertently omitted some further connective which would have increased the expressive power of L . To formulate this question precisely, let $\mathbf{TA}$ denote the set of all truth assignments and to each sentence ϕ assign the function $E_\phi : \mathbf{TA} \rightarrow \{\mathbf{T}, \mathbf{F}\}$ defined by $E_\phi(V) = V(\phi)$. Note that E_ϕ is really a formal representation of the truth table for ϕ : various V correspond to the lines of the table and E_ϕ assigns the appropriate truth value to each line. Thus we have

Proposition 1.3.7 $\phi \models \psi$ iff $E_\phi = E_\psi$. $\square$

Suppose that we construct a new propositional language L' by adding to the language L a new binary (or even 3- or more place) connective and develop the syntax and semantics of L' as we did for L . For example, we might add the symbol $\#$, specify that if ϕ , ψ , and θ are L' -sentences, so is $\#\phi\psi\theta$, and add to our definition of truth assignment the clause

$$V(\#\phi\psi\theta) = \begin{cases} \text{T,} & \text{if at least two of } V(\phi), V(\psi) \text{ and } V(\theta) \text{ are T;} \\ \text{F,} & \text{otherwise.} \end{cases}$$

$\#$ is called the **majority connective**. Suppose further that there exists an L' sentence χ which is not tautologically equivalent to any L -sentence. (This won't happen in the example, since

$$\#\phi\psi\theta \models (\phi \wedge \psi) \vee (\psi \wedge \theta) \vee (\phi \wedge \theta).)$$

Then we would probably want to adopt the language L' in preference to L as expressing a wider range of combinations of truth values. To show that this does not happen in general, we show that every possible such function E_χ is already identical with some E_ϕ . The key point is that Lemma 1.3.1 would also hold for any such L' — that is, any such E_χ would be finitary in the following sense.

Definition 1.3.8 A function $E : \text{TA} \rightarrow \{\text{T}, \text{F}\}$ is **finitary** iff there exists a finite set Q of sentence symbols such that for all truth assignments V and W , if $V \upharpoonright Q = W \upharpoonright Q$, then $E(V) = E(W)$.

Thus Proposition 1.3.1 asserts that every E_ϕ is finitary, and by an analogous proof, the same is true for any extended language L' obtained by adding any number of new n -ary connectives for various n . Thus we can deal with all possible extensions of L at once by proving

Theorem 1.3.9 For every finitary function $E : \text{TA} \rightarrow \{\text{T}, \text{F}\}$, there exists a sentence ϕ such that $E = E_\phi$ — that is, such that for every truth assignment V , $E(V) = V(\phi)$.

Proof. Let $Q = \{q_0, \dots, q_n\}$ witness that E is finitary, and let $\mathcal{F}$ denote the set of all functions $f : Q \rightarrow \{\text{T}, \text{F}\}$; there are 2^{n+1} many such f , which are essentially the lines of a truth table. Let $E^Q : \mathcal{F} \rightarrow \{\text{T}, \text{F}\}$ be the function naturally induced via the definition $E^Q(f) = E(V)$ for some (any) truth assignment V which extends f .

If $E^Q(f) = \text{F}$ for all $f \in \mathcal{F}$, then $E(V) = \text{F}$ for all truth assignments V and we may complete the proof by choosing for ϕ the tautologically false

sentence F . Suppose, then, that $E^Q(f^i) = \mathsf{T}$ for exactly $f^0, \dots, f^k \in \mathcal{F}$. For each $i \leq k$ and $j \leq n$, let

$$q_j^i = \begin{cases} q_j, & \text{if } f^i(q_j) = \mathsf{T}; \\ \neg q_j, & \text{if } f^i(q_j) = \mathsf{F}. \end{cases}$$

Then for any truth assignment V^i which extends f^i , $V^i(q_j^i) = \mathsf{T}$ for all $j \leq n$, so if we set

$$\phi^i = q_0^i \wedge \dots \wedge q_n^i,$$

and

$$\phi = \phi^0 \vee \dots \vee \phi^k,$$

also $V^i(\phi^i) = \mathsf{T}$ so $V^i(\phi) = \mathsf{T} = E(V^i)$. On the other hand, for any $f \in \mathcal{F}$ not among $f^0, \dots, f^k$, for each $i \leq k$ there is some $j_i \leq n$ such that $f(q_{j_i}) \neq f^i(q_{j_i})$, and hence for any truth assignment V extending f , $V(q_{j_i}^i) = \mathsf{F}$ so $V(\phi^i) = \mathsf{F}$. Thus $V(\phi) = \mathsf{F} = E(V)$, and we have shown that for any truth assignment V ,

$$V(\phi) = \mathsf{T} \iff V \text{ extends one of } f^0, \dots, f^k \iff E(V) = \mathsf{T},$$

so that $V(\phi) = E(V)$ and $E = E_\phi$. $\square$

This proof is admittedly somewhat convoluted and is best understood via an example. Suppose that E is the function associated with the sentence $\#p_0p_1p_2$ of L' above. Then

$$E(V) = \begin{cases} \mathsf{T}, & \text{if at least two of } V(p_0), V(p_1) \text{ and } V(p_2) \text{ are } \mathsf{T}; \\ \mathsf{F}, & \text{otherwise.} \end{cases}$$

Clearly $Q = \{p_0, p_1, p_2\}$ and there are $2^3 = 8$ many $f \in \mathcal{F}$. $E^Q(f) = \mathsf{T}$ for the functions f^0, f^1, f^2 and f^3 in $\mathcal{F}$ which assign to p_0, p_1 and p_2 the values $\mathsf{T}\mathsf{T}\mathsf{T}$, $\mathsf{T}\mathsf{T}\mathsf{F}$, $\mathsf{T}\mathsf{F}\mathsf{T}$ and $\mathsf{F}\mathsf{T}\mathsf{T}$, respectively. Then we find

$$\phi = (p_0 \wedge p_1 \wedge p_2) \vee (p_0 \wedge p_1 \wedge \neg p_2) \vee (p_0 \wedge \neg p_1 \wedge p_2) \vee (\neg p_0 \wedge p_1 \wedge p_2).$$

The sentence ϕ constructed in this proof is of a special form which will be of use later:

Definition 1.3.10 A sentence ϕ is said to be in **disjunctive normal form** iff ϕ is a disjunction of conjunctions of sentence symbols and their negations. ϕ is in **strict disjunctive normal form** iff ϕ is a disjunction $\phi^0 \vee \dots \vee \phi^k$, where for some distinct sentence symbols $q_0, \dots, q_n$, each ϕ^i is of the form $q_0^i \wedge \dots \wedge q_n^i$, with each q_j^i either q_j or $\neg q_j$. ϕ is in **conjunctive normal form** or **strict conjunctive normal form** iff the same conditions hold with the roles of $\vee$ and $\wedge$ reversed.

Corollary 1.3.11 *For every L -sentence θ ,*

- (i) *if θ is not tautologically false, then θ is tautologically equivalent to a sentence in strict disjunctive normal form;*
- (ii) *if θ is not a tautology, then θ is tautologically equivalent to a sentence in strict conjunctive normal form.*

Proof. For (i) we apply the proof of the preceding theorem to the function $E = E_\theta$. For (ii) note that the negation of a sentence in (strict) disjunctive normal form is tautologically equivalent to a sentence in (strict) conjunctive normal form. $\square$

Example 1.3.12 The sentence θ whose truth table is displayed following Theorem 1.1.7 has as its (strict) disjunctive normal form

$$(p_1 \wedge \neg p_2 \wedge p_4) \vee (p_1 \wedge \neg p_2 \wedge \neg p_4) \vee (\neg p_1 \wedge p_2 \wedge p_4) \\ \vee (\neg p_1 \wedge \neg p_2 \wedge p_4) \vee (\neg p_1 \wedge \neg p_2 \wedge \neg p_4),$$

with clauses corresponding to the five T-lines of the truth table. Using the three F-lines, we compute the (strict) conjunctive normal form:

$$\neg [(p_1 \wedge p_2 \wedge p_4) \vee (p_1 \wedge p_2 \wedge \neg p_4) \vee (\neg p_1 \wedge p_2 \wedge \neg p_4)]$$

hence

$$(\neg p_1 \vee \neg p_2 \vee \neg p_4) \wedge (\neg p_1 \vee \neg p_2 \vee p_4) \wedge (p_1 \vee \neg p_2 \vee p_4).$$

A common operation in informal mathematics is the conversion of one complex expression into another ‘simpler’ one by replacing some subexpression by another expression which is in the appropriate sense equivalent. In the context of propositional logic, this corresponds to replacing a subsentence χ of a sentence ϕ (cf. Example 1.2.15(iii)) by a tautologically equivalent sentence χ' — that is,

$$\phi = s_0 \dots s_{k-1} \chi t_0 \dots t_{l-1} \mapsto s_0 \dots s_{k-1} \chi' t_0 \dots t_{l-1} = \phi'.$$

The expectation is that ϕ' is tautologically equivalent to ϕ . Of course, if χ occurs more than once as a subsentence in ϕ , we should allow for the replacement of some but not all of these occurrences by χ' . To formulate this precisely, we proceed as follows.

Definition 1.3.13 For any sentences ϕ, ϕ', χ and χ' , we say that ϕ' arises from ϕ by **substitution** of χ' for χ — in symbols, $\phi' \leftarrow \phi_{\chi \mapsto \chi'}$ — iff one of the following holds:

- (i) $\phi' = \phi$;
- (ii) $\phi = \chi$ and $\phi' = \chi'$;
- (iii) for some sentences ψ and ψ' , $\phi = \neg \psi$,

$$\phi' = \neg \psi' \quad \text{and} \quad \psi' \leftarrow \psi_{\chi \mapsto \chi'};$$

- (iv) for some sentences ψ, ψ', θ and θ' , $\phi = \psi \bullet \theta$,

$$\phi' = \psi' \bullet \theta', \quad \psi' \leftarrow \psi_{\chi \mapsto \chi'} \quad \text{and} \quad \theta' \leftarrow \theta_{\chi \mapsto \chi'}.$$

A word of explanation is needed here concerning the form of this definition. Clearly it is a recursive definition, but it is not immediately obvious how it fits the mold of Theorem 1.2.14. Since substitution of χ' for some of the occurrences of χ in ϕ yields generally several different results, a natural formulation of this operation is as a function

$$\text{Sub} : \text{Sent}_L \times \text{Sent}_L \times \text{Sent}_L \rightarrow \wp(\text{Sent}_L),$$

with $\text{Sub}(\phi, \chi, \chi')$ being the set of all possible results of such substitutions. We leave it to an exercise below to give the precise formulation of the definition in this form.

Theorem 1.3.14 For any sentences ϕ, ϕ', χ and χ' , if $\phi' \leftarrow \phi_{\chi \mapsto \chi'}$ and $\models \chi \leftrightarrow \chi'$, then $\models \phi \leftrightarrow \phi'$.

Proof. A straightforward induction on ϕ . $\square$

Exercises

Exercise 1.3.15 Consider the two assertions

- (i) for any sentences ϕ and ψ , $\models \phi \wedge \psi \iff \models \phi \text{ and } \models \psi$;
- (ii) for any sentences ϕ and ψ , $\models \phi \vee \psi \iff \models \phi \text{ or } \models \psi$.

Discover whether one, both or neither of these is true and give proofs or counterexamples.

Definition 1.3.16 A set C of connectives is called **adequate** iff every L -sentence is tautologically equivalent to a sentence using only the connectives in C . Thus Proposition 1.3.6 asserts that the set $\{\neg, \vee\}$ is adequate; the next several exercises explore issues of adequacy for other sets of connectives.

Exercise 1.3.17 Prove that $\{\neg, \wedge\}$ and $\{\neg, \rightarrow\}$ are both adequate.

Exercise 1.3.18 Let $\downarrow$ be the binary connective with semantics

$$V(\phi \downarrow \psi) = \begin{cases} \text{T,} & \text{if } V(\phi) = \text{F and } V(\psi) = \text{F;} \\ \text{F,} & \text{otherwise.} \end{cases}$$

$\downarrow$ is called **nor**, short for “neither $_\$ nor $_\$ ”. Prove that $\{\downarrow\}$ is adequate.

Exercise 1.3.19 Let $|$ be the binary connective with semantics

$$V(\phi | \psi) = \begin{cases} \text{T,} & \text{if } V(\phi) = \text{F or } V(\psi) = \text{F;} \\ \text{F,} & \text{otherwise.} \end{cases}$$

$|$ is called **nand**, short for “not both $_\$ and $_\$ ” and also the **Sheffer stroke**. Prove that $\{| \}$ is adequate.

Exercise 1.3.20 Prove that the set $\{\wedge, \vee, \rightarrow, \leftrightarrow\}$ is not adequate.

Hint: Find a property $\mathcal{P}$ such that every sentence tautologically equivalent to one which uses only these four connectives has $\mathcal{P}$ but not every sentence has $\mathcal{P}$.

Exercise 1.3.21 Prove that the set $\{\neg, \leftrightarrow\}$ is not adequate.

Exercise 1.3.22 Let $\text{Sent}_{\neg, \wedge, \vee}$ denote the set of sentences which contain only these connectives. Give first recursive definitions of mappings $\phi \mapsto \phi^\circ$ from $\text{Sent}_{\neg, \wedge, \vee}$ to itself which interchanges occurrences of $\wedge$ and $\vee$ and $\phi \mapsto \phi^*$ which makes this interchange and in addition replaces each sentence symbol p by its negation $\neg p$. Prove that for all $\phi \in \text{Sent}_{\neg, \wedge, \vee}$

(i) $\phi^* \models \neg \phi$;

(ii) if $\models \neg \phi$ then also $\models \phi^\circ$ (although not generally $\phi^\circ \models \neg \phi$).

Exercise 1.3.23 Give a precise justification of Definition 1.3.13 along the lines suggested in the paragraph following it in the text.

Exercise 1.3.24 (*Propositional Interpolation Theorem*) For any sentence ϕ , let $\text{At}(\phi)$ be the set of sentence symbols which occur in ϕ (as in Example 1.2.15(ii)). Suppose that $\phi \models \psi$ but neither $\models \neg \phi$ nor $\models \psi$ (so $\phi \models \psi$ does not hold for trivial reasons). Show that there exists a sentence θ such that

(i) $\phi \models \theta$;

(ii) $\theta \models \psi$;

(iii) $\text{At}(\theta) \subseteq \text{At}(\phi) \cap \text{At}(\psi)$.

Hint: Show first that $\text{At}(\phi) \cap \text{At}(\psi) \neq \emptyset$.

We shall give in [Section 1.6](#) a proof of the Interpolation Theorem which generalizes to other logics; the proof sought here is a relatively simple application of the ideas of this section.

Exercise 1.3.25 Consider the following refinement of Example 1.2.15(ii) (for simplicity we restrict attention to $\text{Sent}_{\neg \vee}$). Define recursively

$$\begin{aligned} \text{At}^+(p_n) &= \{p_n\}; & \text{At}^-(p_n) &= \emptyset; \\ \text{At}^+(\neg \psi) &= \text{At}^-(\psi); & \text{At}^-(\neg \psi) &= \text{At}^+(\psi); \\ \text{At}^+(\psi \vee \theta) &= \text{At}^+(\psi) \cup \text{At}^+(\theta); & \text{At}^-(\psi \vee \theta) &= \text{At}^-(\psi) \cup \text{At}^-(\theta). \end{aligned}$$

Members of $\text{At}^+(\phi)$ are the sentence symbols which *occur positively* in ϕ ; members of $\text{At}^-(\phi)$ *occur negatively*.

- (i) Explain how the definition of At^+ and At^- is justified by the methods of Section 1.2.
- (ii) Let $\leq$ denote the natural ordering of $\{\mathbf{T}, \mathbf{F}\}$ thinking of $\mathbf{T}$ as 1 and $\mathbf{F}$ as 0. Prove that for all $\phi \in \text{Sent}_{\neg \vee}$ and all truth assignments V and W , if for all propositional symbols p ,

$$\begin{aligned} p \in \text{At}^+(\phi) &\implies V(p) \leq W(p) \quad \text{and} \\ p \in \text{At}^-(\phi) &\implies W(p) \leq V(p), \end{aligned}$$

then $V(\phi) \leq W(\phi)$.

- (iii) Extend the definition and this result to all sentences.

Exercise 1.3.26 (Propositional Lyndon Interpolation Theorem) Use the preceding exercise to establish the following generalization of the Propositional Interpolation Theorem, Exercise 1.3.24. Suppose that $\phi \models \psi$ but neither $\models \neg \phi$ nor $\models \psi$. Show that there exists a sentence θ such that

- (i) $\phi \models \theta$;
- (ii) $\theta \models \psi$;
- (iii) $\text{At}^+(\theta) \subseteq \text{At}^+(\phi) \cap \text{At}^+(\psi)$ and $\text{At}^-(\theta) \subseteq \text{At}^-(\phi) \cap \text{At}^-(\psi)$.

Hint: Show first that either $\text{At}(\phi) \cap \text{At}^+(\psi) \neq \emptyset$ or $\text{At}^-(\phi) \cap \text{At}^-(\psi) \neq \emptyset$.

1.4 Propositional theories

For the remainder of this chapter we suppose that the only connectives of L are $\neg$ and $\vee$ — that is, we replace Sent_L by $\text{Sent}_{\neg \vee}$. The other connectives are regarded as abbreviations:

$$\begin{aligned} \phi \wedge \psi &\text{ is an abbreviation for } \neg(\neg \phi \vee \neg \psi); \\ \phi \rightarrow \psi &\text{ is an abbreviation for } \neg \phi \vee \psi; \\ \phi \leftrightarrow \psi &\text{ is an abbreviation for } \neg(\neg \phi \vee \neg \psi) \vee \neg(\phi \vee \psi). \end{aligned}$$

Thus to show that a function V is a truth assignment, we need only verify that it respects these two connectives and a proof by Sent_L -induction has only three cases: the basis and two induction steps. Recall also that we denote some fixed tautology, say $p_0 \vee \neg p_0$, by $\mathbf{T}$ and some fixed tautologically false sentence, say $p_0 \wedge \neg p_0$, by $\mathbf{F}$.

General properties

One of the hallmarks of modern mathematics is the development of *theories*. A mathematical theory usually consists of a collection of definitions, lemmas, theorems etc. revolving about a common theme. We generally think of the content of the theory as changing over time as new theorems are proved, new concepts introduced and new connections with other theories discovered. One of the goals of the study of mathematical logic is to gain new insights into the structure of mathematical theories. Whereas a working mathematician is immersed in the details of developing a theory, the logician stands outside and views the theory as a whole, studies its global properties, its relationship to other theories etc.

We shall develop in the coming chapters logical frameworks which suffice for the expression of most mathematical theories. Propositional logic is, of course, not sufficient to this task, but it is useful to introduce in this simple context some of the basic ideas.

We shall identify a mathematical theory with the set of its *theorems*. To formulate this precisely, we need to extend the notion of tautological consequence to sets of sentences.

Definition 1.4.1 For any set Γ of sentences,

- (i) a truth assignment V **satisfies** or is a **model** of Γ iff $V(\phi) = \mathbf{T}$ for every $\phi \in \Gamma$; $\text{Mod}(\Gamma)$ is the set of all models of Γ ; we write $\text{Mod}(\phi_0, \dots, \phi_n)$ for $\text{Mod}(\{\phi_0, \dots, \phi_n\})$;
- (ii) for any sentence ψ , ψ is a **tautological consequence** of Γ — in symbols, $\Gamma \models \psi$ — iff $V(\psi) = \mathbf{T}$ for all models V of Γ .

Examples 1.4.2

- (i) If $\Gamma = \{\phi_0, \dots, \phi_n\}$, then $\text{Mod}(\Gamma)$ consists of all truth assignments such that $V(\phi_0 \wedge \dots \wedge \phi_n) = \mathbf{T}$.
- (ii) If $\psi \in \Gamma$, then $\Gamma \models \psi$.
- (iii) If $\Gamma = \{p_{n+1} \rightarrow p_n : n \in \omega\}$, then $\text{Mod}(\Gamma)$ consists of those truth assignments such that either $V(p_n) = \mathbf{T}$ for all n , or for some n , $V(p_i) = \mathbf{T}$ for $i < n$, and $V(p_i) = \mathbf{F}$ for $i \geq n$. Among the tautological consequences of Γ are all sentences of the form $p_m \rightarrow p_n$ for $m \geq n$.

For finite Γ , this notion of tautological consequence reduces to the previous one:

Proposition 1.4.3

- (i) For any ψ , $\models \psi \iff \emptyset \models \psi$.
- (ii) For any $\phi_0, \dots, \phi_n, \psi$, the following are equivalent:
 - (a) $\{\phi_0, \dots, \phi_n\} \models \psi$;
 - (b) $\phi_0 \wedge \dots \wedge \phi_n \models \psi$;
 - (c) for each $i < n$, $\{\phi_0, \dots, \phi_{i-1}\} \models \phi_i \wedge \dots \wedge \phi_n \rightarrow \psi$.

Proof. Immediate from the definitions. $\square$

We sometimes write $\phi_0, \dots, \phi_n \models \psi$ for $\{\phi_0, \dots, \phi_n\} \models \psi$.

Proposition 1.4.4 For any sets Γ and Δ of sentences, and any sentences ϕ and ψ ,

- (i) $\Gamma \subseteq \Delta \implies \text{Mod}(\Delta) \subseteq \text{Mod}(\Gamma)$;
- (ii) $\Gamma \models \psi$ iff $\text{Mod}(\Gamma) \subseteq \text{Mod}(\psi)$;
- (iii) if $\Gamma \subseteq \Delta$ and $\Gamma \models \psi$, then also $\Delta \models \psi$;
- (iv) $\Gamma \cup \{\phi\} \models \psi \iff \Gamma \models \phi \rightarrow \psi$.

Proof. These are easy consequences of the definitions, and the reader is strongly encouraged to work out the proofs. $\square$

The relation $\Gamma \models \psi$ is intended as a precise characterization of the informal notion: ψ follows from the assumptions (axioms) Γ .

Definition 1.4.5 A set T of sentences is a **propositional theory** iff it is closed under tautological consequence — that is, for all sentences ϕ ,

$$T \models \phi \implies \phi \in T.$$

Elements of T are called the **theorems** of T .

Of course always $\phi \in T \implies T \models \phi$. There are three quite distinct ways to describe a propositional theory; we look at them one at a time.

Definition 1.4.6 For any set Γ of sentences, the theory **generated** by Γ is the set

$$\text{Th}(\Gamma) = \{ \psi : \Gamma \models \psi \}.$$

We call Γ a set of **axioms** for $\text{Th}(\Gamma)$.

This is, of course, the formal counterpart of an informal mathematical theory, such as plane geometry or group theory, described axiomatically. Note that $\text{Th}(\Gamma)$ is always infinite.

Proposition 1.4.7 *For any sets Γ and Δ of sentences,*

- (i) $\Gamma \subseteq \text{Th}(\Gamma)$;
- (ii) $\text{Mod}(\Gamma) = \text{Mod}(\text{Th}(\Gamma))$;
- (iii) $\text{Th}(\Gamma)$ is a propositional theory;
- (iv) $\Gamma \subseteq \Delta \implies \text{Th}(\Gamma) \subseteq \text{Th}(\Delta)$;
- (v) Γ is a propositional theory iff $\Gamma = \text{Th}(\Gamma)$;
- (vi) $\text{Th}(\Gamma)$ is the smallest propositional theory T such that $\Gamma \subseteq T$;
- (vii) $\text{Th}(\emptyset) = \{\phi : \phi \text{ is a tautology}\}$ is included in every propositional theory.

Proof. Each part is a straightforward consequence of the definitions and the preceding parts; the reader is strongly encouraged to work out the proofs. $\square$

Definition 1.4.8 For any set Γ of sentences,

- (i) Γ is **consistent** or **satisfiable** iff Γ has at least one model; otherwise, Γ is **inconsistent** or **unsatisfiable**;
- (ii) Γ is **complete** iff for every sentence ϕ , either $\phi \in \Gamma$ or $\neg\phi \in \Gamma$, but not both — that is, $(\neg\phi) \in \Gamma \iff \phi \notin \Gamma$; otherwise Γ is **incomplete**.

Corollary 1.4.9 *For any set Γ of sentences and any sentence ψ ,*

- (i) $\Gamma \models \psi$ iff $\Gamma \cup \{\neg\psi\}$ is inconsistent;
- (ii) $\Gamma \models \neg\psi$ iff $\Gamma \cup \{\psi\}$ is inconsistent.

Proof. The conditions on the right sides are simple translations of the definitions of the left. $\square$

A second way to describe a propositional theory is in terms of a given truth assignment. This corresponds in informal mathematics to the set of sentences true in some mathematical structure, such as the real numbers or a particular group or ring.

Definition 1.4.10 For any truth assignment V , the **theory of V** is the set

$$\text{Th}(V) = \{\psi : V(\psi) = \mathbf{T}\}.$$

Proposition 1.4.11 *For any truth assignment V ,*

- (i) $\text{Th}(V)$ is a complete and consistent propositional theory and V is the unique model of $\text{Th}(V)$;
- (ii) for any truth assignment W , $V = W \iff \text{Th}(V) = \text{Th}(W)$.

Proof. By definition V is a model of $\text{Th}(V)$, so it is consistent. Also by definition, for any ϕ , $V(\neg \phi) = \text{T} \iff V(\phi) \neq \text{T}$, so $\text{Th}(V)$ is complete. To see that $\text{Th}(V)$ is a theory, suppose that $\text{Th}(V) \models \theta$. Since V is a model of $\text{Th}(V)$, $V(\theta) = \text{T}$, so $\theta \in \text{Th}(V)$. Suppose that some truth assignment W is a model of $\text{Th}(V)$. Then for all ϕ , $V(\phi) = \text{T} \implies W(\phi) = \text{T}$, so also

$$V(\phi) = \text{F} \implies V(\neg \phi) = \text{T} \implies W(\neg \phi) = \text{T} \implies W(\phi) = \text{F},$$

so in fact $V = W$. (ii) is now immediate. $\square$

Corollary 1.4.12 *There are uncountably many propositional theories.*

Proof. For each atomic truth assignment $V_0 : \text{Sent}_0 \rightarrow \{\text{T}, \text{F}\}$, there is a unique extension to a truth assignment V and hence a unique propositional theory $\text{Th}(V)$. Distinct V_0 yield distinct $\text{Th}(V)$, and since there are uncountably many V_0 , there are also uncountably many theories. $\square$

Corollary 1.4.13 *For any set Γ of sentences and any propositional theory T , the following are equivalent:*

- (i) *T is a complete propositional theory with $\Gamma \subseteq T$ (a **complete extension** of Γ);*
- (ii) *$T = \text{Th}(V)$ for some model V of Γ .*

Proof. Left as an exercise. $\square$

A variation on the theory of a single truth assignment is the theory of a set of truth assignments:

Definition 1.4.14 For any non-empty class K of truth assignments, the **theory of K** is the set

$$\begin{aligned} \text{Th}(K) &= \{ \psi : V(\psi) = \text{T} \text{ for all } V \in K \} \\ &= \bigcap \{ \text{Th}(V) : V \in K \}. \end{aligned}$$

Proposition 1.4.15 *For any non-empty class K of truth assignments,*

- (i) *$\text{Th}(K)$ is a consistent propositional theory and each $V \in K$ is a model of $\text{Th}(K)$;*
- (ii) *if K has at least two distinct elements, then $\text{Th}(K)$ is not complete.*

Proof. For (i), if $V \in K$, then $\text{Th}(K) \subseteq \text{Th}(V)$ so for any ψ ,

$$\psi \in \text{Th}(K) \implies \psi \in \text{Th}(V) \implies V(\psi) = \text{T},$$

and thus V is a model of $\text{Th}(K)$. In particular, $\text{Th}(K)$ is consistent. If $\text{Th}(K) \models \psi$, then for any $V \in K$, since V is a model of $\text{Th}(K)$, also $V(\psi) = \text{T}$. Hence $\psi \in \text{Th}(K)$; thus $\text{Th}(K)$ is a propositional theory.

For (ii), if $V \neq W$ are both in K , then for some sentence ϕ , $V(\phi) = \mathbf{T}$ and $W(\phi) = \mathbf{F}$, so $\phi \in \text{Th}(V) \sim \text{Th}(W)$ and $(\neg \phi) \in \text{Th}(W) \sim \text{Th}(V)$ and neither ϕ nor $\neg \phi$ is in $\text{Th}(K)$. $\square$

We call a class K of truth assignments **axiomatizable** iff for some set Γ of sentences, $K = \text{Mod}(\Gamma)$.

Corollary 1.4.16

- (i) For any set Γ of sentences, $\Gamma \subseteq \text{Th}(\text{Mod}(\Gamma))$; if Γ is a propositional theory, then $\Gamma = \text{Th}(\text{Mod}(\Gamma))$; in particular, for any set Γ of sentences, $\text{Th}(\Gamma) = \text{Th}(\text{Mod}(\Gamma))$.
- (ii) For any class K of truth assignments, $K \subseteq \text{Mod}(\text{Th}(K))$; if K is axiomatizable, then $K = \text{Mod}(\text{Th}(K))$.

Proof. Left as an exercise. $\square$

We show next that there is only one inconsistent propositional theory:

Proposition 1.4.17 For any set Γ of sentences, the following are equivalent:

- (i) Γ is inconsistent;
- (ii) for all sentences ϕ , $\Gamma \models \phi$;
- (iii) for some (every) sentence ϕ , both $\Gamma \models \phi$ and $\Gamma \models \neg \phi$;
- (iv) $\Gamma \models \mathbf{F}$.

Proof. Suppose that (i) Γ is inconsistent. Then for every sentence ϕ it is vacuously true that $V(\phi) = \mathbf{T}$ for all models V of Γ so that $\Gamma \models \phi$. (iii) follows immediately from (ii), and (iv) follows from (iii) since $\{\phi, \neg \phi\} \models \mathbf{F}$. Finally, (iv) implies (i) since no truth assignment can assign $\mathbf{T}$ to $\mathbf{F}$. $\square$

Interpreting these conditions for theories we have

Corollary 1.4.18 For any propositional theory T , the following are equivalent:

- (i) T is inconsistent;
- (ii) $T = \text{Sent}_L$;
- (iii) for some (every) sentence ϕ , both $\phi \in T$ and $\neg \phi \in T$;
- (iv) $\mathbf{F} \in T$. $\square$

Corollary 1.4.19 For any propositional theory T , T is complete iff T is **maximal consistent** — that is, T is consistent but any properly larger set of sentences is inconsistent. In this case, T has exactly one model V and $T = \text{Th}(V)$.

Proof. If T is complete, then it fails to satisfy (iii) of the preceding corollary so is consistent, and for any sentence $\phi \notin T$, since $\neg\phi \in T$, clearly $T \cup \{\phi\}$ is inconsistent. Conversely, if T is maximal consistent, then for any sentence $\phi \notin T$, $T \cup \{\phi\}$ is inconsistent, so $T \models \neg\phi$ and since T is a theory, $\neg\phi \in T$. The unique model of a complete theory T is the truth assignment V such that for each $n \in \omega$, $V(p_n) = \top$ iff $p_n \in T$. $\square$

Compactness

We turn next to the compactness phenomenon. In everyday mathematics, one typically demonstrates that one assertion ψ follows from some set of assumptions Γ by giving some sort of proof. We shall study this process in greater detail in [Chapter 3](#), but for now we consider just one aspect of any such proof: it is finite and therefore can in some sense depend on or use only finitely much of Γ . The precise statement of this principle in the context of Propositional Logic is the

Propositional Compactness Theorem 1.4.20 (*PCT*) *For any set Γ of sentences and any sentence ψ ,*

$$\Gamma \models \psi \quad \Longleftrightarrow \quad \text{for some finite subset } \Gamma_0 \subseteq \Gamma, \quad \Gamma_0 \models \psi.$$

It will be convenient to write

$$\Gamma \models^* \psi \quad :\Longleftrightarrow \quad \text{for some finite subset } \Gamma_0 \subseteq \Gamma, \quad \Gamma_0 \models \psi,$$

and

$$\text{Th}^*(\Gamma) := \{ \psi : \Gamma \models^* \psi \}.$$

Then the PCT may be expressed as asserting that for all Γ and ψ ,

$$\Gamma \models \psi \quad \Longleftrightarrow \quad \Gamma \models^* \psi \quad \text{or} \quad \text{Th}(\Gamma) = \text{Th}^*(\Gamma).$$

Note that since $\Gamma_0 \subseteq \Gamma$ and $\Gamma_0 \models \psi \implies \Gamma \models \psi$ (by Proposition 1.4.7), the right-to-left implication (inclusion) of the PCT is immediate.

To prove the Propositional Compactness Theorem, it is convenient first to reformulate it.

Definition 1.4.21 A set Γ of sentences is *finitely consistent* iff every finite subset $\Gamma_0 \subseteq \Gamma$ is consistent.

Every consistent set is obviously finitely consistent; the converse is the

Alternative Propositional Compactness Theorem 1.4.22 (*APCT*) *For any set Γ of sentences, if Γ is finitely consistent, then Γ is consistent.*

We shall first show that each of these two versions is easily derivable from the other and then prove the alternative version. Note first

Lemma 1.4.23 *For any set Γ sentences and any sentence ψ ,*

- (i) $\Gamma \models^* \psi$ iff $\Gamma \cup \{\neg \psi\}$ is not finitely consistent;
- (ii) $\Gamma \models^* \neg \psi$ iff $\Gamma \cup \{\psi\}$ is not finitely consistent.

Proof. This follows easily from the closely parallel Proposition 1.4.9. $\square$

Proof. (of PCT from APCT) Assume APCT and that $\Gamma \models \psi$. Then by Corollary 1.4.9, $\Gamma \cup \{\neg \psi\}$ is inconsistent, so by APCT it is not finitely consistent. Hence by the preceding lemma, $\Gamma \models^* \psi$. $\square$

Proof. (of APCT from PCT) Assume PCT and that Γ is finitely consistent. Then for every finite $\Gamma_0 \subseteq \Gamma$, $\Gamma_0 \not\models F$, so by PCT also $\Gamma \not\models F$ and thus Γ is consistent. $\square$

The proof of APCT proceeds by showing that any finitely consistent set Γ can be extended to a finitely consistent and complete set Δ and that any such set is in fact consistent. First, we state a lemma and the analogue of Proposition 1.4.17:

Lemma 1.4.24 *For any set Γ of sentences, $\text{Th}^*(\Gamma)$ is closed under $\models^*$ — that is, for any sentence ψ ,*

$$\text{Th}^*(\Gamma) \models^* \psi \implies \Gamma \models^* \psi.$$

Proof. The hypothesis asserts the existence of $\phi_0, \dots, \phi_{n-1} \in \text{Th}^*(\Gamma)$ such that $\{\phi_0, \dots, \phi_{n-1}\} \models \psi$. Hence, there exist finite subsets $\Gamma_0, \dots, \Gamma_{n-1} \subseteq \Gamma$ such that for $i < n$, $\Gamma_i \models \phi_i$. Then $\Gamma_0 \cup \dots \cup \Gamma_{n-1} \models \psi$ and thus witnesses that $\Gamma \models^* \psi$. $\square$

Note that this is analogous to the fact that $\text{Th}(\Gamma)$ is closed under $\models$ — that is, $\text{Th}(\Gamma)$ is a propositional theory.

Proposition 1.4.25 *For any set Γ of sentences, the following are equivalent:*

- (i) Γ is not finitely consistent;
- (ii) for all sentences ϕ , $\Gamma \models^* \phi$;
- (iii) for some (every) sentence ϕ , both $\Gamma \models^* \phi$ and $\Gamma \models^* \neg \phi$;
- (iv) $\Gamma \models^* F$.

Proof. Given (i), let $\Gamma_0 \subseteq \Gamma$ be a finite inconsistent subset. Then by 1.4.17, for every sentence ϕ , $\Gamma_0 \models \phi$, so $\Gamma \models^* \phi$. (iii) follows trivially from (ii), and (iv) follows from (iii) and the lemma, since $\{\phi, \neg \phi\} \models F$, hence $\text{Th}^*(\Gamma) \models^* F$. Finally, if $\Gamma \models^* F$, then for some finite $\Gamma_0 \subseteq \Gamma$, $\Gamma_0 \models F$, so Γ_0 is inconsistent and hence Γ is not finitely consistent. $\square$

Corollary 1.4.26 *For any set Γ of sentences and any sentence θ , if Γ is finitely consistent then at least one of $\Gamma \cup \{\theta\}$ and $\Gamma \cup \{\neg\theta\}$ is finitely consistent.*

Proof. Immediate from the preceding proposition and Lemma 1.4.23. $\square$

Proposition 1.4.27 *For any finitely consistent set Γ of sentences, there exists a finitely consistent and complete set of sentences $\Delta \supseteq \Gamma$.*

Proof. We showed above in Corollary 1.2.9 that there are denumerably many L -sentences; this also follows from the standard result of Set Theory that the set of finite sequences of elements of a denumerable set (here the symbols) is denumerable. Let $\langle \theta_n : n \in \omega \rangle$ be a complete list of them. Assume that Γ is finitely consistent, set $\Delta_0 := \Gamma$ and for all n ,

$$\Delta_{n+1} := \begin{cases} \Delta_n \cup \{\theta_n\}, & \text{if this set is finitely consistent;} \\ \Delta_n, & \text{otherwise;} \end{cases}$$

$$\Delta := \bigcup_{n \in \omega} \Delta_n.$$

Clearly $\Gamma \subseteq \Delta$ and by induction for all n , Δ_n is finitely consistent. But then also Δ is finitely consistent, since any finite subset of Δ is already a subset of some Δ_n . To check that Δ is complete, note first that if for some sentence ϕ , both $\phi \in \Delta$ and $(\neg\phi) \in \Delta$, then $\{\phi, \neg\phi\}$ would be an inconsistent finite subset of Δ contrary to the finite consistency just established. Suppose that for some n , $\theta_n \notin \Delta$, so by the construction of Δ , $\Delta_n \cup \{\theta_n\}$ is not finitely consistent. Choose p such that θ_p is $\neg\theta_n$ and suppose first that $n < p$. Since $\Delta_n \subseteq \Delta_p$, $\Delta_p \cup \{\theta_n\}$ also fails to be finitely consistent. But Δ_p is finitely consistent, so by the corollary, $\Delta_p \cup \{\neg\theta_n\} = \Delta_p \cup \{\theta_p\}$ is finitely consistent and thus $\neg\theta_n \in \Delta_{p+1} \subseteq \Delta$. A similar argument handles the case $p < n$. $\square$

Lemma 1.4.28 *For any set Δ of sentences which is both finitely consistent and complete,*

- (i) *for any sentence ψ , if $\Delta \models^* \psi$, then $\psi \in \Delta$;*
- (ii) *for any sentences ψ and θ ,*

$$\begin{aligned} \text{(a)} \quad \neg\psi \in \Delta & \iff \psi \notin \Delta; \\ \text{(b)} \quad (\psi \vee \theta) \in \Delta & \iff \psi \in \Delta \text{ or } \theta \in \Delta. \end{aligned}$$

Proof. Suppose that $\Delta \models^* \psi$ but $\psi \notin \Delta$. Then by completeness $(\neg\psi) \in \Delta$ so $\Delta \models^* \neg\psi$, hence by Proposition 1.4.25, Δ is not finitely consistent. For (ii), (a) is the definition of completeness and the implication $(\Leftarrow)$ of (b) is immediate from (i), since $\phi \models \phi \vee \psi$; for $(\Rightarrow)$ if $(\psi \vee \theta) \in \Delta$ but $\psi \notin \Delta$, then by completeness $(\neg\psi) \in \Delta$, so again by (i), $\theta \in \Delta$. $\square$

Proposition 1.4.29 *Every finitely consistent and complete set of sentences is consistent (has a model).*

Proof. Let Δ be finitely consistent and complete, and let V be the unique truth assignment such that for all n ,

$$V(p_n) = \begin{cases} \mathbf{T}, & \text{if } p_n \in \Delta; \\ \mathbf{F}, & \text{otherwise.} \end{cases}$$

We prove by Sent_L -induction that for all ϕ ,

$$\phi \in \Delta \iff V(\phi) = \mathbf{T}.$$

Thus in particular, V is a model of Δ . For atomic ϕ this is immediate from the definition. If ϕ is $\neg\psi$, we have by the definition of completeness and the induction hypothesis

$$\neg\psi \in \Delta \iff \psi \notin \Delta \iff V(\psi) \neq \mathbf{T} \iff V(\neg\psi) = \mathbf{T}.$$

If ϕ is $\psi \vee \theta$, we have by the preceding lemma

$$\begin{aligned} \psi \vee \theta \in \Delta &\iff \psi \in \Delta \text{ or } \theta \in \Delta \\ &\iff V(\psi) = \mathbf{T} \text{ or } V(\theta) = \mathbf{T} \iff V(\psi \vee \theta) = \mathbf{T}. \quad \square \end{aligned}$$

The APCT is an immediate consequence of the two propositions, and hence the PCT also follows.

We have defined a model of a set Γ of sentences to be a truth assignment V such that $V(\phi) = \mathbf{T}$ for *all* $\phi \in \Gamma$. A reader with a penchant for symmetry might well ask: why is there no corresponding term for V such that $V(\phi) = \mathbf{T}$ for *some* $\phi \in \Gamma$. Although this notion seems less useful, it does occasionally arise and we discuss here some notation to deal with it. We introduce two new logical operators $\bigwedge$ (**big and**) and $\bigvee$ (**big or**) which are applied to sets of sentences. $\bigwedge \Gamma$ and $\bigvee \Gamma$ are not actually L -sentences, but may be treated as sentences in certain limited contexts. Their status is similar to that of the symbol ∞ in the theory of real numbers; ∞ is *not* a real number, but may be used meaningfully in some contexts, such as $r < \infty$, as if it were. In [Section 3.7](#) we shall study an extended logic which admits such objects as sentences.

Definition 1.4.30 For any set Γ of sentences and truth assignment V ,

- (i) $V(\bigwedge \Gamma) = \mathbf{T}$ iff $V(\phi) = \mathbf{T}$ for all $\phi \in \Gamma$;
- (ii) $V(\bigvee \Gamma) = \mathbf{T}$ iff $V(\phi) = \mathbf{T}$ for some $\phi \in \Gamma$.

Note that if $\Gamma = \{\phi_0, \dots, \phi_n\}$ is a finite set, $\bigwedge \Gamma$ is essentially the same as the sentence $\phi_0 \wedge \dots \wedge \phi_n$ and $\bigvee \Gamma$ is the same as $\phi_0 \vee \dots \vee \phi_n$; we shall generally identify these. It is now natural to write

$$\models \bigwedge \Gamma \rightarrow \psi \quad \text{with the same meaning as} \quad \Gamma \models \psi,$$

and other similar expressions. For example, we have

$$\Gamma \cup \Delta \models \psi \iff \Gamma \models \bigwedge \Delta \rightarrow \psi.$$

The Propositional Compactness Theorem then has the following

Corollary 1.4.31 *For any set Γ of sentences and any sentence ψ ,*

- (i) *if $\models \bigwedge \Gamma \rightarrow \psi$, then for some finite $\Gamma_0 \subseteq \Gamma$, $\models \bigwedge \Gamma_0 \rightarrow \psi$;*
- (ii) *if $\models \psi \rightarrow \bigvee \Gamma$, then for some finite $\Gamma_0 \subseteq \Gamma$, $\models \psi \rightarrow \bigvee \Gamma_0$;*

in particular

- (iii) *if $\models \bigvee \Gamma$, then for some finite $\Gamma_0 \subseteq \Gamma$, $\models \bigvee \Gamma_0$.*

Proof. Part (i) is just a restatement of the Propositional Compactness Theorem; we leave (ii) and (iii) as an exercise. $\square$

We give in the exercises some other examples of this way of thinking.

Exercises

Exercise 1.4.32 Call two sets of sentences **tautologically equivalent** iff they are satisfied by exactly the same truth assignments, and call Γ **tautologically independent** iff for no $\phi \in \Gamma$ does $\Gamma \sim \{\phi\} \models \phi$. Prove

- (i) every finite set of sentences has an independent equivalent subset;
- (ii) there exists an infinite set of sentences which has no independent equivalent subset;
- (iii) for every (finite or infinite) set of sentences, there exists an independent equivalent set (not necessarily a subset).

Exercise 1.4.33 Show that if Δ is a complete and finitely consistent set of sentences, then for any sentences θ and χ ,

- (i) $(\theta \wedge \chi) \in \Delta \iff \theta \in \Delta \text{ and } \chi \in \Delta$;
- (ii) $(\theta \rightarrow \chi) \in \Delta \iff \theta \notin \Delta \text{ or } \chi \in \Delta$;
- (iii) $(\theta \leftrightarrow \chi) \in \Delta \iff \text{both } \theta, \chi \in \Delta \text{ or both } \theta, \chi \notin \Delta$.

Exercise 1.4.34 Give a proof of Corollary 1.4.31(ii) and (iii).

Exercise 1.4.35 Prove Corollary 1.4.13.

Exercise 1.4.36 Show that for any set Γ of sentences,

$$\Gamma \models \bigvee \left\{ \bigwedge \text{Th}(V) : V \in \text{Mod}(\Gamma) \right\},$$

where $\bigvee$ has the obvious meaning applied here to a set of “pseudosentences”.

Definition 1.4.37 For any propositional theory T and any set Δ of sentences,

- (i) T is **Δ -axiomatizable** iff $T = \text{Th}(\Gamma)$ for some set $\Gamma \subseteq \Delta$;
- (ii) $\Delta^T := \{ \phi : T \models \phi \leftrightarrow \phi^* \text{ for some } \phi^* \in \Delta \}$;
- (iii) T admits **Δ -elimination** iff every sentence belongs to Δ^T ;
- (iv) Δ is **regular** iff it is closed under $\wedge$ and $\vee$ and **regular Boolean** iff it is closed additionally under $\neg$;
- (v) for any set Γ of sentences, $\text{Th}^\Delta(\Gamma) := \text{Th}(\Gamma) \cap \Delta$;
- (vi) for any truth assignment V , $\text{Th}^\Delta(V) := \text{Th}(V) \cap \Delta$.

Exercise 1.4.38 Let T be a consistent propositional theory and Δ a non-empty regular Boolean set of sentences. Show that for any sentence ϕ the following are equivalent:

- (i) $\phi \in \Delta^T$;
- (ii) for all models V and W of T ,

$$\text{if } \text{Th}^\Delta(V) = \text{Th}^\Delta(W), \quad \text{then } V(\phi) = W(\phi);$$
- (iii) $T \cup \text{Th}^\Delta(T \cup \{ \phi \}) \models \phi$.

Hint: For (ii) $\implies$ (iii), suppose that V is a model of $T \cup \text{Th}^\Delta(T \cup \{ \phi \})$ but $V(\phi) = \text{F}$. Use (ii) to show that $T \cup \text{Th}^\Delta(V) \models \neg \phi$, and derive a contradiction. For (iii) $\implies$ (i) use compactness.

Exercise 1.4.39 Let T be a consistent propositional theory and Δ a non-empty regular Boolean set of sentences. Show that the following are equivalent:

- (i) T admits Δ -elimination;
- (ii) for every model V of T , $\text{Th}(V) = \text{Th}(T \cup (\text{Th}^\Delta(V)))$;
- (iii) for all models V and W of T , if $\text{Th}^\Delta(V) = \text{Th}^\Delta(W)$, then $V = W$.

Hint: Use the preceding exercise.

Exercise 1.4.40 Let T be a consistent propositional theory and Δ a regular set of sentences such that for any model V of T , $T \subseteq \text{Th}(\text{Th}^\Delta(V))$. Show that T is Δ -axiomatizable.

Hint: Note first that it suffices to show that $T \subseteq \text{Th}(T \cap \Delta)$. For each $\phi \in T$ and each model V of T , $\text{Th}^\Delta(V) \models \phi$. By Compactness there is a single sentence $\theta_{V,\phi} \in \text{Th}^\Delta(V)$ such that $\models \theta_{V,\phi} \rightarrow \phi$. Show that for some finite list $V_0, \dots, V_k$ of models of T , $\theta_{V_0,\phi} \vee \dots \vee \theta_{V_k,\phi} \in T \cap \Delta$, and conclude that $T \cap \Delta \models \phi$.

Exercise 1.4.41 Prove Corollary 1.4.16 and give examples to show that the two inclusions may be proper.

Exercise 1.4.42 Of the three properties that a set of sentence may have: “consistent”, “complete”, and “is a theory”, which pairs imply the third?

Exercise 1.4.43 Give another proof of the APCT along the following lines. Let Γ be finitely satisfiable, and fix an enumeration $\langle \phi_n : n \in \omega \rangle$ of Γ . For each n , $\{\phi_i : i \leq n\}$ is a finite subset of Γ so has a model — choose one, say V_n . Define an atomic truth assignment W by

$$W(p_k) = \begin{cases} \text{T}, & \text{if } X_k \text{ is infinite;} \\ \text{F}, & \text{otherwise;} \end{cases}$$

where $X_k := \{n : (\forall j < k)[V_n(p_j) = W(p_j)] \text{ and } V_n(p_k) = \text{T}\}$, and let V be the unique truth assignment which extends W . Prove that V is a model of Γ .

Exercise 1.4.44 Call a propositional sentence ϕ **special** iff it is p_0 , $\neg p_0$, or of the form

$$\theta_0 \wedge \dots \wedge \theta_{n-1} \rightarrow \theta_n,$$

where for each $i \leq n$, θ_i is either p_i or $\neg p_i$. For any set Γ of sentences, set

$$\Gamma^* := \{\phi : \phi \text{ is special and } \Gamma \models \phi\}.$$

Prove that Γ^* and Γ are tautologically equivalent.

In [Section 7.5](#) we will establish the following result:

Definition 1.4.45 For any set A and any $n \in \omega$,

- (i) $[A]^n$ denotes the collection of n -element subsets of A ;
- (ii) for and $P \subseteq [A]^n$, a set $H \subseteq A$ is **homogeneous for** P iff either $[H]^n \subseteq P$ or $[H]^n \cap P = \emptyset$.

Ramsey's Theorem 1.4.46 *For any infinite set A , any $n > 0$, and any $P \subseteq [A]^n$, there exists an infinite set $H \subseteq A$ homogeneous for P .*

There is also a finite version:

Finite Ramsey's Theorem 1.4.47 *There exists a function $f : \omega \times \omega \rightarrow \omega$ such that for an $m, n > 0$, any set A with at least $f(m, n)$ elements and any $P \subseteq [A]^n$, there exists a set $H \subseteq A$ with at least m elements which is homogeneous for P .*

Exercise 1.4.48 Use the Propositional Compactness Theorem to deduce the finite version from the infinite one.

1.5 Decidability and effective enumerability

We have already encountered several instances of problems of the following sort: given a set X and a property $\mathcal{P}$, show that all (or none) of the elements of X have the property $\mathcal{P}$. But what can we say about the relationship of $\mathcal{P}$ and X if some elements of X have $\mathcal{P}$ and some do not? For example, all pairs of natural numbers have a greatest common divisor, but not all such pairs are relatively prime. Often in such situations the goal is to characterize $\{x \in X : \mathcal{P}(x)\}$ in various ways — here $\mathcal{P}(x)$ is an abbreviation for “ x has the property $\mathcal{P}$ ”. One such way is to provide an *algorithm* or *effective procedure* for deciding, given $x \in X$, whether or not $\mathcal{P}(x)$ holds. In the example cited the well-known Euclidean algorithm for calculating the greatest common divisor of a pair of natural numbers provides an effective procedure for deciding whether or not the pair is relatively prime. This example makes clear also that we need to consider more generally effective procedures which produce results other than just ‘yes’ or ‘no’.

It is hard to delineate precisely just what should count as an effective procedure, and over the years much has been written attempting to explain the concept. However, there is substantial agreement that an effective procedure should have the following characteristics:

- (i) input to the procedure should consist of finite objects, such as natural numbers or finite strings of symbols drawn from a discrete “alphabet”;
- (ii) the procedure should operate according to a finite set of precisely stated deterministic rules (the program) which cover all cases, proceed in discrete steps and lead to a finite output in finitely many steps.

A common model for an effective procedure is an idealized electronic computer which is not subject to error and has access to unbounded, amounts of memory, time and input/output media. Note that although for each instance of the problem with a particular input only finitely much of each of these resources is used, the size of potential inputs is generally unbounded

and larger inputs will usually require more memory and time and lead to larger outputs.

Clearly the Euclidean algorithm satisfies these provisions as do many other well-known algorithms — for example, those for differentiating a polynomial function or sorting a finite list of names.

For the moment we shall be content with this somewhat imprecise notion of algorithm or effective procedure (we shall generally use these two terms synonymously) and shall from time to time discuss the existence, or seeming non-existence, of procedures for various properties $\mathcal{P}$. Eventually (in Chapter 4), we shall want to be able to *prove* that certain problems do *not* have algorithmic solutions, and for this purpose we shall need to replace this informal notion with a mathematically precise one, that of recursive function. The situation is somewhat analogous to that for many other intuitive notions and their precise mathematical counterparts; for example, the informal characterization of a real number as representing a distance of two points on a line may serve well in many contexts, but eventually the theory requires a more precise notion such as Cauchy sequences or Dedekind cuts.

Definition 1.5.1 A property $\mathcal{P}$ defined over a set X of finite objects is called **decidable in X** iff there exists an effective procedure (**decision procedure**) for deciding, given $x \in X$, whether or not $\mathcal{P}(x)$ holds. A subset $A \subseteq X$ is **decidable in X** iff the property $\mathcal{P}$ defined by $(\mathcal{P}(x) : \iff x \in A)$ is decidable in X . In many cases the set X will be clear from context, and we shall say simply that $\mathcal{P}$ or A is **decidable**. A property or set is **undecidable** (in X) iff it is not decidable. A function $F : X \rightarrow Z$ is called **effectively calculable** iff there is an effective procedure which, given any $x \in X$, calculates the value $F(x)$. For any set $A \subseteq X$, the **characteristic function** $K_A : X \rightarrow \{0, 1\}$ is defined by $K_A(x) := 0$ if $x \notin A$ and $K_A(x) := 1$ if $x \in A$.

Corollary 1.5.2 For any sets $A \subseteq X$,

- (i) A is decidable in X iff its characteristic function K_A is effectively calculable;
- (ii) if A is finite, then A is decidable in X .

Proof. For (i), the procedure for deciding whether or not $x \in A$ is to use the procedure to compute $K_A(x)$ and determine if the value is 1 or 0. If A is finite, then a decision procedure for deciding if $x \in A$ is simply to compare x with a list of the members of A . $\square$

The requirement that the elements of X be ‘finite objects’ is a natural one; before a procedure can begin to decide whether or not an object x has a property $\mathcal{P}$, it must “know” what x is. In the computer model it must be able to read the input. Since the whole process is to take finitely much time, this input must be finite. Objects like natural numbers and L -expressions

are clearly finite in this sense, whereas an irrational number like π is not. Furthermore, if the input x is a set, then to count as finite its elements, elements of elements etc. must also be finite. Formally, if all objects are represented as sets, then the requirement is that x be *hereditarily finite* — that is, its transitive closure is finite (see [Section 6.2](#)).

We want here to focus on some examples of decidable sets of L -expressions. We need to make the reasonable assumption that there is an effective procedure, given one of the L -symbols, to decide whether it is a connective, and if so which one, or a sentence symbol and which one. Many of the procedures we describe are *recursive* ones: in applying the procedure to an input x we are led to apply the same procedure to some finite number of other $x_0, \dots, x_k$ which are in some sense *smaller* than x — for example, if x is an L -expression, the x_i may be subexpressions — so that the whole process terminates in finitely many steps. The Euclidean Algorithm mentioned above is a recursive algorithm; to find the greatest common divisor d of a pair of positive integers (a, b) with $a > b$, we find q such that $a = bq + c$ with $c < b$ and see that d is also the greatest common divisor of (b, c) .

Theorem 1.5.3 *The set Sent_L of L -sentences is decidable in the set Expr_L of L -expressions.*

Proof. We may apply the following algorithm to a given L -expression ϕ . First determine the length ℓ of ϕ by counting the symbols. If $\ell = 1$, then ϕ is a sentence iff the unique symbol is one of the sentence symbols. If $\ell > 1$, then ϕ is not a sentence unless the first symbol, say s_0 , is one of the connectives $\neg$ or $\vee$ (remember that we are treating the other connectives as abbreviations now). Let $s_1 \dots s_{l-1}$ be the remaining symbols of ϕ . If s_0 is $\neg$, then ϕ is a sentence iff the shorter expression $s_1 \dots s_{l-1}$ is a sentence, so apply the algorithm recursively to this expression. If s_0 is $\vee$, then ϕ is a sentence iff $l > 2$ and for some i with $0 < i < l - 1$, both of the expressions $s_1 \dots s_i$ and $s_{i+1} \dots s_{l-1}$ are sentences, and we apply the algorithm recursively to all of these finitely many expressions. $\square$

In dealing with effectiveness in the semantics of L a basic first step is effective calculation of $V(\phi)$ from V and ϕ . On the face of it a truth assignment V is an infinite object — it assigns truth values to all of the infinitely many L -sentences — and thus is not an appropriate input for an effective procedure. However, by Proposition 1.3.1 we know that the value $V(\phi)$ is really determined by only finitely much information about V , namely its values on the sentence symbols which occur in ϕ . When we write V as an input below, it is really this finite function that is meant.

Lemma 1.5.4 *There is an effective procedure which, given an L -sentence ϕ and a truth assignment V , calculates the value $V(\phi)$.*

Proof. This is essentially just the calculation of a line in the truth table for ϕ and may be described as a recursive procedure. If ϕ is atomic, then $V(\phi)$ is determined directly from the input. If ϕ is $\neg\psi$ for some shorter sentence ψ , then this ψ may be effectively determined and $V(\phi)$ calculated as the opposite of $V(\psi)$. Otherwise, ϕ is $\psi \vee \theta$ for some unique shorter sentences ψ and θ . Then as in the preceding proof, ψ and θ may be effectively determined and $V(\phi)$ may be calculated in the obvious way from $V(\psi)$ and $V(\theta)$. $\square$

Theorem 1.5.5 *The set of tautologies is a decidable subset of the set of L -sentences.*

Proof. Given an L -sentence ϕ , we may examine its symbols and generate a finite list $q_0, \dots, q_n$ of the sentence symbols which occur in ϕ . There are then 2^{n+1} distinct functions which assign values **T** or **F** to these sentence symbols. For each of these by the lemma we may calculate the value assigned to ϕ . Then ϕ is a tautology iff in each of the 2^{n+1} cases, ϕ is assigned the value **T**. $\square$

Corollary 1.5.6 *For any finite set Γ of sentences, the propositional theory $\text{Th}(\Gamma) = \{\phi : \Gamma \models \phi\}$ generated by Γ is decidable. In fact, there is a single effective procedure which, given a finite Γ and a sentence ϕ , decides whether or not $\Gamma \models \phi$.*

Proof. Recall that if $\bigwedge \Gamma$ denotes the conjunction of the elements of Γ , then

$$\Gamma \models \phi \quad \Longleftrightarrow \quad \models \bigwedge \Gamma \rightarrow \phi.$$

Thus given Γ and ϕ , we first form the sentence $\bigwedge \Gamma \rightarrow \phi$ and then apply the algorithm of the preceding proof. $\square$

Remark 1.5.7 Note that the second assertion of the corollary really is stronger than the first. The first simply claims that for each finite Γ there is an algorithm, call it $\mathcal{A}_\Gamma$, which, given ϕ , decides whether or not $\Gamma \models \phi$. The second gives a single algorithm which, given both Γ and ϕ , makes this decision. One sometimes describes this by saying that $\{\phi : \Gamma \models \phi\}$ is decidable *uniformly* in Γ . This is equivalent to the existence of the individual algorithms $\mathcal{A}_\Gamma$ together with another procedure which, given Γ , calculates a usable description of $\mathcal{A}_\Gamma$.

Another way of stating the non-uniform version of this result is the following.

Definition 1.5.8 A propositional theory T is called *finitely axiomatizable* iff $T = \text{Th}(\Gamma)$ for some finite set Γ of sentences.

Corollary 1.5.9 *Every finitely axiomatizable propositional theory is decidable. $\square$*

We want next to ask what can be said for theories which are not finitely axiomatizable. Of course, the uniform version is no longer an option, since an infinite Γ is not a suitable input for an algorithm, but it still might happen that for some or all infinite sets Γ of L -sentences, the set $\text{Th}(\Gamma)$ is decidable. In fact we can easily see that not all propositional theories are decidable. Recall that we constructed above uncountably many propositional theories $\text{Th}(V)$. On the other hand,

Proposition 1.5.10 *For any set X , there are only denumerably many decidable subsets of X .*

Proof. We argue that there are altogether only denumerably many effective procedures. Indeed, since any such procedure can be described by a finite set of instructions written in some language, say English, there are at most as many procedures as there are finite sequences of the symbols used to write English. Since there are at most countably many such symbols, there are only denumerably many finite sequences of them and hence only denumerably many effective procedures. $\square$

Corollary 1.5.11 *There exist undecidable propositional theories.*

Proof. Corollary 1.4.12. $\square$

The question at hand now is to what extent can we characterize the axiom sets Γ such that the propositional theory $\text{Th}(\Gamma)$ that it generates is decidable? A natural first conjecture is that if Γ is decidable, then so is $\text{Th}(\Gamma)$. This turns out not to be true; we sketch a proof in the Hint to an exercise below. To analyze the situation further, we need a new concept.

Definition 1.5.12 A set X of finite objects is **effectively enumerable** iff X is empty or there exists an effectively calculable function $F : \omega \rightarrow X$ which enumerates X — that is, $X = \text{Im}(F) := \{ F(n) : n \in \omega \}$, the image of F .

Note that we do not require that F be an injective function, so the enumeration of X may contain repetitions; indeed, if X is finite there will necessarily be infinitely many repetitions. It is useful to think of an effectively enumerable set as one for which there is an infinitely long procedure for generating a complete list of its elements. In finite time the procedure will generate a list of any finite part of the set. An even more useful characterization is the following. Consider a generalized notion of effective procedure called an **effective semi-procedure** obtained by relaxing the requirement that the procedure halt and produce an output for every input; on some inputs the procedure may calculate forever without stopping or giving an output.

Definition 1.5.13 For any sets $A \subseteq X$, A is **semi-decidable in X** iff there exists an effective semi-procedure (**semi-decision procedure**) which when applied to an element x of X halts and produces the output ‘yes’ if $x \in A$ and otherwise calculates forever with no output.

Note that if we think of the procedure as defining a function, then A is exactly the domain of the function.

Proposition 1.5.14 For any sets A and X with $A \subseteq X$,

- (i) if A is decidable in X , then A is semi-decidable in X ;
- (ii) if A is effectively enumerable, then A is semi-decidable in X ;
- (iii) if both A and $X \sim A$ are semi-decidable in X , then A is decidable in X ;
- (iv) if X is effectively enumerable and A is semi-decidable in X , then A is effectively enumerable.

Proof. For (i), from a decision procedure for A we construct a semi-decision procedure by simply adding an instruction to enter an infinite loop whenever the original procedure is about to produce an output ‘no’. For (ii), suppose that A is effectively enumerable. A semi-decision procedure for A in X proceeds as follows: given $x \in X$, begin the listing of A ; if and when the element x appears in the list, halt with output ‘yes’. For (iii), from semi-decision procedures for both A and $X \sim A$, we obtain a decision procedure for A by applying the two procedures in parallel until one produces an output ‘yes’. The final output is ‘yes’ if this comes from the A procedure and otherwise ‘no’.

Finally, for (iv), under the given hypothesis and assuming that A is nonempty, we generate a listing of A as follows. First start generating a listing of X , say $x_0, \dots, x_n, \dots$. At the same time (“in parallel”) begin applying the semi-decision procedure for A to the elements of the list in the following way: carry out 1 step of the procedure with input x_0 ; then 2 steps of the procedure with each of inputs x_0 and x_1 ; then 3 steps with each of inputs x_0, x_1 and x_2 ; and so on. Whenever an output ‘yes’ is obtained with input x_i , add x_i to the list of elements of A . Clearly our list will contain only elements of A , and since for each $x \in X$ eventually any finite number of steps of the procedure will be calculated with input x , every element of A will eventually be detected and put on the list. To ensure that the list is infinite as required by the definition, after each 10 steps, repeat the first element on the list.

It is worth noting that under the stronger hypothesis that A is *decidable* in X , the procedure for (iv) is much simpler: as the elements of X are enumerated, apply the decision procedure for A to each in turn and discard those which are not in A . $\square$

Corollary 1.5.15 *For any effectively enumerable set X and any $A \subseteq X$, A is effectively enumerable iff A is semi-decidable in X . $\square$*

To apply these results to sentences we need to establish that the set of L -sentences is effectively enumerable. This will follow from the following useful general principle.

Proposition 1.5.16 *For any set X , if X is effectively enumerable, then so is the set $X^{<\omega}$ of all finite sequences of elements of X .*

Proof. Let $\langle x_n : n \in \omega \rangle$ be an effective enumeration of X . For each n , let Y_n be the set of all sequences of length at most n with components drawn from among $x_0, \dots, x_{n-1}$. Thus $Y_0 = \emptyset$,

$$Y_1 = \{(x_0)\};$$

$$Y_2 = Y_1 \cup \{(x_1), (x_0, x_0), (x_0, x_1), (x_1, x_0), (x_1, x_1)\};$$

$$Y_3 = Y_2 \cup \{(x_2), (x_0, x_2), (x_1, x_2), (x_2, x_0), (x_2, x_1), (x_2, x_2)\} \\ \cup \{(x_0, x_0, x_0), (x_0, x_0, x_1) \dots\}.$$

The new elements of Y_n are ordered first by length, then **lexicographically** by index (as indicated). Clearly $X^{<\omega} = \bigcup_{n \in \omega} Y_n$, so concatenating the enumerations of the Y_n in the order described gives an effective enumeration of $X^{<\omega}$. $\square$

Corollary 1.5.17 *The set Sent_L of L -sentences is effectively enumerable.*

Proof. Clearly $\neg, \vee, p_0, p_1, \dots, p_n, \dots$ is an effective enumeration of the L -symbols, so by the proposition, the set of all L -expressions is effectively enumerable. Then by Theorem 1.5.3 and Proposition 1.5.14, also Sent_L is effectively enumerable. $\square$

The fundamental result about effective enumerability of propositional theories is

Propositional Enumerability Theorem 1.5.18 (PET) *For any $\Gamma \subseteq \text{Sent}_L$, if Γ is effectively enumerable (in particular if Γ is decidable), then $\text{Th}(\Gamma)$ is semi-decidable and hence effectively enumerable.*

Proof. Let $\theta_0, \dots, \theta_n, \dots$ be an effective listing of Γ . By the Propositional Compactness Theorem, for any L -sentence ϕ ,

$$\Gamma \models \phi \quad \Longleftrightarrow \quad \text{for some } n \in \omega, \quad \{\theta_0, \dots, \theta_n\} \models \phi.$$

By Corollary 1.5.6 — here the “uniform” procedure is needed — we may successively test whether or not $\{\theta_0\} \models \phi$, $\{\theta_0, \theta_1\} \models \phi$, $\dots$. If an output ‘yes’ is encountered, the procedure halts and outputs ‘yes’; otherwise it continues. This is clearly a semi-decision procedure for $\text{Th}(\Gamma)$. $\square$

Again we may reformulate this result in terms of the theory:

Definition 1.5.19 A propositional theory T is called *decidably axiomatizable* iff $T = \text{Th}(\Gamma)$ for some decidable set Γ of sentences.

Corollary 1.5.20 Every decidably axiomatizable propositional theory is effectively enumerable. $\square$

To achieve the stronger conclusion of decidability, we need an additional condition.

Theorem 1.5.21 For any propositional theory T , if T is decidably axiomatizable and complete, then T is decidable.

Proof. Since T is complete, for all ϕ we have $\phi \notin T \iff \neg\phi \in T$. Thus from a semi-decision procedure for T , we have one also for $\text{Sent}_L \sim T$: to semi-decide if $\phi \notin T$, apply the given procedure to $\neg\phi$. Thus both T and $\text{Sent}_L \sim T$ are semi-decidable, hence by Proposition 1.5.14(iii), T is decidable. $\square$

In later chapters and other contexts, we shall use (the analogue of) this result in the following contrapositive forms:

Corollary 1.5.22 For any propositional theory T ,

- (i) if T is complete and undecidable, then T is not decidably axiomatizable;
- (ii) if T is decidably axiomatizable and undecidable, then T is incomplete. $\square$

Collateral Reading. In most logic texts the informal notions of effective calculability and decidability are introduced only as a brief prelude to the more formal definition of the recursive (computable) functions and relations — for example, [Eb-Fl-Th] and [Sh01b]. This approach leads more quickly to precise mathematical results but imposes some technical barriers to the direct intuitive comprehension of these ideas. We have chosen here to use the informal notions through Chapter 3 and Section 4.1 with the development of recursiveness and its application to decidability coming in Sections 4.2–4.4. A similar approach is in [En01]. The reader who is unsatisfied with this informality might read now parts of Chapter 4 or any of many developments of recursion (computability) theory in the literature — for example, Enderton’s [Ba77; Section C1] or parts of the classic [Rog] or [Cut].

Exercises

Exercise 1.5.23 Fix a set X of finite objects and $A, B \subseteq X$.

- (i) Prove that if A and B are decidable in X , then so are $A \cup B$, $A \cap B$ and $A \sim B$;

(ii) which of these operations also preserve semi-decidability?

Exercise 1.5.24 Prove that a set $A \subseteq \omega$ is semi-decidable in ω iff there exists a decidable $R \subseteq \omega \times \omega$ such that for all $m \in \omega$,

$$m \in A \leftrightarrow \exists n (m, n) \in R.$$

Exercise 1.5.25 A set $A \subseteq \omega$ is effectively enumerable in increasing order iff for some effectively calculable function $F : \omega \rightarrow \omega$, $A = \text{Im}(F)$ and for all $m < n$, $F(m) < F(n)$. Show that such an A is actually decidable.

Exercise 1.5.26 Show that for any effectively enumerable set Γ of L -sentences there exists a decidable set Γ^* which is tautologically equivalent to Γ . Conclude that a propositional theory is effectively enumerable iff it is axiomatizable by an effectively enumerable set of axioms iff it is decidable axiomatizable.

Hint: Convert Γ into a tautologically equivalent set Γ^* which is in some sense effectively enumerable in increasing order, and use the method of the preceding exercise.

Exercise 1.5.27 Show that for any sets X and $Y \subseteq Z$ and any effectively calculable function $F : X \rightarrow Z$, if $W = F^{-1}(Y) := \{x \in X : F(x) \in Y\}$, then

- (i) if Y is decidable in Z , then W is decidable in X ;
- (ii) if X and Y are effectively enumerable, then also W is effectively enumerable.

Exercise 1.5.28 Show that for any truth assignment V , $\text{Th}(V)$ is decidable iff $\{n : V(p_n) = \top\}$ is a decidable set of natural numbers.

Exercise 1.5.29 Assume that there exists a set $A \subseteq \omega$ which is effectively enumerable but not decidable (this will be proved in [Section 4.6](#)). Prove that there is a decidable axiomatizable propositional theory T which is not decidable.

Hint: If A is the image of the effectively calculable function F , let T be the theory generated by the set of axioms

$$\Gamma := \{p_{2n} : n \in \omega\} \cup \{p_{2n} \rightarrow p_{2F(n)+1} : n \in \omega\}.$$

Show that Γ is decidable and use the results of the two preceding exercises.

Exercise 1.5.30 Show that if T is decidable, then so is any *finite extension* — that is, any theory of the form $\text{Th}(T \cup \Delta)$ for a finite set Δ of sentences.

Exercise 1.5.31 Show that there is an effectively calculable function DNF such that for any L -sentence ϕ , $\text{DNF}(\phi)$ is a disjunctive normal form of ϕ (or $\mathbf{F}$ if ϕ is tautologically false).

Exercise 1.5.32 For any function $F : X \rightarrow Z$, the *graph* of F is the set

$$\text{Gr}_F = \{ (x, z) : F(x) = z \}.$$

Show that

- (i) if F is effectively calculable, then Gr_F is decidable in $X \times Z$;
- (ii) if Gr_F is effectively enumerable, then F is effectively calculable;
- (iii) if Z (but not necessarily X) is effectively enumerable, and Gr_F is semi-decidable in $X \times Z$, then F is effectively calculable.

1.6 Other constructions

The proof of the Alternative Propositional Compactness Theorem in [Section 1.4](#) is typical of a large class of proofs in logic and indeed in mathematics as a whole — one has some properties and proves that there exists a mathematical object which possesses all of the properties by describing such an object more or less explicitly. Such a proof is usually called a *construction*. We shall often be concerned in the coming chapters with constructions of models for first-order and other theories; we present in this section versions of two of these constructions for propositional logic that contain many of the central ideas, which are more easily comprehended in this simpler setting. However, the reader may find it easier to skip this section on first reading and return to it when the results are needed.

Notions of consistency

The first construction is a generalization of the proof of the Alternative Propositional Compactness Theorem. There we constructed a model for a given finitely consistent set Γ of sentences; here we abstract the properties of finite consistency which are necessary to make the proof work and prove a more general theorem.

Definition 1.6.1 A *propositional notion of consistency* is a set $\mathbb{C}$ of sets of sentences such that if $\Theta \subseteq \Theta' \in \mathbb{C}$, then $\Theta \in \mathbb{C}$, and for all $\Theta \in \mathbb{C}$,

- (i) for any sentence ψ ,
 - (a) $(\neg \psi) \in \Theta \implies \psi \notin \Theta$;
 - (b) $(\neg \neg \psi) \in \Theta \implies \Theta \cup \{\psi\} \in \mathbb{C}$;
- (ii) for any sentences ψ and θ ,
 - (a) $(\psi \vee \theta) \in \Theta \implies$ either $\Theta \cup \{\psi\} \in \mathbb{C}$ or $\Theta \cup \{\theta\} \in \mathbb{C}$;
 - (b) $\neg(\psi \vee \theta) \in \Theta \implies$ both $\Theta \cup \{\neg \psi\} \in \mathbb{C}$ and $\Theta \cup \{\neg \theta\} \in \mathbb{C}$.

Elements of $\mathbb{C}$ are called **$\mathbb{C}$ -consistent**. We note first that this really *is* a generalization:

Examples 1.6.2

- (i) $\{\Theta : \Theta \text{ is consistent}\}$ is a notion of consistency.
- (ii) $\{\Theta : \Theta \text{ is finitely consistent}\}$ is a notion of consistency. Properties (i) and (ii)(b) are immediate. For (ii)(a), suppose that Θ is finitely consistent, $(\psi \vee \theta) \in \Theta$, but neither $\Theta \cup \{\psi\}$ nor $\Theta \cup \{\theta\}$ is finitely consistent. Then there exist finite $\Theta_0, \Theta_1 \subseteq \Theta$ such that $\Theta_0 \cup \{\psi\}$ and $\Theta_1 \cup \{\theta\}$ are inconsistent and $\Theta_0 \cup \Theta_1 \cup \{\psi \vee \theta\}$ is a finite inconsistent subset of Θ , a contradiction.

Of course, given the compactness theorem, these are the same example, but (ii) is used below to give an alternative proof of this theorem.

Definition 1.6.3 A set Δ of sentences is *weakly complete* iff

- (i) for any sentence ψ ,
 - (a) $(\neg \psi) \in \Delta \implies \psi \notin \Delta$;
 - (b) $(\neg \neg \psi) \in \Delta \implies \psi \in \Delta$;
- (ii) for any sentences ψ and θ ,
 - (a) $(\psi \vee \theta) \in \Delta \implies$ either $\psi \in \Delta$ or $\theta \in \Delta$;
 - (b) $\neg(\psi \vee \theta) \in \Delta \implies$ both $(\neg \psi) \in \Delta$ and $(\neg \theta) \in \Delta$.

We note that by Lemma 1.4.28 every complete and finitely consistent set of sentences is weakly complete. Indeed, the main difference from completeness is that for a weakly complete Δ it is generally not the case that $\psi \notin \Delta \implies (\neg \psi) \in \Delta$ (see the remark below).

The main part of the construction consists in the following two propositions.

Proposition 1.6.4 *For any propositional notion of consistency $\mathbb{C}$, every $\mathbb{C}$ -consistent set Θ may be extended to a weakly complete set Δ .*

Proof. Let $\langle \theta_n : n \in \omega \rangle$ be an enumeration of Sent_L , $\mathbb{C}$ a notion of consistency and Θ a $\mathbb{C}$ -consistent set. Set $\Delta_0 := \Theta$ and

$$\Delta_{n+1} := \begin{cases} \Delta_n \cup \{\theta_n\}, & \text{if this set is } \mathbb{C}\text{-consistent;} \\ \Delta_n, & \text{otherwise;} \end{cases}$$

$$\Delta := \bigcup \{ \Delta_n : n \in \omega \}.$$

We claim that Δ is weakly complete. First, a straightforward induction verifies that for all n , Δ_n is $\mathbb{C}$ -consistent. Then condition (i)(a) is immediate, since if both $(\neg \psi), \psi \in \Delta$, then for some n both $(\neg \psi), \psi \in \Delta_n$, contrary to the corresponding clause in the definition of notion of consistency. For (i)(b), suppose that $(\neg \neg \psi) \in \Delta$, say $(\neg \neg \psi) \in \Delta_n$ and choose p such that θ_p is ψ . If $n \leq p$, then $(\neg \neg \psi) \in \Delta_p$, so $\Delta_p \cup \{\psi\}$ is $\mathbb{C}$ -consistent and thus $\psi \in \Delta_{p+1} \subseteq \Delta$. If $p < n$, then $\Delta_p \cup \{\psi\} \subseteq \Delta_n \cup \{\psi\} \in \mathbb{C}$ so is again $\mathbb{C}$ -consistent, and we conclude that $\psi \in \Delta$. The proof of (ii) is similar and is left to the reader. $\square$

Proposition 1.6.5 *Every weakly complete set has a model.*

Proof. Let Δ be weakly complete and V the unique truth assignment such that for all n ,

$$V(p_n) = \begin{cases} \text{T}, & \text{if } p_n \in \Delta; \\ \text{F}, & \text{otherwise.} \end{cases}$$

We prove by Sent_L -induction that for all ϕ

$$\phi \in \Delta \implies V(\phi) = \text{T} \quad \text{and} \quad (\neg \phi) \in \Delta \implies V(\phi) = \text{F}.$$

Thus in particular, V is a model of Δ . For ϕ atomic this is clear, since by (i)(a), if $(\neg p_n) \in \Delta$, then $p_n \notin \Delta$ so $V(p_n) = \text{F}$. Suppose that ϕ is $\neg \psi$ and that the implications hold for ψ . Then by the induction hypothesis,

$$\phi \in \Delta \implies (\neg \psi) \in \Delta \implies V(\psi) = \text{F} \implies V(\phi) = \text{T},$$

and by property (i)(b)

$$\begin{aligned} (\neg \phi) \in \Delta &\implies (\neg \neg \psi) \in \Delta \implies \psi \in \Delta \\ &\implies V(\psi) = \text{T} \implies V(\phi) = \text{F}. \end{aligned}$$

If ϕ is $\psi \vee \theta$, we have by property (ii),

$$\begin{aligned} \phi \in \Delta \quad \implies \quad (\psi \vee \theta) \in \Delta \quad \implies \quad & \left\{ \begin{array}{c} \psi \in \Delta \implies V(\psi) = \top \\ \text{or} \\ \theta \in \Delta \implies V(\theta) = \top \end{array} \right\} \\ & \implies V(\phi) = \top, \end{aligned}$$

and

$$\begin{aligned} (\neg \phi) \in \Delta \quad \implies \quad \neg(\psi \vee \theta) \in \Delta \quad \implies \quad & \neg \psi, \neg \theta \in \Delta \\ \implies \quad V(\psi) = \text{F} = V(\theta) \quad \implies \quad & V(\phi) = \text{F}. \quad \square \end{aligned}$$

Combining these two propositions we have

Propositional Model Existence Theorem 1.6.6 *For any propositional notion of consistency $\mathbb{C}$, every $\mathbb{C}$ -consistent set has a model. $\square$*

The Alternative Propositional Compactness Theorem is an immediate consequence of this corollary together with Example 1.6.2(ii). We illustrate further the use of this construction by applying it to the following theorem, which is a generalization of the result of Exercise 1.3.24. For any set Γ of sentences, let

$$\text{At}(\Gamma) := \{p_n : p_n \text{ occurs in some member of } \Gamma\} = \bigcup \{ \text{At}(\phi) : \phi \in \Gamma \}.$$

Propositional Robinson Interpolation Theorem 1.6.7 *For any consistent sets Γ_0 and Γ_1 of sentences, if $\Gamma_0 \cup \Gamma_1$ is inconsistent, then there exists a sentence χ such that $\Gamma_0 \models \chi$, $\Gamma_1 \models \neg \chi$ and $\text{At}(\chi) \subseteq \text{At}(\Gamma_0) \cap \text{At}(\Gamma_1)$. χ is called a **Robinson interpolant** for Γ_0 and Γ_1 .*

Proof. Fix a pair Γ_0 and Γ_1 of consistent sets. Let

$$\begin{aligned} \mathbb{C} := \{ \Theta_0 \cup \Theta_1 : \Theta_0, \Theta_1 \text{ are consistent, } \text{At}(\Theta_i) \subseteq \text{At}(\Gamma_i) \ (i = 0, 1) \\ \text{and } (\Theta_0, \Theta_1) \text{ has no interpolant } \chi \text{ with} \\ \text{At}(\chi) \subseteq \text{At}(\Gamma_0) \cap \text{At}(\Gamma_1) \}. \end{aligned}$$

We shall prove that $\mathbb{C}$ is a propositional notion of consistency; then by Theorem 1.6.6,

$$\Gamma_0 \cup \Gamma_1 \text{ has no interpolant} \implies \Gamma_0 \cup \Gamma_1 \in \mathbb{C} \implies \Gamma_0 \cup \Gamma_1 \text{ has a model,}$$

which is just what is claimed.

Clearly $\mathbb{C}$ is closed under subset. For (i)(a), suppose that for some ψ both $\psi, (\neg\psi) \in \Theta_0 \cup \Theta_1 \in \mathbb{C}$; we reach a contradiction by showing that (Θ_0, Θ_1) has an interpolant. Since Θ_0 and Θ_1 are individually consistent, it is not the case that both ψ and $\neg\psi$ belong to the same Θ_i . Hence either $\psi \in \Theta_0$ and $(\neg\psi) \in \Theta_1$ or $(\neg\psi) \in \Theta_0$ and $\psi \in \Theta_1$. In either case $\text{At}(\psi) \subseteq \text{At}(\Theta_0) \cap \text{At}(\Theta_1)$ and in the first case ψ is an interpolant and in the second case $\neg\psi$ is an interpolant.

For (i)(b), if $(\neg\neg\psi) \in \Theta_0$, then Θ_0 and $\Theta_0 \cup \{\psi\}$ are tautologically equivalent and have the same sentence symbols, so if $\Theta_0 \cup \Theta_1 \in \mathbb{C}$, also $\Theta_0 \cup \{\psi\} \cup \Theta_1 \in \mathbb{C}$. Similarly if $(\neg\neg\psi) \in \Theta_1$.

For (ii)(a), suppose that $(\psi \vee \theta) \in \Theta_0$, with $\Theta_0 \cup \Theta_1 \in \mathbb{C}$, but both $\Theta_0 \cup \Theta_1 \cup \{\psi\} \notin \mathbb{C}$ and $\Theta_0 \cup \Theta_1 \cup \{\theta\} \notin \mathbb{C}$. It follows that there exist interpolants χ for $(\Theta_0 \cup \{\psi\}, \Theta_1)$ and χ' for $(\Theta_0 \cup \{\theta\}, \Theta_1)$, whence easily $\chi \vee \chi'$ is an interpolant for (Θ_0, Θ_1) , contrary to hypothesis. The proof of (ii)(b) is similar to that of (i)(b). $\square$

Remark 1.6.8 Note that if we apply the construction above of a weakly complete extension of a member of the notion of consistency $\mathbb{C}$ from the preceding proof, the resulting set Δ is certainly *not* in general complete, since it contains only sentences ϕ with $\text{At}(\phi) \subseteq \text{At}(\Gamma_0)$ or $\text{At}(\phi) \subseteq \text{At}(\Gamma_1)$.

Corollary 1.6.9 (Propositional Interpolation Theorem) *For any sentences ϕ and ψ , if $\phi \models \psi$ but neither $\models \neg\phi$ nor $\models \psi$ (so $\phi \models \psi$ does not hold for trivial reasons), there exists a sentence χ such that*

- (i) $\phi \models \chi$ and $\chi \models \psi$;
- (ii) $\text{At}(\chi) \subseteq \text{At}(\phi) \cap \text{At}(\psi)$.

Proof. Apply the preceding theorem with $\Gamma_0 = \{\phi\}$ and $\Gamma_1 = \{\neg\psi\}$. $\square$

Ultraproducts

We turn now to our second construction, the ultraproduct, and again use the Alternative Compactness Theorem for motivation. Suppose that a set Γ of sentences is finitely consistent and for each finite set $i \subseteq \Gamma$, V_i is a truth assignment which is a model of i . To get a single model V for all of Γ , we might try somehow to combine all of these V_i with the general idea that $V(\phi)$ be assigned the value **T** or **F** which occurs ‘most often’ among the values $V_i(\phi)$ as i varies over the finite subsets of Γ . In other words, if we set

$$I_\Gamma = \wp_{<\omega}(\Gamma) = \{i : i \text{ is a finite subset of } \Gamma\},$$

and for each sentence ϕ ,

$$X_\phi = \{i \in I_\Gamma : V_i(\phi) = \mathbf{T}\},$$

then we want a criterion for deciding that exactly one of X_ϕ and $X_{\neg\phi} = I_\Gamma \sim X_\phi$ is ‘large’. Then we will set

$$V(\phi) = \begin{cases} \text{T,} & \text{if } X_\phi \text{ is ‘large’;} \\ \text{F,} & \text{otherwise.} \end{cases}$$

The notion of an ultrafilter is a mechanism for carrying out this idea. Of course generally both X_ϕ and $I_\Gamma \sim X_\phi$ will be countably infinite, so neither is larger than the other in the sense of cardinality; our division must be more subtle. Further, since we want V to be a truth assignment, it must also respect $\neg$ and $\vee$, so that we must have in addition

$$\begin{aligned} X_{\neg\psi} = I_\Gamma \sim X_\psi \text{ is ‘large’} &\iff X_\psi \text{ is not ‘large’} \\ X_{\psi \vee \theta} = X_\psi \cup X_\theta \text{ is ‘large’} &\iff X_\psi \text{ is ‘large’ or } X_\theta \text{ is ‘large’}. \end{aligned}$$

Finally, if V is to be a model of Γ , we must arrange that for each $\phi \in \Gamma$, X_ϕ is ‘large’. The following definition achieves these goals.

Definition 1.6.10 For any set I , a nonempty set $\mathcal{U} \subseteq \wp(I)$ is a **filter over I** iff for all $X, Y \subseteq I$,

- (i) $\emptyset \notin \mathcal{U}$;
- (ii) if $X \in \mathcal{U}$ and $X \subseteq Y$, then $Y \in \mathcal{U}$;
- (iii) if $X \in \mathcal{U}$ and $Y \in \mathcal{U}$, then $X \cap Y \in \mathcal{U}$.

$\mathcal{U}$ is called an **ultrafilter over I** iff in addition for all $X \subseteq I$,

- (iv) $I \sim X \in \mathcal{U}$ iff $X \notin \mathcal{U}$.

The idea is that any ultrafilter $\mathcal{U}$ specifies a notion of ‘largeness’: $X \subseteq I$ is ‘large’ $_{\mathcal{U}}$ iff $X \in \mathcal{U}$. Property (iv) dictates that for each $X \subseteq I$, exactly one of X and $I \sim X$ is ‘large’ $_{\mathcal{U}}$. It is easy to see that there are filters over any set.

Example 1.6.11

- (i) For any fixed $i \in I$, $\{X \subseteq I : i \in X\}$ is an ultrafilter over I , called the **principal ultrafilter** generated by i .
- (ii) A subset $X \subseteq I$ is called **cofinite** iff $I \sim X$ is finite. If I is infinite, the cofinite subsets of I form a filter; however, this is never an ultrafilter.

Lemma 1.6.12 For any infinite set I , any ultrafilter $\mathcal{U}$ over I and any $X, Y \subseteq I$,

$$X \cup Y \in \mathcal{U} \iff X \in \mathcal{U} \text{ or } Y \in \mathcal{U}.$$

Proof. The implication $(\Leftarrow)$ is immediate from (ii) of the definition. Conversely, suppose that $X \cup Y \in \mathcal{U}$, but $X \notin \mathcal{U}$ and $Y \notin \mathcal{U}$. Then by (iv)

$I \sim X \in \mathcal{U}$ and $I \sim Y \in \mathcal{U}$, so by (iii) $I \sim (X \cup Y) \in \mathcal{U}$ and thence $\emptyset = (X \cup Y) \cap (I \sim (X \cup Y)) \in \mathcal{U}$, contrary to (i). $\square$

The similarity of this result with Lemma 1.4.28(ii) will be exploited at the end of the following section.

Definition 1.6.13 For any infinite set I , any ultrafilter $\mathcal{U}$ over I and any family $\langle V_i : i \in I \rangle$ of truth assignments, $V_{\mathcal{U}}$ is the function defined by

$$V_{\mathcal{U}}(\phi) = \begin{cases} \text{T}, & \text{if } X_{\phi} \in \mathcal{U}; \\ \text{F}, & \text{otherwise.} \end{cases}$$

$V_{\mathcal{U}}$ is called the **ultraproduct** of the family $\langle V_i : i \in I \rangle$ with respect to $\mathcal{U}$. (The first-order version of this construction in Section 3.3 will explain the term “product”).

Proposition 1.6.14 *Under the conditions of this definition, $V_{\mathcal{U}}$ is a truth assignment.*

Proof. Since $X_{\neg \phi} = I \sim X_{\phi}$, we have

$$V_{\mathcal{U}}(\neg \phi) = \text{T} \iff X_{\neg \phi} \in \mathcal{U} \iff X_{\phi} \notin \mathcal{U} \iff V_{\mathcal{U}}(\phi) = \text{F};$$

thus $V_{\mathcal{U}}$ respects $\neg$. Similarly, since $X_{\psi \vee \theta} = X_{\psi} \cup X_{\theta}$, using the lemma, we have

$$\begin{aligned} V_{\mathcal{U}}(\psi \vee \theta) = \text{T} &\iff X_{\psi} \in \mathcal{U} \quad \text{or} \quad X_{\theta} \in \mathcal{U} \\ &\iff V_{\mathcal{U}}(\psi) = \text{T} \quad \text{or} \quad V_{\mathcal{U}}(\theta) = \text{T}. \quad \square \end{aligned}$$

Of course, we are aiming to construct truth assignments with special properties — models of a given set Γ , for example. Unfortunately, principal ultrafilters will not do us much good, since if $\mathcal{U}$ is the principal ultrafilter generated by i , then for all ϕ ,

$$V_{\mathcal{U}}(\phi) = \text{T} \iff X_{\phi} \in \mathcal{U} \iff i \in X_{\phi} \iff V_i(\phi) = \text{T},$$

so $V_{\mathcal{U}}$ is the same as V_i . To ensure that $V_{\mathcal{U}}$ is a model of Γ , we need to arrange that for each $\phi \in \Gamma$, $X_{\phi} \in \mathcal{U}$. The following gives a general method for constructing ultrafilters.

Definition 1.6.15 For any infinite set I , a collection $\mathcal{X}$ of subsets of I has the **finite intersection property** iff for any finitely many $X_0, \dots, X_n \in \mathcal{X}$, $X_0 \cap \dots \cap X_n \neq \emptyset$.

Example 1.6.16 Clearly any filter has this property. For any set Γ of L -sentences, let

$$\mathcal{Y}_\Gamma = \{Y_\phi : \phi \in \Gamma\},$$

where $Y_\phi = \{i \in I_\Gamma : \phi \in i\}$. Since for any $\phi_0, \dots, \phi_n \in \Gamma$,

$$\{\phi_0, \dots, \phi_n\} \in Y_{\phi_0} \cap \dots \cap Y_{\phi_n},$$

$\mathcal{Y}_\Gamma$ has the finite intersection property.

Lemma 1.6.17 *For any infinite set I and any collection $\mathcal{X}$ of subsets of I with the finite intersection property, the set*

$$\mathcal{F} := \{Y \subseteq I : \text{for some } n \text{ and some } X_0, \dots, X_n \in \mathcal{X}, \\ X_0 \cap \dots \cap X_n \subseteq Y\}$$

*is a filter over I such that $\mathcal{X} \subseteq \mathcal{F}$. $\mathcal{F}$ is called the **filter generated by $\mathcal{X}$** .*

Proof. Each of the properties (i)–(iii) of the definition is easily verified by inspection. $\square$

Theorem 1.6.18 *For any set I and any $\mathcal{X} \subseteq \wp(I)$ with the finite intersection property, there exists an ultrafilter $\mathcal{U}$ over I such that $\mathcal{X} \subseteq \mathcal{U}$.*

Proof. Fix I and $\mathcal{X}$ as in the hypothesis. Let

$$\mathbb{F} = \{\mathcal{F} : \mathcal{F} \text{ is a filter over } I \text{ such that } \mathcal{X} \subseteq \mathcal{F}\}.$$

By the preceding lemma, $\mathbb{F} \neq \emptyset$; we shall apply Zorn's Lemma (see [Section 6.4](#)) to find a maximal element of $\mathbb{F}$, which will be an ultrafilter. To apply Zorn's Lemma, we need to verify the following: for any $\mathbb{A} \subseteq \mathbb{F}$, if $\mathbb{A}$ is linearly ordered by inclusion ($\subseteq$), then $\bigcup \mathbb{A} \in \mathbb{F}$. Given such an $\mathbb{A}$, it is elementary to check that properties (i)–(iii) hold for $\bigcup \mathbb{A}$ — for example, if $X, Y \in \bigcup \mathbb{A}$, say $X \in \mathcal{F} \in \mathbb{A}$ and $Y \in \mathcal{G} \in \mathbb{A}$, then either $\mathcal{F} \subseteq \mathcal{G}$ or $\mathcal{G} \subseteq \mathcal{F}$, and both X and Y , hence $X \cap Y$ belong to the larger of the two, hence to $\bigcup \mathbb{A}$.

Now by Zorn's Lemma, there exists a maximal element $\mathcal{U}$ of $\mathbb{F}$, and we check that $\mathcal{U}$ has additionally property (iv). Indeed, suppose this is not so and that for some $X \subseteq I$, both $X \notin \mathcal{U}$ and $I \sim X \notin \mathcal{U}$. Then easily $\mathcal{U} \cup \{X\}$ has the finite intersection property, so is included in some $\mathcal{F} \in \mathbb{F}$ contrary to the maximality of $\mathcal{U}$. $\square$

Now we can give the promised third (fourth counting Exercise 1.4.43)

Proof (of the Alternative Propositional Compactness Theorem). Let Γ be a finitely satisfiable set of sentences, I_Γ the set of finite subsets of Γ , and for each $i \in I_\Gamma$ choose a truth assignment V_i which satisfies i . By Example 1.6.16 and the preceding theorem, there exists an ultrafilter $\mathcal{U}$ over I_Γ such that $\mathcal{Y}_\Gamma \subseteq \mathcal{U}$. Since for each $\phi \in \Gamma$, $Y_\phi \in \mathcal{U}$ and easily $Y_\phi \subseteq X_\phi$ (as defined above), also $X_\phi \in \mathcal{U}$. Hence $V_\mathcal{U}$ is a truth assignment such that for each $\phi \in \Gamma$, $V_\mathcal{U}(\phi) = T$, and thus $V_\mathcal{U}$ is a model of Γ . $\square$

Remark 1.6.19 This proof is less elementary than our other proofs in that it uses the Axiom of Choice in the form of Zorn's Lemma. (A second apparent use of Choice in choosing V_i for each $i \in I$ may be avoided by taking V_i to be the first in the lexicographic ordering.)

Collateral Reading. The topics of this section and the next are often omitted from introductory texts. However, they are important for parts of [Chapter 3](#) and [Chapter 7](#) and again benefit from the simpler context of propositional logic for their initial development. Consistency properties were first introduced in [Kei] for infinitary logic — cf. [Section 3.7](#). Ultraproducts of first-order structures have played an important role in Model Theory and have been presented in many more advanced texts, for example [Ch-Ke], or Eklof's [Ba77; Section A.3]. Similarly, there is a large literature on Boolean algebras, both as a part of algebra and in relation to logic; [Mon; Chapter 9] or the beginning parts of [Kop] are good introductions.

Exercises

Exercise 1.6.20 Carry out the proof of condition (ii) for Proposition 1.6.4.

Exercise 1.6.21 Show that the Propositional Robinson Interpolation Theorem can easily be derived from its special case, the Propositional Interpolation Theorem.

Exercise 1.6.22 Let $\mathbb{C}$ be a notion of consistency, and for each $n \in \omega$ let Φ_n be a set of sentences such that for any $\Gamma \in \mathbb{C}$ there is some $\phi \in \Phi_n$ such that $\Gamma \cup \{\phi\} \in \mathbb{C}$. Prove that every $\mathbb{C}$ -consistent set has a model V such that for all n , $V(\bigvee \Phi_n) = \top$ — that is, for some $\phi \in \Phi_n$, $V(\phi) = \top$.

Exercise 1.6.23 Show that for any infinite set I ,

- (i) for any non-principal ultrafilter $\mathcal{U}$ over I , no finite subset of I belongs to $\mathcal{U}$, so $\mathcal{U}$ includes the cofinite filter (see Example 1.6.11);
- (ii) there exists a non-principal ultrafilter over I .

Exercise 1.6.24 Show that for any set I , the following are equivalent:

- (i) I is finite;
- (ii) there are at most finitely many ultrafilters over I ;
- (iii) every ultrafilter over I is principal.

Exercise 1.6.25 Let K be a non-empty set of truth assignments. K is called **axiomatizable** iff for some set Γ of sentences, $K = \text{Mod}(\Gamma)$ and **ultra-product closed** iff for any index set I , any family $\langle V_i : i \in I \rangle$ of members of K and any ultrafilter U over I , the ultraproduct V_U also belongs to K . Prove that these notions are equivalent:

$$K \text{ is axiomatizable} \quad \Longleftrightarrow \quad K \text{ is ultraproduct closed.}$$

Hint: For $(\Leftarrow)$, let $T = \text{Th}(K)$ and show that $\text{Mod}(T) \subseteq K$ (so by Corollary 1.4.16, $\text{Mod}(T) = K$). For any model V of T , let I be the set of finite subsets of $\text{Th}(V)$. For each $i \in I$ show that there is $V_i \in K \cap \text{Mod}(i)$ and find an appropriate ultraproduct of the V_i .

Exercise 1.6.26 Establish the following common generalization of the Propositional Lyndon and Robinson Interpolation Theorems, 1.3.26 and 1.6.7. For a set Γ of sentences, define

$$\text{At}^+(\Gamma) := \bigcup \{ \text{At}^+(\phi) : \phi \in \Gamma \} \quad \text{and} \quad \text{At}^-(\Gamma) := \bigcup \{ \text{At}^-(\phi) : \phi \in \Gamma \}.$$

Prove that for any consistent sets Γ_0 and Γ_1 of sentences, if $\Gamma_0 \cup \Gamma_1$ is inconsistent, then there exists a sentence χ such that $\Gamma_0 \models \chi$, $\Gamma_1 \models \neg \chi$, $\text{At}^+(\chi) \subseteq \text{At}^+(\Gamma_0) \cap \text{At}^-(\Gamma_1)$ and $\text{At}^-(\chi) \subseteq \text{At}^-(\Gamma_0) \cap \text{At}^+(\Gamma_1)$.

Hint: Show that the following set is a propositional notion of consistency:

$$\begin{aligned} \mathbb{C} = \{ \Theta_0 \cup \Theta_1 : \Theta_0, \Theta_1 \text{ are consistent, } \text{At}^\pm(\Theta_i) \subseteq \text{At}^\pm(\Gamma_i) \ (i = 0, 1) \\ \text{and } (\Theta_0, \Theta_1) \text{ has no interpolant } \chi \text{ with} \\ \text{At}^\pm(\chi) \subseteq \text{At}^\pm(\Gamma_0) \cap \text{At}^\mp(\Gamma_1) \}. \end{aligned}$$

1.7 Topology and Boolean algebra

In this section we present two further topics closely connected with propositional logic and especially the compactness theorem. Each provides yet another proof of this theorem, but the focus here is more on developing intuition rather than methods of proof. The ideas of the first part of this section will reappear in later chapters in examples and remarks, but are not crucial to the main development; Boolean algebra will play a central role in [Chapter 7](#). For this reason and to avoid too long a diversion, we shall be somewhat brief and leave many details for the interested reader to pursue with the help of the references. We also assume here of the reader some familiarity with elementary topology and abstract algebra.

Topology

The first topic is really an explanation of the name ‘Compactness Theorem’; we show that this theorem really asserts the compactness of a certain topological space. The underlying set of this space is $\mathbf{TA}$, the set of all truth assignments. A basis for the topology is the collection of all sets of the form

$$\mathcal{B}_\Gamma = \text{Mod}(\Gamma) = \{ V : V(\phi) = \mathbf{T} \text{ for all } \phi \in \Gamma \}$$

for each finite set Γ of sentences. Note that this collection is closed under finite intersection: $\mathcal{B}_{\Gamma_0} \cap \mathcal{B}_{\Gamma_1} = \mathcal{B}_{\Gamma_0 \cup \Gamma_1}$. Furthermore, since $\mathcal{B}_{\{\phi_0, \dots, \phi_n\}} = \mathcal{B}_{\{\phi_0 \wedge \dots \wedge \phi_n\}}$, every basis set is of the form $\mathcal{B}_{\{\phi\}}$ for a single sentence ϕ . It follows that the open sets (unions of basis sets) are exactly those of the form

$$\mathcal{O}_\Gamma = \{ V : V(\phi) = \mathbf{T} \text{ for some } \phi \in \Gamma \}$$

for any set Γ of sentences, and thus the closed sets are exactly the sets $\mathcal{B}_\Gamma$ for arbitrary (not only finite) Γ .

In the following it is useful to note the generalized version of the intersection formula above: for any collection $\mathcal{G}$ of sets of sentences,

$$\bigcap \{ \mathcal{B}_\Gamma : \Gamma \in \mathcal{G} \} = \mathcal{B}_{\bigcup \mathcal{G}}.$$

In particular, when $\mathcal{G} = \{ \{\phi\} : \phi \in \Gamma \}$, we have

$$\bigcap \{ \mathcal{B}_{\{\phi\}} : \phi \in \Gamma \} = \mathcal{B}_\Gamma.$$

The key fact that relates these considerations to the Compactness Theorem in its earlier forms is

$$\Gamma \text{ is satisfiable} \iff \mathcal{B}_\Gamma \neq \emptyset.$$

Proof (of APCT from the topological compactness of $\mathbf{TA}$). Assume that $\mathbf{TA}$ is compact and that a set Γ of sentences is finitely satisfiable. By the preceding equivalence, this asserts exactly that the family of closed sets $\{ \mathcal{B}_{\{\phi\}} : \phi \in \Gamma \}$ has the finite intersection property. By topological compactness, the intersection $\mathcal{B}_\Gamma$ of the whole family is non-empty and thus Γ is satisfiable. $\square$

Proof (of the topological compactness of $\mathbf{TA}$ from APCT). Assume APCT and let $\{ \mathcal{B}_\Gamma : \Gamma \in \mathcal{G} \}$ be a family of closed sets which has the finite intersection property. To show that the family has a non-empty intersection, it suffices to show that $\bigcup \mathcal{G}$ is satisfiable and thus by APCT to show that it is

finitely satisfiable. Let Λ be any finite subset of $\bigcup \mathcal{G}$. Then there is a finite subset $\mathcal{G}_0$ of $\mathcal{G}$ such that $\Lambda \subseteq \bigcup \mathcal{G}_0$. By the finite intersection property,

$$\mathcal{B}_\Lambda \supseteq \mathcal{B}_{\bigcup \mathcal{G}_0} = \bigcap \{ \mathcal{B}_\Gamma : \Gamma \in \mathcal{G}_0 \} \neq \emptyset.$$

Hence Λ is satisfiable as required. $\square$

Finally, we sketch a direct proof of the topological compactness of **TA** thus providing an alternative proof of PCT and APCT. Let **TF** denote the set of all functions $V : \text{Sent}_L \rightarrow \{\mathbf{T}, \mathbf{F}\}$ with the topology generated by basis elements of the form

$$\begin{aligned} \mathcal{C}_{\Gamma, \Delta} = \{ V \in \text{TF} : & V(\phi) = \mathbf{T} \text{ for all } \phi \in \Gamma \\ & \text{and } V(\phi) = \mathbf{F} \text{ for all } \phi \in \Delta \}. \end{aligned}$$

Since for a truth assignment V ,

$$V \in \mathcal{C}_{\Gamma, \Delta} \iff V \in \mathcal{B}_{\Gamma \cup \{ \neg \phi : \phi \in \Delta \}},$$

TA is a subspace of **TF**. The result follows from the following two facts together with a standard theorem of topology:

(i) **TF** is a compact space;

(ii) **TA** is a closed subspace of **TF**.

(i) is left to the reader; it is a simple topological fact which may be proved using either the König Infinity Lemma or the Tychonoff Theorem. (ii) follows from the observation that the complement $\text{TF} \sim \text{TA}$ is the union of infinitely many (basic) open sets of the following forms, for ϕ and ψ any L -sentences:

$$\begin{aligned} & \{ V : V(\phi) = \mathbf{T} \text{ and } V(\neg \phi) = \mathbf{T} \} \\ & \{ V : V(\phi) = \mathbf{F} \text{ and } V(\neg \phi) = \mathbf{F} \} \\ & \{ V : V(\phi \vee \psi) = \mathbf{T}, \quad V(\phi) = \mathbf{F} \text{ and } V(\psi) = \mathbf{F} \} \\ & \{ V : V(\phi \vee \psi) = \mathbf{F} \text{ and } V(\phi) = \mathbf{T} \} \\ & \{ V : V(\phi \vee \psi) = \mathbf{F} \text{ and } V(\psi) = \mathbf{T} \} \end{aligned}$$

together with similar sets for the other connectives.

Boolean algebra

We turn now to Boolean algebras. For any non-empty set X and any set $\mathcal{B} \subseteq \wp(X)$ which is closed under the usual set operations $\cup$ (union), $\cap$ (intersection) and $\sim_X$ (complement with respect to X) and contains the distinguished elements $\emptyset$ and X , we may define the algebraic structure or algebra $(\mathcal{B}, \cup, \cap, \sim_X, \emptyset, X)$. Such an algebra is called a **field of sets**. It turns out, as we shall soon see, that there are many mathematical structures which although not literally fields of sets are isomorphic to fields of sets; the point of the notion of Boolean algebra is to characterize such structures. We give the definition in axiomatic form.

Definition 1.7.1 A **Boolean algebra** is a structure $\mathcal{A} = (A, \oplus, \otimes, \bar{}, \mathbb{O}, \mathbb{I})$ such that

- (i) $\oplus$ and $\otimes$ are binary operations on A ;
- (ii) $\bar{}$ is a unary operation on A ;
- (iii) $\mathbb{O}$ and $\mathbb{I}$ are distinct elements of A ;
- (iv) for all $a, b, c \in A$, the following identities are satisfied:

$$\begin{array}{ll}
 a \oplus a = a & a \otimes a = a \\
 a \oplus b = b \oplus a & a \otimes b = b \otimes a \\
 (a \oplus b) \oplus c = a \oplus (b \oplus c) & (a \otimes b) \otimes c = a \otimes (b \otimes c) \\
 a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c) & a \oplus (b \otimes c) = (a \oplus b) \otimes (a \oplus c) \\
 a \oplus \bar{a} = \mathbb{I} & a \otimes \bar{a} = \mathbb{O} \\
 a \oplus \mathbb{O} = a & a \otimes \mathbb{I} = a.
 \end{array}$$

These are known, respectively, as the **idempotent laws**, **commutative laws**, **associative laws**, **distributive laws**, **complementation laws** and **identity laws**. It is straightforward to verify

Proposition 1.7.2 *Every field of sets is a Boolean algebra.*

Proof. We give as just one example the axiom which may be least familiar, the second distributive law. A field of sets $(\mathcal{B}, \cup, \cap, \sim_X, \emptyset, X)$, with $\mathcal{B} \subseteq X$, satisfies this axiom iff for any $A, B, C \in \mathcal{B}$,

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

To see that this holds, consider any $x \in X$. If $x \in A$, then x belongs to both sets. Otherwise,

$$\begin{aligned}
 x \in A \cup (B \cap C) &\iff x \in B \cap C \iff x \in (A \cup B), (A \cup C) \\
 &\iff x \in (A \cup B) \cap (A \cup C). \quad \square
 \end{aligned}$$

The principal example of interest here of a Boolean algebra which is *not* a field of sets is the following.

Definition 1.7.3 For any set Γ of sentences and any sentences ϕ and ψ ,

- (i) ϕ and ψ are **Γ -tautologically equivalent** — in symbols, $\phi \equiv_{\Gamma} \psi$ — iff $\Gamma \models \phi \leftrightarrow \psi$;
- (ii) $[\phi]_{\Gamma} = \{ \psi : \phi \equiv_{\Gamma} \psi \}$.

The relation $\equiv_\Gamma$ is easily seen to be an equivalence relation on Sent_L , and thus $[\phi]_\Gamma$ is the equivalence class of ϕ . We set

$$\mathcal{S}_\Gamma = \{ [\phi]_\Gamma : \phi \in \text{Sent}_L \}$$

and make $\mathcal{S}_\Gamma$ into an algebra by defining the following operations and distinguished elements:

$$[\phi]_\Gamma \oplus [\psi]_\Gamma = [\phi \vee \psi]_\Gamma$$

$$[\phi]_\Gamma \otimes [\psi]_\Gamma = [\phi \wedge \psi]_\Gamma$$

$$\overline{[\phi]_\Gamma} = [\neg \phi]_\Gamma$$

$$\mathbb{O} = [\mathbf{F}]_\Gamma$$

$$\mathbb{I} = [\mathbf{T}]_\Gamma.$$

Of course, we need to verify that these operations are well-defined on equivalence classes — that is, for example for $\oplus$, that

$$[\phi]_\Gamma = [\phi']_\Gamma \quad \text{and} \quad [\psi]_\Gamma = [\psi']_\Gamma \quad \implies \quad [\phi \vee \psi]_\Gamma = [\phi' \vee \psi']_\Gamma.$$

This is immediate from the fact that

$$\Gamma \models \phi \leftrightarrow \phi' \quad \text{and} \quad \Gamma \models \psi \leftrightarrow \psi' \quad \implies \quad \Gamma \models (\phi \vee \psi) \leftrightarrow (\phi' \vee \psi').$$

The other verifications are done similarly.

Definition 1.7.4 The algebra $\mathcal{L}_\Gamma = (\mathcal{S}_\Gamma, \oplus, \otimes, \overline{}, \mathbb{O}, \mathbb{I})$ is called the **Lindenbaum algebra** of Γ .

Proposition 1.7.5 For any consistent set Γ of sentences, $\mathcal{L}_\Gamma$ is a Boolean algebra.

Proof. Again the verification is straightforward, and we check only one distributive law:

$$[\phi]_\Gamma \oplus ([\psi]_\Gamma \otimes [\theta]_\Gamma) = ([\phi]_\Gamma \oplus [\psi]_\Gamma) \otimes ([\phi]_\Gamma \oplus [\theta]_\Gamma).$$

Unraveling the definitions, we see that this amounts to

$$[\phi \vee (\psi \wedge \theta)]_\Gamma = [(\phi \vee \psi) \wedge (\phi \vee \theta)]_\Gamma,$$

and hence to

$$\Gamma \models \phi \vee (\psi \wedge \theta) \leftrightarrow (\phi \vee \psi) \wedge (\phi \vee \theta),$$

which is true — the sentence occurs as one of the tautologies in the table of [Section 1.3](#). Consistency of Γ guarantees that $\mathbb{O} \neq \mathbb{I}$. $\square$

In the preceding section we introduced the notion of an ultrafilter over a set; essentially the same notion is also a useful tool in the study of Boolean algebras. We need first the analogue of set inclusion:

Definition 1.7.6 For any Boolean algebra $\mathcal{A} = (A, \oplus, \otimes, \bar{}, \mathbb{O}, \mathbb{I})$ and any $a, b \in A$,

$$a \leq b \iff a \oplus b = b.$$

We state first some elementary properties; note that these are all common properties of fields of sets.

Lemma 1.7.7 For any Boolean algebra $\mathcal{A} = (A, \oplus, \otimes, \bar{}, \mathbb{O}, \mathbb{I})$ and any $a, b, c \in A$,

(i) $\leq$ is a partial ordering — that is,

(a) $a \leq a$;

(b) $a \leq b$ and $b \leq a \implies a = b$;

(c) $a \leq b$ and $b \leq c \implies a \leq c$;

(ii) $a \leq b \iff a \oplus b = b \iff \bar{a} \oplus b = \mathbb{I}$;

(iii) $a \leq b \iff a \otimes b = a \iff a \otimes \bar{b} = \mathbb{O}$;

(iv) $a \leq c$ and $b \leq d \implies a \otimes b \leq c \otimes d$;

(v) $\overline{a \oplus b} = \bar{a} \otimes \bar{b}$ and $\overline{a \otimes b} = \bar{a} \oplus \bar{b}$;

(vi) $\mathbb{O} \leq a \leq \mathbb{I}$;

(vii) $a \oplus b$ is the $\leq$ -least upper bound of a and b ;

(viii) $a \otimes b$ is the $\leq$ -greatest lower bound of a and b .

Proof. Left as an exercise. $\square$

Lemma 1.7.8 In $\mathcal{L}_\Gamma$, $[\phi]_\Gamma \leq [\psi]_\Gamma \iff \Gamma \models \phi \rightarrow \psi$.

Proof. This follows from the equivalences

$$\begin{aligned} [\phi]_\Gamma \leq [\psi]_\Gamma &\iff [\neg \phi]_\Gamma \oplus [\psi]_\Gamma = \mathbb{I} \iff \Gamma \models \neg \phi \vee \psi \\ &\iff \Gamma \models \phi \rightarrow \psi. \quad \square \end{aligned}$$

Generalizing Definition 1.6.10,

Definition 1.7.9 For any Boolean algebra $\mathcal{A} = (A, \oplus, \otimes, \bar{}, \mathbb{O}, \mathbb{I})$, a non-empty set $U \subseteq A$ is a **filter on $\mathcal{A}$** iff for all $a, b \in A$,

(i) $\mathbb{O} \notin U$;

(ii) if $a \in U$ and $a \leq b$, then $b \in U$;

(iii) if $a \in U$ and $b \in U$, then $a \otimes b \in U$.

U is called an **ultrafilter on $\mathcal{A}$** iff in addition for all $a \in A$,

(iv) $\bar{a} \in U$ iff $a \notin U$.

Example 1.7.10 Let Γ be any consistent set of sentences and V any model of Γ . Then $U := \{[\phi]_\Gamma : V(\phi) = \top\}$ is an ultrafilter on $\mathcal{L}_\Gamma$.

The existence of ultrafilters on any Boolean algebra is proved exactly as in the preceding section and we simply adapt the terminology.

Definition 1.7.11 For any Boolean algebra $\mathcal{A} = (A, \oplus, \otimes, \bar{}, \mathbb{O}, \mathbb{I})$, a set $B \subseteq A$ is said to have the **finite intersection property** (FIP) iff for any finitely many $a_0, \dots, a_n \in B$, $a_0 \otimes \dots \otimes a_n \neq \mathbb{O}$.

If B has the finite intersection property, it follows easily from the lemma that the set

$$F = \{b \in A : \text{for some } n \text{ and some } a_0, \dots, a_n \in A, \quad a_0 \otimes \dots \otimes a_n \leq b\}$$

is a filter on $\mathcal{A}$ which includes B , and a maximal such filter (with respect to inclusion) is an ultrafilter. Thus we have

Theorem 1.7.12 For any Boolean algebra $\mathcal{A} = (A, \oplus, \otimes, \bar{}, \mathbb{O}, \mathbb{I})$ and any $B \subseteq A$ with the finite intersection property, there exists an ultrafilter U on $\mathcal{A}$ such that $B \subseteq U$. $\square$

We can now sketch the proof of the fact which justifies our choice of axioms:

Lemma 1.7.13 For any Boolean algebra $\mathcal{A}$, any ultrafilter U on $\mathcal{A}$ and any $a, b \in A$,

$$a \oplus b \in U \quad \Longleftrightarrow \quad a \in U \quad \text{or} \quad b \in U.$$

Proof. An easy adaptation of the proof of Lemma 1.6.12. $\square$

Stone Representation Theorem 1.7.14 Every Boolean algebra is isomorphic to a field of sets.

Proof. (sketch) Let $\mathcal{A} = (A, \oplus, \otimes, \bar{}, \mathbb{O}, \mathbb{I})$ be a Boolean algebra and

$$\mathcal{U} := \{U : U \text{ is an ultrafilter on } \mathcal{A}\}.$$

For each $a \in A$, set $a^* := \{U \in \mathcal{U} : a \in U\}$ and $A^* = \{a^* : a \in A\}$. Then $A^* \subseteq \wp(\mathcal{U})$ and it is not hard to check, using the lemma, that for any $a, b \in A$,

$$a^* \cup b^* = (a \oplus b)^*, \quad a^* \cap b^* = (a \otimes b)^* \quad \text{and} \quad \mathcal{U} \sim a^* = (\bar{a})^*,$$

so that $\mathcal{A}^* = (A^*, \cup, \cap, \sim, \emptyset, \mathcal{U})$ is a field of sets. Finally $\mathcal{A}$ and $\mathcal{A}^*$ are isomorphic via the map $a \mapsto a^*$. $\square$

It is instructive to work out explicitly what the Representation Theorem looks like for the Lindenbaum algebra $\mathcal{L}_\Gamma$ of a set of sentences Γ .

Proposition 1.7.15 *For any consistent set Γ of sentences, there is a bijective correspondence between ultrafilters on $\mathcal{L}_\Gamma$ and models of Γ or, equivalently (by Corollary 1.4.19), complete propositional theories which extend Γ .*

Proof. Fix Γ and set $\mathcal{U} = \{U : U \text{ is an ultrafilter on } \mathcal{L}_\Gamma\}$. With each $U \in \mathcal{U}$ we associate the function on $\mathbf{Sent}_L$ defined by:

$$V_U(\phi) = \begin{cases} \mathbf{T}, & \text{if } [\phi]_\Gamma \in U; \\ \mathbf{F}, & \text{otherwise.} \end{cases}$$

We claim first that V_U is a truth assignment; this follows from the equivalences

$$\begin{aligned} V_U(\phi) = \mathbf{T} &\iff [\phi]_\Gamma \in U \iff [\neg \phi]_\Gamma = \overline{[\phi]_\Gamma} \notin U \\ &\iff V_U(\neg \phi) = \mathbf{F}, \end{aligned}$$

and

$$\begin{aligned} V_U(\phi \vee \psi) = \mathbf{T} &\iff [\phi \vee \psi]_\Gamma = [\phi]_\Gamma \oplus [\psi]_\Gamma \in U \\ &\iff [\phi]_\Gamma \in U \text{ or } [\psi]_\Gamma \in U \\ &\iff V_U(\phi) = \mathbf{T} \text{ or } V_U(\psi) = \mathbf{T}. \end{aligned}$$

Next we note that for any sentence ϕ ,

$$\phi \in \Gamma \implies [\phi]_\Gamma = \mathbb{I} \in U \implies V_U(\phi) = \mathbf{T}.$$

Hence for each $U \in \mathcal{U}$, V_U is a model of Γ . Finally, if $U \neq U'$, then for some sentence ϕ , $[\phi]_\Gamma \in U \sim U'$, so $[\neg \phi]_\Gamma \in U'$ and thus $V_U(\phi) \neq V_{U'}(\phi)$, and the mapping is bijective. Conversely, given any model V of Γ , by the example above $U := \{[\phi]_\Gamma : V(\phi) = \mathbf{T}\}$ is an ultrafilter and easily $V = V_U$.

The final clause of the conclusion follows similarly using the mapping

$$U \mapsto \Phi_U := \{\phi : [\phi]_\Gamma \in U\},$$

which is a bijective correspondence between ultrafilters on $\mathcal{L}_\Gamma$ and complete propositional theories which extend Γ . $\square$

With this in mind, we see that the isomorphism described in the proof sketch of the Representation Theorem is essentially

$$[\phi]_\Gamma \mapsto \{V : V \in \mathbf{Mod}(\Gamma) \text{ and } V(\phi) = \mathbf{T}\},$$

and one may easily verify directly that the collection of all such sets of truth assignments form a field of sets isomorphic to $\mathcal{L}_\Gamma$.

So far, this subsection has been purely descriptive: given a set Γ of sentences, we must assume that Γ is consistent to conclude that $\mathcal{L}_\Gamma$ has at least

two elements and hence is a Boolean algebra. Hence the fact that Γ has a model arising from an ultrafilter on $\mathcal{L}_\Gamma$ gives no new information. However, a slight modification of the preceding development leads to yet another proof of the (Alternative) Propositional Compactness Theorem. In place of $\equiv_\Gamma$, consider the relation

$$\phi \equiv_\Gamma^* \psi \quad :\Longleftrightarrow \quad \text{for some finite } \Gamma_0 \subseteq \Gamma, \quad \Gamma_0 \models \phi \leftrightarrow \psi;$$

see the discussion following the statement of Theorem 1.4.20. It is easy to check that $\equiv_\Gamma^*$ is an equivalence relation, and we further set

$$\begin{aligned} [\phi]_\Gamma^* &:= \{ \psi : \phi \equiv_\Gamma^* \psi \} \\ \mathcal{S}_\Gamma^* &:= \{ [\phi]_\Gamma^* : \phi \in \text{Sent}_L \} \\ \mathcal{L}_\Gamma^* &:= (\mathcal{S}_\Gamma^*, \oplus, \otimes, , \mathbb{O}, \mathbb{I}), \end{aligned}$$

where the operations are defined exactly as for $\mathcal{L}_\Gamma$ with the addition of a few obvious asterisks. Now we may check essentially as for $\mathcal{L}_\Gamma$ that $\mathcal{L}_\Gamma^*$ is a Boolean algebra iff Γ is a *finitely* consistent set and that there is a bijective correspondence between ultrafilters on $\mathcal{L}_\Gamma^*$ and models of Γ . Then the existence of ultrafilters on $\mathcal{L}_\Gamma^*$ yields immediately the existence of a model for Γ as claimed by the Alternative Propositional Compactness Theorem.

Exercises

Exercise 1.7.16 Fill in the details of the proof of the topological compactness of TA.

Exercise 1.7.17 Prove Lemma 1.7.7.

Exercise 1.7.18

- (i) Show that any finite set in a Boolean algebra has both a least upper bound and a greatest lower bound.
- (ii) Give an example to show that this is not necessarily true for infinite sets.

Exercise 1.7.19 Fill in the details of the proof of the Stone Representation Theorem.

Exercise 1.7.20 Verify directly as suggested above that for any set Γ of sentences, the Lindenbaum algebra is isomorphic to a field of subsets of $\text{Mod}(\Gamma)$.

Exercise 1.7.21 Show that if U is a filter on $\mathcal{L}_\Gamma$, $\Gamma \models \phi_0 \wedge \cdots \wedge \phi_n \rightarrow \psi$ and for $i \leq n$, $[\phi_i]_\Gamma \in U$, then also $[\psi]_\Gamma \in U$.

Exercise 1.7.22 Fill in the sketch of the proof of the Alternative Propositional Compactness Theorem via Boolean algebra sketched above.

Definition 1.7.23 For any Boolean algebra $\mathcal{A}$,

- (i) $a \in \mathcal{A}$ is an **atom** iff $a \neq \mathbb{0}$ but $\mathbb{0}$ is the only member b of $\mathcal{A}$ such that $b < a$;
- (ii) an ultrafilter U on $\mathcal{A}$ is **principal** iff for some $a \in \mathcal{A}$, $U = U_a := \{b : a \leq b\}$.

Exercise 1.7.24 For any Boolean algebra $\mathcal{A}$ and any $a \in \mathcal{A}$, show

- (i) U_a is a filter on $\mathcal{A}$;
- (ii) U_a is an ultrafilter iff a is an atom;
- (iii) for any ultrafilter U on $\mathcal{A}$, U is non-principal iff $\mathbb{0}$ is the greatest lower bound of U — that is, $\mathbb{0}$ is the unique $c \in \mathcal{A}$ such that $c \leq b$ for all $b \in U$.



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Chapter 2

First-Order Logic

Although the languages of first-order logic are substantially more complex than the propositional language, they share many characteristics. The syntax still consists in specifying a set of symbols and characterizing inductively various sets of meaningful expressions. The semantics provides a notion of interpretation which assigns mathematical meaning to the meaningful expressions.

2.1 Syntax and semantics of first-order languages

The motivation for the form of these languages is that almost every mathematical statement can be considered as referring to some mathematical **structure** $\mathfrak{A}$ of the following sort. There is a non-empty set A of elements, called the *universe* of the structure. Among the elements of the universe there are certain relationships which may be represented by the sets of ordered k -tuples $R \subseteq A^k$ of elements which have the relationship. Additionally, one may want to consider certain *functions* $F : A^k \rightarrow A$ and *constants or distinguished elements* $a \in A$. We write

$$\mathfrak{A} := (A, R, S, \dots, F, G, \dots, a, b, \dots).$$

Examples 2.1.1

- (i) The very simplest structures are simply sets (A) with no relations, functions or distinguished elements.
- (ii) Many interesting mathematical notions concern a set A together with a binary relation $R \subseteq A \times A$. For example, $(\omega, <_\omega)$ is the set of natural numbers with its usual ordering. Here and in all similar contexts we use the same notation $<_\omega$ for both the usual *relationship* between numbers and the *relation* $\{(m, n) : m <_\omega n\}$. More generally, any structure $\mathfrak{A} = (A, R)$ is called a **partial ordering** iff it satisfies the following conditions:

- (a) for all $a \in A$, $(a, a) \in R$; ($\mathfrak{A}$ is **reflexive**);
- (b) for all $a, b, c \in A$, $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$;
($\mathfrak{A}$ is **transitive**);
- (c) for all $a, b \in A$, $(a, b) \in R$ and $(b, a) \in R \implies a = b$
($\mathfrak{A}$ is **antisymmetric**).

$\mathfrak{A}$ is a **linear ordering** iff in addition

- (d) for all $a, b \in A$, $a \neq b \implies (a, b) \in R$ or $(b, a) \in R$;
($\mathfrak{A}$ is **connected** or **total**).

$\mathfrak{A}$ is a **strict partial ordering** iff $\mathfrak{A}$ is transitive and

- (e) for all $a \in A$, $(a, a) \notin R$ ($\mathfrak{A}$ is **irreflexive**);

a **strict linear ordering** iff it is also connected; and a **dense linear ordering** iff additionally

- (f) A has at least two elements and for all $a, b \in A$, if $(a, b) \in R$, then there exists $c \in A$ such that $(a, c) \in R$ and $(c, b) \in R$.

For any set A whose elements are all sets, let $\subseteq_A$ be the subset relation restricted to elements of A :

$$\subseteq_A := \{(X, Y) : X, Y \in A \text{ and } X \subseteq Y\}.$$

Then $(A, \subseteq_A)$ is a partial ordering. If $\mathbb{Q}$ is the set of rational numbers and $<_{\mathbb{Q}}$ its usual ordering, then $(\mathbb{Q}, <_{\mathbb{Q}})$ is a dense linear ordering. Quite different structures of this sort (with just one binary relation) are those of the form $(\mathbb{Z}, \equiv_p)$, the set of integers with the relation of congruence modulo p or $(A, \in_A)$, where A is a set of sets and $\in_A$ is the membership relation restricted to elements of A :

$$\in_A := \{(a, b) : a, b \in A \text{ and } a \in b\}.$$

- (iii) A common definition of a **group** is a structure of the form $(G, *, {}^{-1}, e)$, with $*$ a function $G \times G \rightarrow G$, ${}^{-1}$ a function $G \rightarrow G$ and $e \in G$, which

satisfies the following conditions:

- (a) for all $a, b, c \in G$, $a * (b * c) = (a * b) * c$ (*associative law*);
- (b) for all $a \in G$, $a * e = a = e * a$ (*identity*);
- (c) for all $a \in G$, $a * a^{-1} = e = a^{-1} * a$ (*inverse*).

However, groups are also often regarded as structures of the form $(G, *, e)$ which satisfy (a) and (b) together with

- (d) for each $a \in G$ there exists $d \in G$ such that $a * d = e = d * a$,

or even as structures $(G, *)$ which satisfy (a) together with

- (e) there exists $e \in G$ such that for all $a \in G$, $a * e = a = e * a$ and there exists $d \in G$ such that $a * d = e = d * a$.

Although these formulations are in some senses equivalent, we shall see that there are interesting differences, and one of the advantages of this representation is that it forces us to be precise about such distinctions.

- (iv) A **ring** is a structure $(R, +, -, \times, 0)$, with $+$ and $\times$ binary functions on R and $-$ a unary function on R , such that for all $a, b, c \in R$,

- (a) $a + (b + c) = (a + b) + c$;
- (b) $a + b = b + a$;
- (c) $a + 0 = a = 0 + a$;
- (d) $a + (-a) = 0 = (-a) + a$;
- (e) $a \times (b \times c) = (a \times b) \times c$;
- (f) $a \times (b + c) = (a \times b) + (a \times c)$; $(b + c) \times a = (b \times a) + (c \times a)$.

As for groups there are also formulations without one or both of $-$ and 0 . A **ring with 1** is of the form $(R, +, -, \times, 0, 1)$ with the additional condition that for all $a \in R$,

- (g) $a \times 1 = a = 1 \times a$;

and a **field** satisfies the additional conditions, for all $a, b \in R$,

- (h) $a \times b = b \times a$;
- (i) if $a \neq 0$, then there exists d such that $a \times d = 1$.

- (v) There are many interesting structures with universe ω , the set of natural numbers; for example

$$\Omega := (\omega, <, +, \times, \text{Sc}, 0),$$

with the usual operations (Sc is the function $m \mapsto m + 1$), and variations obtained by omitting one or more of the components or adding new ones, such as the exponentiation function.

Each structure has a characteristic **signature** which specifies for each $n > 0$ how many n -ary relations and functions there are and how many distinguished elements. For each such signature there are first-order languages which contain the corresponding number of relation, function and constant symbols. Thus a language L is identified by its set of symbols together with a function ν_L which assigns a **rank** to each relation and function symbol—a unary symbol has rank 1, a binary symbol rank 2 etc.

Definition 2.1.2 A **first-order language** L is specified by

- (i) a set Rs_L of **L -relation symbols**,
- (ii) a set Fs_L of **L -function symbols**,
- (iii) a set Cs_L of **L -constant symbols**,
- (iv) a function $\nu_L : \text{Rs}_L \cup \text{Fs}_L \rightarrow \omega \setminus \{0\}$.

The sets Rs_L , Fs_L and Cs_L are pairwise disjoint, and members of $\text{Rs}_L \cup \text{Fs}_L \cup \text{Cs}_L$ are called the **non-logical symbols** of L . In addition, every first-order language has the following distinct **logical symbols**:

- (v) a denumerable set $\text{Vb} = \{v_n : n \in \omega\}$ of **variables**,
- (vi) the **equality symbol** $\doteq$,
- (vii) the **connectives** $\neg$ and $\vee$,
- (viii) the **existential quantifier** $\exists$.

Remark 2.1.3 The symbol for equality is written $\doteq$ instead of simply $=$ to make a clear notational distinction between the symbol and the object (the equality relation) that it represents. We shall often use this **dot convention** below to make similar distinctions. Our decision to choose only $\neg$ and $\vee$ as “official” connectives is motivated by a desire to reduce the number of cases in proofs as in the latter sections of [Chapter 1](#). We will, of course, rely on the results of [Chapter 1](#) to be assured that this causes no restriction of expressive power. We shall see that the same applies to our omission of the **universal quantifier** $\forall$ as a logical symbol.

Examples 2.1.4 Corresponding to each of the examples of structures above are the following languages:

- (i) The language $L_=_$ has $\text{Rs}_{L_=} = \text{Fs}_{L_=} = \text{Cs}_{L_=} = \emptyset$; $L_=_$ has no non-logical symbols.
- (ii) Languages L appropriate for structures of the form (A, R) contain one relation symbol p with $\nu_L(p) = 2$. We are free to choose p to be any desired symbol, and this choice is often based on the particular structures of interest in a context. For example, if we plan to consider partial orderings, we might choose $\leq$ as the symbol, while if we are

studying strict partial orderings, $\dot{<}$ is more appropriate. We denote these two essentially equivalent languages by $L_{\leq}$ and $L_{<}$. When we are interested in structures of the form $(A, \in_A)$, we might use the language $L_{\in}$ with the binary relation symbol $\dot{\in}$.

- (iii) A natural language L_G for the study of groups (group theory) has non-logical symbols $\dot{*}$, $\dot{-}^1$ and $\dot{e}$, with $\nu_{L_G}(\dot{*}) = 2$ and $\nu_{L_G}(\dot{-}^1) = 1$. If only Abelian (commutative) groups are under discussion, it is traditional to use $\dot{+}$ instead of $\dot{*}$. For the two variations described above we would omit one or both of the symbols $\dot{-}^1$ and $\dot{e}$.
- (iv) A language L_R for the study of rings might contain non-logical symbols $\dot{+}$, $\dot{-}$, $\dot{\times}$ and $\dot{0}$ or some subset of these. For rings with 1 and fields, we would add an additional constant symbol $\dot{1}$.
- (v) A language L_{Ω} suitable for the first-order theory of arithmetic (the study of Ω) might have the following components:

$$\text{Rs}_{L_{\Omega}} = \{\dot{<}\}; \quad \text{Fs}_{L_{\Omega}} = \{\dot{+}, \dot{\times}, \dot{S}\}; \quad \text{Cs}_{L_{\Omega}} = \{\dot{0}\};$$

with $\nu_{L_{\Omega}}(\dot{<}) = \nu_{L_{\Omega}}(\dot{+}) = \nu_{L_{\Omega}}(\dot{\times}) = 2$ and $\nu_{L_{\Omega}}(\dot{S}) = 1$.

Notational Conventions. When we are discussing an arbitrary language L with no particular class of structures in mind, we use

- $p, q, r, \dots$ to denote elements of Rs_L (relation symbols);
- $f, g, h, \dots$ to denote elements of Fs_L (function symbols);
- $c, d, e, \dots$ to denote elements of Cs_L (constant symbols).

Furthermore, we usually use

- $x, y, z, \dots$ to denote elements of Vb (variables).

We describe the syntax of a first-order language L in several stages. The essential difference from the propositional language is an analysis of atomic propositions into their component parts. As in [Chapter 1](#), an ***L-expression*** is any finite sequence of L -symbols, and the juxtaposition of two expressions denotes the expression which results from their concatenation. The L -terms may be thought of as the *nouns* of the language. These are combined with the *verbs*—the relation symbols and the equality symbol—to form atomic formulas. These are then combined via the connectives and the existential quantifier to form general formulas. Sentences are formulas of a special sort to be defined a bit later. We call an ***L-atomic term*** an expression containing a single symbol which is either a constant symbol or a variable of L .

Definition 2.1.5 The set Te_L of *L-terms* is the smallest set of *L-expressions* such that

- (i) every *L-atomic* term is an *L-term*;
- (ii) for any *L-function* symbol f and any *L-terms* $t_0, \dots, t_{\nu_L(f)-1}$, $ft_0 \dots t_{\nu_L(f)-1}$ is an *L-term*.

Of course, this statement is shorthand for a more formal inductive definition as in [Chapter 1](#). We leave it to the reader to write out this more formal version one last time. Note that we are again “officially” adopting a prefix notation in which the function symbol precedes its arguments and no parentheses or other grouping devices are used. This strategy succeeds for exactly the same reasons as in [Chapter 1](#).

Proposition 2.1.6 (unique readability for L-terms) For every *L-term* t , exactly one of the following holds:

- (i) t is an *L-atomic* term;
- (ii) for some *L-function* symbol f and *L-terms* $u_0, \dots, u_{\nu_L(f)-1}$, t is $fu_0 \dots u_{\nu_L(f)-1}$.

Furthermore, in (ii), f and $u_0, \dots, u_{\nu_L(f)-1}$ are uniquely determined.

Proof. Left as an exercise — compare Theorem 1.1.5. $\square$

Examples 2.1.7

- (i) Since the languages $L_=$, $L_\leq$, $L_<$ and $L_\in$ have no function or constant symbols, the terms are just the variables.
- (ii) Some simple terms of L_G are $\ast x\dot{e}$ and $\dot{\ast}^{-1} \ast xy$. This official notation is not very appealing, and we will normally write these terms as $x\ast\dot{e}$ and $(x\ast y)^{-1}$, with the understanding that these are abbreviations for the official notation. Note that this is a special option for *binary* function symbols, which are the sort most frequently encountered in practical mathematics. Intuitively L_G -terms are what are traditionally called **words**; assuming the group axioms they are (informally) equivalent to expressions of the form $x_0^{\pm 1} \ast \dots \ast x_k^{\pm 1}$.
- (iii) Terms of the languages for rings and fields are essentially multi-variate polynomials.
- (iv) The language L_Ω has a rich assortment of terms. For example, the term $\dot{m} := \dot{S} \dots \dot{S}\dot{0}$ with m -many $\dot{S}$'s is a name in the language for the natural number m , and there are terms representing complex arithmetical combinations of these names and variables.

Definition 2.1.8 The set AtFm_L of *L-atomic formulas* is the set of all *L*-expressions of one of the following forms:

- (i) $\doteq tu$ for *L*-terms t and u ;
- (ii) $pt_0 \dots t_{\nu_L(p)-1}$ for *L*-terms $t_0, \dots, t_{\nu_L(p)-1}$ and an *L*-relation symbol p .

Of course we will normally write $t \doteq u$ instead of $\doteq tu$, but this “official” notation is necessary to ensure unique readability without the (official) use of parentheses. Similarly, when p is a binary relation symbol such as $<$, we write $t < u$ instead of $< tu$.

Examples 2.1.9

- (i) $L_{=}$ -atomic formulas are all of the form $x \doteq y$ for variables x and y . In $L_{<}$ there are additionally atomic formulas of the form $x < y$.
- (ii) The languages for group and ring theory have no relation symbols (other than $\doteq$), so atomic formulas are all of the form $t \doteq u$ for terms t and u .
- (iii) Atomic formulas of L_Ω are of the forms $t \doteq u$ and $t < u$. For example, $\dot{S}x \dot{+} y \doteq \dot{S}(x \dot{+} y)$ expresses a basic relationship between the successor function and addition.

Definition 2.1.10 The set Fm_L of *L-formulas* is the smallest set of *L*-expressions such that

- (i) every *L*-atomic formula is an *L*-formula;
- (ii) if ϕ and ψ are *L*-formulas, then so are $\neg \phi$ and $\vee \phi \psi$;
- (iii) if ϕ is an *L*-formula and x a variable, then $\exists x \phi$ is an *L*-formula.

Proposition 2.1.11 (unique readability for L-formulas) For every *L*-formula θ , exactly one of the following holds:

- (i) θ is an *L*-atomic formula;
- (ii) for some *L*-formula ϕ , θ is $\neg \phi$;
- (iii) for some *L*-formulas ϕ and ψ , θ is $\vee \phi \psi$;
- (iv) for some *L*-formula ϕ and some variable x , θ is $\exists x \phi$.

Furthermore in (i), if θ is $pt_0 \dots t_{\nu_L(p)-1}$, the t_i are uniquely determined, and in (ii)-(iv), ϕ , ψ and x are uniquely determined.

Proof. Left as an exercise. $\square$

Examples 2.1.12 The $L_{=}$ -formula $\exists x \exists y \neg x \dot{=} y$ expresses the idea that there are (at least) two distinct objects. The $L_{\in}$ -formulas $\exists y \neg \exists x (x \dot{\in} y)$ and

$$[\neg \exists z (z \dot{\in} x \wedge \neg z \dot{\in} y) \wedge \neg \exists z (z \dot{\in} y \wedge \neg z \dot{\in} x)] \rightarrow x \dot{=} y$$

express the idea that there is an **empty set** and the principle of **extensionality**: two sets with the same elements are equal. With abbreviations to be introduced shortly, this formula will be written in the more readable form $\forall z (z \dot{\in} x \leftrightarrow z \dot{\in} y) \rightarrow x \dot{=} y$. Similarly, $\forall x [x \dot{<} Sy \rightarrow (x \dot{<} y \vee x \dot{=} y)]$ abbreviates an L_{Ω} -formula which expresses a relationship between the successor function and the ordering.

The two unique readability theorems will be used to justify recursive definitions of functions of terms and formulas using Theorem 1.2.14. For reference, we state here the form of these results that we use most often.

Term Recursion Theorem 2.1.13 *For any set Z , any function $F_0 : \mathbf{Vb} \cup \mathbf{Cs}_L \rightarrow Z$ and any functions $G_f : Z^{\nu_L(f)} \rightarrow Z$ for each $f \in \mathbf{Fs}_L$, there exists a unique function $F : \mathbf{Te}_L \rightarrow Z$ such that F extends F_0 and for all $f \in \mathbf{Fs}_L$ and L -terms $t_0, \dots, t_{\nu_L(f)-1}$,*

$$F(ft_0 \dots t_{\nu_L(f)-1}) = G_f(F(t_0), \dots, F(t_{\nu_L(f)})). \quad \square$$

Formula Recursion Theorem 2.1.14 *For any set Z , any function $F_0 : \mathbf{AtFm}_L \rightarrow Z$ and any functions $G_{\neg} : Z \rightarrow Z$, $G_{\vee} : Z \times Z \rightarrow Z$ and $G_{\exists} : \mathbf{Vb} \times Z \rightarrow Z$, there exists a unique function $F : \mathbf{Fm}_L \rightarrow Z$ such that F extends F_0 and for all L -formulas ϕ and ψ ,*

$$F(\neg \phi) = G_{\neg}(F(\phi)), \quad F(\phi \vee \psi) = G_{\vee}(F(\phi), F(\psi))$$

$$\text{and } F(\exists x \phi) = G_{\exists}(x, F(\phi)). \quad \square$$

In the following, except where specified otherwise, L denotes any first-order language. As in [Chapter 1](#), we adopt a more “user-friendly” style of referring to terms and formulas using parentheses and other devices to facilitate human-readability subject only to the constraints of non-ambiguity. Further, as in [Section 1.4](#), we introduce the following abbreviations:

- $\phi \wedge \psi$ is an abbreviation for $\neg(\neg \phi \vee \neg \psi)$;
- $\phi \rightarrow \psi$ is an abbreviation for $\neg \phi \vee \psi$;
- $\phi \leftrightarrow \psi$ is an abbreviation for $\neg(\neg \phi \vee \neg \psi) \vee \neg(\phi \vee \psi)$;
- $\forall x \phi$ is an abbreviation for $\neg \exists x \neg \phi$.

We used examples of structures to motivate the general definition of language; now we give a general formal definition of the notion of structure for any language.

Definition 2.1.15 An L -structure $\mathfrak{A}$ is specified by

- (i) a non-empty set A , the **universe** $|\mathfrak{A}|$ of $\mathfrak{A}$;
- (ii) for each $p \in \mathbf{Rs}_L$, a relation $p^{\mathfrak{A}} \subseteq A^{\nu_L(p)}$, the **relations** of $\mathfrak{A}$;
- (iii) for each $f \in \mathbf{Fs}_L$, a function $f^{\mathfrak{A}} : A^{\nu_L(f)} \rightarrow A$, the **functions** of $\mathfrak{A}$;
- (iv) for each $c \in \mathbf{Cs}_L$, an element $c^{\mathfrak{A}} \in A$, the **distinguished elements** of $\mathfrak{A}$.

The term $\dot{\mathbf{S}}\dot{\mathbf{S}}\dot{\mathbf{0}}$ is clearly intended to denote the number 2 in the structure Ω , but the term $x \dot{+} \dot{\mathbf{S}}\dot{\mathbf{0}}$ is incomplete; it denotes a particular element only after the denotation of x is given. Thus we define

Definition 2.1.16 For any L -structure $\mathfrak{A}$, an $\mathfrak{A}$ -assignment is a function $\alpha : \mathbf{Vb} \rightarrow A$.

Relative to an $\mathfrak{A}$ -assignment α , we can define the denotation $t^{\mathfrak{A}}[\alpha]$ of a term as follows. Technically, we are defining here by L -term recursion a function $V_{\mathfrak{A},\alpha} : \mathbf{Te}_L \rightarrow A$ with the parameters $\mathfrak{A}$ and α (cf. Theorem 1.2.16).

Definition 2.1.17 For any L -structure $\mathfrak{A}$ and any $\mathfrak{A}$ -assignment α ,

- (i) for every variable x , $x^{\mathfrak{A}}[\alpha] = \alpha(x)$;
- (ii) for every $c \in \mathbf{Cs}_L$, $c^{\mathfrak{A}}[\alpha] = c^{\mathfrak{A}}$;
- (iii) for every $f \in \mathbf{Fs}_L$ and $t_0, \dots, t_{\nu_L(f)-1} \in \mathbf{Te}_L$,

$$(ft_0 \dots t_{\nu_L(f)-1})^{\mathfrak{A}}[\alpha] = f^{\mathfrak{A}}(t_0^{\mathfrak{A}}[\alpha], \dots, t_{\nu_L(f)-1}^{\mathfrak{A}}[\alpha]).$$

It is important to keep straight that c and f are L -symbols, while $c^{\mathfrak{A}}$ is a element of $|\mathfrak{A}|$ and $f^{\mathfrak{A}}$ is a genuine function (not a symbol) on $|\mathfrak{A}|$.

Example 2.1.18 If $\alpha(x) = 3$ and $\alpha(y) = 7$, then

$$\begin{aligned} (\dot{\mathbf{S}}x \dot{\times} \dot{\mathbf{S}}\dot{\mathbf{S}}y)^{\Omega}[\alpha] &= (\dot{\mathbf{S}}x)^{\Omega}[\alpha] \times (\dot{\mathbf{S}}\dot{\mathbf{S}}y)^{\Omega}[\alpha] = \mathbf{Sc}(\alpha(x)) \times \mathbf{Sc}(\mathbf{Sc}(\alpha(y))) \\ &= \mathbf{Sc}(3) \times \mathbf{Sc}(\mathbf{Sc}(7)) = 4 \times 9 = 36. \end{aligned}$$

Once the terms are interpreted, it is straightforward to assign a truth value to atomic formulas.

Definition 2.1.19 For any $\mathfrak{A}$ and α ,

- (i) for any L -terms t and u , $V_{\mathfrak{A},\alpha}(t \doteq u) = \begin{cases} \mathbf{T}, & \text{if } t^{\mathfrak{A}}[\alpha] = u^{\mathfrak{A}}[\alpha]; \\ \mathbf{F}, & \text{otherwise;} \end{cases}$
- (ii) for any $p \in \mathbf{Rs}_L$ and $t_0, \dots, t_{\nu_L(p)-1} \in \mathbf{Te}_L$,

$$V_{\mathfrak{A},\alpha}(pt_0 \dots t_{\nu_L(p)-1}) = \begin{cases} \mathbf{T}, & \text{if } (t_0^{\mathfrak{A}}[\alpha], \dots, t_{\nu_L(p)-1}^{\mathfrak{A}}[\alpha]) \in p^{\mathfrak{A}}; \\ \mathbf{F}, & \text{otherwise.} \end{cases}$$

Remark 2.1.20 That we include the equality symbol $\doteq$ in every language and interpret it always by the equality relation represents a conscious but not a necessary choice. It reflects common mathematical usage.

Example 2.1.21 If $\alpha(x) = 3$ and $\alpha(y) = 5$, then $V_{\Omega, \alpha}(\dot{S}x \dot{<} y) = \mathsf{T}$, since $4 < 5$, that is, $(4, 5) \in \dot{<}^{\Omega}$.

We want now to extend $V_{\mathfrak{A}, \alpha}$ recursively to all of Fm_L . The clauses corresponding to $\neg$ and $\vee$ will be as in [Chapter 1](#) (see following definition), but how should we define $V_{\mathfrak{A}, \alpha}(\exists x \phi)$? For example, we clearly want $V_{\Omega, \alpha}(\exists x (\dot{S}x \doteq \dot{0})) = \mathsf{F}$ regardless of α ; in general, $V_{\mathfrak{A}, \alpha}(\exists x \phi)$ should be independent of $\alpha(x)$ and should be T just in case there is *some* way of assigning x to an element a of A such that together with the rest of α , ϕ becomes true. We implement this idea as follows. For any assignment α , let $\alpha_{x \mapsto a}$ denote the assignment β such that $\beta(x) = a$, and for all other variables y , $\beta(y) = \alpha(y)$. Then we intend to set

$$V_{\mathfrak{A}, \alpha}(\exists x \phi) = \begin{cases} \mathsf{T}, & \text{if for some } a \in A, V_{\mathfrak{A}, \alpha_{x \mapsto a}}(\phi) = \mathsf{T}; \\ \mathsf{F}, & \text{otherwise,} \end{cases}$$

so that, for example,

$$V_{\Omega, \alpha}(\exists x (x \dot{+} y \doteq z)) = \begin{cases} \mathsf{T}, & \text{if for some } m \in \omega, m + \alpha(y) = \alpha(z); \\ \mathsf{F}, & \text{otherwise.} \end{cases}$$

This condition creates a technical problem for our recursive definition — the schema established in Theorem 1.2.16 allows $V_{\mathfrak{A}, \alpha}(\exists x \phi)$ to be defined in terms of $V_{\mathfrak{A}, \alpha}(\phi)$, but not in terms of $V_{\mathfrak{A}, \beta}(\phi)$ for other $\mathfrak{A}$ -assignments β . The solution is to apply the method of Remark 1.2.17; we regard the clauses below as those of a recursive definition of a function $V_{\mathfrak{A}}$ such that $V_{\mathfrak{A}}(\phi)$ is itself a function from assignments to truth values whose value on an assignment α is $V_{\mathfrak{A}, \alpha}(\phi)$ as described above. Thus, formally (iii) below is

$$V_{\mathfrak{A}}(\exists x \phi)(\alpha) = \begin{cases} \mathsf{T}, & \text{if for some } a \in A, V_{\mathfrak{A}}(\phi)(\alpha_{x \mapsto a}) = \mathsf{T}; \\ \mathsf{F}, & \text{otherwise.} \end{cases}$$

Definition 2.1.22 For any L -structure $\mathfrak{A}$, all $\mathfrak{A}$ -assignments α , all L -formulas ϕ and ψ and any variable x ,

- (i) $V_{\mathfrak{A}, \alpha}(\neg \phi) := \begin{cases} \mathsf{T}, & \text{if } V_{\mathfrak{A}, \alpha}(\phi) = \mathsf{F}; \\ \mathsf{F}, & \text{otherwise;} \end{cases}$
- (ii) $V_{\mathfrak{A}, \alpha}(\phi \vee \psi) := \begin{cases} \mathsf{T}, & \text{if } V_{\mathfrak{A}, \alpha}(\phi) = \mathsf{T} \text{ or } V_{\mathfrak{A}, \alpha}(\psi) = \mathsf{T}; \\ \mathsf{F}, & \text{otherwise;} \end{cases}$
- (iii) $V_{\mathfrak{A}, \alpha}(\exists x \phi) := \begin{cases} \mathsf{T}, & \text{if for some } a \in A, V_{\mathfrak{A}, \alpha_{x \mapsto a}}(\phi) = \mathsf{T}; \\ \mathsf{F}, & \text{otherwise.} \end{cases}$

We have written the definitions in this form to emphasize that they are recursive definitions analogous to those of [Chapter 1](#). It is conventional, however, to use a notation which favors the truth of a formula over its falsity.

Definition 2.1.23 For any $\mathfrak{A}$, α and ϕ ,

- (i) $\models_{\mathfrak{A}} \phi[\alpha]$ (α **satisfies** ϕ **in** $\mathfrak{A}$) iff $V_{\mathfrak{A},\alpha}(\phi) = \top$;
- (ii) $\not\models_{\mathfrak{A}} \phi[\alpha]$ iff $V_{\mathfrak{A},\alpha}(\phi) = \text{F}$.

Then it is straightforward to translate the clauses of the preceding definition into this notation.

Corollary 2.1.24 For any $\mathfrak{A}$ and α and any formulas ϕ and ψ ,

- (i) $\models_{\mathfrak{A}} \neg \phi[\alpha] \iff \not\models_{\mathfrak{A}} \phi[\alpha]$;
- (ii) $\models_{\mathfrak{A}} (\phi \vee \psi)[\alpha] \iff \models_{\mathfrak{A}} \phi[\alpha] \text{ or } \models_{\mathfrak{A}} \psi[\alpha]$;
- (iii) $\models_{\mathfrak{A}} (\exists x \phi)[\alpha] \iff \text{for some } a \in A, \models_{\mathfrak{A}} \phi[\alpha_{x \mapsto a}]. \quad \square$

In case (iii) we call an element a such that $\models_{\mathfrak{A}} \phi[\alpha_{x \mapsto a}]$ a **witness** to the fact that $\models_{\mathfrak{A}} (\exists x \phi)[\alpha]$. This is a good point to remark that our abbreviations adopted earlier have the intended effect.

Proposition 2.1.25 For any $\mathfrak{A}$ and α and any formulas ϕ and ψ ,

- (i) $\models_{\mathfrak{A}} (\phi \rightarrow \psi)[\alpha] \iff \not\models_{\mathfrak{A}} \phi[\alpha] \text{ or } \models_{\mathfrak{A}} \psi[\alpha]$;
- (ii) $\models_{\mathfrak{A}} (\phi \wedge \psi)[\alpha] \iff \models_{\mathfrak{A}} \phi[\alpha] \text{ and } \models_{\mathfrak{A}} \psi[\alpha]$;
- (iii) $\models_{\mathfrak{A}} (\phi \leftrightarrow \psi)[\alpha] \iff \models_{\mathfrak{A}} \phi[\alpha] \text{ iff } \models_{\mathfrak{A}} \psi[\alpha]$;
- (iv) $\models_{\mathfrak{A}} (\forall x \phi)[\alpha] \iff \text{for all } a \in A, \models_{\mathfrak{A}} \phi[\alpha_{x \mapsto a}]. \quad \square$

Examples 2.1.26

- (i) For any Ω -assignment α ,

$$\models_{\Omega} x < \dot{S}y \rightarrow (x < y \vee x \dot{=} y)[\alpha];$$

that is, for any $m (= \alpha(x))$ and $n (= \alpha(y)) \in \omega$, if $m < n + 1$, then either $m < n$ or $m = n$.

- (ii) If $\alpha(x) = 1$ or $\alpha(x) = 2$,

$$\models_{\Omega} ((x \dot{\times} x) \dot{+} \dot{S}\dot{S}\dot{0}) \dot{=} [x \dot{+} (x \dot{+} x)][\alpha];$$

that is, 1 and 2 are solutions to the equation $x^2 + 2 = 3x$.

(iii) For any Ω -assignment α ,

$$\models_{\Omega} \exists x [(x \dot{\times} x) \dot{+} \dot{S}\dot{S}\dot{0}] \doteq [x \dot{+} (x \dot{+} x)] [\alpha];$$

that is, the equation $x^2 + 2 = 3x$ has at one least natural number solution.

(iv) For any $(\mathbb{Q}, <)$ -assignment α ,

$$\models_{(\mathbb{Q}, <)} \forall x \forall y (x < y \rightarrow \exists z [x < z \wedge z < y]) [\alpha];$$

that is, the ordering of the rational numbers is dense.

(v) $\models_{(\mathbb{Z}, \geq, s_c, 0)} (x < \dot{S}\dot{0}) [\alpha] \quad \text{iff } \alpha(x) \geq 1. \quad (!)$

(vi) For any structure $\mathfrak{A}$, $\models_{\mathfrak{A}} \exists x \exists y \exists z (\neg x \doteq y \wedge \neg y \doteq z \wedge \neg x \doteq z) [\alpha]$ iff the universe A of $\mathfrak{A}$ has at least three distinct elements.

Much of the material in the remainder of this chapter and beyond will be applicable to any first-order language L , and we will not always write “for any language L ” at the beginning of statements of definitions and results. However, at times we need to make distinctions among various classes of languages. First, we may differentiate on the basis of size:

Definition 2.1.27 A language L is called **finite (countable, denumerable)** iff $\text{Rs}_L \cup \text{Fs}_L \cup \text{Cs}_L$, the set of non-logical symbols of L , is finite (countable, denumerable). For any infinite cardinal κ , L is a **language of cardinality κ** iff $\text{Rs}_L \cup \text{Fs}_L \cup \text{Cs}_L$ is a set of cardinality κ .

Note that even a finite language has infinitely many terms (the variables) and infinitely many formulas ($x \doteq y$ for each pair of variables). More generally,

Proposition 2.1.28 *For any language L ,*

- (i) *if L is countable, then Te_L , AtFm_L and Fm_L are all denumerable sets;*
- (ii) *for any infinite cardinal κ , if L has cardinality κ , then AtFm_L and Fm_L are sets of cardinality κ .*

Proof. By elementary set theory, a language with κ -many symbols has also κ -many expressions, so κ is an upper bound on the sizes of the three sets. In the finite case we noted above that $\aleph_0$ is also a lower bound, and this is clearly also true for a denumerable language. Suppose now that L has κ -many non-logical symbols for some uncountable cardinal κ . Then at least one of the sets Rs_L , Fs_L or Cs_L must alone have cardinality κ . If there are κ -many relation symbols p , then there are κ -many atomic formulas of the form $px_0 \dots x_{\nu_L(p)-1}$. If there are κ -many function symbols f , then there

are κ -many atomic formulas of the form $fx_0 \dots x_{\nu_L(f)-1} \doteq y$. If there are κ -many constant symbols c , then there are κ -many atomic formulas of the form $c \doteq d$. Hence in each case there are at least and hence exactly κ -many atomic formulas and also formulas. $\square$

For countable languages we may also consider questions of effectiveness. These often parallel the corresponding questions for propositional logic but are generally more complex and interesting. For example, if L has an infinite set $\{f_n : n \in \omega\}$ of function symbols, then an expression $fx_0 \dots x_{k-1}$ is a term iff $k = \nu_L(f_n)$. For this to be a decidable property requires that the function $n \mapsto \nu_L(f_n)$ be effectively computable.

To deal with these issues we assume that all countable languages we consider draw their non-logical symbols from a fixed “master list” which is effectively enumerable. This list will contain for each $k > 0$ denumerably many k -ary function and relation symbols and denumerably many constant symbols. We further make the reasonable assumption that the set of variables is a decidable subset of the set of all symbols.

Definition 2.1.29 L is an *effective language* iff

- (i) Rs_L , Fs_L and Cs_L are decidable subsets of the master list;
- (ii) the function ν_L is effectively computable.

In particular, any finite language is effective.

Proposition 2.1.30 *For any effective language L ,*

- (i) *the set of L -expressions is effectively enumerable;*
- (ii) *the sets Te_L , AtFm_L and Fm_L are decidable in the set of all L -expressions, hence are also effectively enumerable.*

Proof. Since the sets of relation, function and constant symbols are by hypothesis decidable subsets of the effectively enumerable master list, they are effectively enumerable by Proposition 1.5.14. Hence, the set of all L -symbols is effectively enumerable, and the effective enumerability of the set of L expressions follows by Proposition 1.5.16.

For (ii), we use arguments parallel to those of the proof of Theorem 1.5.3. For example, to decide if an L -expression is an L -term, first determine the length ℓ of the expression. If $\ell = 1$, then the expression is a term iff the unique symbol is a variable or a constant symbol and this is a decidable property. If $\ell > 1$, then the expression is not a term unless the first symbol is a function symbol f . If so, let $s_1, \dots, s_{l-1}$ be the remaining symbols. By hypothesis, we may effectively compute $n := \nu_L(f)$. Now the expression is a term iff $s_1, \dots, s_{l-1}$ may be divided into n blocks of symbols, each of which is an L -term. There are only finitely many ways to do the division; we apply

the algorithm recursively to each block and return a positive answer iff for one of the divisions all blocks give a positive answer.

The arguments for AtFm_L and Fm_L are similar. The effective enumerability of the sets now follows from (i) and Proposition 1.5.16. $\square$

Collateral Reading. The material of this chapter is for the most part entirely standard, and alternative presentations can be read in almost any logic text, for example [Sh01b], [Eb-Fl-Th], [Men] or [En01].

Exercises

Exercise 2.1.31 Prove the Unique Readability Theorem for L -terms.

Exercise 2.1.32 Prove the Unique Readability Theorem for L -formulas.

Exercise 2.1.33 Prove parts (ii) and (iv) of Proposition 2.1.25.

2.2 Basic semantics

The analogues here of the notions of tautology, tautological consequence and consistency (Definitions 1.3.2, 1.4.1 and 1.4.8) are

Definition 2.2.1 For any formulas ϕ and ψ , and any set Γ of formulas,

(i) ϕ is **logically valid** or a **logical validity** — in symbols, $\models \phi$ — iff for all $\mathfrak{A}$ and α , $\models_{\mathfrak{A}} \phi[\alpha]$;

(ii) ϕ and ψ are **logically equivalent** — in symbols, $\phi \models \psi$ — iff

$$\text{for all } \mathfrak{A} \text{ and } \alpha, \quad \models_{\mathfrak{A}} \phi[\alpha] \iff \models_{\mathfrak{A}} \psi[\alpha];$$

(iii) ψ is a **logical consequence** of ϕ — in symbols, $\phi \models \psi$ — iff

$$\text{for all } \mathfrak{A} \text{ and } \alpha, \quad \models_{\mathfrak{A}} \phi[\alpha] \implies \models_{\mathfrak{A}} \psi[\alpha];$$

(iv) ψ is a **logical consequence** of Γ — in symbols, $\Gamma \models \psi$ — iff

$$\text{for all } \mathfrak{A} \text{ and } \alpha, \quad \text{if for all } \phi \in \Gamma, \models_{\mathfrak{A}} \phi[\alpha], \text{ then } \models_{\mathfrak{A}} \psi[\alpha];$$

(v) Γ is **consistent** iff for some $\mathfrak{A}$ and α , $\models_{\mathfrak{A}} \phi[\alpha]$ for all $\phi \in \Gamma$.

These notions have their most elementary properties in common with the notions of tautology and tautological consequence (see Propositions 1.3.4 and 1.4.3).

Proposition 2.2.2 For any sets Γ and Δ of formulas and any formulas ϕ and ψ ,

- (i) $\phi \models \psi \iff \models \phi \leftrightarrow \psi$;
- (ii) $\phi \models \psi \iff \models \phi \rightarrow \psi$;
- (iii) $\models \psi \iff \emptyset \models \psi$;
- (iv) if $\Gamma \subseteq \Delta$ and $\Gamma \models \psi$, then also $\Delta \models \psi$;
- (v) for any $\phi_0, \dots, \phi_{n-1}$ and $i < n$, the following are equivalent:
 - (a) $\Gamma \cup \{\phi_0, \dots, \phi_n\} \models \psi$
 - (b) $\Gamma \cup \{\phi_0 \wedge \dots \wedge \phi_n\} \models \psi$
 - (c) $\Gamma \cup \{\phi_0, \dots, \phi_{i-1}\} \models \phi_i \wedge \dots \wedge \phi_n \rightarrow \psi$. $\square$

Proposition 2.2.3 For any set Γ of formulas and any formula ψ ,

- (i) $\Gamma \models \psi \iff \Gamma \cup \{\neg \psi\}$ is inconsistent;
- (ii) $\Gamma \models \neg \psi \iff \Gamma \cup \{\psi\}$ is inconsistent;
- (iii) if Γ is consistent, then at least one of $\Gamma \cup \{\psi\}$ and $\Gamma \cup \{\neg \psi\}$ is consistent.

Proof. Immediate from the definitions. $\square$

As in the propositional case we shall sometimes write $\bigwedge \Gamma$ and $\bigvee \Gamma$ for the conjunction and disjunction of all the formulas in Γ , even when Γ is infinite. Thus, for example, (v) has the generalized version

$$\Gamma \cup \Delta \models \psi \iff \Gamma \cup \left\{ \bigwedge \Delta \right\} \models \psi \iff \Gamma \models \bigwedge \Delta \rightarrow \psi.$$

The notions of tautology and tautological equivalence and consequence can also be applied to a first-order language L . An ***L-truth assignment*** is any function $V : \text{Fm}_L \rightarrow \{\mathbf{T}, \mathbf{F}\}$ which respects the connectives $\neg$ and $\vee$ (and therefore also $\wedge$, $\rightarrow$ and $\leftrightarrow$). Note, however, that no constraints are put on the relationship between the values $V(\phi)$ and $V(\exists x \phi)$; from the perspective of truth assignments, a formula $\exists x \phi$ is atomic (see exercises below). Furthermore, equality has no special status, e.g., $V(x \doteq x) = \mathbf{F}$ is permitted.

Then just as for the propositional language, we say that an L -formula is a ***tautology*** — in symbols $\models \phi$ — iff $V(\phi) = \mathbf{T}$ for every L -truth assignment V ; ϕ and ψ are ***tautologically equivalent*** — in symbols $\phi \models \psi$ — iff $V(\phi) = V(\psi)$ for all L -truth assignments V ; and ψ is a ***tautological consequence*** of ψ — in symbols, $\phi \models \psi$ — iff for all L -truth assignments V , if $V(\phi) = \mathbf{T}$, then $V(\psi) = \mathbf{T}$. The notion $\Gamma \models \psi$ is defined similarly. The following proposition asserts that logical validity, equivalence and consequence are generalizations of the corresponding tautological notions.

Proposition 2.2.4 *For any set Γ of formulas and any formulas ϕ and ψ ,*

- (i) $\models \phi \implies \models \phi$;
- (ii) $\phi \models \psi \implies \phi \models \psi$;
- (iii) $\phi \models \psi \implies \phi \models \psi$;
- (iv) $\Gamma \models \psi \implies \Gamma \models \psi$.

Proof. All of these follow easily from the observation (Definition 2.1.22) that for every L -structure $\mathfrak{A}$ and every $\mathfrak{A}$ -assignment α , the function $V_{\mathfrak{A},\alpha}$ is an L -truth assignment. $\square$

The logical validities which are not tautologies arise either from properties of the equality symbol or of the quantifiers. The **equality axioms** below may be summarized by saying that equality is an equivalence relation ((i)–(iii)) with the substitution property for functions and relations ((iv)–(v)).

Proposition 2.2.5 *For any terms $t, u, \dots, f \in \mathbf{Fs}_L$ and $p \in \mathbf{Rs}_L$,*

- (i) $\models t \doteq t$;
- (ii) $\models t \doteq u \rightarrow u \doteq t$;
- (iii) $\models t \doteq u \wedge u \doteq v \rightarrow t \doteq v$;
- (iv) $\models (t_0 \doteq u_0) \wedge \dots \wedge (t_{\nu_L(f)-1} \doteq u_{\nu_L(f)-1}) \rightarrow$
 $f t_0 \dots t_{\nu_L(f)-1} \doteq f u_0 \dots u_{\nu_L(f)-1}$;
- (v) $\models (t_0 \doteq u_0) \wedge \dots \wedge (t_{\nu_L(p)-1} \doteq u_{\nu_L(p)-1}) \rightarrow$
 $[p t_0 \dots t_{\nu_L(p)-1} \leftrightarrow p u_0 \dots u_{\nu_L(p)-1}]$.

Proof. Immediate from the fact that the equality symbol $\doteq$ is always interpreted as the true equality relation on A . $\square$

There are many examples of logical validity, equivalence and consequence which arise from the properties of the quantifiers. On the following page we list some of the most important instances of these. All may be verified by straightforward application of the definitions, but the reader should carry out some of these verifications in detail and do the others mentally.

It is instructive to notice that we have already seen applications of many of these rules. For example, the first distributive law $\forall x (\phi \wedge \psi) \models \forall x \phi \wedge \forall x \psi$ is the principle behind the fact that in propositional logic

$$\models \theta \wedge \chi \iff \models \theta \text{ and } \models \chi,$$

since this translates via the definition of tautology to

$$\begin{aligned} &\text{for all } V[V(\theta) = \mathbf{T} \text{ and } V(\chi) = \mathbf{T}] \\ &\iff \text{for all } V[V(\theta) = \mathbf{T}] \text{ and for all } V[V(\chi) = \mathbf{T}]. \end{aligned}$$

Logical equivalences and logical consequences

$$\neg \exists x \phi \models \forall x \neg \phi$$

quantifier negation laws

$$\neg \forall x \phi \models \exists x \neg \phi$$

$$\forall x (\phi \wedge \psi) \models \forall x \phi \wedge \forall x \psi$$

$\forall$ *distributes over* $\wedge$

$$\forall x \phi \vee \forall x \psi \models \forall x (\phi \vee \psi)$$

$\forall$ *partially distributes over* $\vee$

$$\exists x (\phi \wedge \psi) \models \exists x \phi \wedge \exists x \psi$$

$\exists$ *partially distributes over* $\wedge$

$$\exists x (\phi \vee \psi) \models \exists x \phi \vee \exists x \psi$$

$\exists$ *distributes over* $\vee$

$$\forall x (\phi \rightarrow \psi) \models \forall x \phi \rightarrow \forall x \psi$$

$\forall$ *partially distributes over* $\rightarrow$

$$\forall x (\phi \rightarrow \psi) \models \exists x \phi \rightarrow \exists x \psi$$

$$\forall x \phi \models \exists x \phi$$

universe is non-empty

$$\phi \models \exists x \phi$$

$$\forall x \phi \models \phi$$

$$\forall x \forall y \phi \models \forall y \forall x \phi$$

quantifier interchange

$$\exists x \exists y \phi \models \exists y \exists x \phi$$

$$\exists y \forall x \phi \models \forall x \exists y \phi$$

uniformity

The following hold under the hypothesis that the variable x does not occur free in ϕ :

$$\phi \models \exists x \phi$$

$$\phi \models \forall x \phi$$

$$\forall x (\phi \vee \psi) \models \phi \vee \forall x \psi$$

$$\exists x (\phi \wedge \psi) \models \phi \wedge \exists x \psi$$

$$\forall x (\phi \rightarrow \psi) \models \phi \rightarrow \forall x \psi$$

$$\forall x (\psi \rightarrow \phi) \models \exists x \psi \rightarrow \phi$$

$$\exists x (\phi \rightarrow \psi) \models \phi \rightarrow \exists x \psi$$

$$\exists x (\psi \rightarrow \phi) \models \forall x \psi \rightarrow \phi$$

Similarly, the second distributive law corresponds to

$$\models \theta \quad \text{or} \quad \models \chi \quad \implies \quad \models \theta \vee \chi.$$

Of course, in this case the converse implication fails in general. The difference between these two examples arises from the intuition that the existential quantifier $\exists$ is in effect a generalized “or”; $\models_{\mathfrak{A}} \exists x \phi[\alpha]$ holds iff for any listing $a, b, \dots$ of the elements of $|\mathfrak{A}|$, $\models_{\mathfrak{A}} \phi[\alpha_{x \mapsto a}]$, or $\models_{\mathfrak{A}} \phi[\alpha_{x \mapsto b}]$, or $\dots$. Similarly $\forall$ is a generalized “and”. The reader should find similar examples to illustrate the other assertions from the table.

In propositional logic a sentence may be viewed as a condition which defines the set of its models: $\{V : V(\phi) = \mathbf{T}\}$. Analogously here, a formula represents a condition which defines a set of pairs $(\mathfrak{A}, \alpha)$ of a structure and an assignment. There is a class of formulas, which we will again call sentences, which represent a condition on a structure alone. For example, the sentence $\forall x \forall y (x + y \doteq y + x)$ represents a condition which is satisfied by the structure Ω because addition of natural numbers is commutative, but the formula $\exists x (x + y \doteq z)$ represents a condition which is satisfied by some values assigned to y and z but not for others. The point is that the role of the variable x here is different from that of y and z ; the interpretation of this formula does not depend on the value assigned to x . Indeed, x is merely a placeholder similar to its occurrence in the expression $\int_0^2 f(x) dx$ of calculus. Just as $\int_0^2 f(u) du$ has the same meaning, so also we may write $\exists u (u + y \doteq z)$ without altering the meaning. A sentence is a formula in which every occurrence of a variable is of this latter sort.

We make this precise as follows.

Definition 2.2.6 For any formula ϕ , the set $\text{Fv}(\phi)$ of **free variables** of ϕ is defined recursively by:

- (i) if ϕ is atomic, $\text{Fv}(\phi) = \{x : x \text{ occurs in } \phi\}$;
- (ii) $\text{Fv}(\neg \phi) = \text{Fv}(\phi)$;
- (iii) $\text{Fv}(\phi \vee \psi) = \text{Fv}(\phi) \cup \text{Fv}(\psi)$;
- (iv) $\text{Fv}(\exists x \phi) = \text{Fv}(\phi) \sim \{x\}$.

When $x \in \text{Fv}(\phi)$, we say that x **occurs free** in ϕ . Although it is of less importance, the set of **bound variables** $\text{Bv}(\phi)$ may be defined by a similar recursion with clauses (i) for ϕ atomic, $\text{Bv}(\phi) = \emptyset$, (iv) $\text{Bv}(\exists x \phi) = \text{Bv}(\phi) \cup \{x\}$ and (ii) and (iii) as for Fv . Note that a variable may be both free and bound in a formula.

Definition 2.2.7 An L -formula ϕ is an L -**sentence** iff $\text{Fv}(\phi) = \emptyset$.

Intuitively, a sentence needs no variable assignment to complete its meaning. More generally

Proposition 2.2.8 *For any L -structure $\mathfrak{A}$ and any $\mathfrak{A}$ -assignments α and β ,*

- (i) *for any term t , if for all variables x which occur in t , $\alpha(x) = \beta(x)$, then*

$$t^{\mathfrak{A}}[\alpha] = t^{\mathfrak{A}}[\beta];$$

- (ii) *for any formula ϕ , if for all $x \in \text{Fv}(\phi)$, $\alpha(x) = \beta(x)$, then*

$$\models_{\mathfrak{A}} \phi[\alpha] \iff \models_{\mathfrak{A}} \phi[\beta].$$

Proof. (i) is proved by a straightforward term induction. For atomic terms the result is obvious. For the induction step we have

$$\begin{aligned} (ft_0 \dots t_{\nu_L(f)-1})^{\mathfrak{A}}[\alpha] &= f^{\mathfrak{A}}(t_0^{\mathfrak{A}}[\alpha], \dots, t_{\nu_L(f)-1}^{\mathfrak{A}}[\alpha]) \\ &= f^{\mathfrak{A}}(t_0^{\mathfrak{A}}[\beta], \dots, t_{\nu_L(f)-1}^{\mathfrak{A}}[\beta]) \\ &= (ft_0 \dots t_{\nu_L(f)-1})^{\mathfrak{A}}[\beta]. \end{aligned}$$

The first and third equalities are just the definition; the middle equality relies on the fact that if $\alpha(x) = \beta(x)$ for all variables x which occur in $ft_0 \dots t_{\nu_L(f)-1}$, then also $\alpha(x) = \beta(x)$ for all x which occur in each individual term t_i .

The proof of (ii) is by formula induction. If ϕ is atomic, the result is immediate from (i):

$$\begin{aligned} \models_{\mathfrak{A}} pt_0 \dots t_{\nu_L(p)-1}[\alpha] &\iff (t_0^{\mathfrak{A}}[\alpha], \dots, t_{\nu_L(p)-1}^{\mathfrak{A}}[\alpha]) \in p^{\mathfrak{A}} \\ &\iff (t_0^{\mathfrak{A}}[\beta], \dots, t_{\nu_L(p)-1}^{\mathfrak{A}}[\beta]) \in p^{\mathfrak{A}} \\ &\iff \models_{\mathfrak{A}} pt_0 \dots t_{\nu_L(p)-1}[\beta]. \end{aligned}$$

If ϕ is $\neg \psi$, then since $\text{Fv}(\phi) = \text{Fv}(\psi)$,

$$\models_{\mathfrak{A}} \phi[\alpha] \iff \not\models_{\mathfrak{A}} \psi[\alpha] \iff \not\models_{\mathfrak{A}} \psi[\beta] \iff \models_{\mathfrak{A}} \phi[\beta],$$

where the second equivalence is the induction hypothesis. The $\vee$ case is similar. Suppose, finally, that ϕ is $\exists y \psi$ and that $\alpha(x) = \beta(x)$ for all $x \in \text{Fv}(\phi) = \text{Fv}(\psi) \sim \{y\}$. Then for any $a \in A$, $\alpha_{y \mapsto a}(x) = \beta_{y \mapsto a}(x)$ for all $x \in \text{Fv}(\psi)$, so we have

$$\begin{aligned} \models_{\mathfrak{A}} \phi[\alpha] &\iff \text{for some } a \in A, \models_{\mathfrak{A}} \psi[\alpha_{y \mapsto a}] \\ &\iff \text{for some } a \in A, \models_{\mathfrak{A}} \psi[\beta_{y \mapsto a}] \\ &\iff \models_{\mathfrak{A}} \phi[\beta]. \quad \square \end{aligned}$$

Corollary 2.2.9 For any sentence ϕ and any $\mathfrak{A}$, α and β ,

$$\models_{\mathfrak{A}} \phi[\alpha] \iff \models_{\mathfrak{A}} \phi[\beta].$$

Hence,

$$\text{for some } \alpha, \models_{\mathfrak{A}} \phi[\alpha] \iff \text{for all } \alpha, \models_{\mathfrak{A}} \phi[\alpha].$$

The proposition bears a strong family resemblance to Proposition 1.3.1, and we use it to justify a convention analogous to the finiteness of truth tables.

Definition 2.2.10 For any formula ϕ , if $\text{Fv}(\phi) \subseteq \{x_0, \dots, x_{k-1}\}$, then we write

$$\models_{\mathfrak{A}} \phi_{x_0, \dots, x_{k-1}}[a_0, \dots, a_{k-1}] \quad \text{or simply} \quad \models_{\mathfrak{A}} \phi[a_0, \dots, a_{k-1}]$$

to mean $\models_{\mathfrak{A}} \phi[\alpha]$ for some (equivalently any) assignment α such that for all $i < k$, $\alpha(x_i) = a_i$. In particular, if ϕ is a sentence, we write simply $\models_{\mathfrak{A}} \phi$ and say that ϕ is **true in $\mathfrak{A}$** . When $\not\models_{\mathfrak{A}} \phi$, we say that ϕ is **false in $\mathfrak{A}$** .

The elementary properties of this notion are

Theorem 2.2.11 For any $\mathfrak{A}$, and any sentences ϕ and ψ ,

- (i) $\models_{\mathfrak{A}} \neg \phi \iff \not\models_{\mathfrak{A}} \phi$;
- (ii) $\models_{\mathfrak{A}} \phi \vee \psi \iff \models_{\mathfrak{A}} \phi \text{ or } \models_{\mathfrak{A}} \psi$;
- (iii) $\models_{\mathfrak{A}} \phi \wedge \psi \iff \models_{\mathfrak{A}} \phi \text{ and } \models_{\mathfrak{A}} \psi$;
- (iv) $\models_{\mathfrak{A}} \phi \rightarrow \psi \iff \not\models_{\mathfrak{A}} \phi \text{ or } \models_{\mathfrak{A}} \psi$;
- (v) $\models_{\mathfrak{A}} \phi \leftrightarrow \psi \iff \models_{\mathfrak{A}} \phi \text{ iff } \models_{\mathfrak{A}} \psi$.

Proof. All of these are proved similarly, and we just do two as a sample.

$$\begin{aligned}
 \models_{\mathfrak{A}} \phi \vee \psi &\iff \text{for some } \alpha, \models_{\mathfrak{A}} (\phi \vee \psi)[\alpha] \\
 &\iff \text{for some } \alpha, \models_{\mathfrak{A}} \phi[\alpha] \text{ or for some } \alpha, \models_{\mathfrak{A}} \psi[\alpha] \\
 &\iff \models_{\mathfrak{A}} \phi \text{ or } \models_{\mathfrak{A}} \psi \\
 \models_{\mathfrak{A}} \phi \wedge \psi &\iff \text{for all } \alpha, \models_{\mathfrak{A}} (\phi \wedge \psi)[\alpha] \\
 &\iff \text{for all } \alpha, \models_{\mathfrak{A}} \phi[\alpha] \text{ and for all } \alpha, \models_{\mathfrak{A}} \psi[\alpha] \\
 &\iff \models_{\mathfrak{A}} \phi \text{ and } \models_{\mathfrak{A}} \psi. \quad \square
 \end{aligned}$$

Examples 2.2.12

- (i) In every language there are sentences which are **logically valid** — true in every structure — or **logically false** — false in every structure. If L has at least one constant symbol c , then $c \doteq c$ and $\neg c \doteq c$ are examples; otherwise we may take $\exists x (x \doteq x)$ and $\neg \exists x (x \doteq x)$. In either case we use the symbols **T** and **F** to denote some fixed choice of such sentences.
- (ii) For each $n \geq 2$, denote the following sentence by $\exists^{\geq n}$:

$$\begin{aligned} \exists x_0 \cdots \exists x_{n-1} [\neg (x_0 \doteq x_1) \wedge \neg (x_0 \doteq x_2) \\ \wedge \neg (x_1 \doteq x_2) \wedge \cdots \wedge \neg (x_{n-2} \doteq x_{n-1})], \end{aligned}$$

which we write more compactly and accurately as

$$\exists x_0 \cdots \exists x_{n-1} \bigwedge_{i < j < n} \neg (x_i \doteq x_j).$$

It is easy to verify that

$$\models_{\mathfrak{A}} \exists^{\geq n} \iff A \text{ has at least } n \text{ elements.}$$

For consistency, we write also $\exists^{\geq 1}$ for the sentence $\exists x [x = x]$, which is logically valid because by definition all structures have non-empty universes. Similarly, if $\exists^{\leq n}$ is $\neg \exists^{\geq n+1}$ and $\exists^n$ is $\exists^{\geq n} \wedge \exists^{\leq n}$, then

$$\begin{aligned} \models_{\mathfrak{A}} \exists^{\leq n} &\iff A \text{ has at most } n \text{ elements;} \\ \models_{\mathfrak{A}} \exists^n &\iff A \text{ has exactly } n \text{ elements.} \end{aligned}$$

- (iii) The properties of structures from Example 2.1.1 can be expressed by sentences of the respective languages. For example, in the language $L_{<}$ with one binary relation symbol $\dot{<}$:

$$\begin{aligned} \theta_{\text{irrefl}} &\text{ is } \forall x (\neg x \dot{<} x); \\ \theta_{\text{trans}} &\text{ is } \forall x \forall y \forall z (x \dot{<} y \wedge y \dot{<} z \rightarrow x \dot{<} z); \\ \theta_{\text{conn}} &\text{ is } \forall x \forall y (\neg x \doteq y \rightarrow x \dot{<} y \vee y \dot{<} x); \\ \theta_{\text{dense}} &\text{ is } \forall x \forall y [x \dot{<} y \rightarrow \exists z (x \dot{<} z \wedge z \dot{<} y)]. \end{aligned}$$

Then for any $L_{<}$ -structure $\mathfrak{A} = (A, <_A)$,

$$\begin{aligned} \models_{\mathfrak{A}} \theta_{\text{irrefl}} &\iff \text{for all } a \in A, \quad \models_{\mathfrak{A}} (\neg x \doteq x)[a] \\ &\iff \text{for all } a \in A, \quad \text{not } a <_A a \\ &\iff \mathfrak{A} \text{ is irreflexive.} \end{aligned}$$

Hence $\mathfrak{A}$ is a strict partial ordering iff $\models_{\mathfrak{A}} \theta_{\text{irrefl}}$ and $\models_{\mathfrak{A}} \theta_{\text{trans}}$ or equivalently $\models_{\mathfrak{A}} \theta_{\text{irrefl}} \wedge \theta_{\text{trans}}$, and if θ_{SLO} denotes the sentence $\theta_{\text{irrefl}} \wedge \theta_{\text{trans}} \wedge \theta_{\text{conn}}$, and θ_{DLO} is $\theta_{\text{SLO}} \wedge \exists^{\geq 2} \wedge \theta_{\text{dense}}$, then

$$\models_{\mathfrak{A}} \theta_{\text{SLO}} \iff \mathfrak{A} \text{ is a strict linear ordering,}$$

and

$$\models_{\mathfrak{A}} \theta_{\text{DLO}} \iff \mathfrak{A} \text{ is a dense linear ordering.}$$

As in propositional logic we use the following terminology.

Definition 2.2.13 For any structure $\mathfrak{A}$,

- (i) $\mathfrak{A}$ is a **model** of a sentence ϕ iff ϕ is true in $\mathfrak{A}$;
- (ii) $\mathfrak{A}$ is a **model** of a set Γ of sentences iff each $\phi \in \Gamma$ is true in $\mathfrak{A}$.

Note that as in the preceding examples, for a finite set

$$\mathfrak{A} \text{ is a model of } \{\phi_0, \dots, \phi_n\} \iff \mathfrak{A} \text{ is a model of } \phi_0 \wedge \dots \wedge \phi_n.$$

Example 2.2.14 The various formulations of the notion of group from Examples 2.1.1 may be expressed as follows. A structure $\mathfrak{G} = (G, *, {}^{-1}, e)$ is a group iff $\mathfrak{G}$ is a model of the set

$$\Gamma_{\text{Gp}} := \{\theta_{\text{assoc}}, \theta_{\text{ident}}, \theta_{\text{inverse}}\}$$

of L_G -sentences, where

$$\begin{aligned} \theta_{\text{assoc}} & \text{ is } \forall x \forall y \forall z [x * (y * z) \doteq (x * y) * z] \\ \theta_{\text{ident}} & \text{ is } \forall x (x * e \doteq x \wedge e * x \doteq x) \\ \theta_{\text{inverse}} & \text{ is } \forall x (x * x^{-1} \doteq e \wedge x^{-1} * x \doteq e). \end{aligned}$$

In the alternative version, $\mathfrak{G} = (G, *, e)$ is a group iff $\mathfrak{G}$ is a model of the set

$$\Gamma'_{\text{Gp}} := \{\theta_{\text{assoc}}, \theta_{\text{ident}}, \theta'_{\text{inverse}}\},$$

where

$$\theta'_{\text{inverse}} \text{ is } \forall x \exists y [x * y \doteq e \wedge y * x \doteq e],$$

and if we consider a group to be a structure of the form $\mathfrak{G} = (G, *)$, then $\mathfrak{G}$ is a group iff it is a model of

$$\Gamma''_{\text{Gp}} := \{\theta_{\text{assoc}}, \theta_{\text{identinv}}\},$$

where

$$\theta_{\text{identinv}} \text{ is } \exists z \forall x (x * z \doteq x \wedge z * x \doteq x \wedge \exists y [x * y \doteq z \wedge y * x \doteq z]).$$

Proposition 2.2.15 *For any set Γ of sentences and any sentence ψ ,*

$$\Gamma \models \psi \iff \text{every model of } \Gamma \text{ is a model of } \psi.$$

Proof. This is a simple consequence of the definitions and Corollary 2.2.9. $\square$

Example 2.2.16 Continuing the preceding example, we have that for any L_G -sentence ϕ ,

$$\begin{aligned} \Gamma_{\text{Gp}} \models \phi &\iff \phi \text{ is true in every model of } \Gamma_{\text{Gp}} \\ &\iff \phi \text{ is true in every group.} \end{aligned}$$

Sentences for which this holds are called ***theorems of group theory***.

Remark 2.2.17 It is occasionally convenient to use the notation

$$\models_{\mathfrak{A}} \phi \quad \text{to mean} \quad \text{for all } \mathfrak{A}\text{-assignments } \alpha, \quad \models_{\mathfrak{A}} \phi[\alpha]$$

also for general formulas ϕ with free variables; we say that ϕ is ***universally valid in $\mathfrak{A}$*** . This mirrors traditional usage in elementary algebra where one writes the associative law for addition as $x + (y + z) = (x + y) + z$ rather than the more complete $\forall x \forall y \forall z [x + (y + z) = (x + y) + z]$. Clearly in general we have

Proposition 2.2.18 *For any $\mathfrak{A}$, any formula ϕ and any variable x ,*
 $\models_{\mathfrak{A}} \phi \iff \models_{\mathfrak{A}} \forall x \phi.$ $\square$

We shall generally avoid this notation and write the quantifiers explicitly. Note that most of the equivalences of Theorem 2.2.11 do not hold in general for formulas with free variables under this convention.

Substitution

In the remainder of this section, we consider several other useful facts revolving around the notion of substitution. Although these are essential results for a rigorous development of the subject, they are somewhat technical, and the reader may prefer to postpone a detailed study until they are really needed.

A common operation in informal mathematics is the procedure of replacing a variable by some other expression which denotes an object of the type represented by the variable. The corresponding operation for first-order languages is the substitution of a term for a free variable. We define first $t_x(u)$, which denotes the result of substituting the term u for x in the term t and then analogously $\phi_x(u)$, where the substitution occurs only for the free occurrences of x . Technically, we are defining functions $\text{Subt}_{x,u} : \text{Te}_L \rightarrow \text{Te}_L$ by Te_L -recursion and $\text{Subf}_{x,u} : \text{Fm}_L \rightarrow \text{Fm}_L$ by Fm_L -recursion.

Definition 2.2.19 For any term u and variable x ,

- (i) for any variable y , $y_x(u)$ is $\begin{cases} u, & \text{if } y \text{ is } x; \\ y, & \text{otherwise;} \end{cases}$
- (ii) for any constant symbol c , $c_x(u)$ is c ;
- (iii) for any function symbol f and any terms $t_0, \dots, t_{\nu_L(f)-1}$,
 $(ft_0 \dots t_{\nu_L(f)-1})_x(u)$ is $f(t_0)_x(u) \dots (t_{\nu_L(f)-1})_x(u)$.

Definition 2.2.20 For any term u and variable x ,

- (i) for any terms t_0 and t_1 , $(t_0 \doteq t_1)_x(u)$ is $(t_0)_x(u) \doteq (t_1)_x(u)$;
- (ii) for any relation symbol p and any terms $t_0, \dots, t_{\nu_L(p)-1}$,
 $(pt_0 \dots t_{\nu_L(p)-1})_x(u)$ is $p(t_0)_x(u) \dots (t_{\nu_L(p)-1})_x(u)$;
- (iii) for any formula ϕ , $(\neg \phi)_x(u)$ is $\neg \phi_x(u)$;
- (iv) for any formulas ϕ and ψ , $(\phi \vee \psi)_x(u)$ is $\phi_x(u) \vee \psi_x(u)$;
- (v) for any formula ϕ and any variable y ,

$$(\exists y \phi)_x(u) \text{ is } \begin{cases} \exists y \phi, & \text{if } y \text{ is } x; \\ \exists y \phi_x(u), & \text{otherwise.} \end{cases}$$

The point of the case distinction in (v) is that x is not free in a formula of the form $\exists x \phi$, so a substitution for free occurrences of x is null.

The semantic properties of substitution are not quite so simple as one might hope. Our experience with substitution in informal mathematics leads us to expect formal substitution to have certain properties. For example, in any context where $\forall x \theta$ is true, we expect any “special case” $\theta_x(u)$ to be true as well — in other words, we expect that

$$\models_{\mathfrak{A}} \forall x \theta[\alpha] \implies \models_{\mathfrak{A}} \theta_x(u)[\alpha],$$

or equivalently, $\models \forall x \theta \rightarrow \theta_x(u)$. In other words, if some property holds for all objects in a certain context, then it also holds for any particular object. Analogously, we would expect that $\models \theta_x(u) \rightarrow \exists x \theta$. This is a case where our intuitions are almost but not quite an accurate guide. Consider in L_Ω the substitution $\theta_x(y)$ where θ is $\exists y (\dot{S}x \doteq y)$. The result is $\exists y (\dot{S}y \doteq y)$ and certainly $\models_\Omega \forall x \theta$ but *not* $\models_\Omega \theta_x(y)$. The problem is that the variable y in $\dot{S}y$ has been “captured” by the quantifier $\exists y$. In practice we would, of course simply change the quantified variable and then make the substitution: $\exists z (\dot{S}x \doteq z)_x(y)$ is $\exists z (\dot{S}y \doteq z)$ and has the intended meaning. We shall need to make a careful distinction between situations in which this does or doesn’t happen and verify that changes of bound variable behave as expected.

The intuition governing the following definition is that for a term u to be ‘correctly’ substituted for x , no free occurrence of x should occur in a context of the form $\exists y \dots x \dots$ for any variable y which occurs in u .

Definition 2.2.21 For any formula θ , variable x and term u , u is **freely substitutable** for x or x is **free for** u in θ iff one of the following holds:

- (i) θ is atomic;
- (ii) for some ϕ , θ is $\neg \phi$ and u is freely substitutable for x in ϕ ;
- (iii) for some ϕ and ψ , θ is $\phi \vee \psi$ and u is freely substitutable for x in both ϕ and ψ ;
- (iv) for some ϕ and y , θ is $\exists y \phi$ and either (1) $x \notin \text{Fv}(\theta)$ or (2) u is freely substitutable for x in ϕ and y does not occur in u .

We also say that $\theta_x(u)$ is a **free substitution**. Formally, this should be viewed as a definition by Fm_L -recursion with parameters of a function $\text{FrSub}_{u,x} : \text{Fm}_L \rightarrow \{\top, \text{F}\}$. The point of (2) is that no variable of u is “captured” by any quantifier of ϕ and there is no danger that the leading ‘ $\exists y$ ’ will capture any occurrence of y in u .

The notion of free substitution rules out exactly the problem mentioned above, and we shall show that free substitutions behave as we originally expected. First we need a somewhat technical lemma — the idea is simply that altering an assignment α to assign x to a is equivalent to replacing x with a term u whose value under α is a .

Lemma 2.2.22 For any structure $\mathfrak{A}$, variable x and term u , if $a = u^{\mathfrak{A}}[\alpha]$, then

- (i) for any term t , $(t_x(u))^{\mathfrak{A}}[\alpha] = t^{\mathfrak{A}}[\alpha_{x \mapsto a}]$;
- (ii) for any formula θ , if u is freely substitutable for x in θ , then

$$\models_{\mathfrak{A}} \theta_x(u)[\alpha] \quad \text{iff} \quad \models_{\mathfrak{A}} \theta[\alpha_{x \mapsto a}].$$

Proof. Part (i) is straightforward as is most of (ii) — for example, if t is the variable x , then

$$(t_x(u))^{\mathfrak{A}}[\alpha] = u^{\mathfrak{A}}[\alpha] = a = \alpha_{x \mapsto a}(x) = t^{\mathfrak{A}}[\alpha_{x \mapsto a}].$$

We write out only the case when θ is $\exists y \phi$. Assume that u is freely substitutable for x in θ . If (1) $x \notin \text{Fv}(\theta)$, then $\theta_x(u)$ is θ and for all $z \in \text{Fv}(\theta)$, $\alpha(z) = \alpha_{x \mapsto a}(z)$, so the desired equivalence follows. Suppose now that (2) $x \in \text{Fv}(\theta)$, u is freely substitutable for x in ϕ and y does not occur in u . Then x and y are distinct variables, $\theta_x(u)$ is $\exists y \phi_x(u)$, and

$$\begin{aligned} \models_{\mathfrak{A}} \theta_x(u)[\alpha] &\iff \text{for some } b \in A \quad \models_{\mathfrak{A}} \phi_x(u)[\alpha_{y \mapsto b}] \\ &\iff \text{for some } b \in A \quad \models_{\mathfrak{A}} \phi[\alpha_{y \mapsto b, x \mapsto a}] \\ &\iff \models_{\mathfrak{A}} \exists y \phi[\alpha_{x \mapsto a}]. \end{aligned}$$

Note that we need the hypothesis that y does not occur in u to show that $a = u^{\mathfrak{A}}[\alpha] = u^{\mathfrak{A}}[\alpha_{y \mapsto b}]$ so that the second equivalence is really an instance of the induction hypothesis. $\square$

Corollary 2.2.23 For any terms u_0 and u_1 ,

- (i) for any term t , $\models u_0 \doteq u_1 \rightarrow t_x(u_0) \doteq t_x(u_1)$;
- (ii) for any formula θ such that u_0 and u_1 are freely substitutable for x in θ ,

$$\models u_0 \doteq u_1 \rightarrow [\theta_x(u_0) \leftrightarrow \theta_x(u_1)]. \quad \square$$

Proposition 2.2.24 For all θ , x and u , if u is freely substitutable for x in θ , then

$$\models \forall x \theta \rightarrow \theta_x(u) \quad \text{and} \quad \models \theta_x(u) \rightarrow \exists x \theta.$$

Proof. Note that the second assertion follows from the first applied to $\neg\theta$, so we prove only the first. Fix $\mathfrak{A}$ and α . Then by the preceding lemma,

$$\begin{aligned} \models_{\mathfrak{A}} \forall x \theta[\alpha] &\iff \text{for all } a \in A, \quad \models_{\mathfrak{A}} \theta[\alpha_{x \mapsto a}] \\ &\implies \text{for } a = u^{\mathfrak{A}}[\alpha], \quad \models_{\mathfrak{A}} \theta[\alpha_{x \mapsto a}] \\ &\iff \models_{\mathfrak{A}} \theta_x(u)[\alpha]. \quad \square \end{aligned}$$

As a special case note

Lemma 2.2.25 For any θ , x and u , if no variable other than (possibly) x occurs in u , then the substitution $\theta_x(u)$ is free.

Proof. Left as an exercise. $\square$

Corollary 2.2.26 For any formula θ , variable x and constant symbol c ,

- (i) $\models \forall x \theta \rightarrow \theta$ and $\models \forall x \theta \rightarrow \theta_x(c)$;
- (ii) $\models \theta \rightarrow \exists x \theta$ and $\models \theta_x(c) \rightarrow \exists x \theta$.

Proof. Note that θ is $\theta_x(x)$ and both x and c are freely substitutable for x in θ . $\square$

Proposition 2.2.27 For any set Γ of formulas, formulas ψ and θ and variable x such that $x \notin \text{Fv}(\theta)$ and for all $\phi \in \Gamma$, $x \notin \text{Fv}(\phi)$,

- (i) $\Gamma \models \psi \iff \Gamma \models \forall x \psi$;
- (ii) $\Gamma \models \psi \rightarrow \theta \iff \Gamma \models \exists x \psi \rightarrow \theta$.

Proof. The implication (i) ($\Leftarrow$) is immediate from the preceding corollary. Suppose that $\Gamma \models \psi$ and that $\mathfrak{A}$ and α satisfy Γ . Then for any $a \in A$, also $\models_{\mathfrak{A}} \phi[\alpha_{x \mapsto a}]$ for each $\phi \in \Gamma$ so also $\models_{\mathfrak{A}} \psi[\alpha_{x \mapsto a}]$. Since a is arbitrary, $\models_{\mathfrak{A}} \forall x \psi[\alpha]$. (ii) is proved similarly. $\square$

For sentences, the restrictions on free variables are automatically satisfied, and we may iterate the process of quantifying the variables of a formula ψ to construct a sentence.

Definition 2.2.28 For any formula ψ , if $\text{Fv}(\psi) = \{x_0, \dots, x_{n-1}\}$,

- (i) the **universal closure** of ψ — in symbols, $\overline{\forall\psi}$ — is the sentence $\forall x_0 \cdots \forall x_{n-1} \psi$;
- (ii) the **existential closure** of ψ — in symbols, $\overline{\exists\psi}$ — is the sentence $\exists x_0 \cdots \exists x_{n-1} \psi$;
- (iii) for a set Γ of formulas, $\overline{\forall\Gamma} := \{\overline{\forall\psi} : \psi \in \Gamma\}$.

Corollary 2.2.29 For any set Γ of sentences and any formula ψ ,

$$\Gamma \models \psi \iff \Gamma \models \overline{\forall\psi}. \quad \square$$

We consider next substitutions of one variable for another. Of course in general θ and $\theta_x(y)$ are not logically equivalent — for example,

$$\models_{\Omega} x \dot{<} y[3, 4, 5] \quad \text{but} \quad \not\models_{\Omega} (x \dot{<} y)_x(z)[3, 4, 5],$$

where by convention, 3 is assigned to x , 4 to y and 5 to z — so we cannot in general change free variables without consequence. However, as we remarked above, bound variables are essentially just placeholders and may be changed subject only to the avoidance of conflicts.

Proposition 2.2.30 For any formula θ and any variables x and y such that $y \notin \text{Fv}(\theta)$ and y is freely substitutable for x in θ (in particular if y does not occur at all in θ),

$$\exists x \theta \models \exists y \theta_x(y) \quad \text{and} \quad \forall x \theta \models \forall y \theta_x(y).$$

Proof. For any $\mathfrak{A}$ and α , we have

$$\begin{aligned} \models_{\mathfrak{A}} \exists y \theta_x(y)[\alpha] &\iff \text{for some } a \in A, \quad \models_{\mathfrak{A}} \theta_x(y)[\alpha_{y \mapsto a}] \\ &\iff \text{for some } a \in A, \quad \models_{\mathfrak{A}} \theta[\alpha_{y \mapsto a}, x \mapsto a] \\ &\iff \text{for some } a \in A, \quad \models_{\mathfrak{A}} \theta[\alpha_{x \mapsto a}] \\ &\iff \models_{\mathfrak{A}} \exists x \theta[\alpha]. \end{aligned}$$

The first and fourth equivalences are just the definition of satisfaction for existential formulas, the second is by the technical lemma above, and the third uses the assumption that y is not free in θ . $\square$

This will come in handy on several future occasions; we use it now to show that every formula is logically equivalent to one in which all of the quantifiers come first. For this purpose we think of the universal quantifier $\forall$ as an actual symbol of the language, although officially $\forall x$ is an abbreviation for $\neg \exists x \neg$.

Definition 2.2.31 For any set Γ of formulas, $\text{Bool}(\Gamma)$, the set of **Boolean combinations** of members of Γ , denotes the smallest set of formulas which includes $\Gamma \cup \{\top\}$ and is such that for all $\phi, \psi \in \text{Bool}(\Gamma)$, also $(\neg\phi), (\phi \vee \psi) \in \text{Bool}(\Gamma)$. In particular, $\text{QF} := \text{Bool}(\text{AtFm}_L)$ is the set of **quantifier-free** formulas — those which contain no quantifiers.

Remark 2.2.32

- (i) Note that as pointed out in Examples 2.2.12(i), in the case that L has no constant symbol, the special sentence $\top$ must actually contain a quantifier, so QF is not completely free of quantifiers. This will never cause any problems.
- (ii) There is an obvious method of proof by Boolean induction for establishing that all members of $\text{Bool}(\Gamma)$ have some property $\mathcal{P}$: verify that all members of $\Gamma \cup \{\top\}$ have $\mathcal{P}$ and that the class of formulas that have $\mathcal{P}$ is closed under $\neg$ and $\vee$.
- (iii) Similarly, for many (but not all) choices of Γ , there is a notion of definition by recursion for extending a function $F_0 : \Gamma \cup \{\top\} \rightarrow Z$ (for any set Z) uniquely to a function $F : \text{Bool}(\Gamma) \rightarrow Z$ satisfying relationships of the form $F(\neg\phi) = G_{\neg}(F(\phi))$ and $F(\phi \vee \psi) = G_{\vee}(F(\phi), F(\psi))$. The exceptions involve Γ which contain formulas beginning with $\neg$ or $\vee$ and therefore fail to have the relevant version of unique readability.
- (iv) Let E_L temporarily denote the class of all L -formulas which begin with the symbol $\exists$. Then $\text{Bool}(\text{AtFm}_L \cup \text{E}_L)$ is the class Fm_L of all L -formulas. Although for most purposes this is a strange way of characterizing Fm_L , it is natural, for example, when we are considering truth assignments and tautological consequence. Then by (iii), for any function $V_0 : \text{AtFm}_L \cup \text{E}_L \rightarrow \{\top, \text{F}\}$ there is a unique extension of V_0 to a truth assignment $V : \text{Fm}_L \rightarrow \{\top, \text{F}\}$ in the sense of the discussion following Proposition 2.2.3.
- (v) There are natural notions of disjunctive and conjunctive normal forms for members ϕ of $\text{Bool}(\Gamma)$: $\models \phi \leftrightarrow \psi_0 \vee \dots \vee \psi_k$ such that for each $i \leq k$, ψ_i is $\theta_0^i \wedge \dots \wedge \theta_{l_i}^i$ and each θ_j^i is either a member of $\Gamma \cup \{\top\}$ or the negation of a member.

Definition 2.2.33 A formula θ is in **prenex normal form** or simply **prenex form** iff for some (possibly empty) sequences $\text{Q}_0, \dots, \text{Q}_{n-1}$ of existential and universal quantifiers and $x_0, \dots, x_{n-1}$ of variables, and some quantifier-free formula χ , θ is $\text{Q}_0 x_0 \dots \text{Q}_{n-1} x_{n-1} \chi$. The string of quantifiers and variables is called the **prefix** of θ and χ is called the **matrix** of θ .

Theorem 2.2.34 *For every formula ϕ , there exists a formula ϕ^* in prenex normal form such that $\phi \models \phi^*$. ϕ^* is called a **prenex (normal) form** of ϕ .*

Proof. We proceed by induction on the number of instances of the symbol $\exists$ in ϕ , which will be the same as the number of instances of either $\exists$ or $\forall$ in ϕ^* . If ϕ is quantifier-free, then it is itself in prenex form. Suppose that ϕ is $\neg\psi$. By the induction hypothesis, ψ has a prenex form ψ^* and

$$\psi \models \psi^* \quad \text{so} \quad \phi \models \neg\psi^*.$$

If ψ^* is quantifier-free, then $\neg\psi^*$ is the desired prenex formula. If ψ^* is of the form $\exists x \psi_0^*$, then

$$\phi \models \neg \exists x \psi_0^* \models \forall x \neg \psi_0^*,$$

and by the induction hypothesis again, $\neg\psi_0^*$ has a prenex form χ ; we take ϕ^* to be $\forall x \chi$. If ψ^* begins with $\forall$, we proceed similarly with the obvious changes.

Suppose now that ϕ is $\psi \vee \theta$ and first that θ^* is $\exists x \theta_0^*$. Then choose a variable y which occurs in neither ψ nor θ ; by the preceding proposition we have

$$\phi \models \psi \vee \exists x \theta_0^* \models \psi \vee \exists y \theta_{0x}^*(y) \models \exists y (\psi \vee \theta_{0x}^*(y)).$$

Since $(\psi \vee \theta_{0x}^*(y))$ has fewer quantifiers than ϕ , by the induction hypothesis it has a prenex form χ and we take ϕ^* to be $\exists y \chi$.

There are three more similar subcases: (i) θ^* begins with $\forall$, (ii) and (iii) θ^* is quantifier-free and ψ^* begins with $\exists$ or $\forall$. Finally, if ϕ is $\exists x \psi$, then ϕ^* is $\exists x \psi^*$. $\square$

Another use of such changes of bound variable is, as mentioned above, to make a substitution free. If $\theta_x(t)$ is not a free substitution, we may replace θ by a logically equivalent formula θ' such that $\theta'_x(t)$ is a free substitution.

Definition 2.2.35 For any formula ϕ and any variables $y_0, \dots, y_{k-1}$, the $y_0, \dots, y_{k-1}$ -**free variant** ϕ' of ϕ is defined recursively as follows:

- (i) if ϕ is atomic, then ϕ' is ϕ ;
- (ii) if ϕ is $\neg\psi$, then ϕ' is $\neg\psi'$;
- (iii) if ϕ is $\psi \vee \theta$, then ϕ' is $\psi' \vee \theta'$;
- (iv) if ϕ is $\exists z \psi$, then

$$\phi' \text{ is } \begin{cases} \exists w \psi'_z(w) & \text{if } z \text{ is among } y_0, \dots, y_{k-1}; \\ \exists z \psi' & \text{otherwise;} \end{cases}$$

where w is the first variable in the sequence $v_0, v_1, \dots$ which does not occur in ψ' and is not among $y_0, \dots, y_{k-1}$.

ϕ' is a **variant** of ϕ iff it is a $y_0, \dots, y_{k-1}$ -free variant for some $y_0, \dots, y_{k-1}$.

Corollary 2.2.36

- (i) For any ϕ and ϕ' , if ϕ' is a variant of ϕ , then $\phi \models \phi'$.
- (ii) For any θ and t , if all variables of t are among $y_0, \dots, y_{k-1}$ and θ' is the $y_0, \dots, y_{k-1}$ -free variant of θ , then $\theta'_x(t)$ is a free substitution.

Proof. Left as an exercise. $\square$

Note that in general $\theta_x(t)$ and $\theta'_x(t)$ are not logically equivalent, since the latter has the intended meaning, while the former may not.

We have defined several syntactical notions in this section and for future reference we need to note their effective nature.

Proposition 2.2.37 For any effective language L ,

- (i) the set of sentences is a decidable subset of the set of formulas;
- (ii) the set of tautologies is a decidable subset of the set of sentences;
- (iii) the set of equality axioms is a decidable subset of the set of formulas;
- (iv) the substitution functions $(x, t, u) \mapsto t_x(u)$ and $(x, \phi, u) \mapsto \phi_x(u)$ are effectively calculable;
- (v) the property of a term being freely substitutable for a variable in a formula is decidable;
- (vi) there is an effectively calculable function $\phi \mapsto \phi^*$ such that ϕ^* is in prenex normal form and is logically equivalent to ϕ .

Proof. These all follow from Proposition 2.1.30 by observing that the definitions are in all cases essentially also algorithms. $\square$

Remark 2.2.38 It is important to note that we do not make any claims about the decidability of the set of logically valid sentences. In fact, in [Section 4.6](#) and [Section 5.5](#) we shall establish that for all but the weakest languages this set is *not* decidable. In [Section 2.4](#) below we show that the set of logically valid sentences of the simplest language $L_=_$ is decidable.

Similarly, for a given structure $\mathfrak{A}$, the set of sentences true in $\mathfrak{A}$ is not generally decidable (but see the exercises for the case that $\mathfrak{A}$ is finite).

Exercises

Exercise 2.2.39 Which of the equivalences of Theorem 2.2.11 hold also when ϕ and ψ are formulas which are not necessarily sentences? Prove each of the implications that are true and give counterexamples for the others. Note that these are different from Corollary 2.1.24 and Proposition 2.1.25 because $\models_{\mathfrak{A}} \phi$ is here defined by Remark 2.2.17.

Exercise 2.2.40 Prove Lemma 2.2.25.

Exercise 2.2.41 Suppose that L has only finitely many non-logical symbols and that $\mathfrak{A}$ is an L -structure with finite universe. Show that $\text{Th}(\mathfrak{A}) := \{ \phi : \phi \text{ is a sentence and } \models_{\mathfrak{A}} \phi \}$ is decidable.

Exercise 2.2.42 Give careful proofs of three logical equivalences or consequences from the selected list.

Exercise 2.2.43 Show that if t and u are terms freely substitutable for x in ϕ but do not themselves contain x , then

- (i) $\models \forall x (x \doteq t \rightarrow [\phi \leftrightarrow \phi_x(t)])$;
- (ii) $\models \exists x (x \doteq t \wedge \phi) \leftrightarrow \phi_x(t) \leftrightarrow \forall x (x \doteq t \rightarrow \phi)$;
- (iii) $\models \forall x ([x \doteq t \vee x \doteq u] \rightarrow [\phi \rightarrow (\phi_x(t) \vee \phi_x(u))])$;
- (iv) $\models \forall x ([x \doteq t \vee x \doteq u] \rightarrow [(\phi_x(t) \wedge \phi_x(u)) \rightarrow \phi])$;
- (v) $\models \exists x [(x \doteq t \vee x \doteq u) \wedge \phi] \leftrightarrow \phi_x(t) \vee \phi_x(u)$;
- (vi) $\models \forall x [(x \doteq t \vee x \doteq u) \rightarrow \phi] \leftrightarrow \phi_x(t) \wedge \phi_x(u)$.

Exercise 2.2.44 Prove that for any quantifier-free formula ϕ which does not contain the equality symbol $\doteq$,

$$\models \phi \iff \phi \text{ is a tautology.}$$

Hint: For each truth assignment V , construct an L -structure $\mathfrak{A}_V$ and an $\mathfrak{A}_V$ -assignment α such that

$$\models_{\mathfrak{A}_V} \phi[\alpha] \iff V(\phi) = \mathbf{T}.$$

The universe of $\mathfrak{A}_V$ is the set Te_L of L -terms.

Exercise 2.2.45 Extend the preceding exercise as follows. Call an L -formula a **quasi-tautology** iff it is a tautological consequence of the equality axioms — that is, the formulas (i)–(v) of Proposition 2.2.5. Prove that for any quantifier-free formula ϕ ,

$$\models \phi \iff \phi \text{ is a quasi-tautology.}$$

Hint: Modify the construction of the preceding exercise as follows. For each propositional truth assignment V which assigns $\mathbf{T}$ to all of the equality axioms, define an equivalence relation on L -terms by

$$t \sim u \iff V(t \doteq u) = \mathbf{T};$$

let $\hat{t}$ denote the equivalence class of t . Now construct $\mathfrak{A}_V$ with universe $\{ \hat{t} : t \in \text{Te}_L \}$.

Exercise 2.2.46 At first glance (once you get through the notation) it seems that for any formula θ , distinct variables x and y , term t and constant symbol c , it should be true that

$$(\theta_x(t))_y(c) = (\theta_y(c))_x((t_y(c))).$$

The idea is that we simply interchange the order of substitution.

- (i) Show that equality does hold under the hypothesis: either y does not occur in t or y is freely substitutable for x in θ ;
- (ii) give a counterexample to show that without the hypothesis equality may fail.
- (iii) The situation is still more complicated if we replace c by a complex term u ; can you find any hypotheses which will make the equation true in such cases?

Exercise 2.2.47 Provide a hypothesis and counterexample as in the preceding exercise for the equation

$$(\theta_x(y))_y(t) = \theta_x(t).$$

Exercise 2.2.48 Give a precise definition of a notion: x is **totally bound** for u , which captures the intuition opposite to free substitution: every free occurrence of x is in a context $\exists y \dots x \dots$ for some variable y which occurs in u .

Exercise 2.2.49 Prove Corollary 2.2.36.

Exercise 2.2.50 If x and y are variables and u and v terms, the successive substitution $\phi_x(u)_y(v)$ will replace with v not only the free occurrences of y in ϕ , but also those of instances of u substituted for x which remain free in the result. Define a notion of **simultaneous substitution** which substitutes only for the variables in ϕ and formulate and prove appropriate versions of 2.2.21–2.2.27.

2.3 Structures

In this section we focus on relationships between structures. Many of the notions we introduce here are straightforward generalizations of familiar mathematical concepts: isomorphism, substructure, Cartesian product etc. The main new element here lies in the interaction between these purely algebraic ideas and the corresponding first-order languages.

For most of this section, let L be a fixed first-order language; all structures are L -structures unless otherwise indicated.

Isomorphism and equivalence

Definition 2.3.1 Two structures $\mathfrak{A}$ and $\mathfrak{B}$ are **isomorphic** — in symbols, $\mathfrak{A} \simeq \mathfrak{B}$ — iff there exists a bijection $\eta : |\mathfrak{A}| \rightarrow |\mathfrak{B}|$ such that

(i) for each $p \in \text{Rs}_L$ and all $a_0, \dots, a_{\nu_L(p)-1} \in |\mathfrak{A}|$,

$$p^{\mathfrak{A}}(a_0, \dots, a_{\nu_L(p)-1}) \iff p^{\mathfrak{B}}(\eta(a_0), \dots, \eta(a_{\nu_L(p)-1}));$$

(ii) for each $f \in \text{Fs}_L$ and all $a_0, \dots, a_{\nu_L(f)-1} \in |\mathfrak{A}|$,

$$\eta(f^{\mathfrak{A}}(a_0, \dots, a_{\nu_L(f)-1})) = f^{\mathfrak{B}}(\eta(a_0), \dots, \eta(a_{\nu_L(f)-1}));$$

(iii) for each $c \in \text{Cs}_L$, $\eta(c^{\mathfrak{A}}) = c^{\mathfrak{B}}$.

We also write $\eta : \mathfrak{A} \simeq \mathfrak{B}$ to indicate explicitly the function which realizes the isomorphism. An isomorphism $\eta : \mathfrak{A} \simeq \mathfrak{A}$ is called an **automorphism** of $\mathfrak{A}$.

Example 2.3.2 Let $\mathbb{R}^{>0}$ be the set of positive real numbers and $\eta : \mathbb{R} \rightarrow \mathbb{R}^{>0}$ the function $\eta(r) := 2^r$. Then η is an isomorphism of the structures $\mathfrak{A} := (\mathbb{R}, <, +, 0)$ and $\mathfrak{B} := (\mathbb{R}^{>0}, <, \times, 1)$, since for any $r, s \in \mathbb{R}$,

$$(1) \quad r < s \iff \eta(r) < \eta(s), \quad \eta(r + s) = \eta(r) \times \eta(s) \quad \text{and} \quad \eta(0) = 1.$$

This is simply a property of the structures and does not refer to any language, but we can express it in the form of Definition 2.3.1 as follows. Let L be the language with one binary relation symbol $\dot{<}$, one binary function symbol $*$ and one constant symbol c . Then

$$\dot{<}^{\mathfrak{A}} = <, \quad *^{\mathfrak{A}} = +, \quad \text{and} \quad c^{\mathfrak{A}} = 0$$

while

$$\dot{<}^{\mathfrak{B}} = <, \quad *^{\mathfrak{B}} = \times, \quad \text{and} \quad c^{\mathfrak{B}} = 1.$$

(Technically the two $<$ -relations are different, since they are on different domains.) Hence (1) may be written

$$(2) \quad r \dot{<}^{\mathfrak{A}} s \iff \eta(r) \dot{<}^{\mathfrak{B}} \eta(s), \quad \eta(r *^{\mathfrak{A}} s) = \eta(r) *^{\mathfrak{B}} \eta(s)$$

and $\eta(c^{\mathfrak{A}}) = c^{\mathfrak{B}}$. Of course in this particular case (1) is much more perspicuous, but form (2) is better suited to the general case.

Isomorphic structures should be thought of as having all properties in common except the actual nature of the elements of the universe. In contrast, the following relationship consists in $\mathfrak{A}$ and $\mathfrak{B}$ sharing at least those properties which can be expressed by L -sentences.

Definition 2.3.3 For any $\mathfrak{A}$ and $\mathfrak{B}$, $\mathfrak{A}$ and $\mathfrak{B}$ are *elementarily equivalent* — in symbols, $\mathfrak{A} \equiv \mathfrak{B}$ — iff for all L -sentences ϕ ,

$$\models_{\mathfrak{A}} \phi \iff \models_{\mathfrak{B}} \phi.$$

To substantiate the intuitions mentioned above, we want first to show that if $\mathfrak{A} \simeq \mathfrak{B}$, then $\mathfrak{A} \equiv \mathfrak{B}$. As so often happens, this cannot be proved directly, but only via a more general result.

Definition 2.3.4 For any $\mathfrak{A}$ and $\mathfrak{B}$, and any function $\eta : |\mathfrak{A}| \rightarrow |\mathfrak{B}|$,

- (i) for any $\mathfrak{A}$ -assignment α , the *induced $\mathfrak{B}$ -assignment* is defined by $\alpha^\eta(x) = \eta(\alpha(x))$;
- (ii) η *respects* a term t iff for each $\mathfrak{A}$ -assignment α ,

$$\eta(t^{\mathfrak{A}}[\alpha]) = t^{\mathfrak{B}}[\alpha^\eta];$$

- (iii) η *respects* a formula ϕ iff for each $\mathfrak{A}$ -assignment α ,

$$\models_{\mathfrak{A}} \phi[\alpha] \iff \models_{\mathfrak{B}} \phi[\alpha^\eta].$$

Example 2.3.5 Continuing the example above, if t is the L -term $x * (y * c)$ and α is an assignment such that $\alpha(x) = \pi$ and $\alpha(y) = 7$, then

$$\eta(t^{\mathfrak{A}}[\alpha]) = \eta(\pi + (7 + 0)) = \eta(\pi) \times (\eta(7) \times \eta(0)) = t^{\mathfrak{B}}[\alpha^\eta].$$

Similarly, if ϕ is the L -formula $\exists z (x * z \doteq y)$, then

$$\begin{aligned} \models_{\mathfrak{A}} \phi[\alpha] &\iff \text{for some } r \in \mathbb{R}, \quad \models_{\mathfrak{A}} \phi[\alpha_{z \mapsto r}] \\ &\iff \text{for some } r \in \mathbb{R}, \quad \pi + r = 7 \\ &\iff \text{for some } r \in \mathbb{R}, \quad \eta(\pi) \times \eta(r) = \eta(7) \\ &\iff \text{for some } s \in \mathbb{R}^{>0}, \quad \eta(\pi) \times s = \eta(7) \\ &\iff \models_{\mathfrak{B}} \phi[\alpha^\eta], \end{aligned}$$

where the fourth equivalence follows from the fact that η is surjective. This illustrates the general

Lemma 2.3.6 *For any $\mathfrak{A}$ and $\mathfrak{B}$ and $\eta : \mathfrak{A} \simeq \mathfrak{B}$, η respects all L -terms and formulas.*

Proof. We prove that η respects terms by term induction. If t is a variable x , then

$$\eta(t^{\mathfrak{A}}[\alpha]) = \eta(\alpha(x)) = \alpha^\eta(x) = t^{\mathfrak{B}}[\alpha^\eta].$$

If t is a constant symbol c , then

$$\eta(t^{\mathfrak{A}}[\alpha]) = \eta(c^{\mathfrak{A}}) = c^{\mathfrak{B}} = t^{\mathfrak{B}}[\alpha^\eta].$$

If t is $f t_0, \dots, t_{\nu_L(f)-1}$, then

$$\begin{aligned} \eta(t^{\mathfrak{A}}[\alpha]) &= \eta(f^{\mathfrak{A}}(t_0^{\mathfrak{A}}[\alpha], \dots, t_{\nu_L(f)-1}^{\mathfrak{A}}[\alpha])) \\ &= f^{\mathfrak{B}}(\eta(t_0^{\mathfrak{A}}[\alpha]), \dots, \eta(t_{\nu_L(f)-1}^{\mathfrak{A}}[\alpha])) \\ &= f^{\mathfrak{B}}(t_0^{\mathfrak{B}}[\alpha^\eta], \dots, t_{\nu_L(f)-1}^{\mathfrak{B}}[\alpha^\eta]) \\ &= t^{\mathfrak{B}}[\alpha^\eta]. \end{aligned}$$

That η respects formulas is proved similarly by formula induction. Suppose first that ϕ is an atomic formula $t \doteq u$. Then

$$\begin{aligned} \models_{\mathfrak{A}} \phi[\alpha] &\iff t^{\mathfrak{A}}[\alpha] = u^{\mathfrak{A}}[\alpha] \iff \eta(t^{\mathfrak{A}}[\alpha]) = \eta(u^{\mathfrak{A}}[\alpha]) \\ &\iff t^{\mathfrak{B}}[\alpha^\eta] = u^{\mathfrak{B}}[\alpha^\eta] \iff \models_{\mathfrak{B}} \phi[\alpha^\eta]. \end{aligned}$$

The second equivalence follows from the assumption that η is injective. The calculation for other atomic formulas is similar, and the induction steps for $\neg$ and $\vee$ are straightforward; we leave them to the reader. Finally, suppose that ϕ is $\exists x \psi$. Then

$$\begin{aligned} \models_{\mathfrak{A}} \phi[\alpha] &\iff \text{for some } a \in |\mathfrak{A}|, \models_{\mathfrak{A}} \psi[\alpha_{x \mapsto a}] \\ &\iff \text{for some } a \in |\mathfrak{A}|, \models_{\mathfrak{B}} \psi[\alpha_{x \mapsto \eta(a)}^\eta] \\ &\iff \text{for some } b \in |\mathfrak{B}|, \models_{\mathfrak{B}} \psi[\alpha_{x \mapsto b}^\eta] \\ &\iff \models_{\mathfrak{B}} \phi[\alpha^\eta]. \end{aligned}$$

The second equivalence follows from the fact that $(\alpha_{x \mapsto a})^\eta = \alpha_{x \mapsto \eta(a)}^\eta$ and the third from the hypothesis that η is surjective. $\square$

Theorem 2.3.7 *For any $\mathfrak{A}$ and $\mathfrak{B}$, if $\mathfrak{A} \simeq \mathfrak{B}$, then $\mathfrak{A} \equiv \mathfrak{B}$.*

Proof. If $\eta : \mathfrak{A} \simeq \mathfrak{B}$, then by the lemma, if ϕ is a sentence,

$$\begin{aligned} \models_{\mathfrak{A}} \phi &\iff \text{for some (any) } \mathfrak{A}\text{-assignment } \alpha, \quad \models_{\mathfrak{A}} \phi[\alpha] \\ &\iff \text{for some (any) } \mathfrak{A}\text{-assignment } \alpha, \quad \models_{\mathfrak{B}} \phi[\alpha^\eta] \\ &\iff \models_{\mathfrak{B}} \phi. \quad \square \end{aligned}$$

The natural question to ask at this point is whether or not this implication reverses: does $\mathfrak{A} \equiv \mathfrak{B}$ imply $\mathfrak{A} \simeq \mathfrak{B}$? We shall see that in general the answer is “No”, and only in some very special cases is it “Yes”. We consider first the simplest of all languages, $L_=_$, the language with no non-logical symbols. $L_=_$ -structures are merely sets (A) and an isomorphism is just a bijection; in other words,

Proposition 2.3.8 *For any two $L_=_$ -structures (A) and (B),*

$$(A) \simeq (B) \quad \text{iff} \quad A \text{ and } B \text{ have the same cardinality.} \quad \square$$

Corollary 2.3.9 *For any finite sets A and B , if $(A) \equiv (B)$, then $(A) \simeq (B)$. In fact, for any finite $L_=_$ -structure (A), there exists a single $L_=_$ -sentence $\theta^{(A)}$ such that for any $L_=_$ -structure (B),*

$$(A) \simeq (B) \quad \iff \quad \models_{(B)} \theta^{(A)}.$$

Proof. If A has n elements, let $\theta^{(A)}$ be the sentence $\exists^{=n}$ of Examples 2.2.12(ii). This gives the second clause, hence also the first. $\square$

The second clause of the proposition may be expressed by saying that $\theta^{(A)}$ characterizes (A) **up to isomorphism**. Although this simple result seems to depend heavily on the near triviality of the language $L_=_$, in fact it may be considerably generalized. However, it does depend on the finiteness of A and B ; in Section 2.5 we shall show that $(A) \equiv (B)$ for all infinite A and B , even of different cardinality.

Definition 2.3.10 An L -structure $\mathfrak{A}$ is **finite** (**countable**, **denumerable**) iff its universe $A = |\mathfrak{A}|$ is a finite (countable, denumerable) set.

Theorem 2.3.11 For any finite language L and finite L -structure $\mathfrak{A}$,

- (i) for any L -structure $\mathfrak{B}$, if $\mathfrak{A} \equiv \mathfrak{B}$ then $\mathfrak{A} \simeq \mathfrak{B}$; in fact
- (ii) there is a single L -sentence $\theta^{\mathfrak{A}}$ which characterizes $\mathfrak{A}$ up to isomorphism — that is, for any L -structure $\mathfrak{B}$,

$$\mathfrak{A} \simeq \mathfrak{B} \iff \models_{\mathfrak{B}} \theta^{\mathfrak{A}}.$$

Proof. For ease of exposition we suppose that L has just one binary relation symbol p , one unary function symbol f and one constant symbol c . Let $\mathfrak{A}$ be a finite L -structure, say $\mathfrak{A}$ has k elements $a_0, \dots, a_{k-1}$. We take $\theta^{\mathfrak{A}}$ to be the sentence

$$\begin{aligned} \exists x_0 \cdots \exists x_{k-1} \Big[& \bigwedge_{i < j < k} \neg x_i \dot{=} x_j \wedge \forall y \bigvee_{i < k} y \dot{=} x_i \wedge \bigwedge_{(a_i, a_j) \in p^{\mathfrak{A}}} p x_i x_j \\ & \wedge \bigwedge_{(a_i, a_j) \notin p^{\mathfrak{A}}} \neg p x_i x_j \wedge \bigwedge_{f^{\mathfrak{A}}(a_i) = a_j} f x_i \dot{=} x_j \wedge x_{i_0} \dot{=} c \Big], \end{aligned}$$

where $c^{\mathfrak{A}}$ is a_{i_0} . Now if $\mathfrak{B}$ is any structure which satisfies $\theta^{\mathfrak{A}}$, the matrix of $\theta^{\mathfrak{A}}$ is satisfied by some $\mathfrak{B}$ -assignment β , say $\beta(x_i) = b_i$. Then it is easy to check that the mapping $\eta : a_i \mapsto b_i$ is an isomorphism of $\mathfrak{A}$ and $\mathfrak{B}$. $\square$

We shall see in [Section 2.6](#) that (i) (but not (ii)) still holds even without the assumption that L is finite.

Substructures

We turn next to the notion of substructure. In algebra, a group G is a subgroup of another group H iff $G \subseteq H$ and G is a group with the multiplication, identity and inverse of H . If G is an ordered group, then for H to be an ordered subgroup requires in addition that the ordering on H is simply the restriction of the ordering on G to the elements of H . The corresponding general notion is

Definition 2.3.12 For any L -structures $\mathfrak{A}$ and $\mathfrak{B}$, $\mathfrak{A}$ is a **substructure** of $\mathfrak{B}$ and $\mathfrak{B}$ is an **extension** of $\mathfrak{A}$ — in symbols, $\mathfrak{A} \subseteq \mathfrak{B}$ — iff

- (i) $|\mathfrak{A}|$ is a subset of $|\mathfrak{B}|$;
- (ii) for each $p \in \mathbf{Rs}_L$, $p^{\mathfrak{A}} = p^{\mathfrak{B}} \cap |\mathfrak{A}|^{\nu_L(p)}$;
- (iii) for each $f \in \mathbf{Fs}_L$, $f^{\mathfrak{A}} = f^{\mathfrak{B}} \upharpoonright |\mathfrak{A}|^{\nu_L(f)}$;
- (iv) for each $c \in \mathbf{Cs}_L$, $c^{\mathfrak{A}} = c^{\mathfrak{B}}$.

Remark 2.3.13 Note that since by definition $c^{\mathfrak{A}} \in |\mathfrak{A}|$, clause (iv) implies that $c^{\mathfrak{B}} \in |\mathfrak{A}|$. Similarly, since by definition for $a_0, \dots, a_{\nu_L(f)-1} \in |\mathfrak{A}|$, also $f^{\mathfrak{A}}(a_0, \dots, a_{\nu_L(f)-1}) \in |\mathfrak{A}|$, clause (iii) implies that $|\mathfrak{A}|$ is closed under all functions $f^{\mathfrak{B}}$.

Examples 2.3.14

- (i) $(\omega, <_{\omega})$ is a substructure of $(\mathbb{Z}, <_{\mathbb{Z}})$, but not of $(\mathbb{Z}, \leq_{\mathbb{Z}})$ or of $(\mathbb{Z}, <_{\mathbb{Z}}, 0)$.
- (ii) If $\mathfrak{H} = (H, *, ^{-1}, e)$ is a group, then $\mathfrak{G}$ is a substructure of $\mathfrak{H}$ iff $\mathfrak{G}$ is a subgroup in the usual algebraic sense, since it contains the identity and is closed under the group operation and inverse. We shall see below that this is closely related to the fact that in this language the axioms for group theory are all universal sentences (see Example 2.2.14). However, if $\mathfrak{H} = (H, *, e)$ is a group formulated in the language without a function symbol for the inverse, then a substructure $\mathfrak{G}$ may fail to be a subgroup because it is not closed under the inverse — for example $(\omega, +_{\omega}, 0) \subseteq (\mathbb{Z}, +_{\mathbb{Z}}, 0)$.

Proposition 2.3.15 *For any L -structure $\mathfrak{B}$ and any non-empty subset A of $|\mathfrak{B}|$, A is the universe of a (unique) substructure $\mathfrak{A}$ of $\mathfrak{B}$ iff*

- (i) *for each $f \in \mathbf{Fs}_L$, A is closed under $f^{\mathfrak{B}}$, and*
- (ii) *for each $c \in \mathbf{Cs}_L$, $c^{\mathfrak{B}} \in A$.*

In particular, if $\mathbf{Fs}_L = \emptyset = \mathbf{Cs}_L$, then every non-empty subset of $|\mathfrak{B}|$ is the universe of a unique substructure of $\mathfrak{B}$.

Proof. If we define $|\mathfrak{A}| = A$ and use the clauses of the definition to specify $p^{\mathfrak{A}}$, $f^{\mathfrak{A}}$ and $c^{\mathfrak{A}}$, then under the hypotheses $\mathfrak{A}$ is a structure and clearly $\mathfrak{A} \subseteq \mathfrak{B}$. The uniqueness of $\mathfrak{A}$ is obvious. $\square$

If a subset X of $|\mathfrak{B}|$ does not satisfy (i) and (ii), it is not the universe of a substructure of $\mathfrak{B}$, but there is a unique minimal superset of X which does have the necessary properties.

Proposition 2.3.16 *For any L -structure $\mathfrak{B}$ and any subset X of $|\mathfrak{B}|$, if either $X \neq \emptyset$ or $\mathbf{Cs}_L \neq \emptyset$, then there exists a unique smallest substructure $\langle X \rangle_{\mathfrak{B}} \subseteq \mathfrak{B}$ such that $X \subseteq |\langle X \rangle_{\mathfrak{B}}|$. $\langle X \rangle_{\mathfrak{B}}$ is called the substructure of $\mathfrak{B}$ generated by X . If X and L are both countable, then so is $\langle X \rangle_{\mathfrak{B}}$.*

Proof. Given $X \subseteq |\mathfrak{B}|$ we construct inductively a set A which satisfies the hypotheses of the preceding proposition — namely, let A be the smallest subset of $|\mathfrak{B}|$ such that $X \cup \{c^{\mathfrak{B}} : c \in \mathbf{Cs}_L\} \subseteq A$ and A is closed under all functions $f^{\mathfrak{B}}$ for $f \in \mathbf{Fs}_L$. The hypothesis guarantees that $A \neq \emptyset$. Now let $\langle X \rangle_{\mathfrak{B}}$ be the unique substructure with universe A given by the proposition. The second clause is immediate from Proposition 1.2.8. $\square$

Example 2.3.17 The set $\{30, 42\} \subseteq \mathbb{Z}$ generates the substructure $(\{6n : n \in \mathbb{Z}\}, +, -, 0)$ of the group $(\mathbb{Z}, +, -, 0)$.

Remark 2.3.18 By the results of [Section 1.2](#), we can also characterize the universe of $\langle X \rangle_{\mathfrak{B}}$ as the intersection of all sets $Y \subseteq B$ such that $X \subseteq Y$ and Y is the universe of a substructure of $\mathfrak{B}$. Still another characterization is

Proposition 2.3.19 *For any L -structure $\mathfrak{B}$ and any non-empty set $X \subseteq |\mathfrak{B}|$, the universe of $\langle X \rangle_{\mathfrak{B}}$ is*

$$\{t^{\mathfrak{B}}[\alpha] : t \text{ is an } L\text{-term and } \alpha : \forall \mathbf{b} \rightarrow X\};$$

if L has at least one constant symbol, then the universe of $\langle \emptyset \rangle_{\mathfrak{B}}$ is

$$\{t^{\mathfrak{B}} : t \text{ is a variable-free term}\}.$$

Proof. Left as an exercise. $\square$

Recall that an L -formula ϕ is called quantifier-free iff $\phi \in \text{Bool}(\text{AtFm}_L)$ (Definition 2.2.31). Substructures may be characterized by the fact that they share all properties expressed by quantifier-free formulas as follows. Note that the conditions may also be expressed by saying that the inclusion map respects terms and quantifier-free formulas.

Lemma 2.3.20 *For any structures $\mathfrak{A} \subseteq \mathfrak{B}$, any term t and any $\mathfrak{A}$ -assignment α , $t^{\mathfrak{A}}[\alpha] = t^{\mathfrak{B}}[\alpha]$.*

Proof. We proceed by term induction. If t is a constant symbol c , then $t^{\mathfrak{A}}[\alpha] = c^{\mathfrak{A}} = c^{\mathfrak{B}} = t^{\mathfrak{B}}[\alpha]$ by the definition of substructure. If t is a variable x , then $t^{\mathfrak{A}}[\alpha] = \alpha(x) = t^{\mathfrak{B}}[\alpha]$. If t is a compound term $ft_0, \dots, t_{\nu_L(f)-1}$, then

$$\begin{aligned} t^{\mathfrak{A}}[\alpha] &= f^{\mathfrak{A}}(t_0^{\mathfrak{A}}[\alpha], \dots, t_{\nu_L(f)-1}^{\mathfrak{A}}[\alpha]) \\ &= f^{\mathfrak{B}}(t_0^{\mathfrak{A}}[\alpha], \dots, t_{\nu_L(f)-1}^{\mathfrak{A}}[\alpha]) \\ &= f^{\mathfrak{B}}(t_0^{\mathfrak{B}}[\alpha], \dots, t_{\nu_L(f)-1}^{\mathfrak{B}}[\alpha]) \\ &= t^{\mathfrak{B}}[\alpha], \end{aligned}$$

where the first and fourth equalities are by definition, the second is by the definition of substructure, and the third is the induction hypothesis. $\square$

Proposition 2.3.21 *For any L -structures $\mathfrak{A}$ and $\mathfrak{B}$, $\mathfrak{A} \subseteq \mathfrak{B}$ iff*

- (i) $|\mathfrak{A}|$ is a subset of $|\mathfrak{B}|$, and
- (ii) for any quantifier-free L -formula ϕ and any $\mathfrak{A}$ -assignment α ,

$$\models_{\mathfrak{A}} \phi[\alpha] \iff \models_{\mathfrak{B}} \phi[\alpha].$$

Proof. Suppose first that $\mathfrak{A} \subseteq \mathfrak{B}$. Then (i) is immediate and we prove (ii) by induction on quantifier-free formulas (Remark 2.2.32(ii)). If ϕ is an atomic formula $pt_0 \dots t_{\nu_L(p)-1}$, then

$$\begin{aligned} \models_{\mathfrak{A}} \phi[\alpha] &\iff (t_0^{\mathfrak{A}}[\alpha], \dots, t_{\nu_L(p)-1}^{\mathfrak{A}}[\alpha]) \in p^{\mathfrak{A}} \\ &\iff (t_0^{\mathfrak{A}}[\alpha], \dots, t_{\nu_L(p)-1}^{\mathfrak{A}}[\alpha]) \in p^{\mathfrak{B}} \\ &\iff (t_0^{\mathfrak{B}}[\alpha], \dots, t_{\nu_L(p)-1}^{\mathfrak{B}}[\alpha]) \in p^{\mathfrak{B}} \\ &\iff \models_{\mathfrak{B}} \phi[\alpha], \end{aligned}$$

where the second equivalence follows from property (ii) of the definition of substructure and the third from the lemma. The induction steps for $\neg$ and $\vee$ are easy — for example,

$$\begin{aligned} \models_{\mathfrak{A}} (\psi \vee \theta)[\alpha] &\iff \models_{\mathfrak{A}} \psi[\alpha] \text{ or } \models_{\mathfrak{A}} \theta[\alpha] \\ &\iff \models_{\mathfrak{B}} \psi[\alpha] \text{ or } \models_{\mathfrak{B}} \theta[\alpha] \\ &\iff \models_{\mathfrak{B}} (\psi \vee \theta)[\alpha]. \end{aligned}$$

Conversely, if we assume (i) and (ii), then for any relation symbol p and any $a_0, \dots, a_{\nu_L(p)-1} \in A$,

$$\begin{aligned} (a_0, \dots, a_{\nu_L(p)-1}) \in p^{\mathfrak{A}} &\iff \models_{\mathfrak{A}} px_0 \dots x_{\nu_L(p)-1}[a_0, \dots, a_{\nu_L(p)-1}] \\ &\iff \models_{\mathfrak{B}} px_0 \dots x_{\nu_L(p)-1}[a_0, \dots, a_{\nu_L(p)-1}] \\ &\iff (a_0, \dots, a_{\nu_L(p)-1}) \in p^{\mathfrak{B}}, \end{aligned}$$

which implies condition (ii) of the definition of substructure. The corresponding conditions for function and constant symbols are treated similarly. $\square$

Of course, it is crucial here that the second condition of the proposition was limited to quantifier-free formulas; we could not have handled an $\exists$ induction step. Still, the result can be slightly extended to some special types of formulas in prenex form (Definition 2.2.33).

Definition 2.3.22

- (i) A formula ψ is **universal (existential)** iff for some quantifier-free formula θ and variables $y_0, \dots, y_{n-1}$, ψ is $\forall y_0 \dots \forall y_{n-1} \theta$ ($\exists y_0 \dots \exists y_{n-1} \theta$).
- (ii) A formula ϕ is **logically universal (logically existential)** iff for some universal (existential) formula ψ , $\models \phi \leftrightarrow \psi$.

Lemma 2.3.23 For any formulas ϕ and ψ ,

- (i) if both ϕ and ψ are logically universal (existential), then also $\phi \wedge \psi$ and $\phi \vee \psi$ are logically universal (existential);
- (ii) if ϕ is logically universal (existential), then $\neg \phi$ is logically existential (universal);
- (iii) if ϕ is logically universal (existential), then $\forall x \phi$ ($\exists x \phi$) is logically universal (existential).

Proof. Left as an exercise. $\square$

Definition 2.3.24 A formula ϕ is **upward persistent** iff for any structures $\mathfrak{A} \subseteq \mathfrak{B}$ and any $\mathfrak{A}$ -assignment α ,

$$\models_{\mathfrak{A}} \phi[\alpha] \implies \models_{\mathfrak{B}} \phi[\alpha],$$

and **downward persistent** iff under the same hypotheses,

$$\models_{\mathfrak{B}} \phi[\alpha] \implies \models_{\mathfrak{A}} \phi[\alpha].$$

It is immediate from Proposition 2.3.21 that quantifier-free formulas are both upward and downward persistent.

Proposition 2.3.25 For any formula ϕ ,

- (i) if ϕ is logically existential, then ϕ is upward persistent;
- (ii) if ϕ is logically universal, then ϕ is downward persistent.

Proof. Suppose that ϕ is logically existential, say $\models \phi \leftrightarrow \exists x \theta$ with θ quantifier-free (we assume this simple case for ease of exposition); fix $\mathfrak{A} \subseteq \mathfrak{B}$ and an $\mathfrak{A}$ -assignment α . Then

$$\begin{aligned} \models_{\mathfrak{A}} \phi[\alpha] &\iff \models_{\mathfrak{A}} \exists x \theta[\alpha] \\ &\iff \text{for some } a \in |\mathfrak{A}|, \quad \models_{\mathfrak{A}} \theta[\alpha_{x \mapsto a}] \\ &\iff \text{for some } a \in |\mathfrak{A}|, \quad \models_{\mathfrak{B}} \theta[\alpha_{x \mapsto a}] \\ &\implies \text{for some } a \in |\mathfrak{B}|, \quad \models_{\mathfrak{B}} \theta[\alpha_{x \mapsto a}] \\ &\iff \models_{\mathfrak{B}} \phi[\alpha]. \end{aligned}$$

The third equivalence follows from Proposition 2.3.21, and the implication from the assumption that $|\mathfrak{A}| \subseteq |\mathfrak{B}|$. Thus ϕ is upward persistent. Part (ii) is proved analogously or may be derived from (i) via (ii) of the lemma. $\square$

Thus, for example, since the axioms $\Gamma_{\mathbf{Gp}}$ for groups in the language L_G with symbols for the group operation, identity and inverse (Examples 2.2.14)

are universal sentences, they are downward persistent. This means simply that, as we noted above, any substructure of a group is also a group. We shall prove in [Section 3.5](#) that the converse of the proposition also holds: a formula is logically existential (universal) iff it is upward (downward) persistent.

It is often of interest to consider mappings between structures which fail to be isomorphisms because they are either not injective or not surjective.

Definition 2.3.26 For any structures $\mathfrak{A}$ and $\mathfrak{B}$, and any function $\eta : |\mathfrak{A}| \rightarrow |\mathfrak{B}|$,

- (i) η is a **homomorphism** of $\mathfrak{A}$ into $\mathfrak{B}$ — in symbols, $\eta : \mathfrak{A} \rightarrow \mathfrak{B}$ — iff conditions (i)–(iii) of Definition 2.3.1 are satisfied (but η need not be either injective or surjective);
- (ii) η is an **embedding** of $\mathfrak{A}$ into $\mathfrak{B}$ — in symbols, $\eta : \mathfrak{A} \hookrightarrow_0 \mathfrak{B}$ — iff η is an injective homomorphism of $\mathfrak{A}$ into $\mathfrak{B}$. $\mathfrak{A}$ is **embeddable** in $\mathfrak{B}$ — in symbols $\mathfrak{A} \hookrightarrow_0 \mathfrak{B}$ — iff there exists an embedding.

Remark 2.3.27 By this definition, an isomorphism is exactly a bijective homomorphism or a surjective embedding. For some purposes it is convenient to use a notion of **positive homomorphism** in which the equivalence (i) of Definition 2.3.1 is replaced by the single implication

$$p^{\mathfrak{A}}(a_0, \dots, a_{\nu_L(p)-1}) \implies p^{\mathfrak{B}}(\eta(a_0), \dots, \eta(a_{\nu_L(p)-1})).$$

We state first a refinement of Lemma 2.3.6.

Lemma 2.3.28 For any structures $\mathfrak{A}$ and $\mathfrak{B}$, and any homomorphism $\eta : \mathfrak{A} \rightarrow \mathfrak{B}$,

- (i) η respects every term and every quantifier-free formula which does not contain the equality symbol $\doteq$;
- (ii) if η is an embedding, then η respects every quantifier-free formula;
- (iii) if η is surjective, then η respects every formula which does not contain the equality symbol $\doteq$.

Proof. Similar to that of Lemma 2.3.6; left as an exercise. $\square$

The following is the main consequence we shall use.

Proposition 2.3.29 For any structures $\mathfrak{A}$ and $\mathfrak{B}$, and any function $\eta : |\mathfrak{A}| \rightarrow |\mathfrak{B}|$, the following are equivalent,

- (i) η is an embedding;
- (ii) η respects every quantifier-free formula;
- (iii) for some (unique) substructure $\mathfrak{A}' \subseteq \mathfrak{B}$, $\eta : \mathfrak{A} \simeq \mathfrak{A}'$.

Proof. The implication (i) $\Rightarrow$ (ii) is part (ii) of the Lemma. We prove first that (ii) $\Rightarrow$ (i); assume (ii). Then for any k -ary function symbol f and any $a_0, \dots, a_{k-1}, b \in |\mathfrak{A}|$,

$$\begin{aligned} f^{\mathfrak{A}}(a_0, \dots, a_{k-1}) &= b \\ \iff \models_{\mathfrak{A}} (fx_0, \dots, x_{k-1} \doteq y)[a_0, \dots, a_{k-1}, b] \\ \iff \models_{\mathfrak{B}} (fx_0, \dots, x_{k-1} \doteq y)[\eta(a_0), \dots, \eta(a_{k-1}), \eta(b)] \\ \iff f^{\mathfrak{B}}(\eta(a_0), \dots, \eta(a_{k-1})) &= \eta(b). \end{aligned}$$

The arguments for relation and constant symbols are similar; thus η is a homomorphism. η is injective because it respects the formula $x \doteq y$.

From (iii), we have easily (ii) from the equivalences, for any quantifier-free formula ϕ ,

$$\models_{\mathfrak{A}} \phi[\alpha] \iff \models_{\mathfrak{A}'} \phi[\alpha^\eta] \iff \models_{\mathfrak{B}} \phi[\alpha^\eta],$$

which follow from Lemma 2.3.6 and Proposition 2.3.21. Finally assume (i) and let $A' = \text{Im}(\eta)$. Since for each (k -ary) function symbol f and $\eta(a_0), \dots, \eta(a_{k-1}) \in A'$,

$$f^{\mathfrak{B}}(\eta(a_0), \dots, \eta(a_{k-1})) = \eta(f^{\mathfrak{A}}(a_0, \dots, a_{k-1})) \in A',$$

A' is closed under $f^{\mathfrak{B}}$ and is thus the universe of a unique substructure $\mathfrak{A}'$ of $\mathfrak{B}$. It is straightforward to verify that $\eta : \mathfrak{A} \simeq \mathfrak{A}'$. $\square$

Proposition 2.3.21 suggests a stronger kind of relationship between structures which requires that elements of $\mathfrak{A}$ have all the same properties expressible in the language (not only by quantifier-free formulas) in $\mathfrak{A}$ as in $\mathfrak{B}$.

Definition 2.3.30 For any L -structures $\mathfrak{A}$ and $\mathfrak{B}$, $\mathfrak{A}$ is an **elementary substructure** of $\mathfrak{B}$ and $\mathfrak{B}$ is an **elementary extension** of $\mathfrak{A}$ — in symbols, $\mathfrak{A} \prec \mathfrak{B}$ — iff

- (i) $|\mathfrak{A}|$ is a subset of $|\mathfrak{B}|$, and
- (ii) for any L -formula ϕ and $\mathfrak{A}$ -assignment α ,

$$\models_{\mathfrak{A}} \phi[\alpha] \iff \models_{\mathfrak{B}} \phi[\alpha].$$

We note first

Proposition 2.3.31 *If $\mathfrak{A} \prec \mathfrak{B}$, then $\mathfrak{A} \subseteq \mathfrak{B}$ and $\mathfrak{A} \equiv \mathfrak{B}$.*

Proof. Immediate from property (ii) applied, respectively, to quantifier-free formulas and to sentences. $\square$

Example 2.3.32 At this point we are not in a position to give any non-trivial examples of structures such that $\mathfrak{A} \prec \mathfrak{B}$. On the other hand, it is easy to give negative examples. Let $\mathfrak{A} = (\omega \sim \{0\}, \leq)$ and $\mathfrak{B} = (\omega, \leq)$. Then clearly $\mathfrak{A} \subseteq \mathfrak{B}$ and in fact $\mathfrak{A} \simeq \mathfrak{B}$ so also $\mathfrak{A} \equiv \mathfrak{B}$. However, $\mathfrak{A}$ is not an elementary substructure of $\mathfrak{B}$ because

$$\models_{\mathfrak{A}} \forall y (x \dot{\leq} y)[1] \quad \text{but} \quad \not\models_{\mathfrak{B}} \forall y (x \dot{\leq} y)[1];$$

that is, 1 has the property of being the least element in $\mathfrak{A}$ but not in $\mathfrak{B}$.

For proving that a substructure is elementary, the following is a valuable tool. Note that both sides of the equivalence involve satisfaction in the larger structure $\mathfrak{B}$.

Theorem 2.3.33 (Tarski's Criterion) *If $\mathfrak{A} \subseteq \mathfrak{B}$ and for every L -formula ψ and every $\mathfrak{A}$ -assignment α and variable x ,*

$$\models_{\mathfrak{B}} (\exists x \psi)[\alpha] \quad \Longleftrightarrow \quad \text{for some } a \in |\mathfrak{A}|, \quad \models_{\mathfrak{B}} \psi[\alpha_{x \mapsto a}],$$

then $\mathfrak{A} \prec \mathfrak{B}$.

Proof. Let $\mathfrak{A}$ and $\mathfrak{B}$ satisfy the hypothesis. We prove by Fm_L -induction that for every L -formula ϕ ,

$$\text{for every } \mathfrak{A}\text{-assignment } \alpha, \quad \models_{\mathfrak{A}} \phi[\alpha] \quad \Longleftrightarrow \quad \models_{\mathfrak{B}} \phi[\alpha].$$

For atomic ϕ , this follows from Proposition 2.3.21. The $\neg$ and $\vee$ cases of the induction go as in the proof of Proposition 2.3.21. If ϕ is $\exists x \psi$, then

$$\begin{aligned} \models_{\mathfrak{A}} \phi[\alpha] &\Longleftrightarrow \text{for some } a \in |\mathfrak{A}|, \quad \models_{\mathfrak{A}} \psi[\alpha_{x \mapsto a}] \\ &\Longleftrightarrow \text{for some } a \in |\mathfrak{A}|, \quad \models_{\mathfrak{B}} \psi[\alpha_{x \mapsto a}] \\ &\Longleftrightarrow \models_{\mathfrak{B}} \phi[\alpha]. \quad \square \end{aligned}$$

The second equivalence is the induction hypothesis and the third is the hypothesis.

We can now prove a first version of one of the central results of first-order logic.

Countable Downward Löwenheim-Skolem Theorem 2.3.34 (CLS_↓)

For any countable language L , any L -structure $\mathfrak{B}$ and any countable subset X of $|\mathfrak{B}|$, there exists a countable L -structure $\mathfrak{A}$ such that $X \subseteq |\mathfrak{A}|$ and $\mathfrak{A} \prec \mathfrak{B}$.

Proof. Let X be a countable subset of $|\mathfrak{B}|$. We extend the construction of $\langle X \rangle_{\mathfrak{B}}$ in Proposition 2.3.16 by closing also under functions which provide witnesses for existential statements as required by Tarski's Criterion. To each formula ψ and variable y , we assign a function, called a **Skolem function** $F_{\psi,y}$ as follows. Let $x_0, \dots, x_{k-1}$ be the free variables of $\exists y \psi$. Then $F_{\psi,y} : |\mathfrak{B}|^k \rightarrow |\mathfrak{B}|$ such that for all $b_0, \dots, b_{k-1} \in |\mathfrak{B}|$,

$$\models_{\mathfrak{B}} \exists y \psi[b_0, \dots, b_{k-1}] \implies \models_{\mathfrak{B}} \psi[b_0, \dots, b_{k-1}, F_{\psi,y}(b_0, \dots, b_{k-1})].$$

If $\exists y \psi$ has no free variables, $F_{\psi,y}()$ is simply a single element of $|\mathfrak{B}|$. $F_{\psi,y}$ may be defined using the Axiom of Choice to choose an appropriate witness to the fact that $(b_0, \dots, b_{k-1})$ satisfies $\exists y \psi$ in $\mathfrak{B}$ whenever the set of witnesses is non-empty. Now let A be the smallest subset of $|\mathfrak{B}|$ which includes $X \cup \{c^{\mathfrak{B}} : c \in \text{Cs}_L\}$ and is closed under all $f^{\mathfrak{B}}$ for $f \in \text{Fs}_L$ and all of the functions $F_{\psi,y}$. The countability of L implies that there are only countably many such functions and thus A is countable by Proposition 1.2.8. A is the universe of a unique substructure $\mathfrak{A}$ of $\mathfrak{B}$, and it is immediate from Tarski's Criterion that $\mathfrak{A} \prec \mathfrak{B}$. $\square$

Corollary 2.3.35 For any countable language L and any set Γ of L -sentences, if Γ has any model, then it has a countable model.

Proof. Let $\mathfrak{B}$ be any model of Γ . By the theorem, $\mathfrak{B}$ has a countable elementary substructure $\mathfrak{A}$. Since $\mathfrak{A} \equiv \mathfrak{B}$, $\mathfrak{A}$ is also a model of Γ . $\square$

Example 2.3.36 We shall have many important applications for the Downward Löwenheim-Skolem Theorem, but we note a simple example here. Let $\mathfrak{R} = (\mathbb{R}, <, +, \times, 0, 1)$ be the ordered field of real numbers. Then there is a countable elementary substructure $\mathfrak{A} = (A, <, +, \times, 0, 1)$. Clearly $0, 1, \dots$ belong to A and since $\models_{\mathfrak{A}} \forall x \exists y (x \dot{+} y \dot{=} 0)$, also $-1, -2, \dots$ belong to A . Similarly A is closed under quotients, n -th roots and solutions to algebraic equations — that is, A includes the **real algebraic numbers**.

It is informative to continue this example further. The least upper-bound principle for $\mathfrak{R}$ asserts that every bounded subset of $\mathbb{R}$ has a least upper bound. From the construction of the real numbers via Dedekind cuts, we know that any linear ordering which contains all rational numbers and satisfies the least upper-bound principle is uncountable; hence $\mathfrak{A}$ does *not* satisfy this principle. It follows that there is no sentence of the corresponding first-order language which expresses the least upper-bound principle: if there

were such a sentence θ , then it would be true in $\mathfrak{A}$, hence true in $\mathfrak{A}$, since the two structures are elementarily equivalent.

On the other hand, some bounded subsets of A *do* have least upper bounds; which ones? The answer naturally introduces an important concept.

Definition 2.3.37 For any language L , any L -structure $\mathfrak{A}$ and any L -formula ϕ with free variables $x_0, \dots, x_{m-1}$, the m -ary relation over A

$$\phi^{\mathfrak{A}} := \left\{ (a_0, \dots, a_{m-1}) \in A^m : \models_{\mathfrak{A}} \phi[a_0, \dots, a_{m-1}] \right\}$$

is called the relation **defined over $\mathfrak{A}$** by ϕ . A relation $R \subseteq A^m$ is **definable over $\mathfrak{A}$** iff $R = \phi^{\mathfrak{A}}$ for some formula ϕ with at most m free variables. More generally, if ϕ has free variables among $x_0, \dots, x_{m-1}$ and $y_0, \dots, y_{n-1}$, then we call

$$\phi_{b_0, \dots, b_{n-1}}^{\mathfrak{A}} := \left\{ (a_0, \dots, a_{m-1}) : \models_{\mathfrak{A}} \phi[a_0, \dots, a_{m-1}, b_0, \dots, b_{n-1}] \right\}$$

the relation defined over $\mathfrak{A}$ by ϕ from **parameters** $b_0, \dots, b_{n-1}$. An m -ary function $F : A^m \rightarrow A$ is definable over $\mathfrak{A}$ (from parameters) iff its **graph** $\{(a_0, \dots, a_{m-1}, c) : F(a_0, \dots, a_{m-1}) = c\}$ is so definable. An element of A is definable iff its singleton $\{a\}$ is definable.

Examples 2.3.38

- (i) The usual $<$ relation is defined over $(\omega, +, 0)$ by the formula $\exists z (\neg z \dot{=} 0 \wedge y \dot{=} x \dot{+} z)$.
- (ii) The interval $(3, \infty)$ is definable over $(\mathbb{R}, <)$ from the parameter 3 by the formula $y < x$. We shall see just below that this set is not definable without parameters.
- (iii) The set of prime numbers is definable over Ω by the formula

$$\neg(x \dot{=} 0) \wedge \neg(x \dot{=} \dot{S}0) \wedge \forall y \forall z [y \dot{\times} z \dot{=} x \rightarrow (y \dot{=} x) \vee (z \dot{=} x)].$$

In [Section 4.5](#) we shall see that the class of relations definable over Ω is very rich and includes most familiar relations and functions on natural numbers. Note that since every element n of ω has a name $\dot{n} := \dot{S} \cdots \dot{S}0$ in L_{Ω} , every relation definable over Ω with parameters is also definable without parameters.

Now, to continue Example 2.3.36, for any formula ϕ with free variables x and $y_0, \dots, y_{n-1}$, let ϕ_{ub} be the formula $\forall z (x \dot{<} z \rightarrow \neg \phi_x(z))$. Then for any a and $b_0, \dots, b_{n-1}$ from A ,

$$\models_{\mathfrak{A}} \phi_{\text{ub}}[a, b_0, \dots, b_{n-1}] \iff a \text{ is an upper bound for } \phi_{b_0, \dots, b_{n-1}}^{\mathfrak{A}}.$$

Similarly, if ϕ_{lub} is the formula $\phi_{\text{ub}} \wedge \forall y [y \dot{<} x \rightarrow \neg (\phi_{\text{ub}})_x(y)]$, then

$$\models_{\mathfrak{A}} \phi_{\text{lub}}[a, b_0, \dots, b_{n-1}] \iff a \text{ is the least upper bound for } \phi_{b_0, \dots, b_{n-1}}^{\mathfrak{A}}.$$

Now by the least upper-bound principle, $\models_{\mathfrak{A}} \exists x \phi_{\text{ub}} \rightarrow \exists x \phi_{\text{lub}}$, and since $\mathfrak{A} \prec \mathfrak{R}$, this sentence is also true in $\mathfrak{A}$. In other words, $\mathfrak{A}$ satisfies the principle that every bounded subset definable from parameters from A has a least upper bound.

Note that because the language and A are both denumerable, the number of such sets is only denumerable, so that this principle is far weaker than the full least upper-bound principle, which asserts the existence of least upper bounds for *all* of the uncountably many bounded subsets of $\mathbb{R}$. This is our first example of a concept which is **second-order** — that is, it is expressed only by statements which use not only quantifiers over all elements of a structure but also over all *subsets* of the universe of the structure. We shall encounter several other such examples in the next sections and study this notion systematically in [Section 3.6](#).

The notion of definability of sets and relations over a structure has many other natural relationships with the topics of this section; for example

Proposition 2.3.39 *For any L -structures $\mathfrak{A}$ and $\mathfrak{B}$, if $\eta : \mathfrak{A} \simeq \mathfrak{B}$, then for any formula ϕ with free variables $x_0, \dots, x_{m-1}$,*

$$\phi^{\mathfrak{B}} := \eta(\phi^{\mathfrak{A}}) = \{ (\eta(a_0), \dots, \eta(a_m)) : (a_0, \dots, a_{m-1}) \in \phi^{\mathfrak{A}} \}.$$

Proof. By Lemma 2.3.6. $\square$

In particular, note that if η is an automorphism of $\mathfrak{A}$, then $\phi^{\mathfrak{A}} = \eta(\phi^{\mathfrak{A}})$, which we describe by saying that η **preserves** $\phi^{\mathfrak{A}}$. Thus we have

Corollary 2.3.40 *Every automorphism preserves all relations definable without parameters. $\square$*

Examples 2.3.41

- (i) The interval $(3, \infty)$ is not definable over $(\mathbb{R}, <)$ without parameters as it is not preserved by the automorphism $r \mapsto r + 1$.
- (ii) Neither $+$ nor $<$ is definable over $(\omega, \times, 1)$ without parameters since they are not preserved by the automorphism $0 \mapsto 0, 1 \mapsto 1$ and

$$2^m \times 3^n \times p \mapsto 2^n \times 3^m \times p,$$

where p is not divisible by 2 or 3. Note that this technique does not suffice to show that $\times$ is not definable over $(\omega, +, 0)$, since this structure has no non-trivial automorphisms. A different method for demonstrating this is given in the exercises to [Section 2.6](#).

There is also a natural notion of embedding which respects all first-order properties, and further generalizations will be considered in [Sections 3.4–3.5](#).

Proposition 2.3.42 For any structures $\mathfrak{A}$ and $\mathfrak{B}$ and any function $\eta : |\mathfrak{A}| \rightarrow |\mathfrak{B}|$, the following are equivalent:

- (i) η respects every formula;
- (ii) for some (unique) elementary substructure $\mathfrak{A}' \prec \mathfrak{B}$, $\eta : \mathfrak{A} \simeq \mathfrak{A}'$.

Proof. Essentially the same as that of Proposition 2.3.29; left as an exercise. $\square$

Definition 2.3.43 A function η which satisfies the equivalent conditions of the preceding proposition is called an **elementary embedding** — in symbols, $\eta : \mathfrak{A} \hookrightarrow \mathfrak{B}$. If such an embedding exists, we say that $\mathfrak{A}$ is **elementarily embeddable** in $\mathfrak{B}$ and write $\mathfrak{A} \hookrightarrow \mathfrak{B}$.

In the remainder of this section we study briefly two operations on first-order structures which, again, generalize familiar operations from algebra: cartesian product and unions of chains.

Products and chains

Definition 2.3.44 For any two structures $\mathfrak{A}$ and $\mathfrak{B}$, the **direct product** $\mathfrak{A} \times \mathfrak{B}$ is the structure specified as follows:

- (i) $|\mathfrak{A} \times \mathfrak{B}| = |\mathfrak{A}| \times |\mathfrak{B}| := \{ (a, b) : a \in |\mathfrak{A}| \text{ and } b \in |\mathfrak{B}| \}$;
- (ii) for each $p \in \mathbf{Rs}_L$,

$$p^{\mathfrak{A} \times \mathfrak{B}} := \{ ((a_0, b_0), \dots, (a_{\nu_L(p)-1}, b_{\nu_L(p)-1})) : \\ (a_0, \dots, a_{\nu_L(p)-1}) \in p^{\mathfrak{A}} \text{ and } (b_0, \dots, b_{\nu_L(p)-1}) \in p^{\mathfrak{B}} \};$$

- (iii) for each $f \in \mathbf{Fs}_L$ and any $(a_0, b_0), \dots, (a_{\nu_L(f)-1}, b_{\nu_L(f)-1}) \in |\mathfrak{A} \times \mathfrak{B}|$,

$$f^{\mathfrak{A} \times \mathfrak{B}}((a_0, b_0), \dots, (a_{\nu_L(f)-1}, b_{\nu_L(f)-1})) \\ := (f^{\mathfrak{A}}(a_0, \dots, a_{\nu_L(f)-1}), f^{\mathfrak{B}}(b_0, \dots, b_{\nu_L(f)-1}));$$

- (iv) for each $c \in \mathbf{Cs}_L$, $c^{\mathfrak{A} \times \mathfrak{B}} = (c^{\mathfrak{A}}, c^{\mathfrak{B}})$.

One usually describes this as defining relations and functions **pointwise**.

Example 2.3.45 If $\mathfrak{G} = (G, \dots)$ and $\mathfrak{H} = (H, \dots)$ are groups in any one of the three alternative formulations of Example 2.1.1, then $\mathfrak{G} \times \mathfrak{H}$ is also a group, the usual direct product. This is also true for rings but not for fields.

Our main goal here is to make a preliminary investigation of the question: what is the relationship between truth in $\mathfrak{A} \times \mathfrak{B}$ and truth in $\mathfrak{A}$ and $\mathfrak{B}$? For any $\mathfrak{A}$ -assignment α and any $\mathfrak{B}$ -assignment β , we denote by (α, β) the $\mathfrak{A} \times \mathfrak{B}$ -assignment such that for each variable x , $(\alpha, \beta)(x) = (\alpha(x), \beta(x))$.

Proposition 2.3.46 For all structures $\mathfrak{A}$ and $\mathfrak{B}$, all $\mathfrak{A}$ -assignments α and all $\mathfrak{B}$ -assignments β ,

- (i) for all terms t , $t^{\mathfrak{A} \times \mathfrak{B}}[(\alpha, \beta)] = (t^{\mathfrak{A}}[\alpha], t^{\mathfrak{B}}[\beta])$;
- (ii) for all atomic formulas ϕ ,

$$\models_{\mathfrak{A} \times \mathfrak{B}} \phi[(\alpha, \beta)] \iff \models_{\mathfrak{A}} \phi[\alpha] \text{ and } \models_{\mathfrak{B}} \phi[\beta].$$

Proof. Part (i) is proved by an easy induction; (ii) then follows immediately. $\square$

In contrast to most of our experience to this point, it is not possible to extend (ii) to all formulas or even all quantifier-free formulas. To focus our ideas, we define

Definition 2.3.47 A formula ϕ is **product persistent** iff for all structures $\mathfrak{A}$ and $\mathfrak{B}$, all $\mathfrak{A}$ -assignments α and all $\mathfrak{B}$ -assignments β ,

$$\models_{\mathfrak{A}} \phi[\alpha] \text{ and } \models_{\mathfrak{B}} \phi[\beta] \implies \models_{\mathfrak{A} \times \mathfrak{B}} \phi[(\alpha, \beta)].$$

Lemma 2.3.48

- (i) Atomic and negated atomic formulas are product persistent.
- (ii) If ϕ and ψ are product persistent, then so is $\phi \wedge \psi$.
- (iii) If ϕ is product persistent, then so are $\exists x \phi$ and $\forall x \phi$.

Proof. (i) follows from the preceding proposition and the other parts are straightforward. $\square$

In general the class of product persistent formulas is closed under neither $\neg$ nor $\vee$ (see exercises); the best we can do is

Proposition 2.3.49 For any atomic formulas $\chi_0, \dots, \chi_n$ and θ ,

$$\neg(\chi_0 \wedge \dots \wedge \chi_n) \quad \text{and} \quad \chi_0 \wedge \dots \wedge \chi_n \rightarrow \theta$$

are product persistent.

Proof. Suppose first that ϕ is $\neg(\chi_0 \wedge \dots \wedge \chi_n)$ and $\models_{\mathfrak{A}} \phi[\alpha]$, so for some $i \leq n$, $\not\models_{\mathfrak{A}} \chi_i[\alpha]$. Then also $\not\models_{\mathfrak{A} \times \mathfrak{B}} \chi_i[(\alpha, \beta)]$ and hence $\models_{\mathfrak{A} \times \mathfrak{B}} \phi[(\alpha, \beta)]$.

Now let ϕ be $\chi_0 \wedge \dots \wedge \chi_n \rightarrow \theta$ and suppose both $\models_{\mathfrak{A}} \phi[\alpha]$ and $\models_{\mathfrak{B}} \phi[\beta]$. If either $\not\models_{\mathfrak{A}} \chi_i[\alpha]$ or $\not\models_{\mathfrak{B}} \chi_i[\beta]$ for some $i \leq n$, we argue as before. Otherwise, both $\models_{\mathfrak{A}} \theta[\alpha]$ and $\models_{\mathfrak{B}} \theta[\beta]$, so also $\models_{\mathfrak{A} \times \mathfrak{B}} \theta[(\alpha, \beta)]$ and hence $\models_{\mathfrak{A} \times \mathfrak{B}} \phi[(\alpha, \beta)]$. $\square$

Definition 2.3.50

- (i) The class of **Horn formulas** is the smallest class of formulas which contains all atomic formulas and formulas of the form

$$\neg(\chi_0 \wedge \cdots \wedge \chi_n) \quad \text{and} \quad \chi_0 \wedge \cdots \wedge \chi_n \rightarrow \theta,$$

with $\chi_0, \dots, \chi_n$ and θ atomic, and is closed under $\wedge$, $\exists x$ and $\forall x$.

- (ii) A formula ϕ is **logically Horn** iff ϕ is logically equivalent to a Horn formula.

Note that the displayed condition could also be written

$$\neg\chi_0 \vee \cdots \vee \neg\chi_n \quad \text{and} \quad \neg\chi_0 \vee \cdots \vee \neg\chi_n \vee \theta;$$

that is, logically Horn formulas are those logically equivalent to a disjunction of atomic and negated atomic formulas with at most one unnegated term.

Corollary 2.3.51 *Every logically Horn formula is product persistent.* $\square$

Example 2.3.52 The axioms for group theory as given in Example 2.2.12 are all Horn sentences; this explains the fact that the direct product of groups is a group for all versions of the axiomatization. Similarly the axioms for (strict) partial orderings are Horn sentences, but those for (strict) linear orderings are not, and indeed the class of (strict) linear orderings is not closed under direct product (exercise). It is tempting to conjecture that a formula is product persistent iff it is logically Horn; this is not true, but an exercise below confirms this for logically universal formulas. The axioms for rings (with 1) are Horn sentences, but the axiom asserting the existence of a multiplicative inverse is not Horn; this follows from the fact mentioned above that the direct product of fields is not necessarily a field.

We next generalize these results to arbitrary (infinite) products; the notation is essentially the only new element. We recall from set theory

Definition 2.3.53 For any set I and any family of sets $\langle A_i : i \in I \rangle$,

$$\prod_{i \in I} A_i := \{ F : F \text{ is a function with domain } I \text{ and } (\forall i \in I)[F(i) \in A_i] \}.$$

Note that if $I = \{0, 1\}$, then there is a natural 1-1 correspondence between $\prod_{i \in I} A_i$ and $A_0 \times A_1$.

Definition 2.3.54 For any set I and L -structures $\langle \mathfrak{A}_i : i \in I \rangle$, the **direct product** $\mathfrak{A} := \prod_{i \in I} \mathfrak{A}_i$ is specified as follows:

- (i) $|\mathfrak{A}| := \prod_{i \in I} A_i$;
- (ii) for each $p \in \mathbf{Rs}_L$ and each $F_0, \dots, F_{\nu_L(p)-1} \in |\mathfrak{A}|$,

$$p^{\mathfrak{A}}(F_0, \dots, F_{\nu_L(p)-1}) : \Longleftrightarrow \text{ for all } i \in I, p^{\mathfrak{A}_i}(F_0(i), \dots, F_{\nu_L(p)-1}(i));$$

- (iii) for each $f \in \mathbf{Fs}_L$ and each $F_0, \dots, F_{\nu_L(f)-1} \in A$,

$$\begin{aligned} f^{\mathfrak{A}}(F_0, \dots, F_{\nu_L(f)-1}) &:= G, \quad \text{where for all } i \in I, \\ G(i) &:= f^{\mathfrak{A}_i}(F_0(i), \dots, F_{\nu_L(f)-1}(i)); \end{aligned}$$

- (iv) for each $c \in \mathbf{Cs}_L$, $c^{\mathfrak{A}} := G$, where for all $i \in I$, $G(i) = c^{\mathfrak{A}_i}$.

With the direct product $\mathfrak{A} = \prod_{i \in I} \mathfrak{A}_i$ and any family of assignments $\langle \alpha_i : i \in I \rangle$, we naturally associate an $\mathfrak{A}$ -assignment α defined by $\alpha(x)(i) = \alpha_i(x)$. We have then as before for all terms t

$$t^{\mathfrak{A}}[\alpha] = G \quad \text{where for all } i \in I, \quad G(i) = t^{\mathfrak{A}_i}[\alpha_i],$$

and for all atomic formulas ϕ ,

$$\models_{\mathfrak{A}} \phi[\alpha] \quad \Longleftrightarrow \quad \text{for all } i \in I, \quad \models_{\mathfrak{A}_i} \phi[\alpha_i].$$

Let us now call a formula ϕ **completely product persistent** iff for any I , $\mathfrak{A}_i$ and $\mathfrak{A}$ as above,

$$\text{if for all } i \in I, \models_{\mathfrak{A}_i} \phi[\alpha_i] \quad \text{then} \quad \models_{\mathfrak{A}} \phi[\alpha].$$

Then easily

Corollary 2.3.55 *Every logically Horn formula is completely product persistent.*

Proof. Left as an exercise. $\square$

We shall make use of this construction in [Section 3.3](#) and consider here one other infinitary operation on L -structures.

Definition 2.3.56

- (i) An ***L-chain*** consists of a family $\langle \mathfrak{A}_i : i \in \omega \rangle$ of L -structures such that for all $i < j$, $\mathfrak{A}_i \subseteq \mathfrak{A}_j$. If in addition, $\mathfrak{A}_i \prec \mathfrak{A}_j$ for all $i < j$, then $\langle \mathfrak{A}_i : i \in \omega \rangle$ is called an ***L-elementary chain***.
- (ii) The ***union of an L-chain*** is the L -structure $\mathfrak{A} := \bigcup \langle \mathfrak{A}_i : i \in \omega \rangle$ specified as follows:
 - (a) $|\mathfrak{A}| = \bigcup \{ |\mathfrak{A}_i| : i \in \omega \}$;
 - (b) for each $p \in \text{Rs}_L$, $p^{\mathfrak{A}} = \bigcup \{ p^{\mathfrak{A}_i} : i \in \omega \}$;
 - (c) for each $f \in \text{Fs}_L$ and $a_0, \dots, a_{\nu_L(f)-1}, b \in |\mathfrak{A}|$,

$$f^{\mathfrak{A}}(a_0, \dots, a_{\nu_L(f)-1}) = b \iff \text{for some } i \in \omega, \\ f^{\mathfrak{A}_i}(a_0, \dots, a_{\nu_L(f)-1}) = b;$$

- (d) for each $c \in \text{Cs}_L$ and any $b \in |\mathfrak{A}|$,

$$c^{\mathfrak{A}} = b \iff \text{for some } i \in \omega, \quad c^{\mathfrak{A}_i} = b.$$

Of course, one must check that the structure $\mathfrak{A}$ is well-defined; this follows from the definition of chain. For example, suppose that $c^{\mathfrak{A}}$ were not well-defined — that is, for some $i \neq i' \in \omega$ and some $b \neq b' \in |\mathfrak{A}|$, $c^{\mathfrak{A}_i} = b$ and $c^{\mathfrak{A}_{i'}} = b'$. But then, either $i < i'$ or $i' < i$ so either $\mathfrak{A}_i \subseteq \mathfrak{A}_{i'}$ or $\mathfrak{A}_{i'} \subseteq \mathfrak{A}_i$ and in either case $b = b'$. The justification for function symbols is similar.

Examples 2.3.57

- (i) For each $i \in \omega$, let $\mathfrak{A}_i := ([-i, \infty), <)$, where the interval may be in any of $\mathbb{Z}$, $\mathbb{Q}$ or $\mathbb{R}$ and the ordering is the usual one. Then $\bigcup \langle \mathfrak{A}_i : i \in \omega \rangle = ((-\infty, \infty), <)$.
- (ii) For each $i \in \omega$, let $\mathfrak{G}_i := (\{ \frac{m}{2^i} : m \in \mathbb{Z} \}, +, 0)$. Each $\mathfrak{G}_i$ is a group as is $\bigcup \langle \mathfrak{G}_i : i \in \omega \rangle$, the group of ***dyadic rationals***.
- (iii) If $\mathfrak{A}_i := (\{ \frac{m}{2^i} : m \in \mathbb{Z} \}, <)$, then each $\mathfrak{A}_i$ is a ***discrete ordering*** (every element with a predecessor (successor) has an unique *immediate* predecessor (successor)), but $\bigcup \langle \mathfrak{A}_i : i \in \omega \rangle$ is a dense ordering.

Proposition 2.3.58 *For any L -chain $\langle \mathfrak{A}_i : i \in \omega \rangle$, if $\mathfrak{A} := \bigcup \langle \mathfrak{A}_i : i \in \omega \rangle$, then*

- (i) *for all $i \in \omega$, $\mathfrak{A}_i \subseteq \mathfrak{A}$;*
- (ii) *if the chain is elementary, then for all $i \in \omega$, $\mathfrak{A}_i \prec \mathfrak{A}$.*

Proof. Part (i) is immediate from the definition. For (ii), we prove by formula induction that for all ϕ , all $i \in \omega$ and all $\mathfrak{A}_i$ -assignments α ,

$$\models_{\mathfrak{A}_i} \phi[\alpha] \iff \models_{\mathfrak{A}} \phi[\alpha].$$

The case for ϕ atomic follows from (i), and the $\neg$ and $\vee$ steps are trivial. Suppose that ϕ is $\exists x \psi$. Then

$$\begin{aligned}
\models_{\mathfrak{A}_i} \phi[\alpha] &\iff \text{for some } a \in |\mathfrak{A}_i|, \quad \models_{\mathfrak{A}_i} \psi[\alpha_{x \mapsto a}] \\
&\iff \text{for some } a \in |\mathfrak{A}_i|, \quad \models_{\mathfrak{A}} \psi[\alpha_{x \mapsto a}] \\
&\implies \text{for some } a \in |\mathfrak{A}|, \quad \models_{\mathfrak{A}} \psi[\alpha_{x \mapsto a}] \\
&\iff \models_{\mathfrak{A}} \phi[\alpha] \\
&\implies \text{for some } j > i \text{ and } a \in |\mathfrak{A}_j|, \quad \models_{\mathfrak{A}} \psi[\alpha_{x \mapsto a}] \\
&\iff \text{for some } j > i \text{ and } a \in |\mathfrak{A}_j|, \quad \models_{\mathfrak{A}_j} \psi[\alpha_{x \mapsto a}] \\
&\iff \text{for some } j > i, \quad \models_{\mathfrak{A}_j} \phi[\alpha] \\
&\iff \models_{\mathfrak{A}_i} \phi[\alpha].
\end{aligned}$$

It is in the final equivalence that we use the hypothesis that if $i < j$, then $\mathfrak{A}_i \prec \mathfrak{A}_j$. $\square$

Definition 2.3.59

- (i) A formula ψ is **universal-existential** iff for some quantifier-free formula θ and some variables $y_0, \dots, y_{m-1}, z_0, \dots, z_{n-1}$,

$$\psi \text{ is } \forall y_0 \dots \forall y_{m-1} \exists z_0 \dots \exists z_{n-1} \theta.$$

- (ii) A formula ϕ is **logically universal-existential** iff for some universal-existential formula ψ , $\models \phi \leftrightarrow \psi$.

Example 2.3.60 The formula θ'_{inverse} of Examples 2.2.14 is universal-existential.

Lemma 2.3.61 For any formulas ϕ and ψ ,

- (i) if both ϕ and ψ are logically universal-existential, then also $\phi \wedge \psi$ and $\phi \vee \psi$ are logically universal-existential;
- (ii) if ϕ is logically universal-existential, then also $\forall x \phi$ is logically universal-existential.

Proof. Left as an exercise. $\square$

Definition 2.3.62 A sentence ϕ is **union persistent** iff for any chain

$$\langle \mathfrak{A}_i : i \in \omega \rangle \quad \text{and} \quad \mathfrak{A} := \bigcup \langle \mathfrak{A}_i : i \in \omega \rangle,$$

if for all $i \in \omega$, $\models_{\mathfrak{A}_i} \phi$, then also $\models_{\mathfrak{A}} \phi$.

Proposition 2.3.63 *Every logically universal-existential sentence is union persistent.*

Proof. We show that any universal-existential sentence

$$\psi : \forall y_0 \dots \forall y_{m-1} \exists z_0 \dots \exists z_{n-1} \theta$$

is union persistent. Suppose that for all $i \in \omega$, $\models_{\mathfrak{A}_i} \psi$; we verify that $\models_{\mathfrak{A}} \psi$. Fix any $a_0, \dots, a_{m-1} \in |\mathfrak{A}|$. By the definition of chain, there exists $i \in \omega$ such that $a_0, \dots, a_{m-1} \in |\mathfrak{A}_i|$. Since $\models_{\mathfrak{A}_i} \psi$, in particular $\models_{\mathfrak{A}_i} \exists z_0 \dots \exists z_{n-1} \theta[a_0, \dots, a_{m-1}]$, so as $\mathfrak{A}_i \subseteq \mathfrak{A}$, this existential formula persists and we have also $\models_{\mathfrak{A}} \exists z_0 \dots \exists z_{n-1} \theta[a_0, \dots, a_{m-1}]$. Since $a_0, \dots, a_{m-1}$ were arbitrary elements of $|\mathfrak{A}|$, this shows that $\models_{\mathfrak{A}} \psi$. $\square$

In [Section 3.5](#) we shall show that also conversely every union-persistent sentence is logically universal-existential.

Collateral Reading. The relationship between Horn sentences and products of structures is treated extensively in [Ch-Ke; [Sections 6.2–6.3](#)]; logically Horn sentences are exactly those persistent for *reduced* products, which are closely related to the ultraproducts studied in [Section 3.3](#) below.

Exercises

Exercise 2.3.64 In each case, determine whether or not the structures are isomorphic:

- (i) $(\mathbb{R}, <, +, 0) \stackrel{?}{\simeq} (\mathbb{R}^{>0}, <, \times, 1)$;
- (ii) $(\mathbb{Q}, <, +, 0) \stackrel{?}{\simeq} (\mathbb{Q}^{>0}, <, \times, 1)$;
- (iii) $(\mathbb{Z}, <, +, 0) \stackrel{?}{\simeq} (\mathbb{Z}, >, +, 0)$;
- (iv) $(\mathbb{Z}, <, \times, 0) \stackrel{?}{\simeq} (\mathbb{Z}, >, \times, 0)$.

Exercise 2.3.65 Prove Lemma 2.3.28.

Exercise 2.3.66 Formulate and prove a result analogous to Proposition 2.3.29 for general (not necessarily injective) homomorphisms.

Exercise 2.3.67 Prove Proposition 2.3.19.

Exercise 2.3.68 Prove Proposition 2.3.42.

Exercise 2.3.69 Prove that every logically Horn formula is completely product persistent (Corollary 2.3.55).

Exercise 2.3.70 Prove Lemma 2.3.23.

Exercise 2.3.71 Prove that $+$ is not even definable over $(\omega, \times, 1)$ from parameters (compare Example 2.3.41(ii)).

Exercise 2.3.72 Let $\mathfrak{A}$ and $\mathfrak{B}$ be two L -structures and $\eta : |\mathfrak{A}| \rightarrow |\mathfrak{B}|$ a bijection. Prove that η is an isomorphism iff η respects all quantifier-free L -formulas.

Exercise 2.3.73 Give examples to show that the class of product persistent formulas is closed under neither $\neg$ nor $\vee$.

Exercise 2.3.74 Show that every formula which is both logically universal and product persistent is logically Horn.

Hint: Let ϕ be logically universal and product persistent. Using Remark 2.2.32(v), ϕ is logically equivalent to a conjunction of formulas of the form

$$\forall x_0 \cdots \forall x_{k-1} (\chi_0 \wedge \cdots \wedge \chi_{n-1} \rightarrow \theta_0 \vee \cdots \vee \theta_{p-1}),$$

with all χ_i and θ_j atomic and either list may be empty. If p is 0 or 1, then this is a Horn formula. Use the product persistence of ϕ to show that for those cases with $p \geq 2$ there exists $j < p$ such that

$$\models \phi \rightarrow \forall x_0 \cdots \forall x_{k-1} (\chi_0 \wedge \cdots \wedge \chi_{n-1} \rightarrow \theta_j).$$

Then the conjunction of the resulting formulas witnesses that ϕ is logically Horn.

Exercise 2.3.75 Show that the class of fields is not closed under direct product, and conclude that the axiom asserting the existence of a multiplicative inverse is not logically Horn (Example 2.3.52). Explain.

Exercise 2.3.76 An *integral domain* is a commutative ring with 1 (properties 2.1.1(iv) (a) through (h)) with no **zero divisors** — that is, such that

(j) if $a \times b = 0$, then either $a = 0$ or $b = 0$.

Are the axioms for integral domains all Horn sentences? Prove or disprove and explain.

Exercise 2.3.77 Prove that every formula which is a Boolean combination of existential (equivalently universal) formulas is both universal-existential and existential-universal.

The next series of exercises uses the following notions, which are generalization of familiar ideas from the theory of fields and vector spaces. These results are important in Model Theory and will be used in an exercise to [Section 7.5](#).

Definition 2.3.78 For any structure $\mathfrak{A}$ and any set $X \subseteq A$,

- (i) an element $a \in A$ is **$\mathfrak{A}$ -algebraic over X** iff a belongs to some finite subset of A which is definable from parameters from X (Definition 2.3.37);
- (ii) $\text{acl}_{\mathfrak{A}} X := \{a \in A : a \text{ is } \mathfrak{A}\text{-algebraic over } X\}$ is the **$\mathfrak{A}$ -algebraic closure** of X ;
- (iii) X is **$\mathfrak{A}$ -independent** iff for each $a \in X$, $a \notin \text{acl}_{\mathfrak{A}}(X \sim \{a\})$;
- (iv) for any $X \subseteq C \subseteq A$, X is an **$\mathfrak{A}$ -basis** for C iff X is independent and $C = \text{acl}_{\mathfrak{A}} X$;
- (v) for any formula ϕ , $\exists^{\geq m} x \phi$ denotes the formula

$$\exists x_0 \dots \exists x_{m-1} \left(\bigwedge_{i < m} \phi_x(x_i) \wedge \bigwedge_{i < j < m} \neg x_i \doteq x_j \right);$$

$\exists^{=m} x \phi$ denotes the formula $\exists^{\geq m} x \phi \wedge \neg \exists^{\geq m+1} x \phi$;

- (vi) $\mathfrak{A}$ is **minimal** iff every subset of A definable over $\mathfrak{A}$ from parameters is either finite or cofinite.

Exercise 2.3.79 Prove that for any structure $\mathfrak{A}$ and any sets $X, Y \subseteq A$,

- (i) $X \subseteq \text{acl}_{\mathfrak{A}} X$;
- (ii) $X \subseteq Y \implies \text{acl}_{\mathfrak{A}} X \subseteq \text{acl}_{\mathfrak{A}} Y$;
- (iii) $X \subseteq \text{acl}_{\mathfrak{A}} Y \implies \text{acl}_{\mathfrak{A}} X \subseteq \text{acl}_{\mathfrak{A}} Y$;
- (iv) $\text{acl}_{\mathfrak{A}} \text{acl}_{\mathfrak{A}} X = \text{acl}_{\mathfrak{A}} X$;
- (v) $\text{acl}_{\mathfrak{A}} X = \bigcup \{ \text{acl}_{\mathfrak{A}} X_0 : X_0 \subseteq X \text{ and } X_0 \text{ is finite} \}$.

Hint: For (iii), if $a \in \phi(x, y)_b^{\mathfrak{A}}$ with $b \in X \subseteq \text{acl}_{\mathfrak{A}} Y$, and $b \in \psi(y, z)_c^{\mathfrak{A}}$ with $c \in Y$, then $a \in (\exists y [\phi(x, y) \wedge \psi(y, z)])_c^{\mathfrak{A}}$, but this set is not necessarily finite. Modify this formula to conclude that $a \in \text{acl}_{\mathfrak{A}} Y$.

Exercise 2.3.80 Prove that for any structure $\mathfrak{A}$ and any set $X \subseteq A$,

- (i) $\text{acl}_{\mathfrak{A}} X$ is the universe of a (unique) substructure $\mathfrak{B} \subseteq \mathfrak{A}$ ($\mathfrak{B} \supseteq \langle X \rangle_{\mathfrak{A}}$, but it may be larger);
- (ii) if $\mathfrak{A}$ is minimal and X is infinite, then
 - (a) $\mathfrak{B}$ is an elementary substructure;
 - (b) for any elementary substructure $\mathfrak{C} \prec \mathfrak{A}$, if $X \subseteq C$, then $B \subseteq C$ — that is, $\mathfrak{B}$ is the smallest elementary substructure which includes X .

Exercise 2.3.81 Prove that for any infinite minimal structure $\mathfrak{A}$, the $\mathfrak{A}$ -algebraic closure has the following **exchange property**: for any $X \subseteq A$ and any $a, b \in A$,

$$a \in \text{acl}_{\mathfrak{A}}(X \cup \{b\}) \quad \text{but} \quad a \notin \text{acl}_{\mathfrak{A}} X \implies b \in \text{acl}_{\mathfrak{A}}(X \cup \{a\}).$$

Hint: Fix a formula $\phi(x, y, z)$ and $c \in X$ (in general $c_0, \dots, c_{k-1}$) such that $a \in \phi_{b,c}^{\mathfrak{A}}$ and $\phi_{b,c}^{\mathfrak{A}}$ is finite. Note that then $b \in \phi_{a,c}^{\mathfrak{A}}$ so if this set is finite, we have the desired result immediately. Otherwise, let $\psi(x, y, z)$ be the formula

$$\phi(x, y, z) \wedge \exists^=^m x \phi(x, y, z),$$

where $\phi_{b,c}^{\mathfrak{A}}$ has m elements. Thus also $a \in \psi_{b,c}^{\mathfrak{A}}$ and $b \in \psi_{a,c}^{\mathfrak{A}}$, so again if $\psi_{a,c}^{\mathfrak{A}}$ is finite we are done. Suppose, towards a contradiction, that this is not the case, and therefore by the minimality hypothesis $\neg \psi_{a,c}^{\mathfrak{A}}$ is finite, say with n elements. Let $\theta(x, z)$ be the formula $\exists^=^n y \neg \psi(x, y, z)$, so also $a \in \theta_c^{\mathfrak{A}}$. Since $a \notin \text{acl}_{\mathfrak{A}} X$, $\theta_c^{\mathfrak{A}}$ is infinite. Show that for any $a'_0, \dots, a'_m \in \theta_c^{\mathfrak{A}}$, $\bigcup_{i \leq m} \neg \psi_{a'_i, c}^{\mathfrak{A}}$ is finite and derive a contradiction.

Exercise 2.3.82 Show that for any minimal structure $\mathfrak{A}$ and elementary substructure $\mathfrak{C} \prec \mathfrak{A}$,

- (i) C has an $\mathfrak{A}$ -basis;
- (ii) any two $\mathfrak{A}$ -bases of C have the same cardinality, which we call the **$\mathfrak{A}$ -dimension** of $\mathfrak{C}$;
- (iii) if $\mathfrak{C}$ is countable then it has $\mathfrak{A}$ -dimension either finite or $\aleph_0$;
- (iv) if $\mathfrak{C}$ is uncountable, then it has $\mathfrak{A}$ -dimension $\text{Card}(\mathfrak{C})$.

Hint: These are proved exactly as are the corresponding facts for the theory of vector spaces.

2.4 Theories

In this and the next section we introduce some of the fundamental mathematical theories which deal with equality, orderings, arithmetic and set theory. These theories will continue to play a role throughout the rest of this book. As in propositional logic, we identify a theory with the set of its **theorems**, which we take to be sentences. Unless otherwise stated, all formulas and structures refer to one fixed language L .

Definition 2.4.1 For any set Γ of L -sentences,

- (i) Γ is an *L -theory* iff Γ is closed under logical consequence — that is,

$$\Gamma \models \psi \text{ and } \psi \text{ is a sentence} \implies \psi \in \Gamma;$$

Elements of Γ are called the *theorems* of Γ .

- (ii) $\text{Mod}(\Gamma)$ denotes the class of all models of Γ (Definition 2.2.13); we write

$$\text{Mod}(\phi_0, \dots, \phi_n) \text{ instead of } \text{Mod}(\{\phi_0, \dots, \phi_n\}).$$

Immediately from the definitions we have

Corollary 2.4.2 For any sets Γ and Δ of sentences and any sentence ψ ,

- (i) $\Gamma \models \psi \iff \text{Mod}(\Gamma) \subseteq \text{Mod}(\psi);$
- (ii) $\Gamma \subseteq \Delta \implies \text{Mod}(\Delta) \subseteq \text{Mod}(\Gamma);$
- (iii) $\Gamma \text{ is consistent} \iff \text{Mod}(\Gamma) \neq \emptyset. \quad \square$

Remark 2.4.3 Note particularly that a theory is defined as a set of *sentences*. This is a matter of choice but conforms with mathematical practice. Formal theories are intended to model (in the usual scientific sense of this word) working theories of mathematics. General formulas with free variables are useful in many ways, but the axioms and theorems of mathematical theories are generally sentences. This convention simplifies many results in that interpretations are just structures; no variable assignments are required. For example, recall (Definition 2.2.1) that a set Γ of formulas is consistent iff for some $\mathfrak{A}$ and α , $\models_{\mathfrak{A}} \phi[\alpha]$ for all $\phi \in \Gamma$, whereas a set of sentences is consistent iff it has at least one model. However, in many cases it will be useful to have results in general for formulas, and we continue to state definitions and results in this form whenever possible.

Many of the basic facts about theories in general are essentially the same as in the propositional case in [Section 1.4](#); we sketch them briefly but omit proofs, which are direct translations of the earlier ones. As before, there are three distinct ways to describe a theory.

Definition 2.4.4 For any set Γ of sentences, the *theory generated by* Γ is the set

$$\text{Th}(\Gamma) = \{ \psi : \psi \text{ is a sentence and } \Gamma \models \psi \}.$$

We call Γ a set of *axioms* for $\text{Th}(\Gamma)$.

Note that by Corollary 2.2.29, since we are assuming that Γ is a set of sentences,

$$\text{Th}(\Gamma) = \{ \overline{\forall \psi} : \psi \text{ is a formula and } \Gamma \models \psi \}.$$

Examples 2.4.5

- (i) The models of the $L_{=}$ -theory $\text{Th}(\{\exists \geq^n : n > 0\})$ are exactly the structures (A) with A an infinite set.
- (ii) The models of $T_{\text{SLO}} := \text{Th}(\theta_{\text{SLO}})$ of Example 2.2.12 are exactly the strict linear orderings, and the models of $T_{\text{DLO}} := \text{Th}(\theta_{\text{DLO}})$ are exactly the dense linear orderings.
- (iii) The models of $T_{\text{Gp}} := \text{Th}(\Gamma_{\text{Gp}})$ are exactly the groups of the form $\mathfrak{G} = (G, *, {}^{-1}, e)$. Hence T_{Gp} is often called **group theory**.

Proposition 2.4.6 *For any sets Γ and Δ of sentences,*

- (i) $\text{Mod}(\Gamma) = \text{ModTh}(\Gamma)$;
- (ii) $\text{Th}(\Gamma)$ is a theory;
- (iii) $\Gamma \subseteq \Delta \implies \text{Th}(\Gamma) \subseteq \text{Th}(\Delta)$;
- (iv) Γ is a theory iff $\Gamma = \text{Th}(\Gamma)$;
- (v) $\text{Th}(\Gamma)$ is the smallest theory T such that $\Gamma \subseteq T$;
- (vi) $\text{Th}(\emptyset) = \{\phi : \phi \text{ is logically valid}\}$ is included in every theory.

Proof. Very similar to the propositional version, Proposition 1.4.7. $\square$

Of course generally a theory has many, infinitely many in fact, different sets of axioms. A theory expressed as $\text{Th}(\Gamma)$ is said to be an **axiomatic theory** or a theory presented **axiomatically**. By (iv) of the proposition, any theory T may trivially be given axiomatically as $T = \text{Th}(T)$, but we shall usually be interested in imposing conditions on the set of axioms.

Definition 2.4.7 A theory T is **finitely (decidably) axiomatizable** iff $T = \text{Th}(\Gamma)$ for some finite (decidable) set Γ .

All of the examples above are finitely axiomatizable theories; we shall soon have examples of theories that are decidable but not finitely axiomatizable. A second way to describe a theory is as the set of sentences true in some structure.

Definition 2.4.8 For any structure $\mathfrak{A}$, the **theory of** $\mathfrak{A}$ is the set

$$\text{Th}(\mathfrak{A}) := \left\{ \psi : \psi \text{ is a sentence and } \models_{\mathfrak{A}} \psi \right\}.$$

Thus, for example, we shall be interested in the theories of the structures

$$\Omega_+ := (\omega, <, +, \text{Sc}, 0) \quad \text{and} \quad \Omega := (\omega, <, +, \times, \text{Sc}, 0).$$

In [Section 2.5](#) we shall see that the first of these is decidable axiomatizable and decidable, but the simplest version of the Gödel Incompleteness Theorems ([Section 4.1](#) and [Section 4.5](#)) shows that the second has neither of these properties.

Definition 2.4.9 A set Γ of sentences is **complete** iff for every sentence ϕ , either $\phi \in \Gamma$ or $\neg \phi \in \Gamma$, but not both — that is, $(\neg \phi) \in \Gamma \iff \phi \notin \Gamma$.

Proposition 2.4.10 For any structure $\mathfrak{A}$,

- (i) $\text{Th}(\mathfrak{A})$ is a complete and consistent theory and $\mathfrak{A}$ is a model of $\text{Th}(\mathfrak{A})$;
- (ii) for any structure $\mathfrak{B}$, $\mathfrak{A} \equiv \mathfrak{B} \iff \text{Th}(\mathfrak{A}) = \text{Th}(\mathfrak{B})$.

Proof. Immediate from the definitions. $\square$

Note that (ii) differs significantly from its propositional counterpart Proposition 1.4.11(ii):

$$V = W \iff \text{Th}(V) = \text{Th}(W).$$

By Theorem 2.3.11 we know that for *finite* structures $\mathfrak{A}$,

$$\mathfrak{A} \simeq \mathfrak{B} \iff \text{Th}(\mathfrak{A}) = \text{Th}(\mathfrak{B}),$$

but in general only (ii) holds.

Definition 2.4.11 For any class K of structures, the **theory of K** is the set

$$\begin{aligned} \text{Th}(K) &= \left\{ \psi : \psi \text{ is a sentence and } \models_{\mathfrak{A}} \psi \text{ for all } \mathfrak{A} \in K \right\} \\ &= \bigcap \{ \text{Th}(\mathfrak{A}) : \mathfrak{A} \in K \}. \end{aligned}$$

For example, group theory might as well be (and probably was originally) considered as $\text{Th}(K)$ for K the class of all groups.

Proposition 2.4.12 For any non-empty class K of structures,

- (i) $\text{Th}(K)$ is a consistent theory; each $\mathfrak{A} \in K$ is a model of $\text{Th}(K)$;
- (ii) if K has at least two elements $\mathfrak{A}$ and $\mathfrak{B}$ such that $\mathfrak{A} \not\equiv \mathfrak{B}$, then $\text{Th}(K)$ is not complete.

Proof. Part (i) is immediate. For (ii), if $\mathfrak{A} \not\equiv \mathfrak{B}$, then there is a sentence ψ such that $\models_{\mathfrak{A}} \psi$ but $\not\models_{\mathfrak{B}} \psi$. Then also $\not\models_{\mathfrak{A}} \neg \psi$, so neither ψ nor $\neg \psi$ belongs to $\text{Th}(K)$. $\square$

We call a class K of structures **axiomatizable** iff for some set Γ of sentences, $K = \text{Mod}(\Gamma)$. Parallel to Corollary 1.4.16,

Corollary 2.4.13

- (i) For any set Γ of sentences, $\Gamma \subseteq \text{Th}(\text{Mod}(\Gamma))$; Γ is a theory iff $\Gamma = \text{Th}(\text{Mod}(\Gamma))$; in particular, for any set Γ of sentences, $\text{Th}(\Gamma) = \text{Th}(\text{Mod}(\Gamma))$.
- (ii) For any class K of structures, $K \subseteq \text{Mod}(\text{Th}(K))$; K is axiomatizable iff $K = \text{Mod}(\text{Th}(K))$. $\square$

Note that the inclusion in (ii) may be proper — for example, for any K , $\text{Mod}(\text{Th}(K))$ includes all structures elementarily equivalent to a member of K . We note next that there is only one inconsistent theory.

Proposition 2.4.14 For any set Γ of formulas, the following are equivalent:

- (i) Γ is inconsistent;
- (ii) for all formulas ϕ , $\Gamma \models \phi$;
- (iii) for some (every) formula ϕ , both $\Gamma \models \phi$ and $\Gamma \models \neg \phi$;
- (iv) $\Gamma \models \text{F}$.

Proof. Parallel to that of Proposition 1.4.17. $\square$

Interpreting these conditions for theories we have

Corollary 2.4.15 For any theory T , the following are equivalent:

- (i) T is inconsistent;
- (ii) $T = \{ \phi : \phi \text{ is a sentence} \}$;
- (iii) for some (every) sentence ϕ , both $\phi \in T$ and $\neg \phi \in T$;
- (iv) $\text{F} \in T$. $\square$

Corollary 2.4.16 For any theory T , the following are equivalent:

- (i) T is complete;
- (ii) T is **maximal consistent** — that is, T is consistent but any properly larger set of sentences is inconsistent;
- (iii) T is consistent and for any model $\mathfrak{A}$ of T , $T = \text{Th}(\mathfrak{A})$;
- (iv) T is consistent and for any two models $\mathfrak{A}$ and $\mathfrak{B}$ of T , $\mathfrak{A} \equiv \mathfrak{B}$.

Proof. The equivalence of (i) and (ii) is as for Corollary 1.4.19. Suppose T is complete and let $\mathfrak{A}$ be any model of T , so by definition $T \subseteq \text{Th}(\mathfrak{A})$. For any sentence $\phi \in \text{Th}(\mathfrak{A})$, if $\phi \notin T$, then $\neg \phi \in T$, so also $\neg \phi \in \text{Th}(\mathfrak{A})$, contrary to the consistency of $\text{Th}(\mathfrak{A})$. (iv) follows immediately from (iii), since if T satisfies (iii) and $\mathfrak{A}$ and $\mathfrak{B}$ are any two models of T , then $\text{Th}(\mathfrak{A}) = \text{Th}(\mathfrak{B})$, so by Proposition 2.4.10(ii), $\mathfrak{A} \equiv \mathfrak{B}$.

Suppose now that (iv) holds but, towards a contradiction, T is not complete. Then for some sentence ϕ , neither $\phi \in T$ nor $(\neg \phi) \in T$, so neither $T \models \phi$ nor $T \models \neg \phi$. Hence there exist models $\mathfrak{A}$ and $\mathfrak{B}$ of T such that $\models_{\mathfrak{A}} \neg \phi$ and $\models_{\mathfrak{B}} \phi$; this contradicts $\mathfrak{A} \equiv \mathfrak{B}$. $\square$

We may summarize the content of the corollary by saying that the complete theories of a language are in one-one correspondence with the equivalence classes of structures under the equivalence relation $\equiv$.

If we continued to follow the pattern of [Section 1.4](#), we would next prove the Compactness Theorem. We prefer, however, to postpone this somewhat more difficult proof to the next chapter and spend some time here getting acquainted with a number of interesting and important theories. In particular, we shall consider questions of completeness and decidability. Therefore we assume for the rest of this section that **the language L is effective**.

Some properties of axiomatically given theories $T = \text{Th}(\Gamma)$ can be derived from an analysis of the complexity of the set Γ of axioms. Of course, many different Γ generate the same theory, and usually the goal is to find a set of axioms for T which is optimal by some measure. One way to measure the complexity of the axioms of a theory is to ask if they may all be chosen to be sentences of some special kind, such as existential or universal:

Definition 2.4.17 For any set Δ of formulas, a first-order theory T is **Δ -axiomatizable** iff $T = \text{Th}(\Gamma)$ for some set $\Gamma \subseteq \Delta$.

Most of the classes Δ we have in mind here arise from the following. Here we write $\mathbf{x}$ as an abbreviation for $x_0, \dots, x_{k-1}$ for some $k \geq 0$.

Definition 2.4.18 For any set Δ of formulas,

- (i) $\forall \Delta := \{ \psi(\mathbf{x}) : \text{for some } \theta(\mathbf{x}, \mathbf{y}) \in \Delta, \models \psi(\mathbf{x}) \leftrightarrow \forall \mathbf{y} \theta(\mathbf{x}, \mathbf{y}) \}$;
- (ii) $\exists \Delta := \{ \psi(\mathbf{x}) : \text{for some } \theta(\mathbf{x}, \mathbf{y}) \in \Delta, \models \psi(\mathbf{x}) \leftrightarrow \exists \mathbf{y} \theta(\mathbf{x}, \mathbf{y}) \}$;
- (iii) $\neg \Delta := \{ \psi(\mathbf{x}) : \text{for some } \theta(\mathbf{x}) \in \Delta, \models \psi(\mathbf{x}) \leftrightarrow \neg \theta(\mathbf{x}) \}$;
- (iv) $\forall_0 := \exists_0 := \text{QF}$ and for all $n \in \omega$, $\exists_{n+1} := \exists \forall_n$ and $\forall_{n+1} := \neg \exists_{n+1}$.

In (i) and (ii) we include the possibility that $\mathbf{y}$ is the empty string of variables so that $\Delta \subseteq \forall \Delta, \exists \Delta$.

Thus $\exists_1$ is the set of logically existential formulas, $\forall_1$ is the set of logically universal formulas, and $\forall_2$ is the set of universal-existential formulas (Definitions 2.3.22 and 2.3.59). Hence an $\exists_1$ -axiomatizable ($\forall_1$ -axiomatizable) theory is also called **existentially axiomatizable** (**universally axiomatizable**).

Lemma 2.4.19 For any set Δ of formulas and all $n \in \omega$,

- (i) $\neg \forall \Delta = \exists \neg \Delta$; $\neg \exists \Delta = \forall \neg \Delta$;
- (ii) $\exists_n = \neg \forall_n$ and $\forall_{n+1} = \forall \exists_n$;

- (iii) $\exists_n$ and $\forall_n$ are closed under $\wedge$ and $\vee$;
- (iv) $\text{Bool}(\exists_n) = \text{Bool}(\forall_n) \subseteq \forall_{n+1} \cap \exists_{n+1}$.

Proof. Part (i) is immediate, part (ii) follows by an easy induction, part (iii) then follows by the same techniques used for Lemma 2.3.23, and (iv) follows by induction on $\text{Bool}(\exists_n)$. $\square$

Examples 2.4.20

- (i) The $L_=_$ -theory $\text{Th}(\{\exists^{\geq n} : n > 0\})$ whose models are exactly the infinite sets is $\exists_1$ -axiomatizable, because the sentences $\exists^{\geq n}$ are existential.
- (ii) The $L_<$ -theory of strict partial orderings is $\forall_1$ -axiomatizable because the sentences θ_{irrefl} , θ_{trans} (Examples 2.2.12) are universal sentences; similarly for the theory of strict linear orderings. However, the theory of dense linear orderings appears to be only $\forall_2$ -axiomatizable, because the density axiom θ_{dense} is universal-existential.
- (iii) The question of axiomatizability for the theory of groups is strongly dependent on the language chosen. The sentences belonging to the set Γ_{GP} (Example 2.2.14) are all universal, so the theory of groups of the form $(G, *, ^{-1}, e)$ is $\forall_1$ -axiomatizable. However, the sentence θ'_{inverse} is an $\forall_2$ -sentence, so the theory of groups of the form $(G, *, e)$ seems to be at best $\forall_2$ -axiomatizable. Furthermore, the sentence θ_{identinv} is an $\exists_3$ -sentence, but at least apparently not $\forall_2$ or $\exists_2$. However, in this case appearances are deceiving; it is not hard to show that the following $\forall_2$ sentences are also an axiomatization of group theory in this language:

$$\begin{aligned} & \exists z (z * z \doteq z) \\ & \forall x \forall z (z * z \doteq z \rightarrow x * z \doteq x \wedge z * x \doteq x \\ & \qquad \qquad \qquad \wedge \exists y [x * y \doteq z \wedge y * x \doteq z]). \end{aligned}$$

- (iv) The theory of rings with 1 considered as structures of the form $(R, +, -, \times, 0, 1)$ is $\forall_1$ -axiomatizable, but the theory of fields is (at least apparently) only $\forall_2$ -axiomatizable.

Generalizing Definitions 2.3.24 and 2.3.62 and Propositions 2.3.25 and 2.3.63,

Definition 2.4.21 For any theory T ,

- (i) T is **upward persistent** iff whenever $\mathfrak{A} \subseteq \mathfrak{B}$ and $\mathfrak{A}$ is a model of T , then also $\mathfrak{B}$ is a model of T ;
- (ii) T is **downward persistent** iff whenever $\mathfrak{A} \subseteq \mathfrak{B}$ and $\mathfrak{B}$ is a model of T , then also $\mathfrak{A}$ is a model of T ;
- (iii) T is **union persistent** iff for any chain $\langle \mathfrak{A}_i : i \in \omega \rangle$, if for all $i \in \omega$, $\mathfrak{A}_i$ is a model of T , then also $\bigcup \langle \mathfrak{A}_i : i \in \omega \rangle$ is a model of T .

Corollary 2.4.22 *For any theory T , if T is $\exists_1$ - ($\forall_1$)-axiomatizable, then T is upward (downward) persistent. If T is $\forall_2$ -axiomatizable, then T is union persistent. $\square$*

Examples 2.4.23

- (i) The theories of strict partial and linear orderings are downward persistent, but the theory T_{DLO} of dense linear orderings is easily seen not to be either upward or downward persistent. It follows that this theory is neither $\forall_1$ - or $\exists_1$ -axiomatizable. However, T_{DLO} is $\forall_2$ -axiomatizable, hence is union persistent (which is also easy to see directly). It is natural to wonder whether or not T_{DLO} is $\exists_2$ -axiomatizable. Let θ_{DLO} be as in Examples 2.2.12(iii). It follows from Example 2.3.57(iii) that θ_{DLO} is not an $\exists_2$ sentence, since its negation is not union persistent. However, this alone does not show that T_{DLO} is not $\exists_2$ -axiomatizable, since it might have a completely different set of $\exists_2$ axioms. We shall conclude from the Countable Compactness Theorem in [Section 3.1](#) that this is not the case.
- (ii) Similarly, the theory of groups in the language with both inverse and identity is downward persistent, but this is no longer true in either of the other two languages for groups as we observed in Examples 2.3.14. Hence in these languages, this theory is neither $\forall_1$ nor $\exists_1$ -axiomatizable. The theory of fields is not $\forall_1$ -axiomatizable, since a substructure of a field is a ring with 1 (in fact an integral domain) but is not necessarily a field. However, the theory of fields is $\forall_2$ -axiomatizable and therefore union-persistent.

In [Section 1.5](#) we showed that a finitely axiomatizable propositional theory is decidable. Although this is no longer true for first-order logic, one of the most important consequences of the Compactness and Completeness Theorems of [Sections 3.1–3.2](#) below, parallel to Theorem 1.5.18, is the following, which for now we state without proof.

Enumerability Theorem 2.4.24 *For any effective language L and any L -theory T ,*

- (i) *if T is decidable axiomatizable, then T is effectively enumerable;*
- (ii) *if T is decidable axiomatizable and complete, then T is decidable.*

It is an eminently reasonable requirement that a working mathematical theory be decidable axiomatizable. If one is deriving consequences of a set of axioms, it seems essential to be able to determine, given a statement, whether or not it is one of the axioms!

However for a mathematical theory to be decidable is in a sense evidence that it is trivial. Rather than exercising all the ingenuity and skill of her craft

to determine whether or not some assertion ϕ is a theorem of the theory, a mathematician working in a decidable theory may, in principle, simply apply mechanically (perhaps with the aid of a computer) the decision algorithm to ϕ and read off the results. Even if, because of limitations of time and space, this is not a practical procedure, still the possibility takes some of the luster off the enterprise. For example, many of the famous open questions of number theory can be expressed as sentences in the language L_Ω and would be settled by knowing whether the sentence is true or false in Ω .

It is, therefore, something of an affirmation of the inherent value and interest of mathematics that most of the important theories are undecidable. We shall establish many of these results in [Chapters 4–5](#); here we consider some of the decidable theories.

We give first a simple but powerful tool for demonstrating that a theory is complete and hence, if decidable axiomatizable, also decidable.

Theorem 2.4.25 (Countable Vaught Test) *For any countable consistent theory T , if T has no finite models and any two denumerable models of T are isomorphic (or even elementarily equivalent), then T is complete.*

Proof. Suppose that T satisfies the hypotheses and let $\mathfrak{A}$ and $\mathfrak{B}$ be any two models of T . By the Countable Downward Löwenheim-Skolem Theorem, there exist countable elementary substructures $\mathfrak{A}' \prec \mathfrak{A}$ and $\mathfrak{B}' \prec \mathfrak{B}$. $\mathfrak{A}'$ and $\mathfrak{B}'$ are also models of T , so both are infinite, hence denumerable, and thus $\mathfrak{A}' \simeq \mathfrak{B}'$ (or at least $\mathfrak{A}' \equiv \mathfrak{B}'$). Thus

$$\mathfrak{A} \equiv \mathfrak{A}' \equiv \mathfrak{B}' \equiv \mathfrak{B},$$

and we have shown that any two models of T are elementarily equivalent. It follows by [Corollary 2.4.16](#) that T is complete. $\square$

Another important technique for analyzing a theory T to extract information about completeness and decidability is called **quantifier elimination**.

Definition 2.4.26 For any L -theory T and any set Δ of L -formulas, T admits **quantifier elimination to Δ** or simply **Δ -elimination** iff for each L -formula ϕ there exists a formula $\phi^* \in \Delta$ such that

$$T \models \phi \leftrightarrow \phi^* \quad \text{and} \quad \text{Fv}(\phi^*) \subseteq \text{Fv}(\phi).$$

The terminology arises from the special but important case where Δ is the set **QF** of quantifier-free formulas. The idea of a formula containing quantifiers being equivalent in a theory to a quantifier-free formula is familiar from ordinary algebra. For example, in the (informal) theory of equations, over the real numbers $\mathbb{R}$ we have

$$\begin{aligned} \exists x [ax^2 + bx + c = 0] &\iff [a \neq 0 \wedge b^2 - 4ac \geq 0] \\ &\vee [a = 0 \wedge (b \neq 0 \vee c = 0)], \end{aligned}$$

while over the complex numbers $\mathbb{C}$,

$$\exists x [ax^2 + bx + c = 0] \iff (a \neq 0 \vee b \neq 0 \vee c = 0).$$

Of course, these facts can easily be formulated in the formal language of rings and fields. We shall also be interested in cases when not quite *all* quantifiers can be eliminated — that is, Δ is QF together with some other formulas. The operator $\text{Bool}(\cdot)$ was defined in 2.2.31.

Definition 2.4.27 A set Δ of formulas is **regular Boolean** iff $\Delta = \text{Bool}(\Gamma)$ for some set Γ of formulas such that

- (i) all atomic formulas belong to Γ , and
- (ii) Γ is closed under substitution of variables for free variables.

For example, the set QF of quantifier-free formulas is regular Boolean as is $\text{Bool}(\text{AtFm}_L \cup \Gamma)$ for any set Γ of L -sentences. Of course, the set Fm_L of all formulas is also regular Boolean.

Proposition 2.4.28 For any theory T and regular Boolean set $\Delta = \text{Bool}(\Gamma)$, if T admits Δ -elimination via the mapping $\phi \mapsto \phi^*$, then

- (i) if for every sentence $\psi \in \Gamma$, either $T \models \psi$ or $T \models \neg \psi$, then T is complete;
- (ii) if the mapping $\phi \mapsto \phi^*$ is effectively calculable and $T \cap \Delta$ is decidable, then T is decidable; hence if T is complete and $T \cap \Gamma$ is decidable, then T is decidable.

Proof. The hypothesis of (i) asserts that T is complete for sentences $\psi \in \Gamma$. But then it follows easily by induction that T is also complete for all sentences $\psi \in \Delta$, so T is complete for all sentences ϕ^* and hence for every sentence. Here we use the condition that $\text{Fv}(\phi^*) \subseteq \text{Fv}(\phi)$ so if ϕ is a sentence, so is ϕ^* . For the first part of (ii), to decide if a sentence $\phi \in T$, we need only compute the corresponding ϕ^* and apply the decision algorithm for $T \cap \Delta$ to ϕ^* . The second part follows from the fact that if T is complete, then $T \models \neg \phi$ iff $T \not\models \phi$ and $T \models \phi \vee \psi$ iff $T \models \phi$ or $T \models \psi$. $\square$

The terminology is further explained by the following lemma, which roughly says that we may eliminate one quantifier at a time.

Definition 2.4.29 A **simply Γ -existential** formula is one of the form $\exists x \psi$, where ψ is of the form

$$\theta_0 \wedge \cdots \wedge \theta_{k-1} \wedge \neg \chi_0 \wedge \cdots \wedge \neg \chi_{l-1}$$

with all $\theta_i, \chi_j \in \Gamma \cup \{\top\}$.

Quantifier Elimination Lemma 2.4.30 *For any theory T and any regular Boolean set $\Delta = \text{Bool}(\Gamma)$, if for each simply Γ -existential formula $\exists x \psi$ there exists $\psi^\circ \in \Delta$ such that*

$$T \models \exists x \psi \leftrightarrow \psi^\circ \quad \text{and} \quad \text{Fv}(\psi^\circ) \subseteq \text{Fv}(\exists x \psi),$$

then T admits Δ -elimination.

Proof. Under the hypothesis we prove by formula induction that for all ϕ the required ϕ^* of Definition 2.4.26 exists. For ϕ atomic we may take simply $\phi^* := \phi$ (since $\phi \in \Gamma$). The induction steps for the connectives are trivial; we set $(\neg \psi)^* := \neg \psi^*$ and $(\psi \vee \theta)^* := \psi^* \vee \theta^*$. Suppose now that ϕ is $\exists x \psi$ and as induction hypothesis there exists $\psi^* \in \Delta$ such that $T \models \psi \leftrightarrow \psi^*$. Since ψ^* is a Boolean combination of members of $\Gamma \cup \{\top\}$, by Remark 2.2.32(v) it has a logically equivalent disjunctive normal form; that is, there exist formulas $\psi_0, \dots, \psi_m$ such that $\models \psi^* \leftrightarrow \psi_0 \vee \dots \vee \psi_m$ and for each $i \leq m$

$$\psi_i = \theta_0^i \wedge \dots \wedge \theta_{k_i-1}^i \wedge \neg \chi_0^i \wedge \dots \wedge \neg \chi_{l_i-1}^i,$$

with all $\theta_j^i, \chi_j^i \in \Gamma \cup \{\top\}$. Now $T \models \phi \leftrightarrow \exists x \psi^*$ and

$$\exists x \psi^* \models \exists x (\psi_0 \vee \dots \vee \psi_m) \models \exists x \psi_0 \vee \dots \vee \exists x \psi_m.$$

Each formula $\exists x \psi_i$ is simply Γ -existential, and therefore by hypothesis there exist $\psi_i^\circ \in \Delta$ such that $T \models \exists x \psi_i \leftrightarrow \psi_i^\circ$. Hence

$$T \models \phi \leftrightarrow \psi_0^\circ \wedge \dots \wedge \psi_m^\circ.$$

The formula on the right-hand side belongs to Δ and is thus an appropriate value for ϕ^* . In all cases it is immediate that the condition $\text{Fv}(\phi^*) \subseteq \text{Fv}(\phi)$. $\square$

The language of equality

Let us start with the simplest of all first-order languages, $L_=$, the language with no non-logical symbols. By Corollary 2.3.9,

Proposition 2.4.31 *For any $n \geq 1$ and any n -element set A , $\text{Th}((A)) = \text{Th}(\exists^=n)$; in particular, $\text{Th}(\exists^=n)$ is a complete theory.*

Proof. The inclusion $(\supseteq)$ follows from the fact that $\models_{(A)} \exists^=n$. Conversely, for any $\phi \in \text{Th}((A))$ — that is, any ϕ such that $\models_{(A)} \phi$ — we have $\models \exists^=n \rightarrow \phi$, because for any model (B) of $\exists^=n$, $(A) \simeq (B)$ and thus also $\models_{(B)} \phi$. $\square$

The obvious question now is: what are the possibilities for $\text{Th}((A))$ when A is infinite? If A and B have the same cardinality, then $(A) \simeq (B)$ so $\text{Th}((A)) = \text{Th}((B))$. In fact, the same is true for *any* two infinite sets. Note that any infinite (A) is clearly a model of $\{\exists^{\geq n} : n \geq 1\}$.

Theorem 2.4.32 *The theory $\text{Th}(\{\exists^{\geq n} : n \geq 1\})$ is complete and coincides with $\text{Th}((A))$ for any infinite set A . In particular, $(A) \equiv (B)$ for any infinite sets A and B .*

Proof. Let $T = \text{Th}(\{\exists^{\geq n} : n \geq 1\})$. Clearly T has no finite models and we have just observed that any two denumerable models of T are isomorphic. Hence by the Countable Vaught Test 2.4.25, T is complete. The other conclusions follow from Corollary 2.4.16. $\square$

Corollary 2.4.33 *An $L_{=}$ -theory T is complete iff it is one of $\text{Th}(\exists^=n)$ for some $(n > 0)$ or $\text{Th}(\{\exists^{\geq n} : n \geq 1\})$.*

Proof. If a complete $L_{=}$ -theory T has a finite model (A) , then any two models have the same finite cardinality n and $T = \text{Th}(\exists^=n)$. Otherwise, T has only infinite models and $T = \text{Th}(\{\exists^{\geq n} : n \geq 1\})$. $\square$

If we use the Enumerability Theorem 2.4.24, we may now conclude that each of these theories, being clearly decidable axiomatizable, is decidable. We shall get another proof of this result and more by the method of quantifier elimination below.

The results above suggest that the sentences $\exists^{\geq n}$ and $\exists^=n$ in some sense exhaust the expressive power of $L_{=}$. Of course, that is not literally true, since there exist sentences such as

$$(\exists^{\geq 5} \wedge \exists^{\leq 12}) \vee \exists^=20 \vee \exists^{\geq 100},$$

which is true in a structure (A) just in case A has 5, 6, ..., 12, 20, or at least 100 elements. However, we shall show below that every $L_{=}$ -sentence is logically equivalent to a Boolean combination of the sentences $\exists^{\geq n}$. This will follow from the more general quantifier-elimination result for formulas. Let

$$\Gamma_{=} := \text{AtFm}_{L_{=}} \cup \{\exists^{\geq n} : n \geq 1\} \quad \text{and} \quad \Delta_{=} := \text{Bool}(\Gamma_{=}).$$

We shall prove that for every $L_{=}$ -formula ϕ , there exists a formula $\phi^* \in \Delta_{=}$ with the same free variables such that $\models \phi \leftrightarrow \phi^*$ — in other words, the minimal $L_{=}$ -theory $\text{Th}(\emptyset)$ consisting only of the logically valid sentences admits $\Delta_{=}$ -elimination.

Note that the special sentence $\top$ is logically equivalent to $\exists^{\geq 1}$. We consider first the special case which is the core of the argument.

Lemma 2.4.34 *For any $l > 0$ and any variables $x, z_0, \dots, z_{l-1}$, if*

$$\chi \text{ is the formula } \neg x \doteq z_0 \wedge \dots \wedge \neg x \doteq z_{l-1},$$

then there exists a formula $\chi^{\circ} \in \Delta_{=}$ such that

$$\models \exists x \chi \leftrightarrow \chi^{\circ} \quad \text{and} \quad \text{Fv}(\chi^{\circ}) \subseteq \{z_0, \dots, z_{l-1}\}.$$

Proof. To motivate the proof, consider first the special case $l = 2$. It is easy to see that

$$\models \exists x [\neg x \dot{=} z_0 \wedge \neg x \dot{=} z_1] \leftrightarrow \exists^{\geq 2} \wedge (\neg z_0 \dot{=} z_1 \rightarrow \exists^{> 2}).$$

Now, in general, if x is one of the z_i , let χ° be $\mathbf{F}$. Otherwise, for each set $I \subseteq \{0, 1, \dots, l-1\}$ with, say, m_I elements, if $m_I = 0$ or 1 , let θ_I be $\exists^{\geq 2}$; otherwise let

$$\theta_I \text{ be the formula } \left(\bigwedge_{\substack{i, j \in I \\ i \neq j}} [\neg z_i \dot{=} z_j] \right) \rightarrow \exists^{> m_I},$$

and take χ° to be the formula $\bigwedge_{I \subseteq \{0, 1, \dots, l-1\}} \theta_I$. Then for any structure (A) with, say, m_A elements and any (A) -assignment α ,

$$\begin{aligned} \models_{(A)} \exists x (\neg x \dot{=} z_0 \wedge \dots \wedge \neg x \dot{=} z_{l-1})[\alpha] \\ \iff m_A > \text{Card}(\{\alpha(z_i) : i < l\}) \\ \iff \text{for any } I \subseteq \{0, 1, \dots, l-1\}, \\ \quad \text{if all } \alpha(z_i) \text{ for } i \in I \text{ are distinct, then } m_A > m_I \\ \iff \models_{(A)} \chi^\circ[\alpha]. \quad \square \end{aligned}$$

Lemma 2.4.35 *For any simply Γ -existential formula $\exists x \psi$ there exists $\psi^\circ \in \Delta_-$ such that*

$$T \models \exists x \psi \leftrightarrow \psi^\circ \quad \text{and} \quad \mathbf{Fv}(\psi^\circ) \subseteq \mathbf{Fv}(\exists x \psi).$$

Proof. Let ψ be a formula as described, and let θ denote the conjunction of all the conjuncts of ψ in which x does not occur free together with $\top$ (so that θ is never vacuous). Then ψ is logically equivalent to a formula of the form

$$x \dot{=} y_0 \wedge \dots \wedge x \dot{=} y_{k-1} \wedge \neg x \dot{=} z_0 \wedge \dots \wedge \neg x \dot{=} z_{l-1} \wedge \theta,$$

and we may assume that x is distinct from each of the y_i . If $k > 0$, it suffices by Exercise 2.2.43 to take ψ° to be the formula

$$y_0 \dot{=} y_0 \wedge \dots \wedge y_0 \dot{=} y_{k-1} \wedge \neg y_0 \dot{=} z_0 \wedge \dots \wedge \neg y_0 \dot{=} z_{l-1} \wedge \theta.$$

Otherwise, if $k = 0$, then if $l = 0$ take $\psi^\circ = \theta$, otherwise let χ be $\neg x \dot{=} z_0 \wedge \dots \wedge \neg x \dot{=} z_{l-1}$, so

$$\models \exists x \psi \leftrightarrow \exists x \chi \wedge \theta,$$

and we take $\psi^\circ = \chi^\circ \wedge \theta$, with χ° as in the preceding lemma. $\square$

Theorem 2.4.36 *For every L -formula ϕ there exists $\phi^* \in \Delta_=_$ such that*

$$\models \phi \leftrightarrow \phi^* \quad \text{and} \quad \text{Fv}(\phi^*) \subseteq \text{Fv}(\phi);$$

that is, $\text{Th}(\emptyset)$ admits $\Delta_=-$ elimination.

Proof. Immediate from the Quantifier Elimination Lemma 2.4.30 and the preceding lemma. $\square$

Corollary 2.4.37 *For any L -sentence ϕ , there exists a sentence $\phi^* \in \text{Bool}(\{\exists^{\geq n} : n \geq 1\})$ such that $\models \phi \leftrightarrow \phi^*$.*

Proof. By the second clause of the theorem, if ϕ is a sentence, then so is ϕ^* , and hence it must not contain any atomic formulas, since these all contain free variables which would remain free in ϕ^* . $\square$

This provides a second proof that the theory $T = \text{Th}(\{\exists^{\geq n} : n \geq 1\})$ is complete: clearly also $T \models \phi \leftrightarrow \phi^*$ for each sentence ϕ and $T \models \psi$ for each sentence $\exists^{\geq n}$, so T is complete by Proposition 2.4.28(i). Furthermore

Corollary 2.4.38 *All of the theories $\text{Th}(\{\exists^{\geq n} : n \geq 1\})$ and $\text{Th}(\exists^=n)$, for each $n \geq 1$ are decidable.*

Proof. Immediate from Proposition 2.4.28(ii). $\square$

To get similar results for the incomplete theory $\text{Th}(\emptyset)$, we analyze further what sentences in $\Delta_=_$ can “say”. Let $\exists^{\in \emptyset}$ be F , and for any finite non-empty set $N \subseteq \omega \sim \{0\}$, let

$$\exists^{\in N} \quad \text{denote the sentence} \quad \bigvee_{n \in N} \exists^=n.$$

Thus for any set A , $\models_{(A)} \exists^{\in N} \iff \text{Card}(A) \in N$, and

$$\models_{(A)} \neg \exists^{\in N} \iff (A \text{ is finite and } \text{Card}(A) \notin N) \text{ or } A \text{ is infinite.}$$

Note that $\exists^=n$, $\exists^{\leq n}$ etc. all belong to $\text{Bool}(\{\exists^{\geq n} : n \geq 1\})$.

Theorem 2.4.39 *For every L -sentence ϕ , there exists a finite set $N_\phi \subseteq \omega \sim \{0\}$ such that either*

$$\models \phi \leftrightarrow \exists^{\in N_\phi} \quad \text{or} \quad \models \phi \leftrightarrow \neg \exists^{\in N_\phi}.$$

Proof. By Theorem 2.4.37 it suffices to establish the conclusion for each $\phi \in \text{Bool}(\{\exists^{\geq n} : n \geq 1\})$. Clearly $\models \exists^{\geq n} \leftrightarrow \neg \exists^{\in \{1, \dots, n-1\}}$, and we proceed by induction. If ϕ is $\neg \psi$, then we may take $N_\phi = N_\psi$. If ϕ is $\psi \vee \theta$,

there are four cases depending on which of the alternatives of the induction hypothesis hold:

- (i) if $\models \psi \leftrightarrow \exists^{\in N_\psi}$ and $\models \theta \leftrightarrow \exists^{\in N_\theta}$, then $\models \phi \leftrightarrow \exists^{\in N_\psi \cup N_\theta}$;
- (ii) if $\models \psi \leftrightarrow \exists^{\in N_\psi}$ and $\models \theta \leftrightarrow \neg \exists^{\in N_\theta}$, then $\models \phi \leftrightarrow \neg \exists^{\in N_\theta \sim N_\psi}$;
- (iii) if $\models \psi \leftrightarrow \neg \exists^{\in N_\psi}$ and $\models \theta \leftrightarrow \exists^{\in N_\theta}$, then $\models \phi \leftrightarrow \neg \exists^{\in N_\psi \sim N_\theta}$;
- (iv) if $\models \psi \leftrightarrow \neg \exists^{\in N_\psi}$ and $\models \theta \leftrightarrow \neg \exists^{\in N_\theta}$, then $\models \phi \leftrightarrow \neg \exists^{\in N_\psi \cap N_\theta}$.

Thus in all cases there is an appropriate choice for N_ϕ . $\square$

To apply Proposition 2.4.28(ii), we need first to observe

Proposition 2.4.40 *There is an effective procedure which, given an L_- -sentence ϕ , calculates a finite set $N_\phi \subseteq \omega \sim \{0\}$ and an index $i_\phi = 0$ or 1 such that either*

$$i_\phi = 0 \text{ and } \models \phi \leftrightarrow \exists^{\in N_\phi} \quad \text{or} \quad i_\phi = 1 \text{ and } \models \phi \leftrightarrow \neg \exists^{\in N_\phi}.$$

Proof. All of the steps of the preceding proof may be read as providing instructions for an algorithm which computes the resulting formula. $\square$

Theorem 2.4.41 *The L_- -theory $\text{Th}(\emptyset)$, the set of logical validities of L_- , is decidable.*

Proof. We apply Proposition 2.4.28(ii); among the sentences $\exists^{\in N}$ and $\neg \exists^{\in N}$, the only one which is logically valid is $\neg \exists^{\in \emptyset}$. $\square$

We shall see in [Chapters 4–5](#) that this is a special property of L_- (and a few other very simple languages). In particular, any language which has a binary relation symbol has an undecidable set of logical validities ([Section 5.5](#)).

We recall from Definition 2.3.37 the notion of a relation being definable over a structure. Not much is definable over an L_- -structure.

Proposition 2.4.42 *For any set A , a subset (1-ary relation) R of A is definable without parameters over the L_- -structure (A) iff $R = \emptyset$ or $R = A$.*

Proof. Let ϕ be a formula with the single free variable x . Since the corresponding ϕ^* constructed above can have at most x free, it must be a Boolean combination of $x \doteq x$ and the formulas $\exists^{\geq n}$. It is straightforward to show by induction that every such formula defines either $\emptyset$ or A . $\square$

This thorough analysis of expressability in L_- yields another interesting by-product. Consider first the notions of (elementary) substructure for L_- -structures. Clearly, for any sets A and B , $(A) \subseteq (B) \iff A \subseteq B$. If A and B are of different finite cardinalities or A is finite and B is infinite, then $(A) \not\subseteq (B)$, so $(A) \not\preceq (B)$. However, we have

Proposition 2.4.43 *For any infinite sets A and B , if $A \subseteq B$, then $(A) \prec (B)$.*

Proof. Left as an exercise. $\square$

Dense linear orderings

We turn next to the language $L_{<}$, which has one binary relation symbol $<$ and is the natural language in which to “talk about” orderings. Recall from Examples 2.2.12 (iii) that for an $L_{<}$ -structure $\mathfrak{A} = (A, <_A)$,

$$\models_{\mathfrak{A}} \theta_{\text{DLO}} \iff \mathfrak{A} \text{ is a dense linear ordering,}$$

and as above, let T_{DLO} denote the theory generated by this sentence. Some of the best-known models of T_{DLO} are the standard orderings on various sets of rational and real numbers. From these it is immediate that T_{DLO} is not a complete theory, since it does not specify whether or not a model has a least or greatest element — that is, it does not determine which of

$$\theta_{\text{least}} := \exists x \forall y \neg y < x \quad \text{or} \quad \theta_{\text{greatest}} := \exists x \forall y \neg x < y$$

or their negations are true. For example, the model $(\mathbb{Q}, <_{\mathbb{Q}})$ satisfies neither of these, while $((0, 1], <_{(0, 1]})$ satisfies the second but not the first. We shall see that these are essentially the only undetermined sentences, and any two models of T_{DLO} which agree on them are elementarily equivalent; in particular this entails that $(\mathbb{Q}, <_{\mathbb{Q}}) \equiv (\mathbb{R}, <_{\mathbb{R}})$.

This may be proved by either of the two methods used above for $L_{=}$: use of Vaught’s Test or quantifier elimination. We shall give the first of these and leave the second as an exercise.

Definition 2.4.44

- | | |
|---|---|
| (i) $\theta_{[\text{DLO}]} := \theta_{\text{DLO}} \wedge \theta_{\text{least}} \wedge \theta_{\text{greatest}};$ | $T_{[\text{DLO}]} = \text{Th}(\theta_{[\text{DLO}]})$. |
| (ii) $\theta_{[\text{DLO}]} := \theta_{\text{DLO}} \wedge \theta_{\text{least}} \wedge \neg \theta_{\text{greatest}};$ | $T_{[\text{DLO}]} = \text{Th}(\theta_{[\text{DLO}]})$. |
| (iii) $\theta_{(\text{DLO})} := \theta_{\text{DLO}} \wedge \neg \theta_{\text{least}} \wedge \theta_{\text{greatest}};$ | $T_{(\text{DLO})} = \text{Th}(\theta_{(\text{DLO})})$. |
| (iv) $\theta_{(\text{DLO})} := \theta_{\text{DLO}} \wedge \neg \theta_{\text{least}} \wedge \neg \theta_{\text{greatest}};$ | $T_{(\text{DLO})} = \text{Th}(\theta_{(\text{DLO})})$. |

Both of the structures $(\mathbb{Q}, <_{\mathbb{Q}})$ and $(\mathbb{R}, <_{\mathbb{R}})$ are models of $T_{(\text{DLO})}$, and we shall prove that they are elementarily equivalent by proving that $T_{(\text{DLO})}$ is complete. The key idea is the following classical theorem of Cantor.

Cantor's Theorem 2.4.45 *Any two denumerable models of any one of the theories $T_{[\text{DLO}]}$, $T_{[\text{DLO}]}$, $T_{(\text{DLO})}$ or $T_{(\text{DLO})}$ are isomorphic.*

Proof. Let first $\mathfrak{A} = (A, <_A)$ and $\mathfrak{B} = (B, <_B)$ be two denumerable models of $T_{(\text{DLO})}$ and fix one-one enumerations $\langle a_m : m \in \omega \rangle$ and $\langle b_n : n \in \omega \rangle$ of A and B , respectively. We define a function $\eta : A \rightarrow B$ by $\eta(a_m) = b_{F(m)}$, where

$$F(m) = \text{least } n \text{ such that for all } i < m, (a_i <_A a_m \iff b_{F(i)} <_B b_n).$$

Clearly $F(0) = 0$, and we prove by induction on m that $F(m)$ is well-defined and for all $i, j \leq m$, $a_i <_A a_j \iff b_{F(i)} <_B b_{F(j)}$. If this is true for m then $F(m+1)$ exists by the hypothesis that $\mathfrak{B}$ is a dense linear ordering without endpoints, and the new instances of the induction hypothesis follow directly from the definition of F and the connectedness of the orderings: for $i \leq m$,

$$\begin{aligned} a_{m+1} <_A a_i &\iff a_i \not<_A a_{m+1} \\ &\iff b_{F(i)} \not<_B b_{F(m+1)} \iff b_{F(m+1)} <_B b_{F(i)}. \end{aligned}$$

Hence η is a homomorphism and is clearly injective so is an embedding. Finally, to see that η is surjective, we prove by induction on n that $n \in \text{Im}(F)$. Assume that all $j < n$ are in $\text{Im}(F)$ and that $F(i_j) = j$. By the density of $\mathfrak{A}$, there exists a least $m \geq \max\{i_j : j < n\}$ such that for all $j < n$, $a_{i_j} <_A a_m \iff b_j <_B b_n$. Then either already $F(i) = n$ for some $i < m$ or $F(m) = n$. Thus η is an isomorphism.

If $\mathfrak{A}$ and $\mathfrak{B}$ are models for one of the other theories, we define η by a minor modification. For example, for $T_{[\text{DLO}]}$, if a_{m_0} and b_{n_0} are respectively the least elements of A and B , we set $F(m_0) = n_0$ and otherwise define F as before. $\square$

Theorem 2.4.46 *The theories $T_{[\text{DLO}]}$, $T_{[\text{DLO}]}$, $T_{(\text{DLO})}$ and $T_{(\text{DLO})}$ are complete. In particular, $T_{(\text{DLO})} = \text{Th}((\mathbb{Q}, <_{\mathbb{Q}})) = \text{Th}((\mathbb{R}, <_{\mathbb{R}}))$ so $(\mathbb{Q}, <_{\mathbb{Q}}) \equiv (\mathbb{R}, <_{\mathbb{R}})$.*

Proof. Each of these theories is easily seen to satisfy the first hypothesis of the Countable Vaught Test 2.4.25 — any finite linear ordering is discrete and not dense. Therefore, by Cantor's Theorem, each of the theories is complete. $\square$

Thus we may view this result as providing an axiomatization of the theory $\text{Th}((\mathbb{Q}, <_{\mathbb{Q}}))$. Of course, T_{DLO} is not complete, but we have

Corollary 2.4.47 *The complete extensions of T_{DLO} are exactly the theories $T_{[\text{DLO}]}$, $T_{[\text{DLO})}$, $T_{(\text{DLO}]}$ and $T_{(\text{DLO})}$.*

Proof. Any model of T_{DLO} is a model of exactly one of these four theories.

□

An exercise below uses this to conclude that every $L_{<}$ -sentence is equivalent in T_{DLO} to a Boolean combination of the two sentences θ_{least} and θ_{greatest} . Further information about T_{DLO} and its complete extensions may be obtained by carrying out a quantifier elimination; we sketch the basic ideas here and leave the details to the exercises. Let $\Delta_{\text{DLO}} := \text{Bool}(\Gamma_{\text{DLO}})$ where

$$\Gamma_{\text{DLO}} := \text{AtFm}_{L_{<}} \cup \{\theta_{\text{least}}, \theta_{\text{greatest}}, \forall y \neg (y \dot{<} x), \forall y \neg (x \dot{<} y)\}.$$

We aim to prove that T_{DLO} admits Δ_{DLO} -elimination — that is, for every $L_{<}$ -formula ϕ , there exists $\phi^* \in \Delta_{\text{DLO}}$ such that $T_{\text{DLO}} \models \phi \leftrightarrow \phi^*$ and $\text{Fv}(\phi^*) \subseteq \text{Fv}(\phi)$. It greatly simplifies the proof to note that because of the equivalences

$$\begin{aligned} T_{\text{DLO}} \models \neg (x \dot{<} y) &\leftrightarrow x \dot{=} y \vee y \dot{<} x \quad \text{and} \\ T_{\text{DLO}} \models \neg (x \dot{=} y) &\leftrightarrow x \dot{<} y \vee y \dot{<} x, \end{aligned}$$

every conjunction of atomic formulas and their negations is T_{DLO} -equivalent to a disjunction of conjunctions of atomic formulas without negations. We leave the details to the exercises. The main additional consequences are

Theorem 2.4.48

- (i) *For any models $\mathfrak{A}$ and $\mathfrak{B}$ of $T_{(\text{DLO})}$, if $\mathfrak{A} \subseteq \mathfrak{B}$, then $\mathfrak{A} \prec \mathfrak{B}$.*
- (ii) *The theories T_{DLO} , $T_{[\text{DLO}]}$, $T_{[\text{DLO})}$, $T_{(\text{DLO}]}$ and $T_{(\text{DLO})}$ are decidable.* □

Of course, again the decidability of the four complete theories may also be derived from the Enumerability Theorem 2.4.24, since the theories are finitely, hence decidable axiomatizable.

Exercises

Exercise 2.4.49 The goal of this exercise is a complete catalogue of all (complete or otherwise) $L_{=}$ -theories. For any set A , let m_A be the cardinality of A , if A is finite and ∞ , otherwise. For any $L_{=}$ -theory T , set

$$M_T = \{m_A : (A) \text{ is a model of } T\} \subseteq \{1, 2, \dots, n, \dots, \infty\}.$$

Show that

- (i) for any $L_{=}$ -theories T and U , $T = U \iff M_T = M_U$;

- (ii) for any finite $M \subseteq \{1, 2, \dots, n, \dots, \infty\}$, there is a theory T such that $M_T = M$;
- (iii) for any infinite $M \subseteq \{1, 2, \dots, n, \dots, \infty\}$, there is a theory T such that $M_T = M$ iff $\infty \in M$.

Hint: (for (iii)) For each sentence ϕ , let N_ϕ be as above and set

$$M_\phi = \begin{cases} N_\phi, & \text{if } \models \phi \leftrightarrow \exists \in N_\phi; \\ \omega \sim \{0\} \sim N_\phi, & \text{if } \models \phi \leftrightarrow \neg \exists \in N_\phi. \end{cases}$$

Then $M_T = \bigcap \{M_\phi : \phi \in T\}$.

Note that it follows from (iii) that there is *no* theory T whose models are exactly the finite $L_=-$ structures. We shall see that this is true for other languages as well.

Exercise 2.4.50 Prove Proposition 2.4.43.

Exercise 2.4.51 Characterize the binary relations definable over an $L_=-$ structure (A) without parameters.

Exercise 2.4.52 Let L be $L_=-$ enriched by a denumerable sequence of distinct constant symbols $\{c_n : n \in \omega\}$. Let T be the L -theory generated by the axioms $\{\neg c_m \doteq c_n : m \neq n\}$. Show that T is complete and decidable, although it does not satisfy the hypotheses of Vaught's Test.

Exercise 2.4.53 Let $\Omega_0 = (\omega, \text{Sc}, 0)$; the corresponding language L_0 has one unary function symbol $\dot{S}$ and one constant symbol 0 . An L_0 -term is either $\dot{S} \cdots \dot{S}x$ for some variable x or $\dot{S} \cdots \dot{S}0$, which we write as $\dot{S}^{(m)}x$ and $\dot{m}$, respectively, where m is the number of symbols $\dot{S}$ which occur. Let $\Gamma_0 := \text{AtFm}_{L_0}$ and $\Delta_0 := \text{Bool}(\Gamma_0)$.

- (i) Prove that $\text{Th}(\Omega_0)$ admits Δ_0 -elimination — that is, for every L_0 -formula ϕ there exists a formula $\phi^* \in \Delta_0$ such that $\models_{\Omega_0} \phi \leftrightarrow \phi^*$ and $\text{Fv}(\phi^*) \subseteq \text{Fv}(\phi)$.
- (ii) Conclude that $\text{Th}(\Omega_0)$ is decidable.
- (iii) Show that $R \subseteq \omega$ is definable over Ω_0 iff R is finite or cofinite.
- (iv) Prove that the normal ordering relation on ω is not definable over Ω_0 .

Hint: For (i) follow a modified form of the pattern of 2.4.34–2.4.36. If ψ is a conjunction of atomic formulas and their negations, then $\exists x \psi$ is of the form

$$\exists x \left[\bigwedge_{i < k} \dot{S}^{(m_i)}x \doteq t_i \wedge \bigwedge_{j < l} \neg \dot{S}^{(n_j)}x \doteq u_j \right] \wedge \theta,$$

with x not free in θ . Show first that we may assume also that x is not free in any of the terms t_i and u_j . Let m be the maximum of all the m_i and n_j and use the fact that Sc is injective to rewrite the displayed formula with the same term $\dot{S}^{(m)}$ on the left side of all equalities. Then analyze the possibilities as in the proof of Lemma 2.4.35. Note that if at least one positive equality $\dot{S}^{(m)}x \doteq t$ occurs, then any other equality $\dot{S}^{(n)}x \doteq u$ is equivalent to $\dot{S}^{(n)}t \doteq \dot{S}^{(m)}u$.

Exercise 2.4.54 (cf. Exercise 1.4.40) Let T be a theory (in any language L) such that T has only finitely many complete extensions $T_0, \dots, T_n$ and for each $i \leq n$, there is a sentence χ_i such that $T_i = \text{Th}(T \cup \{\chi_i\})$.

- (i) Prove that for every sentence ϕ there exists a sentence ϕ^* which is a disjunction of some of the sentences $\chi_0, \dots, \chi_n$ such that $T \models \phi \leftrightarrow \phi^*$.
- (ii) Conclude, without doing the full quantifier elimination, that for every $L_{<}$ -sentence ϕ , there exists a sentence $\phi^* \in \text{Bool}(\{\theta_{\text{least}}, \theta_{\text{greatest}}\})$ such that $T_{\text{DLO}} \models \phi \leftrightarrow \phi^*$.
- (iii) Show that in fact the existence of the χ_i follows from the first hypothesis that T has only finitely many complete extensions.

Exercise 2.4.55 For any two L -structures $\mathfrak{A}$ and $\mathfrak{B}$, any $X \subseteq |\mathfrak{A}|$ and any function $\eta : X \rightarrow |\mathfrak{B}|$, η is called a **partial embedding** iff for every quantifier-free formula ϕ with free variables $x_0, \dots, x_{n-1}$ and every $a_0, \dots, a_{n-1} \in X$,

$$(1) \quad \models_{\mathfrak{A}} \phi[a_0, \dots, a_{n-1}] \iff \models_{\mathfrak{B}} \phi[\eta(a_0), \dots, \eta(a_{n-1})].$$

$\mathfrak{A}$ and $\mathfrak{B}$ are called **partially isomorphic** — in symbols, $\mathfrak{A} \simeq_p \mathfrak{B}$ — iff there exists a non-empty set $\mathcal{E}$ of partial embeddings such that the following hold

$$(\forall \eta \in \mathcal{E}) \forall a \in |\mathfrak{A}| \quad (\exists \zeta \in \mathcal{E}) \quad [\zeta \text{ extends } \eta \text{ and } a \in \text{Dom}(\zeta)],$$

and

$$(\forall \eta \in \mathcal{E}) \forall b \in |\mathfrak{B}| \quad (\exists \zeta \in \mathcal{E}) \quad [\zeta \text{ extends } \eta \text{ and } b \in \text{Im}(\zeta)].$$

These two conditions are called the **forward extension** and **backward extension** properties of $\mathcal{E}$ or together the **back-and-forth** property. Note that $\simeq_p$ is a weakening of $\simeq$, since if $\eta : \mathfrak{A} \simeq \mathfrak{B}$, then $\mathcal{E} = \{\eta\}$ has the back-and-forth property. Prove

- (i) $\mathfrak{A} \simeq_p \mathfrak{B} \implies \mathfrak{A} \equiv \mathfrak{B}$;
- (ii) if $\mathfrak{A}$ and $\mathfrak{B}$ are countable, then $\mathfrak{A} \simeq_p \mathfrak{B} \implies \mathfrak{A} \simeq \mathfrak{B}$;
- (iii) if $\mathfrak{A}$ and $\mathfrak{B}$ are any two (not necessarily countable) models of any one of the theories $T_{[\text{DLO}]}$, $T_{(\text{DLO})}$, $T_{[\text{DLO}]}^<$ or $T_{(\text{DLO})}^<$, then $\mathfrak{A} \simeq_p \mathfrak{B}$.

Note that (ii) and (iii) together give another proof of Cantor's Theorem and (i) and (iii) give another proof of the completeness of these theories.

Hint: For (i), if $\mathcal{E}$ witnesses that $\mathfrak{A} \simeq_p \mathfrak{B}$, show by formula induction that for all $\eta \in \mathcal{E}$, (1) holds for all formulas ϕ . For (ii), construct an increasing chain of partial embeddings $\eta_n \in \mathcal{E}$ which include more and more of A (B) in their domain (image). For (iii), let $\mathcal{E}$ be the set of all finite partial embeddings.

Exercise 2.4.56 The theory of undirected loopless graphs without parallel edges is generated by the following axioms in the language with just one binary relation symbol E (for edge):

$$(i) \quad \forall x \forall y (Exy \rightarrow Eyx);$$

$$(ii) \quad \forall x \neg Exx.$$

The theory of random graphs is generated by these together with the additional infinite family of axioms, one for each $k, l \geq 0$:

$$\forall x_0 \cdots \forall x_k \forall y_0 \cdots \forall y_l \left(\bigwedge_{i \leq k} \bigwedge_{j \leq l} \neg x_i \dot{=} y_j \rightarrow \right. \\ \left. \exists z \left[\bigwedge_{i \leq k} Ex_i z \wedge \bigwedge_{j \leq l} (\neg Ey_j z \wedge \neg y_j \dot{=} z) \right] \right).$$

Use the method of the preceding exercise and the Countable Vaught Test to show that the theory of random graphs is complete.

Exercise 2.4.57 Let L be the language with a unary relation symbol p_n for each $n \in \omega$ and T the L -theory generated by the axioms

$$\phi_{X,Y} : \exists x \left[\bigwedge_{m \in X} p_m x \wedge \bigwedge_{n \in Y} \neg p_n x \right]$$

for all disjoint pairs of finite sets $X, Y \subseteq \omega$. Show that T is complete.

Exercise 2.4.58 Carry out the quantifier elimination procedure for T_{DLO} , give a second proof of the completeness of the theories $T_{[\text{DLO}]}$, $T_{(\text{DLO}]}$, $T_{[\text{DLO})}$ and $T_{(\text{DLO})}$, and prove Theorem 2.4.48.

Exercise 2.4.59 Using the results of the preceding exercise, characterize the sets and binary relations definable over $(\mathbb{Q}, <_{\mathbb{Q}})$.

Exercise 2.4.60 Prove that

- (i) for any L -sentence ϕ , if $\models_{(A)} \phi$ for all finite A , then ϕ is logically valid;
- (ii) if $K = \{ (A) : A \text{ is finite} \}$, then $K \subset \text{Mod}(\text{Th}(K))$ is a proper inclusion;
- (iii) for $L_<$ (i) fails but (ii) still holds.

Exercise 2.4.61 Let L be the language with unary relation symbols p_m and constant symbols $c_{m,n}$ for each $m, n \in \omega$. Let T be the L -theory generated by the axioms

- (i) $\forall x \neg (p_m x \wedge p_n x)$ for all $m \neq n$;
- (ii) $p_m c_{m,n}$ for all m, n ;
- (iii) $\neg c_{k,l} \doteq c_{m,n}$ for all $(k, l) \neq (m, n)$.

Prove that every L -formula is T -equivalent to a Boolean combination of atomic formulas and conclude that T is complete.

2.5 Arithmetic

The theories studied in the preceding section are ‘simple’ in a number of ways: they involve few non-logical symbols, they are largely decidable, and they can express relatively few mathematical ideas. In this section we look at some theories with much richer content; we shall also return to some of these theories in later chapters as we develop additional logical tools.

These are usually called theories of arithmetic or number theory, although since they involve variables as well as numbers, algebra might be a better term. We shall eventually want to consider the language L_Ω with non-logical symbols $<, +, \times, \dot{S}$ and $\dot{0}$, but we shall work up to this by way of languages with only some of these symbols. In contrast to the languages $L_<$ and $L_=$, for L_Ω we have in mind a principal or intended structure, the **standard L_Ω -structure** $\Omega := (\omega, <, +, \times, \text{Sc}, 0)$. Thus clearly one theory of interest is the complete theory $\text{Th}(\Omega)$. It is not hard to see that $\text{Th}(\Omega)$ is a quite complex object and that questions about it include some of the deepest problems of number theory. For example, consider the L_Ω -sentence θ_{TP} :

$$\begin{aligned} \forall w \exists z (w < z \wedge \forall x \forall y [x \dot{\times} y \doteq z \rightarrow (x \doteq z \vee y \doteq z)] \\ \wedge \forall x \forall y [x \dot{\times} y \doteq \dot{S}\dot{S}z \rightarrow (x \doteq \dot{S}\dot{S}z \vee y \doteq \dot{S}\dot{S}z)]). \end{aligned}$$

θ_{TP} is true in Ω iff there are infinitely many twin-prime pairs — natural numbers p such that both p and $p + 2$ are prime. This is, of course, a long-standing open question of number theory. Thus, although we know that

by definition $\text{Th}(\Omega)$ is a complete theory, so that either $\theta_{\text{TP}} \in \text{Th}(\Omega)$ or $\neg \theta_{\text{TP}} \in \text{Th}(\Omega)$, we do not know which of these holds.

To ease our way into the complexities of $\text{Th}(\Omega)$, we look first at some simpler subtheories in restricted languages. We consider first $\text{Th}((\omega, \text{Sc}, 0))$ and its associated language $L_{\text{Sc},0}$ with non-logical symbols $\dot{\text{S}}$ and $\dot{0}$. Exercise 2.4.53 shows that this theory is decidable; we want now to carry this a step further and find a natural set of axioms for it.

The problem of axiomatizing parts of mathematics is a very old one, perhaps beginning with Euclid and his axioms for plane geometry. In this tradition, which long predates the formalization of logic, Dedekind [1879] proposed three statements, now known as the **Peano Postulates**, which (in modern terms) characterize $(\omega, \text{Sc}, 0)$ up to isomorphism:

- (P1) 0 is not a successor,
- (P2) the successor function is injective,
- (P3) for any set $X \subseteq \omega$, if $0 \in X$ and X is closed under successor, then $X = \omega$.

In this form, these postulates are not applicable to any other structure, but we can easily rewrite them so that they are. The first two correspond to single $L_{\text{Sc},0}$ -sentences:

- (S1) $\forall x (\neg \dot{0} \dot{=} \dot{\text{S}}x),$
- (S2) $\forall x \forall y (\dot{\text{S}}x \dot{=} \dot{\text{S}}y \rightarrow x \dot{=} y).$

By themselves, (S1) and (S2) have lots of models not isomorphic to the standard one $(\omega, \text{Sc}, 0)$ — for example, $(\omega, \text{Sc}_k, 0)$, where $\text{Sc}_k(m) = m + k$, for any $k > 0$, or similar structures with universe the non-negative rational or real numbers.

(P3) cannot be translated to a $L_{\text{Sc},0}$ -sentence because, like the definition of well-ordering in the last section, it refers to all *sets* of numbers and is therefore a second-order statement. However, it will be interesting to try to understand in greater detail the content of (P3). Without worrying too much about the formalism, we might reformulate (P3) as follows:

- (P3') for all p , $[p\dot{0} \wedge \forall x (px \rightarrow p\dot{\text{S}}x)] \rightarrow \forall x px,$

in which we think of p as a variable unary relation symbol which ranges over all subsets of the universe of an interpretation. This expresses that the induction principle holds for every subset of the universe. Then we have

Proposition 2.5.1 *Every model $(A, \text{Sc}_A, 0_A)$ of (S1), (S2) and (P3') is isomorphic to $(\omega, \text{Sc}, 0)$.*

Proof. Let $\mathfrak{A} = (A, \text{Sc}_A, 0_A)$ be such a model, and let $\eta : \omega \rightarrow A$ be the function recursively defined by

$$\eta(0) = 0_A \quad \text{and} \quad \eta(\text{Sc}(n)) = \text{Sc}_A(\eta(n)).$$

It follows from (S1) and (S2) that η is an injective homomorphism, and we see that η is surjective as follows. Let $B = \{\eta(n) : n \in \omega\}$. Then $0_A \in B$ and B is closed under Sc_A . Since $\mathfrak{A}$ is a model of (P3'), we conclude that $B = A$. $\square$

This property of the ‘theory’ generated by (S1), (S2) and (P3') and variations of it arise in several places and therefore deserve names.

Definition 2.5.2 For any theory T ,

- (i) T is **categorical** iff for any models $\mathfrak{A}$ and $\mathfrak{B}$ of T , $\mathfrak{A} \simeq \mathfrak{B}$;
- (ii) for any cardinal number κ , T is **κ -categorical** iff for any models $\mathfrak{A}$ and $\mathfrak{B}$ of T , if both are of cardinality κ , then $\mathfrak{A} \simeq \mathfrak{B}$.

We can reformulate in these terms several results from the preceding two sections.

Corollary 2.5.3 *For any language L and any L -theory T ,*

- (i) *if T is complete and has a finite model, then T is categorical;*
- (ii) *if L is countable and T has an uncountable model, then T is not categorical;*
- (iii) *if L is countable, T is consistent, has no finite models, and is $\aleph_0$ -categorical, then T is complete;*
- (iv) *every L -theory is κ -categorical for every infinite cardinal κ ;*
- (v) *the theories $T_{[\text{DLO}]}$, $T_{[\text{DLO}]}$, $T_{[\text{DLO}]}$ and $T_{(\text{DLO})}$ are $\aleph_0$ -categorical;*
- (vi) *every L -theory is κ -categorical for each infinite cardinal κ ;*

Proof. For (i), if T is complete, any two models are elementarily equivalent. Hence, if T has a finite model $\mathfrak{A}$, then by Theorem 2.3.11(i) every model is isomorphic to $\mathfrak{A}$ (in the case that L is not finite we use the extension to Theorem 2.3.11 in Section 2.7 below). For (ii), by Corollary 2.3.35 T has a countable model, which cannot be isomorphic to the uncountable model. (iii) is exactly the Countable Vaught Test 2.4.25. For (iv), since L -structures are just sets, two of the same cardinality are automatically isomorphic. (v) is Cantor’s Theorem 2.4.45. $\square$

Note that T_{DLO} is not, say, $2^{\aleph_0}$ -categorical; if we modify $(\mathbb{R}, <_{\mathbb{R}})$ by removing from the interval $[0, 1]$ all irrationals, the result is still of cardinality $2^{\aleph_0}$ but is not isomorphic to $(\mathbb{R}, <_{\mathbb{R}})$.

Proposition 2.5.1 asserts that the ‘theory’ generated by $(S1)$, $(S2)$ and $(P3')$ is categorical, but a consequence of the Upward Löwenheim-Skolem Theorem to be proved in Section 3.4 is the extension of (ii) above that no first-order theory with an infinite model is categorical. Thus there is no way to write $(P3')$ as a set of first-order axioms. Still, it turns out that a significant portion of the content of $(P3')$ can be captured in a first-order way, enough to ensure that any model is at least elementarily equivalent to $(\omega, \text{Sc}, 0)$.

The idea is that we can express that the induction principle holds for every *definable* subset of the universe much as in Example 2.3.36 we could express the definable instances of the least upper-bound principle. For each formula ϕ and variable x , let $I_{\phi, x}$ denote the following formula

$$(\phi_x(\dot{0}) \wedge \forall x [\phi \rightarrow \phi_x(\dot{S}x)]) \rightarrow \forall x \phi.$$

If ϕ has no free variable other than x , then $I_{\phi, x}$ is a sentence and for any $L_{\text{Sc}, 0}$ -structure $\mathfrak{A} = (A, \text{Sc}_A, 0_A)$, if $\phi^{\mathfrak{A}} = \{a \in A : \models_{\mathfrak{A}} \phi[a]\}$ is the set defined over $\mathfrak{A}$ by ϕ , then

$$\models_{\mathfrak{A}} I_{\phi, x} \iff \text{if } 0_A \in \phi^{\mathfrak{A}}, \text{ and } \phi^{\mathfrak{A}} \text{ is closed under } \text{Sc}_A, \text{ then } \phi^{\mathfrak{A}} = A.$$

Note that equivalently $I_{\phi, x}$ expresses that if $A \sim \phi^{\mathfrak{A}} = (\neg \phi)^{\mathfrak{A}}$ is non-empty, then it has a ‘least element’: either 0_A or $\text{Sc}_A(a)$ for some $a \in \phi^{\mathfrak{A}}$.

More generally, if ϕ has free variables $y_0, \dots, y_{k-1}$ other than x , then $\models_{\mathfrak{A}} \forall y_0 \dots \forall y_{k-1} I_{\phi, x}$ means that for each $b_0, \dots, b_{k-1} \in A$, if the set

$$\phi_{b_0, \dots, b_{k-1}}^{\mathfrak{A}} = \{a \in A : \models_{\mathfrak{A}} \phi[a, b_0, \dots, b_{k-1}]\}$$

contains 0_A and is closed under Sc_A , then it equals A .

Definition 2.5.4 For any language L which contains the symbols $\dot{S}$ and $\dot{0}$, the *L -induction schema* is the infinite set of sentences $\forall y_0 \dots \forall y_{k-1} I_{\phi, x}$ for each L -sentence ϕ . Each of these sentences is called an *instance* of the induction schema.

The richer the language L , the stronger the L -induction schema, but note that if $\mathfrak{A}$ and L are denumerable, there are only denumerably many subsets of A of the form $\phi_{b_0, \dots, b_{k-1}}^{\mathfrak{A}}$ and therefore the schema is much weaker than $(P3')$, which refers to all of the uncountably many subsets of A .

Definition 2.5.5 $T_{\text{Sc},0}$ is the $L_{\text{Sc},0}$ -theory generated by (S1), (S2) and the $L_{\text{Sc},0}$ -induction schema. T_0 is the $L_{\text{Sc},0}$ -theory generated by (S1), (S2) and the following sentences:

- (S3) $\forall x (\neg x \doteq \dot{0} \rightarrow \exists y [x \doteq \dot{S}y])$,
 (S4_m) $\forall x (\neg \dot{S}^{(m)}x \doteq x)$ for all $m > 0$.

Lemma 2.5.6 (S3) and (S4_m) for all $m > 0$ are theorems of $T_{\text{Sc},0}$.

Proof. For (S3), let θ be the formula $\neg x \doteq \dot{0} \rightarrow \exists y [x \doteq \dot{S}y]$. Then both $\theta_x(\dot{0})$ and $\forall x (\theta \rightarrow \theta_x(\dot{S}x))$ are logically valid, hence theorems of $T_{\text{Sc},0}$. Then by the instance $I_{\theta,x}$ of the $L_{\text{Sc},0}$ -induction schema, we have $T_{\text{Sc},0} \models \forall x \theta$ as claimed. For (S4_m), let θ^m be the formula $\neg \dot{S}^{(m)}x \doteq x$. Then by (S1), $T_{\text{Sc},0} \models \theta^m_x(\dot{0})$, and it follows easily from (S2) that $T_{\text{Sc},0} \models \forall x (\theta^m \rightarrow \theta^m_x(\dot{S}x))$. Hence by $I_{\theta^m,x}$, $T_{\text{Sc},0} \models \forall x \theta^m$. $\square$

Corollary 2.5.7 $T_0 \subseteq T_{\text{Sc},0} \subseteq \text{Th}((\omega, \text{Sc}, 0))$. $\square$

Theorem 2.5.8 T_0 is complete; hence $T_0 = T_{\text{Sc},0} = \text{Th}((\omega, \text{Sc}, 0))$.

Proof. (Sketch) In Exercise 2.4.53 it is shown that for every $L_{\text{Sc},0}$ -formula ϕ there exists a Boolean combination ϕ^* of atomic formulas such that $\models_{(\omega, \text{Sc}, 0)} \phi \leftrightarrow \phi^*$. A careful analysis of that proof shows that in fact $T_0 \models \phi \leftrightarrow \phi^*$. Now we verify that for each atomic sentence ψ either $T_0 \models \psi$ or $T_0 \models \neg \psi$ and finally appeal to Proposition 2.4.28(i). The atomic sentences here are those of the form $\dot{m} \doteq \dot{n}$ (recall that $\dot{m}$ denotes the term $\dot{S}^{(m)}\dot{0}$). Clearly, $T_0 \models \dot{m} \doteq \dot{m}$, and easily for $m \neq n$, $T_0 \models \neg \dot{m} \doteq \dot{n}$. $\square$

Thus we have found two axiomatizations for $\text{Th}((\omega, \text{Sc}, 0))$. Both are infinite, but both are decidable. This is easy to see for (S1)–(S4_m), but it is also not hard to show that the set of instances of the induction schema is a decidable set of sentences. Hence, by the Enumerability Theorem 2.4.24 we have a second (after that of Exercise 2.4.53) proof that $\text{Th}((\omega, \text{Sc}, 0))$ is decidable.

The axiomatization (S1)–(S4_m) is simpler and leads to a clear picture of what models of this theory look like. Suppose that $\mathfrak{A} = (A, \text{Sc}_A, 0_A)$ is such a model. A has distinct elements $0_A, \text{Sc}_A(0_A), \dots, \text{Sc}_A^{(m)}(0_A), \dots$ called the **standard part** of $\mathfrak{A}$. Note that these elements may also be written as $\dot{0}^{\mathfrak{A}}, \dots, \dot{m}^{\mathfrak{A}}, \dots$

Lemma 2.5.9 $\mathfrak{A} = (A, \text{Sc}_A, 0_A)$ is isomorphic to $(\omega, \text{Sc}, 0)$ iff the elements $\dot{m}^{\mathfrak{A}}$ for $m \in \omega$ exhaust A . $\square$

Definition 2.5.10 A structure $\mathfrak{A} = (A, \text{Sc}_A, 0_A)$ is called **non-standard** iff the elements $m^{\mathfrak{A}}$ for $m \in \omega$ do not exhaust A . Elements not in this standard part are called **non-standard** elements.

Suppose that b is a non-standard element of a model $\mathfrak{A}$ of T_0 . Then easily also $\text{Sc}_A(b)$ and all $\text{Sc}_A^{(m)}(b)$ are non-standard, and by (S2) and (S4_m) these are all distinct. Furthermore, since $b \neq 0_A$, there is by (S3) also an element b' , which we might naturally call $\text{Sc}_A^{-1}(b)$, such that $\text{Sc}_A(b') = b$ — and similarly $\text{Sc}_A^{(-m)}(b)$ for all $m > 0$. Again these must all be distinct from each other and from all $\text{Sc}_A^{(m)}(0_A)$. If there are still more elements of A , the same argument may be applied to them.

This argument leads to the picture of a non-standard model as consisting of a standard part (a substructure) isomorphic to $(\omega, \text{Sc}, 0)$ together with some collection of “blocks”, each of which is isomorphic to the (positive and negative) integers $(\mathbb{Z}, \text{Sc}_{\mathbb{Z}}, 0)$. Any number of blocks are possible, and the number of them determines the isomorphism type of the model. To make this precise, for any set B , let $A_B = \omega \cup (B \times \mathbb{Z})$, define Sc_{A_B} by

$$\text{Sc}_{A_B}(m) := m + 1 \quad \text{and} \quad \text{Sc}_{A_B}((b, s)) := (b, s + 1),$$

and set $\mathfrak{A}_B := (A_B, \text{Sc}_{A_B}, 0)$. Each $b \in B$ determines a block B_b of $\mathfrak{A}_B$ consisting of the elements

$$\dots \rightarrow (b, -2) \rightarrow (b, -1) \rightarrow (b, 0) \rightarrow (b, 1) \rightarrow (b, 2) \rightarrow \dots$$

with the natural successor function on B_b indicated by $\rightarrow$. These structures give a complete “catalog” of the models of $T_{\text{Sc},0}$ up to isomorphism in the following sense.

Proposition 2.5.11

- (i) For any set B , $\mathfrak{A}_B = (A_B, \text{Sc}_{A_B}, 0)$ is a model of $T_{\text{Sc},0}$;
- (ii) for every model $\mathfrak{A}$ of $T_{\text{Sc},0}$ there exists a set B such that $\mathfrak{A} \simeq \mathfrak{A}_B$.

Proof. Left as an exercise. $\square$

From this we may actually prove what we guessed above: the full strength of (P3) or (P3') cannot be captured by any first-order sentence or even by an infinite set Γ of sentences. If Γ were such a set, then all $\phi \in \Gamma$ would be true in $(\omega, \text{Sc}, 0)$, hence by the above true in all of the non-standard models, contrary to Proposition 2.5.1.

Corollary 2.5.12 *$T_{\text{Sc},0}$ is not categorical or even $\aleph_0$ -categorical (Definition 2.5.2), but $T_{\text{Sc},0}$ is κ -categorical for any uncountable cardinal number κ .*

Proof. For B and B' of different finite cardinality, $\mathfrak{A}_B$ and $\mathfrak{A}_{B'}$ are both denumerable but not isomorphic. However, if $\mathfrak{A}$ is any model of $T_{\text{Sc},0}$ of cardinality some uncountable cardinal number κ , then $\mathfrak{A}$ is of the form $\mathfrak{A}_B$ for some set B of cardinality κ and is thus isomorphic to any other such model. $\square$

Note that the first part of this corollary implies that we cannot use the Countable Vaught Test to deduce (again) that $T_{\text{Sc},0}$ is complete. However, in Section 3.4 we shall establish a more general version of the Vaught Test which will allow us to deduce completeness from the second part of the corollary.

Although we shall not go into any detail, a similar argument using elimination of quantifiers applies to the theory of the structure $(\omega, <, \text{Sc}, 0)$; it is decidable and has a natural (in fact finite) axiomatization. The non-standard models of this theory also have a structure determined by some collection of blocks isomorphic to $\mathbb{Z}$, now as a linearly ordered set. Furthermore, since the axioms are ‘local’, the linear ordering of the blocks is completely arbitrary. It follows that the theory is not κ -categorical for any infinite κ . (see exercises)

We proceed directly to the more comprehensive structure

$$\Omega_+ = (\omega, <, +, \text{Sc}, 0)$$

and its theory. It is easy to extend the Peano Postulates to give a second-order categorical axiomatization of this theory. Consider the sentences

- (L1) $\forall x (\neg x < \dot{0})$,
- (L2) $\forall x \forall y [x < \dot{S}y \leftrightarrow (x < y \vee x \doteq y)]$,
- (A1) $\forall x (x \dot{+} \dot{0} \doteq x)$,
- (A2) $\forall x \forall y [x \dot{+} \dot{S}y \doteq \dot{S}(x \dot{+} y)]$.

Proposition 2.5.13 *Each model $(A, <_A, +_A, \text{Sc}_A, 0_A)$ of (S1), (S2), (L1), (L2), (A1), (A2) and (P3') is isomorphic to Ω_+ .*

Proof. An easy exercise. $\square$

Now let T_+ denote the first-order theory generated by (S1), (S2), (L1), (L2), (A1), (A2) and the L_{Ω_+} -induction schema. We aim to show that T_+ is complete and decidable and coincides with $\text{Th}(\Omega_+)$. The method is again the elimination of quantifiers, but is substantially more complicated than earlier

examples. As basic formulas we need more than just the atomic formulas; we first introduce some notation. For any terms t and u and any $r \in \omega \sim \{0\}$

rt denotes $t \dot{+} t \dot{+} \dots \dot{+} t$ with r -many t 's

$t \dot{\equiv}_r u$ denotes $\exists z (rz \dot{+} t \dot{=} u \vee t \dot{=} u \dot{+} rz)$,

where z is a variable not occurring in t or u . For completeness, $0t$ denotes $\dot{0}$. Of course, $\dot{\equiv}_r$ is interpreted in Ω_+ as the relation of congruence modulo r . We shall show that T_+ admits Δ_+ -elimination for $\Delta_+ := \text{Bool}(\Gamma_+)$, where

$$\Gamma_+ := \text{AtFm}_{L_{\Omega_+}} \cup \{t \dot{\equiv}_r u : t, u \text{ } L_{\Omega_+}\text{-terms; } r \in \omega \sim \{0\}\}.$$

The formulas $t \dot{\equiv}_r u$ play here a role analogous to that of the sentences $\exists^{\geq n}$ for L_+ ; quantifiers in general formulas will be eliminated in favor of those in these limited contexts.

Proposition 2.5.14 *The universal closures of the following formulas are theorems of T_+ :*

- (i) $x \dot{<} y \wedge y \dot{<} z \rightarrow x \dot{<} z$,
- (ii) $\neg x \dot{<} x$,
- (iii) $x \dot{+} y \dot{=} x \dot{+} z \leftrightarrow y \dot{=} z$,
- (iv) $x \dot{+} y \dot{<} x \dot{+} z \leftrightarrow y \dot{<} z$,
- (v) $x \dot{+} y \dot{\equiv}_r x \dot{+} z \leftrightarrow y \dot{\equiv}_r z$,
- (vi) $x \dot{\equiv}_r y \leftrightarrow x \dot{+} x \dot{\equiv}_{2r} y \dot{+} y$,
- (vii) $(x \dot{<} y) \vee (x \dot{=} y) \vee (y \dot{<} x)$,
- (viii) $x \dot{+} y \dot{=} y \dot{+} x$,
- (ix) $x \dot{+} (y \dot{+} z) \dot{=} (x \dot{+} y) \dot{+} z$,
- (x) $\neg x \dot{\equiv}_r y \leftrightarrow (x \dot{\equiv}_r y \dot{+} \dot{1}) \vee \dots \vee (x \dot{\equiv}_r y \dot{+} \overbrace{r-1})$.

Proof. An easy if tedious exercise in the use of the induction schema. $\square$

Lemma 2.5.15

- (i) *For any variable x , every member of Γ_+ is T_+ -equivalent to a formula of one of the forms*

$$mx \dot{+} t \dot{=} u, \quad mx \dot{+} t \dot{<} u, \quad t \dot{<} mx \dot{+} u, \quad mx \dot{+} t \dot{\equiv}_r u,$$

where t and u are terms not containing the variable x , $m \in \omega$ and $r \in \omega \sim \{0\}$.

- (ii) *Every conjunction of members of Γ_+ and their negations is T_+ -equivalent to a disjunction of conjunctions of members of Γ_+ (without negations).*

Proof. These follow easily from the preceding lemma, and the proof is left as an exercise. $\square$

Theorem 2.5.16 T_+ admits Δ_+ -elimination — that is, for any L_{Ω_+} -formula ϕ , there exists a formula $\phi^* \in \Delta_+$ such that

$$T_+ \models \phi \leftrightarrow \phi^* \quad \text{and} \quad \text{Fv}(\phi^*) \subseteq \text{Fv}(\phi).$$

Furthermore, ϕ^* is effectively calculable from ϕ .

Proof. By the preceding lemma and the Quantifier Elimination Lemma 2.4.30 we need only consider formulas of the form $\exists x \psi$ where ψ is a formula of the following form. The ellipses ... designate additional formulas of the same type, any one of the types may be missing, and t, u etc. all denote terms not containing the variable x :

$$\begin{aligned} \exists x [(mx \dot{+} t \dot{=} t') \wedge \cdots \wedge (nx \dot{+} u \dot{<} u') \wedge \cdots \\ \wedge (v \dot{<} px \dot{+} v') \wedge \cdots \wedge (qx \dot{+} w \dot{=} r w') \wedge \cdots]. \end{aligned}$$

By the proposition above, this formula is T_+ -equivalent to one of the form

$$\begin{aligned} \exists x [(mx \dot{+} t \dot{=} t') \wedge \cdots \wedge (mx \dot{+} u \dot{<} u') \wedge \cdots \\ \wedge (v \dot{<} mx \dot{+} v') \wedge \cdots \wedge (mx \dot{+} w \dot{=} r w') \wedge \cdots], \end{aligned}$$

where the coefficient m is the least common multiple of the previous ones, and therefore to

$$\begin{aligned} \exists y [(y \dot{+} t \dot{=} t') \wedge \cdots \wedge (y \dot{+} u \dot{<} u') \wedge \cdots \\ \wedge (v \dot{<} y \dot{+} v') \wedge \cdots \wedge (y \dot{+} w \dot{=} r w') \wedge \cdots \wedge y \dot{=} 0]. \end{aligned}$$

For technical reasons explained below, we assume that the redundant disjunct $0 \dot{<} y \dot{+} 1$ is always present. If at least one equality $y \dot{+} t \dot{=} t'$ is present, we may drop the quantifier and replace y everywhere by $t' - t$ — that is, $y \dot{+} u \dot{<} u'$ becomes $t' \dot{+} u \dot{<} u' \dot{+} t$ etc. Thus we assume from now on that no equality is present. Next we make repeated use of the easily checked T_+ -equivalence (here $\dot{\leq}$ has the obvious meaning),

$$\begin{aligned} \exists y \left[\bigwedge_{i < k} y \dot{+} u_i < u'_i \wedge \cdots \right] \leftrightarrow \bigvee_{i < k} (\exists y [y \dot{+} u_i < u'_i \wedge \cdots]) \\ \wedge \bigwedge_{j < k} [u'_i \dot{+} u_j \dot{\leq} u_i \dot{+} u'_j]) \end{aligned}$$

(one among the quantities $u'_i - u_i$ is smallest), and the analogous version for upperbounds,

$$\begin{aligned} \exists y \left[\bigwedge_{i < k} v_i < y \dot{+} v'_i \wedge \cdots \right] \leftrightarrow \bigvee_{i < k} (\exists y [v_i < y \dot{+} v'_i \wedge \cdots]) \\ \wedge \bigwedge_{j < k} [v'_i \dot{+} v_j \dot{\leq} v_i \dot{+} v'_j]) \end{aligned}$$

(one among the quantities $v_i - v'_i$ is largest), to reduce our formula to a Boolean combination of members of Δ_+ and formulas of the form

$$\exists y [(y \dot{+} u \dot{<} u') \wedge (v \dot{<} y \dot{+} v') \wedge (y \dot{+} w \dot{=}_r w') \wedge \cdots \wedge y \dot{=}_m 0]$$

(or to a simpler form in which one of the inequalities is missing). Because of the redundant disjunct added above, we may assume $v' \dot{\leq} v \dot{+} 1$. Now let R be the least common multiple of all of the moduli r which appear. By further use of the induction schema we may derive that the congruences have a solution greater than $v - v'$ iff they have one among the numbers $v - v' + 1, \dots, v - v' + R$. Hence our formula is T_+ -equivalent to

$$\bigvee_{1 \leq s \leq R} [u \dot{+} v \dot{+} s \dot{<} u' \dot{+} v'] \wedge [v \dot{+} s \dot{+} w \dot{=}_r v' \dot{+} w' \wedge \cdots] \wedge [v \dot{+} s \dot{=}_m v'],$$

which belongs to Δ_+ . $\square$

Lemma 2.5.17

- (i) For any m and n , $T_+ \models m \dot{+} n \dot{=} \overbrace{m \dot{+} n}^{\cdot}$.
- (ii) For any variable-free term t there exists a natural number m such that $T_+ \models t \dot{=} m$; furthermore m may be effectively computed from t .

Proof. (i) is proved by induction on n and provides the essential fact needed to prove (ii) by term induction. $\square$

Corollary 2.5.18 T_+ is complete and decidable; hence $T_+ = \text{Th}(\Omega_+)$ and every model of T_+ is elementarily equivalent to Ω_+ .

Proof. We again apply Proposition 2.4.28. The proof of the preceding theorem is effective and provides an algorithm for computing the mapping $\phi \mapsto \phi^*$, so it suffices to show

- (i) for each sentence $\psi \in \Gamma_+$, either $T_+ \models \psi$ or $T_+ \models \neg \psi$;
- (ii) $T_+ \cap \Gamma_+$ is decidable.

The sentences in Γ_+ are of one of the forms $m \dot{=} n$, $m \dot{<} n$ or $m \dot{=}_r n$, and it is easy to verify that for ψ of these forms,

$$\models_{\Omega_+} \psi \implies T_+ \models \psi \quad \text{and} \quad \models_{\Omega_+} \neg \psi \implies T_+ \models \neg \psi.$$

Thus (i) is satisfied and (ii) follows, since there is easily an algorithm for deciding for any such sentence ψ whether or not $\models_{\Omega_+} \psi$. $\square$

Corollary 2.5.19 *The theories of the structures*

$$(\omega, <, +, 0), \quad (\omega, +, \text{Sc}, 0), \quad (\omega, +) \text{ etc.}$$

are all decidable. $\square$

Non-standard models of $\text{Th}(\Omega_+)$ are much more complicated than those of $\text{Th}((\omega, \text{Sc}, 0))$; in fact, we shall have to postpone the proof of their existence to [Section 3.1](#), where we can use the Compactness Theorem. However, we can get some idea of what they have to look like. Suppose that

$$\mathfrak{A} = (A, <_A, +_A, \text{Sc}_A, 0_A)$$

is a non-standard model of $\text{Th}(\Omega_+)$. By the same arguments as before, there is a segment $0_A, 1_A = \text{Sc}_A(0_A), \dots$ isomorphic to ω together with pairwise-disjoint blocks $[a] = \{a \pm_A m_A : m \in \omega\}$. $<_A$ is a strict linear ordering and induces a strict linear ordering $<^*_A$ on these blocks: $[a] <^*_A [b]$ iff $[a] \neq [b]$ and $a <_A b$. We have the rather startling

Proposition 2.5.20 *For any non-standard model*

$$\mathfrak{A} = (A, <_A, +_A, \text{Sc}_A, 0_A)$$

*of $\text{Th}(\Omega_+)$, the ordering $<^*_A$ of the non-standard blocks is a dense linear ordering without greatest or least element.*

Proof. Let a be any non-standard element, so for all $m \in \omega$, $m_A <_A a$. Since the sentence $\forall x \forall y \forall z (y < z \rightarrow x + y < x + z)$ belongs to the theory, it is true in $\mathfrak{A}$ and thus for all m , $a +_A m_A <_A a +_A a$. Thus the block $[a +_A a]$ is distinct from $[a]$ and $[a] <^*_A [a +_A a]$. Hence there is no greatest block.

To see that there is no least block, note that the sentence $\forall x \exists y (y + y \doteq x \vee y + y \doteq \dot{\text{S}}x)$ belongs to $\text{Th}(\Omega_+)$ — that is, every natural number or its successor is even. Hence for a non-standard a there is a b such that $b +_A b = a$ or $b +_A b = \text{Sc}_A(a)$. Similar arguments as above show that b is non-standard and $[b] <^*_A [a]$.

Finally, for the density of the ordering, let a and b be any two non-standard elements such that $[a] <^*_A [b]$ and choose c such that $c +_A c = a +_A b$ or $c +_A c = \text{Sc}_A(a +_A b)$ — c is the average of a and b . Then $[a] <^*_A [c] <^*_A [b]$. $\square$

Of course, we are still far from a complete description of the non-standard model $\mathfrak{A}$, since that would include also a description of the addition function $+_A$.

Corollary 2.5.21 *For any non-standard model $\mathfrak{A}$ of $\text{Th}(\Omega_+)$, the set of standard elements of A is not definable over $\mathfrak{A}$.*

Proof. If the set of standard elements were definable by a formula θ , then since it contains 0_A and is closed under successor Sc_A , the instance $I_{\theta,x}$ of the induction schema would imply that it is the whole universe, contrary to assumption. $\square$

Although a full proof of the following is too long to include here, we give a brief sketch.

Theorem 2.5.22 *$\text{Th}((\omega, \times, 1))$ is decidable.*

Proof. (Sketch) Let $\mathfrak{A}$ denote the direct product of ω -many copies of $(\omega, +, 0)$ and $(\omega, +, 0)^{(\omega)}$ the substructure with universe consisting of those functions $F : \omega \rightarrow \omega$ which are eventually 0. Then it is easy to check that

$$(\omega \sim \{0\}, \times, 1) \simeq (\omega, +, 0)^{(\omega)}$$

via the mapping which sends a number $n = 2^{n_0} \times 3^{n_1} \times \cdots \times p_k^{n_k}$ with its unique prime-power decomposition to the function $(n_0, n_1, \dots, n_k, 0, 0, \dots)$. A general theorem of Feferman and Vaught gives the decidability of

$$\text{Th}((\omega, +, 0)^{(\omega)})$$

from that of $\text{Th}((\omega, +, 0))$, and an easy argument derives the decidability of $\text{Th}((\omega, \times, 1))$ from that of $\text{Th}((\omega \sim \{0\}, \times, 1))$. $\square$

Finally, we turn briefly to the full structure $\Omega = (\omega, <, +, \times, \text{Sc}, 0)$ and its theory. Generalizing on our experience so far, it is an attractive conjecture that by adjoining to T_+ some axioms for multiplication, say

$$(M1) \quad \forall x (x \dot{\times} \dot{0} \doteq \dot{0}),$$

$$(M2) \quad \forall x \forall y [(x \dot{\times} \dot{\text{Sc}} y) \doteq (x \dot{\times} y) \dot{+} x],$$

and extending the induction schema to all L_Ω -formulas, we should get a complete axiomatization for $\text{Th}(\Omega)$. Exactly as before, if we replace the induction schema with the full second-order induction postulate ($P3'$), we do get a categorical description of Ω , but the situation for the first-order theory is much more complicated. The results of [Sections 4.5–4.6](#) show that not only is this particular set of axioms insufficient to generate $\text{Th}(\Omega)$, but there is no reasonable extension of it which will generate any complete theory.

Collateral Reading. Some basic information about subtheories of arithmetic can be found in [En01]; [Kay] is a comprehensive reference. A proof of the theorem of Feferman and Vaught mentioned in the proof of Theorem 2.5.22 may be found in [Mon; Chapter 23].

Exercises

Exercise 2.5.23 Complete the proof of Proposition 2.5.11.

Exercise 2.5.24 Prove Proposition 2.5.13.

Exercise 2.5.25 Prove Proposition 2.5.14(i) and (vii).

Exercise 2.5.26 Prove Lemma 2.5.15.

Exercise 2.5.27 For each of the following T , determine for which infinite cardinals κ the theory T is κ -categorical.

- (i) T is the set of logical validities in a language with just one unary relation symbol p ;
- (ii) T is the theory in the same language as (i) generated by the sentences

$$\{ \exists^{\geq n} x \, px : n \geq 1 \} \cup \{ \exists^{\geq n} x \, \neg px : n \geq 1 \};$$

- (iii) T is the theory generated by the set $\{ \neg c_i \doteq c_j : i \neq j \}$ in the language with constant symbols $\{ c_i : i \in \omega \}$ but no function or relation symbols;
- (iv) T is the theory in the language with one unary function symbol f generated by the following axioms:
 - (a) $\forall x \forall y [fx \doteq fy \rightarrow x \doteq y]$;
 - (b) $\forall x \exists y [fy \doteq x]$;
 - (c) for each $n > 0$, $\forall x \neg \underbrace{ff \cdots f}_n x \doteq x$;

- (v) T is the theory in the language with just one binary relation symbol whose models are exactly structures (A, E) for which E is an equivalence relation on A with infinitely many equivalence classes, each of which is infinite. (First write out the axioms for T !)

Exercise 2.5.28 Carry out the quantifier elimination for the theory $T := \text{Th}(\omega, <, \text{Sc}, 0)$ to prove (parallel to Exercise 2.4.53)

- (i) for every formula ϕ there exists a quantifier-free formula ϕ^* such that

$$\models_{(\omega, <, \text{Sc}, 0)} \phi \leftrightarrow \phi^* \quad \text{and} \quad \text{Fv}(\phi^*) \subseteq \text{Fv}(\phi);$$

- (ii) T is decidable; hence so is $\text{Th}(\omega, <, 0)$;
- (iii) $R \subseteq \omega$ is definable over $(\omega, <, \text{Sc}, 0)$ iff R is finite or cofinite;

- (iv) the normal addition function $+$ on ω is not definable over $(\omega, <, \text{Sc}, 0)$.

Hint: Show first using the technique of the Quantifier Elimination Lemma 2.4.30 and facts about $(\omega, <, \text{Sc}, 0)$ that it suffices to consider formulas of the form

$$\exists x \left[\bigwedge_{i < k} \dot{S}^{(m_i)} x \dot{=} t_i \wedge \bigwedge_{j < l} \dot{S}^{(n_j)} x \dot{<} u_j \wedge \bigwedge_{j < l'} v_j \dot{<} \dot{S}^{(p_j)} x \right].$$

If at least one equality occurs, proceed as in the corresponding part of Exercise 2.4.53. Otherwise consider separately cases in which one or the other type of inequality is missing. For the general case verify and use a generalization of the following equivalence valid in $(\omega, <, \text{Sc}, 0)$:

$$\exists x \left[v \dot{<} \dot{S}^{(p)} x \wedge \dot{S}^{(n)} x \dot{<} u \right] \leftrightarrow \dot{S}^{(n+1)} v \dot{<} \dot{S}^{(p)} u \wedge \dot{n} \dot{<} u.$$

Remark 2.5.29 By analyzing the solution of the preceding exercise in the manner of Theorem 2.5.8, we may construct a finite (!) set of axioms for T .

Exercise 2.5.30 Give an analysis of the isomorphism types of models of $\text{Th}(\omega, <, \text{Sc}, 0)$ analogous to Proposition 2.5.11 (see also 2.5.20).

Hint: If $\mathfrak{A} = (A, <_A, \dot{\text{Sc}}_A, 0_A)$ is a model, $<_A$ will be a strict linear ordering which has an initial segment $0_A = \dot{0}^{\mathfrak{A}} <_A \dot{1}^{\mathfrak{A}} < \dots$ of order-type ω followed by blocks isomorphic to $(\mathbb{Z}, <_{\mathbb{Z}}, \text{Sc}_{\mathbb{Z}}, 0)$.

Exercise 2.5.31 A set X of natural numbers is called *eventually periodic* iff for some m and p , for all $n > m$, $n \in X \iff n + p \in X$. Prove that the definable subsets of Ω_+ are exactly the eventually periodic sets. Conclude that the usual multiplication relation $\{(m, n, p) : m \times n = p\}$ is not definable over Ω_+ .

2.6 Changing languages

Every result we have proved so far has involved only a single language, either a specific one, such as L_{Ω} , or an arbitrary one L . In this section we collect a number of results and techniques all of which involve relationships among two or more languages.

Definition 2.6.1 For any two languages L and L^+ , L is a *sublanguage* of L^+ and L^+ is an *extension* of L — in symbols, $L \subseteq L^+$ — iff $\text{Rs}_L \subseteq \text{Rs}_{L^+}$, $\text{Fs}_L \subseteq \text{Fs}_{L^+}$, $\text{Cs}_L \subseteq \text{Cs}_{L^+}$ and ν_L is the restriction of ν_{L^+} to the non-logical symbols of L . In other words, L^+ has all of the symbols of L with the same syntax plus some additional symbols. In the special case that $\text{Rs}_L = \text{Rs}_{L^+}$ and $\text{Fs}_L = \text{Fs}_{L^+}$, we say that L^+ is an *extension by constants*.

Corollary 2.6.2 *If $L \subseteq L^+$, then every L -term (formula) is an L^+ -term (formula).*

Proof. Obvious (by a straightforward induction). $\square$

Thus any L -formula may also be considered as an L^+ formula for any of the infinitely many extensions of L . This means that our notation is technically ambiguous: if we write $\models \psi$, do we mean ‘for all L -structures $\mathfrak{A}$, $\models_{\mathfrak{A}} \psi$ ’ or ‘for all L^+ -structures $\mathfrak{A}$, $\models_{\mathfrak{A}} \psi$ ’ for various L^+ extending L . Similarly for $\Gamma \models \psi$. Our first job here is to show that these are equivalent so that the ambiguity is harmless.

Definition 2.6.3 For any $L \subseteq L^+$, any L -structure $\mathfrak{A}$ and any L^+ -structure $\mathfrak{A}^+$, $\mathfrak{A}$ is the *L -reduct* of $\mathfrak{A}^+$ and $\mathfrak{A}^+$ is an *L^+ -expansion* of $\mathfrak{A}$ — in symbols, $\mathfrak{A} = \mathfrak{A}^+ \upharpoonright L$ — iff $|\mathfrak{A}| = |\mathfrak{A}^+|$; for all $p \in \text{Rs}_L$, $p^{\mathfrak{A}} = p^{\mathfrak{A}^+}$; for all $f \in \text{Fs}_L$, $f^{\mathfrak{A}} = f^{\mathfrak{A}^+}$; and for all $c \in \text{Cs}_L$, $c^{\mathfrak{A}} = c^{\mathfrak{A}^+}$.

Thus, for example,

$$(\omega, <, +, \times, \text{Sc}, 0) \text{ has reducts } (\omega, +, \times) \text{ and } (\omega, \text{Sc}, 0)$$

(to the obvious languages). Of course, any structure for a given language has a unique reduct to any sublanguage but many expansions to each extension language. Note that since their universes are the same, assignments are the same for structures and their reducts and expansions.

Proposition 2.6.4 *For any $L \subseteq L^+$ and any L^+ -structure $\mathfrak{A}^+$, if $\mathfrak{A} = \mathfrak{A}^+ \upharpoonright L$, then for any $\mathfrak{A}$ -assignment α ,*

- (i) *for each L -term t , $t^{\mathfrak{A}}[\alpha] = t^{\mathfrak{A}^+}[\alpha]$;*
- (ii) *for each L -formula ϕ , $\models_{\mathfrak{A}} \phi[\alpha] \iff \models_{\mathfrak{A}^+} \phi[\alpha]$.*

Proof. Two easy inductions. $\square$

Theorem 2.6.5 *For every $L \subseteq L^+$ and every L -formula ϕ , the following are equivalent:*

- (i) *for all L -structures $\mathfrak{A}$ and $\mathfrak{A}$ -assignments α , $\models_{\mathfrak{A}} \phi[\alpha]$;*
- (ii) *for all L^+ -structures $\mathfrak{A}^+$ and $\mathfrak{A}^+$ -assignments α , $\models_{\mathfrak{A}^+} \phi[\alpha]$.*

Proof. Immediate from the preceding proposition. $\square$

This clears up the technical ambiguity mentioned above (which is unlikely to have bothered the reader!): $\Gamma \models \psi$ has the same meaning viewed as a statement about any language L large enough to include all $\phi \in \Gamma$ and ψ . On the other hand, it does sometimes make sense to include the parameter L in defining

$$\text{Th}_L(\Gamma) = \{ \psi : \psi \text{ is an } L\text{-sentence and } \Gamma \models \psi \}.$$

Of course, if L^+ is an extension of L , $\text{Th}_{L^+}(\Gamma)$ is larger than $\text{Th}_L(\Gamma)$, but they contain the same L -sentences.

There are two other easy corollaries. The first is analogous to Proposition 2.2.27. Note that in each case the implication ($\implies$) is a formal version of a common form of mathematical argument. For (i), to prove that every object (in some class) has a certain property, we often proceed by saying “let c be an arbitrary such object” and then proving that c has the property. The hypothesis that c does not occur in ϕ or any member of Γ is the precise criterion of c being ‘arbitrary’.

Corollary 2.6.6 *For any L , any set Γ of L -formulas, any L -formulas ϕ and ψ and any constant symbol c which does not occur in ϕ , ψ or any member of Γ ,*

- (i) $\Gamma \models \phi_x(c) \iff \Gamma \models \forall x \phi$;
- (ii) $\Gamma \models \phi_x(c) \rightarrow \psi \iff \Gamma \models \exists x \phi \rightarrow \psi$.

Proof. Both implications ($\Leftarrow$) are immediate from Corollary 2.2.26. Suppose that $\Gamma \models \phi_x(c)$, and let L^- be the sublanguage of L obtained by removing c . For each L^- -structure $\mathfrak{A}$ and each $a \in A$, let $\mathfrak{A}_{c \mapsto a}$ denote the expansion of $\mathfrak{A}$ to an L -structure with c interpreted as a . Suppose that $\mathfrak{A}$ and α satisfy Γ . Then by the proposition, $\mathfrak{A}_{c \mapsto a}$ and α still satisfy Γ so they satisfy $\phi_x(c)$. But then by Lemma 2.2.22,

$$\models_{\mathfrak{A}_{c \mapsto a}} \phi[\alpha_{x \mapsto a}] \quad \text{hence also} \quad \models_{\mathfrak{A}} \phi[\alpha_{x \mapsto a}].$$

Since this holds for any $a \in A$, also $\models_{\mathfrak{A}} \forall x \phi$. Now (ii) ($\implies$) follows from the implications

$$\begin{aligned} \Gamma \models \phi_x(c) \rightarrow \psi &\implies \Gamma \cup \{\neg \psi\} \models \neg \phi_x(c) \\ &\implies \Gamma \cup \{\neg \psi\} \models \forall x \neg \phi \implies \Gamma \models \exists x \phi \rightarrow \psi. \quad \square \end{aligned}$$

Corollary 2.6.7 *For any $L^- \subseteq L$ and any L -structures $\mathfrak{A}$ and $\mathfrak{B}$,*

- (i) $\mathfrak{A} \simeq \mathfrak{B} \implies \mathfrak{A} \upharpoonright L^- \simeq \mathfrak{B} \upharpoonright L^-$;
- (ii) $\mathfrak{A} \equiv \mathfrak{B} \implies \mathfrak{A} \upharpoonright L^- \equiv \mathfrak{B} \upharpoonright L^-$. $\square$

An amusing application of this is the promised extension of Theorem 2.3.11.

Theorem 2.6.8 *For any (not necessarily finite) language L , any finite L -structure $\mathfrak{A}$ and any L -structure $\mathfrak{B}$, if $\mathfrak{A} \equiv \mathfrak{B}$ then $\mathfrak{A} \simeq \mathfrak{B}$.*

Proof. Fix $\mathfrak{A}$ and $\mathfrak{B}$, and suppose that $\mathfrak{A} \equiv \mathfrak{B}$ but $\mathfrak{A} \not\simeq \mathfrak{B}$. Clearly $\mathfrak{B}$ has the same finite number of elements as $\mathfrak{A}$, so there are only finitely

many injective functions $\eta : A \rightarrow B$. By hypothesis, each such η fails to be an isomorphism, so for at least one L -symbol, η does not carry its $\mathfrak{A}$ -interpretation to its $\mathfrak{B}$ -interpretation. Choose one such symbol for each η , let L^- be the sublanguage of L with exactly these non-logical symbols, and set $\mathfrak{A}^- = \mathfrak{A} \upharpoonright L^-$ and $\mathfrak{B}^- = \mathfrak{B} \upharpoonright L^-$. Clearly also $\mathfrak{A}^- \not\cong \mathfrak{B}^-$ so, since these are now structures for the finite language L^- , by Theorem 2.3.11, $\mathfrak{A}^- \not\equiv \mathfrak{B}^-$. But then $\mathfrak{A} \not\equiv \mathfrak{B}$, contrary to our assumption. $\square$

In studying a particular structure $\mathfrak{A}$, it is sometimes useful to have a name in the language for each element $a \in A$ — that is, a constant term $\dot{a}$ such that $\dot{a}^{\mathfrak{A}} = a$. Of course, for some structures this is already true; in $(\omega, \text{Sc}, 0)$ and its expansions the term $\dot{n} = \dot{S} \cdots \dot{S} \dot{0}$ is a name for n . In other cases, we may extend the language by adding new constant symbols and expand the structure to interpret them in the intended way.

Definition 2.6.9 For any L , any L -structure $\mathfrak{A}$ and any $X \subseteq A$, L^X is the extension of L obtained by adding, for each $a \in X$, a new constant symbol $\dot{a}$. $\mathfrak{A}_X$ denotes the L^X -expansion of $\mathfrak{A}$ such that for all $a \in X$, $\dot{a}^{\mathfrak{A}_X} = a$. We call $\dot{a}$ a **name** for a .

Remark 2.6.10 We shall see frequent use for this construction in [Chapter 3](#) and [Chapter 7](#), but we sketch briefly here one more elementary use, namely an alternative characterization of the basic notions of satisfaction and truth. The point is that for a structure which has names for all of its elements, the whole machinery of assignments and satisfaction may be avoided in favor of a direct definition of *truth* for sentences. First, we can for any language L and any L -structure $\mathfrak{A}$ give a direct definition of the interpretation $t^{\mathfrak{A}}$ in $\mathfrak{A}$ for all **constant terms** t — that is, terms which contain no variables — by simply restricting the original definition 2.1.17 to the constant ones. That leads immediately to a definition of truth for atomic sentences, and, via the usual clauses, for Boolean combinations of atomic sentences. In general, however, we cannot extend this definition to formulas with quantifiers, since we are led thereby to formulas with free variables. Since, however, every $a \in A$ has a name $\dot{a}$ in L^A , we may define

$$\models_{\mathfrak{A}_A} \exists x \phi \quad \Longleftrightarrow \quad \text{for some } a \in A, \quad \models_{\mathfrak{A}_A} \phi_x(\dot{a}),$$

since if $\exists x \phi$ is a sentence, then so is $\phi_x(\dot{a})$. Note that this cannot be justified as a recursion on formulas, since $\phi_x(\dot{a})$ is not a ‘predecessor’ of $\exists x \phi$ in the inductive generation of Fm_L . The key point is that $\phi_x(\dot{a})$ has the same number of symbols as ϕ so we may cast the definition as one by **recursion on length** of $V_{\mathfrak{A}} : \text{Sent}_{L^A} \rightarrow \{\text{T}, \text{F}\}$. Of course, these two approaches are essentially the same: the correspondence is given by

$$\models_{\mathfrak{A}} \phi[\alpha] \quad \Longleftrightarrow \quad \models_{\mathfrak{A}_A} \phi(\dot{\alpha}),$$

where $\phi(\dot{\alpha})$ denotes the result of replacing each free variable x in ϕ by the constant symbol $\dot{\alpha}(x)$. This is immediate from Lemma 2.2.22.

In informal mathematics, the most common situation in which the language is changed is the introduction of a new symbol by definition. For example, if we characterize an interesting real number (such as the ratio of the circumference of a circle to its diameter) to which we plan to refer often, it is convenient to give it a name (such as π). If we find a useful relationship between integers (such as divisibility) which is definable in terms of other notions for which we already have names (e.g., m divides n iff for some p , $n = m \times p$), we may want to give it a symbolic name (such as $m \mid n$). In each case our intuition is that nothing essential has changed; nothing new can be proved just because we have added a few symbols to our language and their definitions to our set of assumptions. We want now to see how this works out in the formal version. We write here

$$\exists!y \phi \quad \text{as an abbreviation for} \quad \exists y [\phi \wedge \forall z (\phi_y(z) \rightarrow y \dot{=} z)],$$

and read it ‘there **exists a unique** y such that ϕ ’.

Definition 2.6.11 For any language L , any L -theory T and any $p \notin \text{Rs}_L$, $f \notin \text{Fs}_L$ and $c \notin \text{Cs}_L$,

- (i) a **T -definition for p** is a sentence of the form

$$\forall x_0 \dots \forall x_k [px_0 \dots x_k \leftrightarrow \theta_p],$$

where θ_p is an L -formula with at most $x_0, \dots, x_k$ free;

- (ii) a **T -definition for f** is a sentence of the form

$$\forall x_0 \dots \forall x_k \forall y [fx_0 \dots x_k \dot{=} y \leftrightarrow \theta_f],$$

where θ_f is an L -formula with at most $x_0, \dots, x_k, y$ free such that

$$T \models \forall x_0 \dots \forall x_k \exists!y \theta_f;$$

- (iii) a **T -definition for c** is a sentence of the form

$$\forall y [c \dot{=} y \leftrightarrow \theta_c],$$

where θ_c is an L -formula with at most y free such that $T \models \exists!y \theta_c$.

We refer to the extra condition in (ii) and (iii) as the **unique existence condition** and say that T **justifies** the definition.

Definition 2.6.12 An L^+ -theory T^+ is a (finite) **extension by definitions** or **definitional extension** of an L -theory T iff

- (i) L^+ is an extension of L by (finitely many) non-logical symbols;
- (ii) T^+ is the theory generated by T together with a T -definition for each of the new symbols.

Examples 2.6.13

- (i) Let L be the language with only $\dot{+}$ and $\dot{\times}$ as non-logical symbols and L^+ the language L_Ω which has also the additional symbols $\dot{<}$, $\dot{S}$, and $\dot{0}$. Let T be $\text{Th}(\omega, +, \times)$. Then the universal closures of the following are T -definitions:

$$\begin{aligned} \dot{0} \dot{=} y &\leftrightarrow y \dot{+} y \dot{=} y \\ \dot{S}x \dot{=} y &\leftrightarrow \exists z (\neg z \dot{+} z \dot{=} z \wedge z \dot{\times} z \dot{=} z \wedge y \dot{=} x \dot{+} z) \\ x \dot{<} y &\leftrightarrow \exists z (\neg z \dot{+} z \dot{=} z \wedge y \dot{=} x \dot{+} z). \end{aligned}$$

For example, the unique existence condition for $\dot{S}$ is

$$T \models \forall x \exists! y \exists z [(\neg z \dot{+} z \dot{=} z) \wedge (z \dot{\times} z \dot{=} z) \wedge (y \dot{=} x \dot{+} z)],$$

that is, this sentence is true in $(\omega, +, \times)$. This holds since 1 is the unique element n of ω such that $n + n \neq n$ and $n \times n = n$. Thus the theory T^+ generated by T together with these definitions is a finite extension by definitions of T .

- (ii) In Example 2.2.14 we wrote the axioms for group theory in three different languages. Let $T := \text{Th}(\{\theta_{\text{assoc}}, \theta_{\text{identinv}}\})$ be the theory formulated in the language with only the binary function symbol $\dot{*}$. Then (by an easy derivation beloved of all beginning students of group theory)

$$T \models \exists! z \forall x (x \dot{*} z \dot{=} x \wedge z \dot{*} x \dot{=} x),$$

so T justifies the extension by the definition

$$\dot{e} \dot{=} z \leftrightarrow \forall x (x \dot{*} z \dot{=} x \wedge z \dot{*} x \dot{=} x).$$

Similarly, T justifies the definition

$$x \dot{^{-1}} \dot{=} y \leftrightarrow x \dot{*} y \dot{=} \dot{e} \wedge y \dot{*} x \dot{=} \dot{e}.$$

Thus the theory formulated in the richer language is an extension by definitions of T .

- (iii) One of the most important uses of extensions by definitions occurs in set theory. Without defined symbols a sentence expressing even a very simple relationship, such as $x = \{\emptyset, \{\emptyset\}\}$, the usual set-theoretic representation of the number 2, is difficult to grasp:

$$\forall y (y \dot{\in} x \leftrightarrow \forall z [\neg z \dot{\in} y] \vee \forall z [z \dot{\in} y \leftrightarrow \forall u (\neg u \dot{\in} z)])$$

(not to speak of what this would look like if we wrote it also in terms of the primitive logical symbols $\neg$, $\vee$, and $\exists$). Almost any version of set theory is strong enough to justify the definitions of many function and constant symbols — for example

$$\begin{aligned} \dot{\emptyset} \dot{=} y &\leftrightarrow \forall z \neg z \dot{\in} y \\ \{x\} \dot{=} y &\leftrightarrow \forall z (z \dot{\in} y \leftrightarrow z \dot{=} x) \\ x_0 \dot{\cup} x_1 \dot{=} y &\leftrightarrow \forall z (z \dot{\in} y \leftrightarrow z \dot{\in} x_0 \vee z \dot{\in} x_1), \end{aligned}$$

and similarly for $\dot{\cap}$. Note here the use of the dot convention to distinguish formal symbols $\dot{\emptyset}$, $\dot{\cup}$, ... from the objects $\emptyset$, $\cup$, ... that they are intended to denote. The unique existence conditions follow from the relevant axioms (existence) and extensionality (uniqueness). We shall see in [Chapter 6](#) that stronger set theories justify the introduction of defined symbols for most of the common functions and relations of mathematics, although the approach we actually take will be slightly different.

Lemma 2.6.14 *For any L^+ -theory T^+ which is an extension by definitions of an L -theory T and any L -structure $\mathfrak{A}$ which is a model of T , there exists a unique expansion $\mathfrak{A}^+$ of $\mathfrak{A}$ to a L^+ -structure which is a model of T^+ .*

Proof. If a definition of a relation symbol p by a formula θ_p is among the new definitions, we set $p^{\mathfrak{A}^+}$ to be $\theta_p^{\mathfrak{A}}$, the $(k+1)$ -ary relation defined by θ over $\mathfrak{A}$. Then since (by Theorem 2.6.5) $\theta_p^{\mathfrak{A}^+} = \theta_p^{\mathfrak{A}}$, $\mathfrak{A}^+$ satisfies the defining axiom $\forall x_0 \dots \forall x_k [px_0 \dots x_k \leftrightarrow \theta_p]$ for p .

Similarly, if f is defined by θ_f , we define $f^{\mathfrak{A}^+}$ by

$$f^{\mathfrak{A}^+}(a_0, \dots, a_k) := \text{the unique } b \text{ such that } \models_{\mathfrak{A}} \theta_f[a_0, \dots, a_k, b],$$

which exists by the unique existence condition and the assumption that $\mathfrak{A}$ is a model of T . $\square$

Corollary 2.6.15 *T is consistent iff T^+ is consistent. $\square$*

The following is the precise expression of the intuition that definitions don't add anything essential; no sentence written with the original symbols is provable with the definitions but not without them.

Theorem 2.6.16 *For any L^+ -theory T^+ which is an extension by definitions of an L -theory T and any L -sentence ϕ ,*

$$T^+ \models \phi \iff T \models \phi;$$

that is, $T^+ \cap \text{Sent}_L = T$.

Proof. Since T is a subtheory of T^+ , the implication $(\Leftarrow)$ is immediate. Suppose that $T^+ \models \phi$, and let $\mathfrak{A}$ be any model of T . By the lemma, $\mathfrak{A}$ has an expansion to a model $\mathfrak{A}^+$ of T^+ , so by hypothesis, $\mathfrak{A}^+ \models \phi$. But $\mathfrak{A} = \mathfrak{A}^+ \upharpoonright L$, so also $\mathfrak{A} \models \phi$. Thus $T \models \phi$. $\square$

The conclusion of this theorem is summarized by saying that T^+ is **conservative over** T or is a **conservative extension** of T . Note that this would generally not be the case if we did not require the unique existence conditions, since in L^+ it is logically valid for any (say unary) function symbol f that $\forall x \exists! y (fx \doteq y)$. Thus if T^+ includes a definition of f by θ_f , we have also $T^+ \models \forall x \exists! y \theta_f$.

Another sense in which an extension by definitions is an ‘inessential’ extension corresponds to our intuition that defined symbols can always be eliminated by replacing them by their definitions. This is slightly more complicated than it might seem because of the fact that function symbols may be nested arbitrarily deeply in terms. A nice solution to this problem, which will also have other uses, is to show first that every formula is logically equivalent to one without nesting.

Definition 2.6.17 A formula ϕ is **unnested** iff every atomic formula which occurs in ϕ is of one of the forms $x \doteq y$, $px_0 \dots x_k$, $fx_0 \dots x_k \doteq y$ or $c \doteq y$.

Proposition 2.6.18 *For each formula ϕ , there exists a logically equivalent unnested formula ϕ^* .*

Proof. We define first unnested versions for atomic formulas of the special form $t \doteq z$, for z a variable not occurring in t , by term-recursion as follows. If t is a variable or a constant symbol, $t \doteq z$ is already unnested. Otherwise, recursively $(ft_0 \dots t_k \doteq z)^*$ is

$$\exists y_0 \dots \exists y_k [(t_0 \doteq y_0)^* \wedge \dots \wedge (t_k \doteq y_k)^* \wedge fy_0 \dots y_k \doteq z],$$

where, of course, the y_i are new variables. Then for any terms t and u ,

$$(t \doteq u)^* \text{ is } \exists z [(t \doteq z)^* \wedge (u \doteq z)^*] \text{ for a variable } z \text{ not in } t \text{ or } u,$$

and for a general atomic formula, $(pt_0 \dots t_k)^*$ is

$$\exists y_0 \dots \exists y_k [(t_0 \doteq z_0)^* \wedge \dots \wedge (t_k \doteq z_k)^* \wedge pz_0 \dots z_k].$$

The remaining steps are trivial: $(\neg \phi)^*$ is $\neg \phi^*$, $(\phi \vee \psi)^*$ is $\phi^* \vee \psi^*$, and $(\exists x \phi)^*$ is $\exists x \phi^*$. The proof of equivalence is a straightforward, if tedious, induction, which we leave to the devoted reader. $\square$

Proposition 2.6.19 *For any L^+ -theory T^+ which is an extension by definitions of an L -theory T and any L^+ -formula ϕ , there exists an L -formula ϕ^- such that*

$$T^+ \models \phi \leftrightarrow \phi^-.$$

ϕ^- is called the *L -translation* of ϕ .

Proof. For any L^+ -formula ϕ , let ϕ^* be a logically equivalent unnested formula. Then ϕ^- is the result of replacing in ϕ^* for each of the defined symbols

$$\begin{array}{lll} fx_0 \dots x_k \doteq y & \text{by} & \theta'_f(x_0, \dots, x_k, y), \\ px_0 \dots x_k & \text{by} & \theta'_p(x_0, \dots, x_k, y), \\ c \doteq y & \text{by} & \theta'_c(y), \end{array}$$

where θ'_f is an $x_0, \dots, x_k, y$ -variant of θ_f , which ensures that the substitution is free, and similarly for θ'_p and θ'_c . Now it follows by formula induction that $T^+ \models \phi \leftrightarrow \phi^-$. $\square$

Corollary 2.6.20 *For any L^+ -theory T^+ which is an extension by definitions of an L -theory T ,*

- (i) *for any L^+ -sentence ϕ , $T^+ \models \phi \iff T \models \phi^-$;*
- (ii) *for any model $\mathfrak{A}$ of T , any $\mathfrak{A}$ -assignment α and any L^+ -formula ϕ , $\models_{\mathfrak{A}^+} \phi[\alpha] \iff \models_{\mathfrak{A}} \phi^-[\alpha]$;*
- (iii) *for any models $\mathfrak{A}$ and $\mathfrak{B}$ of T , $\mathfrak{A} \equiv \mathfrak{B} \iff \mathfrak{A}^+ \equiv \mathfrak{B}^+$;*
- (iv) *if T is complete, then also T^+ is complete; hence if $\mathfrak{A}$ is any model of T and $\mathfrak{A}^+$ is the unique expansion to a model of T^+ , then $T^+ = \text{Th}(\mathfrak{A}^+)$.*

Proof. For (i), by the proposition and the fact that T^+ is conservative, we have

$$T^+ \models \phi \iff T^+ \models \phi^- \iff T \models \phi^-.$$

For (iv), since $(\neg \phi)^-$ is $\neg \phi^-$, we have for any L^+ -sentence ϕ ,

$$T \models \phi^- \text{ or } T \models \neg \phi^- \implies T^+ \models \phi \text{ or } T^+ \models \neg \phi,$$

so the completeness of T entails that of T^+ . If $\mathfrak{A}$ is a model of T , then $T^+ \subseteq \text{Th}(\mathfrak{A}^+)$, and since T^+ is complete, they must coincide. (ii) and (iii) are left as exercises. $\square$

The proposition (and its proof) have also important, if easy, consequences concerning decidability.

Corollary 2.6.21 *For any L^+ -theory T^+ which is a finite extension by definitions of an L -theory T , T^+ is decidable iff T is decidable.*

Proof. The proof of the proposition is effective in the sense that it provides an algorithm for the effective calculation of ϕ^- from ϕ . Hence, given a decision procedure for T , we have the following procedure for T^+ : given ϕ , compute ϕ^- and apply the T -procedure to ϕ^- . Part (i) of the corollary guarantees that the answer is correct. The other direction is immediate from conservatism: a decision procedure for T^+ works also for T . $\square$

Another consequence of the translation procedure is that sequential definitional extensions (as in the group theory example above) can be reduced to a single extension. Thus, if T^+ is a definitional extension of T and T^{++} a definitional extension of T^+ , we may replace the T^+ -definitions by T -definitions. For example, if $T^+ \models \forall x \exists! y \theta$, where θ may contain some of the new defined symbols of T^+ , then $T \models \forall x \exists! y \theta^-$ and we may make the T -definition $\forall x (fx \doteq y \leftrightarrow \theta^-)$. Since then $T^+ \models \forall x (fx \doteq y \leftrightarrow \theta)$, the effect is the same as the T^+ -definition.

There is another type of definitional extension which corresponds to the common practice in informal mathematics of introducing a name for an arbitrary member of a class of objects when we know that this class is non-empty but may be unable to define explicitly a member.

Definition 2.6.22 For any language L , any L -theory T and any $f \notin \text{Fs}_L$ and $c \notin \text{Cs}_L$,

- (i) a **T -pseudo-definition for f** is a sentence of the form

$$\forall x_0 \dots \forall x_k \theta_y (fx_0 \dots x_k, x_0, \dots, x_k),$$

where θ is an L -formula with at most $x_0, \dots, x_k, y$ free such that $T \models \forall x_0 \dots \forall x_k \exists y \theta$ and y is free for $fx_0 \dots x_k$ in θ ;

- (ii) a **T -pseudo-definition for c** is a sentence of the form $\theta_y(c)$, where θ is an L -formula with at most y free such that $T \models \exists y \theta$;
- (iii) an L^+ -theory T^+ is an (finite) **extension by pseudo-definitions** of an L -theory T iff L^+ is an extension of L by (finitely many) non-logical symbols and T^+ is the theory generated by T together with a T -pseudo-definition for each of the new function and constant symbols and a T -definition for each of the new relation symbols.

Theorem 2.6.23 *For any T and T^+ , if T^+ is an extension of T by pseudo-definitions, then T^+ is conservative over T .*

Proof. If T^+ is an extension of T by pseudo-definitions, then any model $\mathfrak{A}$ of T has a (not generally unique) expansion to a model $\mathfrak{A}^+$ of T^+ : if

$\theta_y(fx_0 \dots x_k)$ is a T -pseudo-definition for f , then we may define $f^{\mathfrak{A}^+}$ by

$$f^{\mathfrak{A}^+}(a_0, \dots, a_k) = \text{some } b \text{ such that } \models_{\mathfrak{A}} \theta[a_0, \dots, a_k, b].$$

Then we argue just as for extensions by definitions. $\square$

Remark 2.6.24

(i) Note that since

$$\models \forall x \exists! y \theta \rightarrow [\forall x \theta_y(fx) \leftrightarrow \forall x \forall y (fx \dot{=} y \leftrightarrow \theta)],$$

every T -definition is also a T -pseudo-definition.

(ii) The preceding proof makes use of the Axiom of Choice (AC) in the meta-theory in choosing ‘some b ’ corresponding to each sequence $a_0, \dots, a_k$. This is appropriate since the informal introduction of such a function clearly also in general requires AC. Note, however, that there is no use of AC in adjoining a new constant symbol by a pseudo-definition $\theta_y(c)$ since by Corollary 2.6.6, if ψ is a sentence in which c does not occur,

$$T \cup \{\theta_y(c)\} \models \psi \implies T \cup \{\exists y \theta\} \models \psi \implies T \models \psi.$$

These ideas have another set of interesting applications. Recall the proof of the Countable Downward Löwenheim-Skolem Theorem 2.3.34; for an L -structure $\mathfrak{B}$ and a formula ψ with at most variables $x_0, \dots, x_{k-1}, y$ free, we introduced Skolem functions $F_{\psi, y} : |\mathfrak{B}|^k \rightarrow |\mathfrak{B}|$ such that for any $b_0, \dots, b_{k-1} \in B$, $F_{\psi, y}(b_0, \dots, b_{k-1})$ is some element c of B such that $\models_{\mathfrak{B}} \psi[b_0, \dots, b_{k-1}, c]$ whenever such a c exists. The proof is then that for any substructure $\mathfrak{A} \subseteq \mathfrak{B}$, if $\mathfrak{A}$ is closed under all of these functions $F_{\psi, y}$, then $\mathfrak{A}$ is in fact an elementary substructure. An alternative way to view this construction is as follows. For each pair ψ, y of an L -formula and a variable, let $f_{\psi, y}$ be a new function symbol and denote by L^+ the extended language which includes all of these **Skolem function symbols**. For an L -structure $\mathfrak{B}$, let $\mathfrak{B}^+$ be the expansion to an L^+ -structure obtained by interpreting $f_{\psi, y}$ as $F_{\psi, y}$ defined as above. Then clearly $\mathfrak{B}^+$ is a model of all sentences of the form $\exists y \psi' \rightarrow \psi'_y(f_{\psi, y}x_0 \dots x_{k-1})$, where ψ' is an $x_0, \dots, x_{k-1}$ -variant of ψ , and the proof of the Downward Löwenheim-Skolem Theorem is essentially the observation that for any L^+ -structure $\mathfrak{A}^+$, if $\mathfrak{A} := \mathfrak{A}^+ \upharpoonright L$,

$$\mathfrak{A}^+ \subseteq \mathfrak{B}^+ \implies \mathfrak{A} \prec \mathfrak{B}.$$

Definition 2.6.25 A theory T is **Skolem complete** iff for every formula $\exists y \psi$ with at most $x_0, \dots, x_{k-1}, y$ free, there exists a term t in the variables $x_0, \dots, x_{k-1}$ freely substitutable for y in ψ such that $T \models \exists y \psi \rightarrow \psi_y(t)$. A structure is Skolem complete iff its theory is.

Proposition 2.6.26 *For any theory T and any models $\mathfrak{A}$ and $\mathfrak{B}$ of T , if T is Skolem complete and $\mathfrak{A} \subseteq \mathfrak{B}$, then $\mathfrak{A} \prec \mathfrak{B}$.*

Proof. We apply Tarski's Criterion 2.3.33. If α is an $\mathfrak{A}$ -assignment and $\models_{\mathfrak{B}} \exists y \psi[\alpha]$, then by the hypothesis for some term t , $\models_{\mathfrak{B}} \psi_y(t)[\alpha]$ (assuming the substitution is already free), so by Lemma 2.2.22, $\models_{\mathfrak{B}} \psi[\alpha_{y \mapsto a}]$ for $a = t^{\mathfrak{B}}[\alpha]$. But since $\mathfrak{A} \subseteq \mathfrak{B}$, also $a = t^{\mathfrak{A}}[\alpha] \in A$ as desired. $\square$

Proposition 2.6.27 *For any countable L -theory T , there exists a countable Skolem complete extension T^* such that every model of T has an expansion to a model of T^* .*

Proof. Let L_0 be L ; for each n , L_{n+1} is L_n^+ containing Skolem function symbols for L_n -formulas. Let T_0 be T ; T_{n+1} is the L_{n+1} -theory generated by T_n together with all formulas of the form $\exists y \psi \rightarrow \psi_y(f_{\psi,y}x_0 \dots x_{k-1})$ for ψ an L_n -formula. Let T^* be the union of the T_n . Clearly T^* is Skolem complete, and as sketched above it is straightforward to expand any model for T_n to a model for T_{n+1} . $\square$

Essentially the same ideas give also the following.

Theorem 2.6.28 *For each L -formula ϕ , there exists a formula ϕ^{Sk} in an extended language L^{Sk} , called a **Skolem normal form** of ϕ , such that*

- (i) ϕ^{Sk} is a universal formula with the same free variables as ϕ ;
- (ii) $\models \phi^{\text{Sk}} \rightarrow \phi$;
- (iii) for any L -theory T , $T \models \neg \phi \iff T \models \neg \phi^{\text{Sk}}$;
- (iv) for any L -theory T , $T \cup \{\phi\}$ is consistent iff $T \cup \{\phi^{\text{Sk}}\}$ is consistent.

Proof. Recall from Theorem 2.2.34 that each formula ϕ is logically equivalent to a formula in prenex normal form with all quantifiers at the beginning. For such a formula we may define recursively on the number of quantifiers

$$\phi^{\text{Sk}} = \begin{cases} \phi, & \text{if } \phi \text{ is quantifier-free;} \\ \forall y \psi^{\text{Sk}}, & \text{if } \phi \text{ is } \forall y \psi; \\ \psi^{\text{Sk}}_y(fx_0 \dots x_{k-1}), & \text{if } \phi \text{ is } \exists y \psi; \end{cases}$$

where in the last case, f is a new k -ary function symbol and the free variables of ψ^{Sk} are among $x_0, \dots, x_{k-1}, y$. For example, if θ is quantifier-free with x, y, z, u, v free, then

$$(\forall y \exists z \forall u \exists v \theta)^{\text{Sk}} = \forall y \forall u \exists v \theta^{\text{Sk}}_z(gxy) = \forall y \forall u \theta_v(fxyzu)_z(gxy).$$

Clearly ϕ^{Sk} is universal and a straightforward induction verifies that $\models \phi^{\text{Sk}} \rightarrow \phi$. It follows that $T \models \neg \phi \implies T \models \neg \phi^{\text{Sk}}$. For the converse,

suppose that $T \not\models \neg \phi$, so there is a model $\mathfrak{B}$ of T and $b_0, \dots, b_{k-1} \in B$ such that $\models_{\mathfrak{B}} \phi[b_0, \dots, b_{k-1}]$. By iterating the argument above, $\mathfrak{B}$ has an expansion $\mathfrak{B}^*$ such that $\models_{\mathfrak{B}^*} \phi^{\text{Sk}}[b_0, \dots, b_{k-1}]$; clearly $\mathfrak{B}^*$ still satisfies T , so $T \not\models \neg \phi^{\text{Sk}}$.

Finally, the implication ($\Leftarrow$) of (iv) follows immediately from (ii), and the converse is proved much as was (iii). Note that (iv) follows from (iii) if ϕ is a sentence, but not in general. $\square$

Corollary 2.6.29 *Every theory T has a **Skoletm extension** T^{Sk} which is universally axiomatizable and conservative over T .*

Proof. Let T^{Sk} be the theory generated by $\{\phi^{\text{Sk}} : \phi \in T\}$. By (i) and (ii) of the theorem, T^{Sk} is universally axiomatized and an extension of T . To see that T^{Sk} is conservative over T , suppose that ψ is an L -sentence such that $T^{\text{Sk}} \models \psi$ and fix a model $\mathfrak{B}$ of T . As in the proof of the Theorem, there exists an expansion $\mathfrak{B}^*$ which is a model of T^{Sk} , hence a model of ψ . But then also $\mathfrak{B}$ is a model of ψ . $\square$

There is also a useful dual form of the Skolem construction.

Corollary 2.6.30 *For any L -formula ϕ there exists a formula ϕ^{Hb} in an extended language L^{Hb} , called a **Herbrand normal form** of ϕ , such that*

- (i) ϕ^{Hb} is a logically existential formula with the same free variables as ϕ ;
- (ii) $\models \phi \rightarrow \phi^{\text{Hb}}$;
- (iii) for any L -theory T , $T \models \phi \iff T \models \phi^{\text{Hb}}$.

Proof. We set $\phi^{\text{Hb}} := \neg (\neg \phi)^{\text{Sk}}$, where $(\neg \phi)^{\text{Sk}}$ is any Skolem normal form of $\neg \phi$. (i)–(iii) are then immediate from the corresponding parts of the preceding theorem. $\square$

We shall have further use for these constructions in [Chapter 3](#) and [Chapter 7](#). In a few contexts it is useful to avoid function and constant symbols altogether; the final result of this subsection shows that these can be replaced with relation symbols.

Definition 2.6.31 For any language L ,

- (i) the **relational translation** L^{rel} of L is the language obtained from L by replacing each k -ary function symbol f by a new $(k+1)$ -ary relation symbol p_f and each constant symbol c by a unary relation symbol p_c ;
- (ii) for any L -structure $\mathfrak{A}$, $\mathfrak{A}^{\text{rel}}$ is the L^{rel} -structure which agrees with $\mathfrak{A}$ on all common symbols and for each L -function symbol f ,

$$p_f^{\mathfrak{A}^{\text{rel}}} = \text{Gr}_{f^{\mathfrak{A}}} := \{ (a_0, \dots, a_{k-1}, b) : f^{\mathfrak{A}}(a_0, \dots, a_{k-1}) = b \},$$

the **graph** of $f^{\mathfrak{A}}$, and for each L -constant symbol c , $p_c^{\mathfrak{A}^{\text{rel}}} = \{ c^{\mathfrak{A}} \}$.

Proposition 2.6.32 *For any language L and every L -formula ϕ , there exists an L^{rel} -formula ϕ^{rel} such that for any L -structure $\mathfrak{A}$ and any $\mathfrak{A}$ -assignment α ,*

$$\models_{\mathfrak{A}} \phi[\alpha] \iff \models_{\mathfrak{A}^{\text{rel}}} \phi^{\text{rel}}[\alpha].$$

Proof. Given ϕ , let ϕ^* denote an unnested formula logically equivalent to ϕ . Then ϕ^{rel} is the result of replacing in ϕ^*

$$\begin{aligned} f x_0 \dots x_{k-1} &\doteq y && \text{by } p_f x_0 \dots x_{k-1} y \\ c &\doteq y && \text{by } p_c y. \end{aligned}$$

The equivalence is then immediate from Proposition 2.6.18. $\square$

Interpretations

Suppose that T_1 and T_2 are theories in languages L_1 and L_2 , respectively. How can we compare the strength of the two theories? For example, $\text{Th}((\omega, \text{Sc}, \times))$ would appear weaker than $\text{Th}((\omega, +, \times))$, since successor is certainly a simpler operation than addition, but is it really? Even less obvious is the comparison of theories from different areas, such as arithmetic and set theory.

The notion that one theory T_1 may be (relatively) interpreted in another T_2 to be developed here provides a precise formulation of ‘ T_2 is at least as strong as T_1 ’. The idea is that T_2 should be able to justify the definitions of the basic non-logical notions of T_1 and derive about them all of the theorems of T_1 . Thus, to interpret say one of the theories T_1 of arithmetic from the preceding section in a system of set theory T_2 , we would need to give definitions of $+$ and $\times$ in terms of sets and membership in such a way that the axioms of T_1 are theorems of T_2 together with the definitions. Of course, $+$ and $\times$ apply to numbers, not sets, so we shall want also to be able to define in T_2 a notion of number.

Definition 2.6.33 An L_1 -theory T_1 is *interpretable* in an L_2 -theory T_2 iff there is a finite extension by definitions T_2^+ of T_2 such that $T_1 \subseteq T_2^+$.

The restriction that the extension be finite is not essential for most results, but is used in connection with decision questions. This notion may also be stated

Proposition 2.6.34 *T_1 is interpretable in T_2 iff there is a finite extension by definitions T_2^+ of T_2 such that for all L_1 -sentences ϕ ,*

$$T_1 \models \phi \implies T_2 \models \phi^-,$$

where ϕ^- is the T_2 -translation of ϕ .

Proof. Immediate from Proposition 2.6.19. $\square$

Note that the extension T_2^+ provides definitions only for the symbols of L_1 not in L_2 ; if some symbols appear in both languages but have ‘different meanings’ in the two theories, we may need to change the symbols of one of the languages.

Examples 2.6.35

- (i) Let T_1 be $\text{Th}(\Omega)$ and T_2 be $\text{Th}((\omega, +, \times))$. In Example 2.6.13(i) we specified an extension T_2^+ , and we claim that $T_1 \subseteq T_2^+$ so that this extension witnesses that T_1 is interpretable in T_2 . In fact, because the definitions are all true in Ω , the expansion $(\omega, +, \times)^+$ is exactly Ω and we have by Corollary 2.6.20 (ii),

$$\text{Th}(\Omega) = \text{Th}((\omega, +, \times)^+) = \text{Th}((\omega, +, \times))^+.$$

- (ii) We can settle the question posed above by showing that

$$\text{Th}((\omega, +, \times)) \text{ is interpretable in } \text{Th}((\omega, \text{Sc}, \times)).$$

By the same argument as in (i), it suffices to give a correct definition of $+$ in terms of Sc and $\times$. This is provided by

$$\begin{aligned} x + y = z &\leftrightarrow \dot{\text{S}}(x \dot{\times} y) = \dot{\text{S}}x \dot{\times} \dot{\text{S}}y \wedge z \dot{\times} \dot{\text{S}}z = z \\ &\vee \dot{\text{S}}(x \dot{\times} z) \dot{\times} \dot{\text{S}}(y \dot{\times} z) = \dot{\text{S}}((z \dot{\times} z) \dot{\times} \dot{\text{S}}(x \dot{\times} y)). \end{aligned}$$

A useful way to regard the interpretation of T_1 in T_2 is as a method for converting a model of T_2 into a model for T_1 .

Proposition 2.6.36 *For any theories T_1 and T_2 , if T_2^+ is an extension by definitions of T_2 such that $T_1 \subseteq T_2^+$, $\mathfrak{A}_2$ is any model of T_2 and $\mathfrak{A}_2^+$ is its expansion to a model of T_2^+ , then $\mathfrak{A}_2^+ \upharpoonright L_1$ is a model of T_1 .*

Proof. We have by Lemma 2.6.14, $T_1 \subseteq T_2^+ \cap \text{Fm}_{L_1} \subseteq \text{Th}(\mathfrak{A}_2^+) \cap \text{Fm}_{L_1} = \text{Th}(\mathfrak{A}_2^+ \upharpoonright L_1)$. $\square$

Corollary 2.6.37 *If T_1 is interpretable in T_2 and T_2 is consistent, then also T_1 is consistent.* $\square$

Note in passing that every theory is interpretable in an inconsistent theory. As regards decidability, we have only the seemingly rather weak

Proposition 2.6.38 *If T_1 is interpretable in T_2 and some consistent extension of T_2 is decidable, then some consistent extension of T_1 is decidable.*

Proof. Let U be a consistent decidable extension of T_2 . Since under the hypotheses $T_1 \subseteq T_2^+ \subseteq U^+$, also $T_1 \subseteq U^+ \cap \text{Fm}_{L_1}$. Easily $U^+ \cap \text{Fm}_{L_1}$ is a consistent theory. Since U^+ is a finite extension by definitions of U , it is decidable and thus by restricting the decision algorithm to Fm_{L_1} , we have a decision procedure for $U^+ \cap \text{Fm}_{L_1}$. $\square$

This result will be applied in [Section 5.1](#) and [Section 5.5](#) with the theory Q (see [Section 4.1](#)) in the role of T_1 . Q has the property that no consistent extension is decidable, so it follows that the same is true for any theory T_2 in which Q is interpretable.

This result also suggests a strengthening of the notion of interpretation.

Definition 2.6.39 T_1 is *faithfully interpretable* in T_2 iff for some finite extension by definitions T_2^+ of T_2 , $T_1 = T_2^+ \cap \text{Fm}_{L_1}$ — equivalently, for any L_1 -sentence ϕ , $T_1 \models \phi \iff T_2 \models \phi^-$.

Note that if T_1 is complete, then any interpretation of T_1 in a consistent theory is faithful.

Corollary 2.6.40 *If T_1 is faithfully interpretable in T_2 and T_2 is decidable, then also T_1 is decidable. $\square$*

Example 2.6.41 Let $T_1 = \text{Th}((\mathbb{Z}, \text{Sc}, +, 0))$ and $T_2 = \text{Th}((\omega, \text{Sc}, +, 0))$. We know (Corollary 2.5.19) that T_2 is decidable, and we now sketch a faithful interpretation of T_1 in T_2 and conclude that also T_1 is decidable. We assume (by changing them if necessary) that the two languages have distinct successor and addition symbols, say L_1 uses $\dot{\text{S}}_{\mathbb{Z}}$ and $\dot{+}_{\mathbb{Z}}$ while L_2 has the usual $\dot{\text{S}}$ and $\dot{+}$. We shall define functions $\text{Sc}_{\mathbb{Z}}$ and $+_{\mathbb{Z}}$ over $(\omega, \text{Sc}, +, 0)$ in such a way that

$$(\mathbb{Z}, \text{Sc}, +, 0) \simeq (\omega, \text{Sc}_{\mathbb{Z}}, +_{\mathbb{Z}}, 0),$$

so in particular these structures are elementarily equivalent. Then if the corresponding definitions are used to extend T_2 to T_2^+ , we have easily $T_1 = T_2^+ \cap \text{Fm}_{L_1}$. Indeed, the definitions are guided by building the isomorphism η . We shall want $\eta(0) = 0$ and for $m > 0$, $\eta(m) = 2m$ and $\eta(-m) = 2m - 1$. Then we are naturally led to define $+_{\mathbb{Z}}$ as follows: $2m +_{\mathbb{Z}} 2n = 2(m + n)$; $(2m - 1) +_{\mathbb{Z}} (2n - 1) = 2(m + n) - 1$; and

$$(2m - 1) +_{\mathbb{Z}} 2n = 2n +_{\mathbb{Z}} 2m - 1 = \begin{cases} 2(n - m), & \text{if } m < n; \\ 0, & \text{if } m = n; \\ 2(m - n) - 1, & \text{if } m > n. \end{cases}$$

Similarly we set $\text{Sc}_{\mathbb{Z}}(1) = 0$, $\text{Sc}_{\mathbb{Z}}(0) = 2$ and, for $m > 1$,

$$\text{Sc}_{\mathbb{Z}}(m) = \begin{cases} m + 2, & \text{if } m \text{ is even;} \\ m - 2, & \text{if } m \text{ is odd.} \end{cases}$$

We leave it to the reader to carry out the remaining details.

We want next to develop the notion of *relative* interpretation, which is somewhat more widely applicable. Roughly speaking, a relative interpretation of T_1 in T_2 provides a method which to any model $\mathfrak{A}_2$ of T_2 assigns a substructure $\mathfrak{A}_1 \subseteq \mathfrak{A}_2^+ \upharpoonright L_1$ which is a model of T_1 . The advantage is that the functions of L_1 do not need to satisfy the requirements of T_1 over the whole universe of $\mathfrak{A}_2$ but only on a subset. For example, if T_1 is a theory of arithmetic, T_2 a set theory and $\mathfrak{A}_2$ some universe of sets, we want to extract a subuniverse consisting of those sets which ‘behave’ like numbers. The technical device for doing this is

Definition 2.6.42 For any language L and any unary relation symbol M not in L , we define $\phi^{(M)}$, the **relativization** of ϕ to M , recursively by

- (i) for ϕ atomic, $\phi^{(M)}$ is ϕ ;
- (ii) $(\neg \psi)^{(M)}$ is $\neg \psi^{(M)}$;
- (iii) $(\psi \vee \theta)^{(M)}$ is $\psi^{(M)} \vee \theta^{(M)}$;
- (iv) $(\exists x \psi)^{(M)}$ is $\exists x (Mx \wedge \psi^{(M)})$.

First note that this definition has the expected effect with respect to the defined connectives: $(\psi \wedge \theta)^{(M)}$ is $(\psi^{(M)} \wedge \theta^{(M)})$, and $(\forall x \psi)^{(M)}$ is logically equivalent to $\forall x (Mx \rightarrow \psi^{(M)})$. We shall sometimes write

$$\exists^M x \psi \quad \text{for} \quad \exists x (Mx \wedge \psi) \quad \text{and} \quad \forall^M x \psi \quad \text{for} \quad \forall x (Mx \rightarrow \psi).$$

Let $L(M)$ denote the extension of L obtained by adjoining the unary relation symbol M .

Definition 2.6.43

- (i) For any L -theory T , $T^{(M)}$ is the $L(M)$ -theory generated by $\{\phi^{(M)} : \phi \in T\}$;
- (ii) for any theories T_1 and T_2 , T_1 is **relatively interpretable** in T_2 iff for a unary relation symbol M not in L_1 or L_2 , $T_1^{(M)}$ is interpretable in T_2 .

If $\mathfrak{A}$ is an L -structure, we write $(\mathfrak{A}; M)$ for its expansion to an $L(M)$ -structure with M interpreted as M . By Proposition 2.3.15, if M is non-empty, contains all $c^{\mathfrak{A}}$ for $c \in \text{Cs}_L$ and is closed under all $f^{\mathfrak{A}}$ for $f \in \text{Fs}_L$,

then M is the universe of a unique substructure $\mathfrak{M}$ of $\mathfrak{A}$. Clearly this is true just in case $(\mathfrak{A}; M)$ is a model of the following set of $L(\mathbf{M})$ -sentences:

$$\begin{aligned} \Delta_{\mathbf{M}} := & \{ \exists x \mathbf{M}x \} \cup \{ \mathbf{M}c : c \in \mathbf{C}s_L \} \\ & \cup \{ \forall x_0 \dots \forall x_{\nu_L(f)-1} [\mathbf{M}x_0 \wedge \dots \wedge \mathbf{M}x_{\nu_L(f)-1} \\ & \quad \rightarrow \mathbf{M}f x_0 \dots x_{\nu_L(f)-1} : f \in \mathbf{F}s_L] \}. \end{aligned}$$

Proposition 2.6.44 *If $(\mathfrak{A}; M)$ is a model of $\Delta_{\mathbf{M}}$, then for any L -formula ϕ and any $\mathfrak{M}$ -assignment α ,*

$$\models_{(\mathfrak{A}; M)} \phi^{(\mathbf{M})}[\alpha] \iff \models_{\mathfrak{M}} \phi[\alpha].$$

Proof. We proceed by induction on ϕ . If ϕ is atomic, we have

$$\models_{(\mathfrak{A}; M)} \phi^{(\mathbf{M})}[\alpha] \iff \models_{\mathfrak{A}} \phi^{(\mathbf{M})}[\alpha] \iff \models_{\mathfrak{M}} \phi[\alpha].$$

If ϕ is $\neg \psi$, then

$$\begin{aligned} \models_{(\mathfrak{A}; M)} \phi^{(\mathbf{M})}[\alpha] & \iff \not\models_{(\mathfrak{A}; M)} \psi^{(\mathbf{M})}[\alpha] \\ & \iff \not\models_{\mathfrak{M}} \psi[\alpha] \iff \models_{\mathfrak{M}} \phi[\alpha], \end{aligned}$$

and similarly for the case that ϕ is $\psi \vee \theta$. Finally, if ϕ is $\exists x \psi$,

$$\begin{aligned} \models_{(\mathfrak{A}; M)} \phi^{(\mathbf{M})}[\alpha] & \iff \models_{(\mathfrak{A}; M)} \exists x (\mathbf{M}x \wedge \psi^{(\mathbf{M})})[\alpha] \\ & \iff \text{for some } a \in M, \models_{(\mathfrak{A}; M)} \psi^{(\mathbf{M})}[\alpha_{x \mapsto a}] \\ & \iff \text{for some } a \in M, \models_{\mathfrak{M}} \psi[\alpha_{x \mapsto a}] \\ & \iff \models_{\mathfrak{M}} \phi[\alpha]. \quad \square \end{aligned}$$

Corollary 2.6.45 *For any L -theory T and any set Γ of axioms for T , $\Delta_{\mathbf{M}} \cup \{ \phi^{(\mathbf{M})} : \phi \in \Gamma \}$ is a set of axioms for $T^{(\mathbf{M})}$.*

Proof. First we note that $\Delta_{\mathbf{M}} \subseteq T^{(\mathbf{M})}$. Since $\exists x (x \doteq x)$ is logically valid and hence belongs to T , its relativization $\exists x (\mathbf{M}x \wedge x \doteq x)$ belongs to $T^{(\mathbf{M})}$, hence so does its logical equivalent $\exists x \mathbf{M}x$. Similarly, for any constant symbol c , $\exists x (x \doteq c)$ belongs to T so $\mathbf{M}c$, which is logically equivalent to its relativization, does also. The argument for function symbols is analogous. Now suppose that $(\mathfrak{A}; M)$ is a model of $\Delta_{\mathbf{M}} \cup \{ \phi^{(\mathbf{M})} : \phi \in \Gamma \}$. By the proposition, $\mathfrak{M}$ is substructure of $\mathfrak{A}$ which is a model of each $\phi \in \Gamma$ and hence is a model of T . Then again by the proposition, $(\mathfrak{A}, M)$ is a model of $T^{(\mathbf{M})}$. $\square$

Corollary 2.6.46 *For any L -theory T and any $L(\mathbf{M})$ -structure $(\mathfrak{A}; M)$, if $(\mathfrak{A}; M)$ is a model of $T^{(\mathbf{M})}$, then M is the universe of a unique substructure $\mathfrak{M} \subseteq \mathfrak{A}$ and $\mathfrak{M}$ is a model of T .*

Proof. Immediate. $\square$

Thus if T_1 is relatively interpretable in T_2 , there is an extension by definitions T_2^{++} of T_2 in a language $L_2^+(\mathbf{M})$ (we think of first extending by defining the L_1 -symbols, then again to define M) such that $T_1^{(\mathbf{M})} \subseteq T_2^{++}$. Now if $\mathfrak{A}_2$ is any model of T_2 , its $L_2^+(\mathbf{M})$ -expansion $(\mathfrak{A}_2^+, M_2)$ is a model of T_2^{++} , so by the corollary, $\mathfrak{M}_2 \upharpoonright L_1$ is a model of T_1 . In particular,

Corollary 2.6.47 *If a theory T_1 is relatively interpretable in a theory T_2 and T_2 is consistent, then also T_1 is consistent.* $\square$

To draw the same decidability conclusions for relative interpretability as we did for interpretability, we need a technical lemma.

Lemma 2.6.48 *For any L -theory T and any $L(\mathbf{M})$ -theory V , if*

$$T^{(\mathbf{M})} \subseteq V \quad \text{and} \quad U = \left\{ \phi : \phi^{(\mathbf{M})} \in V \right\},$$

then

- (i) U is an L -theory extending T ;
- (ii) if $V = T^{(\mathbf{M})}$, then $U = T$;
- (iii) if V is decidable, then also U is decidable.

Proof. For (i), since $\phi \in T \implies \phi^{(\mathbf{M})} \in T^{(\mathbf{M})} \subseteq V \implies \phi \in U$, clearly $T \subseteq U$. To see that U is a theory, suppose that $U \models \psi$. Now if $(\mathfrak{A}; M)$ is any model of V , then $\mathfrak{M}$ is a model of U , so $\models_{\mathfrak{M}} \psi$ and thus $\models_{(\mathfrak{A}; M)} \psi^{(\mathbf{M})}$. Hence $V \models \psi^{(\mathbf{M})}$, so $\psi^{(\mathbf{M})} \in V$ and $\psi \in U$.

For (ii), suppose that $\phi \in U$ and $\mathfrak{A}$ is any model of T . Then $(\mathfrak{A}; A)$ is a model of $T^{(\mathbf{M})} = V$, so $\models_{(\mathfrak{A}; A)} \phi^{(\mathbf{M})}$ and thus $\models_{\mathfrak{A}} \phi$. Hence $T \models \phi$ so $\phi \in U$, and we have $U \subseteq T$. For (iii), if V is decidable, then given an L -sentence ϕ , to decide whether or not $\phi \in U$, we effectively calculate $\phi^{(\mathbf{M})}$ and apply the decision procedure for V . $\square$

Proposition 2.6.49 *If T_1 is relatively interpretable in T_2 and some consistent extension of T_2 is decidable, then also some consistent extension of T_1 is decidable.*

Proof. Suppose that W is a decidable extension of T_2 and $T_1^{(\mathbf{M})} \subseteq T_2^+$. Then $W^+ \cap \text{Fm}_{L_1(\mathbf{M})}$ is a decidable extension of $T_1^{(\mathbf{M})}$, so

$$U := \left\{ \phi : \phi^{(\mathbf{M})} \in W^+ \right\}$$

is a decidable extension of T_1 . $\square$

Exercises

Exercise 2.6.50 Prove Proposition 2.6.4.

Exercise 2.6.51 Complete the proof of Proposition 2.6.19.

Exercise 2.6.52 Let L be any language, L_c the extension by a single constant symbol c , T any L -theory and U any L_c -theory. Show that if for each $\phi(c) \in U$, $T \cup \{\exists x \phi(x)\}$ is consistent, then $T \cup U$ is consistent.

Exercise 2.6.53 Let Γ be any set of L -sentences and ψ an L -sentence. Find a set Δ of L^{rel} -sentences such that

$$\Gamma \models \psi \iff \{ \phi^{\text{rel}} : \phi \in \Gamma \} \cup \Delta \models \psi^{\text{rel}}.$$

Exercise 2.6.54 Prove parts (ii) and (iii) of Corollary 2.6.20.

Exercise 2.6.55 With T and T^+ as in Example 2.6.13 (i), write out the L -translations of the successor axioms (S1) and (S2).

Exercise 2.6.56 Show that if T is a finitely or decidable axiomatizable theory, then so is $T^{(\text{M})}$.

Exercise 2.6.57 Carry out the remaining details of Example 2.6.41. Then give an interpretation of $\text{Th}((\mathbb{Z}, <, +, \times))$ in $\text{Th}((\omega, <, +, \times))$ in the same style.

Exercise 2.6.58 Show that if T_1 is interpretable in T_2 and $T := T_2^+ \cap \text{Fm}_{L_1}$, then T is a theory.

Chapter 3

Completeness and Compactness

In many ways the central focus of at least the elementary parts of Mathematical Logic is the logical consequence relation: $\Gamma \models \psi$. Of course, this relation is designed to mirror the informal mathematical notion that a statement ψ *follows* from a set Γ of *assumptions* or *axioms*. A major goal of this chapter is to establish the two central properties of this relation, the Compactness Theorem (CT) — for any language L , any set Γ of L -formulas and any L -formula ψ ,

$$\Gamma \models \psi \quad \Longleftrightarrow \quad \text{for some finite } \Gamma_0 \subseteq \Gamma, \quad \Gamma_0 \models \psi$$

— and the Enumerability Theorem (ET) — for any effective language L and any set Γ of L -formulas,

$$\Gamma \text{ is effectively enumerable} \implies \text{Th}(\Gamma) \text{ is effectively enumerable.}$$

Before proceeding much further, the reader should review the propositional versions of these results (Theorems 1.4.20 and 1.5.18).

[Section 3.1](#) is devoted to a proof of the Compactness Theorem for countable languages (CCT) and a first proof of the Enumerability Theorem (ET). In [Section 3.2](#) we develop a notion of formal proof and prove another central

result, the Countable Completeness Theorem, from which we get a second proof of both CCT and ET. In [Sections 3.3–3.4](#) we extend the methods of [Sections 1.6–1.7](#) to give other proofs of these results, including a proofs of the full Compactness and Completeness Theorems for arbitrary languages. [Section 3.5](#) explores some applications of the Compactness Theorem in first-order languages, while [Sections 3.6–3.7](#) are devoted to a survey of some logics which go beyond first-order logic in expressive power.

3.1 Countable compactness

Although, as mentioned above, we can (and will) derive the Countable Compactness Theorem from the Countable Completeness Theorem of [Section 3.2](#), it is useful to give a separate proof, because it allow us to develop many of the ideas needed there in a somewhat simpler context. Recall that a language is countable iff the set of non-logical symbols is countable and that this entails that the sets of terms, formulas and sentences are all countable. For the record, we state formally

Countable Compactness Theorem 3.1.1 (CCT) *For any countable language L , any set Γ of L -formulas and any L -formula ψ , $\Gamma \models \psi$ iff for some finite subset $\Gamma_0 \subseteq \Gamma$, $\Gamma_0 \models \psi$.*

Analogously with the presentation in [Section 1.4](#), we set

$$\begin{aligned}\Gamma \models^* \psi &: \Longleftrightarrow \text{ for some finite subset } \Gamma_0 \subseteq \Gamma, \quad \Gamma_0 \models \psi; \\ \text{Th}^*(\Gamma) &:= \{ \psi : \psi \text{ is a sentence and } \Gamma \models^* \psi \}.\end{aligned}$$

Then the CCT asserts that $\Gamma \models \psi \Longleftrightarrow \Gamma \models^* \psi$ and in particular, for sentences, $\text{Th}(\Gamma) = \text{Th}^*(\Gamma)$. We call Γ **finitely consistent** iff every finite subset $\Gamma_0 \subseteq \Gamma$ is consistent. It is immediate from the definitions that every consistent set is finitely consistent, and we approach CCT by way of the

Alternative Countable Compactness Theorem 3.1.2 (ACCT) *For any countable language L and any set Γ of L -formulas, if Γ is finitely consistent, then Γ is consistent.*

We first derive CCT from ACCT much as in [Section 1.4](#). Note first

Lemma 3.1.3 *For any set Γ of formulas and any formula ψ ,*

- (i) $\Gamma \models^* \psi \Longleftrightarrow \Gamma \cup \{\neg \psi\}$ is not finitely consistent;
- (ii) $\Gamma \models^* \neg \psi \Longleftrightarrow \Gamma \cup \{\psi\}$ is not finitely consistent.

Proof. For (i), using Proposition 2.2.3,

$$\begin{aligned}\Gamma \models^* \psi &\Longleftrightarrow \text{ for some finite } \Gamma_0 \subseteq \Gamma, \quad \Gamma_0 \models \psi \\ &\Longleftrightarrow \text{ for some finite } \Gamma_0 \subseteq \Gamma, \quad \Gamma_0 \cup \{\neg \psi\} \text{ is inconsistent} \\ &\Longleftrightarrow \Gamma \cup \{\neg \psi\} \text{ is not finitely consistent.}\end{aligned}$$

Part (ii) is proved similarly. $\square$

Proof. (of CCT from ACCT) Assume ACCT and that $\Gamma \models \psi$. Then by Proposition 2.2.3, $\Gamma \cup \{\neg \psi\}$ is inconsistent, so by ACCT it is not finitely consistent. Hence by the Lemma, $\Gamma \models^* \psi$. Of course, the converse implication is trivial (by Proposition 2.2.2). $\square$

We head now for a proof of ACCT. The first part of the proof is entirely parallel to the corresponding parts of the proof of the propositional version APCT in [Section 1.4](#).

Proposition 3.1.4 *For any sets Γ and Δ of L -formulas and any L -formulas ϕ and ψ ,*

- (i) *if $\phi \in \Gamma$, then $\Gamma \models^* \phi$;*
- (ii) *if $\Gamma \subseteq \Delta$ and $\Gamma \models^* \psi$, then also $\Delta \models^* \psi$;*
- (iii) *if $\text{Th}^*(\Gamma) \models^* \psi$, then $\Gamma \models^* \psi$;*
- (iv) *if $\text{Th}^*(\Gamma) \models \psi$, in particular if $\Gamma \models \psi$, then $\Gamma \models^* \psi$.*

We refer to (iv) as the **Tautology Principle**.

Proof. Parts (i) and (ii) are immediate, and we leave (iii) as an exercise (it is never used). For (iv), if $\text{Th}^*(\Gamma) \models \psi$, then by the Propositional Compactness Theorem 1.4.20 there exist $\phi_0, \dots, \phi_{n-1} \in \text{Th}^*(\Gamma)$ such that $\{\phi_0, \dots, \phi_{n-1}\} \models \psi$. In particular, by Proposition 2.2.4, also $\{\phi_0, \dots, \phi_{n-1}\} \models^* \psi$. By the definition of $\text{Th}^*(\Gamma)$, there exist finite subsets $\Gamma_0, \dots, \Gamma_{n-1} \subseteq \Gamma$ such that for $i < n$, $\Gamma_i \models \phi_i$. Then $\Gamma_0 \cup \dots \cup \Gamma_{n-1} \models \psi$ and thus witnesses that $\Gamma \models^* \psi$. The version with hypothesis $\Gamma \models \psi$ is similar but easier. $\square$

Proposition 3.1.5 *For any set Γ of formulas, the following are equivalent:*

- (i) *Γ is not finitely consistent;*
- (ii) *for all formulas ϕ , $\Gamma \models^* \phi$;*
- (iii) *for some (every) formulas ϕ , both $\Gamma \models^* \phi$ and $\Gamma \models^* \neg \phi$;*
- (iv) *$\Gamma \models^* \text{F}$.*

Proof. Given (i), let $\Gamma_0 \subseteq \Gamma$ be inconsistent. Then by 1.4.17, for every sentence ϕ , $\Gamma_0 \models \phi$, so $\Gamma \models^* \phi$. (iii) follows trivially from (ii), and (iv) follows from (iii) by the Tautology Principle, since $\{\phi, \neg \phi\} \models \text{F}$. Finally, if $\Gamma \models^* \text{F}$, then for some finite $\Gamma_0 \subseteq \Gamma$, $\Gamma_0 \models \text{F}$, so Γ_0 is inconsistent and hence Γ is not finitely consistent. $\square$

Corollary 3.1.6 *For any set Γ of formulas and any formula θ , if Γ is finitely consistent, then at least one of $\Gamma \cup \{\theta\}$ and $\Gamma \cup \{\neg \theta\}$ is finitely consistent.*

Proof. Immediate from the preceding proposition and Lemma 3.1.3. $\square$

In the propositional case, we were at this point able to proceed immediately to extend a finitely consistent set Γ to a finitely consistent and complete set Δ and define directly a model (truth assignment) for Δ and hence for Γ . However, the notion of model here is much more complex, and the construction requires substantially more work. We shall at first restrict attention to (sets of) sentences and introduce first an enhanced notion of completeness.

Definition 3.1.7 A set Δ of sentences is ***L-Henkin complete*** iff Δ is *L*-complete and for every *L*-sentence $\exists x \chi \in \Delta$, there is a $c \in \text{Cs}_L$ such that also $\chi_x(c) \in \Delta$. The constant symbol c is called a ***Henkin witness*** for the existential statement $\exists x \chi$.

Proposition 3.1.8 *For any set Γ of sentences and any sentence of the form $\exists x \chi$ in Γ , if Γ is finitely consistent, then for any constant symbol c which does not occur in any member of Γ , $\Gamma \cup \{\chi_x(c)\}$ is finitely consistent.*

Proof. Suppose that Γ , $\exists x \chi$ and c satisfy the hypotheses but $\Gamma \cup \{\chi_x(c)\}$ is not finitely consistent. By Lemma 3.1.3, $\Gamma \models^* \neg \chi_x(c)$. Then by Corollary 2.6.6, $\Gamma \models^* \forall x \neg \chi$, so $\Gamma \models^* \neg \exists x \chi$. But then $\Gamma = \Gamma \cup \{\exists x \chi\}$ is not finitely consistent, contrary to hypothesis. $\square$

The proof now proceeds in the following two steps: (1) every finitely consistent set Γ of sentences can be extended to a finitely consistent and Henkin complete set Δ , and (2) every Henkin complete set has a model. Generally we cannot hope to carry out step (1) in the original language *L* of Γ , since there may not be enough (or any!) constant symbols in *L*, so we shall need first to extend the language by adjoining a countable set of new constant symbols.

Proposition 3.1.9 *For any countable language *L* and any finitely consistent set Γ of *L*-sentences, there exists a countable extension L^+ of *L* by constants and a set $\Delta \supseteq \Gamma$ of L^+ -sentences which is finitely consistent and L^+ -Henkin complete.*

Proof. Let *L* and Γ be as in the hypothesis, and form L^+ by adjoining a new set of constant symbols $\{e_n : n \in \omega\}$. Fix an enumeration $\langle \theta_n : n \in \omega \rangle$ of the denumerably many L^+ -sentences. We construct recursively a sequence Δ_n of sets of L^+ -sentences such that for each n , at most finitely many of the new constants occur in members of Δ_n . Set $\Delta_0 = \Gamma$, and for each n , let n^* be the smallest number such that the constant symbol e_{n^*} does not occur in

θ_n or in any member of Δ_n . Then recursively

$$\begin{aligned}\Delta_n^0 &:= \begin{cases} \Delta_n \cup \{\theta_n\}, & \text{if this set is finitely consistent;} \\ \Delta_n, & \text{otherwise;} \end{cases} \\ \Delta_{n+1} &:= \begin{cases} \Delta_n^0 \cup \{\chi_x(e_{n^*})\}, & \text{if } \theta_n \in \Delta_n^0 \text{ and } \theta_n \text{ is } \exists x \chi; \\ \Delta_n^0, & \text{otherwise;} \end{cases} \\ \Delta &:= \bigcup_{n \in \omega} \Delta_n.\end{aligned}$$

Clearly by construction $\Gamma \subseteq \Delta$. Δ_0 is finitely consistent by hypothesis, steps $\Delta_n \mapsto \Delta_n^0$ preserve this property by definition, and steps $\Delta_n^0 \mapsto \Delta_{n+1}$ preserve it by the preceding proposition. Hence, by induction, each Δ_n is finitely consistent, whence Δ is also finitely consistent (since any finite subset of Δ is already a subset of some Δ_n). Δ is complete by the same argument as in the propositional case (1.4.27). Furthermore, it is clear that Δ is also L^+ -Henkin complete, since each time a sentence of the form $\exists x \chi$ is added to Δ , or was already there, a sentence of the form $\chi_x(c)$ is also added. Note that this includes such sentences which belong to $\Gamma = \Delta_0$. $\square$

Lemma 3.1.10 *For any language L and any set Δ of L -sentences which is both finitely consistent and L -Henkin complete,*

(i) *for any L -sentence ψ ,*

$$\Delta \models^* \psi \implies \psi \in \Delta; \quad \text{hence} \quad \Delta \models \psi \implies \psi \in \Delta;$$

(ii) *for any L -sentences ψ and θ ,*

$$(a) \quad \neg \psi \in \Delta \iff \psi \notin \Delta;$$

$$(b) \quad (\psi \vee \theta) \in \Delta \iff \psi \in \Delta \text{ or } \theta \in \Delta;$$

(iii) *for any L -formula χ with at most x free,*

$$(\exists x \chi) \in \Delta \iff \text{for some constant symbol } c, \quad \chi_x(c) \in \Delta.$$

Proof. For (i), suppose that $\Delta \models^* \psi$ but $\psi \notin \Delta$. Then by completeness $(\neg \psi) \in \Delta$ so $\Delta \models^* \neg \psi$, and by Proposition 3.1.5, Δ is not finitely consistent. The second clause follows by the Tautology Principle. (ii) follows as in the propositional case 1.4.28. (iii)($\Rightarrow$) is the definition. For (iii)($\Leftarrow$), $\chi_x(c) \rightarrow \exists x \chi$ is logically valid, so by (i) belongs to Δ . Hence if also $\chi_x(c) \in \Delta$, then $\Delta \models \exists x \chi$, so by (i), $\exists x \chi \in \Delta$. $\square$

The remainder of the proof of ACCT consists in the construction of a model for any finitely consistent and Henkin complete set Δ of L -sentences.

We shall construct this model from objects of the language itself, namely from equivalence classes of the constant symbols of L . This method is related to that suggested in the hints for Exercises 2.2.44 and 2.2.45; Henkin completeness allows us here to use only the constant symbols in place of all the constant terms. For constant symbols c and d , set

$$c \sim_{\Delta} d \iff c \doteq d \in \Delta.$$

Proposition 3.1.11 *For any language L and any set Δ of L -sentences which is both finitely consistent and L -Henkin complete,*

- (i) $\sim_{\Delta}$ is an equivalence relation;
- (ii) for any n -ary relation symbol p and constant symbols $c_0, \dots, c_{n-1}$ and $d_0, \dots, d_{n-1}$, if for all $i < n$, $c_i \sim_{\Delta} d_i$, then $pc_0 \dots c_{n-1} \in \Delta$ iff $pd_0 \dots d_{n-1} \in \Delta$;
- (iii) for any n -ary function symbol f and constant symbols $c_0, \dots, c_{n-1}$ and $d_0, \dots, d_{n-1}$, if for all $i < n$, $c_i \sim_{\Delta} d_i$, then $fc_0 \dots c_{n-1} \doteq fd_0 \dots d_{n-1} \in \Delta$.

Proof. These are immediate from Proposition 2.2.5, Proposition 3.1.4 and the preceding lemma. For example, if $c \sim d$ and $pc \in \Delta$, then $(c \doteq d) \in \Delta$ and $(c \doteq d \rightarrow [pc \leftrightarrow pd]) \in \Delta$, so $\Delta \models pd$, and by (i) of the lemma, $pd \in \Delta$. $\square$

Now, for each constant symbol c , denote by $\hat{c}$ its equivalence class under the equivalence relation $\sim_{\Delta}$: $\hat{c} = \{d \in \text{Cs}_L : c \sim_{\Delta} d\}$ and set $A = \{\hat{c} : c \in \text{Cs}_L\}$. To construct a model $\mathfrak{A}$ with universe A requires that we specify the interpretations for the relation, function and constant symbols. The constant symbols are easy — set $c^{\mathfrak{A}} = \hat{c}$. Part (ii) of the proposition provides the justification for defining

$$p^{\mathfrak{A}} = \{(\hat{c}_0, \dots, \hat{c}_{n-1}) : pc_0 \dots c_{n-1} \in \Delta\},$$

since it guarantees that the question of whether $(\hat{c}_0, \dots, \hat{c}_{n-1})$ belongs to $p^{\mathfrak{A}}$ is independent of the choice of representatives for the equivalence classes $\hat{c}_i$. If f is an n -ary function symbol, note that for any $c_0, \dots, c_{n-1}$, $\exists y fc_0 \dots c_{n-1} \doteq y$ is logically valid, hence belongs to Δ , so by Henkin completeness there is a constant symbol e such that $fc_0 \dots c_{n-1} \doteq e \in \Delta$. Furthermore, by (iii) of the preceding proposition and Proposition 3.1.4, $\hat{e}$ is determined by f and $\hat{c}_0, \dots, \hat{c}_{n-1}$. Thus we may define $f^{\mathfrak{A}}$ by

$$f^{\mathfrak{A}}(\hat{c}_0, \dots, \hat{c}_{n-1}) = \hat{e}$$

for any constant symbol e such that $fc_0 \dots c_{n-1} \doteq e \in \Delta$.

Proposition 3.1.12 *For any language L and any set Δ of L -sentences which is both finitely consistent and L -Henkin complete, the structure $\mathfrak{A}$ defined above is a model of Δ with the property that every element of the universe has a name — that is, equals $c^{\mathfrak{A}}$ for some L -constant symbol c . If L is countable, then also $\mathfrak{A}$ is countable.*

Proof. It will suffice to establish the following two facts:

- (1) for any closed L -term t and any $d \in \text{Cs}_L$,

$$t^{\mathfrak{A}} = \hat{d} \iff t \doteq d \in \Delta;$$

- (2) for any L -sentence ϕ , $\models_{\mathfrak{A}} \phi \iff \phi \in \Delta$.

It is immediate from (2) that $\mathfrak{A}$ is a model of Δ .

We prove (1) by closed-term induction. If t is a constant symbol c , the result holds by definition. Now fix $f \in \text{Fs}_L$ and $t = ft_0 \dots t_{n-1}$, assume as induction hypothesis that for $i < n$ the assertion holds for t_i , and choose constant symbols $d_0, \dots, d_{n-1}$ such that both sides are true — that is, for all $i < n$, $t_i^{\mathfrak{A}} = \hat{d}_i$ and $(t_i \doteq d_i) \in \Delta$. This is possible again by Henkin completeness, since $\exists y[t \doteq y]$ is logically valid. Then for any constant symbol d ,

$$\begin{aligned} t^{\mathfrak{A}} = \hat{d} &\iff f^{\mathfrak{A}}(t_0^{\mathfrak{A}}, \dots, t_{n-1}^{\mathfrak{A}}) = \hat{d} \\ &\iff f^{\mathfrak{A}}(\hat{d}_0, \dots, \hat{d}_{n-1}) = \hat{d} \\ &\iff (fd_0 \dots d_{n-1} \doteq d) \in \Delta \\ &\iff (t \doteq d) \in \Delta, \end{aligned}$$

where the last equivalence follows from Lemma 3.1.10(i) and Proposition 2.2.5.

We prove (2) by induction on the length of the sentence ϕ . Suppose first that ϕ is the atomic sentence $pt_0 \dots t_{n-1}$, and by (1) choose constant symbols $d_0, \dots, d_{n-1}$ such that for all $i < n$, $t_i^{\mathfrak{A}} = \hat{d}_i$ and $(t_i \doteq d_i) \in \Delta$. Then

$$\begin{aligned} \models_{\mathfrak{A}} \phi &\iff p^{\mathfrak{A}}(t_0^{\mathfrak{A}}, \dots, t_{n-1}^{\mathfrak{A}}) \\ &\iff p^{\mathfrak{A}}(\hat{d}_0, \dots, \hat{d}_{n-1}) \\ &\iff pd_0 \dots d_{n-1} \in \Delta \\ &\iff \phi \in \Delta. \end{aligned}$$

The induction steps for $\neg$ and $\vee$ are straightforward using the completeness of Δ and Lemma 3.1.10(ii):

$$\models_{\mathfrak{A}} (\neg \phi) \iff \not\models_{\mathfrak{A}} \phi \iff \phi \notin \Delta \iff \neg \phi \in \Delta,$$

and

$$\begin{aligned} \models_{\mathfrak{A}} \phi \vee \psi &\iff \models_{\mathfrak{A}} \phi \text{ or } \models_{\mathfrak{A}} \psi \\ &\iff \phi \in \Delta \text{ or } \psi \in \Delta \iff (\phi \vee \psi) \in \Delta. \end{aligned}$$

Finally, for the quantifier step, we have

$$\begin{aligned} \models_{\mathfrak{A}} (\exists x \phi) &\iff \text{for some } \hat{c} \in A, \models_{\mathfrak{A}} \phi[\hat{c}] \\ &\iff \text{for some } c \in \mathbf{CS}_L, \models_{\mathfrak{A}} \phi_x(c) \\ &\iff \text{for some } c \in \mathbf{CS}_L, \phi_x(c) \in \Delta \\ &\iff \exists x \phi \in \Delta. \end{aligned}$$

The second equivalence follows from Lemma 2.2.22, and the final one from Lemma 3.1.10(iii). $\square$

Thus we have established ACCT for sets of sentences. To complete the proof we proceed as follows. Fix a set $\{d_n : n \in \omega\}$ of new constant symbols, and let $L^\bullet$ be the extension of L to include them. For each formula ϕ , let $\phi^\bullet$ denote the result of replacing each variable v_n in ϕ by d_n . Clearly $\phi^\bullet$ is an $L^\bullet$ -sentence. For any L -structure $\mathfrak{A}$ and any $\mathfrak{A}$ -assignment α , let $\mathfrak{A}_\alpha$ be the $L^\bullet$ -structure such that $d_n^{\mathfrak{A}_\alpha} = \alpha(v_n)$. A straightforward formula induction shows that for all ϕ ,

$$\models_{\mathfrak{A}} \phi[\alpha] \iff \models_{\mathfrak{A}_\alpha} \phi^\bullet.$$

Hence, if we set $\Gamma^\bullet := \{\phi^\bullet : \phi \in \Gamma\}$, we have for any set Γ of formulas

$$\begin{aligned} \Gamma \text{ is finitely consistent} &\implies \Gamma^\bullet \text{ is finitely consistent} \\ &\implies \Gamma^\bullet \text{ is consistent} \implies \Gamma \text{ is consistent,} \end{aligned}$$

where the second implication is by ACCT for sets of sentences.

This completes the proof of ACCT and hence of CCT. Note that the proof of ACCT gives the following additional information:

Corollary 3.1.13 *For any countable language L and any set Γ of L -formulas, if Γ is (finitely) consistent, then Γ has a countable model. $\square$*

Of course, this also follows from ACCT as stated by application of the Countable Downward Löwenheim-Skolem Theorem.

In [Section 3.5](#) and [Chapter 7](#) we shall see many applications of the Compactness Theorem, but we include a sample three here. Recall that in Proposition 2.5.20 we analyzed the structure of a non-standard model for the theory $\text{Th}(\Omega_+)$ of the structure $\Omega_+ = (\omega, <, +, \text{Sc}, 0)$, but were unable to show that they exist. Now this is easy.

Theorem 3.1.14 $\text{Th}(\Omega_+)$ has a countable non-standard model.

Proof. Let L be the language L_{Ω_+} enriched with a constant symbol c , and

$$\Gamma = \text{Th}(\Omega_+) \cup \{ \dot{m} < c : m \in \omega \}.$$

For any finite $\Gamma_0 \subseteq \Gamma$, choose n larger than all m such that $\dot{m} < c$ belongs to Γ_0 . Then (Ω_+, n) is a model of Γ_0 . By the Countable Compactness Theorem, Γ has a countable model

$$\mathfrak{A} = (A, <_A, +_A, \text{Sc}_A, 0_A, e_A).$$

Since $\mathfrak{A}$ is a model of $\text{Th}(\Omega_+)$, so is its reduct $\mathfrak{A}^- = (A, <_A, +_A, \text{Sc}_A, 0_A)$. Since $e_A \in A$ and for all m , $\models_{\mathfrak{A}} \dot{m} < c$, we have $\dot{m}^{\mathfrak{A}} <_A e_A$, so e_A is a non-standard element of $\mathfrak{A}^-$ and therefore $\mathfrak{A}^-$ is a non-standard model. $\square$

Note that the same argument applies also to $\text{Th}(\Omega)$ or any other similar theory. The second application answers a question left open in Examples 2.4.23(i).

Theorem 3.1.15 The theory T_{DLO} of dense linear orderings is not $\exists_2$ -axiomatizable.

Proof. Recall that T_{DLO} is the $L_{<}$ -theory generated by a single sentence θ_{DLO} . Suppose, towards a contradiction, that $T_{\text{DLO}} = \text{Th}(\Gamma)$ for some set Γ of $\exists_2$ -sentences. In particular, $\Gamma \models \theta_{\text{DLO}}$, so by the Countable Compactness Theorem there is a finite subset $\Gamma_0 \subseteq \Gamma$ such that $\Gamma_0 \models \theta_{\text{DLO}}$; let $\chi := \bigwedge \Gamma_0$. χ is a logically $\exists_2$ -sentence, $T_{\text{DLO}} = \text{Th}(\chi)$, and $\neg\chi$ is a logically $\forall_2$ -sentence. Examples 2.3.57(iii) exhibits a chain $\langle \mathfrak{A}_i : i \in \omega \rangle$ of structures which are not dense orderings but whose union $\mathfrak{A}$ is a dense ordering. Hence each $\mathfrak{A}_i$ is a model of $\neg\chi$, so by Proposition 2.3.63, also $\mathfrak{A}$ is a model of $\neg\chi$, a contradiction. $\square$

The final application here yields a first proof of the Enumerability Theorem; a second proof will be given at the end of Section 3.2.

Definition 3.1.16 For any formula ϕ , a **closed instance** of ϕ is the result of substituting closed (variable-free) terms for each free variable of ϕ .

Herbrand's Theorem 3.1.17 For any language L and every L -formula ϕ , there exists a quantifier-free formula ψ of an extended language L^+ such that $\models \phi \iff$

$$\text{for some closed instances } \psi_0, \dots, \psi_n \text{ of } \psi, \bigvee_{i \leq n} \psi_i \text{ is a quasi-tautology}$$

(see Exercise 2.2.45); furthermore, ψ is effectively calculable from ϕ .

Proof. The Herbrand normal form ϕ^{Hb} (Corollary 2.6.30) of ϕ is logically existential and so logically equivalent to a formula $\exists x_0 \cdots \exists x_{k-1} \psi$ with ψ quantifier-free. We may assume that the language of ψ has a constant symbol (add one if it doesn't). Now we establish the following equivalences:

$$\begin{aligned}
 \models \phi &\iff \models \phi^{\text{Hb}} \\
 &\iff \models \exists x_0 \cdots \exists x_{k-1} \psi \\
 &\iff \text{for some closed instances } \psi_0, \dots, \psi_n \text{ of } \psi, \models \bigvee_{i \leq n} \psi_i \\
 &\iff \text{for some closed instances } \psi_0, \dots, \psi_n \text{ of } \psi, \\
 &\qquad \qquad \qquad \bigvee_{i \leq n} \psi_i \text{ is a quasi-tautology.}
 \end{aligned}$$

The first two equivalences and ($\iff$) of the third are now immediate, and the fourth is the result of Exercise 2.2.45. Suppose that no such disjunction of closed instances of ψ is logically valid. It follows that every finite subset of

$$\Gamma := \{ \neg \theta : \theta \text{ is a closed instance of } \psi \}$$

is consistent, so by compactness Γ has a model $\mathfrak{B}$. Let $\mathfrak{A}$ be the unique minimal substructure $\langle \emptyset \rangle_{\mathfrak{B}}$; by Proposition 2.3.19, the elements of A are exactly the values $t^{\mathfrak{B}}$ of all the constant terms. Hence, for each $a_0, \dots, a_{k-1} \in A$, if $a_i = t_i^{\mathfrak{B}}$ and θ is the closed instance $\psi_{x_0, \dots, x_{k-1}}(t_0, \dots, t_{k-1})$,

$$\models_{\mathfrak{B}} \neg \theta \implies \models_{\mathfrak{A}} \neg \theta \implies \not\models_{\mathfrak{A}} \psi[a_0, \dots, a_{k-1}].$$

Hence $\not\models_{\mathfrak{A}} \exists x_0 \cdots \exists x_{k-1} \psi$, so $\exists x_0 \cdots \exists x_{k-1} \psi$ is not logically valid. The effective calculability of ψ follows from the effectiveness of Corollary 2.6.30. $\square$

Enumerability Theorem 3.1.18 *For any effective language L and any L -theory T ,*

- (i) *if T is decidable, or even effectively enumerably, axiomatizable, then T is effectively enumerable;*
- (ii) *if T is also complete, then T is decidable.*

Proof. We begin exactly as in the proof of the propositional version 1.5.18. Let $\theta_0, \dots, \theta_m, \dots$ be an effective listing of a set of axioms Γ for T . By the Countable Compactness Theorem, for any L -sentence ϕ ,

$$\Gamma \models \phi \iff \text{for some } m \in \omega, \models \theta_0 \wedge \cdots \wedge \theta_m \rightarrow \phi.$$

Here, in contrast to 1.5.18, we cannot directly test each of these sentences for validity but instead proceed as follows. For each L -sentence ϕ and each m , let ψ_ϕ^m denote the quantifier-free formula given by Herbrand's Theorem applied to the formula $\theta_0 \wedge \cdots \wedge \theta_m \rightarrow \phi$. A sequence of formulas $(\phi, \psi, \psi_0, \dots, \psi_n)$ is called an **Herbrand proof** from Γ of ϕ iff

- (iii) for some $m \in \omega$, ψ is ψ_ϕ^m ;
- (iv) $\psi_0, \dots, \psi_n$ are closed L^+ -instances of ψ ;
- (v) $\bigvee_{i \leq n} \psi_i$ is a quasi-tautology.

By Herbrand's Theorem,

$$\Gamma \models \phi \iff \text{there exists an Herbrand proof of } \phi \text{ from } \Gamma.$$

We shall show that the set **HP** of all Herbrand proofs from Γ is effectively enumerable. Then we obtain an enumeration of $\text{Th}(\Gamma)$ by composing an enumeration of **HP** with the mapping of an Herbrand proof to its conclusion ϕ .

In turn, it suffices to show that **HP** is semi-decidable. For property (iii), this follows from the effectiveness of Herbrand's Theorem.

Property (iv) is easily decidable using Proposition 2.2.37. Let EqAx^+ denote the set of L^+ -equality axioms (Proposition 2.2.5). Clearly EqAx^+ is decidable and by definition $\bigvee_{i \leq n} \psi_i$ is a quasi-tautology iff it is a tautological consequence of EqAx^+ . Hence the semi-decidability of property (v) is an instance of the Propositional Enumerability Theorem 1.5.18.

Now (ii) follows exactly as for Theorem 1.5.21. $\square$

Note that although we have stated the Enumerability Theorem just for sets of sentences (theories), essentially the same proof shows that the set of all logical consequences of an effectively enumerable set is also effectively enumerable.

Collateral Reading. Most texts give only the proof of the Countable Compactness and Enumerability Theorems via the Countable Completeness Theorem as in the following section; an exception is [Mal], which proceeds much as we do here.

Exercises

Exercise 3.1.19 Prove that $T_{\text{Sc},0}$ (Definition 2.5.5) is not finitely axiomatizable.

Hint: Show first that if $T_{\text{Sc},0}$ were finitely axiomatizable, then some finite subset of $(S1)-(S4_m)$ would be a set of axioms.

Exercise 3.1.20 Let $L \subseteq L^+$ be countable languages, T^+ an L^+ -theory and

$$\mathcal{K} := \{ \mathfrak{A} : \text{for some } \mathfrak{A}^+ \in \text{Mod}(T^+), \mathfrak{A} \equiv \mathfrak{A}^+ \upharpoonright L \}.$$

Prove that there is an L -theory T such that $\mathcal{K} = \text{Mod}(T)$.

Exercise 3.1.21 Let L be a language with at least one constant symbol.

- (i) Let T be a **universally axiomatizable** L -theory — that is, a theory of the form

$$T = \text{Th}(\{ \forall \mathbf{x} \phi(\mathbf{x}) : \phi \in \Gamma \})$$

for some set Γ of quantifier-free formulas, where $\mathbf{x}$ denotes some finite sequence $x_0, \dots, x_{k-1}$ of variables. Prove that if T is inconsistent, then there exist closed instances $\phi_0, \dots, \phi_n$ of members of Γ such that $\bigvee_{i \leq n} \neg \phi_i$ is logically valid.

- (ii) Show that if T_0 and T_1 are two universally axiomatizable L -theories such that $T_0 \cup T_1$ is inconsistent, then there exists a quantifier-free L -sentence θ such that $T_0 \models \theta$ and $T_1 \models \neg \theta$.
- (iii) Let ϕ and ψ be respectively logically universal and logically existential L -sentences such that $\models \phi \rightarrow \psi$. Prove that there exists a quantifier-free L -sentence θ such that $\models \phi \rightarrow \theta$ and $\theta \rightarrow \psi$.
- (iv) Conclude that any L -sentence which is both logically universal and logically existential is logically equivalent to a quantifier-free L -sentence.

Hint: For (i) use the method of proof of Herbrand's Theorem.

Exercise 3.1.22 Prove Proposition 3.1.4(iii).

Exercise 3.1.23 Let L be a finite language and T an L -theory which is both finitely axiomatizable and $\aleph_0$ -categorical. Prove that T is decidable.

3.2 Countable completeness

To establish an instance $\Gamma \models \psi$ of logical consequence requires at some level consideration of all of the infinitely many and arbitrarily large structures $\mathfrak{A}$ and assignments α which satisfy Γ . On the other hand, a mathematical proof of this fact is simply a finite sequence of words and symbols with certain properties. The basic notion to be studied in this section is a formal version of this notion of proof.

A mathematical proof of a statement ψ may be seen as a finite sequence of assertions, each either an assumption or following 'easily' from the assertions before it, ending with ψ . Of course, in practice conventions differ on what

counts as ‘following easily’, and the formal notion must make some choices in specifying precise *rules of inference*. For example, *modus ponens* is the rule which from the **premises** ψ and $\psi \rightarrow \theta$ allows us to infer the **conclusion** θ . We shall say (precisely below), then, that ψ is *formally provable* from Γ — in symbols, $\Gamma \vdash \psi$ — iff there exists a finite sequence $\theta_0, \dots, \theta_n = \psi$ of formulas (a *formal proof*), such that each θ_i is either an element of Γ or one of a special class of sentences called *logical axioms* or follows from $\theta_0, \dots, \theta_{i-1}$ by one of the rules of inference. A natural restriction in the selection of the logical axioms and rules of inference is that only logical consequences should be formally provable — that is, such that for any set Γ of sentences,

$$\Gamma \vdash \psi \implies \Gamma \models \psi.$$

We say that our system of deduction is **sound**. The main content of the Completeness Theorem is that it is possible to select the logical axioms and rules in such a way that they are also **adequate** — that is, also

$$\Gamma \models \psi \implies \Gamma \vdash \psi.$$

Note that these implications do *not* hold in quite this simple form when the members of Γ are arbitrary formulas instead of sentences (see the precise statement below). In view of the results of [Section 3.1](#), we have a natural strategy for achieving this: arrange that the provability relation $\vdash$ has all of the properties of $\models^*$ which were used in the proof of the CCT.

The definition of $\Gamma \vdash \psi$ is an inductive one. We specify certain basic validities (called logical axioms) and some finitary mappings of formulas (called rules of inference). The set of theorems from a set Γ of (non-logical) axioms is the closure of the set of (logical and non-logical) axioms under the rules of inference.

Definition 3.2.1 The set LogAx_L of *L-logical axioms* consists of the following formulas:

- (i) all *L*-tautologies;
- (ii) for any terms $t, u, \dots$, any $f \in \text{Fs}_L$ and any $p \in \text{Rs}_L$, all *L*-formulas of the following forms (**equality axioms**):
 - (a) $t \doteq t$;
 - (b) $t \doteq u \rightarrow u \doteq t$;
 - (c) $t \doteq u \wedge u \doteq v \rightarrow t \doteq v$;
 - (d) $t_0 \doteq u_0 \wedge \dots \wedge t_{\nu_L(f)-1} \doteq u_{\nu_L(f)-1} \rightarrow$

$$ft_0, \dots, t_{\nu_L(f)-1} \doteq fu_0, \dots, u_{\nu_L(f)-1};$$

$$(e) \ t_0 \doteq u_0 \wedge \dots \wedge t_{\nu_L(p)-1} \doteq u_{\nu_L(p)-1} \rightarrow$$

$$[pt_0, \dots, t_{\nu_L(p)-1} \leftrightarrow pu_0, \dots, u_{\nu_L(p)-1}];$$

- (iii) all L -formulas of the form $\phi_x(t) \rightarrow \exists x \phi$, where $\phi_x(t)$ is a free substitution ($\exists$ -*axioms*).

Definition 3.2.2 The *rules of inference* are the following: for any formulas ϕ and ψ ,

- (i) from ϕ and $\phi \rightarrow \psi$, infer ψ (*modus ponens*);
- (ii) if $x \notin \text{Fv}(\psi)$, from $\phi \rightarrow \psi$, infer $\exists x \phi \rightarrow \psi$ ($\exists$ -*rule*).

Definition 3.2.3 For any language L and any set Γ of L -formulas, the set $\text{Thm}_L(\Gamma)$ of Γ -*L-theorems* is the smallest set of L -formulas such that $\text{LogAx}_L \cup \Gamma \subseteq \text{Thm}_L(\Gamma)$ and $\text{Thm}_L(\Gamma)$ is closed under the rules of inference — that is,

- (i) if $\phi, (\phi \rightarrow \psi) \in \text{Thm}_L(\Gamma)$, then $\psi \in \text{Thm}_L(\Gamma)$;
- (ii) if $(\phi \rightarrow \psi) \in \text{Thm}_L(\Gamma)$ and $x \notin \text{Fv}(\psi)$, then $(\exists x \phi \rightarrow \psi) \in \text{Thm}_L(\Gamma)$.

Further, we set

$$\text{Th}_L^-(\Gamma) := \{ \psi : \psi \text{ is an } L\text{-sentence and } \psi \in \text{Thm}_L(\Gamma) \}.$$

Note that $\text{Thm}_L(\Gamma)$ is defined inductively as in [Section 1.2](#); it is the inductive closure of $\text{LogAx}_L \cup \Gamma$ under the functions

$$H_{MP}(\phi, \chi) = \begin{cases} \psi, & \text{if } \chi \text{ is } \phi \rightarrow \psi; \\ \phi, & \text{otherwise;} \end{cases}$$

$$H_{\exists, x}(\chi) = \begin{cases} \exists x \phi \rightarrow \psi, & \text{if } \chi \text{ is } \phi \rightarrow \psi \text{ and } x \notin \text{Fv}(\phi); \\ \chi, & \text{otherwise.} \end{cases}$$

Definition 3.2.4 We write $\Gamma \stackrel{L}{\vdash} \psi$ iff $\psi \in \text{Thm}_L(\Gamma)$ and say that ψ is a Γ -*L-theorem* or is *L-deducible* from Γ . We write $\stackrel{L}{\vdash} \psi$ instead of $\emptyset \stackrel{L}{\vdash} \psi$ and say in this case that ψ is an *L-logical theorem*.

These notions seem heavily dependent on the language; if L^+ is an extension of L , clearly $\text{LogAx}_L \subseteq \text{LogAx}_{L^+}$ so $\text{Thm}_L(\Gamma) \subseteq \text{Thm}_{L^+}(\Gamma)$, but the converse inclusions are not true. We dealt with the analogous problem for the notion $\Gamma \models \psi$ at the beginning of [Section 2.6](#) and return to this one shortly. Note that in light of this we are also for now writing $\text{Th}_L(\Gamma)$ instead of $\text{Th}(\Gamma)$ to emphasize this dependence.

Recall from Proposition 1.2.7 that for any induction system $\mathcal{X}$ and any $z \in X$,

$$z \in \overline{X} \iff \text{there exists an } \mathcal{X}\text{-derivation of } z.$$

For the induction system described above, the requirements to be an $\mathcal{X}$ -derivation are just those in the following.

Definition 3.2.5 For any set Γ of L -formulas, a Γ -*L-proof* or a *formal proof* in L from Γ is a finite sequence $(\theta_0, \dots, \theta_n)$ of L -formulas such that for all $i \leq n$, one of the following holds:

- (i) $\theta_i \in \Gamma$;
- (ii) θ_i is an L -logical axiom;
- (iii) for some $j, k < i$, θ_k is $\theta_j \rightarrow \theta_i$;
- (iv) for some $j < i$ and some x, ϕ and ψ , θ_j is $(\phi \rightarrow \psi)$, $x \notin \text{Fv}(\psi)$, and θ_i is $(\exists x \phi \rightarrow \psi)$.

We say that $(\theta_0, \dots, \theta_n)$ is a Γ -*L-proof of* θ_n .

Then from the general result on induction systems mentioned above we have

Corollary 3.2.6 For any L , Γ and ψ ,

$$\Gamma \vdash^L \psi \iff \text{there exists a } \Gamma\text{-}L\text{-proof of } \psi. \quad \square$$

So the characterization given in the initial sketch turns out to be a corollary rather than the definition, but this is simply a matter of organization.

Remark 3.2.7 The induction system which defines $\text{Thm}_L(\Gamma)$ does *not* have unique readability. For example, if ϕ and ψ are both tautologies, then ψ is both a logical axiom and a consequence of the rule *modus ponens* applied to the two tautologies ϕ and $\phi \rightarrow \psi$. Hence we can use induction to establish properties of members of $\text{Thm}_L(\Gamma)$, but we cannot use recursion to define mappings on this set.

Let us verify first that our deduction system is sound. Recall that $\overline{\forall\Gamma}$ denotes the set of universal closures $\overline{\forall\phi}$ of the members ϕ of Γ (Definition 2.2.28).

Soundness Theorem 3.2.8 For any set Γ of L -formulas and any L -formula ψ ,

$$\Gamma \vdash^L \psi \implies \overline{\forall\Gamma} \models \psi;$$

in particular, if all members of Γ are sentences, then $\Gamma \vdash^L \psi \implies \Gamma \models \psi$.

Proof. We proceed by $\text{Thm}_L(\Gamma)$ -induction. By Propositions 2.2.4, 2.2.5 and 2.2.24, the logical axioms are all logically valid and for $\phi \in \Gamma$, $\overline{\forall\Gamma} \models \phi$ by Corollary 2.2.26. It suffices to verify that $\{\psi : \overline{\forall\Gamma} \models \psi\}$ is closed under the rules of inference. For *modus ponens* this is clear, and closure under the $\exists$ -rule follows from Proposition 2.2.27 because the members of $\overline{\forall\Gamma}$ are sentences. The last clause follows, since if Γ is a set of sentences, then $\overline{\forall\Gamma} = \Gamma$. $\square$

Proposition 3.2.9 *For any sets Γ and Δ of L -formulas and any L formulas ϕ and ψ ,*

- (i) *if $\phi \in \Gamma$, then $\Gamma \vdash^L \phi$;*
- (ii) *if $\Gamma \vdash^L \psi$ and $\Gamma \subseteq \Delta$, then $\Delta \vdash^L \psi$;*
- (iii) *if $\Gamma \vdash^L \psi$ and L^+ is an extension of L , then $\Gamma \vdash^{L^+} \psi$;*
- (iv) *if $\Gamma \vdash^L \psi$, then for some finite $\Gamma_0 \subseteq \Gamma$ and some finite sublanguage L^- of L , $\Gamma_0 \vdash^{L^-} \psi$;*
- (v) *if $\text{Thm}_L(\Gamma) \vdash^L \psi$, then also $\Gamma \vdash^L \psi$;*
- (vi) *if $\text{Thm}_L(\Gamma) \models \psi$, in particular if $\Gamma \models \psi$, then $\Gamma \vdash^L \psi$.*

We refer to (vi) as the **Tautology Principle**.

Proof. Parts (i)–(iii) are immediate from the definitions and the basic properties of inductive definitions. Part (iv) is an application of Proposition 1.2.11 — if ψ belongs to the closure $\text{Thm}_L(\Gamma)$ of $\Gamma \cup \text{LogAx}_L$ under the rules of inference, then it already belongs to the closure of some finite subset $\Gamma_0 \cup Y_0$ of this. The finitely many members of this set contain only finitely many non-logical symbols, so all belong to a finite sublanguage L^- of L , and thus $\psi \in \text{Thm}_{L^-}(\Gamma_0)$.

Part (v) is a general property of inductive definitions: $\text{Thm}_L(\Gamma)$ is a set of formulas which includes $\text{Thm}_L(\Gamma)$ and the logical axioms and is closed under the rules of inference, so it includes $\text{Thm}_L(\text{Thm}_L(\Gamma))$, the smallest such set. For (vi), suppose that $\text{Thm}_L(\Gamma) \models \psi$. Then by the Propositional Compactness Theorem 1.4.20 there exist $\phi_0, \dots, \phi_{n-1} \in \text{Thm}_L(\Gamma)$ such that $\{\phi_0, \dots, \phi_{n-1}\} \models \psi$. In particular,

$$\phi_0 \rightarrow (\dots \phi_{n-2} \rightarrow (\phi_{n-1} \rightarrow \psi) \dots)$$

is a tautology, so belongs to $\text{Thm}_L(\Gamma)$. By n applications of *modus ponens*, also $\psi \in \text{Thm}_L(\Gamma)$. $\square$

We want next to get rid of the pesky superscript on the symbol $\vdash^L$. It will follow from the Completeness Theorem together with Theorem 2.6.5 (see the discussion following that theorem) that if the members of Γ and ψ are L -formulas, then for any extension L^+ of L , $\psi \in \text{Thm}_L(\Gamma)$ iff $\psi \in \text{Thm}_{L^+}(\Gamma)$. However, as a part of the proof we will need to establish first that this is true at least in the case that L^+ has only additional constant symbols (is an extension by constants).

In the following lemma and proposition it will be convenient to write $t_c(y)$ and $\phi_c(y)$ to denote the result of replacing all occurrences of the constant symbol c by the variable y . Of course, a more formal recursive definition of this notion is entirely analogous to the definitions of $t_x(u)$ and $\psi_x(u)$ (2.2.20).

Lemma 3.2.10 *Let L^+ be an extension of L by a single constant symbol c . Then for any set Γ of L -formulas and any L^+ -formula ψ ,*

$$\Gamma \vdash^{L^+} \psi \quad \implies \quad \text{for cofinitely many variables } y, \quad \Gamma \vdash^L \psi_c(y);$$

in particular, if ψ is an L -formula, then

$$\Gamma \vdash^{L^+} \psi \quad \implies \quad \Gamma \vdash^L \psi.$$

Proof. We proceed by $\text{Thm}_{L^+}(\Gamma)$ -induction. If $\psi \in \Gamma$, then c does not occur in ψ so for any y , $\psi_c(y)$ is ψ and thus $\Gamma \vdash^L \psi_c(y)$. If ψ is an L^+ -tautology or an L^+ -equality axiom, it is easy to see that $\psi_c(y)$ is an L -axiom of the same sort. Suppose that ψ is an $\exists$ -axiom $\theta_z(t) \rightarrow \exists z \theta$ and y is any variable different from z and not occurring in θ . Then, by Exercise 2.2.46, $\psi_c(y)$ is $\theta_c(y)_z(t_c(y)) \rightarrow \exists z \theta_c(y)$ and since y does not occur in θ , the substitution on the left-hand side is free, so $\psi_c(y)$ is an L - $\exists$ -axiom.

To check closure under the rules of inference, suppose that for cofinitely many y , $\Gamma \vdash^L \phi_c(y)$ and for cofinitely many y ,

$$\Gamma \vdash^L (\phi \rightarrow \psi)_c(y) \text{ — that is, } \Gamma \vdash^L \phi_c(y) \rightarrow \psi_c(y).$$

Then for the cofinitely many y for which both of these hold, $\Gamma \vdash^L \psi_c(y)$. For the $\exists$ -rule, suppose that $x \notin \text{Fv}(\psi)$ and for cofinitely many y ,

$$\Gamma \vdash^L (\phi \rightarrow \psi)_c(y) \text{ — that is, } \Gamma \vdash^L \phi_c(y) \rightarrow \psi_c(y).$$

For all such y except x , still cofinitely many, by the $\exists$ -rule we have

$$\Gamma \vdash^L \exists x \phi_c(y) \rightarrow \psi_c(y) \text{ — that is, } \Gamma \vdash^L (\exists x \phi \rightarrow \psi)_c(y). \quad \square$$

Proposition 3.2.11 *If L^+ is an extension of L by constants, Γ is a set of L -formulas and ψ is an L -formula, then*

$$\Gamma \vdash^{L^+} \psi \iff \Gamma \vdash^L \psi.$$

Proof. The implication $(\Leftarrow)$ is Proposition 3.2.9 (iii). Suppose that $\Gamma \vdash^{L^+} \psi$. Then by (iv) of that same proposition, there is a sublanguage L^- of L^+ which is an extension of L by finitely many constants such that $\Gamma \vdash^{L^-} \psi$. An easy induction on the size of $\mathbf{Cs}_{L^-} \sim \mathbf{Cs}_L$ using the preceding lemma gives $\Gamma \vdash^L \psi$. $\square$

From now on we shall omit the index L and write simply $\Gamma \vdash \phi$ and $\text{Thm}(\Gamma)$ with the understanding that all languages under discussion differ at most in their constant symbols. Let us now state precisely the

Countable Completeness Theorem 3.2.12 (*CCplT*) *For any countable language L , any set Γ of L -formulas and any L -formula ψ ,*

$$\Gamma \vdash \psi \iff \overline{\forall \Gamma} \models \psi;$$

in particular, if all members of Γ are sentences, then $\Gamma \vdash \psi \iff \Gamma \models \psi$.

The difference between the two cases here (formulas and sentences) is an artifact of the particular definition of formal provability we have chosen and could be avoided. It is important to understand that the use of the word “complete” here is an instance, familiar in all parts of mathematics, of overworking a good word; the meaning here is essentially unrelated to the use of the word as describing a theory which is maximal consistent (Corollary 2.4.16).

Following the outline of [Section 3.1](#), we next define

Definition 3.2.13 A set Γ of L -formulas is $\vdash$ -**inconsistent** or **proof inconsistent** iff $\Gamma \vdash \mathbf{F}$; otherwise Γ is $\vdash$ -**consistent**.

It is immediate from the Soundness Theorem that every consistent set of formulas is $\vdash$ -consistent and we approach CCplT via the

Alternative Countable Completeness Theorem 3.2.14 (*ACCplT*) *For any countable language L and any set Γ of L -formulas, if Γ is $\vdash$ -consistent, then $\overline{\forall \Gamma}$ is consistent.*

Even to derive CCplT from ACCplT is a bit harder than the corresponding part of [Section 3.1](#). One of the key components of the proof which highlights the special role of sentences is the following.

Deduction Theorem 3.2.15 *For any set Γ of formulas, any sentence θ and any formula ψ ,*

$$\Gamma \cup \{\theta\} \vdash \psi \iff \Gamma \vdash \theta \rightarrow \psi.$$

Proof. The implication ($\Leftarrow$) is immediate by 3.2.9(ii) and *modus ponens*. Fix Γ and θ ; we prove ($\Rightarrow$) by $\text{Thm}_L(\Gamma \cup \{\theta\})$ -induction. If ψ is an L -logical axiom or a member of Γ , then $\Gamma \vdash \psi$ so also $\Gamma \vdash \theta \rightarrow \psi$, since $\psi \rightarrow (\theta \rightarrow \psi)$ is a tautology. If ψ is θ , then $\theta \rightarrow \psi$ is a tautology. To verify closure under *modus ponens*, suppose that $\Gamma \vdash \theta \rightarrow \phi$ and $\Gamma \vdash \theta \rightarrow (\phi \rightarrow \psi)$. Then $\Gamma \vdash \theta \rightarrow \psi$ follows by the Tautology Principle. Finally, suppose that $\Gamma \vdash \theta \rightarrow (\phi \rightarrow \psi)$ and $x \notin \text{Fv}(\psi)$. By the Tautology Principle we have $\Gamma \vdash \phi \rightarrow (\theta \rightarrow \psi)$. Since θ is a sentence, $x \notin \text{Fv}(\theta \rightarrow \psi)$ so we may apply the $\exists$ -rule to obtain $\Gamma \vdash \exists x \phi \rightarrow (\theta \rightarrow \psi)$. Then by the Tautology Principle once more, $\Gamma \vdash \theta \rightarrow (\exists x \phi \rightarrow \psi)$. $\square$

Proposition 3.2.16 *For any set Γ of L -formulas, the following are equivalent:*

- (i) Γ is $\vdash$ -inconsistent;
- (ii) for every formula ψ , $\Gamma \vdash \psi$;
- (iii) for some (every) formula ψ , both $\Gamma \vdash \psi$ and $\Gamma \vdash \neg \psi$;
- (iv) there is a finite subset $\Gamma_0 \subseteq \Gamma$ such that Γ_0 is $\vdash$ -inconsistent.

Proof. Since $\mathbf{F} \rightarrow \phi$ is a tautology, (i) $\Rightarrow$ (ii) is immediate from the Tautology Principle. (ii) $\Rightarrow$ (iii) is trivial and (iii) $\Rightarrow$ (i) follows from the Tautology Principle and the fact that $\{\psi, \neg \psi\} \models \mathbf{F}$. The equivalence of (i) and (iv) follows from Proposition 3.2.9(iv). $\square$

Corollary 3.2.17 *For any set Γ of formulas and any sentence ψ ,*

- (i) $\Gamma \vdash \psi \iff \Gamma \cup \{\neg \psi\}$ is $\vdash$ -inconsistent;
- (ii) $\Gamma \vdash \neg \psi \iff \Gamma \cup \{\psi\}$ is $\vdash$ -inconsistent;
- (iii) if Γ is $\vdash$ -consistent, then at least one of $\Gamma \cup \{\psi\}$ and $\Gamma \cup \{\neg \psi\}$ is $\vdash$ -consistent.

Proof. If $\Gamma \vdash \psi$, then $\Gamma \cup \{\neg \psi\} \vdash \psi$, so since also $\Gamma \cup \{\neg \psi\} \vdash \neg \psi$, $\Gamma \cup \{\neg \psi\}$ is $\vdash$ -inconsistent by 3.2.16. Conversely, if $\Gamma \cup \{\neg \psi\} \vdash \mathbf{F}$, then by the Deduction Theorem, $\Gamma \vdash \neg \psi \rightarrow \mathbf{F}$, so since $(\neg \psi \rightarrow \mathbf{F}) \rightarrow \psi$ is a tautology, $\Gamma \vdash \psi$. The proof of (ii) is analogous. Finally, if both $\Gamma \cup \{\psi\}$ and $\Gamma \cup \{\neg \psi\}$ are $\vdash$ -inconsistent, then by part (iii) of the preceding proposition, Γ is $\vdash$ -inconsistent. $\square$

At this point we could derive CCplT for sentences, but for the general case we need to establish some additional properties of formal provability, which will also be needed for the proof of the ACCplT.

Lemma 3.2.18 *For any L -formula ϕ and any L -term t which is freely substitutable for x in ϕ , $\vdash \forall x \phi \rightarrow \phi_x(t)$.*

Proof. This is immediate from the $\exists$ -axioms, the Tautology Principle and the fact that

$$\neg \phi_x(t) \rightarrow \exists x \neg \phi \equiv \forall x \phi \rightarrow \phi_x(t)$$

(remember that $\forall x \phi$ is an abbreviation for $\neg \exists x \neg \phi$). $\square$

Proposition 3.2.19 *For any set Γ of L -formulas, any L -formula ϕ and any constant symbol c which does not occur in ϕ or any member of Γ ,*

$$\Gamma \vdash \phi \iff \Gamma \vdash \forall x \phi \iff \Gamma \vdash \phi_x(c).$$

Proof. By the preceding lemma, $\vdash \forall x \phi \rightarrow \phi$ and $\vdash \forall x \phi \rightarrow \phi_x(c)$, since the substitutions $\phi_x(x)$ and $\phi_x(c)$ are always free. Hence by *modus ponens*

$$\Gamma \vdash \forall x \phi \implies \Gamma \vdash \phi \text{ and } \Gamma \vdash \phi_x(c).$$

Suppose that $\Gamma \vdash \phi$. Then since $\phi \rightarrow (\neg \phi \rightarrow F)$ is a tautology, $\Gamma \vdash \neg \phi \rightarrow F$ and thus by the $\exists$ -rule, $\Gamma \vdash \exists x \neg \phi \rightarrow F$. But $(\exists x \neg \phi \rightarrow F) \rightarrow \forall x \phi$ is a tautology, so $\Gamma \vdash \forall x \phi$.

Suppose now that $\Gamma \vdash \phi_x(c)$. By Lemma 3.2.10, for cofinitely many y , $\Gamma \vdash \phi_x(c)_c(y)$; since c does not occur in ϕ , this is $\Gamma \vdash \phi_x(y)$. By the part just proved, $\Gamma \vdash \forall y \phi_x(y)$, and thus by the preceding lemma, $\Gamma \vdash \phi_x(y)_y(x)$ — that is, $\Gamma \vdash \phi$ — so again by the first part, $\Gamma \vdash \forall x \phi$. $\square$

Corollary 3.2.20 *For any set Γ of formulas and any formula ψ ,*

$$\overline{\forall \Gamma} \vdash \psi \iff \Gamma \vdash \psi \iff \Gamma \vdash \overline{\forall \psi}.$$

In particular, Γ is $\vdash$ -consistent iff $\overline{\forall \Gamma}$ is $\vdash$ -consistent.

Proof. The second equivalence is immediate by successive applications of the preceding proposition. In particular, for each $\phi \in \Gamma$, $\Gamma \vdash \overline{\forall \phi}$; in other words, $\overline{\forall \Gamma} \subseteq \text{Thm}_L(\Gamma)$. Then using Proposition 3.2.9,

$$\text{Thm}_L(\overline{\forall \Gamma}) \subseteq \text{Thm}_L(\text{Thm}_L(\Gamma)) = \text{Thm}_L(\Gamma),$$

which gives the first equivalence. $\square$

Proof. (of CCplT from ACCplT) For any set Γ of formulas and any formula ψ ,

$$\begin{aligned} \overline{\forall \Gamma} \models \psi &\implies \overline{\forall \Gamma} \models \overline{\forall \psi} \\ &\implies \overline{\forall \Gamma} \cup \{\neg \overline{\forall \psi}\} \text{ is inconsistent} \\ &\implies \overline{\forall \Gamma} \cup \{\neg \overline{\forall \psi}\} \text{ is } \vdash\text{-inconsistent} \\ &\implies \overline{\forall \Gamma} \vdash \overline{\forall \psi} \\ &\implies \Gamma \vdash \psi. \end{aligned}$$

The first implication is by Proposition 2.2.27, the second by Proposition 2.2.3, the third by ACCplT (i), the fourth by Corollary 3.2.17, and the fifth by the preceding corollary. $\square$

We proceed now to the proof of ACCplT. Note that we used above only the version for sets of sentences ($\overline{\forall\Gamma}$), but in fact the general version follows from this by the final clause of the preceding corollary.

Proposition 3.2.21 *For any set Γ of formulas and formulas ϕ and ψ ,*

- (i) $\Gamma \vdash \phi \rightarrow \psi \implies \Gamma \vdash \exists x \phi \rightarrow \exists x \psi$;
- (ii) $\vdash \forall x \neg \phi \rightarrow \neg \exists x \phi$.

Proof. Suppose that $\Gamma \vdash \phi \rightarrow \psi$. Since $\psi \rightarrow \exists x \psi$ is an $\exists$ -axiom, by the Tautology Principle also $\Gamma \vdash \phi \rightarrow \exists x \psi$. Then by the $\exists$ -rule, $\Gamma \vdash \exists x \phi \rightarrow \exists x \psi$. For (ii), we have $\vdash \phi \rightarrow \neg \neg \phi$, and hence by (i), $\vdash \exists x \phi \rightarrow \exists x \neg \neg \phi$. Then by the Tautology Principle, $\Gamma \vdash \neg \exists x \neg \neg \phi \rightarrow \neg \exists x \phi$, which is exactly the desired conclusion. $\square$

Proposition 3.2.22 *For any set Γ of formulas and any sentence of the form $\exists x \chi$ in Γ , if Γ is $\vdash$ -consistent, then for any constant symbol c which does not occur in any member of Γ , $\Gamma \cup \{\chi_x(c)\}$ is $\vdash$ -consistent.*

Proof. Suppose that $(\exists x \chi) \in \Gamma$ but $\Gamma \cup \{\chi_x(c)\}$ is $\vdash$ -inconsistent. Then by the propositions above together with the Tautology Principle, we conclude successively that $\Gamma \vdash \neg \chi_x(c)$, $\Gamma \vdash \forall x \neg \chi$ and $\Gamma \vdash \neg \exists x \chi$. But then by Proposition 3.2.16, Γ is $\vdash$ -inconsistent, contrary to hypothesis. $\square$

The remainder of the proof of ACCplT for sets of sentences is virtually identical to the corresponding parts of [Section 3.1](#), and we omit many details.

Proposition 3.2.23 *For any countable language L and any $\vdash$ -consistent set Γ of L -sentences, there exists a countable extension L^+ of L by constants and a set $\Delta \supseteq \Gamma$ of L^+ -sentences which is $\vdash$ -consistent and L^+ -Henkin complete.*

Proof. Let L and Γ be as in the hypothesis and form L^+ by adjoining a new set of constant symbols $\{e_n : n \in \omega\}$. Fix an enumeration $\langle \theta_n : n \in \omega \rangle$ of the denumerably many L^+ -sentences. We define a sequence Δ_n and Δ exactly as in the proof of Proposition 3.1.9, replacing the condition “finitely consistent” by “ $\vdash$ -consistent”. The argument that Δ is $\vdash$ -consistent and L^+ -Henkin complete proceeds exactly as before, now using Corollary 3.2.17 and the preceding proposition. $\square$

Lemma 3.2.24 *For any language L and any set Δ of L -sentences which is both $\vdash$ -consistent and L -Henkin complete,*

- (i) *for any L -sentence ψ ,*

$$\Delta \vdash \psi \implies \psi \in \Delta; \text{ hence } \Delta \models \psi \implies \psi \in \Delta;$$

(ii) for any sentences ψ and θ ,

$$(a) \neg \psi \in \Delta \iff \psi \notin \Delta;$$

$$(b) (\psi \vee \theta) \in \Delta \iff \psi \in \Delta \text{ or } \theta \in \Delta;$$

(iii) for any formula χ with at most x free,

$$(\exists x \chi) \in \Delta \iff \text{for some constant symbol } c, \chi_x(c) \in \Delta.$$

Proof. For (i), suppose that $\Delta \vdash \phi$ but $\phi \notin \Delta$. Then by completeness $(\neg \phi) \in \Delta$ and by Proposition 3.2.16, Δ is $\vdash$ -inconsistent. (ii) follows as in the propositional case. (iii)($\implies$) is the definition and ($\impliedby$) is immediate from (i), the Tautology Principle and the fact that $\chi_x(c) \rightarrow \exists x \chi$ is an $\exists$ -axiom, hence is provable from Δ . $\square$

Now, given a set Δ which is $\vdash$ -consistent and Henkin complete, we may define a structure $\mathfrak{A}$ exactly as in [Section 3.1](#); the definitions are justified by the preceding lemma together with the observation that because of the equality and $\exists$ -axioms, each sentence of the form $\exists y f c_0 \dots c_{n-1} \doteq y$ belongs to Δ . Then we have

Proposition 3.2.25 *For any language L and any set Δ of L -sentences which is both $\vdash$ -consistent and L -Henkin complete, the structure $\mathfrak{A}$ defined above is a model of Δ with the property that every element of the universe has a name — that is, equals $c^{\mathfrak{A}}$ for some L -constant symbol c . If L is countable, then $\mathfrak{A}$ is countable.*

Proof. As before it will suffice to establish the following two facts:

(1) for any closed L -term t and any $d \in \text{Cs}_L$,

$$t^{\mathfrak{A}} = \hat{d} \iff t \doteq d \in \Delta;$$

(2) for any L -sentence ϕ ,

$$\models_{\mathfrak{A}} \phi \iff \phi \in \Delta.$$

Both parts are entirely parallel to the corresponding parts of the proof of Proposition 3.1.12. It is immediate from (2) that $\mathfrak{A}$ is a model of Δ . $\square$

ACCpIT now follows easily: if L is a countable language and Γ a $\vdash$ -consistent set of L -sentences, we first extend L and Γ to L^+ and Δ as in Proposition 3.2.23 so that Δ is $\vdash$ -consistent and L^+ -Henkin complete. By the preceding proposition, Δ has a model $\mathfrak{A}^+$. The desired model of Γ is $\mathfrak{A} = \mathfrak{A}^+ \upharpoonright L$. Since the language L is countable, so is L^+ , and it follows that $\mathfrak{A}$ is countable.

We have also, as promised, second proofs of the Countable Compactness and Enumerability Theorems.

Proof. (of the Countable Compactness Theorem) Suppose that $\Gamma \models \psi$. Then as in the derivation of the ACCT for formulas following Proposition 3.1.12, $\Gamma^\bullet \models \psi^\bullet$, so by the Countable Completeness Theorem $\Gamma^\bullet \vdash \psi^\bullet$, whence by Proposition 3.2.9(iv), for some finite subset $\Gamma_0 \subseteq \Gamma$, $\Gamma_0^\bullet \vdash \psi^\bullet$, so by soundness, $\Gamma_0^\bullet \models \psi^\bullet$ and thus $\Gamma_0 \models \psi$ so $\Gamma \models^* \psi$. $\square$

Proof. (of the Enumerability Theorem) Fix an effective language L and an effectively enumerably axiomatizable L -theory $T := \text{Th}(\Gamma)$. By the Countable Completeness Theorem and Corollary 3.2.6, we have that for any sentence ψ

$$\psi \in T \iff \text{there exists a } \Gamma\text{-proof of } \psi.$$

We aim to show that $\{(\phi_0, \dots, \phi_n) : (\phi_0, \dots, \phi_n) \text{ is a } \Gamma\text{-proof}\}$ is an effectively enumerable set. Then an enumeration of T is obtained by listing proofs but reporting only the ‘last line’ ϕ_n .

Now by the lemma and Proposition 1.5.14, it suffices to show that the set of Γ -proofs is semi-decidable in the set of all finite sequences of L -formulas. Given such a finite sequence, we may effectively check for each formula whether or not it is one of the logical axioms or whether it arises from one or two of the earlier terms of the sequence by one of the rules of inference. This is true simply because the axioms and rules are determined purely by the arrangements of their symbols, not by any ‘meaning’ attached to them. Of course, we are using the result of [Section 1.5](#) that the set of tautologies is decidable. Now if Γ is decidable, we may in addition test each formula in the sequence for membership in Γ and thus effectively decide whether or not the sequence is a Γ -proof. If Γ is only effectively enumerable, we may still semi-decide membership in Γ and thus semi-decide the set of Γ -proofs as required. $\square$

Remark 3.2.26 In choosing the logical axioms, we elected to include all of the tautologies. A common variant is to include only certain ones and prove that all of the remainder are derivable. Although this lends a certain elegance to the axioms, it lengthens the proof and contributes little, since the whole set of tautologies is decidable. Of course, the parallel tactic of including among the logical axioms all of the logical validities would not be feasible, since for most languages this set is not decidable (see [Sections 4.1](#) and [4.6](#)), and we would not be able to carry out the preceding proof.

In the next two sections we shall give still further alternative proofs of some of these results.

Collateral Reading. Had we been completely consistent with the principle of developing ideas first in the simplest possible context, we would have

introduced in [Chapter 1](#) a notion of formal proof for the propositional language and proved there a completeness theorem. Since we didn't, the reader may want to see how this is done in [Men] or [Hod]. Doing this would provide the choice of tautologies referred to in the preceding remark. In the history of logic, the notion of formal proof has played a much larger role than it does in this book. Indeed, there is a whole subarea of logic known as Proof Theory. The handbook [Bus] provides one reasonable starting point to explore this area; another is the more recent text [Tr-Sch].

Exercises

Exercise 3.2.27 Prove directly from the definition of logical theorem that for any terms t , u and v , and any atomic formula ϕ ,

$$\begin{aligned} \vdash t \doteq u &\rightarrow v_x(t) \doteq v_x(u) \\ \vdash t \doteq u &\rightarrow (\phi_x(t) \leftrightarrow \phi_x(u)). \end{aligned}$$

Exercise 3.2.28 Give an example to show that the hypothesis of the Deduction Theorem that θ is a sentence is necessary.

Exercise 3.2.29 Fill in the details of the proof of Proposition 3.2.23.

Exercise 3.2.30 Fill in the details of the proof of Proposition 3.2.25.

3.3 Other constructions

The Completeness, Compactness and Enumerability Theorems are such central results that they have been reworked and reformulated in many different ways. Some of these shed considerable light on the underlying ideas and have other applications. We have developed some of the machinery and motivation already in connection with the propositional compactness theorem.

Notions of consistency

We begin with a first-order version of notions of consistency (cf. [Section 1.6](#)). As in the propositional case, the idea is to abstract from the proofs of the Compactness and Completeness Theorems the minimum conditions that are needed to make the proof work. This provides a tool which will be applied in [Section 3.7](#).

Definition 3.3.1 For any set E of constant symbols, an *E -primitive term* is either a constant symbol (not necessarily in E) or a term of the form $f e_0 \dots e_n$, with all $e_i \in E$.

The main useful feature of a primitive term is that it contains at most one symbol (function or constant) not in E .

Definition 3.3.2 For any language L , an *L -notion of consistency* is a pair $(\mathbb{C}, E)$ with $\mathbb{C}$ a set of sets of L -sentences and E a set of L -constant symbols such such that if $\Theta \subseteq \Delta \in \mathbb{C}$, then $\Theta \in \mathbb{C}$ and for any $\Theta \in \mathbb{C}$,

- (i) for any E -primitive term t and any L -formula ϕ with at most x free,
 - (a) for any $e, e' \in E$, $(e \doteq e') \in \Theta \implies \Theta \cup \{e' \doteq e\} \in \mathbb{C}$;
 - (b) for any $e \in E$, $(t \doteq e), \phi_x(t) \in \Theta \implies \Theta \cup \{\phi_x(e)\} \in \mathbb{C}$;
 - (c) for some $e \in E$, $\Theta \cup \{(t \doteq e)\} \in \mathbb{C}$;
- (ii) for any L -sentence ψ ,
 - (a) $(\neg \psi) \in \Theta \implies \psi \notin \Theta$;
 - (b) $(\neg \neg \psi) \in \Theta \implies \Theta \cup \{\psi\} \in \mathbb{C}$;
- (iii) for any L -sentences ψ and θ ,
 - (a) $(\psi \vee \theta) \in \Theta \implies$ either $\Theta \cup \{\psi\} \in \mathbb{C}$ or $\Theta \cup \{\theta\} \in \mathbb{C}$;
 - (b) $\neg(\psi \vee \theta) \in \Theta \implies$ both $\Theta \cup \{\neg \psi\} \in \mathbb{C}$ and $\Theta \cup \{\neg \theta\} \in \mathbb{C}$;
- (iv) for any L -formula ψ with at most x free,
 - (a) $\exists x \psi \in \Theta \implies$ for some $e \in E$, $\Theta \cup \{\psi_x(e)\} \in \mathbb{C}$;
 - (b) $(\neg \exists x \psi) \in \Theta \implies$ for all $e \in E$, $\Theta \cup \{\neg \psi_x(e)\} \in \mathbb{C}$.

Members of $\mathbb{C}$ are called **$\mathbb{C}$ -consistent**.

Examples 3.3.3 Let L be any language and E any infinite set of constant symbols of L . For the following $\mathbb{C}$, $(\mathbb{C}, E)$ is an L -notion of consistency.

- (i) $\mathbb{C} =$ the set of all Θ such that Θ has a model $\mathfrak{A}$ with universe $|\mathfrak{A}| = \{e^{\mathfrak{A}} : e \in E\}$. All of the conditions are easy to verify.
- (ii) $\mathbb{C} =$ the set of all Θ such that Θ is consistent and there are infinitely many $e \in E$ which occur in no member of Θ (alternatively, only finitely many $e \in E$ occur in members of Θ).
- (iii) $\mathbb{C} =$ the set of all Θ such that Θ is $\vdash$ -consistent and there are infinitely many $e \in E$ which occur in no member of Θ . Most of the conditions are immediate from the results of the preceding section; (i)(b) follows immediately from Exercise 3.2.27. For (iv)(a), suppose that $\exists x \psi \in \Theta \in \mathbb{C}$, and let e be any member of E which occurs in no member of Θ . $\Theta \cup \{\psi_x(e)\}$ is $\vdash$ -consistent, since otherwise we have by the lemmas of the preceding section

$$\begin{aligned} \Theta \cup \{\psi_x(e)\} &\implies \Theta \vdash \neg \psi_x(e) \\ &\implies \Theta \vdash \forall x \neg \psi &\implies \Theta \vdash \neg \exists x \psi, \end{aligned}$$

contrary to the $\vdash$ -consistency of Θ . For (iv)(b), if $(\neg \exists x \psi) \in \Theta \in \mathbb{C}$, then for any $e \in E$, $\Theta \cup \{\neg \psi_x(e)\}$ is $\vdash$ -consistent, since otherwise $\Theta \vdash \psi_x(e)$, so $\Theta \vdash \exists x \psi$, a contradiction.

- (iv) $\mathbb{C}$ = the set of all Θ such that every finite subset of Θ is consistent and there are infinitely many $e \in E$ which occur in no member of Θ .

Of course, for countable L it follows from the Countable Completeness Theorem that (ii) and (iii) coincide and from the Countable Compactness Theorem that (ii) and (iv) coincide, but we state them separately to emphasize that the Countable Model Existence Theorem below provides independent proofs of these two theorems.

Definition 3.3.4 For any language L and any set Δ of L -sentences, Δ is **weakly complete** with respect to the set E of constant symbols iff

- (i) for any E -primitive term t and any L -formula ϕ with at most x free,
 - (a) for any $e, e' \in E$, $(e \doteq e') \in \Delta \implies (e' \doteq e) \in \Delta$;
 - (b) for any $e \in E$, $(t \doteq e), \phi_x(t) \in \Delta \implies \phi_x(e) \in \Delta$;
 - (c) for some $e \in E$, $(t \doteq e) \in \Delta$;
- (ii) for any L -sentence ψ ,
 - (a) $(\neg \psi) \in \Delta \implies \psi \notin \Delta$;
 - (b) $(\neg \neg \psi) \in \Delta \implies \psi \in \Delta$;
- (iii) for any L -sentences ψ and θ ,
 - (a) $(\psi \vee \theta) \in \Delta \implies$ either $\psi \in \Delta$ or $\theta \in \Delta$ (or both);
 - (b) $\neg(\psi \vee \theta) \in \Delta \implies$ both $(\neg \psi) \in \Delta$ and $(\neg \theta) \in \Delta$;
- (iv) for any L -formula ψ with at most x free,
 - (a) $\exists x \psi \in \Delta \implies$ for some $e \in E$, $\psi_x(e) \in \Delta$;
 - (b) $(\neg \exists x \psi) \in \Delta \implies$ for all $e \in E$, $(\neg \psi_x(e)) \in \Delta$.

The main construction is again in two steps parallel to those in the proof of the Alternative Countable Completeness Theorem. The first makes essential use of its hypothesis of countability.

Proposition 3.3.5 For any countable language L and any L -notion of consistency $(\mathbb{C}, E)$, every $\mathbb{C}$ -consistent set Θ may be extended to a set Δ weakly complete with respect to E .

Proof. Let $(\mathbb{C}, E)$ be a notion of consistency, Θ a $\mathbb{C}$ -consistent set, and

$$\langle \theta_n : n \in \omega \rangle, \quad \langle t_n : n \in \omega \rangle \quad \text{and} \quad \langle e_n : n \in \omega \rangle$$

enumerations of Sent_L , the primitive terms of L and E , respectively. Reference to the first $e \in E$ with some property means first in this enumeration. Set $\Delta_0 = \Theta$ and for all n ,

$$\begin{aligned} \Delta_n^0 &= \begin{cases} \Delta_n \cup \{\theta_n\}, & \text{if this set is } \mathbb{C}\text{-consistent;} \\ \Delta_n, & \text{otherwise;} \end{cases} \\ \Delta_n^1 &= \Delta_n^0 \cup \{t_n \doteq e\}, \quad \text{for the first } e \in E \text{ for which this is } \mathbb{C}\text{-consistent;} \\ \Delta_{n+1} &= \begin{cases} \Delta_n^1 \cup \{\chi_x(e')\}, & \text{if } \theta_n \in \Delta_n^1 \text{ and } \theta_n \text{ is } \exists x \chi \text{ for the first } e' \in E \\ & \text{such that this is } \mathbb{C}\text{-consistent;} \\ \Delta_n^1, & \text{otherwise.} \end{cases} \end{aligned}$$

Using the definition of notion of consistency, we can prove by a straightforward induction on n that each of the sets above is $\mathbb{C}$ -consistent and thus such e and e' exist. We then set

$$\Delta := \bigcup \{ \Delta_n : n \in \omega \},$$

and claim that Δ is weakly complete. We have explicitly ensured that conditions (i)(c) and (iv)(a) of the definition of weak completeness are satisfied by the second and third substages of the construction. For (i)(a), suppose that $(e \doteq e') \in \Delta_n$ and $e' \doteq e$ is θ_p . If $p < n$, then since $\Delta_n \cup \{\theta_p\}$ is $\mathbb{C}$ -consistent, so is its subset $\Delta_p \cup \{\theta_p\}$, and hence $\theta_p \in \Delta_p^0$; if $n < p$, then $(e \doteq e') \in \Delta_p$, so again $\Delta_p \cup \{\theta_p\}$ is $\mathbb{C}$ -consistent. (i)(b) and (iv)(b) hold by similar arguments. Properties (ii) and (iii) are proved exactly as in Proposition 1.6.4. $\square$

Lemma 3.3.6 *For any weakly complete set Δ and any $e, e', e'' \in E$,*

- (i) $(e \doteq e'), (e' \doteq e'') \in \Delta \implies (e \doteq e'') \in \Delta$;
- (ii) $(e \doteq e) \in \Delta$.

Proof. Left as an exercise. $\square$

Proposition 3.3.7 *For any language L , any set E of L -constant symbols and any set Δ of L -sentences, if Δ is weakly complete with respect to E , then Δ has a model $\mathfrak{A}$ with universe $|\mathfrak{A}| = \{e^{\mathfrak{A}} : e \in E\}$. In particular, if L is countable, then so is $\mathfrak{A}$.*

Proof. The model is defined similarly as in the proofs of the Countable Compactness and Completeness Theorems. For any $e, e' \in E$, let

$$e \sim e' \iff (e \doteq e') \in \Delta.$$

Property (i)(a) together with the lemma guarantee that $\sim$ is an equivalence relation. We denote by $\hat{e}$ the equivalence class of e and set $A := \{\hat{e} : e \in E\}$. Note carefully that we use here only members of E and not all L -constant symbols. Analogously to the corresponding construction in the preceding section, we define

(i) for each $p \in \mathbf{Rs}_L$,

$$p^{\mathfrak{A}} := \{(\hat{e}_0, \dots, \hat{e}_{\nu_L(p)-1}) : pe_0, \dots, e_{\nu_L(p)-1} \in \Delta\};$$

(ii) for each $f \in \mathbf{Fs}_L$, $f^{\mathfrak{A}}(\hat{e}_0, \dots, \hat{e}_{\nu_L(f)-1}) := \hat{e}$, where e is some member of E such that $(fe_0 \dots e_{\nu_L(f)-1} \doteq e) \in \Delta$;

(iii) for each $c \in \mathbf{Cs}_L$, $c^{\mathfrak{A}} := \hat{e}$, where e is some member of E such that $c \doteq e \in \Delta$.

The existence of such an e is guaranteed by property (i)(c). As before we need to verify that $\mathfrak{A}$ is well-defined. For example, if for all $i < \nu_L(f)$, $e_i \sim e'_i$, and $e, e' \in E$ are such that $(fe_0 \dots e_{\nu_L(f)-1} \doteq e) \in \Delta$ and $(fe'_0 \dots e'_{\nu_L(f)-1} \doteq e') \in \Delta$, then by $\nu_L(f)$ -many applications of (i)(b) we have $(fe_0 \dots e_{\nu_L(f)-1} \doteq e') \in \Delta$ and hence by (i)(a) and the lemma, $(e \doteq e') \in \Delta$ as required. Note that in particular for $e \in E$ we have $e^{\mathfrak{A}} = \hat{e}$.

To show that $\mathfrak{A}$ is a model of Δ , we establish that for any L -sentence ϕ ,

$$(1) \quad \phi \in \Delta \implies \models_{\mathfrak{A}} \phi \quad \text{and} \quad (\neg \phi) \in \Delta \implies \models_{\mathfrak{A}} \neg \phi.$$

For atomic sentences ϕ , we prove (1) by induction on the number n of occurrences in ϕ of function symbols. If $n = 0$, then ϕ is of the form $pc_0 \dots c_{\nu_L(p)-1}$. For each c_i , choose (by (i)(c)) $e_i \in E$ such that $(c_i \doteq e_i) \in \Delta$. Then by (i)(b) $pe_0 \dots e_{\nu_L(p)-1} \in \Delta$, so by definition, $(\hat{e}_0, \dots, \hat{e}_{\nu_L(p)-1}) \in p^{\mathfrak{A}}$, that is, $\models_{\mathfrak{A}} pe_0 \dots e_{\nu_L(p)-1}$. But also $c_i^{\mathfrak{A}} = \hat{e}_i = e_i^{\mathfrak{A}}$, so $\models_{\mathfrak{A}} c_i \doteq e_i$ and hence also $\models_{\mathfrak{A}} pc_0 \dots c_{\nu_L(p)-1}$, as required. A similar proof verifies that

$$\neg pc_0 \dots c_{\nu_L(p)-1} \in \Delta \implies \models_{\mathfrak{A}} \neg pc_0 \dots c_{\nu_L(p)-1},$$

which completes the case $n = 0$ of the induction.

If an atomic sentence ϕ has $n + 1$ function symbols, it may be written as $\psi_x(fc_0 \dots c_{\nu_L(f)-1})$ for an atomic sentence ψ with n function symbols. If we choose $e_i \in E$ such that $(c_i \doteq e_i) \in \Delta$, then again by (i)(b) $\psi_x(fe_0 \dots e_{\nu_L(f)-1}) \in \Delta$. Now $fe_0 \dots e_{\nu_L(f)-1}$ is an E -primitive term, so by (i)(c) there is some $e \in E$ such that $(fe_0 \dots e_{\nu_L(f)-1} \doteq e) \in \Delta$ and thus by (i)(b), $\psi_x(e) \in \Delta$. By the induction hypothesis, $\models_{\mathfrak{A}} \psi_x(e)$ and then easily $\models_{\mathfrak{A}} \phi$.

This establishes (1) for atomic ϕ , and we proceed by induction on the length of ϕ . If ϕ is $\neg\psi$ or $\psi \vee \theta$, the argument is just as in Proposition 1.6.5; for example, if ϕ is $\neg\psi$, then by the induction hypothesis,

$$\phi \in \Delta \implies (\neg\psi) \in \Delta \implies \models_{\mathfrak{A}} \neg\psi \implies \models_{\mathfrak{A}} \phi,$$

and similarly using (ii)(b) we have $(\neg\phi) \in \Delta \implies \models_{\mathfrak{A}} \neg\phi$.

Finally, suppose that ϕ is $\exists x\psi$. Then by (iv)(a),

$$\begin{aligned} \phi \in \Delta &\implies \text{for some } e \in E, \psi_x(e) \in \Delta \\ &\implies \text{for some } e \in E, \models_{\mathfrak{A}} \psi_x(e) \implies \models_{\mathfrak{A}} \phi, \end{aligned}$$

and by (iv)(b),

$$\begin{aligned} \neg\phi \in \Delta &\implies \text{for all } e \in E, \neg\psi_x(e) \in \Delta \\ &\implies \text{for all } e \in E, \models_{\mathfrak{A}} \neg\psi_x(e) \implies \models_{\mathfrak{A}} \neg\phi, \end{aligned}$$

where the last implication uses the fact that every element of A is $e^{\mathfrak{A}}$ for some $e \in E$. $\square$

Combining these two propositions we have

Countable Model Existence Theorem 3.3.8 *For any countable language L and any L -notion of consistency $(\mathbb{C}, E)$, every $\mathbb{C}$ -consistent set has a countable model $\mathfrak{A}$ with universe $|\mathfrak{A}| = \{e^{\mathfrak{A}} : e \in E\}$. $\square$*

Examples 3.3.9

- (i) Let L be a countable language, L^+ the extension by a denumerable set E of new constant symbols and $\mathbb{C}$ the set of sets Θ of L^+ -sentences which are $\vdash$ -consistent and such that there are infinitely many $e \in E$ which occur in no member of Θ . By Examples 3.3.3(iii), $(\mathbb{C}, E)$ is an L^+ -notion of consistency, so by the theorem every $\Theta \in \mathbb{C}$ has a countable model. In particular, this applies to every $\vdash$ -consistent set of L -sentences, and we have a proof of the Alternative Countable Completeness Theorem.
- (ii) With L , E and L^+ as in (i), let $\mathbb{C}$ be the set of sets of L^+ -sentences such that every finite subset is consistent and such that there are infinitely many $e \in E$ which occur in no member of Θ . By Examples 3.3.3(iv), $(\mathbb{C}, E)$ is an L^+ -notion of consistency, and we have thus another proof of the Alternative Countable Compactness Theorem.

The Countable Model Existence Theorem is a quite flexible instrument and we note here a simple generalization.

Definition 3.3.10 For any notion of consistency $(\mathbb{C}, E)$, a set Φ of sentences is $(\mathbb{C}, E)$ -**acceptable** iff for each $\Theta \in \mathbb{C}$ there is at least one $\phi \in \Phi$ such that $\Theta \cup \{\phi\} \in \mathbb{C}$.

Extended Countable Model Existence Theorem 3.3.11 *For any countable language L , any L -notion of consistency $(\mathbb{C}, E)$ and any family $\langle \Phi_n : n \in \omega \rangle$ of $(\mathbb{C}, E)$ -acceptable sets of sentences, every $\mathbb{C}$ -consistent set Θ has a countable model $\mathfrak{A}$ with universe $|\mathfrak{A}| = \{e^{\mathfrak{A}} : e \in E\}$ such that for all $n \in \omega$, $\models_{\mathfrak{A}} \bigvee \Phi_n$.*

Proof. Under the stated hypotheses, we first extend Θ to a weakly complete set Δ with the additional property that for all n , $\Phi_n \cap \Delta \neq \emptyset$. The construction is as before with the added substage

$$\Delta_n^2 = \Delta_n^1 \cup \{\phi\}$$

for the first $\phi \in \Phi_n$ (in some canonical enumeration of the L -sentences) such that this set is $\mathbb{C}$ -consistent. Then the model constructed as before for Δ clearly satisfies each $\bigvee \Phi_n$. $\square$

We apply notions of consistency to prove a countable first-order version of 1.6.7.

Countable Robinson Interpolation Theorem 3.3.12 *For any languages L_0 and L_1 with common part L and any sets Γ_0 of L_0 -sentences and Γ_1 of L_1 -sentences, if $\Gamma_0 \cup \Gamma_1$ is inconsistent, then there exists an L -sentence χ such that $\Gamma_0 \models \chi$ and $\Gamma_1 \models \neg\chi$. χ is called a **Robinson interpolant** for Γ_0 and Γ_1 .*

Proof. With the notation as in the statement, let E be a denumerable set of constant symbols new to all of the languages and let $L(E)$ and $L_i(E)$ denote the extended languages which include E . Clearly $L(E)$ is the common part of $L_0(E)$ and $L_1(E)$. Let $\mathbb{C}$ denote the set of all sets $\Theta_0 \cup \Theta_1$ such that $\Theta_i \subseteq \text{Sent}_{L_i(E)}$ ($i = 0, 1$) are consistent, there are infinitely many elements of E which occur in no member of $\Theta_0 \cup \Theta_1$, and there is no Robinson interpolant for Θ_0 and Θ_1 .

Let L^* denote the smallest language which includes both L_0 and L_1 . The main part of the proof is to establish that $\mathbb{C}$ is an $L^*(E)$ -notion of consistency, and we consider the clauses of the definition. Suppose that $\Theta_0 \cup \Theta_1 \in \mathbb{C}$. (i)(a) is immediate, since if $(e \doteq e') \in \Theta_0$, then $\Theta_0 \cup \{e' \doteq e\}$ is logically equivalent to Θ_0 , so any interpolant for the pair $\Theta_0 \cup \{e' \doteq e\}$, Θ_1 would also be one for Θ_0 , Θ_1 . For (i)(b), if both of $t \doteq e$ and $\phi_x(t)$ belong to the same Θ_i , then the same argument works. Suppose that $(t \doteq e) \in \Theta_0$ and $\phi_x(t) \in \Theta_1$. If x is not free in ϕ , the result is trivial; if it is, then it follows

that t is an $L(E)$ -term. If $\Theta_0 \cup \Theta_1 \cup \{\phi_x(e)\}$ were not $\mathbb{C}$ -consistent, then there would be an interpolant, an $L(E)$ -sentence χ such that

$$\Theta_0 \models \chi \quad \text{and} \quad \Theta_1 \cup \{\phi_x(e)\} \models \neg \chi.$$

But then also $\Theta_1 \models t \doteq e \rightarrow \neg \chi$ so that $(t \doteq e \wedge \chi)$ is an interpolant for Θ_0 and Θ_1 , contrary to assumption. For (i)(c), since the E -primitive term t contains only one function symbol or constant symbol not in E , it is a term of one of the languages $L_i(E)$, say $L_0(E)$. Let e be any member of E which occurs in neither $\Theta_0 \cup \Theta_1$ nor t . If $\Theta_0 \cup \{t \doteq e\} \cup \Theta_1$ were not $\mathbb{C}$ -consistent, then there would be an interpolant, an $L(E)$ -sentence χ such that

$$\Theta_0 \cup \{t \doteq e\} \models \chi \quad \text{and} \quad \Theta_1 \models \neg \chi.$$

Let x be any variable which does not occur in χ . Then since

$$\Theta_0 \models \exists x \chi_e(x) \quad \text{and} \quad \Theta_1 \models \neg \exists x \chi_e(x),$$

Θ_0 and Θ_1 also have an interpolant, contrary to assumption.

The cases (ii) and (iii) for $\neg$ and $\vee$ are exactly as for the propositional version. For (iv)(a), suppose that $(\exists x \psi) \in \Theta_0$ and choose $e \in E$ which occurs in neither Θ_0 nor Θ_1 . If $\Theta_0 \cup \{\psi_x(e)\} \cup \Theta_1$ were not $\mathbb{C}$ -consistent and χ is an interpolant,

$$\Theta_0 \cup \{\psi_x(e)\} \models \chi \quad \text{and} \quad \Theta_1 \models \neg \chi,$$

then easily $\exists x \chi_e(x)$ is an interpolant for Θ_0 and Θ_1 . For (iv)(b), if $(\neg \exists x \psi) \in \Theta_0$, then since $\models \neg \exists x \psi \rightarrow \neg \psi_x(e)$, any interpolant for the pair $\Theta_0 \cup \{\neg \psi_x(e)\}$, Θ_1 would also be one for Θ_0 , Θ_1 .

Since by the Model Existence Theorem, every $\mathbb{C}$ -consistent set has a model, if $\Gamma_0 \cup \Gamma_1$ is inconsistent it does not belong to $\mathbb{C}$. Hence there is an $L(E)$ -sentence χ such that

$$\Gamma_0 \models \chi \quad \text{and} \quad \Gamma_1 \models \neg \chi.$$

Then if $e_0, \dots, e_{k-1}$ is a list of all the members of E which occur in χ and $x_0, \dots, x_{k-1}$ are variables which do not occur in χ , we have also

$$\Gamma_0 \models \exists x_0, \dots, \exists x_{k-1} \chi_{e_0, \dots, e_{k-1}}(x_0, \dots, x_{k-1})$$

and

$$\Gamma_1 \models \neg \exists x_0, \dots, \exists x_{k-1} \chi_{e_0, \dots, e_{k-1}}(x_0, \dots, x_{k-1}),$$

so Γ_0 and Γ_1 have the desired interpolant. $\square$

Countable Craig Interpolation Theorem 3.3.13 *For any countable language L , any L -theory T , any two extensions L_0 and L_1 of L whose common symbols are all symbols of L and any L_0 -formula ϕ and L_1 -formula ψ , if $T \models \phi \rightarrow \psi$, then there exists an L -formula θ such that $T \models \phi \rightarrow \theta$ and $T \models \theta \rightarrow \psi$. θ is called a **Craig interpolant** for ϕ and ψ .*

Proof. Suppose first that ϕ and ψ are sentences. Let T_0 be the L_0 -theory generated by $T \cup \{\phi\}$ and T_1 the L_1 -theory generated by $T \cup \{\neg\psi\}$. The hypothesis implies that $T_0 \cup T_1$ is inconsistent, and the Robinson Interpolant provided by the preceding theorem is clearly sufficient here.

The general case may be reduced to this one by adding a suitable number of new constant symbols to all of the languages and considering the sentences resulting from replacing all of the free variables by constants. $\square$

Ultraproducts

We turn now to another proof of the Compactness Theorem which is purely algebraic in character. It has also the advantage that it provides a uniform proof of Compactness for languages of all cardinality and the disadvantage that it makes essential use of the Axiom of Choice. The idea is similar to that of the ultraproduct proof of the Propositional Compactness Theorem following Theorem 1.6.18. Suppose that Γ is a set of sentences such that each finite subset Γ_0 has a model, say $\mathfrak{A}_{\Gamma_0}$. We shall apply an algebraic construction to the family of L -structures $\langle \mathfrak{A}_{\Gamma_0} : \Gamma_0 \text{ a finite subset of } \Gamma \rangle$ to produce a model for Γ .

We can make substantial use of the machinery developed for the propositional case, and the reader should review this construction. Let I be any infinite set, $\mathcal{U}$ an ultrafilter over I and $\langle \mathfrak{A}_i : i \in I \rangle$ a family of L -structures. The main difference, which complicates the construction here, is that in place of the simple definition of the model (truth assignment) $V_{\mathcal{U}}$ with the property that

$$V_{\mathcal{U}}(\phi) = \mathsf{T} \quad \Longleftrightarrow \quad \{i \in I : V_i(\phi) = \mathsf{T}\} \in \mathcal{U},$$

we need here to build a structure $\mathfrak{A}_{\mathcal{U}}$ such that

$$\models_{\mathfrak{A}_{\mathcal{U}}} \phi \quad \Longleftrightarrow \quad \left\{ i \in I : \models_{\mathfrak{A}_i} \phi \right\} \in \mathcal{U}.$$

We make use of the material on products from [Section 2.4](#): recall that for any set I and any I -indexed family of sets $\langle A_i : i \in I \rangle$,

$$\begin{aligned} A &= \prod_{i \in I} A_i \\ &= \left\{ F : F \text{ is a function with domain } I \text{ and for all } i \in I, [F(i) \in A_i] \right\}, \end{aligned}$$

and if each A_i is the universe of a structure $\mathfrak{A}_i$ (for a fixed language L), we made A the universe of a structure $\mathfrak{A}$ by interpreting all of the non-logical symbols ‘pointwise’ — for example

$$p^{\mathfrak{A}}(F_0, \dots, F_{\nu_L(p)-1}) \iff \text{for all } i \in I, \quad p^{\mathfrak{A}_i}(F_0(i), \dots, F_{\nu_L(p)-1}(i)).$$

This provided a rather weak connection between truth in the structures $\mathfrak{A}_i$ and in $\mathfrak{A}$: only Horn sentences true in all $\mathfrak{A}_i$ are true in $\mathfrak{A}$. Here we will modify this construction by replacing ‘all i ’ by ‘almost all i ’ — that is, we set (almost)

$$p^{\mathfrak{A}}(F_0, \dots, F_{\nu_L(p)-1}) \iff \{i \in I : p^{\mathfrak{A}_i}(F_0(i), \dots, F_{\nu_L(p)-1}(i))\} \in \mathcal{U},$$

where $\mathcal{U}$ is an ultrafilter on I . This doesn’t quite work, since if we apply it to the equality relation, we get

$$F = G \iff \{i \in I : F(i) = G(i)\} \in \mathcal{U},$$

which is false because equal functions agree on *all* arguments. The solution is to regard F as a representative for the set of functions ‘almost equal’ to it.

Definition 3.3.14 For any set I , any I -indexed family of sets $\langle A_i : i \in I \rangle$, any ultrafilter $\mathcal{U}$ over I and any $F, G \in \prod_{i \in I} A_i$,

$$F \sim_{\mathcal{U}} G \iff \{i \in I : F(i) = G(i)\} \in \mathcal{U}.$$

We have immediately by the properties of filters

Lemma 3.3.15 $\sim_{\mathcal{U}}$ is an equivalence relation. $\square$

We denote the equivalence class of a function F by $\hat{F}$.

Definition 3.3.16 For any set I , any I -indexed family of L -structures $\langle \mathfrak{A}_i : i \in I \rangle$ and any ultrafilter $\mathcal{U}$ over I , the **ultraproduct** of $\langle \mathfrak{A}_i : i \in I \rangle$ with respect to $\mathcal{U}$ is the structure $\mathfrak{A}_{\mathcal{U}} := \prod_{i \in I} \mathfrak{A}_i / \mathcal{U} := \prod_{\mathcal{U}} \mathfrak{A}_i$ defined as follows:

- (i) $A_{\mathcal{U}} = |\mathfrak{A}_{\mathcal{U}}| := \left\{ \hat{F} : F \in \prod_{i \in I} |\mathfrak{A}_i| \right\}$;
- (ii) for each $p \in \text{Rs}_L$ and each $\hat{F}_0, \dots, \hat{F}_{\nu_L(p)-1} \in A_{\mathcal{U}}$,

$$p^{\mathfrak{A}_{\mathcal{U}}}(\hat{F}_0, \dots, \hat{F}_{\nu_L(p)-1}) :\iff \{i \in I : p^{\mathfrak{A}_i}(F_0(i), \dots, F_{\nu_L(p)-1}(i))\} \in \mathcal{U};$$

(iii) for each $f \in \mathbf{Fs}_L$ and each $\hat{F}_0, \dots, \hat{F}_{\nu_L(f)-1} \in A_{\mathcal{U}}$,

$$f^{\mathfrak{A}_{\mathcal{U}}}(\hat{F}_0, \dots, \hat{F}_{\nu_L(f)-1}) := \hat{G} \quad \text{where for all } i \in I$$

$$G(i) = f^{\mathfrak{A}_i}(F_0(i), \dots, F_{\nu_L(f)-1}(i));$$

(iv) for each $c \in \mathbf{Cs}_L$, $c^{\mathfrak{A}_{\mathcal{U}}} := \hat{G}$ where for all $i \in I$ $[G(i) = c^{\mathfrak{A}_i}]$.

If for all $i \in I$, the structure $\mathfrak{A}_i$ is the same structure $\mathfrak{A}$, then $\mathfrak{A}_{\mathcal{U}}$ is called the **ultrapower** of $\mathfrak{A}$ with respect to $\mathcal{U}$.

As usual, a definition which specifies relationships among equivalence classes in terms of particular representatives of those classes needs to be justified. As an example, consider a unary relation symbol p and suppose that $p^{\mathfrak{A}_{\mathcal{U}}}(\hat{F})$ and $\hat{F} = \hat{G}$. Then

$$\{i : p^{\mathfrak{A}_i}(F(i))\} \cap \{i : F(i) = G(i)\} \subseteq \{i : p^{\mathfrak{A}_i}(G(i))\}.$$

Both of the two sets on the left belong to $\mathcal{U}$, hence so does their intersection and hence so does the set on the right, so we have also $p^{\mathfrak{A}_{\mathcal{U}}}(\hat{G})$. The other justifications are similar.

Example 3.3.17 Let $I = \omega$ and for each $i \in \omega$, $\mathfrak{A}_i = (\omega, +, \mathbf{Sc}, <, 0)$, so elements of $A_{\mathcal{U}}$ are equivalence classes of infinite sequences $(m_0, m_1, \dots)$ of natural numbers. Two such sequences are equivalent iff they agree on a ‘large’ set of indices, a member of $\mathcal{U}$. The standard elements of $A_{\mathcal{U}}$ are $\hat{m}$, the equivalence class of the constant sequence $(m, m, \dots)$. Let $\hat{\omega}$ be the class of $(0, 1, 2, \dots)$. Then unless $\mathcal{U}$ is principal (Example 1.6.11), it follows from Exercise 1.6.23 that for all $m \in \omega$, $\hat{m} <^{\mathfrak{A}_{\mathcal{U}}} \hat{\omega}$, so $\hat{\omega}$ is a non-standard element. So are the equivalence classes of the sequences $(1, 2, 3, \dots)$ and $(0, 0, 1, 2, \dots)$, which respectively follow and precede $\hat{\omega}$ in the ordering $<^{\mathfrak{A}_{\mathcal{U}}}$. If $2\hat{\omega}$ is the class of $(0, 2, 4, \dots)$, then easily (again unless $\mathcal{U}$ is principal), for all m , $\hat{\omega} +^{\mathfrak{A}_{\mathcal{U}}} \hat{m} <^{\mathfrak{A}_{\mathcal{U}}} 2\hat{\omega}$. From Łoś’s Theorem and its corollary below, it follows that in fact $\mathfrak{A}_{\mathcal{U}}$ is a non-standard model of $\mathbf{Th}(\omega, +, \mathbf{Sc}, <, 0)$, and the reader is encouraged to verify directly that it shares the common features of such models established in Proposition 2.5.20.

Example 3.3.18 Let $I = \omega$ and for each $i \in \omega$, $\mathfrak{B}_i = (\mathbb{R}, +, \times, <, 0)$, so elements of $B_{\mathcal{U}}$ are equivalence classes of infinite sequences $(r_0, r_1, \dots)$ of real numbers. Clearly $\mathfrak{A}_{\mathcal{U}}$ from the preceding example is a substructure of $\mathfrak{B}_{\mathcal{U}}$ and $B_{\mathcal{U}}$ contains additional interesting elements such as the equivalence class of the sequence $(1, \frac{1}{2}, \frac{1}{3}, \dots)$. If $\mathcal{U}$ is non-principal, this element is easily seen to have the property of being larger than $\hat{0}$ but smaller than $\hat{r}$, for all real numbers $r > 0$. Hence it is an **infinitesimal**; this observation led to the development of **non-standard analysis** in which the original arguments used

by Leibnitz in founding the calculus are made precise. See the exercises for [Section 3.5](#) below.

To any family $\langle \alpha_i : i \in I \rangle$ of $\mathfrak{A}_i$ -assignments we associate an assignment α for $\prod_{\mathcal{U}} \mathfrak{A}_i$ as follows: for any variable x , $\alpha(x) := \hat{F}_x$, where F_x is the function defined by $F_x(i) = \alpha_i(x)$.

Łoś's Theorem 3.3.19 *For any ultraproduct $\mathfrak{A}_{\mathcal{U}} = \prod_{\mathcal{U}} \mathfrak{A}_i$ and any family $\langle \alpha_i : i \in I \rangle$ of $\mathfrak{A}_i$ -assignments and its associated $\mathfrak{A}$ -assignment α ,*

- (i) *for every term t , $t^{\mathfrak{A}_{\mathcal{U}}}[\alpha] = \hat{G}$ where $(\forall i \in I) G(i) = t^{\mathfrak{A}_i}[\alpha_i]$;*
- (ii) *for every formula ϕ , $\models_{\mathfrak{A}_{\mathcal{U}}} \phi[\alpha] \iff \left\{ i \in I : \models_{\mathfrak{A}_i} \phi[\alpha_i] \right\} \in \mathcal{U}$.*

Proof. We prove first (i) by term-induction. The cases for t a variable or constant symbol are easy and are left for the reader. Suppose that t is fu (with f unary for simplicity). By the induction hypothesis, $u^{\mathfrak{A}_{\mathcal{U}}}[\alpha] = \hat{H}$, where for all $i \in I$, $H(i) = u^{\mathfrak{A}_i}[\alpha_i]$. Then

$$t^{\mathfrak{A}_{\mathcal{U}}}[\alpha] = f^{\mathfrak{A}_{\mathcal{U}}}(u^{\mathfrak{A}_{\mathcal{U}}}[\alpha]) = f^{\mathfrak{A}_{\mathcal{U}}}(\hat{H}) = \hat{G},$$

where for all $i \in I$,

$$G(i) = f^{\mathfrak{A}_i}(H(i)) = f^{\mathfrak{A}_i}(u^{\mathfrak{A}_i}[\alpha_i]) = t^{\mathfrak{A}_i}[\alpha_i],$$

as desired.

For (ii), the atomic case follows from (i); if p is (for simplicity) unary, then with t and G as in (i),

$$\begin{aligned} \models_{\mathfrak{A}_{\mathcal{U}}} pt[\alpha] &\iff p^{\mathfrak{A}_{\mathcal{U}}}(t^{\mathfrak{A}_{\mathcal{U}}}[\alpha]) \iff p^{\mathfrak{A}_{\mathcal{U}}}(\hat{G}) \\ &\iff \left\{ i : p^{\mathfrak{A}_i}(G(i)) \right\} \in \mathcal{U} \\ &\iff \left\{ i : p^{\mathfrak{A}_i}(t^{\mathfrak{A}_i}[\alpha_i]) \right\} \in \mathcal{U} \\ &\iff \left\{ i : \models_{\mathfrak{A}_i} pt[\alpha_i] \right\} \in \mathcal{U}. \end{aligned}$$

The steps for $\neg$ and $\vee$ go just as in the propositional case; for example,

$$\begin{aligned} \models_{\mathfrak{A}_{\mathcal{U}}} (\psi \vee \theta)[\alpha] &\iff \models_{\mathfrak{A}_{\mathcal{U}}} \psi[\alpha] \text{ or } \models_{\mathfrak{A}_{\mathcal{U}}} \theta[\alpha] \\ &\iff \left\{ i : \models_{\mathfrak{A}_i} \psi[\alpha_i] \right\} \in \mathcal{U} \text{ or } \left\{ i : \models_{\mathfrak{A}_i} \theta[\alpha_i] \right\} \in \mathcal{U} \\ &\iff \left\{ i : \models_{\mathfrak{A}_i} \psi[\alpha_i] \right\} \cup \left\{ i : \models_{\mathfrak{A}_i} \theta[\alpha_i] \right\} \in \mathcal{U} \\ &\iff \left\{ i : \models_{\mathfrak{A}_i} (\psi \vee \theta)[\alpha_i] \right\} \in \mathcal{U}. \end{aligned}$$

Finally, suppose that ϕ is $\exists x \psi$. Then

$$\begin{aligned}
 \models_{\mathfrak{A}_{\mathcal{U}}} \phi[\alpha] &\iff \text{for some } \hat{F} \in A_{\mathcal{U}}, \quad \models_{\mathfrak{A}_{\mathcal{U}}} \psi[\alpha_{x \mapsto \hat{F}}] \\
 &\iff \text{for some } F \in A, \quad \left\{ i : \models_{\mathfrak{A}_i} \psi[(\alpha_i)_{x \mapsto F(i)}] \right\} \in \mathcal{U} \\
 &\iff \left\{ i : \text{for some } a_i \in A_i, \quad \models_{\mathfrak{A}_i} \psi[(\alpha_i)_{x \mapsto a_i}] \right\} \in \mathcal{U} \\
 &\iff \left\{ i : \models_{\mathfrak{A}_i} \phi[\alpha_i] \right\} \in \mathcal{U}.
 \end{aligned}$$

The second equivalence uses the easily checked fact that the $\mathfrak{A}_{\mathcal{U}}$ assignment $\alpha_{x \mapsto \hat{F}}$ is associated with the family of $\mathfrak{A}_i$ -assignments $\langle (\alpha_i)_{x \mapsto F(i)} : i \in I \rangle$, and the third implication ($\Leftarrow$) uses again the Axiom of Choice (which was already used once in obtaining the ultrafilter $\mathcal{U}$). $\square$

Corollary 3.3.20 *For any ultraproduct $\mathfrak{A}_{\mathcal{U}} = \prod_{\mathcal{U}} \mathfrak{A}_i$ and any sentence ϕ ,*

$$\models_{\mathfrak{A}_{\mathcal{U}}} \phi \iff \left\{ i \in I : \models_{\mathfrak{A}_i} \phi \right\} \in \mathcal{U}.$$

Hence if $\mathfrak{A}_{\mathcal{U}}$ is an ultrapower of $\mathfrak{A}$, then $\mathfrak{A}_{\mathcal{U}} \equiv \mathfrak{A}$. $\square$

We now have without any restriction on the size of the language

Compactness Theorem 3.3.21 *For any language L and any set Γ of L -sentences,*

- (i) *for any formula ϕ , if $\Gamma \models \phi$, then for some finite $\Gamma_0 \subseteq \Gamma$, $\Gamma_0 \models \phi$;*
- (ii) *if every finite subset of Γ has a model, then Γ has a model.*

Proof. Just as in the countable case it suffices to prove (ii); (i) is an immediate consequence. Fix a set Γ of sentences, let I_{Γ} denote the set of finite subsets of Γ , and for each $i \in I_{\Gamma}$, choose a model $\mathfrak{A}_i$ of i . Let $\mathcal{Y}$ be the set of all subsets of I_{Γ} of the form $Y_{\phi} = \{i \in I_{\Gamma} : \phi \in i\}$ for $\phi \in \Gamma$. Since for any $\phi_0, \dots, \phi_n \in \Gamma$,

$$\{\phi_0, \dots, \phi_n\} \in Y_{\phi_0} \cap \dots \cap Y_{\phi_n},$$

$\mathcal{Y}$ has the finite intersection property so there is an ultrafilter $\mathcal{U}$ over I_{Γ} such that $\mathcal{Y} \subseteq \mathcal{U}$. Let $X_{\phi} = \left\{ i \in I_{\Gamma} : \models_{\mathfrak{A}_i} \phi \right\}$, so for each $\phi \in \Gamma$, $Y_{\phi} \subseteq X_{\phi}$. Then $Y_{\phi} \in \mathcal{U}$, whence also $X_{\phi} \in \mathcal{U}$, so by the Corollary $\models_{\mathfrak{A}_{\mathcal{U}}} \phi$. Thus $\mathfrak{A}_{\mathcal{U}}$ is a model of Γ . $\square$

Boolean algebra

The Boolean algebraic methods introduced in [Section 1.7](#) can also be adapted to give proofs of the Completeness and Compactness Theorems for countable languages. We shall use the same notation as in that section, here with a different meaning.

Definition 3.3.22 For any language L , any set Γ of L -sentences and any L -formulas ϕ and ψ ,

- (i) ϕ and ψ are Γ -**logically equivalent** — in symbols, $\phi \equiv_\Gamma \psi$ — iff $\Gamma \models \phi \leftrightarrow \psi$;
- (ii) $[\phi]_\Gamma := \{ \psi : \phi \equiv_\Gamma \psi \}$.

The relation $\equiv_\Gamma$ is easily seen as before to be an equivalence relation on the set of L -formulas and thus $[\phi]_\Gamma$ is the equivalence class of ϕ . We set

$$\mathcal{S}_\Gamma := \{ [\phi]_\Gamma : \phi \text{ is an } L\text{-formula} \}$$

and make $\mathcal{S}_\Gamma$ into an algebra exactly as in [Section 1.7](#) by defining the following operations and distinguished elements:

$$\begin{aligned} [\phi]_\Gamma \oplus [\psi]_\Gamma &:= [\phi \vee \psi]_\Gamma \\ [\phi]_\Gamma \otimes [\psi]_\Gamma &:= [\phi \wedge \psi]_\Gamma \\ \overline{[\phi]_\Gamma} &:= [\neg \phi]_\Gamma \\ \mathbb{O} &:= [\mathbf{F}]_\Gamma \\ \mathbb{I} &:= [\mathbf{T}]_\Gamma. \end{aligned}$$

The verification that these operations are well-defined goes exactly as before, and if Γ is consistent, then $\mathcal{L}_\Gamma = (\mathcal{S}_\Gamma, \oplus, \otimes, \overline{}, \mathbb{O}, \mathbb{I})$ is a Boolean algebra, the **Lindenbaum algebra** of Γ .

Our general plan here is to show that certain special sorts of ultrafilters exist and that models can be constructed from such ultrafilters. Since we need to assume that Γ is consistent (has a model) to guarantee that $\mathcal{L}_\Gamma$ has more than one element and is therefore a Boolean algebra, it will only be of interest to construct models with special properties from ultrafilters on $\mathcal{L}_\Gamma$; we will do this in [Section 7.3](#). However, just as in [Section 1.7](#), a slight modification of the definition makes the existence of any model an interesting result. As in that section, let $\phi \equiv_\Gamma^* \psi$ iff for some finite $\Gamma_0 \subseteq \Gamma$, $\Gamma_0 \models \phi \leftrightarrow \psi$, and define starred versions of the other notions above using this equivalence. It is tedious but not difficult to verify that $\equiv^*$ is an equivalence relation, that the operations $\oplus$ and $\otimes$ are well-defined on the equivalence classes and that the set $\mathcal{L}_\Gamma^*$ of these classes is a Boolean algebra iff Γ is *finitely* consistent. Now a proof that $\mathcal{L}_\Gamma^*$ has the required sorts of ultrafilters leads to a model for Γ ; this is a proof of the Countable Compactness Theorem. Similarly, we may define $\phi \equiv_\Gamma^\vdash \psi$ iff $\Gamma \vdash \phi \leftrightarrow \psi$ and verify that with the obvious definitions, $\mathcal{L}_\Gamma^\vdash$ is a Boolean algebra iff Γ is $\vdash$ -consistent. The construction of the requisite ultrafilters on $\mathcal{L}_\Gamma^\vdash$ and from them a model of Γ constitutes a new proof of the Countable Completeness Theorem. In the following exposition,

all results asserted for $\mathcal{L}_\Gamma$ hold also for $\mathcal{L}_\Gamma^*$ and $\mathcal{L}_\Gamma^\perp$ with very similar proofs, which we leave to the exercises.

In the case of propositional logic, to construct a model — that is, a satisfying truth assignment — for a set Γ of propositional sentences, it was enough to find any ultrafilter on $\mathcal{L}_\Gamma$. Here it is a bit more complicated, since the qualifications for being an ultrafilter don't say anything about quantifiers. What we need is a special kind of ultrafilter which takes them into account.

Recall from Lemma 1.7.7 that in any Boolean algebra the Boolean sum $a \oplus b$ is the least upper bound of a and b and $a \otimes b$ is the greatest lower bound. By the associative laws (Exercise 1.7.18) it follows immediately that any finite set $\{a_0, \dots, a_n\}$ of elements has a least upper bound $a_0 \oplus \dots \oplus a_n$ and a greatest lower bound $a_0 \otimes \dots \otimes a_n$, but the same is not necessarily true for infinite sets. On the other hand, sometimes these bounds do exist.

Definition 3.3.23 For any Boolean algebra $\mathcal{A}$, $c \in A$ and $B \subseteq A$,

- (i) if c is the least upper bound of B , we write $\sum B = c$ and call c the *supremum* or *sup* of B ;
- (ii) if c is the greatest lower bound of B , we write $\prod B = c$ and call c the *infimum* or *inf* of B .

Examples 3.3.24

- (i) In the Boolean algebra $\wp(X)$ of all subsets of a set X with the usual operations, for any $B \subseteq \wp(X)$, $\sum B$ is just $\bigcup B$ and $\prod B$ is $\bigcap B$.
- (ii) Let A denote the set of all closed-open (or clopen) subsets of the set $\mathcal{I}$ of irrational numbers viewed as a subspace of the real numbers $\mathbb{R}$ with its usual topology. Note that while $\mathbb{R}$ has no non-trivial clopen subsets, every interval (r, s) in $\mathbb{R}$ with rational endpoints determines a clopen subset $(r, s)_\mathcal{I} := (r, s) \cap \mathcal{I}$ of $\mathcal{I}$. Easily A is a field of sets. The following facts are easy to check:

- (a) $\prod_{n \in \omega} (-\frac{1}{n}, \frac{1}{n})_\mathcal{I} = \emptyset = \bigcap_{n \in \omega} (-\frac{1}{n}, \frac{1}{n})_\mathcal{I}$;
- (b) if $r_0 < \dots < r_n < \dots < \pi < \dots < s_n < \dots < s_0$ are rational sequences which converge to π , then

$$\prod_{n \in \omega} (r_n, s_n)_\mathcal{I} = \emptyset \neq \{\pi\} = \bigcap_{n \in \omega} (r_n, s_n)_\mathcal{I};$$

- (c) if $r_0 < \dots < r_n < \dots < e < \pi < \dots < s_n < \dots < s_0$ are rational sequences which converge to e and π respectively, then $\prod_{n \in \omega} (r_n, s_n)_\mathcal{I}$ does not exist.

- (iii) In the Boolean algebra of all finite and cofinite subsets of a set X , if $Y \subseteq X$ is infinite and coinfinite, then $\sum_{y \in Y} \{y\}$ does not exist.
- (iv) If Φ is an infinite set of formulas and $B = \{[\phi]_\Gamma : \phi \in \Phi\}$, then generally $\sum B$ does not exist in $\mathcal{L}_\Gamma$ because $\bigvee \Phi$ is not a formula.

We shall need below the following generalized associative and distributive laws.

Lemma 3.3.25 *For any Boolean algebra $\mathcal{A}$, $a \in A$ and $B \subseteq A$,*

- (i) *if $\sum B$ exists, then*

$$a \oplus \sum B = \sum_{b \in B} (a \oplus b) \quad \text{and} \quad a \otimes \sum B = \sum_{b \in B} (a \otimes b);$$

- (ii) *if $\prod B$ exists, then*

$$a \oplus \prod B = \prod_{b \in B} (a \oplus b) \quad \text{and} \quad a \otimes \prod B = \prod_{b \in B} (a \otimes b).$$

Proof. We prove the second part of (i) and leave the other parts to the reader. Suppose that $\sum B = c$; we aim to show that $\sum_{b \in B} (a \otimes b) = a \otimes c$ — that is, that $a \otimes c$ is the least upper bound of $\{a \otimes b : b \in B\}$. We use Lemma 1.7.7 several times without individual mention. Since $b \leq c$ for all $b \in B$, also $a \otimes b \leq a \otimes c$, so $a \otimes c$ is an upper bound for this set. If it were not the least upper bound then there would be an upper bound d such that $a \otimes c \not\leq d$. Then we conclude successively $(a \otimes c) \otimes \bar{d} \neq \mathbb{O}$, $c \otimes (a \otimes \bar{d}) \neq \mathbb{O}$ and $c \not\leq a \otimes \bar{d}$. It follows that $\overline{a \otimes \bar{d}}$ is not an upper bound for B , so for some $b \in B$, $b \not\leq \overline{a \otimes \bar{d}}$, whence $b \otimes a \otimes \bar{d} \neq \mathbb{O}$ and thus $a \otimes b \not\leq d$, contrary to the choice of d . $\square$

We are mainly concerned here with sups and infs of certain special sets in the Lindenbaum algebra $\mathcal{L}_\Gamma$. For a formula ϕ , let us temporarily write $\phi'_x(z)$ to denote the result of substituting z for all free occurrences of x in the first (in some ordering) variant ϕ' of ϕ for which this substitution is free.

Proposition 3.3.26 *For any set Γ of sentences and any formula ϕ , in $\mathcal{L}_\Gamma$,*

$$\sum \{ [\phi'_x(z)]_\Gamma : z \in \mathbf{Vb} \} = [\exists x \phi]_\Gamma.$$

Proof. By the definition of ϕ' and Theorem 2.2.24, for each z , $\models \phi'_x(z) \rightarrow \exists x \phi$. Hence $[\exists x \phi]_\Gamma$ is an upper bound for the set $\{ [\phi'_x(z)]_\Gamma : z \in \mathbf{Vb} \}$. To see that it is a *least* upper bound, suppose that a formula ψ is another upper bound — that is, for each variable z , $\Gamma \models \phi'_x(z) \rightarrow \psi$. Then iff z is any variable which does not occur in either ϕ or ψ , it follows that $\Gamma \models \exists x \phi \rightarrow \psi$ — that is, $[\exists x \phi]_\Gamma \leq [\psi]_\Gamma$. $\square$

Definition 3.3.27

- (i) For any Boolean algebra $\mathcal{A}$, any $B \subseteq A$ and $a \in A$ such that $\sum B = a$, and any ultrafilter U on $\mathcal{A}$, U **preserves** $\sum B = a$ iff

$$a \in U \iff \text{for some } b \in B, \quad b \in U.$$

- (ii) An ultrafilter U on $\mathcal{L}_\Gamma$ is **quantifier preserving** iff for all formulas ϕ and variables x , U preserves

$$\sum \{ [\phi'_x(z)]_\Gamma : z \in \mathbf{Vb} \} = [\exists x \phi]_\Gamma.$$

Example 3.3.28 For any set Γ of L -sentences, any model $\mathfrak{A}$ of Γ and any $\mathfrak{A}$ -assignment α , let

$$U_{\mathfrak{A},\alpha} := \{ [\phi]_\Gamma : \models_{\mathfrak{A}} \phi[\alpha] \}.$$

Easily, $U_{\mathfrak{A},\alpha}$ is an ultrafilter on $\mathcal{L}_\Gamma$. If for every $a \in A$ there is at least one variable z such that $\alpha(z) = a$, then $U_{\mathfrak{A},\alpha}$ is quantifier preserving. To see this, suppose that $[\exists x \phi]_\Gamma \in U_{\mathfrak{A},\alpha}$. Then for some $a \in A$, $\models_{\mathfrak{A}} \phi[\alpha_{x \mapsto a}]$. By hypothesis, there is some variable z such that $\alpha(z) = a$, so $\models_{\mathfrak{A}} \phi'_x(z)[\alpha]$ and thus $[\phi'_x(z)]_\Gamma \in U_{\mathfrak{A},\alpha}$ as desired.

The following theorem says that this is essentially the only example.

Theorem 3.3.29 *For any consistent set Γ of L -sentences and any quantifier preserving ultrafilter U on $\mathcal{L}_\Gamma$, there exists a countable L -structure $\mathfrak{A}$ and an $\mathfrak{A}$ -assignment α such that $U = U_{\mathfrak{A},\alpha}$. In particular, $\mathfrak{A}$ is a model of Γ .*

Proof. The construction is a variation of one that we have done in several contexts and will only be sketched. We first define an equivalence relation on $\mathbf{Vb}$ by setting

$$x \sim y \iff [x \doteq y]_\Gamma \in U.$$

We denote by $\tilde{x}$ the equivalence class of x and take as the universe of our structure the set $A = \{ \tilde{x} : x \in \mathbf{Vb} \}$. We define the interpretations of the relation, function and constant symbols by

$$\begin{aligned} p^{\mathfrak{A}}(\tilde{x}_0, \dots, \tilde{x}_n) &: \iff [px_0, \dots, x_n]_\Gamma \in U \\ f^{\mathfrak{A}}(\tilde{x}_0, \dots, \tilde{x}_n) = \tilde{y} &: \iff [fx_0, \dots, x_n \doteq y]_\Gamma \in U \\ c^{\mathfrak{A}} = \tilde{y} &: \iff [c \doteq y]_\Gamma \in U. \end{aligned}$$

To justify these definitions we need to verify that for each constant symbol c (and analogously for function symbols) (1) there is some variable y such that $[c \doteq y]_\Gamma \in U$ and (2) if this holds for any other variable z , then $\tilde{y} = \tilde{z}$. (2) follows from the general fact that $\Gamma \models \phi_0 \wedge \dots \wedge \phi_n \rightarrow \psi$ and for

$i \leq n$, $[\phi_i]_\Gamma \in U$, then also $[\psi]_\Gamma \in U$ (Exercise 1.7.21). (1) is a consequence of quantifier preservation: since $\Gamma \models \exists y (c \dot{=} y)$, $[\exists y (c \dot{=} y)]_\Gamma = \mathbb{I} \in U$, so for some variable z , $[c \dot{=} z]_\Gamma \in U$.

This determines $\mathfrak{A}$, and we set $\alpha(x) = \tilde{x}$. A straightforward induction on terms establishes that

$$t^{\mathfrak{A}}[\alpha] = \tilde{y} \iff [t \dot{=} y]_\Gamma \in U.$$

Then by induction on the length of formulas, we can show

$$\models_{\mathfrak{A}} \phi[\alpha] \iff [\phi]_\Gamma \in U.$$

At the induction step for $\exists$ we use again that U is quantifier preserving:

$$\begin{aligned} \models_{\mathfrak{A}} \exists x \phi[\alpha] &\iff \text{for some } z, \models_{\mathfrak{A}} \phi[\alpha_{x \mapsto \tilde{z}}] \\ &\iff \text{for some } z, \models_{\mathfrak{A}} \phi'_x(z)[\alpha] \\ &\iff \text{for some } z, [\phi'_x(z)]_\Gamma \in U \\ &\iff [\exists x \phi]_\Gamma \in U. \quad \square \end{aligned}$$

For this to be of any use, we need to be able to establish the existence of ultrafilters which preserve quantifiers (and for later applications other sups and infs).

Rasiowa-Sikorski Theorem 3.3.30 *For any Boolean algebra $\mathcal{A}$ and any countable set of sups $\sum B_k = a_k$ ($k \in \omega$) in $\mathcal{A}$, there exists an ultrafilter U on $\mathcal{A}$ which preserves all of them.*

Proof. We define recursively elements $b_k \in B_k$ so that for all $k \in \omega$, $\prod_{j < k} (\bar{a}_j \oplus b_j) \neq \mathbb{O}$. From this it follows that $\{\bar{a}_k \oplus b_k : k \in \omega\}$ has the finite intersection property, and if U is any ultrafilter extending this set, we have

$$a_k \in U \implies b_k = \mathbb{O} \oplus (a_k \otimes b_k) = a_k \otimes (\bar{a}_k \oplus b_k) \in U,$$

as desired.

Given b_j for $j < k$ such that $c_k := \prod_{j < k} (\bar{a}_j \oplus b_j) \neq \mathbb{O}$, suppose that for all $b \in B_k$ we had $c_k \otimes (\bar{a}_k \oplus b) = \mathbb{O}$. Then using Lemma 3.3.25,

$$c_k = c_k \otimes \mathbb{I} = c_k \otimes (\bar{a}_k \oplus \sum B_k) = c_k \otimes \sum_{b \in B_k} (\bar{a}_k \oplus b) = \sum_{b \in B_k} c_k \otimes (\bar{a}_k \oplus b) = \mathbb{O},$$

contrary to the induction hypothesis. Hence b_k exists as desired. $\square$

We shall make further use of this machinery in [Chapter 7](#). As we noted above, it leads also to new proofs for the Countable Completeness and Compactness Theorems, which we leave to the exercises.

Exercises

Exercise 3.3.31 Establish the following refinement of the Countable Craig and Robinson Interpolation Theorems, 3.3.12 and 3.3.13, which is a first-order version of the result of Exercise 1.6.26. Part (ii) is known as the **Lyndon Interpolation Theorem**. Define recursively

$$\begin{aligned} \text{Rs}^+(pt_0 \dots t_{n-1}) &= \{p\}; & \text{Rs}^-(pt_0 \dots t_{n-1}) &= \emptyset; \\ \text{Rs}^+(\neg \psi) &= \text{Rs}^-(\psi); & \text{Rs}^-(\neg \psi) &= \text{Rs}^+(\psi); \\ \text{Rs}^+(\psi \vee \theta) &= \text{Rs}^+(\psi) \cup \text{Rs}^+(\theta); & \text{Rs}^-(\psi \vee \theta) &= \text{Rs}^-(\psi) \cup \text{Rs}^-(\theta); \\ \text{Rs}^+(\exists x \psi) &= \text{Rs}^+(\psi); & \text{Rs}^-(\exists x \psi) &= \text{Rs}^-(\psi). \end{aligned}$$

Members of $\text{Rs}^+(\phi)$ are the relation symbols which **occur positively** in ϕ ; members of $\text{Rs}^-(\phi)$ **occur negatively**. For a set Γ of sentences define

$$\text{Rs}^+(\Gamma) := \bigcup \{ \text{Rs}^+(\phi) : \phi \in \Gamma \} \quad \text{and} \quad \text{Rs}^-(\Gamma) := \bigcup \{ \text{Rs}^-(\phi) : \phi \in \Gamma \}.$$

- (i) Prove that for any consistent sets Γ_0 and Γ_1 of sentences, if $\Gamma_0 \cup \Gamma_1$ is inconsistent, then there exists a sentence χ such that $\Gamma_0 \models \chi$, $\Gamma_1 \models \neg \chi$, $\text{Rs}^+(\chi) \subseteq \text{Rs}^+(\Gamma_0) \cap \text{Rs}^-(\Gamma_1)$ and $\text{Rs}^-(\chi) \subseteq \text{Rs}^-(\Gamma_0) \cap \text{Rs}^+(\Gamma_1)$.
- (ii) Conclude that for any sentences $\models \phi \rightarrow \psi$, there exists a sentence χ such that $\models \phi \rightarrow \chi$, $\models \chi \rightarrow \psi$ and $\text{Rs}^\pm(\chi) \subseteq \text{Rs}^\pm(\phi) \cap \text{Rs}^\pm(\psi)$.

Hint: Modify appropriately the notion of consistency used in the proof of the Countable Robinson Interpolation Theorem.

Exercise 3.3.32 In the preceding exercise we did not claim that occurrences of function or constant symbols or of the equality symbol in an interpolant χ could be similarly correlated with their occurrences in the sets Γ_i .

- (i) State precisely the natural conjectures along these lines and explain why the proof you gave does not establish any of these conjectures.
- (ii) Give counterexamples to show that the conjectures are false.
- (iii) Establish the following weak form of the conjecture involving equality: if the language has no function or constant symbols and the equality symbol does not occur in any member of Γ_0 or Γ_1 , then the result of (i) of the preceding exercise holds with the additional condition that the equality symbol does not occur in χ .

Hint: For (iii), modify further the notion of consistency to avoid the problematic cases exposed in (i).

Exercise 3.3.33 Let I be any infinite set, $\langle \mathfrak{A}_i : i \in I \rangle$ an indexed family of L -structures, i_0 a member of I and $\mathcal{U}$ the principal ultrafilter over I generated by i_0 . Show that $\prod_{\mathcal{U}} \mathfrak{A}_i \simeq \mathfrak{A}_{i_0}$.

Exercise 3.3.34 Prove the following first-order version of the result of Exercise 1.6.25: for any language L and any class $\mathcal{K}$ of L -structures,

$$\mathcal{K} \text{ is axiomatizable} \iff \mathcal{K} \text{ is elementarily and ultraproduct closed.}$$

Here $\mathcal{K}$ is **axiomatizable** iff $\mathcal{K} = \text{Mod}(T)$ for some theory T ; **elementarily closed** iff whenever $\mathfrak{A} \equiv \mathfrak{B} \in \mathcal{K}$, also $\mathfrak{A} \in \mathcal{K}$; and **ultraproduct closed** iff for any index set I , any family $\langle \mathfrak{A}_i : i \in I \rangle$ of members of $\mathcal{K}$ and any ultrafilter $\mathcal{U}$ over I , the ultraproduct $\mathfrak{A}_{\mathcal{U}}$ also belongs to $\mathcal{K}$.

Exercise 3.3.35 Verify the extension of the necessary lemmas and propositions to $\mathcal{L}_{\Gamma}^+$ and $\mathcal{L}_{\Gamma}^*$, and carry out the details of the Boolean algebraic proofs of the

- (i) Countable Completeness Theorem;
- (ii) Countable Compactness Theorem.

Exercise 3.3.36

- (i) Show that in any field of sets A and any $B \subseteq A$, if $\bigcup B \in A$, then $\bigcup B$ is the least upper bound of B .
- (ii) Find an example of a field of sets A and a set $B \subseteq A$ such that B has a least upper bound in A which is different from $\bigcup B$.

Exercise 3.3.37 Work out the results suggested in Example 3.3.17.

Exercise 3.3.38 Let I be any infinite set, $\mathcal{U}$ an ultrafilter over I and $\mathfrak{A}$ a structure. Recall that the $\mathcal{U}$ -**ultrapower** of $\mathfrak{A}$ is the ultraproduct $\prod_{\mathcal{U}} \mathfrak{A} = \prod_{\mathcal{U}} \langle \mathfrak{A} : i \in I \rangle$. Prove that there is an elementary embedding $\eta : \mathfrak{A} \hookrightarrow \prod_{\mathcal{U}} \mathfrak{A}$.

Exercise 3.3.39 A substructure $\mathfrak{A}$ of a structure $\mathfrak{B}$ is **finitely generated** iff for some finite set $X \subseteq |\mathfrak{B}|$, $\mathfrak{A} = \langle X \rangle_{\mathfrak{B}}$. Prove that for any structure $\mathfrak{B}$ there exists an ultraproduct $\mathfrak{C}$ of the finitely generated substructures of $\mathfrak{B}$ such that there is an embedding $\eta : \mathfrak{B} \hookrightarrow_0 \mathfrak{C}$. Give an example to show that there may *not* exist an elementary embedding.

Exercise 3.3.40 An interesting example of a Boolean algebra is the **regular open algebra** $\text{RO}(\mathcal{T})$ of any topological space $(X, \mathcal{T})$. For any $Y \subseteq X$, let $Y^\perp$ denote the complement of the closure of Y and $X^\# = X^{\perp\perp}$, the interior of the closure of Y . Y is called regular open iff $Y = Y^\#$. The regular open sets are not generally closed under union — for example, intervals in $\mathbb{R}$ are regular open, but $(0, 1) \cup (1, 2)$ is not regular open, since 1 belongs to the interior of its closure. Accordingly, we define the Boolean operations on $\text{RO}(\mathcal{T})$ as follows:

$$Y \oplus Z := (Y \cup Z)^\#; \quad Y \otimes Z := Y \cap Z; \quad \overline{Y} := (X \sim Y)^\#.$$

Prove and use the following facts to verify that with these operations, $\text{RO}(\mathcal{T})$ is a Boolean algebra. For all $Y, Z \subseteq X$,

- (i) $Y \subseteq Z \implies Z^\perp \subseteq Y^\perp$;
- (ii) if Y is open, then $Y^\perp = Y^{\perp\perp\perp}$;
- (iii) if Y is open, then $\overline{Y} = Y^\perp$;
- (iv) if Y is open, then $Y \subseteq Y^\# = Y^{\#\#}$;
- (v) if Y is closed, then $Y \supseteq Y^\# = Y^{\#\#}$;
- (vi) if Y and Z are open, then $(Y \cap Z)^\# = Y^\# \cap Z^\#$.

Definition 3.3.41 A Boolean algebra $\mathcal{A}$ is **complete** iff for every subset B of the universe of $\mathcal{A}$, both the sup $\sum B$ and the inf $\prod B$ exist.

Exercise 3.3.42 Continue the preceding exercise to show that the algebra $\text{RO}(\mathcal{T})$ is complete.

Exercise 3.3.43

- (i) Prove that any infinite complete Boolean algebra has cardinality at least $2^{\aleph_0}$.
- (ii) It follows that the Lindenbaum algebra of the logical validities of $L_{=}$, the language with no non-logical symbols, is not complete: find an explicit example of a set with no sup.

3.4 Uncountable languages and structures

We have remarked several times that countable, or even finite, languages suffice for most practical purposes (remember that the size of a language is measured by the number of its non-logical symbols; all languages have infinitely many terms and formulas). Still, there are uses for uncountable languages, as we shall see from time to time below, and we discuss here

the extension of the theory to such languages. One common example of an uncountable language is $L^{\mathfrak{A}}$, the extension of L to include names $\dot{a}$ for all elements of the universe of $\mathfrak{A}$, in the case that $\mathfrak{A}$ is an uncountable structure.

In most of the basic theory of [Chapter 2](#), the size of the language plays no role and all of the particular languages discussed in [Sections 2.4–2.5](#) were finite. In fact, we have needed the hypothesis that L is countable only three times:

- (1) in the proof of the Countable Downward Löwenheim-Skolem Theorem 2.3.34 to evaluate the cardinality of the resulting model;
- (2) in the proofs of Propositions 3.1.9 and 3.2.23 to enumerate the sentences for the construction of a finitely consistent ($\vdash$ -consistent) and Henkin complete extension of any finitely consistent ($\vdash$ -consistent) set;
- (3) in the proof of Proposition 3.3.5 for the Countable Model Existence Theorem to enumerate the sentences, terms and constants for the construction of a weakly complete extension of any $\mathbb{C}$ -consistent set.

We shall discuss how to extend each of these to uncountable languages below, but we first note that there is another simpler path to the Completeness Theorem for arbitrary languages, since we have already proved an unrestricted version of the Compactness Theorem 3.3.21.

Completeness Theorem 3.4.1 *For any language L and any set Γ of L -sentences,*

- (i) *for any L -sentence ψ , $\Gamma \vdash \psi \iff \Gamma \models \psi$;*
- (ii) *if Γ is $\vdash$ -consistent, then Γ is consistent (has a model).*

Proof. Since there was no use of countability in the earlier derivation of (i) from (ii), it suffices to prove (ii). Suppose that Γ is $\vdash$ -consistent and Γ_0 is any finite subset of Γ ; clearly Γ_0 is also $\vdash$ -consistent. Let L_0 be the sublanguage of L which contains only the non-logical symbols which occur in some member of Γ_0 ; clearly L_0 is a finite language. By the Alternative Countable Completeness Theorem, Γ_0 is consistent. This means that it has an L_0 -structure as a model, but this may be arbitrarily expanded to an L -structure which will still be a model of Γ_0 . Thus in L , every finite subset of Γ has a model, so by 3.3.21, Γ is consistent. $\square$

We shall need to evaluate the cardinality of sets generated as inductive closures of large induction systems.

Lemma 3.4.2 *For any infinite cardinal number κ ,*

- (i) *if $\langle X_\sigma : \sigma < \kappa \rangle$ is a family of sets such that for each $\sigma < \kappa$, $\text{Card}(X_\sigma) \leq \kappa$, then also $\text{Card}(\bigcup_{\sigma < \kappa} X_\sigma) \leq \kappa$;*
- (ii) *if $\text{Card}(X) \leq \kappa$, then for all $n \in \omega$, $\text{Card}(X^n) \leq \kappa$.*

Proof. These are standard results of set theory and are proved in [Section 6.4](#). $\square$

Theorem 3.4.3 *For any infinite cardinal κ and any induction system $(X, X_0, \mathcal{H})$, if $\text{Card}(X_0) \leq \kappa$ and $\text{Card}(\mathcal{H}) \leq \kappa$, then $\text{Card}(\overline{X}) \leq \kappa$.*

Proof. The proof is much along the lines of that of Proposition 1.2.8. We prove by induction that for all n , $\text{Card}(X_n) \leq \kappa$. Since $\overline{X} = \bigcup_{n \in \omega} X_n$, it follows from the lemma that $\overline{X}$ is of cardinality at most κ . This is true for X_0 by hypothesis. Assume as induction hypothesis that $\text{Card}(X_n) \leq \kappa$. For each $H \in \mathcal{H}$ let

$$X_{n,H} = \{ H(x_0, \dots, x_{k_H-1}) : x_0, \dots, x_{k_H-1} \in X_n \},$$

so that $X_{n+1} = X_n \cup \bigcup_{H \in \mathcal{H}} X_{n,H}$. By the lemma, for each $H \in \mathcal{H}$, $\text{Card}((X_n)^{k_H}) \leq \kappa$, hence $\text{Card}(X_{n,H}) \leq \kappa$, and thus $\text{Card}(X_{n+1}) \leq \kappa$. $\square$

We now can prove without limitation on the size of the language

Downward Löwenheim-Skolem Theorem 3.4.4 *For any infinite cardinal κ , any language L of cardinality at most κ , any L -structure $\mathfrak{B}$ and any subset X of $|\mathfrak{B}|$ with $\text{Card}(X) \leq \kappa$, there exists an L -structure $\mathfrak{A}$ of cardinality $\leq \kappa$ such that $X \subseteq |\mathfrak{A}|$ and $\mathfrak{A} \prec \mathfrak{B}$.*

Proof. Exactly as before with the preceding theorem in place of Proposition 1.2.8. $\square$

The extensions of the proofs of the Countable Compactness, Completeness, and Model Existence Theorems to the uncountable case rely on replacing the recursive definition of the sequence $\langle \Delta_n : n \in \omega \rangle$ by a definition by transfinite recursion. We present explicitly the Completeness proof; the direct proof of Compactness is very similar.

Proposition 3.4.5 *For any infinite cardinal κ , any language L of cardinality κ and any $\vdash$ -consistent set Γ of L -sentences, there exists an extension L^+ of L by constants of cardinality κ and a set $\Delta \supseteq \Gamma$ of L^+ -sentences which is $\vdash$ -consistent and L^+ -Henkin complete.*

Proof. Let L and Γ be as in the hypothesis and form L^+ by adjoining a new set of constant symbols $\{e_\sigma : \sigma < \kappa\}$. By the lemma L^+ still has cardinality κ and hence so does the set of L^+ -sentences; fix an enumeration $\langle \theta_\sigma : \sigma < \kappa \rangle$ of this set. We construct by transfinite recursion a sequence Δ_σ of sets of formulas such that for each σ , fewer than κ many of the new constants occur in members of Δ_σ . Set $\Delta_0 = \Gamma$, and for all $\sigma < \kappa$, let σ^* be the smallest ordinal such that the constant e_{σ^*} does not occur in θ_σ or in any formula in

Δ_σ . Then for all $\sigma < \kappa$,

$$\begin{aligned}\Delta_\sigma^0 &:= \begin{cases} \Delta_\sigma \cup \{\theta_\sigma\}, & \text{if this set is } \vdash\text{-consistent;} \\ \Delta_\sigma, & \text{otherwise;} \end{cases} \\ \Delta_{\sigma+1} &:= \begin{cases} \Delta_\sigma^0 \cup \{\chi_x(e_{\sigma^*})\}, & \text{if } \theta_\sigma \in \Delta_\sigma^0 \text{ and } \theta_\sigma \text{ is } \exists x \chi; \\ \Delta_\sigma^0, & \text{otherwise;} \end{cases} \\ \Delta_\sigma &:= \bigcup_{\tau < \sigma} \Delta_\tau \quad \text{if } \sigma \text{ is a limit ordinal;} \\ \Delta &:= \bigcup_{\sigma < \kappa} \Delta_\sigma.\end{aligned}$$

The proof that Δ is both $\vdash$ -consistent and L^+ -Henkin complete is essentially the same as before; there are only two new wrinkles. One is verifying that fewer than κ many of the new constants occur in each Δ_σ so that e_{σ^*} is well-defined. In fact, we prove by transfinite induction that for all $\sigma < \kappa$

$$\text{Card}(\{\tau : e_\tau \text{ occurs in some member of } \Delta_\sigma\}) \leq \max\{\text{Card}(\sigma), \aleph_0\}.$$

For $\sigma = 0$ this is clear, and at successor stages we add at most one e_τ . For limit σ , the set of τ which occur in some member of Δ_σ is the union of $\text{Card}(\sigma)$ -many sets each of cardinality at most $\max\{\text{Card}(\sigma), \aleph_0\}$, so this set is also of this size. The other new point is the limit case in the inductive proof that each Δ_σ is $\vdash$ -consistent; this follows immediately from Proposition 3.2.16. $\square$

Alternative Completeness Theorem 3.4.6 *For any infinite cardinal κ , any language L of cardinality κ and any set Γ of L -sentences, if Γ is $\vdash$ -consistent, then Γ is consistent — in fact, Γ has a model of cardinality at most κ .*

Proof. Given the hypotheses, we first extend L and Γ to L^+ and Δ as in the proposition so that Δ is $\vdash$ -consistent and L^+ -Henkin complete. We construct the model $\mathfrak{A}$ exactly as in the proof of the countable version. Since $|\mathfrak{A}|$ consists of equivalence classes of elements of a well-ordered set of cardinality κ , it has cardinality at most κ . $\square$

Note that just as in the countable case, the bound on the size of the model could also have been derived from the Downward Löwenheim-Skolem Theorem.

Finally, we consider notions of consistency for uncountable languages.

Definition 3.4.7 For any infinite cardinal κ and any language L of cardinality κ , an *L -notion of consistency* is a pair $(\mathbb{C}, E)$ which satisfies all of the conditions of Definition 3.3.2 together with

- (v) for any $\sigma < \kappa$ and any family $\langle \Gamma_\tau : \tau < \sigma \rangle$ of sets of sentences, if for all $\tau, \nu < \sigma$, $\Gamma_\tau \in \mathbb{C}$ and $\tau < \nu \implies \Gamma_\tau \subseteq \Gamma_\nu$, then $(\bigcup_{\tau < \sigma} \Gamma_\tau) \in \mathbb{C}$.

The additional condition is exactly what we need to carry through the limit stages of the extension construction. The construction of the model presents no new difficulties and we have

Model Existence Theorem 3.4.8 *For any infinite cardinal κ , any language L of cardinality κ and any L -notion of consistency $(\mathbb{C}, E)$, every $\mathbb{C}$ -consistent set has a model $\mathfrak{A}$ of cardinality at most κ with universe $|\mathfrak{A}| = \{e^{\mathfrak{A}} : e \in E\}$. $\square$*

The Downward Löwenheim-Skolem Theorem implies that a consistent theory in a language of cardinality at most κ has models of power at most κ — in other words, models as small as could be expected. Using compactness we can also derive that consistent theories have large models. Then we shall strengthen this by arranging that the large model is an elementary extension of a given smaller model.

Weak Upward Löwenheim-Skolem Theorem 3.4.9 *For any infinite cardinal κ , any language L of cardinality at most κ and any L -theory T , if T has an infinite model, then for every cardinal $\lambda \geq \kappa$, T has a model of cardinality λ . In particular, if a theory T in a countable language has an infinite model, then T has models of every infinite cardinality.*

Proof. Under the given hypotheses, for any $\lambda \geq \kappa$ let L^+ be the extension of L obtained by adjoining a set C of cardinality λ of new constant symbols and let

$$\Gamma := T \cup \{ \neg c \doteq d : c, d \text{ distinct members of } C \}.$$

Any finite subset Γ_0 of Γ may be satisfied by assigning the finitely many members of C mentioned in Γ_0 to any distinct elements of any infinite model of T . Hence by compactness, Γ has a model $\mathfrak{B}^+$. Since the elements $c^{\mathfrak{B}^+}$ for $c \in C$ must all be distinct, the cardinality of $\mathfrak{B}^+$ is at least λ . It may, in fact, be greater than λ , but since $\lambda \geq \kappa$, by the Downward Löwenheim-Skolem Theorem, $\mathfrak{B}^+$ has an elementary substructure $\mathfrak{A}^+$ of cardinality at most λ which necessarily includes all of the $c^{\mathfrak{B}^+}$ and hence must be of cardinality exactly λ . Of course, since $\mathfrak{A}^+ \equiv \mathfrak{B}^+$, $\mathfrak{A}^+$ is also a model of T , hence so is $\mathfrak{A} := \mathfrak{A}^+ \upharpoonright L$. $\square$

The following generalizes Corollary 2.5.3(v).

Corollary 3.4.10 *For any language L and any L -theory T , if T has an infinite model, then T is not categorical. $\square$*

To prove the general form of the Upward Löwenheim-Skolem Theorem, we need to develop some tools for embedding one structure in another. In the general form in which we prove them, they will also have many other applications in the next section as well as in [Chapter 7](#).

It will be convenient to introduce here an alternative notation for indicating the free variables of a formula and satisfaction. If $\mathbf{x} := x_0, \dots, x_{k-1}$, we write $\phi(\mathbf{x})$ to denote a formula with free variables among $x_0, \dots, x_{k-1}$. For constant symbols $\mathbf{c} := c_0, \dots, c_{k-1}$, $\phi(\mathbf{c})$ denotes the result of substituting each c_i for x_i in $\phi(\mathbf{x})$. Conversely, in a context which specifies a collection of constant symbols, $\phi(\mathbf{c})$ indicates that of these symbols only $c_0, \dots, c_{k-1}$ occur in ϕ and $\phi(\mathbf{x})$ denotes the result of replacing each c_i by x_i . If $\mathfrak{A}$ is a structure and $\mathbf{a} := a_0, \dots, a_{k-1} \in A$, then $\models_{\mathfrak{A}} \phi[\mathbf{a}]$ means that $\models_{\mathfrak{A}} \phi[\alpha]$ for any $\mathfrak{A}$ -assignment such that each $\alpha(x_i) = a_i$. By convention, the sequence $\mathbf{a}$ will always be of the appropriate length. If η is a function with domain including $a_0, \dots, a_{k-1}$, then $\eta(\mathbf{a}) := (\eta(a_0), \dots, \eta(a_{k-1}))$.

Definition 3.4.11 For any set Δ of L -formulas and any L -structures $\mathfrak{A}$ and $\mathfrak{B}$,

- (i) $\mathfrak{A}$ is a Δ -**substructure** of $\mathfrak{B}$ and $\mathfrak{B}$ is a Δ -**extension** of $\mathfrak{A}$ — in symbols, $\mathfrak{A} \prec^{\Delta} \mathfrak{B}$ — iff $|\mathfrak{A}|$ is a subset of $|\mathfrak{B}|$ and for all $\phi(\mathbf{x}) \in \Delta$ and any $\mathbf{a} \in A$,

$$\models_{\mathfrak{A}} \phi[\mathbf{a}] \implies \models_{\mathfrak{B}} \phi[\mathbf{a}];$$

- (ii) for any function $\eta : A \rightarrow B$, η is a Δ -**embedding** of $\mathfrak{A}$ into $\mathfrak{B}$ — in symbols $\eta : \mathfrak{A} \hookrightarrow^{\Delta} \mathfrak{B}$ — iff η is injective and for all $\phi(\mathbf{x}) \in \Delta$ and any $\mathbf{a} \in A$,

$$\models_{\mathfrak{A}} \phi[\mathbf{a}] \implies \models_{\mathfrak{B}} \phi[\eta(\mathbf{a})];$$

- (iii) $\mathfrak{A}$ is Δ -**embeddable** into $\mathfrak{B}$ — in symbols $\mathfrak{A} \hookrightarrow^{\Delta} \mathfrak{B}$ — iff for some η , $\eta : \mathfrak{A} \hookrightarrow^{\Delta} \mathfrak{B}$.

Remark 3.4.12 As the following proposition shows, these notions generalize the notions of (elementary) substructure and (elementary) embedding introduced in [Section 2.3](#). However, it is important to note that the definitions are asymmetric in that they contain implications rather than bi-implications. This is equivalent for Δ closed under negation (see below), but we will be using them also for Δ not so closed. Furthermore, in the literature they are sometimes defined with the implications going the other way.

Recall from Definition 2.2.31 that $\mathbf{QF}$ denotes the set of quantifier-free formulas, and we write $\mathbf{Fm}$ for the set of all formulas (of a fixed language under discussion).

Proposition 3.4.13 *For any L -structures $\mathfrak{A}$ and $\mathfrak{B}$ and any function $\eta : A \rightarrow B$,*

- (i) $\mathfrak{A} \subseteq \mathfrak{B}$ ($\mathfrak{A}$ is a substructure of $\mathfrak{B}$) iff $\mathfrak{A} \prec^{\text{QF}} \mathfrak{B}$;
- (ii) $\mathfrak{A} \hookrightarrow_0 \mathfrak{B}$ ($\mathfrak{A}$ is embeddable in $\mathfrak{B}$) iff $\mathfrak{A} \hookrightarrow^{\text{QF}} \mathfrak{B}$;
- (iii) $\mathfrak{A} \prec \mathfrak{B}$ ($\mathfrak{A}$ is an elementary substructure of $\mathfrak{B}$) iff $\mathfrak{A} \prec^{\text{Fm}} \mathfrak{B}$;
- (iv) $\mathfrak{A} \hookrightarrow \mathfrak{B}$ ($\mathfrak{A}$ is elementarily embeddable into $\mathfrak{B}$) iff $\mathfrak{A} \hookrightarrow^{\text{Fm}} \mathfrak{B}$.

Proof. These are essentially just reformulations of Proposition 2.3.21 and the definitions of (elementary) embedding together with the fact that the sets QF and Fm are regular Boolean (see the following definition and proposition). $\square$

Although the definitions make sense for any set Δ of formulas, the examples of interest share some regularity properties which permit some general results.

Definition 3.4.14 For any language L , set Δ of L -formulas is **regular** iff

- (i) all L -atomic formulas and their negations belong to Δ ;
- (ii) Δ is closed under $\wedge$ and $\vee$;
- (iii) Δ is closed under substitution of variables for free variables.

If Δ is also closed under $\neg$, then it is called (in accord with Definition 2.4.27) **regular Boolean**.

Clearly QF and Fm are regular Boolean; some other important examples are the following (see Definition 2.4.18).

Lemma 3.4.15 *For any regular set Δ of formulas,*

- (i) $\forall\Delta$, $\exists\Delta$ and $\neg\Delta$ are regular;
- (ii) if Δ is regular Boolean, then also $\forall\Delta \cap \exists\Delta$ is regular Boolean;
- (iii) $\text{Bool}(\Delta) \subseteq \forall\Delta \cap \exists\Delta$;
- (iv) for all n , $\exists_n$ and $\forall_n$ are regular and $\exists_n \cap \forall_n$ is regular Boolean.

Proof. Part (i) is proved as was Lemma 2.3.23, and (ii) then follows, since $\neg\forall\Delta = \exists\neg\Delta$ and $\neg\exists\Delta = \forall\neg\Delta$. For (iii) we have first $\Delta \subseteq \forall\Delta \cap \exists\Delta$ because the quantifier strings in Definition 2.4.18 may be empty. Then the result follows by an easy induction. Part (iv) is proved by a straightforward induction on n using (i) and (ii). $\square$

Proposition 3.4.16 *For any regular set Δ of formulas and any structures $\mathfrak{A}$, $\mathfrak{B}$ and $\mathfrak{B}'$,*

- (i) $\mathfrak{A} \prec \mathfrak{B} \implies \mathfrak{A} \prec^\Delta \mathfrak{B} \implies \mathfrak{A} \subseteq \mathfrak{B}$;
- (ii) $\mathfrak{A} \prec^\Delta \mathfrak{B} \iff \mathfrak{A} \prec^{\exists\Delta} \mathfrak{B}$;
- (iii) $\mathfrak{A} \hookrightarrow^\Delta \mathfrak{B}'$ iff for some structure $\mathfrak{A}'$, $\mathfrak{A} \simeq \mathfrak{A}' \prec^\Delta \mathfrak{B}'$;
- (iv) if Δ is regular Boolean, then $\mathfrak{A} \prec^\Delta \mathfrak{B}$ iff $|\mathfrak{A}|$ is a subset of $|\mathfrak{B}|$, and for all $\phi(\mathbf{x}) \in \Delta$ and any $\mathbf{a} \in A$,

$$\models_{\mathfrak{A}} \phi[\mathbf{a}] \iff \models_{\mathfrak{B}} \phi[\mathbf{a}].$$

Proof. The first implication of (i) is immediate; the second follows from the hypothesis that all atomic formulas and their negations belong to Δ . (ii) is a simple generalization of the upward persistence of existential formulas, Proposition 2.3.25. (iii) is a generalization of Propositions 2.3.29 and 2.3.42. Finally (iv) follows from the implications

$$\models_{\mathfrak{B}} \phi[\mathbf{a}] \implies \not\models_{\mathfrak{B}} \neg \phi[\mathbf{a}] \implies \not\models_{\mathfrak{A}} \neg \phi[\mathbf{a}] \implies \models_{\mathfrak{A}} \phi[\mathbf{a}]$$

and the hypothesis that $\phi \in \Delta \implies \neg \phi \in \Delta$. $\square$

For many purposes it is useful to reformulate part (iii) as follows.

Proposition 3.4.17 *For any structures $\mathfrak{A}$ and $\mathfrak{B}'$ and any regular set Δ of formulas,*

$$\mathfrak{A} \hookrightarrow^\Delta \mathfrak{B}' \iff \text{for some structure } \mathfrak{B}, \quad \mathfrak{A} \prec^\Delta \mathfrak{B} \simeq \mathfrak{B}'.$$

Proof. The implication $(\iff)$ is immediate; we prove the implication $(\implies)$. Suppose that $\eta : \mathfrak{A} \hookrightarrow^\Delta \mathfrak{B}'$, and by (iii) of the preceding Proposition, fix $\mathfrak{A}'$ such that $\eta : \mathfrak{A} \simeq \mathfrak{A}' \prec^\Delta \mathfrak{B}'$. Let C be any set such that $C \cap A = \emptyset$ and C has the same cardinality as $B' \sim A'$. Fix a bijective function $F : C \rightarrow B' \sim A'$, let $B' = A \cup C$, and define $\zeta : B \rightarrow B'$ by

$$\zeta(b) = \begin{cases} \eta(b), & \text{if } b \in A; \\ F(b), & \text{if } b \in C. \end{cases}$$

Clearly ζ is bijective. We may now define a structure $\mathfrak{B}$ with universe B in the unique way that makes ζ an isomorphism of $\mathfrak{B}$ with $\mathfrak{B}'$: for any $\mathbf{b} = b_0, \dots, b_{k-1}$ and $\mathbf{b}' := \zeta(\mathbf{b})$, for each of the non-logical symbols of the language, set

$$p^{\mathfrak{B}}(\mathbf{b}) : \iff p^{\mathfrak{B}'}(\mathbf{b}'); \quad f^{\mathfrak{B}}(\mathbf{b}) := \zeta^{-1}(f^{\mathfrak{B}'}(\mathbf{b}')); \quad c^{\mathfrak{B}} := \zeta^{-1}(c^{\mathfrak{B}'}).$$

Now for any formula $\phi(\mathbf{x}) \in \Delta$ and any $\mathbf{a} \in A$,

$$\models_{\mathfrak{A}} \phi[\mathbf{a}] \iff \models_{\mathfrak{A}}, \phi[\eta(\mathbf{a})] \implies \models_{\mathfrak{B}}, \phi[\eta(\mathbf{a})] \iff \models_{\mathfrak{B}}, \phi[\zeta(\mathbf{a})] \iff \models_{\mathfrak{B}} \phi[\mathbf{a}],$$

so $\mathfrak{A} \prec^{\Delta} \mathfrak{B}$. $\square$

Still another characterization of these notions goes as follows. Recall from Definition 2.6.9 that for any L -structure $\mathfrak{A}$ and any $X \subseteq |\mathfrak{A}|$, L^X denotes the extension of L by new constant symbols $\dot{a}$ for each $a \in X$ and $\mathfrak{A}_X$ denotes the canonical expansion of $\mathfrak{A}$ to an L^X -structure obtained by setting $\dot{a}^{\mathfrak{A}_X} = a$. We say that L^X has **names** for all $a \in X$. Then for any structure $\mathfrak{A}$, any formula $\phi(\mathbf{x})$ and any $\mathbf{a} \in X$, $\phi(\dot{\mathbf{a}})$ is a sentence and

$$(4) \quad \models_{\mathfrak{A}} \phi[\mathbf{a}] \iff \models_{\mathfrak{A}_X} \phi(\dot{\mathbf{a}}).$$

If $\eta : |\mathfrak{A}| \rightarrow |\mathfrak{B}|$, then for any $X \subseteq |\mathfrak{A}|$ we denote by $\mathfrak{B}_X^{\eta}$ the expansion of $\mathfrak{B}$ to an L^X -structure obtained by setting $\dot{a}^{\mathfrak{B}_X^{\eta}} := \eta(a)$ for each $a \in X$.

Definition 3.4.18 For any L -structures $\mathfrak{A}$ and $\mathfrak{B}$, any $X \subseteq |\mathfrak{A}|$ and $Y \subseteq |\mathfrak{B}|$ and any set Δ of L -formulas,

- (i) $\text{Th}^{\Delta}(\mathfrak{A}_X) := \left\{ \phi(\dot{\mathbf{a}}) : \mathbf{a} \in X, \quad \phi(\mathbf{x}) \in \Delta, \quad \text{and} \quad \models_{\mathfrak{A}_X} \phi(\dot{\mathbf{a}}) \right\};$
- (ii) $\mathfrak{A}_X \equiv^{\Delta} \mathfrak{B}_Y : \iff \text{Th}^{\Delta}(\mathfrak{A}_X) = \text{Th}^{\Delta}(\mathfrak{B}_Y).$

For the moment we will use the case $X = A$, but other X will play a role below.

Proposition 3.4.19 For any L -structures $\mathfrak{A}$ and $\mathfrak{B}$, any regular set Δ of formulas and any injective function $\eta : |\mathfrak{A}| \rightarrow |\mathfrak{B}|$,

- (i) $\mathfrak{A} \prec^{\Delta} \mathfrak{B} \iff |\mathfrak{A}| \subseteq |\mathfrak{B}| \quad \text{and} \quad \mathfrak{B}_A \text{ is a model of } \text{Th}^{\Delta}(\mathfrak{A}_A);$
- (ii) $\eta : \mathfrak{A} \hookrightarrow^{\Delta} \mathfrak{B} \iff \mathfrak{B}_A^{\eta} \text{ is a model of } \text{Th}^{\Delta}(\mathfrak{A}_A);$
- (iii) if Δ is regular Boolean, then

$$\mathfrak{A} \prec^{\Delta} \mathfrak{B} \iff |\mathfrak{A}| \subseteq |\mathfrak{B}| \quad \text{and} \quad \mathfrak{A}_A \equiv^{\Delta} \mathfrak{B}_A.$$

Proof. Part (i) is immediate from the equivalence (4) above as is (ii) with the observation that since by the hypothesis of regularity the formulas $\neg x \doteq y$ belong to Δ , the right-hand side of (ii) guarantees that η is injective. (iii) follows using the closure of Δ under negation. $\square$

Corollary 3.4.20 *For any L -structures $\mathfrak{A}$ and $\mathfrak{B}$ and any regular set Δ of formulas,*

$$\mathfrak{A} \hookrightarrow^\Delta \mathfrak{B} \iff \text{there exists an expansion of } \mathfrak{B} \text{ to an } L^A\text{-structure} \\ \text{which is a model of } \text{Th}^\Delta(\mathfrak{A}_A).$$

Proof. The implication $(\implies)$ is immediate from (ii) of the preceding proposition. Suppose that the L^A -expansion $\mathfrak{B}^+$ of $\mathfrak{B}$ is a model of $\text{Th}^\Delta(\mathfrak{A}_A)$, and define $\eta : A \rightarrow B$ by $a \mapsto \hat{a}^{\mathfrak{B}^+}$. Then η is injective because the formula $\neg x \doteq y$ is in Δ and $\mathfrak{B}^+ = \mathfrak{B}_A^\eta$, so again by the proposition, $\mathfrak{A} \hookrightarrow^\Delta \mathfrak{B}$. $\square$

The culmination of this sequence of reasoning is the

First Extension Lemma 3.4.21 *For any L -structure $\mathfrak{A}$, any set Γ of L -sentences and any regular set Δ of L -formulas,*

$$\Gamma \cup \text{Th}^\Delta(\mathfrak{A}_A) \text{ is consistent} \\ \iff \text{there exists a model } \mathfrak{B} \text{ of } \Gamma \text{ such that } \mathfrak{A} \prec^\Delta \mathfrak{B}.$$

Proof. Suppose that $\mathfrak{C}^+$ is an L^A -structure which is a model of $\Gamma \cup \text{Th}^\Delta(\mathfrak{A}_A)$. Then by the preceding Corollary, $\mathfrak{A} \hookrightarrow^\Delta \mathfrak{C}$, so by Proposition 3.4.17 there is a structure $\mathfrak{B}$ such that $\mathfrak{A} \prec^\Delta \mathfrak{B} \simeq \mathfrak{C}$. Since $\mathfrak{C}$ is a model of Γ , so is $\mathfrak{B}$. $\square$

Our first application is the

Upward Löwenheim-Skolem Theorem (LS $\uparrow$) 3.4.22 *For any infinite cardinal κ , any language L of cardinality at most κ and any infinite L -structure $\mathfrak{A}$ of cardinality at most κ , for every cardinal $\lambda \geq \kappa$, there exists an L -structure $\mathfrak{B}$ of cardinality λ such that $\mathfrak{A} \prec \mathfrak{B}$.*

Proof. Under the given hypotheses, let L^+ be the extension of L obtained by adjoining a set C of cardinality λ of new constant symbols and let

$$\Gamma := \{ \neg c \doteq d : c, d \text{ distinct members of } C \}.$$

As in the proof of the weak version 3.4.9 above, since $\mathfrak{A}$ is infinite, any finite subset Γ_0 of $\Gamma \cup \text{Th}(\mathfrak{A}_A)$ may be satisfied by an appropriate expansion of $\mathfrak{A}_A$ which assigns distinct elements to the finitely many of the new constant symbols which occur in Γ_0 . Hence by Compactness, $\Gamma \cup \text{Th}(\mathfrak{A}_A)$ is consistent, so by the First Extension Lemma Γ has a model $\mathfrak{C}$ with $\mathfrak{A} \prec \mathfrak{C}$. Clearly $\text{Card}(\mathfrak{C}) \geq \lambda$, so by the Downward Löwenheim-Skolem Theorem there exists a structure $\mathfrak{B}$ such that $A \subseteq B$, $\mathfrak{B} \prec \mathfrak{C}$ and $\text{Card}(\mathfrak{B}) = \lambda$. Then also $\mathfrak{A} \prec \mathfrak{B}$, since for any formula ϕ and any $\mathbf{a} \in A$,

$$\models_{\mathfrak{A}} \phi[\mathbf{a}] \iff \models_{\mathfrak{C}} \phi[\mathbf{a}] \iff \models_{\mathfrak{B}} \phi[\mathbf{a}]. \quad \square$$

Corollary 3.4.23 *For any countable language L and infinite L -structure $\mathfrak{A}$, for every cardinal $\lambda \geq \text{Card}(\mathfrak{A})$ there exists an L -structure $\mathfrak{B}$ of cardinality λ such that $\mathfrak{A} \prec \mathfrak{B}$. $\square$*

One easy consequence is the full version of Vaught's Test 2.4.25. Recall (Definition 2.5.2) that for any infinite cardinal κ , a theory T is called κ -**categorical** iff for any two models $\mathfrak{A}$ and $\mathfrak{B}$ of T of cardinality κ , $\mathfrak{A} \simeq \mathfrak{B}$.

Theorem 3.4.24 (Vaught Test) *For any countable consistent theory T , if T has no finite models and for some infinite cardinal κ , T is κ -categorical, then T is complete.*

Proof. Suppose that T satisfies the hypothesis, and let $\mathfrak{A}$ and $\mathfrak{B}$ be any two models of T . By the Downward Löwenheim-Skolem Theorem there exist countable elementary substructures $\mathfrak{A}' \prec \mathfrak{A}$ and $\mathfrak{B}' \prec \mathfrak{B}$. By the Upward Löwenheim-Skolem Theorem there exist elementary extensions $\mathfrak{A}' \prec \mathfrak{A}''$ and $\mathfrak{B}' \prec \mathfrak{B}''$ each of cardinality κ . Then

$$\mathfrak{A} \equiv \mathfrak{A}' \equiv \mathfrak{A}'' \simeq \mathfrak{B}'' \equiv \mathfrak{B}' \equiv \mathfrak{B},$$

so $\mathfrak{A} \equiv \mathfrak{B}$. Now T is complete by Corollary 2.4.16. $\square$

This result is a powerful tool for establishing completeness. First, recall from Proposition 2.5.12 that the theory T_0 generated by the axioms $(S1)$ – $(S4_m)$ is not $\aleph_0$ -categorical but is κ -categorical for all uncountable cardinals κ . Hence Vaught's Test provides another proof that T_0 is complete.

As a new example, we turn to the theory of algebraically closed fields. We assume that the reader has some familiarity with fields and field extensions, and we shall base our result on several algebraic facts for which proofs may be found in any good advanced algebra text, for example [Hun]. Let L_{FL} be the language of field theory with non-logical symbols $\dot{+}$, $\dot{-}$, $\dot{\times}$, $\dot{0}$ and $\dot{1}$ and T_{FL} the theory of fields — that is, the L_{FL} -theory generated by the L_{FL} -sentences corresponding to conditions (a)–(i) of Example 2.1.1(iv). T_{FL} is, of course, not complete since it fails to decide the characteristic.

Definition 3.4.25 For any prime natural number p ,

- (i) Char_p is the L_{FL} -sentence $\dot{1} \dot{+} \cdots \dot{+} \dot{1} \dot{=} \dot{0}$ with p occurrences of $\dot{1}$;
- (ii) $T_{\text{FL}(p)} := \text{Th}(T_{\text{FL}} \cup \{\text{Char}_p\}$;
- (iii) $T_{\text{FL}(0)} := \text{Th}(T_{\text{FL}} \cup \{\neg \text{Char}_p : p \text{ a prime}\})$.

A model of $T_{\text{FL}(p)}$ is called a field of **characteristic** p .

Fact 3.4.26 Every field has a unique characteristic either 0 or a prime p .

A **polynomial** $P(x)$ with coefficients in a field $\mathfrak{A}$ is an expression of the form

$$P(x) := b_n x^n + \cdots + b_1 x + b_0$$

with $b_0, \dots, b_n \in A$. A **root** of $P(x)$ is an element a of A such that $P(a) = \dot{0}^{\mathfrak{A}}$, where of course the operations are those of $\mathfrak{A}$. For each $n > 0$, let $t_n(x, y_0, \dots, y_n)$ be the L_{FL} -term

$$y_n \dot{\times} \underbrace{x \dot{\times} \cdots \dot{\times} x}_n \dot{+} \cdots \dot{+} y_1 \dot{\times} x \dot{+} y_0.$$

Then for $a, b_0, \dots, b_n, c \in A$,

$$P(a) = c \iff \models_{\mathfrak{A}} (t_n \dot{=} z)[a, b_0, \dots, b_n, c].$$

A field $\mathfrak{A}$ is called **algebraically closed** iff every non-constant polynomial $P(x)$ with coefficients in A has a root in A . Hence, if T_{ACF} is the theory generated by

$$T_{\text{FL}} \cup \{ \forall y_0 \dots \forall y_n \exists x [\neg y_n \dot{=} \dot{0} \rightarrow t_n(x, y_0, \dots, y_n) \dot{=} \dot{0}] : n > 0 \},$$

then the algebraically closed fields are exactly the models of T_{ACF} . Similarly, $T_{\text{ACF}(p)}$ is the extension which includes the axiom for Char_p (for p a prime) or all $\neg \text{Char}_p$ (for $p = 0$).

Fact 3.4.27 If $\mathfrak{A} \subseteq \mathfrak{B}$ are algebraically closed fields and $P(x)$ is a polynomial with coefficients in $\mathfrak{A}$, then every root of $P(x)$ in $\mathfrak{B}$ is already in $\mathfrak{A}$.

When $\mathfrak{A} \subseteq \mathfrak{B}$ are algebraically closed fields, an element $b \in B \sim A$ is called **transcendental** over $\mathfrak{A}$. For any set $X \subseteq B$ of elements transcendental over $\mathfrak{A}$, there is a smallest algebraically closed field $\mathfrak{A}(X) \subseteq \mathfrak{B}$ which includes both A and X . X is called **algebraically independent** over $\mathfrak{A}$ iff for each $b \in X$, $b \notin \mathfrak{A}(X \sim \{b\})$.

Fact 3.4.28 If $\mathfrak{A} \subseteq \mathfrak{B}$ are algebraically closed fields, then there exists a set X algebraically independent over $\mathfrak{A}$ such that $\mathfrak{B} = \mathfrak{A}(X)$. X is called a **transcendence basis** for $\mathfrak{B}$ over $\mathfrak{A}$. Any two transcendence bases for $\mathfrak{B}$ over $\mathfrak{A}$ have the same (finite or infinite) cardinality called the **transcendence degree** of $\mathfrak{B}$ over $\mathfrak{A}$, $\text{TrDg}_{\mathfrak{A}}(\mathfrak{B})$.

Fact 3.4.29 For any algebraically closed fields $\mathfrak{A} \subseteq \mathfrak{B}$ and $\mathfrak{A}' \subseteq \mathfrak{B}'$,

$$\mathfrak{A} \simeq \mathfrak{A}' \quad \text{and} \quad \text{TrDg}_{\mathfrak{A}}(\mathfrak{B}) = \text{TrDg}_{\mathfrak{A}'}(\mathfrak{B}') \implies \mathfrak{B} \simeq \mathfrak{B}'.$$

Fact 3.4.30 For each p a prime or 0, there exists a denumerable algebraically closed field $\mathbb{A}_p$ such that every algebraically closed field of characteristic p has a subfield (which we also denote by $\mathbb{A}_p$) isomorphic to $\mathbb{A}_p$. If $\text{TrDg}_{\mathbb{A}_p}(\mathfrak{B}) \leq \aleph_0$, then $\mathfrak{B}$ is denumerable; if $\text{TrDg}_{\mathbb{A}_p}(\mathfrak{B})$ is an uncountable cardinal κ , then $\text{Card}(\mathfrak{B}) = \kappa$.

Theorem 3.4.31 For each p a prime or 0, $T_{\text{ACF}(p)}$ is not $\aleph_0$ -categorical, but is κ -categorical for all uncountable cardinals κ . Hence $T_{\text{ACF}(p)}$ is complete and therefore decidable.

Proof. There are denumerable algebraically closed fields of all transcendence degrees $n \leq \aleph_0$ over $\mathbb{A}_p$, and it is easy to see that two such of different transcendence degrees cannot be isomorphic. If $\mathfrak{B}$ and $\mathfrak{C}$ are two models of $T_{\text{ACF}(p)}$ of some uncountable cardinality κ , then by 3.4.30 each has transcendence degree κ over $\mathbb{A}_p$, so by 3.4.29, $\mathfrak{B} \simeq \mathfrak{C}$. Every algebraically closed field is infinite since it contains all q -th roots of 1; hence, the completeness of $T_{\text{ACF}(p)}$ now follows by Vaught's Test. Decidability is immediate from the Enumerability Theorem. $\square$

In the next section we will have another proof of the completeness of $T_{\text{ACF}(p)}$ by elimination of quantifiers. In particular, every algebraically closed field of characteristic 0 is elementarily equivalent to the field of complex numbers $\mathbb{C}$, and we may view the axioms for $T_{\text{ACF}(0)}$ as axioms for $\text{Th}(\mathbb{C}, +, -, \times, 0, 1)$.

Collateral Reading. The study of fields from a logical perspective is an active area of research, see [Ma-Me-Pi]. Another algebraic example is the theory of *real closed fields* or equivalently $\text{Th}(\mathbb{R}, +, \times, <)$; a good exposition is [Mar; Section 3.3].

Exercises

Exercise 3.4.32 Let L be a language with one unary function symbol f and no other non-logical symbols. Let T be the L -theory generated by the following infinite set of L -sentences:

$$\{fx \doteq fy \rightarrow x \doteq y, \quad \forall y \exists x [fx \doteq y] \} \cup \{ \neg f^{(m)}x \doteq x : m > 0 \},$$

where $f^{(m)}$ means $ff \cdots f$ with m -many f 's. Prove that T is complete.

Exercise 3.4.33 Give another proof of the (Alternative) Completeness Theorem along the following lines. Let κ be any infinite cardinal and L a language of cardinality κ . Prove the following three lemmas.

- (i) For any set Γ of L -formulas, any L -sentence $\exists x \psi$ and any constant symbol c which does not occur in ψ or any member of Γ , if Γ is $\vdash$ -consistent, then also $\Gamma \cup \{ \exists x \psi \rightarrow \psi_x(c) \}$ is $\vdash$ -consistent.

- (ii) For any $\vdash$ -consistent set Γ of sentences, there exists an extension L^+ of L by constants of cardinality κ and a $\vdash$ -consistent set Γ^+ of L^+ -sentences such that $\Gamma \subseteq \Gamma^+$ and for every L^+ -formula ψ with at most x free, there is an L^+ -constant symbol c such that $(\exists x \psi \rightarrow \psi_x(c)) \in \Gamma^+$.
- (iii) For any $\vdash$ -consistent set Γ of L -sentences, there exists a set Δ of L -sentences such that $\Gamma \subseteq \Delta$ and Δ is $\vdash$ -consistent and L -complete.

Now given a $\vdash$ -consistent set Γ , first extend Γ to Γ^+ as in (ii), then extend Γ^+ to Δ as in (iii). Note that Δ still has the property in (ii), and conclude that Δ is L^+ -Henkin complete and hence has a model of cardinality at most κ .

Hint: (ii) follows reasonably easily from (i) by an iterative procedure: let $L_0 = L$ and $\Gamma_0 = \Gamma$, and construct L_{n+1} and Γ_{n+1} to satisfy the given property for L_n -formulas ϕ . (iii) may be proved by constructing a transfinite sequence of sets Δ_σ as above, but by separating the two parts of the extension in this way we have available a simpler proof using Zorn's Lemma. Let

$$\mathcal{L} = \{ \Lambda : \Gamma \subseteq \Lambda \subseteq \text{Fm}_L \text{ and } \Lambda \text{ is } \vdash\text{-consistent} \}.$$

Verify that

$$\text{for any } \mathcal{L}' \subseteq \mathcal{L}, \text{ if } \mathcal{L}' \text{ is linearly ordered by } \subseteq, \text{ then } \bigcup \mathcal{L}' \in \mathcal{L},$$

so that Zorn's Lemma guarantees the existence of an element Δ of $\mathcal{L}$ which is maximal with respect to $\subseteq$. Show that Δ is L -complete.

3.5 Applications of compactness

Since the statement of the Compactness Theorem explicitly involves a reduction of an infinite set to a finite one, it is not surprising that it has consequences for the expressibility in a first-order language of notions of finiteness and infinity. By the quantifier-elimination result for $L_{=}$ 2.4.37, we know that there is no single sentence of this language whose models are exactly the set $\{ (A) : A \text{ is finite} \}$. It follows that there is also no sentence whose models are exactly the infinite structures. On the other hand, there is a set of $L_{=}$ -sentences $\Gamma = \{ \exists^{\geq n} : n \in \omega \}$ such that the models of Γ are exactly the infinite $L_{=}$ -structures (A) . To make results of this sort easier to state, we introduce some terminology. In the following, unless otherwise stated, L is any first-order language.

Definition 3.5.1 For any theory T and any class $\mathcal{K} \subseteq \text{Mod}_L(T)$ of L -models of T , $\mathcal{K}$ is **axiomatizable over T** iff for some set Γ of L -sentences, $\mathcal{K} = \text{Mod}_L(T \cup \Gamma)$; if Γ may be chosen to be a finite set (or equivalently a single sentence), $\mathcal{K}$ is called **finitely axiomatizable over T** . If T is the minimal theory $\{\phi : \models \phi\}$, we call $\mathcal{K}$ simply **axiomatizable** or **finitely axiomatizable**.

In this terminology, the results on $L_=_$ mentioned above read: the classes $\{(A) : A \text{ is finite}\}$ and $\{(A) : A \text{ is infinite}\}$ are not finitely axiomatizable while $\{(A) : A \text{ is infinite}\}$ is axiomatizable. The same and more is true for all languages.

Proposition 3.5.2 *For any L -theory T , if T has finite models of unbounded finite cardinality — that is, for all $n \in \omega$, T has a finite model with at least n elements — then*

- (i) T has infinite models;
- (ii) $\text{Mod}(T) \cap \{\mathfrak{A} : \mathfrak{A} \text{ is finite}\}$ is not axiomatizable over T ;
- (iii) $\text{Mod}(T) \cap \{\mathfrak{A} : \mathfrak{A} \text{ is infinite}\}$ is axiomatizable over T but not finitely axiomatizable over T .

Proof. For (i), consider the set $\Gamma = T \cup \{\exists^{\geq n} : n \in \omega\}$. For any finite subset $\Gamma_0 \subseteq \Gamma$, there is a natural number $\bar{n}$ such that $\Gamma_0 \subseteq T \cup \{\exists^{\geq n} : n \leq \bar{n}\}$. Then any model of T with at least $\bar{n}$ elements is a model of Γ_0 ; by hypothesis such models exist. By compactness, Γ has a model which clearly is an infinite model of T . For (ii), suppose that $\text{Mod}(T) \cap \{\mathfrak{A} : \mathfrak{A} \text{ is finite}\}$ is axiomatizable over T and so is $\text{Mod}(T \cup \Delta)$ for some set Δ . If $\Gamma := T \cup \Delta \cup \{\exists^{\geq n} : n \in \omega\}$, then again every finite subset of Γ has a model, hence by compactness so does Γ . But such a model would be an infinite member of $\text{Mod}(T) \cap \{\mathfrak{A} : \mathfrak{A} \text{ is finite}\}$, a contradiction. For (iii), clearly $\text{Mod}(T) \cap \{\mathfrak{A} : \mathfrak{A} \text{ is infinite}\}$ is the set of models of $T \cup \{\exists^{\geq n} : n \in \omega\}$, so is axiomatizable over T . If, however, it were finitely axiomatizable over T , it would be of the form $\text{Mod}(T \cup \{\psi\})$ for some sentence ψ . But then $\text{Mod}(T) \cap \{\mathfrak{A} : \mathfrak{A} \text{ is finite}\} = \text{Mod}(T \cup \{\neg\psi\})$, contrary to (ii). $\square$

In [Section 2.4](#) we considered $L_{<}$ -sentences which describe partial orderings, linear orderings, dense orderings etc. A notion which seems at first glance similar is the notion of well-ordering. However

Proposition 3.5.3 *The class of $L_{<}$ -structures $(A, <_A)$ such that $<_A$ well-orders A is not axiomatizable.*

Proof. Suppose towards a contradiction that this class is $\text{Mod}(\Gamma)$. Let L be the language extending $L_{<}$ by adjoining a countable list of constant symbols c_n for $n \in \omega$, and let Δ be the set $\Gamma \cup \{c_{n+1} < c_n : n \in \omega\}$. It is easy to see that every finite subset of Δ has a model, since any infinite well-ordering

has arbitrarily long finite descending sequences. Hence, by compactness, Δ has a model $\mathfrak{A} = (A, <_A, a_0, a_1, \dots, a_n, \dots)$. But then $(A, <_A)$ is not a well-ordering, since the subset $\{a_n : n \in \omega\}$ has no least element. $\square$

A simple general result is

Proposition 3.5.4 *For any class $\mathcal{K}$ of L -structures, if both $\mathcal{K}$ and its complement (in the class of all L -structures) are axiomatizable, then both are actually finitely axiomatizable.*

Proof. Left as an exercise. $\square$

Another result of this kind arises in the context of the theory T_{FL} of fields discussed in the preceding section.

Proposition 3.5.5

- (i) $\{\mathfrak{A} : \mathfrak{A} \text{ is a field of prime characteristic}\}$ is not axiomatizable over T_{FL} .
- (ii) $\{\mathfrak{A} : \mathfrak{A} \text{ is a field of characteristic } 0\}$ is not finitely axiomatizable over T_{FL} .
- (iii) For any L_{FL} -sentence ϕ , if ϕ is true in all fields of characteristic 0, then for some $m_\phi \in \omega$, ϕ is also true in all fields of prime characteristic $p \geq m_\phi$.

Proof. For (i), suppose that $\{\mathfrak{A} : \mathfrak{A} \text{ is a field of prime characteristic}\}$ is the set of models of $T_{\text{FL}} \cup \Gamma$. Then since there exist fields of all prime characteristics (and there are infinitely many primes), every finite subset of $T_{\text{FL}} \cup \Gamma \cup \{\neg \text{Char}_p : p \text{ a prime}\}$ is consistent, so by compactness so is this set. But a model of this set is a model of $T_{\text{FL}} \cup \Gamma$ of characteristic 0, contrary to the hypothesis.

For (ii), if $\{\mathfrak{A} : \mathfrak{A} \text{ is a field of characteristic } 0\}$ were finitely axiomatizable over T_{FL} , then it would be the set of models of $T_{\text{FL}} \cup \{\psi\}$ for some sentence ψ . But then the models of $T_{\text{FL}} \cup \{\neg \psi\}$ would be exactly the fields of prime characteristic, contrary to (i). Finally, suppose that ϕ is a sentence true in all fields of characteristic 0. Then $T_{\text{FL}} \cup \{\neg \text{Char}_p : p \text{ a prime}\} \models \phi$, so by compactness ϕ is also a logical consequence of some finite subset — that is, for some m_ϕ , $T_{\text{FL}} \cup \{\neg \text{Char}_p : p \text{ a prime} < m_\phi\} \models \phi$. Hence ϕ is also true in all fields of characteristic $p \geq m_\phi$. $\square$

We next consider a series of results called **preservation theorems** which establish equivalences between syntactic properties of formulas and theories and algebraic properties of their models. In Corollary 2.4.22 we established that theories whose axioms could be chosen with certain syntactic properties, such as being universal or existential sentences, had corresponding model-theoretic properties, such as being downward or upward persistent. Here we will prove generalized versions of these facts and their converses.

Definition 3.5.6 For any regular set Δ of L -formulas, L -theories $T \subseteq U$ and classes $\mathcal{L} \subseteq \mathcal{K}$ of L -structures,

- (i) U is **Δ -axiomatizable over T** iff there exists a set $\Gamma \subseteq \Delta$ of L -sentences such that $U = \text{Th}(T \cup \Gamma)$;
- (ii) $\mathcal{L}$ is **Δ -substructure closed in $\mathcal{K}$** iff for any L -structures $\mathfrak{A}$ and $\mathfrak{B}$,

$$\mathfrak{A} \prec^\Delta \mathfrak{B} \quad \text{and} \quad \mathfrak{A} \in \mathcal{K} \quad \text{and} \quad \mathfrak{B} \in \mathcal{L} \implies \mathfrak{A} \in \mathcal{L};$$

- (iii) $\mathcal{L}$ is **Δ -superstructure closed in $\mathcal{K}$** iff for any L -structures $\mathfrak{A}$ and $\mathfrak{B}$,

$$\mathfrak{A} \prec^\Delta \mathfrak{B} \quad \text{and} \quad \mathfrak{B} \in \mathcal{K} \quad \text{and} \quad \mathfrak{A} \in \mathcal{L} \implies \mathfrak{B} \in \mathcal{L};$$

- (iv) a formula $\psi(\mathbf{x})$ is **Δ -downward persistent in $\mathcal{K}$** iff for any L -structures $\mathfrak{A}$ and $\mathfrak{B}$ and $\mathbf{a} \in A$,

$$\mathfrak{A}, \mathfrak{B} \in \mathcal{K} \quad \text{and} \quad \mathfrak{A} \prec^\Delta \mathfrak{B} \quad \text{and} \quad \models_{\mathfrak{B}} \psi[\mathbf{a}] \implies \models_{\mathfrak{A}} \psi[\mathbf{a}];$$

- (v) a formula $\psi(\mathbf{x})$ is **Δ -upward persistent in $\mathcal{K}$** iff for any L -structures $\mathfrak{A}$ and $\mathfrak{B}$ and $\mathbf{a} \in A$,

$$\mathfrak{A}, \mathfrak{B} \in \mathcal{K} \quad \text{and} \quad \mathfrak{A} \prec^\Delta \mathfrak{B} \quad \text{and} \quad \models_{\mathfrak{A}} \psi[\mathbf{a}] \implies \models_{\mathfrak{B}} \psi[\mathbf{a}];$$

- (vi) $\Delta^T := \{ \psi : \text{for some } \theta \in \Delta, \quad T \models \psi \leftrightarrow \theta \}$.

As a first simple observation, consider

Corollary 3.5.7 For any regular set Δ of L -formulas and L -theories $T \subseteq U$, U is Δ -axiomatizable over T iff $U \subseteq \text{Th}(T \cup (U \cap \Delta))$ or equivalently $\text{Mod}(T \cup (U \cap \Delta)) \subseteq \text{Mod}(U)$.

Proof. If $U = \text{Th}(T \cup \Gamma)$ for some $\Gamma \subseteq \Delta$, necessarily $\Gamma \subseteq U$ so $\Gamma \subseteq U \cap \Delta$ and thus $U \subseteq \text{Th}(T \cup (U \cap \Delta))$. Conversely, if this inclusion holds, we may take $\Gamma = U \cap \Delta$. $\square$

The idea here is that $U \cap \Delta$ is the largest possible subset of Δ which could serve as a set of Δ -axioms for U over T .

Proposition 3.5.8 For any regular set Δ , formula ψ and theory T ,

- (i) if $\psi \in (\forall \neg \Delta)^T$ then ψ is Δ -downward persistent in $\text{Mod}(T)$;
- (ii) if $\psi \in (\exists \Delta)^T$ then ψ is Δ -upward persistent in $\text{Mod}(T)$;
- (iii) if U is $(\forall \neg \Delta)$ -axiomatizable over T , then $\text{Mod}(U)$ is Δ -substructure closed in $\text{Mod}(T)$;
- (iv) if U is $(\exists \Delta)$ -axiomatizable over T , then $\text{Mod}(U)$ is Δ -superstructure closed in $\text{Mod}(T)$.

Proof. These follow by a refinement of the technique of Proposition 2.3.25; for example, suppose that $T \models \psi(\mathbf{x}) \leftrightarrow \forall y \neg \theta(\mathbf{x}, y)$ with $\theta \in \Delta$ and that $\mathfrak{A}$ and $\mathfrak{B}$ are models of T such that $\mathfrak{A} \prec^\Delta \mathfrak{B}$. Then for any $\mathfrak{A}$ -assignment α ,

$$\begin{aligned}
 \models_{\mathfrak{B}} \psi[\alpha] &\iff \models_{\mathfrak{B}} (\forall y \neg \theta)[\alpha] \\
 &\iff \text{for all } a \in B, \quad \not\models_{\mathfrak{B}} \theta[\alpha_{y \mapsto a}] \\
 &\implies \text{for all } a \in A, \quad \not\models_{\mathfrak{B}} \theta[\alpha_{y \mapsto a}] \\
 &\implies \text{for all } a \in A, \quad \not\models_{\mathfrak{A}} \theta[\alpha_{x \mapsto a}] \\
 &\iff \models_{\mathfrak{A}} (\forall y \neg \theta)[\alpha] \\
 &\iff \models_{\mathfrak{A}} \psi[\alpha].
 \end{aligned}$$

The first and last equivalences follow from the hypothesis that $\mathfrak{A}$ and $\mathfrak{B}$ are models of T and the fourth implication from the hypothesis $\mathfrak{A} \prec^\Delta \mathfrak{B}$. $\square$

Our first goal is to establish the converses of these statements. An important tool is the following.

Proposition 3.5.9 (Consistency Principle) *For any L -theory T and any set Γ of sentences in a language L^+ which extends L by constants, if Γ is closed under conjunction, then $T \cup \Gamma$ is inconsistent iff for some sentence $\psi(\mathbf{c}) \in \Gamma$, $T \models \forall \mathbf{x} \neg \psi(\mathbf{x})$.*

Proof. The condition is clearly sufficient. Assume that $T \cup \Gamma$ is inconsistent. By compactness, there is a finite subset $\Gamma_0 \subseteq \Gamma$ such that $T \cup \Gamma_0$ is inconsistent. Let $\mathbf{c}$ be the new L^+ -constant symbols which occur in members of Γ_0 , and let $\psi(\mathbf{c})$ be the conjunction of the members of Γ_0 ; by the hypothesis, $\psi(\mathbf{c}) \in \Gamma$, and we have that $T \cup \{\psi(\mathbf{c})\}$ is inconsistent so $T \models \neg \psi(\mathbf{c})$. Since the constant symbols $\mathbf{c}$ do not occur in T , by Corollary 2.6.6, $T \models \forall \mathbf{x} \neg \psi(\mathbf{x})$. $\square$

In the next lemma we collect a number of technical results with a common theme. The reader may want to defer a thorough appreciation of each of these until it is used below. It is also helpful to keep in mind the important special case $\Delta = \text{QF}$. Note that $\forall \neg \text{QF}$ is $\forall_1$, the set of universal formulas, and $\forall \exists \text{QF}$ is $\forall_2$, the set of universal-existential formulas (see Definition 2.4.18).

Second Extension Lemma 3.5.10 *For any theory U , any regular set Δ of formulas and any structures $\mathfrak{A} \subseteq \mathfrak{B}$,*

- (i) $\mathfrak{B}$ is a model of $U \cap (\forall \neg \Delta) \iff \mathfrak{B} \prec^\Delta \mathfrak{C}$ for some model $\mathfrak{C}$ of U ;
- (ii) $\mathfrak{A} \prec^{\forall \neg \Delta} \mathfrak{B} \iff \text{for some } \mathfrak{C}, \mathfrak{B} \prec^\Delta \mathfrak{C} \text{ and } \mathfrak{A} \prec \mathfrak{C}$;
- (iii) $\mathfrak{B}$ is a model of $U \cap (\exists \Delta) \iff$

$$\mathfrak{C} \prec^\Delta \mathfrak{D} \text{ for some model } \mathfrak{C} \text{ of } U \text{ and some } \mathfrak{D} \equiv \mathfrak{B};$$

(iv) if Δ is regular Boolean, then $\mathfrak{B}$ is a model of $U \cap (\forall\exists\Delta) \iff$

$\mathfrak{B} \prec^\Delta \mathfrak{C} \prec^\Delta \mathfrak{D}$ for some model $\mathfrak{C}$ of U and some $\mathfrak{D}$ with $\mathfrak{B} \prec \mathfrak{D}$;

(v) if Δ is regular Boolean, then $\mathfrak{A} \prec^{\forall\exists\Delta} \mathfrak{B} \iff$

for some $\mathfrak{C}$ and $\mathfrak{D}$, $\mathfrak{B} \prec^\Delta \mathfrak{C} \prec^\Delta \mathfrak{D}$ and $\mathfrak{A} \prec \mathfrak{C}$ and $\mathfrak{B} \prec \mathfrak{D}$;

(vi) for any theory T and any regular Boolean set Δ^+ such that $\exists\Delta \subseteq \Delta^+$, $\mathfrak{A}$ is a model of $T \cup (U \cap \Delta^+) \implies$

$\mathfrak{A} \prec^\Delta \mathfrak{B}$ for some model $\mathfrak{B}$ of $U \cup (T \cap \Delta^+)$.

Proof. The implications ($\Leftarrow$) of (i) and (iii) are immediate from Proposition 3.5.8. Suppose that $\mathfrak{B}$ is a model of $U \cap (\forall\neg\Delta)$. By the First Extension Lemma 3.4.21 it suffices to show that $U \cup \text{Th}^\Delta(\mathfrak{B}_B)$ is consistent. If it is not, then by the Consistency Principle there is some $\psi(\mathbf{b}) \in \text{Th}^\Delta(\mathfrak{B}_B)$ such that $U \models \forall\mathbf{x}\neg\psi(\mathbf{x})$. But then $\forall\mathbf{x}\neg\psi(\mathbf{x}) \in U \cap (\forall\neg\Delta)$ and is thus by hypothesis true in $\mathfrak{B}$, contrary to the fact that $\psi(\mathbf{b})$ is true in $\mathfrak{B}_B$. Part (ii) is immediate from (i) applied to $U = \text{Th}(\mathfrak{A}_A)$.

For (iii), suppose that $\mathfrak{B}$ is a model of $U \cap (\exists\Delta)$. Consider first the set $\Gamma := U \cup [\text{Th}(\mathfrak{B}) \cap (\forall\neg\Delta)]$. If Γ is inconsistent, then by compactness there is a single sentence $\psi \in (\forall\neg\Delta)$ which is true in $\mathfrak{B}$ such that $U \models \neg\psi$. But then $\neg\psi \in U \cap \exists\Delta$ so by hypothesis is also true in $\mathfrak{B}$, a contradiction. Let $\mathfrak{C}$ be any model of Γ . Then $\mathfrak{C}$ is a model of U , and by (i), $\mathfrak{C} \prec^\Delta \mathfrak{D}$ for some model $\mathfrak{D}$ of $\text{Th}(\mathfrak{B})$ — that is, $\mathfrak{D} \equiv \mathfrak{B}$.

For (iv), suppose that $\mathfrak{B}$ is a model of $U \cap (\forall\exists\Delta)$. Since $\forall\exists\Delta = \forall\neg\forall\neg\Delta$, by (i) there exists a model $\mathfrak{C}$ of U such that $\mathfrak{B} \prec^{\forall\neg\Delta} \mathfrak{C}$. Then the desired $\mathfrak{D}$ exists by (ii). Since $\neg\Delta = \Delta$, also $\mathfrak{B} \prec^\Delta \mathfrak{C}$. We leave the converse as an exercise. Part (v) is immediate from (iv) applied to $U = \text{Th}(\mathfrak{A}_A)$.

Finally, suppose that Δ^+ is as described in (vi) and $\mathfrak{A}$ is a model of $T \cup (U \cap \Delta^+)$. By the First Extension Lemma it will suffice to verify that $U \cup (T \cap \Delta^+) \cup \text{Th}^\Delta(\mathfrak{A}_A)$ is consistent. Suppose, towards a contradiction, that it is not. Then by the Consistency Principle there exists a formula $\psi(\mathbf{a}) \in \text{Th}^\Delta(\mathfrak{A}_A)$ such that $U \cup (T \cap \Delta^+) \models \forall\mathbf{x}\neg\psi(\mathbf{x})$, so by compactness, there is a sentence $\theta \in (T \cap \Delta^+)$ such that $U \models \theta \rightarrow \forall\mathbf{x}\neg\psi(\mathbf{x})$. Since $\psi \in \Delta$, this sentence belongs to Δ^+ and hence to $U \cap \Delta^+$, so is true in $\mathfrak{A}$. But since $\mathfrak{A}$ is a model of T , also θ is true in $\mathfrak{A}$ so $\forall\mathbf{x}\neg\psi(\mathbf{x})$ is true in $\mathfrak{A}$, contradicting the choice of $\psi(\mathbf{a}) \in \text{Th}^\Delta(\mathfrak{A}_A)$. $\square$

Łoś-Tarski Theorem 3.5.11 *For any theories $T \subseteq U$ and any regular set Δ ,*

- (i) *U is $(\forall\neg\Delta)$ -axiomatizable over T iff $\text{Mod}(U)$ is Δ -substructure closed in $\text{Mod}(T)$;*
- (ii) *U is $(\exists\Delta)$ -axiomatizable over T iff $\text{Mod}(U)$ is Δ -superstructure closed in $\text{Mod}(T)$.*

Proof. The implications $(\implies)$ are from Proposition 3.5.8. For (i) suppose that $\text{Mod}(U)$ is Δ -substructure closed in $\text{Mod}(T)$. Then by (i) of the Second Extension Lemma, any model of $T \cup [U \cap (\forall\neg\Delta)]$ is a model of T which is a Δ -substructure of a model of U , hence is also model of U . Thus U is $(\forall\neg\Delta)$ -axiomatizable by Corollary 3.5.7.

For (ii), suppose that $\text{Mod}(U)$ is Δ -superstructure closed in $\text{Mod}(T)$ and fix $\mathfrak{A} \in \text{Mod}(T \cup [U \cap (\exists\Delta)])$. By (iii) of the Second Extension Lemma, there exists a model $\mathfrak{B}$ of U and a structure $\mathfrak{C}$ such that $\mathfrak{B} \prec^\Delta \mathfrak{C} \equiv \mathfrak{A}$. Then since $\mathfrak{A}$ is a model of T , so is $\mathfrak{C}$ so by Δ -superstructure closure in $\text{Mod}(T)$, $\mathfrak{C}$ is in fact a model of U . Hence also $\mathfrak{A}$ is a model of U . $\square$

Corollary 3.5.12 *For any theory U ,*

- (i) *U is universally axiomatizable iff $\text{Mod}(U)$ is substructure closed;*
- (ii) *U is existentially axiomatizable iff $\text{Mod}(U)$ is superstructure closed. $\square$*

We can also easily obtain the corresponding results for single formulas.

Lemma 3.5.13 *For any theory T , any regular set Δ and any sentence ϕ ,*

- (i) *ϕ is Δ -downward persistent in $\text{Mod}(T)$ iff $\text{Mod}(T \cup \{\phi\})$ is Δ -substructure closed in $\text{Mod}(T)$;*
- (ii) *$\phi \in \Delta^T$ iff $\text{Th}(T \cup \{\phi\})$ is Δ -axiomatizable over T .*

Proof. Part (i) and (ii)($\implies$) are immediate. For the implication $(\impliedby)$ of (ii), if $\text{Th}(T \cup \{\phi\}) = \text{Th}(T \cup \Gamma)$ for $\Gamma \subseteq \Delta$, then by compactness there exists a sentence $\psi \in \Delta$, a conjunction of finitely many members of Γ such that $T \models \psi \rightarrow \phi$. But since $\Gamma \subseteq \text{Th}(T \cup \{\phi\})$, $\psi \in \text{Th}(T \cup \{\phi\})$, so also $T \models \phi \rightarrow \psi$ and thus $\phi \in \Delta^T$. $\square$

Tarski-Robinson Theorem 3.5.14 *For any theory T , any regular set Δ and any formula ϕ ,*

- (i) *$\phi \in (\forall\neg\Delta)^T$ iff ϕ is Δ -downward persistent in $\text{Mod}(T)$;*
- (ii) *$\phi \in (\exists\Delta)^T$ iff ϕ is Δ -upward persistent in $\text{Mod}(T)$.*

Proof. When ϕ is a sentence, these are an immediate consequence of the Łoś-Tarski Theorem together with the lemma. Suppose now that $\phi(\mathbf{x})$ is a

formula. Let L^+ be an extension of L by a sequence of new constant symbols $\mathbf{c}$. Let Δ^+ be the set of all substitutions of $\mathbf{c}$ for variables in members of Δ and T^+ the L^+ -theory generated by T . Easily Δ^+ is still regular, so by the version for sentences we have

$$\phi(\mathbf{c}) \in (\forall \neg \Delta^+)^{T^+} \iff \phi(\mathbf{c}) \text{ is } \Delta^+\text{-downward persistent in } \text{Mod}(T^+).$$

By Corollary 2.6.6 for any L -formula ψ ,

$$T^+ \models \phi(\mathbf{c}) \leftrightarrow \psi(\mathbf{c}) \iff T \models \phi(\mathbf{x}) \leftrightarrow \psi(\mathbf{x}),$$

and hence $\phi(\mathbf{c}) \in (\forall \neg \Delta^+)^{T^+} \iff \phi(\mathbf{x}) \in (\forall \neg \Delta)^T$, so it suffices to verify that also

$$\begin{aligned} \phi(\mathbf{c}) \text{ is } \Delta^+\text{-downward persistent in } \text{Mod}(T^+) \\ \iff \phi(\mathbf{x}) \text{ is } \Delta\text{-downward persistent in } \text{Mod}(T). \end{aligned}$$

We leave this and the argument for (ii) as an exercise. $\square$

Corollary 3.5.15 *For any theory T , regular set Δ and formula ϕ ,*

- (i) ϕ is T -universal (belongs to $\forall_1^T$) iff ϕ is downward persistent in $\text{Mod}(T)$; in particular ϕ is logically universal iff ϕ is downward persistent;
- (ii) ϕ is T -existential (belongs to $\exists_1^T$) iff ϕ is upward persistent in $\text{Mod}(T)$; in particular ϕ is logically existential iff ϕ is upward persistent. $\square$

Recall from Definition 2.4.26 that we say that a theory T admits Δ -elimination iff every formula is T -equivalent to one in Δ — that is, every formula belongs to Δ^T . Hence, we have the following.

Corollary 3.5.16 *For any theory T and any regular set Δ of formulas,*

- (i) T admits $(\forall \neg \Delta)$ -elimination iff every formula is Δ -downward persistent in $\text{Mod}(T)$;
- (ii) T admits $(\exists \Delta)$ -elimination iff every formula is Δ -upward persistent in $\text{Mod}(T)$. $\square$

In fact,

Model Completeness Theorem 3.5.17 *For any theory T and any regular set Δ of formulas, the following conditions are equivalent; when they hold we say that T is Δ -model complete.*

- (i) For every model $\mathfrak{A}$ of T , $T \cup \text{Th}^\Delta(\mathfrak{A}_A)$ generates a complete L^A -theory.
- (ii) For any two models $\mathfrak{A}$ and $\mathfrak{B}$ of T , if $\mathfrak{A} \prec^\Delta \mathfrak{B}$, then in fact $\mathfrak{A} \prec \mathfrak{B}$.
- (iii) T admits $(\forall \neg \Delta)$ -elimination.

(iv) T admits $(\exists\Delta)$ -elimination.

Proof. Note in (i) that the complete theory must be exactly $\text{Th}(\mathfrak{A}_A)$. We proceed cyclically. Assume first (i) and that $\mathfrak{A} \prec^\Delta \mathfrak{B}$ are models of T . Then $\mathfrak{B}_A$ is a model of $T \cup \text{Th}^\Delta(\mathfrak{A}_A)$ and hence a model of $\text{Th}(\mathfrak{A}_A)$; that is, $\mathfrak{A}_A \equiv \mathfrak{B}_A$, which means exactly that $\mathfrak{A} \prec \mathfrak{B}$. From (ii) it follows immediately that every formula is Δ -downward persistent in $\text{Mod}(T)$, so (iii) follows by the corollary. (iv) follows from (iii) since for any formula ϕ , if $\neg\phi$ is T -equivalent to $\psi \in (\forall\neg\Delta)$, then ϕ is T -equivalent to $\neg\psi \in (\neg\forall\neg\Delta)$ and hence to some $\theta \in \exists\Delta$.

Finally assume (iv) and fix a model $\mathfrak{A}$ of T . To establish (i) it will suffice to show that for any formula $\phi(\mathbf{x})$ and any $\mathbf{a} \in A$,

$$\models_{\mathfrak{A}} \phi[\mathbf{a}] \implies T \cup \text{Th}^\Delta(\mathfrak{A}_A) \models \phi(\dot{\mathbf{a}}).$$

Fix ϕ and $\mathbf{a}$ such that $\models_{\mathfrak{A}} \phi[\mathbf{a}]$ and a model $\mathfrak{B}^+$ of $T \cup \text{Th}^\Delta(\mathfrak{A}_A)$. $\mathfrak{B}^+$ is an expansion of a model $\mathfrak{B}$ of T , so by Proposition 3.4.17 and Corollary 3.4.20 (and their proofs) there exist a structure $\mathfrak{B}'$ and a function ζ such that $\mathfrak{A} \prec^\Delta \mathfrak{B}'$, $\zeta : \mathfrak{B}' \simeq \mathfrak{B}$, and for each $a \in A$, $\dot{a}^{\mathfrak{B}^+} = \zeta(a)$. Then we have

$$\models_{\mathfrak{A}} \phi[\mathbf{a}] \implies \models_{\mathfrak{B}} \phi[\mathbf{a}] \implies \models_{\mathfrak{B}} \phi[\zeta(\mathbf{a})] \implies \models_{\mathfrak{B}^+} \phi(\dot{\mathbf{a}}),$$

where the first implication uses the Δ -upward persistence of ϕ in $\text{Mod}(T)$ from (iv) and the corollary. $\square$

Corollary 3.5.18 (Robinson's Test) *For any theory T and any regular set Δ of formulas, T is Δ -model complete iff for any two models $\mathfrak{A}$ and $\mathfrak{B}$ of T , if $\mathfrak{A} \prec^\Delta \mathfrak{B}$, then $\mathfrak{A} \prec^{\forall\neg\Delta} \mathfrak{B}$.*

Proof. The condition follows immediately from (ii) of the theorem. Conversely, given this condition we show by formula induction that $\text{Fm} \subseteq (\exists\Delta)^T$. All atomic formulas are in $(\exists\Delta)^T$, and easily this class is closed under $\vee$ and $\exists$. Suppose $\phi \in (\exists\Delta)^T$, say $T \models \phi \leftrightarrow \exists \mathbf{x} \theta$ with $\theta \in \Delta$. Then $T \models \neg\phi \leftrightarrow \forall \mathbf{x} \neg\theta$, so by the condition, $\neg\phi$ is Δ -upward persistent in $\text{Mod}(T)$ and hence in $(\exists\Delta)^T$ by the Tarski-Robinson Theorem. $\square$

Remark 3.5.19 The relationship $\mathfrak{A} \prec^{\forall\neg\Delta} \mathfrak{B}$ is often described by saying that $\mathfrak{A}$ is Δ -**existentially closed** in $\mathfrak{B}$: for any $(\exists\Delta)$ formula $\exists \mathbf{y} \theta(\mathbf{x}, \mathbf{y})$ (with $\theta \in \Delta$) and any $\mathbf{a} \in A$,

$$\models_{\mathfrak{B}} \exists \mathbf{y} \theta(\mathbf{x}, \mathbf{y})[\mathbf{a}] \implies \models_{\mathfrak{A}} \exists \mathbf{y} \theta(\mathbf{x}, \mathbf{y})[\mathbf{a}].$$

In particular, if $\mathfrak{A} \prec^\Delta \mathfrak{B}$, then this is equivalent to

$$\text{for some } \mathbf{b} \in B, \models_{\mathfrak{B}} \theta(\mathbf{x}, \mathbf{y})[\mathbf{a}, \mathbf{b}] \implies \text{for some } \mathbf{b} \in A, \models_{\mathfrak{B}} \theta(\mathbf{x}, \mathbf{y})[\mathbf{a}, \mathbf{b}].$$

If we define $\mathfrak{A}$ to be Δ -existentially closed for T iff for any $\mathfrak{B}$, if $\mathfrak{B}$ is a model of T and $\mathfrak{A} \prec^\Delta \mathfrak{B}$, then $\mathfrak{A} \prec^{\forall\neg\Delta} \mathfrak{B}$ and

$\Delta\text{-ecMod}(T) :=$ class of models of T which are Δ -existentially closed for T ,

then in this terminology Robinson's Test reads:

T is Δ -model complete

$\iff$ every model of T is Δ -existentially closed for T

$\iff \Delta\text{-ecMod}(T) = \text{Mod}(T)$.

This notion will be explored further in the exercises.

The terminology *model complete* comes from (i) of the theorem but also from the following connection with completeness.

Definition 3.5.20 For any theory T and any regular set Δ of formulas, a structure $\mathfrak{A}$ is a Δ -**prime model** of T iff $\mathfrak{A}$ is a model of T and $\mathfrak{A} \hookrightarrow^\Delta \mathfrak{B}$ for every model $\mathfrak{B}$ of T . A QF-prime model is called **algebraically prime** and a Fm-prime model is **elementarily prime**.

Examples 3.5.21 $(\mathbb{Q}, +, \times, 0, 1)$ is an algebraically prime model for the theory of fields of characteristic 0 and the standard model of arithmetic Ω is an elementarily prime model for $\text{Th}(\Omega)$.

Corollary 3.5.22 For any theory T and any regular set Δ of formulas, if T is Δ -model complete and has a Δ -prime model, then T is complete.

Proof. Let $\mathfrak{A}$ be a Δ -prime model for T and $\mathfrak{B}_0$ and $\mathfrak{B}_1$ any two models; we show that $\mathfrak{B}_0 \equiv \mathfrak{B}_1$. For $i = 0, 1$ there exists $\mathfrak{A}_i$ such that $\mathfrak{A} \simeq \mathfrak{A}_i \prec^\Delta \mathfrak{B}_i$. Then $\mathfrak{B}_{i, A_i}$ is a model of $T \cup \text{Th}^\Delta(\mathfrak{A}_{i, A_i})$ which is the same as $T \cup \text{Th}^\Delta(\mathfrak{A}_A)$ and by (i) of the theorem is $T \cup \text{Th}(\mathfrak{A}_A)$. Hence $(\mathfrak{B}_0)_{A_0} \equiv (\mathfrak{B}_1)_{A_1}$, so in particular $\mathfrak{B}_0 \equiv \mathfrak{B}_1$. $\square$

The Model Completeness Theorem gives conditions for $(\forall\neg\Delta)$ - and $(\exists\Delta)$ -elimination; the following theorem gives the stronger Δ -elimination. For $\Delta = \text{QF}$ this really means that all quantifiers can be eliminated.

Lemma 3.5.23 For any theory T , any regular Boolean set Δ of formulas and any formula $\phi(\mathbf{x})$, the following conditions are equivalent:

(i) $\phi \in \Delta^T$;

(ii) for all models $\mathfrak{A}$ and $\mathfrak{B}$ of T and all $\mathbf{a} \in A$ and $\mathbf{b} \in B$,

$$\text{if } (\mathfrak{A}, \mathbf{a}) \equiv^\Delta (\mathfrak{B}, \mathbf{b}) \text{ then } \models_{\mathfrak{A}} \phi[\mathbf{a}] \iff \models_{\mathfrak{B}} \phi[\mathbf{b}].$$

Proof. That (i) $\implies$ (ii) is immediate: if $T \models \phi \leftrightarrow \psi$ for some $\psi \in \Delta$, then

$$\models_{\mathfrak{A}} \phi[\mathbf{a}] \iff \models_{(\mathfrak{A}, \mathbf{a})} \psi(\dot{\mathbf{a}}) \iff \models_{(\mathfrak{B}, \mathbf{b})} \psi(\dot{\mathbf{b}}) \iff \models_{\mathfrak{B}} \phi[\mathbf{b}].$$

Suppose now that (ii) holds. Set

$$\Gamma := \{ \psi(\mathbf{x}) : \psi(\mathbf{x}) \in \Delta \wedge T \models \phi(\mathbf{x}) \rightarrow \psi(\mathbf{x}) \};$$

we show that

$$(1) \quad T \cup \Gamma \models \phi(\mathbf{x}).$$

Then by compactness there exists a formula $\phi^* \in \Gamma$ such that $T \models \phi^*(\mathbf{x}) \rightarrow \phi(\mathbf{x})$ and hence also $T \models \phi^*(\mathbf{x}) \leftrightarrow \phi(\mathbf{x})$. (A propositional version of this method was suggested in Exercise 1.4.39.) To establish (1), suppose towards a contradiction that there is a model $\mathfrak{A}$ of T and a sequence $\mathbf{a} \in A$ which satisfies Γ but $\models_{\mathfrak{A}} \neg \phi[\mathbf{a}]$. Then $T \cup \text{Th}^\Delta(\mathfrak{A}, \mathbf{a}) \models \neg \phi(\dot{\mathbf{a}})$, since if $(\mathfrak{B}, \mathbf{b})$ is any model of $T \cup \text{Th}^\Delta(\mathfrak{A}, \mathbf{a})$, then $(\mathfrak{A}, \mathbf{a}) \equiv^\Delta (\mathfrak{B}, \mathbf{b})$ (since Δ is closed under negation), so by (i), $\models_{\mathfrak{B}} \neg \phi[\mathbf{b}]$. Hence by compactness there exists a formula $\theta \in \text{Th}^\Delta(\mathfrak{A}, \mathbf{a})$ such that $T \models \theta(\dot{\mathbf{a}}) \rightarrow \neg \phi(\dot{\mathbf{a}})$. But then also $T \models \phi(\mathbf{x}) \rightarrow \neg \theta(\mathbf{x})$, so $\neg \theta \in \Gamma$ and hence by hypothesis $\models_{\mathfrak{A}} \neg \theta[\mathbf{a}]$, a contradiction. $\square$

First Quantifier Elimination Theorem 3.5.24 *For any theory T and any regular Boolean set Δ of formulas, the following conditions are equivalent:*

- (i) T admits Δ -elimination;
- (ii) for every model $\mathfrak{A}$ of T and every (finite) set $X \subseteq A$, $T \cup \text{Th}^\Delta(\mathfrak{A}_X)$ generates a complete L^X -theory;
- (iii) for any two models $\mathfrak{A}$ and $\mathfrak{B}$ of T and any (finite) sets $X \subseteq A$ and $Y \subseteq B$,

$$\mathfrak{A}_X \equiv^\Delta \mathfrak{B}_Y \implies \mathfrak{A}_X \equiv \mathfrak{B}_Y.$$

Proof. Suppose that (i) holds, $\mathfrak{A}$ is a model of T and $X \subseteq A$. For any sentence $\phi(\dot{\mathbf{a}})$ with $\mathbf{a} \in X$, there exists $\phi^*(\mathbf{x}) \in \Delta$ such that $T \models \phi(\dot{\mathbf{a}}) \leftrightarrow \phi^*(\dot{\mathbf{a}})$. Then

$$\models_{\mathfrak{A}_X} \phi(\dot{\mathbf{a}}) \implies \phi^*(\dot{\mathbf{a}}) \in \text{Th}^\Delta(\mathfrak{A}_X) \implies T \cup \text{Th}^\Delta(\mathfrak{A}_X) \models \phi(\dot{\mathbf{a}}).$$

Hence $T \cup \text{Th}^\Delta(\mathfrak{A}_X)$ generates the complete theory $\text{Th}(\mathfrak{A}_X)$.

Next assume (ii), that $\mathfrak{A}$ and $\mathfrak{B}$ are models of T and that $X \subseteq A$ and $Y \subseteq B$ are such that $\mathfrak{A}_X \equiv^\Delta \mathfrak{B}_Y$. Then $\mathfrak{B}_Y$ is a model of $T \cup \text{Th}^\Delta(\mathfrak{A}_X)$, hence is a model of the complete theory $\text{Th}(\mathfrak{A}_X)$, so $\mathfrak{A}_X \equiv \mathfrak{B}_Y$. Note that by this argument the versions of (ii) with or without ‘finite’ imply the corresponding versions of (iii).

Finally, the weaker version of (iii) implies (i) by the lemma. $\square$

The most interesting case of quantifier elimination is, of course, $\Delta = \text{QF}$ where *all* quantifiers are eliminated. In this case there are conditions that are easier to verify than those of the First Quantifier Elimination Theorem. Recall as the case $\Gamma = \text{AtFm}_L$ of Definition 2.4.29 that a **simply existential** formula is one of the form

$$\exists x (\theta_0 \wedge \cdots \wedge \theta_{k-1} \wedge \neg \chi_0 \wedge \cdots \wedge \neg \chi_{l-1})$$

with all θ_i, χ_j atomic formulas or $\top$. Let $\exists_1^*$ denote the set of simply existential formulas.

Second Quantifier Elimination Theorem 3.5.25 *For any theory T , if for any models $\mathfrak{A}$ and $\mathfrak{B}$ of T ,*

- (i) *if $\mathfrak{A} \subseteq \mathfrak{B}$, then $\mathfrak{A} \prec^{\neg \exists_1^*} \mathfrak{B}$;*
- (ii) *for any substructures $\mathfrak{C} \subseteq \mathfrak{A}$ and $\mathfrak{D} \subseteq \mathfrak{B}$ and any isomorphism $\eta : \mathfrak{C} \simeq \mathfrak{D}$, there exist models $\overline{\mathfrak{C}}$ and $\overline{\mathfrak{D}}$ of T and an extension ζ of η such that*

$$\mathfrak{C} \subseteq \overline{\mathfrak{C}} \subseteq \mathfrak{A}, \quad \mathfrak{D} \subseteq \overline{\mathfrak{D}} \subseteq \mathfrak{B} \quad \text{and} \quad \zeta : \overline{\mathfrak{C}} \simeq \overline{\mathfrak{D}};$$

then T admits QF-elimination.

Proof. Assume that conditions (i) and (ii) hold for all models of T . We shall establish the following fact: for any models $\mathfrak{A}$ and $\mathfrak{B}$ of T , any simply existential formula $\exists x \psi(x, \mathbf{y})$ and any $\mathbf{a} \in A$ and $\mathbf{b} \in B$,

$$\text{if } (\mathfrak{A}, \mathbf{a}) \equiv^{\text{QF}} (\mathfrak{B}, \mathbf{b}) \quad \text{then} \quad \models_{\mathfrak{A}} \exists x \psi(x, \mathbf{y})[\mathbf{a}] \iff \models_{\mathfrak{B}} \exists x \psi(x, \mathbf{y})[\mathbf{b}].$$

By Lemma 3.5.23 it follows that every simply existential formula belongs to QF^T , and hence by the Quantifier Elimination Lemma 2.4.30 every formula belongs to QF^T . Suppose that $\mathfrak{A}$, $\mathfrak{B}$, $\exists x \psi(x, \mathbf{y})$, $\mathbf{a}$ and $\mathbf{b}$ are as described. Let $\mathfrak{C} := \langle \mathbf{a} \rangle_{\mathfrak{A}}$ and $\mathfrak{D} := \langle \mathbf{b} \rangle_{\mathfrak{B}}$ be the substructures generated by $\mathbf{a}$ and $\mathbf{b}$. Recall from Proposition 2.3.16 that

$$C = \{ t^{\mathfrak{A}}[\mathbf{a}] : t(\mathbf{x}) \text{ is an } L\text{-term} \} \quad \text{and} \quad D = \{ t^{\mathfrak{B}}[\mathbf{b}] : t(\mathbf{x}) \text{ is an } L\text{-term} \}.$$

Suppose that $(\mathfrak{A}, \mathbf{a}) \equiv^{\text{QF}} (\mathfrak{B}, \mathbf{b})$. It follows easily that the mapping $t^{\mathfrak{A}}[\mathbf{a}] \mapsto t^{\mathfrak{B}}[\mathbf{b}]$ is in fact an isomorphism $\mathfrak{C} \simeq \mathfrak{D}$. Let $\overline{\mathfrak{C}}$ and $\overline{\mathfrak{D}}$ be as in (ii), so in

particular $(\overline{\mathfrak{C}}, \mathbf{a}) \equiv (\overline{\mathfrak{D}}, \mathbf{b})$. Then by (i), $\overline{\mathfrak{C}} \prec^{\neg \exists^*} \mathfrak{A}$ and $\overline{\mathfrak{D}} \prec^{\neg \exists^*} \mathfrak{B}$, so

$$\begin{aligned} \models_{\mathfrak{A}} \exists x \psi(x, \mathbf{y})[\mathbf{a}] &\iff \models_{\overline{\mathfrak{C}}} \exists x \psi(x, \mathbf{y})[\mathbf{a}] \\ &\iff \models_{\overline{\mathfrak{D}}} \exists x \psi(x, \mathbf{y})[\mathbf{b}] \\ &\iff \models_{\mathfrak{B}} \exists x \psi(x, \mathbf{y})[\mathbf{b}], \end{aligned}$$

as required. $\square$

Example 3.5.26 The theory T_{ACF} of algebraically closed fields admits QF-elimination. Consider first condition (ii) of the theorem; this is a standard algebraic property of fields. If $\mathfrak{A}$ and $\mathfrak{B}$ are algebraically closed fields and $\mathfrak{C} \subseteq \mathfrak{A}$ and $\mathfrak{D} \subseteq \mathfrak{B}$ are substructures, then $\mathfrak{C}$ and $\mathfrak{D}$ are integral domains. If $\mathfrak{C} \simeq \mathfrak{D}$, then this isomorphism may first be extended to the quotient fields and then to the algebraic closures, which yield the desired $\overline{\mathfrak{C}}$ and $\overline{\mathfrak{D}}$.

Towards (i), because T_{FL} includes the usual commutative, associative and distributive laws of algebra, it is straightforward to verify that for any atomic formula $\theta(x, \mathbf{y})$ there exist $n > 0$ and terms $u_i(\mathbf{y})$ such that

$$T_{\text{FL}} \models \theta(x, \mathbf{y}) \leftrightarrow u_n(\mathbf{y}) \dot{\times} \underbrace{x \dot{\times} \cdots \dot{\times} x}_n \dot{+} \cdots \dot{+} u_1(\mathbf{y}) \dot{\times} x \dot{+} u_0(\mathbf{y}) \dot{=} \dot{0}.$$

Hence in any field $\mathfrak{A}$, for any $a, \mathbf{b} \in A$, there is a polynomial $P_{\theta, \mathbf{b}}$ with coefficients in A such that

$$\models_{\mathfrak{A}} \theta(x, \mathbf{y})[a, \mathbf{b}] \iff P_{\theta, \mathbf{b}}(a) = \dot{0}^{\mathfrak{A}}.$$

In particular, if $\mathfrak{A} \subseteq \mathfrak{B}$, then by Fact 3.4.27, if for some $a \in B$, $P_{\theta, \mathbf{b}}(a) = \dot{0}^{\mathfrak{A}}$, then already $a \in A$ — that is,

$$(2) \quad \models_{\mathfrak{A}} \exists x \theta(x, \mathbf{y})[\mathbf{b}] \iff \models_{\mathfrak{B}} \exists x \theta(x, \mathbf{y})[\mathbf{b}].$$

Now, consider a simply existential formula $\exists x \psi(x, \mathbf{y})$:

$$\exists x (\theta_0 \wedge \cdots \wedge \theta_{k-1} \wedge \neg \chi_0 \wedge \cdots \wedge \neg \chi_{l-1}).$$

If $k > 0$, then it follows from (2) that

$$(3) \quad \models_{\mathfrak{A}} \exists x \psi(x, \mathbf{y})[\mathbf{b}] \iff \models_{\mathfrak{B}} \exists x \psi(x, \mathbf{y})[\mathbf{b}].$$

But (3) also holds if $k = 0$, since each polynomial $P_{\chi_j, \mathbf{b}}$ has only finitely many roots and both $\mathfrak{A}$ and $\mathfrak{B}$ are infinite. Hence (i) is satisfied.

We turn next to a generalization of the notion of a chain introduced in Definition 2.3.56. In Section 2.4 we showed that every (logically) universal-existential sentence ϕ — in the current notation every $\phi \in \forall_2 = \forall \exists \text{QF}$ — is union persistent. Among other results below is a generalization of this result and its converse.

Definition 3.5.27 For any regular set Δ of formulas,

- (i) a Δ -**chain** consists of a family $\langle \mathfrak{A}_i : i \in \omega \rangle$ of L -structures such that $\mathfrak{A}_i \prec^\Delta \mathfrak{A}_j$ for all $i < j$;
- (ii) a **full Δ -chain** is one such that additionally for all $i \in \omega$, $\mathfrak{A}_i \prec^\Delta \bigcup \langle \mathfrak{A}_i : i \in \omega \rangle$;
- (iii) Δ is **chain-friendly** iff every Δ -chain is a full Δ -chain.

Thus Proposition 2.3.58 implies that QF and Fm are chain-friendly; we shall see below that there are many other examples. Note that in contrast to Definition 2.3.56 we are using here only chains indexed by ω ; most of the results below hold also for the general case of chains indexed by any linearly ordered set.

Definition 3.5.28 For any regular set Δ of formulas, any classes $\mathcal{L} \subseteq \mathcal{K}$ of structures and any formula ϕ ,

- (i) $\mathcal{L}$ is Δ -**union closed** in $\mathcal{K}$ iff for any Δ -chain $\langle \mathfrak{A}_i : i \in \omega \rangle$, if all $\mathfrak{A}_i \in \mathcal{L}$ and $\mathfrak{A} := \bigcup \langle \mathfrak{A}_i : i \in \omega \rangle \in \mathcal{K}$, then also $\mathfrak{A} \in \mathcal{L}$;
- (ii) $\phi(\mathbf{x})$ is **(full) Δ -union persistent** in $\mathcal{K}$ iff for any (full) Δ -chain $\langle \mathfrak{A}_i : i \in \omega \rangle$ with all $\mathfrak{A}_i \in \mathcal{K}$, $\mathfrak{A} := \bigcup \langle \mathfrak{A}_i : i \in \omega \rangle \in \mathcal{K}$, and any $\mathbf{a} \in A_0$, if $\models_{\mathfrak{A}_i} \phi[\mathbf{a}]$ for all $i \in \omega$, then also $\models_{\mathfrak{A}} \phi[\mathbf{a}]$.

As usual, we omit mention of $\mathcal{K}$ when it is the class of all structures and of Δ when it is QF.

Proposition 3.5.29 For any regular set Δ of formulas,

- (i) every (full) Δ -chain is also a (full) $(\exists\Delta)$ -chain;
- (ii) every formula $\phi \in (\forall\exists\Delta)$ is full Δ -union persistent;
- (iii) if Δ is chain-friendly, then also both $(\exists\Delta)$ and $(\forall\Delta)$ are chain-friendly.

Proof. Part (i) is immediate from Proposition 3.4.16. For (ii), suppose that $\phi(\mathbf{x})$ is $\forall \mathbf{y} \psi(\mathbf{x}, \mathbf{y})$ with $\psi \in \exists\Delta$, $\langle \mathfrak{A}_i : i \in \omega \rangle$ is a full Δ -chain, $\mathbf{a} \in A_0$ and for all $i \in \omega$, $\models_{\mathfrak{A}_i} \phi[\mathbf{a}]$. If $\mathfrak{A} := \bigcup \langle \mathfrak{A}_i : i \in \omega \rangle$, then for any $\mathbf{b} \in A$, there exists $i \in \omega$ such that $\mathbf{b} \in A_i$, so $\models_{\mathfrak{A}_i} \psi[\mathbf{a}, \mathbf{b}]$, so since by (i) $\mathfrak{A}_i \prec^{\exists\Delta} \mathfrak{A}$, also $\models_{\mathfrak{A}} \psi[\mathbf{a}, \mathbf{b}]$. Hence $\models_{\mathfrak{A}} \phi[\mathbf{a}]$.

For (iii), assume that Δ is chain-friendly. That $(\exists\Delta)$ is chain-friendly is immediate from (i). For the $(\forall\Delta)$ case, suppose that $\langle \mathfrak{A}_i : i \in \omega \rangle$ is a $(\forall\Delta)$ -chain and $\mathfrak{A}$ is its union; we show that for each $j \in \omega$, $\mathfrak{A}_j \prec^{\forall\Delta} \mathfrak{A}$. Fix j , a formula $\phi(\mathbf{x}) \in (\forall\Delta)$ and $\mathbf{a} \in A_j$ such that $\models_{\mathfrak{A}_j} \phi[\mathbf{a}]$. $\mathfrak{A}$ is also the union of the $(\forall\Delta)$ -chain $\langle \mathfrak{A}_i : i \geq j \rangle$ and for $i \geq j$ also $\models_{\mathfrak{A}_i} \phi[\mathbf{a}]$. Since this is also

a Δ -chain and Δ is chain-friendly, it is a full Δ -chain and therefore by (ii) also $\models_{\mathfrak{A}} \phi[\mathbf{a}]$. $\square$

Corollary 3.5.30 *For all $n \in \omega$, $\forall_n$ and $\exists_n$ (Definition 2.4.18) are chain-friendly. $\square$*

Chang-Łoś-Suszko Theorem 3.5.31 *For any theory T and any chain-friendly regular Boolean set Δ ,*

- (i) *for any formula ϕ , $\phi \in (\forall\exists\Delta)^T$ iff ϕ is Δ -union persistent in $\text{Mod}(T)$;*
- (ii) *for any theory $U \supseteq T$, U is $(\forall\exists\Delta)$ -axiomatizable over T iff $\text{Mod}(U)$ is Δ -union closed in $\text{Mod}(T)$.*

Proof. For (i)($\implies$), suppose $\phi(\mathbf{x}) \in (\forall\exists\Delta)^T$, say $T \models \phi \leftrightarrow \psi$ with $\psi \in (\forall\exists\Delta)$. Let $\langle \mathfrak{A}_i : i \in \omega \rangle$ be a Δ -chain such that all $\mathfrak{A}_i$ are models of T and also $\mathfrak{A} := \bigcup \langle \mathfrak{A}_i : i \in \omega \rangle$ is a model of T . Then for any $\mathbf{a} \in |\mathfrak{A}_0|$,

$$\begin{aligned} \text{for all } i \in \omega, \models_{\mathfrak{A}_i} \phi[\mathbf{a}] &\implies \text{for all } i \in \omega, \models_{\mathfrak{A}_i} \psi[\mathbf{a}] \\ &\implies \models_{\mathfrak{A}} \psi[\mathbf{a}] \\ &\implies \models_{\mathfrak{A}} \phi[\mathbf{a}], \end{aligned}$$

where the first and third implications follow from the hypothesis that all $\mathfrak{A}_i$ and $\mathfrak{A}$ are models of T and the second follows from (ii) of the preceding proposition. (ii)($\implies$) now follows immediately by applying this to the $(\forall\exists\Delta)$ axioms for U over T .

For (ii)($\impliedby$), suppose that U is a theory such that $\text{Mod}(U)$ is Δ -union closed in $\text{Mod}(T)$; by Corollary 3.5.7 it suffices to show that every model $\mathfrak{A}$ of $T \cup [U \cap (\forall\exists\Delta)]$ is a model of U . Given such an $\mathfrak{A}$ we construct a Δ -chain

$$\mathfrak{A} = \mathfrak{A}_0 \prec^\Delta \mathfrak{B}_0 \prec^\Delta \dots \prec^\Delta \mathfrak{A}_n \prec^\Delta \mathfrak{B}_n \prec^\Delta \dots$$

such that for all $n \in \omega$,

$$\mathfrak{A}_n \prec \mathfrak{A}_{n+1} \quad \text{and} \quad \mathfrak{B}_n \in \text{Mod}(U).$$

Indeed, given such a chain up to $\mathfrak{A}_n$, since $\mathfrak{A} \prec \mathfrak{A}_n$, also $\mathfrak{A}_n$ is a model of $T \cup [U \cap (\forall\exists\Delta)]$, so by part (iv) of the Second Extension Lemma 3.5.10, there exist a model $\mathfrak{B}_n$ of U and a structure $\mathfrak{A}_{n+1}$ such that $\mathfrak{A}_n \prec^\Delta \mathfrak{B}_n \prec^\Delta \mathfrak{A}_{n+1}$ and $\mathfrak{A}_n \prec \mathfrak{A}_{n+1}$ as desired. This completes the recursion step of the construction.

Now let $\mathfrak{C}$ be the union of this chain. Then also

$$\bigcup \langle \mathfrak{A}_n : n \in \omega \rangle = \mathfrak{C} = \bigcup \langle \mathfrak{B}_n : n \in \omega \rangle.$$

Since $\langle \mathfrak{A}_n : n \in \omega \rangle$ is an elementary chain, $\mathfrak{A} \prec \mathfrak{C}$ and in particular $\mathfrak{C}$ is a model of T . Since $\langle \mathfrak{B}_n : n \in \omega \rangle$ is a Δ -chain of models of U , hence also of T , by the closure hypothesis also $\mathfrak{C}$ is a model of U . Then again because $\mathfrak{A} \prec \mathfrak{C}$, $\mathfrak{A}$ is a model of U .

Part (i)($\Longleftarrow$) for sentences now follows by applying (ii) to the theory $\text{Th}(T \cup \{\phi\})$ and applying Lemma 3.5.13(ii). The general version for formulas may then be derived as in the proof of the Tarski-Robinson Theorem. $\square$

Corollary 3.5.32 *For any formula ϕ and any theory U ,*

(i) $\phi \in \forall_2 \iff \phi$ is union persistent;

(ii) U is $\forall_2$ -axiomatizable $\iff \text{Mod}(U)$ is union closed. $\square$

Corollary 3.5.33 *For any theory T and any regular Boolean set Δ of formulas, if T is Δ -model complete, then T is $(\forall\exists\Delta)$ -axiomatizable.*

Proof. If T is Δ -model complete, then by the Model Completeness Theorem 3.5.17 every Δ -chain of models of T is an elementary chain. Hence T is Δ -union persistent. $\square$

A similar technique yields some interesting facts about consistency and interpolation. In each case the idea is that under certain conditions any inconsistency between two theories or two formulas can be “localized”. The first series generalizes Exercise 3.1.21.

Proposition 3.5.34 *For any L -theories T_0 and T_1 and any chain-friendly regular set Δ of formulas, if both T_0 and T_1 are $\forall\exists\Delta$ -axiomatizable and $T_0 \cup T_1$ is inconsistent, then there exists a sentence $\theta \in \text{Bool}(\exists\Delta)$ such that $T_0 \models \theta$ and $T_1 \models \neg\theta$.*

Proof. We prove this in contrapositive form. Assume that T_0 and T_1 are $\forall\exists\Delta$ -axiomatizable but there is no such sentence θ ; we construct a model $\mathfrak{C}$ for $T_0 \cup T_1$ as follows. From the hypothesis, it follows easily (using compactness) that

$$\Gamma_0 := T_0 \cup (T_1 \cap \text{Bool}(\exists\Delta)) \quad \text{and} \quad \Gamma_1 := T_1 \cup (T_0 \cap \text{Bool}(\exists\Delta))$$

are consistent. We shall construct a Δ -chain

$$\mathfrak{A}_0 \prec^\Delta \mathfrak{B}_0 \prec^\Delta \dots \prec^\Delta \mathfrak{A}_n \prec^\Delta \mathfrak{B}_n \prec^\Delta \dots$$

such that for all $n \in \omega$,

$$\mathfrak{A}_n \text{ is a model of } \Gamma_0 \quad \text{and} \quad \mathfrak{B}_n \text{ is a model of } \Gamma_1.$$

To start, let $\mathfrak{A}_0$ be any model of Γ_0 . Then given such a chain up to $\mathfrak{A}_n$, it follows from part (vi) of the Second Extension Lemma 3.5.10 that there exists a model $\mathfrak{B}_n$ of Γ_1 such that $\mathfrak{A}_n \prec^\Delta \mathfrak{B}_n$ and again for the same reason that there is a model $\mathfrak{A}_{n+1}$ of Γ_0 such that $\mathfrak{B}_n \prec^\Delta \mathfrak{A}_{n+1}$. This completes the recursion step of the construction.

Now let $\mathfrak{C}$ be the union of this chain. Then also

$$\bigcup \langle \mathfrak{A}_n : n \in \omega \rangle = \mathfrak{C} = \bigcup \langle \mathfrak{B}_n : n \in \omega \rangle,$$

and by the easy half of the Chang-Łoś-Suszko Theorem, each of $\text{Mod}(T_0)$ and $\text{Mod}(T_1)$ is Δ -union closed, so $\mathfrak{C}$ is a model of $T_0 \cup T_1$ as desired. $\square$

Boolean Interpolation Theorem 3.5.35 *For any chain-friendly regular set Δ of formulas, any $(\forall\exists\Delta)$ -axiomatizable theory T and any formulas ϕ and ψ , if $\phi \in (\forall\exists\Delta)^T$, $\psi \in (\exists\forall\neg\Delta)^T$ and $T \models \phi \rightarrow \psi$, then there is a formula $\theta \in \text{Bool}(\exists\Delta)$ such that $T \models \phi \rightarrow \theta$ and $T \models \theta \rightarrow \psi$.*

Proof. Given objects as in the hypothesis, suppose first that ϕ and ψ are sentences and let $T_0 := \text{Th}(T \cup \{\phi\})$ and $T_1 := \text{Th}(T \cup \{\neg\psi\})$. Without loss of generality we may suppose that actually $\phi \in (\forall\exists\Delta)$ and $\psi \in (\exists\forall\neg\Delta)$. Then $T_0 \cup T_1$ is inconsistent and both are $(\forall\exists\Delta)$ -axiomatizable, so there exists a sentence $\theta \in \text{Bool}(\exists\Delta)$ such that $T_0 \models \theta$ and $T_1 \models \neg\theta$. But this means that $T \models \phi \rightarrow \theta$ and $T \models \neg\psi \rightarrow \neg\theta$, so θ is as required.

The general case may as usual be reduced to this one by adding a suitable number of new constant symbols to all of the languages and considering the sentences resulting from replacing all of the free variables by constants. $\square$

Corollary 3.5.36 *For any $(\forall\exists\Delta)$ -axiomatizable theory T and any chain-friendly regular set Δ of formulas,*

$$(\forall\exists\Delta)^T \cap (\exists\forall\neg\Delta)^T = (\text{Bool}(\exists\Delta))^T;$$

in particular, for all n and any $\forall_{n+1}$ -axiomatizable theory T ,

$$\forall_{n+1}^T \cap \exists_{n+1}^T = (\text{Bool}(\exists_n))^T.$$

Proof. The inclusions $(\supseteq)$ follow from Lemma 3.4.15. For the inclusions $(\subseteq)$, take $\phi = \psi$ in the theorem. The case $n = 0$ of the second clause is a relativized version of Exercise 3.1.21. $\square$

Remark 3.5.37 The Boolean Interpolation Theorem may also be interpreted as establishing a **separation property** for classes of models. For sentences ϕ and ψ , $\psi \in (\exists\forall\neg\Delta)^T$ means that $\neg\psi \in (\forall\exists\Delta)^T$ and $T \models \phi \rightarrow \psi$ means that $\text{Mod}(T \cup \{\phi\})$ and $\text{Mod}(T \cup \{\neg\psi\})$ are disjoint. The conclusion is that there is a sentence θ in the simpler class $\text{Bool}(\exists\Delta)$ which separates these classes in the sense that $\text{Mod}(T \cup \{\phi\}) \subseteq \text{Mod}(T \cup \{\theta\})$ and $\text{Mod}(T \cup \{\theta\})$ is still disjoint from $\text{Mod}(T \cup \{\neg\psi\})$. In particular we say that the class $\forall_{n+1}$ has the separation property with respect to $\forall_{n+1}$ -axiomatizable theories. Several analogous separation properties in other contexts will be established in [Chapters 4–6](#).

The next result extends Theorem 3.3.12 to arbitrary languages. Phrased contrapositively, it is often called the **Robinson Joint Consistency Theorem**: if $(T_0 \cup T_1) \cap \text{Sent}_L$ is consistent — in particular, if T_0 and T_1 are conservative extensions of a common L -theory T — then also $T_0 \cup T_1$ is consistent.

Robinson Interpolation Theorem 3.5.38 *For any languages L_0 and L_1 with common part L and any L_0 -theory T_0 and L_1 -theory T_1 , if $T_0 \cup T_1$ is inconsistent, then there exists an L -sentence θ such that $T_0 \models \theta$ and $T_1 \models \neg\theta$. θ is called a **Robinson interpolant** for T_0 and T_1 .*

Proof. We prove this in contrapositive form. Assume that T_0 and T_1 have no Robinson interpolant; we construct a model $\mathfrak{C}$ for $T_0 \cup T_1$ as follows. The set $\Delta_L := \text{Fm}_L$ is regular and has the property that $\forall\neg\Delta_L = \Delta_L = \forall\exists\Delta_L$. We shall construct a Δ_L -chain

$$(4) \quad \mathfrak{A}_0 \prec^{\Delta_L} \mathfrak{B}_0 \prec^{\Delta_L} \dots \prec^{\Delta_L} \mathfrak{A}_n \prec^{\Delta_L} \mathfrak{B}_n \prec^{\Delta_L} \dots$$

such that $\mathfrak{A}_0$ is a model of T_0 , $\mathfrak{B}_0$ is a model of T_1 and for all $n \in \omega$,

$$\mathfrak{A}_n \prec \mathfrak{A}_{n+1} \quad \text{and} \quad \mathfrak{B}_n \prec \mathfrak{B}_{n+1}.$$

First, it follows from the hypothesis (and compactness) that $T_0 \cup (T_1 \cap \Delta_L)$ is consistent, so let $\mathfrak{A}_0$ be any model. By part (i) of the Second Extension Lemma 3.5.10 and the remark above, there exists a model $\mathfrak{B}_0$ of T_1 such that $\mathfrak{A}_0 \prec^{\Delta_L} \mathfrak{B}_0$. Now given the chain up to $\mathfrak{B}_n$, in particular $\mathfrak{A}_n \prec^{\forall\exists\Delta_L} \mathfrak{B}_n$, so by (v) of the Second Extension Lemma, there exist $\mathfrak{A}_{n+1}$ and $\mathfrak{B}_{n+1}$ as desired.

Now let $\mathfrak{A} := \bigcup \langle \mathfrak{A}_n : n \in \omega \rangle$ and $\mathfrak{B} := \bigcup \langle \mathfrak{B}_n : n \in \omega \rangle$. Since Δ_L is the set of all L -formulas, (4) may also be expressed as

$$\mathfrak{A}_0 \upharpoonright L \prec \mathfrak{B}_0 \upharpoonright L \prec \dots \prec \mathfrak{A}_n \upharpoonright L \prec \mathfrak{B}_n \upharpoonright L \prec \dots$$

from which it follows that $\mathfrak{A} \upharpoonright L = \mathfrak{B} \upharpoonright L$. Hence we may define an $(L_0 \cup L_1)$ -structure $\mathfrak{C}$ with universe $|\mathfrak{A}| = |\mathfrak{B}|$ as follows. We interpret the non-logical

symbols of L_0 as in $\mathfrak{A}$ and the non-logical symbols of L_1 as in $\mathfrak{B}$; that $\mathfrak{A} \upharpoonright L = \mathfrak{B} \upharpoonright L$ ensures that there is no conflict between these interpretations. Then $\mathfrak{C} \upharpoonright L_0 = \mathfrak{A}$ so $\mathfrak{C}$ is a model of T_0 , and $\mathfrak{C} \upharpoonright L_1 = \mathfrak{B}$ so $\mathfrak{C}$ is a model of T_1 . $\square$

Craig Interpolation Theorem 3.5.39 *For any language L , any L -theory T , any two extensions L_0 and L_1 of L whose common symbols are all symbols of L and any L_0 -formula ϕ and L_1 -formula ψ , if $T \models \phi \rightarrow \psi$, then there exists an L -formula θ such that $T \models \phi \rightarrow \theta$ and $T \models \theta \rightarrow \psi$. θ is called a **Craig interpolant** for ϕ and ψ .*

Proof. Exactly as for the countable case 3.3.13. $\square$

Definition 3.5.40 Let L be any language, p a k -ary relation symbol not in Rs_L , L^+ the language extending L which includes p , and T^+ an L^+ -theory.

- (i) p is **explicitly T^+ - L -definable** iff there exists an L -formula θ with free variables $x_0, \dots, x_{k-1}$ such that

$$T^+ \models px_0 \dots x_{k-1} \leftrightarrow \theta.$$

- (ii) p is **implicitly T^+ - L -definable** iff for any L -structure $\mathfrak{A}$ and any $P, Q \subseteq |\mathfrak{A}|^k$, if both $(\mathfrak{A}, P)$ and $(\mathfrak{A}, Q)$ are models of T^+ , then $P = Q$.

Note that if $\mathfrak{A}$ is an L -structure, $P \subseteq |\mathfrak{A}|^k$ and $T^+ = \text{Th}(\mathfrak{A}, P)$, then p is explicitly T^+ - L -definable iff P is definable over $\mathfrak{A}$. In this case, since $P = \theta^{\mathfrak{A}}$, clearly P is the unique relation $Q \subseteq |\mathfrak{A}|^k$ such that $(\mathfrak{A}, Q)$ is a model of T^+ . Viewing this contrapositively, if there exists a Q different from P such that $(\mathfrak{A}, Q)$ is a model of T^+ , then P is not definable over $\mathfrak{A}$. For example, the solution to Exercise 2.4.53 shows that the usual ordering $<$ on ω is not definable over $(\omega, \times, 1)$ by finding a different relation $<'$ such that even $(\omega, \times, 1, <) \simeq (\omega, \times, 1, <')$. This is called **Padoa's method**; the following theorem shows that it is a general method for establishing non-definability.

Beth Definability Theorem 3.5.41 *Let L be any language, p a k -ary relation symbol not in Rs_L , L^+ the language extending L which includes p , and T^+ an L^+ -theory. Then p is explicitly T^+ - L -definable iff p is implicitly T^+ - L -definable.*

Proof. That explicit definability implies implicit definability is easy and is left to the reader. Assume that p is implicitly T^+ - L -definable. Let L_0 be L^+ and T_0 be T^+ . Let q be a new k -ary relation symbol, L_1 the extension of L which includes q , and T_1 the L_1 -theory which is identical to T_0 except that all occurrences of p are replaced by q . A model of $T_0 \cup T_1$ is of the form $(\mathfrak{A}, P, Q)$, and our hypothesis clearly requires that $P = Q$. Thus we have

$$T_0 \cup T_1 \models px_0 \dots x_{k-1} \rightarrow qx_0 \dots x_{k-1}.$$

By compactness, there are single sentences $\phi_0 \in T_0$ and $\phi_1 \in T_1$ such that

$$\{\phi_0, \phi_1\} \models px_0 \dots x_{k-1} \rightarrow qx_0 \dots x_{k-1}$$

or equivalently

$$\models (\phi_0 \wedge px_0 \dots x_{k-1}) \rightarrow (\phi_1 \rightarrow qx_0 \dots x_{k-1}).$$

By the Craig Interpolation Theorem there exists an L -formula θ such that

$$\models (\phi_0 \wedge px_0 \dots x_{k-1}) \rightarrow \theta \quad \text{and} \quad \models \theta \rightarrow (\phi_1 \rightarrow qx_0 \dots x_{k-1}).$$

Then easily $T^+ \models px_0 \dots x_{k-1} \leftrightarrow \theta$, so p is T^+ - L -definable. $\square$

Collateral Reading. The topics of this section are often classified as elementary model theory; many of them appear in general books devoted to this subject, such as [Ch-Ke], [Rot], [Ho93] or the more approachable [Ho97]. The topic of non-standard analysis, treated in a series of exercises below, has its own extensive literature, for example [Hu-Lo].

Exercises

Exercise 3.5.42 Prove Proposition 3.5.4.

Exercise 3.5.43 Prove the missing implication from the Second Extension Lemma 3.5.10(iv): for any theory U , any regular Boolean set Δ of formulas and any structures $\mathfrak{A}$ and $\mathfrak{B}$, if $\mathfrak{B} \prec^\Delta \mathfrak{C} \prec^\Delta \mathfrak{D}$ for some model $\mathfrak{C}$ of U and some $\mathfrak{D}$ such that $\mathfrak{B} \prec \mathfrak{D}$, then $\mathfrak{B}$ is a model of $U \cap (\forall \exists \Delta)$.

Exercise 3.5.44 Prove that for any theory T and any regular set Δ of formulas, T is Δ -axiomatizable iff for every model $\mathfrak{A}$ of T , $T \subseteq \text{Th}(\text{Th}^\Delta(\mathfrak{A}))$.

Hint: See the hint for Exercise 1.4.40.

Exercise 3.5.45 Give an alternative proof of the unrelativized version of the Łoś-Tarski Theorem 3.5.11 along the following lines. Assume that $\text{Mod}(T)$ is Δ -substructure closed. Show that for any model $\mathfrak{A}$ of T , if $\mathfrak{B}$ is a model of $\text{Th}^{\forall \neg \Delta}(\mathfrak{A})$, then also $\mathfrak{B}$ is a model of T . Apply the result of the preceding exercise to conclude that T is $(\forall \neg \Delta)$ -axiomatizable.

Exercise 3.5.46 Show that for any regular set Δ a theory T is $\forall \exists \Delta$ -axiomatizable iff for any $\mathfrak{A} \subseteq \mathfrak{B} \in \text{Mod}(T)$, if there exists $\mathfrak{C}$ such that $\mathfrak{B} \prec^\Delta \mathfrak{C}$ and $\mathfrak{A} \prec \mathfrak{C}$, then $\mathfrak{A} \in \text{Mod}(T)$.

Exercise 3.5.47 Show that for any theory T and structure $\mathfrak{B}$,

- (i) $\mathfrak{B}$ is a substructure of some model of T iff every finitely generated substructure (Exercise 3.3.39) of $\mathfrak{B}$ is a substructure of a model of T ;
- (ii) if T is universally axiomatizable, then $\mathfrak{B}$ is a model of T iff every finitely generated substructure of $\mathfrak{B}$ is a model of T .

Exercise 3.5.48 Show that for any languages $L \subseteq L^+$, any L^+ -theory T^+ and any L -structure $\mathfrak{B}$,

- (i) $\mathfrak{B}$ is an elementary substructure of an L -reduct of some model of T^+ iff $\mathfrak{B}$ is a model of $T^+ \cap \text{Fm}_L$;

if further T^+ is universally axiomatizable and the symbols of L^+ not in L are all relation symbols, then

- (ii) $\mathfrak{B}$ is a reduct of some model of T^+ iff $\mathfrak{B}$ is a model of $T^+ \cap \text{Fm}_L$;
- (iii) $\mathfrak{B}$ is a reduct of some model of T^+ iff every finitely generated substructure is a reduct of a model of T^+ .

Exercise 3.5.49 Give a proof of the Robinson Interpolation Theorem 3.5.38 using notions of consistency in the style of the proof of the countable case 3.3.12.

Exercise 3.5.50 Prove the last displayed equivalence in the proof of the Tarski-Robinson Theorem 3.5.14 and part (ii).

Exercise 3.5.51 Establish the following variants of the Łoś-Tarski Theorem 3.5.11. Call a class $\mathcal{K}$ of structures **isomorphism closed** iff whenever $\mathfrak{A} \simeq \mathfrak{B} \in \mathcal{K}$, also $\mathfrak{A} \in \mathcal{K}$, **universally axiomatizable** iff $\mathcal{K} = \text{Mod}(T)$ for some universally axiomatizable theory T (see also Exercise 3.3.34), **universal Horn** iff in addition the axioms of T may be taken to be Horn sentences (Definition 2.3.50) and of **finite character** iff whenever all finite substructures of a structure $\mathfrak{A}$ belong to $\mathcal{K}$, then also $\mathfrak{A} \in \mathcal{K}$. Show that for any class $\mathcal{K}$ of structures,

- (i) $\mathcal{K}$ is universally axiomatizable iff $\mathcal{K}$ is substructure, isomorphism and ultraproduct closed;
- (ii) $\mathcal{K}$ is universal Horn iff $\mathcal{K}$ is substructure, isomorphism and ultraproduct closed and closed under finite direct product;
- (iii) if the language has no function or constant symbols and only finitely many relation symbols, then $\mathcal{K}$ is universally axiomatizable iff $\mathcal{K}$ is substructure and isomorphism closed and of finite character.

Hint: For (i)($\Leftarrow$), show that for any model $\mathfrak{A}$ of $\text{Th}(\mathcal{K}) \cap \forall_1$,

- (1) For any finite $i \subseteq \text{Th}^{\text{QF}}(\mathfrak{A}_A)$, there is a $\mathfrak{B}_i \in \mathcal{K}$ which has an expansion $\mathfrak{B}_i^+$ which is a model of i ;
- (2) there is an ultraproduct $\prod_{\mathcal{U}} \mathfrak{B}_i$ which has an expansion which is a model of $\text{Th}^{\text{QF}}(\mathfrak{A}_A)$;
- (3) $\mathfrak{A} \in \mathcal{K}$.

For (ii), use Corollary 2.3.51 and a generalization of Exercise 2.3.74. For (iii), show first that for any finite structure $\mathfrak{B}$, there is a universal sentence $\theta_{\mathfrak{B}}$ such that for any $\mathfrak{A}$, $\models_{\mathfrak{A}} \theta_{\mathfrak{B}}$ iff $\mathfrak{A}$ has no substructure isomorphic to $\mathfrak{B}$.

Exercise 3.5.52 Prove the following “interpolation versions” of the Łoś-Tarski Theorem 3.5.11. Let Δ be a regular set of formulas and T a theory.

- (i) Suppose that T_0 and T_1 are extensions of T such that for any two models $\mathfrak{A}$ and $\mathfrak{B}$ of T ,

$$\mathfrak{A} \prec^{\Delta} \mathfrak{B} \text{ and } \mathfrak{B} \in \text{Mod}(T_1) \implies \mathfrak{A} \notin \text{Mod}(T_0).$$

Show that there exists a sentence $\theta \in (\forall \neg \Delta)$ such that $T_1 \models \theta$ and $T_0 \models \neg \theta$.

- (ii) Suppose that $\phi(\mathbf{x})$ and $\psi(\mathbf{x})$ are formulas such that for any models $\mathfrak{A}$ and $\mathfrak{B}$ of T and any $\mathbf{a} \in A$,

$$\mathfrak{A} \prec^{\Delta} \mathfrak{B} \text{ and } \models_{\mathfrak{B}} \psi[\mathbf{a}] \implies \models_{\mathfrak{A}} \phi[\mathbf{a}].$$

Show that there exists a formula $\theta(\mathbf{x}) \in (\forall \neg \Delta)$ such that

$$T \models \psi(\mathbf{x}) \rightarrow \theta(\mathbf{x}) \quad \text{and} \quad T \models \theta(\mathbf{x}) \rightarrow \phi(\mathbf{x}).$$

Exercise 3.5.53 Show that for any theory T and any regular Boolean set Δ of formulas, if Δ is closed under substitution of terms for free variables, then the following are equivalent:

- (i) T admits Δ -elimination.
- (ii) For every model $\mathfrak{A}$ of T and every (finitely generated) substructure $\mathfrak{C} \subseteq \mathfrak{A}$, $T \cup \text{Th}^{\Delta}(\mathfrak{A}_C)$ generates a complete L^C -theory.
- (iii) For any models $\mathfrak{A}$ and $\mathfrak{B}$ of T and any (finitely generated) substructures $\mathfrak{C} \subseteq \mathfrak{A}$ and $\mathfrak{D} \subseteq \mathfrak{B}$,

$$\mathfrak{A}_C \equiv^{\Delta} \mathfrak{B}_D \implies \mathfrak{A}_C \equiv \mathfrak{B}_D.$$

Hint: Assume that (ii) holds and fix a model $\mathfrak{A}$ of T and $X \subseteq A$. Let $\mathfrak{C}$ be the substructure $\langle X \rangle_{\mathfrak{A}} \subseteq \mathfrak{A}$ generated by X . To show that (ii) of the

First Quantifier Elimination Theorem holds it will suffice to show that for any formula ϕ and any $\mathbf{a} \in X$,

$$T \cup \text{Th}^\Delta(\mathfrak{A}_C) \models \phi(\dot{\mathbf{a}}) \implies T \cup \text{Th}(\mathfrak{A}_X) \models \phi(\dot{\mathbf{a}}).$$

Use compactness to find a single formula $\psi \in \Delta$ and $\mathbf{c} \in C \sim X$ such that $T \models \psi(\dot{\mathbf{a}}, \dot{\mathbf{c}}) \rightarrow \phi(\dot{\mathbf{a}})$. Express each c_j as $t_j^{\mathfrak{A}}(\dot{\mathbf{b}})$ for some term t and $\mathbf{b} \in X$.

Exercise 3.5.54 By the remark following the Boolean Interpolation Theorem 3.5.35, $\forall_2$ has the separation property with respect to any $\forall_2$ -axiomatizable theory, in particular the logical validities. Show that in contrast, $\exists_2$ does *not* have this property.

Hint: Let L be a language with one binary relation symbol p , and consider the sentence $\exists x \forall y \, pxy$.

Exercise 3.5.55 Establish the following version of the Beth Definability Theorem 3.5.41. Let L be any language, p a k -ary relation symbol not in Rs_L , L^+ the language extending L which includes p , and T^+ an L^+ -theory. We say that p is **weakly explicitly T^+ - L -definable** iff there exist L -formulas $\theta_0, \dots, \theta_n$ with free variables $x_0, \dots, x_{k-1}$ such that

$$T^+ \models \bigvee_{i \leq n} p x_0 \dots x_{k-1} \leftrightarrow \theta_i$$

and **weakly implicitly T^+ - L -definable** iff for any L -structure $\mathfrak{A}$ and any $P, Q \subseteq |\mathfrak{A}|^k$, if both $(\mathfrak{A}, P)$ and $(\mathfrak{A}, Q)$ are models of T^+ and $(\mathfrak{A}, P) \equiv (\mathfrak{A}, Q)$, then $P = Q$. Show that these two notions are equivalent and if T^+ is a complete theory they coincide with the earlier versions (without “weakly”).

Hint: If p is not weakly explicitly definable, then the set consisting of T^+ together with all sentences of the form $\neg \forall x_0 \dots \forall x_{k-1} (p x_0 \dots x_{k-1} \leftrightarrow \theta)$ for θ an L -formula is consistent.

An important application of the Compactness Theorem is the construction of non-standard models for various theories; Theorem 3.1.14 was a first example. This is a large and well-developed area, and the next few exercises are only a brief introduction. Let $\mathcal{R} = (\mathbb{R}, +, \times, <, |\cdot|, r)_{r \in \mathbb{R}}$ be a structure based on the real numbers in a language L with a name for each real number. $\mathcal{R}$ is the **standard model** for its theory $\text{Th}(\mathcal{R})$; any L -structure $\mathcal{R}^*$ which is elementarily equivalent but not isomorphic to $\mathcal{R}$ is called a **non-standard model** of $\text{Th}(\mathcal{R})$, and the study of such models is called **non-standard analysis**; see also Example 3.3.18.

Exercise 3.5.56 Prove that non-standard models of $\text{Th}(\mathcal{R})$ exist.

An element a of a non-standard model

$$\mathcal{R}^* = (\mathbb{R}^*, +^*, \times^*, <^*, |\cdot|^*, r^*)_{r \in \mathbb{R}}$$

is called **finite** iff for some $r \in \mathbb{R}$, $|a|^* <^* r^*$, and **infinitesimal** iff $a \neq 0^*$ and for all $r \in \mathbb{R} \setminus \{0\}$, $|a|^* <^* |r|^*$.

Exercise 3.5.57 Prove that in any non-standard model $\mathcal{R}^*$ of $\text{Th}(\mathcal{R})$,

- (i) there exist both infinite and infinitesimal elements;
- (ii) if a and b are finite, then so are $a +^* b$ and $a \times^* b$;
- (iii) if a and b are infinitesimal, then so are $a +^* b$ and $a \times^* b$.

For any function $F : \mathbb{R} \rightarrow \mathbb{R}$, let $\mathcal{R}_F$ be the expanded structure

$$(\mathbb{R}, +, \times, <, |\cdot|, F, r)_{r \in \mathbb{R}}.$$

Exercise 3.5.58

- (i) Extend Exercise 3.5.56 to show that there exist also structures

$$\mathcal{R}_F^* = (\mathbb{R}^*, +^*, \times^*, <^*, |\cdot|^*, F^*, r^*)_{r \in \mathbb{R}}$$

which are elementarily equivalent but not isomorphic to $\mathcal{R}_F$;

- (ii) for $a, b \in \mathbb{R}^*$, write $a \approx b$ to mean that $a = b + i$ for some infinitesimal i (a and b are infinitely close); prove that for any $r, s \in \mathbb{R}$,

$$\lim_{x \rightarrow r} F(x) = s \iff \text{for all } a \neq r^*, \text{ if } a \approx r^*, \text{ then } F^*(a) \approx s.$$

Exercise 3.5.59 Extend the analysis of the preceding exercise to structures $\mathcal{R}_{F,G}$ with two distinguished functions and use the characterization of limit to give a non-standard proof of the (standard) theorem:

$$\lim_{x \rightarrow r} F(x) = s \wedge \lim_{x \rightarrow r} G(x) = t \implies \lim_{x \rightarrow r} (F + G)(x) = s + t.$$

Definition 3.5.60 A theory T^{cpn} is called a **Δ -model companion** of T iff $T^{\text{cpn}} \cap (\forall \neg \Delta) = T \cap (\forall \neg \Delta)$ and T^{cpn} is Δ -model complete.

Exercise 3.5.61 Prove that for any chain-friendly regular set Δ , a theory T has at most one Δ -model companion.

Hint: If U and V are two Δ -model companions of T , show that every model of U is a model of V by constructing a Δ -chain of structures which are alternately models of U and V (use the Second Extension Lemma 3.5.10).

Exercise 3.5.62 Let Δ be a regular set of formulas and T^{cpn} the Δ -model companion of T . Show, using the terminology of Remark 3.5.19, that

- (i) $\Delta\text{-ecMod}(T) \subseteq \text{Mod}(T^{\text{cpn}})$;
- (ii) if T is $(\forall\exists\Delta)$ -axiomatizable, then $\Delta\text{-ecMod}(T) = \text{Mod}(T^{\text{cpn}})$.

Hint: By the Second Extension Lemma 3.5.10 each of T and T^{cpn} has the property that any model of one of them has a Δ -extension which is a model of the other. For (i) use also Corollary 3.5.33 and show that if $\mathfrak{A} \prec^{\forall\neg\Delta} \mathfrak{B}$, then any $(\forall\exists\Delta)$ -sentence true in $\mathfrak{B}$ is also true in $\mathfrak{A}$.

Exercise 3.5.63 Let Δ be a chain-friendly regular set of formulas and T an $(\forall\exists\Delta)$ -axiomatizable theory.

- (i) Show that for any model $\mathfrak{A}$ of T there exists a model $\mathfrak{B}$ of T such that $\mathfrak{A} \prec^\Delta \mathfrak{B}$ and $\mathfrak{B}$ is Δ -existentially closed for T .
- (ii) Show that T has a Δ -model companion (necessarily unique by Exercise 3.5.61) iff $\Delta\text{-ecMod}(T)$ is an axiomatizable class of structures ($=\text{Mod}(T^{\text{cpn}})$ by the preceding exercise).

Hint: For (i), construct a Δ -chain $\mathfrak{A} = \mathfrak{A}_0 \prec^\Delta \mathfrak{A}_1 \dots$ of models of T such that for each n , any $(\exists\Delta)$ sentence with parameters from $\mathfrak{A}_n$ which holds in any model of T which is a Δ -extension of $\mathfrak{A}_{n+1}$ holds already in $\mathfrak{A}_{n+1}$. Show that the union of this chain is the desired structure.

Exercise 3.5.64 Show that for any regular Boolean set Δ of formulas and any theory T , T admits Δ -elimination iff for every model $\mathfrak{A}$ of $T \cap (\forall\neg\Delta)$, $T \cup \text{Th}^\Delta(\mathfrak{A}_A)$ generates a complete L^A -theory. (Compare the Model Completeness Theorem 3.5.17).

Hint: Use the First Quantifier Elimination Theorem 3.5.24.

Definition 3.5.65 For any class Δ of formulas, a theory T has the **Δ -amalgamation property** iff for any models $\mathfrak{A}$, $\mathfrak{B}_0$ and $\mathfrak{B}_1$ of T , if there exist Δ -embeddings $\eta_i : \mathfrak{A} \hookrightarrow^\Delta \mathfrak{B}_i$ ($i=0,1$), then there exists a model $\mathfrak{C}$ of T and Δ -embeddings $\zeta_i : \mathfrak{B}_i \hookrightarrow^\Delta \mathfrak{C}$ such that $\zeta_0 \circ \eta_0 = \zeta_1 \circ \eta_1$.

Exercise 3.5.66 Show that for any theory T ,

- (i) T has the Fm-amalgamation property;
- (ii) for any regular Boolean set Δ of formulas,
 - (a) if T is Δ -model complete, then T has the Δ -amalgamation property;
 - (b) T admits Δ -elimination iff T is Δ -model complete and (the theory generated by) $T \cap (\forall\neg\Delta)$ has the Δ -amalgamation property.

Hint: For (ii)(b) use the preceding exercise. Suppose first that T admits Δ -elimination and $\mathfrak{A}, \mathfrak{B}_0$ and $\mathfrak{B}_1$ are models of $T \cap (\forall \neg \Delta)$ which with η_i satisfy the hypotheses of the amalgamation property. By the Second Extension Lemma 3.5.10 there exist models $\mathfrak{D}_0$ and $\mathfrak{D}_1$ of T such that $\mathfrak{B}_i \prec^\Delta \mathfrak{D}_i$. Use the completeness of $T \cup \text{Th}^\Delta(\mathfrak{A}_A)$ to conclude that there exists a model $\mathfrak{C}$ as required. For the other direction, given a model $\mathfrak{A}$ of $T \cap (\forall \neg \Delta)$ and any two models $\mathfrak{B}_{0,A_0}$ and $\mathfrak{B}_{1,A_1}$ of $T \cup \text{Th}^\Delta(\mathfrak{A}_A)$, use amalgamation and the Δ -model completeness of T to show that $\mathfrak{B}_{0,A_0} \equiv \mathfrak{B}_{1,A_1}$.

In the proof of Corollary 3.4.21 we used essentially the fact that for regular sets Δ the formula $\neg x \doteq y$ belongs to Δ . Without this the mapping η may not be injective and therefore we cannot pass, as in the First Extension Lemma, from $\mathfrak{A} \hookrightarrow^\Delta \mathfrak{B}$ to $\mathfrak{A} \prec^\Delta \mathfrak{B}'$. The following series of exercises (and definitions) develops some of the analogous properties of certain non-regular Δ culminating in a major theorem.

Definition 3.5.67 A set Δ of formulas is *positive regular* iff

- (i) all L -atomic formulas belong to Δ ;
- (ii) Δ is closed under $\wedge$ and $\vee$; and
- (iii) Δ is closed under substitution of variables for free variables;

The set PosQF of *positive quantifier-free* formulas is the smallest positive regular set, and the set Pos of *positive formulas* is the smallest positive regular set closed under both $\exists$ and $\forall$.

Definition 3.5.68 For any set Δ of formulas, any structures $\mathfrak{A}$ and $\mathfrak{B}$ and any function $\eta : A \rightarrow B$, η is a Δ -*homomorphism* of $\mathfrak{A}$ into $\mathfrak{B}$ — in symbols $\eta : \mathfrak{A} \rightarrow^\Delta \mathfrak{B}$ — iff for all $\phi(\mathbf{x}) \in \Delta$ and any $\mathbf{a} \in A$,

$$\models_{\mathfrak{A}} \phi[\mathbf{a}] \implies \models_{\mathfrak{B}} \phi[\eta(\mathbf{a})];$$

$\mathfrak{A}$ is Δ -*homomorphically embeddable* into $\mathfrak{B}$ — in symbols $\mathfrak{A} \rightarrow^\Delta \mathfrak{B}$ — iff for some η , $\eta : \mathfrak{A} \rightarrow^\Delta \mathfrak{B}$.

Thus, for example, a PosQF -homomorphism is exactly a positive homomorphism as defined in Remark 2.3.27, while a QF -homomorphism is an embedding. If Δ contains the negations of all atomic formulas except possibly equalities, then a Δ -homomorphism is a homomorphism as in Definition 2.3.26. Since a Δ -homomorphism $\eta : \mathfrak{A} \rightarrow^\Delta \mathfrak{B}$ is not generally injective, the image of η is a substructure of $\mathfrak{B}$ not generally isomorphic or even elementarily equivalent to $\mathfrak{A}$; it is called a Δ -*homomorphic image* of $\mathfrak{A}$. The following should be compared with the First and Second Extension Lemmas.

Exercise 3.5.69 (First Homomorphism Lemma) For any structure $\mathfrak{A}$, any set Γ of sentences and any positive regular set Δ of formulas,

$\Gamma \cup \text{Th}^\Delta(\mathfrak{A}_A)$ is consistent

$\iff$ there exists a model $\mathfrak{B}$ of Γ such that $\mathfrak{A} \rightarrow^\Delta \mathfrak{B}$.

Exercise 3.5.70 (Second Homomorphism Lemma) For any structures $\mathfrak{A}$ and $\mathfrak{B}$ and any positive regular set of formulas,

- (i) $\text{Th}^{\exists\Delta}(\mathfrak{A}) \subseteq \text{Th}(\mathfrak{B}) \iff$ for some $\mathfrak{C}$, $\mathfrak{A} \rightarrow^\Delta \mathfrak{C}$ and $\mathfrak{B} \prec \mathfrak{C}$;
- (ii) $\text{Th}^{\forall\Delta}(\mathfrak{A}) \subseteq \text{Th}(\mathfrak{B}) \iff$ for some $\mathfrak{D}$, $\mathfrak{B} \rightarrow^{\neg\Delta} \mathfrak{D}$ and $\mathfrak{A} \prec \mathfrak{D}$.

Exercise 3.5.71 Fill in the following outline to give a proof of the **Lyndon Homomorphism Theorem**: for any theory T , T is axiomatizable by a set of positive sentences iff $\text{Mod}(T)$ is closed under positive homomorphic images.

Hint: The implication ($\implies$) is straightforward; for the converse it suffices to show that any model $\mathfrak{B}_0$ of $T \cap \text{Pos}$ is also a model of T . Fix such a $\mathfrak{B}_0$ and use the results of the preceding exercise to show that there exist structures $\mathfrak{A}_n$ and $\mathfrak{B}_n$ and mappings η_n and ζ_n such that for all $n \in \omega$,

- (i) $\mathfrak{A}_n \prec \mathfrak{A}_{n+1}$ and $\mathfrak{B}_n \prec \mathfrak{B}_{n+1}$;
- (ii) $\mathfrak{A}_0 \in \text{Mod}(T)$ and $\text{Th}^{\text{Pos}}(\mathfrak{A}_0) \subseteq \text{Th}(\mathfrak{B}_0)$;
- (iii) $\eta_0 : \mathfrak{A}_0 \rightarrow^{\text{Pos}} \mathfrak{B}_1$;
- (iv) $\zeta_n : (\mathfrak{B}_{n+1})_{A_n^{\eta_n}} \hookrightarrow^{\neg\text{Pos}} (\mathfrak{A}_{n+1})_{A_n}$;
- (v) $\eta_{n+1} : (\mathfrak{A}_{n+1})_{A_n, B_{n+1}^{\zeta_n}} \rightarrow^{\text{Pos}} (\mathfrak{B}_{n+2})_{A_n^{\eta_n}, B_{n+1}}$.

Here the notation $(\mathfrak{C})_{D, E^\zeta}$ denotes the structure $\mathfrak{C}$ expanded to a language with constants $\dot{d}$ for $d \in D$, with $\dot{d}$ interpreted as d , and $\dot{e}$ for $e \in E$, with $\dot{e}$ interpreted as $\zeta(e)$. Let $\mathfrak{A} := \bigcup \langle \mathfrak{A}_n : n \in \omega \rangle$ and $\mathfrak{B} := \bigcup \langle \mathfrak{B}_n : n \in \omega \rangle$. Conclude that $\mathfrak{A}$ is a model of T and for all $n \in \omega$,

- (vi) $\eta_n \subseteq \eta_{n+1}$ and $\zeta_n^{-1} \subseteq \eta_{n+1}$.

Set $\eta := \bigcup \langle \eta_n : n \in \omega \rangle$, show that

- (vii) $\eta : \mathfrak{A} \rightarrow \mathfrak{B}$ is a surjective positive homomorphism,

and conclude that $\mathfrak{B}$ and hence also $\mathfrak{B}_0$ is a model of T .

3.6 Higher-order logic

To this point we have focussed exclusively on first-order logic. In the preceding sections we encountered several examples of natural mathematical notions which cannot be fully and accurately expressed in the relevant first-order language: the finiteness or infiniteness of a structure, the notion of well-ordering, the property of being isomorphic to a given infinite structure and the least-upper bound property. In this section and the next we shall study several extensions of first-order logic which offer greater expressive power with the concomitant sacrifice of some of the elegant structural properties of first-order logic. These extensions arise by introducing more general logical operations such as conjunctions and disjunctions of infinite sets of formulas or quantifiers ranging over sets, functions and other objects.

Monadic second-order logic

The quantifiers of first-order language L are interpreted as referring to the elements of (the universe of) a structure; our first extension L^{2m} adds quantifiers over subsets of the universe.

Definition 3.6.1 For any language L , L^{2m} is defined as follows:

- (i) the symbols of L^{2m} are those of L together with a denumerable set $\mathbf{Vb}^1 = \{V_n : n \in \omega\}$ of **set variables**; the variables $\mathbf{Vb}^0 = \{v_n : n \in \omega\}$ are now called **individual variables**;
- (ii) the terms of L^{2m} are the same as the terms of L ;
- (iii) the L^{2m} -atomic formulas are those of L together with all expressions $V_n t$, where V_n is a set variable and t is a term;
- (iv) the set of L^{2m} -**formulas** is the smallest set $\mathbf{Fm}_L^{2m}$ which includes the L^{2m} -atomic formulas and is such that
 - (a) if $\phi, \psi \in \mathbf{Fm}_L^{2m}$, then also $\neg \phi, (\phi \vee \psi) \in \mathbf{Fm}_L^{2m}$;
 - (b) for all n , $\exists v_n \phi$ and $\exists V_n \phi \in \mathbf{Fm}_L^{2m}$;
- (v) L^{2m} -structures are the same as L -structures;
- (vi) for any structure $\mathfrak{A}$, an L^{2m} - $\mathfrak{A}$ -assignment is a pair of functions

$$\alpha : \mathbf{Vb}^0 \rightarrow |\mathfrak{A}| \quad \text{and} \quad \beta : \mathbf{Vb}^1 \rightarrow \wp(|\mathfrak{A}|);$$

- (vii) for any term t , $t^{\mathfrak{A}}[\alpha, \beta]$ is defined as $t^{\mathfrak{A}}[\alpha]$;

(viii) satisfaction for L^{2m} -atomic formulas is defined as for L with the additional clause

$$\models_{\mathfrak{A}}^{2m} V_n t[\alpha, \beta] \iff t^{\mathfrak{A}}[\alpha, \beta] \in \beta(V_n);$$

(ix) L^{2m} -satisfaction is defined recursively as for L with the additional clause

$$\models_{\mathfrak{A}}^{2m} \exists V_n \phi[\alpha, \beta] \iff \text{for some } B \subseteq A, \models_{\mathfrak{A}}^{2m} \phi[\alpha, \beta_{V_n \mapsto B}].$$

Syntactic notions, such as that of free variable, sentence and substitution, are defined just as for L with modifications to include the set variables. The term ‘monadic’ here refers to the fact that the second-order variables range over subsets of the universe; **full second-order logic** admits also variables which range over arbitrary relations and functions. We shall return to this below.

To make things easier to read, we use letters $X, Y, Z, \dots$ to stand for arbitrary set variables V_n and introduce the usual abbreviations $\wedge, \rightarrow, \leftrightarrow$ and $\forall$. In particular, $\forall X \phi$ is an abbreviation for $\neg \exists X \neg \phi$. Of course, it follows immediately from the definitions that the defined connectives work as expected and that

$$\models_{\mathfrak{A}}^{2m} \forall X \phi[\alpha, \beta] \iff \text{for all } B \subseteq A, \models_{\mathfrak{A}}^{2m} \phi[\alpha, \beta_{X \mapsto B}].$$

To simplify notation, when the context is clear we shall usually write simply $\models$ for $\models^{2m}$.

We begin with several examples of L^{2m} sentences which will serve both to introduce the language and as counterexamples to some of the structural properties of first-order logic.

Example 3.6.2 Let L be the language with just one binary relation symbol $\dot{<}$, and let θ_{WO} be the conjunction of the L -sentence θ_{SLO} of Examples 2.2.12(iii), which says that $\dot{<}$ is a strict linear ordering, together with the L^{2m} -sentence

$$\forall X (\exists y Xy \rightarrow \exists y [Xy \wedge \neg \exists z (Xz \wedge z \dot{<} y)]).$$

Then, for any L -structure $\mathfrak{A} = (A, <_A)$,

$$\models_{\mathfrak{A}} \theta_{\text{WO}} \iff <_A \text{ is a well-ordering of } A.$$

Indeed, if $\mathfrak{A}$ satisfies θ_{WO} , then $\mathfrak{A}$ is a linear ordering and for any $B \subseteq A$, if B is non-empty, then B has an element b which has no proper predecessors. By linearity, b is a least element. Thus $\mathfrak{A}$ is a well-ordering. This fact contrasts with Proposition 3.5.3 to the effect that there is no first-order sentence, or even a set of sentences, which axiomatizes the class of well-ordered structures. More generally, if ϕ is any L^{2m} -formula, let $\theta_{\text{WO};\phi,y}$ be the conjunction of θ_{SLO} with

$$\forall X ((\exists y Xy \wedge \forall y [Xy \rightarrow \phi]) \rightarrow \exists y [Xy \wedge \neg \exists z (Xz \wedge z \dot{<} y)]).$$

Then for any $\mathfrak{A}$ and any $\mathfrak{A}$ -assignment (α, β) , $\models_{\mathfrak{A}} \theta_{\text{WO};\phi,y}[\alpha, \beta]$ iff $<_A$ is a linear ordering of A whose restriction to the set

$$\phi_{\alpha}^{\mathfrak{A},\beta} = \{ a \in A : \models_{\mathfrak{A}} \phi[\alpha_{y \mapsto a}, \beta] \}$$

is a well-ordering.

Example 3.6.3 Let L be any language which includes the unary function symbol $\dot{S}$ and the constant symbol $\dot{0}$, and denote by θ_{Ind} the L^{2m} -sentence

$$\forall X [(X\dot{0} \wedge \forall y [Xy \rightarrow X\dot{S}y]) \rightarrow \forall y Xy],$$

the **induction axiom**. If (as usual) $\dot{n}$ denotes the term $\dot{S}\dot{S}\cdots\dot{S}\dot{0}$ with n occurrences of $\dot{S}$, then for any L -structure $\mathfrak{A}$,

$$\models_{\mathfrak{A}} \theta_{\text{Ind}} \iff |\mathfrak{A}| = \{ \dot{n}^{\mathfrak{A}} : n \in \omega \}.$$

The implication ($\Leftarrow$) is obvious. Suppose that $\models_{\mathfrak{A}} \theta_{\text{Ind}}$, so for any $B \subseteq |\mathfrak{A}|$, if $\dot{0}^{\mathfrak{A}} \in B$ and B is closed under $\dot{S}^{\mathfrak{A}}$, then $B = |\mathfrak{A}|$. The set $B = \{ \dot{n}^{\mathfrak{A}} : n \in \omega \}$ clearly satisfies these hypotheses. In other words, models of θ_{Ind} are standard models (Definition 2.5.10). In particular, models of θ_{Ind} may be finite or denumerable, but not uncountable.

Let θ_{ω} be the conjunction of θ_{Ind} with the two successor axioms:

$$(S1) \quad \forall x (\neg \dot{0} \dot{=} \dot{S}x)$$

$$(S2) \quad \forall x \forall y (\dot{S}x \dot{=} \dot{S}y \rightarrow x \dot{=} y).$$

Then if $\mathfrak{A} = (A, \text{Sc}_A, 0_A, \dots)$ is any model of θ_{ω} , it is straightforward to prove by induction that for all $m < n$, $\dot{m}^{\mathfrak{A}} \neq \dot{n}^{\mathfrak{A}}$. Now the mapping $\eta : n \mapsto \dot{n}^{\mathfrak{A}}$ is bijective $\omega \rightarrow A$. Furthermore, η respects $\dot{S}$ since

$$\eta(\text{Sc}(m)) = \eta(m+1) = \overbrace{m+1}^{\dot{S}^{\mathfrak{A}}} = (\dot{S}\dot{m})^{\mathfrak{A}} = \text{Sc}_A(\dot{m}^{\mathfrak{A}}) = \text{Sc}_A(\eta(m)).$$

Thus we have shown

$$\models_{\mathfrak{A}} \theta_{\omega} \iff (A, \text{Sc}_A, 0_A) \simeq (\omega, \text{Sc}, 0).$$

This is a more formal version of Proposition 2.5.1 and asserts that θ_{ω} is categorical (Definition 2.5.2) in contrast with Corollary 3.4.10.

Example 3.6.4 The preceding example may easily be extended to larger languages: for example, let L_Ω be (as usual) the language of the structure $\Omega = (\omega, <, +, \times, \text{Sc}, 0)$, and let θ_Ω be the conjunction of θ_ω from the preceding example with the following axioms:

- (L1) $\forall x (\neg x \dot{<} 0)$
- (L2) $\forall x \forall y [x \dot{<} \dot{\text{S}}y \leftrightarrow (x \dot{<} y \vee x \dot{=} y)]$
- (A1) $\forall x (x \dot{+} 0 \dot{=} x)$
- (A2) $\forall x \forall y [x \dot{+} \dot{\text{S}}y \dot{=} \dot{\text{S}}(x \dot{+} y)]$
- (M1) $\forall x (x \dot{\times} 0 \dot{=} 0)$
- (M2) $\forall x \forall y [(x \dot{\times} \dot{\text{S}}y) \dot{=} (x \dot{\times} y) \dot{+} x]$.

Then by the same sort of argument as in the preceding example we have that for any L_Ω -structure $\mathfrak{A} = (A, <_A, +_A, \times_A, \text{Sc}_A, 0_A)$,

$$\models_{\mathfrak{A}} \theta_\Omega \iff \mathfrak{A} \simeq \Omega.$$

The implication ($\Leftarrow$) is immediate, so let $\mathfrak{A}$ be a model of θ_Ω . As above, the mapping $\eta : n \mapsto \dot{n}^{\mathfrak{A}}$ is bijective $\omega \rightarrow A$ and respects $\dot{\text{S}}$. To see that η respects $\dot{+}$, we prove by induction on n that for all m ,

$$\models_{\mathfrak{A}} \overbrace{m+n}^{\dot{+}} \dot{=} \dot{m} \dot{+} \dot{n} \quad \text{or equivalently} \quad \overbrace{m+n}^{\dot{+}}^{\mathfrak{A}} = \dot{m}^{\mathfrak{A}} +_A \dot{n}^{\mathfrak{A}}.$$

For $n = 0$, $\overbrace{m+0}^{\dot{+}} \dot{=} \dot{m} \dot{=} \dot{m} \dot{+} \dot{0}$ is true in $\mathfrak{A}$, and given the induction hypothesis,

$$\overbrace{m+(n+1)}^{\dot{+}} \dot{=} \dot{\text{S}} \overbrace{m+n}^{\dot{+}} \dot{=} \dot{\text{S}}(\dot{m} \dot{+} \dot{n}) \dot{=} \dot{m} \dot{+} \dot{\text{S}}\dot{n} \dot{=} \dot{m} \dot{+} \overbrace{n+1}^{\dot{+}}.$$

Then

$$\eta(m+n) = \overbrace{m+n}^{\dot{+}}^{\mathfrak{A}} = (\dot{m} \dot{+} \dot{n})^{\mathfrak{A}} = \dot{m}^{\mathfrak{A}} +_A \dot{n}^{\mathfrak{A}} = \eta(m) +_A \eta(n),$$

as required. The arguments for $\times$ and $<$ are similar.

Example 3.6.5 A well-ordering with no limit points is either finite or of type ω ; thus if we take θ_{CWO} to be the conjunction of θ_{WO} with

$$\forall x [\exists y (y \dot{<} x) \rightarrow \exists y (y \dot{<} x \wedge \forall z [z \dot{<} x \rightarrow z \dot{<} y \vee z \dot{=} y])],$$

which says that any element which has a predecessor has an immediate predecessor, we have for any $\mathfrak{A} = (A, <_A, \dots)$,

$$\models_{\mathfrak{A}} \theta_{\text{CWO}} \iff <_A \text{ well-orders } A \text{ in a finite type or type } \omega.$$

In particular, θ_{CWO} is another L^{2^m} -sentence which has finite and denumerable models but no uncountable models.

Example 3.6.6 Let $x \dot{\leq} y$ abbreviate $x \dot{<} y \vee x \dot{=} y$, and let $\theta_{(\text{CDLO})}$ be the conjunction of the sentence $\theta_{(\text{DLO})}$ of Definition 2.4.44, which says that $\dot{<}$ is a dense linear ordering without endpoints, with the L^{2^m} -sentence

$$\forall X [\exists y Xy \wedge \exists z \forall y (Xy \rightarrow y \dot{\leq} z) \rightarrow \\ \exists z (\forall y (Xy \rightarrow y \dot{\leq} z) \wedge \forall w [\forall y (Xy \rightarrow y \dot{\leq} w) \rightarrow (z \dot{\leq} w)])].$$

It is easy to see that the models of $\theta_{(\text{CDLO})}$ are exactly the structures $\mathfrak{A} = (A, <_A, \dots)$ such that $<_A$ is a **complete dense linear ordering** without endpoints — that is, a dense linear ordering without endpoints in which every non-empty subset of A which is bounded above has a *least* upper bound. (See the discussion following Example 2.3.36). For any such structure there exists an embedding $\eta : (\mathbb{R}, <_{\mathbb{R}}) \hookrightarrow_0 \mathfrak{A}$. Indeed, we may first construct an embedding of $(\mathbb{Q}, <_{\mathbb{Q}})$ into $\mathfrak{A}$ much as in the proof of Cantor's Theorem 2.4.45. η may now be extended to all of $\mathbb{R}$ by completeness: for each irrational $r \in \mathbb{R}$, let $\eta(r)$ be the least upper bound of $\{\eta(q) : q \in \mathbb{Q} \text{ and } q <_{\mathbb{R}} r\}$. It is straightforward to verify that this is the required embedding. Thus $\theta_{(\text{CDLO})}$ has models of power $2^{\aleph_0}$ and larger but none smaller.

We can extract some basic properties of L^{2^m} directly from the examples.

Theorem 3.6.7 *Monadic second-order logic satisfies none of*

- (i) *Compactness Theorem*;
- (ii) *Downward Löwenheim-Skolem Theorem*;
- (iii) *Upward Löwenheim-Skolem Theorem*.

Proof. Left as an exercise. $\square$

In fact, in each case the failure is fairly dramatic; we introduce below some definitions to measure this. First, however, we note how second-order logic provides a natural unification of two of the interpolation theorems 3.5.35 and 3.5.39 of the preceding section. Let $\exists_1^1$ denote the class of L^{2^m} formulas ϕ logically equivalent to one of the form $\exists X_0 \dots \exists X_{k-1} \phi^\circ$, with ϕ° first-order, and $\forall_1^1$ similarly with only universal second-order quantifiers.

Theorem 3.6.8 *For any language L and any L^{2^m} -formulas $\phi \in \exists_1^1$ and $\psi \in \forall_1^1$, if $\models^m \phi \rightarrow \psi$, then there exists a first-order L -formula θ such that $\models^m \phi \rightarrow \theta$ and $\models^m \theta \rightarrow \psi$. Hence $\forall_1^1 \cap \exists_1^1$ is the class of L^{2^m} formulas logically equivalent to a first-order formula.*

Proof. For simplicity, suppose ϕ is $\exists X \phi^\circ$ and ψ is $\forall Y \psi^\circ$. Let L_0 (L_1) be the extension of L by a new unary relations symbol p (q), and ϕ^* (ψ^*) be the result of replacing each atomic formula Xt (Yt) in ϕ° (ψ°) by pt (qt). Then ϕ^* (ψ^*) is a first-order L_0 - (L_1 -) formula, and the hypothesis $\models^m \phi \rightarrow \psi$

easily implies that $\models \phi^* \rightarrow \psi^*$. Hence, by the Craig Interpolation Theorem 3.5.39 there exists a first-order L -formula θ such that $\models \phi^* \rightarrow \theta$ and $\models \theta \rightarrow \psi^*$. But then also $\models^m \phi \rightarrow \theta$ and $\models^m \theta \rightarrow \psi$.

The final clause follows immediately as in similar cases. $\square$

In order that they may apply to other examples, we phrase the following notions in terms of an “abstract logic” $\mathcal{L}$; in Section 7.6 we shall give a formal definition of a logic, but for now it will suffice to think of $\mathcal{L}$ as referring to one of first-order logic $\mathcal{L}^1$, monadic second-order logic $\mathcal{L}^{2m}$, or one of the other logics to be introduced below. By an $\mathcal{L}$ -sentence, we mean a sentence of any of the languages of $\mathcal{L}$, and similarly for other notions. For conciseness of notation we consider ∞ to be a largest cardinal number.

Definition 3.6.9 For any logic $\mathcal{L}$ and any κ and λ , which are either infinite cardinals or ∞ representing a “largest cardinal”,

- (i) $\mathcal{L}$ is (κ, λ) -**compact** iff for any set Γ of $\mathcal{L}$ -sentences, if $\text{Card}(\Gamma) \leq \kappa$ and every subset of Γ of power $< \lambda$ has a model, then Γ has a model;
- (ii) $\mathcal{L}$ has the (κ, λ) -**downward Löwenheim-Skolem property** iff for any set Γ of $\mathcal{L}$ -sentences, if $\text{Card}(\Gamma) \leq \kappa$ and Γ has a model, then Γ has a model of cardinality at most λ ;
- (iii) $\mathcal{L}$ has the (κ, λ) -**upward Löwenheim-Skolem property** iff for every set Γ of $\mathcal{L}$ -sentences, if $\text{Card}(\Gamma) \leq \kappa$ and for each cardinal $\lambda' < \lambda$, Γ has a model of cardinality at least λ' , then for each cardinal $\lambda' \geq \lambda$, Γ has a model of cardinality at least λ' .

Corollary 3.6.10 *First-order logic $\mathcal{L}^1$ is (∞, ω) -compact, for all κ has the (κ, κ) -downward Löwenheim-Skolem property, and has the (∞, ω) -upward Löwenheim-Skolem property. $\square$*

Note that each of the three properties is stronger for larger κ and smaller λ , so the (∞, ω) case implies all of the others. It is sometimes useful to express these properties in terms of the following functions.

Definition 3.6.11 For any logic $\mathcal{L}$ and any cardinal κ ,

- (i) $t_\kappa(\mathcal{L})$ is the smallest λ such that $\mathcal{L}$ is (κ, λ) -compact;
- (ii) $\ell_\kappa(\mathcal{L})$ is the smallest $\lambda \leq \infty$ such that $\mathcal{L}$ has the (κ, λ) -downward Löwenheim-Skolem property;
- (iii) $h_\kappa(\mathcal{L})$ is the smallest $\lambda \leq \infty$ such that $\mathcal{L}$ has the (κ, λ) -upward Löwenheim-Skolem property.

Clearly $t_\kappa(\mathcal{L}) \leq \kappa$, but it is not immediately obvious that $\ell_\kappa(\mathcal{L})$ and $h_\kappa(\mathcal{L})$ always have values less than the trivial ones ∞ . That they in fact do relies only on the mild assumption about $\mathcal{L}$ that for each cardinal κ , up to logical

equivalence there is a bounded (from ∞) number of sets of $\mathcal{L}$ -sentences of cardinality at most κ . We call such a logic **sufficiently nice**. To see that $\mathcal{L}^{2^m}$ is sufficiently nice, for a fixed κ , let $L_\kappa^{2^m}$ be a language with κ -many function and relation symbols of each arity and κ -many constant symbols. Then every set of $\mathcal{L}^{2^m}$ -sentences of cardinality at most κ is logically equivalent to a set of $L_\kappa^{2^m}$ -sentences. Since there are κ -many $L_\kappa^{2^m}$ -sentences, there are 2^κ -many such sets.

Proposition 3.6.12 *For any sufficiently nice logic $\mathcal{L}$, for every infinite cardinal κ , $\ell_\kappa(\mathcal{L}) < \infty$ and $h_\kappa(\mathcal{L}) < \infty$.*

Proof. For each set Γ of at most κ -many $\mathcal{L}$ -sentences, let

$$\lambda_\downarrow(\Gamma) = \begin{cases} \text{least } \lambda [\Gamma \text{ has a model of power } \lambda], & \text{if } \Gamma \text{ is consistent;} \\ 0, & \text{otherwise;} \end{cases}$$

$$\lambda^\uparrow(\Gamma) = \begin{cases} \text{greatest } \lambda [(\forall \lambda' < \lambda) \Gamma \text{ has a model of power } \geq \lambda'], & \text{if any;} \\ 0, & \text{otherwise.} \end{cases}$$

Then easily

$$\ell_\kappa(\mathcal{L}) = \sup \{ \lambda_\downarrow(\Gamma) : \Gamma \text{ is a set of at most } \kappa\text{-many } \mathcal{L}\text{-sentences} \};$$

$$h_\kappa(\mathcal{L}) = \sup^+ \{ \lambda^\uparrow(\Gamma) : \Gamma \text{ is a set of at most } \kappa\text{-many } \mathcal{L}\text{-sentences} \}. \quad \square$$

Definition 3.6.13 $t(\mathcal{L}) := t_\infty(\mathcal{L})$ is called the **Tarski number** of $\mathcal{L}$, $\ell(\mathcal{L}) := \ell_1(\mathcal{L})$ is called the **Löwenheim number** of $\mathcal{L}$, and $h(\mathcal{L}) := h_1(\mathcal{L})$ is called the **Hanf number** of $\mathcal{L}$.

Now for first-order logic we have

Corollary 3.6.14

- (i) $t(\mathcal{L}^1) = \ell(\mathcal{L}^1) = h(\mathcal{L}^1) = \omega$.
- (ii) For each infinite cardinal κ , $\ell_\kappa(\mathcal{L}^1) = \kappa$ and $h_\kappa(\mathcal{L}^1) = \omega$. $\square$

For monadic second-order logic, the examples above yield

Proposition 3.6.15

- (i) $\ell(\mathcal{L}^{2^m}) \geq 2^{\aleph_0}$.
- (ii) $h(\mathcal{L}^{2^m}) > \aleph_1$.

Proof. Part (i) follows from the fact that the sentence $\theta_{(\text{CDLO})}$ has models of cardinality $2^{\aleph_0}$ and larger, but none smaller. Part (ii) follows from the fact that the sentences θ_{CWO} and θ_ω have countable models but none larger. $\square$

The example for (ii) has the following stronger property.

Definition 3.6.16 For any logic $\mathcal{L}$, a cardinal λ is $\mathcal{L}$ -*distinctive* iff there exists a $\mathcal{L}$ -sentence ψ such that for each cardinal $\lambda' < \lambda$, ψ has a model of cardinality at least λ' but ψ has no model of cardinality at least λ .

Thus $\aleph_1$ is $\mathcal{L}^{2m}$ -distinctive. In general we have

Theorem 3.6.17 For any infinite cardinal λ and any logic $\mathcal{L}$, if λ is $\mathcal{L}$ -distinctive, then $h(\mathcal{L}) > \lambda$ and $t(\mathcal{L}) > \lambda$ — in fact, $\mathcal{L}$ is not (λ, λ) -compact so $t_\lambda(\mathcal{L}) > \lambda$.

Proof. Let ψ witness that λ is $\mathcal{L}$ -distinctive. Then ψ is also a witness to the fact that $\mathcal{L}$ does not have the $(1, \lambda)$ -upward Löwenheim-Skolem property so $h(\mathcal{L}) > \lambda$. Let $\{c_\sigma : \sigma < \lambda\}$ be a set of λ -many new constant symbols and $\Gamma := \{\psi\} \cup \{\neg c_\sigma \doteq c_\tau : \sigma < \tau < \lambda\}$. Then any subset of Γ of cardinality $\lambda' < \lambda$ has a model, but Γ does not have a model. Hence Γ witnesses that $t(\mathcal{L}) > \lambda$ and, since Γ has cardinality λ , that $\mathcal{L}$ is not (λ, λ) -compact. $\square$

Corollary 3.6.18 $t(\mathcal{L}^{2m}) > \aleph_1$. $\square$

In fact, all of these values are much larger. As a scale for measuring this, we use

Definition 3.6.19 The *beth numbers* are defined recursively by

$$\beth_0 = \aleph_0; \quad \beth_{\sigma+1} = 2^{\beth_\sigma}; \quad \beth_\sigma = \sup \{ \beth_\tau : \tau < \sigma \}, \text{ for limit } \sigma.$$

Note that for all σ , $\sigma \leq \aleph_\sigma \leq \beth_\sigma$ and the Generalized Continuum Hypothesis is equivalent to the assertion that always $\aleph_\sigma = \beth_\sigma$. However we have also the following. Let $\kappa_0 = \aleph_1$, $\kappa_{n+1} = \beth_{\kappa_n}$ and $\kappa_\omega = \sup \{ \kappa_n : n \in \omega \}$.

Lemma 3.6.20 κ_ω is the smallest ordinal (cardinal) λ such that $\lambda = \beth_\lambda$.

Proof. Left as an exercise. $\square$

Proposition 3.6.21 For any infinite cardinal λ , if λ is $\mathcal{L}^{2m}$ -distinctive, then also $\beth_\lambda$ is $\mathcal{L}^{2m}$ -distinctive.

Proof. By hypothesis there exists a sentence ψ which has models of unbounded cardinality below λ but no model of cardinality $\geq \lambda$. Let L_ψ be the language of ψ , L_ω the language of θ_ω and L the union of these together with two new unary relation symbols q and r and three new binary relation symbols $\dot{<}$, $\dot{=}$ and p . Let ϕ be the conjunction of the following L^{2m} -sentences:

- (i) q and r contain all constants and are closed under all functions of L ;
- (ii) $\psi^{(q)} \wedge \theta_\omega^{(r)}$;
- (iii) $\theta_{\omega_0; qx, x}$;
- (iv) $\forall u \forall x [u \text{ is the } \dot{<}\text{-least element of } q' \wedge pux \rightarrow rx]$;

- (v) $\forall u \forall v \forall x [u \text{ is the } \dot{<}\text{-successor of } v' \wedge pux \rightarrow \forall z (z \dot{\in} x \rightarrow pvz)];$
- (vi) $\forall x \forall y [\forall z (z \dot{\in} x \leftrightarrow z \dot{\in} y) \rightarrow x \dot{=} y];$
- (vii) $\forall u \forall x [u \text{ is a } \dot{<}\text{-limit point}' \wedge pux \rightarrow \exists v (v \dot{<} u \wedge pvx)];$
- (viii) $\forall x \exists u [qu \wedge pux].$

For any model $\mathfrak{A}$ of ϕ , (i) and (ii) ensure that $q^{\mathfrak{A}}$ is the universe of a substructure of $\mathfrak{A}$ which satisfies ψ and hence has cardinality $< \lambda$, and that similarly $r^{\mathfrak{A}}$ is denumerable. Hence by (iii), $\dot{<}^{\mathfrak{A}}$ restricted to $q^{\mathfrak{A}}$ is a well-ordering of order-type some ordinal $\rho < \lambda$. For each $\sigma < \rho$, let c_σ denote the σ -th point in this ordering and set

$$p_\sigma^{\mathfrak{A}} = \{a \in A : (c_\sigma, a) \in p^{\mathfrak{A}}\}.$$

Condition (iv) ensures that $p_0^{\mathfrak{A}} \subseteq p_1^{\mathfrak{A}}$ and is therefore countable by (ii). Conditions (v) and (vi) ensure that for $\sigma + 1 < \rho$, the mapping $a \mapsto \{b : b \dot{\in}^{\mathfrak{A}} a\}$ is an injection of $p_{\sigma+1}^{\mathfrak{A}}$ into $\wp(p_\sigma^{\mathfrak{A}})$, and (vii) ensures that for limit σ , $p_\sigma^{\mathfrak{A}} \subseteq \bigcup \{p_\tau^{\mathfrak{A}} : \tau < \sigma\}$. Thus by a straightforward induction, for all $\sigma < \rho$, $\text{Card}(p_\sigma^{\mathfrak{A}}) \leq \beth_\sigma$, so by (viii), $\text{Card}(A) \leq \beth_\rho < \beth_\lambda$. On the other hand, for any cardinal $\mu < \beth_\lambda$, for some $\rho < \lambda$, $\mu < \beth_\rho$ and it is clear that ϕ has a model of power $\geq \beth_\rho$. Thus ϕ witnesses that $\beth_\lambda$ is $\mathcal{L}^{2m}$ -distinctive. $\square$

Corollary 3.6.22

- (i) $h(\mathcal{L}^{2m}) \geq \kappa_\omega$.
- (ii) $t(\mathcal{L}^{2m}) \geq \kappa_\omega$. $\square$

By similar methods we have also

Proposition 3.6.23 $\ell(\mathcal{L}^{2m}) \geq \kappa_\omega$.

Proof. It suffices to show that for any infinite cardinal λ , if ψ is an $\mathcal{L}^{2m}$ -sentence which has models but none of cardinality $< \lambda$, then there is a $\mathcal{L}^{2m}$ -sentence ϕ which has models but none of cardinality $< \beth_\lambda$. Let ψ be such a sentence, L be as in the preceding proposition, and ϕ be the conjunction of the following $\mathcal{L}^{2m}$ -sentences:

- (i) q and r contain all constants and are closed under all functions of L ;
- (ii) $\psi^{(q)} \wedge \theta_\omega^{(r)}$;
- (iii) $\theta_{\text{WO}; qx, x}$;
- (iv) $\forall u \forall x [u \text{ is the } \dot{<}\text{-least element of } q' \wedge rx \rightarrow pux];$
- (v) $\forall u \forall v [u \text{ is the } \dot{<}\text{-successor of } v' \rightarrow$

$$\forall X (\forall z [Xz \rightarrow pvz] \rightarrow \exists x [pux \wedge \forall z (z \dot{\in} x \leftrightarrow Xz)])];$$

(vi) $\forall u \forall x ['u \text{ is a } \dot{<}\text{-limit point}' \wedge \exists v (v \dot{<} u \wedge pvx) \rightarrow pux]$.

For any model $\mathfrak{A}$ of ϕ , (i) and (ii) ensure that $q^{\mathfrak{A}}$ has cardinality $\geq \lambda$ and $r^{\mathfrak{A}}$ is denumerable. Hence by (iii), $\dot{<}^{\mathfrak{A}}$ restricted to $q^{\mathfrak{A}}$ is a well-ordering of order-type some ordinal $\rho \geq \lambda$. Similarly, with notation as in the preceding proposition, $p_0^{\mathfrak{A}}$ is at least denumerable; for limit $\sigma < \rho$, $\bigcup \{ p_\tau^{\mathfrak{A}} : \tau < \sigma \} \subseteq p_\sigma^{\mathfrak{A}}$; and the mapping $a \mapsto \{ b : b \dot{\in}^{\mathfrak{A}} a \}$ maps $p_{\sigma+1}^{\mathfrak{A}}$ onto $\wp(p_\sigma^{\mathfrak{A}})$. Then by a straightforward induction, for all $\sigma < \rho$, $\text{Card}(p_\sigma^{\mathfrak{A}}) \geq \beth_\sigma$ so $\text{Card}(A) \geq \beth_\lambda$. $\square$

A somewhat different property of first-order logic is the enumerability property (3.1.18): if Γ is a decidable set of sentences, then $\text{Th}(\Gamma)$, the set of its logical consequences, is effectively enumerable. This fails in a strong way for $\mathcal{L}^{2m}$.

Proposition 3.6.24 *The set V_Ω^{2m} of logically valid sentences of L_Ω^{2m} is not effectively enumerable.*

Proof. In Example 3.6.4 we exhibited a sentence θ_Ω which characterizes the standard model Ω up to isomorphism. It follows that for any L_Ω -sentence (in fact, for any L_Ω^{2m} -sentence) ϕ ,

$$\models_\Omega \phi \iff \models \theta_\Omega \rightarrow \phi \iff \not\models \theta_\Omega \rightarrow \neg \phi.$$

Hence, if V_Ω^{2m} were effectively enumerable, then $\text{Th}(\Omega)$ would be decidable; in Section 4.1 and in more detail in Section 4.5, we show that this is not the case. $\square$

The question whether $\mathcal{L}^{2m}$ has a Completeness Theorem is more complex, since it can only be asked relative to a particular deduction system. However, the results on compactness and enumerability above tell us a good deal about the possibilities. Recall the derivation in Section 3.2 of the Countable Compactness Theorem from the Countable Completeness Theorem:

$$\begin{aligned} \Gamma \models \psi &\implies \Gamma \vdash \psi \implies \text{for some finite } \Gamma_0 \subseteq \Gamma, \quad \Gamma_0 \vdash \psi \\ &\implies \text{for some finite } \Gamma_0 \subseteq \Gamma, \quad \Gamma_0 \models \psi. \end{aligned}$$

The first and third implications are the adequacy and soundness properties, respectively, while the second is based on the finiteness of formal proofs and hence ultimately on the finitary character of the rules of inference — that each has only finitely many premises. Thus we have

Proposition 3.6.25 *For any logic $\mathcal{L}$, if there exists a deduction system for $\mathcal{L}$ which is complete for $\mathcal{L}$ -logical consequence and has finitary rules of inference, then $\mathcal{L}$ is (∞, ω) -compact. In particular, there is no such system for $\mathcal{L}^{2m}$. $\square$*

There is a weaker notion of completeness.

Definition 3.6.26 A deduction system for a logic $\mathcal{L}$ is **validity complete** iff for any $\mathcal{L}$ -sentence ψ , $\vdash \psi$ (in this system) iff $\models \psi$.

Of course, any logic trivially has such a system with no rules of inference and all logical validities as logical axioms, so the interesting questions involve notions of effectiveness. The proof of the Enumerability Theorem for first-order logic uses only the fact that the system of Section 3.2 is an **effective deduction system** in having a decidable set of logical axioms and rules of inference which, regarded as functions, are effectively calculable. By Proposition 3.6.24 we have

Corollary 3.6.27 *There is no effective deduction system for $\mathcal{L}^{2^m}$ which is validity complete.* $\square$

To conclude that there was no finitary deduction system complete for $\mathcal{L}^{2^m}$, we made use of the fact that $\mathcal{L}^{2^m}$ is not (∞, ω) -compact, but we have seen that this logic is also not (∞, λ) -compact for many larger cardinals λ . This leads to the conclusion that there is no deduction system complete for $\mathcal{L}^{2^m}$ which has certain **infinitary rules of inference**. Such a rule has infinitely many premises; for example in a language which has the unary function symbol $\dot{S}$ and the constant symbol $\dot{0}$, an infinitary rule is the ω -**rule**:

$$\text{from } \{ \phi_x(\dot{n}) : n \in \omega \} \quad \text{infer} \quad \forall x \phi.$$

We shall show as an exercise of Section 7.3 that the deduction system formed from the standard one for first-order logic by adding the ω -rule is complete for a logic in which all structures $\mathfrak{A}$ have universe $\{ \dot{n}^{\mathfrak{A}} : n \in \omega \}$. There are also natural examples of infinitary rules of inference in the next section of the form

$$\text{from } \{ \phi_\sigma \rightarrow \psi : \sigma < \lambda \} \quad \text{infer} \quad \bigvee_{\sigma < \lambda} \phi_\sigma \rightarrow \psi.$$

In general we have

Definition 3.6.28 For any cardinal λ , a λ -**ary rule of inference** is any rule of the form

$$\text{from } \{ \phi_\sigma : \sigma < \lambda \} \quad \text{infer} \quad \psi.$$

To analyze the properties of infinitary rules of inference, we need to consider infinitary induction systems. Recall that for any cardinal κ and any set X ,

$$\wp_{<\kappa}(X) = \{ Z \subseteq X : \text{Card}(Z) < \kappa \}.$$

Definition 3.6.29 For any infinite cardinal κ , a $< \kappa$ -**induction system** is a triple $\mathcal{X} = (X, X_0, \mathcal{H})$ such that X is a non-empty set, $X_0 \subseteq X$ and $\mathcal{H}$ is a family of functions $H : \wp_{<\kappa}(X) \rightarrow X$. A subset $Y \subseteq X$ is $\mathcal{X}$ -**closed** iff $X_0 \subseteq Y$ and for all H and Z ,

$$H \in \mathcal{H} \text{ and } Z \in \wp_{<\kappa}(Y) \implies H(Z) \in Y.$$

For any $W \subseteq X$, the $\mathcal{X}$ -**closure** of W is the set

$$\overline{\mathcal{X}}(W) := \bigcap \{ Y : W \subseteq Y \subseteq X \text{ and } Y \text{ is } \mathcal{X}\text{-closed} \}.$$

$\overline{\mathcal{X}}$ denotes $\overline{\mathcal{X}}(\emptyset)$. An induction system is **finitary** iff it is a $< \omega$ -induction system and **countable** iff it is a $< \aleph_1$ -induction system.

An exercise below verifies that the notion of finitary induction system includes the previous notion. Exactly as for other induction systems we have

Proposition 3.6.30 For any induction system $\mathcal{H} = (X, X_0, \mathcal{H})$ and any $W \subseteq X$, $\overline{\mathcal{X}}(W)$ is $\mathcal{X}$ -closed and is the smallest $\mathcal{X}$ -closed set which includes W . $\square$

We may also characterize $\overline{\mathcal{X}}(W)$ as the union of a chain of approximations as in the finitary case, but in general we need a chain of transfinite length. For any ordinal σ , let

$$\begin{aligned} X_{(\sigma)}(W) &:= \bigcup \{ X_\tau(W) : \tau < \sigma \} \cup W; \\ X_\sigma(W) &:= X_{(\sigma)}(W) \cup \{ H(Z) : H \in \mathcal{H} \text{ and } Z \in \wp_{<\kappa}(X_{(\sigma)}(W)) \}. \end{aligned}$$

Proposition 3.6.31 For any regular cardinal κ and any $< \kappa$ -induction system $\mathcal{X}$, $\overline{\mathcal{X}}(W) = X_\kappa(W)$.

Proof. Left as an exercise. $\square$

Similar to Proposition 1.2.11, we have

Lemma 3.6.32 For any regular cardinal κ , any $< \kappa$ -induction system $\mathcal{X} = (X, X_0, \mathcal{H})$ and any $W \subseteq X$,

$$\overline{\mathcal{X}}(W) = \bigcup \{ \overline{\mathcal{X}}(V) : V \subseteq W \text{ and } \text{Card}(V) < \kappa \}.$$

Proof. Let Y denote the right-hand set. Clearly $Y \subseteq \overline{\mathcal{X}}(W)$, and $W \subseteq Y$, so it suffices to show that Y is $\mathcal{X}$ -closed. If $Z \in \wp_{<\kappa}(Y)$, then for each $z \in Z$ choose $V_z \subseteq W$ with $\text{Card}(V_z) < \kappa$ such that $z \in \overline{\mathcal{X}}(V_z)$. Let $V = \bigcup \{ V_z : z \in Z \}$. By the regularity of κ , $\text{Card}(V) < \kappa$, and thus $Z \in \wp_{<\kappa}(\overline{\mathcal{X}}(V))$. Thus for any $H \in \mathcal{H}$, also $H(Z) \in \overline{\mathcal{X}}(V) \subseteq Y$. $\square$

From this we have

Theorem 3.6.33 *For any logic $\mathcal{L}$ and any infinite regular cardinal κ , if there exists a deduction system for $\mathcal{L}$ which is complete for $\mathcal{L}$ -logical consequence and has λ -ary rules of inference for $\lambda < \kappa$, then $\mathcal{L}$ is (∞, κ) -compact.*

Proof. Under the given hypotheses, for any set Γ of $\mathcal{L}$ -sentences, the set $\text{Thm}(\Gamma)$ of formal theorems provable in the system from Γ is the closure $\overline{\mathcal{X}}(\Gamma)$ of a suitable $< \kappa$ -induction system. By the preceding lemma we have for any $\mathcal{L}$ -sentence ψ ,

$$\Gamma \vdash \psi \quad \Longleftrightarrow \quad \text{for some } \Gamma_0 \subseteq \Gamma \text{ with } \text{Card}(\Gamma_0) < \kappa, \quad \Gamma_0 \vdash \psi.$$

From this follows (∞, κ) -compactness exactly as in the finitary case. $\square$

Corollary 3.6.34 *For any logic $\mathcal{L}$, there is no deduction system complete for $\mathcal{L}$ -logical consequence which has only λ -ary rules of inference for λ bounded by any $\kappa < t(\mathcal{L})$. In particular, there is no such system for $\mathcal{L}^{2^m}$ with λ bounded below κ_ω .*

To this point we have focussed on monadic second-order logic because most of the features of higher-order logic can be seen in their simplest form there. Our discussion would surely be incomplete, however, without some consideration of still more expressive forms of higher-order logic. We shall be much briefer here.

The first step in this direction is to admit quantification over functions and relations as well as sets. For any first-order language L , the corresponding full second-order language L^2 has as its logical symbols those of L together with additional families of variables:

$$\begin{aligned} \text{Vb}_{\text{Rel}}^{1,k} &= \{ V_n^k : n \in \omega \} && k\text{-ary } \textbf{relation variables} \\ \text{Vb}_{\text{Fn}}^{1,k} &= \{ f_n^k : n \in \omega \} && k\text{-ary } \textbf{function variables}. \end{aligned}$$

The inductive definition of the set of L^2 -terms has the additional clause:

$$\text{if } t_0, \dots, t_{k-1} \text{ are terms, then } f_n^k t_0 \dots t_{k-1} \text{ is a term.}$$

The L^2 -atomic formulas include all expressions of the form $V_n^k t_0 \dots t_{k-1}$, for any terms $t_0, \dots, t_{k-1}$. The inductive definition of the set of L^2 -formulas has the additional clause: if ϕ is an L^2 -formula, then so are $\exists V_n^k \phi$ and $\exists f_n^k \phi$.

The standard interpretation of L^2 is an ordinary first-order L -structure $\mathfrak{A}$ together with an assignment (α, β) such that $\beta(V_n^k)$ is a k -ary relation on $|\mathfrak{A}|$ and $\beta(f_n^k)$ is a k -ary function on $|\mathfrak{A}|$. The definition of satisfaction has the obvious additional clauses; for example,

$$\begin{aligned} (f_n^k t_0, \dots, t_{k-1})^{\mathfrak{A}}[\alpha, \beta] &:= \beta(f_n^k) (t_0^{\mathfrak{A}}[\alpha, \beta], \dots, t_{k-1}^{\mathfrak{A}}[\alpha, \beta]), \\ \models_{\mathfrak{A}} V_n^k t_0 \dots t_{k-1}[\alpha, \beta] &: \Longleftrightarrow (t_0^{\mathfrak{A}}[\alpha, \beta], \dots, t_{k-1}^{\mathfrak{A}}[\alpha, \beta]) \in \beta(V_n^k), \end{aligned}$$

and

$$\models_{\mathfrak{A}} \exists f_n^k \phi[\alpha, \beta] : \Longleftrightarrow \text{for some } k\text{-ary function } F \text{ on } |\mathfrak{A}|, \quad \models_{\mathfrak{A}} \phi[\alpha, \beta_{f_n^k \mapsto F}].$$

Naturally everything expressible in L^{2m} is still expressible in L^2 , but also considerably more.

Example 3.6.35 Let θ_{Fin} denote any one of the following L^2 -sentences:

$$\begin{aligned} & \forall V_0^2 [\text{' } V_0^2 \text{ is a linear ordering' } \rightarrow \exists x \forall y \neg V_0^2 xy], \\ & \forall f_0^1 [\forall x \forall y (f_0^1 x \dot{=} f_0^1 y \rightarrow x \dot{=} y) \rightarrow \forall y \exists x f_0^1 x \dot{=} y], \\ & \exists V_0^2 \theta_{\text{FWO}}, \end{aligned}$$

where θ_{FWO} says that V_0^2 is a well-ordering with no limit points and a greatest element. Then for any structure $\mathfrak{A}$,

$$\models_{\mathfrak{A}} \theta_{\text{Fin}} \quad \Longleftrightarrow \quad A \text{ is finite.}$$

Example 3.6.36 Let $\theta_{\aleph_0}$ be the sentence $\exists V_0^2 \theta_{\text{CWO}}$, where θ_{CWO} is written with V_0^2 in place of $<$. Then

$$\begin{aligned} \models_{\mathfrak{A}} \theta_{\aleph_0} & \Longleftrightarrow A \text{ has a well-ordering with no limit points} \\ & \Longleftrightarrow A \text{ is countable.} \end{aligned}$$

Example 3.6.37 Let $\theta_{\aleph_1}$ be the sentence

$$\exists V_1^2 [\text{' } V_1^2 \text{ is a well-ordering of the universe' } \wedge \forall x \exists V_0^2 \theta_{\text{CWO}; V_1^2 yx, y}],$$

which expresses that the universe has a well-ordering such that each proper initial segment is countable. Then

$$\models_{\mathfrak{A}} \theta_{\aleph_1} \quad \Longleftrightarrow \quad \text{Card}(A) \leq \aleph_1.$$

There are many other similar examples, some of which we suggest as exercises.

The next step is **third-order logic**, which introduces variables for sets, relations and functions whose arguments are themselves sets, relations and functions over the universe. We shall not consider this separately but only as a (small) fragment of the logical extension of this line of development, the **theory of types**. The idea is that our language should have variables ranging over all types of objects which can be generated from a domain of individuals by the notions of *relation* and *function*. A convenient way to describe these object types is

Definition 3.6.38 The set of **type symbols**, or simply **types**, is the smallest set $\mathcal{T}$ such that

- (i) $0 \in \mathcal{T}$;
 - (ii) if $\sigma_0, \dots, \sigma_k \in \mathcal{T}$, then $(\sigma_0, \dots, \sigma_k) \in \mathcal{T}$;
 - (iii) if $\sigma_0, \dots, \sigma_k, \tau \in \mathcal{T}$, then $(\sigma_0, \dots, \sigma_k) \rightarrow \tau \in \mathcal{T}$.
- $(\sigma_0, \dots, \sigma_k)$ is called a **relation type** and $(\sigma_0, \dots, \sigma_k) \rightarrow \tau$ is a **function type**.

The standard domain of each type is given by

Definition 3.6.39 For any set A , to each type σ we assign A^σ , the σ -**domain over** A , recursively as follows:

- (i) $A^0 = A$;
- (ii) $A^{(\sigma_0, \dots, \sigma_k)} = \{ R : R \subseteq A^{\sigma_0} \times \dots \times A^{\sigma_k} \}$;
- (iii) $A^{(\sigma_0, \dots, \sigma_k) \rightarrow \tau} = \{ F : F \text{ is a function } A^{\sigma_0} \times \dots \times A^{\sigma_k} \rightarrow A^\tau \}$.

An element of A^σ is said to be of **type** σ over A .

Thus, for example, $A^{(0)} = \wp(A)$ and $A^{((0,0),0) \rightarrow (0)}$ is the set of functions from $\wp(A \times A) \times A$ to $\wp(A)$. A first-order structure includes generally relations of types $(0, 0, \dots, 0)$ and functions of types $(0, 0, \dots, 0) \rightarrow 0$. In the context of type theory, it is natural to consider structures and languages which admit (constant) functions and relations of arbitrary type σ . Thus languages are more complex sorts of objects. This actually simplifies the formal definitions in that relation, function and constant symbols and variables are all just special sorts of terms.

The corresponding **typed language** is specified by, for each type σ , a set Cs^σ of **constant symbols of type** σ . The other symbols consist of, for each σ , a set $\text{Vb}^\sigma = \{ v_n^\sigma : n \in \omega \}$ of **variables of type** σ and the usual logical symbols $\doteq, \neg, \vee$ and $\exists$. We leave it to the reader to formulate the definitions of term and formula, assignment and satisfaction appropriate to this language.

Of course, these languages are enormously expressive and indeed suffice to capture most everyday mathematics. This is because most everyday mathematical objects belong to the finite type structure over an appropriate underlying set. For example, consider the finite type structure over the real numbers $\mathbb{R}$. Infinite sequences of reals $\langle r_n : n \in \omega \rangle$ are functions $\omega \rightarrow \mathbb{R}$, which may be considered as functions $\mathbb{R} \rightarrow \mathbb{R}$ by setting $r_a = 0$ for $a \notin \omega$, so belong to $\mathbb{R}^{(0 \rightarrow 0)}$. The limit and summation operations thus belong to $\mathbb{R}^{(0 \rightarrow 0) \rightarrow 0}$, again with default values assigned for non-convergent sequences or non-summable series. The definite integral (Riemann or Lebesgue) over an interval is a member of $\mathbb{R}^{(0,0,(0 \rightarrow 0)) \rightarrow 0}$. It is thus clear that any theorem of real analysis can be expressed in a typed language interpreted over the universe $\mathbb{R}$. Since all of the usual functions and relations of analysis are

definable in this language from the field operations $+$ and $\times$, it suffices to have constant symbols $+$ and $\times$.

For all of these higher-order logics we have been using the **standard interpretation** in which the higher-order variables range over the largest possible domains as described in the preceding definition. There are alternative interpretations in which these domains are limited. For example, in interpreting an L^{2m} -sentence in a structure $\mathfrak{A}$, we might restrict the range of the second-order variables X to the *finite* subsets of A rather than all subsets. This variant is called **weak monadic second-order** logic and is as expected midway in expressive power between first-order logic and monadic second-order logic (see exercises below).

More generally with any structure $\mathfrak{A}$ and any set $\mathcal{S} \subseteq \wp(A)$ we associate the monadic second-order structure $(\mathfrak{A}, \mathcal{S})$ and interpret monadic second-order formulas in $(\mathfrak{A}, \mathcal{S})$ by treating second-order variables as ranging over $\mathcal{S}$:

$$\models_{(\mathfrak{A}, \mathcal{S})} \exists X \phi[\alpha, \beta] \iff \text{for some } B \in \mathcal{S}, \quad \models_{(\mathfrak{A}, \mathcal{S})} \phi[\alpha, \beta_{X \mapsto B}].$$

A logic $\mathcal{L}^F$ is specified by a function F which associates with each structure $\mathfrak{A}$ a family of sets $\mathcal{S} \subseteq \wp(A)$. Then the notion of logical consequence of $\mathcal{L}^F$ is defined by $\Gamma \models^F \phi$ iff for all $\mathfrak{A}$ and all $\mathcal{S} \in F(\mathfrak{A})$, if $(\mathfrak{A}, \mathcal{S})$ is a model of Γ , then also ϕ is true in $(\mathfrak{A}, \mathcal{S})$. In the standard and weak monadic second-order logics, $F(\mathfrak{A})$ is a singleton. For an example in which it is not, let Δ be a set of **comprehension axioms** — that is, sentences of the form $\forall \mathbf{x} \exists X \forall y [Xy \leftrightarrow \psi]$ — and take $F_\Delta(\mathfrak{A})$ to be the set of all $\mathcal{S}$ such that $(\mathfrak{A}, \mathcal{S})$ is a model of Δ . Such comprehension axioms guarantee that $\mathcal{S}$ is a ‘rich’ collection of sets and hence that quantification over $\mathcal{S}$ has the intended meaning. In this case, the consequence relation is in fact reducible to a first-order one in the following sense: for any monadic second-order language L^{2m} there is a first-order language L^* , a set of L^* -sentences Φ and a mapping $\phi \mapsto \phi^*$ of L^{2m} formulas to L^* -formulas such that for any L^{2m} -formula ϕ ,

$$\Gamma \models^{F_\Delta} \phi \iff \Gamma^* \cup \Delta^* \cup \Phi \models \phi^*.$$

Similar considerations apply to all of the higher-order languages; for the theory of types, this approach is essentially equivalent to a weak version of set theory.

Collateral Reading. A comprehensive survey of logics which go beyond first-order is the handbook [Ba-Fe]. Some texts include a short treatment of second-order logic, for example [En01] and [Eb-Fl-Th]. Second-order logic, and other extensions such as fixed-point logics, have played an important role in theoretical computer science, largely as interpreted over finite structures;

good references for these topics are [Eb-Fl], [Ott] and [Var]. Another area in which second-order notions play a big role is reverse mathematics, which attempts to measure the strength of mathematical assertions by analyzing carefully the axioms needed for their proof. The basic reference here is [Sim].

Exercises

Exercise 3.6.40 Prove Lemma 3.6.20.

Exercise 3.6.41 Prove Proposition 3.6.31.

Exercise 3.6.42 Prove Theorem 3.6.7.

Exercise 3.6.43 Show that the notion of finitary induction system defined here does include the earlier one. If $\mathcal{X}' = (X, X_0, \mathcal{H}')$ is a (finitary) induction system in the sense of Section 1.2, find a finitary induction system $\mathcal{X} = (X, X_0, \mathcal{H})$ as defined in this section such that $\overline{\mathcal{X}}(\emptyset) = \overline{\mathcal{X}'}$.

Let $\mathcal{L}^{w2m}$ be weak monadic second-order logic as defined above.

Exercise 3.6.44 Formulate and prove a Downward Löwenheim-Skolem Theorem for $\mathcal{L}^{w2m}$ and show that $\ell(\mathcal{L}^{w2m}) = \omega$.

Exercise 3.6.45 Let L be a language with the unary function symbol $\dot{S}$, the binary relation symbol $\dot{<}$ and the constant symbol $\dot{0}$. Show that there exists an L^{w2m} -sentence θ_ω such that for any L -structure $\mathfrak{A} = (A, <_A, \text{Sc}_A, 0_A)$,

$$\models_{\mathfrak{A}} \theta_\omega \iff \mathfrak{A} \simeq (\omega, <, \text{Sc}, 0).$$

Exercise 3.6.46 Show that $h(\mathcal{L}^{w2m}) > \beth_\omega^+$ by showing that $\beth_\omega^+$ is $\mathcal{L}^{w2m}$ -distinctive.

Hint: Use the result of the preceding exercise.

Exercise 3.6.47 Extending Exercise 2.4.55, call a function η a **2m-partial embedding** from $\mathfrak{A}$ into $\mathfrak{B}$ iff it has domain $X \cup \mathcal{X}$ with $X \subseteq |\mathfrak{A}|$ and $\mathcal{X} \subseteq \wp(|\mathfrak{A}|)$; for $a \in X$, $\eta(a) \in |\mathfrak{B}|$ and for $C \in \mathcal{X}$, $\eta(C) \subseteq |\mathfrak{B}|$; and for every quantifier-free L^{2m} -formula ϕ with free variables $x_0, \dots, x_{m-1}$ and $X_0, \dots, X_{n-1}$ and every $a_0, \dots, a_{m-1} \in X$ and $C_0, \dots, C_{n-1} \in \mathcal{X}$,

$$\models_{\mathfrak{A}} \phi[a_0, \dots, a_{m-1}, C_0, \dots, C_{n-1}] \iff \models_{\mathfrak{B}} \phi[\eta(a_0), \dots, \eta(a_{m-1}), \eta(C_0), \dots, \eta(C_{n-1})].$$

$\mathfrak{A}$ and $\mathfrak{B}$ are called **2m-partially isomorphic** — in symbols, $\mathfrak{A} \simeq_p^{2m} \mathfrak{B}$ — iff there exists a non-empty set $\mathcal{E}$ of 2m-partial embeddings such that in

addition to the forward and backward extension properties of Exercise 2.4.55 the following hold:

$$(\forall \eta \in \mathcal{E}) \forall C \subseteq |\mathfrak{A}| \quad (\exists \zeta \in \mathcal{E}) \quad [\zeta \text{ extends } \eta \text{ and } C \in \text{Dom}(\zeta)]$$

and

$$(\forall \eta \in \mathcal{E}) \forall D \subseteq |\mathfrak{B}| \quad (\exists \zeta \in \mathcal{E}) \quad [\zeta \text{ extends } \eta \text{ and } D \in \text{Im}(\zeta)].$$

Further, define $\mathfrak{A} \equiv^{2m} \mathfrak{B}$ iff they satisfy the same L^{2m} -sentences.

- (i) Show that $\mathfrak{A} \simeq_p^{2m} \mathfrak{B} \implies \mathfrak{A} \equiv^{2m} \mathfrak{B}$.
- (ii) Apply this result to show that for any structures $\mathfrak{A} = (A, U)$ and $\mathfrak{B} = (B, V)$ with $U \subseteq A$, $V \subseteq B$ and all of U , V , $A \sim U$ and $B \sim V$ infinite, $\mathfrak{A} \equiv^{2m} \mathfrak{B}$.

Exercise 3.6.48 Show along the following lines that the analogue of the Craig Interpolation Theorem 3.5.39 fails for monadic second-order logic. Let L be the language with just one unary relation symbol $\dot{U}$, L_0 the extension of L with additionally a binary relation symbol $\dot{E}_0$ and unary function symbols f_0 and g_0 , and L_1 similarly with $\dot{E}_1$, f_1 and g_1 . Construct an L_0^{2m} -sentence ϕ which guarantees that in any model (A, U) , both U and $A \sim U$ are infinite and U is of smaller cardinality than $A \sim U$, and construct an analogous L_1^{2m} -sentence ψ such that $\models \phi \rightarrow \neg \psi$. Use the result of the preceding exercise to show that there is no Craig interpolant.

3.7 Infinitary logic

A second extension of first-order logic is **infinitary logic** in which we admit conjunctions and disjunctions of infinite sets of formulas and/or simultaneous quantification of infinitely many variables. We have already used in [Chapter 2](#) expressions of the form $\bigvee \Gamma$ and $\bigwedge \Gamma$, for Γ a set of formulas, as a suggestive notation. In infinitary logic we treat such expressions as full-fledged formulas which may be further combined using other connectives and quantifiers.

Definition 3.7.1 For any language L and any infinite cardinal κ , the language $L_{\kappa\omega}$ is defined as follows:

- (i) the symbols of $L_{\kappa\omega}$ are those of L together with the **infinite disjunction symbol** $\bigvee$ and a set $\text{Vb}^\kappa = \{v_\sigma : \sigma < \kappa\}$ of κ -many variables;
- (ii) the terms and atomic formulas of $L_{\kappa\omega}$ are the same as those of L using the larger set of variables;
- (iii) the set of $L_{\kappa\omega}$ -**formulas** is the smallest set $\text{Fm}_{L_{\kappa\omega}}$ which includes the $L_{\kappa\omega}$ -atomic formulas and is such that if $\phi, \psi \in \text{Fm}_{L_{\kappa\omega}}$, then also

$\neg \phi, (\phi \vee \psi) \in \mathbf{Fm}_{L_{\kappa\omega}}$; for all $\sigma < \kappa$, $\exists v_\sigma \phi \in \mathbf{Fm}_{L_{\kappa\omega}}$; and for any set $\Gamma \subseteq \mathbf{Fm}_{L_{\kappa\omega}}$ with $\text{Card}(\Gamma) < \kappa$, $\bigvee \Gamma \in \mathbf{Fm}_{L_{\kappa\omega}}$;

- (iv) $L_{\kappa\omega}$ -structures and assignments are the same as L -structures and assignments using the larger set of variables;
- (v) $L_{\kappa\omega}$ -satisfaction is defined recursively as for L with the additional clause

$$\models_{\mathfrak{A}} \bigvee \Gamma[\alpha] \iff \text{for some } \phi \in \Gamma, \models_{\mathfrak{A}} \phi[\alpha].$$

Technically we should use a different symbol for satisfaction here, but the meaning is almost always clear from context and we forego this nicety in the interest of simplicity.

We introduce the **infinite conjunction symbol** $\bigwedge$ as an abbreviation: for any set Γ of fewer than κ -many $L_{\kappa\omega}$ -formulas,

$$\bigwedge \Gamma \text{ is an abbreviation for } \neg \bigvee \{ \neg \phi : \phi \in \Gamma \}.$$

It then follows immediately that

$$\models_{\mathfrak{A}} \bigwedge \Gamma[\alpha] \iff \text{for all } \phi \in \Gamma, \models_{\mathfrak{A}} \phi[\alpha].$$

We write sometimes $\bigvee_{i \in I} \phi_i$ in place of $\bigvee \{ \phi_i : i \in I \}$ and similarly for $\bigwedge$.

Although the language $L_{\omega\omega}$ is technically different from L , there is an obvious translation from each to the other. For $\kappa \leq \kappa'$, every $L_{\kappa\omega}$ -formula or sentence is also one of $L_{\kappa'\omega}$. The ω in $L_{\kappa\omega}$ is included to differentiate these languages from a broader class of languages $L_{\kappa\lambda}$ that we shall not consider here; formulas of $L_{\kappa\lambda}$ admit simultaneous quantification over any set of fewer than λ -many variables. We shall also occasionally refer to ‘languages’ $L_{\infty\omega}$ which admit disjunction of arbitrary sets Γ of formulas. Note that every $L_{\infty\omega}$ -formula is an $L_{\kappa\omega}$ -formula for sufficiently large κ and that the sets of $L_{\infty\omega}$ -variables, terms and formulas are proper classes rather than sets. To designate the group of languages $L_{\kappa\omega}$ as a whole, we refer to the **logic** $\mathcal{L}_{\kappa\omega}$.

Although we have stated the definition of $L_{\kappa\omega}$ -formula above in a form closely parallel to the previous definitions, we are forced here to a radical change in our formal notion of formula. If Γ is an infinite set, then $\bigvee \Gamma$ is inherently an infinite object, and it no longer makes sense to regard formulas as finite sequences of symbols. Rather, we must consider $L_{\kappa\omega}$ -formulas as abstract mathematical objects — as we do real numbers and other infinite constructs. This, in turn, demands some changes in a few basic concepts. In finitary languages, that a symbol *occurs* in a formula just means that it is a component of the finite sequence of symbols; here it should be given

by a recursive definition. For ϕ atomic, $\mathbf{NlogSymb}(\phi)$ is the set of non-logical symbols which occur in ϕ in the former sense, $\mathbf{NlogSymb}(\phi \vee \psi) = \mathbf{NlogSymb}(\phi) \cup \mathbf{NlogSymb}(\psi)$,

$$\mathbf{NlogSymb}\left(\bigvee \Gamma\right) = \bigcup \{ \mathbf{NlogSymb}(\phi) : \phi \in \Gamma \},$$

and similar clauses for $\neg$ and $\exists$. Analogous remarks apply to the notions of free and bound variables; we leave details to the reader. The set $\mathbf{SubFm}(\phi)$ of *subformulas* of ϕ is defined as for $\mathcal{L}^1$ with the additional clause

$$\mathbf{SubFm}\left(\bigvee \Gamma\right) = \left\{ \bigvee \Gamma \right\} \cup \bigcup \{ \mathbf{SubFm}(\phi) : \phi \in \Gamma \}.$$

The good news here is that a few features of first-order logic carry over to $\mathcal{L}_{\kappa\omega}$ with only minor modifications: the Downward Löwenheim-Skolem Theorem and a version of the Completeness Theorem for $L_{\omega_1\omega}$. However, before we look at these we shall give some examples of $L_{\kappa\omega}$ -sentences which witness the failure of other properties. For most we shall need only the smallest interesting value $\aleph_1$ of κ ; for historical reasons we call these languages $L_{\omega_1\omega}$ rather than $L_{\aleph_1\omega}$.

Example 3.7.2 Let θ_{fin} be the $L_{\omega_1\omega}$ -sentence $\bigvee_{n \in \omega} \exists^{\leq n}$, where $\exists^{\leq n}$ asserts that the universe has at most n elements. Then clearly $\models_{\mathfrak{A}} \theta_{\text{fin}}$ iff $\mathfrak{A}$ is finite. Hence, ω is $\mathcal{L}_{\omega_1\omega}$ -distinctive, so by Theorem 3.6.17, $\mathcal{L}_{\omega_1\omega}$ is not (ω, ω) -compact, $t(\mathcal{L}_{\omega_1\omega}) \geq \aleph_1$ and $h(\mathcal{L}_{\omega_1\omega}) \geq \aleph_1$.

Example 3.7.3 If L has constant symbols c_n ($n < \omega$), $\forall x \bigvee_{n < \omega} x \doteq c_n$ is an $L_{\omega_1\omega}$ -sentence which has countable models but no uncountable model. It witnesses, therefore, that $\aleph_1$ is $\mathcal{L}_{\omega_1\omega}$ -distinctive, so $t(\mathcal{L}_{\omega_1\omega}) > \aleph_1$ and $h(\mathcal{L}_{\omega_1\omega}) > \aleph_1$.

Example 3.7.4 Let L be any language which includes the unary function symbol $\dot{S}$ and the constant symbol $\dot{0}$. Then,

$$\models_{\mathfrak{A}} \forall x \bigvee \{ x \doteq \dot{n} : n \in \omega \} \iff |\mathfrak{A}| = \{ \dot{n}^{\mathfrak{A}} : n \in \omega \}.$$

Hence, just as in the case of monadic second-order logic (Example 3.6.3), if we take θ_{ω} to be the conjunction of this sentence with the $\dot{S}$ -axioms (S1) and (S2), and θ_{Ω} to be the conjunction of θ_{ω} with the axioms (L1), (L2), (A1), (A2), (M1) and (M2), we have, for any structures for the appropriate languages,

$$\models_{\mathfrak{A}} \theta_{\omega} \iff \mathfrak{A} \simeq (\omega, \text{Sc}, \dot{0})$$

and

$$\models_{\mathfrak{A}} \theta_{\Omega} \iff \mathfrak{A} \simeq \Omega.$$

That is, these sentences are categorical.

We shall see in [Section 7.1](#) below that these sentences are typical in the sense that for *any* countable structure $\mathfrak{A}$, there is an $\mathcal{L}_{\omega_1\omega}$ -sentence $\chi_{\mathfrak{A}}$ which characterizes $\mathfrak{A}$ up to isomorphism, at least among countable structures. Here are some more instances of this phenomenon.

Example 3.7.5 Let L contain the binary relation symbol $\dot{<}$. Let ψ_0 be the formula $\forall y \neg y \dot{<} x$, and for each ordinal $\sigma > 0$, by transfinite recursion, let ψ_σ be the conjunction of the sentence ‘ $\dot{<}$ is a linear ordering’ with the formula

$$\forall y [y \dot{<} x \rightarrow \exists x (x \dot{=} y \wedge \bigvee_{\tau < \sigma} \psi_\tau)] \wedge \bigwedge_{\tau < \sigma} \exists y [y \dot{<} x \wedge \exists x (x \dot{=} y \wedge \psi_\tau)].$$

The rather strange syntax is a device to avoid variable conflicts. For $\sigma < \kappa$, ψ_σ is an $L_{\kappa\omega}$ -formula and a straightforward induction verifies that

$$\models_{\mathfrak{A}} \psi_\sigma[a] \iff \left\{ b : b \dot{<}^{\mathfrak{A}} a \right\} \text{ is well-ordered in type } \sigma.$$

Hence if θ_σ denotes the sentence $\forall x \bigvee_{\tau < \sigma} \psi_\tau \wedge \bigwedge_{\tau < \sigma} \exists x \psi_\tau$,

$$\models_{\mathfrak{A}} \theta_\sigma \iff \mathfrak{A} \simeq (\sigma, <_\sigma).$$

In terms of Definitions 3.6.9 and 3.6.13,

Proposition 3.7.6 *For all infinite cardinals κ ,*

- (i) $\ell(\mathcal{L}_{\kappa^+\omega}) \geq \kappa$, $h(\mathcal{L}_{\kappa^+\omega}) > \kappa^+$ and $t(\mathcal{L}_{\kappa^+\omega}) > \kappa^+$; $\mathcal{L}_{\kappa^+\omega}$ is not (κ^+, κ^+) -compact;
- (ii) if κ is a limit cardinal, then $\ell(\mathcal{L}_{\kappa\omega}) \geq \kappa$, $h(\mathcal{L}_{\kappa\omega}) \geq \kappa$ and $t(\mathcal{L}_{\kappa\omega}) \geq \kappa$.

Proof. The $\mathcal{L}_{\kappa^+\omega}$ -sentence θ_κ from the preceding example has no model of cardinality $< \kappa$, so witnesses that $\ell(\mathcal{L}_{\kappa^+\omega}) \geq \kappa$. This same sentence witnesses that κ^+ is $\mathcal{L}_{\kappa^+\omega}$ -distinctive, so the second and third parts of (i) follow from Theorem 3.6.17.

For (ii), if κ is a limit cardinal, then we have from (i) that for all $\lambda < \kappa$, $\ell(\mathcal{L}_{\kappa\omega}) \geq \ell(\mathcal{L}_{\lambda+\omega}) \geq \lambda$; hence $\ell(\mathcal{L}_{\kappa\omega}) \geq \kappa$. The other parts follow similarly. $\square$

These bounds for the Hanf and Tarski numbers can be much improved by techniques similar to those used in the preceding section for $\mathcal{L}^{2m}$.

Example 3.7.7 Let L be any language which includes the non-logical symbols $\dot{S}$, $\dot{0}$ and $\dot{\in}$. Let ψ_0 be the formula $\bigvee_{n \in \omega} x \dot{=} n$ and for limit ordinals σ , let ψ_σ be $\bigvee_{\tau < \sigma} \psi_\tau$. Let $\psi_{\sigma+1}$ be the conjunction of the formulas

- (i) $\neg \psi_0$,
- (ii) $\forall x \forall y [\neg \psi_0 \wedge \neg (\psi_0)_x(y) \wedge \forall z (z \dot{\in} x \leftrightarrow z \dot{\in} y) \rightarrow x \dot{=} y]$,
- (iii) $\forall z [z \dot{\in} x \rightarrow \exists x (x \dot{=} z \wedge \psi_\sigma)]$.

For $\sigma < \kappa$, ψ_σ is an $L_{\kappa\omega}$ -formula, and as in the corresponding example for L^{2^m} , for any L -structure $\mathfrak{A}$, the mapping $a \mapsto \{b : b \dot{\in}^{\mathfrak{A}} a\}$ is an injection of $\psi_{\sigma+1}^{\mathfrak{A}}$ into $\wp(\psi_\sigma^{\mathfrak{A}})$. Since $\text{Card}(\psi_0^{\mathfrak{A}}) \leq \omega$, it follows by induction that for all σ , $\text{Card}(\psi_\sigma^{\mathfrak{A}}) \leq \beth_\sigma$. In particular, the sentence $\forall x \psi_\sigma$ has models of all cardinalities $\leq \beth_\sigma$ but no model of larger cardinality and thus witnesses that for $\sigma < \kappa$, $\beth_\sigma^+$ is $\mathcal{L}_{\kappa\omega}$ -distinctive. It follows that for all $\sigma < \kappa$, $\mathcal{L}_{\kappa\omega}$ is not $(\infty, \beth_\sigma^+)$ -compact, so for all $\lambda < \beth_\kappa$, $\mathcal{L}_{\kappa\omega}$ is not (∞, λ) -compact, $t(\mathcal{L}_{\kappa\omega}) \geq \beth_\kappa$ and $h(\mathcal{L}_{\kappa\omega}) \geq \beth_\kappa$.

Remark 3.7.8 It follows immediately from the definition that always $\kappa \leq \beth_\kappa$ and ‘usually’ κ is much smaller than $\beth_\kappa$ (cf. Lemma 3.6.20) — in particular, also $\kappa^+, \kappa^{++}, \dots < \beth_\kappa$ — so that this lower bound on the Hanf number of $\mathcal{L}_{\kappa\omega}$ is an improvement on the results above. Note that these examples do not provide large lower bounds for $\ell(\mathcal{L}_{\kappa\omega})$; the Downward Löwenheim-Skolem Theorem below shows why.

We give one further example which will be of use below.

Example 3.7.9 Fix an infinite cardinal κ , and let L contain one unary relation symbol p and a family $\langle c_\sigma : \sigma < \kappa \rangle$ of constant symbols. Fix $\lambda < \kappa$; for each subset $X \subseteq \lambda$, let ψ_X be the $L_{\kappa\omega}$ -sentence

$$\bigwedge_{\sigma \in X} p c_\sigma \wedge \bigwedge_{\sigma \notin X} \neg p c_\sigma,$$

so that for any L -structure $\mathfrak{A}$,

$$\models_{\mathfrak{A}} \psi_X \iff X = \{ \sigma < \lambda : c_\sigma^{\mathfrak{A}} \in p^{\mathfrak{A}} \}.$$

Thus $\{ \neg \psi_X : X \subseteq \lambda \}$ is a set of cardinality 2^λ which is inconsistent, while every proper subset is consistent. In particular, every set of smaller cardinality is consistent; thus $\mathcal{L}_{\kappa\omega}$ is not $(\infty, 2^\lambda)$ -compact and $t(\mathcal{L}_{\kappa\omega}) > 2^\lambda$.

The counterexamples above place severe limitations on possible positive results. With regard to compactness, we discuss briefly the following natural question: does there exist any uncountable cardinal κ such that $\mathcal{L}_{\kappa\omega}$ has compactness properties analogous to those of $\mathcal{L}_{\omega\omega}$?

Definition 3.7.10 For any infinite cardinal κ ,

- (i) κ is **weakly compact** iff $\mathcal{L}_{\kappa\omega}$ is (κ, κ) -compact;
- (ii) κ is **strongly compact** iff $\mathcal{L}_{\kappa\omega}$ is (∞, κ) -compact.

Of course, ω is strongly compact and every strongly compact cardinal is weakly compact. It is natural to ask if there any uncountable weakly or strongly compact cardinals. From the results above (see [Sections 6.4](#) and [6.10](#)) we have

Theorem 3.7.11

- (i) Any weakly compact cardinal is a regular limit cardinal — that is, **weakly inaccessible**.
- (ii) Any strongly compact cardinal is a regular strong limit cardinal — that is, **strongly inaccessible**.

Proof. Proposition 3.7.6 and its extension in the exercises show that if κ is either a successor cardinal or a singular limit cardinal, then $\mathcal{L}_{\kappa\omega}$ is not (κ, κ) -compact. If κ is strongly compact, then κ is certainly weakly inaccessible. Suppose that κ is not a strong limit — that is, for some $\lambda < \kappa$, $\kappa \leq 2^\lambda$. Then since by the final example above, $\mathcal{L}_{\kappa\omega}$ is not $(2^\lambda, 2^\lambda)$ -compact, it is also not (∞, κ) -compact. $\square$

We shall see in [Section 6.10](#) that it is not provable in the axiomatic set theory ZFC that there exist weakly inaccessible cardinals (unless ZFC is inconsistent!); hence, it is also not provable that weakly or strongly compact cardinals exist. Although it is not known to be impossible that it is provable in ZFC that no weakly or strongly compact cardinals exist, this is generally considered unlikely. It is also known that the existence of weakly compact cardinals is a strictly stronger set-theoretic hypothesis than the existence of strongly inaccessible cardinals.

With respect to upward Löwenheim-Skolem properties, much more is known about the the values of the various Hanf numbers $h(\mathcal{L}_{\kappa\omega})$, but it is beyond the scope of this book to give the proofs of these results. We state them here and refer the reader to [Dic; [Chapter 4](#)] for details.

Theorem 3.7.12 For any limit cardinal κ ,

- (i) $h(\mathcal{L}_{\omega_1\omega}) = \beth_{\omega_1}$; more generally, assuming the GCH, if $\text{cf}(\kappa) = \omega$, then $h(\mathcal{L}_{\kappa^+\omega}) = \beth_{\kappa^+}$;
- (ii) if $\text{cf}(\kappa) > \omega$, then $\beth_{\kappa^+} < h(\mathcal{L}_{\kappa^+\omega}) < \beth_{(2^\kappa)^+}$; in general no better bounds can be established on the basis of ZFC.

Finally, we consider downward Löwenheim-Skolem properties. Here, the results are much more similar to those for first-order logic. Note the restriction that formulas in Δ have only finitely many free variables.

Downward $\mathcal{L}_{\kappa+\omega}$ -Löwenheim-Skolem Theorem 3.7.13 *For any infinite cardinals $\lambda \geq \kappa$, any language L of cardinality $\leq \lambda$ and L -structure $\mathfrak{B}$, any set $X \subseteq |\mathfrak{B}|$ with $\text{Card}(X) \leq \lambda$ and any regular Boolean set Δ of $\mathcal{L}_{\kappa+\omega}$ -formulas with finitely many free variables with $\text{Card}(\Delta) \leq \lambda$, there exists an L -structure $\mathfrak{A}$ such that $X \subseteq |\mathfrak{A}|$, $\text{Card}(\mathfrak{A}) \leq \lambda$ and $\mathfrak{A} \prec^\Delta \mathfrak{B}$.*

Proof. Let $\kappa, \lambda, \mathfrak{B}, X$ and Δ be given as in the hypothesis. By a simple cardinality argument and the assumption that $\lambda \geq \kappa$, $\text{SubFm}(\Delta)$ has power $\leq \lambda$. We follow the pattern of earlier proofs by assigning to each $\psi \in \text{SubFm}(\Delta)$ and each variable x a function $F_{\psi,x} : B^k \rightarrow B$, where k is the number (finite by assumption) of free variables of ψ , such that

$$\models_{\mathfrak{B}} \exists x \psi[b_0, \dots, b_{k-1}] \implies \models_{\mathfrak{B}} \psi[b_0, \dots, b_{k-1}, F_{\psi,x}(b_0, \dots, b_{k-1})].$$

Then, if A is the closure of X under all of the $\leq \lambda$ -many functions $f^{\mathfrak{B}}$ for $f \in \text{Fs}_L$ and $F_{\psi,x}$ for $\psi \in \text{SubFm}(\Delta)$ and $x \in \text{Vb}$, then just as in the first-order case, A is the universe of a unique substructure $\mathfrak{A}$ of $\mathfrak{B}$ and $\text{Card}(A) \leq \lambda$. That $\mathfrak{A} \prec^\Delta \mathfrak{B}$ is easily proved by an induction on $\text{SubFm}(\Delta)$. $\square$

Corollary 3.7.14 *For any infinite cardinal κ , $\ell(\mathcal{L}_{\kappa+\omega}) = \kappa$. $\square$*

Finally we sketch a version of the Completeness and Interpolation Theorems for $\mathcal{L}_{\omega_1\omega}$.

The logical axioms and rules of inference for first-order logic are naturally seen as capturing in a concise way the meanings of the various logical symbols; for example, to explain the existential quantifier, we have the

$$\begin{aligned} \exists\text{-axioms : } & \phi_x(t) \rightarrow \exists x \phi \quad \text{if } t \text{ is freely substitutable for } x \text{ in } \phi; \\ \exists\text{-rule : } & \text{from } \phi \rightarrow \psi \text{ infer } \exists x \phi \rightarrow \psi \quad \text{if } x \text{ is not free in } \psi. \end{aligned}$$

The Completeness Theorem verifies that the meaning of $\exists$ is fully captured by these two schemas.

If we are to have any sort of completeness theorem for another logic $\mathcal{L}$ with other logical symbols, we must expect to include axioms and rules to explain these symbols. In particular, for $\mathcal{L}_{\kappa\omega}$ we need to account for the infinite disjunction symbol $\bigvee$. Given the analogy between $\exists$ and $\bigvee$ — both express that at least one of many possible statements holds — the following are natural choices: for any set Θ of $\mathcal{L}_{\kappa\omega}$ -formulas of cardinality less than κ ,

$$\begin{aligned} \bigvee\text{-axioms : } & \phi \rightarrow \bigvee \Theta \quad \text{for any } \phi \in \Theta; \\ \bigvee\text{-rule : } & \text{from } \{ \phi \rightarrow \psi : \phi \in \Theta \} \text{ infer } \bigvee \Theta \rightarrow \psi. \end{aligned}$$

The $\bigvee$ -rule is a collection of λ -ary rules of inference in the sense of Definition 3.6.28.

Definition 3.7.15 For any language L and any infinite cardinal κ , the (κ, ω) -deduction system is specified as follows. The logical axioms are the tautologies, equality axioms and $\exists$ -axioms as for L (but for arbitrary $L_{\kappa\omega}$ -formulas) together with the $\bigvee$ -axioms. The rules of inference are *modus ponens*, the $\exists$ -rule and the $\bigvee$ -rule. For any set Γ of $L_{\kappa\omega}$ -formulas, $\text{Thm}_L^{\kappa\omega}(\Gamma)$ is the smallest set of $L_{\kappa\omega}$ -formulas which includes Γ and the logical axioms and is closed under the rules of inference. We write

$$\Gamma \vdash_{\kappa\omega, L} \psi \iff \psi \in \text{Thm}_L^{\kappa\omega}(\Gamma).$$

It is easy to show that the Soundness Theorem holds for the (κ, ω) -deduction system described above, but there is no Completeness Theorem.

Theorem 3.7.16 *Unless κ is a strongly compact cardinal, the (κ, ω) -deduction system above is not complete for $\mathcal{L}_{\kappa\omega}$ — that is, there exists a set Γ of $L_{\kappa\omega}$ -sentences and an $L_{\kappa\omega}$ -sentence ψ such that $\Gamma \models \psi$ but it is not the case that $\Gamma \vdash_{\kappa\omega} \psi$.*

Proof. This is immediate from Theorem 3.6.33 and the results above. $\square$

This does not settle the question of validity completeness. It turns out that the system described is not even validity complete except in the one case $\kappa = \omega_1$. For proofs of these negative results and for more complex systems which *are* validity complete in some cases, the reader is referred to [Kar] and [Dic; Appendix C]. We sketch here the proof that the (ω_1, ω) -deduction system is validity complete for $L_{\omega_1\omega}$.

The proof can be arranged in various ways parallel to the different proofs of the Completeness Theorem for $\mathcal{L}^1$; we choose to formulate it in terms of notions of consistency. In any form, the crucial idea is to avoid the uncountability of the set of all $L_{\omega_1\omega}$ -formulas by restricting attention to appropriate countable subsets.

Definition 3.7.17 For any language L , an $L_{\omega_1\omega}$ -*notion of consistency* is a pair $(\mathbb{C}, E)$, with $\mathbb{C}$ a set of countable sets of $L_{\omega_1\omega}$ -sentences and $E \subseteq \text{Cs}_L$ a denumerable set of constant symbols such that if $\Theta \subseteq \Theta' \in \mathbb{C}$, then $\Theta \in \mathbb{C}$, and for any $\Theta \in \mathbb{C}$,

(i)–(iv) as in Definition 3.3.2;

(v) for any countable set Γ of $L_{\omega_1\omega}$ -sentences,

- (a) $\bigvee \Gamma \in \Theta \implies$ for some $\phi \in \Gamma$, $\Theta \cup \{\phi\} \in \mathbb{C}$;
- (b) $(\neg \bigvee \Gamma) \in \Theta \implies$ for all $\phi \in \Gamma$, $\Theta \cup \{\phi\} \in \mathbb{C}$.

Examples 3.7.18 The first two of Examples 3.3.3 for $\mathcal{L}^1$ carry over without change. The third example there involves the deduction system and is postponed. The fourth example $\mathbb{C}$ consisting of the finitely consistent sets is *not* an example of an $L_{\omega_1\omega}$ -notion of consistency, since

$$\Theta = \left\{ \bigvee_{n \in \omega} \exists^{\leq n} \right\} \cup \{ \exists^{\geq n} : n \in \omega \} \in \mathbb{C},$$

but for no n is $\Theta \cup \{ \exists^{\leq n} \} \in \mathbb{C}$ and thus condition (v)(a) fails.

The extension of each member of an L -notion of consistency to a weakly complete set in Proposition 3.3.5 depends crucially on the countability of the language. To carry out the analogous step here we need to isolate a countable set of formulas which can substitute for the uncountable set of all $L_{\omega_1\omega}$ -formulas.

Definition 3.7.19 For any language L and any set Φ of $L_{\omega_1\omega}$ -formulas, let L_Φ denote the sublanguage containing exactly the non-logical symbols which occur in some member of Φ . A Φ -**variable** is one which occurs free or bound in some member of Φ . A Φ -**term** (Φ -**atomic formula**) is an L_Φ -term (L_Φ -atomic formula) all of whose variables are Φ -variables. Φ is an $L_{\omega_1\omega}$ -**fragment** iff

- (i) Φ contains all Φ -atomic formulas;
- (ii) Φ is closed under $\neg$, $\vee$ and $\exists x$ for all Φ -variables x ;
- (iii) for all $\phi \in \Phi$, $\text{SubFm}(\phi) \subseteq \Phi$;
- (iv) for any $L_{\omega_1\omega}$ -formula ϕ , any variable x and any Φ -terms t and u , if $\phi_x(t) \in \Phi$, then $\phi_x(u) \in \Phi$.

Examples 3.7.20 For any $L' \subseteq L$, $L'_{\omega\omega}$ and $L'_{\omega_1\omega}$ are fragments. Similarly, for any $\mathcal{V} \subseteq \text{Vb}^\kappa$, the set of $L_{\omega_1\omega}$ -formulas which contain only variables in $\mathcal{V}$ is a fragment. The most important example is

Proposition 3.7.21 *For any set Θ of $L_{\omega_1\omega}$ -formulas, there is a smallest fragment $\text{Frag}(\Theta)$ which includes Θ . If Θ is countable, so is $\text{Frag}(\Theta)$.*

Proof. $\text{Frag}(\Theta)$ is the set inductively generated from the set of Θ -atomic formulas by the countably many finitary operations corresponding to (ii)–(iv) of the definition, using in (ii) all Θ -variables. $\square$

We shall write $\text{Frag}(\Theta, E)$ for $\text{Frag}(\Theta \cup \{ e \doteq e : e \in E \})$, the smallest fragment which contains Θ and all $e \in E$.

Definition 3.7.22 For any language L , any $L_{\omega_1\omega}$ -fragment Φ , any set $\Delta \subseteq \Phi$ of $L_{\omega_1\omega}$ -sentences and any denumerable set E of Φ -constant symbols, Δ is **weakly complete** with respect to (Φ, E) iff

- (i) for any E -primitive Φ -term t and any formula $\phi \in \Phi$ with at most x free,
 - (a) for any $e, e' \in E$, $(e \doteq e') \in \Delta \implies (e' \doteq e) \in \Delta$;
 - (b) for any $e \in E$, $(t \doteq e), \phi_x(t) \in \Delta \implies \phi_x(e) \in \Delta$;
 - (c) for some $e \in E$, $(t \doteq e) \in \Delta$;
- (ii) for any sentence $\psi \in \Phi$,
 - (a) $(\neg \psi) \in \Delta \implies \psi \notin \Delta$;
 - (b) $(\neg \neg \psi) \in \Delta \implies \psi \in \Delta$;
- (iii) for any sentences $\psi, \theta \in \Phi$,
 - (a) $(\psi \vee \theta) \in \Delta \implies$ either $\psi \in \Delta$ or $\theta \in \Delta$ (or both);
 - (b) $\neg(\psi \vee \theta) \in \Delta \implies$ both $(\neg \psi) \in \Delta$ and $(\neg \theta) \in \Delta$;
- (iv) for any formula $\psi \in \Phi$ with at most x free,
 - (a) $\exists x \psi \in \Delta \implies$ for some $e \in E$, $\psi_x(e) \in \Delta$;
 - (b) $(\neg \exists x \psi) \in \Delta \implies$ for all $e \in E$, $(\neg \psi_x(e)) \in \Delta$;
- (v) for any countable set $\Gamma \subseteq \Phi$,
 - (a) $\bigvee \Gamma \in \Delta \implies$ for some $\phi \in \Gamma$, $\phi \in \Delta$;
 - (b) $(\neg \bigvee \Gamma) \in \Delta \implies$ for all $\phi \in \Gamma$, $\phi \in \Delta$.

We can now establish two propositions entirely parallel to those of [Section 3.3](#).

Proposition 3.7.23 *For any language L , any $L_{\omega_1\omega}$ -notion of consistency $(\mathbb{C}, E)$ and any $\Theta \in \mathbb{C}$, there exists a countable set $\Delta \subseteq \text{Frag}(\Theta, E)$ of $L_{\omega_1\omega}$ -sentences such that $\Theta \subseteq \Delta$ and Δ is weakly complete with respect to $(\text{Frag}(\Theta, E), E)$.*

Proof. Let $(\mathbb{C}, E)$ be an $L_{\omega_1\omega}$ -notion of consistency and Θ a $\mathbb{C}$ -consistent set. Since Θ and E are countable, so is $\text{Frag}(\Theta, E)$, so we may fix enumerations $\langle \theta_n : n \in \omega \rangle$ of the set of $\text{Frag}(\Theta, E)$ -sentences, $\langle t_n : n \in \omega \rangle$ of the set of $\text{Frag}(\Theta, E)$ -terms and $\langle e_n : n \in \omega \rangle$ of E . Reference below to the *first* $e \in E$ or $\phi \in \Phi$ means the first in these enumerations. Set $\Delta_0 = \Theta$ and for all n ,

$$\Delta_n^0 = \begin{cases} \Delta_n \cup \{\theta_n\}, & \text{if this set is } \mathbb{C}\text{-consistent;} \\ \Delta_n, & \text{otherwise;} \end{cases}$$

$$\begin{aligned} \Delta_n^1 &= \Delta_n^0 \cup \{t_n \doteq e\}, \quad \text{for the first } e \in E \text{ for which this is } \mathbb{C}\text{-consistent;} \\ \Delta_{n+1}^2 &= \begin{cases} \Delta_n^1 \cup \{\chi_x(e')\}, & \text{if } \theta_n \in \Delta_n^1 \text{ and } \theta_n \text{ is } \exists x \chi \text{ for the first } e' \in E \\ & \text{such that this is } \mathbb{C}\text{-consistent;} \\ \Delta_n^1, & \text{otherwise;} \end{cases} \\ \Delta_{n+1} &= \begin{cases} \Delta_n^2 \cup \{\phi\}, & \text{if } \theta_n \in \Delta_n^2 \text{ is } \bigvee \Gamma \text{ and } \phi \text{ is the first element of } \Gamma \\ & \text{such that this is } \mathbb{C}\text{-consistent;} \\ \Delta_n^2, & \text{otherwise.} \end{cases} \end{aligned}$$

As before we prove by a straightforward induction on n that each of the sets Δ_n^i is $\mathbb{C}$ -consistent and thus such e , e' and ϕ exist. We then set

$$\Delta := \bigcup \{ \Delta_n : n \in \omega \}$$

and claim that Δ is weakly complete with respect to $(\text{Frag}(\Theta, E), E)$. That Δ satisfies conditions (i)–(iv) goes just as in Proposition 3.3.5; by the definition of fragment, the relevant sentences all belong to $\text{Frag}(\Theta, E)$ and thus appear in the enumeration. Condition (v)(a) is satisfied explicitly by the third substage of the construction. For (v)(b), suppose that $(\neg \bigvee \Gamma) \in \Delta$, say $\neg \bigvee \Gamma$ is θ_n . Any $\phi \in \Gamma$ is a subformula of $\neg \bigvee \Gamma$ and hence is θ_p for some p . If $n \leq p$, then $(\neg \bigvee \Gamma) \in \Delta_p$ so by clause (v)(b) of the definition of notion of consistency, $\Delta_p \cup \{\phi\}$ is $\mathbb{C}$ -consistent and thus $\phi \in \Delta_{p+1} \subseteq \Delta$. If $p < n$, then $\Delta_p \cup \{\phi\} \subseteq \Delta_n \cup \{\neg \bigvee \Gamma\} \cup \{\phi\}$ and is thus $\mathbb{C}$ -consistent, so again $\phi \in \Delta_{p+1}$. $\square$

Proposition 3.7.24 *For any language L , any $L_{\omega_1\omega}$ -fragment Φ , any countable set E of Φ -constant symbols and any countable set $\Delta \subseteq \Phi$ of $L_{\omega_1\omega}$ -sentences, if Δ is weakly complete with respect to (Φ, E) , then Δ has a countable model $\mathfrak{A}$ with $|\mathfrak{A}| = \{e^{\mathfrak{A}} : e \in E\}$.*

Proof. We may assume that $L = L_\Phi$ and is thus a countable language. We define a L_Φ -structure $\mathfrak{A}$ exactly as in the proof of Proposition 3.3.7 and show that $\mathfrak{A}$ is a model of Δ by proving by induction on sentences $\phi \in \Phi$ that

$$\phi \in \Delta \implies \models_{\mathfrak{A}} \phi \quad \text{and} \quad (\neg \phi) \in \Delta \implies \models_{\mathfrak{A}} \neg \phi.$$

The basis for Φ -atomic formulas and the induction steps for $\neg$, $\bigvee$ and $\exists$ go just as before; the properties of fragments ensure that the requisite formulas all belong to Φ . The new step for $\bigvee$ is also straightforward: if ϕ is $\bigvee \Gamma$, then

$$\begin{aligned} \phi \in \Delta &\implies \text{for some } \psi \in \Gamma, \quad \psi \in \Delta \\ &\implies \text{for some } \psi \in \Gamma, \quad \models_{\mathfrak{A}} \psi \implies \models_{\mathfrak{A}} \phi, \end{aligned}$$

and

$$\begin{aligned}
 (\neg \phi) \in \Delta &\implies \text{for all } \psi \in \Gamma, \quad (\neg \psi) \in \Delta \\
 &\implies \text{for all } \psi \in \Gamma, \quad \frac{}{\vdash_{\mathfrak{A}} \neg \psi} \implies \frac{}{\vdash_{\mathfrak{A}} \neg \phi}. \quad \square
 \end{aligned}$$

Combining these two propositions, we have the

$\mathcal{L}_{\omega_1\omega}$ -Model Existence Theorem 3.7.25 *For any language L and any $L_{\omega_1\omega}$ -notion of consistency $(\mathbb{C}, E)$, every $\mathbb{C}$ -consistent set has a countable model with universe $|\mathfrak{A}| = \{e^{\mathfrak{A}} : e \in E\}$. $\square$*

We proceed now to the $\mathcal{L}_{\omega_1\omega}$ -Completeness Theorem by showing that the set of countable $\frac{\omega_1\omega, L}{\vdash}$ -consistent sets is an $L_{\omega_1\omega}$ -notion of consistency. We need first to establish a few properties of this deduction system. We follow closely the development of [Section 3.2](#) and indicate only the differences.

The Soundness Theorem — for any set $\Gamma \cup \{\psi\}$ of $L_{\omega_1\omega}$ -sentences,

$$\Gamma \frac{\omega_1\omega, L}{\vdash} \psi \implies \Gamma \models \psi$$

— is proved just as before by $\text{Thm}_L^{\omega_1\omega}$ -induction; the new case for the $\bigvee$ -rule just requires the observation the this rule is sound — that is,

$$\text{if for all } \phi \in \Theta \quad \Gamma \models \phi \rightarrow \psi, \quad \text{then} \quad \Gamma \models \bigvee \Theta \rightarrow \psi.$$

All but part (iv) of Proposition 3.2.9 extend immediately as stated, and in place of (iv) we have if $\Gamma \frac{\omega_1\omega, L}{\vdash} \psi$, then

$$\text{for some countable } \Gamma_0 \subseteq \Gamma \text{ and } L^- \subseteq L, \quad \Gamma_0 \frac{\omega_1\omega, L^-}{\vdash} \psi.$$

If L^+ is an extension by constants, we can prove much as before that for any set $\Gamma \cup \{\psi\}$ of $L_{\omega_1\omega}$ -formulas,

$$\Gamma \frac{\omega_1\omega, L^+}{\vdash} \psi \implies \Gamma \frac{\omega_1\omega, L}{\vdash} \psi,$$

by replacing new constant symbols by variables. Since $L_{\omega_1\omega}^+$ -formulas may contain denumerably many variables and constant symbols, it will not suffice to replace constants one at a time. Note that it is important here that each language of $\mathcal{L}_{\omega_1\omega}$ has uncountably many variables. We leave details to the reader. From now on we omit the parameter L in most contexts where all languages under consideration differ at most in their constant symbols.

The Tautology Lemma and the next several corollaries and propositions of [Section 3.2](#) follow essentially unchanged. Note that by (iv) above we have

$$\Gamma \text{ is } \frac{\omega_1\omega}{\text{---}} \text{-inconsistent} \iff \text{for some countable } \Gamma_0 \subseteq \Gamma, \quad \Gamma_0 \text{ is } \frac{\omega_1\omega}{\text{---}} \text{-inconsistent.}$$

The proof of the Deduction Theorem here involves checking one additional case corresponding to the $\bigvee$ -rule:

$$\text{if for all } \phi \in \Theta, \quad \Gamma \frac{\omega_1\omega}{\text{---}} \theta \rightarrow (\phi \rightarrow \psi), \quad \text{then} \quad \Gamma \frac{\omega_1\omega}{\text{---}} \theta \rightarrow \left(\bigvee \Gamma \rightarrow \psi \right);$$

this is handled in the same way as the $\exists$ -rule. Finally, we have as before

Corollary 3.7.26 *For any language L , any set Γ of $L_{\omega_1\omega}$ -formulas and any $L_{\omega_1\omega}$ -sentence θ ,*

- (i) $\Gamma \frac{\omega_1\omega}{\text{---}} \theta \iff \Gamma \cup \{\neg \theta\} \text{ is } \frac{\omega_1\omega}{\text{---}} \text{-inconsistent};$
- (ii) $\Gamma \frac{\omega_1\omega}{\text{---}} \neg \theta \iff \Gamma \cup \{\theta\} \text{ is } \frac{\omega_1\omega}{\text{---}} \text{-inconsistent.} \quad \square$

We can now establish

Theorem 3.7.27 *For any language L and any denumerable set $E \subseteq \text{Cs}_L$,*

$$\mathbb{C} = \left\{ \Theta : \begin{array}{l} \Theta \text{ is a countable } \frac{\omega_1\omega}{\text{---}} \text{-consistent set of } L_{\omega_1\omega}\text{-sentences} \\ \text{such that infinitely many } e \in E \text{ occur in no member of } \Theta \end{array} \right\}$$

is an $L_{\omega_1\omega}$ -notion of consistency.

Proof. Let $\mathbb{C}$ be as described. That $\mathbb{C}$ has properties (i)–(iv) of Definition 3.7.17 follows exactly as in the first-order case. For (v)(a), suppose that $\bigvee \Gamma \in \Theta \in \mathbb{C}$ but for every $\phi \in \Gamma$, $\Theta \cup \{\phi\} \notin \mathbb{C}$ — that is, $\Theta \cup \phi$ is $\frac{\omega_1\omega}{\text{---}}$ -inconsistent, so by the corollary, $\Theta \frac{\omega_1\omega}{\text{---}} \neg \phi$. Then for each $\phi \in \Gamma$, we have $\Theta \frac{\omega_1\omega}{\text{---}} \phi \rightarrow \text{F}$, hence by the $\bigvee$ -rule, $\Theta \frac{\omega_1\omega}{\text{---}} \bigvee \Gamma \rightarrow \text{F}$, that is $\Theta \frac{\omega_1\omega}{\text{---}} \neg \bigvee \Gamma$, so Θ would be $\frac{\omega_1\omega}{\text{---}}$ -inconsistent, contrary to hypothesis.

For (v)(b), suppose $(\neg \bigvee \Gamma) \in \Theta \in \mathbb{C}$, but for some $\phi \in \Gamma$, $(\Theta \cup \{\neg \phi\}) \notin \mathbb{C}$. Then $\Theta \frac{\omega_1\omega}{\text{---}} \phi$, so by the $\bigvee$ -axiom (and *modus ponens*) $\Theta \frac{\omega_1\omega}{\text{---}} \bigvee \Gamma$, and again Θ is $\frac{\omega_1\omega}{\text{---}}$ -inconsistent. $\square$

Corollary 3.7.28 *For any language L and any countable set Γ of $L_{\omega_1\omega}$ -sentences, if Γ is $\vdash^{\omega_1\omega}$ -consistent, then Γ has a model.*

Proof. Let Γ be as described, E be a denumerable set of new constant symbols and $\mathbb{C}$ be as described in the theorem for the language L^+ which includes E . Γ is still $\vdash^{\omega_1\omega, L^+}$ -consistent and contains no $e \in E$, so $\Gamma \in \mathbb{C}$. By the $\mathcal{L}_{\omega_1\omega}$ -Model Existence Theorem, Γ has a model. $\square$

$\mathcal{L}_{\omega_1\omega}$ -Completeness Theorem 3.7.29 *For any countable set $\Gamma \cup \{\psi\}$ of $\mathcal{L}_{\omega_1\omega}$ -sentences,*

$$\Gamma \vdash^{\omega_1\omega} \psi \iff \Gamma \models \psi.$$

Proof. Suppose that not $\Gamma \vdash^{\omega_1\omega} \psi$. Then $\Gamma \cup \{\psi\}$ is $\vdash^{\omega_1\omega}$ -consistent, hence has a model, and therefore not $\Gamma \models \psi$. $\square$

As we have observed above, the restriction to countable Γ is crucial here, since completeness for arbitrary Γ would imply that $\mathcal{L}_{\omega_1\omega}$ is (∞, ω_1) -compact, contrary to the results above. For $\Gamma = \emptyset$, we have validity completeness, which is in fact equivalent to the full theorem, since

$$\Gamma \vdash^{\omega_1\omega} \psi \iff \vdash^{\omega_1\omega} \bigwedge \Gamma \rightarrow \psi \iff \models \bigwedge \Gamma \rightarrow \psi \iff \Gamma \models \psi.$$

We leave the first equivalence as an easy exercise.

Using the same techniques, we have also the

$\mathcal{L}_{\omega_1\omega}$ -Robinson Interpolation Theorem 3.7.30 *For any languages L_0 and L_1 with common part L and any consistent countable sets Γ_0 of $L_{0,\omega_1\omega}$ -sentences and Γ_1 of $L_{1,\omega_1\omega}$ -sentences, if $\Gamma_0 \cup \Gamma_1$ is inconsistent, then there exists an $L_{\omega_1\omega}$ -sentence χ such that $\Gamma_0 \models \chi$ and $\Gamma_1 \models \neg \chi$. χ is called a **Robinson interpolant** for Γ_0 and Γ_1 .*

Proof. We proceed essentially as in the first-order case 3.3.12 by adjoining a denumerable set E of new constant symbols to each of the languages to form $L_0(E)$ and $L_1(E)$ and setting $\mathbb{C}$ to be the set of all countable sets of the form $\Theta_0 \cup \Theta_1$, with Θ_i a consistent set of $L_{i,\omega_1\omega}(E)$ sentences such that infinitely many $e \in E$ occur in no member of $\Theta_0 \cup \Theta_1$ and there is no Robinson Interpolant χ for (Θ_0, Θ_1) which contains at most finitely many $e \in E$. We show that $\mathbb{C}$ is an $(L_0 \cup L_1)_{\omega_1\omega}(E)$ -notion of consistency. Properties (i)–(iv) go just as before. For (v)(a), suppose that $\bigvee \Gamma \in \Theta_0$, but for each $\phi \in \Gamma$, $\Theta_0 \cup \{\phi\} \cup \Theta_1 \notin \mathbb{C}$ — that is, there are $L_{\omega_1\omega}(E)$ -sentences χ_ϕ such that

$$\Theta_0 \cup \{\phi\} \models \chi_\phi \quad \text{and} \quad \Theta_1 \models \neg \chi_\phi.$$

Then it is easy to compute that

$$\Theta_0 \models \bigvee_{\phi \in \Gamma} \chi_\phi \quad \text{and} \quad \Theta_1 \models \neg \bigvee_{\phi \in \Gamma} \chi_\phi,$$

so that (Θ_0, Θ_1) has an interpolant, contrary to assumption.

For (v)(b), suppose that $(\neg\Gamma) \in \Theta_0$ but for some $\phi \in \Gamma$, $\Theta_0 \cup \{\neg\phi\} \cup \Theta_1 \notin \mathbb{C}$ — that is, there is an $L_{\omega_1\omega}(E)$ -sentence χ such that

$$\Theta_0 \cup \{\neg\phi\} \models \chi \quad \text{and} \quad \Theta_1 \models \neg\chi.$$

Since $\models \neg \bigvee \Gamma \rightarrow \neg\phi$, also $\Theta_0 \models \chi$, so χ is an interpolant for (Θ_0, Θ_1) .

Now, if $\Gamma_0 \cup \Gamma_1$ is inconsistent, by the $\mathcal{L}_{\omega_1\omega}$ -Model Existence Theorem, $\Gamma_0 \cup \Gamma_1 \notin \mathbb{C}$, so there is an $L_{\omega_1\omega}(E)$ -sentence χ , which contains only finitely many $e \in E$, say $e_0, \dots, e_{k-1}$, such that

$$\Gamma_0 \models \chi \quad \text{and} \quad \Gamma_1 \models \neg\chi.$$

Then as before,

$$\exists x_0 \dots \exists x_{k-1} \chi_{e_0, \dots, e_{k-1}}(x_0, \dots, x_{k-1})$$

is an $L_{\omega_1\omega}$ -interpolant for (Γ_0, Γ_1) . $\square$

$\mathcal{L}_{\omega_1\omega}$ -Interpolation Theorem 3.7.31 *For any language L , any countable set Γ of $L_{\omega_1\omega}$ -sentences, any two extensions L_0 and L_1 of L whose common symbols are all symbols of L and any $L_{0,\omega_1\omega}$ -sentence θ_0 and $L_{1,\omega_1\omega}$ -sentence θ_1 , if $\Gamma \models \theta_0 \rightarrow \theta_1$, then there exists an $L_{\omega_1\omega}$ -formula χ such that $\Gamma \models \theta_0 \rightarrow \chi$ and $\Gamma \models \chi \rightarrow \theta_1$.*

Proof. This follows from the $\mathcal{L}_{\omega_1\omega}$ -Robinson Interpolation Theorem as the Craig Interpolation Theorem 3.3.13 did from the (first-order) Robinson Interpolation Theorem. $\square$

The following example shows that the restriction to countable sets in the Robinson Interpolation Theorem is necessary. Let (as usual) $L_=_$ denote the language with no non-logical symbols. Recall that a $L_=_$ -structure is simply a set (A) and that two $L_=_$ -structures (A) and (B) are isomorphic iff they have the same cardinality.

Lemma 3.7.32 *For any two infinite sets A and B , (A) and (B) satisfy the same $L_{=,\infty\omega}$ -sentences.*

Proof. It suffices to prove that if $A \subseteq B$ are infinite, then for any (A) -assignment α and any $L_{=,\infty\omega}$ -formula ϕ ,

$$\models_{(A)} \phi[\alpha] \iff \models_{(B)} \phi[\alpha].$$

We proceed by formula induction; the atomic, $\neg$, $\vee$ and $\bigvee$ steps are immediate. Suppose that $\models_{(B)} \exists x \psi[\alpha]$, so for some $b \in B$, $\models_{(B)} \psi[\alpha_{x \mapsto b}]$. Then by the induction hypothesis, also $\models_{(A \cup \{b\})} \psi[\alpha_{x \mapsto b}]$. But $(A \cup \{b\}) \simeq (A)$, and if a is the image of b under the isomorphism, then $\models_{(A)} \psi[\alpha_{x \mapsto a}]$, so $\models_{(A)} \exists x \psi$. $\square$

Now let L_0 be the language which includes no relation or function symbols and constant symbols $\{c_n : n \in \omega\}$, while L_1 has constant symbols $\{c_\sigma : \omega \leq \sigma < \omega_1\}$. Clearly $L_0 \cap L_1 = L_{=}$. Let

$$\Gamma_0 = \left\{ \forall x \bigvee_{n \in \omega} x \doteq c_n \right\};$$

$$\Gamma_1 = \{ \neg c_\sigma \doteq c_\tau : \omega \leq \sigma < \tau < \omega_1 \}.$$

Any model of Γ_0 is countable while any model of Γ_1 is uncountable, so $\Gamma_0 \cup \Gamma_1$ is inconsistent. But (Γ_0, Γ_1) has no $L_{=,\omega_1\omega}$ -interpolant, since if $\Gamma_0 \models \chi$, then χ is true in some, hence all infinite sets, so also $\Gamma_1 \models \chi$.

Note that this same example shows that for $\kappa > \omega_1$ even the interpolation property fails for $\mathcal{L}_{\kappa\omega}$, since then $\models \bigwedge \Gamma_0 \rightarrow \neg \bigwedge \Gamma_1$, but there is no interpolant.

Collateral Reading. Infinitary logic will be a basic tool in parts of [Chapter 7](#), and many texts include at least a mention of $L_{\kappa\omega}$ and a few examples of its use, for example, [Eb-Fl-Th] and [Mon]. The classic references are [Kei] (particularly for $L_{\omega_1\omega}$) and [Dic] for large languages including the languages $L_{\kappa\lambda}$ with infinitary quantification. Infinitary languages in the context of admissible sets (small set-theoretic universes) are treated in [Ba75] and [Ba77; Chapter A.7].

Exercises

Exercise 3.7.33 Show that for every infinite cardinal κ ,

- (i) every $\lambda < \kappa$ is $\mathcal{L}_{\kappa\omega}$ -distinctive;
- (ii) if κ is a singular limit cardinal, then also both κ and κ^+ are $\mathcal{L}_{\kappa\omega}$ -distinctive.

Exercise 3.7.34 Compute, for a countable language L and any infinite cardinal κ , the number of $L_{\kappa\omega}$ -formulas.

Chapter 4

Incompleteness and Undecidability

In [Sections 2.4–2.5](#) we worked out solutions to several instances of the problem: given a structure $\mathfrak{A}$, find a set Γ of sentences in the language of $\mathfrak{A}$ which axiomatize the theory $\text{Th}(\mathfrak{A})$ of $\mathfrak{A}$ — that is, such that $\text{Th}(\mathfrak{A}) = \text{Th}(\Gamma)$. This has been considered a natural question throughout the history of mathematics because it provides a deductive structure for a theory of mathematical interest. Of course, before the development of the techniques of logic, the problem could not be stated in this sharp form, but we may see even Euclid’s axiomatization of geometry as a first example.

We discussed at the end of the Introduction some of the historical context in the first two decades of the 20th century which heightened interest in axiomatizing mathematics. In particular, the Hilbert Program had as its goal to carry out the following steps for increasingly large areas of mathematics:

- (A) make precise the fundamental notions of the area and set up a formal language which provides the means to express these notions;
- (B) codify the rules of logic;
- (C) find a reasonable set of axioms which express the basic truths about the fundamental notions;
- (D) show that the set of theorems provable from these axioms using these rules either (a) coincides with the set of true statements about the

- notions (assuming that there is a secure definition of truth) or (b) could be taken as the *definition* of this set of true statements (thus avoiding the potentially problematic notion of truth altogether);
- (E) prove in a completely secure way that no contradiction could be derived from the axioms.

It was exactly this program that provided much of the impetus for the development of the subject matter of this book, and at this stage we have dealt thoroughly with a general framework for steps (A) and (B). Hilbert saw arithmetic (number theory) as the natural starting point; not only are integers the mathematical objects most directly accessible to human intuition but essentially every part of mathematics depends to some extent on integer arithmetic. Hence he sought to provide an axiomatic framework (C) for arithmetic that would satisfy both (D) and (E). Once this was done, he planned to extend the framework to larger portions of mathematics. The results of Gödel, amplified by Tarski, Rosser and others, showed that no such system exists. We give a proof of these results in [Section 4.1](#) based on the intuitive notions of decidability and effective calculability and therefore on two unproved “hypotheses”. In the remainder of this chapter and part of the next we develop the precise mathematical notion of recursiveness for sets, relations and functions and adapt the methods of the first section to give complete proofs of the results.

4.1 A first look

We have already described in [Section 2.5](#) an appropriate language L_Ω for arithmetic with the nonlogical symbols $\dot{0}$, $\dot{S}$, $\dot{+}$, $\dot{\times}$ and $\dot{<}$. The notion of truth for statements of L_Ω is characterized by truth in the standard model $\Omega = (\omega, <, +, \times, \text{Sc}, 0)$, so version (a) of the task that Hilbert sets is exactly that of finding a “reasonable” set of axioms for $\text{Th}(\Omega)$.

In [Section 2.5](#) we were able to analyze in this way the structure $\Omega_+ = (\omega, <, +, \text{Sc}, 0)$, which differs from Ω ‘merely’ in leaving out multiplication, and we mentioned that a similar analysis could be done for the structure $\Omega_\times = (\omega, <, \times, \text{Sc}, 0)$ which leaves out addition. We found a natural axiom system for the theory of Ω_+ consisting of the sentences

- (S1) $\forall x (\neg \dot{0} \dot{=} \dot{S}x)$
 (S2) $\forall x \forall y (\dot{S}x \dot{=} \dot{S}y \rightarrow x \dot{=} y)$
 (L1) $\forall x (\neg x \dot{<} \dot{0})$
 (L2) $\forall x \forall y [x \dot{<} \dot{S}y \leftrightarrow (x \dot{<} y \vee x \dot{=} y)]$
 (A1) $\forall x (x \dot{+} \dot{0} \dot{=} x)$
 (A2) $\forall x \forall y [x \dot{+} \dot{S}y \dot{=} \dot{S}(x \dot{+} y)]$

together with the L_{Ω_+} -induction schema.

We also suggested in [Section 2.5](#) the attractive conjecture that by adjoining to this axiom system the recursion equations for multiplication

$$(M1) \quad \forall x (x \dot{\times} \dot{0} \doteq \dot{0})$$

$$(M2) \quad \forall x \forall y [(x \dot{\times} \dot{S}y) \doteq (x \dot{\times} y) \dot{+} x]$$

and extending the induction schema to all L_Ω -formulas, we might get similarly an axiomatization for the structure Ω . This turns out to be far from the case, but these axioms will still play a significant role, so for future reference we define

Definition 4.1.1

- (i) **Q** (*Robinson Arithmetic*) is the theory generated by the axioms (S1), (S2), (L1), (L2), (L3), (A1), (A2), (M1) and (M2), where (L3) denotes the sentence

$$\forall x \forall y [(x < y) \vee (x \doteq y) \vee (y < x)].$$

- (ii) **P** (*Peano Arithmetic*) is the extension of **Q** generated by the L_Ω -**induction schema**, the set of L_Ω -sentences

$$\forall y_0 \dots \forall y_{k-1} [(\phi_x(\dot{0}) \wedge \forall x [\phi \rightarrow \phi_x(\dot{S}x)]) \rightarrow \forall x \phi],$$

where ϕ has free variables among $x, y_0, \dots, y_{k-1}$.

Note that by Proposition 2.5.14(vii), (L3) is redundant as an axiom of **P**, but it will be needed as an axiom of **Q**.

Hilbert's condition (C) requires a “reasonable” set of axioms. We have made the point several times that decidability is a very natural minimal condition to impose on a set of axioms; it is hard to understand how one could in practice deduce theorems from a set of axioms not equipped with any method for determining whether a given statement is an axiom. If we take the position that any reasonable set of axioms must be decidable, then to show that a theory cannot be reasonably axiomatized it suffices to show that it cannot be decidable axiomatized.

At the same time, it is much less natural to expect that a whole theory, whether generated by a set of axioms or determined by a structure, should be decidable; for this to be true means that the theory is trivial in the sense that a machine is, in principle, capable of determining which sentences are and are not theorems of the theory. Nevertheless, in each example in [Sections 2.4–2.5](#) for which we found a decidable axiomatization of the theory of a structure, we also determined that in fact this theory is decidable. We now know that this was no coincidence; since by its definition $\text{Th}(\mathfrak{A})$ is complete, it follows from the Enumerability Theorem 3.1.18 that if it is decidable axiomatizable then it is decidable. Since this result is so central here, we restate it in the form in which we use it.

Enumerability Theorem for L_Ω 4.1.2 For any L_Ω -theory T ,

- (i) if T is decidable axiomatizable, then T is effectively enumerable;
- (ii) if T is decidable axiomatizable and complete, then T is decidable.

Thus if there were to exist a decidable set Γ of axioms for arithmetic — for example those of $\mathbf{P}$ or some extension — such that

$$\text{for all } L_\Omega\text{-sentences } \psi, \quad \Gamma \vdash \psi \iff \models_\Omega \psi,$$

or more compactly, $\text{Th}(\Gamma) = \text{Th}(\Omega)$, we could conclude that $\text{Th}(\Omega)$ itself is decidable and hence trivial in the sense described above. The simplest of the results of this chapter is that this is not the case.

The method of quantifier elimination used in [Sections 2.4–2.5](#) gave in some cases additional information about the structures involved, in particular a characterization of the sets and relations definable over the structure. Recall that a subset $X \subseteq \omega$ is **definable over Ω** iff for some formula ψ with one free variable x ,

$$\text{for all } m, \quad m \in X \iff \models_\Omega \psi[m].$$

The structure Ω has the special property that every element m has a name $\dot{m} := \underbrace{\dot{S}\dot{S}\dots\dot{S}}_m\dot{0}$ in the language, so by Lemma 2.2.22 (see also Remark 2.6.10), this can equivalently be written

$$\text{for all } m, \quad m \in X \iff \models_\Omega \psi(\dot{m}) \iff \psi(\dot{m}) \in \text{Th}(\Omega),$$

since $\psi(\dot{m})$, the result of substituting $\dot{m}$ for all free occurrences of x , is a sentence. We say that X is **defined by ψ** . Similarly, we say that a function $F : \omega \rightarrow \omega$ is definable over Ω iff for some formula ψ with free variables x and y ,

$$\text{for all } m \text{ and } n, \quad F(m) = n \iff \psi(\dot{m}, \dot{n}) \in \text{Th}(\Omega).$$

The result of Exercise 2.5.31 is that over the structure Ω_+ the definable sets are exactly the eventually periodic ones. The key fact here is that the sets definable over Ω form a much richer class. We state this in the form of a hypothesis, since it cannot be proved using only the intuitive notion of decidability; we will prove the corresponding fact for the more precise notion of recursiveness in [Section 4.5](#) below.

Definability Hypothesis 4.1.3 (DH)

- (i) Every effectively calculable function $F : \omega \rightarrow \omega$ is definable over Ω .
- (ii) Every decidable set $X \subseteq \omega$ is definable over Ω .
- (iii) Every effectively enumerable set $X \subseteq \omega$ is definable over Ω . $\square$

We shall see in [Section 4.5](#) that parts (ii) and (iii) follow easily from (i). To derive from this that $\text{Th}(\Omega)$ is undecidable, we need a few more observations.

Lemma 4.1.4 *The set Fm_{L_Ω} of L_Ω -formulas is effectively enumerable and the set Sent_{L_Ω} of L_Ω -sentences is a decidable subset of Fm_{L_Ω} .*

Proof. Since L_Ω is clearly an effective language, this follows immediately from Propositions 2.1.30 and 2.2.37. $\square$

This lemma guarantees the existence of an effective enumeration

$$\chi_0, \chi_1, \dots, \chi_m, \dots$$

of Fm_{L_Ω} . Fix for the rest of this section such an enumeration and fix a variable x . We often write simply $\phi(t)$ in place of $\phi_x(t)$, the result of substituting the term t for all free occurrences of x . For any L_Ω -formula ϕ , let $\#(\phi)$ denote the least number p such that χ_p is ϕ .

Corollary 4.1.5 *The mappings $n \mapsto \chi_n(\dot{n})$ and $\phi \mapsto \#(\phi)$ are effectively calculable.*

Proof. The first assertion is immediate from Proposition 2.2.37(iv). For the second, given ϕ , enumerate the list $\chi_0, \chi_1, \dots$ comparing each χ_i to ϕ until one is identical. This will, of course, eventually happen since the list contains all L_Ω -formulas. $\square$

Finally we need the following general fact which was Exercise 1.5.27.

Proposition 4.1.6 *For any sets X and $Y \subseteq Z$ and any effectively calculable function $F : X \rightarrow Z$, if $W = F^{-1}(Y) := \{x \in X : F(x) \in Y\}$, then*

- (i) *if Y is decidable in Z , then W is decidable in X ;*
- (ii) *if X and Y are effectively enumerable, then also W is effectively enumerable.* $\square$

Theorem 4.1.7 (assuming DH) *$\text{Th}(\Omega)$ is undecidable, hence not decidablely axiomatizable.*

Proof. Suppose towards a contradiction that $\text{Th}(\Omega)$ is decidable. Let $X \subseteq \omega$ be the set defined by

$$(A.1) \quad \text{for all } m, \quad m \in X \iff \chi_m(\dot{m}) \in \text{Th}(\Omega).$$

By 4.1.5 and 4.1.6, X is a decidable set, and therefore by DH(ii) it is definable over Ω . Hence for some formula ψ we have

$$(A.2) \quad \text{for all } m, \quad m \in X \iff \psi(\dot{m}) \in \text{Th}(\Omega).$$

Since $\text{Th}(\Omega)$ is consistent and complete, we have also

$$(A.3) \quad \text{for all } m, \quad m \in X \iff \neg \psi(\dot{m}) \notin \text{Th}(\Omega).$$

Let $p := \#(\neg \psi)$ so in particular $\chi_p(\dot{p})$ is $\neg \psi(\dot{p})$. Then by (A.1) and (A.3), respectively,

$$\begin{aligned} p \in X &\iff \neg \psi(\dot{p}) \in \text{Th}(\Omega) \\ p \in X &\iff \neg \psi(\dot{p}) \notin \text{Th}(\Omega). \end{aligned}$$

Hence if θ is the sentence $\neg \psi(\dot{p})$,

$$(A.4) \quad \theta \in \text{Th}(\Omega) \iff \theta \notin \text{Th}(\Omega),$$

a contradiction. $\square$

Theorem 4.1.8 (assuming DH) *For any L_Ω -theory $T \subseteq \text{Th}(\Omega)$, if T is decidablely axiomatizable, then T is incomplete.*

Proof. We give two proofs. First, note that if $T \subseteq \text{Th}(\Omega)$ and is complete, then necessarily $T = \text{Th}(\Omega)$: otherwise there is some sentence $\phi \in \text{Th}(\Omega) \sim T$, hence $\neg \phi \in T$, so also $\neg \phi \in \text{Th}(\Omega)$, a contradiction. Hence by the preceding theorem, such a T cannot be decidablely axiomatizable.

The second proof is a modification of the proof of the preceding theorem and has the advantage of being *constructive*: rather than concluding by contradiction that each T satisfying the hypothesis is incomplete we can describe a particular sentence θ_T which witnesses the incompleteness in that neither θ_T nor $\neg \theta_T$ belongs to T . Fix a decidablely axiomatizable theory $T \subseteq \text{Th}(\Omega)$, and let $X_T \subseteq \omega$ be the set defined by

$$(B.1) \quad \text{for all } m, \quad m \in X_T \iff \chi_m(\dot{m}) \in T.$$

By the Enumerability Theorem T is effectively enumerable, so by 4.1.5 and 4.1.6, also X_T is effectively enumerable and hence by DH(iii) definable over Ω , say by the formula ψ_T

$$(B.2) \quad \text{for all } m, \quad m \in X_T \iff \psi_T(\dot{m}) \in \text{Th}(\Omega),$$

from which we have immediately

$$(B.3) \quad \text{for all } m, \quad m \in X_T \iff \neg \neg \psi_T(\dot{m}) \in \text{Th}(\Omega).$$

Let $p_T := \#(\neg\psi_T)$ so in particular $\chi_{p_T}(\dot{p}_T)$ is $\neg\psi_T(\dot{p}_T)$. Then by (B.1) and (B.3), respectively,

$$\begin{aligned} p_T \in X_T &\iff \neg\psi_T(\dot{p}_T) \in T \\ p_T \in X_T &\iff \neg\neg\psi_T(\dot{p}_T) \in \text{Th}(\Omega). \end{aligned}$$

Hence if θ_T is the sentence $\neg\psi_T(\dot{p}_T)$,

$$(B.4) \quad \theta_T \in T \iff \neg\theta_T \in \text{Th}(\Omega).$$

Suppose first that $\theta_T \in T$. Then since $T \subseteq \text{Th}(\Omega)$, also $\theta_T \in \text{Th}(\Omega)$, but by (B.4) also $\neg\theta_T \in \text{Th}(\Omega)$, a contradiction; hence $\theta_T \notin T$. Now suppose that $\neg\theta_T \in T$. Again since $T \subseteq \text{Th}(\Omega)$ also $\neg\theta_T \in \text{Th}(\Omega)$, hence by (B.4) $\theta_T \in T$, contrary to what we just established. Hence also $\neg\theta_T \notin T$. $\square$

Remark 4.1.9 Of course this result applies to the theories Q and P above, since Q has a finite set of axioms and it is not hard to see that the given set of axioms for P is decidable. However, the result applies also to any extension of P by a decidable set of axioms — in other words, the incompleteness of the theory P is not due just to the inadvertant omission of a few axioms; there is no way to make the theory complete by adjoining any decidable set of new axioms.

This (along with its extensions below) may be one of the most misquoted theorems in mathematics. There are many published descriptions of it along the lines of “there are some truths which cannot be proved”. Note that we did derive in the proof above that $\theta_T \notin T$, so by (B.4) $\neg\theta_T \notin \text{Th}(\Omega)$ and thus $\theta_T \in \text{Th}(\Omega)$ — that is, θ_T is indeed true but not provable in T . However, θ_T depends on T and is in no absolute sense unprovable. For each T to which the theorem applies, there is a consistent proper extension $T' := \text{Th}(T \cup \{\theta_T\})$ in which θ_T is provable, but since T' is still decidable axiomatizable there is a new sentence $\theta_{T'}$ which is true but not provable in T' . A better one-line summary of the theorem is “mathematics is too complicated to be completely captured by a decidable set of axioms”.

It is also interesting to consider the intuitive content of θ_T . Since we have $\theta_T \notin T \iff \theta_T \in \text{Th}(\Omega)$, in a sense θ_T expresses the idea “I am not a theorem of T ”. This sentence is closely related to the paradoxical sentence (liar paradox) “this sentence is false” or “I am false”, but as we have seen, is not paradoxical but simply true but unprovable in T .

We have now established, assuming DH, that version (a) of Hilbert’s program for arithmetic cannot be carried out, and we consider now version (b). The idea here would be to find a decidable set Γ of L_Ω -sentences such that we could take $T := \text{Th}(\Gamma)$ as a *definition* of the set of theorems of arithmetic

without worrying whether or not this coincides exactly with $\text{Th}(\Omega)$. This imposes two obvious conditions. First, T should include standard theorems of arithmetic, certainly at least the (axioms of the) theory $\mathbf{Q}$. Second, to serve as a definition of truth, T should be *complete* — that is, for each sentence ϕ contain either ϕ or $\neg\phi$. The central result here is that this too is impossible. The key notion is a refinement of definability over a structure.

Definition 4.1.10 For any L_Ω -theory T and any set $X \subseteq \omega$,

- (i) X is ***T-representable*** iff there is some formula $\psi(x)$ with one free variable x such that for all $m \in \omega$,

$$m \in X \implies \psi(\dot{m}) \in T \quad \text{and} \quad m \notin X \implies \neg\psi(\dot{m}) \in T;$$

- (ii) X is ***weakly T-representable*** iff there exists a formula ψ with one free variable x such that for all $m \in \omega$,

$$m \in X \iff \psi(\dot{m}) \in T;$$

- (iii) A function $F : \omega \rightarrow \omega$ is ***T-representable*** iff there exists a formula $\psi(x, y)$ such that for all $m \in \omega$,

$$\forall y [\psi(\dot{m}, y) \leftrightarrow y \doteq \overbrace{F(m)}^{\cdot}] \in T.$$

Note that the condition of (ii) may equivalently be expressed: for all $m \in \omega$,

$$m \in X \implies \psi(\dot{m}) \in T \quad \text{and} \quad m \notin X \implies \psi(\dot{m}) \notin T.$$

Now, parallel to the Definability Hypothesis above, we have the

Representability Hypothesis 4.1.11 (RH) For any consistent L_Ω -theory $T \supseteq \mathbf{Q}$,

- (i) every effectively calculable function $F : \omega \rightarrow \omega$ is *T-representable*;
- (ii) every decidable set $X \subseteq \omega$ is *T-representable*;
- (iii) if T is decidable axiomatizable, then every effectively enumerable set $X \subseteq \omega$ is weakly *T-representable*; in fact, for each such X there exists a formula ψ such that for all m ,

$$m \in X \iff \psi(\dot{m}) \in T \iff \psi(\dot{m}) \in \text{Th}(\Omega).$$

Remark 4.1.12 Note that if T is complete, then T -representability and weak T -representability are equivalent. In particular, a set X is (weakly) $\text{Th}(\Omega)$ -representable iff it is definable over Ω . Although representability of a set X is preserved in passing from a theory T to any extension T' , this is not true for weak representability, since T' may contain a sentence $\psi(\dot{m})$ for $m \notin X$. Hence it is to be expected in (iii) that there be some restriction on T .

Again, the more precise version of this using the notion of recursiveness in place of intuitive decidability will be proved below in [Section 4.6](#) where we shall also see that parts (ii) and (iii) follow from (i).

Theorem 4.1.13 (assuming RH) *Every consistent L_Ω -theory $T \supseteq \mathbf{Q}$ is undecidable.*

Proof. Suppose that T is a consistent L_Ω -theory such that $\mathbf{Q} \subseteq T$ and, towards a contradiction, that T is decidable. Let X be the set defined by

$$(C.1) \quad \text{for all } m, \quad m \in X \iff \chi_m(\dot{m}) \in T.$$

As before, by 4.1.5 and 4.1.6, X is a decidable set, hence by RH(ii) is T -representable, so for some formula ψ ,

$$(C.2) \quad \begin{array}{ll} \text{for all } m, & m \in X \implies \psi(\dot{m}) \in T \\ \text{and} & m \notin X \implies \neg \psi(\dot{m}) \in T. \end{array}$$

Since T is consistent it follows that

$$(C.3) \quad \text{for all } m, \quad m \in X \iff \neg \psi(\dot{m}) \notin T.$$

Now as before let $p := \#(\neg \psi)$, so $\chi_p(\dot{p})$ is $\neg \psi(\dot{p})$ and by (C.1) and (C.3) respectively we have

$$\begin{array}{ll} p \in X & \iff \neg \psi(\dot{p}) \in T \\ p \in X & \iff \neg \psi(\dot{p}) \notin T. \end{array}$$

Hence if θ is the sentence $\neg \psi(\dot{p})$,

$$(C.4) \quad \theta \in T \iff \theta \notin T,$$

a contradiction. $\square$

First Incompleteness Theorem 4.1.14 (assuming RH) *For any consistent L_Ω -theory $T \supseteq \mathbf{Q}$, if T is decidable axiomatizable, then T is incomplete.*

Proof. This follows immediately from the preceding theorem together with the Enumerability Theorem. However, as for Theorem 4.1.8, we can give another proof which is constructive in providing an explicit witness for the incompleteness of each theory T . Suppose that T is a consistent decidable axiomatizable L_Ω -theory $T \supseteq \mathbf{Q}$, and let $X_T \subseteq \omega$ be the set defined by

$$(D.1) \quad \text{for all } m, \quad m \in X_T \iff \chi_m(\dot{m}) \in T.$$

By the Enumerability Theorem T is effectively enumerable, so again by 4.1.5 and 4.1.6, also X_T is effectively enumerable and hence by RH(iii) weakly T -representable, say by the formula ψ_T ,

$$(D.2) \quad \text{for all } m, \quad m \in X_T \iff \psi_T(\dot{m}) \in T,$$

hence also

$$(D.3) \quad \text{for all } m, \quad m \in X_T \iff \neg \neg \psi_T(\dot{m}) \in T.$$

As before, let $p_T := \#(\neg \psi_T)$, $\chi_{p_T}(\dot{p}_T)$ is $\neg \psi_T(\dot{p}_T)$ and by (D.1) and (D.3) respectively,

$$\begin{aligned} p_T \in X_T &\iff \neg \psi_T(\dot{p}_T) \in T \\ p_T \in X_T &\iff \neg \neg \psi_T(\dot{p}_T) \in T. \end{aligned}$$

Hence, if θ_T is the sentence $\neg \psi_T(\dot{p}_T)$,

$$(D.4) \quad \theta_T \in T \iff \neg \theta_T \in T.$$

It follows from the consistency of T that neither $\theta_T \in T$ nor $\neg \theta_T \in T$ as desired. $\square$

Note that θ_T again has the intuitive meaning “I am not a theorem of T ”, and since $\neg \theta_T \notin T$, also $\psi_T(\dot{p}_T) \notin T$, so by the last clause of RH(iii) $\psi_T(\dot{p}_T) \notin \text{Th}(\Omega)$ and hence $\theta_T \in \text{Th}(\Omega)$ — that is, θ_T is true but unprovable in T .

These results have one important consequence that has nothing to do with theories or incompleteness.

Corollary 4.1.15 (assuming RH) *There exist sets which are effectively enumerable but not decidable.*

Proof. Any consistent decidable axiomatizable L_Ω -theory $T \supseteq \mathbf{Q}$ — in particular $\mathbf{Q}$ itself — is an example of such a set. $\square$

The fact that $\mathbf{Q}$ is finitely axiomatizable leads to a surprising strengthening of these results.

Definition 4.1.16 Two theories T and U are **compatible** iff $T \cup U$ is consistent.

Extended First Incompleteness Theorem 4.1.17 (assuming RH) For any L_Ω -theory T compatible with $\mathbf{Q}$,

- (i) T is undecidable;
- (ii) if T is decidable axiomatizable, then T is incomplete.

Proof. Fix an L_Ω theory T compatible with $\mathbf{Q}$, and let $\chi_{\mathbf{Q}}$ be the conjunction of the finitely many axioms of $\mathbf{Q}$. Let T^+ be the theory generated by $T \cup \mathbf{Q}$; by hypothesis T^+ is a consistent extension of $\mathbf{Q}$ and hence by Theorem 4.1.13 is undecidable. But easily, for any formula θ ,

$$\theta \in T^+ \iff (\chi_{\mathbf{Q}} \rightarrow \theta) \in T.$$

Now if T were decidable, since the mapping $\theta \mapsto (\chi_{\mathbf{Q}} \rightarrow \theta)$ is effectively calculable, it would follow by Proposition 4.1.6 that also T^+ is decidable, a contradiction.

For (ii), suppose that T is decidable axiomatizable, say $T = \text{Th}(\Gamma)$ for a decidable set Γ . Then $\Gamma \cup \{\chi_{\mathbf{Q}}\}$ is a decidable set of axioms for T^+ , so by the First Incompleteness Theorem, T^+ is incomplete. But then so is any subtheory, in particular T . $\square$

Corollary 4.1.18 (assuming RH) The validity problem for L_Ω is undecidable — that is, there is no algorithm which given an L_Ω -sentence θ decides whether or not θ is logically valid.

Proof. Let T be the set of logical validities of L_Ω . Clearly T is compatible with $\mathbf{Q}$, so by the Extended First Incompleteness Theorem is undecidable. $\square$

Note that this is in stark contrast to the results of [Sections 1.5](#) and [2.4](#) that the sets of tautologies of propositional logic and of logical validities of $L_=$ are decidable.

If these proofs strike the reader as somewhat mysterious, you are in good company; they were poorly understood by the mathematical community for some time after their publication in 1931 and as mentioned above are misinterpreted even today. However, the central idea is actually very simple, and it is helpful to see it in abstract form. It is known as a **diagonal argument** and will be applied many times in the the remainder of this chapter and the next.

Definition 4.1.19 For any set A and any relation $U \subseteq A \times A$,

- (i) for each $a \in A$, $U_a := \{b : U(a, b)\}$ is called a **U -section**; a is called a **U -index** for U_a ;
- (ii) U is **universal** for a class $\mathcal{C} \subseteq \wp(A)$ iff every member of $\mathcal{C}$ is a U -section;
- (iii) $D_U := \{a : U(a, a)\}$ is the **diagonal set** of U .

Example 4.1.20 For any L_Ω -theory T , let

$$U_T := \{(a, m) : \chi_a(\dot{m}) \in T\}.$$

Thus $U_{T,a}$ is exactly the set weakly T -represented by χ_a , and since $\chi_0, \chi_1, \dots$ is a complete list of L_Ω -formulas, U_T is universal for the class of weakly T -representable sets. The diagonal set

$$D_{U_T} := \{m : \chi_m(\dot{m}) \in T\}$$

is, of course, exactly the set X of the proofs of Theorems 4.1.7 and 4.1.13. In these proofs, the equivalences numbered (A.3) and (C.3) could be written

$$\text{for all } m, \quad m \notin X \iff \neg \psi(\dot{m}) \in T,$$

which asserts that the complement $\omega \sim X$ of X is weakly T -represented by the formula $\neg \psi$. Since this leads to a contradiction, we can see the core of the proof as establishing that $\omega \sim X$ is *not* weakly T -representable. This is an instance of the following general principle.

Diagonal Lemma 4.1.21 For any sets $U \subseteq A \times A$, $A \sim D_U$ is not a U -section.

Proof. Suppose the contrary, so for some $a \in A$, $(A \sim D_U) = U_a$. Then we have the contradiction

$$a \in D_U \iff U(a, a) \iff a \in U_a \iff a \notin D_U. \quad \square$$

We can now rephrase the proof of Theorem 4.1.13 (or 4.1.7 with $\text{Th}(\Omega)$ for T) as follows: given a consistent L_Ω -theory $T \supseteq \mathbf{Q}$, with U_T as above, by the Diagonal Lemma, $\omega \sim D_{U_T}$ is not a U_T -section, hence is not weakly T -representable. Hence D_{U_T} is not T -representable, so by RH(ii) not decidable, and thus by 4.1.5 and 4.1.6, also T is not decidable.

The first use of this method was Cantor's proof of

Proposition 4.1.22 *For any set A , there is no surjective mapping $F : A \rightarrow \wp(A)$.*

Proof. For any mapping $F : A \rightarrow \wp(A)$, set $U_F = \{ (a, b) : b \in F(a) \}$. Thus the collection of U_F -sections is exactly the image of F . By the Diagonal Lemma, $A \sim D_{U_F}$ is a subset of A which is *not* a U_F section and thus witnesses that F is not surjective. $\square$

We can also use this method to prove a strengthening of Theorem 4.1.7 which is generally referred to as the **undefinability of truth**. Let

$$\text{Th}(\Omega)^\# := \{ p : \chi_p \in \text{Th}(\Omega) \}$$

and $\text{Sb}(m) := \#(\chi_m(\dot{m}))$. By Corollary 4.1.5, **Sb** is an effectively calculable function.

Tarski's Theorem 4.1.23 *(assuming DH) $\text{Th}(\Omega)^\#$ is not definable over Ω . In particular, $\text{Th}(\Omega)$ is not effectively enumerable.*

Proof. With

$$U := U_{\text{Th}(\Omega)} = \{ (a, m) : \chi_a(\dot{m}) \in \text{Th}(\Omega) \},$$

by Remark 4.1.12 and Example 4.1.20, U is universal for the class of sets definable over Ω . By the Diagonal Lemma, D_U is not a U -section, hence is not definable over Ω .

Now suppose towards a contradiction that $\text{Th}(\Omega)^\#$ is definable, so there exists a formula $\phi(y)$ such that for all p ,

$$p \in \text{Th}(\Omega)^\# \iff \phi(\dot{p}) \in \text{Th}(\Omega).$$

Since **Sb** is effectively calculable, by DH(i) it is also definable, so for some formula $\psi(x, y)$,

$$\text{Sb}(m) = p \iff \psi(\dot{m}, \dot{p}) \in \text{Th}(\Omega).$$

But then

$$\begin{aligned} m \in D_U &\iff \exists p [\phi(\dot{p}) \in \text{Th}(\Omega) \text{ and } \psi(\dot{m}, \dot{p}) \in \text{Th}(\Omega)] \\ &\iff \exists y [\phi(y) \wedge \psi(\dot{m}, y)] \in \text{Th}(\Omega), \end{aligned}$$

so D_U is definable, contrary to what we just established. Hence $\text{Th}(\Omega)^\#$ is not definable, so by DH(iii) not effectively enumerable; it follows from Proposition 4.1.6 that also $\text{Th}(\Omega)$ is not effectively enumerable. $\square$

Finally, we give a very brief sketch, to be expanded in [Section 5.3](#), of the central ideas of Gödel's Second Incompleteness Theorem which shows that Step (E) of Hilbert's Program is also unrealizable. The key step is a formalization of the observation in Remark 4.1.9 that the sentence θ_T intuitively expresses the idea "I am not a theorem of T ", and thus θ_T describes a property of itself and is **self-referent**. The following lemma exhibits another way that a sentence ϕ can describe itself: if ϕ is χ_p , the p -th sentence in the fixed enumeration, we think of p as a name for ϕ and a sentence $\theta_y(\dot{p})$ as expressing that ϕ has the property described by θ . Then if ϕ is provably equivalent to $\theta_y(\dot{p})$, ϕ expresses the idea "I have the property θ ".

Self-reference Lemma 4.1.24 (assuming RH) *For any L_Ω -formula θ with at most y free, there exists a number p such that χ_p is a sentence and*

$$\mathbf{Q} \vdash \chi_p \leftrightarrow \theta_y(\dot{p}).$$

Proof. Fix a formula $\theta(y)$. By Corollary 4.1.5 and RH(i), Sb is $\mathbf{Q}$ -representable — that is, there exists a formula $\psi(x, y)$ such that

$$\text{for all } m, \quad \mathbf{Q} \vdash \psi(\dot{m}, y) \leftrightarrow y \doteq \overbrace{\text{Sb}(m)}^{\dot{}}.$$

Choose m such that the formula $\chi_m(x)$ is $\forall y[\psi(x, y) \rightarrow \theta(y)]$. Then $\chi_{\text{Sb}(m)}$ is the sentence $\forall y[\psi(\dot{m}, y) \rightarrow \theta(y)]$. Hence

$$\mathbf{Q} \vdash \chi_{\text{Sb}(m)} \leftrightarrow \forall y[y \doteq \overbrace{\text{Sb}(m)}^{\dot{}} \rightarrow \theta(y)] \quad \text{so} \quad \mathbf{Q} \vdash \chi_{\text{Sb}(m)} \leftrightarrow \theta(\overbrace{\text{Sb}(m)}^{\dot{}}),$$

and $p := \text{Sb}(m)$ is as required. $\square$

Now fix a decidable axiomatizable theory T such that $\mathbf{Q} \subseteq T$, and let $T^\# := \{m : \chi_m \in T\}$. Since (by the Enumerability Theorem) T is effectively enumerable, by Proposition 4.1.6, so is $T^\#$. By RH(iii) there exists a formula $\dot{T}(y)$ which weakly $\mathbf{Q}$ -represents $T^\#$ — more precisely, such that for all m ,

$$(E.1) \quad \chi_m \in T \iff \dot{T}(\dot{m}) \in \mathbf{Q} \iff \dot{T}(\dot{m}) \in \text{Th}(\Omega).$$

We think of $\dot{T}$ as a name for T . Because $\text{Th}(\Omega)$ is a complete theory, (E.1) entails that for all m ,

$$(E.2) \quad \chi_m \notin T \iff \neg \dot{T}(\dot{m}) \in \text{Th}(\Omega),$$

and, since $\mathbf{Q} \subseteq T$,

$$(E.3) \quad \chi_m \in T \implies \dot{T}(\dot{m}) \in T.$$

The Self-reference Lemma guarantees the existence of a number q_T such that

$$\mathbf{Q} \vdash \chi_{q_T} \leftrightarrow \neg \dot{T}(\dot{q}_T);$$

the sentence χ_{q_T} intuitively expresses “I am not a theorem of T ”. In particular, again because $\mathbf{Q} \subseteq T$,

$$(E.4.1) \quad \dot{T}(\dot{q}_T) \in T \implies \neg \chi_{q_T} \in T$$

$$(E.4.2) \quad \neg \dot{T}(\dot{q}_T) \in T \implies \chi_{q_T} \in T.$$

Now

$$(E.5) \quad T \text{ is consistent} \implies \chi_{q_T} \notin T,$$

because if $\chi_{q_T} \in T$, then by (E.3), $\dot{T}(\dot{q}_T) \in T$, hence by (E.4.1), $\neg \chi_{q_T} \in T$, so T is inconsistent.

Let Con_T be the sentence $\neg \dot{T}(\dot{r})$, where χ_r is the sentence $\dot{0} \doteq \dot{1}$. Since $\neg (\dot{0} \doteq \dot{1}) \in \mathbf{Q} \subseteq T$,

$$T \text{ is consistent} \iff \chi_r \notin T,$$

so by (E.2),

$$(E.6) \quad T \text{ is consistent} \iff \text{Con}_T \in \text{Th}(\Omega).$$

Thus by (E.2), (E.5) and (E.6),

$$\text{Con}_T \in \text{Th}(\Omega) \implies \neg \dot{T}(\dot{q}_T) \in \text{Th}(\Omega),$$

which is simply

$$(E.7) \quad \models_{\Omega} \text{Con}_T \rightarrow \neg \dot{T}(\dot{q}_T).$$

The core of the argument is that this sentence is not only true in Ω but in fact a theorem of $\mathbf{P}$:

$$(E.8) \quad \mathbf{P} \vdash \text{Con}_T \rightarrow \neg \dot{T}(\dot{q}_T).$$

The proof of (E.8) is a very long and difficult analysis of the argument leading to (E.7) which shows that all of the steps can be carried out using only the axioms of $\mathbf{P}$. Gödel in his original paper did not make any pretense of writing out all (or almost any) of the details but merely remarked that as the argument used only methods of classical mathematics they could be formalized in $\mathbf{P}$. However, later authors have carefully verified that this is true and we will go much more deeply into the argument in [Section 5.3](#).

Second Incompleteness Theorem 4.1.25 *For any consistent decidable axiomatizable L_Ω -theory $T \supseteq P$, $T \not\vdash \text{Con}_T$.*

Proof. Suppose that T is consistent but $T \vdash \text{Con}_T$. Since $P \subseteq T$, we have from (E.8) that also $T \vdash \text{Con}_T \rightarrow \neg \dot{T}(\dot{q}_T)$, whence $T \vdash \neg \dot{T}(\dot{q}_T)$. But then by (E.4.2) $\chi_{q_T} \in T$, contrary to (E.5). $\square$

It is perhaps not obvious to the reader why this technical result actually shows that Hilbert's task (E) is not achievable. Hilbert envisioned a secure proof of the consistency of a theory T as one which used only the simplest mathematical tools and thus was insulated from any of the paradoxes and contradictions which had led to the crisis of confidence. More precisely, the goal was a consistency proof for T which ignored completely any meaning or interpretation of the (formal) language and was concerned only with the finite configurations of symbols which represent (formal) proofs from the axioms of T . If one could show at this level that there could be no such configuration ending with a statement such as $\dot{0} \doteq \dot{S}\dot{0}$, which directly contradicts the axioms, then one would be assured that T is consistent. However we now know that any such proof could be described in terms of the numbers associated with the formulas involved and thus carried out in any sufficiently strong arithmetic theory, in particular in P . But this would mean that $P \vdash \text{Con}_T$ and hence, if $P \subseteq T$, that $T \vdash \text{Con}_T$, contrary to the Second Incompleteness Theorem.

Remark 4.1.26 In the sketch above, note that by the choice of q_T ,

$$\dot{T}(\dot{q}_T) \in T \iff \neg \chi_{q_T} \in T,$$

and by (E.1),

$$\dot{T}(q_T) \in T \iff \chi_{q_T} \in T.$$

Hence $\chi_{q_T} \in T \iff \neg \chi_{q_T} \in T$ and thus χ_{q_T} is another witness to the incompleteness of T .

In [Sections 5.5](#) and [6.6](#) we shall discuss how these results can be extended to other languages and theories, in particular to a comprehensive system of set theory such as ZFC (see [Chapter 6](#), particularly [Section 6.6](#)). Since essentially all of standard mathematics can be formalized in ZFC, *any* consistency proof for any theory T could be formalized in ZFC. In particular, this would be true for a consistency proof of ZFC itself and we would have $\text{ZFC} \vdash \text{Con}_{\text{ZFC}}$. However, the mentioned extension shows that this is false (unless ZFC actually is inconsistent!), and we conclude that we can never expect to prove in any satisfactory way the consistency of ZFC or any other comprehensive axiomatic system.

Collateral Reading. The incompleteness theorems appear in some form in all but the most elementary logic texts. [En01] has a nice version, and

Smoryński's [Ba77; Chapter D.1] is particularly detailed. It is also interesting to read at least part of Gödel's original paper reprinted in both [vHe] and [Dav].

Exercises

Exercise 4.1.27 Show that the following are theorems of $\mathbf{P}$ but not theorems of $\mathbf{Q}$:

- | | |
|--|--|
| (i) $\neg x \doteq \dot{\Sigma}x$ | (vi) $(x \dot{+} y) \dot{+} z \doteq x \dot{+} (y \dot{+} z)$ |
| (ii) $\dot{0} \dot{+} x \doteq x$ | (vii) $(x \dot{\times} y) \dot{\times} z \doteq x \dot{\times} (y \dot{\times} z)$ |
| (iii) $\dot{0} \dot{\times} x \doteq \dot{0}$ | (viii) $x \dot{\times} (y \dot{+} z) \doteq (x \dot{\times} y) \dot{+} (x \dot{\times} z)$ |
| (iv) $x \dot{+} y \doteq y \dot{+} x$ | (ix) $(x \dot{+} y) \dot{\times} z \doteq (x \dot{\times} z) \dot{+} (y \dot{\times} z)$ |
| (v) $x \dot{\times} y \doteq y \dot{\times} x$ | |

Hint: To show that a sentence ϕ is not a theorem of $\mathbf{Q}$, it is necessary to construct a model $\mathfrak{A}$ of $\mathbf{Q}$ in which ϕ is false. Since these ϕ are all true in the standard model, $\mathfrak{A}$ must be a non-standard model. Recall the discussion of non-standard models from [Section 2.5](#) but note that models of $\mathbf{Q}$ can be much simpler than those described in Proposition 2.5.20.

Exercise 4.1.28 Prove that parts (ii) and (iii) of the Definability Hypothesis 4.1.3 follow from (i).

Exercise 4.1.29 Show that for any L_Ω -theory T and any $X \subseteq \omega$,

- (i) if $T \vdash \neg \dot{0} \doteq \dot{1}$, then X is T -representable iff its characteristic function K_X is T -representable;
- (ii) X is T -representable iff its complement $\omega \sim X$ is T -representable;
- (iii) if X is T -representable, then for any theory $T' \supseteq T$, also X is T' -representable;
- (iv) if T is consistent and X is T -representable, then X is also weakly T -representable;
- (v) if T is complete and X is weakly T -representable, then X is also T -representable;
- (vi) X is $\text{Th}(\Omega)$ -(weakly)-representable iff X is definable over Ω .

Exercise 4.1.30 Assume (this will be proved in [Section 4.6](#)) that

$$\text{for all } m \neq n, \quad \mathbf{Q} \vdash \neg \dot{m} \doteq \dot{n}.$$

Use some of the results of the preceding exercise to show that under the additional hypothesis that $T \subseteq \text{Th}(\Omega)$ parts (ii) and (iii) of the Representability Hypothesis 4.1.11 follow from (i).

Exercise 4.1.31 Prove the following partial converse of the Representability Hypothesis: for any decidable axiomatizable L_Ω -theory T ,

- (i) every T -representable function $F : \omega \rightarrow \omega$ is effectively calculable;
- (ii) every T -representable set is decidable;
- (iii) every weakly T -representable set is effectively enumerable.

4.2 Recursive functions and relations

Up to this point we have been using the word *decidable* to refer to an intuitively understood notion defined in terms of the notions of *algorithm* and *effectively calculable function*. These concepts are sufficiently precise for the proofs of positive results, but it is hard to imagine how one might demonstrate in this framework that a set or theory is *not* decidable. While a proof of decidability of a theory requires only the specification of a single algorithm and a verification that it correctly decides membership in the theory, a proof of undecidability must establish that *no* algorithm is adequate to this task and requires, therefore, a careful delineation of the whole class of allowable algorithms.

This difficulty arises similarly around the Definability and Representability Hypotheses assumed in [Section 4.1](#): to prove them would require a clear understanding of the entire class of decidable sets and effectively calculable functions of natural numbers.

This section is devoted to the study of the precise mathematical counterpart of the intuitive notions of decidability and effective calculability: recursive functions, sets, and relations. By replacing the intuitively understood notion of decidability with a precise mathematical one, we both sharpen our earlier positive results and are able to carry out precise proofs of undecidability. Although the intuitive notions are directly applicable to all sorts of finite objects, recursiveness will be defined directly only for functions, sets, and relations of natural numbers and then extended to other sorts of objects via enumerations.

There are many instances in mathematics of the replacement of an intuitively understood notion with a precise mathematical one. One familiar example is the continuity of functions on the real line. The intuitive notion of a graph which can be drawn without lifting the pen from the paper has been replaced by that of a function satisfying a well-known precise condition for continuity. A common feature of all such replacements is that we cannot expect to prove that the intuitive and mathematical notions coincide for exactly the reasons that make the intuitive notion unsuitable for some types of proof; although individual instances of the intuitive notion are generally clear, the extent of the class of objects defined is less so.

We shall tailor the definition so that it is fairly obvious that every recursive function is effectively calculable. The proof of this will be in the spirit of the decidability results of [Chapter 2](#); we shall show that for each recursive function F there exists an intuitive algorithm which calculates F . This will be relatively easy because of the inductive form of the definition of recursiveness.

The converse is known as **Church's Thesis**: every effectively calculable function of natural numbers is recursive. We cannot expect to give a mathematical proof of this for the same reasons that we cannot prove the Definability and Representability Hypotheses of [Section 4.1](#); by its nature the concept of effective calculability is intuitive and not sufficiently precisely given. However, for any *particular* well-defined effectively calculable function F we can verify that F is recursive, and by a sequence of such proofs we can establish with complete rigor that each of the intuitively decidable theories of [Chapter 2](#) — and by extension, others like them — remain decidable under the new definition. We shall consider some examples of this in [Section 4.4](#).

Although we cannot *prove* Church's Thesis, we can give evidence for its truth. The evidence is of two kinds, not unlike that which one might give to support the claim that the modern definition of continuity captures correctly the intuitive notion:

- (1) every known example of an effectively calculable function is recursive; despite many years of searching, no counterexample has been found;
- (2) formalizations of various aspects of the intuitive notion of effective calculability all lead to the same class of recursive functions.

Of course, the success of the translation of particular decision procedures contributes evidence of type (1); we consider in [Section 5.4](#) some of the evidence of type (2).

The reader should note that the term 'recursive' will have here very specific meaning related to but not coinciding with other more general meanings in the preceding chapters, particularly [Section 1.2](#). This is a result of the way that terminology develops. There is some tendency in recent years to replace 'recursive' with 'computable', but we will not follow this line.

As we shall study them, recursive functions are **finitary functions** on ω — that is, functions F with $\text{Dom}(F) \subseteq \omega^k$, for some $k \in \omega$, and $\text{Im}(F) \subseteq \omega$. In this chapter, letters $F, G, H, \dots$ always denote finitary functions on ω , and in this section they further denote **total functions** — that is, functions F such that $\text{Dom}(F) = \omega^k$ for some k . Beginning in [Section 4.3](#) we consider also **partial functions** F with $\text{Dom}(F)$ a possibly proper subset of some ω^k .

We define first the smaller class of primitive recursive functions inductively as the smallest class of functions which includes certain initial functions and is closed under two operations on functions.

Definition 4.2.1 The *initial functions* are the following for all $k, n \in \omega$ and all $i < k$:

$$\begin{aligned} \text{Cs}_n^k(m_0, \dots, m_{k-1}) &= n && \text{(constant functions);} \\ \text{Pr}_i^k(m_0, \dots, m_{k-1}) &= m_i && \text{(projection functions);} \\ \text{Sc}_i^k(m_0, \dots, m_{k-1}) &= m_i + 1 && \text{(successor functions).} \end{aligned}$$

Definition 4.2.2 For all $k, l \in \omega$,

- (i) for any functions $G, H_0, \dots, H_{l-1}$, $\text{Cmp}_l^k(G, H_0, \dots, H_{l-1})$ is the function F such that for all $m_0, \dots, m_{k-1}$,
 - (a) if G is l -ary and $H_0, \dots, H_{l-1}$ are all k -ary, then

$$\begin{aligned} F(m_0, \dots, m_{k-1}) &= \\ &G(H_0(m_0, \dots, m_{k-1}), \dots, H_{l-1}(m_0, \dots, m_{k-1})); \end{aligned}$$

- (b) otherwise, $F(m_0, \dots, m_{k-1}) = 0$.

We say that F arises from $G, H_0, \dots, H_{l-1}$ by **composition**.

- (ii) For any functions G and H , $\text{Rec}^{k+1}(G, H)$ is the function F such that for all $m_0, \dots, m_{k-1}$ and all $n \in \omega$,
 - (a) if G is k -ary and H is $(k+2)$ -ary, then

$$\begin{aligned} F(m_0, \dots, m_{k-1}, 0) &= G(m_0, \dots, m_{k-1}); \\ F(m_0, \dots, m_{k-1}, n+1) &= \\ &H(F(m_0, \dots, m_{k-1}, n), m_0, \dots, m_{k-1}, n); \end{aligned}$$

- (b) otherwise, $F(m_0, \dots, m_{k-1}, n) = 0$.

We say that F arises from G and H by **primitive recursion**.

The point of clause (b) in each case is simply to make sure that Cmp_l^k and Rec^{k+1} are totally defined operations, often called **functionals**, on the class of (partial) functions.

Definition 4.2.3 The class Prim of **primitive recursive** functions is the smallest class of functions which contains the initial functions and is closed under composition and primitive recursion.

The class of recursive functions is obtained by imposing an additional closure condition.

Definition 4.2.4 For any $k \in \omega$ and any function G , $\text{Se}^k(G)$ is the function F such that

- (i) if G is $(k+1)$ -ary and for all $m_0, \dots, m_{k-1}$ there exists n such that $G(m_0, \dots, m_{k-1}, n) = 0$, then for all $m_0, \dots, m_{k-1}$,

$$F(m_0, \dots, m_{k-1}) = \text{least } n [G(m_0, \dots, m_{k-1}, n) = 0];$$

- (ii) otherwise for all $m_0, \dots, m_{k-1}$, $F(m_0, \dots, m_{k-1}) = 0$.

F is said to arise from G by **(unbounded) search**. We call the condition

$$\forall m_0 \cdots \forall m_{k-1} \exists n [G(m_0, \dots, m_{k-1}, n) = 0]$$

the **search condition** for G .

Definition 4.2.5 The class Rec of **recursive functions** is the smallest class of functions which contains the initial functions and is closed under composition, primitive recursion and search.

Since each of the definitions of Prim and Rec are inductive in the sense of [Section 1.2](#), we have available for each a method of inductive proof to establish that all (primitive) recursive functions have some property or belong to some set. This is, of course, exactly what was missing for the intuitive notion of effective calculability. Note, however, that these inductive definitions do not have unique readability, so we cannot make recursive (in the sense of [Section 1.2](#)) definitions over these classes.

Corollary 4.2.6 $\text{Prim} \subseteq \text{Rec}$.

Proof. Immediate by Prim -induction. $\square$

Remark 4.2.7 Up to this point we have reserved the quantifier symbols $\exists$ and $\forall$ for use in formal languages and used the corresponding English words in the metalanguage. In this chapter we shall have relatively little to do with formal languages but much to do with quantification over natural numbers. Accordingly we shall from now on use these symbols freely in the metalanguage and write $\exists p[\dots]$ to mean that there is at least one natural number p such that $[\dots]$. Furthermore, we shall write $(\exists p < q) \dots$ to mean that $\dots$ holds for some $p < q$ and $(\forall p < q) \dots$ to mean that $\dots$ holds for all $p < q$. We also occasionally use the connectives $\wedge$ and $\vee$ to abbreviate metalanguage assertions.

In the spirit of our earlier arguments concerning decidability we have

Proposition 4.2.8 *Every recursive function is effectively calculable.*

Proof. We proceed by Rec-induction. Clearly the initial functions are effectively calculable. If $G, H_0, \dots, H_{l-1}$ are all effectively calculable, then an algorithm for $\text{Cmp}_l^k(G, H_0, \dots, H_{l-1})$ is obtained in the obvious way from algorithms for $G, H_0, \dots, H_{l-1}$. If G and H are effectively calculable, then the algorithm for $F = \text{Rec}^{k+1}(G, H)$ goes as follows: to compute $F(m, n)$, apply first the algorithm for G to compute $F(m, 0)$. If $n > 0$, apply the algorithm for H successively to the result of the preceding computation to compute $F(m, 1), F(m, 2), \dots, F(m, n)$. Finally, if G is effectively calculable, we may compute $F = \text{Se}^k(G)$ by computing successively $G(m, 0), G(m, 1), \dots$ until a first value $G(m, n) = 0$ is obtained. The value of F is simply the number n at which we succeed. The search condition guarantees that this procedure will terminate after finitely many steps. $\square$

We argued in [Section 1.5](#) that there are only denumerably many effectively calculable functions over any domain. Here we have rigorously

Proposition 4.2.9 *The classes Prim and Rec are denumerable.*

Proof. Immediate from Proposition 1.2.8, since each of these classes is the closure of a denumerable set of functions under denumerably many operations. $\square$

In the remainder of this section we develop some of the basic properties of these two classes of functions. In many cases the proofs for the two classes are identical, and we write (primitive) recursive to indicate that the statement may be read either with or without the adjective ‘primitive’. Of course, we have not yet shown that this is a meaningful distinction — that is, that there are functions which are recursive but not primitive recursive. This is true and will eventually be given two proofs, one in the exercises for [Section 4.4](#) and another in [Section 5.2](#).

We first note that the composition schema allows for rearranging arguments — for example,

$$\begin{aligned} F'(m_0, m_1, m_2) &= F(\text{Pr}_2^3(m_0, m_1, m_2), \text{Pr}_0^3(m_0, m_1, m_2), \text{Pr}_1^3(m_0, m_1, m_2)) \\ &= F(m_2, m_0, m_1). \end{aligned}$$

Hence if the function F is (primitive) recursive, so is the function F' with the arguments in a different order. In particular the schemas of primitive recursion and search may be applied with the special argument n in any position.

Proposition 4.2.10 *The following functions are primitive recursive:*

- (i) $+, \times, m^n$ and $!$ (factorial);

(ii) $\text{Pred}(0) = 0$; $\text{Pred}(n+1) = n$ (**predecessor function**);

(iii) $m \overset{\circ}{-} n = \begin{cases} m - n, & \text{if } n \leq m; \\ 0, & \text{otherwise} \end{cases}$ (**truncated subtraction**);

(iv) $|m - n|$;

(v) $\text{Sg}(0) = 0$; $\text{Sg}(m+1) = 1$ (**signum function**).

Proof. The computations are all straightforward, so we give only a few examples. The function $+$ is characterized by the equations

$$m + 0 = m \quad \text{and} \quad m + (n+1) = (m+n) + 1.$$

Thus if we set $G(m) = m$ and $H(p, m, n) = p + 1$, the function $F(m, n) = m + n$ arises from G and H by primitive recursion. Since G is Pr_0^1 and H is Sc_0^3 , they are both primitive recursive and thus so is F . For multiplication, we have similarly

$$m \times 0 = 0 \quad \text{and} \quad m \times (n+1) = (m \times n) + m,$$

so we take $G(m) = 0$ and $H(p, m, n) = p + m$. The other parts of (i) are proved analogously. For (ii), if $G() = 0$ (G is Cs_0^0) and $H(p, n) = n$ (H is Pr_1^2), then Pred arises from G and H by primitive recursion. For (iii), note that $m \overset{\circ}{-} 0 = m$ and $m \overset{\circ}{-} (n+1) = \text{Pred}(m \overset{\circ}{-} n)$. Parts (iv) and (v) follow similarly. $\square$

It is convenient to treat relations along with functions.

Definition 4.2.11 For any $k \in \omega$ and any relation $R \subseteq \omega^k$,

(i) for any $m_0, \dots, m_{k-1} \in \omega$,

$$\mathbf{K}_R(m_0, \dots, m_{k-1}) = \begin{cases} 0, & \text{if } \sim R(m_0, \dots, m_{k-1}); \\ 1, & \text{if } R(m_0, \dots, m_{k-1}). \end{cases}$$

$\mathbf{K}_R$ is called the **characteristic function** of R ;

(ii) R is a (**primitive**) **recursive relation** iff $\mathbf{K}_R$ is a (primitive) recursive function.

Proposition 4.2.12 (Boolean operations) For any $k \in \omega$ and any $R, S \subseteq \omega^k$, if R and S are (primitive) recursive, then also $R \cup S$, $R \cap S$ and $R \sim S$ are (primitive) recursive.

Proof. This follows from the equations

$$\begin{aligned} \mathbf{K}_{R \cap S}(m_0, \dots, m_{k-1}) &= \mathbf{K}_R(m_0, \dots, m_{k-1}) \times \mathbf{K}_S(m_0, \dots, m_{k-1}); \\ \mathbf{K}_{R \cup S}(m_0, \dots, m_{k-1}) &= \text{Sg}(\mathbf{K}_R(m_0, \dots, m_{k-1}) + \mathbf{K}_S(m_0, \dots, m_{k-1})); \\ \mathbf{K}_{R \sim S}(m_0, \dots, m_{k-1}) &= \mathbf{K}_R(m_0, \dots, m_{k-1}) \overset{\circ}{-} \mathbf{K}_S(m_0, \dots, m_{k-1}); \end{aligned}$$

which are easily verified by cases. $\square$

Proposition 4.2.13 *The relations $<$, $>$, $\leq$, $\geq$ and $=$ are all primitive recursive.*

Proof. Clearly $K_{<}(m, n) = \text{Sg}(n \overset{\circ}{-} m)$ and $K_{>}(m, n) = \text{Sg}(m \overset{\circ}{-} n)$. Then

$$K_{\leq}(m, n) = K_{<}(m, n + 1), \quad K_{\geq}(m, n) = K_{>}(m + 1, n),$$

and $K_{=}(m, n) = K_{\leq}(m, n) \times K_{\geq}(m, n)$. $\square$

Definition 4.2.14 For any $k \in \omega$ and any $(k + 2)$ -ary function G , we write

$$F(m_0, \dots, m_{k-1}, n) = \text{least } p < n [G(m_0, \dots, m_{k-1}, n, p) = 0]$$

to mean

$$F(m_0, \dots, m_{k-1}, n) = \begin{cases} p, & \text{if } p < n, \quad G(m_0, \dots, m_{k-1}, n, p) = 0, \\ & \text{and } (\forall q < p) \, G(m_0, \dots, m_{k-1}, n, q) \neq 0; \\ n, & \text{if there is no such } p. \end{cases}$$

We say that F arises from G by **bounded search**.

Proposition 4.2.15 *For any functions F and G , if G is (primitive) recursive and F arises from G by bounded search, then also F is (primitive) recursive.*

Proof. It is straightforward to verify by induction on n that if F and G are as in the statement, then $F(m_0, \dots, m_{k-1}, n) = H(m_0, \dots, m_{k-1}, n, n)$, where

$$\begin{aligned} H(m_0, \dots, m_{k-1}, r, 0) &= 0; \\ H(m_0, \dots, m_{k-1}, r, n + 1) &= H(m_0, \dots, m_{k-1}, r, n) \\ &\quad + \text{Sg}(G(m_0, \dots, m_{k-1}, r, H(m_0, \dots, m_{k-1}, r, n))). \end{aligned} \quad \square$$

Corollary 4.2.16 *For any (primitive) recursive function G and any (primitive) recursive relation R , the functions defined by*

$$\begin{aligned} \text{least } p \leq n [G(m_0, \dots, m_{k-1}, n, p) = 0] \\ &:= \text{least } p < n + 1 [G(m_0, \dots, m_{k-1}, n, p) = 0]; \\ \text{least } p < n [R(m_0, \dots, m_{k-1}, n, p)] \\ &:= \text{least } p < n [1 - K_R(m_0, \dots, m_{k-1}, n, p) = 0]; \\ \text{least } p \leq n [R(m_0, \dots, m_{k-1}, n, p)] \\ &:= \text{least } p < n + 1 [1 - K_R(m_0, \dots, m_{k-1}, n, p) = 0]; \end{aligned}$$

are all (primitive) recursive. $\square$

Proposition 4.2.17 (definition by cases) For any $l \in \omega$ and any (primitive) recursive functions $G_0, \dots, G_l$ and relations $R_0, \dots, R_{l-1}$ such that for all $i < l$, $i \neq j \implies R_i \cap R_j = \emptyset$, the function F defined by

$$F(m_0, \dots, m_{k-1}) := \begin{cases} G_0(m_0, \dots, m_{k-1}), & \text{if } R_0(m_0, \dots, m_{k-1}); \\ \vdots & \vdots \\ G_{l-1}(m_0, \dots, m_{k-1}), & \text{if } R_{l-1}(m_0, \dots, m_{k-1}); \\ G_l(m_0, \dots, m_{k-1}), & \text{otherwise;} \end{cases}$$

is (primitive) recursive.

Proof. This follows from the easily verified fact that

$$\begin{aligned} F(m_0, \dots, m_{k-1}) = & [G_0(m_0, \dots, m_{k-1}) \times \mathbf{K}_{R_0}(m_0, \dots, m_{k-1})] + \dots \\ & + [G_{l-1}(m_0, \dots, m_{k-1}) \times \mathbf{K}_{R_{l-1}}(m_0, \dots, m_{k-1})] \\ & + [G_l(m_0, \dots, m_{k-1}) \times \prod_{i < l} (1 - \mathbf{K}_{R_i}(m_0, \dots, m_{k-1}))]. \quad \square \end{aligned}$$

Proposition 4.2.18 (bounded quantification) For any (primitive) recursive relation S , the relations P and Q defined by

$$\begin{aligned} P(m_0, \dots, m_{k-1}, p) &: \iff (\exists q < p) S(m_0, \dots, m_{k-1}, p, q); \\ Q(m_0, \dots, m_{k-1}, p) &: \iff (\forall q < p) S(m_0, \dots, m_{k-1}, p, q); \end{aligned}$$

are also (primitive) recursive.

Proof. It suffices to observe that

$$\begin{aligned} P(m_0, \dots, m_{k-1}, p) &\iff \text{least } q < p [S(m_0, \dots, m_{k-1}, p, q)] < p \quad \text{and} \\ Q(m_0, \dots, m_{k-1}, p) &\iff \sim (\exists q < p) \sim S(m_0, \dots, m_{k-1}, p, q). \quad \square \end{aligned}$$

Proposition 4.2.19 (relational composition) For any $k, l \in \omega$, any l -ary (primitive) recursive relation S and any k -ary (primitive) recursive functions $H_0, \dots, H_{l-1}$, the relation P defined by

$$P(m_0, \dots, m_{k-1}) : \iff S(H_0(m_0, \dots, m_{k-1}), \dots, H_{l-1}(m_0, \dots, m_{k-1}))$$

is also (primitive) recursive.

Proof. This is immediate from the fact that under the stated conditions

$$\mathbf{K}_P(m_0, \dots, m_{k-1}) = \mathbf{K}_S(H_0(m_0, \dots, m_{k-1}), \dots, H_{l-1}(m_0, \dots, m_{k-1})). \quad \square$$

Proposition 4.2.20 *The relation m **divides** n — in symbols, $m|n$ — is primitive recursive.*

Proof. Note that $m|n \iff (\exists p < n+1) [m \times p = n]$. This is primitive recursive by several of the preceding results. $\square$

Definition 4.2.21 p_n denotes the n -th prime number: $p_0 = 2$, $p_1 = 3$, $p_2 = 5$, $\dots$

Proposition 4.2.22 *The function $n \mapsto p_n$ is primitive recursive.*

Proof. We recall Euclid's proof that there are infinitely many primes: given $p_0, \dots, p_n$, any prime divisor q of $(p_0 \times \dots \times p_n) + 1$ cannot be one of $p_0, \dots, p_n$, so must be greater than p_n . In particular, $p_{n+1} \leq q \leq p_n! + 1$, so the sequence p_n satisfies the following primitive recursion:

$$\begin{aligned} p_0 &= 2; \\ p_{n+1} &= \text{least } q < p_n! + 2 \text{ } (p_n < q \wedge \sim (\exists r < q) (\exists s < q) [r \times s = q]). \quad \square \end{aligned}$$

We are now in a position to introduce one of the most fundamental definitions, the coding of finite sequences. Let $<^\omega \omega := \bigcup \{ \omega^k : k \in \omega \}$ denote the set of all finite sequences of natural numbers. By elementary set theory, $<^\omega \omega$ is denumerable, hence can be mapped injectively into ω . What we need here is a refinement of this result which asserts essentially that this map and its inverses are primitive recursive. This can be done in many ways, and for most purposes the details of the next definition are important only to the extent that they guarantee the provisions of the following two propositions.

Definition 4.2.23

(i) $\langle \rangle^0 = 1$, and for any k and any $m_0, \dots, m_k$,

$$\langle m_0, \dots, m_k \rangle^{k+1} = p_0^{m_0+1} \times \dots \times p_k^{m_k+1}.$$

(ii) For any $s, i \in \omega$, $(s)_i = \text{least } m < s \text{ } (p_i^{m+2} \nmid s)$.

Proposition 4.2.24

(i) *The function $(s, i) \mapsto (s)_i$ and each of the functions $\langle \rangle^k$ is primitive recursive.*

(ii) *For any k , any $i < k$ and any $m_0, \dots, m_{k-1} \in \omega$,*

$$i < \langle m_0, \dots, m_{k-1} \rangle^k, \quad \text{each } m_i < \langle m_0, \dots, m_{k-1} \rangle^k \text{ and}$$

$$(\langle m_0, \dots, m_{k-1} \rangle^k)_i = m_i.$$

Proof. Immediate. $\square$

Remark 4.2.25 Note that numbers of the form $\langle m_0, \dots, m_{k-1} \rangle^k$ (**sequence numbers**) are of a very special form as the product of positive powers of an initial segment of the sequence of primes; $s = 2^2 \times 3^4 \times 5$ is such a number ($s = \langle 1, 3, 0 \rangle^3$), but $t = 2^3 \times 5^6 \times 7$ is not. Although $(s)_i$ is defined for all s and i , it has the intended meaning only for sequence numbers. Because it is generally determined by context, we shall from now on usually omit the superscript k on $\langle m_0, \dots, m_{k-1} \rangle^k$.

Definition 4.2.26 For all s and t ,

- (i) $\lg(s) = \text{least } k < s \text{ (} p_k \nmid s \text{)}$ (**length function**);
- (ii) $\text{Sq}(s) \iff \forall i < s [p_i | s \implies i < \lg(s)]$;
- (iii) $s * t = s \times p_{\lg(s)}^{(t)_0+1} \times \dots \times p_{\lg(s)+\lg(t)-1}^{(t)_{\lg(t)-1}+1}$
(**concatenation**).

Example 4.2.27 With s and t as in the preceding remark, we have $\lg(s) = 3$, $\lg(t) = 1$, $\text{Sq}(s) \sim \text{Sq}(t)$, and if $u = 2^3 \times 3^6 = \langle 2, 5 \rangle$, then $s * u = 2^2 \times 3^4 \times 5 \times 7^3 \times 11^6 = \langle 1, 3, 0, 2, 5 \rangle$.

Proposition 4.2.28 For any k, l , $m_0, \dots, m_{k-1}$ and $n_0, \dots, n_{l-1}$,

- (i) $\text{Sq}(\langle m_0, \dots, m_{k-1} \rangle)$ and $\lg(\langle m_0, \dots, m_{k-1} \rangle) = k$;
- (ii) $\langle m_0, \dots, m_{k-1} \rangle * \langle n_0, \dots, n_{l-1} \rangle = \langle m_0, \dots, m_{k-1}, n_0, \dots, n_{l-1} \rangle$.

Furthermore, the functions $\lg$ and $*$ and the relation Sq are primitive recursive. $\square$

We may now show that (primitive) recursiveness is preserved by a stronger, and much more useful, sort of recursion. If F arises from G and H by primitive recursion, $F(m_0, \dots, m_{k-1}, n+1)$ is calculated from the single value $F(m_0, \dots, m_{k-1}, n)$; in the scheme of course-of-values recursion, $F(m_0, \dots, m_{k-1}, n+1)$ may depend on the whole sequence of values

$$F(m_0, \dots, m_{k-1}, 0), \dots, F(m_0, \dots, m_{k-1}, n).$$

This is arranged by coding this sequence into a single number.

Definition 4.2.29 $\overline{F}(m_0, \dots, m_{k-1}, n) :=$

$$\langle F(m_0, \dots, m_{k-1}, 0), \dots, F(m_0, \dots, m_{k-1}, n-1) \rangle.$$

Theorem 4.2.30 (course-of-values recursion) For any function G , there exists a unique function F such that for all n and $m_0, \dots, m_{k-1}$,

$$F(m_0, \dots, m_{k-1}, n) = G(\overline{F}(m_0, \dots, m_{k-1}, n), m_0, \dots, m_{k-1}, n).$$

Furthermore, if G is (primitive) recursive, then also F is (primitive) recursive.

Proof. First note that the function $\overline{F}$ satisfies the following primitive recursion:

$$\begin{aligned} \overline{F}(m_0, \dots, m_{k-1}, 0) &= \langle \quad \rangle^0 \\ \overline{F}(m_0, \dots, m_{k-1}, n+1) &= \overline{F}(m_0, \dots, m_{k-1}, n) \\ &\quad * \langle G(\overline{F}(m_0, \dots, m_{k-1}, n), m_0, \dots, m_{k-1}, n) \rangle^1. \end{aligned}$$

Thus, if G is (primitive) recursive, so is $\overline{F}$. The (primitive) recursiveness of F follows from the fact that $F(m_0, \dots, m_{k-1}, n) = (\overline{F}(m_0, \dots, m_{k-1}, n+1))_n$. $\square$

Example 4.2.31 The *Fibonacci sequence* defined by $F(0) = F(1) = 1$ and $F(n+2) = F(n) + F(n+1)$ is primitive recursive. Let

$$G(s, n) := \begin{cases} 1, & \text{if } n < 2; \\ (s)_{n-2} + (s)_{n-1}, & \text{otherwise.} \end{cases}$$

Then easily G is primitive recursive and for all n , $F(n) = G(\overline{F}(n), n)$.

For various purposes it will be convenient to have a characterization of the class Rec of recursive functions which avoids the explicit closure under primitive recursion. The idea is that by including three particular functions defined by recursion as new initial functions, we can eliminate the schema.

Definition 4.2.32 Rec' is the smallest class of functions which includes the initial functions and the functions $+$, $\times$ and $\text{K}_<$ and is closed under composition and search. A relation R is Rec' iff K_R belongs to Rec' .

Proposition 4.2.33 $\text{Rec}' \subseteq \text{Rec}$.

Proof. Immediate by Rec' -induction. $\square$

To establish the converse inclusion, it will suffice to show that Rec' is closed under primitive recursion. This will require some development of the properties of Rec' . Note that membership in Rec' is not obvious for some very simple functions, such as Pred .

Lemma 4.2.34

- (i) The class of Rec' relations is closed under the Boolean operations.
- (ii) The relations $<$, $>$, $\leq$, $\geq$, $=$, and $|$ are in Rec' .
- (iii) Rec' is closed under bounded search and bounded quantification.

Proof. For (i) note first that $\text{Sg}(m) = K_{<}(0, m)$ so $\text{Sg} \in \text{Rec}'$. Then the equations

$$\begin{aligned} K_{R \cap S}(m_0, \dots, m_{k-1}) &= K_R(m_0, \dots, m_{k-1}) \times K_S(m_0, \dots, m_{k-1}); \\ K_{R \cup S}(m_0, \dots, m_{k-1}) &= \text{Sg}(K_R(m_0, \dots, m_{k-1}) + K_S(m_0, \dots, m_{k-1})); \\ K_{R \sim S}(m_0, \dots, m_{k-1}) &= K_R(m_0, \dots, m_{k-1}) \times K_{<}(K_S(m_0, \dots, m_{k-1}), 1); \end{aligned}$$

verify that the class of Rec' relations is closed under the Boolean operations. For (ii), the relation $<$ is Rec' by definition. Then since

$$m \leq n \iff m < (n + 1) \quad \text{so} \quad K_{\leq}(m, n) = K_{<}(m, \text{Sc}_0^1(n)),$$

also $\leq$ is Rec' . The relations $\geq$, $>$ and $=$ are Boolean combinations of these two and we apply (i). For (iii), let G be in Rec' and

$$F(m_0, \dots, m_{k-1}, n) = \text{least } p < n [G(m_0, \dots, m_{k-1}, n, p) = 0].$$

Let $H(m_0, \dots, m_{k-1}, n, p) = G(m_0, \dots, m_{k-1}, n, p) \times K_{<}(p, n)$. Then $H \in \text{Rec}'$, H satisfies the search condition

$$\forall m_0 \dots \forall m_{k-1} \forall n \exists p [H(m_0, \dots, m_{k-1}, n, p) = 0],$$

and

$$F(m_0, \dots, m_{k-1}, n) = \text{least } p [H(m_0, \dots, m_{k-1}, n, p) = 0].$$

Thus $F \in \text{Rec}'$. Finally, closure under bounded quantification and that the divisibility relation $|$ is Rec' follow as for Rec above. $\square$

In order to prove that Rec' is closed under primitive recursion, we shall need some facility for dealing with the encoding and decoding of finite sequences using Rec' functions. The functions $\langle \quad \rangle^k$ and $(s, i) \mapsto (s)_i$ make heavy use of primitive recursion and are therefore not suitable at the moment, although after we have finished we shall know that these functions are in Rec' . We must find new functions Sq^k and Cp (**component function**) for these operations; surprisingly we shall need only that $\text{Cp} \in \text{Rec}'$. We start with a Rec' function for coding ordered pairs.

Definition 4.2.35 $\text{Op}(m, n) := (m + n)^2 + m$.

Proposition 4.2.36 *Op is in Rec' and for any m, n, p and q ,*

$$\text{Op}(m, n) = \text{Op}(p, q) \implies m = p \text{ and } n = q.$$

Proof. Suppose that $\text{Op}(m, n) = \text{Op}(p, q)$. Then $m + n = p + q$, since if, say, $m + n < p + q$, we have

$$\text{Op}(m, n) < (m + n + 1)^2 \leq (p + q)^2 \leq \text{Op}(p, q),$$

contrary to hypothesis. It is then immediate that $m = p$ and $n = q$. $\square$

Definition 4.2.37 For any $m_0, \dots, m_{k-1}$,

$$\text{Sq}^k(m_0, \dots, m_{k-1}) = \text{Op}(u, n!),$$

where

$$q_i = \text{Op}(m_i, i) + 1; \quad n = \max\{q_i : i < k\}; \quad u = \prod_{i < k} (1 + [q_i \times n!]).$$

The properties of Sq^k and Cp depend on some simple facts from number theory. Fix $k, m_0, \dots, m_{k-1}$ and q_i, n and u defined as above.

Lemma 4.2.38 *If $q < r \leq n$, then $1 + [q \times n!]$ and $1 + [r \times n!]$ are relatively prime.*

Proof. Suppose that $q < r \leq n$ and a prime p divides both $1 + [q \times n!]$ and $1 + [r \times n!]$. Then p divides their difference $(r - q) \times n!$. Clearly p cannot divide $n!$, so p divides $(r - q)$ and thus $p \leq r$. But then $p \leq n$ so p does divide $n!$, a contradiction. $\square$

Lemma 4.2.39 *For any $q \leq n$, $(1 + [q \times n!]) \mid u$ iff for some $j < k$, $q = q_j$.*

Proof. If $q \leq n$ but q is distinct from all of the q_j ($j < k$), then by the preceding lemma, $1 + [q \times n!]$ is relatively prime to all of the numbers $1 + [q_j \times n!]$ ($j < k$), and hence to u ; in particular, $1 + [q \times n!]$ does not divide u . $\square$

Lemma 4.2.40 *For all $i < k$, $m_i = \text{least } m \left[(1 + [\text{Op}(m, i) + 1] \times n!) \mid u \right]$.*

Proof. Fix $i < k$ and note that m_i satisfies the condition on the right:

$$1 + [(\text{Op}(m_i, i) + 1) \times n!] = 1 + [q_i \times n!],$$

which does divide u . Suppose towards a contradiction that the condition is satisfied for some $m < m_i$. Let $q = \text{Op}(m, i) + 1$. Then $q < q_i \leq n$ and by hypothesis, $1 + [q \times n!]$ divides u . Thus by the preceding lemma, for some $j < k$, $q = q_j$. But then $\text{Op}(m, i) = \text{Op}(m_j, j)$ so $i = j$ and $m = m_j$ so $m = m_i$, contrary to assumption. Thus m_i is indeed the least m which satisfies the condition. $\square$

Definition 4.2.41

$$\text{Cp}(t, i) := \text{least } m < t \left(\exists u < t \exists v < t \right. \\ \left. \left[t = \text{Op}(u, v) \wedge (1 + [(\text{Op}(m, i) + 1) \times v]) \mid u \right] \right).$$

Lemma 4.2.42 $\text{Cp} \in \text{Rec}'$ and for all $k \in \omega$, all $m_0, \dots, m_{k-1} \in \omega$ and all $i < k$,

$$\text{Cp}(\text{Sq}^k(m_0, \dots, m_{k-1}), i) = m_i.$$

Proof. That $\text{Cp} \in \text{Rec}'$ is straightforward from the closure conditions for Rec' established above. Suppose that for some $m, k, m_0, \dots, m_{k-1}$ and $i < k$, $\text{Cp}(\text{Sq}^k(m_0, \dots, m_{k-1}), i) = m$. Fix $u, v < \text{Sq}^k(m_0, \dots, m_{k-1})$ such that $\text{Sq}^k(m_0, \dots, m_{k-1}) = \text{Op}(u, v)$ and $(1 + [(\text{Op}(m, i) + 1) \times v])$ divides u . Then with q_i and n as above, this u is also as above and $v = n!$. Thus m is as in the preceding lemma, so $m = m_i$. $\square$

The idea for the proof of the following theorem is really just that of the proof of Theorem 1.2.12.

Theorem 4.2.43 Rec' is closed under primitive recursion.

Proof. Suppose F arises from G and H by primitive recursion. Now

$$F(m_0, \dots, m_{k-1}, n) = p \iff \exists q_0 \dots \exists q_n [q_0 = G(m_0, \dots, m_{k-1}) \wedge \\ (\forall i < n) q_{i+1} = H(q_i, m_0, \dots, m_{k-1}, i) \wedge q_n = p]$$

(think of q_i as $F(m_0, \dots, m_{k-1}, i)$). Hence

$$F(m_0, \dots, m_{k-1}, n) = p \iff \exists s [\text{Cp}(s, 0) = G(m_0, \dots, m_{k-1}) \wedge \\ (\forall i < n) \text{Cp}(s, i+1) = H(\text{Cp}(s, i), m_0, \dots, m_{k-1}, i) \wedge \text{Cp}(s, n) = p].$$

Thus if we set

$$I(m_0, \dots, m_{k-1}, n) = \text{least } s [\text{Cp}(s, 0) = G(m_0, \dots, m_{k-1}) \wedge \\ (\forall i < n) \text{Cp}(s, i+1) = H(\text{Cp}(s, i), m_0, \dots, m_{k-1}, i)],$$

then the search condition holds for the relation in square brackets and $F(m_0, \dots, m_{k-1}, n) = \text{Cp}(I(m_0, \dots, m_{k-1}, n), n)$. By the closure conditions established above, I is in Rec' , hence so is F . Note that although we need the existence of the function Sq^{n+1} in order to show that when $F(m_0, \dots, m_{k-1}, n) = p$, there exists s as above, it is not necessary to know at this stage that Sq^{n+1} is in Rec' . $\square$

Theorem 4.2.44 $\text{Rec} = \text{Rec}'$.

Proof. By Rec-induction, $\text{Rec} \subseteq \text{Rec}'$, and the converse inclusion was established above. $\square$

Collateral Reading. There are dozens, perhaps hundreds, of books which present the basic definitions and results of recursion (computability) theory contained in this and the next section. These include all of the logic texts which cover the Gödel theorems, but also many more specialized ones which go much further in this direction including much of the material from our [Chapter 8](#) and a great deal more. The classic text is [Rog], and a favorite of mine has always been [Cut]. [Hin] develops recursiveness for functions with both number and function arguments along lines very similar to the presentation here. The most encyclopedic version is [Od89] and [Od99] (with a third volume promised). In recent years there has been a movement to use the term “computable” in place of “recursive” and “computably enumerable” in place of “recursively enumerable”; see Soare’s contribution to [Gri] for some interesting history and a rationale for the change.

Exercises

Exercise 4.2.45 Prove that the inductive definitions of Prim and Rec do not have unique readability.

Exercise 4.2.46 A finite set $A = \{m_0, \dots, m_k\}$ may be represented by the number $\text{code}(A) = 2^{m_0} + \dots + 2^{m_k}$; $\text{code}(\emptyset) = 0$. Show that there are primitive recursive functions un and int such that for all finite A and B ,

$$\begin{aligned}\text{code}(A \cup B) &= un(\text{code}(A), \text{code}(B)) \quad \text{and} \\ \text{code}(A \cap B) &= int(\text{code}(A), \text{code}(B)).\end{aligned}$$

Exercise 4.2.47 Evaluate $(17)_0$, $(432)_1$ and $(750)_2$.

Exercise 4.2.48 Let

$$F(m) = \begin{cases} 0, & \text{if the decimal expansion of } \pi \text{ has a run of at least } m \text{ 7s;} \\ 1, & \text{otherwise.} \end{cases}$$

Prove that F is primitive recursive.

Exercise 4.2.49 Show that if G is (primitive) recursive and $F(m, n) = \max_{p \leq n} G(m, p)$, then also F is (primitive) recursive.

Exercise 4.2.50 Show that if G is (primitive) recursive and $F(m, n) = \sum_{p \leq n} G(m, p)$, then also F is (primitive) recursive.

Exercise 4.2.51 The function Op above does not have all natural numbers in its image. Show that the functions

$$\begin{aligned}\text{Op}'(m, n) &:= \frac{(m+n)(m+n+1)}{2} + m \quad \text{and} \\ \text{Op}''(m, n) &:= \frac{(m+n)(m+n+1)}{2} + n\end{aligned}$$

are alternative ordered pair functions in Rec' which are onto ω .

Exercise 4.2.52 (*simultaneous recursion*) Show that for any functions G and H ,

- (i) there exist unique functions F_0 and F_1 such that for all m and n

$$\begin{aligned}F_0(m, n) &= G(\overline{F}_0(m, n), \overline{F}_1(m, n), m, n) \quad \text{and} \\ F_1(m, n) &= H(\overline{F}_0(m, n), \overline{F}_1(m, n), m, n);\end{aligned}$$

- (ii) if G and H are (primitive) recursive, then also both F_0 and F_1 are (primitive) recursive.

Exercise 4.2.53 Prove that for any functions G , H and I ,

- (i) there exists a unique function F such that for all m and n ,

$$F(m, 0) = G(m) \quad \text{and} \quad F(m, n+1) = H(F(I(m), n), m, n);$$

- (ii) if G , H and I are (primitive) recursive, also F is (primitive) recursive.

Note that primitive recursion is this schema with $I(m) = m$.

Exercise 4.2.54 Let $G^{(p)}$ denote the function $\underbrace{G \circ G \circ \cdots \circ G}_p$ — that is, G composed with itself p times. Let $F(m) = G^{(H(m))}(m)$. Show that if G and H are both (primitive) recursive, then so is F .

4.3 Recursively enumerable sets and relations

We turn next to the precise counterpart of the intuitive notion of effective enumerability.

Definition 4.3.1 A set $X \subseteq \omega$ is **recursively enumerable** or **r.e.** iff $X = \emptyset$ or for some unary recursive function F , $X = \text{Im}(F)$.

The name is due, of course, to the fact that the function F **enumerates** X . Note that it is not required that the enumeration be 1-1 or in increasing order. It is also easy to see that the class of sets is unchanged if we remove the adjective “unary” from the definition, since if F is binary and $X = \text{Im}(F)$, then also X is the image of the unary function $G(m) := F((m)_0, (m)_1)$. We first establish some simple properties of this notion, several of which correspond closely to the parts of Proposition 1.5.14.

Proposition 4.3.2 Every recursive set is recursively enumerable.

Proof. If $X \neq \emptyset$ is recursive, let p be a fixed element of X and let F be the function defined by

$$F(m) = \begin{cases} m, & \text{if } m \in X; \\ p, & \text{otherwise.} \end{cases}$$

Then F is recursive (definition by cases) and $X = \text{Im}(F)$, so X is recursively enumerable. $\square$

Proposition 4.3.3 For any $X \subseteq \omega$, X is recursively enumerable iff for some recursive binary relation R , $X = \{m : \exists p R(m, p)\}$.

Proof. Suppose first that X is recursively enumerable. If $X = \emptyset$, we may take $R = \emptyset$. If $X = \text{Im}(F)$, take R to be the converse graph of F : $R = \{(m, p) : F(p) = m\}$. Conversely, if $X = \{m : \exists p R(m, p)\}$, with R recursive and $X \neq \emptyset$, let q be a fixed element of X and set

$$F(m) = \begin{cases} (m)_0, & \text{if } R((m)_0, (m)_1); \\ q, & \text{otherwise.} \end{cases}$$

Then F is recursive and $X = \text{Im}(F)$, so X is recursively enumerable. $\square$

When $m \in X$ because $R(m, p)$, we say that p is a **witness** for m ; of course, an element may have many witnesses. This representation of a recursively enumerable set is closely related to the intuitive notion of semi-decidability. An algorithm to semi-decide whether or not $\exists p R(m, p)$ proceeds by checking successively whether $R(m, 0)$, $R(m, 1)$ etc. If the correct answer is ‘yes’, this procedure will discover it, but if it is ‘no’, the computation will go on forever. Furthermore, it suggests a corresponding notion for relations.

Definition 4.3.4 A relation $P \subseteq \omega^k$ is **semi-recursive** iff for some recursive relation R and all $m_0, \dots, m_{k-1}$,

$$P(m_0, \dots, m_{k-1}) \iff \exists p R(m_0, \dots, m_{k-1}, p).$$

We say that P is a **projection** of R and write it also as $\exists R$. P is **co-semi-recursive** iff its complement $\omega^k \sim P$ is semi-recursive.

Corollary 4.3.5 *A set $X \subseteq \omega$ is recursively enumerable iff it is semi-recursive. $\square$*

We shall generally use the term ‘recursively enumerable’ for sets and ‘semi-recursive’ to include both sets and relations, but at times we shall use the term ‘recursively enumerable’ for both; this is justified by the first exercise below. The classes of semi-recursive sets and relations share some but far from all properties with the classes of (primitive) recursive relations. Note, however, that the proofs are considerably different.

Proposition 4.3.6

- (i) *Every recursive relation is semi-recursive.*
- (ii) *If P and Q are semi-recursive, then so are $P \cup Q$ and $P \cap Q$.*
- (iii) *The class of semi-recursive relations is closed under*
 - (a) *bounded existential and bounded universal quantification;*
 - (b) *relational composition with recursive functions;*
 - (c) *unbounded existential quantification.*

Proof. For (i), suppose that R is a recursive k -ary relation and let S be a $(k+1)$ -ary relation defined by

$$S(m_0, \dots, m_{k-1}, p) \iff R(m_0, \dots, m_{k-1}).$$

Clearly (by composition) S is still recursive and R is the projection of S , so R is semi-recursive. For (ii), let P and Q be semi-recursive k -ary relations. By the preceding proposition, there exist recursive relations R and S such that

$$\begin{aligned} P &= \{ (m_0, \dots, m_{k-1}) : \exists p R(m_0, \dots, m_{k-1}, p) \} \quad \text{and} \\ Q &= \{ (m_0, \dots, m_{k-1}) : \exists q S(m_0, \dots, m_{k-1}, q) \}. \end{aligned}$$

Then easily

$$\begin{aligned} P \cup Q &= \{ (m_0, \dots, m_{k-1}) : \exists p (R \cup S)(m_0, \dots, m_{k-1}, p) \} \quad \text{and} \\ P \cap Q &= \{ (m_0, \dots, m_{k-1}) : \exists p [R(m_0, \dots, m_{k-1}, (p)_0) \text{ and} \\ &\quad S(m_0, \dots, m_{k-1}, (p)_1)] \}, \end{aligned}$$

so $P \cup Q$ and $P \cap Q$ are semi-recursive.

For (iii), suppose that P is semi-recursive with

$$P(m_0, \dots, m_{k-1}, n) \iff \exists p R(m_0, \dots, m_{k-1}, n, p)$$

for a recursive relation R . Then the desired conclusion follows from the equivalences

$$\begin{aligned} (\exists n < q) P(m_0, \dots, m_{k-1}, n) &\iff \exists p (\exists n < q) R(m_0, \dots, m_{k-1}, n, p); \\ (\forall n < q) P(m_0, \dots, m_{k-1}, n) &\iff \exists r (\forall n < q) R(m_0, \dots, m_{k-1}, n, (r)_n); \\ P(H_0(m_0, \dots, m_{k-1}), \dots) &\iff \exists p R(H_0(m_0, \dots, m_{k-1}), \dots, p); \\ \exists n P(m_0, \dots, m_{k-1}, n) &\iff \exists r R(m_0, \dots, m_{k-1}, (r)_0, (r)_1). \end{aligned}$$

These are all easy to verify — for example, if

$$(\forall n < q) P(m_0, \dots, m_{k-1}, n) \quad \text{so} \quad (\forall n < q) \exists p R(m_0, \dots, m_{k-1}, n, p),$$

for each $n < q$, let p_n be least such that $R(m_0, \dots, m_{k-1}, n, p_n)$ and take $r = \langle p_0, \dots, p_{q-1} \rangle$. By the closure properties of the class of recursive relations, the right-hand expressions define semi-recursive relations. $\square$

Proposition 4.3.7 *A relation P is recursive iff P is both semi-recursive and co-semi-recursive.*

Proof. If P is recursive, then so is $\sim P$, so both are semi-recursive. Suppose now that both P and $\sim P$ are semi-recursive, and let R and S be recursive relations such that

$$\begin{aligned} P &= \{ (m_0, \dots, m_{k-1}) : \exists p R(m_0, \dots, m_{k-1}, p) \} \quad \text{and} \\ \sim P &= \{ (m_0, \dots, m_{k-1}) : \exists p S(m_0, \dots, m_{k-1}, p) \}. \end{aligned}$$

Then the function F defined by

$$F(m_0, \dots, m_{k-1}) = \text{least } p [R(m_0, \dots, m_{k-1}, p) \text{ or } S(m_0, \dots, m_{k-1}, p)]$$

is recursive and

$$P(m_0, \dots, m_{k-1}) \iff R(m_0, \dots, m_{k-1}, F(m_0, \dots, m_{k-1})),$$

so P is recursive. $\square$

We have not yet answered a very natural question: is every semi-recursive relation also recursive? If this were true, then the three classes of recursive, semi-recursive and co-semi-recursive relations all coincide and the preceding proposition would be trivial. Of course, this is not the case, and we showed the corresponding fact for effective enumerability in Corollary 4.1.15 assuming the Representability Hypothesis, but the final proof will have to wait for the developments of [Section 4.6](#), and we state it here as

Fact 4.3.8 For every $k > 0$ there exist k -ary relations which are semi-recursive but not recursive. $\square$

Corollary 4.3.9

- (i) *The class of semi-recursive relations is not closed under complementation.*
- (ii) *The class of recursive relations is not closed under unbounded existential quantification.*
- (iii) *Neither class is closed under unbounded universal quantification.*

Proof. Left as an exercise. $\square$

Corollary 4.3.10 *There exists a relation P such that $\forall m (\exists n < 2) P(m, n)$ and P is semi-recursive but the function $F(m) = \text{least } n < 2[P(m, n)]$ is not recursive.*

Proof. Let A be any set which is recursively enumerable but not recursive, hence also $\omega \sim A$ is not recursive. Set

$$P(m, n) \iff [m \in A \text{ and } n = 0] \text{ or } n = 1.$$

P is semi-recursive and $\forall m (\exists n < 2) P(m, n)$, but F is the characteristic function of $\omega \sim A$ so is not recursive. $\square$

The ‘least’ operator may be thought of as a sort of effective choice principle. Given a recursive relation $P \subseteq \omega \times \omega$, let $A_m = \{n : P(m, n)\}$. Then $F(m) = \text{least } n P(m, n)$ effectively selects an element (in fact the smallest element) of A_m . The preceding corollary tells us that when P is only semi-recursive, this is no longer possible. However if we require only that $F(m)$ be *some* element of A_m , not necessarily the least, then it is possible. The idea is to select n for which there exists a witness p such that the pair $\langle n, p \rangle$ is the least among all such pairs.

Selection Theorem 4.3.11 *For any semi-recursive relation P such that*

$$\forall m_0 \cdots \forall m_{k-1} \exists n P(m_0, \dots, m_{k-1}, n),$$

there exists a recursive function F such that

$$\forall m_0 \cdots \forall m_{k-1} P(m_0, \dots, m_{k-1}, F(m_0, \dots, m_{k-1})).$$

Proof. If R is a recursive relation such that

$$P(m_0, \dots, m_{k-1}, n) \iff \exists p R(m_0, \dots, m_{k-1}, n, p),$$

we set

$$F(m_0, \dots, m_{k-1}) := (\text{least } q [R(m_0, \dots, m_{k-1}, (q)_0, (q)_1)])_0. \square$$

Definition 4.3.12 The **graph** of a k -ary function F is the $(k+1)$ -ary relation

$$\text{Graph}(F) = \{ (m_0, \dots, m_{k-1}, n) : F(m_0, \dots, m_{k-1}) = n \}.$$

Corollary 4.3.13 For any (total) function F , the following are equivalent:

- (i) F is a recursive function;
- (ii) $\text{Graph}(F)$ is a recursive relation;
- (iii) $\text{Graph}(F)$ is a semi-recursive relation.

Proof. The implications (i) $\implies$ (ii) $\implies$ (iii) are immediate and (iii) $\implies$ (i) follows from the Selection Theorem. $\square$

Another natural question to ask here is which relations are representable in the form

$$\exists p R(m_0, \dots, m_{k-1}, p) \text{ with } R \text{ primitive recursive.}$$

Perhaps surprisingly, all semi-recursive relations are so expressible, but we shall not be in a position to prove this until [Section 4.4](#).

In defining the class of recursive functions, we restricted attention to **total functions** — those with domain equal to some ω^k — as opposed to **partial functions** whose domain is a possibly proper subset of ω^k . For primitive recursion this was automatic; the initial functions are total and the closure operations of composition and primitive recursion applied to total functions produce total functions. The same remarks apply to the class of recursive functions, but only because of the requirement that the search operator be applied only to functions which satisfy the search condition. From the perspective of (intuitive) effective calculability, this is somewhat unnatural; given any effectively calculable (total) function $G : \omega \times \omega \rightarrow \omega$, for each m we may successively compute $G(m, 0), G(m, 1), \dots$ stopping only when we find n such that $G(m, n) = 0$. If $\exists n[G(m, n) = 0]$, this procedure will discover the least such n , but otherwise will go on forever. We view this as an algorithm for computing a partial function. This notion is common in practical computer programming; many programs will terminate with an intended result only for input data of certain type, but for other inputs will enter an infinite loop and thus in principle never terminate.

The class of partial recursive functions is the mathematical counterpart of this intuitive notion. To give the definition, we modify the definition of the class of recursive functions to allow for such partially defined functions in a way which ensures that the partial recursive functions are all effectively calculable. This involves a slight reinterpretation of the operations of composition and search. For the rest of this section $F, G, \dots$ denote functions with domain a subset of some ω^k and image a subset of ω . We write

$$F(m_0, \dots, m_{k-1}) \downarrow \text{ to denote that } (m_0, \dots, m_{k-1}) \in \text{Dom}(F)$$

and say that F is **defined** on $(m_0, \dots, m_{k-1})$; otherwise, $F(m_0, \dots, m_{k-1}) \uparrow$. We write

$$F(m_0, \dots, m_{k-1}) \simeq G(n_0, \dots, n_{l-1})$$

to mean that either both values are defined and are equal or both are undefined. In particular, $F(m_0, \dots, m_{k-1}) \simeq q$ implies that the value is defined.

Consider first a composition

$$F(m_0, \dots, m_{k-1}) \simeq G(H_0(m_0, \dots, m_{k-1}), \dots, H_{l-1}(m_0, \dots, m_{k-1}))$$

of partial functions. By the natural algorithm $F(m_0, \dots, m_{k-1})$ is defined with value q iff for some $p_0, \dots, p_{l-1}$,

$$\text{for all } j < l, \quad H_j(m_0, \dots, m_{k-1}) \simeq p_j \quad \text{and} \quad G(p_0, \dots, p_{l-1}) \simeq q.$$

Hence we take $\text{Cmp}_l^k(G, H_0, \dots, H_{l-1})$ to be the partial function with this domain and value.

The primitive recursion operator Rec^{k+1} needs no change except to interpret the composition in its definition in the manner just specified. Note that this implies that when $F = \text{Rec}^{k+1}(G, H)$, for each $m_0, \dots, m_{k-1}$, $F(m_0, \dots, m_{k-1}, n)$ is defined either for all n or only for n less than some n_0 .

Finally, we will say that F arises from a partial function G by search — in symbols, $F = \text{Se}^k(G)$ or

$$F(m_0, \dots, m_{k-1}) \simeq \text{least } n [G(m_0, \dots, m_{k-1}, n) \simeq 0]$$

— iff for each $m_0, \dots, m_{k-1}$, $F(m_0, \dots, m_{k-1}) \simeq q$ exactly when

$$G(m_0, \dots, m_{k-1}, q) \simeq 0 \text{ and for all } p < q,$$

$$G(m_0, \dots, m_{k-1}, p) \text{ is defined with a non-zero value.}$$

Note that the second clause is necessary to ensure that the intuitive computation procedure discussed above terminates.

With these conventions, we have

Definition 4.3.14 The class of **partial recursive functions** is the smallest class of partial functions which contains the initial functions and is closed under composition, primitive recursion and search.

It is clear that each partial recursive function is intuitively effectively calculable in the natural sense that there exists an algorithm which leads to a value exactly for elements of the domain. Of course, the conventions concerning composition and search were chosen partially so that this would be true.

One reason for introducing this class here is that there is a very strong connection between partial recursive functions and semi-recursive relations.

Partial Selection Theorem 4.3.15 *For any semi-recursive relation P , there exists a partial recursive function F such that*

$$\text{Dom}(F) = \{ (m_0, \dots, m_{k-1}) : \exists n P(m_0, \dots, m_{k-1}, n) \},$$

and for all $(m_0, \dots, m_{k-1}) \in \text{Dom}(F)$, $F(m_0, \dots, m_{k-1}) = F(m_0, \dots, m_{k-1})$.

Proof. If R is a recursive relation such that

$$P(m_0, \dots, m_{k-1}, n) \iff \exists p R(m_0, \dots, m_{k-1}, n, p),$$

we set as before

$$F(m_0, \dots, m_{k-1}) := (\text{least } q [R(m_0, \dots, m_{k-1}, (q)_0, (q)_1)])_0. \quad \square$$

Extending the notation above, the graph of a partial function F is the set

$$\text{Graph}(F) = \{ (m_0, \dots, m_{k-1}, n) : F(m_0, \dots, m_{k-1}) \simeq n \}.$$

Theorem 4.3.16 *For any relation P and any partial function F ,*

- (i) F is partial recursive iff $\text{Graph}(F)$ is semi-recursive;
- (ii) P is semi-recursive iff for some partial recursive function G , $P = \text{Dom}(G)$.

Proof. We first prove (i)($\implies$) by induction on the class of partial recursive functions. If F is one of the initial functions, F is (total) recursive, so $\text{Graph}(F)$ is semi-recursive by Corollary 4.3.13. Suppose that $F = \text{Cmp}_l^k(G, H_0, \dots, H_{l-1})$, and assume as induction hypothesis that $\text{Graph}(G)$ and all $\text{Graph}(H_j)$ are semi-recursive. Then since

$$F(m_0, \dots, m_{k-1}) \simeq n \iff \exists p_0 \cdots \exists p_{l-1} \left[\bigwedge_{j < l} H_j(m_0, \dots, m_{k-1}) \simeq p_j \text{ and } G(p_0, \dots, p_{l-1}) \simeq n \right],$$

it follows from the closure properties that $\text{Graph}(F)$ is also semi-recursive. If $F = \text{Rec}^{k+1}(G, H)$, then the semi-recursive nature of $\text{Graph}(F)$ follows from the characterization

$$\begin{aligned} F(m_0, \dots, m_{k-1}, n) \simeq p &\iff \\ \exists s [\lg(s) = n + 1 \text{ and } G(m_0, \dots, m_{k-1}) \simeq (s)_0 \text{ and} \\ (\forall i < n) H((s)_i, m_0, \dots, m_{k-1}, i) \simeq (s)_{i+1} \wedge (s)_n = p]. \end{aligned}$$

If $F = \text{Se}^k(G)$, we have similarly

$$F(m_0, \dots, m_{k-1}) \simeq n \iff G(m_0, \dots, m_{k-1}, n) \simeq 0 \quad \text{and} \\ (\forall i < n) \exists p [G(m_0, \dots, m_{k-1}, i) \simeq p \quad \text{and} \quad p \neq 0].$$

(i)($\Leftarrow$) and (ii)($\Rightarrow$) are immediate from the Partial Selection Theorem, and (ii)($\Leftarrow$) follows from (i) via the closure properties of the class of semi-recursive relations, since

$$(m_0, \dots, m_{k-1}) \in \text{Dom}(G) \iff \exists n G(m_0, \dots, m_{k-1}, n) \simeq n \\ \iff \exists n [(m_0, \dots, m_{k-1}, n) \in \text{Graph}(G)]. \quad \square$$

Corollary 4.3.17 *The class of semi-recursive relations has the **uniformization property** — that is, for any $(l+1)$ -ary semi-recursive relation R , there exists a semi-recursive relation $\hat{R} \subseteq R$ with the same domain which is the graph of a partial function. $\square$*

Corollary 4.3.18 *For any partial function F , F is recursive iff F is partial recursive and total.*

Proof. The implication ($\Rightarrow$) is immediate by Rec-induction. For ($\Leftarrow$), by (i)($\Rightarrow$) of the theorem, $\text{Graph}(F)$ is semi-recursive, hence by Corollary 4.3.13, F is recursive. $\square$

Corollary 4.3.19 *For any set $X \subseteq \omega$, the following are equivalent:*

- (i) X is recursively enumerable;
- (ii) $X = \text{Im}(F)$ for some partial recursive function F ;
- (iii) $X = \text{Dom}(F)$ for some partial recursive unary function F .

Proof. (i) $\Rightarrow$ (ii) and (i) $\Leftarrow$ (iii) are immediate from Definition 4.3.1, Corollary 4.3.5, the preceding theorem and the observation that the empty set $\emptyset$ is $\text{Im}(F)$ for the partial recursive completely undefined function. For (ii) $\Rightarrow$ (i), if $X = \text{Im}(F)$, then $X = \{n : \exists m F(m) \simeq n\}$, which is semi-recursive by (i) of the Theorem, hence recursively enumerable. $\square$

Corollary 4.3.20 *The class of semi-recursive relations is closed under relational composition with partial recursive functions.*

Proof. Let S be a semi-recursive relation, $H_0, \dots, H_{l-1}$ partial recursive functions and

$$P(m_0, \dots, m_{k-1}) : \iff S(H_0(m_0, \dots, m_{k-1}), \dots, H_{l-1}(m_0, \dots, m_{k-1})).$$

Fix a partial recursive function G such that $S = \text{Dom}(G)$ and set

$$F(m_0, \dots, m_{k-1}) \simeq G(H_0(m_0, \dots, m_{k-1}), \dots, H_{l-1}(m_0, \dots, m_{k-1})).$$

Then $P = \text{Dom}(F)$ and is therefore semi-recursive. Alternatively note that $P(m_0, \dots, m_{k-1})$ is equivalent to

$$\begin{aligned} \exists p_0 \cdots \exists p_{l-1} [H_0(m_0, \dots, m_{k-1}) \simeq p_0 \wedge \cdots \wedge \\ H_{l-1}(m_0, \dots, m_{k-1}) \simeq p_{l-1} \wedge S(p_0, \dots, p_{l-1})]. \quad \square \end{aligned}$$

Exercises

Exercise 4.3.21 Show that the notions of recursive enumerability and semi-recursive are really the same: a k -ary relation R is semi-recursive iff

$$\langle R \rangle := \{ \langle m_0, \dots, m_{k-1} \rangle : R(m_0, \dots, m_{k-1}) \}$$

is a recursively enumerable set.

Exercise 4.3.22 Prove Corollary 4.3.9.

The following is a simple but very useful notion.

Definition 4.3.23 For any $X, Y \subseteq \omega$, X is **many-one reducible** to Y — in symbols, $X \leq_m Y$ — iff for some recursive function F , $X = F^{-1}(Y)$ — that is, $X = \{ m : F(m) \in Y \}$.

The terminology here arises from the idea that the function F allows questions $m \in X?$ about membership in X to be *reduced* to questions $F(m) \in Y?$ about membership in Y . A parallel notion we do not consider here is **one-one reducibility** for which the function F is required to be injective.

Exercise 4.3.24 Prove that $\leq_m$ is reflexive and transitive but not antisymmetric.

Exercise 4.3.25 Show that if $X \leq_m Y$, then

- (i) if Y is recursive, then also X is recursive;
- (ii) if Y is semi-recursive, then also X is semi-recursive.

Compare Proposition 4.1.6.

Exercise 4.3.26 For $X, Y \subseteq \omega$, let $X \equiv_m Y \iff X \leq_m Y$ and $Y \leq_m X$. We say that X is **many-one equivalent** to Y . Prove that $\equiv_m$ is an equivalence relation.

Exercise 4.3.27 The equivalence classes $\text{dg}_m(X)$ of subsets X of ω under the relation $\equiv_m$ of the preceding exercise are called **many-one degrees** or **m-degrees**. They inherit a natural partial ordering defined by

$$\text{dg}_m(X) \leq \text{dg}_m(Y) \iff X \leq_m Y.$$

Show that

- (i) $\leq$ is a partial ordering well-defined on m-degrees;
- (ii) there are exactly two incomparable minimal m-degrees;
- (iii) any two m-degrees have a least upper bound;
- (iv) each m-degree has at most countably many predecessors;
- (v) there is no maximal m-degree;
- (vi) even excluding the minimal m-degrees, $\leq$ is not a linear ordering;
- (vii) there exist $2^{\aleph_0}$ m-degrees.

The structure of the m-degrees will be further studied in [Section 8.1](#).

Exercise 4.3.28 Show that for any infinite recursively enumerable set X ,

- (i) X is the image of some injective (one-to-one) recursive function;
- (ii) if X is the image of some strictly increasing recursive function, then X is recursive;
- (iii) X has an infinite recursive subset.

Exercise 4.3.29 For any relation $R \subseteq \omega^k$ and any $m_0, \dots, m_{k-1}, n \in \omega$, let

$$K_R^+(m_0, \dots, m_{k-1}) \simeq n \iff R(m_0, \dots, m_{k-1}) \text{ and } n = 1.$$

K_R^+ is called the **positive characteristic function** of R . Show that R is semi-recursive iff K_R^+ is partial recursive.

Exercise 4.3.30 A class $\mathcal{C}$ of sets has the **separation property** iff for any $X, Y \in \mathcal{C}$ with $X \cap Y = \emptyset$, there exists Z such that both Z and $\omega \setminus Z$ belong to $\mathcal{C}$ and Z **separates** X and Y — that is, $X \subseteq Z$ and $Y \cap Z = \emptyset$. $\mathcal{C}$ has the **reduction property** iff for any $X, Y \in \mathcal{C}$, there exist $X^- \subseteq X$ and $Y^- \subseteq Y$ such that also $X^-, Y^- \in \mathcal{C}$, $X^- \cup Y^- = X \cup Y$, and $X^- \cap Y^- = \emptyset$. Show that

- (i) the class of r.e. sets has the reduction property;
- (ii) the class of co-r.e. sets has the separation property.

Hint: For (ii), note that under the hypotheses, the sets $\omega \setminus X$ and $\omega \setminus Y$ are r.e. sets whose union is all of ω and apply (i). Compare the remark following the Boolean Interpolation Theorem 3.5.35.

4.4 Gödel numbering

The theory of recursive functions and relations is a fascinating and intricate theory in its own right, and we shall continue its development in [Chapter 8](#). However, our focus here is on using the theory to provide a precise and rigorous definition to replace the intuitive notions of decidability and effective enumerability for formal languages and theories.

We have set the stage for this in [Section 4.1](#), and it seems at first reasonable to proceed as follows. Given a fixed effective enumeration

$$\chi_0, \chi_1, \dots, \chi_m, \dots$$

of the set of L_Ω -formulas, for any set Γ of formulas, set

$$\Gamma^\# := \{m : \chi_m \in \Gamma\}.$$

Now call an L_Ω -theory T **recursively decidable** (**recursively enumerable**) (temporary definitions!) iff $T^\#$ is a recursive (recursively enumerable) set of natural numbers and **recursively axiomatizable** iff $T = \text{Th}(\Gamma)$ for some set Γ of sentences such that $\Gamma^\#$ is recursive. The two mappings

$$p \mapsto \chi_p \quad \text{and} \quad \phi \mapsto \#(\phi) := \text{least } p[\chi_p = \phi]$$

are intuitively calculable so T is intuitively decidable iff $T^\#$ is intuitively decidable. Hence, by Proposition 4.2.8,

$$T \text{ recursively decidable} \implies T \text{ intuitively decidable,}$$

and assuming Church's Thesis, the two notions are equivalent.

Consider how we might now try to prove a new version of Theorem 4.1.7 to the effect that $\text{Th}(\Omega)$ is recursively undecidable and hence not recursively axiomatizable. The first task is to establish, corresponding to DH(ii), that every recursive subset of ω is definable over Ω ; this is relatively simple and will be done at the beginning of [Section 4.5](#). Now, following the proof of Theorem 4.1.7, suppose, towards a contradiction, that $\text{Th}(\Omega)$ is recursively decidable and define a set X by

$$\text{for all } m, \quad m \in X \iff \chi_m(\dot{m}) \in \text{Th}(\Omega).$$

To make use of the hypothesis, we need to reformulate this as

$$\text{for all } m, \quad m \in X \iff \text{Sb}(m) \in \text{Th}(\Omega)^\#,$$

where $\text{Sb}(m) = \#(\chi_m(\dot{m}))$. Now we can conclude that X is recursive and finish the proof much as before if we can show that Sb is a recursive function.

Of course, we did argue in Corollary 4.1.5 that **Sb** is effectively calculable, and we might hope that this argument could be converted into a proof that it is actually recursive, but it turns out that this is not possible without imposing additional conditions on the enumeration $\chi_0, \chi_1, \dots$. Upon reflection this is not surprising; the enumeration of the formulas was assumed to be intuitively effective but no explicit algorithm was provided. This is our first task below.

Next, to conclude that $\text{Th}(\Omega)$ is not recursively axiomatizable, we need a recursive version of the Enumerability Theorem:

$$\begin{aligned} T \text{ is recursively axiomatizable (and complete)} \\ \implies T \text{ is recursively enumerable (recursively decidable).} \end{aligned}$$

Since if Γ is a recursively decidable set of axioms for T ,

$$m \in T^\# \iff \exists n_0 \cdots \exists n_{l-1} [(\chi_{n_0}, \dots, \chi_{n_{l-1}}, \chi_m) \text{ is a } \Gamma\text{-proof}],$$

the core of this argument will be to establish that for our new enumeration of formulas the relation R defined by the expression in square brackets is recursive. This places additional demands on the enumeration and occupies the remainder of the section.

There are many ways to enumerate the formulas in a way that will serve these purposes, but the standard method, due to Gödel, seems at first backward: rather than assigning formulas to numbers we assign numbers to formulas. To each L_Ω -formula ϕ we assign a number $\text{gn}(\phi)$, called its **Gödel number**, in such a way that the two mappings

$$p \mapsto \text{unique } \phi \text{ such that } \text{gn}(\phi) = p \quad \text{and} \quad \phi \mapsto \text{gn}(\phi)$$

are effectively calculable. However, a major difference is that the first of these is only a partial mapping: not every natural number is the Gödel number of a formula, and a key fact is that the set of numbers which *are* Gödel numbers is a recursive set. Note that this is parallel to the assignment of codes $\langle m_0, \dots, m_{k-1} \rangle$ to finite sequences (Definition 4.2.23) and indeed is based on this mechanism. From this we have just as before that T is intuitively decidable iff $\text{gn}(T) := \{\text{gn}(\phi) : \phi \in T\}$ is intuitively decidable and, assuming Church's Thesis, iff $\text{gn}(T)$ is recursive.

The key idea for achieving the goals discussed above is to arrange that the numerical value $\text{gn}(\phi)$ in some sense reflects the structure of the formula ϕ . We shall deal here explicitly only with the language L_Ω , but the method is completely general and can be applied to any finite language. Furthermore, the particular definition we give is based on many arbitrary choices that affect the details of the proofs but not the results.

We first assign a natural number to each of the symbols of L_Ω according to the following table:

symbol	$\dot{0}$	$\dot{S}$	$\dot{+}$	$\dot{\times}$	$\dot{<}$	$\dot{=}$	$\neg$	$\vee$	$\exists$	$v_i \ (i \in \omega)$
symbol number	1	3	5	7	9	11	13	15	17	$2i$

Since a formula ϕ is a finite sequence of symbols, we might assign to ϕ simply the code $\langle n_0, \dots, n_{k-1} \rangle$, where n_i is the number assigned to the i -th symbol of ϕ . However, the theory is easier to develop under the following definitions.

Definition 4.4.1

- (i) The assignment of Gödel numbers to L_Ω -terms is defined recursively according to the following table:

t	$\dot{0}$	v_i	$\dot{S}u$	$u \dot{+} u'$	$u \dot{\times} u'$
$\text{gn}(t)$	$\langle 1 \rangle$	$\langle 2i \rangle$	$\langle 3, \text{gn}(u) \rangle$	$\langle 5, \text{gn}(u), \text{gn}(u') \rangle$	$\langle 7, \text{gn}(u), \text{gn}(u') \rangle$

- (ii) The assignment of Gödel numbers to L_Ω -formulas is defined recursively according to the following table:

ϕ	$t \dot{<} u$	$t \dot{=} u$	$\neg \psi$
$\text{gn}(\phi)$	$\langle 9, \text{gn}(t), \text{gn}(u) \rangle$	$\langle 11, \text{gn}(t), \text{gn}(u) \rangle$	$\langle 13, \text{gn}(\psi) \rangle$

ϕ	$\psi \vee \theta$	$\exists v_i \psi$
$\text{gn}(\phi)$	$\langle 15, \text{gn}(\psi), \text{gn}(\theta) \rangle$	$\langle 17, 2i, \text{gn}(\psi) \rangle$

- (iii) For any set Γ of L_Ω -formulas, $\text{gn}(\Gamma) = \{ \text{gn}(\phi) : \phi \in \Gamma \}$.

Example 4.4.2 If ϕ is the formula $\forall v_3 (\dot{0} \dot{<} v_3 \vee \dot{0} \dot{=} v_3)$, then

$$\begin{aligned} \text{gn}(\phi) &= \langle 13, \langle 17, 6, \langle 13, \langle 15, \langle 9, \langle 1, \langle 6 \rangle \rangle \langle 11, \langle 1, \langle 6 \rangle \rangle \rangle \rangle \rangle \rangle \\ &= 2^{14} \times 3^{2^{18} \times 3^7 \times 5^{2^{14}} \times \dots}, \end{aligned}$$

a rather large number! Fortunately, we never need to compute individual Gödel numbers.

Definition 4.4.3 A set Γ of L_Ω -formulas is **recursive** or **decidable** iff $\text{gn}(\Gamma)$ is a recursive set of natural numbers and **recursively enumerable** iff $\text{gn}(\Gamma)$ is recursively enumerable. A L_Ω -theory T is **recursively axiomatizable** iff it has a recursive set of axioms — that is, there is a recursive set Γ of sentences such that $T = \text{Th}(\Gamma)$.

To study the conditions under which a theory T is decidable or undecidable — in particular to establish the recursive version of the Enumerability Theorem — we need to establish a sequence of results which assert that various syntactic relationships correspond to recursive relations of their Gödel numbers. For purposes to be explained later, we shall in many cases establish the stronger conclusion that these relations are primitive recursive.

Proposition 4.4.4 *The following sets are primitive recursive:*

- (i) $\text{Te} := \{ m : m \text{ is } \text{gn}(t) \text{ for some } L_\Omega\text{-term } t \};$
- (ii) $\text{Afo} := \{ m : m \text{ is } \text{gn}(\phi) \text{ for some } L_\Omega\text{-atomic formula } \phi \};$
- (iii) $\text{Fo} := \{ m : m \text{ is } \text{gn}(\phi) \text{ for some } L_\Omega\text{-formula } \phi \}.$

Proof. This proof, as well as several others below, is an exercise in the use of course-of-values recursion Theorem 4.2.30). For (i) we define a primitive recursive function G such that $\text{K}_{\text{Te}}(m) = G(\overline{\text{K}_{\text{Te}}}(m), m)$. The definition of G is based on the recursive definition of gn applied to L_Ω -terms. A number m is the Gödel number of an L_Ω -term iff it is $\langle 1 \rangle$ or $\langle 2i \rangle$, for some i , or it is one of the forms $\langle 3, \text{gn}(u) \rangle$, $\langle 5, \text{gn}(u), \text{gn}(u') \rangle$ or $\langle 7, \text{gn}(u), \text{gn}(u') \rangle$ for some L_Ω -terms u and u' . Since for any p , $p < \langle 3, p \rangle$, we have

$$\text{K}_{\text{Te}}(\langle 3, p \rangle) = 1 \iff \text{K}_{\text{Te}}(p) = 1 \iff (\overline{\text{K}_{\text{Te}}}(\langle 3, p \rangle))_p = 1.$$

Similarly,

$$\text{K}_{\text{Te}}(\langle 5, p, q \rangle) = 1 \iff (\overline{\text{K}_{\text{Te}}}(\langle 5, p, q \rangle))_p = 1 = (\overline{\text{K}_{\text{Te}}}(\langle 5, p, q \rangle))_q.$$

To put this into the correct form, we define G by cases as follows. Note that in the final application, the variable s will be replaced by $\overline{\text{K}_{\text{Te}}}(m)$, and if $\text{lg}(m) = 2$ and $\text{Sq}(m)$, then $m = \langle (m)_0, (m)_1 \rangle$.

$$G(s, m) := \begin{cases} 1, & \text{if } m = \langle 1 \rangle \text{ or } (\exists i < m) m = \langle 2i \rangle; \\ (s)_{(m)_1}, & \text{if } m = \langle 3, (m)_1 \rangle; \\ (s)_{(m)_1} \times (s)_{(m)_2}, & \text{if } (m)_0 = 5 \text{ or } 7, \text{Sq}(m) \text{ and } \text{lg}(m) = 3; \\ 0, & \text{otherwise.} \end{cases}$$

Since the conditions and functions are clearly primitive recursive, G is primitive recursive. That $\text{K}_{\text{Te}}(m) = G(\overline{\text{K}_{\text{Te}}}(m), m)$ is easily proved by induction

on m , and it follows that $\mathbf{K}_{\mathbf{Te}}$ is a primitive recursive function and thus that $\mathbf{Te}$ is a primitive recursive set.

Part (ii) follows from the easily verified equivalences:

$$\begin{aligned} \mathbf{Afo}(m) \iff \mathbf{Sq}(m) \text{ and } \lg(m) = 3 \text{ and } (m)_0 = 9 \text{ or } 11 \\ \text{and } \mathbf{Te}((m)_1) \text{ and } \mathbf{Te}((m)_2). \end{aligned}$$

Part (iii) is proved by a course-of-values recursion very similar to that used for $\mathbf{Te}$; details are left as an exercise. $\square$

Proposition 4.4.5 *There exists a primitive recursive function $\mathbf{Sub}$ such that for any L_Ω -terms t and u , any L_Ω -formula ϕ and any $i \in \omega$,*

$$(i) \quad \mathbf{Sub}(\mathbf{gn}(u), i, \mathbf{gn}(t)) = \mathbf{gn}(u_{v_i}(t));$$

$$(ii) \quad \mathbf{Sub}(\mathbf{gn}(\phi), i, \mathbf{gn}(t)) = \mathbf{gn}(\phi_{v_i}(t)).$$

Furthermore, for any m, i and q , if $q \in \mathbf{Te}$, then

$$\begin{aligned} \mathbf{Sub}(m, i, q) \in \mathbf{Te} &\iff m \in \mathbf{Te} \quad \text{and} \\ \mathbf{Sub}(m, i, q) \in \mathbf{Fo} &\iff m \in \mathbf{Fo}. \end{aligned}$$

Proof. For (i), let G be defined by cases as follows:

$$G(s, i, q, m) = \begin{cases} q, & \text{if } m = \langle 2i \rangle; \\ \langle (m)_0, (s)_{(m)_1} \rangle, & \text{if } (m)_0 = 3, \mathbf{Sq}(m) \\ & \text{and } \lg(m) = 2; \\ \langle (m)_0, (s)_{(m)_1}, (s)_{(m)_2} \rangle, & \text{if } (m)_0 = 5 \text{ or } 7, \mathbf{Sq}(m) \\ & \text{and } \lg(m) = 3; \\ m, & \text{otherwise.} \end{cases}$$

Let $F(i, q, m) = G(\overline{F}(i, q, m), i, q, m)$; we prove by term-induction (on u) that for all terms t and u ,

$$F(i, \mathbf{gn}(t), \mathbf{gn}(u)) = \mathbf{gn}(u_{v_i}(t)),$$

and hence take $\mathbf{Sub}(m, i, q) := F(i, q, m)$, which is primitive recursive by course-of-values recursion. If u is v_i , then

$$\begin{aligned} F(i, \mathbf{gn}(t), \mathbf{gn}(u)) &= G(\overline{F}(i, \mathbf{gn}(t), \mathbf{gn}(u)), i, \mathbf{gn}(t), \langle 2i \rangle) \\ &= \mathbf{gn}(t) = \mathbf{gn}(u_{v_i}(t)). \end{aligned}$$

For a term of the form $\dot{S}u$ we have

$$\begin{aligned}
 & F(i, \text{gn}(t), \text{gn}(\dot{S}u)) \\
 &= G\left(\overline{F}(i, \text{gn}(t), \langle 3, \text{gn}(u) \rangle), i, \text{gn}(t), \langle 3, \text{gn}(u) \rangle\right) \\
 &= \left\langle (\langle 3, \text{gn}(u) \rangle)_0, \left(\overline{F}(i, \text{gn}(t), \langle 3, \text{gn}(u) \rangle)\right)_{(\langle 3, \text{gn}(u) \rangle)_1} \right\rangle \\
 &= \left\langle 3, \left(\overline{F}(i, \text{gn}(t), \langle 3, \text{gn}(u) \rangle)\right)_{\text{gn}(u)} \right\rangle \\
 &= \langle 3, F(i, \text{gn}(t), \text{gn}(u)) \rangle \\
 &= \langle 3, \text{gn}(u_{v_i}(t)) \rangle \\
 &= \text{gn}((\dot{S}u)_{v_i}(t)).
 \end{aligned}$$

The other induction steps are similar, and we leave them, as well as part (ii) and the final assertion, as an exercise. $\square$

We shall have frequent occasion below to make multiple substitutions of a special sort; we introduce here some notation.

Corollary 4.4.6 *For each $k > 0$ there exists a primitive recursive function Sb^k such that for any L_Ω -formula ϕ and all $m_0, \dots, m_{k-1}$,*

$$\text{Sb}^k(\text{gn}(\phi), m_0, \dots, m_{k-1}) = \text{gn}(\phi_{v_0, \dots, v_{k-1}}(\dot{m}_0, \dots, \dot{m}_{k-1})),$$

and for any $e \in \omega$,

$$\text{Sb}^k(e, m_0, \dots, m_{k-1}) \in \text{Fo} \iff e \in \text{Fo}.$$

Proof. We need first to observe that the function $m \mapsto \text{gn}(\dot{m})$ is primitive recursive because it satisfies the recursion

$$\text{gn}(\dot{0}) = \langle 1 \rangle; \quad \text{gn}(\overbrace{\dot{m} + 1}^{\cdot}) = \langle 3, \text{gn}(\dot{m}) \rangle.$$

Then we set

$$\begin{aligned}
 & \text{Sb}^1(e, m_0) := \text{Sub}(e, 0, \text{gn}(\dot{m}_0)) \quad \text{and} \\
 & \text{Sb}^{k+1}(e, m_0, \dots, m_k) := \text{Sub}(\text{Sb}^k(e, m_0, \dots, m_{k-1}), k, \text{gn}(\dot{m}_k)). \quad \square
 \end{aligned}$$

With this we have accomplished the first of the goals described above.

Proposition 4.4.7 *The following two relations are primitive recursive:*

- (i) $\text{Free}(m, i) : \Longleftrightarrow$ for some formula ϕ , $m = \text{gn}(\phi)$ and $v_i \in \text{Fv}(\phi)$;
- (ii) $\text{Freesub}(m, i, q) : \Longleftrightarrow$ for some term u and some formula ϕ , $m = \text{gn}(\phi)$, $q = \text{gn}(u)$, and u is freely substitutable for v_i in ϕ .

Proof. For (i), let G be the primitive recursive function defined by

$$G(s, i, m) := \begin{cases} 1, & \text{if } m \in \text{Afo}; \\ (s)_{(m)_1}, & \text{if } (m)_0 = 13, \text{Sq}(m) \\ & \text{and } \text{lg}(m) = 2; \\ \text{Sg}((s)_{(m)_1} + (s)_{(m)_2}), & \text{if } (m)_0 = 15, \text{Sq}(m) \\ & \text{and } \text{lg}(m) = 3; \\ (s)_{(m)_2}, & \text{if } (m)_0 = 17, \text{Sq}(m), \\ & \text{lg}(m) = 3 \text{ and } (m)_1 \neq 2i; \\ 0, & \text{otherwise.} \end{cases}$$

Let $F(i, m) = G(\overline{F}(i, m), i, m)$; we establish by formula induction that for all formulas ϕ ,

$$F(i, \text{gn}(\phi)) = \text{K}_{\text{Free}}(\text{gn}(\phi), i).$$

We give only the quantifier step and leave the remainder as an exercise.

$$\begin{aligned} F(i, \text{gn}(\exists v_j \phi)) &= G(\overline{F}(i, \text{gn}(\exists v_j \phi)), i, \text{gn}(\exists v_j \phi)) \\ &= G(\overline{F}(i, \langle 17, 2j, \text{gn}(\phi) \rangle), i, \langle 17, 2j, \text{gn}(\phi) \rangle) \\ &= \begin{cases} \overline{F}(i, \langle 17, 2j, \text{gn}(\phi) \rangle)_{(\langle 17, 2j, \text{gn}(\phi) \rangle)_2}, & \text{if } 2j \neq 2i; \\ 0, & \text{otherwise;} \end{cases} \\ &= \begin{cases} F(i, \text{gn}(\phi)), & \text{if } j \neq i; \\ 0, & \text{otherwise;} \end{cases} \\ &= \text{K}_{\text{Free}}(\text{gn}(\exists v_j \phi), i). \quad \square \end{aligned}$$

Corollary 4.4.8 *The set*

$$\text{Sent} := \{m : m \text{ is the Gödel number of an } L_\Omega \text{ sentence}\}$$

is primitive recursive.

Proof. $\text{Sent}(m) \Longleftrightarrow \text{Fo}(m) \wedge (\forall i < m) \sim \text{Free}(m, i). \quad \square$

With these tools at hand, we could now return to each of the decidability results of [Chapters 1–2](#) and prove a precise version which asserts the recursiveness, or sometimes even primitive recursiveness, of the set in question. In fact, to do this in any detail would be an extremely tedious exercise, and we shall only indicate the outline of the process.

Let us take as an example Theorem 2.4.41(i) which asserts that $\text{Th}(\emptyset)$, the set of logical validities of $L_=$, is decidable. Working backwards from the proof of that theorem, we see that it will suffice to produce a recursive function H such that for any $L_=-$ sentence ϕ , $H(\text{gn}(\phi)) = \langle i, s \rangle$, where if $N_s = \{(s)_0, \dots, (s)_{\text{lg}(s)-1}\}$, then

$$i = 0 \quad \text{and} \quad \models \phi \leftrightarrow \exists \in^{N_s} \quad \text{or} \quad i = 1 \quad \text{and} \quad \models \phi \leftrightarrow \neg \exists \in^{N_s}.$$

Then m is the Gödel number of a logical validity iff $H(m) = \langle 1, 1 \rangle$. For m which are Gödel numbers of Boolean combinations of the sentences $\exists^{\geq n}$, we can extract a definition of an appropriate H from the proof of Proposition 2.4.39. For example, we would set

$$H(\text{gn}(\neg \psi)) = \langle 1 - \overset{\circ}{(H(\text{gn}(\psi)))_0}, (H(\text{gn}(\psi)))_1 \rangle.$$

Handling $\psi \vee \theta$ would involve a definition by cases depending on the two i values $(H(\text{gn}(\psi)))_0$ and $(H(\text{gn}(\theta)))_0$.

To define H for Gödel numbers of arbitrary formulas, we must construct an auxiliary primitive recursive function G such that for any $L_=-$ formula ϕ , $G(\text{gn}(\phi))$ is the Gödel number of a formula $\phi^* \in \Delta_=$ such that $\models \phi \leftrightarrow \phi^*$ and $\text{Fv}(\phi^*) \subseteq \text{Fv}(\phi)$ as in Theorem 2.4.36. This, in turn, is best viewed as the end result of two functions G_0 and G_1 corresponding to the two lemmas preceding 2.4.36. It is clear that all of this can be done, but it is hardly very inviting to write out in detail. In many ways this may be regarded as a problem of coding a complex algorithm in a rudimentary programming language. The clauses in the definition of the class of (primitive) recursive functions and other derived schemas are the constructs of the programming language, and we are to verify that the decision algorithm can indeed be coded using just these constructs.

We turn now to the Enumerability Theorem 3.1.18 and in particular 4.1.2 stated in terms of recursiveness instead of intuitive calculability. Our goal is to show that if Γ is any recursive set of sentences — that is, $\text{gn}(\Gamma)$ is recursive — then the theory $\text{Th}(\Gamma) = \{\phi : \Gamma \models \phi\}$ generated by Γ is recursively enumerable. Again for simplicity we treat explicitly only the language L_Ω , but the method is general. We follow the second proof at the end of [Section 3.2](#) and consider the relations (see Definition 3.2.5)

Definition 4.4.9 For any set Γ of sentences,

(i) $\text{Proof}_\Gamma :=$

$$\{ \langle \text{gn}(\phi_0), \dots, \text{gn}(\phi_l) \rangle : l \in \omega \text{ and } (\phi_0, \dots, \phi_l) \text{ is a proof from } \Gamma \};$$

(ii) $\text{Proofof}_\Gamma := \{ (s, m) : \text{Proof}_\Gamma(s) \text{ and } m := (s)_{\text{lg}(s)-1} \}.$

It will suffice to show that if $\text{gn}(\Gamma)$ is (primitive) recursive, then Proof_Γ and Proofof_Γ are also (primitive) recursive. Recall from Definitions 3.2.1 and 3.2.2 that the logical axioms fall into three groups and there are two rules of inference. We define first the following sets and relations.

Definition 4.4.10

(i) $\text{Taut} := \{ m : m \text{ is the Gödel number of an } L_\Omega\text{-tautology} \}.$

(ii) $\text{Eqax} := \{ m : m \text{ is the Gödel number of an } L_\Omega\text{-equality axiom} \}.$

(iii) $\exists\text{-ax} := \{ m : m \text{ is the Gödel number of an } L_\Omega\text{-}\exists\text{-axiom} \}.$

(iv) $\text{ModPon} := \left\{ (m, n, p) : \begin{array}{l} \text{for some } L_\Omega\text{-formulas } \phi \text{ and } \psi, \\ m = \text{gn}(\phi), n = \text{gn}(\phi \rightarrow \psi) \\ \text{and } p = \text{gn}(\psi) \end{array} \right\}.$

(v) $\exists\text{-rule} := \left\{ (m, n) : \begin{array}{l} \text{for some } L_\Omega\text{-formulas } \phi \text{ and } \psi \\ \text{and some variable } x \text{ not free in } \psi, \\ m = \text{gn}(\phi \rightarrow \psi) \text{ and } n = \text{gn}(\exists x \phi \rightarrow \psi) \end{array} \right\}.$

Proposition 4.4.11 *Taut is primitive recursive.*

Proof. Recall that an L_Ω -formula ϕ is a tautology iff $V(\phi) = \mathbf{T}$ for all truth valuations V and that a truth valuation is a function $V : \text{Fm}_{L_\Omega} \rightarrow \{\mathbf{T}, \mathbf{F}\}$ such that for all ψ and θ , $V(\neg \psi) \neq V(\psi)$ and $V(\psi \vee \theta) = \mathbf{T} \iff [V(\psi) = \mathbf{T}] \text{ or } [V(\theta) = \mathbf{T}]$. The key point here is that in evaluating this condition for a fixed formula ϕ it suffices to consider finitely many truth valuations which assume all possible combinations of values on the atomic and existential subformulas of ϕ and that only values on subformulas of ϕ are relevant (see Remark 2.2.32(iv)). Accordingly, for any $s \in \omega$, let V_s be defined by

$$V_s(\phi) := \begin{cases} \mathbf{T}, & \text{if } \text{gn}(\phi) < \text{lg}(s) \wedge (s)_{\text{gn}(\phi)} = 1; \\ \mathbf{F}, & \text{otherwise.} \end{cases}$$

V_s is, of course, not a truth valuation, but we can decide primitive recursively whether or not the “interesting part” of V_s satisfies the requirements for being a truth valuation. Set

$$\text{Val}(s) : \iff \forall i, j < \text{lg}(s)$$

$$\left(\langle 13, i \rangle < \text{lg}(s) \implies [(s)_{\langle 13, i \rangle} = 1 \iff (s)_i \neq 1] \right.$$

$$\left. \wedge \langle 15, i, j \rangle < \text{lg}(s) \implies [(s)_{\langle 15, i, j \rangle} = 1 \iff (s)_i = 1 \vee (s)_j = 1] \right).$$

Then clearly Val is a primitive recursive relation and

$$\mathbf{Taut}(n) \iff \mathbf{Fo}(n) \wedge \forall s [Val(s) \wedge n < \lg(s) \implies (s)_n = 1].$$

To complete the proof we need to find a bound for the quantifier in this expression. To check whether or not $\mathbf{Taut}(n)$, it clearly suffices to examine all s with $\lg(s) \leq n+1$ and $(\forall i < \lg(s)) [(s)_i \leq 1]$. Among these the numerically largest is $\langle 1^{n+1} \rangle$, the code for the sequence of $n+1$ many 1s. The function $n \mapsto \langle 1^{n+1} \rangle$ is easily primitive recursive, and we have

$$\mathbf{Taut}(n) \iff \mathbf{Fo}(n) \wedge (\forall s < \langle 1^{n+1} \rangle) [Val(s) \wedge n < \lg(s) \implies (s)_n = 1],$$

from which it follows that $\mathbf{Taut}$ is primitive recursive. $\square$

Proposition 4.4.12 *Eqax is primitive recursive.*

Proof. The equality axioms for L_Ω are all formulas of one of the forms

$$\begin{aligned} t &\doteq t, & t &\doteq u \rightarrow u \doteq t, & t &\doteq u \wedge u \doteq v \rightarrow t \doteq v, \\ t &\doteq u \rightarrow \dot{S}t \doteq \dot{S}u, \\ t_0 &\doteq u_0 \wedge t_1 \doteq u_1 \rightarrow t_0 \dot{+} t_1 \doteq u_0 \dot{+} u_1, \\ t_0 &\doteq u_0 \wedge t_1 \doteq u_1 \rightarrow t_0 \dot{\times} t_1 \doteq u_0 \dot{\times} u_1. \end{aligned}$$

It is a straightforward computation that the set of Gödel numbers of such formulas is primitive recursive. $\square$

Proposition 4.4.13 *$\exists$ -ax is primitive recursive.*

Proof. The $\exists$ -axioms are all formulas of the form $\phi_x(t) \rightarrow \exists x\phi$ such that the substitution $\phi_x(t)$ is free. Hence their Gödel numbers are exactly all numbers of the form

$$\langle 15, \langle 13, \mathbf{Sub}(m, i, q) \rangle, \langle 17, 2i, m \rangle \rangle$$

such that $\mathbf{Fo}(m)$, $\mathbf{Te}(q)$ and $\mathbf{Freesub}(m, i, q)$. $\square$

Proposition 4.4.14 *ModPon is primitive recursive.*

Proof. We have simply

$$\mathbf{ModPon}(m, n, p) \iff \mathbf{Fo}(m) \wedge \mathbf{Fo}(p) \wedge n = \langle 15, \langle 13, m \rangle, p \rangle. \square$$

Proposition 4.4.15 *$\exists$ -rule is primitive recursive.*

Proof. The relation $\exists$ -rule consists of pairs

$$\langle 15, \langle 13, p \rangle, q \rangle \quad \text{and} \quad \langle 15, \langle 13, \langle 17, 2i, p \rangle \rangle, q \rangle$$

such that $\mathbf{Fo}(p)$, $\mathbf{Fo}(q)$, and $\sim \mathbf{Free}(q, i)$. $\square$

Theorem 4.4.16 *For any set Γ of sentences, if Γ is (primitive) recursive, then Proof_Γ and Proof_Γ are (primitive) recursive.*

Proof. We simply write out the definition of a proof from Γ in terms of the Gödel numbers:

$$\begin{aligned} \text{Proof}_\Gamma(s) \iff & \text{Sq}(s) \wedge \forall i < \text{lg}(s) [\text{Taut}((s)_i) \vee \text{Eqax}((s)_i) \vee \exists \text{-ax}((s)_i) \\ & \vee (s)_i \in \text{gn}(\Gamma) \vee (\exists j, k < i) \text{ModPon}((s)_j, (s)_k, (s)_i) \\ & \vee (\exists j < i) \exists \text{-rule}((s)_j, (s)_i)]. \end{aligned}$$

From this equivalence, it is clear that if $\text{gn}(\Gamma)$ is (primitive) recursive, so are Proof_Γ and Proof_Γ . $\square$

Enumerability Theorem 4.4.17 *For any finite language L — in particular L_Ω — and any L -theory T , if T is recursively axiomatizable, then T is recursively enumerable; hence if T is also complete, then T is decidable.*

Proof. Let Γ be a recursive set of axioms for T . Clearly

$$\text{gn}(T) = \{ m : \text{Sent}(m) \wedge \exists s \text{Proof}_\Gamma(s, m) \}.$$

Then by Proposition 4.3.3, $\text{gn}(T)$ is recursively enumerable. Now if T is also complete, for any m ,

$$m \notin \text{gn}(T) \iff \sim \text{Sent}(m) \text{ or } \langle 13, m \rangle \in \text{gn}(T).$$

Thus $\sim \text{gn}(T)$ is also recursively enumerable, so by Proposition 4.3.7, $\text{gn}(T)$ is recursive. $\square$

In particular for the case $\Gamma = \emptyset$ we have

Corollary 4.4.18 *For any finite language L , the set of logical validities of L is recursively enumerable.*

We shall see in [Sections 4.6](#) and [5.6](#) that this set is not always recursive.

Exercises

Exercise 4.4.19 Write out the proof that Fo is primitive recursive (Proposition 4.4.4).

Exercise 4.4.20 Complete the proof of Proposition 4.4.5 by extending the definition of Sub to include formulas.

Exercise 4.4.21 Finish the proof of Proposition 4.4.7 by showing that the relation **Freesub** is primitive recursive.

For the next three exercises let A be the unique binary function such that for all p and n , $A(0, n) = n + 1$,

$$A(p + 1, 0) = A(p, 1) \quad \text{and} \quad A(p + 1, n + 1) = A(p, A(p + 1, n)).$$

For each p , let A_p denote the unary function $A_p(n) = A(p, n)$. A is called the **Ackermann function** and was discussed in Exercise 1.2.23.

Exercise 4.4.22 Prove that for all p and n ,

- (i) A_p is primitive recursive;
- (ii) $n < A(p, n)$;
- (iii) A is properly increasing in both arguments: for all p and n ,

$$A(p, n) < A(p, n + 1) \quad \text{and} \quad A(p, n) < A(p + 1, n);$$

- (iv) if $p, n > 0$, then $A(p, n + 1) < A(p + 1, n)$.

Exercise 4.4.23 For any (binary) function F , say that F is **dominated** by A_p iff for all m, n , $F(m, n) \leq A_p(\max(m, n))$. Prove that every primitive recursive function is dominated by A_p for some $p \in \omega$. Conclude that A is not primitive recursive.

Hint: Proceed by Prim-induction. The only difficult step is primitive recursion. Consider first the case with no parameters, say $F(0) = s$ and $F(n + 1) = H(F(n), n)$. Show that if H is dominated by A_r , then F is dominated by A_p with $p = \max(s, r + 1)$. For the case with a parameter, say $F(m, 0) = G(m)$ and $F(m, n + 1) = H(F(m, n), m, n)$, show that if G and H are dominated by A_q and A_r , respectively, then F is dominated by A_p for $p = \max(q, r + 2)$. As an intermediate step show that for all m, n ,

$$F(m, n) \leq A_{p-1}(\max(m, n) + n).$$

Exercise 4.4.24 Prove that A is a recursive function. Hence, by the preceding exercise, there exist functions which are recursive but not primitive recursive.

Hint: It suffices to show that $\text{Graph}(A)$ is a semi-recursive relation. Think of triples $\langle m, n, p \rangle$ as ‘formulas’ and devise a notion of ‘proof’ such that the ‘theorems’ are exactly those triples such that $A(m, n) = p$ and use the method of the Enumerability Theorem 4.4.17.

Exercise 4.4.25 Let G , H , and I be primitive recursive functions such that $\forall n \exists p [I^{(p)}(n) = 0]$. Consider the condition

$$\begin{aligned} F(m_0, \dots, m_{k-1}, 0) &= G(m_0, \dots, m_{k-1}) \quad \text{and for } n > 0, \\ F(m_0, \dots, m_{k-1}, n) &= H(F(m_0, \dots, m_{k-1}, I(n)), m_0, \dots, m_{k-1}, n). \end{aligned}$$

Note that for $I(n) = \text{Pred}(n)$ this is essentially primitive recursion.

- (i) Prove that there is a unique function F which satisfies the condition.
- (ii) Show that F is not necessarily primitive recursive (compare Exercise 4.2.53).
- (iii) Find a condition on I which will guarantee that F is primitive recursive.

Hint: For (ii) construct a function F such that the least p such that $F^{(p)}(n) = 0$ is the value of the Ackermann function at $((n)_0, (n)_1)$.

Exercise 4.4.26 Verify that the theories $\mathbf{Q}$ and $\mathbf{P}$ are recursively axiomatizable.

4.5 Definability in arithmetic I

We have seen in Exercise 2.5.31 that the sets of natural numbers definable over the structure $\Omega_+ = (\omega, <, +, \text{Sc}, 0)$ are exactly the eventually periodic ones and in particular that the multiplication relation is not definable over Ω_+ . The goal of this section and [Section 5.1](#) is to explore the enormous difference that adjoining multiplication makes: over $\Omega = (\omega, <, +, \times, \text{Sc}, 0)$ are definable not only all recursive and semi-recursive sets and relations but much more as well.

We recall first the following definitions. A relation $R \subseteq \omega^k$ is **definable** over Ω iff there exists a formula $\phi(x_0, \dots, x_{k-1})$ such that for all $m_0, \dots, m_{k-1}$,

$$\begin{aligned} R(m_0, \dots, m_{k-1}) &\iff \models_{\Omega} \phi_{x_0, \dots, x_{k-1}}[m_0, \dots, m_{k-1}] \\ &\iff \models_{\Omega} \phi_{x_0, \dots, x_{k-1}}(\dot{m}_0, \dots, \dot{m}_{k-1}) \\ &\iff \phi_{x_0, \dots, x_{k-1}}(\dot{m}_0, \dots, \dot{m}_{k-1}) \in \text{Th}(\Omega). \end{aligned}$$

Note that the equivalence of the second and third formulations is a special feature of definability over a structure in which each element is distinguished — that is, is named by a constant term of the language; we shall generally use this version. A function $F : \omega^k \rightarrow \omega$ is **definable** over Ω iff its graph $\text{Graph}(F)$

is definable over Ω — that is, there exists a formula $\phi(x_0, \dots, x_{k-1}, y)$ such that for all $m_0, \dots, m_{k-1}$ and n ,

$$\begin{aligned} F(m_0, \dots, m_{k-1}) = n &\iff \models_{\Omega} \phi_{x_0, \dots, x_{k-1}, y}[m_0, \dots, m_{k-1}, n] \\ &\iff \models_{\Omega} \phi_{x_0, \dots, x_{k-1}, y}(\dot{m}_0, \dots, \dot{m}_{k-1}, \dot{n}) \\ &\iff \phi_{x_0, \dots, x_{k-1}, y}(\dot{m}_0, \dots, \dot{m}_{k-1}, \dot{n}) \in \text{Th}(\Omega). \end{aligned}$$

Lemma 4.5.1 *For any relation R ,*

- (i) *R is definable over Ω iff its characteristic function K_R is definable over Ω ;*
- (ii) *R is definable over Ω iff its complement is definable over Ω ;*
- (iii) *if R is definable over Ω , then so is its projection*

$$\{(m_0, \dots, m_{k-1}) : \exists p R(m_0, \dots, m_{k-1}, p)\}.$$

Proof. Suppose first that K_R is definable, say

$$K_R(m_0, \dots, m_{k-1}) = n \iff \models_{\Omega} \phi_{x_0, \dots, x_{k-1}, y}(\dot{m}_0, \dots, \dot{m}_{k-1}, \dot{n}).$$

Then clearly R is defined by the formula $\phi_y(\dot{1})$. Conversely, if R is defined by a formula ψ , then

$$\begin{aligned} K_R(m_0, \dots, m_{k-1}) = n &\iff \models_{\Omega} (\psi \wedge y \doteq \dot{1}) \vee (\neg \psi \wedge y \doteq \dot{0})(\dot{m}_0, \dots, \dot{m}_{k-1}, \dot{n}), \end{aligned}$$

so also K_R is definable. Parts (ii) and (iii) are immediate: if R is defined by ϕ , then its complement is defined by $\neg \phi$ and its projection is defined by $\exists x_k \phi$. $\square$

Now we can replace the Definability Hypothesis 4.1.3 with the

Definability Theorem 4.5.2 *For all $k > 0$,*

- (i) *every recursive function $F : \omega^k \rightarrow \omega$ is definable over Ω ;*
- (ii) *every recursive relation $R \subseteq \omega^k$ is definable over Ω ;*
- (iii) *every semi-recursive relation $R \subseteq \omega^k$ is definable over Ω .*

Proof. Parts (ii) and (iii) follow immediately from (i) and the lemma. For (i) we proceed by Rec' -induction. The initial functions are definable:

$$\begin{aligned} \text{Cs}_n^k &\text{ is defined by } y \doteq \dot{n}; \\ \text{Pr}_i^k &\text{ is defined by } y \doteq x_i; \\ \text{Sc}_i^k &\text{ is defined by } y \doteq \dot{S}x_i. \end{aligned}$$

Furthermore,

$$\begin{aligned} + & \text{ is defined by } x_0 \dot{+} x_1 \doteq y; \\ \times & \text{ is defined by } x_0 \dot{\times} x_1 \doteq y; \\ K_{<} & \text{ is defined by } (x_0 \dot{<} x_1 \wedge y \doteq \dot{1}) \vee (\neg x_0 \dot{<} x_1 \wedge y \doteq \dot{0}). \end{aligned}$$

These are all simply checked — for example,

$$\begin{aligned} m_0 + m_1 = n & \iff \models_{\Omega} \dot{m}_0 \dot{+} \dot{m}_1 \doteq \dot{n} \\ & \iff \models_{\Omega} (x_0 \dot{+} x_1 \doteq y)_{x_0, x_1, y} (\dot{m}_0, \dot{m}_1, \dot{n}). \end{aligned}$$

It remains to show that the set of definable functions is closed under composition and search. Suppose that F arises from G, H_0, H_1 by composition,

$$F(m) = G(H_0(m), H_1(m)),$$

and assume that G, H_0 and H_1 are defined over Ω by formulas

$$\psi(x_0, x_1, z), \quad \theta^0(x, y_0) \quad \text{and} \quad \theta^1(x, y_1),$$

respectively: for all m, p_0, p_1, n and $i = 0, 1$,

$$\begin{aligned} G(p_0, p_1) = n & \iff \models_{\Omega} \psi(\dot{p}_0, \dot{p}_1, \dot{n}) \quad \text{and} \\ H_i(m) = p_i & \iff \models_{\Omega} \theta^i(\dot{m}, \dot{p}_i). \end{aligned}$$

Then

$$\begin{aligned} F(m) = n & \iff \exists p_0 \exists p_1 [H_0(m) = p_0 \wedge H_1(m) = p_1 \wedge G(p_0, p_1) = n] \\ & \iff \exists p_0 \exists p_1 \left(\models_{\Omega} \theta^0(\dot{m}, \dot{p}_0) \wedge \models_{\Omega} \theta^1(\dot{m}, \dot{p}_1) \wedge \models_{\Omega} \psi(\dot{p}_0, \dot{p}_1, \dot{n}) \right) \\ & \iff \models_{\Omega} \exists y_0 \exists y_1 (\theta^0(x, y_0) \wedge \theta^1(x, y_1) \wedge \psi(y_0, y_1, z)) (\dot{m}, \dot{n}), \end{aligned}$$

so that F is definable. This is, of course, a special case, but the general case is an obvious generalization.

Finally, suppose that F arises from G by search, so G satisfies the search condition

$$\forall m \exists n G(m, n) = 0$$

and $F(m) = \text{least } n [G(m, n) = 0]$. Suppose also that G is defined over Ω by the formula $\psi(x, y, z)$. Then F is definable because

$$\begin{aligned} F(m) = n & \iff (\forall q < n) G(m, q) \neq 0 \quad \text{and} \quad G(m, n) = 0 \\ & \iff (\forall q < n) \models_{\Omega} \neg \psi(\dot{m}, \dot{q}, \dot{0}) \quad \text{and} \quad \models_{\Omega} \psi[\dot{m}, \dot{n}, \dot{0}] \\ & \iff \models_{\Omega} ((\forall w \dot{<} y) \neg \psi(x, w, \dot{0}) \wedge \psi(x, y, \dot{0}))_{x, y} (\dot{m}, \dot{n}). \end{aligned}$$

Now (ii) follows immediately from the lemma. Suppose that P is a semi-recursive k -ary relation, so there is a $(k+1)$ -ary relation R and a formula $\phi(x_0, \dots, x_{k-1}, y)$ such that

$$\begin{aligned} P(m_0, \dots, m_{k-1}) &\iff \exists n R(m_0, \dots, m_{k-1}, n) \\ &\iff \exists n \models_{\Omega} \phi(\dot{m}_0, \dots, \dot{m}_{k-1}, \dot{n}) \\ &\iff \models_{\Omega} (\exists y \phi)(\dot{m}_0, \dots, \dot{m}_{k-1}). \end{aligned}$$

Hence the formula $\exists y \phi$ defines P . $\square$

Now we can prove the final version of Theorem 4.1.7; the numbering of displays matches that of the earlier proof.

Theorem 4.5.3 (*Undecidability of Arithmetic*) $\text{Th}(\Omega)$ is undecidable, hence not recursively axiomatizable.

Proof. Suppose towards a contradiction that $\text{Th}(\Omega)$ is decidable — that is, $\text{gn}(\text{Th}(\Omega))$ is a recursive set. Let $X \subseteq \omega$ be the set defined by

$$(A.1) \quad \text{for all } m, \quad m \in X \iff \text{Sb}^1(m, m) \in \text{gn}(\text{Th}(\Omega)).$$

By Proposition 4.2.19, X is also recursive so by the Definability Theorem (i) is definable over Ω , and for some formula ψ we have

$$(A.2) \quad \text{for all } m, \quad m \in X \iff \psi(\dot{m}) \in \text{Th}(\Omega).$$

Since $\text{Th}(\Omega)$ is consistent and complete, also

$$(A.3) \quad \text{for all } m, \quad m \in X \iff \neg \psi(\dot{m}) \notin \text{Th}(\Omega).$$

Let $p := \text{gn}(\neg \psi)$, so by its definition, $\text{gn}(\neg \psi(\dot{p})) = \text{Sb}^1(p, p)$, and by (A.1) and (A.3), respectively,

$$\begin{aligned} p \in X &\iff \text{Sb}^1(p, p) \in \text{gn}(\text{Th}(\Omega)) \iff \neg \psi(\dot{p}) \in \text{Th}(\Omega) \\ p \in X &\iff \neg \psi(\dot{p}) \notin \text{Th}(\Omega). \end{aligned}$$

Hence, if θ is the sentence $\neg \psi(\dot{p})$,

$$(A.4) \quad \theta \in \text{Th}(\Omega) \iff \theta \notin \text{Th}(\Omega),$$

a contradiction. Hence $\text{Th}(\Omega)$ is undecidable and by the Enumerability Theorem 4.4.17 also not recursively axiomatizable. $\square$

Theorem 4.5.4 *For any L_Ω -theory $T \subseteq \text{Th}(\Omega)$, if T is recursively axiomatizable, then T is incomplete.*

Proof. As for Theorem 4.1.8, we can derive this simply as a corollary of the preceding theorem or adapt the proof of that theorem to give a constructive proof. For the first proof note that if $T \subseteq \text{Th}(\Omega)$ and is complete, then necessarily $T = \text{Th}(\Omega)$: otherwise there is some sentence $\phi \in \text{Th}(\Omega) \sim T$, hence $\neg\phi \in T$, so also $\neg\phi \in \text{Th}(\Omega)$, a contradiction. Hence by the preceding theorem, such a T cannot be recursively axiomatizable.

For the second proof, fix a recursively axiomatizable theory $T \subseteq \text{Th}(\Omega)$, and let $X_T \subseteq \omega$ be the set defined by

$$(B.1) \quad \text{for all } m, \quad m \in X_T \iff \text{Sb}^1(m, m) \in \text{gn}(T).$$

By the Enumerability Theorem $\text{gn}(T)$ is recursively enumerable, so by Proposition 4.3.20 X_T is recursively enumerable and hence by the Definability Theorem (iii) is definable over Ω , say by the formula ψ_T ,

$$(B.2) \quad \text{for all } m, \quad m \in X_T \iff \psi_T(\dot{m}) \in \text{Th}(\Omega),$$

from which we have immediately

$$(B.3) \quad \text{for all } m, \quad m \in X_T \iff \neg\neg\psi_T(\dot{m}) \in \text{Th}(\Omega).$$

Let $p_T := \text{gn}(\neg\psi_T)$ so $\text{gn}(\neg\psi_T(\dot{p}_T)) = \text{Sb}^1(p_T, p_T)$, and from (B.1) and (B.3), respectively,

$$\begin{aligned} p_T \in X_T &\iff \text{Sb}^1(p_T, p_T) \in \text{gn}(T) \iff \neg\psi(\dot{p}_T) \in T \\ p_T \in X_T &\iff \neg\neg\psi_T(\dot{p}_T) \in \text{Th}(\Omega). \end{aligned}$$

Hence if θ_T is the sentence $\neg\psi_T(\dot{p}_T)$, we have

$$(B.4) \quad \theta_T \in T \iff \neg\theta_T \in \text{Th}(\Omega).$$

Suppose first that $\theta_T \in T$. Then since $T \subseteq \text{Th}(\Omega)$, also $\theta_T \in \text{Th}(\Omega)$, but by (B.4) also $\neg\theta_T \in \text{Th}(\Omega)$, a contradiction; hence $\theta_T \notin T$. Now suppose that $\neg\theta_T \in T$. Again since $T \subseteq \text{Th}(\Omega)$, also $\neg\theta_T \in \text{Th}(\Omega)$, hence by (B.4) $\theta_T \in T$, contrary to what we just established. Hence also $\neg\theta_T \notin T$. $\square$

Note that just as in Remark 4.1.9, θ_T is true ($\theta_T \in \text{Th}(\Omega)$) but unprovable in T . We have also the updated version of 4.1.23.

Tarski's Theorem 4.5.5 $\text{gn}(\text{Th}(\Omega))$ is not definable over Ω .

Proof. Suppose towards a contradiction that $\text{gn}(\text{Th}(\Omega))$ is definable. Then by Lemma 4.5.1 so is its characteristic function $K_{\text{gn}(\text{Th}(\Omega))}$. The function $n \mapsto \text{Sb}^1(n, n)$ is recursive, hence definable, so by the closure of the class of definable functions under composition established in the proof of the Definability Theorem, so is $n \mapsto K_{\text{gn}(\text{Th}(\Omega))}(\text{Sb}^1(n, n))$. Then again using Lemma 4.5.1, the set X defined by (A.1) in the proof of Theorem 4.5.3 is definable and we reach a contradiction exactly as in that proof. $\square$

4.6 Representability: First Incompleteness Theorem

The main goal of this section is the proof of the Representability Theorem, the precise version of the Representability Hypothesis 4.1.11. This will allow us to finish the proof of the First Incompleteness Theorem. We extend first the definitions to all relations and functions.

Definition 4.6.1 For any L_Ω -theory T and any $k > 0$,

- (i) a relation $R \subseteq \omega^k$ is ***T-representable*** iff for some L_Ω -formula $\phi(x_0, \dots, x_{k-1})$ and all $m_0, \dots, m_{k-1}$,

$$\begin{aligned} R(m_0, \dots, m_{k-1}) &\implies T \vdash \phi(\dot{m}_0, \dots, \dot{m}_{k-1}), \quad \text{and} \\ \sim R(m_0, \dots, m_{k-1}) &\implies T \vdash \neg \phi(\dot{m}_0, \dots, \dot{m}_{k-1}); \end{aligned}$$

- (ii) a relation $R \subseteq \omega^k$ is ***weakly T-representable*** iff for some L_Ω -formula $\phi(x_0, \dots, x_{k-1})$ and all $m_0, \dots, m_{k-1}$,

$$R(m_0, \dots, m_{k-1}) \iff T \vdash \phi(\dot{m}_0, \dots, \dot{m}_{k-1}),$$

or equivalently,

$$\begin{aligned} R(m_0, \dots, m_{k-1}) &\implies T \vdash \phi(\dot{m}_0, \dots, \dot{m}_{k-1}), \quad \text{and} \\ \sim R(m_0, \dots, m_{k-1}) &\implies T \nVdash \phi(\dot{m}_0, \dots, \dot{m}_{k-1}); \end{aligned}$$

- (iii) a function $F : \omega^k \rightarrow \omega$ is ***T-representable*** iff for some L_Ω -formula $\phi(x_0, \dots, x_{k-1}, y)$ and all $m_0, \dots, m_{k-1}$,

$$T \vdash \phi(\dot{m}_0, \dots, \dot{m}_{k-1}, y) \leftrightarrow y \doteq \overbrace{F(m_0, \dots, m_{k-1})}^{\cdot}.$$

In each case we say that ϕ (***weakly***) ***T-represents*** R or F .

Remark 4.6.2

- (i) Note that this extends Definition 4.1.10 in including relations and functions of arbitrary ranks.
- (ii) The reader may note that in this section, and occasionally already in [Section 4.1](#), we usually write $T \vdash \theta$ in place of its equivalent (when θ is a sentence) $\theta \in T$, especially when θ is a relatively long expression. The $\vdash$ version often seems easier to read and it emphasizes the role of formal proof as reflected in the Enumerability Theorem 4.4.17. This is also the reason for not using the third equivalent (by the Completeness Theorem) alternative $T \models \theta$.
- (iii) See the discussion at the end of this section of alternative notions of representability for functions.

Proposition 4.6.3 *For any consistent theory T and any relation R ,*

- (i) *if R is T -representable, then R is weakly T -representable;*
- (ii) *if T is complete and R is weakly T -representable, then R is T -representable;*
- (iii) *if R is T -representable, then for any theory $T' \supseteq T$, R is T' -representable;*
- (iv) *if $T \vdash \neg \dot{0} \doteq \dot{1}$, then R is T -representable iff its characteristic function K_R is T -representable;*
- (v) *R is T -representable iff its complement $\sim R$ is T -representable.*

Proof. These are all straightforward applications of the definitions and are left as exercises. $\square$

Corollary 4.6.4 *For any relation R ,*

$$\begin{aligned}
 R \text{ is weakly } \text{Th}(\Omega)\text{-representable} &\iff R \text{ is } \text{Th}(\Omega)\text{-representable} \\
 &\iff R \text{ is definable over } \Omega. \quad \square
 \end{aligned}$$

In understanding these notions it helps to notice what does *not* appear in the preceding proposition. In particular from the hypothesis that a relation R is weakly T -representable, it does not follow either that $\sim R$ is weakly T -representable or that R is weakly T' -representable for every theory $T' \supseteq T$. Counterexamples will be given once we have the necessary tools.

Our goal now is to show that all recursive functions — and hence all recursive relations — are $\mathbf{Q}$ -representable and hence T -representable for every theory $T \supseteq \mathbf{Q}$. Again we shall use the characterization Rec' and show that the initial functions are $\mathbf{Q}$ -representable and that the class of $\mathbf{Q}$ -representable functions is closed under composition and search. First we have

Lemma 4.6.5 For any L_Ω -theory T and any function F ,

- (i) F is T -representable iff for some L_Ω -formula $\phi(x_0, \dots, x_{k-1}, y)$ and all $m_0, \dots, m_{k-1}$,

$$T \vdash \phi(\dot{m}_0, \dots, \dot{m}_{k-1}, \overbrace{F(m_0, \dots, m_{k-1})}) \quad \text{and} \\ T \vdash \exists! y \phi(\dot{m}_0, \dots, \dot{m}_{k-1}, y);$$

- (ii) if there exists an L_Ω -term $t(x_0, \dots, x_{k-1})$ such that for all $m_0, \dots, m_{k-1}$,

$$T \vdash t(\dot{m}_0, \dots, \dot{m}_{k-1}) \doteq \overbrace{F(m_0, \dots, m_{k-1})},$$

then F is T -represented by the formula $y \doteq t$.

Proof. Let $\psi(y)$ be $\phi(\dot{m}_0, \dots, \dot{m}_{k-1}, y)$ and t be $\overbrace{F(m_0, \dots, m_{k-1})}$. Then (i) asserts

$$T \vdash \psi(y) \leftrightarrow y \doteq t \iff T \vdash \psi(t) \text{ and } T \vdash \exists! y \psi(y).$$

But for any formula ψ and any term t ,

$$(\psi(y) \leftrightarrow y \doteq t) \leftrightarrow [\psi(t) \wedge \exists! y \psi(y)]$$

is a logically valid formula, so (i) follows immediately. For (ii) we just observe that

$$(y \doteq t)(\dot{m}_0, \dots, \dot{m}_{k-1}) \text{ is } y \doteq t(\dot{m}_0, \dots, \dot{m}_{k-1}),$$

and under the hypothesis clearly

$$T \vdash y \doteq t(\dot{m}_0, \dots, \dot{m}_{k-1}) \leftrightarrow y \doteq \overbrace{F(m_0, \dots, m_{k-1})}. \quad \square$$

Proposition 4.6.6 The initial functions (4.2.1) are Q-representable.

Proof. We apply (ii) of the lemma; the terms required to witness the Q-representability of Cs_n^k , Pr_i^k and Sc_i^k are, respectively, $\dot{n}$, x_i and $\dot{\text{S}}x_i$. Thus we need to verify that

$$\begin{aligned} \text{Q} \vdash \dot{n}_{x_0, \dots, x_{k-1}}(\dot{m}_0, \dots, \dot{m}_{k-1}) &\doteq \overbrace{\text{Cs}_n^k(m_0, \dots, m_{k-1})}, \\ \text{Q} \vdash (x_i)_{x_0, \dots, x_{k-1}}(\dot{m}_0, \dots, \dot{m}_{k-1}) &\doteq \overbrace{\text{Pr}_i^k(m_0, \dots, m_{k-1})} \quad \text{and} \\ \text{Q} \vdash (\dot{\text{S}}x_i)_{x_0, \dots, x_{k-1}}(\dot{m}_0, \dots, \dot{m}_{k-1}) &\doteq \overbrace{\text{Sc}_i^k(m_0, \dots, m_{k-1})}. \end{aligned}$$

By the rules of substitution, these reduce to

$$\text{Q} \vdash \dot{n} \doteq \dot{n}, \quad \text{Q} \vdash \dot{m}_i \doteq \dot{m}_i \quad \text{and} \quad \text{Q} \vdash \dot{\text{S}}\dot{m}_i \doteq \overbrace{\dot{m}_i + 1},$$

which are all substitution instances of the logical validity $x \doteq x$. $\square$

Proposition 4.6.7 *The functions $+$ and $\times$ are $\mathbf{Q}$ -representable.*

Proof. We again apply Lemma 4.6.5(ii). The required terms are $x_0 \dot{+} x_1$ and $x_0 \dot{\times} x_1$. For addition we need to verify that for all m and n ,

$$\mathbf{Q} \vdash (x_0 \dot{+} x_1)_{x_0, x_1}(\dot{m}, \dot{n}) \doteq \overbrace{m+n}^{\dot{}}$$

or equivalently that

$$\mathbf{Q} \vdash \dot{m} \dot{+} \dot{n} \doteq \overbrace{m+n}^{\dot{}}.$$

This we do by induction on n . For $n = 0$, the equation is just a substitution instance of axiom (A1). Assume now as induction hypothesis that the assertion holds of n . Since $\overbrace{n+1}^{\dot{}}$ is exactly $\dot{S}n$, by axiom (A2) we have

$$\mathbf{Q} \vdash \dot{m} \dot{+} \overbrace{n+1}^{\dot{}} \doteq \dot{S}(\dot{m} \dot{+} \dot{n}).$$

Then by the induction hypothesis,

$$\mathbf{Q} \vdash \dot{m} \dot{+} \overbrace{n+1}^{\dot{}} \doteq \dot{S}(\overbrace{m+n}^{\dot{}}),$$

and the right-hand side of this is exactly $\overbrace{m+(n+1)}^{\dot{}}$ as required. The argument for multiplication is similar using axioms (M1) and (M2). $\square$

Proposition 4.6.8 *The relation $=$ is $\mathbf{Q}$ -representable.*

Proof. We need to verify that for all m and n ,

$$m = n \implies \mathbf{Q} \vdash \dot{m} \doteq \dot{n} \quad \text{and} \quad m \neq n \implies \mathbf{Q} \vdash \neg \dot{m} \doteq \dot{n}.$$

Then $=$ is $\mathbf{Q}$ -represented by the formula $x_0 \doteq x_1$. The first implication is immediate: if $m = n$, then $\dot{m} \doteq \dot{n}$ is a substitution instance of $x \doteq x$. For the second implication we prove by induction on m that for all n ,

$$m < n \implies \mathbf{Q} \vdash \neg \dot{m} \doteq \dot{n}.$$

If $m = 0 < n$, then $\dot{n}$ is $\dot{S}\overbrace{n-1}^{\dot{}}$ and $\mathbf{Q} \vdash \neg \dot{0} \doteq \dot{n}$ is an instance of axiom (S1). Suppose that $m+1 < n$. Then $m < n-1$ so by the induction hypothesis, $\mathbf{Q} \vdash \neg \dot{m} \doteq \overbrace{n-1}^{\dot{}}$. Then axiom (S2) guarantees that $\mathbf{Q} \vdash \neg \overbrace{m+1}^{\dot{}} \doteq \dot{n}$ as required. $\square$

Proposition 4.6.9 *The relation $<$ and its characteristic function $K_<$ are Q-representable.*

Proof. The representing formula is the obvious one, $x_0 \dot{<} x_1$, so we need to verify that for all m and n ,

$$m < n \implies Q \vdash \dot{m} \dot{<} \dot{n} \quad \text{and} \quad m \not< n \implies Q \vdash \neg \dot{m} \dot{<} \dot{n}.$$

Both of these are proved by induction on n . The first is vacuous for $n = 0$. Suppose that $m < (n + 1)$, so either $m < n$ or $m = n$. By, respectively, the induction hypothesis and the (proof of the) preceding proposition, this gives $Q \vdash \dot{m} \dot{<} \dot{n}$ or $Q \vdash \dot{m} \dot{=} \dot{n}$. Thus $Q \vdash (\dot{m} \dot{<} \dot{n} \vee \dot{m} \dot{=} \dot{n})$, so by axiom (L2), $Q \vdash \dot{m} \dot{<} \overbrace{\dot{n} + 1}^{\dot{}}$ as required.

The second implication with $n = 0$ is an instance of axiom (L1). Suppose $m \not< (n + 1)$, so both $m \not< n$ and $m \neq n$. By, respectively, the induction hypothesis and the preceding proposition, $Q \vdash \neg \dot{m} \dot{<} \dot{n}$ and $Q \vdash \neg \dot{m} \dot{=} \dot{n}$. Thus $Q \vdash \neg (\dot{m} \dot{<} \dot{n} \vee \dot{m} \dot{=} \dot{n})$, so again by axiom (L2), $Q \vdash \neg \dot{m} \dot{<} \overbrace{\dot{n} + 1}^{\dot{}}$ as required.

That $K_<$ is Q-representable now follows from Proposition 4.6.3(iv). $\square$

Proposition 4.6.10 *The class of Q-representable functions is closed under composition.*

Proof. Suppose that F arises from G, H_0, H_1 by composition,

$$F(m) = G(H_0(m), H_1(m)),$$

and assume that G, H_0 and H_1 are Q-representable by formulas

$$\psi(x_0, x_1, z), \quad \theta^0(x, y_0) \quad \text{and} \quad \theta^1(x, y_1),$$

respectively. Let $\phi(x, z)$ be the formula

$$\exists y_0 \exists y_1 [\theta^0(x, y_0) \wedge \theta^1(x, y_1) \wedge \psi(y_0, y_1, z)].$$

Fix m and let $p_i := H_i(m)$ ($i = 0, 1$) and $n := F(m) = G(p_0, p_1)$. By hypothesis,

$$\begin{aligned} Q \vdash \psi(\dot{p}_0, \dot{p}_1, z) &\leftrightarrow z \dot{=} \overbrace{G(p_0, p_1)}^{\dot{}} \quad \text{and} \\ Q \vdash \theta^i(\dot{m}, y_i) &\leftrightarrow y_i \dot{=} \overbrace{H_i(m)}^{\dot{}}. \end{aligned}$$

Then also all of the following equivalences are theorems of $\mathbf{Q}$:

$$\begin{aligned}
 \phi_x(\dot{m}, z) &\leftrightarrow \exists y_0 \exists y_1 [\theta^0(\dot{m}, y_0) \wedge \theta^1(\dot{m}, y_1) \wedge \psi(y_0, y_1, z)] \\
 &\leftrightarrow \exists y_0 \exists y_1 [y_0 \doteq p_0 \wedge y_1 \doteq p_1 \wedge \psi(y_0, y_1, z)] \\
 &\leftrightarrow \psi(\dot{p}_0, \dot{p}_1, z) \\
 &\leftrightarrow z \doteq \dot{n}.
 \end{aligned}$$

It follows that ϕ $\mathbf{Q}$ -represents F . $\square$

The next two lemmas are technical preparation for the proof that the $\mathbf{Q}$ -representable functions are closed under search and will also have other applications. To improve the readability of the notation we introduce in L_Ω abbreviations parallel to those in [Section 4.2](#):

$$\begin{aligned}
 (\exists x \dot{<} t)\phi &\text{ is an abbreviation for } \exists x(x \dot{<} t \wedge \phi) \\
 (\forall x \dot{<} t)\phi &\text{ is an abbreviation for } \forall x(x \dot{<} t \rightarrow \phi).
 \end{aligned}$$

Lemma 4.6.11 *For any $n > 0$,*

$$(i) \quad \mathbf{Q} \vdash y \dot{<} \dot{n} \rightarrow (y \doteq \dot{0} \vee \cdots \vee y \doteq \overbrace{\dot{n}-1}^{\cdot});$$

hence, for any formula $\phi(y)$,

$$(ii) \quad \mathbf{Q} \vdash (\forall y \dot{<} \dot{n})\phi \leftrightarrow \phi(\dot{0}) \wedge \cdots \wedge \phi(\overbrace{\dot{n}-1}^{\cdot});$$

$$(iii) \quad \mathbf{Q} \vdash (\exists y \dot{<} \dot{n})\phi \leftrightarrow \phi(\dot{0}) \vee \cdots \vee \phi(\overbrace{\dot{n}-1}^{\cdot});$$

$$(iv) \quad \text{if for all } p < n, \mathbf{Q} \vdash \phi_y(\dot{p}), \text{ then } \mathbf{Q} \vdash (\forall y \dot{<} \dot{n})\phi.$$

Proof. For (i) we proceed by induction on n . The case $n = 0$ is immediate from axiom (L1) (the empty disjunction is false). By axiom (L2) we have

$$\mathbf{Q} \vdash y \dot{<} \overbrace{\dot{n}+1}^{\cdot} \rightarrow (y \dot{<} \dot{n} \vee y \doteq \dot{n}),$$

and from this the induction step is also immediate. Then (ii) and (iii) follow from Exercise 2.2.43 and (iv) is immediate from (ii). $\square$

Remark 4.6.12 Part (i) of the lemma has a natural interpretation in any model $\mathfrak{A}$ of $\mathbf{Q}$: the predecessors of each $\dot{n}^{\mathfrak{A}}$ under the relation $\dot{<}^{\mathfrak{A}}$ are exactly the elements $\dot{0}^{\mathfrak{A}}, \dots, \overbrace{\dot{n}-1}^{\cdot}^{\mathfrak{A}}$. In other words, the set $\{\dot{n}^{\mathfrak{A}} : n \in \omega\}$ of standard elements of $\mathfrak{A}$ forms an initial segment of the universe of the model. We say that $\mathfrak{A}$ is an **end extension** of the model consisting of the standard elements.

Lemma 4.6.13 *For any n , any formula $\phi(y)$ and any variable w not occurring in ϕ , if for all $p < n$, $\mathbf{Q} \vdash \neg \phi(\dot{p})$, but $\mathbf{Q} \vdash \phi(\dot{n})$, then*

$$\mathbf{Q} \vdash [\phi(y) \wedge (\forall w \dot{<} y) \neg \phi(w)] \leftrightarrow y \doteq \dot{n}.$$

Proof. The idea of the proof is that in any model $\mathfrak{A}$ of $\mathbf{Q}$, under the given hypotheses, $\dot{n}^{\mathfrak{A}}$ is the unique element which satisfies the left side of the equivalence. Let $\theta(y)$ denote the formula $\phi(y) \wedge (\forall w \dot{<} y) \neg \phi(w)$. By (iii) of the preceding lemma and the hypothesis, we have $\mathbf{Q} \vdash \theta(\dot{n})$. Hence (by Exercise 2.2.43)

$$(1) \quad \mathbf{Q} \vdash y \doteq \dot{n} \rightarrow \theta(y).$$

Since $\mathbf{Q} \vdash y \dot{<} \dot{n} \rightarrow \neg \phi(y)$, also $\mathbf{Q} \vdash \phi(y) \rightarrow \neg y \dot{<} \dot{n}$ and thus

$$(2) \quad \mathbf{Q} \vdash \theta(y) \rightarrow \neg y \dot{<} \dot{n}.$$

Furthermore, $\mathbf{Q} \vdash \theta(y) \rightarrow (\dot{n} \dot{<} y \rightarrow \neg \phi(\dot{n}))$, so since by hypothesis $\mathbf{Q} \vdash \phi(\dot{n})$,

$$(3) \quad \mathbf{Q} \vdash \theta(y) \rightarrow \neg \dot{n} \dot{<} y.$$

Then from (2), (3) and axiom (L3) (its first use) we have $\mathbf{Q} \vdash \theta(y) \rightarrow y \doteq \dot{n}$, which together with (1) gives the result. $\square$

Proposition 4.6.14 *The class of $\mathbf{Q}$ -representable functions is closed under search.*

Proof. Suppose that the unary function F arises from G by search, so G satisfies the search condition

$$\forall m \exists n [G(m, n) = 0],$$

and $F(m) = \text{least } n [G(m, n) = 0]$. Suppose also that G is $\mathbf{Q}$ -represented by the formula $\psi(x, y, z)$: for all m and n ,

$$\mathbf{Q} \vdash \psi(\dot{m}, \dot{n}, z) \leftrightarrow z \doteq \overbrace{G(m, n)}^{\dot{}}.$$

Let $\theta(x, y)$ be the formula $(\forall w \dot{<} y) \neg \psi(x, w, \dot{0}) \wedge \psi(x, y, \dot{0})$. We shall show that F is $\mathbf{Q}$ -represented by θ . Fix m and $n = F(m)$. We need to verify that

$$(4) \quad \mathbf{Q} \vdash \theta_x(\dot{m}, y) \leftrightarrow y \doteq \dot{n},$$

or, putting in the definition of θ ,

$$(5) \quad \mathbf{Q} \vdash ((\forall w \dot{<} y) \neg \psi(\dot{m}, w, \dot{0}) \wedge \psi(\dot{m}, y, \dot{0})) \leftrightarrow y \doteq \dot{n}.$$

Let $\phi(y)$ denote the formula $\psi(\dot{m}, y, \dot{0})$. Then the desired result (5) becomes

$$(6) \quad \mathbf{Q} \vdash [\phi(y) \wedge (\forall w \prec y) \neg \phi(w)] \leftrightarrow y \doteq \dot{n}.$$

By the preceding lemma, to establish (6) it suffices to show

$$(7) \quad \text{for all } p < n, \quad \mathbf{Q} \vdash \neg \phi(\dot{p}) \quad \text{and} \quad \mathbf{Q} \vdash \phi(\dot{n}).$$

Suppose first that p is some number less than n . Then $G(m, p) \neq 0$, so, since ψ represents G , $\mathbf{Q} \vdash \neg \psi(\dot{m}, \dot{p}, \dot{0})$ — in other words, $\mathbf{Q} \vdash \neg \phi(\dot{p})$ as desired. On the other hand, $G(m, n) = 0$, so in the same way, $\mathbf{Q} \vdash \psi(\dot{m}, \dot{n}, \dot{0})$, and hence $\mathbf{Q} \vdash \phi(\dot{n})$. $\square$

Representability Theorem 4.6.15 *For any consistent L_Ω -theory $T \supseteq \mathbf{Q}$ and all $k \in \omega$,*

- (i) *every recursive function $F : \omega^k \rightarrow \omega$ is T -representable;*
- (ii) *every recursive relation $R \subseteq \omega^k$ is T -representable.*

Proof. We have now shown that the class of $\mathbf{Q}$ -representable functions contains the initial functions and satisfies the closure conditions for $\mathbf{Rec}'$. It follows by $\mathbf{Rec}'$ -induction that $\mathbf{Rec}'$ is included in this class. Thus by Theorem 4.2.44, the same is true of $\mathbf{Rec}$. The extension to all $T \supseteq \mathbf{Q}$ is immediate: if the representing equivalence is a theorem of $\mathbf{Q}$, then it is also a theorem of any $T \supseteq \mathbf{Q}$. $\square$

Note that as compared with the Representability Hypothesis 4.1.11 there is no mention here of the weak representability of semi-recursive relations. A version of this does hold but will take some additional work, and we don't need it for the first proof below. Again the numbering matches that of the proof of Theorem 4.1.13.

First Incompleteness Theorem 4.6.16 *For any L_Ω -theory T compatible with $\mathbf{Q}$,*

- (i) *T is undecidable;*
- (ii) *if T is recursively axiomatizable, then T is incomplete.*

Proof. Suppose first that T is consistent and $T \supseteq \mathbf{Q}$ and, towards a contradiction, that T is decidable — that is, $\mathbf{gn}(T)$ is a recursive set. Let $X \subseteq \omega$ be the set defined by

$$(C.1) \quad \text{for all } m, \quad m \in X \iff \mathbf{Sb}^1(m, m) \in \mathbf{gn}(T).$$

By Proposition 4.2.19, X is a recursive set, so by (ii) of the Representability Theorem is T -representable and for some formula ψ ,

$$(C.2) \quad \begin{array}{lll} \text{for all } m, & m \in X & \implies \psi(\dot{m}) \in T \\ \text{and} & m \notin X & \implies \neg \psi(\dot{m}) \in T. \end{array}$$

Since T is consistent it follows that

$$(C.3) \quad \text{for all } m, \quad m \in X \iff \neg \psi(\dot{m}) \notin T.$$

Let $p := \mathbf{gn}(\neg \psi)$, so by its definition $\mathbf{gn}(\neg \psi(\dot{p})) = \mathbf{Sb}^1(p, p)$, and by (C.1) and (C.3) we have

$$\begin{aligned} p \in X &\iff \mathbf{Sb}^1(p, p) \in \mathbf{gn}(T) \iff \neg \psi(\dot{p}) \in T \\ p \in X &\iff \neg \psi(\dot{p}) \notin T. \end{aligned}$$

Hence if θ is the sentence $\neg \psi(\dot{p})$,

$$(C.4) \quad \theta \in T \iff \theta \notin T,$$

a contradiction. Thus T is undecidable and it follows immediately from the Enumerability Theorem 4.4.17 that if T is recursively axiomatizable then T cannot be complete. Note that this corresponds to the first proof of Theorem 4.1.14; that corresponding to the second must wait until we establish part (iii) of the Representability Theorem below.

Now if T is only compatible with $\mathbf{Q}$, proceed as in the proof of Theorem 4.1.17. Let T^+ be the theory generated by $T \cup \mathbf{Q}$, so by hypothesis T^+ is a consistent extension of $\mathbf{Q}$ and hence by the first part is undecidable. But easily, if $\chi_{\mathbf{Q}}$ is the conjunction of the (finitely many) axioms of $\mathbf{Q}$, then for any formula θ ,

$$\theta \in T^+ \iff (\chi_{\mathbf{Q}} \rightarrow \theta) \in T,$$

and the function $\mathbf{gn}(\theta) \mapsto \mathbf{gn}(\chi_{\mathbf{Q}} \rightarrow \theta)$ is obviously recursive. Thus if $\mathbf{gn}(T)$ were recursive, then so would be $\mathbf{gn}(T^+)$. $\square$

Corollary 4.6.17

- (i) $\mathbf{Q}$ and $\mathbf{P}$ are undecidable.
- (ii) The validity problem for L_{Ω} is undecidable — that is, the set of Gödel numbers of logically valid sentences of L_{Ω} is not recursive. $\square$

We shall see in Sections 5.6 and 6.4 that the same is true for languages much simpler than L_{Ω} , although it is not true for $L_{=}$ (Theorem 2.4.41). From this we have also a result which has nothing (explicitly) to do with undecidability or incompleteness of theories:

Corollary 4.6.18 *There exist sets of natural numbers which are recursively enumerable but not recursive.*

Proof. For any recursively axiomatizable theory $T \supseteq \mathbf{Q}$, $\mathbf{gn}(T)$ is such a set. $\square$

Despite the simplicity of this proof, it is instructive and useful for later application to give another longer one based on the Diagonal Lemma 4.1.21.

Definition 4.6.19 For any L_Ω -theory T and each $k > 0$,

$$U_T^k := \left\{ (a, m_0, \dots, m_{k-1}) : \mathbf{Sb}^k(a, m_0, \dots, m_{k-1}) \in \mathbf{gn}(T) \right\}.$$

Proposition 4.6.20 For any consistent L_Ω -theory $T \supseteq \mathbf{Q}$ and each $k > 0$, U_T^k is universal for the class of k -ary recursive relations. If T is recursively axiomatizable, then U_T^k is semi-recursive.

Proof. Fix T as in the hypothesis. By the Representability Theorem for any k -ary recursive relation R , there is a formula ψ which T -represents R and hence by Proposition 4.6.3 also weakly represents R — that is, for all $m_0, \dots, m_{k-1}$,

$$R(m_0, \dots, m_{k-1}) \iff \psi(\dot{m}_0, \dots, \dot{m}_{k-1}) \in T.$$

Hence

$$\begin{aligned} R(m_0, \dots, m_{k-1}) &\iff \mathbf{gn}(\psi(\dot{m}_0, \dots, \dot{m}_{k-1})) \in \mathbf{gn}(T) \\ &\iff \mathbf{Sb}^k(\mathbf{gn}(\psi), m_0, \dots, m_{k-1}) \in \mathbf{gn}(T) \\ &\iff U_T^k(\mathbf{gn}(\psi), m_0, \dots, m_{k-1}). \end{aligned}$$

Thus $\mathbf{gn}(\psi)$ is a U_T^k -index for R , and every k -ary recursive relation is a U_T^k -section.

If T is recursively axiomatizable, then by the Enumerability Theorem $\mathbf{gn}(T)$ is a recursively enumerable set so U_T^k is semi-recursive. $\square$

Corollary 4.6.21 For any consistent L_Ω -theory $T \supseteq \mathbf{Q}$ and each $k > 0$, U_T^k is not recursive.

Proof. For $k = 1$ this is a direct application of the Diagonal Lemma 4.1.21; for arbitrary k we use a slight modification of the proof of 4.1.21. Suppose towards a contradiction that U_T^k is recursive. Let $\mathbf{m}$ denote a sequence $(m, m, \dots, m)$ of length k and set

$$D_T^k := \{ \mathbf{m} : U_T^k(m, \mathbf{m}) \}.$$

Then also $\sim D_T^k$ is recursive, so is a U_T^k -section, say with index a . Then

$$\mathbf{a} \in D_T^k \iff U_T^k(a, \mathbf{a}) \iff \mathbf{a} \in D_{T,a}^k \iff \mathbf{a} \notin D_T^k,$$

a contradiction. $\square$

Of course this provides another proof that T as above is undecidable, since if T were decidable, it would follow that U_T^k is recursive. Note that D_T^1 is exactly the set X of the proof of the First Incompleteness Theorem.

For recursively axiomatizable T , the relation U_T^k is a semi-recursive relation which is universal for the class of k -ary recursive relations. The essence of the preceding proof is that there can be no *recursive* relation universal for this class. We shall see below that in fact U_T^k is in fact universal for the larger class of k -ary semi-recursive relations.

This proof of the First Incompleteness Theorem is not the original one of Gödel but is due to Tarski some years later. Gödel's proof was a constructive one, providing an explicit sentence θ_T such that neither θ_T nor $\neg\theta_T$ belongs to T as in the second proof of the version 4.1.14. However it was weaker in applying only to ω -consistent theories. The first version without this restriction is due to Rosser; these proofs are sketched in the exercises.

Definition 4.6.22 An L_Ω -theory T is ω -**consistent** iff for every L_Ω -formula $\theta(x)$, if for all $p \in \omega$, $\neg\theta(\dot{p}) \in T$, then $\exists x \theta \notin T$.

Remark 4.6.23 Note first that an ω -consistent theory is also consistent, since *every* sentence belongs to the inconsistent theory. T is ω -consistent iff whenever for all $p \in \omega$, $\neg\theta(\dot{p}) \in T$, then the theory generated by $T \cup \{\forall x \neg\theta\}$ is consistent. A stronger property called ω -**completeness** requires that if for all $p \in \omega$, $\neg\theta(\dot{p}) \in T$, then $\forall x \neg\theta \in T$. Any subtheory of $\text{Th}(\Omega)$, in particular $\mathbf{Q}$ and $\mathbf{P}$, is ω -consistent and $\text{Th}(\Omega)$ itself is ω -complete. If θ is a witness to the non- ω -completeness of a consistent theory T , then the theory generated by $T \cup \{\exists x \theta\}$ is consistent but not ω -consistent.

Proposition 4.6.24 For any ω -consistent L_Ω -theory $T \supseteq \mathbf{Q}$, every semi-recursive relation is weakly T -representable — in fact, for each k -ary semi-recursive relation R there exists an L_Ω -formula $\psi(x_0, \dots, x_{k-1})$ such that for all $m_0, \dots, m_{k-1}$,

$$\begin{aligned} R(m_0, \dots, m_{k-1}) &\iff \psi(\dot{m}_0, \dots, \dot{m}_{k-1}) \in T \\ &\iff \psi(\dot{m}_0, \dots, \dot{m}_{k-1}) \in \text{Th}(\Omega). \end{aligned}$$

Proof. To simplify notation we do the proof for sets ($k = 1$). Fix T as in the hypothesis, a recursively enumerable set X , a recursive relation S such that $X = \{m : \exists n S(m, n)\}$, and a formula $\phi(x, y)$ which $\mathbf{Q}$ -represents S — that is for all m and n ,

$$S(m, n) \implies \phi(\dot{m}, \dot{n}) \in \mathbf{Q} \quad \text{and} \quad \sim S(m, n) \implies \neg\phi(\dot{m}, \dot{n}) \in \mathbf{Q}.$$

Let $\psi(x)$ be $\exists y \phi(x, y)$; we show that for all m ,

$$m \in X \iff \psi(\dot{m}) \in T \iff \psi(\dot{m}) \in \text{Th}(\Omega).$$

This follows immediately from the implications

$$m \in X \implies \exists n S(m, n) \implies \exists n [\phi(\dot{m}, \dot{n}) \in \mathbf{Q}] \xrightarrow{(8)} \begin{cases} \psi(\dot{m}) \in T \\ \psi(\dot{m}) \in \text{Th}(\Omega), \end{cases}$$

and

$$\begin{aligned}
 m \notin X &\implies \forall n \sim S(m, n) \\
 &\implies \forall n [\neg \phi(\dot{m}, \dot{n}) \in \mathbf{Q}] \stackrel{(9)}{\implies} \begin{cases} \psi(\dot{m}) \notin T \\ \psi(\dot{m}) \notin \text{Th}(\Omega). \end{cases}
 \end{aligned}$$

Here the implications (8) are simply logical, using the hypothesis that $\mathbf{Q} \subseteq T$ and $\mathbf{Q} \subseteq \text{Th}(\Omega)$, while (9) uses additionally the ω -consistency of T and $\text{Th}(\Omega)$, respectively. $\square$

Now we can answer a question mentioned following Corollary 4.3.13.

Corollary 4.6.25 *Every semi-recursive relation is expressible in the form $\exists p R(m_0, \dots, m_{k-1}, p)$ with R primitive recursive.*

Proof. Let Γ be the finite set of axioms for $\mathbf{Q}$. For any semi-recursive relation P , fix a formula ψ which weakly $\mathbf{Q}$ -represents it. Then by the proof of the Enumerability Theorem 4.4.17, for all $m_0, \dots, m_{k-1}$,

$$\begin{aligned}
 P(m_0, \dots, m_{k-1}) &\iff \psi(\dot{m}_0, \dots, \dot{m}_{k-1}) \in \mathbf{Q} \\
 &\iff \exists s \text{Proofof}_\Gamma(\text{gn}(\psi(\dot{m}_0, \dots, \dot{m}_{k-1})), s) \\
 &\iff \exists s \text{Proofof}_\Gamma(\text{Sb}^k(\text{gn}(\psi), m_0, \dots, m_{k-1}), s).
 \end{aligned}$$

Since both Proofof_Γ and Sb^k are primitive recursive, this is of the desired form. $\square$

The proof of the preceding proposition under the hypothesis of simple consistency was given (essentially) by Rosser and uses a generalization of 4.1.24. Since we can not expect the implications (9) to hold, the formula ψ must be derived from ϕ in a more complex way.

Definition 4.6.26 For any formula ϕ , $\ulcorner \phi \urcorner := \overbrace{\text{gn}(\phi)}^{\cdot}$.

Think of $\ulcorner \phi \urcorner$ as being a name for ϕ in the language L_Ω . Note that if Γ is a recursive set of formulas, then for any $T \supseteq \mathbf{Q}$, $\text{gn}(\Gamma)$ is $\mathbf{Q}$ -representable, say for all m ,

$$m \in \text{gn}(\Gamma) \implies T \vdash \theta(\dot{m}) \quad \text{and} \quad m \notin \text{gn}(\Gamma) \implies T \vdash \neg \theta(\dot{m}),$$

hence for all formulas ϕ ,

$$\phi \in \Gamma \implies T \vdash \theta(\ulcorner \phi \urcorner) \quad \text{and} \quad \phi \notin \Gamma \implies T \vdash \neg \theta(\ulcorner \phi \urcorner).$$

For the rest of the section let x, y and z denote the fixed variables v_0, v_1 and v_2 . Extending 4.1.24 we have the

Self-reference Lemma 4.6.27 *For any L_Ω -formula $\theta(y)$, there exists a sentence η such that*

$$\mathbf{Q} \vdash \eta \leftrightarrow \theta(\ulcorner \eta \urcorner).$$

More generally, for any formula $\theta(y, z)$, there exists a formula $\eta(z)$ such that for all $n \in \omega$,

$$\mathbf{Q} \vdash \eta(\dot{n}) \leftrightarrow \theta(\ulcorner \eta(\dot{n}) \urcorner, \dot{n}),$$

and similarly for more free variables.

Proof. Fix a formula $\theta(y)$ and let δ be the recursive function, $\delta(m) = \text{Sb}^1(m, m)$. Note that (just by unraveling the definitions) for any formula ϕ ,

$$\delta(\text{gn}(\phi)) = \text{gn}(\phi(\ulcorner \phi \urcorner)).$$

Choose a formula $\psi(x, y)$ which $\mathbf{Q}$ -represents δ . In particular,

$$\mathbf{Q} \vdash \psi(\ulcorner \phi \urcorner, y) \leftrightarrow y \doteq \phi(\ulcorner \phi \urcorner).$$

Let $\chi(x)$ denote the formula $\forall y (\psi(x, y) \rightarrow \theta(y))$ and η the sentence $\chi(\ulcorner \chi \urcorner)$. Then in particular,

$$\mathbf{Q} \vdash \psi(\ulcorner \chi \urcorner, y) \leftrightarrow y \doteq \ulcorner \eta \urcorner.$$

Now η is the sentence $\forall y (\psi(\ulcorner \chi \urcorner, y) \rightarrow \theta(y))$, and by substitution using the last equivalence we have

$$\mathbf{Q} \vdash \eta \leftrightarrow \forall y (y \doteq \ulcorner \eta \urcorner \rightarrow \theta(y)) \leftrightarrow \theta(\ulcorner \eta \urcorner),$$

as desired.

For the second version, let δ be a recursive function such that for any formula $\phi(x, z)$ and any n ,

$$\delta(\text{gn}(\phi), n) = \text{gn}(\phi(\ulcorner \phi \urcorner, \dot{n})),$$

and choose a formula $\psi(x, y, z)$ which $\mathbf{Q}$ -represents δ , so for any ϕ and n ,

$$\mathbf{Q} \vdash \psi(\ulcorner \phi \urcorner, y, \dot{n}) \leftrightarrow y \doteq \phi(\ulcorner \phi \urcorner, \dot{n}).$$

Let $\chi(x, z)$ denote the formula $\forall y (\psi(x, y, z) \rightarrow \theta(y, z))$ and $\eta(z)$ the formula $\chi(\ulcorner \chi \urcorner, z)$. Then in particular, for any n ,

$$\mathbf{Q} \vdash \psi(\ulcorner \chi \urcorner, y, \dot{n}) \leftrightarrow y \doteq \ulcorner \eta(\dot{n}) \urcorner.$$

Now $\eta(\dot{n})$ is the formula $\forall y (\psi(\ulcorner \chi \urcorner, y, \dot{n}) \rightarrow \theta(y, \dot{n}))$ and by substitution using the last equivalence we have

$$\mathbf{Q} \vdash \eta(\dot{n}) \leftrightarrow \forall y (y \doteq \ulcorner \eta(\dot{n}) \urcorner \rightarrow \theta(y, \dot{n})) \leftrightarrow \theta(\ulcorner \eta(\dot{n}) \urcorner, \dot{n}),$$

as desired. $\square$

Definition 4.6.28 For any L_Ω -theory T , a formula $\phi(y)$ is called ***T -definite (below n)*** iff for all p (all $p < n$) either $T \vdash \phi(\dot{p})$ or $T \vdash \neg \phi(\dot{p})$.

Note that if ϕ is a formula which T -represents a set, then ϕ is T -definite.

Lemma 4.6.29 For any consistent L_Ω -theory T , any $n \in \omega$ and any formulas $\phi(y)$ and $\psi(y)$,

- (i) if ϕ and ψ are T -definite (below n), then so are $\neg \phi$, $\phi \wedge \psi$ and $\phi \vee \psi$;
- (ii) if $\psi(y)$ is T -definite (below n), then so are $(\exists y \dot{<} z) \psi$ and $(\forall y \dot{<} z) \psi$;
- (iii) if $T \supseteq Q$, $\phi(y)$ is Q -definite below n , and $T \vdash (\exists y \dot{<} \dot{n}) \phi$, then for some $p < n$, $Q \vdash \phi(\dot{p})$.

Proof. Part (i) is immediate. For (ii), suppose that ψ is T -definite below p . If for some $q < p$, $T \vdash \psi(\dot{q})$, then by Lemma 4.6.11(ii), also

$$T \vdash (\exists y \dot{<} \dot{p}) \psi(y).$$

Otherwise, for all $q < p$, $T \vdash \neg \psi(\dot{q})$, so by Lemma 4.6.11(iii),

$$T \vdash (\forall y \dot{<} \dot{p}) \neg \psi(y) \quad \text{so} \quad T \not\vdash (\exists y \dot{<} \dot{p}) \psi(y),$$

since T is consistent. Finally, for (iii), suppose that for all $p < n$, $Q \not\vdash \phi(\dot{p})$. Then for all $p < n$, $Q \vdash \neg \phi(\dot{p})$, so by Lemma 4.6.11(iii),

$$Q \vdash (\forall y \dot{<} \dot{n}) \neg \phi(y).$$

Since $T \subseteq Q$,

$$T \vdash (\forall y \dot{<} \dot{n}) \neg \phi(y) \quad \text{so} \quad T \not\vdash (\exists y \dot{<} \dot{n}) \phi(y),$$

again by the consistency of T . $\square$

Representability Theorem 4.6.30 (Part (iii)) For any consistent recursively axiomatizable L_Ω -theory $T \supseteq Q$, every semi-recursive relation is weakly T -representable — in fact, for each k -ary semi-recursive relation R there exists an L_Ω -formula $\psi(x_0, \dots, x_{k-1})$ such that for all $m_0, \dots, m_{k-1}$,

$$\begin{aligned} R(m_0, \dots, m_{k-1}) &\iff \psi(\dot{m}_0, \dots, \dot{m}_{k-1}) \in T \\ &\iff \psi(\dot{m}_0, \dots, \dot{m}_{k-1}) \in \text{Th}(\Omega). \end{aligned}$$

Proof. Fix a consistent recursively axiomatizable theory $T \supseteq Q$, a recursively enumerable set X (to simplify notation we do the proof for $k = 1$), a recursive relation S such that

$$(10) \quad X = \{ m : \exists n S(m, n) \},$$

and an L_Ω -formula $\phi(z, u)$ which $\mathbf{Q}$ -represents S — that is, such that for all m and n ,

$$S(m, n) \implies \mathbf{Q} \vdash \phi(\dot{m}, \dot{n}) \quad \text{and} \quad \sim S(m, n) \implies \mathbf{Q} \vdash \neg \phi(\dot{m}, \dot{n}).$$

Because T is recursively axiomatizable, $\mathbf{gn}(T)$ is recursively enumerable and hence there is also a recursive relation S' such that

$$\mathbf{gn}(T) = \{ p : \exists q S'(p, q) \}$$

and an L_Ω -formula $\chi(y, w)$ which $\mathbf{Q}$ -represents S' — that is, such that for all p and q ,

$$(11) \quad S'(p, q) \implies \mathbf{Q} \vdash \chi(\dot{p}, \dot{q}) \quad \text{and} \quad \sim S'(p, q) \implies \mathbf{Q} \vdash \neg \chi(\dot{p}, \dot{q}).$$

By the Self-reference Lemma there is an L_Ω -formula $\psi(z)$ such that for all m ,

$$(12) \quad \mathbf{Q} \vdash \psi(\dot{m}) \leftrightarrow \exists u [\phi(\dot{m}, u) \wedge (\forall w \dot{<} u) \neg \chi(\ulcorner \psi(\dot{m}) \urcorner, w)].$$

We shall show that for all m ,

$$(13) \quad m \in X \implies \psi(\dot{m}) \in T \implies \psi(\dot{m}) \in \mathbf{Th}(\Omega) \implies m \in X.$$

As in the proof of Proposition 4.6.24, by (10), for all m

$$(14) \quad m \in X \implies \exists n [\phi(\dot{m}, \dot{n}) \in \mathbf{Q}],$$

$$(15) \quad m \notin X \implies \forall n [\neg \phi(\dot{m}, \dot{n}) \in \mathbf{Q}],$$

and as in that proof by (15), here using also (12),

$$m \notin X \implies \exists u \phi(\dot{m}, u) \notin \mathbf{Th}(\Omega) \implies \psi(\dot{m}) \notin \mathbf{Th}(\Omega).$$

This gives the third implication of (13). Next, by (11), for all m ,

$$(16) \quad \psi(\dot{m}) \in T \implies \exists q [\chi(\ulcorner \psi(\dot{m}) \urcorner, \dot{q}) \in \mathbf{Q}];$$

$$(17) \quad \psi(\dot{m}) \notin T \implies \forall q [\neg \chi(\ulcorner \psi(\dot{m}) \urcorner, \dot{q}) \in \mathbf{Q}].$$

For the first implication of (13), assume that $m \in X$ and by (14) fix n such that $\phi(\dot{m}, \dot{n}) \in \mathbf{Q}$; towards a contradiction, suppose that $\psi(\dot{m}) \notin T$. Then by (17) and Lemma 4.6.11(iii),

$$(\forall w \dot{<} \dot{n}) \neg \chi(\ulcorner \psi(\dot{m}) \urcorner, w) \in \mathbf{Q},$$

and we conclude

$$(18) \quad \exists u [\phi(\dot{m}, u) \wedge (\forall w \dot{<} u) \neg \chi(\ulcorner \psi(\dot{m}) \urcorner, w)] \in \mathbf{Q}.$$

But then, by (12), also $\psi(\dot{m}) \in \mathbf{Q} \subseteq T$, contrary to hypothesis.

Finally, for the second implication of (13), assume that $\psi(\dot{m}) \in T$, so by (12) also

$$(19) \quad \exists u [\phi(\dot{m}, u) \wedge (\forall w \dot{<} u) \neg \chi(\ulcorner \psi(\dot{m}) \urcorner, w)] \in T.$$

Also, by (16) we may choose q such that

$$(\exists w \dot{<} \dot{q}) \chi(\ulcorner \psi(\dot{m}) \urcorner, w) \in \mathbf{Q} \subseteq T.$$

It follows, using axiom (L3), that

$$T \vdash (\forall w \dot{<} u) \neg \chi(\ulcorner \psi(\dot{m}) \urcorner, w) \rightarrow u \dot{<} \dot{q},$$

and therefore by (19) that

$$(20) \quad (\exists u \dot{<} \dot{q}) [\phi(\dot{m}, u) \wedge (\forall w \dot{<} u) \neg \chi(\ulcorner \psi(\dot{m}) \urcorner, w)] \in T.$$

Since ϕ and χ are $\mathbf{Q}$ -definite, by (i) and (ii) of the preceding lemma so is the formula inside the square brackets in (20), so by (iii) of the lemma, for some $n < q$,

$$(21) \quad \phi(\dot{m}, \dot{n}) \wedge (\forall w \dot{<} \dot{n}) \neg \chi(\ulcorner \psi(\dot{m}) \urcorner, w) \in \mathbf{Q}.$$

Hence also again (18) holds, so by (12), $\psi(\dot{m}) \in \mathbf{Q} \subseteq \text{Th}(\Omega)$. $\square$

Now we can give (compare the proof of Theorem 4.1.14) the

Constructive Proof of the First Incompleteness Theorem

Fix a consistent recursively axiomatizable theory $T \supseteq \mathbf{Q}$, and let $X_T \subseteq \omega$ be the set defined by

$$(D.1) \quad \text{for all } m, \quad m \in X_T \iff \text{Sb}^1(m, m) \in \text{gn}(T).$$

By the Enumerability Theorem, $\text{gn}(T)$ is recursively enumerable, by Exercise 4.3.25 (or Corollary 4.3.20), X_T is recursively enumerable and hence by the Representability Theorem (iii) X is weakly T -representable, say by the formula ψ_T ,

$$(D.2) \quad \text{for all } m, \quad m \in X_T \iff \psi_T(\dot{m}) \in T,$$

hence also

$$(D.3) \quad \text{for all } m, \quad m \in X_T \iff \neg \neg \psi_T(\dot{m}) \in T.$$

Let $p_T := \text{gn}(\neg \psi_T)$, so $\text{gn}(\neg \psi_T(\dot{p}_T)) = \text{Sb}^1(p_T, p_T)$, and by (D.1) and (D.3), respectively,

$$\begin{aligned} p_T \in X_T &\iff \text{Sb}^1(p_T, p_T) \in \text{gn}(T) \iff \neg \psi_T(\dot{p}_T) \in T; \\ p_T \in X_T &\iff \neg \neg \psi_T(\dot{p}_T) \in T. \end{aligned}$$

Hence if θ_T is the sentence $\neg \psi_T(\dot{p}_T)$, we have

$$(D.4) \quad \theta_T \in T \iff \neg \theta_T \in T,$$

and it follows immediately from the consistency of T that neither θ_T nor $\neg \theta_T$ belongs to T . $\square$

Remark 4.6.31

- (i) In the preceding proof, we have by (D.2) that $p_T \notin X_T$ and therefore by the second equivalence of 4.6.30, $\psi(\dot{p}_T) \notin \text{Th}(\Omega)$. Hence $\theta_T \in \text{Th}(\Omega)$, so as in Theorem 4.5.4, θ_T is true but not a theorem of T .
- (ii) By (12) in the proof of 4.6.30,

$$\mathbf{Q} \vdash \theta_T \leftrightarrow \forall u [\neg \phi(\dot{p}_T, u) \vee (\exists w \dot{<} u) \neg \chi(\ulcorner \psi(\dot{p}_T) \urcorner, w)].$$

Since $p_T \notin X_T$, by (15), for all n , $\neg \phi(p_T, \dot{n}) \in \mathbf{Q}$ and hence

$$\neg \phi(\dot{p}_T, \dot{n}) \vee (\exists w \dot{<} \dot{n}) \neg \chi(\ulcorner \psi(\dot{p}_T) \urcorner, w) \in T.$$

But since $\theta_T \notin T$,

$$\forall u [\neg \phi(\dot{p}_T, u) \vee (\exists w \dot{<} u) \neg \chi(\ulcorner \psi(\dot{p}_T) \urcorner, w)] \notin T.$$

Thus this formula witnesses that T is not ω -complete and the method of Remark 4.6.23 produces an example of a theory which is consistent but not ω -consistent.

- (iii) Although this proof of the First Incompleteness Theorem provides an algorithm which on input T produces a witness θ_T to the incompleteness of T , it is natural to ask if for particular theories such as $\mathbf{Q}$ or $\mathbf{P}$ we might be able to find witnesses with more interesting mathematical content — that is, questions which arose independently and turn out to be independent of the theory. For $\mathbf{Q}$ there are many such examples; some are given in Exercise 4.1.27. However, it took many years after Gödel's work to come across such examples for $\mathbf{P}$, and it is still not universally agreed that they are “natural” — see the Collateral Reading below. In [Chapter 6](#) we shall see that for set theory there are very natural examples to witness the incompleteness of $\mathbf{ZF}$ and $\mathbf{ZFC}$.

Theorem 4.6.30 also gives a strengthening of Proposition 4.6.20.

Corollary 4.6.32 *For any recursively axiomatizable consistent L_Ω -theory $T \supseteq \mathbf{Q}$ and each $k > 0$, U_T^k is a semi-recursive relation universal for the class of k -ary semi-recursive relations.*

Proof. The proof of 4.6.20 showed that every k -ary relation which is weakly T -representable is a U_T^k -section. $\square$

This finishes the main objective of this section and chapter, but there are a few remarks which may help to clarify the situation. First we note the following converse to two of our main results.

Theorem 4.6.33 *For any consistent recursively axiomatizable L_Ω -theory T , any total function F and any relation R ,*

- (i) *if $R(F)$ is T -representable, then $R(F)$ is recursive; hence if $T \supseteq \mathbf{Q}$, then $R(F)$ is recursive iff $R(F)$ is T -representable;*
- (ii) *if R is weakly T -representable, then R is semi-recursive; hence if $T \supseteq \mathbf{Q}$, then R is semi-recursive iff R is weakly T -representable.*

Proof. For simplicity we deal with sets. Suppose that a set X is weakly T -representable, say by the formula ψ . Then for all n ,

$$n \in X \iff \psi(\dot{n}) \in T \iff \text{Sb}^1(\text{gn}(\psi), n) \in \text{gn}(T).$$

Since $\text{gn}(T)$ is recursively enumerable (by the Enumerability Theorem), also X is recursively enumerable (by Proposition 4.2.19). If X is T -representable, then both X and $\omega \sim X$ are weakly T -representable, hence recursively enumerable, so X is recursive. $\square$

Corollary 4.6.34 *For any consistent recursively axiomatizable L_Ω -theory $T \supseteq \mathbf{Q}$, there exist sets X such that X is weakly T -representable but X is not T -representable and $\omega \sim X$ is not weakly T -representable.* $\square$

Finally, for use in [Chapter 5](#), we want to make some observations about the notion of T -representability for functions that we have used. There are at least two other natural notions of representability for functions.

Definition 4.6.35 For any L_Ω -theory T and any function $F : \omega^k \rightarrow \omega$,

- (i) F is **T -graph-representable** iff its graph

$$\text{Graph}(F) = \{ (m_0, \dots, m_{k-1}, n) : F(m_0, \dots, m_{k-1}) = n \}$$

is T -representable (as a relation) — that is, for some formula ϕ and all $m_0, \dots, m_{k-1}, n$,

$$\begin{aligned} F(m_0, \dots, m_{k-1}) = n &\implies \phi(\dot{m}_0, \dots, \dot{m}_{k-1}, \dot{n}) \in T; \\ F(m_0, \dots, m_{k-1}) \neq n &\implies \neg \phi(\dot{m}_0, \dots, \dot{m}_{k-1}, \dot{n}) \in T; \end{aligned}$$

(ii) F is T -**provable** iff for some L_Ω -formula ϕ and all $m_0, \dots, m_{k-1}$,

$$\begin{aligned} T \vdash \phi(\dot{m}_0, \dots, \dot{m}_{k-1}, \overbrace{F(m_0, \dots, m_{k-1})}) \quad \text{and} \\ T \vdash \forall x_0 \dots \forall x_{k-1} \exists! y \phi. \end{aligned}$$

Proposition 4.6.36 *For any L_Ω -theory T and function F ,*

- (i) *if F satisfies the hypotheses of Lemma 4.6.5(ii), then F is T -provable;*
- (ii) *if F is T -provable, then F is T -representable;*
- (iii) *if the relation $=$ is T -representable and F is T -representable, then F is T -graph representable.*

Proof. (i) is immediate, since $\exists! y (y \doteq t)$ is logically valid, and (ii) is immediate from Lemma 4.6.5(ii). For (iii), suppose that F is T -represented by ϕ . Then by Lemma 4.6.5(i),

$$F(m_0, \dots, m_{k-1}) = n \implies T \vdash \phi(\dot{m}_0, \dots, \dot{m}_{k-1}, \dot{n}).$$

Furthermore, if $F(m_0, \dots, m_{k-1}) \neq n$, then by the T -representability of $=$,

$$T \vdash \neg \dot{n} \doteq \overbrace{F(m_0, \dots, m_{k-1})} \quad \text{so} \quad T \vdash \neg \phi(\dot{m}_0, \dots, \dot{m}_{k-1}, \dot{n}). \quad \square$$

To grasp the difference between these notions, it is instructive to consider what they imply about the natural extension of F to a possibly non-standard model of T . Let

$$\Omega^* := (\omega^*, +^*, \times^*, S^*, <^*, 0^*)$$

be such a model, and assume that T extends $\mathbf{Q}$, so that ω^* has a $<^*$ -initial segment consisting of the (standard) elements n^* which interpret the terms $\dot{n}$ in Ω^* ; for notational simplicity, we identify this set with ω (see Remark 4.6.12). Fix a (for simplicity unary) function F , let $\phi(x, y)$ be as in any of the notions above, and set

$$F^* = \left\{ (a, b) \in \omega^* \times \omega^* : \models_{\Omega^*} \phi_{x,y}[a, b] \right\},$$

the binary relation defined by ϕ over Ω^* . Suppose first that ϕ T -graph-represents F . Then for all m and n ,

$$\begin{aligned} F(m) = n &\implies \phi(\dot{m}, \dot{n}) \in T \implies \models_{\Omega^*} \phi(\dot{m}, \dot{n}) \\ &\implies \models_{\Omega^*} \phi_{x,y}[m^*, n^*] \implies (m^*, n^*) \in F^*, \end{aligned}$$

and

$$\begin{aligned} F(m) \neq n &\implies \neg \phi(\dot{m}, \dot{n}) \in T \implies \models_{\Omega^*} \neg \phi(\dot{m}, \dot{n}) \\ &\implies \not\models_{\Omega^*} \phi_{x,y}[m^*, n^*] \implies (m^*, n^*) \notin F^*. \end{aligned}$$

Thus (by the identification of m^* with m), $F \subseteq F^*$ and $F \cap (\omega \times \omega) = F^* \cap (\omega \times \omega)$. However, it is not ruled out that for some non-standard $b \in \omega^*$, also $(m^*, b) \in F^*$ — that is, F^* and even $F^* \upharpoonright \omega$ may not be a function.

Suppose next that ϕ T -represents F . Then

$$\models_{\Omega^*} y \neq \dot{n} \rightarrow \neg \phi(\dot{m}, y) \quad \text{so} \quad b \neq n \implies (m^*, b) \notin F^*$$

and thus $F \cap (\omega \times \omega^*) = F^* \cap (\omega \times \omega^*)$ and $F^* \upharpoonright \omega$ is a function, but it is not guaranteed that $\text{Dom}(F^*) = \omega^*$ or that F^* is a function. However, this distinction is only one of the particular formulas, not the function, since if $\phi(x, y)$ T -graph-represents F , then easily the formula $\phi(x, y) \wedge \forall z [\phi(x, z) \rightarrow y \leq z]$ T -represents F .

Finally, if ϕ witnesses that F is T -provable, then F^* itself is a function with $\text{Dom}(F^*) = \omega^*$. This is a real distinction and will play an important role in [Section 5.3](#). Note that T -provability is exactly the condition that we would require to extend the theory by defining a new function symbol $\dot{F}$ with defining formula ϕ (Definition 2.6.11).

Collateral Reading. In addition to the references cited at the end of [Section 4.1](#), the reader might now want to explore [Kay] for a wealth of information on non-standard models of arithmetic and in particular for a proof of the Paris-Harrington Theorem, which provides a L_{Ω} -sentence independent of $\mathbf{P}$ which is in some sense more “natural” than the sentence $\theta_{\mathbf{P}}$ provided by the constructive proof of the First Incompleteness Theorem (see Remark 4.6.31). Another proof of this can be found in [Mar; [Section 5.4](#)].

Exercises

Exercise 4.6.37 Prove Proposition 4.6.3 (see also Proposition 4.5.1 and Exercise 4.1.29).

Exercise 4.6.38 Proposition 4.6.20 established that the semi-recursive relation $U_{\mathbf{Q}}^k$ is universal for the class of k -ary recursive relations and improved this in Corollary 4.6.32 (or Proposition 4.6.24) to show that $U_{\mathbf{Q}}^k$ is universal for the class of k -ary semi-recursive relations. Prove that there does not exist any *recursive* relation which is universal for the class of k -ary recursive relations.

Exercise 4.6.39 Verify that for $T = \text{Th}(\Omega)$, the three notions of representability of functions coincide.

Exercise 4.6.40 Let L_{Ω}^{-} be L_{Ω} with the symbol $\dot{<}$ removed and $\mathbf{Q}^{-}$ the L_{Ω}^{-} -theory generated by axioms $(S1)$, $(S2)$, $(A1)$, $(A2)$, $(M1)$ and $(M2)$ of $\mathbf{Q}$ (from the beginning of this chapter) together with $(S3)$ from Definition 2.5.5. Prove that all recursive functions and relations are $\mathbf{Q}^{-}$ -representable.

Hint: Let $x \dot{<} y$ be an abbreviation for $\exists z (\dot{S}z \dot{+} x \dot{=} y)$. Establish first that all of the following are theorems of $\mathbf{Q}^{-}$:

- (i) $\forall x [\dot{0} \dot{<} \dot{S}x]$;
- (ii) $\forall x \forall y [\dot{S}x \dot{<} \dot{S}y \leftrightarrow x \dot{<} y]$;
- (iii) $\forall x \neg [x \dot{<} \dot{0}]$;
- (iv) for each $m \in \omega$, $\forall x [x \dot{<} \dot{S}m \leftrightarrow x \dot{<} m \vee x \dot{=} m]$;
- (v) for each $m \in \omega$, $\forall x [x \dot{<} m \vee x \dot{=} m \vee m \dot{<} x]$.

Exercise 4.6.41 This exercise outlines Gödel's original proof of the First Incompleteness Theorem. Let T be a consistent recursively axiomatizable L_{Ω} -theory such that $\mathbf{Q} \subseteq T$. Let S be a recursive relation such that

$$\text{gn}(T) = \{ q : \exists p S(p, q) \},$$

fix an L_{Ω} -formula $\chi(x, y)$ which $\mathbf{Q}$ -represents S , and let η be a sentence guaranteed by the Self-reference Lemma that satisfies

$$\mathbf{Q} \vdash \eta \leftrightarrow \neg \exists x \chi(x, \ulcorner \eta \urcorner).$$

Note that η expresses the idea “I am not a theorem of T ”. Prove

- (i) $\eta \notin T$;
- (ii) if T is ω -consistent, then also $\neg \eta \notin T$, so η is a witness to the incompleteness of T .

Exercise 4.6.42 This exercise outlines Rosser's improvement of Gödel's method to avoid the extra assumption of ω -consistency in (ii) of the preceding exercise. Let T , S and χ be as above. Show first that there is a recursive relation $\bar{S}$ such that

$$\{ \text{gn}(\phi) : \neg \phi \in T \} = \{ q : \exists p \bar{S}(p, q) \},$$

fix an L_{Ω} -formula $\bar{\chi}(x, y)$ which $\mathbf{Q}$ -represents $\bar{S}$, and let η be a sentence guaranteed by the Self-reference Lemma that satisfies

$$\mathbf{Q} \vdash \eta \leftrightarrow \forall x [\chi(x, \ulcorner \eta \urcorner) \rightarrow (\exists w \dot{<} x) \bar{\chi}(w, \ulcorner \eta \urcorner)].$$

Note that η expresses the idea “for any witness to my being a theorem of T there is a smaller witness to my negation being a theorem of T ”. Prove

- (i) $\eta \notin T$;
- (ii) $\neg \eta \notin T$.

Hint. Establish first that for any $m \in \omega$ and any sentence ϕ ,

$$\begin{aligned} (\exists x \prec m) \chi(x, \ulcorner \phi \urcorner) \in T &\implies \phi \in T \quad \text{and} \\ (\exists x \prec m) \bar{\chi}(x, \ulcorner \phi \urcorner) \in T &\implies \neg \phi \in T. \end{aligned}$$

Exercise 4.6.43 Prove that for no consistent extension T of $\mathbf{Q}$ does there exist an L_Ω -formula $\dot{V}(x)$ (for “Valid”) such that for all L_Ω -sentences ϕ and ψ , all of the following hold:

- (i) $\models \phi \implies T \vdash \dot{V}(\ulcorner \phi \urcorner)$;
- (ii) $T \vdash \dot{V}(\ulcorner \phi \urcorner) \rightarrow \phi$;
- (iii) $T \vdash \dot{V}(\ulcorner \dot{V}(\ulcorner \phi \urcorner) \rightarrow \phi \urcorner)$;
- (iv) $T \vdash \dot{V}(\ulcorner \phi \rightarrow \psi \urcorner) \rightarrow [\dot{V}(\ulcorner \phi \urcorner) \rightarrow \dot{V}(\ulcorner \psi \urcorner)]$.

Hint: Let θ be the conjunction of the axioms of $\mathbf{Q}$, and argue that the Self-reference Lemma guarantees that there exists a sentence η such that

$$\mathbf{Q} \vdash \eta \leftrightarrow \dot{V}(\ulcorner \theta \rightarrow \neg \eta \urcorner).$$

Show that both $T \vdash \eta$ and $T \vdash \neg \eta$.

Exercise 4.6.44 Verify all of the assertions of Remark 4.6.23.

Exercise 4.6.45 For any theory T , let $\text{Neg}(T) := \{\phi : \neg \phi \in T\}$. Prove that for any theory $T \supseteq \mathbf{Q}$, T and $\text{Neg}(T)$ are **recursively inseparable** — that is, there is no recursive set Z such that $\text{gn}(T) \subseteq Z$ and $Z \cap \text{gn}(\text{Neg}(T)) = \emptyset$. Explain how this contrasts with Exercise 4.3.30, derive a result concerning the reduction property, and compare with the result of Exercise 3.5.54.

Exercise 4.6.46 For any two sets $X, Y \subseteq \omega$ such that $X \cap Y = \emptyset$ and any L_Ω -theory T , we say that X and Y are **T -separable** iff there exists an L_Ω -formula $\phi(x)$ such that for all m ,

$$m \in X \implies T \vdash \phi(\dot{m}) \quad \text{and} \quad m \in Y \implies T \vdash \neg \phi(\dot{m}).$$

Thus X is T -representable iff X and $\sim X$ are T -separable. Note that if X and Y are recursively inseparable (preceding exercise) then the formula ϕ is not T -definite. Prove that for any $T \supseteq \mathbf{Q}$,

- (i) every disjoint pair of recursively enumerable sets is T -separable;
- (ii) if T is consistent, then some non-recursive set is weakly T -representable.

Exercise 4.6.47 Call a set $C \subseteq \omega$ *many-one complete* iff C is recursively enumerable and for every recursively enumerable set $A \subseteq \omega$, $A \leq_m C$ (Definition 4.3.23). Show that

- (i) for any consistent recursively axiomatizable theory $T \supseteq Q$, $\text{gn}(T)$ is m -complete;
- (ii) $K := \{ m : \text{Sb}^1(m, m) \in \text{gn}(Q) \}$ is m -complete.

Hint: For (ii), given a recursively enumerable set A , consider a formula $\phi(x, y)$ which weakly Q -represents $A \times \omega$.

Exercise 4.6.48 Complete the proof of Theorem 4.6.33 by proving the parenthetical part of (i) (about functions).

Exercise 4.6.49 Let A be any non-recursive r.e. set, T a recursively axiomatizable consistent L_Ω -theory extending Q and ψ a formula which weakly T -represents A . Show that for some $m \in \omega$, $\psi(\dot{m})$ is a witness to the incompleteness of T : neither $\psi(\dot{m})$ nor $\neg \psi(\dot{m})$ belongs to T .



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Chapter 5

Topics in Definability

5.1 Definability in arithmetic II

The Definability Theorem 4.5.2 tells us that the recursive and semi-recursive relations are definable over Ω . The theme of this section is that many more relations than these are also definable and that we can usefully classify the definable relations by looking at the complexity of the formulas needed to define them. This is again in contrast with the situation for structures like Ω_+ . The quantifier-elimination Theorem 2.5.16 for $T_+ = \text{Th}(\Omega_+)$ implies that every L_{Ω_+} -formula is T_+ -equivalent to one which is a Boolean combination of existential formulas, but we shall see that relative to $\text{Th}(\Omega)$ generally formulas with more quantifiers are more expressive in the sense that they are not equivalent to formulas with fewer quantifiers.

Recall from Definition 4.3.4 that for any relation $R \subseteq \omega^{k+1}$, the **projection** of R is the relation

$$\exists R := \{ (m_0, \dots, m_{k-1}) : \exists p R(m_0, \dots, m_{k-1}, p) \},$$

and that the semi-recursive relations are exactly the projections of recursive relations. Furthermore, by Proposition 4.3.6(iii)(b), the projection of a semi-recursive relation is still semi-recursive. We shall also write

$$\forall R := \{ (m_0, \dots, m_{k-1}) : \forall p R(m_0, \dots, m_{k-1}, p) \} \text{ and } \sim R := \omega^k \sim R$$

and occasionally call $\forall R$ the **co-projection** of R . We shall always be working with classes of relations closed under permutation of arguments, so the quantification can equivalently be over an argument place other than the last one. The key idea is that successive application of the operations of complementation, projection and co-projection in all possible orders starting with recursive relations generates the class of relations definable over Ω (cf. Definition 2.4.18).

Definition 5.1.1

(i) $\Sigma_0 = \Pi_0 = \Delta_0$ is the class of recursive relations.

For all $n \in \omega$,

(ii) Σ_{n+1} is the class of projections of members of Π_n — in symbols

$$\Sigma_{n+1} := \{ \exists R : R \in \Pi_n \};$$

(iii) Π_{n+1} is the class of complements of members of Σ_{n+1} — in symbols,

$$\Pi_{n+1} := \{ \sim R : R \in \Sigma_{n+1} \};$$

(iv) $\Delta_{n+1} := \Sigma_{n+1} \cap \Pi_{n+1}$;

(v) a relation R is **arithmetical** iff for some n , $R \in \Sigma_n$ or $R \in \Pi_n$.

We usually write ‘ P is (a) Σ_n (relation)’ instead of the more formal ‘ $P \in \Sigma_n$ ’.

Thus, Σ_1 is the class of semi-recursive relations, Π_1 is the class of co-semi-recursive relations, and by Proposition 4.3.7, $\Delta_0 = \Delta_1$ is the class of recursive relations. For example, a relation R is Σ_3 iff for some recursive relation S , $R = \exists \sim \exists \sim \exists S$ — that is,

$$\begin{aligned} R(m_0, \dots, m_{k-1}) &\iff \exists p \sim \exists q \sim \exists r S(m_0, \dots, m_{k-1}, p, q, r) \\ &\iff \exists p \forall q \exists r S(m_0, \dots, m_{k-1}, p, q, r). \end{aligned}$$

We exhibit more concrete examples below.

Lemma 5.1.2 *For all n ,*

- (i) Σ_n , Π_n and Δ_n are closed under relational composition with recursive functions;
- (ii) Σ_n , Π_n and Δ_n are closed under $\cup$ and $\cap$;
- (iii) for $n > 0$, Σ_n is closed under $\exists$ and bounded $\forall$; Π_n is closed under $\forall$ and bounded $\exists$; Δ_n is closed under complementation, bounded $\exists$ and bounded $\forall$.

Proof. Each of these results for Σ_n and Π_n is proved by induction on n ; in each case the results for Δ_n are then immediate. For (i), the basis case $n = 0$ is Proposition 4.2.19, so assume the result for n as induction hypothesis. Let R be a Σ_{n+1} relation, $S \in \Pi_n$ such that $R = \exists S$ and $F_0, \dots, F_{l-1}$ recursive functions. Then

$$\begin{aligned} P(m_0, \dots, m_{k-1}) &\iff R(F_0(m_0, \dots, m_{k-1}), \dots, F_{l-1}(m_0, \dots, m_{k-1})) \\ &\iff \exists p S(F_0(m_0, \dots, m_{k-1}), \dots, F_{l-1}(m_0, \dots, m_{k-1}), p) \\ &\iff \exists p Q(m_0, \dots, m_{k-1}, p), \end{aligned}$$

where

$$Q(m_0, \dots, m_{k-1}, p) \iff S(F_0(m_0, \dots, m_{k-1}), \dots, F_{l-1}(m_0, \dots, m_{k-1}), p).$$

By the induction hypothesis, $Q \in \Pi_n$, so $P = \exists Q \in \Sigma_{n+1}$. The induction step for Π_{n+1} follows immediately.

For (ii), the basis case is Proposition 4.2.12, so assume the result for n and fix $P, Q \in \Sigma_{n+1}$ and $R, S \in \Pi_n$ such that $P = \exists R$ and $Q = \exists S$. Then just as in the proof of Proposition 4.3.6(ii),

$$\begin{aligned} (P \cup Q)(m_0, \dots, m_{k-1}) &\iff \exists p [R(m_0, \dots, m_{k-1}, p) \\ &\quad \vee S(m_0, \dots, m_{k-1}, p)] \\ (P \cap Q)(m_0, \dots, m_{k-1}) &\iff \exists p [R(m_0, \dots, m_{k-1}, (p)_0) \\ &\quad \wedge S(m_0, \dots, m_{k-1}, (p)_1)]. \end{aligned}$$

The relations in square brackets are Π_n by the induction hypothesis and (i), so also $P \cup Q$ and $P \cap Q$ are Σ_{n+1} . Again the induction step for Π_{n+1} is immediate.

The proof of (iii) follows similarly the pattern of 4.3.6 and is left as an exercise. $\square$

Proposition 5.1.3 *For all n ,*

- (i) $\Sigma_n = \{ \sim R : R \in \Pi_n \};$
- (ii) $\Pi_{n+1} = \{ \forall R : R \in \Sigma_n \};$
- (iii) $\Sigma_n \cup \Pi_n \subseteq \Delta_{n+1}.$

Proof. Parts (i) and (ii) are an easy exercise. (iii) for $n = 0$ is just the fact that every recursive relation is both semi-recursive and co-semi-recursive (Proposition 4.3.6), and we proceed by induction on n . Assume the result for n as induction hypothesis and let P be a Σ_{n+1} relation, say the projection of a Π_n relation R . Then by the induction hypothesis, $R \in \Pi_{n+1}$, hence $P \in \Sigma_{n+2}$. By the lemma, if P' is defined by

$$P'(m_0, \dots, m_{k-1}, q) \iff P(m_0, \dots, m_{k-1}),$$

then also $P' \in \Sigma_{n+1}$, so since $P = \forall P'$, also $P \in \Pi_{n+2}$, so $P \in \Delta_{n+2}$. The result for $P \in \Pi_{n+1}$ follows by symmetry. $\square$

At this point we know (from Corollary 4.6.18) that Δ_1 is a proper subset of both Σ_1 and Π_1 , and we shall soon see that each of these is a proper subclass of Δ_2 . To extend these relationships to larger indices, we shall again use the diagonal method of 4.1.21 and expand on the second proof of Corollary 4.6.18.

Definition 5.1.4 For each $k > 0$,

$$U^k := U_Q^k = \left\{ (a, m_0, \dots, m_{k-1}) : \text{Sb}^k(a, m_0, \dots, m_{k-1}) \in \text{gn}(\mathbb{Q}) \right\}.$$

Recall from Corollary 4.6.32 that for each $k > 0$, U^k is a $(k+1)$ -ary semi-recursive relation universal for the class of k -ary semi-recursive relations — that is, $\langle U_a^k : a \in \omega \rangle$ is a complete list of the k -ary semi-recursive relations. It follows that $\sim U^k$ is a $(k+1)$ -ary co-semi-recursive relation such that $\langle \sim U_a^k : a \in \omega \rangle$ is a complete list of the k -ary co-semi-recursive relations. These lead to some interesting examples.

Examples 5.1.5

$$\begin{aligned} \text{Emp} &:= \{ a : U_a^1 = \emptyset \} \in \Pi_1, & \text{Fin} &:= \{ a : U_a^1 \text{ is finite} \} \in \Sigma_2, \\ \text{Inf} &:= \{ a : U_a^1 \text{ is infinite} \} \in \Pi_2, & \text{Rec} &:= \{ a : U_a^1 \text{ is recursive} \} \in \Sigma_3. \end{aligned}$$

These are easy to verify — for example,

$$\begin{aligned} a \in \text{Fin} &\iff \exists n \forall m [U^1(a, m) \implies m < n] \\ &\iff \exists n \forall m [\sim U^1(a, m) \vee m < n]. \end{aligned}$$

The relation in square brackets is Π_1 , hence so is that defined by $\forall m [\dots]$, so Fin is Σ_2 . Similarly, since a recursively enumerable set is recursive just in case its complement is also recursively enumerable,

$$\begin{aligned} a \in \text{Rec} &\iff \exists b [\sim U_a^1 = U_b^1] \\ &\iff \exists b \forall m [(\sim U^1(a, m) \implies U^1(b, m)) \\ &\quad \wedge (U^1(b, m) \implies \sim U^1(a, m))] \\ &\iff \exists b \forall m [(U^1(a, m) \vee U^1(b, m)) \\ &\quad \wedge (\sim U^1(b, m) \vee \sim U^1(a, m))]. \end{aligned}$$

The relation defined by the expression in square brackets is a Boolean combination of Δ_2 relations, hence is Δ_2 and in particular Π_2 . Hence so is that defined by $\forall m [\dots]$, and hence Rec is Σ_3 .

Note that we have not established that these evaluations of the complexity of these sets is the best possible — for example that Fin is *not* Π_2 or Σ_1 . This question is strongly connected with the more general question of proper inclusions among the arithmetical classes.

Definition 5.1.6 ${}_1U^k := U^k$ and for each $n > 0$,

$${}_{n+1}U^k := \{ (a, m_0, \dots, m_{k-1}) : \exists p \sim {}_nU^{k+1}(a, m_0, \dots, m_{k-1}, p) \}.$$

Proposition 5.1.7 *For all $k, n > 0$,*

- (i) ${}_nU^k$ is a $(k+1)$ -ary Σ_n relation universal for the class of k -ary Σ_n relations;
- (ii) $\sim {}_nU^k$ is a $(k+1)$ -ary Π_n relation universal for the class of k -ary Π_n relations.

Proof. We proceed by induction on n . The basis case is done; assume as induction hypothesis the result for n . It is immediate from the definitions that ${}_{n+1}U^k$ is Σ_{n+1} and $\sim {}_{n+1}U^k$ is Π_{n+1} . Let P be any Σ_{n+1} relation, so for some Π_n relation R

$$P(m_0, \dots, m_{k-1}) \iff \exists p R(m_0, \dots, m_{k-1}, p).$$

By the induction hypothesis, for some $a \in \omega$, $R = \sim {}_nU_a^{k+1}$, and thus

$$\begin{aligned} P(m_0, \dots, m_{k-1}) &\iff \exists p \sim {}_nU_a^{k+1}(m_0, \dots, m_{k-1}, p) \\ &\iff {}_{n+1}U_a^k(m_0, \dots, m_{k-1}), \end{aligned}$$

that is, $P = {}_{n+1}U_a^k$. Thus every Σ_{n+1} relation is a ${}_{n+1}U^k$ -section, and hence every Π_{n+1} relation is a $\sim {}_{n+1}U^k$ -section as desired. $\square$

Corollary 5.1.8 *For all $k, n > 0$, $\langle {}_nU_a^k : a \in \omega \rangle$ is a complete list of the k -ary Σ_n relations and $\langle \sim {}_nU_a^k : a \in \omega \rangle$ is a complete list of the k -ary Π_n relations.*

Proof. By the preceding proposition every Σ_n (Π_n) relation appears on this list, and using Lemma 5.1.2 every member of the list is Σ_n (Π_n). $\square$

Arithmetical Hierarchy Theorem 5.1.9 *For all $n > 0$,*

- (i) Δ_n is a proper subset of both Σ_n and Π_n ;
- (ii) $\Sigma_n \cup \Pi_n$ is a proper subset of Δ_{n+1} .

Proof. For (i), Δ_n is a subset of both Σ_n and Π_n by definition. Let $D_n = \{m : {}_nU^1(m, m)\}$, the diagonal set of ${}_nU^1$. By the proposition, $D_n \in \Sigma_n$, but by the Diagonal Lemma 4.1.21, $\sim D_n$ is not a ${}_nU^1$ -section, hence is not Σ_n . Hence $D_n \notin \Pi_n$, so also $D_n \notin \Delta_n$. Thus $\sim D_n$ is an example of a set in Π_n but not in Δ_n .

By Proposition 5.1.3 we need only show that the inclusion is proper. Let

$$C_n = \{2m : m \in D_n\} \cup \{2m+1 : m \notin D_n\}.$$

By Lemma 5.1.2, these two sets are respectively Σ_n and Π_n , hence both are Δ_{n+1} and hence so is their union C_n . Suppose, towards a contradiction that $C_n \in \Sigma_n$. Then since $m \notin D_n \iff 2m+1 \in C_n$, also $\sim D_n \in \Sigma_n$, contrary to our conclusion above; hence $C_n \notin \Sigma_n$. Similarly, $C_n \notin \Pi_n$. $\square$

The content of this theorem may be summarized in the diagram below. The classes Σ_1 and Π_1 are represented by the two triangles whose intersection is $\Delta_0 = \Delta_1$, Π_2 by the left-most trapezoid with longer lower base etc. The Arithmetical Hierarchy Theorem says exactly that this is an accurate picture.

Examples 5.1.10 Now that we have established that there exist sets which belong to certain arithmetical classes but not to others, it is natural to ask if the evaluations of the examples above are optimal. For example, could **Emp** be also Σ_1 or **Inf** also Σ_2 ? The proof that these are not the case is based on the following simple principle.

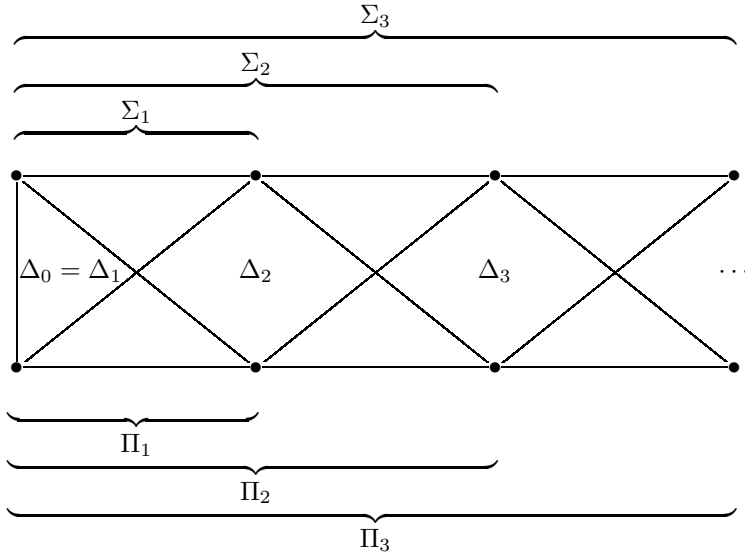
Lemma 5.1.11 *For $\mathcal{C}$ any of the classes Σ_n , Π_n or Δ_n and any sets $X, Y \subseteq \omega$, if $X \leq_m Y$ (Definition 4.3.23) and $Y \in \mathcal{C}$, then also $X \in \mathcal{C}$.*

Proof. Immediate from Lemma 5.1.2(i). $\square$

Now we will show the following: for all $X \subseteq \omega$,

$$(1) \quad X \in \Pi_1 \implies X \leq_m \mathbf{Emp} \quad \text{and} \quad X \in \Pi_2 \implies X \leq_m \mathbf{Inf}.$$

Hence, if $\mathbf{Emp} \in \Sigma_1$ ($\mathbf{Inf} \in \Sigma_2$), then by the lemma, every Π_1 set is also Σ_1 (every Π_2 set is also Σ_2) contrary to the Arithmetical Hierarchy Theorem.



Arithmetical Hierarchy

It follows immediately that $\text{Fin} \notin \Pi_2$; a similar but more involved argument establishes that $\text{Rec} \notin \Pi_3$. An exercise below shows that at least $\text{Rec} \notin \Sigma_2$.

To establish (1), let X be any Π_1 set and choose a recursive relation R such that $X = \{m : \forall p R(m, p)\}$. For each m , set $Y^m := \{p : \sim R(m, p)\}$, so for all m ,

$$m \in X \iff Y^m = \emptyset.$$

Now fix an L_Ω -formula $\psi(x, y)$ which $\mathbf{Q}$ -represents the recursive relation $\sim R$. Then for each m , $\psi(\dot{m}, y)$ $\mathbf{Q}$ -represents Y^m , so $Y^m = U_{\text{gn}(\psi(\dot{m}, y))}^1$. Thus

$$m \in X \iff U_{\text{gn}(\psi(\dot{m}, y))}^1 = \emptyset \iff \text{gn}(\psi(\dot{m}, y)) \in \text{Emp}.$$

Since the function $m \mapsto \text{gn}(\psi(\dot{m}, y))$ is recursive, this shows that $X \leq_m \text{Emp}$.

Similarly, let X be any Π_2 set, choose a recursive relation S such that

$$X = \{m : \forall p \exists q S(m, p, q)\},$$

and for each m set $Z^m := \{r : (\forall p < r) \exists q S(m, p, q)\}$. Then for all m ,

$$m \in X \iff Z^m \text{ is infinite}.$$

Let $\theta(x, y)$ be an L_Ω -formula which weakly $\mathbf{Q}$ -represents the semi-recursive relation

$$\{(m, r) : (\forall p < r) \exists q S(m, p, r)\}.$$

Then for each m , $\theta(\dot{m}, y)$ weakly Q-represents Z^m so $Z^m = U_{\text{gn}(\theta(\dot{m}, y))}^1$ and thus

$$m \in X \iff U_{\text{gn}(\theta(\dot{m}, y))}^1 \text{ is infinite} \iff \text{gn}(\theta(\dot{m}, y)) \in \text{Inf},$$

so $X \leq_m \text{Inf}$.

The first connection of these ideas with definability is

Theorem 5.1.12 *A relation R is arithmetical iff it is definable over Ω .*

Proof. For the implication $(\implies)$ we prove by induction on n that

$$R \in \Sigma_n \cup \Pi_n \implies R \text{ is definable over } \Omega.$$

The basis cases $n = 0, 1$ are the Definability Theorem 4.5.2, and the induction step is easy:

if $\phi(x_0, \dots, x_{k-1}, y)$ defines R , then $\neg \phi$ defines $\sim R$ and $\exists y \phi$ defines $\exists R$.

For $(\impliedby)$, we prove by formula induction that for all ϕ , the relation ϕ^Ω defined by ϕ is arithmetical. Again each of the steps is immediate. $\square$

Corollary 5.1.13 *$\text{gn}(\text{Th}(\Omega))$ is not arithmetical.*

Proof. Immediate from Tarski's Theorem 4.5.5. $\square$

We now want to show how the stratification of the arithmetical relations into levels Σ_n and Π_n is mirrored by the complexity of the formulas needed to define them. Recall from Definition 2.3.22 that in any first-order language a universal (existential) formula is one of the form $\forall x_0 \dots \forall x_{k-1} \theta$ ($\exists x_0 \dots \exists x_{k-1} \theta$) where θ is a quantifier-free formula — that is, a Boolean combination of atomic formulas. A natural conjecture is that for any $R \subseteq \omega^k$,

$$R \in \Sigma_1 \stackrel{?}{\iff} R \text{ is definable over } \Omega \text{ by an existential formula.}$$

The implication $(\impliedby)$ is easy to verify: easily any relation defined by a quantifier-free formula is recursive, so any relation defined by an existential formula is the projection of a recursive relation and hence is semi-recursive.

However, the implication $(\implies)$, although true, turns out to be surprisingly deep and difficult to prove. It is related to a famous problem, the tenth in a list of twenty-three stated by Hilbert in the year 1900. A set $X \subseteq \omega$ is called **Diophantine** iff for some l there is a polynomial $F(x, y_0, \dots, y_{l-1})$ with integer coefficients such that for all $m \in \omega$, $m \in X$ iff the equation $F(m, y_0, \dots, y_{l-1}) = 0$ has a solution $n_0, \dots, n_{l-1} \in \omega$. By writing F as the

difference $G - H$ of two polynomials with natural number coefficients, we have

$$m \in X \iff \exists n_0 \cdots \exists n_{l-1} [G(m, n_0, \dots, n_{l-1}) = H(m, n_0, \dots, n_{l-1})].$$

The equation $G = H$ can be expressed by a quantifier-free L_Ω -formula $\theta(x)$, so

$$m \in X \iff \models_{\Omega} \exists y_0 \cdots \exists y_{l-1} \theta(\dot{m}).$$

Thus every Diophantine set is definable over Ω by an existential L_Ω -formula.

Hilbert's question was, in slightly altered form: is every Diophantine set decidable? In 1900 he was, of course, thinking of the intuitive notion of decidability, but by the 1930s the question was being interpreted to mean: is every Diophantine set recursive? The question was a focal point of research in the 1950s and 1960s, but was not finally solved until 1970.

Matiyasevich's Theorem 5.1.14 *Every recursively enumerable set is Diophantine; hence every recursively enumerable set is definable over Ω by an existential L_Ω -formula and there exist non-recursive Diophantine sets.*

Of course this immediately implies the conjecture above. The proof involves a considerable amount of Number Theory and would lead us far from our current direction, so we shall replace the existential formulas by a larger class for which the proof is easy.

We temporarily call a **simple term** one which is either a variable or a numeral $\dot{m}$. Recall that we use the abbreviations

$$(\forall x \dot{<} t) \phi \text{ for } \forall x (x \dot{<} t \rightarrow \phi) \quad \text{and} \quad (\exists x \dot{<} t) \phi \text{ for } \exists x (x \dot{<} t \wedge \phi).$$

Definition 5.1.15 The class of Σ formulas is the smallest class $\mathcal{C}$ of L_Ω -formulas such that

- (i) for any simple terms t and u and any variable y , the following **simple formulas** belong to $\mathcal{C}$:
 - (a) $t \doteq u$ and $\neg t \doteq u$,
 - (b) $t \dot{<} u$ and $\neg t \dot{<} u$,
 - (c) $\dot{S}t \doteq y$, $t \dot{+} u \doteq y$ and $t \dot{\times} u \doteq y$;
- (ii) $\mathcal{C}$ is closed under $\wedge$ and $\vee$;
- (iii) $\psi \in \mathcal{C} \implies (\forall y \dot{<} t) \psi \in \mathcal{C}$ for any simple term t in which y does not occur;
- (iv) $\psi \in \mathcal{C} \implies \exists y \psi \in \mathcal{C}$.

The class of Π formulas is defined similarly with $\exists$ and $\forall$ interchanged in (iii) and (iv).

The basis clause of the definition admitted only atomic formulas of very special kinds for technical reasons; the second part of the following lemma shows that there is no real restriction.

Lemma 5.1.16

- (i) Every Π (Σ) formula is logically equivalent to the negation of a Σ (Π) formula. Hence, the negation of any Π (Σ) formula is logically equivalent to a Σ (Π) formula.
- (ii) Every quantifier-free L_Ω -formula is logically equivalent to both a Σ formula and a (possibly different) Π formula. Hence every existential (universal) L_Ω -formula is logically equivalent to a Σ (Π) formula.

Proof. For (i), to each Π formula ϕ associate recursively a formula ϕ^* as follows: for any simple terms t and u and any variable y ,

$$\begin{aligned}
 (t \doteq u)^* & \text{ is } \neg t \dot{=} u & \text{ and } & (\neg t \dot{=} u)^* & \text{ is } t \dot{=} u; \\
 (t \dot{<} u)^* & \text{ is } \neg t \dot{<} u & \text{ and } & (\neg t \dot{<} u)^* & \text{ is } t \dot{<} u; \\
 (\dot{S}t \dot{=} y)^* & \text{ is } \exists z [\dot{S}t \dot{=} z \wedge \neg z \dot{=} y]; \\
 (t \dot{+} u \dot{=} y)^* & \text{ is } \exists z [t \dot{+} u \dot{=} z \wedge \neg z \dot{=} y]; \\
 (t \dot{\times} u \dot{=} y)^* & \text{ is } \exists z [t \dot{\times} u \dot{=} z \wedge \neg z \dot{=} y].
 \end{aligned}$$

Then $(\psi \wedge \theta)^*$ is $(\psi^* \vee \theta^*)$, $(\psi \vee \theta)^*$ is $(\psi^* \wedge \theta^*)$, $((\exists x \dot{<} t) \psi)^*$ is $(\forall x \dot{<} t) \psi^*$, and $(\forall x \psi)^*$ is $\exists x \psi^*$. It is straightforward to prove by induction that for all Π formulas ϕ , ϕ^* is a Σ formula and ϕ is logically equivalent to $\neg \phi^*$.

For (ii) we show first that the statement is true for every atomic formula using the same technique as in Proposition 2.6.18. First, by term induction we show that every atomic formula of the form $t \dot{=} y$ (for any term t) is logically equivalent to a Σ formula. One of the induction steps reads

$$\models t \dot{+} u \dot{=} y \quad \leftrightarrow \quad \exists z_0 \exists z_1 [t \dot{=} z_0 \wedge u \dot{=} z_1 \wedge z_0 \dot{+} z_1 \dot{=} y].$$

Then every atomic formula and negated atomic formula is logically equivalent to a Σ formula: for example, for any terms t and u ,

$$\begin{aligned}
 \models t \dot{<} u & \quad \leftrightarrow \quad \exists z_0 \exists z_1 [t \dot{=} z_0 \wedge u \dot{=} z_1 \wedge z_0 \dot{<} z_1] \\
 \models \neg t \dot{<} u & \quad \leftrightarrow \quad \exists z_0 \exists z_1 [t \dot{=} z_0 \wedge u \dot{=} z_1 \wedge \neg z_0 \dot{<} z_1],
 \end{aligned}$$

so by (i), every atomic formula is logically equivalent to both a Σ and a Π formula. The result now follows by a straightforward induction on quantifier-free formulas. $\square$

Now we can refine the Definability and Representability Theorems 4.5.2 and 4.6.15.

Theorem 5.1.17 *For any consistent L_Ω -theory $T \supseteq \mathbf{Q}$ and all $k \in \omega$,*

- (i) *every recursive function $F : \omega^k \rightarrow \omega$ is definable over Ω and T -representable by both a Σ formula and a (generally different) Π formula;*
- (ii) *every recursive relation $R \subseteq \omega^k$ is definable over Ω and T -representable by both a Σ formula and a (generally different) Π formula.*

Proof. We follow the earlier inductive proofs. The formulas used to define the initial functions are quantifier-free so by the lemma logically equivalent to both Σ and Π formulas. It will therefore suffice to show that both composition and search preserve definability by both sorts of formulas.

We treat the same special cases as before. Suppose that $F(m) = G(H_0(m), H_1(m))$ and that $\psi^\Sigma(x_0, x_1, z)$, $\theta^0(x, y_0)$ and $\theta^1(x, y_1)$ are Σ formulas which, respectively, define (T -represent) G , H_0 and H_1 . Then the formula

$$\exists y_0 \exists y_1 [\theta^0(x, y_0) \wedge \theta^1(x, y_1) \wedge \psi^\Sigma(y_0, y_1, z)]$$

shown to define (T -represent) F is also a Σ formula. On the other hand, if ψ^Π is now taken to be a Π formula which defines (T -represents) G , then the same argument shows that the formula

$$\forall y_0 \forall y_1 [\theta^0(x, y_0) \wedge \theta^1(x, y_1) \rightarrow \psi^\Pi(y_0, y_1, z)]$$

also defines (T -represents) F and by the lemma is logically equivalent to a Π formula.

Next, suppose that

$$\forall m \exists n G(m, n) = 0, \quad F(m) = \text{least } n [G(m, n) = 0],$$

and G is defined over Ω (T -represented) by the Σ and Π formulas $\psi^\Sigma(x, y, z)$ and $\psi^\Pi(x, y, z)$. Then as in the proofs of Theorems 4.5.2 and 4.6.14, F is defined (T -represented) by each of the formulas

$$\begin{aligned} (\forall w \dot{<} y) \neg \psi^\Pi(x, w, \dot{0}) \wedge \psi^\Sigma(x, y, \dot{0}) \quad \text{and} \\ (\forall w \dot{<} y) \neg \psi^\Sigma(x, w, \dot{0}) \wedge \psi^\Pi(x, y, \dot{0}), \end{aligned}$$

which by the lemma are logically equivalent, respectively, to a Σ and a Π formula.

For (ii), if $R \subseteq \omega^k$ is a recursive relation and $\mathbf{K}_R$ is defined (T -represented) by formulas

$$\phi^\Sigma(x_0, \dots, x_{k-1}, y) \quad \text{and} \quad \phi^\Pi(x_0, \dots, x_{k-1}, y),$$

respectively Σ and Π , then R is defined (T -represented) by the formulas $\phi_y^\Sigma(\dot{1})$ and $\phi_y^\Pi(\dot{1})$, which are also respectively Σ and Π . $\square$

Corollary 5.1.18 *For any relation R ,*

- (i) *R is (co-)semi-recursive iff R is definable over Ω by a Σ (Π) formula;*
- (ii) *R is recursive iff R is definable over Ω by both a Σ formula and a (generally different) Π formula.*

Proof. For (i), suppose that S is recursive, $X := \{m : \exists p S(m, p)\}$ and $\phi(x, y)$ is a Σ formula which defines S . Then $\exists y \phi(x, y)$ is a Σ formula which defines X . The converse is proved by a straightforward induction on Σ formulas using the closure properties of the class of semi-recursive relations expressed in Proposition 4.3.6. Then (ii) is immediate by Proposition 4.3.7. $\square$

Next is a refinement of part (iii) of the Representability Theorem 4.6.30.

Corollary 5.1.19 *For any consistent recursively axiomatizable L_Ω -theory $T \supseteq Q$, every semi-recursive relation is weakly T -representable by a Σ formula — in fact, for each k -ary semi-recursive relation R there exists a Σ formula $\psi(x_0, \dots, x_{k-1})$ such that for all $m_0, \dots, m_{k-1}$,*

$$\begin{aligned} R(m_0, \dots, m_{k-1}) &\iff \psi(\dot{m}_0, \dots, \dot{m}_{k-1}) \in T \\ &\iff \psi(\dot{m}_0, \dots, \dot{m}_{k-1}) \in \text{Th}(\Omega). \end{aligned}$$

Proof. We note first the following refinement of the Self-reference Lemma 4.6.27: for any Σ formula $\theta(y, z)$, there exists a Σ formula $\eta(z)$ such that for all $n \in \omega$,

$$Q \vdash \eta(\dot{n}) \leftrightarrow \theta(\ulcorner \eta(\dot{n}) \urcorner, \dot{n}).$$

Following the earlier proof, let $\psi^\Sigma(x, y, z)$ be a Σ formula which Q -represents δ , but now set $\chi(x, z)$ to be the Σ formula $\exists y (\psi^\Sigma(x, y, z) \wedge \theta(y, z))$. Then with $\eta(z)$ the formula $\chi(\ulcorner \chi \urcorner, z)$, $\eta(z)$ is also a Σ formula and by the same proof is as desired. Note that the original version of χ shows that if θ is a Π formula, then also η is a Π formula.

Now with notation as in the proof of part (iii) of the Representability Theorem 4.6.30, choose $\phi^\Sigma(z, u)$ a Σ formula which Q -represents S and $\chi^\Pi(y, w)$ a Π formula which Q -represents S' . Then applying the Self-reference Lemma to the Σ formula logically equivalent to

$$\theta(y, z) := \exists u [\phi^\Sigma(z, u) \wedge (\forall w \dot{<} u) \neg \chi^\Pi(y, w)]$$

produces a Σ formula $\psi(z)$ which is as desired. $\square$

It is now quite easy to define a notion of Σ_n and Π_n formula and show that the Σ_n (Π_n) relations are exactly those definable by Σ_n (Π_n) formulas; we leave the details to an exercise. The notion of Σ formula leads also to another example of (for each $k > 0$) a $(k+1)$ -ary semi-recursive relation universal for the class of k -ary semi-recursive relations (cf. Corollary 4.6.32). We consider first the case $k = 1$.

Definition 5.1.20 $\text{Th}^\Sigma(\Omega) := \left\{ \phi : \phi \text{ is a } \Sigma \text{ sentence and } \models_\Omega \phi \right\};$

$$\begin{aligned} U^\Sigma &:= \{ (a, m) : \text{for some } \Sigma \text{ formula } \phi, a = \text{gn}(\phi) \text{ and } \phi(\dot{m}) \in \text{Th}(\Omega) \} \\ &= \left\{ (a, m) : \text{Sb}^1(a, m) \in \text{gn}(\text{Th}^\Sigma(\Omega)) \right\}; \end{aligned}$$

and

$$D_\Sigma := \{ m : U^\Sigma(m, m) \}.$$

Corollary 5.1.21 U^Σ is universal for the class of recursively enumerable sets. Hence U^Σ and D_Σ are not recursive.

Proof. For any recursively enumerable set X , let $\psi(x)$ be a Σ formula which defines X . Then for all m

$$\begin{aligned} m \in X &\iff \psi(\dot{m}) \in \text{Th}(\Omega) \\ &\iff \psi(\dot{m}) \in \text{Th}^\Sigma(\Omega) \iff U^\Sigma(\text{gn}(\psi), m). \end{aligned}$$

By the Diagonal Lemma 4.1.21, $\sim D_\Sigma$ is not recursively enumerable so D_Σ is not recursive, and therefore also U^Σ is not recursive. $\square$

We now want to verify that U^Σ is semi-recursive and hence D^Σ is recursively enumerable; clearly it suffices to prove

Proposition 5.1.22 $\text{Th}^\Sigma(\Omega)$ is recursively enumerable.

Proof. Although there is no natural inductive definition of the whole set of Σ sentences (as opposed to general formulas), the subset consisting of those true in Ω is inductively definable: $\text{Th}^\Sigma(\Omega)$ is the smallest set $\mathcal{C}$ of L_Ω -sentences such that for any n

- (i) all simple sentences true in Ω belong to $\mathcal{C}$;
- (ii) $\mathcal{C}$ is closed under $\wedge$;
- (iii) if $\psi \in \mathcal{C}$ and θ is any Σ sentence, then $\psi \vee \theta \in \mathcal{C}$ and $\theta \vee \psi \in \mathcal{C}$;
- (iv) for any n , if $\psi_y(\dot{0}), \dots, \psi_y(\overbrace{\dot{n}-1}^{\text{---}})$ belong to $\mathcal{C}$, then also $(\forall y \prec \dot{n}) \psi$ is in $\mathcal{C}$;
- (v) if $\psi_y(\dot{n})$ is in $\mathcal{C}$, then also $\exists y \psi$ is in $\mathcal{C}$.

It follows by a simple induction that every element of $\mathcal{C}$ is a Σ sentence true in Ω so belongs to $\text{Th}^\Sigma(\Omega)$. Towards the converse inclusion, for any Σ formula ϕ with, say, free variables $x_0, \dots, x_{k-1}$, we call any sentence of the form $\phi_{x_0, \dots, x_{k-1}}(\dot{m}_0, \dots, \dot{m}_{k-1})$ an **instance** of ϕ . Then we prove by induction that for every Σ formula ϕ , every instance of ϕ which is true in Ω belongs to $\mathcal{C}$; this class clearly includes every true Σ sentence. If ϕ is a simple formula,

then any instance is a simple sentence, so if true belongs to $\mathcal{C}$ by clause (i). Suppose that ϕ is $\psi \wedge \theta$ and $\phi_x(\dot{m})$ is a true instance (with $k = 1$ for simplicity). Then both of $\psi_x(\dot{m})$ and $\theta_x(\dot{m})$ are true in Ω , so belong to $\mathcal{C}$ by the induction hypothesis. Hence so does $\phi_x(\dot{m})$ by (ii). Similarly, if ϕ is $\psi \vee \theta$ and $\phi_x(\dot{m})$ is a true instance, then one of $\psi_x(\dot{m})$ or $\theta_x(\dot{m})$ is true, so belongs to $\mathcal{C}$; hence so does $\phi_x(\dot{m})$ by (iii). Next, suppose that ϕ is $(\forall y < t) \psi$ and $\phi_x(\dot{m})$ is a true instance. Since we assume that x is the only free variable and t is a simple term, t is either x or a numeral $\dot{p}$. In either case, $\phi_x(\dot{m})$ is of the form $(\forall y < \dot{p}) \psi_x(\dot{m})$. The hypothesis that this is true means that each of the instances $\psi_{x,y}(\dot{m}, \dot{q})$, for $q < p$ is true, so by the induction hypothesis belongs to $\mathcal{C}$. But then also $\phi_x(\dot{m}) \in \mathcal{C}$ by clause (iv). The unbounded existential case is similar.

Now we use the technique of [Section 4.4](#). Call a sequence $(\phi_0, \dots, \phi_l)$ of formulas a Σ -derivation iff it is a derivation for this inductive definition in the sense of Definition 1.2.6, and set

$$\text{Deriv}^\Sigma = \{ \langle \text{gn}(\phi_0), \dots, \text{gn}(\phi_l) \rangle : l \in \omega \text{ and } (\phi_0, \dots, \phi_l) \text{ is a } \Sigma\text{-derivation} \}$$

$$\text{Derivof}^\Sigma = \{ (s, a) : \text{Deriv}^\Sigma(s) \text{ and } a = (s)_{|\text{g}(s)-1} \}.$$

Then it is easy to verify that Deriv^Σ and Derivof^Σ are primitive recursive relations and that for any a ,

$$a \in \text{gn}(\text{Th}^\Sigma(\Omega)) \iff \exists s \text{ Derivof}^\Sigma(s, a).$$

Hence $\text{Th}^\Sigma(\Omega)$ is recursively enumerable. $\square$

Corollary 5.1.23 *$\text{Th}^\Sigma(\Omega)$ and D_Ω^Σ are recursively enumerable but not recursive. $\square$*

We have carried through the arguments for sets rather than arbitrary relations for simplicity, but if we now set

$$U^{\Sigma,k}(a, m_0, \dots, m_{k-1}) : \iff \text{Sb}^k(a, m_0, \dots, m_{k-1}) \in \text{gn}(\text{Th}^\Sigma(\Omega)),$$

we have immediately

Corollary 5.1.24 *For all $k > 0$, $U^{\Sigma,k}$ is a $(k+1)$ -ary semi-recursive relation universal for the class of k -ary semi-recursive relations. $\square$*

Now we could also use $U^{\Sigma,k}$ in place of U_Q^k in Definition 5.1.6.

Collateral Reading. All of the references at the end of [Section 4.2](#) except [Cut] include material on the Arithmetical Hierarchy. [Mat] provides a proof of Matiyasevich's Theorem 5.1.14.

Exercises

Exercise 5.1.25 Complete the proof of Proposition 5.1.2.

Exercise 5.1.26 Define natural notions of Σ_n formula and Π_n formula, and show for that all $n > 0$,

- (i) a relation is Σ_n (Π_n) iff it is definable over Ω by a Σ_n (Π_n) formula;
- (ii) the set $\text{Th}^{\Sigma_n}(\Omega)$ of Gödel numbers of true Σ_n sentences is a Σ_n set.

Exercise 5.1.27 Prove that Rec (Examples 5.1.5) is not Σ_2 . (In fact, $\text{Rec} \notin \Pi_3$, but this is much harder to prove.)

Hint: Find a semi-recursive binary relation V such that for all a ,

$$\begin{aligned} a \in \text{Inf} &\implies V_a := \{m : V(a, m)\} = \omega; \\ a \in \text{Fin} &\implies D_1 \subseteq V_a \text{ and } V_a \sim D_1 \text{ is finite.} \end{aligned}$$

Use the technique of Examples 5.1.10 to show that if $\text{Rec} \in \Sigma_2$, then also $\text{Inf} \in \Sigma_2$.

Exercise 5.1.28 Show that despite the preceding result there does exist a binary semi-recursive relation V such that $\langle V_a : a \in \omega \rangle$ is a list (with repetitions) of exactly the recursive sets.

Exercise 5.1.29 A function F is Σ_n , Π_n or Δ_n iff its graph is a relation in the corresponding class. Prove that for all n

- (i) the class of Δ_n functions is closed under composition;
- (ii) the class of Δ_n functions is closed under primitive recursion;
- (iii) a function is Σ_n iff it is Δ_n .

Exercise 5.1.30 Say that T is an *arithmetical theory* iff $\text{gn}(T)$ is an arithmetical set of numbers and *arithmetically axiomatizable* iff $T = \text{Th}(\Gamma)$ for some set Γ such that $\text{gn}(\Gamma)$ is arithmetical.

- (i) Show that T is arithmetical iff T is arithmetically axiomatizable and in particular, if $\text{gn}(\Gamma) \in \Delta_n$, then $\text{gn}(\text{Th}(\Gamma)) \in \Sigma_n$.
- (ii) Show that any recursively or even arithmetically axiomatizable theory T has a complete arithmetical extension T' . Where in the arithmetical hierarchy may we expect to find T' ?

Hint: For (ii), recall that part of the proofs of the Countable Completeness Theorem and Countable Compactness Theorem involved the construction of a complete extension of any set of sentences (ignore the addition of Henkin witnesses). Show that this construction applied to an arithmetical set produces an arithmetical complete theory.

Exercise 5.1.31 This is a version of *Post's Theorem*; see also the exercises to [Section 8.3](#). For any set $Y \subseteq \omega$, let

$$\Omega[Y] := (\omega, <, +, \times, \text{Sc}, 0, Y),$$

with the associated language $L_\Omega[\dot{Y}]$ containing a new unary relation symbol $\dot{Y}$. Define the class of $\Sigma[\dot{Y}]$ formulas as in Definition 5.1.15 with the additional clause that for any simple term t both $\dot{Y}t$ and $\neg \dot{Y}t$ belong to $\mathcal{C}$. For each $n > 0$ let

$$Y_n := \{ \langle a, m \rangle : {}_nU^1(a, m) \}$$

be the set-encoding of the universal Σ_n relation of Definition 5.1.6.

- (i) Verify that Lemma 5.1.16 holds for the extended language $L_\Omega[\dot{Y}]$.
- (ii) Show that for any $n > 0$, if $Y \in \Sigma_n$, then for any relation R ,

$$R \text{ is definable over } \Omega[Y] \text{ by a } \Sigma[\dot{Y}] \text{ formula} \implies R \in \Sigma_{n+1}.$$

- (iii) $R \in \Sigma_{n+1} \implies R$ is definable over $\Omega[Y_n]$ by a $\Sigma[\dot{Y}_n]$ formula.
- (iv) $R \in \Delta_{n+1} \iff R$ is definable over $\Omega[Y_n]$ by both a $\Sigma[\dot{Y}_n]$ formula and a (generally different) $\Pi[\dot{Y}_n]$ formula.

Exercise 5.1.32 This is a version of the *Limit Lemma* of [Section 8.3](#). Prove that for any set $X \subseteq \omega$, $X \in \Delta_2$ iff for some recursive function G ,

$$K_X(m) = \lim_{s \rightarrow \infty} G(m, s).$$

Hint: Suppose that for some recursive relations R and S ,

$$\exists n \forall p R(m, n, p) \iff m \in X \iff \forall n \exists p S(m, n, p).$$

Let

$$\begin{aligned} F(m, s) &\simeq \text{least } n \left[(\forall p \leq s) P(m, n, p) \vee (\forall p \leq s) \neg Q(m, n, p) \right]; \\ G(m, s) &\simeq \begin{cases} 0, & \text{if } (\forall p \leq s) \neg Q(m, F(m, s), p); \\ 1, & \text{otherwise.} \end{cases} \end{aligned}$$

Exercise 5.1.33 Show that $\text{gn}(\text{Th}^\Sigma(\Omega))$ is many-one complete (cf. Exercise 4.6.47).

Exercise 5.1.34 Let ${}_nU^1$ be the Σ_n universal relation of Definition 5.1.6, and set

$${}_\infty U^1 := \{ (\langle n, a \rangle, m) : {}_nU^1(a, m) \}.$$

Prove that ${}_\infty U^1$ is universal for the class of unary arithmetical relations and is therefore not itself arithmetical.

Exercise 5.1.35 Fill in the details in the proof of Proposition 5.1.22. In particular give a more complete definition of the notion of Σ -derivation and a proof that Deriv^Σ is primitive recursive.

Exercise 5.1.36 Generalize Exercises 4.3.30 and 4.6.45 by showing that for all $n > 0$,

- (i) Σ_n has the reduction property but not the separation property;
- (ii) Π_n has the separation property but not the reduction property.

Hint: For the second part of (i), let

$$X := \{ m : {}_nU^1((m)_0, m) \} \quad \text{and} \quad Y := \{ m : {}_nU^1((m)_1, m) \}.$$

Apply the first part of (i) to obtain disjoint Σ_n sets X^- and Y^- as in the definition and derive a contradiction from the hypothesis that there exists $Z \in \Delta_n$ which separates X^- and Y^- . Note that for such a Z there exist a and b such that $Z = {}_nU_a^1$ and $\omega \sim Z = {}_nU_b^1$.

5.2 Indexing

Any countably infinite set A has, by definition, an enumeration by natural numbers, but we have seen by now several examples that show that certain enumerations are much more interesting and useful than others. We discussed this first at the beginning of [Section 4.4](#) in introducing the mapping $\phi \mapsto \mathbf{gn}(\phi)$ as a replacement for the enumeration $m \mapsto \chi_m$ of [Section 4.1](#). The advantage is that the arithmetical structure of $\mathbf{gn}(\phi)$ reflects the structure of ϕ as a formula — for example, so that the mapping $\mathbf{gn}(\phi \vee \psi) \mapsto \mathbf{gn}(\phi)$ is the primitive recursive mapping $\langle 15, m, n \rangle \mapsto m$. This led to the Enumerability Theorem and thence to the very useful enumeration of the recursively enumerable sets (and semi-recursive relations)

$$a \mapsto U_a^1 := \{ m : \mathbf{Sb}^1(a, m) \in \mathbf{gn}(\mathbf{Q}) \}.$$

Of course, if we just used the denumerability of the class of recursively enumerable sets to produce a list $X_0, X_1, \dots, X_a, \dots$ of them, we would not expect that the relation $m \in X_a$ would itself be semi-recursive. Then, in the preceding section, we produced further enumerations of the classes Σ_n and Π_n .

In this section we examine further this process and most importantly describe some useful enumerations of the classes of primitive and partial recursive functions. We shall call any assignment of natural numbers to a class of objects an **indexing** of the objects; in each case we shall only be

interested in indexings which somehow reflect the structure and properties of the objects. Although all of the indexings we have used so far are ultimately derived from the Gödel numbering of formulas, some of those considered below have other origins. The material of this section will be used in the following sections of this chapter and heavily in [Chapter 8](#).

Let us begin by reexamining the process by which we assigned Gödel numbers to the terms of L_Ω in Definition 4.4.1. The function $\mathbf{gn} : \mathbf{Te}_{L_\Omega} \rightarrow \omega$ was defined by term-recursion, a procedure which is valid because the inductive definition of $\mathbf{Te}_{L_\Omega}$ has unique readability. We could have achieved the same effect by reversing the process and defining the Gödel numbers first: let $\mathbf{TE}$ denote the smallest set of natural numbers such that for all m and n ,

- (i) $\langle 1 \rangle \in \mathbf{TE}$ and for all $i \in \omega$, $\langle 2i \rangle \in \mathbf{TE}$;
- (ii) $m \in \mathbf{TE} \implies \langle 3, m \rangle \in \mathbf{TE}$;
- (iii) $m, n \in \mathbf{TE} \implies \langle 5, m, n \rangle, \langle 7, m, n \rangle \in \mathbf{TE}$.

It is easy to see that this inductive definition also has unique readability, so we may define recursively a function $m \mapsto t_m$ from $\mathbf{TE}$ to $\mathbf{Te}_{L_\Omega}$ by the clauses $t_{\langle 1 \rangle} = \dot{0}$, $t_{\langle 2i \rangle} = v_i$, $t_{\langle 3, m \rangle} = \dot{S}t_m$ etc. It is pretty clear (formally by induction) that this function is simply the inverse of $\mathbf{gn}$ and that $\mathbf{TE}$ is exactly the set of Gödel numbers of L_Ω -terms. All of the same remarks apply equally to the relationship between formulas and their Gödel numbers.

Let us contrast this situation with the inductive definition of the set $\mathbf{Thm}(\Gamma)$ of theorems provable from some recursive set of axioms Γ . Of course in this case there is no unique readability (Remark 3.2.7), so we cannot hope to define functions recursively on theorems, but we already have the function $\mathbf{gn}$, hence the set $\mathbf{gn}(\mathbf{Thm}(\Gamma))$. Here, too, we could give an independent inductive definition of this set as the smallest set $\mathbf{THM}_\Gamma$ of natural numbers such that (see Definition 4.4.10)

- (i) $\mathbf{Taut} \cup \mathbf{Eqax} \cup \exists\text{-ax} \cup \mathbf{gn}(\Gamma) \subseteq \mathbf{THM}_\Gamma$;
- (ii) $m, n \in \mathbf{THM}_\Gamma$ and $\mathbf{ModPon}(m, n, p) \implies p \in \mathbf{THM}_\Gamma$;
- (iii) $m \in \mathbf{THM}_\Gamma$ and $\exists\text{-rule}(m, n) \implies n \in \mathbf{THM}_\Gamma$.

This definition also fails to have unique readability, but it is straightforward to verify by two inductions that $\mathbf{THM}_\Gamma = \mathbf{gn}(\mathbf{Thm}(\Gamma))$.

So far these two cases seem reasonably parallel, but there is a very significant difference: we showed in [Section 4.4](#) that $\mathbf{TE}$ is primitive recursive while $\mathbf{THM}_\Gamma$ is in general only recursively enumerable, for example, if Γ is the set of axioms of $\mathbf{Q}$. This difference can be traced to a fact mentioned in Exercise 4.3.28: a set of natural numbers which can be recursively enumerated in increasing order is recursive. Each of the closure functions $m \mapsto \langle 3, m \rangle$ etc. involved in the definition of $\mathbf{TE}$ is increasing while those for $\mathbf{THM}_\Gamma$ are not:

if $\text{ModPon}(m, n, p)$ then for some formulas $m = \text{gn}(\phi)$, $n = \text{gn}(\phi \rightarrow \psi)$ and $p = \text{gn}(\psi)$, so necessarily $n > p$ and m may be larger or smaller than p . The following makes this more precise.

Proposition 5.2.1 *For any induction system $\mathcal{X} = (\omega, X_0, \mathcal{H})$ over ω (Definition 1.2.1),*

- (i) *if X_0 is (primitive) recursive and $\mathcal{H}$ is a finite set of (primitive) recursive functions, then $\overline{\mathcal{X}}$ is recursively enumerable;*
- (ii) *if in addition each $H \in \mathcal{H}$ is increasing in all arguments — that is, for each $i < k_H$, $m_i < H(m_0, \dots, m_{k_H-1})$ — then $\overline{\mathcal{X}}$ is (primitive) recursive.*

Proof. Let $\mathcal{X}$ be as in (i). Recall that an $\mathcal{X}$ -derivation is a finite sequence $m_0, \dots, m_{k-1}$ of (in this case) natural numbers such that for each $i < k$, $m_i \in X_0$ or for some $H \in \mathcal{H}$ and $j_0, \dots, j_{k_H-1} < i$, $m_i = H(m_{j_0}, \dots, m_{j_{k_H-1}})$. Now set

$$\begin{aligned} \text{Deriv}^{\mathcal{X}} &= \left\{ s : ((s)_0, \dots, (s)_{\text{lg}(s)-1}) \text{ is an } \mathcal{X}\text{-derivation} \right\} \\ \text{Derivof}^{\mathcal{X}} &= \left\{ (s, m) : \text{Deriv}^{\mathcal{X}}(s) \text{ and } (s)_{\text{lg}(s)-1} = m \right\}. \end{aligned}$$

Then $\text{Deriv}^{\mathcal{X}}$ is (primitive) recursive since the criterion for membership is

$$\begin{aligned} &(\forall i < \text{lg}(s)) \left((s)_i \in X_0 \quad \text{or} \right. \\ &\quad \left. \bigvee_{H \in \mathcal{H}} (\exists j_0 < i) \cdots (\exists j_{k_H-1} < i) \left[(s)_i = H((s)_{j_0}, \dots, (s)_{j_{k_H-1}}) \right] \right), \end{aligned}$$

whence also $\text{Derivof}^{\mathcal{X}}$ is (primitive) recursive. Then for any m ,

$$m \in \overline{\mathcal{X}} \iff \exists s \text{Derivof}^{\mathcal{X}}(s, m),$$

which implies that $\overline{\mathcal{X}}$ is recursively enumerable.

The proof of (ii) is simply an abstract version of the proof of Proposition 4.4.4; we define a (primitive) recursive function G such that for all m , $\text{K}_{\overline{\mathcal{X}}}(m) = G(\overline{\text{K}}_{\overline{\mathcal{X}}}(m), m)$. It suffices to set $G(s, m) = 1$ if $m \in X_0$ or for some $H \in \mathcal{H}$,

$$\begin{aligned} &(\exists p_0 < m) \cdots (\exists p_{k_H-1} < m) \left[(s)_{p_0} = \cdots = (s)_{p_{k_H-1}} = 1 \right. \\ &\quad \left. \text{and } H(p_0, \dots, p_{k_H-1}) = m \right], \end{aligned}$$

otherwise, $G(s, m) = 0$. The point is that the hypothesis that H is increasing allows the quantifiers to be bounded. $\square$

Although this version suffices to conclude that TE is primitive recursive (which we already knew), it is not general enough for most of the other applications we want because they in fact require infinitely many closure functions. Although it is not immediately obvious from the statement above, this is even true for THM_Γ : closure under the $\exists$ -rule is actually closure under an infinite family of functions $(\phi \rightarrow \psi) \mapsto (\exists x\phi \rightarrow \psi)$, one for each variable x not free in ψ .

Before giving the more general form of the theorem which handles these cases, we present another important example of indexing. Consider the set Prim of primitive recursive functions. Prim is inductively defined as the smallest class of functions which contains the (infinitely many) initial functions Cs_n^k , Pr_i^k and Sc_i^k and is closed under the operations Cmp_l^k and Rec^{k+1} . We observed in Exercise 4.2.45 that this inductive definition does not have unique readability, and thus it is not obvious how we might assign indices to the primitive recursive functions. This is a case in which the idea of generating the indices separately and then assigning objects (in this case functions) to indices becomes essential.

Definition 5.2.2 PrimInd , the set of *primitive recursive indices*, is the smallest set of natural numbers such that for all k ,

- (i) for all n and $i < k$, $\langle 0, k, n \rangle$, $\langle 1, k, i \rangle$, and $\langle 2, k, i \rangle$ belong to PrimInd ;
- (ii) for all l , if $b, c_0, \dots, c_{l-1} \in \text{PrimInd}$, then $\langle 3, k, b, c_0, \dots, c_{l-1} \rangle \in \text{PrimInd}$;
- (iii) if $b, c \in \text{PrimInd}$, then $\langle 4, k+1, b, c \rangle \in \text{PrimInd}$.

This definition also requires infinitely many closure functions

$$(b, c_0, \dots, c_{l-1}) \mapsto \langle 3, k, b, c_0, \dots, c_{l-1} \rangle,$$

one for each k and l . The definition of PrimInd has unique readability simply because of the properties of the sequence coding function $\langle \cdot \rangle$, and we may assign primitive recursive functions to indices (rather than the other way around) as follows.

Definition 5.2.3 The function $a \mapsto [a]$ from PrimInd to Prim is defined recursively by

- (i) $[\langle 0, k, n \rangle] = \text{Cs}_n^k$, $[\langle 1, k, i \rangle] = \text{Pr}_i^k$ and $[\langle 2, k, i \rangle] = \text{Sc}_i^k$;
- (ii) $[\langle 3, k, b, c_0, \dots, c_{l-1} \rangle] = \text{Cmp}_l^k([b], [c_0], \dots, [c_{l-1}])$;
- (iii) $[\langle 4, k+1, b, c \rangle] = \text{Rec}^{k+1}([b], [c])$.

We set $\text{PrimInd}^k := \{a \in \text{PrimInd} : (a)_1 = k\}$.

Proposition 5.2.4 *The function $a \mapsto [a]$ maps PrimInd^k onto the set of k -ary primitive recursive functions and hence maps PrimInd onto Prim .*

Proof. That the mapping is into is proved by a straightforward induction over PrimInd . Then a proof by induction over Prim easily establishes that for each k -ary primitive recursive function F there exists $a \in \text{PrimInd}$ such that $(a)_1 = k$ and $F = [a]$. $\square$

Note that the index assignment $a \mapsto [a]$ is certainly not injective. Indeed, it is easy to see that every primitive recursive function has infinitely many indices. An index a for a primitive recursive function F should be thought of as a specific set of instructions or **program** for computing F . The non-unique readability of the inductive definition of Prim is due exactly to the fact that there are (infinitely) many distinct programs for computing each primitive recursive function, just as each $\phi \in \text{Thm}(\Gamma)$ has infinitely many distinct proofs from Γ .

Definition 5.2.5 For each k and all $a, m_0, \dots, m_{k-1}$,

$$U_{\text{Prim}}^k(a, m_0, \dots, m_{k-1}) = \begin{cases} [a](m_0, \dots, m_{k-1}), & \text{if } a \in \text{PrimInd}^k; \\ 0, & \text{otherwise.} \end{cases}$$

In a sense quite analogous to the use of the term in the preceding section, U_{Prim}^k is a function **universal** for the class of k -ary primitive recursive functions. A version of the diagonal argument gives

Proposition 5.2.6 *For each k , U_{Prim}^k is not primitive recursive.*

Proof. We do the proof for $k = 1$. Suppose that U_{Prim}^1 is primitive recursive; then so is the function $G(m) = U_{\text{Prim}}^1(m, m) + 1$. Hence, if b is any index for G , then $b \in \text{PrimInd}^1$, and for all m , $[b](m) = U_{\text{Prim}}^1(m, m) + 1$. In particular, for $m = b$ we have

$$b = U_{\text{Prim}}^1(b, b) + 1 = b + 1,$$

a contradiction. $\square$

Although the inductive definition of PrimInd involves an infinite family of closure functions, there is a good deal of uniformity which we exploit to extend Proposition 5.2.1.

Definition 5.2.7 For any binary function I on ω and any k and l , let $I^{*k,l}$ be the k -ary function

$$I^{*k,l}(m_0, \dots, m_{l-1}) = I(\langle m_0, \dots, m_{l-1} \rangle, k),$$

and for a set $\mathcal{I}$ of such functions, $\mathcal{I}^* = \{ I^{*k,l} : I \in \mathcal{I} \text{ and } k, l \in \omega \}$. We call a binary function I **suitable** iff for all k, l and $m_0, \dots, m_{l-1}$, $k, l < I^{*k,l}(m_0, \dots, m_{l-1})$.

Definition 5.2.8 A **(primitive) recursive induction system** is one of the form $(\omega, X_0, \mathcal{H} \cup \mathcal{I}^*)$, where X_0 is (primitive) recursive, $\mathcal{H}$ is a finite set of (primitive) recursive functions and $\mathcal{I}$ is a finite set of suitable binary (primitive) recursive functions.

Examples 5.2.9

- (i) The induction system above for PrimInd is primitive recursive with $\mathcal{I} = \{I_0, I_1\}$ and

$$\begin{aligned} I_0(\langle b, c_0, \dots, c_{l-1} \rangle, k) &= \langle 3, k, b, c_0, \dots, c_{l-1} \rangle; \\ I_1(\langle b, c, \dots \rangle, k) &= \langle 4, k+1, b, c \rangle. \end{aligned}$$

- (ii) Similarly, the induction system for THM_Γ is primitive recursive with

$$I(\langle m, \dots \rangle, k) = \begin{cases} \langle 15, \langle 13, \langle 17, 2k, p \rangle \rangle, q \rangle, & \text{if } \sim \text{Free}(q, k) \text{ and} \\ & m = \langle 15, \langle 13, p \rangle, q \rangle; \\ \langle 13, \langle 17, 0, \langle 13, \langle 11, \langle 0 \rangle, \langle 0 \rangle \rangle \rangle \rangle, & \text{otherwise,} \end{cases}$$

which means that for any formulas ϕ and ψ ,

$$I^{*k,l}(\text{gn}(\phi \rightarrow \psi), \dots) = \begin{cases} \text{gn}(\exists v_k \phi \rightarrow \psi), & \text{if } v_k \text{ is not free in } \psi; \\ \text{gn}(\forall v_0 (v_0 \doteq v_0)), & \text{otherwise.} \end{cases}$$

Recursive Induction Theorem 5.2.10 For any (primitive) recursive induction system $\mathcal{X} = (\omega, X_0, \mathcal{H} \cup \mathcal{I}^*)$,

- (i) the closure set $\overline{\mathcal{X}}$ is recursively enumerable;
(ii) if in addition each $H \in \mathcal{H}$ and $I^{*k,l} \in \mathcal{I}^*$ is increasing in all arguments, then $\overline{\mathcal{X}}$ is (primitive) recursive.

Proof. For (i) we need only modify the definition of $\text{Derivof}^{\mathcal{X}}$ from the previous proof to include the disjunct

$$\bigvee_{I \in \mathcal{I}} (\exists k < s) (\exists l < s) (\exists j_0 < i) \cdots (\exists j_{l-1} < i) \left[(s)_i = I \left(\langle (s)_{j_0}, \dots, (s)_{j_{l-1}} \rangle, k \right) \right],$$

or more formally

$$\bigvee_{I \in \mathcal{I}} (\exists t < s^*) (\exists k < s) \left(\text{Sq}(t) \text{ and } (\forall p < \text{lg}(t)) (\exists j < i) \left[((t)_p = (s)_j) \text{ and } (s)_i = I(t, k) \right] \right).$$

Here s^* is a number primitive recursively computable from s such that for any l and $m_0, \dots, m_{l-1} < s$, $\langle m_0, \dots, m_{k-1} \rangle < s^*$ — for example, s^* could be the code for the sequence of s -many numbers all equal to s . The other bounds are derived from the assumption that each I is suitable. We have as before that $\text{Deriv}^{\mathcal{X}}$ and $\text{Derivof}^{\mathcal{X}}$ are (primitive) recursive and thus $\mathcal{X}$ is recursively enumerable. The proof of (ii) is a similar straightforward modification of the earlier proof. $\square$

Corollary 5.2.11 *PrimInd is primitive recursive.*

Proof. Immediate from the theorem and preceding example. $\square$

Our next application is to give a second proof of the result of Exercise 4.4.24 that there exist functions which are recursive but not primitive recursive by showing that each function U_{Prim}^k is recursive. The idea is to show that its graph is semi-recursive by providing a recursive inductive definition for it.

Definition 5.2.12 *PrimComp*, the set of **primitive recursive computations**, is the smallest set of natural numbers such that for all k and $m_0, \dots, m_{k-1}$,

(i) for all n and $i < k$,

$$\begin{aligned} \langle \langle 0, k, n \rangle, m_0, \dots, m_{k-1}, n \rangle &\in \text{PrimComp}; \\ \langle \langle 1, k, i \rangle, m_0, \dots, m_{k-1}, m_i \rangle &\in \text{PrimComp}; \\ \langle \langle 2, k, i \rangle, m_0, \dots, m_{k-1}, m_i + 1 \rangle &\in \text{PrimComp}; \end{aligned}$$

(ii) for all $l, p_0, \dots, p_{l-1}$ and q ,

$$\begin{aligned} &\text{if for all } i < l, \quad \langle c_i, m_0, \dots, m_{k-1}, p_i \rangle \in \text{PrimComp} \\ &\quad \text{and } \langle b, p_0, \dots, p_{l-1}, q \rangle \in \text{PrimComp}, \\ &\text{then } \langle \langle 3, k, b, c_0, \dots, c_{l-1} \rangle, m_0, \dots, m_{k-1}, q \rangle \in \text{PrimComp}; \end{aligned}$$

(iii) for all q ,

(a) if $\langle b, m_0, \dots, m_{k-1}, q \rangle \in \text{PrimComp}$, then

$$\langle \langle 4, k + 1, b, c \rangle, m_0, \dots, m_{k-1}, 0, q \rangle \in \text{PrimComp};$$

(b) for all n ,

$$\begin{aligned} &\text{if } \langle \langle 4, k + 1, b, c \rangle, m_0, \dots, m_{k-1}, n, p \rangle \in \text{PrimComp} \\ &\quad \text{and } \langle c, p, m_0, \dots, m_{k-1}, n, q \rangle \in \text{PrimComp}, \\ &\text{then } \langle \langle 4, k + 1, b, c \rangle, m_0, \dots, m_{k-1}, n + 1, q \rangle \in \text{PrimComp}. \end{aligned}$$

Proposition 5.2.13 *For each k , $a \in \text{PrimInd}^k$ and $m_0, \dots, m_{k-1}$,*

$$[a](m_0, \dots, m_{k-1}) = \text{unique } q \langle a, m_0, \dots, m_{k-1}, q \rangle \in \text{PrimComp}.$$

Furthermore, if $a \notin \text{PrimInd}^k$, then for all q ,

$$\langle a, m_0, \dots, m_{k-1}, q \rangle \notin \text{PrimComp}.$$

Proof. It suffices to prove by induction that

$$\begin{aligned} a \in \text{PrimInd}^k \text{ and } [a](m_0, \dots, m_{k-1}) = q \\ \iff \langle a, m_0, \dots, m_{k-1}, q \rangle \in \text{PrimComp}. \end{aligned}$$

Both inductions are straightforward and are left as an exercise. $\square$

Proposition 5.2.14 *PrimComp is recursively enumerable.*

Proof. It is easy to see that the inductive definition of PrimComp is primitive recursive, so this follows by the Recursive Induction Theorem 5.2.10. $\square$

Of course, the closure functions here are not in general increasing: in clause (ii), the p_i may be much larger than q . Hence we cannot conclude that PrimComp is primitive recursive, which is fortunate, since this would contradict the conclusion above that the functions U_{Prim}^k are not primitive recursive. However,

Theorem 5.2.15 *For each k , the universal primitive recursive function U_{Prim}^k is recursive.*

Proof. For each k , a , $m_0, \dots, m_{k-1}$ and q ,

$$\begin{aligned} U_{\text{Prim}}^k(a, m_0, \dots, m_{k-1}) = q \iff \\ [(a)_1 = k \text{ and } \langle a, m_0, \dots, m_{k-1}, q \rangle \in \text{PrimComp}] \\ \text{or } [a \notin \text{PrimInd}^k \text{ and } q = 0]. \end{aligned}$$

Hence the graph of U_{Prim}^k is semi-recursive, so by Proposition 4.3.13, U_{Prim}^k is recursive. $\square$

Remark 5.2.16 It is worth unraveling the proof of the preceding proposition a bit to get a clearer idea of why this result is true. We have for $(a)_1 = 1$,

$$[a](m) = q \iff \exists s [\text{Derivof}^{\mathcal{X}}(s, \langle a, m, q \rangle)],$$

where $\mathcal{X}$ is the induction system which defines **PrimComp**. The witness s may naturally be seen as coding the scratchwork for a computation of this value. For example, if $a = \langle 3, 1, b, c_0, c_1 \rangle$, so

$$[a](m) = \text{Cmp}_2^1([b], [c_0], [c_1])(m) = [b]([c_0](m), [c_1](m)),$$

then if $\text{Derivof}^{\mathcal{X}}(s, \langle a, m, q \rangle)$ holds, there exist j_0, j_1 and k less than the length of s and p_0, p_1 such that

$$(s)_{j_0} = \langle c_0, m, p_0 \rangle, \quad (s)_{j_1} = \langle c_1, m, p_1 \rangle \quad \text{and} \quad (s)_k = \langle b, p_0, p_1, q \rangle.$$

Thus s codes a sequence of tuples of the form $\langle d, n_0, \dots, n_{l-1}, r \rangle$ which represent computations $[d](n_0, \dots, n_{l-1}) = r$ which are **subcomputations** of the computation of $[a](m) = q$.

We turn next to the application of these ideas to the classes of (partial) recursive functions. In both cases the inductive definition differs from that of the definition of the class of primitive recursive functions by the inclusion of the search schema

$$F(m_0, \dots, m_{k-1}) = \text{least } n [G(m_0, \dots, m_{k-1}, n) = 0]$$

applied either unconditionally to partial functions or to total functions restricted to those which satisfy the search condition

$$\forall m_0 \dots \forall m_{k-1} \exists n G(m_0, \dots, m_{k-1}, n) = 0.$$

Consider first the total case; to define a set **Reclnd** of indices of (total) recursive functions, we would like to introduce for each index b of a recursive function G which satisfies the search condition an index $\langle 5, k, b \rangle$ for the corresponding F . The search condition is, however, not a property of the indices but a property of the functions, and there seems no natural way to phrase the definition. For the case of the partial recursive functions, this is not a problem, and we might well define a set **Partlnd** by adjoining to Definition 5.2.2 of **Primlnd** a clause

(iv) if $b \in \text{Partlnd}$, then $\langle 5, k, b \rangle \in \text{Partlnd}$.

It turns out that the resulting class of indices is not very interesting or useful, since properties of the partial function $\{a\}$ associated with an index a , in particular its domain, cannot be determined effectively from a . We can, however, get considerable information from the second approach developed above, the inductive generation of a set of codes for computations of partial recursive functions.

Definition 5.2.17 PartComp , the set of *partial recursive computations*, is the smallest set of natural numbers such that for all k and $m_0, \dots, m_{k-1}$,

- (i)–(iii) as for PrimComp ;
- (iv) for all b and n , if $\langle b, m_0, \dots, m_{k-1}, n, 0 \rangle \in \text{PartComp}$ and for each $p < n$, there is some $q_p > 0$ such that

$$\langle b, m_0, \dots, m_{k-1}, p, q_p \rangle \in \text{PartComp},$$

then $\langle \langle 5, k, b \rangle, m_0, \dots, m_{k-1}, n \rangle \in \text{PartComp}$.

Proposition 5.2.18 PartComp is recursively enumerable.

Proof. Easily PartComp is the closure of a recursive induction system so by the Recursive Induction Theorem 5.2.10 is recursively enumerable. $\square$

Remark 5.2.19 Note that this technique still fails for the (total) recursive functions. The corresponding definition of a set RecComp would need to include in clause (iv) the condition

$$\forall m_0 \dots \forall m_{k-1} \exists n \langle b, m_0, \dots, m_{k-1}, n, 0 \rangle \in \text{RecComp}.$$

The resulting induction system is no longer recursive, so the resulting set will not be recursively enumerable.

Proposition 5.2.20 For all a, k and $m_0, \dots, m_{k-1}$, there is at most one number n such that $\langle a, m_0, \dots, m_{k-1}, n \rangle \in \text{PartComp}$; further, for $k \neq (a)_1$ there are no such n .

Proof. A straightforward induction on PartComp . $\square$

Because of this second proposition it makes sense to assign to every number a a partial function, which we denote by $\{a\}$.

Definition 5.2.21 For each k , all a such that $k = (a)_1$ and all $m_0, \dots, m_{k-1}$ and n ,

- (i) $\{a\}(m_0, \dots, m_{k-1}) \simeq n \iff \langle a, m_0, \dots, m_{k-1}, n \rangle \in \text{PartComp}$;
- (ii) $U^k(a, m_0, \dots, m_{k-1}) \simeq \{a\}(m_0, \dots, m_{k-1})$
 $\simeq \text{least } n (\langle a, m_0, \dots, m_{k-1}, n \rangle \in \text{PartComp})$;

U^k is called the *universal k -ary partial recursive function*.

Corollary 5.2.22 For each k , U^k is partial recursive. $\square$

Of course, for infinitely many a , $\{a\}$ is the empty (totally undefined) function. The following should be compared with Corollary 4.3.13.

Theorem 5.2.23 *For any partial function F , the following are equivalent:*

- (i) F is partial recursive;
- (ii) for some $a \in \omega$, $F = \{a\}$;
- (iii) $\text{Graph}(F)$ is semi-recursive.

Proof. The implication (i) $\implies$ (ii) is a straightforward induction on the class of partial recursive functions. Given (ii), we have

$$F(m_0, \dots, m_{k-1}) \simeq n \iff \langle a, m_0, \dots, m_{k-1}, n \rangle \in \text{PartComp},$$

and since **PartComp** is recursively enumerable, $\text{Graph}(F)$ is semi-recursive. That (iii) $\implies$ (i) was proved as part of Theorem 4.3.16. $\square$

Parallel to Remark 5.2.16, we may also view a witness to $\langle a, m, n \rangle \in \text{PartComp}$ as coding the scratchwork for a computation of $\{a\}(m) \simeq n$. This computation is defined iff there is such a code in **PartComp**, and we may think of this as providing a uniform algorithm for computing $\{a\}(m)$: search for the least pair $\langle s, n \rangle$ such that s witnesses that $\langle a, m, n \rangle \in \text{PartComp}$ and output n .

It is interesting to observe what happens if we try to carry out the diagonal argument with U^1 in place of U^1_{Prim} . Let G be the partial recursive function defined by $G(m) \simeq U^1(m, m) + 1$ and choose an index b for G . Then as before

$$\{b\}(b) \simeq U^1(b, b) + 1 \simeq \{b\}(b) + 1.$$

This is no contradiction, however; we conclude merely that $b \notin \text{Dom}(\{b\})$! We can, however, still get something interesting from this argument.

Definition 5.2.24 The k -ary **halting problem** is the set

$$W^k = \{ (a, m_0, \dots, m_{k-1}) : (a)_1 = k \text{ and } \{a\}(m_0, \dots, m_{k-1}) \downarrow \},$$

where we recall from Section 4.3 that $\{a\}(m_0, \dots, m_{k-1}) \downarrow$ means simply that $(m_0, \dots, m_{k-1}) \in \text{Dom}(\{a\})$. The name comes from thinking of $\{a\}$ as a procedure or program which, when applied to arguments $m_0, \dots, m_{k-1}$, will halt just in case the function is defined on those arguments and otherwise go on forever trying to compute a value.

Theorem 5.2.25 *For all k , the k -ary halting problem W^k is universal for the class of k -ary semi-recursive relations and is semi-recursive but not recursive.*

Proof. Clearly W^k is universal for the class of domains of k -ary partial recursive functions, which by Theorem 4.3.16 is exactly the class of k -ary

semi-recursive relations. Hence by the Diagonal Lemma W^k is not recursive. That W^k is semi-recursive follows immediately from the equation

$$W^k = \{ (a, m_0, \dots, m_{k-1}) : \exists n U^k(a, m_0, \dots, m_{k-1}) \simeq n \}. \quad \square$$

This theorem is the starting point for the theory of semi-recursive relations and particularly recursively enumerable sets which we shall explore further in the exercises and much more extensively in [Chapter 8](#).

Exercises

Exercise 5.2.26 Show that the set $\text{gn}(\text{Th}^\Sigma(\Omega))$ is the closure of a primitive recursive induction system. Does this system satisfy the hypothesis of (ii) of the Recursive Induction Theorem 5.2.10?

Exercise 5.2.27 Give another proof of Proposition 5.2.1 along the following lines. Let X_n be (as usual) the elements of $\overline{\mathcal{X}}$ generated by stage n . Define Y_n by $Y_0 = X_0$ and

$$Y_{n+1} = Y_n \cup \{ m : (\exists p_0 < n) \cdots (\exists p_{k_H-1} < n) (\exists H \in \mathcal{H}) \\ [p_0, \dots, p_{k_H-1} \in Y_n \quad \text{and} \quad H(p_0, \dots, p_{k_H-1}) = m] \}.$$

Verify that

- (i) for all n , $Y_n \subseteq X_n$ and $\overline{\mathcal{X}} = \bigcup_{n \in \omega} Y_n$;
- (ii) $\{ (m, n) : m \in Y_n \}$ is (primitive) recursive;
- (iii) $\overline{\mathcal{X}}$ is recursively enumerable;
- (iv) under the hypotheses of part (ii), $\overline{\mathcal{X}}$ is (primitive) recursive;
- (v) if for each $H \in \mathcal{H}$, $\text{Im}(H)$ is recursive and H is injective, then $\{ (m, n) : m \in X_n \}$ is recursive so there is no need to introduce the sets Y_n . Note that these hypotheses are satisfied for TE.

Exercise 5.2.28 Carry out the proof of Proposition 5.2.13.

Exercise 5.2.29 Let $\mathcal{X}$ be the recursive induction system given above in the definition of PartComp. Then

$$\{ a \}(m_0, \dots, m_{k-1}) \simeq n \leftrightarrow \exists s [\text{Derivof}^\mathcal{X}(s, \langle a, m_0, \dots, m_{k-1}, n \rangle)].$$

Show that

- (i) the relation in square brackets is primitive recursive and we obtain a second proof of Corollary 4.6.25;

- (ii) a set X is recursively enumerable iff $X = \emptyset$ or for some unary *primitive* recursive function F , $X = \text{Im}(F)$.

Is it also true that

- (iii) a set X is recursive iff X is finite or for some unary *primitive* recursive function F , $X = \text{Im}(F)$ and F is strictly increasing? Why or why not? (Compare Exercise 4.3.28.)

5.3 Second Incompleteness Theorem

The goal of this section is to expand the sketch of a proof of this theorem given at the end of [Section 4.1](#). In words, the theorem says: no theory which extends Peano arithmetic $\mathbf{P}$ can prove its own consistency. In particular, this means that for any such theory there can be no finitary proof of its consistency, since (it seems to be true that) all finitary methods can be formalized in $\mathbf{P}$.

Fix a recursively axiomatizable theory $T \supseteq \mathbf{Q}$; we follow the numbering of [Section 4.1](#), and the reader should refer frequently to the sketch there. Since by the Enumerability Theorem $\text{gn}(T)$ is recursively enumerable, by Proposition 4.6.24 there exists an L_Ω -formula $\dot{T}(y)$ such that for all formulas ϕ ,

$$(E.1) \quad \phi \in T \iff \dot{T}(\ulcorner \phi \urcorner) \in \mathbf{Q} \iff \dot{T}(\ulcorner \phi \urcorner) \in \text{Th}(\Omega).$$

Furthermore, by the proof of that proposition, $\dot{T}(y)$ may be taken to be $\exists u \chi_\Gamma(u, y)$, where Γ is a recursive set of axioms for T and χ_Γ strongly $\mathbf{Q}$ -represents the relation **Proof** _{Γ} of Definition 4.4.9, and by Corollary 5.1.19 we may assume that χ_Γ and hence $\dot{T}$ is a Σ formula. Because $\text{Th}(\Omega)$ is a complete theory, (E.1) entails that for all ϕ ,

$$(E.2) \quad \phi \notin T \iff \neg \dot{T}(\ulcorner \phi \urcorner) \in \text{Th}(\Omega),$$

and since $\mathbf{Q} \subseteq T$,

$$(E.3) \quad \phi \in T \implies \dot{T}(\ulcorner \phi \urcorner) \in T.$$

The Self-reference Lemma 4.6.27 guarantees the existence of a sentence η such that

$$\mathbf{Q} \vdash \eta \leftrightarrow \neg \dot{T}(\ulcorner \eta \urcorner).$$

In particular, again because $\mathbf{Q} \subseteq T$,

$$(E.4.1) \quad \dot{T}(\ulcorner \eta \urcorner) \in T \implies \neg \eta \in T$$

$$(E.4.2) \quad \neg \dot{T}(\ulcorner \eta \urcorner) \in T \implies \eta \in T$$

$$(E.4.3) \quad (\dot{T}(\ulcorner \eta \urcorner) \rightarrow \neg \eta) \in T.$$

Now

$$(E.5) \quad T \text{ is consistent} \implies \eta \notin T,$$

because if $\eta \in T$, then by (E.3), $\dot{T}(\ulcorner \eta \urcorner) \in T$, hence by (E.4.1), $\neg \eta \in T$, so T is inconsistent.

Let Con_T be the sentence $\neg \dot{T}(\ulcorner \dot{0} \doteq \dot{1} \urcorner)$. Since $\neg (\dot{0} \doteq \dot{1}) \in Q \subseteq T$,

$$T \text{ is consistent} \iff (\dot{0} \doteq \dot{1}) \notin T,$$

so by (E.2),

$$(E.6) \quad T \text{ is consistent} \iff \text{Con}_T \in \text{Th}(\Omega).$$

Thus by (E.2), (E.5) and (E.6),

$$\text{Con}_T \in \text{Th}(\Omega) \implies \neg \dot{T}(\ulcorner \eta \urcorner) \in \text{Th}(\Omega),$$

which is simply

$$(E.7) \quad \models_{\Omega} \text{Con}_T \rightarrow \neg \dot{T}(\ulcorner \eta \urcorner),$$

a formal version of (E.5).

The remainder of the argument consists in showing that this sentence is not only true in Ω but in fact a theorem of P :

$$(E.8) \quad P \vdash \text{Con}_T \rightarrow \neg \dot{T}(\ulcorner \eta \urcorner).$$

As a start towards this goal, we write down more formally the main steps in the proof of (E.5):

$$(E.5.1) \quad \eta \in T \implies \dot{T}(\ulcorner \eta \urcorner) \in T$$

$$(E.5.2) \quad \dot{T}(\ulcorner \eta \urcorner) \in T \implies \neg \eta \in T$$

$$(E.5.3) \quad \eta \in T \implies \left[\neg \eta \in T \implies (\dot{0} \doteq \dot{1}) \in T \right].$$

By (E.1) we may view these as asserting the truth in Ω of certain L_{Ω} -sentences:

$$(E.7.1) \quad \dot{T}(\ulcorner \eta \urcorner) \rightarrow \dot{T}(\ulcorner \dot{T}(\ulcorner \eta \urcorner) \urcorner)$$

$$(E.7.2) \quad \dot{T}(\ulcorner \dot{T}(\ulcorner \eta \urcorner) \urcorner) \rightarrow \dot{T}(\ulcorner \neg \eta \urcorner)$$

$$(E.7.3) \quad \dot{T}(\ulcorner \eta \urcorner) \rightarrow \left[\dot{T}(\ulcorner \neg \eta \urcorner) \rightarrow \dot{T}(\ulcorner \dot{0} \doteq \dot{1} \urcorner) \right].$$

Indeed, $\text{Con}_T \rightarrow \neg \dot{T}(\ulcorner \eta \urcorner)$ is a tautological consequence of (E.7.1)–(E.7.3), so (E.7) is a direct consequence of the truth of (E.7.1)–(E.7.3), and (E.8) follows similarly from the provability of (E.7.1)–(E.7.3) in $\mathbf{P}$. This, in turn, can be reduced to two general conditions known as the **derivability conditions**: for all formulas ϕ and ψ ,

$$(\mathbf{DC1}) \quad \mathbf{P} \vdash \dot{T}(\ulcorner \phi \urcorner) \rightarrow \dot{T}(\ulcorner \dot{T}(\ulcorner \phi \urcorner) \urcorner)$$

$$(\mathbf{DC2}) \quad \mathbf{P} \vdash \dot{T}(\ulcorner \phi \rightarrow \psi \urcorner) \rightarrow [\dot{T}(\ulcorner \phi \urcorner) \rightarrow \dot{T}(\ulcorner \psi \urcorner)].$$

Now assuming **(DC1)** and **(DC2)** we have the

Second Incompleteness Theorem 5.3.1 *For any consistent recursively axiomatizable theory $T \supseteq \mathbf{P}$, $T \not\vdash \text{Con}_T$.*

Proof. That $(\text{E.7.1}) \in \mathbf{P}$ is an instance of **(DC1)**. By (E.4.3) and (E.1),

$$\dot{T}(\ulcorner \dot{T}(\ulcorner \eta \urcorner) \rightarrow \neg \eta \urcorner) \in \mathbf{Q} \subseteq \mathbf{P}.$$

Then that $(\text{E.7.2}) \in \mathbf{P}$ follows by **(DC2)**. Finally, $(\text{E.7.3}) \in \mathbf{P}$ follows similarly by (E.1) and two further uses of **(DC2)** from the fact that the tautology $\eta \rightarrow (\neg \eta \rightarrow \dot{0} \doteq \dot{1})$ belongs to T . Now, $\text{Con}_T \rightarrow \neg \dot{T}(\ulcorner \eta \urcorner)$ is a tautological consequence of (E.7.1)–(E.7.3) so is also a member of $\mathbf{P} \subseteq T$. Then if $T \vdash \text{Con}_T$, we have $T \vdash \neg \dot{T}(\ulcorner \eta \urcorner)$, so by (E.4.2), $T \vdash \eta$, contrary to (E.5). $\square$

The remainder of the proof of the Second Incompleteness Theorem consists in verifying the derivability conditions. It is only here that we need the extra strength of $\mathbf{P}$ (beyond $\mathbf{Q}$), the induction axioms. We shall not give a complete proof, but try to illustrate the main ideas.

Let us consider first **(DC2)**. Of course, by (E.1), the sentences that we aim to prove in $\mathbf{P}$ are true in Ω and simply reflect the fact that T is closed under *modus ponens*. A precise analysis of why this is so might look as follows. Let Γ be a fixed recursive set of axioms for T :

$$\begin{aligned} (\phi_0, \dots, \phi_k) \text{ is a } \Gamma\text{-proof of } \phi \quad \text{and} \quad (\psi_0, \dots, \psi_l) \text{ is a } \Gamma\text{-proof of } \phi \rightarrow \psi \\ \implies (\phi_0, \dots, \phi_k, \psi_0, \dots, \psi_l, \psi) \text{ is a } \Gamma\text{-proof of } \psi. \end{aligned}$$

In terms of Gödel numbers and codes, this is

$$\begin{aligned} (\text{E.9}) \quad \text{Proofof}_\Gamma(s, \text{gn}(\phi)) \quad \text{and} \quad \text{Proofof}_\Gamma(t, \text{gn}(\phi \rightarrow \psi)) \\ \implies \text{Proofof}_\Gamma(s * t * (\text{gn}(\psi)), \text{gn}(\psi)). \end{aligned}$$

Now χ_Γ was chosen to Q-(hence also P-)represent **Proof** $_{\Gamma}$, so we have

$$P \vdash \chi_\Gamma(\dot{s}, \ulcorner \phi \urcorner) \wedge \chi_\Gamma(\dot{t}, \ulcorner \phi \rightarrow \psi \urcorner) \rightarrow \chi_\Gamma(\overbrace{s * t * (\text{gn}(\psi))}^{\cdot}, \ulcorner \psi \urcorner).$$

However, since $\dot{T}$ is $\exists u \chi_\Gamma$, what we actually need to show is

$$P \vdash \exists u \chi_\Gamma(u, \ulcorner \phi \urcorner) \wedge \exists v \chi_\Gamma(v, \ulcorner \phi \rightarrow \psi \urcorner) \rightarrow \exists w \chi_\Gamma(w, \ulcorner \psi \urcorner).$$

The difference between these may best be understood by remembering that P has non-standard models and that a theorem of P must be true in these as well as in Ω . In such a model, $\exists u \chi_\Gamma(u, \ulcorner \phi \urcorner)$ may be witnessed by a non-standard element — intuitively, a code for a non-standard “proof” — which, if T is not ω -consistent, may exist even if $\phi \notin T$ because there is no standard proof. To handle this what we would like is a term (or function symbol) ConCat (for ‘concatenate’) such that

$$(E.10) \quad P \vdash \chi_\Gamma(u, \ulcorner \phi \urcorner) \wedge \chi_\Gamma(v, \ulcorner \phi \rightarrow \psi \urcorner) \rightarrow \chi_\Gamma(\text{ConCat}(u, v, \ulcorner \psi \urcorner), \ulcorner \psi \urcorner).$$

That is, in any model, ConCat defines a function which maps even non-standard “proofs” of ϕ and $\phi \rightarrow \psi$ to a “proof” of ψ . We shall see below that the key to the proofs of both of the derivability conditions is the construction of an extension by definitions P^+ of P which has such a function symbol among many others.

A similar, but more complex, problem arises with **(DC1)**. The sentences to be proved are true by (E.1), since they say that $\phi \in T \implies \dot{T}(\ulcorner \phi \urcorner) \in T$. They are instances of a more general schema

$$(E.11) \quad \theta \rightarrow \dot{T}(\ulcorner \theta \urcorner),$$

and our approach to **(DC1)** will be to show that for certain θ , including all those of the form $\dot{T}(\ulcorner \phi \urcorner)$, (E.11) is a theorem of P . Of course, we can only hope to prove instances of (E.11) which are true — that is, for which

$$\models_{\Omega} \theta \rightarrow \dot{T}(\ulcorner \theta \urcorner) \quad \text{or equivalently} \quad \models_{\Omega} \theta \implies T \vdash \theta,$$

and we show next that this holds for all Σ sentences. This will suffice for **(DC1)**, since as noted above we may assume that $\dot{T}$ is a Σ formula. It will be convenient to prove this in the following form.

Proposition 5.3.2 *Every true instance of a Σ formula is a theorem of any theory $T \supseteq Q$ — that is, for every Σ formula $\theta(x_0, \dots, x_{k-1})$ and every $m_0, \dots, m_{k-1}$,*

$$\models_{\Omega} \theta(\dot{m}_0, \dots, \dot{m}_{k-1}) \implies T \vdash \theta(\dot{m}_0, \dots, \dot{m}_{k-1}).$$

In particular, every true Σ sentence is a theorem of T .

Proof. We consider only two cases of the definition of Σ formulas (5.1.15). Suppose, for example, that θ is $\dot{0} \doteq y$. The only true instance is $\dot{0} \doteq \dot{0}$, which is a theorem of Q . The other atomic and negated atomic cases follow similarly from the propositions of [Section 4.6](#). The cases for $\vee$ and $\wedge$ are trivial. Suppose that θ is $\exists z \phi$. We have (suppressing the variables $x_0, \dots, x_{k-1}$)

$$\begin{aligned} \models_{\Omega} \exists z \phi(\dot{m}_0, \dots, \dot{m}_{k-1}) &\implies \exists n \models_{\Omega} \phi(\dot{n}, \dot{m}_0, \dots, \dot{m}_{k-1}) \\ &\implies \exists n T \vdash \phi(\dot{n}, \dot{m}_0, \dots, \dot{m}_{k-1}) \\ &\implies T \vdash \exists z \phi(\dot{m}_0, \dots, \dot{m}_{k-1}). \end{aligned}$$

The bounded quantifier case is similar: using Lemma 4.6.11,

$$\begin{aligned} \models_{\Omega} (\forall z < \dot{p}) \phi(\dot{m}_0, \dots, \dot{m}_{k-1}) &\implies (\forall n < p) \models_{\Omega} \phi(\dot{n}, \dot{m}_0, \dots, \dot{m}_{k-1}) \\ &\implies (\forall n < p) T \vdash \phi(\dot{n}, \dot{m}_0, \dots, \dot{m}_{k-1}) \\ &\implies T \vdash (\forall z < \dot{p}) \phi(\dot{m}_0, \dots, \dot{m}_{k-1}). \quad \square \end{aligned}$$

Now, we want to formalize this proof in P . At first glance, it seems that this means proving (for simplicity, with $k = 1$)

$$(E.12) \quad P \vdash \theta(\dot{m}) \rightarrow \dot{T} \left(\ulcorner \theta(\dot{m}) \urcorner \right)$$

for all Σ formulas θ , but this does not quite work for roughly the same reason we encountered above for **(DC2)**. At the $\exists$ -induction step we would need to conclude from the two-variable analogue of (E.12) as induction hypothesis that

$$P \vdash \exists y \theta(\dot{m}) \rightarrow \dot{T} \left(\ulcorner \exists y \theta(\dot{m}) \urcorner \right),$$

but to use the induction hypothesis we would need an interim step of the form $\theta(\dot{m}, \dot{n})$, for some n . This does not follow from $\exists y \theta(\dot{m})$ again because in some models of P the witness may be non-standard. Here, again, what is required are terms or function symbols Num and Sb_1 which define the functions

$$\text{Num}(\dot{m}) = \text{gn}(\dot{m}) \quad \text{and} \quad \text{Sb}_1(\text{gn}(\theta), \text{gn}(\dot{m})) = \text{gn}(\theta(\dot{m}))$$

in the sense that

$$\mathsf{P} \vdash \mathsf{Num}(\dot{m}) \doteq \ulcorner \dot{m} \urcorner \quad \text{and} \quad \mathsf{P} \vdash \mathsf{Sb}_1(\ulcorner \theta \urcorner, \ulcorner \dot{m} \urcorner) \doteq \ulcorner \theta(\dot{m}) \urcorner,$$

and for which we can prove

$$(E.13) \quad \mathsf{P} \vdash \forall x \left[\theta(x) \rightarrow \dot{T} \left(\mathsf{Sb}(\ulcorner \theta \urcorner, \mathsf{Num}(x)) \right) \right].$$

We now turn to the process mentioned above of carrying out an extension by definitions as defined in [Section 2.6](#) to obtain a theory P^+ in a language L^+ in which there exist function symbols ConCat , Sb^1 and Num as well as many other similar functions. Since extensions by definition are conservative (Theorem 2.6.16), to establish **(DC1)** and **(DC2)** it is good enough to carry out the proofs in P^+ .

Recall from Definition 4.6.35 that a k -ary function F is T -provable iff it is T -representable by a formula ψ_F such that $T \vdash \forall x_0 \dots \forall x_{k-1} \exists! y \psi_F$. For any such function F we may extend T by the definition

$$\dot{F}x_0, \dots, x_{k-1} \doteq y \leftrightarrow \psi_F$$

and in the extension T^+ , represent F by the formula $\dot{F}x_0, \dots, x_{k-1} \doteq y$. We shall show that every primitive recursive function F is P -provable.

However, in order to carry out the proofs of (E.10) and (E.13) and more general versions, we need even more. In verifying (informally) that (E.9) is true, we use many facts about the concatenation operation and the relation Proof_Γ . These are generally obvious and passed over without notice — for example in checking that if s and t code proofs, so does $s * t$, we use that the component values $(s * t)_i$ are $(s)_i$ for $i < \mathsf{lg}(s)$ and $(t)_{i-k}$ for $\mathsf{lg}(s) \leq i < \mathsf{lg}(s) + \mathsf{lg}(t)$. This is implicit in the definition of $*$, but to show rigorously that (E.10) holds, we would need to show that formal versions of these facts are theorems of P . Clearly it would be extremely long and tedious even to write out all of the statements of these results, much less their proofs.

Instead, we shall rely on the general principle that the needed properties of these functions flow simply from their definitions as primitive recursive functions and hence if the function symbols are introduced with their natural defining formulas, the needed properties will be theorems of P . Note that many of these will be universal properties and the induction schema of P will be used in the proofs. Recall that the inductive definition of the class of primitive recursive functions does not have unique readability and each such function has infinitely many derivations, or equivalently by the results of [Section 5.2](#), indices. This means that we must specify the definitions for the new function symbols recursively on the indices — different indices will lead to different defining formulas — and expect that the needed properties will be provable only for the function symbol with the “natural” definition which captures the intended use of the function.

Theorem 5.3.3 *For each $a \in \text{PrimInd}$, there exists a L_Ω -formula ψ^a such that*

- (i) *if $(a)_1 = k$, then $\text{Fv}(\psi^a) \subseteq \{x_0, \dots, x_{k-1}, y\}$;*
- (ii) *$\text{P} \vdash \forall x_0 \dots \forall x_{k-1} \exists! y \psi^a$;*
- (iii) *for all $m_0, \dots, m_{k-1} \in \omega$,*

$$\text{P} \vdash \psi^a(\dot{m}_0, \dots, \dot{m}_{k-1}, \overbrace{[a](m_0, \dots, m_{k-1})}).$$

Hence P has an extension by definitions P^+ with new function symbols $\dot{F}_a$ for each $a \in \text{PrimInd}$ such that $[a]$ is P^+ -represented by the formula $\dot{F}_a x_0, \dots, x_{k-1} \doteq y$.

Proof. (sketch) We proceed by recursion on PrimInd ; to improve legibility we carry out the extension as we go. If a is one of the initial indices,

$$\psi^{\langle 0, k, n \rangle} \text{ is } y \doteq \dot{n}; \quad \psi^{\langle 1, k, i \rangle} \text{ is } y \doteq x_i; \quad \psi^{\langle 2, k, i \rangle} \text{ is } y \doteq \dot{S}x_i.$$

If $a = \langle 3, k, b, c_0, \dots, c_{l-1} \rangle$, we may take ψ^a to be

$$y \doteq \dot{F}_b(\dot{F}_{c_0}(x_0, \dots, x_{k-1}), \dots, \dot{F}_{c_{l-1}}(x_0, \dots, x_{k-1})).$$

To handle the recursion case, we need first to verify that the function Cp of Definition 4.2.41 is P -provable and introduce a function symbol $\dot{\text{Cp}}$ for it. Then if $a = \langle 4, k+1, b, c \rangle$, take ψ^a to be

$$\begin{aligned} \exists w \left[\dot{\text{Cp}}(w, \dot{0}) \doteq \dot{F}_b(x_0, \dots, x_{k-1}) \wedge \dot{\text{Cp}}(w, x_k) \doteq y \right. \\ \left. \wedge (\forall u \prec x_k) \dot{\text{Cp}}(w, \dot{S}u) \doteq \dot{F}_c(\dot{\text{Cp}}(w, u), x_0, \dots, x_{k-1}, u) \right]. \end{aligned}$$

The induction schema of P is needed to verify (ii) for this case. $\square$

Once we have in P^+ a symbol $\dot{F}$ for each primitive recursive function F , we can also adjoin a symbol $\dot{R}$ for each primitive recursive relation R . Now the argument for **(DC2)** goes as follows. First, we replace the formula $\chi_\Gamma(u, y)$ by an atomic formula $\text{Proofof}_\Gamma(u, y)$, where the definition of Proofof_Γ reflects the definition of Proofof_Γ from Theorem 4.4.16 — that is, we expect that

$$\begin{aligned} \text{P} \vdash \text{Proofof}_\Gamma(u, y) \quad \leftrightarrow \\ \dot{S}q(u) \wedge (\forall z \prec \dot{I}g(u)) \left[\dot{\text{Taut}}(\dot{(u)}_z) \vee \dot{\text{Eqax}}(\dot{(u)}_z) \vee \dots \right] \end{aligned}$$

where the other new relation symbols are similarly given with their natural definitions. Then $\dot{T}$ is now viewed as a relation symbol with the definition

$$\dot{T}(y) \leftrightarrow \exists u \text{Proofof}_\Gamma(u, y)$$

and (ambiguously) as the L_Ω -formula which is its L_Ω -translation (Definition 2.6.19). Now (DC2) follows once we show that for any ϕ and ψ ,

$$\begin{aligned} \text{P} \vdash \text{Proofof}_\Gamma(u, \ulcorner \phi \urcorner) \wedge \text{Proofof}_\Gamma(v, \ulcorner \phi \rightarrow \psi \urcorner) \\ \rightarrow \text{Proofof}_\Gamma(\text{ConCat}(u, v, \ulcorner \psi \urcorner), \ulcorner \psi \urcorner). \end{aligned}$$

This, in turn, follows from a long list of other theorems of P, such as

$$\begin{aligned} \dot{\text{Sq}}(u) \wedge \dot{\text{Sq}}(v) &\rightarrow \dot{\text{Sq}}(\text{ConCat}(u, v, w)), \\ \dot{\text{lg}}(\text{ConCat}(u, v, w)) &\doteq \dot{\text{lg}}(u) \dot{+} \dot{\text{lg}}(v) \dot{+} \dot{1}, \\ z < \dot{\text{lg}}(u) &\rightarrow (\text{ConCat}(u, v, w))_z \doteq (\dot{u})_z \end{aligned}$$

and so on, each of which can be laboriously verified from the definitions of the relevant symbols.

Towards establishing (DC1), let Sb^k be the primitive recursive functions of Definition 4.4.6 and let $\dot{\text{Sb}}^k$ be $\dot{F}_{a_k}$ for a “natural” index a_k for Sb_k — that is, an index which reflects the intended use of the function. Then (E.11) is a special case of the following.

Theorem 5.3.4 *For every Σ formula $\theta(x_0, \dots, x_{k-1})$,*

$$\begin{aligned} \text{P}^+ \vdash \forall x_0 \dots \forall x_{k-1} \left[\theta(x_0, \dots, x_{k-1}) \right. \\ \left. \rightarrow \dot{T}(\dot{\text{Sb}}^k(\ulcorner \theta \urcorner, \text{Num}(x_0), \dots, \text{Num}(x_{k-1}))) \right]. \end{aligned}$$

In particular, if θ is a Σ sentence, then

$$\text{P}^+ \vdash \theta \rightarrow \dot{T}(\ulcorner \theta \urcorner) \quad \text{hence} \quad \text{P} \vdash \theta \rightarrow \dot{T}(\ulcorner \theta \urcorner).$$

Proof. (sketch) We proceed by induction on Σ formulas. Consider first the Σ formula $\dot{0} \doteq x$. In this case we need to establish

$$\text{P}^+ \vdash \forall x \left[\dot{0} \doteq x \rightarrow \dot{T}(\dot{\text{Sb}}^1(\ulcorner \dot{0} \doteq x \urcorner, \text{Num}(x))) \right].$$

Let $\phi(x)$ denote this formula; we will use the induction schema of P and show that

$$\text{P}^+ \vdash \phi(\dot{0}) \quad \text{and} \quad \text{P}^+ \vdash \phi \rightarrow \phi(\dot{\text{Sb}}).$$

Since the properties of $\dot{\mathbf{Sb}}^1$ ensure that

$$\mathbf{P}^+ \vdash \dot{\mathbf{Sb}}^1 \left(\ulcorner \dot{0} \dot{=} x \urcorner, \text{Num}(\dot{0}) \right) \dot{=} \ulcorner \dot{0} \dot{=} \dot{0} \urcorner,$$

the first of these reduces to $\mathbf{P}^+ \vdash \dot{T} \left(\ulcorner \dot{0} \dot{=} \dot{0} \urcorner \right)$, which holds by Proposition 5.3.2. The second of these holds simply because $\mathbf{P}^+ \vdash \neg \dot{0} \dot{=} \dot{S}x$, so $\mathbf{P}^+ \vdash \phi(\dot{S}x)$ (with no use of the induction hypothesis ϕ).

As another sample initial case, consider the Σ formula $\dot{m} \dot{<} x$ and let $\phi(x)$ denote the formula $\dot{m} \dot{<} x \rightarrow \dot{T}(\dot{\mathbf{Sb}}^1(\ulcorner \dot{m} \dot{<} x \urcorner, \text{Num}(x)))$. Again we use induction in $\mathbf{P}^+$. $\mathbf{P}^+ \vdash \phi(\dot{0})$ because $\mathbf{P}^+ \vdash \neg \dot{m} \dot{<} \dot{0}$. Now

$$\mathbf{P}^+ \vdash \dot{m} \dot{<} \dot{S}x \rightarrow \dot{m} \dot{=} x \vee \dot{m} \dot{<} x.$$

Since

$$\mathbf{P}^+ \vdash \dot{m} \dot{=} x \rightarrow \dot{\mathbf{Sb}}^1 \left(\ulcorner \dot{m} \dot{<} x \urcorner, \text{Num}(\dot{S}x) \right) \dot{=} \ulcorner \dot{m} \dot{<} \dot{S}m \urcorner,$$

and $\mathbf{P}^+ \vdash \dot{T} \left(\ulcorner \dot{m} \dot{<} \dot{S}m \urcorner \right)$ by Proposition 5.3.2, also

$$\mathbf{P}^+ \vdash \dot{m} \dot{=} x \rightarrow \dot{T} \left(\dot{\mathbf{Sb}}^1 \left(\ulcorner \dot{m} \dot{<} x \urcorner, \text{Num}(\dot{S}x) \right) \right).$$

Using the other disjunct, the induction hypothesis is

$$\mathbf{P}^+ \vdash \dot{m} \dot{<} x \rightarrow \dot{T} \left(\dot{\mathbf{Sb}}^1 \left(\ulcorner \dot{m} \dot{<} x \urcorner, \text{Num}(x) \right) \right),$$

so it suffices to establish

$$\mathbf{P}^+ \vdash \dot{T} \left(\dot{\mathbf{Sb}}^1 \left(\ulcorner \dot{m} \dot{<} x \urcorner, \text{Num}(x) \right) \right) \rightarrow \dot{T} \left(\dot{\mathbf{Sb}}^1 \left(\ulcorner \dot{m} \dot{<} x \urcorner, \text{Num}(\dot{S}x) \right) \right).$$

This follows similarly as in the proof above of **(DC2)** — there is a primitive recursive function F mapping a proof of $\dot{m} \dot{<} x$ to a proof of $\dot{m} \dot{<} \dot{S}x$, so with the natural definitions

$$\begin{aligned} \mathbf{P}^+ \vdash \text{Proofof}_\Gamma \left(u, \dot{\mathbf{Sb}}^1 \left(\ulcorner \dot{m} \dot{<} x \urcorner, \text{Num}(x) \right) \right) \\ \rightarrow \text{Proofof}_\Gamma \left(\dot{F}(u), \dot{\mathbf{Sb}}^1 \left(\ulcorner \dot{m} \dot{<} x \urcorner, \text{Num}(\dot{S}x) \right) \right). \end{aligned}$$

As a final case, consider the induction step for the quantifier case — say we want to prove

$$\mathbf{P}^+ \vdash \exists y \theta \rightarrow \dot{T} \left(\dot{\mathbf{Sb}}^1 \left(\ulcorner \exists y \theta \urcorner, \text{Num}(x) \right) \right)$$

under the (induction) hypothesis that

$$P^+ \vdash \theta \rightarrow \dot{T}(\dot{S}b^2(\ulcorner \theta \urcorner, \text{Num}(x), \text{Num}(y))).$$

The crucial fact to be verified is that

$$P^+ \vdash \exists y \dot{T}(\dot{S}b^2(\ulcorner \theta \urcorner, \text{Num}(x), \text{Num}(y))) \rightarrow \dot{T}(\dot{S}b^1(\ulcorner \exists y \theta \urcorner, \text{Num}(x))).$$

This, again, may be accomplished by a careful analysis of the relationship between the two proofs and their codes. $\square$

Collateral Reading. As far as we know, there is no accessible complete proof of the Second Incompleteness Theorem in print. As we noted in [Section 4.1](#), Gödel in his original paper [Da; pp. 36-37] says “For, we have used throughout only the ordinary methods of definition and proof of classical mathematics, as they are formalized in the system P ”. The proof that comes closest to completeness is in the two-volume *Grundlagen der Mathematik* of Hilbert and Bernays, but a modern reader will find this very difficult going. Even the venerable [Kl52] makes no attempt to carry out the transition from (E.7) to (E.8), remarking only (p. 211) “we shall not take the space to carry [it] through in this book”. [Mon; p. 303] aspires to “sketch the high points”, and Smoryński in [Ba77; Chapter D.1, p. 839] asserts that “the proof, which is too long to be included here, proceeds ...”. However, these references do contain much interesting and relevant material. The reader should note, however, that in [Section 6.6](#) below we do give an essentially complete proof of the Second Incompleteness Theorem as it applies to Zermelo-Fraenkel set theory ZF.

Exercises

Exercise 5.3.5 State the main lemmas needed to prove the existence conditions for the function symbol $\dot{C}p$ needed to introduce symbols for functions defined by primitive recursion.

Exercise 5.3.6 Let T be a recursively axiomatizable extension of P and η a sentence which asserts its own unprovability in T (cf Exercise 4.6.41: $T \vdash \eta \leftrightarrow \neg \dot{T}(\ulcorner \eta \urcorner)$). Show that $T \vdash \eta \leftrightarrow \text{Con}_T$.

Exercise 5.3.7 (Löb’s Theorem) Show that for any consistent recursively axiomatizable extension T of P and any sentence ϕ ,

$$T \vdash (\dot{T}(\ulcorner \phi \urcorner) \rightarrow \phi) \implies T \vdash \phi.$$

In particular if, in contrast to the preceding exercise, $T \vdash \eta \leftrightarrow \dot{T}(\ulcorner \eta \urcorner)$, then $T \vdash \eta$. Give a short proof of the Second Incompleteness Theorem from this result.

Hint: If $T \not\vdash \phi$, then $T \cup \{\neg \phi\}$ generates a consistent recursively axiomatizable theory. Alternatively, by the Self-reference Lemma 4.6.27 there exists a sentence ξ such that $T \vdash \xi \leftrightarrow (\dot{T}(\ulcorner \xi \urcorner) \rightarrow \phi)$.

Exercise 5.3.8 Let T be a consistent recursively axiomatizable extension of P . Prove that

- (i) for no formula ϕ , $T \vdash \neg \dot{T}(\ulcorner \phi \urcorner)$;
- (ii) there is a consistent recursively axiomatizable theory $T^* \supseteq T$ and a sentence ϕ such that

$$T^* \vdash \dot{T}^*(\ulcorner \phi \urcorner) \quad \text{but not} \quad T^* \vdash \phi;$$

- (iii) $T \vdash \text{Con}_T \rightarrow \neg \dot{T}(\ulcorner \text{Con}_T \urcorner)$.
- (iv) for Rosser, J. B. Rosser's version of η (Exercise 4.6.42),
 - (a) $T \vdash \text{Con}_T \rightarrow \neg \dot{T}(\ulcorner \eta \urcorner)$;
 - (b) $T \vdash \text{Con}_T \rightarrow \neg \dot{T}(\ulcorner \neg \eta \urcorner)$;
 - (c) $T \vdash \text{Con}_T \rightarrow \eta$;
 - (d) $T \not\vdash \eta \rightarrow \text{Con}_T$.

Note that T^* in (ii) must not be ω -consistent and that (iii) is a formalized version of the Second Incompleteness Theorem. Compare (iv) with Exercise 5.3.6 above.

Exercise 5.3.9 Let T be a consistent recursively axiomatizable extension of P and η a sentence which asserts its own refutability in T : $T \vdash \eta \leftrightarrow \dot{T}(\ulcorner \neg \eta \urcorner)$. Show that

- (i) $T \not\vdash \neg \eta$;
- (ii) if T is ω -consistent, then also $T \not\vdash \eta$;
- (iii) find an example of T such that $T \vdash \eta$.
- (iv) Suppose that $T \vdash \eta \leftrightarrow \dot{T}(\ulcorner \dot{T}(\ulcorner \neg \eta \urcorner) \urcorner)$. What can you conclude about η ?

5.4 Church's Thesis

At the beginning of [Section 4.2](#) we promised a further discussion of the evidence for Church's Thesis: every intuitively effectively calculable function is recursive. We reiterate that this assertion is not susceptible of mathematical proof, since one of its terms is defined intuitively and not mathematically. On the other hand, none of the mathematical results after [Section 4.1](#) depend

on the truth of the Thesis, since they deal exclusively with recursiveness. We use Church's Thesis only when we argue that if $\mathbf{gn}(T)$ is not recursive, then T is also not intuitively decidable or if T is decidable axiomatizable, then it is recursively axiomatizable; without it our undecidability and incompleteness results are true but less interesting.

We focus here on the second sort of evidence for Church's Thesis, that the class of recursive functions and relations is naturally described in many distinct ways. Note that we already have some weak evidence of this sort from the characterizations of recursiveness given in Theorem 4.6.33, Corollary 5.1.18 and Theorem 5.2.23. The characterizations given in this section arise more directly from the intuitive notion of effective calculability.

Recursion equations

Let us look first at the idea of *recursion*. The basis of the method developed in Section 1.2 is that a recursive definition consists of a number of functional equations together with a proof that they have a unique solution. A primitive recursion

$$\begin{aligned} F(m_0, \dots, m_{k-1}, 0) &= G(m_0, \dots, m_{k-1}); \\ F(m_0, \dots, m_{k-1}, n+1) &= H(F(m_0, \dots, m_{k-1}, n), m_0, \dots, m_{k-1}, n); \end{aligned}$$

is the simplest of these in the sense that the uniqueness of the solution is an immediate consequence of the form of the equations. Only slightly more complex are the equations which define the Ackermann function (Exercises 4.4.22–4.4.24), $A(0, n) = n + 1$,

$$A(p+1, 0) = A(p, 1) \quad \text{and} \quad A(p+1, n+1) = A(p, A(p+1, n)),$$

but we know from the cited exercises that A is not primitive recursive so cannot be generated by any sequence of primitive recursions. This fact led mathematicians in the early 1930s to try to characterize the most general notion of recursion.

The first proposal was by Herbrand, who devised a simple formal language for writing equations and considered the functions which arise as unique solutions to a finite set of these equations. A version of this goes as follows.

Definition 5.4.1 An *equational language* L is specified by

- (i) the symbols of L : $\doteq, \dot{0}, \dot{S}$, a denumerable list $\langle v_n : n \in \omega \rangle$ of variables and a finite list $f_0, \dots, f_{l-1}$ of function symbols;
- (ii) a function ν which assigns a finite rank to each function symbol.

The set of terms of L is defined as for a first-order language, and the formulas of L are all equations $t \doteq u$ for any terms t and u . Note that L has

no connectives or quantifiers, hence only atomic formulas. We consider only L -structures of the special form

$$\mathfrak{A} = (\omega, \text{Sc}, 0, F_0, \dots, F_{l-1}),$$

which we abbreviate as simply $(F_0, \dots, F_{l-1})$, and say that such an $\mathfrak{A}$ is a model of a set Γ of equations iff the function symbols which occur in members of Γ are among $f_0, \dots, f_{l-1}$ and $\models_{\mathfrak{A}} t \doteq u$ for all $(t \doteq u) \in \Gamma$ in the usual sense — that is, if f_i is interpreted as F_i , then $t^{\mathfrak{A}}[\alpha] = u^{\mathfrak{A}}[\alpha]$ for all assignments α .

Definition 5.4.2 A function $F : \omega^k \rightarrow \omega$ is **equationally implicit** iff for some equational language L — say with function symbols $f, f_0, \dots, f_{l-1}$ — and some set Γ of L -equations, F is the unique function such that for some functions $F_0, \dots, F_{l-1}$, $(F, F_0, \dots, F_{l-1})$ is a model of Γ . The symbol f is called the **principal function symbol** and $f_0, \dots, f_{l-1}$ are the **auxiliary function symbols**. We say that Γ **implicitly defines** F .

Note that the functions $F_0, \dots, F_{l-1}$ need not be unique, although in many instances they are. Rather than give examples, we proceed directly to

Theorem 5.4.3 *Every recursive function is equationally implicit.*

Proof. We proceed by Rec' -induction, although here we could as easily use Rec -induction. We first note that the initial functions Cs_n^k , Pr_i^k and Sc_i^k are equationally implicit — for example, (Pr_i^k) is the unique model of the singleton set $\{fx_0 \cdots x_{k-1} \doteq x_i\}$.

Next consider the function $+$. Let

$$\Gamma_+ := \left\{ fx\dot{0} \doteq x, \quad fx\dot{\text{S}}y \doteq \dot{\text{S}}fxy \right\}.$$

Clearly $(+)$ is a model of Γ_+ and if (F) is any model, then for all m , $F(m, 0) = m$ and $F(m, n+1) = F(m, n) + 1$, so F is $+$. For multiplication, let

$$\Gamma_{\times} := \left\{ f_0x\dot{0} \doteq x, \quad f_0x\dot{\text{S}}y \doteq \dot{\text{S}}f_0xy, \quad fx\dot{0} = \dot{0}, \quad fx\dot{\text{S}}y = f_0fxy \right\};$$

$\Gamma_{\times}$ has the unique model $(\times, +)$. $\text{K}_{<}$ is similarly implicitly defined by

$$\Gamma_{<} := \left\{ fx\dot{0} \doteq \dot{0}, \quad f\dot{0}\dot{\text{S}}y \doteq \dot{1}, \quad f\dot{\text{S}}x\dot{\text{S}}y \doteq fxy \right\}.$$

To see that the class of equationally implicit functions is closed under composition, suppose that

$$F(m_0, \dots, m_{k-1}) = G(H_0(m_0, \dots, m_{k-1}), \dots, H_{l-1}(m_0, \dots, m_{k-1})),$$

that Γ_G implicitly defines G , and for $i < l$, that Γ_{H_i} implicitly defines H_i . By substitution if necessary we may assume that the principal function symbols are $g, h_0, \dots, h_{l-1}$, respectively, and that the auxiliary function symbols used in these sets are all distinct and do not include any of $f, g, h_0, \dots, h_{l-1}$. Then the finite set

$$\Gamma_G \cup \Gamma_{H_0} \cup \dots \cup \Gamma_{H_{l-1}} \cup \{f x_0 \dots x_{k-1} \doteq g h_0 x_0 \dots x_{k-1} \dots h_{l-1} x_0 \dots x_{k-1}\}$$

implicitly defines F .

Finally, we show that the class of equationally implicit functions is closed under search. Let

$$F(m_0, \dots, m_{k-1}) = \text{least } n [G(m_0, \dots, m_{k-1}, n) = 0],$$

and suppose that Γ_G implicitly defines G with principal symbol g . We assume that the auxiliary function symbols of Γ_G do not include f or h . Then F is implicitly defined by Γ_G together with the following three equations:

$$\begin{aligned} h\dot{0}y_0 \dots y_{k-1}z &\doteq z; \\ h\dot{S}xy_0 \dots y_{k-1}z &\doteq hgy_0 \dots y_{k-1}\dot{S}zy_0 \dots y_{k-1}\dot{S}z; \\ fy_0 \dots y_{k-1} &\doteq hgy_0 \dots y_{k-1}\dot{0}y_0 \dots y_{k-1}\dot{0}. \quad \square \end{aligned}$$

It is tempting now to believe that the converse might hold, that every equationally implicit function is recursive. This is not the case.

Proposition 5.4.4 *There exist non-recursive equationally implicit functions.*

Proof. From [Section 4.6](#) we have several examples of sets A which are recursively enumerable but not recursive; choose any such A and a recursive relation R such that $A = \{m : \exists n R(m, n)\}$. It is clear that K_A is the unique function F such that for all m ,

$$[F(m) = 1 \text{ and } \exists n R(m, n)] \quad \text{or} \quad [F(m) = 0 \text{ and } \forall p \sim R(m, p)].$$

Let G be the function defined by

$$G(m) = \left\langle K_A(m), \begin{cases} \text{least } n R(m, n), & \text{if } m \in A; \\ 0, & \text{otherwise.} \end{cases} \right\rangle$$

Clearly G is also non-recursive, and it is not hard to see that G is the unique function such that for all m and p , $G(m) = \langle (G(m))_0, (G(m))_1 \rangle$ and

$$\begin{aligned} [((G(m))_0 = 1 \text{ and } R(m, (G(m))_1) \text{ and } (\forall q < (G(m))_1) \sim R(m, q))] \text{ or} \\ [(G(m))_0 = 0 \text{ and } \sim R(m, p) \text{ and } (G(m))_1 = 0]. \end{aligned}$$

Hence, there exists a recursive function H such that G is the unique function such that for all m and p , $H(m, p, G(m)) = 0$.

By the preceding theorem there is a finite set Γ_H of equations which implicitly defines H ; clearly the set $\Gamma_H \cup \{hxygx \doteq \dot{0}\}$ implicitly defines G . $\square$

We sketch in an exercise below how to extend this technique to show that there are even non-arithmetical equationally implicit functions. The class of equationally implicit functions is known as the class of **hyperarithmetical functions** and is an interesting and important class — see [Hin; Section IV.4].

Gödel observed that in all of the standard examples of recursively defined functions another element is present; each value of the function depends in some explicit way on a finite sequence of other values. There are at least two ways to make this intuition precise; we develop one here and refer the reader to [Od89; I.2] for the other. We define a notion of formal provability for any equational language L . There are no logical axioms. The rules of inference are the following; for any formulas (equations) ϕ and ψ , terms t and u and variable x ,

- (i) from ϕ infer $\phi_x(t)$;
- (ii) from ϕ and $t \doteq u$ infer ψ , where ψ is ϕ with one occurrence of t replaced by u .

For any set Γ of L -equations, the set $\text{Thm}_L(\Gamma)$ of **theorems from** Γ is the smallest set of L -equations such that $\Gamma \subseteq \text{Thm}_L(\Gamma)$ and $\text{Thm}_L(\Gamma)$ is closed under the rules of inference. We write $\Gamma \vdash \phi$ iff $\phi \in \text{Thm}_L(\Gamma)$. As for first-order languages, there is a corresponding notion of proof from Γ such that for any ϕ , ϕ is a Γ -theorem iff there is a Γ -proof of ϕ .

Definition 5.4.5 A partial function $F : \omega^k \rightarrow \omega$ is **equationally derivable** iff for some equational language L — say with function symbols $f, f_0, \dots, f_{l-1}$ — and some finite set Γ of L -equations, for all $m_0, \dots, m_{k-1}$ and n ,

$$F(m_0, \dots, m_{k-1}) \simeq n \iff \Gamma \vdash f\dot{m}_0 \dots \dot{m}_{k-1} \doteq \dot{n}.$$

Theorem 5.4.6 A partial function is partial recursive iff it is equationally derivable.

Proof. (Sketch) For the implication from right to left, we proceed by Rec' -induction in parallel with the preceding theorem; indeed in each case the same set Γ of equations suffices. It is relatively clear that at most “correct” equations are derivable (the implication $\Leftarrow$ in the definition) and we concentrate on showing that *all* correct equations are derivable (the implication $\Rightarrow$ of the definition). For the initial functions this is evident. As before, let

$$\Gamma_+ := \left\{ fx\dot{0} \doteq x, \quad fx\dot{S}y \doteq \dot{S}fxy \right\}.$$

Then for all m , $\Gamma_+ \vdash f\dot{m}\dot{0} \doteq m$ from the first equation of Γ and rule (i). If $\Gamma_+ \vdash f\dot{m}\dot{n} \doteq \overbrace{m+n}^{\cdot}$, then by the second equation of Γ and rule (i), $\Gamma_+ \vdash f\dot{m}\dot{n}+1 \doteq \dot{S}f\dot{m}\dot{n}$, so by rule (ii), $\Gamma_+ \vdash f\dot{m}\dot{n}+1 \doteq \dot{S}\overbrace{m+n}^{\cdot}$ as required. The arguments for all of the other cases are similar.

The proof that every equationally derivable function is recursive follows a by now familiar pattern. Let L and Γ be the language and set of equations which witness that F is equationally derivable. We assign Gödel numbers to the terms and equations of L . Since the rules are simple syntactical functions, they correspond to recursive functions of the Gödel numbers. Hence the set $\mathbf{gn}(\mathbf{Thm}_L(\Gamma))$ is generated by a recursive induction system and is therefore recursively enumerable. The definition of equational derivability immediately implies that the graph of F is many-one reducible to $\mathbf{gn}(\mathbf{Thm}_L(\Gamma))$, and we conclude that $\mathbf{Graph}(F)$ is semi-recursive, hence F is recursive. $\square$

Abstract machines

We turn now to a quite different set of intuitions based on a direct analysis of the notion of effective calculability. The underlying idea, first developed by Alan Turing in the 1930s, is that since an algorithm which guides the calculation of a function F should determine completely every step of the passage from argument m to value $F(m)$, the process should be analyzable into a series of completely elementary steps. He envisioned a totally obedient but unimaginative clerk working with a pencil in a workspace consisting of a linear tape divided into discrete squares, each blank (0) or containing a single mark (1). The clerk is capable only of carrying out such commands as ‘make a mark and move one square to the right’ or ‘erase a mark and move one square to the left’. The instructions for the clerk should specify precisely which action should be taken based on the ‘current configuration’. Since there are potentially infinitely many arrangements of 0s and 1s on the tape, it would not be possible to specify in a finite way a distinct action for each such pattern. Instead, he proposed that the clerk be capable of recognizing a fixed finite number of abstract ‘states’ and the next action be determined by a combination of the current state together with the contents of the single square of the tape then being considered. In addition to moving and writing, our clerk might at each step undergo a change of state. The clerk would be said to compute the function F if whenever he is presented with a tape containing some representation of the input $m_0, \dots, m_{k-1}$, following his instructions would lead him to construct in finitely many steps a tape containing the value $F(m_0, \dots, m_{k-1})$.

We shall sketch below a proof that the (partial) functions computable in this manner are exactly the (partial) recursive ones. Turing called his

construct a *machine* because once the clerk is given his instructions, he proceeds mechanically without further intelligent intervention. Of course, at the time the only physical computing machines in existence were calculators which performed simple arithmetic operations with gears and levers; Turing's conception of a machine which could perform any algorithmic calculation was a crucial step in the development of modern digital computers in the 1940s.

In 1963 Shepherdson and Sturgis introduced a variant of the Turing machine, called a *register machine*, which permits squares of the tape, now called registers, to contain any natural number and has a somewhat more complex instruction set. Although the basic intuition is very similar to Turing's, register machines have the practical advantage that the proof that all (partial) recursive functions are computable by register machines is much simpler and more natural, and the technical advantage that the workspace — the number of registers used — can be specified in advance of the computation.

Of course, both Turing and register machines are easily recognizable by a reader of the 21st century as abstract notions of a digital computer. Magnetic or optical input and output media are very similar to Turing's tape or the registers of a register machine. A computer is naturally seen as being in one of a finite number of states corresponding to the contents of its internal memory, and a (machine language) program can be thought of as a transition function from one pair (state, scanned character) to another, together with writing and erasing. Thus, a Turing machine may be thought of as a very elemental computer equipped with a very simplistic programming language. In the case of a register machine, the state of the machine is the contents of the registers together with the program and the number of the next instruction to be executed (see the precise definition below).

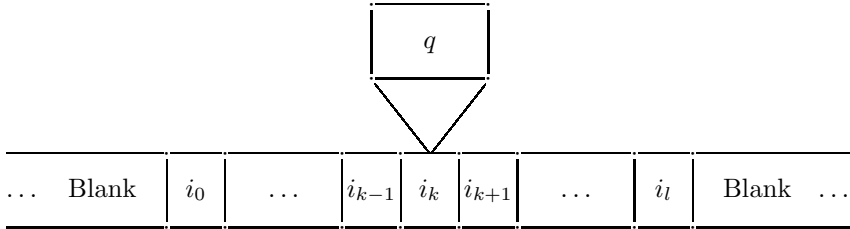
There is one way in which both of these models are unrealistic: although the sets of states, commands, tape-squares or registers and the number of steps in the computation are all required to be finite, there is no *a priori* bound on any of these quantities. Indeed, it is not difficult to describe a finite computation of either sort of machine which at, say, 10^{12} steps per second will take longer than the current age of the universe. Thus we should describe these abstract machines as *idealized* computers, freed from physical limitations on time and space.

The most important aspect of the Turing and Shepherdson-Sturgis analyses is that they can easily be made precise enough to be converted into a real mathematical definition. Thus, we shall define a Turing machine to be a mathematical model of a device consisting of four components:

- (i) a control unit which can be in any one of a finite number of *states* $q_0, \dots, q_l$;
- (ii) a finite stored program;

- (iii) a two-way infinite tape divided into discrete squares, each of which can contain a 0 (blank) or a 1 (mark);
- (iv) a read-write-erase head.

We may picture the machine thus:



An atomic step of the machine consists in reading the contents of a single square of the tape and on the basis of the result and the current state (a) marking or erasing the square, (b) moving one square to the right or left and (c) changing to a (possibly) new state.

Definition 5.4.7 A **Turing machine** is a triple $\mathcal{M} = (M, Q, q_I)$ such that

- (i) Q is a finite set and $q_I \in Q$;
- (ii) M is a finite partial function

$$M : Q \times \{0, 1\} \rightarrow \{\text{Write, Erase}\} \times \{\text{Left, Right}\} \times Q.$$

Informally, we think of $\mathcal{M}$ being started in the initial state q_I with some specification of tape contents and which square is being scanned. $\mathcal{M}$ runs as long as the current state-symbol pair is in the domain of M ; if this fails it stops. More precisely,

Definition 5.4.8 A **configuration** of a Turing machine $\mathcal{M} = (M, Q, q_I)$ is a finite sequence

$$\sigma = (i_0, \dots, i_{k-1}, q, i_k, \dots, i_l)$$

such that each i_j is 0 or 1 and $q \in Q$. With $\mathcal{M}$ is associated a partial **transition function** M^* from configurations to configurations generated by regarding the symbol i_k as the scanned symbol. A **computation** of $\mathcal{M}$ is a (finite or infinite) sequence $(\sigma_0, \sigma_1, \dots)$ such that for all i , if σ_i is in the domain of M^* , then σ_{i+1} exists and $\sigma_{i+1} = M^*(\sigma_i)$.

For example, if $M(q, 1) = (\text{Erase, Right}, q')$, then

$$M^*(i_0, \dots, i_{k-1}, q, 1, i_{k+1}, \dots, i_l) = (i_0, \dots, i_{k-1}, 0, q', i_{k+1}, \dots, i_l).$$

Note that for each configuration σ , there is a unique computation with $\sigma_0 = \sigma$. A computation **halts** iff it is finite.

Definition 5.4.9 A Turing machine $\mathcal{M}$ **computes** a k -ary partial function F iff for each $m_0, \dots, m_{k-1}$, the computation of $\mathcal{M}$ which begins with the configuration

$$\text{input}(m_0, \dots, m_{k-1}) = (q_I \underbrace{11 \dots 1}_{m_0+1} 0 \underbrace{11 \dots 1}_{m_1+1} 0 \dots 0 \underbrace{11 \dots 1}_{m_{k-1}+1})$$

is finite and ends with the configuration

$$\text{output}(F(m_0, \dots, m_{k-1})) = (q \quad \underbrace{11 \dots 1}_{F(m_0, \dots, m_{k-1})+1})$$

iff $(m_0, \dots, m_{k-1}) \in \text{Dom}(F)$; otherwise it does not halt. A partial function is **Turing computable** iff it is computed by some Turing machine.

Theorem 5.4.10 A partial function is Turing computable iff it is partial recursive.

Proof. (Sketch) To show that all partial recursive functions are Turing computable requires a very tedious explicit construction of machines corresponding to the initial functions, composition and search. We shall spare the reader this unpleasant exercise in programming which is carried out in more or less detail in the books [Od], [Da] and [He]. The corresponding result for register machines below is much simpler and will be done in greater detail.

The converse inclusion is proved by a slight modification of our earlier proofs of this type. Suppose that the Turing machine $\mathcal{M}$ computes F . First we assign Gödel numbers to the states of $\mathcal{M}$ and thus to the configurations and finite sequences of configurations. It is not hard to see that the set

$$\text{Comp}_{\mathcal{M}} = \{s : s \text{ codes a finite computation of } \mathcal{M}\}$$

is primitive recursive. But

$$\begin{aligned} F(m_0, \dots, m_{k-1}) \simeq n &\iff \exists s [\text{Comp}_{\mathcal{M}}(s) \text{ and} \\ &(s)_0 = \text{gn}(\text{input}(m_0, \dots, m_{k-1})) \text{ and } (s)_{\text{lg}(s)-1} = \text{gn}(\text{output}(n)). \end{aligned}$$

Thus $\text{Graph}(F)$ is semi-recursive, so F is partial recursive. $\square$

We turn now to **register machines**. Such a machine has a (potentially) infinite set $R_0, R_1, \dots$ of **registers**. At any moment during the computation, each R_p holds the natural number r_p , and for all but finitely many p , $r_p = 0$. A computation is directed by a **program** $\pi = (I_0, \dots, I_l)$ consisting of a list of **instructions**, each of one of the following four types, with $j \leq l$:

In other words, if the register machine is started with $m_0, \dots, m_{k-1}$ in the first k registers, then $F_{\pi,k}(m_0, \dots, m_{k-1})$ is defined iff the computation directed by π terminates and the value is the content of the first register R_0 upon termination.

Definition 5.4.11 A k -ary partial function is **register computable** iff for some program π , $F = F_{\pi,k}$.

Theorem 5.4.12 A partial function is register computable iff it is partial recursive.

Proof. We argue first by Rec' -induction that every partial recursive function is register computable by showing that the initial functions, $+$, $\times$ and $K_<$ are register computable and that the class of register computable functions is closed under composition and search. For each of the initial functions F , we have the following simple explicit programs π_F :

$$\begin{aligned}\pi_{\text{Cs}_n^k} &:= (r_0 := 0, \underbrace{r_0 := r_0 + 1, \dots, r_0 := r_0 + 1}_n); \\ \pi_{\text{Pr}_i^k} &:= (r_0 := r_i); \\ \pi_{\text{Sc}_i^k} &:= (r_0 := r_i, r_0 := r_0 + 1).\end{aligned}$$

For $+$, we have the program

(if $r_1 = r_2$, go to I_4 ; $r_0 := r_0 + 1$; $r_2 := r_2 + 1$; if $r_1 = r_1$, go to I_0 ; $r_3 := 0$).

The idea is that r_2 is a counter which guarantees that r_0 is incremented r_1 many times. The last instruction I_4 is a dummy instruction to end the computation. We leave as an exercise the construction of similar programs for $\times$ and $K_<$. Suppose, now, that G , H_0 and H_1 are register computable functions and F is the composition

$$F(m_0, m_1) \simeq G(H_0(m_0, m_1), H_1(m_0, m_1));$$

it will be obvious how to generalize from this special case to the general one. Fix programs π_G , π_{H_0} and π_{H_1} which compute these functions, and let $u \geq 1$ be the smallest number such that none of these programs accesses any register R_p with $p \geq u$. Then the following program is easily seen to compute F :

$r_u := r_0, r_{u+1} := r_1$	save m_0 and m_1 in the first unused registers
π_{H_1}	compute $r_0 := H_1(m_0, m_1)$
$r_{u+2} := r_0$	store this value in the first unused register
$r_0 := r_u, r_1 := r_{u+1}$	restore m_0 and m_1 to the first two registers

$r_2 := 0, \dots, r_{u-1} := 0$	clear workspace registers
π_{H_0}	compute $r_0 := H_0(m_0, m_1)$
$r_1 := r_{u+2}$	retrieve computed value $r_1 := H_1(m_0, m_1)$
$r_2 := 0, \dots, r_{u-1} := 0$	clear workspace registers
π_G	compute final value.

Now suppose that G is register computable by the program π_G , and

$$F(m_0, m_1) \simeq \text{least } n [G(m_0, m_1, n) \simeq 0].$$

Let $u \geq 3$ be the smallest number such that π_G accesses no register R_p with $p \geq u$. Then the following program computes F by successively computing $G(m_0, m_1, 0)$, $G(m_0, m_1, 1) \dots$ until a 0 value is found:

$r_u := r_0, r_{u+1} := r_1$	save m_0 and m_1 in the first unused registers
$I_1 \pi_G$	compute $r_0 := G(m_0, m_1, n)$ ($n = r_2$ is initially 0)
if $r_0 = r_{u+3}$, go to I_s	finish if $r_0 = 0$ (r_{u+3} is always 0)
$r_{u+2} := r_{u+2} + 1$	increment n
$r_i := r_{u+i}, i = 0, 1, 2$	restore m_0, m_1, n to the first three registers
$r_3 := 0, \dots, r_{u-1} := 0$	clear workspace registers
if $r_0 = r_0$, go to I_1	compute next value of G
$I_s r_0 := r_{u+2}$	set r_0 to the final value of n .

To show that all register computable functions are partial recursive, we proceed as follows. A configuration $\sigma = (j, (r_0, \dots, r_{u-1}))$ may be coded by the natural number (its Gödel number or index) $\text{gn}(\sigma) = \langle j, \langle r_0, \dots, r_{u-1} \rangle \rangle$. Then a π computation may also be coded as a natural number

$$\text{gn}(\sigma_0, \dots, \sigma_t) = \langle \text{gn}(\sigma_0), \dots, \text{gn}(\sigma_t) \rangle.$$

Let Comp_π denote the set of all such codes of π -computations. Then much as in the proof of Theorem 4.4.16, we can show that for each register machine program π , Comp_π is primitive recursive: if $\pi = (I_0, \dots, I_l)$ and $u = u(\pi)$, then

$$\begin{aligned} \text{Comp}_\pi(s) \iff & \text{Sq}(s) \wedge (s_0)_0 = 0 \wedge (\forall i < \text{lg}(s) - 1) [((s_i)_0 \leq l \\ & \wedge \text{lg}(((s)_i)_1) = u \wedge \pi\text{-Succ}((s)_i, (s)_{i+1})] \wedge \pi\text{-Term}((s)_{\text{lg}(s)-1}), \end{aligned}$$

where $\pi\text{-Succ}$ holds between the codes of two configurations σ and τ iff τ is the π -successor of σ and $\pi\text{-Term}$ holds exactly of codes of terminal π -configurations. These are easily verified to be primitive recursive relations.

Now we have

$$F_{\pi,k}(m_0, \dots, m_{k-1}) \simeq n \iff \exists s \left[\text{Comp}_{\pi}(s) \text{ and } (s)_0 = \langle 0, \langle m_0, \dots, m_{k-1}, 0, \dots, 0 \rangle \rangle \text{ and } (((s)_{\lg(s)-1})_1)_0 = n \right].$$

Thus the graph of each register-computable function $F_{\pi,k}$ is semi-recursive, so $F_{\pi,k}$ is partial recursive by Theorem 4.3.16.

Collateral Reading. There are many discussions of Church's Thesis or the equivalent Turing's Thesis in the literature. Historically the first is Church's article [Dav; p.100] in which he first proposes the thesis. Some of the most thorough are [Kl53; Section 70], Soare's [Gri; Section 1] and [Od89; Section I.8]. [Cut; Chapter 3] has an approachable discussion of several alternative characterizations of the class of recursive functions.

Exercises

Exercise 5.4.13 Show directly that the Ackermann function is equationally derivable.

Exercise 5.4.14 Carry out (some of) the details of the proof of the second part of Theorem 5.4.6.

Exercise 5.4.15 Use the technique of Proposition 5.4.4 to prove that all arithmetical functions are equationally implicit.

Exercise 5.4.16 Show that the characteristic function of the set ${}_{\infty}U^1$ from Exercise 5.1.34 (which is not arithmetical) is equationally implicit.

Exercise 5.4.17 Construct programs to show that $\times$ and $K_{<}$ are register computable.

5.5 Applications to other languages and theories

Although the incompleteness and undecidability results seem to depend heavily on the language L_{Ω} of arithmetic and the theories $\mathbf{Q}$ and $\mathbf{P}$, we shall see that they can be transferred to many other contexts. The principal vehicle for this is the notion of an *interpretation* of one theory in another developed in Section 2.6. In the following we consider only finite languages and assume that each language under discussion has been Gödel-numbered. We recall first some definitions.

Definition 5.5.1 For any languages L_1 and L_2 and any L_1 -theory T_1 and L_2 -theory T_2 ,

- (i) T_1 is **interpretable** in T_2 iff there is a finite extension by definitions (Definition 2.6.12) T_2^+ of T_2 such that $T_1 \subseteq T_2^+$;
- (ii) T_1 is **faithfully interpretable** in T_2 iff additionally $T_1 = T_2^+ \cap \text{Fm}_{L_1}$.

Intuitively, T_1 is interpretable in T_2 iff there are L_2 -definitions of the primitive concepts (non-logical symbols) of L_1 that are not in L_2 such that all of their T_1 -properties are provable in T_2 . The interpretation is faithful if no additional properties are provable. These definitions provide a construction which from any model $\mathfrak{A}_2$ of T_2 leads to a model $\mathfrak{A}_1$ (with the same universe) of T_1 by first adjoining the new defined objects corresponding to symbols in L_1 but not L_2 and then discarding the superfluous objects corresponding to symbols in L_2 but not L_1 (Proposition 2.6.36): $\mathfrak{A}_1 := \mathfrak{A}_2^+ \upharpoonright L_1$.

By Examples 2.6.35, $\mathbf{Q}$, $\mathbf{P}$ and indeed the full $\text{Th}(\Omega)$ are interpretable in each of $\text{Th}(\omega, +, \times)$ and $\text{Th}(\omega, \text{Sc}, \times)$, since it is possible to define 0 , $<$ and Sc in terms of $+$ and $\times$; and 0 , $<$ and $+$ in terms of Sc and $\times$. By Example 2.6.41, $\text{Th}(\mathbb{Z}, \text{Sc}, +, 0)$ is faithfully interpretable in $\text{Th}(\omega, \text{Sc}, +, 0)$.

To state concisely the effects of interpretations on undecidability, we first introduce some further terminology. Recall that a theory T is decidable iff $\text{gn}(T)$ is a recursive set of natural numbers; otherwise T is undecidable.

Definition 5.5.2 For any language L and any L -theory T ,

- (i) T is **essentially undecidable** iff every consistent L -theory $T' \supseteq T$ is undecidable;
- (ii) T is **contagiously undecidable** iff every L -theory T' compatible with T (Definition 4.1.16) is undecidable;
- (iii) T is **contagiously incomplete** iff every recursively axiomatizable L -theory T' compatible with T is incomplete.

The First Incompleteness Theorem 4.6.16 asserts that $\mathbf{Q}$ is contagiously and therefore essentially undecidable and contagiously incomplete. For complete theories, essential undecidability is the same as simple undecidability.

Lemma 5.5.3 *For any finite language L and any L -theory T , if T is essentially undecidable and finitely axiomatizable, then T is contagiously undecidable and contagiously incomplete.*

Proof. This is just a general form of the last part of the proof of the First Incompleteness Theorem 4.6.16. Suppose that Γ is a finite set of axioms for the essentially undecidable theory T and let $\bigwedge \Gamma$ be the conjunction of the members of Γ . Let U be a theory compatible with T and T^+ the (consistent)

theory generated by $T \cup U$ or equivalently $\Gamma \cup U$. T^+ is an extension of T so is undecidable by hypothesis. For any L -sentence θ ,

$$\theta \in T^+ \iff (\bigwedge \Gamma \rightarrow \theta) \in U.$$

The function $\mathbf{gn}(\theta) \mapsto \mathbf{gn}(\bigwedge \Gamma \rightarrow \theta)$ is obviously recursive, so it follows that also U is undecidable. $\square$

Proposition 5.5.4 *For any theories T_1 and T_2 ,*

- (i) *if T_1 is faithfully interpretable in T_2 , and T_1 is undecidable, then also T_2 is undecidable;*
- (ii) *if T_1 is interpretable in T_2 , and T_1 is essentially undecidable, then also T_2 is essentially undecidable;*
- (iii) *if in addition T_1 is finitely axiomatizable, then T_2 has a finitely axiomatizable subtheory which is essentially and hence contagiously undecidable.*

Proof. Parts (i) and (ii) are simply the translations of Corollary 2.6.40 and Proposition 2.6.38 from intuitive decidability to recursiveness. For example, under the conditions of (i), for any L_1 -formula ϕ ,

$$\phi \in T_1 \iff \phi \in T_2^+ \iff \phi^- \in T_2.$$

Analysis of the proof of Proposition 2.6.19 leads easily to the conclusion that the function $\mathbf{gn}(\phi) \mapsto \mathbf{gn}(\phi^-)$ is recursive. Hence if $\mathbf{gn}(T_2)$ were a recursive set, then so would be $\mathbf{gn}(T_1)$.

Suppose now that T_1 is finitely axiomatizable, say $T_1 = \mathbf{Th}(\Gamma)$ for a finite set of sentences Γ . Let Δ denote the (finite) set of definitions used for the interpretation of T_1 in T_2 : $T_1 \subseteq T_2^+ = \mathbf{Th}(T_2 \cup \Delta)$. By compactness, for each $\phi \in \Gamma$ there exists a sentence $\phi^* \in T_2$ such that $\Delta \models \phi^* \rightarrow \phi$. Hence if $U := \mathbf{Th}(\{\phi^* : \phi \in \Gamma\})$, then $T_1 \subseteq U^+ := \mathbf{Th}(U \cup \Delta)$ — that is, T_1 is interpretable in U . Finally, U is contagiously undecidable by the lemma. $\square$

Now applying these results to the example mentioned above,

Theorem 5.5.5 *Both $\mathbf{Th}(\omega, +, \times)$ and $\mathbf{Th}(\omega, \mathbf{Sc}, \times)$ are undecidable, are not recursively axiomatizable and have finitely axiomatizable contagiously undecidable subtheories.*

Proof. $\mathbf{Q}$ is a finitely axiomatizable and essentially undecidable theory interpretable in each of these theories. $\square$

Corollary 5.5.6 *For any language L with at least two binary function symbols or one unary and one binary function symbol, the set of L -logical validities is undecidable.*

Proof. The set of L -logical validities is an L -theory compatible with any consistent L -theory. $\square$

In Section 2.6, we also developed the notion of relative interpretation, which can be applied here to expand greatly the scope of undecidability results. Recall that we defined there (2.6.43) the relativization $T^{(M)}$ of a theory T to a new unary relation symbol M as the theory generated by the sentences $\phi^{(M)}$ for $\phi \in T$. Then T_1 is called **relatively interpretable** in T_2 iff $T_1^{(M)}$ is interpretable in T_2 . This is best thought of in terms of models; as discussed following Definition 2.6.43, the interpretation provides a canonical construction which leads from any model $\mathfrak{A}_2$ of T_2 to a unique model $(\mathfrak{A}_1; M)$ of $T_1^{(M)}$, and thence to a unique substructure $\mathfrak{M}$, which is a model of T_1 . The advantage is that the definitions of the L_1 -symbols need only satisfy the requirements of T_1 on this substructure. For example, we have

Theorem 5.5.7 *Q , P and $\text{Th}(\Omega)$ are relatively interpretable in $T := \text{Th}(\mathbb{Z}, +, \times)$.*

Proof. The required extension T^+ of T is generated by the definitions of $\dot{0}$ and $\dot{S}$ used in Example 2.6.13 together with the two definitions

$$\begin{aligned} Mx &\leftrightarrow \exists y \exists z \exists u \exists v [x \dot{=} (y \dot{\times} y) \dot{+} (z \dot{\times} z) \dot{+} (u \dot{\times} u) \dot{+} (v \dot{\times} v)] \\ x < y &\leftrightarrow \exists z (Mz \wedge \neg z \dot{+} z \dot{=} z \wedge y \dot{=} x \dot{+} z). \end{aligned}$$

The success of the interpretation depends on the classical theorem of number theory, due to Lagrange, that every natural number is the sum of four squares (see [Mon; Theorem 16.6]). Then for any L_Ω -sentence ϕ , using Corollary 2.6.20 we have:

$$\phi \in \text{Th}(\Omega) \implies \models_{(\mathbb{Z}, +, \times)^+} \phi^{(M)} \implies \phi^{(M)} \in \text{Th}((\mathbb{Z}, +, \times)^+) = T^+. \quad \square$$

Proposition 5.5.8 *For any theories T_1 and T_2 ,*

- (i) *if T_1 is relatively interpretable in T_2 , and T_1 is essentially undecidable, then also T_2 is essentially undecidable;*
- (ii) *if in addition T_1 is finitely axiomatizable, then T_2 has a finitely axiomatizable subtheory which is essentially and hence contagiously undecidable.*

Proof. Part (i) is the recursive translation of Proposition 2.6.49. For (ii), by Corollary 2.6.45, if T_1 is finitely axiomatizable, so is $T_1^{(M)}$, so the result follows from Proposition 5.5.4. $\square$

Corollary 5.5.9 $\text{Th}(\mathbb{Z}, +, \times)$ is undecidable, is not recursively axiomatizable and has a finitely axiomatizable contagiously undecidable subtheory. $\square$

Corollary 5.5.10 The algebraic theories of rings, commutative rings, ordered rings, rings with unit and integral domains are all undecidable.

Proof. These are all theories compatible with $\text{Th}(\mathbb{Z}, +, \times)$. $\square$

To extend these results to theories of fields we have

Theorem 5.5.11 $\mathbb{Q}$ is relatively interpretable in $\text{Th}(\mathbb{Q}, +, \times)$, where $\mathbb{Q}$ is the set of rational numbers.

Proof. (Sketch) As above it suffices to show that the set of natural numbers is definable over the rational numbers using $+$ and $\times$. This is accomplished as follows (using obvious abbreviations). Let

ψ be the formula $\exists u \exists v \exists w [x^2 yz + zw^2 + 2 \doteq u^2 + yv^2]$;

θ be the formula $\psi_x(\dot{0}) \wedge \forall x [\psi \rightarrow \psi_x(\dot{S}x)]$; and

ϕ be the formula $\forall y \forall z (\theta \rightarrow \psi)$.

Then $Mx \leftrightarrow \phi$ is the required definition.

To unravel this slightly, let

$$A_{b,c} = \left\{ a \in \mathbb{Q} : \models_{(\mathbb{Q}, +, \times)} \psi[a, b, c] \right\},$$

and call $A_{b,c}$ inductive iff it contains 0 and is closed under successor. Then the theorem to be proved is that

$$\omega = \bigcap \{ A_{b,c} : b, c \in \mathbb{Q} \text{ and } A_{b,c} \text{ is inductive} \}. \quad \square$$

Corollary 5.5.12 The theories of fields, fields of characteristic 0 and ordered fields are undecidable.

In the next chapter we shall develop the set theory ZF, **Zermelo-Fraenkel set theory**. Much of the interest in axiomatic theories of sets stems from the fact that most of the concepts of mathematics can be defined in terms of sets, so that set theory provides a uniform framework for the axiomatic development of mathematics. The aims of Hilbert's program would have been best realized by a complete recursively axiomatizable theory T of sets together with a finitary consistency proof for T . Of course, the Gödel incompleteness theorems show that neither of these goals is attainable. We shall prove in [Section 6.2](#)

Theorem 5.5.13 *$\mathbf{Q}$ and $\mathbf{P}$ are relatively interpretable in $\mathbf{ZF}$. Hence $\mathbf{ZF}$ is contagiously undecidable and has a finitely axiomatizable subtheory T which is also contagiously undecidable. Any recursively axiomatizable theory compatible with T is incomplete.*

Corollary 5.5.14 *For any language L with at least one binary relation symbol, the set of L -logical validities is undecidable. $\square$*

Remark 5.5.15 Since we know that in the language $L_=_$ with no non-logical symbols the set of logical validities is decidable, it is natural to ask if this property is preserved in any more complex languages. It turns out that languages with any number of unary relation symbols and at most one unary function symbol have decidable sets of logical validities, while the presence of any binary relation or function symbol or two or more unary function symbols causes the set of logical validities to be undecidable [Mon; 16.51–16.58].

A version of the Second Incompleteness Theorem holds for any consistent recursively axiomatizable theory T such that $\mathbf{P}$ is relatively interpretable in T . We sketch one version of this here and in [Section 6.6](#) develop the results much more thoroughly for the case of $\mathbf{ZF}$. Fix a language L , a consistent recursively axiomatizable L -theory T and a finite extension by definitions T^+ such that $\mathbf{P}^{(\mathbf{M})} \subseteq T^+$. Once we have assigned Gödel numbers to L -formulas of the language of T , $\mathbf{gn}(T)$ is a recursively enumerable set and there exists an L_Ω -formula $\dot{T}(y)$ satisfying (1) of [Section 5.3](#). Let $\mathbf{Con}_T$ be the L_Ω -sentence $\neg \dot{T}(\ulcorner \neg \forall x (x \doteq x) \urcorner)$. In the same way as in [Section 5.3](#), $\mathbf{Con}_T$ expresses the consistency of T . Let

$$U := \left\{ \phi \in \mathbf{Sent}_{L_\Omega} : T^+ \vdash \phi^{(\mathbf{M})} \right\}.$$

As in the proof of Lemma 2.6.48, U is a consistent L_Ω -theory, and by hypothesis $\mathbf{P} \subseteq U$. Then

Theorem 5.5.16 *With notation and hypotheses as above,*

- (i) $U \not\vdash \mathbf{Con}_U$;
- (ii) $\mathbf{P} \vdash \mathbf{Con}_{T^+} \rightarrow \mathbf{Con}_U$;
- (iii) $\mathbf{P} \vdash \mathbf{Con}_T \rightarrow \mathbf{Con}_{T^+}$;
- (iv) $T^+ \not\vdash \mathbf{Con}_T^{(\mathbf{M})}$,
- (v) $T \not\vdash \mathbf{Con}_T^{(\mathbf{M})^-}$.

Proof. (i) is an application of the Second Incompleteness Theorem. (ii) is a formal version of Proposition 2.6.37. (iii) is a formal version of Corollary

2.6.15. (iv) is immediate from (i)–(iii) and the definition of T , and (v) follows by Proposition 2.6.19. $\square$

In Section 6.6 we shall sketch the development of first-order logic in ZF . All of the main notions of Chapters 1–3 will have set-theoretic counterparts described in the language $L_{\in}$ of set theory. In particular, if $T = Th(T)$ is a recursively axiomatizable $L_{\in}$ -theory extending ZF , we can represent these by set-theoretic objects $\dot{T}$ and $\dot{T}$ and show

$$ZF \vdash \exists a \text{ ModelOf}(a, \dot{T}) \leftrightarrow \neg \exists y \text{ ProofOf}(\dot{T}, \dot{F}, y),$$

where $\dot{F}$ is a set-theoretic logically false sentence. Then we may take either of these equivalent $L_{\in}$ -sentences as $\dot{\text{Con}}(\dot{T})$ and show that

$$T \text{ is consistent} \implies T \not\vdash \dot{\text{Con}}(\dot{T}).$$

This result for ZF itself is considerably more important than the corresponding result for P for two reasons. First, since, as we have mentioned several times and will pursue in greater depth in Sections 6.2 and 6.6, most of mathematics can be embedded in ZF , the consistency of this theory is more crucial to our belief in the consistency of mathematics as a whole. If in the worst scenario it were at some future time discovered that ZF is actually inconsistent, then it would be necessary to formulate an alternative set theory (or other foundational theory) and verify step by step that each argument of mathematics can still be carried out in this theory. Note that an inconsistency in ZF , as in any axiomatic theory, would be an explicit formal proof of some contradiction from the axioms. From such a proof we could isolate the offending axioms and avoid them.

The second reason is that the assertion that $\dot{\text{Con}}(ZF)$ is not a theorem of ZF is, by the same argument, essentially an assertion that there can be no mathematical proof of the consistency of ZF , since any such proof should itself be embeddable in ZF . In the case of a recursively axiomatizable extension T of P , we might dismiss the significance by saying that since T is concerned with arithmetic and number theory, it is not surprising that a statement such as Con_T is not a theorem of T . Even though technically it is phrased in terms of numbers, Con_T is really not “about” numbers, so perhaps T should not be expected to prove it. For ZF this argument is not available, since ZF is “about” sets and hence about all of mathematics.

We conclude this section with a result of a somewhat different flavor which will play a crucial role in Section 7.6. For any language L , let

$$\begin{aligned} \text{Val}_L &:= \{ \chi : \chi \in \text{Sent}_L \quad \text{and} \quad \text{for all } L\text{-structures } \mathfrak{A}, \models_{\mathfrak{A}} \chi \}, \\ \text{Sat}_L &:= \{ \chi : \chi \in \text{Sent}_L \quad \text{and} \quad \text{for some } L\text{-structure } \mathfrak{A}, \models_{\mathfrak{A}} \chi \}, \end{aligned}$$

the sets of the logically valid and satisfiable L -sentences. By the Enumeration Theorem and Corollary 5.5.14, we know that as long as L contains at least one binary relation symbol,

Val_L is r.e. but not recursive, hence not co-r.e.

Since $\chi \in \text{Val}_L \iff \neg \chi \notin \text{Sat}_L$, it follows immediately that

Sat_L is co-r.e. but not recursive, hence not r.e.

This situation changes when we restrict attention to finite structures. Let

$$\begin{aligned} \text{FinVal}_L &:= \{ \chi : \chi \in \text{Sent}_L \text{ and for all finite } L\text{-structures } \mathfrak{A}, \models_{\mathfrak{A}} \chi \}, \\ \text{FinSat}_L &:= \{ \chi : \chi \in \text{Sent}_L \text{ and for some finite } L\text{-structure } \mathfrak{A}, \models_{\mathfrak{A}} \chi \}. \end{aligned}$$

Proposition 5.5.17 *For any finite language L , FinSat_L is r.e. and FinVal_L is co-r.e.*

Proof. We give here an outline and leave the details to an exercise. For simplicity, suppose that L has only one binary relation symbol p . Any finite L -structure is isomorphic to one of the form $\mathfrak{A} = (\{0, 1, \dots, n\}, P)$. Any such structure can be coded by a single natural number a in many ways — for example, by representing P by a sequence of length $n + 1$ of 0s and 1s. It is then not hard to see that the relation

$$\begin{aligned} R(a, m) : & \iff a \text{ codes a finite structure } \mathfrak{A} \text{ and} \\ & m \text{ is the Gödel number of a sentence } \chi \text{ such that } \models_{\mathfrak{A}} \chi \end{aligned}$$

is (primitive) recursive. Then since $\chi \in \text{FinSat}_L \iff \exists a R(a, \text{gn}(\chi))$, it is immediate that FinSat_L is r.e. and hence that FinVal_L is co-r.e. $\square$

Trachtenbrot's Theorem 5.5.18 *There exists a finite language L such that FinSat_L and FinVal_L are not recursive, hence FinVal_L is not r.e. and FinSat_L is not co-r.e.*

Proof. We shall show that for each register machine program (Definition 5.4.11) π which computes a unary partial recursive function F , there exists a finite language L_π and an L_π -formula $\psi(x)$ such that for each $m \in \omega$,

$$F(m) \downarrow \iff \psi(\dot{m}) \in \text{FinSat}_{L_\pi}.$$

The result then follows from the fact that every r.e. set, including some non-recursive ones, is the domain of a partial recursive function.

Fix $\pi = (I_0, \dots, I_l)$ which computes the partial recursive function F and $u = u(\pi)$, the least strict upper-bound for the the indices of registers accessed by π . The language L_π has as its symbols one constant symbol $\dot{0}$, one unary relation symbol $\dot{S}$, one binary relation symbol $\dot{<}$ and one $(u+2)$ -ary relation symbol p . Suppose that $(\sigma_0, \dots, \sigma_t)$ is a π -computation of a value for $F(m)$, and let $\mathfrak{A}_m$ be the L_π -structure

$$(\{0, 1, \dots, r\}, 0, \text{Sc}, <, P_m),$$

where r is a number larger than l and t , Sc is the normal successor operation, except that $\text{Sc}(r) = r$, $<$ is the normal ordering and $P_m(s, j, r_0, \dots, r_{u-1})$ holds iff

after s steps in the computation, the configuration is $(j, (r_0, \dots, r_{u-1}))$.

The formula ψ will be designed to satisfy the following two conditions, which imply the desired result:

- (i) $F(m) \downarrow \implies \models_{\mathfrak{A}_m} \psi(\dot{m}); \quad \text{and}$
- (ii) $F(m) \uparrow \implies \text{any model of } \psi(\dot{m}) \text{ is infinite.}$

Let ψ be the conjunction of L_π formulas expressing the following:

- (1) $\dot{<}$ is a discrete linear ordering with least element $\dot{0}$;
- (2) $\dot{S}$ is the immediate successor function for $\dot{<}$;
- (3) if $\dot{<}$ has a greatest element w , then $\dot{S}w \doteq w$;
- (4) $p\dot{0}\dot{0}x\dot{0}\dots\dot{0}$;
- (5) $\forall y\forall z_0\dots\forall z_{u-1}(\theta_0, \dots, \theta_l) \quad (\theta_k \text{ to be described below}).$

Clauses (1), (2) and (3) have an obvious role, and (4) describes the initial configuration at step 0. For each instruction I_k , θ_k describes the action of I_k as follows:

if I_k is $r_p := 0$, then θ_k is

$$py\dot{k}z_0\dots z_p\dots z_{u-1} \wedge \dot{k} \dot{<} \dot{l} \rightarrow y \dot{<} \dot{S}y \wedge p\dot{S}y\dot{S}\dot{k}z_0\dots\dot{0}\dots z_{u-1};$$

if I_k is $r_p := r_p + 1$, then θ_k is

$$py\dot{k}z_0\dots z_p\dots z_{u-1} \wedge \dot{k} \dot{<} \dot{l} \rightarrow y \dot{<} \dot{S}y \wedge p\dot{S}y\dot{S}\dot{k}z_0\dots\dot{S}z_p\dots z_{u-1};$$

if I_k is $r_p := r_q$, then θ_k is

$$py\dot{k}z_0\dots z_p\dots z_q\dots z_{u-1} \wedge \dot{k} \dot{<} \dot{l} \rightarrow y \dot{<} \dot{S}y \wedge p\dot{S}y\dot{S}\dot{k}z_0\dots z_q\dots z_q\dots z_{u-1};$$

if I_k is if $r_p = r_q$ go to $I_{k'}$, then θ_k is

$$py\dot{k}z_0 \dots z_p \dots z_q \dots z_{u-1} \wedge \dot{k} \dot{<} \dot{l} \rightarrow y \dot{<} \dot{S}y \wedge \\ \left[(z_p \dot{=} z_q \wedge p\dot{S}y\dot{k}'z_0 \dots z_p \dots z_q \dots z_{u-1}) \right. \\ \left. \vee (\neg z_p \dot{=} z_q \wedge p\dot{S}y\dot{S}\dot{k}z_0 \dots z_p \dots z_q \dots z_{u-1}) \right].$$

To see that (i) holds, let $(\sigma_0, \dots, \sigma_t)$ be a π -computation for $F(m)$. Clearly (1), (2) and (3) are true in $\mathfrak{A}_m$, and since $\sigma_0 = (0, (m, 0, \dots, 0))$, also $\models_{\mathfrak{A}_m} p\dot{0}\dot{0}\dot{m}\dot{0} \dots \dot{0}$. That the formulas θ_k are universally valid in $\mathfrak{A}_m$ is immediate from their relationship to the instructions I_k — for example, if $k < l$ and I_k is $r_p := r_p + 1$, and for some $s < t$, $P_m(s, k, r_0, \dots, r_p, \dots, r_{u-1})$, then σ_s is $(k, (r_0, \dots, r_p, \dots, r_{u-1}))$, so σ_{s+1} is $(k+1, (r_0, \dots, r_p+1, \dots, r_{u-1}))$ and hence $P_m(s+1, k+1, r_0, \dots, r_p+1, \dots, r_{u-1})$. Thus $\models_{\mathfrak{A}_m} \theta_k$, and by analogous arguments we have $\models_{\mathfrak{A}_m} \psi(\dot{m})$ as required.

For (ii), suppose that $F(m) \uparrow$ and $\models_{\mathfrak{A}} \psi(\dot{m})$. Let $(\sigma_0, \sigma_1, \dots, \sigma_s, \dots)$ be the infinite sequence of configurations generated by π from

$$\sigma_0 = (0, (m, 0, \dots, 0)).$$

It is easy to show by induction that if σ_s is $(k_s, (r_0^s, \dots, r_{u-1}^s))$, then

$$\dot{0}^{\mathfrak{A}}, \dots, \dot{s}^{\mathfrak{A}} \text{ are pairwise distinct and } \models_{\mathfrak{A}} p\dot{s}\dot{k}_s\dot{r}_0^s \dots \dot{r}_{u-1}^s.$$

It follows that $\mathfrak{A}$ is infinite. $\square$

Collateral Reading. There is a large literature on the general topic of decidability and undecidability for theories and other sets of sentences. Rabin's [Ba77; Chapter C.3] focuses on decidable theories, while [Mon; Part III] has a rich collection of both sorts of results. The book [Bo-Gr-Gu] is a comprehensive survey of decision problems involving sets of sentences defined by syntactic criteria, most often the number and type of quantifiers occurring in the prenex normal forms of the sentences.

Exercises

Exercise 5.5.19 Fill in the details of the proof of Proposition 5.5.17.

Exercise 5.5.20 Let Sq denote the squaring function $x \mapsto x^2$ on the integers $\mathbb{Z}$ and $|$ the relation of divisibility. Show that

- (i) $\text{Th}(\mathbb{Z}, +, \times)$ is interpretable in $\text{Th}(\mathbb{Z}, +, \text{Sq})$;
- (ii) $\text{Th}(\mathbb{Z}, +, \text{Sq})$ is interpretable in $\text{Th}(\mathbb{Z}, +, |, 1)$.

Conclude any relevant results on undecidability and axiomatizability.

Exercise 5.5.21 Let Sq denote the squaring function $x \mapsto x^2$ on the natural numbers and Sqr denote the set of squares of natural numbers. Show that

- (i) $\text{Th}(\omega, +, \times)$ is interpretable in $\text{Th}(\omega, +, \text{Sq})$;
- (ii) $\text{Th}(\omega, +, \text{Sq})$ is interpretable in $\text{Th}(\omega, +, \text{Sqr})$.

Conclude any relevant results on undecidability and axiomatizability.



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Chapter 6

Set Theory

Among all first-order theories, the theory of sets plays a very special role; this is why it deserves a chapter to itself. Sets have been part of mathematical thinking for centuries, but it was only in the second half of the 19th century that the general abstract notion of set was formulated and studied. Earlier uses were usually in the context of sets *of* some particular type of objects and were merely a tool in the service of some other interest. In the 1870s Georg Cantor began to investigate the concept of set for its own sake and soon found that it is an enormously rich and fertile idea. Since then set theory has become such an integral part of mathematics that students encounter parts of it throughout the mathematics curriculum from elementary school on.

Indeed, set-theoretic notions have been part of the background of all of the preceding material. Some of these, such as those of the algebra of sets (union, intersection, complement), functions (image, injective, surjective, composition, inverse) and relations (orderings, equivalence relations) are so universal that every student of abstract mathematics is familiar with most of them. Others, such as those concerned with infinite cardinalities (cardinal numbers) and well-orderings (ordinal numbers) may be less familiar to some but still important in a wide range of mathematical topics. At the level we

have been using them, these ideas constitute what we shall call Informal Set Theory (IST).

The first goal of this chapter is to develop a first-order Axiomatic Set Theory (AST) with sufficient power to represent the notions of IST and deduce their most important properties. There are many possibilities here; we choose the theory known as ZF (**Zermelo-Fraenkel**). The language is $L_{\in}$ with only the binary relation symbols $\in$ and $\doteq$. Whereas IST is “about” many different sorts of objects — sets, functions, relations, numbers etc. — ZF is a theory of sets only and succeeds because each of these other objects can be represented as a set. For example, the essence of the informal notion of function as mapping can be captured by its representation as a set of ordered pairs. This development occupies [Sections 6.1–6.4](#).

The centrality of AST to mathematics lies in the fact that not only these informal set-theoretic notions can be reduced to the single notion of set, but also most of the other notions of mathematics, those of algebra, analysis, topology etc. Most of this is far from the domain of this book, but it will be important to see how some of the material of logic and finally AST itself can be formalized in AST. Thus we will have a third version Formalized Axiomatic Set Theory (FAST), which is developed in [Section 6.6](#). In the literature there is often insufficient (in my view) attention paid to making clear at each point which of these three levels of the theory is being used; the notation adopted here is intended to address this issue (see Remark 6.1.1).

As discussed at the end of [Section 5.5](#), the results of [Section 6.2](#) include the relative interpretability of the theories Q and P in ZF and hence allow the conclusion that ZF and any recursively axiomatizable extension of it is incomplete. However, ZF is incomplete in a more interesting way in that it fails to decide some very natural questions. In particular, in [Section 6.5](#) and [Sections 6.7–6.9](#) we establish that if ZF itself is consistent, then neither the **Continuum Hypothesis** CH nor its negation is a theorem of ZF or indeed of the theory ZFC obtained by adding the Axiom of Choice AC to ZF. We also establish that $\neg AC$ is not a theorem of ZF and sketch briefly how to extend the methods to prove that AC itself is not a theorem of ZF.

Finally, in [Sections 6.10–6.11](#) we consider some possible extensions of ZF.

6.1 Zermelo-Fraenkel set theory

The key idea of set theory is the notion of a set as a single abstract object apart from the elements of which it is composed. That a set is completely determined by its elements is expressed by the principle of **extensionality**: two sets with the same elements are identical.

For the formalization of set theory we shall use a first-order language $L_{\in}$ with just one binary relation symbol $\in$. This principle expressed in $L_{\in}$ is

the first axiom of the theory ZF:

Extensionality $\forall x \forall y [\forall z (z \in x \leftrightarrow z \in y) \rightarrow x = y].$

Cantor based his development on just one additional principle which asserted that for any well-defined property Φ of sets, possibly defined in terms of other sets, there exists a set whose members are all and exactly the sets which have the property Φ . Informally we might express this as

for all $x_0, \dots, x_{k-1}$, $\{z : \Phi(x_0, \dots, x_{k-1}, z)\}$ exists.

From a modern perspective this principle is imprecise because we have not delimited the class of properties Φ which are to be admitted. A natural choice, which was not available to Cantor, is to accept exactly those properties which are expressible in $L_{\in}$. This may be expressed as an $L_{\in}$ -schema

Global Comprehension $\forall x_0 \dots \forall x_{k-1} \exists y \forall z [z \in y \leftrightarrow \phi(x_0, \dots, x_{k-1}, z)],$

for each $L_{\in}$ -formula ϕ which has at most the variables $x_0, \dots, x_{k-1}, z$ but not the variable y free.

The theory generated by this schema is indeed sufficient to develop all of set theory and indeed all of mathematics, but unfortunately this turns out to be true for a trivial reason — the theory is inconsistent. The simplest demonstration of this is just the so-called **Russell paradox** mentioned in the Introduction. The following is an instance of Global Comprehension:

$$\exists y \forall z (z \in y \leftrightarrow \neg z \in z),$$

but the existence of such a y is logically contradictory, since either $y \in y$ or $\neg y \in y$, but each of these leads to the opposite conclusion. Of course, this is a paradox only to someone who believes in Global Comprehension.

In the years following the discovery that Cantor's system is inconsistent, there were many diverse attempts to extract a consistent subtheory which would preserve the essence of the idea. The result, which is today not the only theory of sets, but certainly the most widely used, is known for its two main formulators as Zermelo-Fraenkel (ZF), although many others, particularly Skolem, made important contributions.

The key idea is that the universe of sets is a big place and some collections of sets are “too big” to be themselves sets. In particular, to avoid the Russell contradiction, the collection of all sets which are not members of themselves should not be a set. Thus the axioms of ZF are mainly special cases of Global Comprehension and assert that certain collections of sets are themselves also sets.

Before writing these down we develop some notation for discussing collections of sets which may or may not turn out to be themselves sets. Such a collection is called a **class** and will be denoted by double brackets:

$$\{\{ z : \phi(x_0, \dots, x_{k-1}, z) \}\}.$$

As usual, we may give particular classes special names; for example,

$$\begin{aligned} \dot{\emptyset} &:= \{\{ z : z \neq z \}\} && \textbf{empty class;} \\ \dot{V} &:= \{\{ z : z \doteq z \}\} && \textbf{universal class;} \\ \text{Ru} &:= \{\{ z : \neg z \in z \}\} && \textbf{Russell class.} \end{aligned}$$

We write ‘ $\{\{ z : \phi(x_0, \dots, x_{k-1}, z) \}\}$ is a set’ as an abbreviation for the $L_{\in}$ -formula

$$\exists y \forall z [z \in y \leftrightarrow \phi(x_0, \dots, x_{k-1}, z)]$$

for a variable y not free in ϕ . Thus the argument above shows that ‘Ru is a set’ is logically contradictory, and the Global Comprehension principle asserts that for all $L_{\in}$ -formulas ϕ ,

$$\forall x_0 \dots \forall x_{k-1} [\{\{ z : \phi(x_0, \dots, x_{k-1}, z) \}\} \text{ is a set}].$$

We also write, for any class $B := \{\{ z : \phi(x_0, \dots, x_{k-1}, z) \}\}$,

$$B \text{ is a proper class} \quad \text{for} \quad \neg (B \text{ is a set}).$$

When the formula ϕ has additional free variables $x_0, \dots, x_{k-1}$, we regard the class expression as defining a mapping

$$F : (x_0, \dots, x_{k-1}) \mapsto \{\{ z : \phi(x_0, \dots, x_{k-1}, z) \}\}$$

called a **class operation**. For example,

$$\begin{aligned} x \dot{\cup} y &:= \{\{ z : z \in x \vee z \in y \}\} && \textbf{union} \\ x \dot{\cap} y &:= \{\{ z : z \in x \wedge z \in y \}\} && \textbf{intersection} \\ x \setminus y &:= \{\{ z : z \in x \wedge z \notin y \}\} && \textbf{set difference} \\ \dot{S}x &:= \{\{ z : z \in x \vee z \doteq x \}\} && \textbf{successor} \\ \dot{\bigcup} x &:= \{\{ z : \exists v [v \in x \wedge z \in v] \}\} && \textbf{Union} \\ \dot{\bigcap} x &:= \{\{ z : \forall v [v \in x \rightarrow z \in v] \}\} && \textbf{Intersection} \\ \dot{\wp} x &:= \{\{ z : \forall u [u \in z \rightarrow u \in x] \}\} =: \{\{ z : z \subseteq x \}\} && \textbf{powerclass} \end{aligned}$$

We write F *is a set operation* as an abbreviation for

$$\forall x_0 \cdots \forall x_{k-1} [F(x_0, \dots, x_{k-1}) \text{ is a set}].$$

That most of these examples are set operations will be theorems of **ZF** (see below). Now we have a still more concise way of expressing the Global Comprehension principle:

for all class operations F , F is a set operation.

Note that any statement which quantifies over class operations refers implicitly to all $L_{\in}$ -formulas and is therefore not generally expressible by a single $L_{\in}$ -formula.

We emphasize that class expressions are not formal symbols of $L_{\in}$ and any expression containing one is an abbreviation for an $L_{\in}$ -formula described as follows. If $F(x) := \{ \{ z : \phi(x, z) \} \}$, then

$$\begin{aligned} z \dot{\in} F(x) & \text{ is an abbreviation for } \phi(x, z); \\ F(x) \dot{=} y \text{ or } y \dot{=} F(x) & \text{ is an abbreviation for } \forall z [z \dot{\in} y \leftrightarrow \phi(x, z)]; \\ F(x) \dot{\in} u & \text{ is an abbreviation for } \exists y [y \dot{=} F(x) \wedge y \dot{\in} u]. \end{aligned}$$

Note in particular that ' $F(x) \dot{\in} u \rightarrow F(x)$ is a set' is logically valid. More generally, if ψ is a non-atomic $L_{\in}$ -formula, then $\psi_v(F(x))$ is an abbreviation for $\exists y [y \dot{=} F(x) \wedge \psi_v(y)]$ for the first variable y which occurs in neither ϕ nor ψ .

Remark 6.1.1 Throughout this chapter we shall make extensive use of the *dot convention* to clarify the distinction between statements of our informal mathematical language (IST) and the formal language of **ZF** (AST). Because set-theoretic notions are used constantly in mathematical discourse, there is always a danger of confusion. Generally, the convention here is that any statement involving dotted symbols represents an $L_{\in}$ -formula. Thus we write 'is a set' rather than 'is a set' and use symbols such as $\dot{\phi}$ and $\dot{\bigcup}$ rather than their informal and undotted counterparts. Furthermore, when we have shown (or are about to show) that $\mathbf{ZF} \vdash \{ \{ z : \phi(x, z) \} \}$ is a set, we will use the notation $\{ z : \phi(x, z) \}$ to remind the reader of this fact. For example, the axioms of **ZF** below justify immediately writing

$$\dot{\emptyset} \dot{=} \{ z : z \neq z \} \quad \text{and} \quad \dot{\bigcup} x \dot{=} \{ z : \exists v [v \dot{\in} x \wedge z \dot{\in} v] \}.$$

That said, the reader should avoid being overly distracted by the dots and for the most part ignore them when reading except when confusion arises. Note that there are other useful clues which indicate when an expression

represents an $L_{\in}$ -formula; prime among these is the use of the formal single arrow $\rightarrow$ rather than the informal $\implies$. This distinction is maintained throughout the book.

It is useful to think of classes and class operations in connection with their interpretations in $L_{\in}$ -structures. Since this is a language with just one binary relation symbol, a structure is of the form $\mathfrak{A} = (A, E)$ with A any non-empty set and $E \subseteq A \times A$.

Definition 6.1.2 For any set Γ of sentences and any $L_{\in}$ -structure $\mathfrak{A} = (A, E)$,

- (i) for any class $B = \{ \{ z : \phi(z) \} \}$,

$$B^{\mathfrak{A}} := \{ \{ z : \phi(z) \} \}^{\mathfrak{A}} = \{ c \in A : \models_{\mathfrak{A}} \phi[c] \};$$

- (ii) for any class operation

$$F(x_0, \dots, x_{k-1}) = \{ \{ z : \phi(x_0, \dots, x_{k-1}, z) \} \}$$

and each $a_0, \dots, a_{k-1} \in A$,

$$\begin{aligned} F^{\mathfrak{A}}(a_0, \dots, a_{k-1}) &:= \{ \{ z : \phi(x_0, \dots, x_{k-1}, z) \} \}^{\mathfrak{A}}_{a_0, \dots, a_{k-1}} \\ &= \{ c \in A : \models_{\mathfrak{A}} \phi[a_0, \dots, a_{k-1}, c] \}. \end{aligned}$$

Thus the sets $B^{\mathfrak{A}}$ are exactly the subsets of A definable over $\mathfrak{A}$ without parameters, and sets of the form $F^{\mathfrak{A}}(a_0, \dots, a_{k-1})$ are those definable from the parameters $a_0, \dots, a_{k-1}$. Now

$$\begin{aligned} \models_{\mathfrak{A}} B \text{ is a set} &\iff \models_{\mathfrak{A}} \exists y \forall z [z \dot{\in} y \leftrightarrow \phi(z)] \\ &\iff (\text{exists } b \in A)(\text{for all } c \in A) \\ &\quad \models_{\mathfrak{A}} (z \dot{\in} y)[b, c] \iff \models_{\mathfrak{A}} \phi(z)[c] \\ &\iff (\text{exists } b \in A) B^{\mathfrak{A}} = \{ c \in A : (c, b) \in E \}. \end{aligned}$$

The meaning is clearer in the following special case.

Definition 6.1.3

- (i) A set A is **transitive** iff for every $b \in A$ and $a \in b$, also $a \in A$ — that is, for every $b \in A$, also $b \subseteq A$;
(ii) an $L_{\in}$ -structure $\mathfrak{A} = (A, E)$ is a **transitive $\in$ -structure** iff A is a transitive set and E is

$$\in_A := \{ (c, b) : c, b \in A \text{ and } c \in b \}.$$

Proposition 6.1.4 *For any transitive $\in$ -structure $\mathfrak{A} = (A, \in_A)$,*

- (i) *for any class B , $\models_{\mathfrak{A}} B$ is a set iff $B^{\mathfrak{A}} \in A$;*
- (ii) *for any k -ary class operation F , $\models_{\mathfrak{A}} F$ is a set operation iff $F^{\mathfrak{A}} : A^k \rightarrow A$.*

Proof. If $\mathfrak{A} = (A, \in_A)$, then as above $\models_{\mathfrak{A}} B$ is a set iff for some $b \in A$,

$$B^{\mathfrak{A}} = \{c \in A : (c, b) \in \in_A\} = \{c \in A : c \in b\} = \{c : c \in b\} = b,$$

where the third equality follows from the transitivity of A . Similarly, if F is a k -ary class operation, then $\models_{\mathfrak{A}} F$ is a set operation iff for all $a_0, \dots, a_{k-1} \in A$,

$$F^{\mathfrak{A}}(a_0, \dots, a_{k-1}) = \{c \in A : \models_{\mathfrak{A}} \phi[a_0, \dots, a_{k-1}, c]\} \in A. \quad \square$$

Although we shall deal explicitly with such interpretations only occasionally, it is often helpful to keep them in mind. One of the advantages of classes and class operations over their defining formulas is that it is natural to apply many set-theoretic operations to them. For example, if $B = \{z : \phi(z)\}$ and $C = \{z : \psi(z)\}$, then

$$\begin{aligned} B \dot{\cup} C &:= \{z : \phi(z) \vee \psi(z)\} = \{z : z \in B \vee z \in C\} \\ B \dot{\cap} C &:= \{z : \phi(z) \wedge \psi(z)\} = \{z : z \in B \wedge z \in C\} \\ \dot{\bigcup} B &:= \{y : \exists z (\phi(z) \wedge y \in z)\} = \{y : \exists z (z \in B \wedge y \in z)\} \\ \dot{\bigcap} B &:= \{y : \forall z (\phi(z) \rightarrow y \in z)\} = \{y : \forall z (z \in B \rightarrow y \in z)\}. \end{aligned}$$

Also, $B \dot{\subseteq} C$ is an abbreviation for

$$\forall z [\phi(z) \rightarrow \psi(z)] \text{ or equivalently } \forall z [z \in B \rightarrow z \in C].$$

Note that $\bigcap \dot{\emptyset} = \dot{V}$, the universal class. Composition of class operations is defined by

$$F(Gx) := \{z : \exists y [y \dot{=} Gx \wedge z \in F(y)]\},$$

where z is a variable not free in the formulas defining F or G . In particular, it is logically valid that

$$Gx \text{ is a proper class} \rightarrow F(Gx) \dot{=} \dot{\emptyset},$$

but $F(Gx)$ has its usual meaning under the hypothesis that G is a set operation. Similarly, we write $G(x) := \{F(x, y) : \phi(x, y)\}$ for

$$G(x) := \{z : \exists y [z \dot{=} F(x, y) \wedge \phi(x, y)]\}.$$

In particular,

$$F^*(u, x) := \{ \{ F(x, y) : y \in u \} \doteq \{ \{ z : \exists y [z \doteq F(x, y) \wedge y \in u] \} \}$$

is the (parameterized) **image** of u . Note that generally

$$\{ \{ F(x, y) : \phi(x, y) \} \doteq \{ \{ F(x, y) : \phi(x, y) \wedge F(x, y) \text{ is a set} \} \}.$$

In preparation for the statement of the axioms of ZF, we need two more definitions.

Definition 6.1.5

- (i) $\dot{\omega} := \bigcap \{ \{ u : \dot{\emptyset} \in u \wedge \forall z (z \in u \rightarrow \dot{S}z \in u) \} \}.$
- (ii) z is $\dot{\in}$ -**minimal in** y is an abbreviation for $z \in y \wedge \forall u (u \in z \rightarrow u \notin y)$
— that is, $z \in y \wedge y \cap z \doteq \dot{\emptyset}.$

Definition 6.1.6 ZF is the $L_{\in}$ -theory generated by the following sentences:

- (i) (**Extensionality**) $\forall x \forall y [\forall z (z \in x \leftrightarrow z \in y) \rightarrow x \doteq y];$
- (ii) (**Union**) $\dot{\bigcup}$ is a set operation;
- (iii) (**Powerset**) $\dot{\wp}$ is a set operation;
- (iv) (**Infinity**) $\dot{\omega}$ is a set;
- (v) (**Replacement**) for every class operation F , F^* is a set operation;
- (vi) (**Foundation**) $\forall y (y \neq \dot{\emptyset} \rightarrow \exists z [z \text{ is } \dot{\in}\text{-minimal in } y]).$

As we said at the beginning, Extensionality expresses the fundamental notion that a set is completely determined by its elements. Union and Powerset are simple instances of Global Comprehension. To understand the meaning of Infinity, let $B := \{ \{ u : \dot{\emptyset} \in u \wedge \forall z (z \in u \rightarrow \dot{S}z \in u) \} \}$ so $\dot{\omega} := \bigcap B$. Intuitively, any member of B must be infinite, and the thrust of the Infinity axiom is to assert that there exists such a set, namely $\dot{\omega}$. Since $\dot{\emptyset} \in u$ for each $u \in B$, by the definition of $\bigcap$, also $\dot{\emptyset} \in \dot{\omega}$. Similarly, for any $z \in \dot{\omega}$, $z \in u$ for each $u \in B$, so also $\dot{S}z \in u$ for each $u \in B$ and hence $\dot{S}z \in \dot{\omega}$. It follows that $\dot{\omega} \in B$. Of course, $\dot{\omega}$ has the additional property of being the *smallest* member of B in the sense of being a subset of any other member. $\dot{\omega}$ is the formal representation of the set of natural numbers and its definition leads directly to the principle of proof by induction (see below).

Replacement gets its name from the intuition that

$$F^*(u, x) := \{ \{ F(x, y) : y \in u \} \}$$

arises from u by *replacing* each element y of u by the corresponding image $F(x, y)$ as long as this is a set; intuitively there are at most as many of these images as there are $y \in u$ so the resulting collection is of ‘set size’. Foundation arises from intuitions about the nature of the intuitive universe of sets which we develop at the end of this section. In particular, it implies that $\forall x (x \notin x)$ (so $\text{Ru} \doteq \dot{\mathbf{V}}$).

Remark 6.1.7 We have stated the axioms in this form to emphasize their intuitive meaning. It is a simple exercise to replace all the abbreviations with their $L_{\in}$ -equivalents to obtain the sentences which are the actual axioms. For example, Powerset is the sentence

$$\forall x \exists y \forall z [z \in y \leftrightarrow \forall u (u \in z \rightarrow u \in x)].$$

We leave the other translations as exercises.

The axioms were known in roughly the form that we have given them already by 1922. They arose gradually as a codification of our intuitions about the informal “universe” of sets. This is the “standard model” for the axioms, but of course it is much less precisely given than the standard model Ω for the various theories of arithmetic we have studied. We shall return to this topic at the end of this section, but the reader may want to give this a first reading now.

Remark 6.1.8 The economy of formulating ZF in a language with only the one binary relation symbol $\in$ is a major inconvenience when it comes to expressing anything even mildly sophisticated. Many expositions of Set Theory use the results of [Section 2.6](#) to expand the language $L_{\in}$ by introducing defined relation, function and constant symbols for frequently used notions. We get the same effect here by consistent use of symbols for classes and class operations. For example, we show just below that $\text{ZF} \vdash \dot{\emptyset}$ is a set — that is, $\text{ZF} \vdash \exists y \forall z [z \in y \leftrightarrow z \neq z]$. By Extensionality, there can be at most one such y , so also $\text{ZF} \vdash \exists! y \forall z [z \in y \leftrightarrow z \neq z]$. This is, of course, just the condition required for the introduction of a constant symbol (also called) $\dot{\emptyset}$ with the definition $y = \dot{\emptyset} \leftrightarrow \forall z [z \in y \leftrightarrow z \neq z]$. Any formula of this expanded language $L_{\in}^+$ can always be thought of as an $L_{\in}$ -formula in which occurrences of $\dot{\emptyset}$ are replaced by its definition; to verify this in detail is slightly more complicated than it might seem, and we leave it as an exercise. Similarly, since $\text{ZF} \vdash \dot{\phi}$ is a set operation, using again Extensionality we have that also $\text{ZF} \vdash \forall x \exists! y \forall z [z \in y \leftrightarrow z \subseteq x]$, and we would be justified in introducing a new unary function symbol (also called) $\dot{\phi}$ with the definition

$$\forall x (\dot{\phi}x \doteq y \leftrightarrow \forall z [z \in y \leftrightarrow z \subseteq x]).$$

Of course, the introduction of relation symbols is much less problematic, but here also we formally regard all defined relations as abbreviations. We have

already introduced the abbreviation

$$x \subseteq y \quad \text{for} \quad \forall z [z \in x \rightarrow z \in y]$$

and will define many others below, often using English phrases rather than symbols. In most cases at least one word of such a phrase will be dotted to remind the reader that it does stand for an $L_\in$ -formula.

Standard presentations of ZF often include other axioms, which in fact follow from the ones we have given. Let

$$\dot{\{x, y\}} := \{\{z : z \dot{=} x \vee z \dot{=} y\}\}$$

and

$$F^\cap(x_0, \dots, x_{k-1}, u) := F(x_0, \dots, x_{k-1}) \dot{\cap} u,$$

for any k -ary class operation F .

Theorem 6.1.9 *The following are ZF-theorems:*

- (i) $\dot{\emptyset}$ is a set;
- (ii) (**Pair**) $\dot{\{\cdot, \cdot\}}$ is a set operation;
- (iii) $\dot{\cup}$ and $\dot{S}$ are set operations;
- (iv) (**Comprehension**) for every class operation F , $F^\cap$ is a set operation.

Proof. For (i), we argue in ZF and let F denote the class operation such that for all x , $Fx := Ru$. Since Ru is not a set, for any u , $F^*u \dot{=} \dot{\emptyset}$, which is therefore a set by Replacement. For (ii), first observe that it follows from the Powerset axiom that there are sets $\dot{\wp}\dot{\emptyset} \dot{=} \{\dot{\emptyset}\}$, $\dot{\wp}\dot{\wp}\dot{\emptyset} \dot{=} \{\dot{\emptyset}, \{\dot{\emptyset}\}\}$. Now if F is the class operation

$$F(w) = \left\{ \left\{ z : (w \dot{=} \dot{\emptyset} \wedge z \in x) \vee (w \neq \dot{\emptyset} \wedge z \in y) \right\} \right\} = \begin{cases} x, & \text{if } w \dot{=} \dot{\emptyset}; \\ y, & \text{otherwise;} \end{cases}$$

clearly F is a set operation. By Replacement F^* is a set operation, so $F^*(\dot{\wp}\dot{\wp}\dot{\emptyset})$ is a set and has exactly x and y as elements as desired. For (iii), $x \dot{\cup} y \dot{=} \dot{\cup} \{x, y\}$, $\{x\} \dot{=} \{x, x\}$ and $\dot{S}x \dot{=} x \dot{\cup} \{x\}$.

For (iv), given (say unary) F , let

$$\begin{aligned} G(x, y) &:= \left\{ \left\{ z : (y \in F(x) \wedge z \in y) \vee (y \notin F(x) \wedge z \notin z) \right\} \right\} \\ &= \begin{cases} y, & \text{if } y \in F(x); \\ Ru, & \text{otherwise.} \end{cases} \end{aligned}$$

Then

$$G^*(u, x) = \{ \dot{G}(x, y) : y \dot{\in} u \} = F(x) \dot{\cap} u.$$

By Replacement, G^* is a set operation, so $F(x) \dot{\cap} u$ is a set. $\square$

Note that Comprehension can also be stated: for all $L_{\in}$ -formulas ϕ ,

$$ZF \vdash \forall x_0 \cdots \forall x_{k-1} \{ \{ z : z \dot{\in} u \wedge \phi(x_0, \dots, x_{k-1}, z) \} \} \text{ is a set,}$$

and we can use without further explanation the notation

$$\{ z \dot{\in} u : \phi(x_0, \dots, x_{k-1}, z) \}.$$

Remark 6.1.10 In more detailed axiomatic studies one considers subtheories of ZF which omit one or both of the axioms Powerset and Replacement. In such cases these theorems, which are essential for any reasonable theory of sets, must be included as axioms.

Corollary 6.1.11 *The following are ZF-theorems:*

- (i) $\dot{\cap}$ and $\setminus$ are set operations;
- (ii) for each $k \in \omega$, $\{ \cdot, \dots, \cdot \}_k$ is a set operation, where

$$\{ x_0, \dots, x_{k-1} \}_k := \{ \{ z : z \dot{=} x_0 \vee \cdots \vee z \dot{=} x_{k-1} \} \};$$

- (iii) $\forall x [x \neq \dot{\emptyset} \rightarrow \dot{\bigcap} x \text{ is a set}]$.

Proof. We leave these simple proofs as an exercise. $\square$

Note that part (iii) falls short of saying that $\dot{\bigcap}$ is a set operation; of course this is because $\dot{\bigcap} \dot{\emptyset} \dot{=} \dot{V}$ and we show just below that $\dot{V}$ is a proper class. Correspondingly, for classes we have

Corollary 6.1.12 *For any classes B and C, the following are ZF-theorems:*

- (i) $B \text{ is a set} \wedge C \text{ is a set} \rightarrow$

$$B \dot{\cup} C \text{ is a set} \wedge B \dot{\cap} C \text{ is a set} \wedge B \setminus C \text{ is a set;}$$

- (ii) $B \text{ is a set} \rightarrow \dot{\bigcup} B \text{ is a set;}$

- (iii) $B \neq \dot{\emptyset} \rightarrow \dot{\bigcap} B \text{ is a set. } \square$

Remark 6.1.13 We are frequently faced with the task of showing that some particular $L_{\in}$ -sentence θ is a theorem of ZF. Of course, the straightforward way to do this is to write down a formal proof of θ from the axioms. However, exactly because such a proof is formal, it is hard to read and does not convey well the underlying ideas. In many such cases, as in the proof of Theorem 6.1.9 above, we “argue in ZF” — that is, give an outline of a formal proof intended to convince the reader that such a proof exists and to allow her to fill in the details of this proof to the desired extent. Sometimes we say this explicitly, but any argument which involves $L_{\in}$ -expressions — for example, ones with dots or (most frequently) the symbol $\rightarrow$ (instead of $\implies$) — is such an argument. Indeed, in many cases we shall not even say explicitly $\text{ZF} \vdash \theta$ but simply θ , when this is clearly (an abbreviation for) an $L_{\in}$ -sentence.

Corollary 6.1.14 $\forall x (x \notin x)$.

Proof. For any x , $\{x\} \neq \emptyset$, so by Foundation $\{x\}$ has a minimal element, which must be x . This means $\forall u [u \in x \rightarrow u \notin \{x\}]$ or equivalently, $\forall u [u \in x \rightarrow u \neq x]$. In particular, this is true for $u = x$. $\square$

Proposition 6.1.15 (*no universal set*) $\dot{V}$ is a proper class.

Proof. Note that for any class B , $B \cap \dot{V} = B$. Hence from ‘ $\dot{V}$ is a set’ follows by Comprehension for each class B also B is a set. In particular, this is true for the Russell class R_u , which is a contradiction. Alternatively, apply the preceding corollary. $\square$

Note that in a transitive $\in$ -model $\mathfrak{A}$ of ZF, this just means that $A \notin A$. We conclude this section by returning to the question: where do the axioms of ZF come from and why were they chosen in this way? Of course, part of the answer is that they evolved over the years as set-theoretic intuitions were developed and refined. To a certain extent the axioms may be described as a codification of our understanding of a natural universe V of sets much as the axioms of P describe a good part of what we know about the natural number structure Ω .

This is not an exact parallel because we do not have nearly as firm a grasp on V as we do on Ω ; not only have the natural numbers been part of human experience for a much longer time, but they are inherently a much simpler structure. We observed in Remark 4.6.31 that although P is incomplete, it is difficult to find natural examples ϕ of sentences independent of P . Furthermore, when we do find such examples, we usually have a clear sense of which of ϕ or $\neg\phi$ is true in Ω . On the other hand we shall spend much of this chapter proving that a very natural sentence CH is independent of ZF, and there is no general agreement whether CH or $\neg\text{CH}$ is true in V . The same is true of many other sentences of set theory.

Before Cantor conceived of and developed the notion of abstract set, sets were always sets *of* something: sets of natural or real numbers, sets of points in the plane, sets of solutions to an equation etc. The new idea was to look at sets whose elements were sets whose elements were sets etc. This seems circular; where do we start? The (eventually) obvious answer is to start with the empty set $\emptyset$. Once we have this we have an object so we can form the set $\{\emptyset\}$ with that object as its (only) element. Now we have two objects, so there are two new sets available: $\{\{\emptyset\}\}$ and $\{\emptyset, \{\emptyset\}\}$. Note that we *are* always forming sets *of* something, namely the objects that we have constructed so far. In general, if we have a collection A of objects, then the collection of sets with members from A is just $\wp A$. In particular

$$\{\emptyset\} = \wp\emptyset, \quad \{\emptyset, \{\emptyset\}\} = \wp\{\emptyset\} = \wp\wp\emptyset,$$

and

$$\{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\} = \wp\{\emptyset, \{\emptyset\}\} = \wp\wp\{\emptyset\} = \wp\wp\wp\emptyset.$$

Note that each of these collections contains not only the new sets at that level but the old ones as well. This suggests that we continue the construction as follows: for all natural numbers n , set

$$V_0 := \emptyset \quad \text{and} \quad V_{n+1} := \wp V_n,$$

and collect what we have as $V_\omega := \bigcup_{n \in \omega} V_n$. Could V_ω serve as our universe of sets? Clearly not if we expect such a universe to encompass all intuitive sets, since V_ω itself is a perfectly reasonable set which obviously is

not a member of this collection. Note that each V_n is finite, with $2^{2^{\dots^2}}\}_{n-1}^{n-1}$ members (for $n > 1$), while V_ω is the first infinite set we encounter in this process. By the principle of forming all sets with members among what we have, the next step should be $\wp V_\omega$, which we call $V_{\omega+1}$, and recursively set $V_{\omega+n+1} := \wp V_{\omega+n}$. Then we can collect what we have so far as $V_{\omega+\omega} := \bigcup_{n \in \omega} V_{\omega+n}$ and continue with $V_{\omega+\omega+n}$. Note that since V_ω is countably infinite, $V_{\omega+1}$ is uncountable, of cardinality $2^{\aleph_0}$, and generally the infinite size of $V_{\omega+n}$ increases with n , so $V_{\omega+\omega}$ is already a quite large set, and our ability to grasp its structure intuitively is fairly weak. Furthermore, there is no natural stopping point; as long as we can collect what we have constructed, applying $\wp$ once more produces new sets to add to the collection.

One sometimes says, with a high level of vagueness, that the universe of sets is what you get by carrying on this process “forever” with the universe V being the totality of all sets that arise. This can be made somewhat more precise by appealing to the intuitive notion of ordinal number to define the following **cumulative hierarchy** of sets.

Definition 6.1.16 For all ordinal numbers σ and ρ ,

- (i) $V_0 := \emptyset$;
- (ii) $V_{\sigma+1} := \wp V_\sigma$;
- (iii) $V_\rho := \bigcup_{\sigma < \rho} V_\sigma$, if ρ is a limit ordinal;
- (iv) $V := \bigcup_{\sigma} V_\sigma$ is an ordinal number V_σ .

Clearly, the further one goes in this sequence, the less direct are the intuitions, and it seems hardly surprising that there are easily formulated questions without clear answers. However, despite this it *is* possible to convince ourselves, and is generally agreed, that the axioms of ZF are all true in V when $\in$ is interpreted as membership $\in$ — that is, $(V, \in)$ is a “model” of ZF, where the quotation marks refer to the awkward fact that V is not a set. We consider these fairly briefly. It will be useful to state some simple properties of the stages V_σ ; we shall not prove these here but refer the reader to the corresponding formal development of the hierarchy $\dot{V}$ in [Section 6.3](#). In all of the following, we use informal properties of ordinal numbers whose formal counterparts will be shown in [Section 6.3](#) to be theorems of ZF.

Proposition 6.1.17 For all ordinals σ ,

- (i) V_σ is transitive; V is transitive;
- (ii) for all $\sigma \leq \tau$, $V_\sigma \subseteq V_\tau$;
- (iii) $V_\sigma \in V_{\sigma+1}$;
- (iv) for all $z \in y \in V_\sigma$, $z \in V_\tau$ for some $\tau < \sigma$. $\square$

Extensionality is true in $(V, \in)$ by the very nature of the notion of set and the transitivity of V . Towards Foundation, suppose $y \in V_\sigma$ and $y \neq \emptyset$. Then by (iv), for each $z \in y$, there is a smallest $\tau_z < \sigma$ such that $z \in V_{\tau_z}$. Let $\bar{z}$ be such that $\tau_{\bar{z}}$ is smallest among such τ_z ; $\bar{z}$ exists because the ordinals are well-founded. Then for any $u \in \bar{z}$, $u \in V_v$ for some $v < \tau_{\bar{z}}$ and thus by the choice of $\bar{z}$, $u \notin y$. Hence $\bar{z}$ is $\in$ -minimal in y .

The remaining axioms all make assertions of the form ‘ F is a set (operation)’. By Proposition 6.1.4,

$$\begin{aligned} \models_{(V, \in)} B \text{ is a set} &\iff B^{(V, \in)} \in V; \\ \models_{(V, \in)} F \text{ is a set operation} &\iff F^{(V, \in)} : V \rightarrow V \end{aligned}$$

(for unary F). It is easy to see that

$$\dot{\bigcup}^{(V, \in)} \text{ is } \bigcup, \quad \dot{\wp}^{(V, \in)} \text{ is } \wp, \quad \dot{\emptyset}^{(V, \in)} \text{ is } \emptyset \quad \text{and} \quad \dot{S}^{(V, \in)} \text{ is } S_c,$$

where $\text{Sc } x = x \cup \{x\}$. Furthermore, for any class operation F ,

$$\begin{aligned} (F^*)^{(\mathbf{V}, \in)}(u, x) &= \{F^{(\mathbf{V}, \in)}(x, y) : y \in u\} \\ &= \{z \in \mathbf{V} : z = F^{(\mathbf{V}, \in)}(x, y) \text{ for some } y \in u\}. \end{aligned}$$

Of course, $\dot{\omega}^{(\mathbf{V}, \in)}$ is *not* ω , since the informal natural numbers are not sets at all; it is rather an isomorphic copy ω' of ω with members

$$0' = \emptyset, 1' = \{0'\}, \dots, (k+1)' = \{0', 1', \dots, k'\}, \dots$$

In view of these facts, it will suffice to verify that for all x and u ,

- (1) $x \in \mathbf{V} \implies \bigcup x \in \mathbf{V}$;
- (2) $x \in \mathbf{V} \implies \wp x \in \mathbf{V}$;
- (3) $\omega' \in \mathbf{V}$;
- (4) for every (binary) class operation F ,

$$x, u \in \mathbf{V} \implies (F^*)^{(\mathbf{V}, \in)}(u, x) \in \mathbf{V}.$$

Most of these are straightforward calculations using the preceding proposition; for any ordinal σ ,

- (1) $x \in \mathbf{V}_\sigma \implies x \subseteq \mathbf{V}_\sigma \implies \bigcup x \subseteq \mathbf{V}_\sigma \implies \bigcup x \in \mathbf{V}_{\sigma+1}$;
- (2) $x \in \mathbf{V}_\sigma \implies x \subseteq \mathbf{V}_\sigma \implies \wp x \subseteq \mathbf{V}_{\sigma+1} \implies \wp x \in \mathbf{V}_{\sigma+2}$;
- (3) easily $\omega' \subseteq \mathbf{V}_\omega$, so $\omega' \in \mathbf{V}_{\omega+1}$.

The verification of (4), and therefore the Replacement schema, depends on a principle about the construction of $\mathbf{V}$, or alternatively about the extent of the ordinal numbers, that we have not yet discussed. It is based on the idea that we want our universe of sets to be as rich and inclusive as possible. More specifically, if to each element y of a set u we can associate a stage $\mathbf{V}^y$, then there should be a stage after all of the stages $\mathbf{V}^y$. If we are thinking of the stages as indexed by ordinals, then this says simply that if to each $y \in u$ is associated an ordinal τ_y , then there is an ordinal greater than all τ_y . In fact, this is the same as saying that $\{\tau_y : y \in u\}$ exists and for any set Z of ordinals there is an ordinal larger than all members of Z . This will, of course, be true about the formal notion of ordinal number introduced in the next section. Using this we have the following.

Let F be a binary class operation and fix x and u . For each $y \in u$ there is an ordinal τ_y such that $F^{(\mathbf{V}, \in)}(x, y) \in \mathbf{V}_{\tau_y}$. By the preceding discussion, there exists an ordinal v such that for all $y \in u$, $\tau_y \leq v$. By (iii) of the

proposition, for each $y \in u$, $F^{(V, \in)}(x, y) \in V_v$ — that is, $(F^*)^{(V, \in)}(u, x) \subseteq V_v$, so $(F^*)^{(V, \in)}(u, x) \in V_{v+1}$.

Collateral Reading. Just about any text on axiomatic set theory has a discussion of how the axioms came to be and how they are intuitively justified — for example, [En77], [Jec] or [Mo94]. Historically the first is Zermelo's [Dav; pp. 199-215]. Shoenfield's [Ba77; Chapter B.1] has an interesting discussion of the intuitive justification of the axioms and possible extensions.

Exercises

Exercise 6.1.18 Write all of the ZF-axioms in their primitive form as $L_{\in}$ -sentences.

Exercise 6.1.19 Let $L_{\in}^+$ be the extension of $L_{\in}$ by the introduction of a constant symbol $\dot{\emptyset}$ and a unary function symbol $\dot{\phi}$ and ZF^+ the expansion of ZF by the definitions for these symbols discussed in Remark 6.1.8. For each $L_{\in}^+$ -formula ϕ , let ϕ^* denote the $L_{\in}$ -formula obtained by interpreting the symbols $\dot{\emptyset}$ and $\dot{\phi}$ as a class and a class operation. Prove that for all such ϕ , $ZF^+ \vdash \phi \leftrightarrow \phi^*$.

Hint: Use Proposition 2.6.18.

Exercise 6.1.20 Prove Corollary 6.1.11.

Exercise 6.1.21 Show that the following are theorems of ZF:

- (i) $\forall x \forall y \neg [x \dot{\in} y \wedge y \dot{\in} x]$;
- (ii) $\forall x \forall y \forall z \neg [x \dot{\in} y \wedge y \dot{\in} z \wedge z \dot{\in} x]$.

Exercise 6.1.22 Let $E \subseteq \omega \times \omega$ be the relation

$$\{ (m, n) : \text{the } m\text{-th digit of the binary representation of } n \text{ is } 1 \}.$$

Determine which of the axioms of ZF are true in the structure (ω, E) . Analyze this further to show that a large part of ZF is interpretable in P.

Exercise 6.1.23 Let $ZF - \text{Inf}$ be the theory generated by all of the axioms of ZF except Infinity and set

$$\dot{\omega}' := \left\{ \left\{ y : \forall u [y \dot{\in} u \wedge \forall z (\dot{S}z \dot{\in} u \rightarrow z \dot{\in} u)] \rightarrow \dot{\emptyset} \dot{\in} u \right\} \right\}.$$

Show that the following are theorems of $ZF - \text{Inf}$:

- (i) $\dot{\emptyset} \dot{\in} \dot{\omega}'$;

- (ii) $\forall z (z \in \dot{\omega}' \leftrightarrow \dot{S}z \in \dot{\omega}')$;
- (iii) $\dot{\omega}' \subseteq \dot{\omega}$;
- (iv) $\dot{\omega}' \text{ is a set} \rightarrow \dot{\omega} \doteq \dot{\omega}'$;
- (v) $\dot{\omega} \text{ is a set} \rightarrow \dot{\omega} \doteq \dot{\omega}'$.

Hence on the basis of the other axioms, Infinity is equivalent to ' $\dot{\omega}'$ is a set'.

Exercise 6.1.24 Show that the following are theorems of ZF: for $u, v \neq \emptyset$

- (i) $u \dot{\cup} \dot{\bigcup} v \doteq \dot{\bigcup} \{u \dot{\cup} y : y \in v\}$;
- (ii) $\dot{\bigcup} u \dot{\cup} \dot{\bigcup} v \doteq \dot{\bigcup} \{x \dot{\cup} y : x \in u \wedge y \in v\} \doteq \dot{\bigcup} (u \dot{\cup} v)$;
- (iii) $u \dot{\cap} \dot{\bigcap} v \doteq \dot{\bigcap} \{u \dot{\cap} y : y \in v\}$;
- (iv) $\dot{\bigcap} u \dot{\cap} \dot{\bigcap} v \doteq \dot{\bigcap} \{x \dot{\cap} y : x \in u \wedge y \in v\} \doteq \dot{\bigcap} (u \dot{\cup} v)$.

Exercise 6.1.25 Show that the following are theorems of ZF:

- (i) $u \dot{\cup} \dot{\bigcap} v \doteq \dot{\bigcap} \{u \dot{\cup} y : y \in v\}$ if $v \neq \emptyset$;
- (ii) $\dot{\bigcap} u \dot{\cup} \dot{\bigcap} v \doteq \dot{\bigcap} \{x \dot{\cup} y : x \in u \wedge y \in v\}$ if $u, v \neq \emptyset$;
- (iii) $u \dot{\cap} \dot{\bigcup} v \doteq \dot{\bigcup} \{u \dot{\cap} y : y \in v\}$;
- (iv) $\dot{\bigcup} u \dot{\cap} \dot{\bigcup} v \doteq \dot{\bigcup} \{x \dot{\cap} y : x \in u \wedge y \in v\}$.

Exercise 6.1.26 Show that the following are theorems of ZF:

- (i) $\dot{\bigcup} u \setminus v \doteq \dot{\bigcup} \{x \setminus v : x \in u\}$; $\dot{\bigcap} u \setminus v \doteq \dot{\bigcap} \{x \setminus v : x \in u\}$ if $u \neq \emptyset$;
- (ii) $u \setminus \dot{\bigcup} v \doteq \dot{\bigcap} \{u \setminus y : y \in v\}$; $u \setminus \dot{\bigcap} v \doteq \dot{\bigcup} \{u \setminus y : y \in v\}$ if $v \neq \emptyset$;
- (iii) $\dot{\bigcup} u \doteq w \leftrightarrow \forall y (y \in u \rightarrow y \subseteq w)$
 $\wedge \forall z [(\forall y (y \in u \rightarrow y \subseteq z)) \rightarrow w \subseteq z]$;
- (iv) $\dot{\bigcap} u \doteq w \leftrightarrow \forall y (y \in u \rightarrow w \subseteq y)$
 $\wedge \forall z [(\forall y (y \in u \rightarrow z \subseteq y)) \rightarrow z \subseteq w]$.

Exercise 6.1.27 Show that $\text{ZF} \vdash \forall x [\dot{\bigcap} \{\{z : z \notin x\} \doteq \emptyset]$.

6.2 Mathematics in set theory I

As we have said, the notion of set, imbued with the properties expressed by the axioms of ZF, is sufficiently rich to represent most other mathematical notions. Therefore, although set theory is nominally just about sets, under this interpretation it is about all of mathematics.

It is (far) outside the scope of this book to develop large parts of mathematics in ZF, and in this section and its successor [Section 6.6](#) we focus mainly on topics central to logic and set theory itself. However, these examples should help the reader to appreciate the larger picture and provide guidelines for exploring how to represent other mathematical notions from areas such as algebra, analysis and topology.

We begin with the notions of function and relation which have been part of mathematics from its earliest days and play a role in essentially every branch of the subject. The representation of any mathematical notion in ZF involves two steps: finding the appropriate formal definition and showing that the essential properties of the informal objects may be translated to provable formal properties of this definition. This process is best appreciated by examples, and we start with the notion of ordered pair, and more generally ordered k -tuple. The ordered pair $\langle a, b \rangle$ of two objects (of any sort) is an object determined by a and b together with the order in which they occur. In other words, it is characterized by the property

$$\langle a, b \rangle = \langle c, d \rangle \iff a = c \text{ and } b = d.$$

Thus the task of representing the ordered pair in ZF is to find a set operation $\langle \cdot \rangle$ which satisfies the proposition below. This was, in fact, the subject of considerable research in the early 20th century, and several solutions were proposed before the following, due to Kuratowski and now standard, were adopted.

Definition 6.2.1

- (i) $\langle x, y \rangle := \{ \{x\}, \{x, y\} \}.$
- (ii) $\langle x_0, \dots, x_k \rangle := \langle \langle x_0, \dots, x_{k-1} \rangle, x_k \rangle \quad \text{for } k \geq 2.$

Formally we are defining for each $k > 1$ a set operation $\langle \cdot, \dots, \cdot \rangle_k$. That these are (provably) set operations is immediate.

Proposition 6.2.2 $\forall x \forall y \forall u \forall v [\langle x, y \rangle \doteq \langle u, v \rangle \leftrightarrow x \doteq u \wedge y \doteq v].$

Proof. This follows by a simple examination of several cases and is left as an exercise. $\square$

A more subtle but crucial property of the ordered pair is the following.

Definition 6.2.3 For all u and v , $u \dot{\times} v \doteq \{ \langle x, y \rangle : x \dot{\in} u \wedge y \dot{\in} v \}$ (*Cartesian product*).

To justify use of the notation $\{ \dot{} : \dot{} \}$ we need

Proposition 6.2.4 *Cartesian product is a set-operation.*

Proof. This follows by Comprehension from the straightforward calculation that

$$x \dot{\in} u \wedge y \dot{\in} v \rightarrow \{ \dot{x} \}, \{ \dot{x}, y \} \dot{\in} \dot{\phi}(u \dot{\cup} v) \rightarrow \langle x, y \rangle \dot{\in} \dot{\phi} \dot{\phi}(u \dot{\cup} v).$$

Hence,

$$u \dot{\times} v \doteq \{ w \dot{\in} \dot{\phi} \dot{\phi}(u \dot{\cup} v) : \exists x \exists y (x \dot{\in} u \wedge y \dot{\in} v \wedge w \doteq \langle x, y \rangle) \}. \quad \square$$

Remark 6.2.5 Although for most purposes the only important properties of the set operation $\langle \cdot, \cdot \rangle$ are those expressed by the two propositions, it will occasionally be convenient to make use of other “unintended” properties which follow from the particular choice of definition — for example that $\{x\} \dot{\in} \langle x, y \rangle$. If we chose an alternative definition — for example, $\{ \{ \dot{\emptyset}, x \}, \{ \{ \dot{\emptyset} \}, y \} \}$ — then the two propositions would still hold (exercise), but the unintended properties, and hence some minor details of a few proofs, would be different. Similar considerations apply to many other formalizations of mathematical notions.

The idea of a relation is already an abstraction from the more primitive notion of *relationship*. Thus from the natural relationship “smaller than” between natural numbers, we make an object $<$ which represents this relationship. The natural way to do this is as a set of ordered pairs:

$$< = \{ \langle m, n \rangle : m, n \in \omega \text{ and } m \text{ is smaller than } n \}.$$

This is, of course, exactly what we have been doing throughout this book and is already in terms of sets so provides no challenge to formalize along with some other notions related to relations.

Definition 6.2.6 For all r and z ,

- (i) $\dot{\text{Op}}(z) : \leftrightarrow \exists x \exists y [z \doteq \langle x, y \rangle]$, z is an *ordered pair*;
- (ii) $\dot{\text{Rel}}(r) : \leftrightarrow \forall z (z \dot{\in} r \rightarrow \dot{\text{Op}}(z))$, r is a *relation*;
- (iii) $\dot{\text{Dom}}(r) := \{ x : \exists y [\langle x, y \rangle \dot{\in} r] \}$, *domain* of r ;

$$(iv) \quad \dot{\text{Im}}(r) := \left\{ y : \exists x [\langle x, y \rangle \dot{\in} r] \right\}, \quad \text{\textit{image} of } r;$$

$$(v) \quad \dot{\text{Fld}}(r) := \dot{\text{Dom}}(r) \dot{\cup} \dot{\text{Im}}(r), \quad \text{\textit{field} of } r.$$

Note that we write here $:=$ rather than $\doteq$ since these are definitions of classes. Similarly, the symbol $\dot{\leftrightarrow}$ indicates that the left side is being defined as an abbreviation for the right.

Proposition 6.2.7 $\dot{\text{Dom}}$, $\dot{\text{Im}}$ and $\dot{\text{Fld}}$ are set operations.

Proof. These follow by Comprehension from the observation that $\langle x, y \rangle \dot{\in} r \rightarrow x, y \dot{\in} \dot{\bigcup} \dot{\bigcup} r$. $\square$

The representation of functions as a special sort of relation has become such a standard part of mathematics that it is easy to forget that it *is* a representation. Although calculus students generally think of functions as graphs, which are essentially sets of ordered pairs of real numbers, in more advanced mathematics functions are more commonly regarded dynamically as mappings which carry out an action of associating one object with another. Thus it is not immediately obvious that the set-theoretic representation of a function as a set of ordered pairs will in fact have all of the properties we expect. We indicate the most important of these below, but leave it to the reader to verify others as they arise.

Definition 6.2.8 For any f , x and y ,

$$(i) \quad \dot{\text{Fn}}(f) : \dot{\leftrightarrow} \dot{\text{Rel}}(f) \wedge$$

$$\forall x \forall y \forall z \left([\langle x, y \rangle \dot{\in} f \wedge \langle x, z \rangle \dot{\in} f] \rightarrow y \dot{=} z \right), \quad f \text{ is a } \text{\textit{function}};$$

$$(ii) \quad \dot{\text{Apl}}(f, x) := \dot{\bigcup} \left\{ y : \langle x, y \rangle \dot{\in} f \right\}, \quad \text{\textit{function application}};$$

$$(iii) \quad \dot{\text{Inj}}(f) : \dot{\leftrightarrow} \dot{\text{Fn}}(f) \wedge$$

$$\forall x \forall y \forall z \left([\langle x, z \rangle \dot{\in} f \wedge \langle y, z \rangle \dot{\in} f] \rightarrow x \dot{=} y \right), \quad f \text{ is } \text{\textit{injective}};$$

$$(iv) \quad f : x \dot{\rightarrow} y : \dot{\leftrightarrow} \dot{\text{Fn}}(f) \wedge \dot{\text{Dom}}(f) \dot{=} x \wedge \dot{\text{Im}}(f) \dot{\subseteq} y;$$

$$(v) \quad x y := \left\{ f : f : x \dot{\rightarrow} y \right\}.$$

The definition of $\dot{\text{Apl}}$ is slightly awkward because class operations must be totally defined, but we have

Proposition 6.2.9

- (i) $\mathbf{Apl}$ is a set operation.
- (ii) $\forall f \forall x [\mathbf{Fn}(f) \wedge x \in \mathbf{Dom}(f) \rightarrow \langle x, \mathbf{Apl}(f, x) \rangle \in f]$.
- (iii) $(x, y) \mapsto {}^x y$ is a set operation.

Proof. Left as an exercise. $\square$

Of course we usually write $f(x)$ for $\mathbf{Apl}(f, x)$; (ii) tells us that when f is a function and $x \in \mathbf{Dom}(f)$, this has the intended meaning and for $x \notin \mathbf{Dom}(f)$, $f(x) \doteq \emptyset$. Also, as usual, when $f : x \rightarrow y$ we say that f is **onto** y when $\mathbf{Im}(f) \doteq y$ and that f is a **bijection** when f is an injection and onto y .

Definition 6.2.10 For any f, g and u ,

- (i) $g \circ f \doteq \left\{ \langle x, z \rangle : \exists y [\langle x, y \rangle \in f \wedge \langle y, z \rangle \in g] \right\}$, **composition**;
- (ii) $f^{-1} \doteq \left\{ \langle y, x \rangle : \langle x, y \rangle \in f \right\}$, **inverse** of f ;
- (iii) $f \upharpoonright u \doteq \left\{ \langle x, y \rangle : \langle x, y \rangle \in f \wedge x \in u \right\}$, **restriction** of f to u ;
- (iv) $f^*u \doteq \mathbf{Im}(f \upharpoonright u)$, **f -image** of u .

Here and in many future examples we leave to the reader without comment the simple verifications that all of these are set operations.

Proposition 6.2.11 For any functions f and g ,

- (i) if $\mathbf{Dom}(f) \doteq \mathbf{Dom}(g)$ and $\forall x [x \in \mathbf{Dom}(f) \rightarrow f(x) \doteq g(x)]$, then $f \doteq g$;
- (ii) $g \circ f$ is a function and

$$\mathbf{Dom}(g \circ f) \doteq \left\{ x : x \in \mathbf{Dom}(f) \wedge f(x) \in \mathbf{Dom}(g) \right\};$$

- (iii) if $x \in \mathbf{Dom}(g \circ f)$, then $(g \circ f)(x) \doteq g(f(x))$;
- (iv) if f is injective, then

- (a) f^{-1} is a function;
- (b) $\mathbf{Dom}(f^{-1}) \doteq \mathbf{Im}(f)$;
- (c) $\mathbf{Im}(f^{-1}) \doteq \mathbf{Dom}(f)$;

- (d) $\forall x \left[x \in \dot{\text{Dom}}(f) \rightarrow f^{-1}(f(x)) \dot{=} x \right];$
 (e) $\forall y \left[y \in \dot{\text{Im}}(f) \rightarrow f(f^{-1}(y)) \dot{=} y \right];$
 (v) for any u , $\dot{\text{Dom}}(f \upharpoonright u) \dot{=} \dot{\text{Dom}}(f) \cap u$.

Proof. Again left as an exercise. $\square$

Some of this terminology is also naturally applicable to classes and class operations. Thus we write

$$\begin{aligned} R \text{ is a } \mathbf{relation-class} &: \leftrightarrow \forall z (z \in R \rightarrow \dot{\text{Op}}(z)); \\ \dot{\text{Dom}}(R) &:= \{ \{ x : \exists y (\langle x, y \rangle \in R) \} \}; \\ B \dot{\times} C &:= \{ \{ \langle x, y \rangle : x \in B \wedge y \in C \} \}; \\ F \text{ is } \mathbf{injective} &: \leftrightarrow \forall x \forall y [F(x) \dot{=} F(y) \rightarrow x \dot{=} y]; \\ \dot{\text{Im}}(F) &:= \{ \{ y : \exists x [F(x) \dot{=} y] \} \}; \\ F^{-1}(y) &:= \{ \{ z : \exists x [F(x) \dot{=} y \wedge z \in x] \} \} \dot{=} \bigcup \{ \{ x : F(x) \dot{=} y \} \}; \\ F \upharpoonright y &:= \{ \{ \langle z, F(z) \rangle : z \in y \} \} \dot{=} \{ \{ \langle z, v \rangle : F(z) \dot{=} v \wedge z \in y \} \}; \\ F \upharpoonright_x y &:= \{ \{ \langle z, F(x, z) \rangle : z \in y \} \} \dot{=} \{ \{ \langle z, v \rangle : F(x, z) \dot{=} v \wedge z \in y \} \}; \\ F^*A &:= \dot{\text{Im}}(F \upharpoonright A) \dot{=} \{ \{ y : \exists x [F(x) \dot{=} y \wedge x \in A] \} \}; \end{aligned}$$

and other similar expressions. Note that by our definitions every (unary) class operation F has domain V .

These generally have the expected meaning, but it is worth spelling out a few special properties.

Proposition 6.2.12 For any (unary) class operation F ,

(i) the mappings $u \mapsto F \upharpoonright u$ and $(x, u) \mapsto F \upharpoonright_x u$ are set operations;
 if F is injective, then

- (ii) F^{-1} is a set operation;
 (iii) $\forall x [F(x) \text{ is a set} \rightarrow F^{-1}(F(x)) \dot{=} x];$
 (iv) $\forall y [y \in \dot{\text{Im}}(F) \rightarrow F(F^{-1}(y)) = y]. \quad \square$

Corollary 6.2.13 For any (unary) class operation F and any class A ,

- (i) $A \text{ is a set} \rightarrow F^*A \text{ is a set};$
 (ii) $F \text{ is injective} \wedge A \text{ is a proper class} \rightarrow F^*A \text{ is a proper class}.$

Proof. Part (i) is a direct consequence of Replacement. For (ii), by the proposition $(F^{-1})^*(F^*(A)) \doteq A$. Thus by Replacement, $F^*(A)$ is a set $\rightarrow$ A is a set. $\square$

We shall need in the following sections many special sorts of relations, and we summarize these here for reference. All are simply formal versions of familiar properties that we have used regularly in the preceding chapters, and the main point here is to observe that they are all easily expressible in $L_{\in}$.

Definition 6.2.14 For any relations r and s ,

$$(i) \ r \text{ is } \mathbf{reflexive} : \leftrightarrow \forall x \left[x \in \text{Fld}(r) \rightarrow \langle x, x \rangle \in r \right];$$

$$(ii) \ r \text{ is } \mathbf{irreflexive} : \leftrightarrow \forall x \left[\langle x, x \rangle \notin r \right];$$

$$(iii) \ r \text{ is } \mathbf{transitive} : \leftrightarrow$$

$$\forall x \forall y \forall z \left[\langle x, y \rangle \in r \wedge \langle y, z \rangle \in r \rightarrow \langle x, z \rangle \in r \right];$$

$$(iv) \ r \text{ is a } \mathbf{strict\ partial\ ordering} : \leftrightarrow \ r \text{ is irreflexive and transitive;}$$

$$(v) \ r \text{ is } \mathbf{connected} : \leftrightarrow$$

$$\forall x \forall y \left[x, y \in \text{Fld}(r) \rightarrow \langle x, y \rangle \in r \vee x = y \vee \langle y, x \rangle \in r \right];$$

$$(vi) \ r \text{ is a } \mathbf{linear\ ordering} : \leftrightarrow$$

$$r \text{ is a strict partial ordering and is connected;}$$

$$(vii) \ r \text{ is } \mathbf{well-founded} : \leftrightarrow$$

$$\forall u \left[\emptyset \neq u \subseteq \text{Fld}(r) \rightarrow \exists x (x \in u \wedge \forall y [y \in u \rightarrow \langle y, x \rangle \notin r]) \right];$$

$$(viii) \ r \text{ is a } \mathbf{well-ordering} : \leftrightarrow$$

$$r \text{ is a linear ordering and is well-founded;}$$

$$(ix) \ r \text{ is } \mathbf{symmetric} : \leftrightarrow \forall x \forall y [\langle x, y \rangle \in r \rightarrow \langle y, x \rangle \in r];$$

$$(x) \ r \text{ is an } \mathbf{equivalence\ relation} : \leftrightarrow$$

$$r \text{ is reflexive, symmetric and transitive;}$$

$$(xi) \quad r \simeq_f s : \leftrightarrow \text{Inj}(f) \wedge \text{Dom}(f) \doteq \text{Fid}(r) \wedge \text{Im}(f) \doteq \text{Fid}(s) \\ \wedge \forall x \forall y [\langle x, y \rangle \in r \leftrightarrow \langle f(x), f(y) \rangle \in s];$$

$$(xii) \quad r \simeq s : \leftrightarrow \exists f [r \simeq_f s].$$

Again, when appropriate we apply the same terminology to relation-classes — for example,

$$R \text{ is } \textbf{reflexive} : \leftrightarrow \forall x [x \in \text{Fid}(R) \rightarrow \langle x, x \rangle \in R]$$

$$R \text{ is } \textbf{transitive} : \leftrightarrow \forall x \forall y \forall z [\langle x, y \rangle \in R \wedge \langle y, z \rangle \in R \rightarrow \langle x, z \rangle \in R].$$

In particular, we say that

$$R \text{ is a } \textbf{class partial ordering} : \leftrightarrow R \text{ is irreflexive and transitive,}$$

and similarly for linear and well-orderings.

Note that $R \simeq S$ is generally *not* expressible by an $L_{\in}$ -sentence, since the existential quantifier would have to be over class operations. However, for a given class operation F we have

$$R \simeq_F S : \leftrightarrow \text{Inj}(F) \wedge \text{Fid}(S) \doteq F^*(\text{Fid}(R)) \\ \wedge \forall x \forall y (x, y \in \text{Fid}(R) \rightarrow [\langle x, y \rangle \in R \leftrightarrow \langle F(x), F(y) \rangle \in S]).$$

We turn next to the representation $\dot{\omega}$ of the natural numbers in ZF.

Definition 6.2.15 $\dot{0} := \emptyset$, $\dot{1} := \{\dot{0}\}$, and generally $\overbrace{k+1}^{\dot{}} := \dot{S}k \doteq k \cup \{\dot{k}\}$.

Thus more informally, $\dot{k} := \{\dot{0}, \dots, \overbrace{k-1}^{\dot{}}\}$. Note that the brace over $k-1$ is not formally part of the notation but only a device for indicating the scope of the dot.

Corollary 6.2.16 For all $k \in \omega$, $\dot{k} \in \dot{\omega}$. $\square$

Note that this asserts the ZF-provability of an infinite list of sentences — for example, $\dot{2} \in \dot{\omega}$ is the sentence

$$\exists y (\forall w (w \in y \leftrightarrow w \in \dot{1} \vee w \doteq \dot{1}) \\ \wedge \forall u ((\dot{0} \in u \wedge \forall z [z \in u \rightarrow \dot{S}z \in u]) \rightarrow y \in u)),$$

with further replacements for $\dot{0}$ and $\dot{1}$. There is no way in $L_{\in}$ to express the idea that $\dot{\omega} \doteq \{\dot{0}, \dots, \dot{k}, \dots\}$. In a model $\mathfrak{A}$ of ZF it may or may not be true that $\dot{\omega}^{\mathfrak{A}} = \{\dot{k}^{\mathfrak{A}} : k \in \omega\}$. This is, of course, similar to the existence of both standard and non-standard models of the various theories of arithmetic. Built in to the definition of $\dot{\omega}$ is the principle of induction.

$\dot{\omega}$ -Induction Theorem 6.2.17 *For any class B,*

$$\left[\dot{0} \in B \wedge \forall z \left(z \in B \dot{\cap} \dot{\omega} \rightarrow \dot{S}z \in B \right) \right] \rightarrow \dot{\omega} \subseteq B.$$

Proof. Given B satisfying the hypothesis, by Comprehension $B \dot{\cap} \dot{\omega}$ is a set and clearly

$$\dot{0} \in B \dot{\cap} \dot{\omega} \wedge \forall z \left(z \in B \dot{\cap} \dot{\omega} \rightarrow \dot{S}z \in B \dot{\cap} \dot{\omega} \right).$$

Hence by its definition, $\dot{\omega} \subseteq B \dot{\cap} \dot{\omega}$, so also $\dot{\omega} \subseteq B$. $\square$

Applications are typically to $B \subseteq \dot{\omega}$, and as usual we call the assertion $z \in B \dot{\cap} \dot{\omega}$ the **induction hypothesis**. We have stated this in terms of classes, but of course it can equivalently be stated in terms of formulas: for any $L_{\in}$ -formula ϕ ,

$$\left[\phi(\dot{0}) \wedge \forall z \left(\phi(z) \wedge z \in \dot{\omega} \rightarrow \phi(\dot{S}z) \right) \right] \rightarrow \forall z [z \in \dot{\omega} \rightarrow \phi(z)],$$

and similarly for ϕ with additional free variables. To see that $\dot{\omega}$ has the properties required to consider it a faithful representation of the set of natural numbers ω , it is natural to start with the axioms of P (Definition 4.1.1). Precisely, we shall show that P is relatively interpretable in ZF. As in Definition 2.6.43, this means that we should provide $L_{\in}$ -definitions for the non-logical symbols of P such that if ZF^+ is the extension of ZF generated by these definitions, then $\text{P}^{(\dot{\omega})} \subseteq \text{ZF}^+$, or equivalently, for each axiom ϕ of P, $\text{ZF} \vdash \phi^{(\dot{\omega})-}$, the relativized $L_{\in}$ -translation of ϕ obtained by replacing each non-logical symbol by its definition. Here we get the same effect by simply regarding these symbols as abbreviations. Since we already have definitions for $\dot{\omega}$, $\dot{0}$ and $\dot{S}$, we can start immediately with

Proposition 6.2.18

- (i) $\text{ZF} \vdash (S1)^{(\dot{\omega})}$ — in fact, $\text{ZF} \vdash \forall x \neg \dot{0} \doteq \dot{S}x$.
- (ii) $\text{ZF} \vdash (S2)^{(\dot{\omega})}$ — in fact, $\text{ZF} \vdash \forall x \forall y [\dot{S}x \doteq \dot{S}y \rightarrow x \doteq y]$.

Proof. (i) is immediate, since $x \in \dot{S}x$ but $x \notin \dot{0}$. For (ii), suppose that $\dot{S}x \doteq \dot{S}y$ — that is, $x \dot{\cup} \{x\} \doteq y \dot{\cup} \{y\}$ — but $x \neq y$. Then $x \notin \{y\}$, so since $x \in \dot{S}y$ we must have $x \in y$. Similarly, $y \in x$. This contradicts Exercise 6.1.21. $\square$

To go further requires providing definitions for $\dot{<}$, $\dot{+}$ and $\dot{\times}$. It is a happy accident of our choice of $\dot{\omega}$ (see Remark 6.2.5) that $\dot{<}$ can be given the following simple definition: $x \dot{<} y :\leftrightarrow x \in y$. As usual we write $x \dot{\leq} y$ for $x \dot{<} y \vee x \doteq y$ — that is, $x \in \dot{S}y$. This is another instance of the phenomenon discussed in Remark 6.2.5. Then we have immediately

Proposition 6.2.19

- (i) $\text{ZF} \vdash (L1)^{(\dot{\omega})} \text{ — in fact, } \text{ZF} \vdash \forall x \neg x \dot{<} \dot{0}$.
(ii) $\text{ZF} \vdash (L2)^{(\dot{\omega})} \text{ — in fact, } \text{ZF} \vdash \forall x \forall y \left[x \dot{<} \dot{S}y \leftrightarrow x \dot{<} y \vee x \dot{=} y \right]$. $\square$

To establish $(L3)^{(\dot{\omega})}$, we need first

Definition 6.2.20

- (i) x is **transitive** : $\leftrightarrow \forall y [y \dot{\in} x \rightarrow y \dot{\subseteq} x]$.
(ii) $\dot{<}_{\dot{\omega}} := \{ \langle \dot{x}, \dot{y} \rangle : x, y \dot{\in} \dot{\omega} \wedge x \dot{<} y \}$.
(iii) $\dot{\leq}_{\dot{\omega}} := \{ \langle \dot{x}, \dot{y} \rangle : x, y \dot{\in} \dot{\omega} \wedge x \dot{\leq} y \}$.

Note that x is transitive $\leftrightarrow \bigcup x \dot{\subseteq} x \leftrightarrow x \dot{\subseteq} \dot{\phi} x$.

Lemma 6.2.21 For all $x, y \dot{\in} \dot{\omega}$,

- (i) x is transitive $\wedge x \dot{\subseteq} \dot{\omega}$; hence $\dot{\omega}$ is transitive;
(ii) $\dot{<}_{\dot{\omega}}$ and $\dot{\leq}_{\dot{\omega}}$ are transitive relations;
(iii) $x \dot{\subseteq} y \leftrightarrow x \dot{\leq} y$;
(iv) $x \dot{<} y \leftrightarrow \dot{S}x \dot{\leq} y$.

Proof. For (i), let $\mathbf{B} := \{ x \dot{\in} \dot{\omega} : x \text{ is transitive} \wedge x \dot{\subseteq} \dot{\omega} \}$; we show by $\dot{\omega}$ -induction that $\dot{\omega} \dot{\subseteq} \mathbf{B}$. Clearly $\dot{0} \dot{\in} \mathbf{B}$. Suppose $x \dot{\in} \mathbf{B}$ and $y \dot{\in} \dot{S}x$. Then either $y \dot{\in} x$ or $y \dot{=} x$. In either case $y \dot{\subseteq} x \dot{\subseteq} \dot{S}x$ and $y \dot{\in} \dot{\omega}$. Hence $\dot{S}x$ is transitive and $\dot{S}x \dot{\subseteq} \dot{\omega}$, so $\dot{S}x \dot{\in} \mathbf{B}$. Parts (ii) and (iii)($\leftarrow$) now follow immediately. For (iii)($\rightarrow$), let

$$\mathbf{C} := \{ y \dot{\in} \dot{\omega} : \forall x (x \dot{\in} \dot{\omega} \wedge x \dot{\subseteq} y \rightarrow x \dot{\leq} y) \};$$

we show by $\dot{\omega}$ -induction that $\dot{\omega} \dot{\subseteq} \mathbf{C}$. $\dot{0} \dot{\in} \mathbf{C}$, since $x \dot{\subseteq} \dot{0} \rightarrow x \dot{=} \dot{0}$ and $\dot{0} \dot{\leq} \dot{0}$. Assume as induction hypothesis that $y \dot{\in} \mathbf{C}$ and that $x \dot{\subseteq} \dot{S}y$. If $x \dot{\subseteq} y$, then $x \dot{\leq} y \dot{\leq} \dot{S}y$. Otherwise, $y \dot{\in} x$, so by (i), $y \dot{\subseteq} x$ and therefore $x \dot{=} \dot{S}y$ so again $x \dot{\leq} \dot{S}y$. Thus $\dot{S}y \dot{\in} \mathbf{C}$. Part (iv)($\leftarrow$) is straightforward; ($\rightarrow$) is proved by another induction. Let

$$\mathbf{D} := \{ y \dot{\in} \dot{\omega} : \forall x (x \dot{\in} y \rightarrow \dot{S}x \dot{\in} y \vee \dot{S}x \dot{=} y) \}.$$

Clearly $\dot{0} \dot{\in} \mathbf{D}$. Assume as induction hypothesis that $y \dot{\in} \mathbf{D}$ and that $x \dot{\in} \dot{S}y$, so $x \dot{\in} y$ or $x \dot{=} y$. If $x \dot{\in} y$, then by the induction hypothesis, $\dot{S}x \dot{\in} y$ or $\dot{S}x \dot{=} y$ so by transitivity, $\dot{S}x \dot{\in} \dot{S}y$. If $x \dot{=} y$, then $\dot{S}x \dot{=} \dot{S}y$, so in either case $\dot{S}y \dot{\in} \mathbf{D}$. $\square$

Now we can establish the third order-axiom of P.

Proposition 6.2.22 $\text{ZF} \vdash (L3)^{(\dot{\omega})}$ — that is,

$$\text{ZF} \vdash \forall x \forall y [x \dot{\in} \dot{\omega} \wedge y \dot{\in} \dot{\omega} \rightarrow (x \dot{<} y) \vee (x \dot{=} y) \vee (y \dot{<} x)].$$

Proof. Let

$$\mathbf{E} := \{x \dot{\in} \dot{\omega} : \forall y [y \dot{\in} \dot{\omega} \rightarrow (x \dot{<} y) \vee (x \dot{=} y) \vee (y \dot{<} x)]\};$$

we prove by $\dot{\omega}$ -induction that $\dot{\omega} \subseteq \mathbf{E}$. $\dot{0} \dot{\in} \mathbf{E}$, since for any y , $\dot{0} \subseteq y$, so by (iii) of the lemma, $\dot{0} \dot{\in} y \vee \dot{0} \dot{=} y$. Assume as induction hypothesis that $x \dot{\in} \mathbf{E}$ and that $y \dot{\in} \dot{\omega}$ is such that $y \notin \dot{S}x$. Then $y \notin x \wedge y \neq x$, so $x \dot{\in} y$. Then by (iv) of the lemma, $\dot{S}x \dot{\in} y \vee \dot{S}x \dot{=} y$, so $\dot{S}x \dot{\in} \mathbf{E}$. $\square$

Corollary 6.2.23 Every nonempty subset of $\dot{\omega}$ has a unique least element — that is,

$$\forall u [\dot{0} \neq u \subseteq \dot{\omega} \rightarrow \exists! y (y \dot{\in} u \wedge \forall z [z \dot{\in} u \rightarrow y \dot{\leq} z])].$$

Proof. Given $\dot{0} \neq u \subseteq \dot{\omega}$, by Foundation u has an $\dot{\in}$ -minimal element y — that is, $y \dot{\in} u \wedge \forall z [z \dot{<} y \rightarrow z \notin u]$. Hence, by the proposition, $z \dot{\in} u \rightarrow z \not\dot{<} y \rightarrow y \dot{\leq} z$. Uniqueness of y follows also from the proposition. $\square$

It now remains to find appropriate definitions for $\dot{+}$ and $\dot{\times}$. This requires the following general principle, which formalizes the notion of recursive definition that we have used throughout the book.

$\dot{\omega}$ -Recursion Theorem 6.2.24 For any set operations $\mathbf{G}$ (unary) and $\mathbf{H}$ (ternary), there exists a set operation $\mathbf{F}$ such that for all x and all $y \dot{\in} \dot{\omega}$,

- (i) $\mathbf{F}(x, \dot{0}) \dot{=} \mathbf{G}(x)$;
- (ii) $\mathbf{F}(x, \dot{S}y) \dot{=} \mathbf{H}(\mathbf{F}(x, y), x, y)$.

Furthermore, $\mathbf{F}$ is **unique on** $\dot{\omega}$ in the sense that for any other class operation $\mathbf{F}'$ which satisfies (i) and (ii), $\forall x (\forall y \dot{\in} \dot{\omega}) \mathbf{F}(x, y) \dot{=} \mathbf{F}'(x, y)$.

Proof. Given $\mathbf{G}$ and $\mathbf{H}$, let $\theta(f, x, y)$ denote the following $L_{\in}$ formula:

$$\begin{aligned} \text{Fn}(f) \wedge y \dot{\in} \dot{\omega} \wedge \text{Dom}(f) \dot{=} \dot{S}y \wedge f(\dot{0}) \dot{=} \mathbf{G}(x) \\ \wedge \forall u [u \dot{<} y \rightarrow f(\dot{S}u) \dot{=} \mathbf{H}(f(u), x, u)], \end{aligned}$$

and let $\mathbf{F}$ be the class operation

$$\mathbf{F}(x, y) := \{z : \exists f [\theta(f, x, y) \wedge z \dot{\in} f(y)]\}.$$

Intuitively, a function f which satisfies $\theta(f, x, y)$ is an approximation to F valid for all u in its domain $\dot{S}y$, in particular for $u \doteq y$. We establish first that such approximations exist and are unique — that is,

- (1) $\forall x (\forall y \dot{\in} \dot{\omega}) (\forall z \dot{<} y) \forall f \forall g [\theta(f, x, y) \wedge \theta(g, x, z) \rightarrow f \upharpoonright \dot{S}z \doteq g];$
- (2) $\forall x (\forall y \dot{\in} \dot{\omega}) \exists f \theta(f, x, y).$

For (1), let

$$J(x) := \{u \dot{\in} \dot{\omega} : (\forall y \dot{\in} \dot{\omega}) (\forall z \dot{<} y) \forall f \forall g [\theta(f, x, y) \wedge \theta(g, x, z) \wedge u \dot{\leq} z \rightarrow f(u) \doteq g(u)]\};$$

it suffices to show by $\dot{\omega}$ -induction that $\forall x [J(x) \dot{\in} \dot{\omega}]$. $\dot{0} \dot{\in} J(x)$ because if $\theta(f, x, y)$ and $\theta(g, x, z)$, then $f(\dot{0}) \doteq G(x) \doteq g(\dot{0})$. Assume as induction hypothesis that $u \dot{\in} J(x)$, and fix $z \dot{<} y \dot{\in} \dot{\omega}$, f and g such that $\theta(f, x, y)$, $\theta(g, x, z)$ and $\dot{S}u \dot{\leq} z$. Then $u \dot{<} z$, so $f(u) \doteq g(u)$ and hence

$$f(\dot{S}u) \doteq H(f(u), x, u) \doteq H(g(u), x, u) \doteq g(\dot{S}u),$$

so also $\dot{S}u \dot{\in} J(x)$. For (2), let

$$K(x) := \{y : y \dot{\in} \dot{\omega} \wedge \exists f \theta(f, x, y)\};$$

it suffices to show by $\dot{\omega}$ -induction that $\forall x [K(x) \dot{\in} \dot{\omega}]$. $\dot{0} \dot{\in} K(x)$ because if $f \doteq \{\langle \dot{0}, G(x) \rangle\}$, then $\theta(f, x, \dot{0})$. Assume as induction hypothesis that $y \dot{\in} K(x)$, and fix h such that $\theta(h, x, y)$. Then if $f \doteq h \cup \{\langle \dot{S}y, H(h(y), x, y) \rangle\}$, easily $\theta(f, x, \dot{S}y)$, so $\dot{S}y \dot{\in} K(x)$.

Now, it follows from (1) and (2) that $\forall x (\forall y \dot{\in} \dot{\omega}) \exists! f \theta(f, x, y)$; hence there is a set operation $y \mapsto f_{x,y}$ such that $\theta(f_{x,y}, x, y)$. Then by (1) and the definition of F , for all $y \dot{\in} \dot{\omega}$, $F(x, y) \doteq f_{x,y}(y) \doteq f_{x, \dot{S}y}(y)$, so $F(x, \dot{0}) \doteq G(x)$ and

$$\begin{aligned} F(x, \dot{S}y) &\doteq f_{x, \dot{S}y}(\dot{S}y) \doteq H(f_{x, \dot{S}y}(y), x, y) \\ &\doteq H(f_{x,y}(y), x, y) \doteq H(F(x, y), x, y). \end{aligned}$$

The uniqueness of F on $\dot{\omega}$ follows by a straightforward induction. $\square$

As a first application, we have

Definition 6.2.25 $\dot{+}$ and $\dot{\times}$ are the set operations unique on $\dot{\omega}$ such that for all $x, y \dot{\in} \dot{\omega}$,

- (i) $x \dot{+} \dot{0} \doteq x$ and $x \dot{+} \dot{S}y \doteq \dot{S}(x \dot{+} y);$
- (ii) $x \dot{\times} \dot{0} \doteq \dot{0}$ and $x \dot{\times} \dot{S}y \doteq (x \dot{\times} y) \dot{+} x.$

Corollary 6.2.26 $\text{ZF} \vdash (A1)^{(\dot{\omega})} \wedge (A2)^{(\dot{\omega})} \wedge (M1)^{(\dot{\omega})} \wedge (M2)^{(\dot{\omega})}$. $\square$

Theorem 6.2.27 $\mathcal{P}$ is relatively interpretable in ZF.

Proof. We have shown explicitly for each axiom θ of $\mathcal{Q}$ that $\text{ZF} \vdash \theta^{(\dot{\omega})}$; that this holds also for each instance $\bar{I}_{\phi,x}$ of the L_{Ω} -induction schema is immediate from the $\dot{\omega}$ -induction Theorem 6.2.17. $\square$

We have thus the fact announced in [Section 5.5](#).

First Incompleteness Theorem for ZF 6.2.28 ZF is contagiously undecidable and contagiously incomplete — that is, for any $L_{\in}$ -theory T compatible with ZF,

- (i) T is undecidable;
- (ii) if T is recursively axiomatizable, then T is incomplete. $\square$

Remark 6.2.29 By Proposition 5.5.4, there is a finitely axiomatizable subtheory $U \subseteq \text{ZF}$ such that $\mathcal{Q}$ is relatively interpretable in U and thus U is also contagiously undecidable and incomplete. One description of such a theory could be extracted from a careful analysis of just which of the axioms of ZF are needed to verify that for each of the finitely many axioms θ of $\mathcal{Q}$, the relativization $\theta^{(\dot{\omega})}$ is a theorem of ZF. However, it is known that in fact the seemingly very weak theory $\mathcal{U}$ generated by the three axioms

Extensionality $\forall z (z \in x \leftrightarrow z \in y) \rightarrow x = y$

Nullset $\exists z \forall u (u \notin z)$

Successor $\forall x \forall y \exists z \forall u [u \dot{\in} z \leftrightarrow u \dot{\in} x \vee u \dot{=} y]$

is already contagiously undecidable and incomplete. In outline, the proof of this goes as follows. Let $\mathcal{Q}^*$ denote the subtheory of $\mathcal{Q}$ generated by the axioms $(S1)$, $(S2)$, $(L1)$, $(A1)$, $(A2)$, $(M1)$ and $(M2)$ together with the following infinite schemas which are consequences of $(L2)$ and $(L3)$: for all $n \in \omega$,

$(L2)_n^* \quad \forall x [x \dot{<} \dot{S}n \leftrightarrow (x \dot{<} \dot{n} \vee x \dot{=} \dot{n})]$

$(L3)_n^* \quad \forall x [(x \dot{<} \dot{n}) \vee (x \dot{=} \dot{n}) \vee (\dot{n} \dot{<} x)].$

Then we have

- (3) every recursive function is $\mathcal{Q}^*$ -representable;
- (4) $\mathcal{Q}^*$ is interpretable in $\mathcal{U}$.

The proof of (3) simply involves examining carefully the proof for $\mathcal{Q}$ to notice that these instances of $(L2)$ and $(L3)$ suffice. The proof of (4) is quite

intricate. Let Q^{**} denote the theory in the language without $\dot{<}$ generated by the axioms (S1), (S2), (S3), (A1), (A2), (M1) and (M2), where, as in [Section 2.5](#), (S3) is

$$\forall x(\neg x \dot{=} \dot{0} \rightarrow \exists y[x \dot{=} \dot{S}y]).$$

Then one shows

- (5) Q^* is interpretable in Q^{**} ;
- (6) Q^{**} is interpretable in U .

(5) is relatively simple and involves defining $x \dot{<} y$ by $\exists z(\dot{S}z \dot{+} x \dot{=} y)$ and verifying that all of the instances $(L2)_n^*$ and $(L3)_n^*$ are then provable in Q^{**} . The proof of (6) is quite long and delicate; see [Mon; 16.8-16.48].

We shall see in [Section 6.6](#) that proper extensions of P are also relatively interpretable in ZF , since also $ZF \vdash \text{Con}_P^{(\dot{\omega})}$, although by the Second Incompleteness Theorem $P \not\vdash \text{Con}_P$. On the other hand

Proposition 6.2.30 *$\text{Th}(\Omega)$ is not relatively interpretable in ZF .*

Proof. By the remark following Definition 2.6.39, if $\text{Th}(\Omega)$ were relatively interpretable in ZF then $\text{Th}(\Omega)^{(\dot{\omega})}$ would be faithfully interpretable in ZF . Since ZF is effectively axiomatizable and hence effectively enumerable, it would follow from Proposition 4.1.6 that $\text{Th}(\Omega)$ is also effectively enumerable, contrary to Theorem 4.1.7. Of course, the same argument could be made using recursive enumerability. $\square$

Let T denote the set of L_Ω -sentences ϕ such that $ZF \vdash \phi^{(\dot{\omega})}$. Then T is an L_Ω -theory which properly extends P but $T \neq \text{Th}(\Omega)$. Can we at least conclude that $T \subseteq \text{Th}(\Omega)$? If we assume that ZF is consistent and has a transitive $\in$ -model, then it is possible to show that ZF has an ω -**standard** model — that is a model $\mathfrak{A}$ such that $\dot{\omega}^\mathfrak{A} = \omega$. In such a model the set of L_Ω -sentences ϕ such that $\phi^{(\dot{\omega})}$ is true in $\mathfrak{A}$ is exactly $\text{Th}(\Omega)$ — hence $T \subseteq \text{Th}(\Omega)$. However, the extension of the Second Incompleteness Theorem to ZF , which is discussed in [Section 6.6](#), implies that Con_{ZF} is not a theorem of ZF and hence the consistency of ZF is not provable in ordinary mathematics. Note that if ZF is *not* consistent, then T is the entire set of L_Ω -sentences.

As in [Section 4.2](#), we can also derive the alternative **course-of-values** version of $\dot{\omega}$ -recursion. Note that in the set-theoretic context it is more natural, since there is no need for coding of sequences.

Proposition 6.2.31 *For any ternary set operation G , there exists a set operation F such that for all x and all $y \in \dot{\omega}$,*

$$F(x, y) \dot{=} G(F \upharpoonright_x y, x, y).$$

Furthermore, F is unique on $\dot{\omega}$.

Proof. By $\dot{\omega}$ -recursion there exists a set operation $\bar{F}$ such that for all x and all $y \in \dot{\omega}$,

- (i) $\bar{F}(x, \dot{0}) \doteq \dot{\emptyset}$;
- (ii) $\bar{F}(x, \dot{S}y) \doteq \bar{F}(x, y) \dot{\cup} \{ \dot{\langle y, G(\bar{F}(x, y), x, y) \rangle} \}$.

It is easy to prove by induction that for all $y \in \dot{\omega}$,

$$\begin{aligned} \dot{F}n(\bar{F}(x, y)) \wedge \dot{\text{Dom}}(\bar{F}(x, y)) &\doteq y \wedge \\ &\forall w [y \dot{\leq} w \in \dot{\omega} \rightarrow \bar{F}(x, y) \doteq \bar{F}(x, w) \upharpoonright y]. \end{aligned}$$

Hence if we set $F(x, y) := \bar{F}(x, \dot{S}y)(y)$, it follows that $\bar{F}(x, y) \doteq F \upharpoonright_x y$, so

$$F(x, y) \doteq G(\bar{F}(x, y), x, y) \doteq G(F \upharpoonright_x y, x, y),$$

as desired. $\square$

Intuitively, a finite set is one which has k elements for some $k \in \omega$; the natural formalization of this is

Definition 6.2.32 For any x, y and f ,

- (i) $x \dot{\approx}_f y \leftrightarrow \dot{\text{Inj}}(f) \wedge \dot{\text{Dom}}(f) \doteq x \wedge \dot{\text{Im}}(f) \doteq y$;
- (ii) $x \dot{\approx} y \leftrightarrow \exists f [x \dot{\approx}_f y]$, x and y are **equipotent**;
- (iii) x is **finite** $\leftrightarrow \exists y [y \in \dot{\omega} \wedge x \dot{\approx} y]$; otherwise x is **infinite**;
- (iv) x is **denumerable** $\leftrightarrow x \dot{\approx} \dot{\omega}$;
- (v) x is **countable** $\leftrightarrow x$ is finite or denumerable; otherwise x is **uncountable**.

Proposition 6.2.33 For all u, y and z ,

- (i) $\dot{\approx}$ is a class equivalence relation;
- (ii) $\dot{\emptyset}$ is **finite**;
- (iii) u is **finite** $\rightarrow u \dot{\cup} \{z\}$ is **finite**;
- (iv) $y \in \dot{\omega} \rightarrow y$ is **finite**;
- (v) u is **finite** $\rightarrow \exists! y [y \in \dot{\omega} \wedge u \dot{\approx} y]$;
- (vi) $u \dot{\approx} \dot{S}y \rightarrow \exists v \exists w [v \dot{\subseteq} u \wedge v \dot{\approx} y \wedge \neg w \in v \wedge u \doteq v \dot{\cup} \{w\}]$;
- (vii) u is **finite** $\wedge u \dot{\subseteq} \dot{\omega} \rightarrow \exists y (y \in \dot{\omega} \wedge u \dot{\subseteq} y)$;
- (viii) y is **finite** $\wedge u \dot{\subseteq} y \rightarrow u$ is **finite**;

(ix) for any class operation F , u is finite $\rightarrow F^*u$ is finite.

Proof. Most of these are simple and are left to the exercises. For (i), $x \approx x$ via the function $\{\langle \dot{u}, u \rangle : u \in x\}$. If $x \approx y$ via f , then $y \approx x$ via f^{-1} . If $x \approx y$ and $y \approx z$ via, respectively, f and g , then $x \approx z$ via the composition $g \circ f$. For (vii), let

$$B := \{ \dot{z} \in \dot{\omega} : \forall u [u \approx z \wedge u \subseteq \dot{\omega} \rightarrow \exists y (y \in \dot{\omega} \wedge u \subseteq y)] \};$$

it suffices to prove by $\dot{\omega}$ -induction that $\dot{\omega} \subseteq B$. $\dot{0} \in B$ since if $u \approx \dot{0}$, then $u = \dot{0}$, so $u \subseteq \dot{0}$. Assume as induction hypothesis that $z \in B$, and fix $u \approx \dot{S}z$. By (vi), there is $v \subseteq u$ and $w \in \dot{\omega}$ such that $v \approx z$ and $u = v \dot{\cup} \{w\}$. By the induction hypothesis, $v \subseteq y$ for some $y \in \dot{\omega}$. If $w \in y$, then also $u \subseteq y$. Otherwise $y \dot{\subset} w$, and it follows that $u \subseteq \dot{S}w$. Hence $\dot{S}z \in B$. $\square$

Of particular importance when we come to representing first-order logic in ZF will be the notion of finite sequence. Recall from Definition 6.2.8 that yu is the set of all functions $f : y \rightarrow u$. In particular, for $y \in \dot{\omega}$ we call such an f a **finite sequence** from u of **length** y , and we set

Definition 6.2.34 For any u , ${}^{<\dot{\omega}}u := \bigcup \{ {}^yu : y \in \dot{\omega} \}$.

Note that although for $k \in \omega$, ku is not the same as the k -fold Cartesian product of u , there is a natural bijection $f \mapsto \langle f(\dot{0}), \dots, f(\overbrace{k-1}) \rangle$ and occasionally we shall identify these. In some models, ${}^{<\dot{\omega}}u$ includes finite sequences of both standard ($k \in \omega$) and non-standard lengths. Note also that for any u , ${}^{\dot{0}}u = \{ \emptyset \}$.

We have dealt with the interpretation of the arithmetic of natural numbers in ZF in some detail so that it may serve as a model for other theories that we shall treat more briefly. First we sketch how to represent other number systems in set theory. A natural way to represent integers is as pairs of natural numbers. For example, -7 could be represented as any of the pairs $\langle \dot{5}, \dot{12} \rangle$, $\langle \dot{3}, \dot{10} \rangle$, $\langle \dot{0}, \dot{7} \rangle$ etc. Rather than choosing one of these, it will make the definitions easier if we instead use the set of all of them — that is, we define

$$\langle x, y \rangle \sim_{\mathbb{Z}} \langle u, v \rangle : \leftrightarrow x \dot{+} v = y \dot{+} u,$$

verify that $\text{ZF} \vdash \sim_{\mathbb{Z}}$ is an equivalence relation, and define

$$[\langle x, y \rangle] := \{ \langle u, v \rangle : u, v \in \dot{\omega} \wedge \langle x, y \rangle \sim_{\mathbb{Z}} \langle u, v \rangle \},$$

the equivalence class of $\langle x, y \rangle$. Then, for example, $-\dot{7} := [\langle \dot{5}, \dot{12} \rangle] = [\langle \dot{0}, \dot{7} \rangle]$ and

$$\dot{\mathbb{Z}} := \{ [\langle x, y \rangle] : x, y \in \dot{\omega} \}.$$

Easily $\text{ZF} \vdash -7 \in \dot{\mathbb{Z}}$, and similarly for any integer, but in general a model $\mathfrak{A}$ of ZF will have both these standard as well as non-standard members of $\dot{\mathbb{Z}}^{\mathfrak{A}}$. The arithmetical operations are easily defined by

$$[\dot{\langle} x, y \rangle] \dot{+}_{\mathbb{Z}} [\dot{\langle} u, v \rangle] := [\dot{\langle} x \dot{+} u, y \dot{+} v \rangle]$$

and

$$[\dot{\langle} x, y \rangle] \dot{\times}_{\mathbb{Z}} [\dot{\langle} u, v \rangle] := [\dot{\langle} (x \dot{\times} u) \dot{+} (y \dot{\times} v), (x \dot{\times} v) \dot{+} (y \dot{\times} u) \rangle].$$

Of course, this requires the usual verification that these are (provably) well-defined operations on equivalence classes; then it is straightforward to show that the standard facts of integer arithmetic are theorems of ZF .

Similarly, the rational numbers can be represented as equivalence classes of pairs of integers:

$$\dot{\mathbb{Q}} := \{ [\dot{\langle} x, y \rangle] : x, y \in \dot{\mathbb{Z}} \wedge y \neq \dot{0} \},$$

where for $x, y \in \dot{\mathbb{Z}}$,

$$[\dot{\langle} x, y \rangle] := \{ \dot{\langle} u, v \rangle : \dot{\langle} u, v \rangle \in \dot{\mathbb{Q}} \wedge x \dot{\times}_{\mathbb{Z}} v = y \dot{\times}_{\mathbb{Z}} u \}.$$

Again it is straightforward to define the arithmetic operations $\dot{+}_{\mathbb{Q}}$ and $\dot{\times}_{\mathbb{Q}}$ and verify that they have provably the usual properties — in particular that $\dot{\mathbb{Q}}$ is a field. The ordering $\dot{<}_{\mathbb{Q}}$ may be defined by

$$[\dot{\langle} x, y \rangle] \dot{<}_{\mathbb{Q}} [\dot{\langle} u, v \rangle] :\leftrightarrow x \dot{\times}_{\mathbb{Z}} v \dot{<}_{\mathbb{Z}} y \dot{\times}_{\mathbb{Z}} u$$

for positive y and v and analogously for other combinations. Then we may verify that the properties of an ordered field are ZF -theorems and that this ordering is dense:

$$\begin{aligned} [\dot{\langle} x, y \rangle] \dot{<}_{\mathbb{Q}} [\dot{\langle} u, v \rangle] &\rightarrow \\ [\dot{\langle} x, y \rangle] \dot{<}_{\mathbb{Q}} [\dot{\langle} (x \dot{\times}_{\mathbb{Z}} v) \dot{+}_{\mathbb{Z}} (y \dot{\times}_{\mathbb{Z}} u), (y \dot{\times}_{\mathbb{Z}} v) \dot{+}_{\mathbb{Z}} (y \dot{\times}_{\mathbb{Z}} v) \rangle] &\dot{<}_{\mathbb{Q}} [\dot{\langle} u, v \rangle]. \end{aligned}$$

The next step in this development is the representation of the real numbers as sets. This can be done in any of the standard ways, such as Cauchy sequences or Dedekind cuts of rationals, and the development of the properties of $\dot{\mathbb{R}}$ in ZF is entirely parallel to the less formal one given in books and courses in analysis. Then the complex numbers $\dot{\mathbb{C}}$ may be introduced as ordered pairs of reals in the usual way. One minor problem with all of this is that using these definitions clearly

$$\text{ZF} \not\vdash \omega \subseteq \dot{\mathbb{Z}} \subseteq \dot{\mathbb{Q}} \subseteq \dot{\mathbb{R}} \subseteq \dot{\mathbb{C}}.$$

An easy solution is to regard the definitions of all of the number systems except $\dot{\mathbb{C}}$ as provisional and finally to replace them with isomorphic copies which are subsets of $\dot{\mathbb{C}}$.

We turn now to the topic of **inductive definitions** which plays a central role in many aspects of logic as well as in other parts of mathematics. In [Section 1.2](#) we developed (in IST) the notion of an induction system $\mathcal{X} = (X, X_0, \mathcal{H})$ and the corresponding set

$$\overline{X} = \bigcup_{n \in \omega} X_n = \bigcap \{ Y \subseteq X : Y \text{ is } \mathcal{X}\text{-closed} \}$$

called the inductive closure of X_0 under $\mathcal{H}$. In the process of representing these structures in ZF, we will reformulate and generalize the original notion.

Definition 6.2.35 For any (binary) set operation I ,

- (i) I is **monotone** $\leftrightarrow \forall x \forall u \forall v [u \dot{\subseteq} v \rightarrow \mathsf{I}(u, x) \dot{\subseteq} \mathsf{I}(v, x)]$;
- (ii) $\bar{\mathsf{I}}(x) := \bigcap \{ \{ u : \mathsf{I}(u, x) \dot{\subseteq} u \} \}$;
- (iii) u is a **fixed-point** of I at $x \leftrightarrow \mathsf{I}(u, x) \dot{=} u$;
- (iv) u is the **least fixed-point** of I at $x \leftrightarrow$

$$\mathsf{I}(u, x) \dot{=} u \wedge \forall v [\mathsf{I}(v, x) \dot{=} v \rightarrow u \dot{\subseteq} v].$$

Note that by Corollary 6.1.12(iii), $\bar{\mathsf{I}}(x)$ is a set iff $\exists u [\mathsf{I}(u, x) \dot{\subseteq} u]$; otherwise $\bar{\mathsf{I}}(x) \dot{=} \dot{V}$. As the examples below illustrate, this condition may be immediate, in need of proof or false.

Remark 6.2.36 In this and many other definitions and theorems we simplify the notation by using a single parameter x to stand for a list $x_0, \dots, x_{k-1}$ for $k \geq 0$. Thus (ii) above represents the more general

$$\bar{\mathsf{I}}(x_0, \dots, x_{k-1}) := \bigcap \{ \{ u : \mathsf{I}(u, x_0, \dots, x_{k-1}) \dot{\subseteq} u \} \}$$

which includes the case $k \dot{=} 0$:

$$\bar{\mathsf{I}} := \bigcap \{ \{ u : \mathsf{I}(u) \dot{\subseteq} u \} \}.$$

We sometimes remind the reader of this convention by a parenthesized descriptor, such as “(binary)” in the hypothesis.

Examples 6.2.37

- (i) Let $\mathcal{H}$ be a set of **finitary functions** on a set X — that is, for each $h \in \mathcal{H}$, there is $y_h \in \dot{\omega}$ such that $h : {}^{y_h}X \rightarrow X$, and

$$l(u) := \begin{cases} \dot{\bigcup} \{ h^*(y_h u) : h \in \mathcal{H} \}, & \text{if } u \subseteq X; \\ X, & \text{otherwise.} \end{cases}$$

Then

$$\bar{l} \doteq \dot{\bigcap} \{ u \subseteq X : (\forall h \in \mathcal{H}) h^*(y_h u) \subseteq u \}$$

is the formal analogue of $\bar{X}$ from [Section 1.2](#). Note that the role of X_0 is played here by $\{ h(\emptyset) : h \in \mathcal{H} \wedge h : {}^0X \rightarrow X \}$; if this set is empty, then also $\bar{l} \doteq \emptyset$. Such an l is called a **finitary function closure operation**; it is clearly monotone and $\bar{l}$ is a set, since by definition $\bar{l} \subseteq X$.

- (ii) Let $l_{\dot{\omega}}(u) := \{ \emptyset \} \dot{\bigcup} \{ \dot{S}z : z \in u \}$ so

$$l_{\dot{\omega}}(u) \subseteq u \quad \leftrightarrow \quad \emptyset \in u \wedge \forall z (z \in u \rightarrow \dot{S}z \in u),$$

and $\bar{l} \doteq \dot{\bigcap} \{ \{ u : l_{\dot{\omega}}(u) \subseteq u \} \} \doteq \dot{\omega}$. Of course, the Infinity axiom asserts that $\dot{\omega}$ is a set.

- (iii) Let $l_{TC}(u, x) := x \dot{\bigcup} \dot{\bigcup} u$. l_{TC} is a set operation, is monotone, and

$$l_{TC}(u, x) \subseteq u \quad \leftrightarrow \quad x \subseteq u \wedge \dot{\bigcup} u \subseteq u \quad \leftrightarrow \quad x \subseteq u \wedge u \text{ is transitive.}$$

In this example it is not entirely obvious that any such u exist; we show below that they do.

- (iv) Let l be any set operation such that $u \in l(u)$ — for example, $l(u) = \dot{\varphi}(u)$. Then by Corollary 6.1.14 for no u is $l(u) \subseteq u$.

Monotone Fixed-point Theorem 6.2.38 *For any (binary) monotone set operation l ,*

$$\bar{l}(x) \text{ is a set} \rightarrow \bar{l}(x) \text{ is the least fixed-point of } l \text{ at } x.$$

Proof. Arguing in ZF, suppose that $\bar{l}(x)$ is a set and let

$$F(x) := \{ \{ u : l(u, x) \subseteq u \} \} \quad \text{so} \quad \bar{l}(x) \doteq \bigcap F(x).$$

Now for any u , using monotonicity,

$$u \in F(x) \quad \rightarrow \quad \bar{l}(x) \subseteq u \quad \rightarrow \quad l(\bar{l}(x), x) \subseteq l(u, x) \subseteq u.$$

Hence (1) $I(\bar{I}(x), x) \dot{\subseteq} \bar{I}(x)$. Then by another application of monotonicity, $I(I(\bar{I}(x), x), x) \dot{\subseteq} I(\bar{I}(x), x)$ — that is, $I(\bar{I}(x), x) \dot{\in} F(x)$, so (2) $\bar{I}(x) \dot{\subseteq} I(\bar{I}(x), x)$. (1) and (2) imply that $\bar{I}(x)$ is a fixed-point of I , and it is the least such because for any u ,

$$I(u, x) \dot{=} u \quad \rightarrow \quad u \dot{\in} F(x) \quad \rightarrow \quad \bar{I}(x) \dot{\subseteq} u. \quad \square$$

The other description of $\bar{X}$ in [Section 1.2](#) involved iteration and is a useful technique for establishing that $\bar{I}(x)$ is a set.

Definition 6.2.39 For any (binary) set operation I and $y \dot{\in} \dot{\omega}$, we denote by $I^y(x)$ the set operation such that for all x ,

$$(i) \quad I^{\dot{0}}(x) := \dot{\emptyset};$$

$$(ii) \quad I^{\dot{S}y}(x) := I(I^y(x), x);$$

we call I^y the y -th *iterate* of I and set

$$(iii) \quad I^{\dot{\omega}}(x) := \dot{\bigcup} \{ I^y(x) : y \dot{\in} \dot{\omega} \}.$$

Formally we are defining here by $\dot{\omega}$ -recursion a set operation $(x, y) \mapsto I^y(x)$. Note that we use Replacement in (iii).

Proposition 6.2.40 For any monotone set operation I , all x and u and all $y, z \dot{\in} \dot{\omega}$,

$$(i) \quad y \dot{\leq} z \rightarrow I^y(x) \dot{\subseteq} I^z(x);$$

$$(ii) \quad I(u, x) \dot{\subseteq} u \rightarrow I^{\dot{\omega}}(x) \dot{\subseteq} u \text{ — that is } I^{\dot{\omega}}(x) \dot{\subseteq} \bar{I}(x);$$

$$(iii) \quad v \dot{\subseteq} I^{\dot{\omega}}(x) \wedge v \text{ is finite} \rightarrow (\exists y \dot{\in} \dot{\omega}) v \dot{\subseteq} I^y(x).$$

Proof. For (i), let $F(x) := \{ y \dot{\in} \dot{\omega} : \forall z [y \dot{\leq} z \rightarrow I^y(x) \dot{\subseteq} I^z(x)] \}$; we show by $\dot{\omega}$ -induction that $\forall x [\dot{\omega} \dot{\subseteq} F(x)]$. Fix x ; clearly $\dot{0} \dot{\in} F(x)$. Assume as induction hypothesis that $y \dot{\in} F(x)$ and that $\dot{S}y \dot{\leq} z$. Then for some w , $z \dot{=} \dot{S}w$ and $y \dot{\leq} w$, so by the induction hypothesis, $I^y(x) \dot{\subseteq} I^w(x)$. Then by monotonicity,

$$I^{\dot{S}y}(x) \dot{=} I(I^y(x), x) \dot{\subseteq} I(I^w(x), x) \dot{=} I^z(x).$$

For (ii), fix x and u such that $I(u, x) \dot{\subseteq} u$ and let

$$G(x) := \{ y \dot{\in} \dot{\omega} : I^y(x) \dot{\subseteq} u \}.$$

Obviously $\dot{0} \dot{\in} G(x)$, and given $y \dot{\in} G(x)$, we have by monotonicity $I^{\dot{S}y}(x) \dot{=} I(I^y(x), x) \dot{\subseteq} I(u, x) \dot{\subseteq} u$.

Finally, for (iii), let $v \dot{\subseteq} I^{\dot{\omega}}(x)$ be finite and let H be a set operation such that for $z \dot{\in} v$, $H(x, z) := \text{least } y [z \dot{\in} I^y(x)]$. By Proposition 6.2.33(vii) and (ix), for some $\bar{y} \dot{\in} \dot{\omega}$, $\forall z [z \dot{\in} v \rightarrow H(x, z) \dot{<} \bar{y}]$. But then by (i), $v \dot{\subseteq} I^{\bar{y}}(x)$. $\square$

Definition 6.2.41 A (binary) set operation l is **finitary** iff for all u and x ,

$$\forall z (z \in l(u, x) \rightarrow \exists v [v \subseteq u \wedge v \text{ is finite} \wedge z \in l(v, x)]).$$

For monotone l this condition may be stated equivalently as

$$l(u, x) = \bigcup \{ l(v, x) : v \subseteq u \wedge v \text{ is finite} \}.$$

Finitary Fixed-point Theorem 6.2.42 For any (binary) finitary monotone set-operation l ,

$$\forall x \left[l^\omega(x) = \bar{l}(x) \text{ is a set and the least fixed-point of } l \text{ at } x \right].$$

Proof. By (iii) of the preceding proposition,

$$\begin{aligned} l(l^\omega, x) &= \bigcup \{ l(v) : v \subseteq l^\omega \wedge v \text{ is finite} \} \\ &\subseteq \bigcup \{ l(l^y, x) : y \in \omega \} \\ &= l^\omega(x). \end{aligned}$$

Thus as the smallest such set, $\bar{l}(x) \subseteq l^\omega(x)$; the converse inclusion is (ii) of the preceding proposition. $\square$

Of course, a finitary function closure operation is finitary; another important example is l_{TC} .

Corollary 6.2.43 The class operation

$$TC(x) := \bigcap \{ \{ u : x \subseteq u \wedge u \text{ is transitive} \} \}$$

is a set operation and $TC(x)$ is the smallest transitive set which includes x , the **transitive closure** of x .

Proof. This follows immediately by applying the preceding theorem to the operation $l_{TC}(u, x) = x \cup \bigcup u$. Note that l_{TC} is monotone and finitary since

$$l_{TC}(u, x) = \bigcup \{ l_{TC}(\{z\}, x) : z \in u \}. \quad \square$$

Definition 6.2.44 $TC^+(x) := TC(x) \cup \{x\}$, the smallest transitive set of which x is an element.

Corollary 6.2.45 *For all x ,*

- (i) $\text{TC}^+(x) := \bigcap \{ \{ u : x \in u \wedge u \text{ is transitive} \} \};$
- (ii) $\text{TC}(x) \doteq \bigcup \{ \text{TC}^+(y) : y \in x \}.$ $\square$

As an important consequence, we derive a generalization of Foundation and some very general induction and recursion principles.

Proposition 6.2.46 *For any class B ,*

- (i) $B \neq \emptyset \rightarrow \exists z [z \text{ is } \in\text{-minimal in } B];$
- (ii) $B \neq \emptyset \rightarrow \exists z [z \in B \wedge \text{TC}(z) \cap B \doteq \emptyset].$

Proof. Although (i) follows immediately from (ii), we prove (i) first and use it in the proof of (ii). Given $B \neq \emptyset$, fix $x \in B$ and let $y := B \cap \text{TC}(x)$. If $y \doteq \emptyset$, then x is $\in$ -minimal in B . Otherwise, by Foundation there exists $z \in y$ which is $\in$ -minimal in y — that is, $y \cap z \doteq \emptyset$. Then since $z \in \text{TC}(x)$ also $z \subseteq \text{TC}(x)$, so $B \cap z \subseteq y \cap z$ and thus also $B \cap z \doteq \emptyset$, so z is $\in$ -minimal in B .

For (ii), let

$$B^+ := \{ \{ z : \text{TC}^+(z) \cap B \neq \emptyset \} \}.$$

Clearly $B \subseteq B^+$ and in particular $B^+ \neq \emptyset$, so by (i) let $z \in B^+$ be $\in$ -minimal in B^+ — that is, for each $y \in z$, $\text{TC}^+(y) \cap B \doteq \emptyset$, so

$$\text{TC}(z) \cap B \doteq \bigcup \{ \text{TC}^+(y) : y \in z \} \cap B \doteq \bigcup \{ \text{TC}^+(y) \cap B : y \in z \} \doteq \emptyset.$$

But since $\text{TC}^+(z) \cap B \neq \emptyset$, necessarily $z \in B$. $\square$

Set Induction Theorem 6.2.47 *For any classes A and C ,*

$$\forall x ([x \in C \wedge \text{TC}(x) \cap C \subseteq A] \rightarrow x \in A) \rightarrow C \subseteq A.$$

Proof. Suppose A and C are as in the hypothesis of the implication, but $C \not\subseteq A$, and let $B = C \setminus A$. Thus $B \neq \emptyset$, so B has an element z such that $\text{TC}(z) \cap B \doteq \emptyset$. Then $z \in C$, and $\text{TC}(z) \cap C \subseteq A$, so $z \in A$, a contradiction. $\square$

Remark 6.2.48 Many applications of Set Induction take the following simpler forms:

$$\forall x ([x \in C \wedge x \cap C \subseteq A] \rightarrow x \in A) \rightarrow C \subseteq A,$$

or for the case $C \doteq \dot{V}$,

$$\forall x (\text{TC}(x) \subseteq A \rightarrow x \in A) \rightarrow A \doteq \dot{V} \quad \text{or}$$

$$\forall x (x \subseteq A \rightarrow x \in A) \rightarrow A \doteq \dot{V}.$$

These should be seen as corresponding to course-of-values and ordinary induction on ω .

Set Recursion Theorem 6.2.49 *For any (ternary) set operation G , there exists a unique set operation F such that*

$$\forall x \forall y [F(x, y) \doteq G(f \upharpoonright_x \text{TC}(y), x, y)].$$

Proof. Given G , let $\theta(f, x, y)$ denote the formula

$$\text{Fn}(f) \wedge \text{Dom}(f) \doteq \text{TC}^+(y) \wedge (\forall u \in \text{TC}^+(y)) f(u) \doteq G(f \upharpoonright \text{TC}(u), x, u),$$

and let F be the class operation

$$F(x, y) = \{ \{ z : \exists f [\theta(f, x, y) \wedge z \in f(y)] \} \}.$$

As in the parallel $\dot{\omega}$ -Recursion Theorem 6.2.24, we establish first that for all x and y ,

- (1) $\forall z \forall f \forall g [\theta(f, x, y) \wedge \theta(g, x, z) \rightarrow$
 $f \upharpoonright \text{TC}^+(y) \dot{\cap} \text{TC}^+(z) \doteq g \upharpoonright \text{TC}^+(y) \dot{\cap} \text{TC}^+(z)];$
- (2) $\exists f \theta(f, x, y).$

For (1), let

$$J(x) := \{ \{ u : \forall y \forall z \forall f \forall g [\theta(f, x, y) \wedge \theta(g, x, z) \wedge u \in \text{TC}^+(y) \dot{\cap} \text{TC}^+(z) \rightarrow f(u) \doteq g(u)] \} \}.$$

We show by set induction that $\forall x [J(x) \doteq \dot{V}]$. Assume as induction hypothesis that $\text{TC}(u) \subseteq J(x)$, and fix y, z, f and g such that $\theta(f, x, y), \theta(g, x, z)$ and $u \in \text{TC}^+(y) \dot{\cap} \text{TC}^+(z)$. Then for each $v \in \text{TC}(u)$, $v \in J(x)$ and also $v \in \text{TC}^+(y) \dot{\cap} \text{TC}^+(z)$, so $f(v) \doteq g(v)$. Hence $f \upharpoonright \text{TC}(u) \doteq g \upharpoonright \text{TC}(u)$ and therefore by the definition of θ ,

$$f(u) \doteq G(f \upharpoonright \text{TC}(u), x, u) \doteq G(g \upharpoonright \text{TC}(u), x, u) \doteq g(u),$$

so also $u \in J(x)$.

For (2), let

$$K(x) := \{ \{ y : \exists f \theta(f, x, y) \} \};$$

we again use set induction to establish that $\forall x [K(x) \doteq \dot{V}]$. Assume as induction hypothesis that $\text{TC}(y) \subseteq K(x)$ so in particular $y \subseteq K(x)$. It follows from (1) that $(\forall z \in y) \exists! f \theta(f, x, z)$ and thus that there is a set operation $(x, z) \mapsto f_{x,z}$ such that $(\forall z \in y) \theta(f_{x,z}, x, z)$. By replacement,

$h_{x,y} := \dot{\bigcup} \{ \dot{f}_{x,z} : z \in y \}$ is a set, it follows from (1) that $h_{x,y}$ is a function, and

$$\text{Dom}(h_{x,y}) \doteq \dot{\bigcup} \left\{ \text{Dom}(f_{x,z}) : z \in y \right\} \doteq \dot{\bigcup} \left\{ \text{TC}^+(z) : z \in y \right\} \doteq \text{TC}(y).$$

Hence, if $f_{x,y} := h_{x,y} \dot{\cup} \left\{ \langle y, G(h_{x,y}, x, y) \rangle \right\}$, easily $\theta(f_{x,y}, x, y)$ and thus $y \in K(x)$.

From (1) and (2) we conclude first that $\forall x \forall y \exists! f \theta(f, x, y)$ and hence there exists a set operation $(x, y) \mapsto f_{x,y}$ such that $\theta(f_{x,y}, x, y)$. Then by (1), for all x and y and all $z \in \text{TC}^+(y)$, $F(x, z) \doteq f_{x,y}(z)$, and

$$F(x, y) \doteq f_{x,y}(y) \doteq G(f_{x,y} \upharpoonright \text{TC}(y), x, y) \doteq G(F \upharpoonright_x \text{TC}(y), x, y).$$

The uniqueness of F is established by a straightforward set induction. $\square$

Remark 6.2.50 Many applications of Set Recursion are of the following form: for any set operation G there exists a unique set operation F such that

$$\forall x \forall y [F(x, y) \doteq G(F \upharpoonright_x y, x, y)].$$

Another useful example of inductive definition is the following.

Definition 6.2.51

- (i) $\text{I}_{\text{HF}}(u) := \left\{ \dot{\emptyset} \right\} \dot{\cup} \left\{ y \dot{\cup} \{z\} : y, z \in u \right\}$;
- (ii) $\text{I}_{\text{HFO}}(u, x) := \left\{ \dot{\emptyset} \right\} \dot{\cup} \left\{ y \dot{\cup} \{z\} : y \in u \wedge z \in u \dot{\cup} x \right\}$;
- (iii) $\text{HF} := \bar{\text{I}}_{\text{HF}}$, the *hereditarily finite* sets;
- (iv) $\text{HFO}(x) := \bar{\text{I}}_{\text{HFO}}(x)$, the sets *hereditarily finite over* x .

Clearly both I_{HF} and I_{HFO} are finitary, so HF and $\text{HFO}(x)$ are sets and are the least fixed-points of the operations. We leave the proof of the following to the exercises.

Proposition 6.2.52 For all x ,

- (i) $\dot{\omega} \subseteq \text{HF} \subseteq \text{HFO}(x)$;
- (ii) $\text{HF} \doteq \dot{V}_{\dot{\omega}} \doteq \dot{\wp}^{\dot{\omega}}$;
- (iii) $\forall v [v \in \text{HF} \leftrightarrow \text{TC}(v) \text{ is finite}]$;
- (iv) $\forall v (v \subseteq \text{HFO}(x) \dot{\cup} x \rightarrow [v \in \text{HFO}(x) \leftrightarrow v \text{ is finite}])$;
- (v) $(\forall v \in \text{HFO}(x)) \forall f [(f : v \rightarrow \text{HFO}(x)) \rightarrow f \in \text{HFO}(x)]$. $\square$

In [Section 1.2](#) we established also the method of recursive definition for functions with domain an inductively defined set. This method does not extend to the generalized notions of inductive definition, but for use in [Section 6.6](#) below, we note the ZF version of the original. A finitary function closure operation I has *unique readability* iff each $h \in \mathcal{H}$ is injective, and for all $g, h \in \mathcal{H}$, $\text{Im}(g) \cap \text{Im}(h) \doteq \dot{\emptyset}$. Then we have

Finitary Closure Recursion Theorem 6.2.53 *For any finitary closure operation l as above with unique readability, any set W and any set operation G such that for each $h \in \mathcal{H}$, $\mathsf{G}(h)$ is a function satisfying*

$$\forall x (\forall s \in {}^{y_h}U) (\forall t \in {}^{y_h}W) \mathsf{G}(h)(t, x, s) \in W,$$

there exists a set operation F such that for all x , all $h \in \mathcal{H}$ and all $s \in {}^{y_h}\bar{\mathsf{l}}$,

$$\mathsf{F}(x, h(s)) \doteq \mathsf{G}(h)(\mathsf{F}_x[s], x, s), \quad \text{where}$$

$$\mathsf{F}_x[s] \doteq t \in {}^{y_h}W \quad \text{such that} \quad (\forall z \prec_{y_h} t)(z) \doteq \mathsf{F}(x, s(z)). \quad \square$$

Collateral Reading. Representations of the rational and real numbers in terms of sets are often described in analysis texts, but with less attention to the axiomatic foundation. Some set-theory texts such as [En77] and [Mo94] also cover this material.

Exercises

Exercise 6.2.54 Prove that each of the following is a proper class (operation):

- (i) $\{ \{ \dot{x} \} : x = x \}$;
- (ii) $\{ \{ z : z \notin x \} \}$;
- (iii) $\{ \{ r : \mathsf{Rel}(r) \} \}$;
- (iv) $\{ \{ \dot{\langle x, y \rangle} : y = \dot{\bigcup} x \} \}$.

Exercise 6.2.55 Prove Proposition 6.2.2.

Exercise 6.2.56 Show that the alternative definition of ordered pair mentioned in Remark 6.2.5

$$\dot{\langle x, y \rangle} \doteq \left\{ \left\{ \dot{\emptyset}, x \right\}, \left\{ \left\{ \dot{\emptyset} \right\}, y \right\} \right\}$$

satisfies Propositions 6.2.2 and 6.2.4.

Exercise 6.2.57 Prove Proposition 6.2.9.

Exercise 6.2.58 Prove Proposition 6.2.11.

Exercise 6.2.59 Show that for any class B , if

$$\emptyset \in B \wedge \forall y \forall z (y \in B \rightarrow y \cup \{z\} \in B),$$

then every finite set is an element of B .

Exercise 6.2.60 Prove that for any x , x is finite iff

$$\forall u ([u \subseteq \dot{x} \wedge \emptyset \in u \wedge \forall y \forall z (y \in u \wedge z \in x \rightarrow y \cup \{z\} \in u)] \rightarrow x \in u).$$

Exercise 6.2.61 Show that

- (i) $\forall x \forall y [x \subseteq y \in \dot{\omega} \wedge x \text{ is transitive} \rightarrow x \in \dot{\omega} \wedge x \subseteq y]$;
- (ii) $\forall u [u \subseteq \dot{\omega} \wedge u \neq \emptyset \rightarrow \bigcap u \text{ is the least element of } u]$.

Exercise 6.2.62 Give proofs for the remaining parts of Proposition 6.2.33.

Exercise 6.2.63 Prove Proposition 6.2.52.

Exercise 6.2.64 Suppose that F and G are monotone unary set operations and x and y sets such that $Fx \subseteq y$ and $Gy \subseteq x$. Prove that there exist subsets $a \subseteq x$ and $b \subseteq y$ such that $F(a) = b$ and $G(y \setminus b) = x \setminus a$.

Hint: Consider the set operation $h_u := F(x \setminus G(y \setminus u))$.

Exercise 6.2.65 Prove the following variant of the Set Induction Theorem 6.2.47: for any classes A , C and D , if

$$\forall x \forall y ([x \in C \wedge y \in D \wedge (TC(x) \cap C) \times (TC(y) \cap D) \subseteq A] \rightarrow \langle x, y \rangle \in A),$$

then $C \times D \subseteq A$.

Exercise 6.2.66 Prove the following variant of the Set Recursion Theorem 6.2.49: for any set operation G , there exists a unique set operation F such that

$$\forall x \forall y \forall z [F(x, y, z) = G(F \restriction_x^2 (TC(y), TC(z)), x, y, z)],$$

where

$$F \restriction_x^2 (u, v) := \left\{ \langle \langle y, z \rangle, F(x, y, z) \rangle : y \in u \wedge z \in v \right\}.$$

Exercise 6.2.67 Suppose that u and v are transitive sets and $f : u \rightarrow v$ is a function from u onto v such that

$$(\forall x \in u)(\forall y \in u)[x \in y \leftrightarrow f(x) \in f(y)].$$

Prove that for all $x \in u$, $f(x) = x$.

Exercise 6.2.68 Let $\dot{\varphi}_{<\omega}(u) \doteq \{z : z \dot{\subseteq} u \wedge z \text{ is finite}\}$. Show that $\text{HF} \doteq \dot{\varphi}_{<\omega}^\infty \doteq \dot{\varphi}_{<\omega}^\omega$.

6.3 Ordinal numbers: induction and recursion

In this section and the next we develop in ZF the basic facts about ordinal and cardinal numbers. The emphasis is on those parts of the theory which will be needed in later sections and many important and interesting topics will be treated very briefly or only in the exercises.

In ordinary usage the natural numbers $0, 1, \dots$ play a dual role as ordinal numbers (indicating a place in a sequence or a type of ordering) and as cardinal numbers (indicating a size or number of objects). This will be reflected in our formalizations by the fact that the sets of natural numbers, finite ordinals and finite cardinals will all coincide. Beyond the finite, however, there are in a sense more (well-)orderings than there are sizes, and the roles of ordinal and cardinal numbers diverge. In fact, we will follow standard usage and use a subclass of the infinite ordinal numbers as the class of infinite cardinal numbers. This choice will have convenient consequences in the sense of Remark 6.2.5.

Increasingly we shall be using expressions containing bounded quantifiers, and we introduce the abbreviations

$$\begin{aligned} (\exists x \dot{\in} y) \phi & \text{ for } \exists x [x \dot{\in} y \wedge \phi] & \text{ and} \\ (\forall x \dot{\in} y) \phi & \text{ for } \forall x [x \dot{\in} y \rightarrow \phi]. \end{aligned}$$

Definition 6.3.1

(i) y is an **ordinal** (number) : $\leftrightarrow$

y is transitive and $(\forall x \dot{\in} y) x$ is transitive.

(ii) $\dot{\text{Or}} := \{\{y : y \text{ is an ordinal}\}\}$.

As a notational convention we shall use lowercase Greek letters $\rho, \sigma, \tau, \nu, \dots$ as variables which denote ordinals. This means, for example, that the formula $\exists \sigma \phi(\sigma)$ is an abbreviation for $\exists x [x \text{ is an ordinal} \wedge \phi(x)]$, and similarly in other contexts. It is far from obvious that the objects satisfying this definition have anything to do with ordering or place in a sequence, and our first task is to clarify this.

Proposition 6.3.2 $\dot{\text{Or}}$ is transitive and a proper class.

Proof. For any σ , each $x \dot{\in} \sigma$ is transitive by definition, and for any $y \dot{\in} x$ also $y \dot{\in} \sigma$, so also y is transitive. Hence x is an ordinal. Now towards a contradiction suppose that $\dot{\text{Or}}$ is a set. Then $\dot{\text{Or}}$ is transitive and every element of $\dot{\text{Or}}$ is transitive (being an ordinal), so $\dot{\text{Or}}$ is an ordinal and we conclude $\dot{\text{Or}} \dot{\in} \dot{\text{Or}}$, contrary to Lemma 6.1.14. $\square$

Remark 6.3.3 Historically, in the context of belief in Global Comprehension, this result was seen as a paradox, the so-called **Burali-Forti paradox**. Hence it may be seen as an alternative to Russell's paradox for demonstrating the inconsistency of Global Comprehension.

Proposition 6.3.4 $\dot{\omega}$ is an ordinal; every $y \in \dot{\omega}$ is an ordinal.

Proof. This is exactly the content of Lemma 6.2.21. $\square$

As for members of $\dot{\omega}$, we write $\sigma < \tau$ for $\sigma \in \tau$ and $\sigma \leq \tau$ for $\sigma \in \tau \vee \sigma = \tau$, or equivalently $\sigma \in \dot{S}\tau$. Let also

$$<_{\dot{\omega}} := \{ \{ \langle \sigma, \tau \rangle : \sigma < \tau \} \},$$

and for any $u \subseteq \dot{\omega}$, in particular for $u \in \dot{\omega}$,

$$<_u := \{ \langle \sigma, \tau \rangle : \sigma < \tau \wedge \sigma, \tau \in u \}.$$

Lemma 6.3.5

- (i) $<_{\dot{\omega}}$ is a class partial ordering.
- (ii) For all $u \subseteq \dot{\omega}$, $<_u$ is a partial ordering.

Proof. These relations are irreflexive by Proposition 6.1.14 and transitive by the definition of ordinal. $\square$

Proposition 6.3.6

- (i) $<_{\dot{\omega}}$ is well-founded — more generally, for any class B ,

$$B \subseteq \dot{\omega} \wedge B \neq \emptyset \rightarrow \exists \sigma [\sigma \in B \wedge (\forall \tau < \sigma) \tau \notin B].$$

- (ii) For all $u \subseteq \dot{\omega}$, $<_u$ is well-founded.

Proof. This is immediate from Proposition 6.2.46. $\square$

Ordinal Induction Theorem 6.3.7 For any class A ,

$$\forall \sigma [\sigma \subseteq A \rightarrow \sigma \in A] \rightarrow \dot{\omega} \subseteq A.$$

Proof. This is a special case of the Set Induction Theorem 6.2.47. $\square$

As in the case of $\dot{\omega}$ induction, it is often useful to think of this in terms of formulas: for any formula ϕ in which τ is not free,

$$\forall \sigma \left([(\forall \tau < \sigma) \phi(\tau)] \rightarrow \phi(\sigma) \right) \rightarrow \forall \sigma \phi(\sigma).$$

We shall generally write simply $<$ for $<_{\dot{\omega}}$.

Theorem 6.3.8

- (i) $\dot{<}$ is connected and hence is a class well-ordering.
- (ii) For all $u \subseteq \dot{\mathcal{O}}r$, $\dot{<}_u$ is connected and hence is a well-ordering.

Proof. Suppose that the assertion of (i) is false and set

$$B = \{ \{ \sigma : \exists \tau \neg (\sigma \dot{<} \tau \vee \sigma \dot{=} \tau \vee \tau \dot{<} \sigma) \} \}.$$

Thus $B \neq \dot{\emptyset}$ so has a minimal element $\hat{\sigma}$. Now let

$$C = \{ \{ \tau : \neg (\hat{\sigma} \dot{<} \tau \vee \hat{\sigma} \dot{=} \tau \vee \tau \dot{<} \hat{\sigma}) \} \}.$$

By the choice of $\hat{\sigma}$, $C \neq \dot{\emptyset}$, so C has a minimal element $\hat{\tau}$. To reach a contradiction, we show that $\hat{\sigma} \dot{=} \hat{\tau}$. Let ρ be any element of $\hat{\sigma}$. Then $\rho \dot{<} \hat{\sigma}$ so by the minimality of $\hat{\sigma}$, we have $\rho \dot{<} \hat{\tau} \vee \rho \dot{=} \hat{\tau} \vee \hat{\tau} \dot{<} \rho$. Both of the options $\rho \dot{=} \hat{\tau}$ and $\hat{\tau} \dot{<} \rho$ lead to the conclusion $\hat{\tau} \dot{<} \hat{\sigma}$, contrary to the choice of $\hat{\tau}$, and we conclude $\rho \dot{<} \hat{\tau}$. Thus we have proved $\hat{\sigma} \subseteq \hat{\tau}$. A similar argument gives $\hat{\tau} \subseteq \hat{\sigma}$, so $\hat{\sigma} \dot{=} \hat{\tau}$, a contradiction. (ii) follows immediately. $\square$

Corollary 6.3.9 For any class $B \subseteq \dot{\mathcal{O}}r$,

$$B \neq \dot{\emptyset} \rightarrow \exists! \sigma [\sigma \in B \wedge \forall \tau (\tau \in B \rightarrow \sigma \dot{\leq} \tau)],$$

that is, B has a unique least element. Furthermore, this least element is exactly $\bigcap B$.

Proof. The first part is immediate from Proposition 6.3.6 and Theorem 6.3.8. Let σ be the least element of B . Then $\bigcap B \subseteq \sigma$ by the definition of $\bigcap$, and since for each $\tau \in B$, $\sigma \dot{\leq} \tau$, also $\sigma \subseteq \tau$. Thus $\sigma \subseteq \bigcap B$. $\square$

Definition 6.3.10 For all ordinals σ ,

- (i) σ is a **successor ordinal** $:\leftrightarrow \exists \tau (\sigma \dot{=} \dot{S}\tau)$;
- (ii) σ is a **limit ordinal** $:\leftrightarrow \sigma \neq \dot{0} \wedge \forall \tau (\tau \dot{<} \sigma \rightarrow \dot{S}\tau \dot{<} \sigma)$.

Corollary 6.3.11

- (i) $(\forall y \dot{<} \dot{\omega}) y \dot{=} \dot{0} \vee y$ is a successor.
- (ii) $\dot{\omega}$ is a limit ordinal. $\square$

Proposition 6.3.12 *For all ordinals σ and τ and any u ,*

- (i) $\dot{S}\sigma$ is an ordinal;
- (ii) $\sigma \subseteq \tau \leftrightarrow \dot{S}\sigma \subseteq \dot{S}\tau$;
- (iii) $\sigma < \tau \leftrightarrow \dot{S}\sigma < \dot{S}\tau$;
- (iv) $u \subseteq \dot{O}r \wedge u$ is transitive $\rightarrow u \in \dot{O}r$;
- (v) $u \subseteq \dot{O}r \rightarrow \dot{\bigcup} u \in \dot{O}r$; in particular, $\dot{\bigcup} \sigma \in \dot{O}r$;
- (vi) $\dot{\bigcup} \dot{S}\sigma \doteq \sigma$;
- (vii) σ is a limit ordinal iff $\sigma \neq \dot{0}$ and $\sigma \doteq \dot{\bigcup} \sigma$;
- (viii) σ is a successor ordinal iff $\sigma \doteq \dot{S}(\dot{\bigcup} \sigma)$;
- (ix) $(\sigma \doteq \dot{0}) \vee (\sigma \text{ is a successor}) \vee (\sigma \text{ is a limit})$, and exactly one of these holds.

Proof. (i) is straightforward as is (ii)($\leftarrow$). Suppose that $\sigma \subseteq \tau$, but $\sigma \not\subseteq \tau$. Then by connectedness, $\tau \in \sigma$, hence $\tau \in \tau$, contrary to Proposition 6.1.14. (iii) needs only transitivity, and (iv) is immediate from the definition. (v) follows from (iv), since if $u \subseteq \dot{O}r$, then also $\dot{\bigcup} u \subseteq \dot{O}r$, and $\dot{\bigcup} u$ is transitive, since for any x ,

$$x \in \dot{\bigcup} u \rightarrow \exists \sigma [x \in \sigma \in u] \rightarrow \exists \sigma [x \subseteq \sigma \in u] \rightarrow x \subseteq \dot{\bigcup} u.$$

(vi) uses only transitivity. For (vii), if σ is a limit ordinal, then for any $\tau < \sigma$, also $\tau < \dot{S}\tau < \sigma$, so $\tau < \dot{\bigcup} \sigma$. Conversely, if $\sigma \doteq \dot{\bigcup} \sigma$, then for any $\tau < \sigma$ there is some ρ such that $\tau < \rho < \sigma$, so $\dot{S}\tau \leq \rho < \sigma$ and thus $\dot{S}\tau < \sigma$. Hence σ is a limit. (viii) follows from (vi). For (ix), suppose that $\sigma \neq \dot{0}$ and σ is not a successor. Then for any $\tau < \sigma$, $\sigma \neq \dot{S}\tau$, and if $\sigma < \dot{S}\tau$, then $\sigma \subseteq \tau$, so $\tau \in \tau$, a contradiction. Hence by connectedness, $\dot{S}\tau < \sigma$. Thus σ is a limit. $\square$

The partition of the ordinals into three distinct types as in (ix) above leads to an alternative version of ordinal induction.

Proposition 6.3.13 *For any class A ,*

$$[\dot{0} \in A \wedge \forall \sigma (\sigma \in A \rightarrow \dot{S}\sigma \in A) \\ \wedge \forall \sigma (\sigma \text{ is a limit} \wedge \sigma \subseteq A \rightarrow \sigma \in A)] \rightarrow \dot{O}r \subseteq A.$$

Proof. Show by cases that given the hypothesis, $\forall \sigma [\sigma \subseteq A \rightarrow \sigma \in A]$. $\square$

Beyond $\dot{\omega}$ we have clearly ordinals $\dot{S}\dot{\omega}$, $\dot{S}\dot{S}\dot{\omega}$ etc. More generally, using $\dot{\omega}$ -recursion, let H be a set operation such that for all x and all $y \in \dot{\omega}$,

$$H(x, 0) \doteq x \quad \text{and} \quad H(x, \dot{S}y) \doteq \dot{S}H(x, y).$$

The conditions for H are exactly those for addition, so it is natural to write $x \dot{+} y$ for $H(x, y)$. For example, $H(\dot{\omega}, \dot{3}) \doteq \dot{S}\dot{S}\dot{S}\dot{\omega}$ or $\dot{\omega} \dot{+} \dot{3}$. Furthermore, if we set

$$G(x) := \dot{\bigcup} \{ H(x, y) : y \in \dot{\omega} \},$$

then by Replacement, G is a set operation, and by Proposition 6.3.12(iv), $G(\dot{\omega})$ is an ordinal, which we might naturally write $\dot{\omega} \dot{+} \dot{\omega}$. Then $G(\dot{\omega} \dot{+} \dot{\omega})$ is an ordinal, which we write $\dot{\omega} \dot{+} \dot{\omega} \dot{+} \dot{\omega}$. It is easy to check that all of these ordinals are countable, and clearly we can carry this much further, but the orderly and precise way to do this involves ordinal recursion.

Ordinal Recursion Theorem 6.3.14 *For any ternary set operation G , there exists a set operation F such that*

$$\forall x \forall \sigma [F(x, \sigma) \doteq G(F \upharpoonright_x \sigma, x, \sigma)].$$

Furthermore, F is **unique on $\dot{O}r$** in the sense that for any other set operation F' which satisfies this condition, $\forall x \forall \sigma [F(x, \sigma) \doteq F'(x, \sigma)]$.

Proof. Immediate from the Set Recursion Theorem 6.2.49. Note that since an ordinal σ is transitive, $TC(\sigma) = \sigma$. $\square$

Many applications of this principle are of the following form.

Corollary 6.3.15 *For any set operations H and I , there exists a set operation F such that for all x and σ ,*

- (i) $F(x, \dot{0}) \doteq H(x)$;
- (ii) $F(x, \dot{S}\sigma) \doteq I(F(x, \sigma), x, \sigma)$;
- (iii) $F(x, \sigma) \doteq \dot{\bigcup} \{ F(x, \tau) : \tau \dot{<} \sigma \}$ if σ is a limit ordinal.

Furthermore, F is unique on $\dot{O}r$.

Proof. This is an instance of the theorem with

$$G(f, x, \sigma) := \begin{cases} H(x), & \text{if } \sigma \doteq \dot{0}; \\ I(f(\dot{\bigcup} \sigma), x, \sigma), & \text{if } \sigma \text{ is a successor ordinal}; \\ \dot{\bigcup} \dot{\text{Im}}(f), & \text{if } \sigma \text{ is a limit ordinal.} \end{cases}$$

Then for $F(x, \sigma) \doteq G(F \restriction_x \sigma, x, \sigma)$, clearly $F(x, \dot{0}) \doteq H(x)$, for any σ ,

$$F(x, \dot{\Sigma}\sigma) \doteq G(F \restriction_x \dot{\Sigma}\sigma, x, \dot{\Sigma}\sigma) \doteq I(F(x, \bigcup \dot{\Sigma}\sigma), x, \sigma) \doteq I(F(x, \sigma), x, \sigma),$$

and for σ a limit ordinal,

$$F(x, \sigma) \doteq G(F \restriction_x \sigma, x, \sigma) \doteq \bigcup \text{Im}(F \restriction_x \sigma) \doteq \bigcup \{F(x, \tau) : \tau \dot{<} \sigma\}. \quad \square$$

As a first application of this principle, we introduce the operations of **ordinal arithmetic**.

Proposition 6.3.16 *There exist set-operations $\dot{+}_o$ and $\dot{\times}_o$ unique on $\dot{\text{Or}}$ such that for all ordinals ρ, τ and σ ,*

- (i) $\rho \dot{+}_o \dot{0} \doteq \rho$; $\rho \dot{+}_o \dot{\Sigma}\tau \doteq \dot{\Sigma}(\rho \dot{+}_o \tau)$; $\rho \dot{+}_o \sigma \doteq \bigcup \{\rho \dot{+}_o \tau : \tau \dot{<} \sigma\}$
for σ a limit ordinal;
- (ii) $\rho \dot{\times}_o \dot{0} \doteq \dot{0}$; $\rho \dot{\times}_o \dot{\Sigma}\tau \doteq (\rho \dot{\times}_o \tau) \dot{+}_o \rho$; $\rho \dot{\times}_o \sigma \doteq \bigcup \{\rho \dot{\times}_o \tau : \tau \dot{<} \sigma\}$
for σ a limit ordinal. $\square$

Now we have a precise meaning for arbitrary arithmetical expressions involving ordinals. Note, however, that although these operations extend the operations of the same names on $\dot{\omega}$, they behave somewhat differently. For example, $\dot{\omega} \dot{+}_o \dot{2} \doteq \dot{\Sigma}\dot{\Sigma}\dot{\omega}$, but $\dot{2} \dot{+}_o \dot{\omega} \doteq \dot{\omega}$. Similarly, $\dot{\omega} \dot{+}_o \dot{\omega} \dot{+}_o \dot{\omega} \doteq \dot{\omega} \dot{\times}_o \dot{3}$, but $\dot{3} \dot{\times}_o \dot{\omega} \doteq \dot{\omega}$. The exercises explore some of these facts. Often from now on we write $\sigma \dot{+}_o \dot{1}$ in place of $\dot{\Sigma}\sigma$. The exercises also show how we may characterize $\rho \dot{+}_o \sigma$ and $\rho \dot{\times}_o \sigma$ in terms of the well-orderings $\dot{<}_\rho$ and $\dot{<}_\sigma$. These characterizations show also

Corollary 6.3.17 *For any countable ordinals ρ and σ , also $\rho \dot{+}_o \sigma$ and $\rho \dot{\times}_o \sigma$ are countable.*

Proof. Left as an exercise $\square$

Thus to reach an uncountable ordinal requires additional tools; in the next section we show that the smallest uncountable ordinal is simply the set of all countable ordinals.

In Theorem 6.3.8 we showed that the membership relation $\dot{<}_\sigma$ on any ordinal σ is a well-ordering. Another easy and important consequence of ordinal recursion is the fact that *every* well-ordering is isomorphic to exactly one of these.

Theorem 6.3.18 $\forall r (r \text{ is a well-ordering} \leftrightarrow \exists \rho [r \dot{\prec}_\rho])$.

Proof. Fix a well-ordering r . Any nonempty $u \subseteq \text{Fld}(r)$ has a unique r -least element which we denote by $\text{lst}_r(u)$. Thus, $\text{lst}_r(u) \in u$ and $(\forall y \in u) \langle \text{lst}_r(u), y \rangle \in r \vee y \doteq \text{lst}_r(u)$. Now let F be a set operation such that

$$F(\sigma) := \begin{cases} \text{lst}_r(\text{Fld}(r) \setminus \text{Im}(F \upharpoonright \sigma)), & \text{if } \text{Fld}(r) \not\subseteq \text{Im}(F \upharpoonright \sigma); \\ r, & \text{otherwise.} \end{cases}$$

The ‘otherwise’ value r is chosen only such that it is not a member of $\text{Fld}(r)$. First, there must exist a (least) ordinal ρ such that $F(\rho) \doteq r$. If this were not the case then F would be an injective mapping of $\dot{\text{Or}}$ into $\text{Fld}(r)$ contrary to Corollary 6.2.13. Now easily $\sigma < \tau < \rho \rightarrow \langle F(\sigma), F(\tau) \rangle \in r$, so $F \upharpoonright \rho$ witnesses the isomorphism $\dot{\prec}_\rho \simeq r$. $\square$

At the end of Section 6.2 we established some facts about the iteration of monotone set operations. Using ordinal recursion we can now generalize these results in two ways.

Definition 6.3.19 A (binary) set operation l is *inclusive* iff $\forall u \forall x [u \subseteq l(u, x)]$.

Definition 6.3.20 For any binary set operation l , we denote by $l^\sigma(x)$ the set operation such that for all x and σ ,

- (i) $l^0(x) \doteq \emptyset$;
- (ii) $l^{\sigma+1}(x) \doteq l(l^\sigma(x), x)$;
- (iii) $l^\sigma(x) \doteq \bigcup \{ l^\tau(x) : \tau < \sigma \}$ if σ is a limit ordinal;

We call l^σ the σ -th *iterate* of l (cf. Definition 6.2.39). Further,

- (iv) $l^\infty(x) := \bigcup \{ l^\sigma(x) : \sigma \in \dot{\text{Or}} \}$;
- (v) l *stops by σ at x* $\leftrightarrow l^\sigma(x) = l^{\sigma+1}(x)$.

The Finitary Fixed-point Theorem 6.2.42 shows that every finitary monotone operation stops by ω . Examples of operations which stop at larger σ will be given in the next section.

Proposition 6.3.21 For any (binary) set operation l , if l is either inclusive or inonotone, then for all x ,

- (i) $\forall \sigma [l^\sigma(x) \subseteq l^{\sigma+1}(x)]$;
- (ii) $\forall \sigma \forall \tau [\sigma \leq \tau \rightarrow l^\sigma(x) \subseteq l^\tau(x)]$;

- (iii) if $\mathbf{l}$ is monotone, then $\forall u \ [\mathbf{l}(u, x) \dot{\subseteq} u \rightarrow \mathbf{l}^\infty(x) \dot{\subseteq} u]$ — that is, $\mathbf{l}^\infty(x) \dot{\subseteq} \dot{\mathbf{l}}(x)$;
- (iv) $\forall \sigma \ (\mathbf{l} \text{ stops by } \sigma \text{ at } x \rightarrow \forall \tau \ [\sigma \dot{\leq} \tau \rightarrow \mathbf{l}^\sigma(x) \dot{=} \mathbf{l}^\tau(x) \dot{=} \mathbf{l}^\infty(x)])$;
- (v) $\forall u \ (u \dot{\subseteq} \mathbf{l}^\infty(x) \rightarrow \exists \sigma \ [u \dot{\subseteq} \mathbf{l}^\sigma(x)])$.

Proof. For inclusive $\mathbf{l}$, (i) is immediate; for monotone $\mathbf{l}$, we show by ordinal induction that

$$\dot{\mathbf{O}}\mathbf{r} \dot{\subseteq} \mathbf{G}(x) := \{ \{ \sigma : \mathbf{l}^\sigma(x) \dot{\subseteq} \mathbf{l}^{\sigma+\dot{1}}(x) \} \}.$$

Assume as induction hypothesis that $\sigma \dot{\subseteq} \mathbf{G}(x)$. Clearly $\dot{0} \dot{\in} \mathbf{G}(x)$. Suppose that σ is a successor $\tau + \dot{1}$. Then since $\tau \dot{\in} \mathbf{G}(x)$,

$$\mathbf{l}^\sigma(x) \dot{=} \mathbf{l}(\mathbf{l}^\tau(x), x) \dot{\subseteq} \mathbf{l}(\mathbf{l}^{\tau+\dot{1}}(x), x) \dot{=} \mathbf{l}^{\sigma+\dot{1}}(x).$$

Finally, if σ is a limit, then since for all $\tau \dot{<} \sigma$ we have $\mathbf{l}^\tau(x) \dot{\subseteq} \mathbf{l}^{\tau+\dot{1}}(x)$, and by definition of $\mathbf{l}^\sigma$ also $\mathbf{l}^\tau(x) \dot{\subseteq} \mathbf{l}^\sigma(x)$, again using monotonicity,

$$\mathbf{l}^\tau(x) \dot{\subseteq} \mathbf{l}^{\tau+\dot{1}}(x) \dot{\subseteq} \mathbf{l}^{\sigma+\dot{1}}(x), \quad \text{so} \quad \mathbf{l}^\sigma(x) \dot{=} \bigcup \{ \mathbf{l}^\tau(x) : \tau \dot{<} \sigma \} \dot{\subseteq} \mathbf{l}^{\sigma+\dot{0}\dot{1}}(x).$$

For (ii) it will suffice to show that

$$\dot{\mathbf{O}}\mathbf{r} \dot{\subseteq} \mathbf{F}(x) := \{ \{ \rho : \forall \sigma \forall \tau \ [\sigma \dot{<} \tau \leq \rho \rightarrow \mathbf{l}^\sigma(x) \dot{\subseteq} \mathbf{l}^\tau(x)] \} \}.$$

Assume as induction hypothesis that $\rho \dot{\subseteq} \mathbf{F}(x)$ and fix $\sigma \dot{<} \tau \dot{\leq} \rho$. If $\tau \dot{<} \rho$, the induction hypothesis gives immediately $\mathbf{l}^\sigma(x) \dot{\subseteq} \mathbf{l}^\tau(x)$, so we need only show $\sigma \dot{<} \rho \rightarrow \mathbf{l}^\sigma(x) \dot{\subseteq} \mathbf{l}^\rho(x)$. This is true vacuously if $\rho \dot{=} \dot{0}$. If ρ is a successor $v + \dot{1}$, and $\sigma \dot{<} \rho$, then by (i), $\mathbf{l}^\sigma(x) \dot{\subseteq} \mathbf{l}^v(x) \dot{\subseteq} \mathbf{l}^\rho(x)$. If ρ is a limit, then

$$\mathbf{l}^\sigma(x) \dot{\subseteq} \bigcup \{ \mathbf{l}^v(x) : v \dot{<} \rho \} \dot{=} \mathbf{l}^\rho(x).$$

For (iii) a straightforward ordinal induction shows that for monotone $\mathbf{l}$, if $\mathbf{l}(u, x) \dot{\subseteq} u$, then $\forall \sigma \ [\mathbf{l}^\sigma(x) \dot{\subseteq} u]$, so immediately $\mathbf{l}^\infty(x) \dot{\subseteq} u$. For (iv), suppose that $\mathbf{l}$ stops by σ at x ; we show by ordinal induction that

$$\dot{\mathbf{O}}\mathbf{r} \dot{\subseteq} \mathbf{H}(x) := \{ \{ \tau : \sigma \dot{\leq} \tau \rightarrow \mathbf{l}^\sigma(x) \dot{=} \mathbf{l}^\tau(x) \} \}.$$

Assume as induction hypothesis that $\tau \dot{\subseteq} \mathbf{H}(x)$ and that $\sigma \dot{<} \tau$. If τ is a successor $v + \dot{1}$, then $v \dot{\in} \mathbf{H}(x)$, so

$$\mathbf{l}^\sigma(x) \dot{=} \mathbf{l}^{\sigma+\dot{0}\dot{1}}(x) \dot{=} \mathbf{l}(\mathbf{l}^\sigma(x), x) \dot{=} \mathbf{l}(\mathbf{l}^v(x), x) \dot{=} \mathbf{l}^\tau(x),$$

and hence also $\tau \in H(x)$. Otherwise, τ is a limit and

$$I^\tau(x) \doteq \bigcup \{I^v(x) : v \dot{<} \tau\}.$$

For $v \dot{\leq} \sigma$, $I^v(x) \subseteq I^\sigma(x)$; otherwise $I^v(x) \doteq I^\sigma(x)$, so this union is exactly $I^\sigma(x)$.

Finally, for (v) suppose that $u \subseteq I^\infty(x)$, so for each $z \in u$, $\exists \sigma [z \in I^\sigma(x)]$. Therefore, there exists a set operation F such that $F(z)$ is the least such σ and hence $z \in I^{F(z)}(x)$. By Replacement, $v := \{F(z) : z \in u\}$ is a set of ordinals, so $\sigma := \bigcup v$ is an ordinal. Each $F(z) \leq \sigma$ so by (i), $z \in I^\sigma(x)$. Hence $u \subseteq I^\sigma(x)$. $\square$

In part (iii) we are using that for monotone I , $\dot{\varphi}(u)$ is closed in the following sense.

Definition 6.3.22 For any (binary) set operation I , any class U , any $w \subseteq U$ and any x ,

- (i) w is a **chain** in $U \leftrightarrow (\forall x \in w) (\forall y \in w) x \subseteq y \vee y \subseteq x$;
- (ii) U is **chain closed** $\leftrightarrow (\forall w \subseteq U) [w \text{ is a chain} \rightarrow \bigcup w \in U]$;
- (iii) U is **I -inductively closed** at $x \leftrightarrow U \text{ is chain closed} \wedge \forall u [u \in U \rightarrow I(u, x) \in U]$.

Note that since $\subseteq$ is always a partial order on any set, a chain is simply a set which is linearly ordered by $\subseteq$.

Lemma 6.3.23 For any (binary) set operation I which is either monotone or inclusive, and any class U , if U is I -inductively closed at x , then $\forall \sigma [I^\sigma(x) \in U]$.

Proof. A straightforward induction using the preceding proposition and the fact that $\dot{\bigcup} \emptyset \doteq \bigcup \emptyset \in U$. $\square$

Fixed-point Theorem 6.3.24 For any (binary) set operation I which is either monotone or inclusive, any x and any class U , if U is I -inductively closed at x and U is a set, then

- (i) $\exists \sigma [I \text{ stops by } \sigma \text{ at } x]$;
- (ii) if I is monotone, then $\dot{I}(x) \doteq I^\infty(x)$ is a set, is a member of U and is the least fixed point of I ;
- (iii) if I is inclusive, then $I^\infty(x)$ is a set, is a member of U and is a fixed point of I .

Proof. For (i) assume the hypotheses but, towards a contradiction, that there is no σ such that I stops by σ at x . It follows from the preceding

proposition that for all $\sigma < \tau$, $I^\sigma(x)$ is a proper subset of $I^\tau(x)$. In other words, the class operation $\sigma \mapsto I^\sigma(x)$ is injective. By Corollary 6.2.13, the image of $\dot{O}r$ under this mapping $\{\{I^\sigma(x) : \sigma \in \dot{O}r\}\}$ is a proper class. But by the lemma this class is included in U , so is a set.

Now fix σ such that I stops by σ at x . $I^\infty(x) = I^\sigma(x)$ so is a set. $I^\infty(x)$ is a fixed-point of I since

$$I(I^\infty(x), x) = I(I^\sigma(x), x) \doteq I^{\sigma+1}(x) = I^\infty(x).$$

Furthermore, if I is monotone, then by Proposition 6.3.21(ii), $I^\infty(x)$ is included in any other fixed-point so is the least fixed point $\bar{I}(x)$. Finally, $I^\infty(x) \doteq I^\sigma(x) \in U$. $\square$

As a first application of these ideas, we show that in a certain sense, the axioms of ZF characterize a formal universe of sets modeled after the informal universe V described at the end of Section 6.1. We consider the iteration of the set operation $\dot{\varphi}$ and let $\dot{V}_\sigma$ denote the stages $\dot{\varphi}^\sigma$ — that is we have

Lemma 6.3.25

- (i) $\dot{V}_0 \doteq \emptyset$.
- (ii) $\dot{V}_{\sigma+1} \doteq \dot{\varphi}(\dot{V}_\sigma)$.
- (iii) $\dot{V}_\sigma \doteq \dot{\bigcup} \{\dot{V}_\tau : \tau \dot{<} \sigma\}$, for limit σ .

Proposition 6.3.26 For all σ and τ ,

- (i) $\sigma \dot{\leq} \tau \rightarrow \dot{V}_\sigma \dot{\subseteq} \dot{V}_\tau$;
- (ii) $\dot{V}_\sigma$ is transitive;
- (iii) $\dot{V}_\sigma \in \dot{V}_{\sigma+1}$;
- (iv) $\dot{V}_\sigma \dot{\cap} \dot{O}r \doteq \sigma$.

Proof. Since $\dot{\varphi}$ is monotone, (ii) is immediate from Proposition 6.3.21(i); (ii) follows since $\dot{V}_\sigma \dot{\subseteq} \dot{V}_{\sigma+o1} \doteq \dot{\varphi}(\dot{V}_\sigma)$. (iii) is also immediate: for all x , $x \in \dot{\varphi}(x)$, so in particular, $\dot{V}_\sigma \in \dot{\varphi}(\dot{V}_\sigma) \doteq \dot{V}_{\sigma+1}$. (iv) is clear for $\sigma \doteq \dot{0}$. If $\sigma \doteq \tau +_o 1$, then assume as induction hypothesis that $\dot{V}_\tau \dot{\cap} \dot{O}r \doteq \tau$. Then $\tau \dot{\subseteq} \dot{V}_\tau$, so $\tau \in \dot{\varphi}(\dot{V}_\tau) \doteq \dot{V}_\sigma$ and by (i), $\sigma \doteq \tau \dot{\cup} \{\tau\} \dot{\subseteq} \dot{V}_\sigma$. Conversely, for any $\rho \dot{\geq} \sigma$, $\rho \not\dot{\subseteq} \dot{V}_\tau$, since $\tau \notin \dot{V}_\tau$, so $\rho \not\dot{\subseteq} \dot{V}_\tau$ and thus also $\dot{V}_\sigma \dot{\cap} \dot{O}r \dot{\subseteq} \sigma$. Finally, if σ is a limit, we have

$$\dot{V}_\sigma \dot{\cap} \dot{O}r \doteq \dot{\bigcup} \{\dot{V}_\tau \dot{\cap} \dot{O}r : \tau \dot{<} \sigma\} \doteq \dot{\bigcup} \{\tau : \tau \dot{<} \sigma\} \doteq \dot{\bigcup} \sigma \doteq \sigma. \quad \square$$

Theorem 6.3.27 $\dot{\varphi}^\infty \doteq \dot{V}$ — that is, $\dot{V} \doteq \bigcup \{ \dot{V}_\sigma : \sigma \in \dot{O}r \}$.

Proof. By the Set Induction Theorem 6.2.47, it suffices to show that

$$\forall u [u \subseteq \dot{\varphi}^\infty \rightarrow u \in \dot{\varphi}^\infty].$$

But if $u \subseteq \dot{\varphi}^\infty$, then by Proposition 6.3.21(v), for some σ , $u \subseteq \dot{\varphi}^\sigma$. Then $u \in \dot{\varphi}^{\sigma+1} \subseteq \dot{\varphi}^\infty$. $\square$

The following alternative characterization of the sets $\dot{V}_\sigma$ will be useful in [Section 6.5](#).

Definition 6.3.28 rank is the unique set operation such that for all y ,

$$\text{rank}(y) \doteq \bigcup \{ \text{rank}(z) \dot{+}_o 1 : z \in y \}.$$

Proposition 6.3.29

- (i) $\forall y [\text{rank}(y) \in \dot{O}r]$.
- (ii) $\forall \sigma [\dot{V}_\sigma \doteq \{ y : \text{rank}(y) < \sigma \}]$.
- (iii) $\forall y [\text{rank}(y) \doteq \text{least } \sigma (y \in \dot{V}_{\sigma+o1})]$.

Proof. (i) is established by a simple set induction using Proposition 6.3.12. We prove (ii) by ordinal induction. First,

$$\dot{V}_0 \doteq \emptyset \doteq \{ y : \text{rank}(y) < 0 \}.$$

The successor step follows from the equivalences

$$\begin{aligned} y \in \dot{V}_{\sigma+o1} &\leftrightarrow y \subseteq \dot{V}_\sigma \\ &\leftrightarrow (\forall z \in y) \text{rank}(z) < \sigma \\ &\leftrightarrow (\forall z \in y) \text{rank}(z) \dot{+}_o 1 \subseteq \sigma \\ &\leftrightarrow \text{rank}(y) \subseteq \sigma \\ &\leftrightarrow \text{rank}(y) < \sigma \dot{+}_o 1. \end{aligned}$$

Similarly, for σ a limit ordinal,

$$y \in \dot{V}_\sigma \leftrightarrow (\exists \tau < \sigma) y \in \dot{V}_\tau \leftrightarrow (\exists \tau < \sigma) \text{rank}(y) < \tau \leftrightarrow \text{rank}(y) < \sigma.$$

Finally, (iii) is immediate from (ii). $\square$

A useful consequence of Theorem 6.3.27 is

Theorem 6.3.30 (Collection Principle) For any class operation F , there exists a set operation G such that

$$\forall x \forall u (\forall y \in u) [F(x, y) \neq \emptyset \rightarrow F(x, y) \cap G(u, x) \neq \emptyset].$$

Proof. Given F , let

$$H(x, y) := \begin{cases} \text{least } \sigma [F(x, y) \cap \dot{V}_\sigma \neq \emptyset, & \text{if } F(x, y) \neq \emptyset; \\ 0, & \text{otherwise;} \end{cases}$$

$$G(u, x) := \dot{V}_{\bigcup \{H(x, y) : y \in u\}}.$$

Easily G is as required. $\square$

Remark 6.3.31 It is sometimes more natural to think of this principle in the following equivalent form: for any $L_\in$ -formula $\psi(x, y, z)$, there exists a set operation G such that

$$\forall x \forall u (\forall y \in u) [\exists z \psi(x, y, z) \rightarrow (\exists z \in G(u, x)) \psi(x, y, z)].$$

The set $G(u, x)$ collects witnesses z for each $y \in u$.

Collateral Reading. Most of the material of this and the following section is to be found in practically any basic text on Set Theory, for example, [Jec], [Vau] or [En77].

Exercises

Exercise 6.3.32 Show that there is no set w such that

$$\forall z [z \in w \leftrightarrow \dot{0} \in z \wedge \forall x (x \in z \rightarrow \dot{S}x \in z)].$$

Exercise 6.3.33 Show that for all ordinals ρ , σ and τ ,

- (i) $\rho \dot{+}_o (\sigma \dot{+}_o \tau) \doteq (\rho \dot{+}_o \sigma) \dot{+}_o \tau$; $\rho \dot{\times}_o (\sigma \dot{\times}_o \tau) \doteq (\rho \dot{\times}_o \sigma) \dot{\times}_o \tau$;
- (ii) $\rho \dot{\times}_o (\sigma \dot{+}_o \tau) \doteq (\rho \dot{\times}_o \sigma) \dot{+}_o (\rho \dot{\times}_o \tau)$;
- (iii) $\dot{0} \dot{+}_o \rho \doteq \rho$; $\dot{0} \dot{\times}_o \rho \doteq \dot{0}$; $\dot{1} \dot{\times}_o \rho \doteq \rho \doteq \rho \dot{\times}_o \dot{1}$;
- (iv) σ is a limit ordinal $\leftrightarrow \exists \rho [\sigma \doteq \dot{\omega} \dot{\times}_o \rho]$;
- (v) $\sigma \dot{<} \tau \leftrightarrow \rho \dot{+}_o \sigma \dot{<} \rho \dot{+}_o \tau$;
- (vi) if $\dot{0} \dot{<} \rho$, then $\sigma \dot{<} \tau \leftrightarrow \rho \times \sigma \dot{<} \rho \dot{\times}_o \tau$;
- (vii) $\rho \dot{\leq} \sigma \rightarrow \rho \dot{+}_o \tau \dot{\leq} \sigma \dot{+}_o \tau \wedge \rho \dot{\times}_o \tau \dot{\leq} \sigma \dot{\times}_o \tau$;
- (viii) $\rho \dot{\leq} \sigma \rightarrow \exists! \tau [\rho \dot{+}_o \tau \doteq \sigma]$ (**subtraction lemma**);
- (ix) $\exists! \tau \exists! v [\rho \doteq (\sigma \dot{\times}_o \tau) \dot{+}_o v \wedge v \dot{<} \sigma]$ (**division lemma**);
- (x) give an example to show that generally $(\rho \dot{+}_o \sigma) \dot{\times}_o \tau \neq (\rho \dot{\times}_o \tau) \dot{+}_o (\sigma \dot{\times}_o \tau)$.

Exercise 6.3.34 There exists a unique set operation $(\rho, \sigma) \mapsto \rho^\sigma$ such that

$$\rho^{\dot{0}} \doteq \dot{1}; \quad \rho^{\dot{\sigma}^\tau} \doteq \rho^\tau \dot{\times}_o \rho; \quad \rho^\sigma \doteq \bigcup \{ \rho^\tau : \tau \dot{<} \sigma \} \quad \text{for } \sigma \text{ a limit ordinal.}$$

Establish the following for all ordinals ρ, σ and τ ,

- (i) $\rho^{\sigma \dot{+}_o \tau} \doteq \rho^\sigma \dot{\times}_o \rho^\tau; \quad \rho^{\sigma \dot{\times}_o \tau} \doteq (\rho^\sigma)^\tau;$
- (ii) if $\dot{1} \dot{<} \rho$, then $\sigma \dot{<} \tau \leftrightarrow \rho^\sigma \dot{<} \rho^\tau;$
- (iii) $\rho \dot{\leq} \sigma \rightarrow \rho^\tau \dot{\leq} \sigma^\tau;$
- (iv) $\dot{0} \dot{<} \rho \wedge \dot{1} \dot{<} \sigma \rightarrow$

$$\exists! \tau \exists! v \exists! v' [\rho \doteq (\sigma^\tau \dot{\times}_o v) \dot{+}_o v' \wedge \dot{0} \dot{<} v \dot{<} \sigma \wedge v' \dot{<} \sigma^\tau];$$

(*logarithm lemma*).

Exercise 6.3.35 Prove that the ordinal ρ of Theorem 6.3.18 is uniquely determined.

Exercise 6.3.36 Prove that for any relation r , r is well-founded iff

$$\exists f (f : \text{Fld}(r) \rightarrow \dot{\mathbf{O}}r \wedge \forall x \forall y [\langle x, y \rangle \in r \rightarrow f(x) \dot{<} f(y)]).$$

Exercise 6.3.37 Let $\dot{\mathbf{V}}^a$ denote the closure of the set operation $x \mapsto \dot{\varphi}(x \dot{\times}_o a)$. Define an appropriate set operation rank^a such that for all x ,

$$\forall x \forall \sigma [x \in \dot{\mathbf{V}}_\sigma^a \leftrightarrow \text{rank}^a(x) \dot{<} \sigma].$$

Exercise 6.3.38 Show that for all ordinals ρ and σ ,

- (i) $\rho \dot{+}_o \sigma \doteq \rho \dot{\cup} \{ \rho \dot{+}_o \tau : \tau \dot{<} \sigma \};$
- (ii) $\rho \dot{+}_o \sigma$ is the unique ordinal such that $\dot{<}_{\rho \dot{+}_o \sigma}$ is isomorphic to

$$r_{\rho, \sigma}^+ := \left\{ \langle \langle \dot{\tau}, i \rangle, \langle \dot{\tau}', j \rangle \rangle : (i \doteq j \doteq \dot{0} \wedge \tau \dot{<} \tau' \dot{<} \rho) \right. \\ \left. \vee (i \doteq \dot{0} \wedge j \doteq \dot{1}) \vee (i \dot{<} j \doteq \dot{1} \wedge \tau \dot{<} \tau' \dot{<} \sigma) \right\};$$

- (iii) $\rho \dot{\times}_o \sigma$ is the unique ordinal such that $\dot{<}_{\rho \dot{\times}_o \sigma}$ is isomorphic to

$$r_{\rho, \sigma}^\times := \left\{ \langle \langle \dot{\tau}, v \rangle, \langle \dot{\tau}', v' \rangle \rangle : (v \dot{<} v' \dot{<} \sigma) \right. \\ \left. \vee (v \doteq v' \dot{<} \sigma \wedge \tau \dot{<} \tau' \dot{<} \rho) \right\}.$$

Exercise 6.3.39 Prove Corollary 6.3.17.

Exercise 6.3.40 Give examples of sets y and z such that

- (i) y is transitive but y is not an ordinal;
- (ii) every element of z is transitive but z is not an ordinal.

Exercise 6.3.41 Show that $\forall x [\text{rank}(x) \doteq \{\text{rank}(y) : y \in \text{TC}(x)\}]$.

Exercise 6.3.42 Show that for any monotone binary set operation I , for all σ ,

$$\mathsf{I}^\sigma \doteq \bigcup \{ \mathsf{I}^\tau(x), x : \tau < \sigma \}.$$

6.4 Cardinal numbers

In this section we develop the main properties of cardinality and cardinal numbers. Some of this will again take place in **ZF**, but many parts of the theory of cardinality require the Axiom of Choice.

Definition 6.4.1 The *Axiom of Choice* (AC) is the following sentence:

$$\forall x \exists c \left(\text{Fn}(c) \wedge x \subseteq \text{Dom}(c) \wedge \forall y \in x [y \neq \emptyset \rightarrow c(y) \in y] \right);$$

c is called a *choice function* for x .

We denote by **ZFC** the theory generated by the axioms of **ZF** together with AC, and put this as a label on those theorems which make use of AC.

In Definition 6.2.32 we introduced the notion of equipotence:

$$x \approx y \leftrightarrow \exists f [\text{Inj}(f) \wedge \text{Dom}(f) \doteq x \wedge \text{Im}(f) \doteq y].$$

Intuitively, two sets are the “same size” iff there is a way to pair off their elements. Analogously,

Definition 6.4.2 For any x and y ,

- (i) $x \preceq y : \leftrightarrow \exists f [\text{Inj}(f) \wedge \text{Dom}(f) \doteq x \wedge \text{Im}(f) \subseteq y]$;
- (ii) $x \prec y : \leftrightarrow [x \preceq y \wedge \neg (y \approx x)]$.

Thus $x \preceq y$ iff there is a pairing of the elements of x with some of the elements of y — that is, $x \approx z$ for some $z \subseteq y$.

Proposition 6.4.3

- (i) $\dot{\preceq}$ is reflexive and transitive.
- (ii) $\dot{\preceq}$ is a class strict partial ordering.

Proof. For (i), $x \dot{\preceq} x$ via the identity function $\{ \langle u, u \rangle : u \in x \}$. If $x \dot{\preceq} y$ and $y \dot{\preceq} z$ via, respectively, f and g , then $x \dot{\preceq} z$ via the composition $g \circ f$. For (ii), irreflexivity is immediate from the definition. Suppose that $x \dot{\preceq} y$ and $y \dot{\preceq} z$. Then $x \dot{\preceq} z$ by (i), but if $z \approx x$, then again by (i), $z \dot{\preceq} y$, contrary to hypothesis. $\square$

The next result really justifies our choice of notation.

Schröder-Bernstein Theorem 6.4.4 $\forall x \forall y [x \dot{\preceq} y \wedge y \dot{\preceq} x \rightarrow x \approx y]$.

Proof. Fix injective functions f and g with

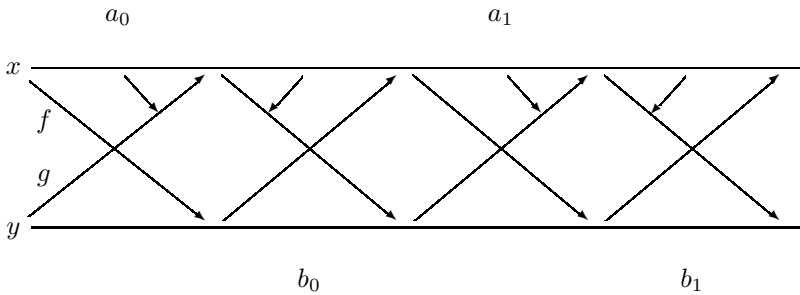
$$\text{Dom}(f) \doteq x, \quad \text{Im}(f) \subseteq y, \quad \text{Dom}(g) \doteq y \quad \text{and} \quad \text{Im}(g) \subseteq x.$$

By $\dot{\omega}$ -recursion, there exist set operations $z \mapsto a_z$ and $z \mapsto b_z$ such that for all $z \in \dot{\omega}$,

$$a_0 \doteq x \setminus \text{Im}(g); \quad b_z \doteq f^* a_z; \quad a_{z+\dot{o}1} \doteq g^* b_z.$$

Set

$$a \doteq \bigcup \{ a_z : z \in \dot{\omega} \} \quad \text{and} \quad b \doteq \bigcup \{ b_z : z \in \dot{\omega} \}.$$



To establish that $x \approx y$ it will suffice to show

- (i) $f^* a \doteq b$ and
- (ii) $g^*(y \setminus b) \doteq x \setminus a$,

since then easily $f \upharpoonright a \cup [g^{-1} \upharpoonright (x \setminus a)]$ is a bijective function from x to y . (i) is immediate from the definitions. $g^*(y \setminus b) \subseteq x \setminus a$ because for any $v \in y$,

$$v \notin b \rightarrow (\forall z \in \dot{\omega}) v \notin b_z \rightarrow (\forall z \in \dot{\omega}) g(v) \notin a_z \rightarrow g(v) \notin a.$$

Similarly, $x \setminus a \subseteq g^*(y \setminus b)$ because for any $u \in x$,

$$\begin{aligned} u \notin a &\rightarrow u \in \dot{\text{Im}}(g) \wedge (\forall z \in \dot{\omega}) u \notin a_{z+o1} \\ &\rightarrow u \in \dot{\text{Im}}(g) \wedge (\forall z \in \dot{\omega}) g^{-1}(u) \notin b_z \rightarrow u \in \dot{\text{Im}}(g) \wedge g^{-1}(u) \notin b, \end{aligned}$$

so if $u \doteq g(w)$, then $w \notin b$ and hence $w \in y \setminus b$. $\square$

Remark 6.4.5 A shorter proof (which is essentially the same proof) uses the result of Exercise 6.2.64: with $F := f^*$ and $G := g^*$, the a and b of that exercise are the same as the a and b above.

Corollary 6.4.6 $x \prec y \leftrightarrow [x \preceq y \wedge \neg(y \preceq x)]$. $\square$

Note that we have not (yet) claimed the principle of **cardinal comparability**: $\forall x \forall y [x \preceq y \vee y \preceq x]$. This will follow below, but only as a theorem of ZFC.

Definition 6.4.7 For any x ,

$$x \text{ is } \dot{\text{Dedekind infinite}} : \leftrightarrow \exists z [z \subseteq x \wedge x \neq z \wedge x \preceq z];$$

otherwise x is ***Dedekind finite***. A Dedekind infinite set is one that can be paired up with a proper subset of itself.

Proposition 6.4.8 For any x and w ,

- (i) x is *Dedekind infinite* $\rightarrow x$ is infinite;
- (ii) w is *Dedekind infinite* $\wedge w \preceq x \rightarrow x$ is *Dedekind infinite*;
- (iii) x is *Dedekind infinite* $\leftrightarrow \dot{\omega} \preceq x$;
- (iv) x is *denumerable* $\rightarrow x$ is infinite;
- (v) x is *countable* $\leftrightarrow x \preceq \dot{\omega}$.

Proof. We only sketch the proofs and leave some details to the reader. For (i) it suffices to prove by $\dot{\omega}$ -induction that $(\forall y \in \dot{\omega}) y$ is *Dedekind finite*. Clearly $\dot{0}$ is *Dedekind finite*. For the induction step prove that

$$\dot{S}y \text{ is } \dot{\text{Dedekind infinite}} \rightarrow y \text{ is } \dot{\text{Dedekind infinite}}.$$

For (ii), fix injective functions $g : w \rightarrow x$ and $h : w \rightarrow y \subseteq w$ with $y \neq w$. Let

$$f(u) \doteq \begin{cases} g(h(g^{-1}(u))), & \text{if } u \in \dot{\text{Im}}(g); \\ u, & \text{otherwise.} \end{cases}$$

Then $f : x \rightarrow x$ is injective, and $\dot{\text{Im}}(f) \subseteq g^*y \cup (x \setminus \dot{\text{Im}}(g)) \neq x$, so f witnesses that x is *Dedekind infinite*. (iii)($\leftarrow$) is now immediate; for (iii)($\rightarrow$),

suppose that x is Dedekind infinite and fix $f : x \rightarrow y \subseteq x$ and $u \in x \setminus y$. By $\dot{\omega}$ -recursion, let g be a function such that $g(0) \doteq u$ and for all $z \in \dot{\omega}$, $g(\dot{S}z) \doteq f(g(z))$. Then $g : \dot{\omega} \rightarrow x$ and g is injective. (iv) is immediate from (i) and (iii).

Finally, for (v)($\rightarrow$), if x is finite, say $x \approx y \in \dot{\omega}$, then since $y \subseteq \dot{\omega}$, also $x \preceq \dot{\omega}$. If x is denumerable, $x \approx \dot{\omega}$ so again $x \preceq \dot{\omega}$. Suppose now towards (v)($\leftarrow$) that $x \preceq \dot{\omega}$, and fix an injective function $f : x \rightarrow \dot{\omega}$. If $x \doteq \emptyset$, then x is countable; otherwise let $g(\dot{0}) \doteq \bar{z}$ be the least element of $\text{Im}(f)$ and by $\dot{\omega}$ -recursion

$$g(\dot{S}y) \doteq \begin{cases} \text{least } z > g(y) [z \in \text{Im}(f)], & \text{if such } z \text{ exists;} \\ \dot{\omega}, & \text{otherwise.} \end{cases}$$

Then $\{y \in \dot{\omega} : g(y) \in \dot{\omega}\}$ is either some $z \in \dot{\omega}$, in which case $x \approx z$ via $(g^{-1} \upharpoonright \text{Im}(f)) \circ f$ and x is finite, or is $\dot{\omega}$, in which case $x \approx \omega$ via the same function. $\square$

Intuitively one expects also the implication (i)($\leftarrow$) in the preceding proposition; surprisingly this requires AC.

Proposition 6.4.9 (ZFC) *For all x ,*

- (i) x is infinite $\leftrightarrow x$ is Dedekind infinite;
- (ii) x is uncountable $\leftrightarrow \dot{\omega} \prec x$.

Proof. It suffices to prove that x is infinite $\rightarrow \dot{\omega} \preceq x$. Fix an infinite set x , an element w of x and a choice function c for $\wp x$. By $\dot{\omega}$ -recursion there exists a function f such that $f(\dot{0}) \doteq w$ and for all $y \in \dot{\omega}$,

$$f(\dot{S}y) = c(x \setminus \bigcup \{f(z) : z \leq y\}).$$

By $\dot{\omega}$ -induction, for all $y \in \dot{\omega}$, $f \upharpoonright y$ is injective and hence

$$x \setminus \bigcup \{f(z) : z \leq y\} \neq \emptyset,$$

since otherwise $x \approx y$ and is finite, contrary to hypothesis. Hence $f : \dot{\omega} \rightarrow x$ is injective and witnesses that $\dot{\omega} \preceq x$. $\square$

The idea of comparing sizes of infinite sets is of interest only because of Cantor's discovery that infinite sets come in many sizes. The simplest expression of this is that there exist uncountable sets. Cantor proved that the set $\mathbb{R}$ of real numbers is uncountable. Because we have not developed in ZF enough of the properties of $\mathbb{R}$ to carry out this proof without some preparation, we give an essentially equivalent version (cf. Proposition 4.1.22).

Cantor's Theorem 6.4.10 $x \prec \wp(x)$.

Proof. Clearly $x \succeq \wp(x)$ via the function $\{\langle u, \{u\} \rangle : u \in x\}$. Suppose that $x \approx \wp(x)$ via some function f . Let $u \doteq \{y \in x : y \notin f(y)\}$. Since $u \in \wp(x) = \text{Im}(f)$, for some $z \in x$, $u \doteq f(z)$. Then $z \in f(z) \leftrightarrow z \notin f(z)$, a contradiction. $\square$

Corollary 6.4.11 $\wp\omega$ is uncountable. $\square$

This is equivalent to Cantor's original result also because $\mathbb{R} \approx \wp\omega$, and it has the advantage that it can be iterated: $\omega \prec \wp\omega \prec \wp\wp\omega \cdots$ is an infinite sequence of sets of ever increasing size. A natural question is whether or not $\wp\omega$ is of minimal size among uncountable sets — that is, is the following (first version of the) **Continuum Hypothesis** true?

$\text{CH}_1 : \neg \exists u [\omega \prec u \wedge u \prec \wp\omega]$.

Cantor conjectured that CH_1 is true and spent considerable effort trying to prove it. Although he was, of course, not working in our strict axiomatic framework, which was introduced much later, unless he had used methods still unknown today which could not be formalized in ZFC, his success would have in fact shown that $\text{ZFC} \vdash \text{CH}_1$. Since we will show in [Section 6.8](#) that this is false (unless ZF is inconsistent), his failure to prove CH_1 was inevitable. Cantor believed also the **Generalized Continuum Hypothesis**:

$\text{GCH}_1 : \forall x \neg \exists u [x \prec u \wedge u \prec \wp x]$.

Note that in ZFC, where cardinal comparability holds, GCH_1 may be equivalently stated $\forall x \forall u [x \prec u \rightarrow \wp x \preceq u]$.

Early in the 20th century was discovered a second method for producing an uncountable set and more generally a set of size larger than that of a given one.

Definition 6.4.12 (*Hartogs' operation*) $\text{Ha}(x) := \{\{\sigma : \sigma \preceq x\}\}$.

For example, by Proposition 6.4.8(v), $\text{Ha}(\omega)$ is the class of all countable ordinals and includes the finite ordinals, ω , $\omega +_o \omega$ and many more.

Proposition 6.4.13 Ha is a set operation, and for all x ,

- (i) $\text{Ha}(x)$ is a ordinal;
- (ii) $\text{Ha}(x) \not\preceq x$;
- (iii) $x \in \dot{\text{Or}} \rightarrow x \prec \text{Ha}(x)$.

Proof. By Theorem 6.3.18 there is a set operation F such that for every well-ordering r , $F(r)$ is an ordinal and $r \simeq \dot{\prec}_{F(r)}$. To verify that each $\text{Ha}(x)$ is a set, it will suffice (by Powerset and Replacement) to verify that for all x ,

$$\text{Ha}(x) \doteq F^* \left\{ r : r \subseteq x \times x \wedge r \text{ is a well-ordering} \right\}.$$

Let w denote the set on the right and suppose first that $\sigma \in w$, say $\sigma \doteq \mathbf{F}(r)$. Then $\dot{\prec}_\sigma \dot{\simeq} r$, say via f , and in particular f is an injection of σ into $\mathbf{Fld}(r) \subseteq x$. Hence $\sigma \dot{\preceq} x$. Conversely, if $\sigma \dot{\preceq} x$ via g , let

$$r = \{ \langle g(\rho), g(\tau) \rangle : \rho \dot{\prec} \tau \dot{\prec} \sigma \}.$$

Easily r is a well-ordering of $\mathbf{Fld}(r) \doteq \mathbf{Im}(g) \subseteq x$ and $\mathbf{F}(r) \doteq \sigma$, so $\sigma \in w$.

For (i), $\mathbf{Ha}(x)$ is a set of ordinals and is transitive, since if $\tau \dot{\prec} \sigma \in \mathbf{Ha}(x)$, then $\tau \subseteq \sigma \dot{\preceq} x$, so also $\tau \dot{\preceq} x$ and thus $\tau \in \mathbf{Ha}(x)$. Therefore $\mathbf{Ha}(x)$ is an ordinal. Now (ii) follows, since otherwise $\mathbf{Ha}(x) \in \mathbf{Ha}(x)$, contrary to Lemma 6.1.14. (iii) is now immediate, since if x is an ordinal, then $x \subseteq \mathbf{Ha}(x)$, so $x \dot{\preceq} \mathbf{Ha}(x)$. $\square$

In particular, $\dot{\omega} \dot{\prec} \mathbf{Ha}(\dot{\omega}) \dot{\prec} \mathbf{Ha}(\mathbf{Ha}(\dot{\omega})) \cdots$. We shall see below that $\mathbf{Ha}(x)$ is minimal above x in the sense that it is a theorem of ZFC that

$$\forall x \neg \exists u [x \dot{\prec} u \wedge u \dot{\prec} \mathbf{Ha}(x)]; \quad \text{in fact, } \forall x \forall u [x \dot{\prec} u \rightarrow \mathbf{Ha}(x) \dot{\preceq} u],$$

so in particular, $\text{ZFC} \vdash \forall x [\mathbf{Ha}(x) \dot{\preceq} \dot{\phi}x]$. Hence we have the following alternative versions of the (G)CH:

$$\text{CH}_2 : \quad \dot{\phi}\dot{\omega} \dot{\preceq} \mathbf{Ha}(\dot{\omega})$$

$$\text{GCH}_2 : \quad \forall x [\dot{\phi}x \dot{\preceq} \mathbf{Ha}(x)].$$

These questions, and many others, are greatly clarified by the notion of cardinal number. Let

$$\text{ep}(x) := \{ \{ y : x \dot{\approx} y \} \}, \quad \text{the \textit{equipotency class} of } x,$$

the equivalence class of x under the class equivalence relation $\dot{\approx}$. Except for $x \doteq \emptyset$, $\text{ep}(x)$ is a proper class (exercise). For $k \in \omega$, $\text{ep}(\dot{k})$ is the class of sets with k elements, and $\text{ep}(\dot{\omega})$ is the class of denumerable sets. By Proposition 6.2.33(ii), these are all distinct and hence pairwise disjoint. By the above, the same is true for the classes $\text{ep}(\dot{\phi}\dot{\omega})$, $\text{ep}(\dot{\phi}\dot{\phi}\dot{\omega})$ etc. The role of cardinal numbers is to provide a canonical representative element of each equivalence class $\text{ep}(x)$ — that is, we want to define a set operation $|\cdot|$ such that for all x and y ,

$$(1) \quad |x| \dot{\approx} x \quad \text{and} \quad x \dot{\approx} y \leftrightarrow |x| \doteq |y|.$$

For example, a natural representative for the class of 3-element sets is $\dot{3} \doteq \{ \dot{0}, \dot{1}, \dot{2} \}$, and for the class $\text{ep}(\dot{\omega})$ of denumerable sets, the obvious choice is $\dot{\omega}$ itself.

Definition 6.4.14 For all x , $\dot{|}x| := \dot{\bigcap} [\text{ep}(x) \dot{\cap} \dot{\text{Or}}]$.

Thus if $\text{ep}(x) \dot{\cap} \dot{\text{Or}} \neq \emptyset$, $\dot{|}x|$ is the smallest ordinal σ such that $x \approx \sigma$; otherwise, $\dot{|}x| \doteq \dot{\mathbf{V}}$.

Corollary 6.4.15 $(\forall y \dot{\in} \dot{\omega}) \dot{|}y| \doteq y; \quad \dot{|}\dot{\omega}| \doteq \dot{\omega}$.

Proof. For $y \dot{\in} \dot{\omega}$, by Proposition 6.2.33(v), $\text{ep}(y) \dot{\cap} \dot{\text{Or}} \doteq \{y\}$, so $\dot{|}y| \doteq \dot{\bigcap} \{y\} \doteq y$. Clearly $\dot{\omega} \dot{\in} \text{ep}(\dot{\omega}) \dot{\cap} \dot{\text{Or}}$. Although this set has many other ordinal members, clearly $\dot{\omega}$ is the smallest among them. $\square$

To show that $\dot{|} \cdot |$ is a set operation, we need to establish that $\forall x [\text{ep}(x) \dot{\cap} \dot{\text{Or}} \neq \emptyset]$. This will again require AC, and indeed is equivalent to it.

Definition 6.4.16 For any x ,

$$x \text{ is } \textbf{well-orderable} \quad \leftrightarrow \quad \exists r [r \text{ is a well-ordering} \wedge \text{Fid}(r) \doteq x].$$

Such an r is called a well-ordering of x .

Proposition 6.4.17 For all x , the following are equivalent:

- (i) $x \dot{\preceq} \text{Ha}(x)$;
- (ii) x is well-orderable;
- (iii) $\exists \sigma [x \approx \sigma]$.

Proof. Assuming (i), let $f : x \dot{\rightarrow} \text{Ha}(x)$ be an injection. Then it is easy to check that $\{\langle y, z \rangle : y, z \dot{\in} x \wedge f(y) \dot{<} f(z)\}$ is a well-ordering of x . Given any well-ordering r of x , by Theorem 6.3.18, for some ordinal σ , $r \dot{\simeq} \dot{<}_\sigma$, so in particular, $x \approx \sigma$. Finally, given $x \approx \sigma$, $\sigma \dot{\in} \text{Ha}(x)$, so $\sigma \dot{\subseteq} \text{Ha}(x)$ and hence $x \dot{\preceq} \text{Ha}(x)$. $\square$

Well-ordering Theorem 6.4.18 The following are equivalent:

- (i) AC;
- (ii) $\forall x [x \dot{\preceq} \text{Ha}(x)]$;
- (iii) $\forall x [x \text{ is well-orderable}]$;
- (iv) $\forall x \exists \sigma [x \approx \sigma]$, so $\dot{|} \cdot |$ is a set operation;
- (v) $\forall x \exists \sigma \exists f [\text{Fn}(f) \wedge \text{Dom}(f) \doteq \sigma \wedge x \dot{\subseteq} \text{Im}(f)]$.

Proof. By the preceding proposition, (ii) $\leftrightarrow$ (iii) $\leftrightarrow$ (iv) $\rightarrow$ (v), so it will suffice to prove (v) $\rightarrow$ (iii) $\rightarrow$ (i) $\rightarrow$ (iv). First, suppose that x , σ and f are as in (v). Then the following is a well-ordering of x :

$$\{\langle y, z \rangle : y, z \dot{\in} x \wedge (\exists \tau \dot{<} \sigma) [f(\tau) \doteq y \wedge (\forall v \dot{<} \sigma)(f(v) \doteq z \rightarrow \tau \dot{<} v)]\}.$$

Next, assuming (iii), for any x , let r be a well-ordering of $\dot{\bigcup} x$. Then for $y \in x$ set

$$c(y) := \begin{cases} \text{the } r\text{-least element of } y, & \text{if } y \neq \dot{\emptyset}; \\ \dot{\emptyset}, & \text{otherwise.} \end{cases}$$

Clearly c is a choice function for x .

Finally, given a set x , let c be a choice function for $\dot{\varphi}(x)$. By ordinal recursion there exists a set-operation F such that for all σ

$$F(\sigma) \doteq \begin{cases} c(x \setminus F^*(\sigma)), & \text{if } x \setminus F^*(\sigma) \neq \dot{\emptyset}; \\ x, & \text{otherwise.} \end{cases}$$

Suppose that for all σ , $F(\sigma) \neq x$. Then F is an injection of $\dot{\mathcal{O}}r$ into x , contrary to Corollary 6.2.13. Hence, if $\bar{\sigma} = \text{least } \sigma [F(\sigma) = x]$, then easily $F \upharpoonright \bar{\sigma}$ is a set-operation which witnesses that $\bar{\sigma} \approx x$. $\square$

Corollary 6.4.19 (ZFC) $\dot{\mid} \cdot \mid$ is a set operation and satisfies (1). Further, for all x and y ,

- (i) $x \dot{\preceq} y \iff \mid x \mid \dot{\preceq} \mid y \mid$;
- (ii) $x \dot{\prec} y \iff \mid x \mid \dot{\prec} \mid y \mid$;
- (iii) $x \dot{\preceq} y \vee y \dot{\preceq} x$.

Proof. The first statement is now immediate and parts (i)–(iii) are left as exercises. $\square$

Definition 6.4.20 $\dot{\mathcal{C}}a := \dot{\text{lm}}(\dot{\mid} \cdot \mid) = \{ \{ \sigma : \exists x (\mid x \mid = \sigma) \} \}$ is the class of **cardinal numbers** or simply **cardinals**.

We use letters κ , λ , μ and ν as variables to represent (usually infinite) cardinal numbers. Then extending Corollary 6.4.15,

Corollary 6.4.21

- (i) $(\forall y \in \dot{\omega}) y$ is a cardinal.
- (ii) $\dot{\omega}$ is a cardinal.
- (iii) For every cardinal κ , $\mid \kappa \mid \doteq \kappa$.

Proof. Parts (i) and (ii) are immediate. For (iii), if κ is a cardinal, then $\kappa \doteq \mid x \mid$ for some x . Hence $x \approx \kappa$, but for no $\sigma \dot{<} \kappa$ is $x \approx \sigma$. Hence for no $\sigma \dot{<} \kappa$ is $\kappa \approx \sigma$ and thus $\mid \kappa \mid \doteq \kappa$. $\square$

Of course, $\dot{\omega} \dot{+}_o \dot{1}$ is not a cardinal because the function

$$\{ \langle \dot{y}, \dot{S}y \rangle : y \in \dot{\omega} \} \cup \{ \langle \dot{\omega}, \dot{0} \rangle \}$$

witnesses that $\dot{\omega} \dot{+}_o \dot{1} \approx \dot{\omega}$. We have the following simple characterization of $\dot{\mathcal{C}}a$ as a subclass of $\dot{\mathcal{O}}r$.

Proposition 6.4.22 $\dot{\mathcal{C}}a \doteq \{ \{ \sigma : \neg (\exists \tau \dot{<} \sigma) \sigma \dot{\leq} \tau \} \}$.

Proof. If σ satisfies the condition, clearly $|\sigma| = \sigma$, so $\sigma \in \dot{\mathcal{C}}a$. Conversely, suppose that $\kappa \in \dot{\mathcal{C}}a$ but, towards a contradiction, there exists $\sigma \dot{<} \kappa$ such that $\kappa \dot{\leq} \sigma$. But also $\sigma \subseteq \kappa$ so $\sigma \dot{\leq} \kappa$, and thus by the Schröder-Bernstein Theorem, $\kappa \approx \sigma$ so $|\kappa| \leq \sigma$, contradicting (iii) of the preceding corollary. $\square$

Corollary 6.4.23 For all x ,

(i) $\text{Ha}(x)$ is a cardinal and if x is well-orderable, then

$$\text{Ha}(x) \doteq \text{Ha}(|x|);$$

(ii) for all cardinals κ , $\text{Ha}(\kappa)$ is the least cardinal λ such that $\kappa \dot{<} \lambda$.

Proof. $\text{Ha}(x)$ is a cardinal because for each $\tau \dot{<} \text{Ha}(x)$, $\tau \dot{\leq} x$ so if $\text{Ha}(x) \dot{\leq} \tau$, then $\text{Ha}(x) \dot{\leq} x$, contrary to Proposition 6.4.13(ii). For the second part of (i), note that for any $x \approx y$, $\sigma \dot{\leq} x \leftrightarrow \sigma \dot{\leq} y$. Clearly $\kappa \dot{<} \text{Ha}(\kappa)$, since $\neg \text{Ha}(\kappa) \dot{\leq} \kappa$. For any λ , if $\kappa \dot{\leq} \lambda \dot{<} \text{Ha}(\kappa)$, then $\lambda \dot{\leq} \kappa$, so since λ is a cardinal, $\lambda \dot{\leq} \kappa$, so $\kappa \doteq \lambda$. $\square$

Definition 6.4.24 For every cardinal κ ,

(i) $\kappa^+ := \text{Ha}(\kappa)$;

(ii) $\dot{2}^\kappa := |\dot{\wp} \kappa|$.

A cardinal of the form κ^+ is called a **successor cardinal**. Note that since, for example, $|\dot{\wp} \dot{3}| \doteq \dot{8}$, the notation of (ii) is consistent with the normal meaning of exponentiation. A more general version of this is given in the exercises.

Corollary 6.4.25 (ZFC) $\forall \kappa [\kappa^+ \dot{\leq} \dot{2}^\kappa]$. $\square$

Thus, another version of the (G)CH is

$$\text{CH}_3 : \quad \dot{2}^\omega \doteq \omega^+$$

$$\text{GCH}_3 : \quad \forall \kappa [\dot{2}^\kappa \doteq \kappa^+].$$

Lemma 6.4.26 $\forall x [x \subseteq \dot{\mathcal{C}}a \rightarrow \bigcup x \in \dot{\mathcal{C}}a]$.

Proof. Fix $x \subseteq \dot{\mathcal{C}}a$ — that is, x is a set of cardinals. In particular, x is a set of ordinals, so $\bigcup x$ is an ordinal. Suppose that for some $\sigma \dot{<} \bigcup x$, $\bigcup x \dot{\leq} \sigma$, say via a function f . Since $\sigma \in \bigcup x$, for some $\kappa \in x$, $\sigma \dot{<} \kappa$. But $\kappa \subseteq \bigcup x$ and thus $f \upharpoonright \kappa$ is an injection of κ into σ , contrary to the assumption that κ is a cardinal. Hence $\bigcup x$ is a cardinal. $\square$

It is traditional to use the first letter $\aleph$ (aleph) of the Hebrew alphabet to enumerate the infinite cardinals.

Definition 6.4.27 $\dot{\aleph}$ is the set-operation defined by the following ordinal recursion:

- (i) $\dot{\aleph}_0 = \omega$;
- (ii) $\dot{\aleph}_{\sigma+1} = \dot{\aleph}_\sigma^+$;
- (iii) $\dot{\aleph}_\sigma = \dot{\bigcup} \{ \dot{\aleph}_\tau : \tau < \sigma \}$ for limit σ .

Theorem 6.4.28 $\dot{\mathcal{C}}a \doteq \dot{\omega} \dot{\bigcup} \{ \{ \dot{\aleph}_\sigma : \sigma \in \dot{\mathcal{O}}r \} \}$.

Proof. By Corollary 6.4.21, $\dot{\omega} \dot{\bigcup} \{ \dot{\omega} \} \subseteq \dot{\mathcal{C}}a$, and it follows by a straightforward ordinal induction that $\forall \sigma [\dot{\aleph}_\sigma \in \dot{\mathcal{C}}a]$. For the converse inclusion, another induction shows that $\forall \sigma [\sigma \leq \dot{\aleph}_\sigma]$. Now if κ is any infinite cardinal, $\kappa \leq \dot{\aleph}_\kappa$, so there exists a least σ such that $\kappa \leq \dot{\aleph}_\sigma$. Suppose that $\kappa < \dot{\aleph}_\sigma$. Since κ is infinite, $\sigma \neq \dot{0}$. If $\sigma = \tau + 1$, then $\dot{\aleph}_\sigma = \text{Ha } \dot{\aleph}_\tau$ so $\kappa \leq \dot{\aleph}_\tau$. But by the minimality of σ , $\dot{\aleph}_\tau < \kappa$, so this contradicts the assumption that κ is a cardinal. Finally, if σ is a limit ordinal, we have $\kappa \in \dot{\aleph}_\sigma = \dot{\bigcup} \{ \dot{\aleph}_\tau : \tau < \sigma \}$ so for some $\tau < \sigma$, $\kappa \in \dot{\aleph}_\tau$, contrary to the choice of σ . Hence $\kappa = \dot{\aleph}_\sigma$. $\square$

Our final version version of the (G)CH is now

$$\text{CH}_4 : \quad \dot{2}^{\dot{\aleph}_0} \doteq \dot{\aleph}_1$$

$$\text{GCH}_4 : \quad \forall \sigma [\dot{2}^{\dot{\aleph}_\sigma} \doteq \dot{\aleph}_{\sigma+o1}].$$

We leave to an exercise the proof that in ZFC the four versions are equivalent.

In the following sections we shall have many occasions to compute the cardinality of sets formed from other sets by operations such as union and Cartesian product. To this end it is efficient to introduce operations of addition and multiplication for cardinals; again we take the finite sets as our model.

Definition 6.4.29 For any cardinals κ and λ ,

- (i) $\kappa \dot{+}_c \lambda \doteq \left| (\kappa \dot{\times} \{ \dot{0} \}) \dot{\bigcup} (\lambda \dot{\times} \{ \dot{1} \}) \right|$;
- (ii) $\kappa \dot{\times}_c \lambda \doteq \left| \kappa \dot{\times} \lambda \right|$.

The $\dot{0}$ and $\dot{1}$ of (i) are simply a device for making the union disjoint, and we have, for example,

$$\dot{3} \dot{+}_c \dot{5} \doteq \dot{8} \quad \text{and} \quad \dot{3} \dot{\times}_c \dot{5} \doteq \dot{15}.$$

Thus $\dot{+}_c$ and $\dot{\times}_c$ agree with $\dot{+}$ and $\dot{\times}$, and hence with $\dot{+}_o$ and $\dot{\times}_o$ on $\dot{\omega}$, but it is easy to see that this does not in general extend to larger cardinals — for example,

$$\dot{\omega} \dot{+}_c \dot{1} \doteq \dot{\omega} \neq \dot{\omega} \dot{+}_o \dot{1} \quad \text{and} \quad \dot{\omega} \dot{\times}_c \dot{2} \doteq \dot{\omega} \neq \dot{\omega} \dot{\times}_o \dot{2} \doteq \dot{\omega} \dot{+}_o \dot{\omega}.$$

However, many properties from ordinary arithmetic do carry over easily to cardinal arithmetic.

Proposition 6.4.30 *For any cardinals κ , λ and μ ,*

- (i) $\kappa \dot{+}_c \dot{0} \doteq \kappa$; $\kappa \dot{\times}_c \dot{0} \doteq \dot{0}$; $\kappa \dot{\times}_c \dot{1} \doteq \kappa$; $\kappa \dot{\times}_c \dot{2} \doteq \kappa \dot{+}_c \kappa$;
- (ii) $\kappa \dot{+}_c \lambda \doteq \lambda \dot{+}_c \kappa$; $\kappa \dot{\times}_c \lambda \doteq \lambda \dot{\times}_c \kappa$;
- (iii) $\kappa \dot{+}_c (\lambda \dot{+}_c \mu) \doteq (\kappa \dot{+}_c \lambda) \dot{+}_c \mu$; $\kappa \dot{\times}_c (\lambda \dot{\times}_c \mu) \doteq (\kappa \dot{\times}_c \lambda) \dot{\times}_c \mu$;
- (iv) $\kappa \dot{\times}_c (\lambda \dot{+}_c \mu) \doteq (\kappa \dot{\times}_c \lambda) \dot{+}_c (\kappa \dot{\times}_c \mu)$;
- (v) $\kappa \leq \lambda \rightarrow \kappa \dot{+}_c \mu \leq \lambda \dot{+}_c \mu \wedge \kappa \dot{\times}_c \mu \leq \lambda \dot{\times}_c \mu$;
- (vi) $\kappa \dot{+}_c \lambda \doteq |\kappa \dot{+}_o \lambda|$; $\kappa \dot{\times}_c \lambda \doteq |\kappa \dot{\times}_o \lambda|$;

Proof. Left as an exercise. $\square$

Proposition 6.4.31 $\aleph_0 \dot{+}_c \aleph_0 \doteq \aleph_0 \doteq \aleph_0 \dot{\times}_c \aleph_0$.

Proof. By the preceding propositions, it suffices to show that $\aleph_0 \dot{\times}_c \aleph_0 \leq \aleph_0$. Since $\aleph_0 \doteq \dot{\omega}$, we may appeal to (the provability in ZF of) facts of ordinary arithmetic; the function $\langle x, y \rangle \mapsto (x \dot{+} y)^2 \dot{+} x$ of Section 4.2 is an injection of $\dot{\omega} \dot{\times} \dot{\omega}$ into $\dot{\omega}$. $\square$

These properties of cardinals do not require AC for their proof, but many more sophisticated results of cardinal arithmetic do need it.

Proposition 6.4.32 (ZFC) *For any infinite cardinal κ and any family of sets $\langle a_\sigma : \sigma \dot{<} \kappa \rangle$ such that $|a_\sigma| \dot{\leq} \lambda$ for all $\sigma \dot{<} \kappa$,*

$$\left| \dot{\bigcup} \{a_\sigma : \sigma \dot{<} \kappa\} \right| \dot{\leq} \kappa \dot{\times}_c \lambda.$$

Proof. Since for each $\sigma \dot{<} \kappa$, $|a_\sigma| \dot{\leq} \lambda$, there is an injective function $f : a_\sigma \dot{\rightarrow} \lambda$. By AC there exists a function $\sigma \mapsto f_\sigma$ such that for all $\sigma \dot{<} \kappa$, f_σ is such an injection. We define a function $g : \dot{\bigcup} \{a_\sigma : \sigma \dot{<} \kappa\} \dot{\rightarrow} \kappa \dot{\times} \lambda$ as follows. For each $x \in \dot{\bigcup} \{a_\sigma : \sigma \dot{<} \kappa\}$, let $\sigma_x \doteq$ least $\sigma[x \in a_\sigma]$. Then set $g(x) \doteq \langle \sigma_x, f_{\sigma_x}(x) \rangle$. Easily g is injective so $\left| \dot{\bigcup} \{a_\sigma : \sigma \dot{<} \kappa\} \right| \dot{\leq} |\kappa \dot{\times} \lambda| \doteq \kappa \dot{\times}_c \lambda$. $\square$

Corollary 6.4.33 (ZFC) *For any family of sets $\langle a_y : y \in \dot{\omega} \rangle$ such that for all $y \in \dot{\omega}$, a_y is countable, $\dot{\bigcup} \{a_y : y \in \dot{\omega}\}$ is countable.*

Proof. This is the instance of Proposition 6.4.32 for $\kappa \doteq \lambda \doteq \aleph_0$. $\square$

To prove the results corresponding to this corollary and Proposition 6.4.31 for all infinite cardinals κ in place of $\aleph_0$, we shall apply a consequence (actually, another equivalent, see exercises) of AC.

Theorem 6.4.34 (Zorn's Lemma) (ZFC) For any set $U \neq \emptyset$, if U is chain closed (Definition 6.3.22), then U contains a **maximal** element — that is, $m \in U$ such that for no $b \in U$ is $m \subseteq b$ but $m \neq b$.

Proof. Let c be a choice function for $\dot{\varnothing}U$, and set

$$l(a) := \begin{cases} c(\{b \in U : a \subseteq b \wedge a \neq b\}), & \text{if } a \in U \text{ but not maximal in } U; \\ a, & \text{otherwise.} \end{cases}$$

By hypothesis, $U \cup \{\emptyset\}$ is l -inductively closed (Definition 6.3.22) and l is clearly inclusive, so by the Fixed Point Theorem 6.3.24, $l^\infty \in U$ and is a fixed-point of l . Thus $l(l^\infty) = l^\infty$, which means exactly that l^∞ is maximal in U . $\square$

Theorem 6.4.35 (ZFC) For all infinite cardinals κ , $\kappa \dot{+}_c \kappa \doteq \kappa \doteq \kappa \dot{\times}_c \kappa$.

Proof. Let κ be an infinite cardinal; as before it suffices to prove that $\kappa \dot{\times}_c \kappa \leq \kappa$. Let

$$U := \left\{ f : \dot{\text{Fn}}(f) \wedge \dot{\text{Inj}}(f) \wedge \exists a \subseteq \kappa [(\dot{\text{Dom}}(f) \doteq a \wedge \dot{\text{Im}}(f) \doteq a \dot{\times} a)] \right\}.$$

In words, $f \in U$ iff f witnesses that for some $a \subseteq \kappa$, $a \approx a \dot{\times} a$. Since $\dot{\omega} \subseteq \kappa$ and $\dot{\omega} \approx \dot{\omega} \dot{\times} \dot{\omega}$, $U \neq \emptyset$. Let $V \subseteq U$ be a chain. Then if $b \doteq \dot{\bigcup} \left\{ \dot{\text{Dom}}(f) : f \in V \right\}$, easily $\dot{\bigcup} V$ is a function which witnesses that $b \approx b \dot{\times} b$, so $\dot{\bigcup} V \in U$. Hence by Zorn's Lemma, U contains a maximal element $\bar{f}$. Let $\bar{a}$ denote $\dot{\text{Dom}}(\bar{f})$, so $\bar{f}$ witnesses that $\bar{a} \approx \bar{a} \dot{\times} \bar{a}$, and set $\lambda \doteq |\bar{a}|$. Since

$$\lambda \doteq |\bar{a}| \doteq |\bar{a} \dot{\times} \bar{a}| \doteq |\lambda \dot{\times} \lambda| \doteq \lambda \dot{\times}_c \lambda,$$

if $\lambda \doteq \kappa$ we are done. Otherwise, $\lambda < \kappa$ and we shall show that this is impossible. Suppose that $\lambda < \kappa$, and let $\mu \doteq |\kappa \setminus \lambda|$. Clearly $\lambda \dot{+}_c \mu \doteq \kappa$, and it follows that $\lambda < \mu$, since if $\mu \leq \lambda$ we have

$$\kappa \doteq \lambda \dot{+}_c \mu \leq \lambda \dot{+}_c \lambda \doteq \lambda \dot{\times}_c 2 \leq \lambda \dot{\times}_c \lambda \doteq \lambda,$$

contrary to the current hypothesis. Hence there exists a set $\bar{b} \subseteq \kappa \setminus \lambda$ with $|\bar{b}| \doteq \lambda$. Now

$$(\bar{a} \dot{\cup} \bar{b}) \dot{\times} (\bar{a} \dot{\cup} \bar{b}) \doteq (\bar{a} \dot{\times} \bar{a}) \dot{\cup} [(\bar{a} \dot{\times} \bar{b}) \dot{\cup} (\bar{b} \dot{\times} \bar{a}) \dot{\cup} (\bar{b} \dot{\times} \bar{b})],$$

and

$$\begin{aligned} |(\bar{a} \dot{\times} \bar{b}) \dot{\cup} (\bar{b} \dot{\times} \bar{a}) \dot{\cup} (\bar{b} \dot{\times} \bar{b})| &\leq (\lambda \dot{\times}_c \lambda) \dot{+}_c (\lambda \dot{\times}_c \lambda) \dot{+}_c (\lambda \dot{\times}_c \lambda) \\ &\doteq \lambda \dot{+}_c \lambda \dot{+}_c \lambda \doteq \lambda \dot{\times}_c 3 \leq \lambda \dot{\times}_c \lambda \doteq \lambda, \end{aligned}$$

so

$$\bar{b} \dot{\approx} (\bar{a} \dot{\times} \bar{b}) \dot{\cup} (\bar{b} \dot{\times} \bar{a}) \dot{\cup} (\bar{b} \dot{\times} \bar{b})$$

via some function $\bar{g}$. Since the domains and images of $\bar{f}$ and $\bar{g}$ are disjoint, $\bar{f} \dot{\cup} \bar{g}$ is an injective function which witnesses that

$$\bar{a} \dot{\cup} \bar{b} \dot{\approx} (\bar{a} \dot{\cup} \bar{b}) \dot{\times} (\bar{a} \dot{\cup} \bar{b}).$$

But then $\bar{f} \dot{\cup} \bar{g} \in U$ and properly includes $\bar{f}$, contrary to the maximality of $\bar{f}$. $\square$

Corollary 6.4.36 (ZFC) *For all infinite cardinals κ and λ ,*

$$\kappa \dot{+}_c \lambda \dot{=} \kappa \dot{\times}_c \lambda \dot{=} \max\{\kappa, \lambda\}. \quad \square$$

Corollary 6.4.37 (ZFC) *For any infinite cardinal κ and any family of sets $\langle a_\sigma : \sigma \dot{<} \kappa \rangle$ such that $|a_\sigma| \dot{\leq} \kappa$ for all $\sigma \dot{<} \kappa$, also*

$$\left| \dot{\bigcup} \{a_\sigma : \sigma \dot{<} \kappa\} \right| \dot{\leq} \kappa. \quad \square$$

Corollary 6.4.38 *For any infinite cardinal κ and any x ,*

$$|x| \dot{\leq} \kappa \wedge (\forall v \in x) |v| \dot{\leq} \kappa \rightarrow \left| \dot{\bigcup} x \right| \dot{\leq} \kappa. \quad \square$$

Proposition 6.4.39 (ZFC) *For any x and y , $|y| \dot{\leq} |x|$ iff there exists a function $f : x \dot{\rightarrow} y$ with $y = \text{lm}(f)$ — that is, f maps x onto y .*

Proof. If $|y| \dot{\leq} |x|$, so there exists an injective function $g : y \dot{\rightarrow} x$, then easily there is a function $f : x \dot{\rightarrow} y$ which extends g^{-1} and is onto y . Conversely, given such an f , let $\mathbf{c}$ be a choice function for $\wp(x)$ and define $g : y \dot{\rightarrow} x$ by

$$g(v) \dot{=} \mathbf{c}(\{u : f(u) \dot{=} v\}).$$

Clearly g is injective, so $y \dot{\leq} x$. $\square$

We end this section with some definitions which will be explored somewhat in the exercises and more fully in [Section 6.10](#).

Definition 6.4.40 For any ordinals $\sigma \dot{\leq} \rho$,

- (i) σ is **cofinal** with ρ iff there exists a function $f : \sigma \dot{\rightarrow} \rho$ such that $\rho \dot{=} \dot{\bigcup} \text{lm}(f)$;
- (ii) the **cofinality** $\text{cf}(\rho)$ of a limit ordinal ρ is the least ordinal σ cofinal with ρ .

Definition 6.4.41 For any infinite cardinal κ ,

- (i) κ is a **regular cardinal** iff $\text{cf}(\kappa) \doteq \kappa$; otherwise, κ is a **singular cardinal**;
- (ii) κ is a **limit cardinal** iff $\forall \lambda (\lambda \dot{<} \kappa \rightarrow \lambda^+ \dot{<} \kappa)$;
- (iii) κ is **weakly inaccessible** iff κ is a regular limit cardinal.

Collateral Reading. The Axiom of Choice has been the subject of immense interest, suspicion and study ever since Zermelo isolated it as a principle early in the 20th century. [Moo] is a fascinating comprehensive history of all this, while [Ho-Ru] and [Ru-Ru] focus on the mathematical consequences and equivalents of AC. Other good discussions are in [Lev; Chapter V], [Ba77; Chapter B.2] and [Jec].

Exercises

Exercise 6.4.42 Prove in ZF that if x and y are Dedekind finite, then so are $x \dot{\cup} y$ and $x \dot{\times} y$.

Exercise 6.4.43 Say that x **absorbs** y iff $x \dot{\cup} y \approx x$. Show in ZF that

- (i) x is Dedekind infinite iff $\exists y [y \setminus x \neq \emptyset \wedge x \text{ absorbs } y]$;
- (ii) y is countable iff $\forall x [x \text{ is Dedekind infinite} \rightarrow x \text{ absorbs } y]$;
- (iii) if x is countable, then x is infinite iff x is Dedekind infinite;
- (iv) x is infinite iff $\dot{\phi} \dot{\phi} x$ is Dedekind infinite.

Exercise 6.4.44 Prove that for $x, y \dot{\in} \dot{\omega}$, $x \dot{+}_c y \doteq x \dot{+} y$ and $x \dot{\times}_c y \doteq x \dot{\times} y$.

Exercise 6.4.45 Prove Proposition 6.4.30.

Exercise 6.4.46 We define **cardinal exponentiation** by: for any cardinals κ and λ , $\kappa^\lambda \doteq |\lambda^\kappa|$. Show that this notion has the following familiar properties: for all cardinals κ , λ and μ ,

- (i) $\kappa^{\dot{0}} \doteq \dot{1}$; $\kappa^{\dot{1}} \doteq \kappa$; $\kappa^{\dot{2}} \doteq \kappa \dot{\times}_c \kappa$;
- (ii) $\kappa^{\lambda \dot{+}_c \mu} \doteq \kappa^\lambda \dot{\times}_c \kappa^\mu$; $\kappa^{\lambda \dot{\times}_c \mu} \doteq (\kappa^\lambda)^\mu$;
- (iii) $(\kappa \dot{\times}_c \lambda)^\mu \doteq \kappa^\mu \dot{\times}_c \lambda^\mu$;
- (iv) $\kappa \dot{\leq} \lambda \rightarrow \kappa^\mu \dot{\leq} \lambda^\mu$; $\lambda \dot{\leq} \mu \rightarrow \kappa^\lambda \dot{\leq} \kappa^\mu$;
- (v) give examples to show that generally $\kappa^\lambda \neq \lambda^\kappa$ and $(\kappa^\lambda)^\mu \neq \kappa^{(\lambda^\mu)}$.

Exercise 6.4.47 (König's Theorem) Show that for any infinite cardinal κ ,

- (i) $\kappa < \kappa^{\text{cf}(\kappa)}$;
- (ii) $\text{cf}(2^\kappa) > \kappa$.

Hint: For (i), fix $f : \text{cf}(\kappa) \rightarrow \kappa$ such that $\kappa = \bigcup \text{Im}(f)$. Show that any function $g : \kappa \rightarrow \text{cf}(\kappa)$ fails to be surjective along the following lines. For each $\sigma < \text{cf}(\kappa)$, let $a_\sigma := \{g(\rho)(\sigma) : \rho < f(\sigma)\}$. Show that there is a function h such that $(\forall \sigma < \text{cf}(\kappa)) h(\sigma) \in \kappa \setminus a_\sigma$, and conclude that $h \in \text{cf}(\kappa)^\kappa \setminus \text{Im}(g)$.

Definition 6.4.48

- (i) A set operation $F : \dot{\mathcal{O}}r \rightarrow \dot{\mathcal{O}}r$ is **normal** iff for all σ and τ
 - (a) $\sigma < \tau \rightarrow F(\sigma) < F(\tau)$;
 - (b) σ a limit $\rightarrow F(\sigma) = \bigcup \{F(\tau) : \tau < \sigma\}$.
- (ii) A class $C \subseteq \dot{\mathcal{O}}r$ is **closed unbounded** iff
 - (a) $\forall x [\emptyset \neq x \subseteq C \rightarrow \bigcup x \in C]$;
 - (b) $\forall \sigma (\exists \tau \geq \sigma) \tau \in C$.

Exercise 6.4.49 Prove that a class C is closed unbounded iff for some (unique) normal set operation F , $C = \text{Im}(F)$.

Exercise 6.4.50 Show that for any normal set operation F , the set of **fixed points** of F , $\{\sigma : F(\sigma) = \sigma\}$, is closed unbounded.

Exercise 6.4.51 Let G be a class operation such that for all τ , $G(\tau)$ is a closed unbounded class. Prove that

- (i) $\forall \rho [\bigcap_{\tau < \rho} G(\tau) \text{ is closed unbounded}]$;
- (ii) the **diagonal intersection** $\{\sigma : \sigma \in \bigcap_{\tau < \sigma} G(\tau)\}$ is closed unbounded.

Definition 6.4.52 Let κ be a regular cardinal.

- (i) A function $f : \kappa \rightarrow \kappa$ is **normal** iff for all $\sigma, \tau < \kappa$,
 - (a) $\sigma < \tau \rightarrow f(\sigma) < f(\tau)$;
 - (b) σ a limit $\rightarrow f(\sigma) = \bigcup \{f(\tau) : \tau < \sigma\}$.
- (ii) A set $c \subseteq \kappa$ is **closed unbounded in κ** iff
 - (a) $(\forall \dot{\sigma} < \sigma < \kappa)(\forall f : \sigma \rightarrow c) \bigcup \text{Im} f \in c$;
 - (b) $(\forall \sigma < \kappa)(\exists \tau \geq \sigma) \tau \in c$.

Exercise 6.4.53 Prove that for any regular cardinal κ , a set $c \subseteq \kappa$ is closed unbounded in κ iff for some (unique) normal function $f : \kappa \rightarrow \kappa$, $c \equiv \text{Im} f$.

Exercise 6.4.54 Show that for any uncountable regular cardinal κ and any normal function $f : \kappa \rightarrow \kappa$, $\{\sigma < \kappa : f(\sigma) \leq \sigma\}$ is closed unbounded in κ .

Exercise 6.4.55 Let $g : \kappa \rightarrow \kappa$ be a function such that for all $\tau < \kappa$, $g(\tau)$ is closed unbounded in κ . Prove that

- (i) $(\forall \rho < \kappa)[\bigcap_{\tau < \rho} g(\tau) \text{ is closed unbounded in } \kappa]$;
- (ii) the **diagonal intersection** $\{\sigma < \kappa : \sigma \in \bigcap_{\tau < \sigma} g(\tau)\}$ is closed unbounded in κ .

Exercise 6.4.56 Prove (in ZF) that the following are equivalent:

- (i) AC;
- (ii) Zorn's Lemma;
- (iii) the principle of cardinal comparability: $\forall x \forall y [x \preceq y \vee y \preceq x]$;
- (iv) $\forall x [x \text{ is well-orderable} \rightarrow \wp(x) \text{ is well-orderable}]$;
- (v) $\forall x [x \text{ is linearly orderable} \rightarrow x \text{ is well-orderable}]$.

Exercise 6.4.57 Prove (in ZF) that $\text{AC} \leftrightarrow \forall z [z \times z \approx z]$.

Hint: For the implication $(\leftarrow)$, establish first that for all x and y ,

$$x \times y \preceq x \cup y \rightarrow x \preceq y \vee y \preceq x,$$

and use the preceding exercise.

Exercise 6.4.58 Prove in ZFC that

- (i) $\forall \rho [\text{cf}(\text{cf}(\rho)) = \text{cf}(\rho)]$;
- (ii) for all limit ρ , $\text{cf}(\rho)$ is a cardinal; hence any regular ordinal is a regular cardinal;
- (iii) for all infinite cardinals κ , κ^+ is regular;
- (iv) if $\kappa > \omega$ is weakly inaccessible, then $\kappa = \aleph_\kappa$.

We shall see in [Section 6.10](#) that it is consistent with ZFC that no weakly inaccessible cardinals exist (unless ZFC is inconsistent).

Exercise 6.4.59 In ZFC let H be a class operation such that for all infinite cardinals κ ,

$$H_\kappa \doteq \{ \{ x : |TC(x)| < \kappa \} \}.$$

Members of H_κ are said to have **hereditary cardinal** less than κ . $H_{\aleph_0}$ is HF (Definition 6.2.51).

- (i) Prove that for all κ , $H_\kappa \subseteq \dot{V}_\kappa$; conclude that H is a set operation.
- (ii) Let $\dot{\varphi}_{<\kappa}(u) \doteq \{ \{ z : z \subseteq u \wedge |z| < \kappa \} \}$; show that $H_\kappa \doteq \dot{\varphi}_{<\kappa}^\infty \doteq \dot{\varphi}_{<\kappa}^\kappa$ (compare Exercise 6.2.68).

Hint: For (i) use Exercise 6.3.41 or prove (ii) first.

Exercise 6.4.60 Show that for every infinite cardinal κ ,

- (i) H_κ is transitive;
- (ii) $\forall x [x \in H_\kappa \rightarrow \bigcup x \in H_\kappa]$;
- (iii) if κ is regular, then $\forall x [x \in H_\kappa \leftrightarrow x \subseteq H_\kappa \wedge |x| < \kappa]$.

Exercise 6.4.61 A cardinal κ is called **strongly inaccessible** iff κ is regular and a **strong limit cardinal** — that is, $\forall \lambda (\lambda < \kappa \rightarrow 2^\lambda < \kappa)$. Show that if κ is strongly inaccessible, then $H_\kappa \doteq \dot{V}_\kappa$ and $\kappa \doteq \aleph_\kappa$.

Exercise 6.4.62 Prove that in ZFC the four versions (G)CH₁–(G)CH₄ of the (Generalized) Continuum Hypothesis are equivalent.

Exercise 6.4.63 Prove that $\forall x [x \neq \emptyset \rightarrow \text{ep}(x) \text{ is a proper class}]$.

Exercise 6.4.64 Consider the first-order language L which extends $L_\in$ by a single unary function symbol f and the L -theory T generated by the axioms of ZF together with the axiom

$$\forall x (y \neq \emptyset \rightarrow f y \in y).$$

- (i) Use Theorem 2.6.23 to argue that T is a conservative extension of ZF.
- (ii) It would seem to follow in T that for any x , $f \upharpoonright x$ is a choice function for x and therefore that $T \vdash AC$, so by (i), $ZF \vdash AC$; explain why this is not a valid argument.

Several weaker versions of the Axiom of Choice have been widely studied; the most interesting and important are the following.

Countable Axiom of Choice (CAC)

$$\forall x [x \text{ is countable} \rightarrow$$

$$\exists c \left(\text{Fn}(c) \wedge x \subseteq \text{Dom}(c) \wedge \forall y \in x [y \neq \emptyset \rightarrow c(y) \in y] \right)].$$

Axiom of Dependent Choice (ADC)

$$\forall y \forall r \left[y \neq \emptyset \wedge r \subseteq y \times y \wedge (\forall u \in y)(\exists v \in y) \langle u, v \rangle \in r \rightarrow \right. \\ \left. (\forall u \in y) \exists c \left(c : \dot{\omega} \rightarrow x \wedge c(\dot{0}) = u \wedge (\forall z \in \dot{\omega}) \langle c(z), c(\dot{S}z) \rangle \in r \right) \right].$$

Exercise 6.4.65 Show that

- (i) the conclusions of Proposition 6.4.9 and Corollary 6.4.33 are theorems of ZF + CAC;
- (ii) ZF + ADC $\vdash$ CAC.

Hint: In proving that $\dot{\omega} \preceq x$ for any infinite x , consider the family $\langle u_y : y \in \dot{\omega} \rangle$, where u_y is the set of sequences of length y of elements of x .

6.5 Models and independence

Since ZF and ZFC are recursively axiomatizable, the First Incompleteness Theorem for ZF 6.2.28 guarantees that

- (1) ZF consistent $\implies$ for some θ [neither $\theta \in \text{ZF}$ nor $\neg \theta \in \text{ZF}$];

and similarly for ZFC. We say that such a θ is **independent** of ZF; equivalently

$$\theta \text{ independent of ZF} \iff \text{both } \text{ZF} \cup \{\neg \theta\} \text{ and } \text{ZF} \cup \{\theta\} \text{ are consistent.}$$

In contrast with the situation for arithmetic, there are natural and important sentences which are independent of ZF and ZFC; in [Sections 6.5–6.9](#) we shall establish (most of)

- (2) ZF is consistent $\implies$ AC is independent of ZF;
- (3) ZFC is consistent $\implies$ CH and GCH are independent of ZFC.

Here CH and GCH are any one of the four versions of the preceding section, which are all equivalent by Exercise 6.4.62. As we shall explain at the end of [Section 6.6](#), by the Second Incompleteness Theorem we cannot expect here to prove the conclusion without the hypothesis “ZF(C) is consistent”, since the conclusion implies the consistency of ZF so cannot be proved without this hypothesis by any methods formalizable in ZF.

Cantor believed that CH was provable in his theory and devoted considerable effort to proving it. As intuitions about these notions became sharper

in the period 1900–1930, some began to consider the notion that AC and (G)CH might be independent. Gödel’s proof of the Incompleteness Theorems in 1931 focussed attention on the notion of independence, and it was Gödel himself who made the first step by showing in 1939

$$(4) \quad \text{ZF is consistent} \implies \text{ZF} \cup \{\text{AC}, \text{GCH}\} \text{ is consistent.}$$

Cohen completed the proof of (2) and (3) in 1963 by showing

$$(5) \quad \text{ZF is consistent} \implies \text{ZF} \cup \{\neg \text{AC}\} \text{ is consistent;}$$

$$(6) \quad \text{ZFC is consistent} \implies \text{ZFC} \cup \{\neg \text{CH}\} \text{ is consistent.}$$

Of course, by (4) the hypothesis of (6) can be weakened to “ZF is consistent”. We shall establish (4) in [Section 6.7](#) and (6) in [Sections 6.8](#) and [6.9](#); the proof of (5) will only be sketched.

The method for each of these proofs is that of Corollaries 2.6.45 and 2.6.47; we shall show that

$$(7) \quad \text{ZF} \cup \{\text{AC}, \text{GCH}\} \text{ is relatively interpretable in ZF;}$$

$$(8) \quad \text{ZFC} \cup \{\neg \text{CH}\} \text{ is relatively interpretable in } \text{ZFL}_m,$$

where ZFL_m is a theory to be defined in [Section 6.8](#) whose consistency follows from that of ZF.

For now we focus on (7). We need an extension ZF^+ by definitions of ZF such that

$$(\text{ZF} \cup \{\text{AC}, \text{GCH}\})^{(M)} \subseteq \text{ZF}^+.$$

Since the language of both theories is $L_\in$, the only new symbol to be introduced and defined in ZF^+ is the unary relation symbol M . Furthermore, in the spirit of the development of ZF we shall not actually introduce M as a relation symbol but regard it as a class. Technically, then, $\phi^{(M)}$ is $\phi^{(M)} -$ (Proposition 2.6.19) in which M is replaced by its definition, but we shall write it simply $\phi^{(M)}$. Hence we aim to show that

(9) there exists a class M such that

$$(i) \text{ for each axiom } \phi \text{ of ZF, } \text{ZF} \vdash \phi^{(M)};$$

$$(ii) \text{ ZF} \vdash \text{AC}^{(M)} \wedge \text{GCH}^{(M)}.$$

By Corollary 2.6.46, this implies

$$(10) \quad \text{each model } \mathfrak{A} \text{ of ZF has a substructure } \mathfrak{M} \\ \text{which is a model of } \text{ZF} \cup \{\text{AC}, \text{GCH}\}.$$

Of course, (10) alone is sufficient to conclude (4); it is helpful to think of (7) as a *uniform* version of (10) in that the submodel $\mathfrak{M}$ is defined by the same formula for each model $\mathfrak{A}$ of ZF. Because of (10), M is called an *inner model*.

We consider first (9)(i) and give it a name for reference.

Definition 6.5.1 A class M is a *class model* of ZF iff for all axioms ϕ of ZF, $ZF \vdash \phi^{(M)}$. If also $ZF \vdash M$ is a set, we also call M a *set model* of ZF.

By Corollary 2.6.45, if M is a class model of ZF, then also $ZF \vdash \phi^{(M)}$ for every ZF-theorem ϕ . Note that being a class model is a statement about all the infinitely many axioms of ZF so is not expressible as a single $L_\in$ -sentence. Recall that we use the following notation:

$$\exists^M x \phi \quad \text{for} \quad \exists x (x \in M \wedge \phi), \quad \text{and} \quad \forall^M x \phi \quad \text{for} \quad \forall x (x \in M \rightarrow \phi).$$

Let us start with the simplest of the axioms, the Extensionality axiom. By applying the rules for quantifier relativization, $\text{Extensionality}^{(M)}$ is the sentence

$$\forall^M x \forall^M y [\forall^M z (z \in x \leftrightarrow z \in y) \rightarrow x \dot{=} y].$$

Arguing in ZF, fix x and y in M and suppose that $\forall^M z (z \in x \leftrightarrow z \in y)$ — that is, $x \dot{\cap} M \dot{=} y \dot{\cap} M$. Without further hypotheses there is no reason to expect this to entail that $x \dot{=} y$. However, if M is transitive, then $x, y \dot{\subseteq} M$, so $x \dot{\cap} M \dot{=} x$ and $y \dot{\cap} M \dot{=} y$, and the conclusion is clear.

Foundation is similarly easy; $\text{Foundation}^{(M)}$ is the sentence

$$\forall^M y [\exists^M z (z \in y) \rightarrow \exists^M z (z \in y \wedge \forall^M u [u \in z \rightarrow u \notin y])].$$

To see that this is a theorem of ZF, fix $y \in M$ such that $\exists^M z (z \in y)$ — that is, $y \dot{\cap} M \neq \emptyset$. By Foundation, $y \dot{\cap} M$ has an $\in$ -minimal element z — that is,

$$\exists^M z (z \in y \wedge \forall u [u \in z \rightarrow u \notin y \dot{\cap} M]).$$

But then for any $u \in M$, if $u \in z$, then $u \notin y$. Thus we have shown

Proposition 6.5.2 For any transitive class M , $ZF \vdash \text{Extensionality}^{(M)}$ and $ZF \vdash \text{Foundation}^{(M)}$. $\square$

Of course, we cannot expect the other axioms to be so easy or to hold under such weak conditions. Since they all involve sentences of the form $(F \text{ is a set operation})^{(M)}$, we develop first some tools for proving such sentences. As usual we deal explicitly with unary F with the intent that the results apply also to 0-ary F (classes) and k -ary F .

Definition 6.5.3 For any class M and any (unary) class operation $F(x) \dot{=} \{\{z : \phi(x, z)\}\}$, F^M is the class operation defined by

$$F^M(x) := \{\{z \in M : \phi^{(M)}(x, z)\}\}.$$

$F^M(x)$ is what an inhabitant of M thinks $F(x)$ is; she interprets the definition ϕ of $F(x)$ using the objects in the world she knows as the range of the

quantifiers in ϕ and then takes as the value of $F^M(x)$ the class of objects z in her world M which satisfy $\phi^{(M)}$. Technically $F^M(x)$ is defined for any x , but the intended arguments are $x \in M$. Although by definition $F^M(x) \subseteq M$, it may or may not happen that $F^M(x) \in M$. As the following examples show, for even the simplest F , F^M can have unexpected values unless M is transitive; we shall see later that this is no real restriction in the sense that any class of interest (technically, which is extensional) is isomorphic to a transitive one. For more complex F , more is required of M for F^M to be anything like F .

Examples 6.5.4 For any transitive class M and any $x, y \in M$,

$$(i) \quad \dot{\emptyset}^M := \{ z \in M : (z \neq z)^{(M)} \} \doteq \{ z \in M : z \neq z \} \doteq \dot{\emptyset};$$

$$(ii) \quad \{x, y\}^M := \{ z \in M : (z \doteq x \vee z \doteq y)^{(M)} \} \\ \doteq \{x, y\} \dot{\cap} M \doteq \{x, y\};$$

$$(iii) \quad \dot{S}^M_x := \{ z \in M : z \in x \vee z \doteq x \} \doteq \dot{S}x \dot{\cap} M \doteq \dot{S}x;$$

$$(iv) \quad \dot{\bigcup}^M x := \{ z \in M : \exists^M v (z \in v \wedge v \in x) \} \\ \doteq \dot{\bigcup} (x \dot{\cap} M) \dot{\cap} M \doteq (\dot{\bigcup} x) \dot{\cap} M \doteq \dot{\bigcup} x;$$

$$(v) \quad \dot{\wp}^M x := \{ z \in M : \forall^M v (v \in z \rightarrow v \in x) \} \\ \doteq \{ z \in M : z \dot{\cap} M \subseteq x \} \doteq \dot{\wp} x \dot{\cap} M;$$

$$(vi) \quad \dot{O}r^M := \left\{ \left\{ y \in M : (y \text{ is transitive})^{(M)} \wedge \right. \right. \\ \left. \left. \forall^M x \left[x \in y \rightarrow (x \text{ is transitive})^{(M)} \right] \right\} \right\}.$$

For $y \in M$,

$$(y \text{ is transitive})^{(M)} \leftrightarrow \forall^M x \forall^M z (z \in x \wedge x \in y \rightarrow z \in y) \\ \leftrightarrow \forall x \forall z (z \in x \wedge x \in y \rightarrow z \in y) \\ \leftrightarrow y \text{ is transitive.}$$

In the second equivalence, $(\leftrightarrow)$ holds simply by logic and $(\rightarrow)$ holds because if $z \in x \in y \in M$, then $x \in M$ and hence $z \in M$ by transitivity. Thus $\dot{O}r^M \doteq \dot{O}r \dot{\cap} M$.

The examples illustrate the following two important ways that F and F^M can be related.

Definition 6.5.5 For any class M and any (unary) class operation F ,

- (i) F is **M-*élément absolute*** $:\leftrightarrow \forall^M x (F^M x \doteq Fx \dot{\cap} M)$;
- (ii) F is **M-*absolute*** $:\leftrightarrow \forall^M x (F^M x \doteq Fx)$.

Note that since by definition $F^M x \subseteq M$,

F is M-absolute $\leftrightarrow F$ is M-élément absolute and $\forall^M x (Fx \subseteq M)$.

Proposition 6.5.6 For any transitive class M ,

- (i) $\emptyset, \{\cdot, \cdot\}, \dot{S}$ and $\dot{\cup}$ are M-absolute;
- (ii) $\dot{\cap}$ and $\dot{O}r$ are M-élément absolute.

Proof. Immediate from the examples. $\square$

Definition 6.5.7 ‘ F is an **M-set operation**’ is an abbreviation for the sentence $\forall^M x [F(x) \in M]$ — that is, $\forall^M x \exists^M y [y \doteq F(x)]$.

Note that generally neither of the sentences ‘ F is a set operation’ and ‘ F^M is a set operation’ implies the other, and neither implies that F or F^M is an M-set operation.

Lemma 6.5.8 For any transitive class M , all $x, y, z \in M$ and all u ,

- (i) $(z \in Fx)^{(M)} \leftrightarrow z \in F^M x$;
- (ii) $(y \doteq Fx)^{(M)} \leftrightarrow y \doteq F^M x$;
- (iii) $[F^M \text{ is an M-set operation} \vee u \in M] \rightarrow$

$$[(Fx \in u)^{(M)} \leftrightarrow F^M x \in u].$$

Proof. Part (i) is immediate from the definition. For (ii), using (i), the left-hand side is

$$\forall^M z (z \in y \leftrightarrow z \in F^M x), \quad \text{that is,} \quad y \dot{\cap} M \doteq F^M x \dot{\cap} M.$$

But since $y \in M$, also $y \subseteq M$ and $F^M x \subseteq M$ by definition; hence this is equivalent to $y \doteq F^M x$. For (iii), using (ii), the left-hand side is $\exists^M y (y \doteq F^M x \wedge y \in u)$. Since by hypothesis either $F^M x \in M$ or $u \in M$, this is equivalent to $\exists y (y \doteq F^M x \wedge y \in u)$, which is exactly $F^M x \in u$. $\square$

Proposition 6.5.9 *For any class operation F and any transitive class M ,*

- (i) F is a set operation $\wedge F$ is M -element absolute
 $\rightarrow F^M$ is a set operation;
- (ii) $(F \text{ is a set operation})^{(M)} \leftrightarrow F^M$ is an M -set operation;
- (iii) if M is a class model of ZF , then

$$ZF \vdash F \text{ is an } M\text{-absolute set operation} \implies ZF \vdash F \text{ is an } M\text{-set operation.}$$

Proof. Part (i) is immediate by Comprehension. Let

$$F(x) = \{ \{ z : \phi(x, z) \} \}.$$

The left side of (ii) is

$$\forall^M x \exists^M y \forall^M z [z \in y \leftrightarrow \phi^{(M)}(x, z)].$$

By the transitivity of M , this is equivalent to

$$\forall^M x \exists^M y \forall z [z \in y \leftrightarrow z \in M \wedge \phi^{(M)}(x, z)],$$

which is exactly the sentence F^M is an M -set operation. For (iii), the hypothesis implies that

$$ZF \vdash \forall^M x (F^M x \doteq Fx) \wedge \forall x \exists y \forall z (z \dot{\in} y \leftrightarrow z \dot{\in} Fx).$$

Since M is a class model, using (i) of the lemma,

$$ZF \vdash \forall^M x \exists^M y \forall^M z (z \dot{\in} y \leftrightarrow z \dot{\in} F^M x),$$

and using transitivity we conclude that $ZF \vdash \forall^M x \exists^M y (y \doteq Fx)$ as desired. $\square$

Note that (iii) does not make the stronger claim that

$$ZF \vdash F \text{ is an } M\text{-absolute set operation} \rightarrow F \text{ is an } M\text{-set operation.}$$

Proposition 6.5.10 *For any transitive class M , if $\dot{\omega} \in M$, then $\dot{\omega}$ is M -absolute.*

Proof. Recall that $\dot{\omega} \doteq \dot{\bigcap} B$, where

$$B := \left\{ \left\{ u : \dot{\emptyset} \in u \wedge \forall z \left[z \in u \rightarrow \dot{S}z \in u \right] \right\} \right\}.$$

Now, using (iii) of the lemma, the M -absoluteness of $\dot{\emptyset}$ and $\dot{S}$ and the transitivity of M ,

$$\begin{aligned} B^M &:= \left\{ \left\{ u \in M : (\dot{\emptyset} \in u)^{(M)} \wedge \forall^M z \left[z \in u \rightarrow (\dot{S}z \in u)^{(M)} \right] \right\} \right\} \\ &\doteq \left\{ \left\{ u \in M : \dot{\emptyset} \in u \wedge \forall^M z \left[z \in u \rightarrow \dot{S}z \in u \right] \right\} \right\} \\ &\doteq \left\{ \left\{ u \in M : \dot{\emptyset} \in u \wedge \forall z \left[z \in u \rightarrow \dot{S}z \in u \right] \right\} \right\} \doteq B \dot{\cap} M. \end{aligned}$$

In particular, $B^M \neq \dot{\emptyset}$, so

$$\begin{aligned} \dot{\omega}^M &\doteq \left(\dot{\bigcap} B \right)^M \doteq \left\{ v \in M : \forall^M u \left(u \in B^M \rightarrow v \in u \right) \right\} \\ &\doteq \left\{ v : \forall u \left(u \in B^M \rightarrow v \in u \right) \right\} \doteq \dot{\bigcap} B^M. \end{aligned}$$

Then $\dot{\omega} \doteq \dot{\bigcap} B \subseteq \dot{\bigcap} (B \dot{\cap} M) \doteq \dot{\bigcap} B^M \doteq \dot{\omega}^M$. But by hypothesis, $\dot{\omega} \in M$, so $\dot{\omega} \in B \dot{\cap} M$ and thus also $\dot{\omega}^M \subseteq \dot{\omega}$ so $\dot{\omega}^M \doteq \dot{\omega}$. $\square$

Definition 6.5.11 A class M is **rich** iff $ZF \vdash M$ is transitive $\wedge \dot{\omega} \in M$, and for every class operation F ,

$$ZF \vdash F^M \text{ is a set operation} \quad \rightarrow \quad F^M \text{ is an } M\text{-set operation.}$$

Theorem 6.5.12 *For any class M , if M is rich, then M is a class model of ZF .*

Proof. Let M be a rich class. By Proposition 6.5.2,

$$ZF \vdash \text{Extensionality}^{(M)} \quad \text{and} \quad ZF \vdash \text{Foundation}^{(M)}.$$

Let F be any class operation such that

$$ZF \vdash F \text{ is a set operation} \wedge F \text{ is } M\text{-element absolute.}$$

In particular by Propositions 6.5.6 and 6.5.10 this is true for F any of $\dot{\bigcup}$, $\dot{\cap}$, $\dot{S}$, $\emptyset$ or $\dot{\omega}$. Then by Proposition 6.5.9(i) together with the second clause of the definition of rich,

$$\begin{aligned} ZF \vdash F^M \text{ is an } M\text{-set operation,} \quad \text{and hence} \\ ZF \vdash (F \text{ is a set operation})^{(M)} \end{aligned}$$

by Proposition 6.5.9(ii). In particular,

$$ZF \vdash \text{Union}^{(M)}, \quad ZF \vdash \text{Powerset}^{(M)} \quad \text{and} \quad ZF \vdash \text{Infinity}^{(M)}.$$

Towards Replacement^(M) let F be any class operation. Using (ii) of the lemma, for any u ,

$$\begin{aligned} F^{*M}(u) &:= \left\{ \left\{ y \in M : \exists^M x \left[x \in u \wedge (y \dot{=} Fx)^{(M)} \right] \right\} \right\} \\ &\dot{=} \left\{ \left\{ y \in M : \exists^M x \left[x \in u \wedge y \dot{=} F^M x \right] \right\} \right\} \\ &\dot{=} F^{M*}(u \dot{\cap} M) \dot{\cap} M. \end{aligned}$$

By Replacement, F^{M*} is a set operation, and by Comprehension, the definition of rich and Proposition 6.5.9(ii), we conclude that

$$F^{*M} \text{ is a set operation,} \quad F^{*M} \text{ is an } M\text{-set operation}$$

and hence $(F^* \text{ is a set operation})^{(M)}$, as required. $\square$

At this point we cannot apply the preceding theorem in any useful way because we have not established that any class, other than trivially $\dot{V}$, is rich. The next series of results constitutes a reduction but not a solution of this problem; we will derive a general method for constructing rich classes, but will not be in a position to give any concrete examples until [Section 6.7](#).

Definition 6.5.13 For any (unary) class operation $F(x) = \{ \{ z : \phi(x, z) \} \}$, the set operation $(x, u) \mapsto F^u(x)$ is defined by

$$F^u(x) := \{ \{ z \in u : \phi^{(u)}(x, z) \} \},$$

where $\phi^{(u)}$ results from ϕ by relativizing all of the quantifiers to u .

This is, of course, the same as F^M but now with the relativizing class as a set variable. Note that by Comprehension, F^u is always a set operation. Intuitively (and formally in the next section) $F^u(x)$ is the subset of u defined by the formula ϕ with the parameter x . Hence (i) below intuitively requires that $I(u)$ contain all of the subsets of u definable from parameters in u .

Definition 6.5.14

- (i) A unary set operation l is a **weak power operation** iff for all u ,
- (a) $lu \subseteq \wp u$;
 - (b) for every class operation F ,

$$(\forall x_0 \in u) \cdots (\forall x_{k-1} \in u) F^u(x_0, \dots, x_{k-1}) \in lu.$$

- (ii) A class M is a **universe** iff $M \doteq l^\infty$ for some weak power operation l — that is, $M \doteq \bigcup \{ M_\sigma : \sigma \in \mathring{O}r \}$, where $M_{\dot{0}} \doteq \emptyset$ and for all σ ,

$$\begin{aligned} M_{\sigma \dot{+}_o 1} &:= l(M_\sigma); \\ M_\sigma &:= \bigcup \{ M_\tau : \tau \dot{<} \sigma \} \quad \text{if } \sigma \text{ is a limit ordinal.} \end{aligned}$$

At this point $\dot{\wp}$ is the only weak power operation we know and $\dot{V}$ the only universe; others will be constructed in [Section 6.7](#). Our next goal is to establish that every universe M is a class model of **ZF** because it is rich. The key idea of the proof, which motivates the definition, is that whenever F^M is a (unary) set operation, for each $x \in M$ there exists an ordinal σ such that $x \in M_\sigma$ and

$$F^M(x) \doteq F^{M_\sigma}(x) \in l(M_\sigma) \doteq M_{\sigma \dot{+}_o 1} \subseteq M,$$

so F^M is an M -set operation. This will require some preparation. First we see that any universe shares some simple properties with $\dot{V}$ (6.3.26).

Lemma 6.5.15 *For any weak power operation l and any u and τ ,*

- (i) $u \in lu$;
- (ii) *if u is transitive, then*
 - (a) $u \subseteq lu$ and lu is transitive;
 - (b) $u \dot{\cap} \mathring{O}r \doteq \tau \rightarrow lu \dot{\cap} \mathring{O}r \doteq \tau \dot{+}_o 1$.

Proof. For (i), note that $u \doteq \dot{V}^u \in lu$. If u is transitive, then for each $x \in u$, $x \doteq \{ z : z \in x \} \doteq \{ z \in u : (z \in x)^u \} \in lu$. Then $lu \subseteq \dot{\wp} u \subseteq \dot{\wp} lu$, so lu is transitive. Finally, if $u \dot{\cap} \mathring{O}r \doteq \tau$, then $\tau \subseteq u \subseteq lu$, and also $\tau \doteq \mathring{O}r^u \in lu$, so $\tau \dot{+}_o 1 \subseteq lu$. But obviously for $\rho \dot{>} \tau$, $\rho \not\subseteq u$, so $\rho \notin lu$. $\square$

Proposition 6.5.16 *For any universe M and all ordinals σ and τ ,*

- (i) M_σ is transitive and $M_\sigma \subseteq M_{\sigma+o1}$; M is transitive;
- (ii) $\sigma \leq \tau \rightarrow M_\sigma \subseteq M_\tau$;
- (iii) $M_\sigma \in M_{\sigma+o1}$;
- (iv) $M_\sigma \cap \dot{O}r \doteq \sigma$; $\dot{O}r \subseteq M$.

Proof. Part (iii) is immediate from (i) of the lemma, and by (ii)(a) of the lemma,

$$M_\sigma \text{ is transitive} \rightarrow M_\sigma \subseteq M_{\sigma+o1} \wedge M_{\sigma+o1} \text{ is transitive.}$$

Then (i) and (ii) follow by easy inductions. Similarly (iv) follows by induction from (ii)(b) of the lemma. $\square$

Definition 6.5.17 For any classes $M \subseteq N$ and any formula ϕ with free variables $x_0, \dots, x_{k-1}$, ϕ is (M, N) -absolute iff

$$\forall^M x_0 \dots \forall^M x_{k-1} [\phi^{(M)}(x_0, \dots, x_{k-1}) \leftrightarrow \phi^{(N)}(x_0, \dots, x_{k-1})];$$

ϕ is ***M-absolute*** iff ϕ is $(M, \dot{V})$ -absolute.

The connection between this and the earlier notion of absoluteness for class operations is

Corollary 6.5.18 *For any class M and any (unary) class operation F ,*

- (i) F is *M-element absolute* $\leftrightarrow$ ' $z \in Fx$ ' is *M-absolute*;
- (ii) if F^M is an *M-set operation*, then
 - (a) F is *M-absolute* $\leftrightarrow$ ' $y \doteq Fx$ ' is *M-absolute*;
 - (b) F is *M-absolute* $\rightarrow$ ' $Fx \in u$ ' is *M-absolute*.

Proof. For (i),

$$\begin{aligned} F \text{ is M-element absolute} &\leftrightarrow \forall^M x [F^M x = Fx \cap M] \\ &\leftrightarrow \forall^M x \forall^M z [z \in F^M x \leftrightarrow z \in Fx] \\ &\leftrightarrow \forall^M x \forall^M z [(z \in Fx)^{(M)} \leftrightarrow z \in Fx], \end{aligned}$$

which is just what is claimed. Now assume that F^M is an *M-set operation*. Then

$$\begin{aligned} F \text{ is M-absolute} &\leftrightarrow \forall^M x [F^M x \doteq Fx] \\ &\leftrightarrow \forall^M x \forall y [y \doteq F^M x \leftrightarrow y \doteq Fx] \\ &\leftrightarrow \forall^M x \forall^M y [y \doteq F^M x \leftrightarrow y \doteq Fx] \\ &\leftrightarrow \forall^M x \forall^M y [(y \doteq Fx)^{(M)} \leftrightarrow y \doteq Fx], \end{aligned}$$

by Lemma 6.5.8(ii). Finally, assuming that F is M -absolute and $x \in M$,

$$\begin{aligned}
 (Fx \in u)^{(M)} &\leftrightarrow (\exists y [y \doteq Fx \wedge y \in u])^{(M)} \\
 &\leftrightarrow \exists^M y [(y \doteq Fx)^{(M)} \wedge y \in u] \\
 &\leftrightarrow \exists y [y \doteq Fx \wedge y \in u] \\
 &\leftrightarrow Fx \in u. \quad \square
 \end{aligned}$$

Recall that a set Φ of formulas is subformula-closed iff all subformulas of members of Φ are also members of Φ . For classes $M \subseteq N$, let us say that M has Φ -*witnesses* for N iff for any formula of the form $\exists y \psi(x) \in \Phi$,

$$\forall^M x \left[(\exists y \psi(x))^{(N)} \rightarrow \exists^M y \psi^{(N)}(x) \right],$$

and similarly for more free variables. Note the similarity of this condition with that of Tarski's Criterion 2.3.33; it plays the same role here:

Proposition 6.5.19 *For any subformula-closed set Φ of $L_\in$ -formulas and any classes $M \subseteq N$, if M has Φ -witnesses for N , then all $\phi \in \Phi$ are (M, N) -absolute.*

Proof. Given Φ , M , and N as in the hypothesis, we prove by formula induction that for all $\phi \in \Phi$,

$$(11) \quad \forall^M x_0 \cdots \forall^M x_{k-1} [\phi^{(M)}(x_0, \dots, x_{k-1}) \leftrightarrow \phi^{(N)}(x_0, \dots, x_{k-1})].$$

(11) holds for atomic formulas since they are not affected by relativization. The induction steps for $\neg$ and $\vee$ are immediate, using that Φ is subformula closed. Suppose that $\phi(x)$ is $\exists y \psi(x, y)$. Then $\psi(x, y) \in \Phi$, so by the induction hypothesis and the hypothesis,

$$\begin{aligned}
 \phi^{(M)}(x) &\leftrightarrow \exists^M y \psi^{(M)}(x, y) \leftrightarrow \exists^M y \psi^{(N)}(x, y) \\
 &\leftrightarrow \exists^N y \psi^{(N)}(x, y) \leftrightarrow \phi^{(N)}(x). \quad \square
 \end{aligned}$$

The following is a version of the Downward Löwenheim-Skolem Theorem 3.4.4.

Reflection Theorem 6.5.20 *For any universe M and any $L_\in$ -formula ϕ ,*

$$\forall \sigma (\exists \rho \geq \sigma) \phi \text{ is } (M_\rho, M)\text{-absolute.}$$

Proof. Fix M and ϕ , and let Φ be the set of all subformulas of ϕ . Φ is finite and subformula closed. For each formula of the form $\exists y \psi(x_0, \dots, x_{k-1}) \in \Phi$,

define

$$\begin{aligned}
 F_{\psi,y}(x_0, \dots, x_{k-1}) &:= \begin{cases} \text{least } \tau(\exists y \in M_\tau) \psi^{(M)}(x_0, \dots, x_{k-1}), & \text{if } (\exists y \psi)^{(M)}; \\ 0, & \text{otherwise;} \end{cases} \\
 G_{\psi,y}(v) &:= \bigcup \{ F_{\psi,y}(x_0, \dots, x_{k-1}) : x_0, \dots, x_{k-1} \in M_v \}; \\
 H(\sigma, 0) &:= \sigma; \\
 H(\sigma, z \dot{+}_o \dot{1}) &:= \max\{ G_{\psi,y}(H(\sigma, z)) : \exists y \psi \in \Phi \} \dot{\cup} \{ H(\sigma, z) \dot{+}_o \dot{1} \}; \\
 I(\sigma) &:= \bigcup \{ H(\sigma, z) : z \in \dot{\omega} \}.
 \end{aligned}$$

By the preceding proposition, it suffices to show that for all σ , $M_{I(\sigma)}$ has Φ -witnesses for M . Given $x_0, \dots, x_{k-1} \in M_{I(\sigma)}$, there is $z \in \dot{\omega}$ such that $x_0, \dots, x_{k-1} \in M_{H(\sigma, z)}$, so

$$(\exists y \psi(x_0, \dots, x_{k-1}, y))^{(M)} \rightarrow (\exists y \in M_{G_{\psi,z}(H(\sigma, z))}) \psi(x_0, \dots, x_{k-1}, y)^{(M)},$$

so such a y can be found in $M_{H(\sigma, z \dot{+}_o \dot{1})}$ and hence in $M_{I(\sigma)}$ as desired. $\square$

We can express this in terms of class operations by

Corollary 6.5.21 *For any universe M and any (unary) class operation F ,*

$$\forall^M x \forall \sigma (\exists \rho \geq \sigma) [x \in M_\rho \wedge F^{M_\rho}(x) \dot{=} F^M(x) \dot{\cap} M_\rho]. \quad \square$$

Universe Theorem 6.5.22 *Every universe M is a transitive class model of ZF such that $\dot{O}r \subseteq M$.*

Proof. Fix a universe M ; we verify that M is rich and apply Theorem 6.5.12. M is transitive by Proposition 6.5.16 and

$$\dot{\omega} \dot{=} M_{\dot{\omega}} \dot{\cap} \dot{O}r \dot{=} \dot{O}r^{(M_{\dot{\omega}})} \in M_{\dot{\omega} \dot{+}_o \dot{1}} \subseteq M.$$

Now fix a (unary) class operation F and assume that F^M is a set operation. For any $x \in M$, $F^M(x) \subseteq M$ and is a set, so exactly as in the proof of Proposition 6.3.21(v), for some σ , $F^M(x) \subseteq M_\sigma$. By the last corollary, for some $\rho \geq \sigma$, $F^M(x) \dot{=} F^M(x) \dot{\cap} M_\rho \dot{=} F^{M_\rho}(x) \in M_{\rho \dot{+}_o \dot{1}} \subseteq M$. Hence F^M is an M -set operation. $\square$

In the next two sections it will be important that many formulas and class operations are absolute. In Proposition 6.5.6 we needed only the transitivity of M , but more complex instances will need stronger hypotheses. Although for some purposes it is useful to keep track precisely of just what is required, here we shall regularly assume the relatively strong hypothesis that M is a transitive class model of ZF. This has several simplifying consequences.

Proposition 6.5.23 *For any class model M of ZF and any (unary) class operations F and G ,*

- (i) *if $ZF \vdash F$ is a set operation, then $ZF \vdash F^M$ is an M -set operation and*
 - (a) F is M -absolute $\leftrightarrow$ ' $y \doteq Fx$ ' is M -absolute;
 - (b) F is M -absolute $\rightarrow$ ' $Fx \dot{\in} u$ ' is M -absolute $\wedge \forall^M x [Fx \dot{\in} M]$;
- (ii) *if $ZF \vdash \forall x [Fx \doteq Gx]$, then also $ZF \vdash \forall^M x [F^M x \doteq G^M x]$; hence*

$$F \text{ is } M\text{-absolute} \quad \leftrightarrow \quad G \text{ is } M\text{-absolute}.$$

Proof. For (i), since $ZF \vdash \phi^{(M)}$ for every theorem ϕ of ZF,

$$ZF \vdash F \text{ is a set operation} \quad \implies \quad ZF \vdash (F \text{ is a set operation})^{(M)},$$

hence $ZF \vdash F^M$ is an M -set operation by Proposition 6.5.9. Now (a) and (b) are immediate from Corollary 6.5.18. For (ii), let ϕ and ψ be the defining formulas for F and G , respectively. The hypothesis is

$$\begin{aligned} ZF \vdash \forall x \forall z [\phi(x, z) \leftrightarrow \psi(x, z)] \quad \text{hence} \\ ZF \vdash \forall^M x \forall^M z [\phi^{(M)}(x, z) \leftrightarrow \psi^{(M)}(x, z)], \end{aligned}$$

and thus $ZF \vdash \forall^M x [F^M x \doteq G^M x]$. Hence,

$$\forall^M x [F^M x \doteq Fx] \quad \leftrightarrow \quad \forall^M x [G^M x \doteq Gx]. \quad \square$$

Remark 6.5.24 The impact of (ii) is that in assessing the absoluteness of a class operation F , we may choose any provably equivalent formula in place of the original definition of F .

Each of the instances of Proposition 6.5.6 relies on the same basic fact: if M is transitive and $u \dot{\in} M$, then for any formula ϕ ,

$$(\exists x \dot{\in} u) \phi \leftrightarrow (\exists^M x \dot{\in} u) \phi \quad \text{and} \quad (\forall x \dot{\in} u) \phi \leftrightarrow (\forall^M x \dot{\in} u) \phi,$$

since elements of u are in M anyway. We formalize and extend this principle to obtain a syntactic criterion for absoluteness (compare Definition 5.1.15). Here, as usual, the one free variable x is intended to represent any finite number of free variables. Compare also Definition 2.3.22 and the following lemma.

Definition 6.5.25

- (i) The class of Δ_0 formulas of $L_\in$ is the smallest class of formulas which contains the atomic formulas and is closed under $\vee$, $\neg$ and the bounded quantifiers $(\exists x \dot{\in} y)$ and $(\forall x \dot{\in} y)$.
- (ii) For any $L_\in$ -formula $\phi(x)$,
 - (a) $\phi(x)$ is a Σ^{ZF} formula iff for some Δ_0 formula ψ ,

$$\text{ZF} \vdash \forall x [\phi(x) \leftrightarrow \exists y_0 \cdots \exists y_{k-1} \psi(x, y_0, \dots, y_{k-1})];$$

- (b) $\phi(x)$ is a Π^{ZF} formula iff for some Δ_0 formula θ ,

$$\text{ZF} \vdash \forall x [\phi(x) \leftrightarrow \forall z_0 \cdots \forall z_{k-1} \theta(x, z_0, \dots, z_{k-1})];$$

- (c) $\phi(x)$ is a Δ^{ZF} formula iff $\phi(x)$ is both a Σ^{ZF} formula and a Π^{ZF} formula.

Note that in each case the quantifier string may be empty so $\Delta_0 \subseteq \Delta^{\text{ZF}}$. On rare occasions we will also want to consider formulas ϕ such that the equivalences are theorems of some other theory T in place of ZF, and we will use the notations Σ^T , Π^T and Δ^T .

Lemma 6.5.26

- (i) Every Π^{ZF} (Σ^{ZF}) formula is ZF-provably equivalent to the negation of a Σ^{ZF} (Π^{ZF}) formula.
- (ii) The class of Σ^{ZF} formulas is closed under $\wedge$, $\vee$, $\exists u$, and $(\forall u \dot{\in} v)$.
- (iii) The class of Π^{ZF} formulas is closed under $\wedge$, $\vee$, $\forall u$, and $(\exists u \dot{\in} v)$.
- (iv) The class of Δ^{ZF} formulas is closed under $\wedge$, $\vee$, $\neg$, $(\exists u \dot{\in} v)$, and $(\forall u \dot{\in} v)$.

Proof. Suppose that ϕ is a Σ^{ZF} formula, say $\text{ZF} \vdash \phi \leftrightarrow \exists y_0 \cdots \exists y_{k-1} \psi$ with ψ a Δ_0 formula. Then $\text{ZF} \vdash \neg \phi \leftrightarrow \forall y_0 \cdots \forall y_{k-1} \neg \psi$ and $\neg \psi$ is also a Δ_0 formula, so $\neg \phi$ is a Π^{ZF} formula.

For (ii), closure of the class of Σ^{ZF} formulas under $\wedge$, $\vee$ and $\exists u$ is easy. For closure under $(\forall u \dot{\in} v)$, we apply the Collection Principle 6.3.30:

$$\begin{aligned} (\forall u \dot{\in} v) \exists y_0, \dots, \exists y_{k-1} \psi &\leftrightarrow (\forall u \dot{\in} v) \exists w (\exists x_0 \dot{\in} w) \cdots (\exists x_{k-1} \dot{\in} w) \psi \\ &\leftrightarrow \exists z (\forall u \dot{\in} z) (\exists w \dot{\in} z) (\exists x_0 \dot{\in} w) \cdots (\exists x_{k-1} \dot{\in} w) \psi. \end{aligned}$$

- (iii) and (iv) are immediate from (ii). $\square$

Definition 6.5.27 A (unary) class operation F is Δ_0 (Σ^{ZF} , Π^{ZF} , Δ^{ZF}) iff the formula $y \dot{=} Fx$ is Δ_0 (Σ^{ZF} , Π^{ZF} , Δ^{ZF}) and (*element*-) Δ_0 iff the formula $z \dot{\in} Fx$ is Δ_0 (Σ^{ZF} , Π^{ZF} , Δ^{ZF}).

Proposition 6.5.28

- (i) $\dot{\emptyset}$, $\dot{\cup}$, $\dot{\cap}$, $\dot{\setminus}$, $\dot{\bigcup}$ and $\dot{\bigcap}$ are Δ^{ZF} .
- (ii) $\dot{\{ \cdot \}}$, $\dot{\{ \cdot, \cdot \}}$ and $\dot{S}$ are Δ^{ZF} .
- (iii) $\{ \{ z : z \text{ is transitive} \} \}$, $\dot{O}r$, $\{ \{ \sigma : \sigma \text{ is a successor ordinal} \} \}$ and $\{ \{ \sigma : \sigma \text{ is a limit ordinal} \} \}$ are *element*- Δ^{ZF} .
- (iv) $\dot{\omega}$ is Δ^{ZF} .

Proof. We write out some of these and leave the rest as exercises. Note that in each case we actually have equivalence with a Δ_0 formula.

$$\begin{aligned}
 y \dot{=} \dot{\emptyset} &\leftrightarrow (\forall z \dot{\in} y) z \neq z \\
 u \dot{=} x \dot{\cup} y &\leftrightarrow (\forall z \dot{\in} u) [z \dot{\in} x \vee z \dot{\in} y] \wedge \\
 &\quad (\forall z \dot{\in} x) z \dot{\in} u \wedge (\forall z \dot{\in} y) z \dot{\in} u \\
 y \dot{=} \dot{\bigcup} x &\leftrightarrow (\forall z \dot{\in} y) (\exists v \dot{\in} x) z \dot{\in} v \wedge (\forall v \dot{\in} x) (\forall z \dot{\in} v) z \dot{\in} y \\
 u \dot{=} \dot{\{ x, y \}} &\leftrightarrow (\forall z \dot{\in} u) [z \dot{=} x \vee z \dot{=} y] \wedge x \dot{\in} u \wedge y \dot{\in} u \\
 y \dot{=} \dot{S}x &\leftrightarrow (\forall z \dot{\in} y) [z \dot{\in} x \vee z \dot{=} x] \wedge (\forall z \dot{\in} x) z \dot{\in} y \wedge x \dot{\in} y \\
 z \text{ is transitive} &\leftrightarrow (\forall y \dot{\in} z) (\forall x \dot{\in} y) x \dot{\in} z \\
 z \dot{\in} \dot{O}r &\leftrightarrow z \text{ is transitive} \wedge (\forall y \dot{\in} z) y \text{ is transitive} \\
 y \dot{=} \dot{\omega} &\leftrightarrow y \text{ is a limit ordinal} \wedge (\forall z \dot{\in} y) [z \text{ is not a limit ordinal}].
 \end{aligned}$$

Note particularly the last equivalence, which is just Corollary 6.3.11. The original definition of $\dot{\omega}$ seems far from being a Δ^{ZF} or even a Σ^{ZF} or Π^{ZF} formula. $\square$

The importance of a formula or operation being Δ^{ZF} arises from the following theorem.

Lemma 6.5.29 For any transitive class M and every Δ_0 formula ϕ , $\text{ZF} \vdash \phi$ is M -absolute.

Proof. We proceed by induction on Δ_0 formulas. Atomic formulas are M -absolute for every M , and the induction steps for $\vee$ and $\neg$ are immediate. Suppose ϕ is M -absolute; then under the hypothesis $x, z \in M$,

$$\begin{aligned} [(\forall y \dot{\in} z) \phi(x, y)]^{(M)} &\leftrightarrow (\forall^M y \dot{\in} z) \phi^{(M)}(x, y) \\ &\leftrightarrow (\forall^M y \dot{\in} z) \phi(x, y) \\ &\leftrightarrow (\forall y \dot{\in} z) \phi(x, y), \end{aligned}$$

where the third equivalence follows from the transitivity of M . $\square$

Theorem 6.5.30 *For any transitive class model M of ZF and every Δ^{ZF} formula ϕ ,*

$$\text{ZF} \vdash \phi \text{ is } M\text{-absolute}.$$

Hence for every Δ^{ZF} class operation F , if $\text{ZF} \vdash F$ is a set operation, then

$$\text{ZF} \vdash F \text{ is } M\text{-absolute} \wedge F \text{ is an } M\text{-set operation}.$$

Proof. Suppose that ϕ is a Δ^{ZF} formula, and fix Δ_0 formulas ψ and θ such that

$$\text{ZF} \vdash \forall x [\phi(x) \leftrightarrow \exists y \psi(x, y)] \quad \text{and} \quad \text{ZF} \vdash \forall x [\phi(x) \leftrightarrow \forall z \theta(x, z)].$$

Since M is a class-model of ZF, we have also

$$\begin{aligned} \text{ZF} \vdash \forall^M x [\phi^{(M)}(x) \leftrightarrow \exists^M y \psi^{(M)}(x, y)] \quad \text{and} \\ \text{ZF} \vdash \forall^M x [\phi^{(M)}(x) \leftrightarrow \forall^M z \theta^{(M)}(x, z)]. \end{aligned}$$

Hence, since by the lemma both ψ and θ are M -absolute, under the hypothesis that $x \in M$,

$$\phi^{(M)}(x) \rightarrow \exists^M y \psi^{(M)}(x, y) \rightarrow \exists^M y \psi(x, y) \rightarrow \exists y \psi(x, y) \rightarrow \phi(x),$$

and

$$\phi(x) \rightarrow \forall z \theta(x, z) \rightarrow \forall^M z \theta(x, z) \rightarrow \forall^M z \theta^{(M)}(x, z) \rightarrow \phi^{(M)}(x).$$

The remaining claims now follow by Proposition 6.5.23(i). $\square$

In view of this result, the following provides tools for establishing the absoluteness of many formulas and operations.

Theorem 6.5.31 *For any class operations G, H and I and any $L_{\in}$ -formulas ψ and θ , if*

$$ZF \vdash G, H \text{ and } I \text{ are set operations,}$$

and G, H, I, ψ and θ are Δ^{ZF} , then in (i) ϕ is Δ^{ZF} , and in (ii)–(iv), F is Δ^{ZF} and $ZF \vdash F$ is a set operation:

- (i) $\phi(x) : \leftrightarrow \psi(Gx, Hx);$
- (ii) $Fx := G(Hx);$
- (iii) $Fx := \begin{cases} Gx, & \text{if } \theta(x); \\ Hx, & \text{otherwise;} \end{cases}$
- (iv) $Fx := I^*(Hx, x).$

Proof. For (i) we have

$$\begin{aligned} \phi(x) &\leftrightarrow \exists y \exists z [y \dot{=} Gx \wedge z \dot{=} Hx \wedge \psi(y, z)] \\ &\leftrightarrow \forall y \forall z [y \dot{=} Gx \wedge z \dot{=} Hx \rightarrow \psi(y, z)], \end{aligned}$$

which provides Σ^{ZF} and Π^{ZF} equivalents for ϕ . The key point in this and the following parts of this proof is that because $ZF \vdash G$ is a set operation, $ZF \vdash \exists! y[y \dot{=} Gx]$, and therefore the existential and universal quantifiers have the same scope. Similarly, for (ii),

$$\begin{aligned} y \dot{=} Fx &\leftrightarrow \exists z [z \dot{=} Hx \wedge y \dot{=} Gz] \\ &\leftrightarrow \forall z [z \dot{=} Hx \rightarrow y \dot{=} Gz]. \end{aligned}$$

For (iii),

$$y \dot{=} Fx \leftrightarrow [\theta(x) \wedge y \dot{=} Gx] \vee [\neg \theta(x) \wedge y \dot{=} Hx],$$

and for (iv),

$$y \dot{=} Fx \leftrightarrow (\forall z \dot{=} y)(\exists v \dot{=} Hx)z \dot{=} I(v, x) \wedge (\forall v \dot{=} Hx)I(v, x) \dot{=} y.$$

This is Δ^{ZF} by (i). $\square$

Now we have another large group of formulas and class operations which are Δ^{ZF} . Generally the proper classes are element- Δ^{ZF} , while sets and set operations are Δ^{ZF} .

Proposition 6.5.32 *Each of the following is Δ^{ZF} :*

- (i) $\dot{\langle} \cdot, \cdot \rangle$ and $\dot{\times}$ (Cartesian product);
- (ii) all of the parts of Definitions 6.2.6, 6.2.8, 6.2.10, 6.2.14, 6.2.15 and 6.2.20 except ${}^x y$ and $r \dot{\simeq} s$.

Proof. Many of these are straightforward, and we give the more interesting cases and leave the other parts as exercises.

$\dot{\langle} x, y \rangle := \{ \dot{\langle} x \rangle, \dot{\langle} x, y \rangle \}$ is a composition of Δ^{ZF} operations.

Because $x, y \dot{\in} \dot{\bigcup} \dot{\langle} x, y \rangle$ (see Remark 6.2.5),

$$\dot{\text{Op}}(z) \leftrightarrow (\exists x \dot{\in} \dot{\bigcup} z) (\exists y \dot{\in} \dot{\bigcup} z) z \dot{=} \dot{\langle} x, y \rangle.$$

For the next few cases we write

$$\begin{aligned} (\forall \dot{\langle} x, y \rangle \dot{\in} u) \phi & \text{ for } (\forall x \dot{\in} \dot{\bigcup} \dot{\bigcup} u) (\forall y \dot{\in} \dot{\bigcup} \dot{\bigcup} u) [\dot{\langle} x, y \rangle \dot{\in} u \rightarrow \phi] \\ (\exists \dot{\langle} x, y \rangle \dot{\in} u) \phi & \text{ for } (\exists x \dot{\in} \dot{\bigcup} \dot{\bigcup} u) (\exists y \dot{\in} \dot{\bigcup} \dot{\bigcup} u) [\dot{\langle} x, y \rangle \dot{\in} u \wedge \phi]. \end{aligned}$$

Now we have

$$\begin{aligned} u \dot{=} \dot{\text{Dom}}(r) & \leftrightarrow (\forall x \dot{\in} u) (\exists \dot{\langle} v, y \rangle \dot{\in} r) x \dot{=} v \wedge (\forall \dot{\langle} x, y \rangle \dot{\in} r) x \dot{\in} u \\ \dot{\text{Fn}}(f) & \leftrightarrow \dot{\text{Rel}}(f) \wedge (\forall \dot{\langle} x, y \rangle \dot{\in} f) (\forall \dot{\langle} v, w \rangle \dot{\in} f) [x \dot{=} v \rightarrow y \dot{=} w] \\ h \dot{=} g \circ f & \leftrightarrow \dot{\text{Rel}}(h) \\ & \wedge (\forall \dot{\langle} x, z \rangle \dot{\in} h) (\exists y \dot{\in} \dot{\text{Im}}(f)) [\dot{\langle} x, y \rangle \dot{\in} f \wedge \dot{\langle} y, z \rangle \dot{\in} g] \\ & \wedge (\forall \dot{\langle} x, y \rangle \dot{\in} f) (\forall \dot{\langle} y', z \rangle \dot{\in} g) [y \dot{=} y' \rightarrow \dot{\langle} x, z \rangle \dot{\in} h]. \end{aligned}$$

The relations ‘is well-founded’ and ‘is a well-ordering’ are more interesting. The definition is easily a Π^{ZF} formula:

$$r \text{ is well-founded} \leftrightarrow \forall u \left[\dot{\emptyset} \not\dot{\subseteq} u \subseteq \dot{\text{Fid}}(r) \rightarrow (\exists x \dot{\in} u) (\forall y \dot{\in} u) \dot{\langle} y, x \rangle \not\dot{\in} r \right].$$

Hence also ‘ r is a well-ordering’ is Π^{ZF} . But by Theorem 6.3.18,

$$r \text{ is a well-ordering} \leftrightarrow \exists \rho \exists f [r \dot{\simeq}_f \dot{<}_\rho],$$

which is easily a Σ^{ZF} formula. The corresponding fact for well-founded is Exercise 6.3.36. $\square$

There is one more very important general principle for showing that class operations are Δ^{ZF} .

Proposition 6.5.33 *For any Δ^{ZF} class operation G , if $\text{ZF} \vdash G$ is a set operation and F is the unique set operation such that for all x and y , $F(x, y) \doteq G(F \upharpoonright_x \text{TC}(y), x, y)$, then also*

$$\text{ZF} \vdash F \text{ is a set operation, and } F \text{ is } \Delta^{\text{ZF}}.$$

Proof. Recall from the proof of the Set Recursion Theorem 6.2.49 that

$$F(x, y) := \{ \{ z : \exists f [\theta(f, x, y) \wedge z \dot{\in} f(y)] \} \},$$

where $\theta(f, x, y)$ is the formula

$$\text{Fn}(f) \wedge \text{Dom}(f) \doteq \text{TC}^+(y) \wedge (\forall u \in \text{TC}^+(y)) f(u) \doteq G(f \upharpoonright \text{TC}(u), x, u).$$

Furthermore, in that proof we established that

$$\text{ZF} \vdash \forall x \forall y \exists! f \theta(f, x, y),$$

and we have

$$z \dot{\in} F(x, y) \leftrightarrow \exists f [\theta(f, x, y) \wedge z \dot{\in} f(y)] \leftrightarrow \forall f [\theta(f, x, y) \rightarrow z \dot{\in} f(y)].$$

We would like now to claim that θ is a Δ^{ZF} formula and hence that these equivalences verify that F is Δ^{ZF} . Unfortunately, this is not entirely straightforward since TC is itself given by a recursive definition. To get around this, we first show that if F is given by the simpler form of Remark 6.2.50,

$$\forall x \forall y [F(x, y) \doteq G(F \upharpoonright_x y, x, y)],$$

then if G is a Δ^{ZF} set operation, so is F . For this it suffices to note that in the proof of 6.2.49 we did not really need the domain of the approximation f to be exactly $\text{TC}^+(y)$ but only to include it — that is, we could carry out a version of that proof for this form of recursion using the following formula θ' in place of θ :

$$\begin{aligned} \text{Fn}(f) \wedge \text{Dom}(f) \text{ is transitive} \wedge y \in \text{Dom}(f) \\ \wedge (\forall u \in \text{Dom}(f)) f(u) \doteq G(f \upharpoonright u, x, u). \end{aligned}$$

Then θ' is clearly Δ^{ZF} , hence by the sketch above so is F . In particular, the operations TC and TC^+ are Δ^{ZF} , and we can complete the proof of the full version as sketched above. $\square$

With this we have

Proposition 6.5.34 *The following are Δ^{ZF} and hence $\mathbf{M}$ -absolute for any transitive class model $\mathbf{M}$ of ZF :*

- (i) $\dot{+}_o$ and $\dot{\times}_o$;
- (ii) $(\sigma, x) \mapsto \mathbf{l}^\sigma(x)$ for any Δ^{ZF} set operation $\mathbf{l}$;
- (iii) rank ;
- (iv) $\mathbf{l}^\omega$ for any Δ^{ZF} set operation $\mathbf{l}$;
- (v) $\bar{\mathbf{l}}$ for any finitary Δ^{ZF} set operation $\mathbf{l}$;
- (vi) TC , HF and HFO ;
- (vii) ‘is finite’ and $u \mapsto <^\omega u$.

Proof. Parts (i)–(iii) are simply applications of the preceding proposition and (iv) is a composition with the Δ^{ZF} set ω . Then (v) is immediate from the Finitary Fixed-point Theorem 6.2.42. Now (vi) follows, since each of these is of the form $\bar{\mathbf{l}}$ for a finitary Δ^{ZF} set operation $\mathbf{l}$. For (vii), using the results of Proposition 6.2.52,

$$x \text{ is finite} \leftrightarrow (\exists y \in \omega) (\exists f \in \text{HFO}(x)) [x \approx_f y]$$

$$y \doteq <^\omega u \leftrightarrow (\forall f \in y) (\exists z \in \omega) [f : z \dot{\rightarrow} u]$$

$$\wedge (\forall f \in \text{HFO}(u)) ((\exists z \in \omega) [f : z \dot{\rightarrow} u] \rightarrow f \in y). \quad \square$$

Corollary 6.5.35 *For any transitive class model $\mathbf{M}$ of ZF , the operation $\sigma \mapsto \dot{\mathbf{V}}_\sigma$ is $\mathbf{M}$ -element absolute. Hence*

$$\mathbf{M} \doteq \dot{\mathbf{V}}^{\mathbf{M}} \doteq \bigcup \{ \{ \dot{\mathbf{V}}_\sigma \dot{\cap} \mathbf{M} : \sigma \in \dot{\text{Or}} \dot{\cap} \mathbf{M} \} \}.$$

Proof. The first assertion follows from the equivalences (Proposition 6.3.29)

$$y \in \dot{\mathbf{V}}_\sigma^{\mathbf{M}} \leftrightarrow y \in \mathbf{M} \wedge \text{rank}^{\mathbf{M}}(y) \dot{<} \sigma \leftrightarrow y \in \dot{\mathbf{V}}_\sigma \dot{\cap} \mathbf{M}.$$

For the second, note that since $\text{ZF} \vdash \forall x \exists \sigma (x \in \dot{\mathbf{V}}_\sigma)$, this is also true relativized $\mathbf{M}$, so by the first part

$$\mathbf{M} \doteq \bigcup \{ \{ \dot{\mathbf{V}}_\sigma^{\mathbf{M}} : \sigma \in \dot{\text{Or}}^{\mathbf{M}} \} \} \doteq \bigcup \{ \{ \dot{\mathbf{V}}_\sigma \dot{\cap} \mathbf{M} : \sigma \in \dot{\text{Or}} \dot{\cap} \mathbf{M} \} \}. \quad \square$$

To complete our survey of absoluteness, we consider what can be said about some notions which are (at least apparently) not Δ^{ZF} and not absolute. Some important examples are

$$y \doteq \dot{\phi} x \leftrightarrow (\forall u \in y) u \dot{\subseteq} x \wedge \forall u [u \dot{\subseteq} x \rightarrow u \in y] \quad \text{is } \Pi^{\text{ZF}}$$

$$x \dot{\preceq} y \leftrightarrow \exists f [\dot{\text{Fn}}(f) \wedge \dot{\text{Inj}}(f) \wedge \dot{\text{Dom}} f \doteq x \wedge \dot{\text{Im}} f \dot{\subseteq} y] \quad \text{is } \Sigma^{\text{ZF}}$$

$$\sigma \text{ a cardinal} \leftrightarrow \neg (\exists \tau \dot{<} \sigma) \sigma \dot{\preceq} \tau \quad \text{is } \Pi^{\text{ZF}}$$

$$\kappa \doteq \text{Ha}(x) \leftrightarrow (\forall \sigma \dot{<} \kappa) \sigma \dot{\preceq} x \wedge \neg \kappa \dot{\preceq} x.$$

From the proof of Theorem 6.5.30 it is clear that

Corollary 6.5.36 *For any class model M of ZF,*

$$\phi(x) \text{ is } \Sigma^{\text{ZF}} \implies \text{ZF} \vdash \forall^M x [\phi^{(M)}(x) \rightarrow \phi(x)],$$

$$\phi(x) \text{ is } \Pi^{\text{ZF}} \implies \text{ZF} \vdash \forall^M x [\phi(x) \rightarrow \phi^{(M)}(x)],$$

and similarly for classes and class operations. $\square$

This is, of course, for the same basic reasons as the upward (downward) persistence of existential (universal) formulas (2.3.25). In the course of establishing the independence of CH we shall be interested in finding particular class models M for which $\text{ZFC} \vdash \text{CH}^{(M)}$ or $\text{ZFC} \vdash \neg \text{CH}^{(M)}$. Since $\text{ZFC} \vdash \aleph_1 \leq 2^{\aleph_0}$, we have (in ZFC; this is essentially CH_2 from Section 6.4)

$$\text{CH} \leftrightarrow \exists f [\text{Inj}(f) \wedge f : \dot{\varphi}(\dot{\omega}) \dot{\rightarrow} \dot{\aleph}_1],$$

and hence, if M is a transitive class model of ZFC such that $\dot{\omega} \in M$, using the absoluteness results above,

$$\begin{aligned} \text{CH}^{(M)} &\leftrightarrow (\exists f \in M) [\text{Inj}(f)^{(M)} \wedge (f : \dot{\varphi}(\dot{\omega}) \dot{\rightarrow} \dot{\aleph}_1)^{(M)}] \\ &\leftrightarrow (\exists f \in M) [\text{Inj}(f) \wedge f : \dot{\varphi}(\dot{\omega}) \dot{\cap} M \dot{\rightarrow} \dot{\aleph}_1^M]. \end{aligned}$$

Of course generally $\dot{\varphi}(\dot{\omega}) \dot{\cap} M$ may be much smaller than $\dot{\varphi}(\dot{\omega})$ and by the above,

$$\dot{\aleph}_1^M \doteq \left\{ \sigma : (\sigma \dot{\preceq} \dot{\omega})^{(M)} \right\} \subseteq \left\{ \sigma : \sigma \dot{\preceq} \dot{\omega} \right\} \doteq \dot{\aleph}_1.$$

that is, $\dot{\aleph}_1^M$ is an ordinal which may be countable (in $\dot{V}$) but cannot be counted by any function $f \in M$. In particular, for the class L defined in Section 6.7, we will show that $\dot{\aleph}_1^L$ takes on very different values in different extensions of ZFC: in the theory ZFL (defined in Section 6.7), $\dot{\aleph}_1^L \doteq \dot{\aleph}_1$, while in the theory $\text{ZF} + \exists \text{MC}$ described in Section 6.9, $\dot{\aleph}_1^L < \dot{\aleph}_1$; in fact, for every countable ordinal σ , $\dot{\aleph}_\sigma^L < \dot{\aleph}_1$ and $\dot{\aleph}_{\aleph_1}^L \doteq \dot{\aleph}_1$.

We conclude this section with a result of a somewhat different flavor.

Theorem 6.5.37 *If ZF is consistent, then it is not finitely axiomatizable.*

Proof. Suppose that ZF is finitely axiomatizable, and let χ be the conjunction of some finite set of axioms. By the Reflection Theorem 6.5.20 applied to $\dot{V}$,

$$\text{ZF} \vdash \exists \rho [\chi^{(\dot{V}_\rho)} \leftrightarrow \chi^{(\dot{V})}].$$

Since $\mathbf{ZF} \vdash \chi$ and $\chi \leftrightarrow \chi^{(\dot{V})}$, also

$$\mathbf{ZF} \vdash \exists \rho \left[\chi^{(\dot{V}_\rho)} \wedge (\forall \sigma < \rho) \neg \chi^{(\dot{V}_\sigma)} \right].$$

Arguing in $\mathbf{ZF}$, fix such a ρ . Since χ axiomatizes $\mathbf{ZF}$, $\dot{V}_\rho$ is a set model of $\mathbf{ZF}$, so for any theorem ϕ of $\mathbf{ZF}$, $\phi^{(\dot{V}_\rho)}$ — in particular

$$(\exists \sigma \chi^{(\dot{V}_\sigma)})^{(\dot{V}_\rho)} \quad \text{— that is, } (\exists \sigma \in \dot{V}_\rho) \chi^{(\dot{V}_{\sigma^\rho})}.$$

But $\sigma \in \dot{V}_\rho \rightarrow \sigma < \rho$ and since $\mathbf{rank}$ is $\dot{V}_\rho$ -absolute (6.5.34(iii)), $\dot{V}_{\sigma^\rho} \doteq \dot{V}_\sigma \cap \dot{V}_\rho \doteq \dot{V}_\sigma$. Hence $(\exists \sigma < \rho) \chi^{(\dot{V}_\sigma)}$, contrary to the choice of ρ , and $\mathbf{ZF}$ is inconsistent. $\square$

To simplify the discussion here we have focussed on the notion of a class model as in Definition 6.5.1, but for some purposes a more general notion is needed: for any two $L_\in$ -theories T and U , we say that a class $\mathbf{M}$ is a **class model** of U in T iff $T \vdash \phi^{(\mathbf{M})}$ for all $\phi \in U$. By the same arguments as above, this implies that

$$T \text{ is consistent} \implies U \text{ is consistent,}$$

and in this language the assertion (9) at the beginning of the section is: there exists $\mathbf{M}$ such that

$$\mathbf{M} \text{ is a class model of } \mathbf{ZF} \cup \{\mathbf{AC}, \mathbf{GCH}\} \text{ in } \mathbf{ZF}.$$

There are other interesting examples of this phenomenon; we explore in the exercises below several of the following sort. Let θ be one of the axioms of $\mathbf{ZF}$. By $\mathbf{ZF} - \theta$ we mean the theory generated by all of the axioms except θ and by $\mathbf{ZF} - \theta + (\neg \theta)$ the theory generated by additionally adding $\neg \theta$ as an axiom. Suppose that there is a class $\mathbf{M}$ such that

$$\mathbf{M} \text{ is a class model of } \mathbf{ZF} - \theta + (\neg \theta) \text{ in } \mathbf{ZF}.$$

Then we conclude

$$(12) \quad \mathbf{ZF} \text{ is consistent} \implies \mathbf{ZF} - \theta + (\neg \theta) \text{ is consistent,}$$

or equivalently

$$(13) \quad \mathbf{ZF} - \theta \text{ is consistent} \implies \mathbf{ZF} - \theta \not\vdash \theta,$$

since if $\mathbf{ZF} - \theta \vdash \theta$, then $\mathbf{ZF} = \mathbf{ZF} - \theta$. The import of (13) is that the axiom θ is not redundant; $\mathbf{ZF} - \theta$ is a theory weaker than $\mathbf{ZF}$.

Collateral Reading. [Kun; Chapter IV] and [Jec; Chapter 12] are good sources for alternative presentations of some of the material of this section.

Exercises

Exercise 6.5.38 Show that $\dot{V}_{\dot{\omega}}$ is a class model of $\text{ZF} - \text{Infinity} + (\neg \text{Infinity})$ in ZF.

Exercise 6.5.39 Using the results of Exercise 6.4.60, show that for every axiom ϕ of ZFC – Powerset,

$$\text{ZFC} \vdash \forall \kappa [\kappa \text{ is a regular uncountable cardinal} \rightarrow \phi^{(H_\kappa)}].$$

Conclude that $H_{\aleph_1}$ is a class model of $\text{ZF} - \text{Powerset} + (\neg \text{Powerset})$ in ZF.

Exercise 6.5.40 Show that for every axiom ϕ of ZF – Replacement (ZF with all instances of Replacement removed),

$$\text{ZF} \vdash \forall \sigma [\sigma \text{ is a limit ordinal} \rightarrow \phi^{(\dot{V}_\sigma)}].$$

Find a particular instance θ of Replacement such that $\dot{V}_{\dot{\omega} + \dot{\omega}}$ is a class model of $\text{ZF} - \text{Replacement} + (\neg \theta)$ in ZF.

Exercise 6.5.41 Show that if M is a transitive class-model of ZF,

- (i) $\text{ZF} \vdash F$ is a set operation $\implies \text{ZF} \vdash F^M$ is an M -set operation;
- (ii) $\text{ZF} \vdash F^M$ is a set operation $\implies \text{ZF} \vdash F^M$ is an M -set operation.

Exercise 6.5.42 Show that if M is a proper class model of ZF, then $\dot{O}r \subseteq M$.

Exercise 6.5.43 Let M be a transitive class such that

- (a) $(\forall v \subseteq M) (\exists w \in M) v \subseteq w$, (M is *almost universal*);
- (b) for every element- Δ_0 class operation F ,

$$\text{ZF} \vdash F^\cap \text{ is an } M\text{-set operation, } (\Delta_0\text{-Comprehension}),$$

where (as for Theorem 6.1.9), $F^\cap(x, u) \doteq F(x) \dot{\cap} u$. Show that

- (i) M is a proper class and $\dot{\omega} \in M$;
- (ii) for each $L_\in$ -formula $\phi(x)$ there exists a Δ_0 formula ϕ_0 and a set operation H_ϕ such that for all $u, v \in M$ and $x \in u$,

$$H_\phi(u) \subseteq w \rightarrow [\phi^{(M)}(x) \leftrightarrow \phi_0^{(M)}(x, w)];$$

- (iii) M is rich, hence a class model of ZF.

Hint: Prove (ii) by formula induction; for the $\exists$ step use the Collection Principle (6.3.30).

6.6 Mathematics in set theory II

In [Section 6.2](#) we showed how many fundamental mathematical notions can be developed within the confines of ZF. Here we continue this theme with a focus on our main subject, logic. After representing first-order languages, structures and theories as set-theoretic objects, we can establish as theorems of ZF many of the main results of [Chapters 2–3](#). Of course, to work this out in full detail would take as much space as the original, and we shall be very brief in many instances. This development will have here two main consequences: (1) a version of Gödel's Second Incompleteness Theorem for set theory, and in the next section (2) the existence of many universes (Definition 6.5.14), which by the Universe Theorem 6.5.22 are class models of ZF.

Let us start at the beginning of [Chapter 2](#). Recall that a finite sequence is a function s with $\text{Dom}(s) \in \dot{\omega}$. We use juxtaposition to indicate concatenation and ignore the distinction noted following Definition 6.2.34, so, for example, $\langle x \rangle_{st} := u$ such that for all $y < \text{Dom}(s)$ and $z < \text{Dom}(t)$,

$$u(\dot{0}) \doteq x, \quad u(y \dot{+} \dot{1}) \doteq s(y), \quad \text{and} \quad u(z \dot{+} \text{Dom}(s) \dot{+} \dot{1}) \doteq t(z).$$

To represent first-order languages as sets, we need first to assign to each logical symbol a a particular set $\dot{a}$; as for the assignment of Gödel numbers in [Section 4.4](#), the particular choice is of little consequence. We take the sets

$$\dot{0}, \dot{2}, \dot{4}, \dot{6} \quad \text{as} \quad \ddot{=}, \dot{\neg}, \dot{\vee}, \dot{\exists},$$

and set

$$\text{vb} := \text{Even}_{\omega} \setminus \{\dot{0}, \dot{2}, \dot{4}, \dot{6}\},$$

where

$$\begin{aligned} \text{Even}_{\omega} &\doteq \{x \in \dot{\omega} : \exists z(z \dot{+} z \doteq x)\} \quad \text{and} \\ \text{Odd}_{\omega} &\doteq \{x \in \dot{\omega} : \exists z(z \dot{+} z \dot{+} \dot{1} \doteq x)\}. \end{aligned}$$

In particular, for each $n \in \omega$, $\dot{v}_n := \overbrace{2n+8}^{\dot{=}}$. Note that for a symbol like $\ddot{=}$, which is already dotted, we write $\ddot{=}$ instead of $\dot{\ddot{=}}$.

Definition 6.6.1 A *first-order language* is a quadruple $\ell \doteq \langle \text{rs}, \text{fs}, \text{cs}, \nu \rangle$ such that rs , fs and cs are pairwise disjoint sets and $\nu : \text{rs} \dot{\cup} \text{fs} \rightarrow \dot{\omega} \setminus \{\dot{0}\}$. The cardinality $\|\ell\|$ of ℓ is $|\text{rs} \dot{\cup} \text{fs} \dot{\cup} \text{cs}|$. For a countable language, it is natural to take rs , fs and cs to be subsets of Odd_{ω} . Let $\ell \doteq \langle \text{rs}, \text{fs}, \text{cs}, \nu \rangle$ be a fixed first-order language. The set of *atomic terms* of ℓ is

$$\text{AtTe}(\ell) := \{\langle x \rangle : x \in \text{cs} \vee x \in \text{vb}\}.$$

It will be slightly more convenient to represent a compound term (and similarly formulas below) not as in [Chapter 2](#) as a finite sequence of symbols but as a finite sequence starting with a function symbol and followed by the appropriate number of terms. Thus the set of ℓ -terms is defined inductively (via Example 6.2.37(i)) as

$$\dot{\mathbf{Te}}(\ell) := \bigcap \left\{ \{ z : \mathbf{AtTe}(\ell) \dot{\subseteq} z \wedge (\forall f \in \mathbf{fs})(\forall s \in {}^{\nu(f)}z) \langle \dot{f} \rangle s \in z \} \right\}.$$

Of course, $\dot{\mathbf{Te}}$ is a set operation because such z exist, for example, the closure of $\mathbf{fs} \cup \mathbf{AtTe}$ under formation of finite sequences. Similarly, the set of ℓ -**atomic formulas** of ℓ is

$$\mathbf{AtFm}(\ell) := \left\{ \langle \dot{p} \rangle s : p \in \mathbf{rs} \wedge s \in {}^{\nu(p)}\dot{\mathbf{Te}}(\ell) \right\},$$

and the set of ℓ -**formulas** is

$$\mathbf{Fm}(\ell) := \bigcap \left\{ \{ z : \mathbf{AtFm}(\ell) \dot{\subseteq} z \wedge \mathbf{Bool}(z) \dot{\subseteq} z \wedge \exists(z) \dot{\subseteq} z \} \right\},$$

where

$$\mathbf{Bool}(z) := \left\{ \langle \dot{\dot{\neg}}, x \rangle : x \in z \right\} \dot{\cup} \left\{ \langle \dot{\dot{\vee}}, x, y \rangle : x, y \in z \right\}$$

and

$$\exists(z) := \left\{ \langle \dot{\dot{\exists}}, v, x \rangle : x \in z \wedge v \in \mathbf{vb} \right\}.$$

Formalization of the notions of free and bound variable (left as an exercise) leads easily to the set operation $\mathbf{Sent}(\ell)$.

Among the languages are representations $\dot{L}$ of particular languages L . For example,

$$\dot{L}_\Omega := \left\langle \langle \dot{\dot{<}} \rangle, \langle \dot{\dot{S}}, \dot{+}, \dot{\times} \rangle, \langle \dot{\dot{0}} \rangle, \dot{\nu}_\Omega \right\rangle := \left\langle \langle \dot{\dot{1}} \rangle, \langle \dot{\dot{3}}, \dot{5}, \dot{7} \rangle, \langle \dot{\dot{9}} \rangle, \dot{\nu}_\Omega \right\rangle,$$

where

$$\dot{\nu}_\Omega := \left\{ \langle \dot{\dot{<}}, \dot{2} \rangle, \langle \dot{\dot{S}}, \dot{1} \rangle, \langle \dot{\dot{+}}, \dot{2} \rangle, \langle \dot{\dot{\times}}, \dot{2} \rangle \right\}.$$

Similarly,

$$\dot{L}_\in := \left\langle \langle \dot{\dot{\in}} \rangle, \dot{\emptyset}, \dot{\emptyset}, \dot{\nu}_\in \right\rangle := \left\langle \langle \dot{\dot{1}} \rangle, \dot{\emptyset}, \dot{\emptyset}, \dot{\nu}_\in \right\rangle,$$

with $\dot{\nu}_\in := \left\{ \langle \dot{\dot{\in}}, \dot{2} \rangle \right\}$.

Recursively, to each term $t \in \mathbf{Te}_{L_\Omega}$ (and similarly for other languages) we assign a set $\dot{t}$ by

$$\overbrace{\dot{S}t} := \langle \dot{\dot{S}} \rangle \dot{t}, \quad \overbrace{\dot{+}tu} := \langle \dot{\dot{+}}, \dot{t}, \dot{u} \rangle \quad \text{and} \quad \overbrace{\dot{\times}tu} := \langle \dot{\dot{\times}}, \dot{t}, \dot{u} \rangle,$$

so, for example, if t is $\dot{2}$, then $\dot{t} \doteq \ddot{2} \doteq \langle \dot{\ddot{S}}, \langle \dot{\ddot{S}}, \ddot{0} \rangle \rangle$. Similarly, to each formula $\phi \in \mathbf{Fm}_{L_\Omega}$, we assign a set $\dot{\phi}$ by

$$\overbrace{\langle \dot{t} u \rangle} \doteq \langle \dot{\ddot{<}}, \dot{t}, \dot{u} \rangle, \quad \neg \dot{\psi} \doteq \langle \dot{\neg}, \dot{\psi} \rangle \quad \text{etc.}$$

It is straightforward to prove by term- or formula-induction that for any language L ,

$$\begin{aligned} t \in \mathbf{Te}_L &\implies \mathbf{ZF} \vdash \dot{t} \in \dot{\mathbf{Te}}(\dot{L}) \quad \text{and} \\ \phi \in \mathbf{Fm}_L &\implies \mathbf{ZF} \vdash \dot{\phi} \in \dot{\mathbf{Fm}}(\dot{L}). \end{aligned}$$

Note that we can neither state nor prove a sentence of $\mathbf{ZF}$ which asserts that all of the elements of $\dot{\mathbf{Fm}}(\dot{L})$ are of the form $\dot{\phi}$ for some $\phi \in \mathbf{Fm}_L$; in general for $\mathfrak{A}$ a model of $\mathbf{ZF}$, the set $\dot{\mathbf{Fm}}(\dot{L})^\mathfrak{A}$ will contain non-standard “formulas” in addition to the standard formulas $\dot{\phi}^\mathfrak{A}$.

For any language $\ell \doteq \langle \mathbf{rs}, \mathbf{fs}, \mathbf{cs}, \nu \rangle$, an ℓ -**structure** $\mathbf{a}$ is a pair $\mathbf{a} = \langle a, \Xi \rangle$ such that a is a non-empty set, $\mathbf{Dom}(\Xi) \doteq \mathbf{rs} \cup \mathbf{fs} \cup \mathbf{cs}$, for each $p \in \mathbf{rs}$, $\Xi(p) \subseteq {}^{\nu(p)}a$, for each $f \in \mathbf{fs}$, $\Xi(f) : {}^{\nu(f)}a \rightarrow a$, and for each $c \in \mathbf{cs}$, $\Xi(c) \in a$. For example, the standard $\dot{L}_\Omega$ structure is $\dot{\Omega} \doteq \langle \dot{\omega}, \Xi_\Omega \rangle$, where

$$\Xi_\Omega(\ddot{S}) \doteq \dot{S}, \quad \Xi_\Omega(\ddot{+}) \doteq \dot{+} \quad \text{etc.}$$

The machinery of satisfaction is formalized using the Finitary Closure Recursion Theorem 6.2.53. First, for a structure $\mathbf{a} \doteq \langle a, \Xi \rangle$, an $\mathbf{a}$ -**assignment** is any function $\alpha : \mathbf{vb} \rightarrow a$. Then, illustrating with $\dot{L}_\Omega$, there is a unique set operation $\mathbf{v}$ such that

$$\begin{aligned} \mathbf{v}(\mathbf{a}, \langle \dot{v} \rangle, \alpha) &\doteq \alpha(v) \quad \text{for each } v \in \mathbf{vb}; \\ \mathbf{v}(\mathbf{a}, \langle \dot{\ddot{0}} \rangle, \alpha) &\doteq \Xi(\ddot{0}); \\ \mathbf{v}(\mathbf{a}, \langle \dot{\ddot{+}}, x, y \rangle, \alpha) &\doteq \Xi(\ddot{+})(\mathbf{v}(\mathbf{a}, x, \alpha), \mathbf{v}(\mathbf{a}, y, \alpha)); \end{aligned}$$

etc. Similarly, there is a unique set operation $\mathbf{w}$ such that for $x, y \in \dot{\mathbf{Te}}(\dot{L}_\Omega)$ and any $\mathbf{a}$ and α ,

$$\mathbf{w}(\mathbf{a}, \langle \dot{\ddot{=}}, x, y \rangle)(\alpha) \doteq \begin{cases} \dot{1}, & \text{if } \mathbf{v}(\mathbf{a}, x, \alpha) \doteq \mathbf{v}(\mathbf{a}, y, \alpha); \\ \dot{0}, & \text{otherwise;} \end{cases}$$

and for $x, y \in \dot{\mathbf{Fm}}(\dot{L}_\Omega)$,

$$\begin{aligned} \mathbf{w}(\mathbf{a}, \langle \dot{\neg}, x \rangle)(\alpha) &\doteq \begin{cases} \dot{1}, & \text{if } \mathbf{w}(\mathbf{a}, x)(\alpha) \doteq \dot{0}; \\ \dot{0}, & \text{otherwise;} \end{cases} \\ \mathbf{w}(\mathbf{a}, \langle \dot{\vee}, x, y \rangle)(\alpha) &\doteq \begin{cases} \dot{1}, & \text{if } \mathbf{w}(\mathbf{a}, x)(\alpha) \doteq \dot{1} \vee \mathbf{w}(\mathbf{a}, y)(\alpha) \doteq \dot{1}; \\ \dot{0}, & \text{otherwise;} \end{cases} \\ \mathbf{w}(\mathbf{a}, \langle \dot{\exists}, v, x \rangle)(\alpha) &\doteq \begin{cases} \dot{1}, & \text{if } (\exists z \in a) [\mathbf{w}(\mathbf{a}, x)(\alpha_{v \mapsto z}) \doteq \dot{1}]; \\ \dot{0}, & \text{otherwise.} \end{cases} \end{aligned}$$

The definition of w is in the form discussed just before Definition 2.1.22. Then we write

$$\dot{\models}_{\mathbf{a}} x[\alpha] : \leftrightarrow w(\mathbf{a}, x)(\alpha) \dot{=} \dot{1},$$

and for $\gamma \dot{\subseteq} \dot{\mathbf{Fm}}(\dot{L}_\Omega)$,

$$\gamma \dot{\models} y : \leftrightarrow \forall \mathbf{a} \forall \alpha [(\forall x \in \gamma) \dot{\models}_{\mathbf{a}} x[\alpha] \rightarrow \dot{\models}_{\mathbf{a}} y[\alpha]].$$

When $\gamma \dot{\cup} \{\dot{y}\} \dot{\subseteq} \dot{\mathbf{Sent}}(\dot{L}_\Omega)$, these simplify to

$$\gamma \dot{\models} y : \leftrightarrow \forall \mathbf{a} [\mathbf{ModelOf}(\mathbf{a}, \gamma) \rightarrow \dot{\models}_{\mathbf{a}} y],$$

where

$$\mathbf{ModelOf}(\mathbf{a}, \gamma) : \leftrightarrow (\forall x \in \gamma) \dot{\models}_{\mathbf{a}} x.$$

Then it is straightforward to establish some elementary model theory (from Sections 2.3–2.4) in ZF. We leave it to the reader to write out formal definitions, for any ℓ -structures, of the basic notions

$$\begin{array}{ll} \mathbf{a} \dot{\simeq}_f \mathbf{b} & f \text{ is an } \textbf{isomorphism} \text{ of } \mathbf{a} \text{ and } \mathbf{b}; \\ \mathbf{a} \dot{\prec} \mathbf{b} & \mathbf{a} \text{ is an } \textbf{elementary substructure} \text{ of } \mathbf{b}. \end{array}$$

In particular, we will make use below of the following two theorems of ZF and ZFC, respectively, formal versions of Lemma 2.3.6 and Theorem 3.4.4.

Theorem 6.6.2 *For any language ℓ and any ℓ -structures $\mathbf{a}$ and $\mathbf{b}$,*

$$\mathbf{a} \dot{\simeq}_f \mathbf{b} \rightarrow [\forall x \in \dot{\mathbf{Fm}}(\ell) (\dot{\models}_{\mathbf{a}} x[\alpha] \leftrightarrow \dot{\models}_{\mathbf{b}} x[\alpha^f])],$$

where α^f is the $\mathbf{a}$ -assignment such that for all $v \in \mathbf{vb}$, $\alpha^f(v) \dot{=} f(\alpha(v))$. $\square$

Downward Löwenheim-Skolem Theorem 6.6.3 (ZFC) *For any appropriate values of the variables,*

$$w \dot{\subseteq} \mathbf{b} \wedge \|\ell\| \dot{\leq} |w| \rightarrow \exists \mathbf{a} [w \dot{\subseteq} \mathbf{a} \dot{\subseteq} \mathbf{b} \wedge |\mathbf{a}| \dot{=} |w| \wedge \mathbf{a} \dot{\prec} \mathbf{b}]. \quad \square$$

For any language ℓ , an ℓ -**theory** is a set $t \subseteq \dot{\mathbf{Sent}}(\ell)$ such that for all $y \in \dot{\mathbf{Sent}}(\ell)$, $t \dot{\models} y \rightarrow y \in t$. In particular, for any $\gamma \dot{\subseteq} \dot{\mathbf{Sent}}(\ell)$,

$$\mathbf{Th}(\gamma, \ell) := \left\{ y \in \dot{\mathbf{Sent}}(\ell) : \gamma \dot{\models} y \right\},$$

the ℓ -**theory generated by** γ . $\dot{\mathbf{Con}}(t)$ is the formula $\exists \mathbf{a} \mathbf{ModelOf}(\mathbf{a}, t)$ which expresses the consistency of t . Then with F any logically false sentence, a formalized version of part of Corollary 2.4.15 is

$$\mathbf{ZF} \vdash \neg \dot{\mathbf{Con}}(t) \leftrightarrow \dot{\mathbf{F}} \dot{\in} t.$$

The most important remaining topic for formalization in ZF is the Countable Completeness Theorem. It is straightforward to define a set operation LogAx which for a language ℓ gives the set $\text{LogAx}(\ell)$ of logical axioms for ℓ . The rules of inference of ℓ formulated as functions are

$$\begin{aligned} \text{MP}(x, y) &:= \begin{cases} z, & \text{if } y \doteq \langle \dot{\forall}, \langle \dot{\neg}, x \rangle, z \rangle; \\ \overbrace{\forall v_0 (v_0 \doteq v_0)}, & \text{otherwise;} \end{cases} \\ \dot{\exists}\text{-rule}(x, v) &:= \begin{cases} \langle \dot{\forall}, \langle \dot{\neg}, \langle \dot{\exists}, v, y \rangle \rangle, z \rangle, & \text{if } x \doteq \langle \dot{\forall}, \langle \dot{\neg}, y \rangle, z \rangle \\ & \wedge v \in \text{vb} \setminus \dot{\text{Fv}}(z); \\ \overbrace{\forall v_0 (v_0 \doteq v_0)}, & \text{otherwise.} \end{cases} \end{aligned}$$

Then for any $\gamma \subseteq \dot{\text{Fm}}(\ell)$,

$$\begin{aligned} \dot{\text{Thm}}(\gamma, \ell) &:= \left\{ x \in \dot{\text{Fm}}(\ell) : \exists y \text{ Proof}(\gamma, x, y) \right\} \\ &= \bigcap \left\{ \left\{ \delta \subseteq \dot{\text{Fm}}(\ell) : \gamma \subseteq \delta \wedge \text{LogAx}(\ell) \subseteq \delta \wedge \text{Infr}(\delta) \subseteq \delta \right\} \right\}, \end{aligned}$$

where

$$\text{Infr}(\delta) := \text{MP}^*(\delta \dot{\times} \delta) \dot{\cup} \dot{\exists}\text{-rule}^*(\delta \dot{\times} \text{vb}).$$

Then if $\dot{\text{Th}}^+(\gamma, \ell) := \dot{\text{Thm}}(\gamma, \ell) \dot{\cap} \dot{\text{Sent}}(\ell)$, we have

Countable Completeness Theorem 6.6.4 *For any countable language ℓ and any $\gamma \subseteq \dot{\text{Sent}}(\ell)$,*

$$\dot{\text{Th}}(\gamma, \ell) \doteq \dot{\text{Th}}^+(\gamma, \ell). \quad \square$$

Recall that in [Section 6.2](#) we defined a mapping which to each L_Ω -formula ϕ assigned an $L_{\dot{\epsilon}}$ -formula $\phi^{(\dot{\omega})}$ and showed that this is a relative interpretation of $\mathbf{P}$ in $\mathbf{ZF}$: for all ϕ ,

$$\phi \in \mathbf{P} \implies \mathbf{ZF} \vdash \phi^{(\dot{\omega})}.$$

The next theorem tells us that this interpretation agrees with the formalized notion of truth in $\dot{\Omega}$. We have used a similar notation with a closely related meaning in [Section 6.5](#): $\phi \mapsto \phi^{(a)}$ is a mapping of $L_{\dot{\epsilon}}$ -formulas to $L_{\dot{\epsilon}}$ -formulas defined by relativizing quantifiers to the variable a . Here we are in effect interpreting ϕ in the $\dot{L}_{\dot{\epsilon}}$ -structure $\langle a, \dot{\epsilon}_a \rangle$, where

$$\dot{\epsilon}_a := \{ \langle \dot{x}, y \rangle : x, y \in a \wedge x \dot{\in} y \}.$$

More formally, this is $\langle a, \Xi_a \rangle$, where Ξ_a is the function which assigns $\dot{\epsilon}_a$ to the only non-logical symbol $\dot{\epsilon}$ of $\dot{L}_{\dot{\epsilon}}$.

Theorem 6.6.5

(i) For every L_Ω -formula $\phi(v_0, \dots, v_{k-1})$,

$$\text{ZF} \vdash (\forall x_0 \dot{\in} \dot{\omega}) \cdots (\forall x_{k-1} \dot{\in} \dot{\omega}) \\ \left(\phi^{(\dot{\omega})}(x_0, \dots, x_{k-1}) \leftrightarrow \models_{\dot{\omega}} \dot{\phi}[x_0, \dots, x_{k-1}] \right).$$

(ii) For every $L_{\in}$ -formula $\phi(v_0, \dots, v_{k-1})$,

$$\text{ZF} \vdash \forall a (\forall x_0 \dot{\in} a) \cdots (\forall x_{k-1} \dot{\in} a) \\ \left(\phi^{(a)}(x_0, \dots, x_{k-1}) \leftrightarrow \models_{\langle a, \dot{\in}_a \rangle} \dot{\phi}[x_0, \dots, x_{k-1}] \right).$$

Here $x_0, \dots, x_{k-1}$ means the (any) assignment α such that for each $i < k$, $\alpha(\dot{v}_i) \dot{\in} x_i$.

Proof. We prove (ii) by formula induction. If ϕ is an atomic formula, say $v_0 \dot{=} v_1$, then

$$(v_0 \dot{=} v_1)^{(a)}(x_0, x_1) \leftrightarrow x_0 \dot{=} x_1 \leftrightarrow \models_{\langle a, \dot{\in}_a \rangle} \overbrace{v_0 \dot{=} v_1}^{\dot{=}}[x_0, x_1].$$

Also, easily (for sentences to simplify notation),

$$\begin{aligned} (\neg \phi)^{(a)} &\leftrightarrow \neg \phi^{(a)} \leftrightarrow \neg \models_{\langle a, \dot{\in}_a \rangle} \dot{\phi} \leftrightarrow \models_{\langle a, \dot{\in}_a \rangle} \overbrace{\neg \phi}^{\dot{=}}, \\ (\phi \vee \psi)^{(a)} &\leftrightarrow \phi^{(a)} \vee \psi^{(a)} \leftrightarrow \models_{\langle a, \dot{\in}_a \rangle} \dot{\phi} \vee \models_{\langle a, \dot{\in}_a \rangle} \dot{\psi} \\ &\leftrightarrow \models_{\langle a, \dot{\in}_a \rangle} \overbrace{\phi \vee \psi}^{\dot{=}}, \end{aligned}$$

and

$$\begin{aligned} (\exists v_0 \phi)^{(a)} &\leftrightarrow (\exists v_0 \dot{\in} a) \phi^{(a)} \leftrightarrow (\exists x_0 \dot{\in} a) \phi_{v_0}^{(a)}(x_0) \\ &\leftrightarrow (\exists x_0 \dot{\in} a) \models_{\langle a, \dot{\in}_a \rangle} \dot{\phi}[x_0] \leftrightarrow \models_{\langle a, \dot{\in}_a \rangle} \overbrace{\exists v_0 \phi}^{\dot{=}}. \quad \square \end{aligned}$$

Corollary 6.6.6 *For any $L_{\in}$ -formula $\phi(v_0, \dots, v_{k-1})$, the following are theorems of ZF: $(\forall x_0 \in a) \cdots (\forall x_{k-1} \in a)$*

- (i) $\langle \dot{a}, \dot{\in}_a \rangle \simeq_f \langle \dot{b}, \dot{\in}_b \rangle \rightarrow$

$$[\phi_{v_0, \dots, v_{k-1}}^{(a)}(x_0, \dots, x_{k-1}) \leftrightarrow \phi_{v_0, \dots, v_{k-1}}^{(b)}(f(x_0), \dots, f(x_{k-1}))];$$
- (ii) $\langle \dot{a}, \dot{\in}_a \rangle \prec \langle \dot{b}, \dot{\in}_b \rangle \rightarrow$

$$[\phi_{v_0, \dots, v_{k-1}}^{(a)}(x_0, \dots, x_{k-1}) \leftrightarrow \phi_{v_0, \dots, v_{k-1}}^{(b)}(x_0, \dots, x_{k-1})]. \quad \square$$

Remark 6.6.7 Since we are often concerned with relativizations $\phi^{(M)}$ of $L_{\in}$ -formulas ϕ to (possibly proper) classes M , it is natural to ask if for each class M it is possible to define a relation-class $\models_M$ such that for all ϕ ,

$$\text{ZF} \vdash (\forall x_0 \in M) \cdots (\forall x_{k-1} \in M) \\
(\phi^{(M)}(x_0, \dots, x_{k-1}) \leftrightarrow \models_M \dot{\phi}[x_0, \dots, x_{k-1}]).$$

Note that the method of definition of $\models_{\langle \dot{a}, \dot{\in}_a \rangle}$ above fails, since the corresponding w^M from the definition of satisfaction should be a function such that for each $x \in \text{Fm}(\dot{L}_{\in})$, $w^M(x) : {}^{\text{vb}}M \rightarrow \{0, 1\}$. But if M is a proper class, ${}^{\text{vb}}M$, the class of M -assignments, is also proper, so w^M is not a set operation and the recursion is not an instance of the Set Recursion Theorem. We shall prove later in this section that in fact there is no such class relation for $M = \dot{V}$.

Among theories, of particular interest are the formalized versions $\dot{P}$, $\dot{ZF}$ and $\dot{ZFC}$. In general, for a theory T given axiomatically as $\text{Th}(\text{Ax}_T)$, we would expect to define $\dot{T}$ as $\dot{\text{Th}}(\dot{\text{Ax}}_T)$ for an appropriate formalization of the set Ax_T of axioms for T . If $\text{Ax}_T = \{\phi_0, \dots, \phi_n\}$ is finite, the obvious choice is simply $\dot{\text{Ax}}_T := \{\dot{\phi}_0, \dots, \dot{\phi}_n\}$. However, the theories P , ZF and ZFC are not finitely axiomatized (or even finitely axiomatizable (6.5.37)) and the collection $\{\dot{\phi} : \phi \in \text{Ax}_T\}$ is not a class, since it is not defined by an $L_{\in}$ -formula. In such cases we must rather use a definition of the formal axiom set $\dot{\text{Ax}}_T$ parallel to the definition of Ax_T . The axioms of P are a finite set together with the L_{Ω} -induction schema, the set of L_{Ω} -sentences $\bar{I}_{\phi, x}$:

$$\forall y_0 \dots \forall y_{k-1} [(\phi_x(\dot{0}) \wedge \forall x [\phi \rightarrow \phi_x(\dot{S}x)]) \rightarrow \forall x \phi],$$

where ϕ has free variables among $x, y_0, \dots, y_{k-1}$. We can view this as a mapping $\text{Ind} : \text{Fm}_{L_{\Omega}} \times \text{Vb} \rightarrow \text{Fm}_{L_{\Omega}}$, with $\text{Ind}(\phi, x) := \bar{I}_{\phi, x}$. Then we can define in ZF the corresponding set operation $\text{Ind} : \text{Fm}(\dot{L}_{\Omega}) \times \text{vb} \rightarrow \text{Fm}(\dot{L}_{\Omega})$ and set

Definition 6.6.8

- (i) $\dot{A}x_P := \{ \dot{\text{Ind}}(u, v) : u \in \dot{\text{Fm}}(\dot{L}_\Omega) \wedge v \in \dot{\text{vb}} \}$
 $\dot{\cup} \{ \dot{\phi} : \phi \text{ is one of } (S1), (S2), (L1), (L2), (A1), (A2), (M1), (M2) \}.$
- (ii) $\dot{P} := \dot{\text{Th}}(\dot{A}x_P, \dot{L}_\Omega).$

It is easy to verify that for each $\phi \in P$, $\text{ZF} \vdash \dot{\phi} \in \dot{P}$. Furthermore, we have

Theorem 6.6.9 $\text{ZF} \vdash \text{ModelOf}(\dot{\Omega}, \dot{P})$, hence $\text{ZF} \vdash \dot{\text{Con}}(\dot{P})$.

Proof. We need to show that

$$\text{ZF} \vdash \forall u [u \in \dot{A}x_P \rightarrow \dot{\models}_{\dot{\Omega}} u].$$

For each such u of the form $\dot{\phi}$ with ϕ an axiom of P , by the results of [Section 6.2](#) we have $\text{ZF} \vdash \phi^{(\dot{\omega})}$, and therefore by the preceding theorem, $\dot{\models}_{\dot{\Omega}} \dot{\phi}$. We still need to prove in general that

$$\text{ZF} \vdash \forall u \in \dot{\text{Fm}}(\dot{L}_\Omega) (\forall v \in \dot{\text{vb}}) \dot{\models}_{\dot{\Omega}} \dot{\text{Ind}}(u, v).$$

This follows by modeling in ZF the proof that each $\bar{I}_{\phi, x}$ is true in Ω . If, for example, u has only the variable $\dot{v}$ free, then we apply $\dot{\omega}$ -induction to the set

$$y := \left\{ x \in \dot{\omega} : \dot{\models}_{\dot{\Omega}} u[x] \right\}.$$

Under the hypothesis of $\dot{\text{Ind}}(u, v)$, clearly $\dot{0} \in y \wedge \forall x (x \in y \rightarrow \dot{S}x \in y)$, and we conclude that $y \doteq \dot{\omega}$. $\square$

Of course, the $L_\in$ -sentence $\dot{\text{Con}}(\dot{P})$ is not identical with the L_Ω -sentence Con_P of [Section 5.3](#) or even with its translation $\text{Con}_P^{(\dot{\omega})}$ to an $L_\in$ -sentence given by the relative interpretation of P in ZF . However, by the general principle that standard mathematical arguments may be formalized in ZF , it is reasonable to expect that

$$\text{ZF} \vdash \dot{\text{Con}}(\dot{P}) \leftrightarrow \text{Con}_P^{(\dot{\omega})},$$

and thus to conclude that the proper extension $P \cup \{\text{Con}_P\}$ of P is also relatively interpretable in ZF .

To define the formal theories $\dot{\text{ZF}}$ and $\dot{\text{ZFC}}$, note that for each $L_\in$ -formula $\psi(x, y, z)$ there is a Replacement axiom asserting that for all u and x , $\text{F}_\psi^*(u, x)$ is a set, where

$$\text{F}_\psi^*(u, x) := \{ \text{F}_\psi(x, y) : y \in u \} \quad \text{and} \quad \text{F}_\psi(x, y) := \{ \{ z : \psi(x, y, z) \} \}.$$

Expanded to an $L_{\in}$ -formula this is

$$\text{Repl}(\psi, u, y, z) := \forall x \exists w \forall v [v \dot{\in} w \leftrightarrow \exists y (y \dot{\in} u \wedge \forall z [z \dot{\in} v \leftrightarrow \psi(x, y, z)])],$$

where for definiteness we take w and v to be the first available variables, and as usual, x is standing for the sequence of the remaining free variables of ψ . Thus Repl is a function $\text{Fm}_{L_{\in}} \times \text{Vb} \times \text{Vb} \times \text{Vb} \rightarrow \text{Fm}_{L_{\in}}$ whose image is the Replacement schema. Then it is easy to define a formalized version $\dot{\text{Repl}} : \dot{\text{Fm}}(\dot{L}_{\in}) \dot{\times} \text{vb} \dot{\times} \text{vb} \dot{\times} \text{vb} \dot{\rightarrow} \dot{\text{Fm}}(\dot{L}_{\in})$, and we set

Definition 6.6.10

- (i) $\dot{\text{Ax}}_{\text{ZF}} := \{ \dot{\text{Repl}}(t, u, y, z) : t \dot{\in} \dot{\text{Fm}}(\dot{L}_{\in}) \wedge u, y, z \dot{\in} \text{vb} \} \dot{\cup} \{ \dot{\phi} : \phi \text{ one of Extensionality, Union, Powerset, Infinity, Foundation} \}.$
- (ii) $\dot{\text{ZF}} := \dot{\text{Th}}(\dot{\text{Ax}}_{\text{ZF}}).$
- (iii) $\dot{\text{Ax}}_{\text{ZFC}} := \dot{\text{Ax}}_{\text{ZF}} \dot{\cup} \{ \dot{\text{AC}} \}.$
- (iv) $\dot{\text{ZFC}} := \dot{\text{Th}}(\dot{\text{Ax}}_{\text{ZFC}}).$

It is again easy to verify that for each $\phi \in \text{Ax}_{\text{ZF}}$, $\text{ZF} \vdash \dot{\phi} \dot{\in} \dot{\text{Ax}}_{\text{ZF}}$. However, the situation regarding models for these theories is very different from that for P ; not only is there no likely candidate for a specific set $\mathfrak{m}$ such that $\text{ZF} \vdash \text{ModelOf}(\langle \mathfrak{m}, \dot{\in}_{\mathfrak{m}} \rangle, \dot{\text{Ax}}_{\text{ZF}})$, it follows from the Second Incompleteness Theorem for ZF below that the sentence $\exists \mathfrak{a} \text{ModelOf}(\mathfrak{a}, \dot{\text{Ax}}_{\text{ZF}})$ is not a theorem of ZF (unless ZF is inconsistent). In [Section 5.5](#) we sketched a proof of this using the interpretability of P in ZF , established in [Section 6.2](#). Here we present a different proof due to Thomas Jech which is considerably simplified by the greater expressiveness of the language of set theory, in particular the fact that in ZF the consistency of a theory T may be expressed in the natural form ‘ T has a model’. We begin with a natural set-theoretic version of Lemma 4.6.27.

Self-reference Lemma for ZF 6.6.11 *There exists a finitely axiomatizable theory $T_0 \subseteq \text{ZF}$ such that for any $L_{\in}$ -formula $\theta(y)$, there exists an $L_{\in}$ -sentence η such that*

$$T_0 \vdash \eta \leftrightarrow \theta(\dot{\eta}).$$

Proof. Since substitution is defined recursively on the structure of formulas, there is a set operation Sub such that

$$\text{ZF} \vdash (\forall x \dot{\in} \dot{\text{Fm}}(L_{\in})) \forall z [\text{Sub}(x, z) \dot{\in} \dot{\text{Fm}}(\dot{L}_{\in})],$$

and for any $L_{\in}$ -formula ϕ and any set $\mathbf{b}$,

$$\text{ZF} \vdash \text{Sub}(\dot{\phi}, \mathbf{b}) \doteq \overbrace{\phi(\mathbf{b})}^{\cdot}.$$

Let T_0 be a finitely axiomatizable subtheory of ZF sufficient to prove these statements; note that although the second one refers to all formulas, it is really only the inductive clauses in the definition of formula that need to be proved. Given $\theta(y)$, let $\chi(x)$ be the formula $\theta(\text{Sub}(x, x))$ and η the sentence

$\chi(\dot{\chi})$ — that is, $\theta(\text{Sub}(\dot{\chi}, \dot{\chi}))$. Since $T_0 \vdash \text{Sub}(\dot{\chi}, \dot{\chi}) \doteq \overbrace{\chi(\dot{\chi})}^{\cdot} \doteq \dot{\eta}$, also $T_0 \vdash \eta \leftrightarrow \theta(\dot{\eta})$. $\square$

One simple consequence verifies the claim of Remark 6.6.7 that satisfaction and truth cannot generally be defined for proper classes — compare 4.5.5.

Tarski's Theorem for ZF 6.6.12 *There exists no class Tr such that for all $L_{\in}$ -sentences ϕ ,*

$$\text{ZF} \vdash \phi \leftrightarrow \dot{\phi} \in \text{Tr}.$$

Proof. If $\text{Tr} \doteq \{ \{ y : \theta(y) \} \}$, apply the Self-reference Lemma to the formula $\neg \theta(y)$. $\square$

Definition 6.6.13 For any theory T and any sentence ϕ , $T + \phi$ denotes the theory $\text{Th}(T \cup \{ \phi \})$.

Proposition 6.6.14 *There exists a finitely axiomatizable theory $T_1 \subseteq \text{ZF}$ such that for any finitely axiomatizable $L_{\in}$ -theories $T \supseteq T_1$ and U ,*

(i) $T + \text{Con}(\dot{U})$ is consistent $\implies U$ is consistent;

(ii) $T \vdash \text{Con}(\overbrace{T + \text{Con}(\dot{U})}^{\cdot}) \rightarrow \text{Con}(\dot{U})$;

(iii) for any $L_{\in}$ -sentences ϕ and ψ ,

$$(a) \quad T \vdash \text{Con}(\dot{U}) \rightarrow \left[\text{Con}(\overbrace{U + \phi}^{\cdot}) \vee \text{Con}(\overbrace{U + \neg \phi}^{\cdot}) \right];$$

$$(b) \quad T \vdash \phi \rightarrow \psi \implies T \vdash \text{Con}(\overbrace{U + \phi}^{\cdot}) \rightarrow \text{Con}(\overbrace{U + \psi}^{\cdot}).$$

Proof. We proceed first with $T_1 = \text{ZF}$; at the end we may observe that we have used only finitely many theorems of ZF and thus may replace ZF by a finitely axiomatizable subtheory T_1 . For all of the parts of the theorem, we need the following technical results (1) and (2). Let $\mathfrak{A} = \langle A, E \rangle$ be a model of ZF (as first discussed in Section 6.1). Among the members of A are objects $\mathbf{b}$ such that

$$\models_{\mathfrak{A}} \mathbf{b} \text{ is an } \dot{L}_{\in}\text{-structure},$$

and therefore objects b and f such that

$$\models_{\mathfrak{A}} \mathfrak{b} \doteq \langle \dot{b}, f \rangle \wedge f \subseteq b \dot{\times} b.$$

Of course, technically we should write these

$$\models_{\mathfrak{A}} v_0 \text{ is an } \dot{L}_{\in}\text{-structure}[\mathfrak{b}] \quad \text{and} \quad \models_{\mathfrak{A}} v_0 \doteq \langle \dot{v}_1, v_2 \rangle \wedge v_2 \subseteq v_1 \dot{\times} v_1[\mathfrak{b}, b, f],$$

but we adopt the first way as clearer. Still more technically, instead of $f \subseteq b \dot{\times} b$, from the definition of $\dot{\text{structure}}$, we should write

$$\text{Fn}(f) \wedge \text{Dom}(f) \doteq \{ \dot{\epsilon} \} \wedge f(\dot{\epsilon}) \subseteq b \dot{\times} b,$$

but again we adopt the simpler expression. Note that if $\mathfrak{A}$ is a transitive $\in$ -structure (6.1.3), then in fact $\mathfrak{b} = \langle b, f \rangle$, b is a set, $f \subseteq b \times b$ and $\mathfrak{b}$ is actually an $L_{\in}$ -structure. In the general case this may not be true, but still there is a natural $L_{\in}$ -structure $\mathfrak{b}^* = \langle b^*, f^* \rangle$ associated with $\mathfrak{b}$, where

$$\begin{aligned} b^* &:= \{ u \in A : \langle u, b \rangle \in E \} \\ &= \left\{ u \in A : \models_{\mathfrak{A}} u \dot{\in} b \right\}, \\ f^* &:= \left\{ \langle u, v \rangle \in b^* \times b^* : \models_{\mathfrak{A}} \langle \dot{u}, v \rangle \dot{\in} f \right\}. \end{aligned}$$

Now we claim that for any such $\mathfrak{b}$ and all $L_{\in}$ -sentences ϕ ,

$$(1) \quad \models_{\mathfrak{b}^*} \phi \iff \models_{\mathfrak{A}} \left(\models_{\mathfrak{b}} \dot{\phi} \right),$$

and more generally, for all $L_{\in}$ -formulas ϕ and any $\alpha \in A$ such that $\models_{\mathfrak{A}} \alpha$ is a $\mathfrak{b}$ -assignment,

$$\models_{\mathfrak{b}^*} \phi[\alpha^*] \iff \models_{\mathfrak{A}} \left(\models_{\mathfrak{b}} \dot{\phi}[\alpha] \right),$$

where α^* is the $\mathfrak{b}^*$ -assignment such that for each $n \in \omega$, $\alpha^*(v_n) := u$ such that $\models_{\mathfrak{A}} \alpha(\dot{v}_n) \doteq u$. This is proved by a straightforward $L_{\in}$ -formula induction. One of the basis steps is

$$\begin{aligned} \models_{\mathfrak{b}^*} (v_0 \dot{\in} v_1)[\alpha^*] &\iff \langle \alpha^*(v_0), \alpha^*(v_1) \rangle \in f^* \\ &\iff \models_{\mathfrak{A}} \langle \dot{\alpha}(\dot{v}_0), \alpha(\dot{v}_1) \rangle \dot{\in} f \\ &\iff \models_{\mathfrak{A}} \left(\models_{\mathfrak{b}} \overbrace{v_0 \dot{\in} v_1}^{\dot{\phi}} [\alpha] \right), \end{aligned}$$

and the induction step for $\vee$ is

$$\begin{aligned}
 \models_{\mathfrak{b}^*} (\psi \vee \theta)[\alpha^*] &\iff \models_{\mathfrak{b}^*} \psi[\alpha^*] \text{ or } \models_{\mathfrak{b}^*} \theta[\alpha^*] \\
 &\iff \models_{\mathfrak{A}} (\models_{\mathfrak{b}} \dot{\psi}[\alpha] \vee \models_{\mathfrak{b}} \dot{\theta}[\alpha]) \\
 &\iff \models_{\mathfrak{A}} (\models_{\mathfrak{b}} \overbrace{\psi \vee \theta}^{\dot{}}[\alpha]).
 \end{aligned}$$

Here we are using that $\mathfrak{A}$ is a model of (some finite part of) **ZF** to guarantee that the (characteristic function of the) satisfaction relation is a set operation with the usual properties. Finally, the $\exists$ -step is

$$\begin{aligned}
 \models_{\mathfrak{b}^*} \exists x \psi[\alpha^*] &\iff \text{for some } u \in \mathfrak{b}^*, \models_{\mathfrak{b}^*} \psi[\alpha_{x \mapsto u}^*] \\
 &\iff \text{for some } u \in A, \models_{\mathfrak{A}} u \in b \text{ and } \models_{\mathfrak{b}^*} \psi[(\alpha_{x \mapsto u})^*] \\
 &\iff \models_{\mathfrak{A}} (\exists u \in b) \models_{\mathfrak{b}} \dot{\psi}[\alpha_{x \mapsto u}] \\
 &\iff \models_{\mathfrak{A}} (\models_{\mathfrak{b}} \overbrace{\exists x \psi}^{\dot{}}[\alpha]).
 \end{aligned}$$

From (1) follows immediately that for any finitely axiomatizable $L_{\in}$ -theory U ,

$$(2) \quad \mathfrak{b}^* \text{ is a model of } U \iff \models_{\mathfrak{A}} \text{ModelOf}(\mathfrak{b}, \dot{U}).$$

Note that the implication $(\implies)$ requires the finite axiomatizability of U , since (1) applies only to standard sentences ϕ . Now (i) is immediate, since if $\mathfrak{A}$ is a model of $T + \text{Con}(\dot{U})$, then $\mathfrak{A}$ is a model of **ZF** and $\models_{\mathfrak{A}} \exists \mathfrak{b} [\text{ModelOf}(\mathfrak{b}, \dot{U})]$, so for some $\mathfrak{b} \in A$, by (2), $\mathfrak{b}^*$ is a model of U and thus U is consistent.

For (ii), suppose $\mathfrak{A}$ is a model of T such that $\models_{\mathfrak{A}} \overbrace{\text{Con}(T + \text{Con}(\dot{U}))}^{\dot{\phantom{\text{Con}(T + \text{Con}(\dot{U}))}}}$ — that is,

$$\models_{\mathfrak{A}} \exists \mathfrak{b} [\text{ModelOf}(\mathfrak{b}, \dot{T}) \wedge \models_{\mathfrak{b}} \text{Con}(\dot{U})].$$

We aim to show that $\models_{\mathfrak{A}} \text{Con}(\dot{U})$. For some $\mathfrak{b} \in A$, by (2) $\mathfrak{b}^*$ is a model of T , hence of **ZF**, and $\models_{\mathfrak{b}^*} \exists \mathfrak{c} [\text{ModelOf}(\mathfrak{c}, \dot{U})]$. Hence, again by (2), for some $\mathfrak{c} \in \mathfrak{b}^*$, $\mathfrak{c}^*$ is a model of $\dot{U}$. We claim that there exists $\mathfrak{d} \in A$ such that $\mathfrak{c}^* = \mathfrak{d}^*$. From this, it follows immediately by (2) that $\models_{\mathfrak{A}} \exists \mathfrak{d} [\text{ModelOf}(\mathfrak{d}, \dot{U})]$, hence $\models_{\mathfrak{A}} \text{Con}(\dot{U})$ as desired.

To define $\mathfrak{d}$, fix $c, g \in b^*$ such that $\mathfrak{c}^* = \langle c^*, g^* \rangle$. Then

$$\begin{aligned} c^* &= \{ u \in b^* : \langle u, c \rangle \in f^* \} \\ &= \left\{ u \in b^* : \models_{\mathfrak{A}} \langle u, c \rangle \dot{\in} f \right\} \\ &= \left\{ u \in A : \models_{\mathfrak{A}} u \dot{\in} b \wedge \langle u, c \rangle \dot{\in} f \right\} \\ &= \left\{ u \in A : \models_{\mathfrak{A}} u \dot{\in} d \right\} = d^*, \end{aligned}$$

where d is the unique element of A such that

$$\models_{\mathfrak{A}} d \dot{=} \left\{ x : x \dot{\in} b \wedge \langle x, c \rangle \dot{\in} f \right\}.$$

Similarly,

$$\begin{aligned} g^* &= \left\{ \langle u, v \rangle \in c^* \times c^* : \models_{\mathfrak{b}^*} \langle u, v \rangle \dot{\in} g \right\} \\ &= \left\{ \langle u, v \rangle \in c^* \times c^* : \models_{\mathfrak{A}} \langle \langle u, v \rangle^{\mathfrak{b}^*}, g \rangle \dot{\in} f \right\} \\ &= \left\{ \langle u, v \rangle \in A \times A : \langle u, v \rangle \dot{\in} h \right\} = h^*, \end{aligned}$$

where $\langle u, v \rangle^{\mathfrak{b}^*}$ is the unique element of b^* such that

$$\begin{aligned} \models_{\mathfrak{A}} \forall z [z \dot{\in} \langle u, v \rangle^{\mathfrak{b}^*} \leftrightarrow \forall x (\langle x, z \rangle \dot{\in} f \leftrightarrow x \dot{=} u) \vee \\ \forall x (\langle x, z \rangle \dot{\in} f \leftrightarrow x \dot{=} u \vee x \dot{=} v)], \end{aligned}$$

and h is the unique element of A such that

$$\models_{\mathfrak{A}} h \dot{=} \left\{ \langle u, v \rangle : u \dot{\in} d \wedge v \dot{\in} d \wedge \exists w [w \dot{=} \langle u, v \rangle^{\mathfrak{b}^*} \wedge \langle w, g \rangle \dot{\in} f] \right\}.$$

Thus for $\mathfrak{d}$ the unique element of A such that $\models_{\mathfrak{A}} \mathfrak{d} \dot{=} \langle d, h \rangle$, $\mathfrak{c}^* = \langle d^*, h^* \rangle = \mathfrak{d}^*$.

For (iii)(a), suppose that $\mathfrak{A}$ is a model of $T + \dot{\text{Con}}(\dot{U})$, so

$$\models_{\mathfrak{A}} \exists \mathfrak{b} [\text{ModelOf}(\mathfrak{b}, \dot{U})],$$

and hence there is $\mathfrak{b} \in A$ such that, by (2), $\mathfrak{b}^*$ is a model of U . Then either $\models_{\mathfrak{b}^*} \phi$ or $\models_{\mathfrak{b}^*} \neg \phi$, so $\mathfrak{b}^*$ is a model of either $U + \phi$ or $U + \neg \phi$ and hence by

(2), $\models_{\mathfrak{A}} \text{ModelOf}(\mathfrak{b}, \overbrace{U + \phi})$ or $\models_{\mathfrak{A}} \text{ModelOf}(\mathfrak{b}, \overbrace{U + \neg \phi})$ so $\models_{\mathfrak{A}} \dot{\text{Con}}(\overbrace{U + \phi})$

or $\models_{\mathfrak{A}} \dot{\text{Con}}(\overbrace{U + \neg \phi})$. (iii)(b) is proved similarly. $\square$

Second Incompleteness Theorem for ZF 6.6.15 *For any consistent recursively axiomatizable $L_{\in}$ -theory $T \supseteq \text{ZF}$, $T \not\vdash \text{Con}(\dot{T})$.*

Proof. Towards a contradiction, fix a recursively axiomatizable theory $T' \supseteq \text{ZF}$ such that T' is consistent and $T' \vdash \text{Con}(\dot{T}')$. Then there exists a finitely axiomatizable subtheory T of T' such that $T_0 \cup T_1 \subseteq T$ and

(3) T is consistent;

(4) $T \vdash \text{Con}(\dot{T})$.

By the Self-reference Lemma, fix an $L_{\in}$ -sentence η such that

(5) $T \vdash \eta \leftrightarrow \text{Con}(\overbrace{T + \neg \eta}^{\dot{}})$.

Now by (4) and (iii)(a) (of the preceding proposition),

(6) $T \vdash \text{Con}(\overbrace{T + \eta}^{\dot{}}) \vee \text{Con}(\overbrace{T + \neg \eta}^{\dot{}})$.

By (5)($\rightarrow$) and (iii)(b),

(7) $T \vdash \text{Con}(\overbrace{T + \eta}^{\dot{}}) \rightarrow \text{Con}(\overbrace{T + \text{Con}(\overbrace{T + \neg \eta}^{\dot{}})}^{\dot{}})$,

so by (ii)

(8) $T \vdash \text{Con}(\overbrace{T + \eta}^{\dot{}}) \rightarrow \text{Con}(\overbrace{T + \neg \eta}^{\dot{}})$,

and hence by (6),

(9) $T \vdash \text{Con}(\overbrace{T + \neg \eta}^{\dot{}})$.

But this has the following two contradictory consequences: by (5)($\leftarrow$),

(10) $T \vdash \eta$,

and by (3) and (i),

(11) $T + \neg \eta$ is consistent. $\square$

The Second Incompleteness Theorem puts some practical limits on what can be a theorem of **ZF** or **ZFC** (if, as we believe, these theories are consistent). It follows that there are similar limits to what can be proved by informal mathematical methods which are formalizable in these theories — that is, essentially all of classical mathematics. In particular, we cannot expect to prove ‘**ZF** is consistent’ without invoking methods which go beyond those formalizable in **ZF**. Hence, we cannot prove for any particular $L_{\in}$ -sentence ϕ

ϕ is independent of **ZF**,

since this implies that **ZF** is consistent. This explains why the claims made at the beginning of [Section 6.5](#) to be established in the following three sections are all about *relative* consistency.

In the discussion at the end of [Section 6.5](#) and the exercises following, we considered the implication

$$\text{ZF} - \theta \text{ is consistent} \implies \text{ZF} - \theta \not\vdash \theta$$

for several θ among the axioms of ZF. A parallel question is whether

$$\text{ZF} - \theta \text{ is consistent} \implies \text{ZF} - \theta \not\vdash \neg \theta,$$

or equivalently

$$(12) \quad \text{ZF} - \theta \text{ is consistent} \implies \text{ZF} \text{ is consistent.}$$

It is a consequence of the Second Incompleteness Theorem that for the θ of Exercises 6.5.38–6.5.40 (Infinity, Powerset and Replacement), we cannot expect to find a proof of (12) in standard informal mathematics. First, any such proof could be formalized in ZF so that

$$(13) \quad \text{ZF} \vdash \text{Con}(\text{ZF} - \dot{\theta}) \rightarrow \text{Con}(\text{ZF}).$$

Second, in each of the cited exercises, the class model $\mathbf{m}$ exhibited is actually a set and formalizing the argument that

$$\text{for all axioms } \phi \text{ of } \text{ZF} - \theta, \quad \text{ZF} \vdash \phi^{(\mathbf{m})}$$

gives

$$\text{ZF} \vdash \forall u [u \text{ is an axiom of } \text{ZF} - \dot{\theta} \rightarrow \models_{\mathbf{m}} u]$$

and hence $\text{ZF} \vdash \text{Con}(\text{ZF} - \dot{\theta})$. But then we could conclude from (13) that $\text{ZF} \vdash \text{Con}(\text{ZF})$, contrary to the Second Incompleteness Theorem.

We conclude this section with two results on the existence of ‘standard’ — that is transitive models for finite parts of ZF. These will play a major role in all of the following sections. The first result tells us that a set which satisfies Extensionality is isomorphic to a transitive set.

Definition 6.6.16 A set b is *extensional* iff $\text{Extensionality}^{(b)}$ — that is, for all $x, y \in b$, $x \dot{\cap} b \doteq y \dot{\cap} b \rightarrow x \doteq y$.

Mostowski Collapsing Theorem 6.6.17 For any extensional set b and any transitive $w \subseteq b$,

$$\exists a \exists f [w \subseteq a \wedge a \text{ is transitive} \wedge \langle \dot{b}, \dot{\in}_b \rangle \simeq_f \langle \dot{a}, \dot{\in}_a \rangle \wedge (\forall x \in w) f(x) \doteq x].$$

Proof. Given b and w as in the hypothesis, let F be the unique set operation which (by set recursion) satisfies

$$F(x) \doteq \begin{cases} \dot{\{F(z) : z \in x \dot{\cap} b\}}, & \text{if } x \in b; \\ \dot{\emptyset}, & \text{otherwise;} \end{cases}$$

and set $f := F \dot{\restriction} b$ and $a := \dot{\text{Im}}(f)$. To establish that f is injective, we show by set induction that if

$$B := \{ \{ x : x \in b \rightarrow (\forall y \in b) [f(x) \dot{=} f(y) \rightarrow x \dot{=} y] \} \},$$

then $B \dot{=} \dot{V}$. Assume as induction hypothesis that $x \subseteq B$ and that $x, y \in b$ and $f(x) \dot{=} f(y)$ — that is,

$$\dot{\{f(z) : z \in x \dot{\cap} b\}} \dot{=} \dot{\{f(z) : z \in y \dot{\cap} b\}}.$$

From this we conclude that $x \dot{\cap} b \dot{=} y \dot{\cap} b$ as follows. For any $z \in x \dot{\cap} b$, there exists $z' \in y \dot{\cap} b$ such that $f(z) \dot{=} f(z')$. By the induction hypothesis $z \in B$, we have $z \dot{=} z'$, so $z \in y \dot{\cap} b$. Thus $x \dot{\cap} b \subseteq y \dot{\cap} b$, and the converse inclusion is established similarly. Now by the extensionality of b , $x \dot{=} y$; thus $x \in B$.

Now the definition gives immediately that for $x, y \in b$, $x \dot{=} y \rightarrow f(x) \dot{=} f(y)$, and the converse implication follows from the injectivity just established. Thus f is an isomorphism: $\langle b, \dot{\in}_b \rangle \simeq_f \langle a, \dot{\in}_a \rangle$. Furthermore,

$$\begin{aligned} u \in a &\rightarrow (\exists x \in b) u \dot{=} f(x) \rightarrow (\exists x \in b) u \dot{=} \dot{\{f(z) : z \in x \dot{\cap} b\}} \\ &\rightarrow u \subseteq \dot{\text{Im}}(f) \dot{=} a, \end{aligned}$$

so a is transitive. Finally, by a straightforward set induction, $x \in w \rightarrow f(x) \dot{=} x$, so also $w \subseteq \dot{\text{Im}}(f) \dot{=} a$. $\square$

Transitive Model Theorem 6.6.18 (ZFC) For any $L_{\in}$ -formula $\phi(x)$,

$$\forall w \left[w \text{ is infinite and transitive} \rightarrow \exists a \left(w \subseteq a \wedge \right. \right. \\ \left. \left. \dot{|} a \dot{=} \dot{|} w \wedge a \text{ is transitive} \wedge (\forall x \in w) [\phi(x) \leftrightarrow \phi^{(a)}(x)] \right) \right].$$

In particular, for any sentence ϕ , if $\text{ZFC} \vdash \phi$, then

$$\text{ZFC} \vdash (\forall \tau \dot{\geq} \dot{\omega}) \exists a [\tau \subseteq a \wedge \dot{|} a \dot{=} \dot{|} \tau \wedge a \text{ is transitive} \wedge \phi^{(a)}].$$

Proof. Arguing in ZFC, given $\phi(x)$ and a transitive set w , by the Reflection Theorem 6.5.20 there is an ordinal ρ such that

$$w \subseteq \dot{V}_\rho \wedge (\forall x \in \dot{V}_\rho) [\phi(x) \leftrightarrow \phi^{\dot{V}_\rho}(x)].$$

Then, by the Downward Löwenheim-Skolem Theorem 6.6.3, there exists b such that

$$w \subseteq b \subseteq \dot{V}_\rho \wedge |b| = |w| \wedge \langle \dot{b}, \dot{\in}_b \rangle \prec \langle \dot{V}_\rho, \dot{\in}_{\dot{V}_\rho} \rangle,$$

so in particular,

$$(\forall x \in b) [\phi^{\dot{V}_\rho}(x) \leftrightarrow \phi^{(b)}(x)].$$

Further, since $\dot{V}_\rho$ is transitive, Extensionality $^{\dot{V}_\rho}$, so also Extensionality $^{(b)}$ and b is extensional. Hence by the Mostowski Collapsing Theorem, there exist a and f such that

$$w \subseteq a \wedge a \text{ is transitive} \wedge \langle \dot{b}, \dot{\in}_b \rangle \simeq_f \langle \dot{a}, \dot{\in}_a \rangle \wedge (\forall x \in w) f(x) = x.$$

In particular,

$$(\forall x \in b) [\phi^{(b)}(x) \leftrightarrow \phi^{(a)}(f(x))] \quad \text{so} \quad (\forall x \in w) [\phi^{(b)}(x) \leftrightarrow \phi^{(a)}(x)],$$

and by combining the equivalences we conclude

$$(\forall x \in w) [\phi(x) \leftrightarrow \phi^{(a)}(x)]$$

as desired. $\square$

Exercises

Exercise 6.6.19 The Second Incompleteness Theorem for ZF establishes that if ZF is consistent, then $\text{ZF} \not\vdash \exists a \text{ ModelOf}(a, \dot{\text{ZF}})$.

- (i) Give a much simpler argument to prove the apparently weaker statement: if ZF is consistent, then $\text{ZF} \not\vdash \exists a \text{ ModelOf}(\langle \dot{a}, \dot{\in}_a \rangle, \dot{\text{ZF}})$.
- (ii) Verify informally that this really is weaker by showing that if ZF is consistent, then there exists a consistent theory $T \supseteq \text{ZF}$ such that T has no transitive $\in$ -model.
- (iii) Verify formally that this is weaker by showing that if ZF is consistent, then

$$\text{ZF} \not\vdash \text{Con}(\dot{\text{ZF}}) \rightarrow \exists a \text{ ModelOf}(\langle \dot{a}, \dot{\in}_a \rangle, \dot{\text{ZF}}).$$

Hint: The formulas $\text{ModelOf}(a, \dot{\text{ZF}})$ and $\exists a \text{ ModelOf}(a, \dot{\text{ZF}})$ are Δ^{ZF} . The first of these follows by analysis of the definitions, the second by the Countable Completeness Theorem 6.6.4.

6.7 The constructible universe

In this section we complete the first part of the program described at the beginning of [Section 6.5](#) by defining a class L such that

- (1) L is a universe, hence a class model of ZF ;
- (2) $ZF \vdash AC^{(L)} \wedge GCH^{(L)}$.

It follows that

$$ZF \text{ is consistent} \implies ZF \cup \{AC, GCH\} \text{ is consistent.}$$

Step (2) will be accomplished by establishing

- (3) L is absolute for any universe, in particular for L itself;
- (4) $ZF \vdash (\dot{V} \doteq L)^{(L)}$;
- (5) $ZF \cup \{\dot{V} \doteq L\} \vdash AC \wedge GCH$.

L will be the universe generated by a weak power operation (Definition 6.5.14) $\dot{\wp}_{\text{def}}$ which (in a natural sense) is as small as possible. Condition (i)(b) of 6.5.14 requires that for each (unary) class operation $F(x) = \{\{z : \phi_{v_0, v_1}(x, z)\}\}$ and each $x \in u$, to be a weak power operation $\dot{\wp}_{\text{def}}(u)$ must contain the set

$$F^u(x) \doteq \{z \in u : \phi^{(u)}(z, x)\} \doteq \left\{z \in u : \models_{\langle u, \dot{\in}_u \rangle} \dot{\phi}[z, x]\right\},$$

where the last equality is by Theorem 6.6.5. In a natural sense we call such a set **definable** from the parameter x over u . We cannot write in L_∞ a definition of $\dot{\wp}_{\text{def}}$ which contains exactly these sets but must include also sets definable by possibly non-standard formulas — that is, a first candidate for $\dot{\wp}_{\text{def}}$ is

$$\dot{\wp}_{\text{def}}(u) := \left\{ \dot{\text{def}}(u, y, \alpha) : y \in \dot{\text{Fm}}_{L_\infty} \wedge \alpha \text{ is a } u\text{-assignment} \right\},$$

where

$$\dot{\text{def}}(u, y, \alpha) := \left\{ z \in u : \models_{\langle u, \dot{\in}_u \rangle} y[\alpha_{v_0 \mapsto z}] \right\}.$$

However, to simplify the proofs below it will be better to formulate the definition slightly differently using a formal version of the idea of Remark 2.6.10. That is, to each set u we shall associate a language $\dot{L}_\infty^u$ which has a constant symbol $\dot{z} := \langle \dot{0}, z \rangle$ for each element z of u . Then the sets definable over $\langle u, \dot{\in}_u \rangle$ are exactly those definable by $\dot{L}_\infty^u$ -formulas with a single free variable $\dot{v}_0$ and each $\dot{z}$ interpreted as z .

Definition 6.7.1 For any u , $\dot{L}_\epsilon^u := \langle \dot{\{ \ddot{x} \}}, \emptyset, \dot{\{ \dot{v} : v \in u \}}, \dot{\nu}_\epsilon \rangle$.

Thus

$$\begin{aligned} \dot{\text{Te}}(\dot{L}_\epsilon^u) &\doteq \text{At}\dot{\text{Te}}(\dot{L}_\epsilon^u) \doteq \dot{\{ \langle \dot{x} \rangle : x \in \text{vb} \}} \cup \dot{\{ \langle \dot{z} \rangle : z \in u \}}; \\ \text{At}\dot{\text{Fm}}(\dot{L}_\epsilon^u) &\doteq \dot{\{ \langle \dot{\ddot{x}}, x, y \rangle : x, y \in \dot{\text{Te}}(\dot{L}_\epsilon^u) \}} \cup \dot{\{ \langle \dot{\ddot{z}}, x, y \rangle : x, y \in \dot{\text{Te}}(\dot{L}_\epsilon^u) \}}; \\ \dot{\text{Fm}}(\dot{L}_\epsilon^u) &\doteq \bigcup_{t \in \dot{\omega}} \dot{\text{Fm}}_t(\dot{L}_\epsilon^u); \end{aligned}$$

where

$$\begin{aligned} \dot{\text{Fm}}_{\dot{0}}(\dot{L}_\epsilon^u) &\doteq \text{At}\dot{\text{Fm}}(\dot{L}_\epsilon^u); \\ \dot{\text{Fm}}_{t+\dot{o}1}(\dot{L}_\epsilon^u) &\doteq \dot{\text{Fm}}_t(\dot{L}_\epsilon^u) \cup \dot{\text{Bool}}(\dot{\text{Fm}}_t(\dot{L}_\epsilon^u)) \cup \exists(\dot{\text{Fm}}_t(\dot{L}_\epsilon^u)); \end{aligned}$$

and again we choose this description to facilitate details below (cf. Theorem 6.2.42). Now we set

Definition 6.7.2 $\dot{\wp}_{\text{def}}(u) := \dot{\{ \text{def}(u, y) : y \in \dot{\text{Fm}}(\dot{L}_\epsilon^u) \}}$, where

$$\text{def}(u, y) := \dot{\{ z \in u : \Vdash_{\langle u, \dot{\epsilon}_u \rangle} y(\dot{z}) \}},$$

and $y(\dot{z})$ is the result of substituting $\dot{z}$ for all free occurrences of $\dot{v}_0$ in y .

Proposition 6.7.3 $\dot{\wp}_{\text{def}}$ is a weak power operation.

Proof. Clearly $\dot{\wp}_{\text{def}}(u) \subseteq \dot{\wp}(u)$, and for any (unary) class operation $F(x) = \{\{ z : \phi_{v_0, v_1}(z, x) \}\}$, for $x \in u$,

$$F^u(x) \doteq \dot{\{ z \in u : \Vdash_{\langle u, \dot{\epsilon}_u \rangle} \dot{\phi}(\dot{z}, \dot{x}) \}} \doteq \text{def}(u, \dot{\phi}(\dot{x})) \in \dot{\wp}_{\text{def}}(u),$$

since $\dot{\phi}(\dot{x}) \in \dot{\text{Fm}}(\dot{L}_\epsilon^u)$. $\square$

Definition 6.7.4 $\mathbf{L} := \wp_{\text{def}}^\infty$ — that is, $\mathbf{L} := \bigcup \{\{ \mathbf{L}_\sigma : \sigma \in \dot{\text{Or}} \}\}$, where $\mathbf{L}_{\dot{0}} := \dot{\emptyset}$ and for all σ ,

$$\begin{aligned} \mathbf{L}_{\sigma+\dot{o}1} &:= \dot{\wp}_{\text{def}}(\mathbf{L}_\sigma) \quad \text{and} \\ \mathbf{L}_\sigma &:= \bigcup \dot{\{ \mathbf{L}_\tau : \tau \dot{<} \sigma \}} \quad \text{if } \sigma \text{ is a limit ordinal.} \end{aligned}$$

Corollary 6.7.5 L is a universe. For all ordinals σ and τ ,

- (i) L_σ is transitive and $L_\sigma \subseteq L_{\sigma+1}$; L is transitive;
- (ii) $\sigma \leq \tau \rightarrow L_\sigma \subseteq L_\tau$;
- (iii) $L_\sigma \in L_{\sigma+1}$;
- (iv) $L_\sigma \cap \dot{O}r \doteq \sigma$; $\dot{O}r \subseteq L$.
- (v) L is a transitive class model of ZF such that $\dot{O}r \subseteq L$.

Proof. Immediate from the Universe Theorem 6.5.22 and Proposition 6.5.16. $\square$

Theorem 6.7.6 For any transitive class model M of ZF,

- (i) $\dot{\phi}_{\text{def}}$ is a Δ^{ZF} set operation;
- (ii) the set operation $\sigma \mapsto L_\sigma$ is Δ^{ZF} , hence M -absolute and an M -set operation;
- (iii) $L^M \doteq \begin{cases} L, & \text{if } \dot{O}r \subseteq M; \\ L_\rho, & \text{if } \dot{O}r \cap M \doteq \rho; \end{cases}$
- (iv) L is L -absolute;
- (v) $\text{ZF} \vdash (\dot{V} \doteq L)^{(L)}$.

Proof. Part (i) follows from Proposition 6.5.34 and a careful analysis of the definitions of the syntactic and semantic notions for first-order languages at the beginning of Section 6.6. The key point that is easy to miss is that in defining satisfaction over a structure $\langle a, \dots \rangle$, the quantifier case uses a bounded quantifier $\exists x \in a$; compare the proof of Theorem 6.6.5.

Now (ii) follows by 6.5.34(ii). For (iii) we have

$$L^M \doteq \{ x \in M : \exists^M \sigma (x \in L_\sigma^M) \} \doteq \{ x : \exists^M \sigma (x \in L_\sigma) \},$$

where the second equality follows from (ii). Now (iv) is immediate from (ii) and the preceding corollary, and for (v), $(\dot{V} \doteq L)^{(L)}$ is equivalent to $\dot{V}^L \doteq L^L$, and by (iv) and Corollary 6.5.35, both sides are equal to L . $\square$

Definition 6.7.7 ZFL is the $L_\in$ -theory generated by $\text{ZF} \cup \{ \dot{V} \doteq L \}$.

Corollary 6.7.8 ZF is consistent $\implies$ ZFL is consistent.

Proof. By the preceding corollary and part (v) of the preceding theorem, ZFL is relatively interpretable in ZF, so the result follows from Corollary 2.6.47. $\square$

Corollary 6.7.9 *For any $L_{\in}$ -sentence ϕ , if $\text{ZFL} \vdash \phi$, then $\text{ZF} \vdash \phi^{(\mathbf{L})}$ — in other words, $\mathbf{L}$ is a class model of ZFL in ZF. $\square$*

Corollary 6.7.10 *For any Σ^{ZF} sentence ϕ , if $\text{ZFL} \vdash \phi$, then $\text{ZF} \vdash \phi$. In particular, this applies to any sentence $\psi^{(\dot{\omega})}$ which is the translation of an L_{Ω} -sentence ψ . $\square$*

The remainder of the proof as sketched above is to show that

$$\text{ZFL} \vdash \text{AC} \wedge \text{GCH}.$$

We consider first AC — we shall show that there is a set operation $\mathbf{C}$ such that

$$\text{ZFL} \vdash \forall y \left[y \neq \dot{\emptyset} \rightarrow \mathbf{C}(y) \in y \right],$$

that is, $\mathbf{C}$ is a **global choice** operation. Then in particular, for any x , $\mathbf{C} \upharpoonright x$ is a choice function for x , so AC holds. $\mathbf{C}$, in turn, arises from a relation class $<_{\mathbf{L}}$ such that

$$\text{ZFL} \vdash <_{\mathbf{L}} \text{ is a class well-ordering of } \dot{\mathbf{V}};$$

given $<_{\mathbf{L}}$ we may define simply

$$\mathbf{C}(y) := \begin{cases} \text{the } <_{\mathbf{L}}\text{-least element of } y, & \text{if } y \neq \dot{\emptyset}; \\ \dot{\emptyset}, & \text{otherwise.} \end{cases}$$

First, analogously to the rank operation introduced in 6.3.28, we have

Definition 6.7.11 $(\text{ZFL}) \text{rank}_{\mathbf{L}}(x) := \text{least } \sigma [x \in \mathbf{L}_{\sigma+1}]$.

It is immediate that $\mathbf{L}_{\sigma} \doteq \{x : \text{rank}_{\mathbf{L}}(x) < \sigma\}$ and $\text{rank}(x) \leq \text{rank}_{\mathbf{L}}(x)$. We shall define a set operation $\sigma \mapsto <_{\mathbf{L}_{\sigma}}$ such that

$$<_{\mathbf{L}_{\sigma}} \text{ is a well-ordering of } \mathbf{L}_{\sigma}.$$

The key fact is the following (recall that by the general properties of weak power operations (Lemma 6.5.15), if u is transitive, then $u \subseteq \dot{\wp}_{\text{def}}(u)$).

Theorem 6.7.12 *There is a Δ^{ZF} set operation $<_u \mapsto <_{\dot{\wp}_{\text{def}}(u)}$ such that for all transitive sets u , if $<_u$ is a well-ordering of u , then*

- (i) $<_{\dot{\wp}_{\text{def}}(u)}$ is a well-ordering of $\dot{\wp}_{\text{def}}(u)$;
- (ii) $<_{\dot{\wp}_{\text{def}}(u)}$ is an **end extension** of $<_u$ — that is, for all $x \in u$ and $y \in \dot{\wp}_{\text{def}}(u)$,
 - (a) if $y \in u$, then $x <_{\dot{\wp}_{\text{def}}(u)} y \leftrightarrow x <_u y$;
 - (b) if $y \notin u$, then $x <_{\dot{\wp}_{\text{def}}(u)} y$.

Proof. Let u and $<_u$ be fixed with $<_u$ a well-ordering of u . We define first a well-ordering $<_u^{\text{Te}}$ of $\text{Te}(\dot{L}_\epsilon^u)$ by

$$\begin{aligned} \langle \dot{x} \rangle <_u^{\text{Te}} \langle \dot{x}' \rangle \leftrightarrow & (x, x' \in \mathbf{vb} \wedge x <_\omega x') \vee (x \in \mathbf{vb} \wedge x' \notin \mathbf{vb}) \\ & \vee (\exists z \in u)(\exists z' \in u)[x \dot{=} z \wedge x' \dot{=} z' \wedge z <_u z'] \end{aligned}$$

(recall that $\mathbf{vb} \subseteq \dot{\omega}$). This is perhaps best understood in the following informal form:

$$\dots \langle \dot{v}_y \rangle \dots \langle \dot{v}_{y'} \rangle \dots \langle \dot{z} \rangle \dots \langle \dot{z}' \rangle \dots$$

where $y <_\omega y'$ and $z <_u z'$. We leave to an exercise the simple verification that this is a well-ordering and present other orderings in this form. $\text{AtFm}(\dot{L}_\epsilon^u)$ now has the well-ordering $<_u^{\text{AtFm}}$

$$\dots \langle \dot{\ddot{=}}, y, z \rangle \dots \langle \dot{\ddot{=}}, y', z' \rangle \dots \langle \dot{\ddot{=}}, y, z \rangle \dots \langle \dot{\ddot{=}}, y', z' \rangle \dots$$

where $(y <_u^{\text{Te}} y') \vee (y \dot{=} y' \wedge z <_u^{\text{Te}} z')$. Now we define a well-ordering of $\text{Fm}(\dot{L}_\epsilon^u)$ as follows. For any $x \in \text{Fm}(\dot{L}_\epsilon^u)$ let $\text{lev}(x)$ denote the least $t \in \omega$ such that $x \in \text{Fm}_t(\dot{L}_\epsilon^u)$, and for $x, x' \in \text{Fm}(\dot{L}_\epsilon^u)$,

$$x <_u^{\text{Fm}} x' \leftrightarrow [\text{lev}(x) <_\omega \text{lev}(x')] \vee (\exists t \in \omega)[\text{lev}(x) \dot{=} \text{lev}(x') \dot{=} t \wedge x <_u^{\text{Fm}, t} x'],$$

where the orderings $<_u^{\text{Fm}, t}$ of $\text{Fm}_t(\dot{L}_\epsilon^u)$ are defined recursively as follows. Given $<_u^{\text{Fm}, t}$, we order $\text{Fm}_{t+1}(\dot{L}_\epsilon^u)$ according to the following schema

$$\begin{aligned} \dots \langle \dot{\dot{=}}, y \rangle \dots \langle \dot{\dot{=}}, y' \rangle \dots \langle \dot{\dot{=}}, y, z \rangle \dots \langle \dot{\dot{=}}, y', z' \rangle \\ \dots \langle \dot{\dot{=}}, v, y \rangle \dots \langle \dot{\dot{=}}, v', y' \rangle \dots \end{aligned}$$

where in the first case $y <_u^{\text{Fm}, t} y'$, in the second $y <_u^{\text{Fm}, t} y'$ or $y \dot{=} y'$ and $z <_u^{\text{Fm}, t} z'$ and in the third $v <_\omega v'$ or $v \dot{=} v'$ and $y <_u^{\text{Fm}, t} y'$. Again it is straightforward to verify that $<_u^{\text{Fm}}$ is a well-ordering of $\text{Fm}(\dot{L}_\epsilon^u)$. Finally for $x, x' \in \dot{\phi}_{\text{def}}(u)$ we set

$$\begin{aligned} x <_{\dot{\phi}_{\text{def}}(u)} x' \leftrightarrow & [x \dot{=} u \wedge (x' \dot{=} u \rightarrow x <_u x')] \\ & \vee \left[x, x' \notin u \wedge \exists y \in \text{Fm}(\dot{L}_\epsilon^u) \left(x \dot{=} \text{def}(u, y) \right. \right. \\ & \left. \left. \wedge \forall y' \in \text{Fm}(\dot{L}_\epsilon^u) [x' \dot{=} \text{def}(u, y') \rightarrow y <_u^{\text{Fm}} y'] \right) \right]. \end{aligned}$$

That the operation $<_u \mapsto <_{\dot{\phi}_{\text{def}}(u)}$ is a Δ^{ZF} set operation follows as usual from the fact that all of the syntactic operations used in the definition are Δ^{ZF} together with an application of Proposition 6.5.33. $\square$

Definition 6.7.13

(i) $<_{\mathbf{L}_0}$ is the empty ordering;
for all σ ,

(ii) $<_{\mathbf{L}_{\sigma+\mathbf{o}1}} \doteqdot <_{\dot{\varphi}_{\text{def}}(\mathbf{L}_\sigma)}$;

(iii) if σ is a limit ordinal, $x <_{\mathbf{L}_\sigma} y \leftrightarrow (\exists \tau < \sigma) x <_{\mathbf{L}_\tau} y$;

(iv) $x <_{\mathbf{L}} y \leftrightarrow \exists \sigma [x <_{\mathbf{L}_\sigma} y]$.

Theorem 6.7.14 *For any transitive class model $\mathbf{M}$ of ZF,*

(i) *the set operation $\sigma \mapsto <_{\mathbf{L}_\sigma}$ is Δ^{ZF} , hence $\mathbf{M}$ -absolute and an $\mathbf{M}$ -set operation;*

(ii) $<_{\mathbf{L}}^{\mathbf{M}} \doteqdot \begin{cases} <_{\mathbf{L}}, & \text{if } \dot{\mathbf{O}}r \subseteq \mathbf{M}; \\ <_{\mathbf{L}_\rho}, & \text{if } \dot{\mathbf{O}}r \cap \mathbf{M} \doteqdot \rho; \end{cases}$

(iii) $\text{ZF} \vdash <_{\mathbf{L}}$ *is a class well-ordering of $\mathbf{L}$; hence $\text{ZFL} \vdash <_{\mathbf{L}}$ is a class well-ordering of $\mathbf{V}$.*

Proof. This is closely parallel to the proof of Theorem 6.7.6. The main new point for (iii) is that for any $x \subseteq \mathbf{L}$, there exists σ such that $x \subseteq \mathbf{L}_\sigma$ so x has a least element in the ordering $<_{\mathbf{L}_\sigma}$; the end extension property guarantees that this is also a least element in the ordering $<_{\mathbf{L}}$. $\square$

Theorem 6.7.15 $\text{ZFL} \vdash \text{AC}$; hence $\text{ZF} \vdash \text{AC}^{(\mathbf{L})}$.

Proof. As described above and using Corollary 6.7.9. $\square$

That $\text{ZFL} \vdash \text{GCH}$ will follow easily from two propositions.

Lemma 6.7.16 $(\text{ZFC}) \forall u (u \text{ is infinite} \rightarrow |\dot{\varphi}_{\text{def}}(u)| \doteqdot |u|)$.

Proof. Arguing in ZFC, let u be infinite. $\dot{\mathbf{T}}e(\dot{L}_\epsilon^u)$ is the disjoint union of a countable set and one of cardinality $|u|$, hence $|\dot{\mathbf{T}}e(\dot{L}_\epsilon^u)| \doteqdot |u|$. Similarly, $|\dot{\mathbf{A}}t\dot{\mathbf{F}}m(\dot{L}_\epsilon^u)| \doteqdot |u|$, and by a straightforward induction, for all $t \in \omega$, $|\dot{\mathbf{F}}m_t(\dot{L}_\epsilon^u)| \doteqdot |u|$, so also (by Corollary 6.4.37) $|\dot{\mathbf{F}}m(\dot{L}_\epsilon^u)| \doteqdot |u|$. Finally, the correspondence $y \mapsto \text{def}(u, y)$ is a surjection of $\dot{\mathbf{F}}m(\dot{L}_\epsilon^u)$ onto $\dot{\varphi}_{\text{def}}(u)$, so by Proposition 6.4.39, $|\dot{\varphi}_{\text{def}}(u)| \leq |u|$; the opposite inequality is obvious. $\square$

Proposition 6.7.17 $\text{ZFC} \vdash (\forall \sigma \geq \omega) |\mathbf{L}_\sigma| \doteqdot |\sigma|$.

Proof. We proceed by ordinal induction. For $\sigma \in \omega$, $\mathbf{L}_{\sigma+\mathbf{o}1} \doteqdot \dot{\varphi}_{\text{def}}(\mathbf{L}_\sigma) \doteqdot \varnothing(\mathbf{L}_\sigma)$, so by ω -induction, $\mathbf{L}_\sigma$ is finite. Hence $|\mathbf{L}_\omega| \doteqdot \omega \doteqdot |\omega|$. By the preceding lemma, for infinite σ ,

$$|\mathbf{L}_{\sigma+\mathbf{o}1}| \doteqdot |\dot{\varphi}_{\text{def}}(\mathbf{L}_\sigma)| \doteqdot |\mathbf{L}_\sigma| \doteqdot |\sigma| \doteqdot |\sigma + \mathbf{o}1|.$$

Finally, for limit σ , by the induction hypothesis, for each $\tau \dot{<} \sigma$, $|\dot{\mathbf{L}}_\tau| \dot{=} |\dot{\tau}| \dot{\leq} |\dot{\sigma}|$, so

$$|\dot{\mathbf{L}}_\sigma| \dot{=} \left| \dot{\bigcup} \{ \dot{\mathbf{L}}_\tau : \tau \dot{<} \sigma \} \right| \dot{=} |\dot{\sigma}|$$

again by Corollary 6.4.37. $\square$

The second proposition puts a limit on the relative size of $\text{rank}(x)$ and $\text{rank}_\mathbf{L}(x)$. For example, since $\dot{\omega} \dot{\subseteq} \dot{\mathbf{V}}_{\dot{\omega}}$, for every $x \dot{\subseteq} \dot{\omega}$, $x \in \wp(\dot{\mathbf{V}}_{\dot{\omega}}) \dot{=} \dot{\mathbf{V}}_{\dot{\omega}+\dot{o}1}$ — that is, $\text{rank}(x) \dot{\leq} \dot{\omega}$. But although also $x \dot{\subseteq} \dot{\mathbf{L}}_{\dot{\omega}}$, it is not necessarily true that $x \in \dot{\mathbf{L}}_{\dot{\omega}+\dot{o}1}$. The hypothesis $\dot{\mathbf{V}} \dot{=} \dot{\mathbf{L}}$ implies that for *some* ordinal σ , $x \in \dot{\mathbf{L}}_\sigma$, and the following proposition tells us that $\sigma < \aleph_1$ — that is, $\text{rank}_\mathbf{L}(x) < \aleph_1$.

Proposition 6.7.18 (ZFL) *For any infinite cardinal κ , $(\forall \tau \dot{<} \kappa^+) \wp(\dot{\mathbf{L}}_\tau) \dot{\subseteq} \dot{\mathbf{L}}_{\kappa^+}$; in particular, $\wp(\kappa) \dot{\subseteq} \dot{\mathbf{L}}_{\kappa^+}$.*

Proof. Arguing in ZFL, fix an infinite cardinal κ , $\tau \dot{<} \kappa^+$ and $x \dot{\subseteq} \dot{\mathbf{L}}_\tau$; we show that $x \in \dot{\mathbf{L}}_{\kappa^+}$. By Theorem 6.7.6 there exists a Δ_0 formula ψ such that

$$x \in \dot{\mathbf{L}}_\sigma \leftrightarrow \exists y \psi(x, y, \sigma),$$

and by the hypothesis $\dot{\mathbf{V}} \dot{=} \dot{\mathbf{L}}$ we have $\exists \sigma \exists y \psi(x, y, \sigma)$. Since $|\dot{\mathbf{L}}_\tau| \dot{=} \kappa$, by the Transitive Model Theorem 6.6.18 there exists a such that

$$\dot{\mathbf{L}}_\tau \dot{\cup} \{x\} \dot{\subseteq} a \wedge |a| \dot{=} \kappa \wedge a \text{ is transitive} \wedge (\exists \sigma \exists y \psi(x, y, \sigma))^{(a)}.$$

Hence $(\exists \sigma \dot{\in} a)(\exists y \dot{\in} a) \psi^{(a)}(x, y, \sigma)$, so because ψ is Δ_0 ,

$$(\exists \sigma \dot{\in} a) \exists y \psi(x, y, \sigma) \text{ and thus } (\exists \sigma \dot{\in} a) x \in \dot{\mathbf{L}}_\sigma.$$

But

$$\sigma \dot{\in} a \rightarrow \sigma \dot{\subseteq} a \rightarrow |\dot{\sigma}| \dot{\leq} \kappa \rightarrow \sigma \dot{<} \kappa^+,$$

so $(\exists \sigma \dot{<} \kappa^+) x \in \dot{\mathbf{L}}_\sigma$ and hence $x \in \dot{\bigcup} \{ \dot{\mathbf{L}}_\sigma : \sigma \dot{<} \kappa^+ \} \dot{=} \dot{\mathbf{L}}_{\kappa^+}$. $\square$

Theorem 6.7.19 $\text{ZFL} \vdash \text{GCH}$; hence, $\text{ZF} \vdash \text{GCH}^{(\mathbf{L})}$.

Proof. By the two preceding propositions, for each infinite cardinal κ ,

$$2^\kappa \dot{=} |\wp(\kappa)| \dot{\leq} |\dot{\mathbf{L}}_{\kappa^+}| \dot{=} \kappa^+. \quad \square$$

It is instructive to recall the discussion at the end of [Section 6.5](#), in particular

$$\text{CH}^{(\mathbf{L})} \leftrightarrow (\exists f \dot{\in} \dot{\mathbf{L}}) [\text{Inj}(f) \wedge f : \wp(\dot{\omega}) \dot{\cap} \dot{\mathbf{L}} \dot{\rightarrow} \aleph_1^{\mathbf{L}}].$$

In [Section 6.11](#) we discuss relatively natural set-theoretic assertions θ such that apparently $\text{ZFC} \cup \{\theta\}$ is consistent and

$$\text{ZFC} \cup \{\theta\} \vdash (\forall \sigma < \dot{\aleph}_1) \dot{\aleph}_\sigma^L < \dot{\aleph}_1,$$

in particular $\dot{\aleph}_1^L < \dot{\aleph}_1$; in this theory $\text{CH}^{(L)}$ is a statement very different from CH.

With this we have completed the proof that

$$\text{ZF is consistent} \implies \text{ZF} \cup \{\text{AC}, \text{GCH}\} \text{ is consistent,}$$

and we close this section with two ideas that will be used in [Section 6.11](#), a strengthening of the second clause of the Transitive Model Theorem 6.6.18 and a generalization of the notion of constructibility.

Strong Transitive Model Theorem 6.7.20 *For any L_ϵ -sentence ϕ , if $\text{ZFL} \vdash \phi$, then*

$$\text{ZF} \vdash (\forall \tau \geq \dot{\omega}) \exists a [\tau \subseteq a \wedge |a| \doteq |\tau| \wedge a \text{ is transitive} \wedge \phi^{(a)}].$$

Proof. Since by the preceding $\text{ZFC} \subseteq \text{ZFL}$, by the Transitive Model Theorem 6.6.18, if $\text{ZFL} \vdash \phi$, then

$$\text{ZFL} \vdash (\forall \tau \geq \dot{\omega}) \exists a [\tau \subseteq a \wedge |a| \doteq |\tau| \wedge a \text{ is transitive} \wedge \phi^{(a)}].$$

Then by Corollary 6.7.9,

$$\text{ZF} \vdash ((\forall \tau \geq \dot{\omega}) \exists a [\tau \subseteq a \wedge |a| \doteq |\tau| \wedge a \text{ is transitive} \wedge \phi^{(a)}])^{(L)}.$$

Since $\dot{\aleph}_1 \subseteq L$, $\text{ZF} \vdash (|a| \doteq |\tau|)^{(L)} \rightarrow |a| \doteq |\tau|$ and $\phi^{(a)}$ is a Δ_0 formula, the result follows. $\square$

The notion of constructibility can be modified in a number of ways; we mention only one here which will be of use in [Section 6.11](#). For each set a , we define the class $L[a]$ of sets **constructible from a** . Formally, we are defining a class operation $a \mapsto L[a]$. The construction is closely parallel to that of L , and we only sketch the definitions and results that we need. First, for each a , let $\dot{L}_\epsilon^{u;a}$ be a language like $\dot{L}_\epsilon^u$ with one additional unary relation symbol $\dot{a}$. The set of formulas $\text{Fm}(\dot{L}_\epsilon^{u;a})$ is defined in the obvious way as is the notion of satisfaction in the structure $\langle u, \dot{\in}_u \rangle$ with the additional clause that $\dot{a}$ is interpreted as $a \dot{\cap} u$:

$$\dot{\varphi}_{\text{def}}(u; a) := \{ \dot{\text{def}}(u, y; a) : y \in \text{Fm}(\dot{L}_\epsilon^{u;a}) \},$$

where

$$\text{def}(u, y; a) := \left\{ z \in u : \dot{\Vdash}_{\langle u, \dot{\in}_u, a \dot{\cap} u \rangle} y(\dot{z}) \right\}.$$

Then $\mathsf{L}[a] := \dot{\bigcup} \{ \mathsf{L}_\sigma[a] : \sigma \in \dot{\mathsf{Or}} \}$, where $\mathsf{L}_0[a] \doteq \dot{\emptyset}$, and for all σ ,

$$\mathsf{L}_{\sigma \dot{+} o \dot{+} i}[a] := \dot{\wp}_{\text{def}}(\mathsf{L}_\sigma[a]; a) \quad \text{and}$$

$$\mathsf{L}_\sigma[a] := \dot{\bigcup} \{ \mathsf{L}_\tau[a] : \tau \dot{<} \sigma \} \quad \text{if } \sigma \text{ is a limit ordinal.}$$

With essentially no changes in the proof, $\mathsf{L}[a]$ is a universe and hence a transitive class model of ZF such that $\dot{\mathsf{Or}} \dot{\subseteq} \mathsf{L}$. Furthermore, the set operation $(\sigma, a) \mapsto \mathsf{L}_\sigma[a]$ is Δ^{ZF} so M -absolute for any transitive class model M of ZF . Similarly, there is a Δ^{ZF} set operation $(\sigma, a) \mapsto <_{\mathsf{L}_\sigma[a]}$ such that if we set

$$x <_{\mathsf{L}[a]} y \leftrightarrow \exists \sigma (x, y \in \mathsf{L}_\sigma[a] \wedge x <_{\mathsf{L}_\sigma[a]} y),$$

then $\mathsf{ZF} \vdash <_{\mathsf{L}[a]}$ is a class well-ordering of $\mathsf{L}[a]$. In particular, $\mathsf{ZF} \vdash (\mathsf{AC})^{\mathsf{L}[a]}$.

We leave to the exercises proofs of the following facts.

Theorem 6.7.21 *For all A , let $\bar{a} := a \dot{\cap} \mathsf{L}[a]$;*

- (i) $\bar{a} \in \mathsf{L}[a]$ and $\mathsf{L}[a] \doteq \mathsf{L}[\bar{a}]$;
- (ii) $(\dot{\mathsf{V}} \doteq \mathsf{L}[a])^{(\mathsf{L}[a])}$;
- (iii) $\dot{\mathsf{V}} \doteq \mathsf{L}[a] \rightarrow \mathsf{AC}$;
- (iv) for any ordinal σ ,

$$\dot{\mathsf{V}} \doteq \mathsf{L}[a] \wedge \bar{a} \dot{\subseteq} \mathsf{L}_\sigma[a] \rightarrow (\forall \kappa \dot{\geq} |\sigma|) 2^\kappa \doteq \kappa^+.$$

Corollary 6.7.22

- (i) For any $a \dot{\subseteq} \mathsf{L}$, $a \in \mathsf{L}[a]$.
- (ii) $a \dot{\subseteq} \dot{\mathsf{V}}_\omega \wedge \dot{\mathsf{V}} \doteq \mathsf{L}[a] \rightarrow \mathsf{GCH}$.

Collateral Reading. The standard reference for constructibility, which goes very far beyond the material of this section, is [De84]. [Jec; Chapter 13] and [Kun; Chapter VI] are slightly more comprehensive than this section.

Exercises

Exercise 6.7.23 Give proofs for Theorem 6.7.21 and its corollary.

In the following series of exercises we develop another class model, HOD , the class of **hereditarily ordinal definable** sets. For each ordinal σ , $\dot{L}_\epsilon^\sigma$ is as in Definition 6.7.1 with constant symbols $\dot{\tau}$ for all $\tau < \sigma$. Set

$$\dot{\phi}_{\text{odef}}(\dot{V}_\sigma) := \left\{ \text{def}(\dot{V}_\sigma, y) : y \in \text{Fm}(\dot{L}_\epsilon^\sigma) \right\}.$$

Intuitively, $\dot{\phi}_{\text{odef}}(\dot{V}_\sigma)$ is the set of subsets of $\dot{V}_\sigma$ definable using ordinals as parameters. Then

$$\begin{aligned} \text{OD} &:= \bigcup \left\{ \dot{\phi}_{\text{odef}}(\dot{V}_\sigma) : \sigma \in \dot{\text{Or}} \right\}; \\ \text{HOD} &:= \left\{ x : \text{TC}^+(x) \subseteq \text{OD} \right\}. \end{aligned}$$

Note that OD is not (obviously) transitive and HOD is *not* a universe.

Exercise 6.7.24 Show in ZF that

- (i) $\dot{\text{Or}} \subseteq \text{HOD} \subseteq \text{OD}$;
- (ii) HOD is transitive;
- (iii) $\forall x [x \in \text{OD} \wedge x \subseteq \text{HOD} \rightarrow x \in \text{HOD}]$;
- (iv) for every class operation F ,

$$\forall \tau [\text{F}\tau \text{ is a set} \rightarrow \text{F}\tau \in \text{OD}];$$

and hence

$$\forall \tau [\text{F}\tau \text{ is a set} \rightarrow \text{F}\tau \cap \text{HOD} \in \text{HOD}].$$

Hint: For (iv), use the corollary to the Reflection Theorem 6.5.20 and the technique of the proof of Proposition 6.7.3.

Exercise 6.7.25

- (i) Show that there exists a class well-ordering $<_{\text{OD}}$ of OD .
- (ii) Conclude that there exists a bijective set operation $\text{I} : \dot{\text{Or}} \rightarrow \text{OD}$.

Hint: For (i) use the technique of the proof of Theorem 6.7.12 to construct a well-ordering of each $\text{Fm}(\dot{L}_\epsilon^\sigma)$ and set

$$x <_{\text{OD}} x' \leftrightarrow \exists \sigma (\exists y \in \text{Fm}(\dot{L}_\epsilon^\sigma))$$

$$[x \doteq \text{def}(\dot{V}_\sigma, y) \wedge \forall \sigma' (\forall y \in \text{Fm}(\dot{L}_\epsilon^{\sigma'})) [x' \doteq \text{def}(\dot{V}_{\sigma'}, y') \rightarrow \dots].$$

Exercise 6.7.26 Combine the results of the preceding two exercises to show that for any class operation F ,

$$\forall x \in \text{OD} [Fx \text{ is a set} \rightarrow Fx \dot{\cap} \text{HOD} \in \text{HOD}].$$

Conclude that HOD is rich (Definition 6.5.11) and hence a class model of ZF .

Exercise 6.7.27 Show that $\text{ZF} \vdash \text{AC}^{(\text{HOD})}$.

Hint: For $x \in \text{HOD}$, show that $r := <_{\text{OD}} \dot{\cap} (x \dot{\times} x)$ belongs to HOD and well-orders x . Then appeal to the absoluteness of well-ordering (Theorem 6.5.30).

Note that these results give an alternative proof that $\dot{V} \dot{=} \text{L} \rightarrow \text{AC}$, since $\text{L} \subseteq \text{HOD} \subseteq \dot{V}$, so $\dot{V} \dot{=} \text{L} \rightarrow \dot{V} \dot{=} \text{HOD}$. However, the definition of HOD is not absolute so it is not provable that $\text{HOD}^{\text{HOD}} \dot{=} \text{HOD}$.

6.8 Generic extensions

The next two sections are devoted to the proof that

$$(1) \quad \text{ZF is consistent} \implies \text{ZFC} \cup \{\neg \text{CH}\} \text{ is consistent.}$$

Together with the results of [Section 6.7](#), this will complete the proof that

$$\text{ZF is consistent} \implies \text{CH and GCH are independent of ZFC.}$$

As outlined at the beginning of [Section 6.5](#), the basic strategy for establishing (1) is to describe a theory ZFL_m such that

$$(2) \quad \text{ZF is consistent} \implies \text{ZFL}_m \text{ is consistent;}$$

$$(3) \quad \text{ZFC} \cup \{\neg \text{CH}\} \text{ is relatively interpretable in } \text{ZFL}_m.$$

The first result explains why we are forced to adopt an approach different from that of the preceding section.

Proposition 6.8.1 *There does not exist a proper transitive class model M such that $\text{ZFC} \vdash (\neg \text{CH})^{(M)}$.*

Proof. Suppose that M were such a class. By Theorem 6.7.6(iii) together with Exercise 6.5.42, arguing in ZFC , $\text{L} \dot{=} \text{L}^M \subseteq M$, but since $\text{ZFC} \vdash \text{CH}^{(\text{L})}$, necessarily $\text{L} \neq M$ so also $\dot{V} \neq \text{L}$. But then $\text{ZFC} \vdash \dot{V} \neq \text{L}$, contrary to Corollary 6.7.8. $\square$

Definition 6.8.2 $L_{\in}^m$ is the language extending $L_{\in}$ by a single constant symbol m . ZFL_m is the $L_{\in}^m$ -theory with the following axioms:

- (i) ϕ for each $L_{\in}$ -axiom ϕ of ZFL;
- (ii) m is countable and transitive;
- (iii) $\phi^{(m)}$ for each $L_{\in}$ -axiom ϕ of ZFL.

The restriction to $L_{\in}$ -axioms means that in the Replacement schema no formulas (defining class operations) which contain the new constant symbol m are admitted. For almost all purposes it would suffice to use ZFC in place of ZFL in clauses (i) and (iii) and work with the resulting theory ZFC_m .

Theorem 6.8.3 *If ZF is consistent, then ZFL_m is consistent.*

Proof. Suppose that ZFL_m is inconsistent, so there exist finitely many axioms $\phi_0, \dots, \phi_k$ of ZFL such that

$$ZFL \vdash \neg [m \text{ is countable and transitive} \wedge (\phi_0 \wedge \dots \wedge \phi_k)^{(m)}].$$

But then by Corollary 2.6.6, since the axioms (i) do not contain the symbol m ,

$$ZFL \vdash \neg \exists a [a \text{ is countable and transitive} \wedge (\phi_0 \wedge \dots \wedge \phi_k)^{(a)}].$$

Then by the Transitive Model Theorem 6.6.18 we conclude that ZFL is inconsistent, and hence by Corollary 6.7.8 that also ZF is inconsistent. Alternatively we could appeal to the Strong Transitive Model Theorem 6.7.20 to conclude directly the inconsistency of ZF. $\square$

For most of this and the next section we shall be working in ZFL_m ; when not otherwise specified the assertion of an $L_{\in}^m$ -sentence means that it is a theorem of ZFL_m .

The strategy for (3) is now to define in ZFL_m a class n such that the following are theorems of ZFL_m :

- (4) n is countable and transitive $\wedge m \subseteq n$;
- (5) $\phi^{(n)}$ for each $L_{\in}$ -axiom ϕ of ZFC;
- (6) $(\neg CH)^{(n)}$.

We consider first the methods for satisfying (6). By hypothesis and the fact that $ZFL \vdash CH$ (Theorem 6.7.19), $ZFL_m \vdash CH^{(m)}$, so in particular, arguing in ZFL_m , $(\dot{2}^{\aleph_0} \leq \aleph_1)^{(m)}$ — that is, by the absoluteness results of [Section 6.5](#),

$$(\exists f \in m) [\text{Inj}(f) \wedge f : \dot{\varphi}(\dot{\omega}) \dot{\cap} m \dot{\rightarrow} \dot{\aleph}_1^m],$$

and our goal is to ensure that

$$\neg (\exists f \in \mathfrak{n}) [\text{Inj}(f) \wedge f : \dot{\wp}(\dot{\omega}) \dot{\cap} \mathfrak{n} \dot{\rightarrow} \dot{\aleph}_1^n].$$

We shall accomplish this by arranging that

$$(7) \quad (\exists g \in \mathfrak{n}) [\text{Inj}(g) \wedge g : \dot{\aleph}_2^{\mathfrak{m}} \dot{\rightarrow} \dot{\wp}(\dot{\omega}) \dot{\cap} \mathfrak{n}];$$

$$(8) \quad \dot{\aleph}_2^{\mathfrak{n}} \doteq \dot{\aleph}_2^{\mathfrak{m}}$$

(see the discussion at the end of [Section 6.5](#)). Loosely speaking, $\mathfrak{n}$ will contain ‘many’ more subsets of $\dot{\omega}$ than does $\mathfrak{m}$ but ‘few’ more well-orderings. It follows that $(\dot{\aleph}_2 \leq |\dot{\wp}(\dot{\omega})|)^{(n)}$ or equivalently $(\dot{\aleph}_2 \leq \dot{2}^{\dot{\aleph}_0})^{(n)}$ and hence $(\neg \text{CH})^{(n)}$. In fact, by altering details of the construction, we can arrange that $(\dot{\aleph}_2 \doteq \dot{2}^{\dot{\aleph}_0})^{(n)}$ or indeed $(\dot{\aleph}_7 \doteq \dot{2}^{\dot{\aleph}_0})^{(n)}$ or one of many other possibilities.

We begin now preparations for the construction of a suitable $\mathfrak{n}$.

Definition 6.8.4

- (i) A **notion of forcing** (**n.o.f.**) is a reflexive partial ordering $\mathbb{P} \doteq \langle P, \leq, \mathbb{I} \rangle$ with largest element $(\forall p \in P) p \leq \mathbb{I}$; elements of $\mathbb{P}$ are called (**forcing**) **conditions**;

for any n.o.f. $\mathbb{P}$,

- (ii) a set $G \subseteq P$ is a **$\mathbb{P}$ -filter** iff

- (a) $(\forall p, q \in G) (\exists r \in G) r \leq p \wedge r \leq q$, G is **coherent**;
 (b) $(\forall p \in G) \forall q [p \leq q \rightarrow q \in G]$, G is **upward closed**;

- (iii) a set $D \subseteq P$ is **$\mathbb{P}$ -dense** iff $(\forall p \in P) (\exists q \in D) q \leq p$;

- (iv) for any $\mathbb{P}$ -filter G and any set $\mathbb{D}$ of $\mathbb{P}$ -dense sets, G is **$\mathbb{D}$ -generic** iff $(\forall D \in \mathbb{D}) D \dot{\cap} G \neq \emptyset$.

Example 6.8.5 Our primary examples of n.o.f. are the reflexive partial orderings

$$\mathbb{FF}(u, v) \doteq \langle \mathbb{FF}(u, v), \leq_{\mathbb{FF}}, \emptyset \rangle,$$

of **finite functions**, where

$$\mathbb{FF}(u, v) := \left\{ p : p \text{ is a finite function} \wedge \text{Dom}(p) \subseteq u \wedge \text{Im}(p) \subseteq v \right\}$$

and

$$p \leq_{\mathbb{FF}} q \leftrightarrow q \subseteq p \leftrightarrow \{ f : p \subseteq f \} \subseteq \{ f : q \subseteq f \}.$$

To avoid triviality, we consider only $|v| \geq 2$ and u infinite. For any $x \in u$, $y \in v$ and $h : u \rightarrow v$, the sets

$$\begin{aligned} A_x &\doteq \left\{ p \in \text{FF}(u, v) : x \in \text{Dom}(p) \right\} \\ B^y &\doteq \left\{ p \in \text{FF}(u, v) : y \in \text{Im}(p) \right\} \\ C_h &\doteq \left\{ p \in \text{FF}(u, v) : (\exists z \in \text{Dom}(p)) \, p(z) \neq h(z) \right\} \end{aligned}$$

are easily seen to be $\text{FF}(u, v)$ -dense. For example, any $p \in \text{FF}(u, v)$ can be extended to $q \in \text{FF}(u, v)$ with $x \in \text{Dom}(q)$. For any function $f : u \rightarrow v$, the set

$$\text{Seg}_f := \left\{ p \in \text{FF}(u, v) : p \subseteq f \right\}$$

is clearly a $\text{FF}(u, v)$ -filter. Conversely, if G is any $\text{FF}(u, v)$ -filter, then $f_G := \bigcup G$ is a (not necessarily finite) function with $\text{Dom}(f_G) \subseteq u$ and $\text{Im}(f_G) \subseteq v$. In general, these may be proper inclusions, but if $\mathbb{D}$ is a set of $\mathbb{P}$ -dense sets and G is $\mathbb{D}$ -generic, then

$$\begin{aligned} (\forall x \in u) \, A_x \in \mathbb{D} &\implies \text{Dom}(f_G) = u; \\ (\forall y \in v) \, B^y \in \mathbb{D} &\implies \text{Im}(f_G) = v; \\ C_h \in \mathbb{D} &\implies f_G \neq h. \end{aligned}$$

For the central family of examples, for $u, v \in \mathfrak{m}$ let

$$\mathbb{D}_{\mathfrak{m}}^{\text{FF}}(u, v) := \left\{ D \subseteq \text{FF}(u, v) : D \text{ is } \text{FF}(u, v)\text{-dense} \wedge D \in \mathfrak{m} \right\}.$$

Then for all $x \in u$, $y \in v$ and $h \in \mathfrak{m}$, A_x , B^y and C_h do belong to $\mathbb{D}_{\mathfrak{m}}^{\text{FF}}(u, v)$, so if G is $\mathbb{D}_{\mathfrak{m}}^{\text{FF}}(u, v)$ -generic, then $f_G : u \rightarrow v$ is a surjective function *not* in $\mathfrak{m}$. Since $\mathfrak{m}$ is a set model of ZF, also $G \notin \mathfrak{m}$.

Thus, if G is $\mathbb{D}_{\mathfrak{m}}^{\text{FF}}(\dot{\omega}, \dot{2})$ -generic, then f_G is the characteristic function of a set $z \subseteq \dot{\omega}$. Since f_G is easily definable from z , and $\mathfrak{m}$ is a set model of ZFC, also $z \notin \mathfrak{m}$. Hence, if $\mathfrak{n}$ is a class such that

$$(9) \quad \mathfrak{m} \subseteq \mathfrak{n}, \quad \mathfrak{n} \text{ is a transitive set model of ZFC} \quad \text{and} \quad G \in \mathfrak{n},$$

then $\dot{\varphi}(\dot{\omega}) \cap \mathfrak{m}$ is a proper subset of $\dot{\varphi}(\dot{\omega}) \cap \mathfrak{n}$.

Similarly, if G is $\mathbb{D}_{\mathfrak{m}}^{\text{FF}}(\dot{\omega}, \aleph_1^{\mathfrak{m}})$ -generic, then f_G is a surjection of $\dot{\omega}$ onto $\aleph_1^{\mathfrak{m}}$. Hence for any class $\mathfrak{n}$ satisfying (9) for such a G , $\aleph_1^{\mathfrak{m}}$ is countable in $\mathfrak{n}$ so $\aleph_1^{\mathfrak{m}} \in \aleph_1^{\mathfrak{n}}$.

Of course, for all this to be of any interest we need to know that generic filters exist.

Generic Existence Theorem 6.8.6 (ZFC) *For any n.o.f. $\mathbb{P}$, any $p \in P$ and any countable set $\mathbb{D}$ of $\mathbb{P}$ -dense sets, there exists a $\mathbb{D}$ -generic $\mathbb{P}$ -filter G such that $p \in G$.*

Proof. Fix an enumeration $\mathbb{D} \doteq \{D_z : z \in \dot{\omega}\}$, and by AC let c be a function such that for all $z \in \dot{\omega}$ and $q \in P$, $c(q, z) \in D_z \wedge c(q, z) \leq q$. Then recursively set

$$p_0 := p \quad \text{and} \quad p_{z+\dot{i}} := c(p_z, z).$$

Then $G := \{q \in P : (\exists z \in \dot{\omega}) p_z \leq q\}$ is easily seen to be the required generic filter. $\square$

Remark 6.8.7 We think of a condition p as expressing information about or a property of any set G such that $p \in G$. For example, if $p \in \text{FF}(u, v)$, then $p \in G$ guarantees that $p \subseteq f_G$ — the function values specified by p are also correct for f_G . A set D of conditions may be thought of as expressing the information or property which is the common content of its members, which will be true about G if $G \cap D \neq \emptyset$. For example, $D \doteq \{p \in \text{FF}(u, v) : x \in \text{Dom}(p)\}$ doesn't specify any particular value for $f_G(x)$, only that it has *some* value. For D to be dense means that its information is consistent with that of any single condition or that the property it expresses is never precluded by a single condition. Thus the process of constructing the sequence $\langle p_z : z \in \dot{\omega} \rangle$ in the preceding proof is one of progressively increasing the information which, by the definition, will be true of G . By meeting every dense set, we give G every property which is never precluded by a single condition. This is one origin of the descriptor “generic”.

Since $(u, v) \mapsto \mathbb{D}_m^{\text{FF}}(u, v)$ is ZF-provably an m -absolute set operation, by Proposition 6.5.9(iii) it is an m -set operation — that is, for $u, v \in m$ also $\mathbb{D}_m^{\text{FF}}(u, v) \in m$. Since m is countable, also $\mathbb{D}_m^{\text{FF}}(u, v)$ is countable. Hence there always exist $\mathbb{D}_m^{\text{FF}}(u, v)$ -generic filters. Furthermore, by absoluteness,

$$(\text{FF}(\dot{\aleph}_2 \dot{\times} \dot{\omega}, 2))^m \doteq \text{FF}(\dot{\aleph}_2^m \dot{\times} \dot{\omega}, 2).$$

These facts are key to satisfying condition (7).

Theorem 6.8.8 *For any class n , if $m \subseteq n$, n is a transitive set model of ZFC, and there is a $\mathbb{D}_m^{\text{FF}}(\dot{\aleph}_2^m \dot{\times} \dot{\omega}, 2)$ -generic filter $G \in n$, then*

$$(\exists g \in n) [\text{Inj}(g) \wedge g : \dot{\aleph}_2^m \dot{\rightarrow} \wp^n(\dot{\omega})].$$

Proof. For each $\sigma \in \dot{\aleph}_2^m$ and $y \in \dot{\omega}$,

$$A_{\sigma, y} := \left\{ p \in \text{FF}(\dot{\aleph}_2^m \dot{\times} \dot{\omega}, 2) : \langle \sigma, y \rangle \in \text{Dom}(p) \right\}$$

is $\mathbb{FF}(\aleph_2^m \dot{\times} \dot{\omega}, \dot{2})$ -dense and belongs to $\mathfrak{m}$; hence by the definition of genericity, $G \dot{\cap} A_{\sigma, y} \neq \emptyset$. Thus as in Example 6.8.5, $f_G := \dot{\bigcup} G$ has domain $\aleph_2^m \dot{\times} \dot{\omega}$. For each $\sigma < \aleph_2^m$, set $g(\sigma) := \{y \in \dot{\omega} : f_G(\sigma, y) \dot{=} \dot{1}\}$. For each $\sigma < \tau < \aleph_2^m$, the set

$$D_{\sigma, \tau} := \{p \in \mathbb{FF}(\aleph_2^m \dot{\times} \dot{\omega}, \dot{2}) : (\exists y \in \dot{\omega}) [\langle \sigma, y \rangle, \langle \tau, y \rangle \in \text{Dom}(p) \wedge p(\sigma, y) \neq p(\tau, y)]\}$$

is $\mathbb{FF}(\aleph_2^m \dot{\times} \dot{\omega}, \dot{2})$ -dense and belongs to $\mathfrak{m}$, so again $G \dot{\cap} D_{\sigma, \tau} \neq \emptyset$ and hence for some $y \in \dot{\omega}$, $f_G(\sigma, y) \neq f_G(\tau, y)$ — that is, $g(\sigma) \neq g(\tau)$. Thus g is injective. Finally, $g \in \mathfrak{n}$, since it is simply defined from $G \in \mathfrak{n}$. It follows that $\text{Im}(g) \subseteq \wp^n(\dot{\omega})$. $\square$

For the remainder of this section, let $\mathbb{P} \dot{=} \langle P, \leq, \mathbb{I} \rangle$ be any fixed n.o.f. in $\mathfrak{m}$ and

$$\mathbb{D}_{\mathfrak{m}} := \{D \subseteq P : D \text{ is } \mathbb{P}\text{-dense and } D \in \mathfrak{m}\}.$$

Since $\mathfrak{m}$ is countable, also $\mathbb{D}_{\mathfrak{m}}$ is countable and hence by the Generic Existence Theorem there exist $\mathbb{D}_{\mathfrak{m}}$ -generic filters. We shall show that for any $\mathbb{D}_{\mathfrak{m}}$ -generic filter G , there exists a set $\mathfrak{m}[G]$ such that the following are theorems of $\text{ZFL}_{\mathfrak{m}}$:

$$(10) \quad \mathfrak{m} \subseteq \mathfrak{m}[G], \quad \mathfrak{m}[G] \text{ is countable and transitive, and } G \in \mathfrak{m}[G];$$

$$(11) \quad \phi^{(\mathfrak{m}[G])} \quad \text{for each } L_{\in}\text{-axiom } \phi \text{ of ZFC.}$$

Thus we may satisfy (4) and (5) with $\mathfrak{n} = \mathfrak{m}[G]$. Furthermore, $\mathfrak{m}[G]$ is minimal in the sense that if $\mathfrak{n}$ is any transitive set model of ZFC, then

$$(12) \quad \mathfrak{m} \subseteq \mathfrak{n} \wedge G \in \mathfrak{n} \rightarrow \mathfrak{m}[G] \subseteq \mathfrak{n}.$$

Each element of $\mathfrak{m}[G]$ will be the G -interpretation a_G of a **name** $a \in \mathfrak{m}$. A name a is a set with elements of the form $\langle b, q \rangle$ with b a name and $p \in P$. Intuitively, $\langle b, p \rangle \in a$ means that the set named by b belongs to the set named by a under any G -interpretation for which $p \in G$. Since $\mathbb{I} \in G$ for every filter G , if $\langle b, \mathbb{I} \rangle \in a$, then $b_G \in a_G$ for every G .

Definition 6.8.9 The class $\text{Na}^{\mathbb{P}}$ of $\mathbb{P}$ -names is $\dot{\bigcup} \{\{\text{Na}_{\sigma}^{\mathbb{P}} : \sigma \in \dot{\text{Or}}\}\}$, where $\text{Na}_0^{\mathbb{P}} \dot{=} \emptyset$ and for all σ ,

$$\text{Na}_{\sigma+\dot{o}1}^{\mathbb{P}} := \dot{\wp}(\text{Na}_{\sigma}^{\mathbb{P}} \dot{\times} P);$$

$$\text{Na}_{\sigma}^{\mathbb{P}} := \dot{\bigcup} \{\dot{\text{Na}}_{\tau}^{\mathbb{P}} : \tau < \sigma\}, \quad \text{if } \sigma \text{ is a limit ordinal.}$$

Lemma 6.8.10 *The set operation $\sigma \mapsto \mathbf{Na}_\sigma^\mathbb{P}$ and the class $\mathbf{Na}^\mathbb{P}$ are $\mathfrak{m}$ -element absolute — that is, for all $\sigma \in \mathfrak{m}$,*

$$\begin{aligned} (\mathbf{Na}_\sigma^\mathbb{P})^\mathfrak{m} &\doteq \mathbf{Na}_\sigma^\mathbb{P} \dot{\cap} \mathfrak{m}; \\ (\mathbf{Na}^\mathbb{P})^\mathfrak{m} &\doteq \mathbf{Na}^\mathbb{P} \dot{\cap} \mathfrak{m} \doteq \bigcup \left\{ \mathbf{Na}_\sigma^\mathbb{P} \dot{\cap} \mathfrak{m} : \sigma \in \mathfrak{m} \right\}. \end{aligned}$$

Proof. Parallel to the corresponding result for $\dot{\mathbf{V}}$ (6.5.35), we define by set recursion (6.2.49) a set operation

$$\mathbf{rank}_\mathbb{P}(b) := \bigcup \left\{ \mathbf{rank}_\mathbb{P}(c) \dot{+}_o \dot{\mathbf{i}} : c \subseteq \dot{\mathbf{V}} \dot{\times} P \wedge c \in \mathbf{Dom}(b) \right\}$$

(cf. Exercise 6.3.37). It is straightforward to prove by induction that for all σ ,

$$\mathbf{Na}_\sigma^\mathbb{P} \doteq \left\{ b : b \subseteq \dot{\mathbf{V}} \dot{\times} P \wedge \mathbf{rank}_\mathbb{P}(b) < \sigma \right\}.$$

Then the result follows from the absoluteness of the operation $\mathbf{rank}_\mathbb{P}$. $\square$

Definition 6.8.11 For any $\mathbb{D}_\mathfrak{m}$ -generic filter G ,

- (i) the operation $a \mapsto a_G$ for $a \in \mathbf{Na}^\mathbb{P}$ is defined by set recursion:

$$a_G := \left\{ b_G : (\exists p \in G) \langle b, p \rangle \in a \right\};$$

- (ii) $\mathfrak{m}[G] := \left\{ a_G : a \in \mathbf{Na}^\mathbb{P} \dot{\cap} \mathfrak{m} \right\}$.

Remark 6.8.12 An important and easily overlooked point is that

$$b_G \in a_G \rightarrow \exists b' (\exists q \in G) \langle b', q \rangle \in a \wedge b_G \doteq b'_G,$$

but it does not follow that $b' \doteq b$ or even that $b \in \mathbf{Dom}(a)$. In general, members of $\mathfrak{m}[G]$ do not have unique names.

Now it is relatively easy to establish conditions (10) and (12).

Definition 6.8.13

- (i) For each $x \in \mathfrak{m}$, $\check{x} := \left\{ \langle \check{y}, \mathbb{I} \rangle : y \in x \right\};$
(ii) $\check{G} := \left\{ \langle \check{p}, p \rangle : p \in P \right\}.$

We read $\check{x}$ as “**x check**” and regard it as a canonical name for x . This idea will be extended below with $\check{\mathbf{F}}$ being (in a slightly different sense) a canonical name for a class operation $\mathbf{F}$.

Lemma 6.8.14 *For any $\mathbb{D}_m$ -generic filter G ,*

- (i) *for all $x \in m$, $(\check{x})_G \doteq x$;*
- (ii) *$(\check{G})_G \doteq G$.*

Proof. Part (i) is easy to prove by set induction. Then (ii) follows from

$$(\check{G})_G \doteq \{ \check{p}_G : (\exists q \in G) \langle \check{p}, q \rangle \in \check{G} \} \doteq \{ \check{p}_G : p \in G \} \doteq G. \quad \square$$

Proposition 6.8.15 *For any $\mathbb{D}_m$ -generic filter G , $m \subseteq m[G]$, $G \in m[G]$ and $m[G]$ is countable and transitive.*

Proof. The first two assertions are immediate from the lemma. $m[G]$ is the image under the operation $a \mapsto a_G$ of the countable set $\text{Na}^{\mathbb{P}} \dot{\cap} m$ so is countable. By definition, any element of any $a_G \in m[G]$ is of the form b_G so belongs to $m[G]$; thus $m[G]$ is transitive. $\square$

Proposition 6.8.16 *For any transitive set model n of ZFC, if $m \subseteq n$ and $G \in n$, then $m[G] \subseteq n$.*

Proof. Under the hypotheses, the operation $a \mapsto a_G$ is n -absolute, so for each $a \in \text{Na}^{\mathbb{P}} \dot{\cap} m$, $a_G \doteq a_G^n \in n$. $\square$

The remaining task of showing (11) that $m[G]$ is a set model of ZFC is a bit more difficult. First we have

Proposition 6.8.17 *For any $\mathbb{D}_m$ -generic filter G ,*

$$\text{ZFL}_m \vdash \text{Extensionality}^{(m[G])} \quad \text{and} \quad \text{ZFL}_m \vdash \text{Foundation}^{(m[G])}.$$

Proof. Immediate by Propositions 6.5.2 and 6.8.15. $\square$

For the remaining axioms we need to establish sentences of the form

$$\text{ZFL}_m \vdash (\text{F is a set operation})^{(m[G])},$$

or equivalently, by Proposition 6.5.9(ii),

$$\text{ZFL}_m \vdash F^{m[G]} \text{ is an } m[G]\text{-set operation},$$

for F each of $\dot{\bigcup}$, $\dot{\omega}$, $\dot{\wp}$ or G^* for some class operation G . The key notion here is

Definition 6.8.18 A (unary) class operation F is **defined on names** iff there exists a set operation $\check{F}$ such that for any $\mathbb{D}_m$ -generic filter G and all $a \in \text{Na}^{\mathbb{P}} \dot{\cap} m$,

$$\check{F}a \in \text{Na}^{\mathbb{P}} \dot{\cap} m \quad \text{and} \quad (\check{F}a)_G \doteq F^{m[G]}(a_G).$$

We remind the reader that, as always, notions are stated for unary class operations as a shorthand for a more cumbersome statement for k -ary class operations including $k = 0$. In particular, a class B is defined on names iff there exists $\check{B} \in \text{Na}^{\mathbb{P}} \dot{\cap} m$ such that $(\check{B})_G \doteq B^{m[G]}$.

Corollary 6.8.19 *For any class operation F , if F is defined on names, then $F^{m[G]}$ is an $m[G]$ -set operation, so also $(F \text{ is a set operation})^{(m[G])}$. $\square$*

Thus it suffices to show that the operations mentioned above are defined on names. A few of these are easy.

Proposition 6.8.20 *The following definitions witness that the indicated operations are defined on names:*

- (i) $\check{\{a, b\}} := \{ \langle \check{a}, \mathbb{I} \rangle, \langle \check{b}, \mathbb{I} \rangle \};$
- (ii) $\langle \check{a}, b \rangle := \{ \check{\{a\}}, \check{\{a, b\}} \};$
- (iii) $a \check{\cup} b := a \cup b;$
- (iv) $\check{S}a := a \check{\cup} \check{\{a\}};$
- (v) $\check{\bigcup} a := \{ \langle \check{c}, r \rangle : \exists b \exists p \exists q [\langle \check{b}, p \rangle \in a \wedge \langle \check{c}, q \rangle \in b \wedge r \leq p \wedge r \leq q] \}.$

Proof. Parts (i)–(iv) are easy to verify and are left to the reader. Note that it follows easily that for $x, y \in m$,

$$\check{\{x, y\}} \doteq \overbrace{\{x, y\}} \quad \text{and} \quad \langle \check{x}, \check{y} \rangle \doteq \overbrace{\langle x, y \rangle}.$$

For (v) we need to establish that $(\check{\bigcup} a)_G \doteq \bigcup a_G$ (since $\bigcup$ is $m[G]$ -absolute by Proposition 6.5.6). Suppose first that $x \in \bigcup a_G$ so for some y , $x \in y \in a_G$. Then

$$\begin{aligned} y \in a_G &\rightarrow \text{for some } b \text{ and } p, \quad y \doteq b_G \wedge p \in G \wedge \langle \check{b}, p \rangle \in a; \\ x \in b_G &\rightarrow \text{for some } c \text{ and } q, \quad x \doteq c_G \wedge q \in G \wedge \langle \check{c}, q \rangle \in b. \end{aligned}$$

By coherence of filters, there exists $r \in G$ such that $r \leq p \wedge r \leq q$. Then $\langle \check{c}, r \rangle \in \bigcup a$ so $x \doteq c_G \in (\bigcup a)_G$. Conversely,

$$x \in (\bigcup a)_G \rightarrow \text{for some } c \text{ and } r, \quad x \doteq c_G \wedge r \in G \wedge \langle \check{c}, r \rangle \in \bigcup a.$$

Hence there exist b, p and q as in (v). By upward closure of filters, $p, q \in G$ so $b_G \in a_G$ and $c_G \in b_G$, so $x \in \bigcup a_G$. $\square$

Corollary 6.8.21 *For any $\mathbb{D}_m$ -generic filter G ,*

$$\text{ZFL}_m \vdash \text{Union}^{(m[G])} \quad \text{and} \quad \text{ZFL}_m \vdash \text{Infinity}^{(m[G])}.$$

Proof. The Union axiom is immediate from the preceding two results, and by Proposition 6.5.10,

$$(\check{\omega})_G \doteq \dot{\omega} \doteq \dot{\omega}^{m[G]};$$

so also $\dot{\omega}$ as a class is defined on names, and the result holds by Corollary 6.8.19. $\square$

Note that we have used so far only that G is a filter; for verifying the Powerset and Replacement axioms the genericity of G will play a crucial role via the following central notion.

Definition 6.8.22 For any $L_{\in}$ -formula $\phi(x_0, \dots, x_{k-1})$ and any $p, a_0, \dots, a_{k-1}, p \Vdash \phi(a_0, \dots, a_{k-1})$ is the $L_{\in}^m$ -formula

$$p \dot{\in} P \wedge a_0, \dots, a_{k-1} \dot{\in} \mathbf{Na}^{\mathbb{P}} \cap \mathbf{m} \\ \wedge \forall G [G \text{ is a } \mathbb{D}_{\mathbf{m}}\text{-generic filter} \wedge p \dot{\in} G \rightarrow \phi^{(\mathbf{m}[G])}((a_0)_G, \dots, (a_{k-1})_G)].$$

We say that p **forces** $\phi(a_0, \dots, a_{k-1})$. The intuition is that p contains enough information about any of the generic G 's with $p \dot{\in} G$ to ensure that $\phi(a_0, \dots, a_{k-1})$ is true in $\mathbf{m}[G]$.

Examples 6.8.23 If $\langle b, p \rangle \dot{\in} a$, then for any generic filter G such that $p \dot{\in} G, b_G \dot{\in} a_G$ — that is, $p \Vdash b \dot{\in} a$. In particular, if $y \dot{\in} x \dot{\in} \mathbf{m}$, then for *any* generic filter $G, \check{y}_G \dot{\in} \check{x}_G$, so *every* $p \Vdash \check{y} \dot{\in} \check{x}$. Note that by upward closure of filters,

$$q \Vdash \phi(a_0, \dots, a_{k-1}) \wedge p \leq q \rightarrow p \Vdash \phi(a_0, \dots, a_{k-1}),$$

so in particular,

$$\mathbb{I} \Vdash \phi(a_0, \dots, a_{k-1}) \leftrightarrow (\forall p \dot{\in} P) p \Vdash \phi(a_0, \dots, a_{k-1}).$$

It follows immediately from the definition that

$$\begin{aligned} p \Vdash \phi(a) \wedge p \Vdash \psi(a) &\leftrightarrow p \Vdash \phi(a) \wedge \psi(a); \\ p \Vdash \phi(a) \vee p \Vdash \psi(a) &\rightarrow p \Vdash \phi(a) \vee \psi(a); \\ p \Vdash \neg \phi(a) &\rightarrow \neg [p \Vdash \phi(a)]; \end{aligned}$$

but generally the second and third implications do not reverse. The relationship between forcing and the logical connectives is fairly complex and will be important in [Section 6.9](#); see also the exercises below. If ϕ and ψ are logically equivalent, then

$$p \Vdash \phi(a_0, \dots, a_{k-1}) \leftrightarrow p \Vdash \psi(a_0, \dots, a_{k-1}).$$

Definition 6.8.24 For any $a, c \in \mathbf{Na}^{\mathbb{P}}$ and $p \in P$,

$$c \dot{\in}_p a \leftrightarrow \exists r [p \leq r \wedge \langle c, r \rangle \dot{\in} a].$$

Then it is immediate from the upward closure of filters that for any $\mathbb{D}_m$ -generic filter G ,

$$p \dot{\in} G \wedge c \dot{\in}_p a \rightarrow c_G \dot{\in} a_G,$$

that is, $c \dot{\in}_p a \rightarrow p \Vdash c \dot{\in} a$, and we think of $c \dot{\in}_p a$ as a strong form of $p \Vdash c \dot{\in} a$. However, the converse implication is not true; we shall see eventually that actually

$$p \Vdash c \dot{\in} a \leftrightarrow (\forall q \leq p)(\exists r \leq q)\exists d [d \dot{\in}_r a \wedge r \Vdash c \dot{=} d].$$

This relationship is not very helpful in understanding the notion, but fortunately it and others like it are necessary only when we come in the next section to proving the following technical facts about forcing. Everything needed to complete the proof of the independence of CH follows from these.

Forcing Definability Theorem 6.8.25 *For each $L_{\in}$ -formula $\phi(x_0, \dots, x_{k-1})$, there exists an $L_{\in}$ -formula $p \Vdash \phi$ such that for all $p \in P$ and all $a_0, \dots, a_{k-1} \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} m$,*

$$\mathbf{ZFL}_m \vdash p \Vdash \phi(a_0, \dots, a_{k-1}) \leftrightarrow (p \Vdash \phi)^{(m)}(a_0, \dots, a_{k-1}).$$

Forcing Completeness Theorem 6.8.26 *For each $L_{\in}$ -formula $\phi(x_0, \dots, x_{k-1})$, any $a_0, \dots, a_{k-1} \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} m$ and any $\mathbb{D}_m$ -generic filter G ,*

$$\mathbf{ZFL}_m \vdash \phi^{(m[G])}((a_0)_G, \dots, (a_{k-1})_G) \leftrightarrow (\exists p \dot{\in} G) p \Vdash \phi(a_0, \dots, a_{k-1}).$$

Note that the implication ($\leftarrow$) is part of the definition of forcing. Proofs of these two results will be given in the next section, but it should be noted that taken together they are a bit surprising. Although the definition of $p \Vdash \phi$ makes reference to all $\mathbb{D}_m$ -generic G , which are generally not members of m , Forcing Definability asserts that the truth of this assertion can be evaluated in m . The (intuitive) point is that a statement which is true for *all* generic filters is really independent of any of them and really a property of m . Forcing Completeness tells us that every property of any generic extension $m[G]$ is already forced by some $p \dot{\in} G$. Intuitively this means that $m[G]$ does not have any “strange” properties; this is another explanation of the descriptor “generic”.

Here we will use these two results to complete the verification that $m[G]$ is a set model of ZFC.

Definition 6.8.27 A (unary) class operation F is *name bounded* iff there exists a class operation $\mathring{F}$ such that

- (i) $\text{ZFL}_m \vdash \mathring{F}^m$ is an m -set operation;
- (ii) $\text{ZFL}_m \vdash (\forall a \in \text{Na}^{\mathbb{P}} \dot{\cap} m) \mathring{F}^m(a) \subseteq \text{Na}^{\mathbb{P}} \dot{\cap} m$;
- (iii) for any $\mathbb{D}_m$ -generic filter G and any $a \in \text{Na}^{\mathbb{P}} \dot{\cap} m$,

$$F^{m[G]}(a_G) \subseteq \{c_G : c \in \mathring{F}^m(a)\}.$$

Even though for different G , the set $F^{m[G]}(a_G)$ will have different elements, name boundedness ensures that there is an m -set of names sufficient to name all of them. We shall show below that every name bounded operation is defined on names, but to understand the notion let us first observe that the examples 6.8.20 are name bounded.

Examples 6.8.28 We may take

$$\begin{aligned} \mathring{\circ}a, b \} &:= \{a, b\}; & a \mathring{\cup} b &:= \text{D}\dot{\text{om}}(a) \cup \text{D}\dot{\text{om}}(b); \\ \bigcup a &:= \bigcup \{ \text{D}\dot{\text{om}}(b) : b \in \text{D}\dot{\text{om}}(a) \}. \end{aligned}$$

To see how $\bigcup$ works, suppose that $x \in \bigcup a_G$, so for some b and $p \in G$, $\langle b, p \rangle \in a$ and $x \in b_G$. Then there exist c and $q \in G$ such that $\langle c, q \rangle \in b$ and $x \in c_G$. Clearly $c \in \bigcup a$.

A slightly more complicated example is

Proposition 6.8.29 $\mathring{\wp}$ is name bounded.

Proof. For any a , set

$$\mathring{\wp}a := \{c \in \text{Na}^{\mathbb{P}} : \text{D}\dot{\text{om}}(c) \subseteq \text{D}\dot{\text{om}}(a)\}.$$

By Lemma 6.8.10, $\mathring{\wp}^m(a) \doteq \mathring{\wp}a \dot{\cap} m$, so by m -Comprehension, $\mathring{\wp}^m$ is an m -set operation. Arguing in ZFL_m , fix a $\mathbb{D}_m$ -generic filter G and $a \in \text{Na}^{\mathbb{P}} \dot{\cap} m$. Any element of $\mathring{\wp}^{m[G]}(a_G)$ is of the form b_G for some name b such that $\text{D}\dot{\text{om}}(b) \subseteq \text{D}\dot{\text{om}}(a)$. Hence by Forcing Completeness,

$$\begin{aligned} b_G &\doteq \{e_G : e \in \text{D}\dot{\text{om}}(a) \wedge e_G \in b_G\} \\ &\doteq \{e_G : e \in \text{D}\dot{\text{om}}(a) \wedge (\exists p \in G) p \Vdash e \in b\} \doteq c_G, \end{aligned}$$

for $c := \{ \langle e, p \rangle \in \text{D}\dot{\text{om}}(a) \times P : p \Vdash e \in b \}$. By Forcing Definability, $c \in m$, so $c \in \mathring{\wp}^m(a)$. Thus $\mathring{\wp}^{m[G]}(a_G) \subseteq \{c_G : c \in \mathring{\wp}^m(a)\}$ as desired. $\square$

The main reason for interest in name boundedness is

Theorem 6.8.30 *For any class operation F , if F is name bounded, then F is defined on names.*

Proof. Suppose that (unary) F is name bounded by $\mathring{F}$ and set

$$\check{F}a := \{ \langle \dot{c}, p \rangle : c \in \mathring{F}^m a \wedge p \Vdash c \in Fa \}.$$

By Forcing Definability and $\mathfrak{m}$ -Comprehension, $\check{F}$ is an $\mathfrak{m}$ -set operation so $\check{F}a \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathfrak{m}$. Let G be any $\mathbb{D}_{\mathfrak{m}}$ -generic filter and $a \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathfrak{m}$. Then by Forcing Definability and the definition of name boundedness,

$$\begin{aligned} (\check{F})_G &\doteq \{ c_G : (\exists p \in G) \langle \dot{c}, p \rangle \in \check{F}a \} \\ &\doteq \{ c_G : c \in \mathring{F}^m a \wedge (\exists p \in G) p \Vdash c \in Fa \} \\ &\doteq \{ c_G : c \in \mathring{F}^m a \wedge c_G \in F^{m[G]}(a_G) \} \\ &\doteq F^{m[G]}(a_G). \quad \square \end{aligned}$$

Proposition 6.8.31 *For any $\mathbb{D}_{\mathfrak{m}}$ -generic filter G ,*

$$\mathbf{ZFL}_{\mathfrak{m}} \vdash \mathbf{Powerset}^{(m[G])}.$$

Proof. Immediate from the two preceding results and Corollary 6.8.19. $\square$

To show that $m[G]$ is a set model of ZFC, it remains only to verify Replacement and Choice.

Proposition 6.8.32 *For any class operation F , F^* is name bounded.*

Proof. To avoid awkward notation, we write H for F^* . For any class operation F , by the Collection Principle 6.3.30 there exists a set operation $\mathring{H}$ such that

$$\forall a (\forall e \in \mathring{\text{Dom}} a) (\forall p \in P) [\exists b (p \Vdash b \doteq Fe) \rightarrow \exists c \in \mathring{H}a (p \Vdash c \doteq Fe)]$$

is a theorem of ZF. Since in $\mathbf{ZFL}_{\mathfrak{m}}$ $\mathfrak{m}$ is a set model of ZF, this sentence relativized to $\mathfrak{m}$ is also a theorem — that is, taking account of absoluteness and Forcing Definability,

$$\forall^m a (\forall e \in \mathring{\text{Dom}} a) (\forall p \in P) [\exists^m b (p \Vdash b \doteq Fe) \rightarrow \exists c \in \mathring{H}^m a (p \Vdash c \doteq Fe)]$$

is a theorem of $\mathbf{ZFL}_{\mathfrak{m}}$. Then

$$\begin{aligned} H^{m[G]}(a_G) &\doteq F^{m[G]*}(a_G) \dot{\cap} m[G] \\ &\doteq \{ F^{m[G]}(e_G) : e_G \in a_G \} \dot{\cap} m[G] \\ &\subseteq \{ b_G : b \in \mathfrak{m} \wedge b_G \doteq F^{m[G]}(e_G) \wedge e \in \mathring{\text{Dom}}(a) \} \\ &\doteq \{ b_G : b \in \mathfrak{m} \wedge (\exists p \in G) p \Vdash b \doteq Fe \wedge e \in \mathring{\text{Dom}}(a) \} \\ &\subseteq \{ c_G : c \in \mathring{H}^m(a) \}. \end{aligned}$$

Here, the first equality is as in the proof of Theorem 6.5.12, the third equality by Forcing Completeness and the final inclusion by the fact that if $p \in G$, $p \Vdash b \dot{=} Fe$ and $p \Vdash c \dot{=} Fe$, then $b_G \dot{=} c_G$. $\square$

Theorem 6.8.33 *For any n.o.f. $\mathbb{P} \in \mathfrak{m}$ and any $\mathbb{D}_{\mathfrak{m}}$ -generic $\mathbb{P}$ -filter G ,*

$$\text{ZFL}_{\mathfrak{m}} \vdash \phi^{(\mathfrak{m}[G])}$$

for each $L_{\in}$ -axiom ϕ of ZFC, hence for each $L_{\in}$ -theorem of ZFC.

Proof. Combining Corollaries 6.8.17, 6.8.21 and 6.8.31 and the preceding proposition, we have established $\phi^{(\mathfrak{m}[G])}$ for each of the axioms ϕ of ZFC except for AC itself. By Corollary 6.4.18 it suffices to show that

$$\text{ZFL}_{\mathfrak{m}} \vdash \left(\forall x \exists \sigma \exists f (\dot{\text{Fn}}(f) \wedge \dot{\text{Dom}}(f) \dot{=} \sigma \wedge x \dot{\subseteq} \dot{\text{Im}}(f)) \right)^{(\mathfrak{m}[G])}.$$

Arguing in $\text{ZFL}_{\mathfrak{m}}$, fix $x \in \mathfrak{m}[G]$ and $a \in \text{Na}^{\mathbb{P}} \dot{\cap} \mathfrak{m}$ such that $x \dot{=} a_G$. By $\text{AC}^{(\mathfrak{m})}$, there exist $\sigma, g \in \mathfrak{m}$ such that $\sigma \approx_g \dot{\text{Dom}}(a)$. Then if

$$c := \left\{ \langle \check{\tau}, g(\tau) \rangle : \tau \dot{<} \sigma \right\} \dot{\times} \check{\mathbb{I}},$$

easily $c \in \text{Na}^{\mathbb{P}} \dot{\cap} \mathfrak{m}$ and $c_G \dot{=} \left\{ \langle \tau, g(\tau)_G \rangle : \tau \dot{<} \sigma \right\}$ is a function $f \in \mathfrak{m}[G]$ such that $\dot{\text{Dom}}(f) \dot{=} \sigma$ and

$$x \dot{=} a_G \dot{\subseteq} \left\{ b_G : b \in \dot{\text{Dom}}(a) \right\} \dot{=} \dot{\text{Im}}(f). \quad \square$$

In particular, for $\mathbb{P} \dot{=} \text{FF}(\dot{\aleph}_2^{\mathfrak{m}} \dot{\times} \dot{\omega}, \dot{2})$ and

$$\mathbb{D}_{\mathfrak{m}}^{\text{FF}}(\dot{\aleph}_2^{\mathfrak{m}} \dot{\times} \dot{\omega}, \dot{2}) := \left\{ D \dot{\subseteq} \text{FF}(\dot{\aleph}_2^{\mathfrak{m}} \dot{\times} \dot{\omega}, \dot{2}) : D \text{ is } \mathbb{P}\text{-dense} \wedge D \in \mathfrak{m} \right\},$$

for any $\mathbb{D}_{\mathfrak{m}}^{\text{FF}}(\dot{\aleph}_2^{\mathfrak{m}} \dot{\times} \dot{\omega}, \dot{2})$ -generic filter G , $\mathfrak{m}[G]$ satisfies not only ZFC but also by Theorem 6.8.8,

$$(7) \quad \exists g \in \mathfrak{m}[G] \left[\dot{\text{Inj}}(g) \wedge g : \dot{\aleph}_2^{\mathfrak{m}} \dot{\rightarrow} \wp^{\mathfrak{m}[G]}(\dot{\omega}) \right].$$

To complete the proof that $(\neg \text{CH})^{(\mathfrak{m}[G])}$, it remains to establish that (8) $\dot{\aleph}_2^{\mathfrak{m}[G]} \dot{=} \dot{\aleph}_2^{\mathfrak{m}}$, since then $(\dot{\aleph}_2 \dot{\leq} \dot{\aleph}(\dot{\omega}))^{(\mathfrak{m}[G])}$. First, however, we note that we have already

Corollary 6.8.34 ZF is consistent $\implies \text{ZFC} \cup \{\neg \dot{V} \dot{=} \mathbb{L}\}$ is consistent.

Proof. Let G be any $\mathbb{D}_m^{\text{FF}}(\dot{\omega}, 2)$ -generic filter. By Example 6.8.5, $\dot{m} \neq m[G]$. However, it is easy to see that $\dot{\text{Or}} \dot{\cap} \dot{m} \dot{=} \dot{\text{Or}} \dot{\cap} m[G]$, since by set induction, for all $a \in \text{Na}^{\mathbb{P}} \dot{\cap} \dot{m}$, $\text{rank}(a_G) \leq \text{rank}(a)$. Hence by absoluteness of the function $\sigma \mapsto \mathbb{L}_\sigma$ and $\dot{\text{Or}}$,

$$\mathbb{L}^m \dot{=} \bigcup \left\{ \mathbb{L}_\sigma : \sigma \in \dot{\text{Or}} \dot{\cap} \dot{m} \right\} \dot{=} \bigcup \left\{ \mathbb{L}_\sigma : \sigma \in \dot{\text{Or}} \dot{\cap} m[G] \right\} \dot{=} \mathbb{L}^{m[G]}.$$

In particular, $\mathbb{L}^{m[G]} \subseteq m$ so $\mathbb{L}^{m[G]} \neq \dot{V}^{m[G]}$ — that is, $(\neg \dot{V} \dot{=} \mathbb{L})^{(m[G])}$. Thus there is an interpretation of $\text{ZFC} \cup \{\neg \dot{V} \dot{=} \mathbb{L}\}$ in ZF . $\square$

Definition 6.8.35 For any n.o.f. $\mathbb{P} \dot{=} \langle P, \leq, \mathbb{I} \rangle$,

- (i) $p, q \in P$ are **compatible** iff $(\exists r \in P) r \leq p \wedge r \leq q$;
- (ii) $Q \subseteq P$ is an **antichain** iff $(\forall p, q \in Q)$ if $p \neq q$ then p, q are not compatible;
- (iii) $\mathbb{P}$ has the **countable chain condition** (**c.c.c.**) iff there is no uncountable antichain $Q \subseteq P$.

Examples 6.8.36 $p, q \in \text{FF}(u, v)$ are compatible iff

$$p \upharpoonright (\dot{\text{Dom}}(p) \dot{\cap} \dot{\text{Dom}}(q)) \dot{=} q \upharpoonright (\dot{\text{Dom}}(p) \dot{\cap} \dot{\text{Dom}}(q)).$$

$\text{FF}(\dot{\omega}, \dot{\aleph}_1^m)$ from Example 6.8.5 does *not* have the c.c.c. in m , since the set of $\langle \dot{0}, \sigma \rangle$ for $\sigma \in \dot{\aleph}_1^m$ is an antichain which is uncountable in m . This is consistent with the proposition below and the fact clear from that example that if G is $\mathbb{D}_m^{\text{FF}}(\dot{\omega}, \dot{\aleph}_1^m)$ -generic, then m and $m[G]$ do not agree on cardinals.

Proposition 6.8.37 (ZFC) For any u and v , if v is countable, then $\text{FF}(u, v)$ has the c.c.c.

Proof. Let v be countable and $Q \subseteq \text{FF}(u, v)$ an antichain. We construct recursively a countable sequence of countable sets $Q_0 \subseteq Q_1 \subseteq \dots \subseteq Q$ and show that $Q \dot{=} \bigcup_{y \in \dot{\omega}} Q_y$, so also Q is countable.

We say that $p \in \text{FF}(u, v)$ is **Q -extendable** iff there exists $q \in Q$ with $p \subseteq q$. Let c be a choice function such that when p is Q -extendable, $p \subseteq c(p) \in Q$. Fix any $q_0 \in Q$, and set

$$Q_0 := \{q_0\} \quad \text{and} \quad w_0 := \dot{\text{Dom}}(q_0).$$

Given Q_y and w_y , let

$$P_y := \left\{ p \in \text{FF}(u, v) : \dot{\text{Dom}}(p) \subseteq w_y \wedge p \text{ is } Q\text{-extendable} \right\},$$

and set

$$Q_{y \dot{+}_o i} := Q_y \dot{\cup} \{ \dot{c}(p) : p \in P_y \} \quad \text{and} \\ w_{y \dot{+}_o i} := \dot{\bigcup} \{ \dot{\text{Dom}}(q) : q \in Q_{y \dot{+}_o i} \}.$$

By $\dot{\omega}$ -induction, using the countability of v , it is easy to check that for all $y \in \dot{\omega}$, both Q_y and w_y are countable; hence $\dot{\bigcup} \{ Q_y : y \in \dot{\omega} \} \subseteq Q$ is countable, and it suffices to show that this set includes Q .

Let $w := \dot{\bigcup}_{y \in \dot{\omega}} w_y$. For any $q \in Q$, because q is a finite function, for some $y \in \dot{\omega}$, $q \upharpoonright w \dot{=} q \upharpoonright w_y$. Since by definition $q \upharpoonright w_y$ is Q -extendable,

$$r := \dot{c}(q \upharpoonright w_y) \in Q_{y \dot{+}_o i}, \quad q \upharpoonright w_y \subseteq r \quad \text{and} \quad \dot{\text{Dom}}(r) \subseteq q_{y \dot{+}_o i} \subseteq w.$$

Hence $q \upharpoonright \dot{\text{Dom}}(r) \subseteq r$ and thus by the example, q and r are compatible. But since also $r \in Q$ and Q is an antichain, this implies that $q \dot{=} r$, so actually $q \in Q_{y \dot{+}_o i}$. $\square$

Corollary 6.8.38 *For any $u, v \in \mathfrak{m}$,*

$$\text{if } (v \text{ is countable})^{(\mathfrak{m})}, \text{ then } (\mathbb{FF}(u, v) \text{ has c.c.c.})^{(\mathfrak{m})},$$

that is, for any antichain $Q \subseteq \mathbb{FF}(u, v)$, $(Q \text{ is countable})^{(\mathfrak{m})}$.

Proof. Since the preceding proposition is a theorem of ZFC, it is also valid relativized to $\mathfrak{m}$. $\square$

Proposition 6.8.39 *For any n.o.f. $\mathbb{P} \in \mathfrak{m}$ and any $\mathbb{D}_{\mathfrak{m}}$ -generic $\mathbb{P}$ -filter G , if $(\mathbb{P} \text{ has c.c.c.})^{(\mathfrak{m})}$, then $\mathfrak{m}$ and $\mathfrak{m}[G]$ agree on cardinals — that is,*

$$\text{ZFL}_{\mathfrak{m}} \vdash \forall^{\mathfrak{m}} \sigma [(\sigma \text{ is a cardinal})^{(\mathfrak{m})} \leftrightarrow (\sigma \text{ is a cardinal})^{(\mathfrak{m}[G])}].$$

Hence

$$\text{ZFL}_{\mathfrak{m}} \vdash \forall^{\mathfrak{m}} \rho \left(\dot{\aleph}_{\rho}^{\mathfrak{m}} \dot{=} \dot{\aleph}_{\rho}^{\mathfrak{m}[G]} \right).$$

Proof. By absoluteness of the notions involved, $(\sigma \text{ is a cardinal})^{(\mathfrak{m})}$ iff

$$\neg \exists^{\mathfrak{m}} f (\exists \tau < \sigma) [\dot{\text{Fn}}(f) \wedge \dot{\text{Dom}}(f) \dot{=} \tau \wedge \sigma \subseteq \dot{\text{Im}}(f)],$$

and similarly for $\mathfrak{m}[G]$, so just because $\mathfrak{m} \subseteq \mathfrak{m}[G]$,

$$(\sigma \text{ is a cardinal})^{(\mathfrak{m}[G])} \rightarrow (\sigma \text{ is a cardinal})^{(\mathfrak{m})}.$$

Furthermore, since $\dot{\aleph}_0^{\mathfrak{m}} \dot{=} \dot{\omega} \dot{=} \dot{\aleph}_0^{\mathfrak{m}[G]}$, we need to establish the converse only for infinite $\sigma \in \mathfrak{m}$.

Suppose that $\neg(\sigma \text{ is a cardinal})^{(\mathfrak{m}[G])}$, and fix a function $f \in \mathfrak{m}[G]$ and $\tau < \sigma$ such that $\text{Dom}(f) \dot{=} \tau \wedge \sigma \dot{\subseteq} \text{Im}(f)$; clearly also τ is infinite. By Forcing Completeness there exist a name a and $p \in G$ such that

$$p \Vdash \dot{\text{Fn}}(a) \wedge \dot{\text{Dom}}(a) \dot{=} \dot{\tau} \wedge \dot{\sigma} \dot{\subseteq} \dot{\text{Im}}(a).$$

For $\rho < \tau$ set

$$g(\rho) := \{ v < \sigma : (\exists q \leq p) q \Vdash a(\dot{\rho}) \dot{=} \dot{v} \}.$$

By Forcing Definability and $\mathfrak{m}$ -Comprehension, $g \in \mathfrak{m}$ and we will show that

$$(13) \quad \forall \rho < \tau [f(\rho) \in g(\rho) \wedge (g(\rho) \text{ is countable})^{(\mathfrak{m})}].$$

Then

$$\sigma \dot{\subseteq} \{ f(\rho) : \rho < \tau \} \dot{\subseteq} \bigcup \{ g(\rho) : \rho < \tau \},$$

so $|\sigma|^{\mathfrak{m}} \leq \aleph_0 \dot{\times}_c |\tau|^{\mathfrak{m}} \dot{=} |\tau|^{\mathfrak{m}}$. Hence also $\neg(\sigma \text{ is a cardinal})^{(\mathfrak{m})}$.

To establish (13), note first that if $f(\rho) \dot{=} v$, then by Forcing Completeness for some $r \in G$, $r \Vdash a(\dot{\rho}) \dot{=} \dot{v}$, and by coherence of filters there exists $q \in G$ such that $q \leq p \wedge q \leq r$, so $v \in g(\rho)$. For the second part of (13), by $\text{AC}^{(\mathfrak{m})}$, for any $\rho < \tau$ there exists a function $v \mapsto q_v$ such that

$$\forall v \in g(\rho) [q_v \leq p \wedge q_v \Vdash a(\dot{\rho}) \dot{=} \dot{v}].$$

The set $Q := \{ q_v : v \in g(\rho) \}$ is an antichain in $\mathbb{P}$, since if q_v and $q_{v'}$ have a common extension r , then there exists a $\mathbb{D}_{\mathfrak{m}}$ -generic filter H with $r \in H$, hence also $q_v, q_{v'} \in H$, so $v \dot{=} a_H(\rho) \dot{=} v'$. Hence, by the hypothesis $(Q \text{ is countable})^{(\mathfrak{m})}$, so also $(g(\rho) \text{ is countable})^{(\mathfrak{m})}$. $\square$

Theorem 6.8.40 *For any $u, v \in \mathfrak{m}$, if $(v \text{ is countable})^{(\mathfrak{m})}$, then for any $\mathbb{D}_{\mathfrak{m}}^{\mathbb{R}\mathbb{F}}(u, v)$ -generic filter G , $\mathfrak{m}$ and $\mathfrak{m}[G]$ agree on cardinals.*

Proof. Immediate from the two preceding results. $\square$

Corollary 6.8.41 *For any $\mathbb{D}_{\mathfrak{m}}^{\mathbb{R}\mathbb{F}}(\aleph_2^{\mathfrak{m}} \dot{\times} \dot{\omega}, \dot{2})$ -generic filter G , $\mathfrak{m}$ and $\mathfrak{m}[G]$ agree on cardinals. Hence $(\neg \text{CH})^{(\mathfrak{m}[G])}$. $\square$*

Collateral Reading. Two standard references for forcing and generic extensions are [Jec] and [Kun]. Both provide extensions of the method far beyond what is covered here.

The following series of exercises develops some properties of the relation $p \Vdash \phi$ which are technically never needed. However, they provide the motivation for the definition of the relation $p \Vdash \phi$ in the next section. Of course, since some of them rely on the Forcing Completeness Theorem, which will be proved in the next section, they cannot actually be used there.

Exercises

Exercise 6.8.42 Show that for any $a, b \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathbf{m}$ and any $p \in P$,
 $p \Vdash a \dot{=} b$ iff

$$\forall c (\forall q \leq p) [c \dot{\in}_q a \rightarrow q \Vdash c \dot{\in} b] \wedge \forall d (\forall q \leq p) [d \dot{\in}_q b \rightarrow q \Vdash d \dot{\in} a].$$

Exercise 6.8.43 Show that for any $b, c \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathbf{m}$ and any $p \in P$,

$$p \Vdash c \dot{\in} b \leftrightarrow (\forall q \leq p)(\exists r \leq q) \exists d [d \dot{\in}_r b \wedge r \Vdash c \dot{=} d].$$

Exercise 6.8.44 Show that for any $a, b \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathbf{m}$ and any $p \in P$,

$$\begin{aligned} p \Vdash a \dot{=} b &\leftrightarrow \forall c (\forall q \leq p)(\exists r \leq q) (c \dot{\in}_r a \rightarrow \exists d [d \dot{\in}_r b \wedge r \Vdash c \dot{=} d]) \\ &\wedge \forall d (\forall q \leq p)(\exists r \leq q) (d \dot{\in}_r b \rightarrow \exists c [c \dot{\in}_r a \wedge r \Vdash c \dot{=} d]). \end{aligned}$$

Hint: First combine the results of the two preceding exercises. Then as an intermediate step show that

$$\begin{aligned} p \Vdash a \dot{=} b &\leftrightarrow \forall c (\forall q \leq p) (c \dot{\in}_q a \rightarrow (\exists r \leq q) \exists d [d \dot{\in}_r b \wedge r \Vdash c \dot{=} d]) \\ &\wedge \forall d (\forall q \leq p) (d \dot{\in}_q b \rightarrow (\exists r \leq q) \exists c [c \dot{\in}_r a \wedge r \Vdash c \dot{=} d]). \end{aligned}$$

Exercise 6.8.45 Show that for any $a, b \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathbf{m}$ and any $p \in P$,
 $p \Vdash a \dot{=} b$ iff

$$(\forall c \in \dot{\text{Dom}}(a) \dot{\cup} \dot{\text{Dom}}(b)) (\forall q \leq p) [q \Vdash c \dot{\in} a \leftrightarrow q \Vdash c \dot{\in} b].$$

Exercise 6.8.46 Show that for any formula $\phi(x)$, any $p \in P$ and any $a \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathbf{m}$,

$$p \Vdash \neg \phi(a) \leftrightarrow \neg (\exists q \leq p) q \Vdash \phi(a).$$

Exercise 6.8.47 Show that for any formulas $\phi(x)$ and $\psi(x)$, any $p \in P$ and any $a \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathbf{m}$,

$$p \Vdash \phi(a) \vee \psi(a) \leftrightarrow (\forall q \leq p)(\exists r \leq q) [r \Vdash \phi(a) \vee r \Vdash \psi(a)].$$

Exercise 6.8.48 Show that for any formula $\phi(x, y)$, any $p \in P$ and any $a \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathbf{m}$,

$$p \Vdash \exists y \phi(a, y) \leftrightarrow (\forall q \leq p)(\exists r \leq q)(\exists b \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathbf{m}) r \Vdash \phi(a, b).$$

Exercise 6.8.49 Improve the characterization of the preceding exercise by showing that

$$p \Vdash \exists y \phi(a, y) \leftrightarrow (\exists b \in \text{Na}^{\mathbb{P}} \dot{\cap} \mathfrak{m}) p \Vdash \phi(a, b).$$

Hint: Fix $p \in P$ and ϕ , and set

$$Q := \dot{\bigcup} \{ q \leq p : (\exists b \in \text{Na}^{\mathbb{P}} \dot{\cap} \mathfrak{m}) q \Vdash \phi(a, b) \}.$$

Using AC, fix a mapping $q \mapsto b^q$ such that $(\forall q \in Q) q \Vdash \phi(a, b^q)$. For any $A \subseteq Q$, let

$$b^A := \dot{\bigcup} \{ \langle c, r \rangle : (\exists q \in A) [r \leq q \wedge c \in \text{Dom}(b^q) \wedge r \Vdash c \in b^q] \}.$$

Show

- (i) if $A \subseteq Q$ is an antichain, then $(\forall q \in A) q \Vdash b^A \in b^q$, hence $(\forall q \in A) q \Vdash \phi(a, b^A)$;
- (ii) if $A \subseteq Q$ is a maximal antichain, then

$$p \Vdash \exists y \phi(a, y) \rightarrow p \Vdash \phi(a, b^A).$$

Exercise 6.8.50 Show that for any formula $\phi(x, y)$, any $p \in P$ and any $a \in \text{Na}^{\mathbb{P}} \dot{\cap} \mathfrak{m}$,

$$p \Vdash \forall y \phi(a, y) \leftrightarrow (\forall b \in \text{Na}^{\mathbb{P}} \dot{\cap} \mathfrak{m}) p \Vdash \phi(a, b).$$

Exercise 6.8.51 Give a direct proof of Comprehension^(m[G]) by showing that for every (unary) class operation F , the operation $F^\cap(x, u) := F(x) \cap u$ is name bounded.

Exercise 6.8.52 Suppose that $\mathcal{B} \dot{=} \langle B, \oplus, \otimes, \bar{}, \mathbb{O}, \mathbb{I} \rangle \in \mathfrak{m}$ and

$$(\mathcal{B} \text{ is a complete Boolean algebra})^{(m)}$$

(Definition 3.3.41). Then $\mathbb{B} := \dot{\bigcup} \langle B \setminus \{\mathbb{O}\}, \leq, \mathbb{I} \rangle$ (cf. [Section 1.7](#)) is a n.o.f. in $\mathfrak{m}$. For $G \subseteq B \setminus \{\mathbb{O}\}$ the two definitions 6.8.4 and 1.7.9 of filter agree. By Forcing Definability, for any $L_\in$ -formula $\phi(x_0, \dots, x_{k-1})$ and any $a_0, \dots, a_{k-1} \in \text{Na}^{\mathbb{P}}$,

$$\llbracket \phi(a_0, \dots, a_{k-1}) \rrbracket := \sum \dot{\bigcup} \{ p \in B : p \Vdash \phi(a_0, \dots, a_{k-1}) \} \in B$$

is well-defined and called the $\mathbb{B}$ -*truth value* of $\phi(a_0, \dots, a_{k-1})$. Prove that for any ϕ, ψ and $p \in B$,

- (i) $\llbracket \phi \rrbracket \doteq \mathbb{I} \leftrightarrow \phi$ is true in every $\mathfrak{m}[G]$;
- (ii) $\llbracket \phi \rrbracket \doteq \mathbb{O} \leftrightarrow \phi$ is false in every $\mathfrak{m}[G]$;
- (iii) $p \Vdash \phi \leftrightarrow p \leq \llbracket \phi \rrbracket$;
- (iv) $\llbracket \neg \phi \rrbracket \doteq \overline{\llbracket \phi \rrbracket}$;
- (v) $\llbracket \phi \wedge \psi \rrbracket \doteq \llbracket \phi \rrbracket \otimes \llbracket \psi \rrbracket$;
- (vi) $\llbracket \phi \vee \psi \rrbracket \doteq \llbracket \phi \rrbracket \oplus \llbracket \psi \rrbracket$;
- (vii) $\llbracket \exists x \phi \rrbracket \doteq \sum \{ \llbracket \phi(a) \rrbracket : a \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathfrak{m} \}$.

Boolean truth values are in some ways better behaved than forcing and can be used for an alternative presentation of the methods of this and the following section.

6.9 Forcing

The main tasks of this section are the proofs of the Forcing Definability and Completeness Theorems from the preceding section. As in the preceding section we fix a notion of forcing $\mathbb{P} \in \mathfrak{m}$ and work generally in $\mathbf{ZFL}_{\mathfrak{m}}$. We shall deduce both theorems from the

Forcing Theorem 6.9.1 *For each $L_{\in}$ -formula $\phi(x_0, \dots, x_{k-1})$, there exists an $L_{\in}$ -formula $p \Vdash \phi$ such that for all $a_0, \dots, a_{k-1} \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathfrak{m}$ and any $\mathbb{D}_{\mathfrak{m}}$ -generic filter G ,*

$$\phi^{(\mathfrak{m}[G])}((a_0)_G, \dots, (a_{k-1})_G) \leftrightarrow (\exists p \in G)(p \Vdash \phi)^{(\mathfrak{m})}(a_0, \dots, a_{k-1}).$$

The definition of $p \Vdash \phi$ will be modeled on the properties of the formula $p \Vdash \phi$ developed in the series of Exercises 6.8.42–6.8.50, and we recommend strongly to the reader to work at least some of these exercises out along with the reading of this section. As a very fine technical point we note that as $\Vdash$ is formally defined as a mapping from $L_{\in}$ -formulas to $L_{\in}$ -formulas, then by our usual conventions $\Vdash$ should be a mapping from $\mathbf{Fm}(\dot{L}_{\in})$ to $\mathbf{Fm}(\dot{L}_{\in})$ and we should write $p \Vdash \dot{\phi}(a_0, \dots, a_{k-1})$. The definition given below may indeed be so interpreted, but we have no need for $\Vdash$ applied to non-standard formulas and we use the simpler notation. Of course, $\Vdash$ deviates from the standard dot-convention also in that the definition is not a straightforward translation of the definition of $\Vdash$ into $L_{\in}$.

We begin with the definition of $p \Vdash a \doteq b$. This will be given formally by the version of set recursion of Exercise 6.2.66: $p \Vdash a \doteq b$ is described

in terms of $p \Vdash c \dot{=} d$ for $c \dot{\in} \text{Dom}(a) \dot{\subseteq} \text{TC}(a)$ and $d \dot{\in} \text{Dom}(b) \dot{\subseteq} \text{TC}(b)$. Still more formally, we are defining recursively a set operation $F(a, b) \doteq \{p : p \Vdash a \dot{=} b\}$. With this in mind we introduce the following notation:

$$q \Vdash^* c \dot{=} b \quad \text{is an abbreviation for} \quad \exists d (d \dot{\in}_q b \wedge q \Vdash c \dot{=} d).$$

Recall from the discussion following Definition 6.8.24 that for any $\mathbb{D}_m$ -generic filter G

$$(1) \quad q \dot{\in} G \wedge c \dot{\in}_q a \rightarrow c_G \dot{\in} a_G;$$

we shall see in (5) below that under suitable hypotheses the relation $\Vdash^*$ also has a natural meaning.

Definition 6.9.2 For any $p \dot{\in} P$ and $D \dot{\subseteq} P$, D is **dense below** p iff $(\forall q \leq p)(\exists r \leq q) r \dot{\in} D$.

Proposition 6.9.3 For any $\mathbb{D}_m$ -generic filter G and any $D \dot{\subseteq} P$ in $\mathfrak{m}$,

(i) for each $p \dot{\in} P$

$$p \dot{\in} G \wedge D \text{ is dense below } p \rightarrow \exists r (r \leq p \wedge r \dot{\in} D \dot{\cap} G);$$

(ii) if D is **closed downward**, $q \leq p \dot{\in} D \rightarrow q \dot{\in} D$, then

$$(\exists p \dot{\in} G) D \text{ is dense below } p \leftrightarrow D \dot{\cap} G \neq \emptyset.$$

Proof. Given G and D as in the hypothesis, towards (i) for any $p \dot{\in} P$ let

$$E := \{s : (\exists r \dot{\in} D)[s \leq r \leq p]\} \cup \{s : (\forall r \dot{\in} D)[r \leq p \rightarrow r \not\leq s]\}.$$

Clearly $E \dot{\in} \mathfrak{m}$ and E is dense: for any q , either $q \dot{\in} E$ itself or (by the second clause) there exists $r \dot{\in} D$ such that $r \leq p$ and $r \leq q$, so $r \dot{\in} E$ by the first clause. Hence $E \dot{\in} \mathbb{D}_m$ and therefore $E \dot{\cap} G \neq \emptyset$. If

$$(\exists s \dot{\in} G)(\exists r \dot{\in} D) s \leq r \leq p,$$

then also $r \dot{\in} G$ as desired (since filters are closed upward). Otherwise,

$$(\exists s \dot{\in} G)(\forall r \dot{\in} D)[r \leq p \rightarrow r \not\leq s].$$

Fix such an s and by coherence of filters a $q \dot{\in} G$ such that $q \leq p \wedge q \leq s$. Since D is dense below p , there exists $r \dot{\in} D$ with $r \leq q$. But then $r \leq p \wedge r \leq s$, a contradiction. Now (ii) is immediate. $\square$

Definition 6.9.4 For any $p \in P$ and $a, b \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathbf{m}$,

$$p \Vdash a \dot{=} b \leftrightarrow \forall c \left(\left\{ r : c \dot{\in}_r a \rightarrow r \Vdash^* c \dot{\in} b \right\} \text{ is dense below } p \right) \\ \wedge \forall d \left(\left\{ r : d \dot{\in}_r b \rightarrow r \Vdash^* d \dot{\in} a \right\} \text{ is dense below } p \right).$$

Note that once we eliminate the definitions of $\Vdash^*$ and ‘dense below’ this is exactly the relationship of Exercise 6.8.44 with $\Vdash$ in place of $\Vdash$.

First we have two simple observations which rely only on the definition of ‘dense below’ and whose proof we leave to the reader:

$$(2) \quad p \Vdash a \dot{=} b \wedge q \leq p \rightarrow q \Vdash a \dot{=} b;$$

$$(3) \quad q \Vdash^* c \dot{\in} b \wedge r \leq q \rightarrow r \Vdash^* c \dot{\in} b.$$

We can now establish the first part of the Forcing Theorem for equality formulas. For this part because of absoluteness, no relativization to $\mathbf{m}$ is required.

Proposition 6.9.5 For all $a, b \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} \mathbf{m}$ and any $\mathbb{D}_{\mathbf{m}}$ -generic filter G ,

$$a_G \dot{=} b_G \leftrightarrow (\exists p \in G) p \Vdash a \dot{=} b.$$

Proof. We proceed by the version of set induction described in Exercise 6.2.65; fix a, b and G as in the hypothesis, and assume as induction hypothesis that for all $c \in \mathbf{Dom}(a)$ and $d \in \mathbf{Dom}(b)$,

$$(4) \quad c_G \dot{=} d_G \leftrightarrow (\exists q \in G) q \Vdash c \dot{=} d.$$

We first establish the following facts:

$$(5) \quad (\forall c \in \mathbf{Dom}(a)) [c_G \dot{\in} b_G \leftrightarrow (\exists r \in G) r \Vdash^* c \dot{\in} b];$$

$$(6) \quad q \in G \wedge a_G \dot{\subseteq} b_G \rightarrow (\forall c \dot{\in}_q a) (\exists r \leq q) r \Vdash^* c \dot{\in} b.$$

For (5), suppose first that $c_G \dot{\in} b_G$ so for some d and some $s \in G$, $c_G \dot{=} d_G$ and $\langle d, s \rangle \dot{\in} b$. By the induction hypothesis (4) there exists $q \in G$ such that $q \Vdash c \dot{=} d$, and by coherence of filters there exists $r \in G$ such that $r \leq q \wedge r \leq s$. Then $d \dot{\in}_r b$, and by (2) $r \Vdash c \dot{=} d$ so $r \Vdash^* c \dot{\in} b$. Conversely, if for some $r \in G$, $r \Vdash^* c \dot{\in} b$, choose d such that $d \dot{\in}_r b$ and $r \Vdash c \dot{=} d$. Then by (1) $d_G \dot{\in} b_G$ and by (4) $c_G \dot{=} d_G$, so $c_G \dot{\in} b_G$.

For (6), fix $q \in G$ and assume $a_G \dot{\subseteq} b_G$. Then if $c \dot{\in}_q a$, by (1) $c_G \dot{\in} a_G \dot{\subseteq} b_G$ so by (5) there exists $s \in G$ such that $s \Vdash^* c \dot{\in} b$. By coherence there exists $r \in G$ such that $r \leq q \wedge r \leq s$, and by (3), $r \Vdash^* c \dot{\in} b$.

Now for the main equivalence, suppose first that for some $p \in G$, $p \Vdash a \dot{=} b$, and towards establishing that $a_G \dot{\subseteq} b_G$, fix $x \in a_G$. Then there is $\langle c, s \rangle \in a$ with $s \in G$ such that $x \dot{=} c_G$. By coherence there exists $q \in G$ such that $q \leq p \wedge q \leq s$, and by (2) also $q \Vdash a \dot{=} b$. Hence by the definition of $\Vdash$,

$$\{ r : c \dot{\in}_r a \rightarrow r \Vdash^* c \dot{\in} b \} \text{ is dense below } q,$$

so by Proposition 6.9.3(i) there exists $r \leq q$ with $r \in G$ and $c \dot{\in}_r a \rightarrow r \Vdash^* c \dot{\in} b$. But $r \leq s$ so $c \dot{\in}_r a$ and thus $r \Vdash^* c \dot{\in} b$, hence $x \dot{=} c_G \dot{\in} b_G$ by (5). Thus $a_G \dot{\subseteq} b_G$; the converse inclusion follows by a parallel argument.

Now suppose that $a_G \dot{=} b_G$, and let

$$D := \{ q : q \Vdash a \dot{=} b \vee (\exists c \dot{\in}_q a)(\forall r \leq q) \neg r \Vdash^* c \dot{\in} b \\ \vee (\exists d \dot{\in}_q b)(\forall r \leq q) \neg r \Vdash^* d \dot{\in} a \}.$$

We show first that D is dense; fix $p \in P$. If $p \Vdash a \dot{=} b$, then $p \in D$ itself. Otherwise, either

$$\begin{aligned} \exists c \{ r : c \dot{\in}_r a \rightarrow r \Vdash^* c \dot{\in} b \} & \text{ is not dense below } p \quad \text{or} \\ \exists d \{ r : d \dot{\in}_r b \rightarrow r \Vdash^* d \dot{\in} a \} & \text{ is not dense below } p. \end{aligned}$$

In the first case, for some c and some $q \leq p$, for every $r \leq q$, $c \dot{\in}_r a \wedge \neg r \Vdash^* c \dot{\in} b$. In particular, $c \dot{\in}_q a \wedge (\forall r \leq q) \neg r \Vdash^* c \dot{\in} b$ so $q \in D$. The second case is parallel.

Now $D \in \mathbb{D}_m$, so there exists $q \in D \dot{\cap} G$. By (6), q cannot satisfy the second or third clauses of D , so $q \Vdash a \dot{=} b$. $\square$

Next we define the $\Vdash$ relation for the other type of atomic formula, membership statements. Note that now we may take the relation $r \Vdash^* c \dot{\in} b$ as defined for all r , c and b and (5) and (6) universally valid. Thus the following is not a recursive definition. Note also the parallel with Exercise 6.8.43.

Definition 6.9.6 For any $b, c \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} m$ and any $p \in P$,

$$p \Vdash c \dot{\in} b \leftrightarrow \{ r : r \Vdash^* c \dot{\in} b \} \text{ is dense below } p.$$

Proposition 6.9.7 For all $b, c \in \mathbf{Na}^{\mathbb{P}} \dot{\cap} m$ and any $\mathbb{D}_m$ -generic filter G ,

$$c_G \dot{\in} b_G \leftrightarrow (\exists p \in G) p \Vdash c \dot{\in} b.$$

Proof. The result follows from the following chain of equivalences:

$$\begin{aligned}
 c_G \dot{\in} b_G &\leftrightarrow (\exists r \dot{\in} G) r \Vdash^* c \dot{\in} b \\
 &\leftrightarrow (\exists p \dot{\in} G) \dot{\{ r : r \Vdash^* c \dot{\in} b \}} \text{ is dense below } p \\
 &\leftrightarrow (\exists p \dot{\in} G) p \Vdash c \dot{\in} b.
 \end{aligned}$$

The first equivalence is by (5), the second by Proposition 6.9.3(ii) and the third by the definition. $\square$

This takes care of the cases of the Forcing Theorem for atomic formulas, and we proceed to extend the definitions and results to all L_∞ -formulas by formula recursion and induction. For clarity we write the formulas with a single free variable x .

Definition 6.9.8 For all L_∞ -formulas $\phi(x)$, $p \dot{\in} P$ and $a \dot{\in} \mathbf{Na}^\mathbb{P}$,

$$\begin{aligned}
 p \Vdash \neg \phi(a) &:\leftrightarrow \neg (\exists q \leq p) q \Vdash \phi(a); \\
 p \Vdash \phi(a) \vee \psi(a) &:\leftrightarrow \dot{\{ r : r \Vdash \phi(a) \vee r \Vdash \psi(a) \}} \text{ is dense below } p; \\
 p \Vdash \exists y \phi(a, y) &:\leftrightarrow \dot{\{ r : (\exists b \dot{\in} \mathbf{Na}^\mathbb{P}) r \Vdash \phi(a, b) \}} \text{ is dense below } p.
 \end{aligned}$$

Of course it is in the last part of this definition that relativization to $\mathfrak{m}$ has its effect.

Lemma 6.9.9 For all L_∞ -formulas $\phi(x)$ and $a \dot{\in} \mathbf{Na}^\mathbb{P} \dot{\cap} \mathfrak{m}$ and any $\mathbb{D}_\mathfrak{m}$ -generic filter G ,

$$(\exists p \dot{\in} G) (p \Vdash \neg \phi(a))^{(\mathfrak{m})} \leftrightarrow \neg (\exists p \dot{\in} G) (p \Vdash \phi(a))^{(\mathfrak{m})}.$$

Proof. It follows from the definition of $p \Vdash \neg \phi(a)$ that the set

$$D := \dot{\{ p : p \Vdash \phi(a) \vee p \Vdash \neg \phi(a) \}}$$

is dense. Since this is a theorem of ZFC, it holds also in $\mathfrak{m}$ — that is, also

$$D^\mathfrak{m} := \dot{\{ r : (p \Vdash \phi(a))^{(\mathfrak{m})} \vee (p \Vdash \neg \phi(a))^{(\mathfrak{m})} \}}$$

is dense. Hence $D^\mathfrak{m} \dot{\cap} G \neq \emptyset$, so

$$(\exists p \dot{\in} G) (p \Vdash \phi(a))^{(\mathfrak{m})} \vee (\exists p \dot{\in} G) (p \Vdash \neg \phi(a))^{(\mathfrak{m})}.$$

But also these cannot both be the case, since by coherence this would imply the existence of $q \dot{\in} G$ such that

$$(q \Vdash \phi(a))^{(\mathfrak{m})} \wedge (q \Vdash \neg \phi(a))^{(\mathfrak{m})}$$

in immediate contradiction with the definition. $\square$

Lemma 6.9.10 *For each $L_{\in}$ -formula $\phi(x)$ and any $a \in \mathbf{Na}^{\mathbb{P}}$ and $p \in P$, the following are equivalent:*

- (i) $p \Vdash \phi(a)$;
- (ii) $(\forall r \leq p) r \Vdash \phi(a)$;
- (iii) $\{ r : r \Vdash \phi(a) \}$ is dense below p .

Proof. That (ii) implies both (i) and (iii) is immediate from the definitions. For all formulas ϕ not of the form $\neg \psi$, the condition for $p \Vdash \phi(a)$ involves only statements of the form “ D is dense below p ”, so for such ϕ the implications (iii) $\rightarrow$ (i) $\rightarrow$ (ii) follow from the easily verified fact that for any $D \subseteq P$,

$$\begin{aligned} \{ r : D \text{ is dense below } r \} \text{ is dense below } p &\rightarrow D \text{ is dense below } p \\ &\rightarrow (\forall r \leq p) D \text{ is dense below } r. \end{aligned}$$

However, to deal with negated formulas we must still carry out a formula induction. All of the steps for non-negated formulas are as above and make no use of the induction hypothesis. For the negation step we have for (iii) $\rightarrow$ (i)

$$\begin{aligned} \{ r : r \Vdash \neg \psi(a) \} \text{ is dense below } p &\rightarrow (\forall q \leq p)(\exists r \leq q) \neg (\exists s \leq r) s \Vdash \psi(a) \\ &\rightarrow (\forall q \leq p)(\exists r \leq q) \neg r \Vdash \psi(a) \\ &\rightarrow (\forall q \leq p) \neg q \Vdash \psi(a) \\ &\rightarrow p \Vdash \neg \psi(a), \end{aligned}$$

where the third implication uses (i) $\rightarrow$ (ii) for ψ . Similarly, for (i) $\rightarrow$ (ii),

$$\begin{aligned} (\exists r \leq p) \neg r \Vdash \neg \psi(a) &\rightarrow (\exists r \leq p)(\exists q \leq r) q \Vdash \psi(a) \\ &\rightarrow (\exists q \leq p) q \Vdash \psi(a) \\ &\rightarrow \neg p \Vdash \neg \psi(a). \quad \square \end{aligned}$$

Corollary 6.9.11 *For each $L_{\in}$ -formula $\phi(x)$ and any $a \in \mathbf{Na}^{\mathbb{P}}$ and $p \in P$,*

$$p \Vdash \phi(a) \rightarrow \neg (\exists q \leq p) q \Vdash \neg \phi(a).$$

Proof. By the lemma we have

$$\begin{aligned} (\exists q \leq p) q \Vdash \neg \phi(a) &\rightarrow (\exists q \leq p)(\forall r \leq q) \neg r \Vdash \phi(a) \\ &\rightarrow \neg (\forall r \leq p) r \Vdash \phi \rightarrow \neg p \Vdash \phi. \quad \square \end{aligned}$$

Forcing Theorem 6.9.12 For each $L_\in$ -formula $\phi(x)$, all $a \in \mathbf{Na}^\mathbb{P} \dot{\cap} \mathfrak{m}$ and any $\mathbb{D}_\mathfrak{m}$ -generic filter G ,

$$\phi^{(\mathfrak{m}[G])}(a_G) \leftrightarrow (\exists p \in G)(p \Vdash \phi(a))^{(\mathfrak{m})}.$$

Proof. As noted we have established this for atomic formulas, and we proceed by formula induction. The induction step for $\neg$ follows from Lemma 6.9.9:

$$\begin{aligned} (\neg \phi(a_G))^{(\mathfrak{m}[G])} &\leftrightarrow \neg (\phi(a_G))^{(\mathfrak{m}[G])} \\ &\leftrightarrow \neg (\exists p \in G)(p \Vdash \phi(a))^{(\mathfrak{m})} \\ &\leftrightarrow (\exists p \in G)(p \Vdash \neg \phi(a))^{(\mathfrak{m})}. \end{aligned}$$

For the $\vee$ step, given $\phi(x)$, $\psi(x)$ and a , let

$$D := \{ r : r \Vdash \phi(a) \} \quad \text{and} \quad E := \{ r : r \Vdash \psi(a) \}.$$

By Lemma 6.9.10 D and E are closed downward, so by Proposition 6.9.3(ii) and the induction hypothesis

$$\begin{aligned} \phi^{(\mathfrak{m}[G])}(a_G) \vee \psi^{(\mathfrak{m}[G])}(a_G) &\leftrightarrow D^\mathfrak{m} \dot{\cap} G \neq \emptyset \vee E^\mathfrak{m} \dot{\cap} G \neq \emptyset \\ &\leftrightarrow (D^\mathfrak{m} \dot{\cup} E^\mathfrak{m}) \dot{\cap} G \neq \emptyset \\ &\leftrightarrow (\exists p \in G) (D^\mathfrak{m} \dot{\cup} E^\mathfrak{m}) \text{ is dense below } p \\ &\leftrightarrow (\exists p \in G)(p \Vdash \phi(a) \vee \psi(a))^{(\mathfrak{m})}. \end{aligned}$$

Finally, for the $\exists$ step, given $\phi(x, y)$ and a , for each $b \in \mathbf{Na}^\mathbb{P}$ let

$$D_b := \{ r : r \Vdash \phi(a, b) \}.$$

Then similarly we have

$$\begin{aligned} (\exists y \phi(a_G, y))^{(\mathfrak{m}[G])} &\leftrightarrow (\exists b \in \mathbf{Na}^\mathbb{P} \dot{\cap} \mathfrak{m}) \phi^{(\mathfrak{m}[G])}(a_G, b_G) \\ &\leftrightarrow (\exists b \in \mathbf{Na}^\mathbb{P} \dot{\cap} \mathfrak{m}) D_b^\mathfrak{m} \dot{\cap} G \neq \emptyset \\ &\leftrightarrow \bigcup \{ D_b^\mathfrak{m} : b \in \mathbf{Na}^\mathbb{P} \dot{\cap} \mathfrak{m} \} \dot{\cap} G \neq \emptyset \\ &\leftrightarrow (\exists p \in G) \bigcup \{ D_b^\mathfrak{m} : b \in \mathbf{Na}^\mathbb{P} \dot{\cap} \mathfrak{m} \} \text{ is dense below } p \\ &\leftrightarrow (\exists p \in G) (p \Vdash \exists y \phi(a, y))^{(\mathfrak{m})}. \quad \square \end{aligned}$$

It remains to conclude

Forcing Definability Theorem 6.9.13 *For each $L_\in$ -formula $\phi(x)$, all p and all $a \in \mathcal{N}a^{\mathbb{P}} \dot{\cap} \mathfrak{m}$,*

$$\text{ZFL}_{\mathfrak{m}} \vdash p \Vdash \phi(a) \leftrightarrow (p \Vdash \phi(a))^{(\mathfrak{m})}.$$

Proof. Arguing in $\text{ZFL}_{\mathfrak{m}}$, suppose first that $(p \Vdash \phi(a))^{(\mathfrak{m})}$. Then by the implication $(\leftarrow)$ of the Forcing Theorem, for any $\mathbb{D}_{\mathfrak{m}}$ -generic filter G , if $p \in G$, then $\phi^{(\mathfrak{m}[G])}(a_G)$ — that is, $p \Vdash \phi(a)$.

For the converse, assume that $p \Vdash \phi(a)$. By Lemma 6.9.10 (interpreted in $\mathfrak{m}$), it suffices to show that $\left\{ r : (r \Vdash \phi(a))^{(\mathfrak{m})} \right\}$ is dense below p . If this is not the case, then for some $q \leq p$ there is no $r \leq q$ such that $(r \Vdash \phi(a))^{(\mathfrak{m})}$ and hence $(q \Vdash \neg \phi(a))^{(\mathfrak{m})}$, so by the first part, $q \Vdash \neg \phi(a)$. But $p \Vdash \phi(a) \rightarrow q \Vdash \phi(a)$, so we have a contradiction. $\square$

Forcing Completeness Theorem 6.9.14 *For each $L_\in$ -formula $\phi(x)$, any $a \in \mathcal{N}a^{\mathbb{P}} \dot{\cap} \mathfrak{m}$ and any $\mathbb{D}_{\mathfrak{m}}$ -generic filter G ,*

$$\text{ZFL}_{\mathfrak{m}} \vdash \phi^{(\mathfrak{m}[G])}(a_G) \rightarrow (\exists p \in G) p \Vdash \phi(a).$$

Proof. This is now immediate from the Forcing and Forcing Definability Theorems. $\square$

The methods of forcing and generic extensions are very flexible and have been adapted to the solution of many set-theoretic problems. We can only hint here at the richness of this field and recommend to the interested reader the two books [Jec] and [Kun]. First, we mention that by closer analysis of the argument that $(\text{Powerset})^{(\mathfrak{m}[G])}$, it can be seen that for any $\mathbb{D}_{\mathfrak{m}}^{\mathbb{R}\mathbb{F}}(\aleph_2^{\mathfrak{m}} \dot{\times} \dot{\omega}, \dot{2})$ -generic filter actually $(2^{\aleph_0} \dot{=} \aleph_2^{(\mathfrak{m}[G])})$. Therefore we have the implication

$$\text{ZF is consistent} \implies \text{ZFC} + 2^{\aleph_0} \dot{=} \aleph_2 \text{ is consistent.}$$

Furthermore, there is nothing special about $\aleph_2$ here; it can be replaced by $\aleph_3$, $\aleph_{100}$, $\aleph_{\aleph_1}$ and most other possibilities. The only restriction is given by König's Theorem (Exercise 6.4.47); since $\text{cf}(2^{\aleph_0}) \dot{\geq} \aleph_0$, $2^{\aleph_0}$ cannot have value $\aleph_{\dot{\omega}}$ or generally any $\aleph_{\sigma}$ for σ an ordinal of cofinality $\dot{\omega}$.

It is then natural to ask about the consistency of statements like $2^{\aleph_1} \dot{=} \aleph_7$, either by itself or together with some of the statements above concerning $2^{\aleph_0}$. One obvious limitation is that always $2^{\aleph_0} \dot{\leq} 2^{\aleph_1}$. The appropriate notion of forcing here is

$$\mathbb{P} := \left\{ p \in \mathfrak{m} : p \text{ is a function with } \text{Dom}(p) \dot{=} \aleph_7^{\mathfrak{m}} \dot{\times} \aleph_1^{\mathfrak{m}} \right. \\ \left. \wedge \text{Im}(p) \dot{\subseteq} 2 \wedge |p|^{\mathfrak{m}} < \aleph_1^{\mathfrak{m}} \right\}.$$

Note that since the forcing conditions are no longer finite, $\mathbb{P}$ is not absolute. If $\mathbb{D}$ is the set of $\mathbb{P}$ -dense sets in $\mathfrak{m}$ and G is $\mathbb{D}$ -generic, then $(2^{\aleph_1} \dot{=} \aleph_7)^{(\mathfrak{m}[G])}$. A new argument is needed here to show that $\mathfrak{m}$ and $\mathfrak{m}[G]$ agree on cardinals, since $\mathbb{P}$ does not have the c.c.c. It turns out that in this model, automatically $(2^{\aleph_0} \dot{=} \aleph_1)^{(\mathfrak{m}[G])}$, even without assuming $\text{CH}^{(\mathfrak{m})}$, so this also provides an alternative proof of the consistency of CH. Again by König's Theorem, allowable values for $2^{\aleph_1}$ are $\aleph_\sigma$ of cofinality greater than $\aleph_1$.

The next step is to construct a model for, say, $2^{\aleph_0} \dot{=} \aleph_3 \wedge 2^{\aleph_1} \dot{=} \aleph_7$. This requires two successive generic extensions, and some effort is required to verify that they don't interfere with each other. Still more complex techniques lead to models in which, for example, $(\forall n \in \omega)[2^{\aleph_n} \dot{=} \aleph_{n+3}]$ and further generalizations.

Of course, forcing methods have been used to construct models of many other kinds of mathematical statements. Perhaps the most important of these is $\neg \text{AC}$. Since as we saw above in Theorem 6.8.33, $\text{AC}^{(\mathfrak{m}[G])}$ holds under very general conditions, something different is required to construct $\mathfrak{n}$ such that $(\neg \text{AC})^{(\mathfrak{n})}$. The idea that works is to find a suitable notion of forcing $\mathbb{P}$ and show that for any $\mathbb{D}_{\mathfrak{m}}$ -generic filter G , there exists a model $\mathfrak{n}$ such that $\mathfrak{m} \subsetneq \mathfrak{n} \subsetneq \mathfrak{m}[G]$ and $(\neg \text{AC})^{(\mathfrak{n})}$. There are various methods for doing this, some of which also satisfy stronger assertions which imply $\neg \text{AC}$ — for example

$$(\mathbb{R} \text{ is not well-orderable})^{(\mathfrak{n})} \quad \text{or}$$

$$(\forall x [x \subseteq \mathbb{R} \rightarrow x \text{ is Lebesgue measurable}])^{(\mathfrak{n})}.$$

Many other assertions have been shown relatively consistent with ZF or ZFC by forcing methods (see the references cited above).

Exercises

Exercise 6.9.15 Give an alternative proof that

$$\text{ZFC is consistent} \implies \text{ZFC} \cup \{\text{CH}\} \text{ is consistent}$$

along the following lines. Work in the theory $\text{ZFC}_{\mathfrak{m}}$ described following Definition 6.8.2. Let

$$P := \left\{ p \in \mathfrak{m} : p \text{ is a function} \wedge (\text{Dom}(p) \text{ is a countable ordinal})^{(\mathfrak{m})} \right. \\ \left. \wedge \text{Im}(p) \subseteq \dot{\varnothing}(\dot{\omega}) \dot{\cap} \mathfrak{m} \right\}.$$

$\mathbb{P} := \langle P, \supseteq, \dot{\emptyset} \rangle$ is a n.o.f. in $\mathfrak{m}$. Show first that for any $\mathbb{D}_{\mathfrak{m}}$ -generic $\mathbb{P}$ -filter G , if $f_G := \bigcup G$, then

(i) f_G is a function $\wedge \mathsf{Dom}(f_G) \doteq \aleph_1^m \wedge \mathsf{Im}(f_G) \doteq \dot{\varphi}(\dot{\omega}) \dot{\cap} m$.

Since $\aleph_1^m \leq \aleph_1^{m[G]}$, to show $(\mathsf{CH})^{(m[G])}$ it suffices to show that $\dot{\varphi}(\dot{\omega}) \dot{\cap} m[G] \doteq \dot{\varphi}(\dot{\omega}) \dot{\cap} m$. Towards this, for any $a \in \mathsf{Na}^{\mathbb{P}} \dot{\cap} m$, let

$$D^a := \{ q \in P : (\exists^m x \subseteq \dot{\omega}) q \Vdash a \doteq \check{x} \}.$$

Now show

(ii) $q \Vdash a \subseteq \dot{\omega} \rightarrow (\forall n \in \dot{\omega})(\exists r \leq q)[r \Vdash n \in a \vee r \Vdash \neg n \in a]$;

(iii) $p \Vdash a \subseteq \dot{\omega} \rightarrow D^a$ is dense below p ;

and use (iii) to complete the proof.

6.10 Large cardinals

The discovery that the theories ZF and ZFC are incomplete in the practical sense that they don't settle natural mathematical questions such as GCH has led to a search for extensions of these theories — that is, for additional axioms. There are two main criteria for such axioms: they should agree with our intuitions about sets and they should have interesting consequences. Over the past forty years many axioms have been proposed, but none have gained anything like the acceptance of the axioms of ZF or ZFC . Many of the proposals satisfy the second criterion but all founder on the first; they are not seen as necessary or obvious properties of the intuitive concept of set.

We shall be able here to give only a brief introduction to this large and technically difficult field of inquiry and consider in this section a group of so-called **large cardinal axioms**. These are assertions of the form “there exists a cardinal number such that ...”, where ... is a property which implies, either immediately or after some effort, that the cardinal must be intuitively very large, well beyond any cardinal which can be directly defined, such as $\aleph_{\aleph_1}$. The second group of extensions considered in the following section involve various versions of the **Axiom of Determinacy**.

We shall focus here on the two notions of inaccessible and measurable cardinals. Many other notions which have been studied are generalizations of these. As usual when dealing with cardinals, we work in ZFC . We introduced some of the notions below in Definitions 6.4.40 and 6.4.41 and in the exercises to [Section 6.4](#), but for completeness we repeat the definitions here. Recall that letters $\kappa, \lambda, \dots$ always denote cardinals and λ^+ is the smallest cardinal larger than λ .

Definition 6.10.1 For any ordinals $\sigma \leq \rho$,

- (i) σ is **cofinal** with $\rho \leftrightarrow$ there exists a function $f : \sigma \rightarrow \rho$ such that $\rho = \bigcup \text{Im}(f)$;
- (ii) the **cofinality** $\text{cf}(\rho)$ of a limit ordinal ρ is the least ordinal σ cofinal with ρ .

Recall from Exercise 6.4.58 that $\text{cf}(\rho)$ is always a cardinal; hence the property of regularity in the next definition holds only for cardinals.

Definition 6.10.2 For any infinite cardinal κ ,

- (i) κ is a **regular cardinal** $\leftrightarrow \text{cf}(\kappa) = \kappa$; otherwise, κ is a **singular cardinal**;
- (ii) κ is a **limit cardinal** iff $\forall \lambda (\lambda < \kappa \rightarrow \lambda^+ < \kappa)$;
- (iii) κ is a **strong limit cardinal** $\leftrightarrow \forall \lambda (\lambda < \kappa \rightarrow 2^\lambda < \kappa)$;
- (iv) κ is **weakly inaccessible** $\leftrightarrow \kappa$ is a regular limit cardinal;
- (v) κ is **strongly inaccessible** $\leftrightarrow \kappa$ is a regular strong limit cardinal.

The descriptor “inaccessible” arises from the property that such a cardinal cannot be reached from smaller cardinals using certain operations. In fact, we see below that when $\kappa \geq \omega$ is strongly inaccessible, $\dot{V}_\kappa$ is a set model of ZF, so for any class operation F ,

$$\text{ZF} \vdash F \text{ is a set operation} \implies \text{ZF} \vdash F^{\dot{V}_\kappa} \text{ is a } \dot{V}_\kappa\text{-set operation,}$$

and hence κ , which is outside $\dot{V}_\kappa$, cannot be reached by any such operation from elements of $\dot{V}_\kappa$.

Easily ω is both weakly and strongly inaccessible (and $\dot{V}_\omega$ is a set model of $\text{ZF} - \text{Infinity}$ by Exercise 6.5.38), and indeed it is this fact that suggests that there might be other infinite cardinals with these properties. Statements asserting the existence of inaccessible (or other large) cardinals are sometimes called **generalized axioms of infinity**.

The following proposition shows that uncountable cardinals with any *one* of the properties regular, limit and strong limit are easy to come by, but these examples do not provide instances of either sort of inaccessibility for an uncountable cardinal. As the results below show, this is not accidental; unless ZFC is inconsistent, it cannot prove the existence of uncountable inaccessible cardinals.

Proposition 6.10.3 (ZFC) For any infinite cardinal κ ,

- (i) κ is a limit cardinal iff $\kappa = \aleph_\sigma$ for some limit ordinal σ ;
- (ii) for all ordinals σ , $\aleph_{\sigma+1}$ is regular;

(iii) for all $\kappa \dot{>} \aleph_0$,

$$\kappa \text{ is weakly inaccessible} \leftrightarrow \kappa \text{ is regular and } \kappa \dot{=} \aleph_\kappa;$$

(iv) there exists a strong limit cardinal $\lambda > \kappa$;

(v) if κ is strongly inaccessible, then κ is weakly inaccessible;

(vi) $\text{GCH} \rightarrow [\kappa \text{ is weakly inaccessible} \leftrightarrow \kappa \text{ is strongly inaccessible}]$.

Proof. Part (i) is immediate and (ii) follows from Corollary 6.4.38: if $\tau < \aleph_{\sigma+\omega}$ and $f : \tau \rightarrow \aleph_{\sigma+\omega}$, then $|\tau| \leq \aleph_\sigma$ and $(\forall v < \tau) |f(v)| \leq \aleph_\sigma$, so $|\bigcup \text{Im}(f)| \leq \aleph_\sigma$ and hence $\bigcup \text{Im}(f) < \aleph_{\sigma+\omega}$. (iii)($\leftarrow$) is immediate from (i), since every cardinal is a limit ordinal. For ($\rightarrow$), always $\tau \leq \aleph_\tau$; if $\kappa < \aleph_\kappa$ is a limit cardinal, then $\kappa \dot{=} \aleph_\sigma$ for some limit ordinal $\sigma < \kappa$ and $f(\tau) := \aleph_\tau$ for $\tau < \sigma$ is a function $f : \sigma \rightarrow \kappa$ such that $\kappa \dot{=} \bigcup \text{Im}(f)$. Thus $\text{cf}(\kappa) \leq \sigma < \kappa$, so κ is singular.

For (iv), let $\lambda_0 := \kappa$, $\lambda_{y+\omega} := 2^{\lambda_y}$ and $\lambda := \bigcup \{\lambda_y : y \in \omega\}$. Easily λ is a strong limit cardinal $> \kappa$. Parts (v) and (vi) are obvious from the definitions. $\square$

Thus the smallest uncountable limit cardinal is $\aleph_\omega$, which has cofinality ω , so is singular. Similarly, for any countable limit ordinal σ , $\text{cf}(\aleph_\sigma) \dot{=} \text{cf}(\sigma) \dot{=} \omega$, and for limit σ such that $\aleph_1 \leq \sigma < \aleph_2$, $\text{cf}(\aleph_\sigma) \dot{=} \aleph_1$, so again $\aleph_\sigma$ is singular. Since always $\sigma \leq \aleph_\sigma$ (by an easy induction), (iii) says that $\aleph_\sigma$ is weakly inaccessible only when the index σ “catches up” with the cardinal it is indexing *and* the result is regular. Without this last condition, we have the example

$$\kappa_0 \dot{=} \aleph_0, \quad \text{for all } y \in \omega, \quad \kappa_{\aleph_y} \dot{=} \aleph_{\kappa_y}, \quad \text{and} \quad \kappa \dot{=} \bigcup \{\kappa_y : y \in \omega\}.$$

Easily $\kappa \dot{=} \aleph_\kappa$, but by its very definition, $\text{cf}(\kappa) \dot{=} \omega$, so κ is singular.

We will need some alternative characterizations of regularity.

Proposition 6.10.4 (ZFC) *For any infinite cardinal κ , the following are equivalent:*

- (i) κ is regular;
- (ii) for any x and any function $f : x \rightarrow \kappa$,

$$|x| < \kappa \rightarrow (\exists \rho < \kappa) [\text{Im}(f) \subseteq \rho];$$

- (iii) for any x , $(|x| < \kappa \wedge (\forall v \in x) |v| < \kappa) \rightarrow \left| \bigcup x \right| < \kappa$.

Proof. For (i)→(ii), suppose that κ is a regular cardinal, $|x| < \kappa$ and $f : x \rightarrow \kappa$. Then for some cardinal $\lambda < \kappa$ and some function g , $\lambda \approx_g x$, so $f \circ g : \lambda \rightarrow \kappa$. If $\bigcup \text{lm}(f) = \bigcup \text{lm}(f \circ g) = \kappa$, then $\text{cf}(\kappa) \leq \lambda < \kappa$, contrary to the regularity of κ ; hence, since $\bigcup \text{lm}(f)$ is an ordinal ρ , we conclude $\rho < \kappa$.

Now, assuming (ii), suppose that $|x| < \kappa$ and $(\forall v \in x)|v| < \kappa$. Then $f(v) = |v|$ is a function $f : x \rightarrow \kappa$, so by (ii), for some $\rho < \kappa$, $\text{lm}(f) \subseteq \rho$. Since $\text{lm}(f)$ is a set of cardinals, it follows that $\bigcup \text{lm}(f)$ is a cardinal $\lambda < \kappa$. In particular, for all $v \in x$, $|v| \leq \lambda$, so if $\mu = \max\{|x|, \lambda\}$, then $\mu < \kappa$, and by Corollary 6.4.38, $|\bigcup x| \leq \mu$.

Finally, assuming (iii), suppose that $\sigma \leq \kappa$ is cofinal with κ , so there is a function $f : \sigma \rightarrow \kappa$ such that $\kappa = \bigcup \text{lm}(f)$. Since for all $\tau < \sigma$, $f(\tau) < \kappa$, so $|f(\tau)| < \kappa$, if $\sigma < \kappa$, (iii) implies $|\bigcup \text{lm}(f)| < \kappa$, a contradiction. Hence, $\sigma = \kappa$, and thus $\text{cf}(\kappa) = \kappa$. $\square$

Proposition 6.10.5 *For any infinite cardinal κ , if κ is strongly inaccessible, then for all x ,*

$$x \in \dot{V}_\kappa \leftrightarrow x \subseteq \dot{V}_\kappa \wedge |x| < \kappa.$$

Proof. This follows immediately from Exercises 6.4.60 and 6.4.61, but we give an independent proof. Suppose that κ is strongly inaccessible. We prove first by induction that $(\forall \sigma < \kappa)|\dot{V}_\sigma| < \kappa$. The successor step follows from the fact that κ is a strong limit:

$$|\dot{V}_\sigma| < \kappa \rightarrow |\dot{V}_{\sigma+1}| = 2^{|\dot{V}_\sigma|} < \kappa.$$

The limit step follows from regularity in the form of (iii) of the preceding proposition:

$$(\forall \tau < \sigma)|\dot{V}_\tau| < \kappa \rightarrow |\dot{V}_\sigma| = |\bigcup \{\dot{V}_\tau : \tau < \sigma\}| < \kappa.$$

Now, for any x , $x \in \dot{V}_\kappa \rightarrow x \subseteq \dot{V}_\kappa$, because $\dot{V}_\kappa$ is transitive, and

$$x \in \dot{V}_\kappa \rightarrow (\exists \sigma < \kappa) x \in \dot{V}_\sigma \rightarrow (\exists \sigma < \kappa) x \subseteq \dot{V}_\sigma \rightarrow |x| < \kappa.$$

For the converse, suppose that $x \subseteq \dot{V}_\kappa$ and $|x| < \kappa$. For each $y \in x$, set $f(y) = \text{rank}(y)$, so $f : x \rightarrow \kappa$ and hence by (ii) of the proposition, for some $\rho < \kappa$, $\text{lm}(f) \subseteq \rho$ — that is, $(\forall y \in x) \text{rank}(y) < \rho$, so $x \subseteq \dot{V}_\rho$ and hence $x \in \dot{V}_{\rho+1} \subseteq \dot{V}_\kappa$. $\square$

Definition 6.10.6

- (i) $\text{WI} \doteq \{ \kappa : \kappa \text{ is an uncountable weakly inaccessible cardinal} \}$.
- (ii) $\text{SI} \doteq \{ \kappa : \kappa \text{ is an uncountable strongly inaccessible cardinal} \}$.
- (iii) $\exists \text{WI}$ is the sentence $\text{WI} \neq \emptyset$; $\exists \text{SI}$ is the sentence $\text{SI} \neq \emptyset$.

Proposition 6.10.7 *For every axiom ϕ of ZFC,*

$$\text{ZFC} \vdash \forall \kappa [\kappa \in \text{SI} \rightarrow \phi^{\dot{V}_\kappa}].$$

Proof. By Proposition 6.5.2, arguing in ZFC, for any $\kappa \in \text{SI}$,

$$\text{Extensionality}^{(\dot{V}_\kappa)} \quad \text{and} \quad \text{Foundation}^{(\dot{V}_\kappa)}.$$

The set operations $\dot{\bigcup}$, $\dot{\omega}$ and $\dot{\wp}$ are all $\dot{V}_\kappa$ -absolute by Propositions 6.5.6 and 6.5.10 together with the fact that $x \in \dot{V}_\kappa \rightarrow \dot{\wp} x \subseteq \dot{V}_\kappa$. Then, because each of these is clearly a $\dot{V}_\kappa$ -set operation, we have immediately

$$\text{Union}^{(\dot{V}_\kappa)}, \quad \text{Powerset}^{(\dot{V}_\kappa)} \quad \text{and} \quad \text{Infinity}^{(\dot{V}_\kappa)}.$$

For Replacement, for any (unary) class operation F and $u \in \dot{V}_\kappa$, as in the proof of Theorem 6.5.12,

$$F^{*\dot{V}_\kappa}(u) \doteq F^{\dot{V}_\kappa*}(u \dot{\cap} \dot{V}_\kappa) \dot{\cap} \dot{V}_\kappa \doteq F^{\dot{V}_\kappa*}(u) \dot{\cap} \dot{V}_\kappa.$$

Now by $(\rightarrow)$ of the preceding proposition, this is a subset of $\dot{V}_\kappa$ of cardinality at most $|u| < \kappa$, so by $(\leftarrow)$ of the proposition is an element of $\dot{V}_\kappa$ as required. Finally, for AC, for any $x \in \dot{V}_\kappa$, because we are working in ZFC, there exists a choice function $c \in \dot{V}$ for x . But since $c \subseteq x \times \dot{\bigcup} x$, also $c \in \dot{V}_\kappa$. Hence $\text{AC}^{(\dot{V}_\kappa)}$. $\square$

Proposition 6.10.8 *SI and WI are element- Π^{ZF} classes, so for any transitive class model M ,*

$$\text{ZF} \vdash \text{WI} \cap M \subseteq \text{WI}^M \wedge \text{SI} \cap M \subseteq \text{SI}^M.$$

Moreover,

$$\text{ZF} \vdash \forall \kappa [\kappa \in \text{SI} \rightarrow \text{WI} \cap \dot{V}_\kappa \doteq \text{WI}^{\dot{V}_\kappa} \wedge \text{SI} \cap \dot{V}_\kappa \doteq \text{SI}^{\dot{V}_\kappa}].$$

Proof. Just about any natural explicit version of the definitions of WI and SI is a Π^{ZF} formula — for example, $\kappa \in \text{WI}$ iff $\dot{\omega} < \kappa$ and

$$(\forall \sigma < \kappa)(\forall f : \sigma \rightarrow \kappa)(\exists \rho < \kappa) \dot{\text{lim}}(f) \subseteq \rho \\ \wedge (\forall \sigma < \kappa)(\exists \tau < \kappa) \neg \exists g [\dot{\text{Inj}}(g) \wedge g : \tau \rightarrow \sigma],$$

and for SI the same formula with the final $g : \tau \dot{\rightarrow} \sigma$ replaced by $(\forall v \dot{<} \tau)g(v) \dot{\subseteq} \sigma$, which guarantees that $2^{|\sigma|} \dot{<} \tau$.

These imply the stated inclusions by Corollary 6.5.36. Now fix $\kappa \in \text{SI}$; By what we have just established together with the preceding proposition, $\text{WI} \cap \dot{V}_\kappa \dot{\subseteq} \text{WI}^{\dot{V}_\kappa} \wedge \text{SI} \cap \dot{V}_\kappa \dot{\subseteq} \text{SI}^{\dot{V}_\kappa}$. Suppose for some $\lambda \dot{<} \kappa$ that $\lambda \notin \text{WI}$, so λ fails to satisfy one of the two clauses above, say

$$(\exists \sigma \dot{<} \lambda)(\exists f : \sigma \dot{\rightarrow} \lambda)(\forall \rho \dot{<} \lambda) \text{Im}(f) \dot{\not\subseteq} \rho.$$

But any such $f \dot{\subseteq} \sigma \dot{\times} \lambda \dot{\subseteq} \dot{V}_{\max\{\sigma, \lambda\} + \omega \cdot 2}$, so $f \in \dot{V}_\kappa$ and hence also $\lambda \notin \text{WI}^{\dot{V}_\kappa}$. The other cases are handled similarly. $\square$

Definition 6.10.9 $\dot{V}_{\text{SI}} := \bigcap \{ \dot{V}_\kappa : \kappa \in \text{SI} \}$.

Thus in ZFC, if $\exists \text{SI}$, then $\dot{V}_{\text{SI}} \dot{=} \dot{V}_{\bar{\kappa}}$ for $\bar{\kappa}$ the smallest element of SI , otherwise, $\dot{V}_{\text{SI}} \dot{=} \bigcap \emptyset \dot{=} \dot{V}$. In either case,

Theorem 6.10.10 (ZFC) $\dot{V}_{\text{SI}}$ is a class model of $\text{ZFC} + \neg \exists \text{SI}$. Hence,

$$\text{ZF is consistent} \implies \text{ZFC} + \neg \exists \text{SI is consistent};$$

equivalently, $\text{ZF is consistent} \implies \text{ZFC} \not\models \exists \text{SI}$.

Proof. Arguing in ZFC, if $\neg \exists \text{SI}$, then $\dot{V}_{\text{SI}} \dot{=} \dot{V}$ so $\phi^{(\dot{V}_{\text{SI}})}$ for each axiom ϕ of ZFC, and also $(\neg \exists \text{SI})^{(\dot{V}_{\text{SI}})}$. Otherwise, $\exists \text{SI}$ and $\dot{V}_{\text{SI}} \dot{=} \dot{V}_{\bar{\kappa}}$ for $\bar{\kappa}$ the smallest element of SI . Then by Proposition 6.10.7, $\phi^{(\dot{V}_{\text{SI}})}$ for each axiom ϕ of ZFC, and by the preceding proposition

$$(\exists \text{SI})^{(\dot{V}_{\bar{\kappa}})} \leftrightarrow (\exists \lambda \dot{<} \bar{\kappa}) \lambda \in \text{SI}^{\dot{V}_{\bar{\kappa}}} \leftrightarrow (\exists \lambda \dot{<} \bar{\kappa}) \lambda \in \text{SI},$$

so by the choice of $\bar{\kappa}$, $(\neg \exists \text{SI})^{(\dot{V}_{\bar{\kappa}})}$, that is $(\neg \exists \text{SI})^{(\dot{V}_{\text{SI}})}$. In the second clause we are using the result of Section 6.7 that the consistency of ZFC follows from that of ZF. $\square$

To prove the corresponding result for weak inaccessibility, we invoke some of the properties of the constructible universe $\mathbf{L}$ established in Section 6.7.

Proposition 6.10.11

- (i) $\text{ZFL} \vdash \text{WI} \dot{=} \text{SI}$; hence, $\text{ZF} \vdash \text{WI}^{\mathbf{L}} \dot{=} \text{SI}^{\mathbf{L}}$.
- (ii) For any infinite cardinal κ , $\text{ZFL} \vdash \forall \kappa [\kappa \in \text{WI} \rightarrow \dot{V}_\kappa \dot{=} \mathbf{L}_\kappa]$; hence

$$\text{ZF} \vdash \forall \kappa [\kappa \in \text{WI}^{\mathbf{L}} \rightarrow \dot{V}_\kappa^{\mathbf{L}} \dot{=} \mathbf{L}_\kappa].$$

Proof. Part (i) is immediate from Theorem 6.7.19, Proposition 6.10.3(vi) and Corollary 6.7.9. For (ii), arguing in ZFL, fix $\kappa \in \text{WI} \dot{=} \text{SI}$. An easy

induction shows that for all σ , $L_\sigma \subseteq \dot{V}_\sigma$; we show that $\dot{V}_\kappa \subseteq L_\kappa$ by set induction. Suppose as induction hypothesis that $(\forall y \in x)[y \in \dot{V}_\kappa \rightarrow y \in L_\kappa]$ and that $x \in \dot{V}_\kappa$. Then $x \subseteq L_\kappa$ and $|x| < \kappa$. For each $y \in x$, set $f(y) := \text{rank}_L(y)$, so $f : x \rightarrow \kappa$ and hence by regularity for some $\rho < \kappa$, $\text{Im}(f) \subseteq \rho$, so $x \subseteq L_\rho$. By Proposition 6.7.18, if $|\rho| = \lambda$, then $x \in L_{\lambda^+} \subseteq L_\kappa$. The second clause uses the absoluteness of the set operation $\sigma \mapsto L_\sigma$ (6.7.6). $\square$

Definition 6.10.12 $L_{\text{WI}} := \bigcap \{ \{ L_\kappa : \kappa \in \text{WI}^L \} \cap L$.

Thus in ZFC, if $(\exists \text{WI})^{(L)}$, then $L_{\text{WI}} = L_{\bar{\kappa}}$ for $\bar{\kappa}$ the smallest element of WI^L , otherwise, $L_{\text{WI}} = L$. In either case,

Theorem 6.10.13 L_{WI} is a class model of $\text{ZFC} + \neg \exists \text{WI}$. Hence,

$$\text{ZF is consistent} \implies \text{ZFC} + \neg \exists \text{WI is consistent};$$

equivalently, $\text{ZF is consistent} \implies \text{ZFC} \not\vdash \exists \text{WI}$.

Proof. Arguing in ZF, if $(\neg \exists \text{WI})^{(L)}$, then $L_{\text{WI}} = L$, so since L is a class model of ZFC, $\phi^{(L_{\text{WI}})}$ for each axiom ϕ of ZFC, and also $(\neg \exists \text{WI})^{(L_{\text{WI}})}$. Otherwise, $L_{\text{WI}} = L_{\bar{\kappa}}$ for $\bar{\kappa}$ the smallest element of WI^L . By Propositions 6.10.7 and 6.10.11, for each axiom ϕ of ZFC,

$$\text{ZFL} \vdash \forall \kappa [\kappa \in \text{WI} \rightarrow \phi^{(L_\kappa)}] \quad \text{hence} \quad \text{ZF} \vdash \left(\forall \kappa [\kappa \in \text{WI} \rightarrow \phi^{(L_\kappa)}] \right)^{(L)},$$

and thus, by the absoluteness of the set operation $\sigma \mapsto L_\sigma$,

$$\text{ZF} \vdash \forall \kappa [\kappa \in \text{WI}^L \rightarrow \phi^{(L_\kappa)}], \quad \text{and in particular} \quad \phi^{(L_{\bar{\kappa}})},$$

so $\phi^{(L_{\text{WI}})}$. Similarly, by Propositions 6.10.8 and 6.10.11,

$$\text{ZFL} \vdash \text{WI} \cap L_{\bar{\kappa}} = \text{WI}^{L_{\bar{\kappa}}} \quad \text{so} \quad \text{ZF} \vdash \text{WI}^L \cap L_{\bar{\kappa}} = \text{WI}^{L_{\bar{\kappa}}},$$

and hence

$$(\exists \text{WI})^{(L_{\bar{\kappa}})} \leftrightarrow (\exists \lambda < \bar{\kappa}) \lambda \in \text{WI}^{L_{\bar{\kappa}}} \leftrightarrow (\exists \lambda < \bar{\kappa}) \lambda \in \text{WI}^L,$$

so by the choice of $\bar{\kappa}$, $(\neg \exists \text{WI})^{(L_{\bar{\kappa}})}$, that is $(\neg \exists \text{WI})^{(L_{\text{WI}})}$. $\square$

Proposition 6.10.14 $\text{ZF} + \exists \text{WI} \vdash \dot{\text{Con}}(\text{Z}\dot{\text{F}}\text{C})$.

Proof. We first observe that formalizing Proposition 6.10.7 (cf. Definition 6.6.10) we have

$$\text{ZFC} \vdash \forall \kappa [\kappa \in \text{SI} \rightarrow \text{ModelOf}(\dot{\text{V}}_\kappa, \text{Z}\dot{\text{F}}\text{C})].$$

It follows that this same sentence relativized to L is a theorem of ZF , that is,

$$\text{ZF} \vdash \forall \kappa [\kappa \in \text{SI}^{\text{L}} \rightarrow \text{ModelOf}^{\text{L}}(\dot{\text{V}}_\kappa^{\text{L}}, \text{Z}\dot{\text{F}}\text{C})],$$

and since ModelOf is absolute and by Proposition 6.10.11, $\dot{\text{V}}_\kappa^{\text{L}} \doteq \text{L}_\kappa$ and $\text{WI}^{\text{L}} \doteq \text{SI}^{\text{L}}$,

$$\text{ZF} \vdash \forall \kappa [\kappa \in \text{WI}^{\text{L}} \rightarrow \text{ModelOf}(\text{L}_\kappa, \text{Z}\dot{\text{F}}\text{C})].$$

Finally, since by Proposition 6.10.8, $\text{WI} \subseteq \text{WI}^{\text{L}}$,

$$\text{ZF} \vdash \forall \kappa [\kappa \in \text{WI} \rightarrow \text{ModelOf}(\text{L}_\kappa, \text{Z}\dot{\text{F}}\text{C})]. \quad \square$$

Applying to the preceding theorem the general principle that informal mathematical arguments can be formalized in ZFC tells us that

$$\text{ZFC} \vdash \dot{\text{Con}}(\text{Z}\dot{\text{F}}) \rightarrow \dot{\text{Con}}(\text{Z}\dot{\text{F}}\text{C} + \neg \exists \text{WI}).$$

However, it follows from the proposition that there can be no such relative consistency proof for the existence of inaccessibles.

Theorem 6.10.15 *Unless ZF is actually inconsistent, there is no proof formalizable in ZFC that*

$$\text{ZF is consistent} \implies \text{ZFC} + \exists \text{WI is consistent}.$$

Proof. If there were such a proof, we would have

$$\text{ZFC} \vdash \dot{\text{Con}}(\text{Z}\dot{\text{F}}) \rightarrow \dot{\text{Con}}(\text{Z}\dot{\text{F}}\text{C} + \exists \dot{\text{WI}}),$$

so by the preceding proposition,

$$\text{ZFC} + \exists \text{WI} \vdash \dot{\text{Con}}(\text{Z}\dot{\text{F}}\text{C} + \exists \dot{\text{WI}}).$$

But then by the Second Incompleteness Theorem, $\text{ZFC} + \exists \text{WI}$ is inconsistent, and thus by the hypothesis so is ZF . $\square$

We turn now to the notion of measurable cardinal. The notion of ultrafilter was introduced in Definition 1.6.10; we repeat it here in formal form.

Definition 6.10.16 For any cardinal κ , an **ultrafilter** on κ is a non-empty set $U \subseteq \wp(\kappa)$ such that for all $x, y \subseteq \kappa$,

- (i) $\emptyset \notin U$;
- (ii) $x \in U \wedge x \subseteq y \rightarrow y \in U$;
- (iii) $x \in U \wedge y \in U \rightarrow x \dot{\cap} y \in U$;
- (iv) $(\kappa \setminus x) \in U \leftrightarrow x \notin U$.

An ultrafilter U is **non-principal** iff

- (v) for all $\sigma < \kappa$, $\{\sigma\} \notin U$,

and $< \kappa$ -**complete** iff

- (vi) for any $v \subseteq U$, if $|v| < \kappa$, then $\bigcap v \in U$.

Definition 6.10.17 A cardinal κ is **measurable** iff there exists a non-principal, $< \kappa$ -complete ultrafilter U on κ . U is called a **measure ultrafilter**.

$\text{MC} := \{\{\kappa : \kappa \text{ is an uncountable measurable cardinal}\}\};$

$\exists \text{MC}$ is the sentence $\text{MC} \neq \emptyset$.

Again, ω is an example of a measurable cardinal (because closure of an ultrafilter under intersection of two sets implies closure under finite intersections, so every ultrafilter is $< \omega$ -complete), so the interesting question concerns the existence of uncountable ones. The term “measurable” (from measure theory in analysis) arises from thinking of an ultrafilter on κ as determining a function $\mu : \wp(\kappa) \rightarrow \dot{2}$, the characteristic function of U which assigns a “size”, 0 or 1, to every subset of κ . It follows from the next lemma that μ is finitely additive — $\mu(x \dot{\cup} y) = \mu(x) + \mu(y)$ when $x \dot{\cap} y = \emptyset$ — and even $< \kappa$ -additive — for $\rho < \kappa$, $\mu(\bigcup_{\sigma < \rho} x_\sigma) = \sum_{\sigma < \rho} \mu(x_\sigma)$ when the x_σ are pairwise disjoint.

The following proposition shows that we cannot expect to prove in ZFC that any uncountable measurable cardinals exist.

Lemma 6.10.18 For any measure ultrafilter U on a cardinal κ ,

- (i) $\kappa \in U$;
- (ii) $(\forall y \subseteq \wp(\kappa)) \left(|y| < \kappa \rightarrow \left[\bigcup y \in U \leftrightarrow (\exists z \in y) z \in U \right] \right)$;
- (iii) $(\forall y \subseteq \kappa) \left(|y| < \kappa \rightarrow y \notin U \right)$.

Proof. (i) follows from the fact that U is non-empty and closed under superset. The implication ($\leftarrow$) of (ii) is immediate, since $z \in y \rightarrow z \subseteq \bigcup y$.

Suppose that $(\forall z \in y)z \notin U$, so $(\forall z \in y)[\kappa \setminus z] \in U$. Then by $< \kappa$ -completeness,

$$\emptyset \doteq \bigcup y \cap \bigcap \{ \kappa \setminus z : z \in y \} \in U,$$

contrary to hypothesis. Finally, (iii) follows from the observation that $y \doteq \bigcup \{ \{ \sigma \} : \sigma \in y \}$. $\square$

Proposition 6.10.19 $\text{MC} \subseteq \text{SI}$ — that is, every measurable cardinal is strongly inaccessible.

Proof. Suppose that κ is a measurable cardinal with measure ultrafilter U . For any $\sigma < \kappa$ and $f : \sigma \rightarrow \kappa$, for each $\tau < \sigma$, $|f(\tau)| < \kappa$, so by (iii) of the lemma, $f(\tau) \notin U$. Hence by (ii), $\bigcup \text{Im}(f) \notin U$, so by (i), $\bigcup \text{Im}(f) \neq \kappa$. Thus κ is regular. To see that it is also a strong limit, suppose to the contrary that for some λ , $\lambda < \kappa \leq 2^\lambda$, so there exists an injective function $g : \kappa \rightarrow {}^\lambda 2$. For each $\tau < \lambda$ and $i < 2$, set

$$y_{\tau,i} := \{ \sigma < \kappa : g(\sigma)(\tau) = i \}.$$

Since $y_{\tau,0} \cup y_{\tau,1} \doteq \kappa \in U$, for each $\tau < \lambda$ there exists i_τ such that $y_{\tau,i_\tau} \in U$. Then also, $z := \bigcap \{ y_{\tau,i_\tau} : \tau < \lambda \} \in U$. But $\sigma \in z \rightarrow (\forall \tau < \lambda) g(\sigma)(\tau) = i_\tau$, and since g is injective, there is at most one such σ — that is, z is either $\emptyset$ or $\{ \sigma \}$, a contradiction. $\square$

Theorem 6.10.20 There exist class operations j and N such that for any measure ultrafilter U (on the cardinal $\kappa \doteq \bigcup U$ and writing $j(x)$ for $j(x, U)$ and N for $N(U)$),

- (i) N is transitive, $\forall x [j(x) \in N]$ and $\text{Or} \subseteq N$;
- (ii) $(\forall \sigma < \kappa) j(\sigma) = \sigma$ but $\kappa < j(\kappa)$;
- (iii) for every $L_\in$ -formula $\phi(v_0, \dots, v_{k-1})$,

$$\forall x_0 \dots \forall x_{k-1} \left[\phi(x_0, \dots, x_{k-1}) \leftrightarrow \phi^{(N)}(j(x_0), \dots, j(x_{k-1})) \right].$$

The conclusion is usually expressed (in part) as ‘ j is an elementary embedding of V into N ’, but of course this is a statement which, by Tarski’s Theorem 6.6.12, cannot (at least apparently) be expressed by a single $L_\in$ -sentence. We postpone the proof of this result to draw two important consequences. Note that it follows immediately from (i)–(iii) that N is a class model of ZFC and j is injective.

Proposition 6.10.21 $\exists MC \rightarrow \neg \dot{V} \dot{=} L$.

Proof. Arguing in ZFC, assume $\exists MC$ and fix κ as the smallest uncountable measurable cardinal, U a measure ultrafilter on κ and j and N as in the theorem. Let $\phi(v)$ be the formula expressing ' v is the smallest uncountable measurable cardinal'. Then $\phi(\kappa)$, so $\phi^{(N)}(j(\kappa))$. Since $\kappa < j(\kappa)$, $j(\kappa)$ is *not* the smallest uncountable measurable cardinal — that is, $\neg \phi^{(\dot{V})}(j(\kappa))$ — from which it follows that $N \neq \dot{V}$. But N is a class model of ZFC and $\dot{O}r \subseteq N$, whence by Theorem 6.7.6 $L \subseteq N \subseteq \dot{V}$, so also $L \neq \dot{V}$. $\square$

As another consequence we see not only is a measurable cardinal itself strongly inaccessible, but in fact there must be many strongly inaccessible cardinals below it.

Proposition 6.10.22 For any cardinal κ , $\kappa \in MC \rightarrow |SI \cap \kappa| \dot{=} \kappa$.

Proof. Arguing in ZFC, fix $\kappa \in MC$ and let $\phi(v, w)$ be the formula

$$(\exists \lambda < v) [w \dot{\leq} \lambda \wedge \lambda \in SI].$$

It suffices to show that $\phi(\kappa, \sigma)$ for each $\sigma < \kappa$, since then $SI \cap \kappa$ is unbounded in κ so is of cardinality κ by the regularity of κ . But $\phi(\kappa, \sigma) \leftrightarrow \phi^{(N)}(j(\kappa), j(\sigma))$, which by absoluteness and the fact that $j(\sigma) \dot{=} \sigma$ is equivalent to

$$(\exists \lambda < j(\kappa)) [\sigma \dot{\leq} \lambda \wedge \lambda \in SI^N].$$

This assertion is true as witnessed by $\lambda \dot{=} \kappa$, since $\kappa \in SI \subseteq SI^N$. $\square$

For the proof of the theorem, we shall need a more general notion of class model in which the membership relation is also reinterpreted.

Definition 6.10.23 For any classes M and $E \subseteq M \times M$ and any L_E -formula ϕ , $\phi^{(M, E)}$ is the formula defined recursively as follows:

$$\begin{aligned} (x \dot{=} y)^{(M, E)} & \text{ is } x \dot{=} y; \\ (x \dot{\in} y)^{(M, E)} & \text{ is } \langle x, y \rangle \in E; \\ (\neg \psi)^{(M, E)} & \text{ is } \neg \psi^{(M, E)}; \\ (\psi \vee \theta)^{(M, E)} & \text{ is } \psi^{(M, E)} \vee \theta^{(M, E)}; \\ (\exists x \psi)^{(M, E)} & \text{ is } \exists^M x \psi^{(M, E)}. \end{aligned}$$

Of course, the previous notion $\phi^{(M)}$ is equivalent to $\phi^{(M, \in_M)}$, where

$$\in_M \dot{=} \{ \langle x, y \rangle : x, y \in M \wedge x \dot{\in} y \}.$$

The main fact we need about such a **class structure** (M, E) is a generalization of the Mostowski Collapsing Theorem 6.6.17.

Definition 6.10.24 For any classes M and $E \subseteq M \dot{\times} M$,

- (i) for each $x \in M$, $x_E := \{ \langle z, x \rangle \in E \}$;
- (ii) (M, E) is **extensional** iff $(\text{Extensionality})^{(M, E)}$ — that is,

$$\forall^M x \forall^M y [x_E \dot{=} y_E \rightarrow x \dot{=} y];$$

- (iii) (M, E) is **narrow** iff $\forall^M x [x_E \text{ is a set}]$;
- (iv) (M, E) is **well-founded** iff

$$\forall u [\emptyset \dot{\neq} u \subseteq M \rightarrow (\exists x \in u) x_E \cap u \dot{=} \emptyset].$$

Note that (iv) is stronger than $(\text{Foundation})^{(M)}$, which is the same statement with $\forall u$ replaced by $\forall^M u$. Note also that easily any transitive class structure $(M, \in_M)$ has all of these properties. It is straightforward (and left to the exercises) to generalize 6.2.43–6.2.49 and obtain the following principles, stated here only in the simple forms we need. If (M, E) is extensional, narrow and well-founded, then the following are theorems of ZF.

E-Induction: for any class A , $\forall^M x (x_E \subseteq A \rightarrow x \in A) \rightarrow M \subseteq A$.

E-Recursion: for any set operation G , there exists a set operation F , unique on M , such that

$$\forall^M x \forall^M y [F(x, y) \dot{=} G(F \upharpoonright_{x_E} y, x, y)],$$

$$\text{where } F \upharpoonright_{x_E} y := \{ \langle z, F(x, z) \rangle : z \in y_E \}.$$

Mostowski Collapsing Theorem 6.10.25 For any classes M and $E \subseteq M \dot{\times} M$, if (M, E) is extensional, narrow and well-founded, there exists a unique transitive class N and a unique set operation F such that $(M, E) \simeq_F (N, \in_N)$ — that is,

- (i) $F : M \dot{\rightarrow} N$ is a bijection;
- (ii) $\forall^M x \forall^M y [\langle x, y \rangle \in E \leftrightarrow F(x) \in F(y)]$.

Proof. We follow closely the proof of the earlier version 6.6.17. Given M and E as in the hypothesis, let F be the unique set operation (by E-Recursion) such that

$$F(x) \dot{=} \begin{cases} \{ F(z) : z \in x_E \}, & \text{if } x \in M; \\ \emptyset, & \text{otherwise;} \end{cases}$$

and set $N := \dot{\text{Im}}(F)$. To establish that F is injective on M , we show by E -Induction that if

$$B := \{ \{ x : x \in M \rightarrow (\forall y \in M) [F(x) \dot{=} F(y) \rightarrow x \dot{=} y] \} \},$$

then $M \dot{\subseteq} B$. Assume as induction hypothesis that $x_E \dot{\subseteq} B$ and that $x, y \in M$ and $F(x) \dot{=} F(y)$ — that is,

$$\{ F(z) : z \in x_E \} \dot{=} \{ F(z) : z \in y_E \}.$$

From this we conclude that $x_E \dot{=} y_E$ as follows. For any $z \in x_E$ there exists $z' \in y_E$ such that $F(z) \dot{=} F(z')$. Since by the induction hypothesis $z \in B$, we conclude that $z \dot{=} z'$ so $z \in y_E$. Thus $x_E \dot{\subseteq} y_E$, and the converse inclusion is established similarly. Finally, by the extensionality of E , $x \dot{=} y$. Now the definition gives immediately that $\langle z, x \rangle \in E \rightarrow F(z) \in F(x)$, and the converse implication follows from the injectivity just established. To see that N is transitive,

$$\begin{aligned} u \in N &\rightarrow (\exists x \in M) u \dot{=} F(x) \rightarrow (\exists x \in M) u \dot{=} \{ F(z) : z \in x_E \} \\ &\rightarrow u \dot{\subseteq} \dot{\text{Im}}(F) \dot{=} N. \quad \square \end{aligned}$$

Proof. (of Theorem 6.10.20) Arguing in ZFC, assume that U is a measure ultrafilter on κ . We define first a class structure (M, E) as a “class ultrapower” of the universe $\dot{V}$ and then apply the Mostowski Collapse. The development follows closely the material of the middle part of [Section 3.3](#). For any two functions $f, g : \kappa \dot{\rightarrow} \dot{V}$, set

$$f \sim_U g \leftrightarrow \{ \sigma \dot{<} \kappa : f(\sigma) \dot{=} g(\sigma) \} \in U.$$

Easily $\sim_U$ is a class equivalence relation with its equivalence classes proper classes. To work with them we select from each a set consisting of the members of minimal rank:

$$\hat{f} := \left\{ g : f \sim_U g \wedge \exists \sigma [g \in \dot{V}_\sigma \wedge (\forall \tau \dot{<} \sigma) \neg (\exists h \in \dot{V}_\tau) f \sim_U h] \right\}.$$

Then

$$\begin{aligned} M &:= \{ \{ \hat{f} : \text{Fn}(f) \wedge \text{Dom}(f) \dot{=} \kappa \} \}; \\ E &:= \{ \{ \langle \hat{f}, \hat{g} \rangle : \hat{f}, \hat{g} \in M \wedge \{ \sigma \dot{<} \kappa : f(\sigma) \in g(\sigma) \} \in U \} \}. \end{aligned}$$

As in [Section 3.3](#), E is well-defined, since if $\langle \hat{f}, \hat{g} \rangle \in E$ and $\hat{f} \dot{=} \hat{f}'$, then

$$\{ \sigma \dot{<} \kappa : f(\sigma) \in g(\sigma) \} \cap \{ \sigma \dot{<} \kappa : f(\sigma) \dot{=} f'(\sigma) \} \dot{\subseteq} \{ \sigma \dot{<} \kappa : f'(\sigma) \in g(\sigma) \},$$

so also $\langle \dot{f}', \dot{g} \rangle \in \mathbf{E}$. The proof of Łoś's Theorem 3.3.19 by formula-induction goes through in this context with no significant change, and we have that for each $L_{\mathbf{E}}$ -formula $\phi(v_0, \dots, v_{k-1})$,

$$\forall f_0 \dots \forall f_{k-1} \left[\phi^{(\mathbf{M}, \mathbf{E})}(\hat{f}_0, \dots, \hat{f}_{k-1}) \leftrightarrow \left\{ \sigma \prec \kappa : \phi(f_0(\sigma), \dots, f_{k-1}(\sigma)) \right\} \in U \right].$$

In particular, if for each x , $h_x : \kappa \rightarrow \{x\}$ is the constant function, then

$$(1) \quad \forall x_0 \dots \forall x_{k-1} \left[\phi^{(\mathbf{M}, \mathbf{E})}(\hat{h}_{x_0}, \dots, \hat{h}_{x_{k-1}}) \leftrightarrow \phi(x_0, \dots, x_{k-1}) \right],$$

since $\{ \sigma \prec \kappa : \phi(h_{x_0}(\sigma), \dots, h_{x_{k-1}}(\sigma)) \}$ is either $\emptyset$ or κ . Now to apply the Mostowski Collapsing Theorem, we need to establish that $(\mathbf{M}, \mathbf{E})$ is extensional, narrow and well-founded. Extensionality is immediate from (1) applied to the sentence ϕ which is the axiom of extensionality. For any $g : \kappa \rightarrow \dot{\mathbf{V}}$,

$$\{ \{ \hat{f} : \langle \dot{f}, \dot{g} \rangle \in \mathbf{E} \} \} \subseteq \{ \hat{f} : \text{Im}(f) \subseteq \bigcup \text{Im}(g) \cup \{ \emptyset \} \},$$

since if $v := \{ \sigma \prec \kappa : f(\sigma) \in g(\sigma) \} \in U$, then set

$$f'(\sigma) := \begin{cases} f(\sigma), & \text{if } \sigma \in v; \\ \emptyset, & \text{otherwise;} \end{cases}$$

so $\hat{f} \doteq \hat{f}'$ and $\text{Im}(f') \subseteq \bigcup \text{Im}(g) \cup \{ \emptyset \}$. Hence $(\mathbf{M}, \mathbf{E})$ is narrow. Finally, suppose that $(\mathbf{M}, \mathbf{E})$ is not well-founded so some nonempty $u \subseteq \mathbf{M}$ has no $\mathbf{E}$ -minimal element. Let c be a choice function for $\{ \hat{f}_{\mathbf{E}} \cap u : \hat{f} \in u \}$; by hypothesis each such $\hat{f}_{\mathbf{E}} \cap u \neq \emptyset$. Let $\hat{f}_0$ be any element of u and define recursively $\hat{f}_{y+\omega 1} := c((\hat{f}_y)_{\mathbf{E}} \cap u)$. Then $(\forall y \in \omega) \{ \sigma \prec \kappa : f_{y+\omega 1}(\sigma) \in f_y(\sigma) \} \in U$, so by $< \kappa$ -completeness also

$$w := \bigcap \left\{ \{ \sigma \prec \kappa : f_{y+\omega 1}(\sigma) \in f_y(\sigma) \} : y \in \omega \right\} \in U.$$

In particular, $w \neq \emptyset$. But for any $\sigma \in w$, $\{ f_y(\sigma) : y \in \omega \}$ is a set with no $\in$ -minimal element, contrary to regularity.

Now fix $\mathbf{N}$ and $\mathbf{F}$ as in the Mostowski Collapsing Theorem. Since $\mathbf{F}$ is an isomorphism, for each $L_{\mathbf{E}}$ -formula $\phi(v_0, \dots, v_{k-1})$,

$$(2) \quad \forall f_0 \dots \forall f_{k-1} \left[\phi^{(\mathbf{M}, \mathbf{E})}(\hat{f}_0, \dots, \hat{f}_{k-1}) \leftrightarrow \phi^{(\mathbf{N})}(\mathbf{F}(\hat{f}_0), \dots, \mathbf{F}(\hat{f}_{k-1})) \right].$$

Then defining $j(x) := F(\hat{h}_x)$, we have from (1) and (2),

$$\forall x_0 \cdots \forall x_{k-1} \left[\phi(x_0, \dots, x_{k-1}) \leftrightarrow \phi^{(N)}(j(x_0), \dots, j(x_{k-1})) \right].$$

Clearly N is transitive and $\forall x [j(x) \in N]$.

To establish the remaining claims of the theorem, we have first for all σ ,

$$\sigma \text{ is an ordinal} \rightarrow (j(\sigma) \text{ is an ordinal})^{(N)} \rightarrow j(\sigma) \text{ is an ordinal}$$

and

$$\sigma < \tau \rightarrow (j(\sigma) < j(\tau))^{(N)} \rightarrow j(\sigma) < j(\tau).$$

It follows by ordinal induction that $\forall \sigma [\sigma \leq j(\sigma)]$, so by transitivity of N , $\text{Ord} \subseteq N$. Suppose, towards a contradiction, that there is some least $\sigma < \kappa$ such that $\sigma < j(\sigma)$, and fix $\hat{f} \in M$ such that $F(\hat{f}) \doteq \sigma$. Then

$$\begin{aligned} \sigma < j(\sigma) &\rightarrow F(\hat{f}) \in F(\hat{h}_\sigma) \\ &\rightarrow \langle \hat{f}, \hat{h}_\sigma \rangle \in E \\ &\rightarrow \{ \tau < \kappa : f(\tau) < \sigma \} \in U \\ &\rightarrow (\exists \rho < \sigma) \{ \tau < \kappa : f(\tau) \doteq \rho \} \in U \\ &\rightarrow (\exists \rho < \sigma) \hat{f} \doteq \hat{h}_\rho \\ &\rightarrow (\exists \rho < \sigma) F(\hat{f}) \doteq \rho, \end{aligned}$$

a contradiction. The fourth implication is an application of Lemma 6.10.18(ii).

Finally, to see that $\kappa < j(\kappa)$, let $i : \kappa \rightarrow \kappa$ be the identity function $i(\sigma) \doteq \sigma$. Then

$$\{ \sigma < \kappa : i(\sigma) \in h_\kappa(\sigma) \} \doteq \kappa \in U \rightarrow \langle \hat{i}, \hat{h}_\kappa \rangle \in E \rightarrow F(\hat{i}) < j(\kappa),$$

and for each $\tau < \kappa$, $\{ \sigma < \kappa : h_\tau(\sigma) < i(\sigma) \} \doteq \{ \sigma < \kappa : \tau < \sigma \} \in U$, so $\langle \hat{h}_\tau, \hat{i} \rangle \in E$ and thus $\tau < F(\hat{i})$. Hence $\kappa \leq F(\hat{i}) < j(\kappa)$. $\square$

Collateral Reading. There is a large literature on the subject of large cardinals. Good starting points are [Dra], [Jec] and [Kan].

Exercises

Exercise 6.10.26 Assume $\exists\text{SI}$ and let $\bar{\kappa}$ be the smallest uncountable strongly inaccessible cardinal. Prove that there exists a cardinal $\lambda < \bar{\kappa}$ such that $\dot{V}_\lambda$ is a set model of ZFC.

Hint: It is necessary to prove the stronger assertion that $\dot{V}_\lambda$ is an elementary substructure of $\dot{V}_{\bar{\kappa}}$. Use the techniques of the proofs of the Reflection Theorem 6.5.20 and Proposition 2.3.58 to show that for each $y \in \dot{\text{Fm}}(\dot{L}_\in)$ there is a closed unbounded (Definition 6.4.52) set $X_y \subseteq \bar{\kappa}$ such that for each $\sigma \in X_y$, y is $(\dot{V}_\sigma, \dot{V}_{\bar{\kappa}})$ -absolute. Then use the result of Exercise 6.4.55 to conclude that the desired λ exists.

Exercise 6.10.27 Let U be a non-principal ultrafilter on a cardinal κ . Show that for any cardinal $\lambda < \kappa$ the following three conditions are equivalent (U is $< \lambda$ -complete):

- (i) $(\forall v \subseteq U) \left(|v| < \lambda \rightarrow \bigcap v \in U \right)$;
- (ii) $(\forall y \subseteq \wp \kappa) \left(|y| < \lambda \wedge \bigcup y \neq \kappa \rightarrow (\exists z \in y) z \in U \right)$;
- (iii) $(\forall y \subseteq \wp \kappa) \left(|y| < \lambda \rightarrow \left[\bigcup y \in U \leftrightarrow (\exists z \in y) z \in U \right] \right)$.

Exercise 6.10.28 Call a cardinal κ **weakly measurable** iff there exists a non-principal $< \aleph_1$ -complete (also called **countably complete**) ultrafilter on κ . Assume that there exists a weakly measurable cardinal. Show that the smallest weakly measurable cardinal is in fact measurable.

Hint: Use the preceding exercise. Suppose that V is a non-principal countably complete ultrafilter on κ and that there exist $\lambda < \kappa$ and $y \subseteq \wp \kappa$ such that $|y| \neq \lambda$ and $\bigcup y \neq \kappa$ but $(\forall z \in y) z \notin V$. Say $y = \{z_\sigma : \sigma < \kappa\}$. Show that

$$U := \left\{ x \subseteq \lambda : \bigcup \{z_\sigma : \sigma \in x\} \in V \right\}$$

is a non-principal countably complete ultrafilter on λ .

Exercise 6.10.29 Show that every measurable cardinal is weakly compact (Definition 3.7.10).

Hint: Suppose that κ is measurable, Γ is a set of $L_{\kappa\omega}$ -sentences of power at most κ , say $\Gamma = \{\phi_\sigma : \sigma < \kappa\}$, and every subset of Γ of power $< \kappa$ has a model; in particular, for each $\rho < \kappa$ there exists a model $\mathfrak{A}_\rho$ of $\{\phi_\sigma : \sigma < \rho\}$. Show that the ultrapower of the $\mathfrak{A}_\rho$ by the measure ultrafilter is a model of Γ .

Exercise 6.10.30 Suppose that there exists a class operation j and a class N such that

- (i) $\forall x [j(x) \dot{\in} N]$;
- (ii) $\exists z [j(z) \dot{\neq} z]$;
- (iii) for every $L_{\in}$ -formula $\phi(v_0, \dots, v_{k-1})$,

$$\forall x_0 \cdots \forall x_{k-1} \left[\phi(x_0, \dots, x_{k-1}) \leftrightarrow \phi^{(N)}(j(x_0), \dots, j(x_{k-1})) \right].$$

Prove that there exists an uncountable measurable cardinal.

Hint: Show first that there is a least ordinal ρ such that $j(\rho) \dot{\neq} \rho$; otherwise, using (iii), for all z , $\text{rank}(j(z)) \dot{=} \text{rank}(z)$, and this leads to a contradiction with (ii). Next show that ρ is an uncountable cardinal. Finally, show that

$$U := \{ x \dot{\subseteq} \rho : \rho \in j(x) \}$$

is a measure ultrafilter on ρ and thus that ρ is a measurable cardinal.

Exercise 6.10.31 With notation as in the statement and proof of Theorem 6.10.20, show that for each Δ^{ZF} $L_{\in}$ -formula $\phi(v_0, \dots, v_{k-1})$,

$$\begin{aligned} \forall f_0 \cdots \forall f_{k-1} \left[\phi(F(\hat{f}_0), \dots, F(\hat{f}_{k-1})) \right. \\ \left. \leftrightarrow \{ \sigma \dot{<} \kappa : \phi(f_0(\sigma), \dots, f_{k-1}(\sigma)) \} \in U \right]. \end{aligned}$$

Exercise 6.10.32 With notation as in the statement and proof of Theorem 6.10.20, show that

- (i) $(\forall x \dot{\in} \dot{V}_{\kappa}) j(x) \dot{=} x$;
- (ii) $\dot{V}_{\kappa} \dot{\subseteq} N$; hence $\dot{V}_{\kappa} \dot{\in} N$;
- (iii) ${}^{\kappa}N \dot{\subseteq} N$;
- (iv) $\dot{V}_{\kappa+\circ 1} \dot{\subseteq} N$; hence $\dot{V}_{\kappa+\circ 1} \dot{\in} N$;
- (v) $U \not\dot{\subseteq} N$; hence $\dot{V}_{\kappa+\circ 2} \not\dot{\subseteq} N$.

Hint: Prove (i) by set induction along the lines of the proof that $j(\sigma) \dot{=} \sigma$ for $\sigma \dot{<} \kappa$. For the second part of (ii) and (iv) use Corollary 6.5.35. For (iii), given $f \dot{\in} {}^{\kappa}N$, choose functions g_{σ} such that $F(\hat{g}_{\sigma}) \dot{=} f(\sigma)$ and define $h \dot{\in} {}^{\kappa}(\dot{V})$ by $h(\tau)(\sigma) \dot{=} g_{\sigma}(\tau)$. Use the result of the preceding exercise to conclude that $F(\hat{h}) \dot{\in} j^{(\kappa)}\dot{V}$ and that for all $\sigma \dot{<} \kappa$, $F(\hat{h})(\sigma) \dot{=} f(\sigma)$, so $f \dot{=} F(\hat{h}) \upharpoonright \kappa \dot{\in} N$. For (iv), use Propositions 6.10.5 and 6.10.19. For (v), show that if $U \dot{\subseteq} N$, then $(\kappa \text{ is a measurable cardinal})^{(N)}$ and derive a contradiction.

Exercise 6.10.33 With notation as in the statement and proof of Theorem 6.10.20, show that $2^\kappa < j(\kappa) < (2^\kappa)^+$ so $j(\kappa)$ is not a cardinal.

Hint: For the first inequality, note that by (iii) of the preceding exercise, $(2^\kappa)^N \doteq 2^\kappa$ and that $(j(\kappa) \text{ is strongly inaccessible})^{(N)}$. For the second, show that $j(\kappa) \doteq \{F(\hat{f}) : f \in {}^\kappa\kappa\}$.

6.11 Determinacy

We turn now to the notion of determinacy of infinite games, which at first glance seems an unlikely principle to have profound set-theoretic consequences. The reader may have encountered at some point the theory of finite games of perfect information. Such a game is played by two players: I, who plays first, and II, moving alternately according to unambiguous rules. Each player has full knowledge of the sequence of moves. If the rules preclude ties, so in every play of the game one player wins and the other loses, then (once the notions are made sufficiently precise) it can be proved that one of I and II has a *winning strategy*, a set of rules which if followed guarantee a win for that player no matter how the opponent plays. The basic idea of determinacy is the extension of this idea to *infinite games*.

To make this into an effective set-theoretic principle, we make the games as simple as possible: each move consists in simply choosing an element x of a given set u . Thus, if I chooses x_0 , II chooses x_1 , I chooses x_2 , the play proceeds forever and generates an infinite sequence $h = \langle x_k : k \in \omega \rangle \in {}^\omega u$. The outcome of the game is specified by a subset $A \subseteq {}^\omega u$: player I wins if $h \in A$ and player II wins if $h \notin A$. A strategy for player I is a function $f : {}^{<\omega}u \rightarrow u$; following the strategy f means that after the elements $x_0, \dots, x_{2k-1}$ have been played, I plays $f(x_0, \dots, x_{2k-1})$ as his next move x_{2k} . Similarly, II follows a strategy g by playing $g(x_0, \dots, x_{2k})$ as his move x_{2k+1} . Thus, if I follows f and II follows g , they generate a uniquely determined play of the game written $f \# g$ and read *f clash g* . f is a winning strategy for I iff f wins against every g — that is, for all g , $(f \# g) \in A$. g is a winning strategy for II iff for all f , $(f \# g) \notin A$. Clearly at most one player can have a winning strategy and A is determined iff one of them does.

For our purposes here, it will usually suffice to take $u \doteq \dot{\omega}$: players choose a natural number at each turn; and we formulate the definitions for this case. It will be convenient to use codings of finite sequences of natural numbers as natural numbers as in Definition 4.2.23. We write $\langle \dot{m}, \dot{n} \rangle$ etc. for the formal versions of these functions and use the notations $(s)_i$ and $\lg(s)$ as in Section 4.2. For technical convenience we set the code $\langle \rangle$ for the empty sequence equal to $\dot{0}$. We use $\frown$ for the formal concatenation operator: $\langle \dot{m}, \dot{n} \rangle \frown \langle \dot{p}, \dot{q}, \dot{r} \rangle \doteq \langle \dot{m}, \dot{n}, \dot{p}, \dot{q}, \dot{r} \rangle$. Letters $k, l, m, n, \dots$ will always

denote elements of $\dot{\omega}$, letters $\alpha, \beta, \gamma, \dots$ denote elements of $\dot{\omega}$, and as in Definition 4.2.29, $\bar{\alpha}(k) := \langle \alpha(0), \dots, \alpha(k-1) \rangle$. $\dot{\text{Sq}}$ will denote the set of natural numbers which code finite sequences, $s \subseteq t$ means that the sequence coded by t extends that coded by s , $s \dot{\subset} t$ is strict extension and $s \dot{\subset} \alpha$ means that $s \doteq \bar{\alpha}(\lg(s))$ — that is, s codes an initial segment of α . s and t are **compatible** — in symbols, $s \sim t$ — iff either $s \subseteq t$ or $t \subseteq s$; otherwise s and t are **incompatible** — in symbols, $s \not\sim t$.

Definition 6.11.1

- (i) For any $\gamma, \delta \in \dot{\omega}$, $\gamma \# \delta$ is the function ε such that for all $k \in \dot{\omega}$,

$$\varepsilon(2k) \doteq \gamma(\bar{\varepsilon}(2k)) \quad \text{and} \quad \varepsilon(2k+1) \doteq \delta(\bar{\varepsilon}(2k+1)).$$

- (ii) For any $A \subseteq \dot{\omega}$,

$$\text{WS}_\text{I}(A) := \{ \dot{\gamma} : \forall \delta [(\gamma \# \delta) \in A] \} \quad \text{and}$$

$$\text{WS}_\text{II}(A) := \{ \dot{\delta} : \forall \gamma [(\gamma \# \delta) \notin A] \};$$

an element of $\text{WS}_\text{I}(A)$ ($\text{WS}_\text{II}(A)$) is a **winning strategy** for player I (II); A is **determined** iff $\text{WS}_\text{I}(A) \dot{\cup} \text{WS}_\text{II}(A) \neq \emptyset$.

- (iii) For any $\mathcal{C} \subseteq \dot{\wp}(\dot{\omega})$, $\text{AD}[\mathcal{C}]$ is the sentence

$$(\forall A \in \mathcal{C}) A \text{ is determined};$$

AD is $\text{AD}[\dot{\wp}(\dot{\omega})]$, the **Axiom of Determinacy**.

AD can be thought of as a **uniformity principle**:

$$\begin{aligned} \text{WS}_\text{I}(A) \doteq \emptyset \rightarrow \text{WS}_\text{II}(A) \dot{\neq} \emptyset \quad \text{means} \\ \forall \gamma \exists \delta [(\gamma \# \delta) \notin A] \rightarrow \exists \delta \forall \gamma [(\gamma \# \delta) \notin A]. \end{aligned}$$

We first show that AD is incompatible with AC in a strong way. For the next few results we work in ZF .

Proposition 6.11.2 $\text{AD} \rightarrow \neg \text{AC}$; in fact,

$$\text{AD} \rightarrow \dot{\omega} \text{ is not well-orderable.}$$

Proof. Suppose that $\dot{\omega}$ is well-orderable; we construct a set $A \subseteq \dot{\omega}$ which is not determined. By the Well-Ordering Theorem 6.4.18 there is a smallest ordinal κ such that $\kappa \approx \dot{\omega}$; fix a function $\sigma \mapsto \gamma_\sigma$ which witnesses this. We define recursively sequences $\langle A_\sigma : \sigma < \kappa \rangle$ and $\langle B_\sigma : \sigma < \kappa \rangle$ of subsets of $\dot{\omega}$ such that for all $\sigma \leq \kappa$ if

$$A_{<\sigma} := \bigcup \{ A_\tau : \tau < \sigma \} \quad \text{and} \quad B_{<\sigma} := \bigcup \{ B_\tau : \tau < \sigma \},$$

then

- (i) $A_{<\sigma} \dot{\cap} B_{<\sigma} \dot{=} \emptyset$;
- (ii) $A_{<\sigma} \dot{\supseteq} \sigma$ and $B_{<\sigma} \dot{\supseteq} \sigma$;
- (iii) $A_{<\kappa}$ is not determined.

Set $A_0 \dot{=} \emptyset \dot{=} B_0$, and suppose that $A_{<\sigma}$ and $B_{<\sigma}$ have been defined to satisfy (i) and (ii). Set

$$\begin{aligned} \beta_\sigma &\dot{=} \text{first member of } \langle \gamma_\sigma \# \gamma_\rho : \rho \dot{<} \kappa \rangle \text{ not in } A_{<\sigma}; \\ B_\sigma &\dot{=} B_{<\sigma} \dot{\cup} \{ \beta_\sigma \}; \\ \alpha_\sigma &\dot{=} \text{first member of } \langle \gamma_\rho \# \gamma_\sigma : \rho \dot{<} \kappa \rangle \text{ not in } B_\sigma; \\ A_\sigma &\dot{=} A_{<\sigma} \dot{\cup} \{ \alpha_\sigma \}. \end{aligned}$$

Property (ii) guarantees that α_σ and β_σ exist, and the construction clearly preserves both (i) and (ii) because of the choice of κ . Finally, $A_{<\kappa}$ is not determined since for each $\sigma \dot{<} \kappa$,

$$\begin{aligned} \exists \delta [(\gamma_\sigma \# \delta) \notin A_{<\kappa}] \quad \text{so} \quad \gamma_\sigma \notin \text{WS}_I(A_{<\kappa}); \\ \exists \delta [(\delta \# \gamma_\sigma) \in A_{<\kappa}] \quad \text{so} \quad \gamma_\sigma \notin \text{WS}_{II}(A_{<\kappa}). \quad \square \end{aligned}$$

This suggests that AD is perhaps too strong a principle and we should look at $\text{AD}[\mathcal{C}]$ for smaller classes $\mathcal{C}$. We discuss this option below, but first we examine a bit further the consequences of the full axiom AD. First, although AD contradicts the full AC, it actually implies some restricted choice principles and a version of the Continuum Hypothesis.

Proposition 6.11.3 (ZF + AD) *For any countable set $x \dot{=} \{y_k : k \dot{\in} \dot{\omega}\}$ of non-empty subsets of $\dot{\omega}$,*

$$\exists c (\text{Fn}(c) \wedge \dot{\omega} \dot{=} \text{Dom}(c) \wedge (\forall k \dot{\in} \dot{\omega}) c(k) \dot{\in} y_k).$$

Proof. Fix such an x and let

$$A \dot{=} \{ \varepsilon : \varepsilon \in \Pi \notin y_{\varepsilon_I(0)} \} \quad \text{where} \quad \varepsilon_I(k) \dot{=} \varepsilon(2k) \quad \text{and} \quad \varepsilon_{II}(k) \dot{=} \varepsilon(2k+1).$$

Clearly player I cannot have a winning strategy for A , so by AD, player II has a winning strategy δ . For each k , let γ_k be any strategy which makes initial move k and for $k \dot{\in} \dot{\omega}$, set $c(k) \dot{=} (\gamma_k \# \delta)_{II}$. $\square$

Definition 6.11.4 For each $\alpha \in \dot{\omega}\dot{\omega}$,

- (i) $<_\alpha := \{ \langle \dot{m}, n \rangle : \alpha(\langle \dot{m}, n \rangle) \doteq 1 \}$;
- (ii) $\text{WO} := \{ \alpha : <_\alpha \text{ is a well-ordering} \}$;
- (iii) $\|\alpha\| := \begin{cases} \text{the unique countable ordinal } \sigma \\ \text{such that } <_\alpha \simeq \dot{<}_\sigma, \end{cases} \quad \begin{array}{l} \text{if } \alpha \in \text{WO}; \\ \aleph_1, \end{array} \quad \begin{array}{l} \\ \text{otherwise.} \end{array}$

Lemma 6.11.5 $\aleph_1 \doteq \{ \|\alpha\| : \alpha \in \text{WO} \}$.

Proof. The countable ordinals are exactly those which can be represented by a well-ordering of a subset of $\dot{\omega}$. $\square$

Recall from Proposition 6.10.3 that in ZFC every successor cardinal is regular. Replacing AC by AD we have at least

Proposition 6.11.6 $(\text{ZF} + \text{AD}) \aleph_1 \text{ is a regular cardinal.}$

Proof. It suffices to show that $\bigcup \text{Im}(f) \dot{<} \aleph_1$ for any function $f : \dot{\omega} \rightarrow \aleph_1$. For each $m \in \dot{\omega}$, let $y_m := \{ \alpha \in \text{WO} : \|\alpha\| \doteq f(m) \}$, and by the preceding proposition let c be a choice function such that for all m , $c(m) \in y_m$. Define a function β by

$$\beta(\langle \langle \dot{m}, p \rangle, \langle \dot{n}, q \rangle \rangle) \doteq 1 \leftrightarrow m \dot{<} n \vee (m \doteq n \wedge c(m)(\langle \dot{p}, q \rangle) \doteq 1).$$

Easily $\beta \in \text{WO}$ and since β codes a well-ordering which is the sum of those coded by the $c(m)$, for all m , $f(m) \leq \|\beta\|$. Hence since $\|\beta\| \dot{<} \aleph_1$, also $\bigcup \text{Im}(f) \dot{<} \aleph_1$. $\square$

Proposition 6.11.7 $(\text{ZF} + \text{AD})$ For any $A \subseteq \dot{\omega}^2$, if A is uncountable, then $A \approx \dot{\omega}^2$.

Proof. Fix $A \subseteq \dot{\omega}^2$. We define a game A^* as follows. The idea of A^* is that each move of player I is interpreted as a finite sequence of moves in A , while player II makes single moves. For any $\varepsilon \in \dot{\omega}\dot{\omega}$,

$$\varepsilon^* := (\varepsilon(0))_0, \dots, (\varepsilon(0))_{\text{lg}(\varepsilon(0))-1}, \varepsilon(1), (\varepsilon(2))_0, \dots, (\varepsilon(2))_{\text{lg}(\varepsilon(2))-1}, \varepsilon(3), \dots$$

and

$$A^* := \{ \varepsilon : \varepsilon^* \in A \vee \exists k [\varepsilon(2k+1) \geq 2] \}.$$

Intuitively the game is heavily slanted in favor of player I; he may make any finite number of moves at each turn and automatically wins if player II ever

plays any number larger than 1. Hence, II can win only when A is a very small set, and it will suffice to show that

- (i) $\text{WS}_I(A^*) \neq \emptyset \rightarrow A \approx \omega \dot{2}$;
- (ii) $\text{WS}_{II}(A^*) \neq \emptyset \rightarrow A$ is countable.

Suppose first that player I has a winning strategy γ . Then

$$\{\dot{\gamma} \# \delta : \delta \in \dot{\omega} \dot{2}\} \subseteq A^* \quad \text{so} \quad \dot{\{(\gamma \# \delta)^* : \delta \in \dot{\omega} \dot{2}\}} \subseteq A.$$

For each $\alpha \in \dot{\omega} \dot{2}$, let δ_α be a function such that for any t with $\text{lg}(t) \doteq 2k+1$, $\delta_\alpha(t) \doteq \alpha(k)$ — that is, δ_α is a strategy that plays $\alpha(k)$ at player II's k -th move regardless of the preceding play. Now if $\varepsilon_\alpha \doteq \gamma \# \delta_\alpha$, for all $k \in \dot{\omega}$,

$$\varepsilon_\alpha(2k+1) \doteq \delta_\alpha(\bar{\varepsilon}_\alpha(2k+1)) \doteq \alpha(k).$$

If $\alpha \neq \beta$, then for some k , $\alpha \upharpoonright k \doteq \beta \upharpoonright k$ but $\alpha(k) \neq \beta(k)$, so

$$\bar{\varepsilon}_\alpha(2k+1) \doteq \bar{\varepsilon}_\beta(2k+1) \quad \text{but} \quad \varepsilon_\alpha(2k+1) \neq \varepsilon_\beta(2k+1),$$

and thus the mapping $\alpha \mapsto \varepsilon_\alpha^*$ is injective and witnesses that $\dot{\omega} \dot{2} \preceq A$. Then $A \approx \dot{\omega} \dot{2}$ by the Schröder-Bernstein Theorem 6.4.4.

For (ii), suppose that player II has a winning strategy δ for A^* ; clearly by the second clause in the definition of A^* , we may assume that $\delta \in \dot{\omega} \dot{2}$. Let

$$B \doteq \{\dot{\alpha} \in \dot{\omega} \dot{2} : \forall k (\exists l \geq k) \alpha(l) \doteq \delta(\bar{\alpha}(l))\}.$$

For any $\alpha \in B$, let γ_α be a function such that for all k ,

$$\gamma_\alpha(\bar{\alpha}(k)) \doteq \langle \alpha(k), \dots, \alpha(l-1) \rangle \quad \text{for } l \geq k \text{ such that } \alpha(l) \doteq \delta(\bar{\alpha}(l)).$$

Then clearly $(\gamma_\alpha \# \delta)^* \doteq \alpha$, and since δ is a winning strategy for II, $\alpha \notin A$; hence $A \subseteq \dot{\omega} \dot{2} \setminus B$, and we finish by showing that this set is countable. For any $\alpha \in \dot{\omega} \dot{2}$,

$$\begin{aligned} \alpha \notin B &\rightarrow \exists k (\forall l \geq k) \alpha(l) \neq \delta(\bar{\alpha}(l)) \\ &\rightarrow \exists k (\forall l \geq k) \alpha(l) \doteq 1 - \delta(\bar{\alpha}(l)). \end{aligned}$$

For each k , there are at most 2^k many such α corresponding to the possible initial segments of length k ; hence in all there are at most countably many such α . $\square$

Corollary 6.11.8 (ZF+AD) *For any set $X \approx \dot{\omega}2$, any well-orderable subset of X is countable.*

Proof. Suppose towards a contradiction that there exists an uncountable set $A \subseteq X \approx_f \dot{\omega}2$ which is well-orderable. Then $\dot{\text{lm}}(f \upharpoonright A)$ is a well-orderable uncountable subset of $\dot{\omega}2$. By the preceding proposition, $\dot{\text{lm}}(f \upharpoonright A) \approx \dot{\omega}2 \approx \dot{\omega}\dot{\omega}$, which implies that $\dot{\omega}\dot{\omega}$ is well-orderable, contrary to Proposition 6.11.2. $\square$

This result has strong consequences for the constructible universe L .

Theorem 6.11.9 (ZF + AD) $\aleph_1^L < \aleph_1$; in fact, $\aleph_1 \in \text{SI}^L$, so $\aleph_1 \doteq \aleph_{\aleph_1}^L$.

Proof. We argue in ZF+AD. By Proposition 6.10.11 $\text{WI}^L \doteq \text{SI}^L$, so it suffices to prove that $\aleph_1 \in \text{WI}^L$. By Proposition 6.11.6, $\aleph_1$ is a regular cardinal, and by the analysis in the proof of Proposition 6.10.8, this is a Π^{ZF} property so also by Proposition 6.5.36, $\aleph_1$ (is a regular cardinal) $^{(L)}$. Hence it suffices to show that $\aleph_1$ (is a limit cardinal) $^{(L)}$. If this is not the case, then there exists $\sigma < \aleph_1$ such that

$$(\sigma \text{ is a cardinal})^{(L)} \quad \text{but} \quad \forall \tau [\sigma < \tau < \aleph_1 \rightarrow \neg (\tau \text{ is a cardinal})^{(L)}].$$

Hence $(\forall \tau < \aleph_1) \exists^L f [\text{Inj}(f) \wedge f : \tau \rightarrow \sigma]$. By Theorem 6.7.14, there is a function $\tau \mapsto f_\tau$ with f_τ the $<_L$ -least such function, and we define

$$r_\tau := \left\{ \langle v, v' \rangle : f_\tau^{-1}(v) < f_\tau^{-1}(v') \right\}.$$

Thus $\{r_\tau : \tau < \aleph_1\}$ is an uncountable well-ordered subset of $\dot{\wp}(\sigma \times \sigma)$, but since σ is countable, $\dot{\wp}(\sigma \times \sigma) \approx \dot{\omega}2$ and we have a contradiction with the corollary. $\square$

Corollary 6.11.10 (ZF + AD) $\text{Con}(\text{ZFC})$.

Proof. Recall that in the proof of Theorem 6.10.14 we showed that

$$\text{ZF} \vdash \forall \kappa [\kappa \in \text{SI}^L \rightarrow \text{ModelOf}(L_\kappa, \text{ZFC})].$$

By the preceding theorem we have thus

$$\text{ZF} + \text{AD} \vdash \text{ModelOf}(L_{\aleph_1}, \text{ZFC}). \quad \square$$

It follows that, as for the other principles we have discussed, there can be no relative consistency proof.

Corollary 6.11.11 *Unless ZF is actually inconsistent, there is no proof formalizable in ZFC that*

$$\text{ZF is consistent} \implies \text{ZF} + \text{AD is consistent.} \quad \square$$

We turn now to consideration of $\text{AD}[\mathcal{C}]$ for various $\mathcal{C}$. It turns out that the classes of interest are described in terms of a natural topology on ${}^\omega\omega$ and fall into hierarchies which share many of the properties of the Arithmetical Hierarchy discussed in [Section 5.1](#) — see Definition 5.1.1 and the following results; indeed the sources mentioned for Collateral Reading below explain how these hierarchies are actually generalizations and extensions of the Arithmetical Hierarchy. As in [Section 5.1](#) we shall need to consider relations as well as sets (which may thought of as unary relations), and we write

$$R(\alpha_0, \dots, \alpha_{l-1}) \quad \text{for} \quad \langle \alpha_0, \dots, \alpha_{l-1} \rangle \dot{\in} R.$$

Letters $R, S, \dots$ always denote subsets of $({}^\omega\omega)^l$ for some l . In this section we shall often use AC, since it simplifies things and seems to be consistent with the restricted principles $\text{AD}[\mathcal{C}]$ we consider. Note that these are really much weaker than the full AD, since the classes $\mathcal{C}$ we consider all have cardinality $2^{\aleph_0}$ in contrast to $|\dot{\varnothing}({}^\omega\omega)| \doteq 2^{2^{\aleph_0}}$.

Definition 6.11.12

- (i) A **tree** is a set $T \subseteq {}^\omega\omega$ such that for all s and t ,

$$s \dot{\in} T \rightarrow [t \subseteq s \rightarrow t \dot{\in} T];$$

$[T] := \{ \alpha : \forall k [\bar{\alpha}(k) \dot{\in} T] \}$ is the set of (infinite) **branches** of T .

- (ii) For $l \dot{\in} \omega$, an **l -tree** is a set $T \subseteq {}^\omega\omega^l$ such that (for $l \dot{\in} 2$) for all s_0, s_1, t_0 and t_1 ,

$$\begin{aligned} \langle s_0, s_1 \rangle \dot{\in} T &\rightarrow [\text{lg}(s_0) \doteq \text{lg}(s_1)] \wedge \\ &[t_0 \subseteq s_0, \wedge t_1 \subseteq s_1 \wedge \text{lg}(t_0) \doteq \text{lg}(t_1) \rightarrow \langle t_0, t_1 \rangle \dot{\in} T]; \end{aligned}$$

$$[T] := \left\{ \langle \alpha_0, \dots, \alpha_{l-1} \rangle : \forall k [\langle \bar{\alpha}_0(k), \dots, \bar{\alpha}_{l-1}(k) \rangle \dot{\in} T] \right\}.$$

Definition 6.11.13

- (i) A set $A \subseteq {}^\omega\omega$ is **closed** iff $A \doteq [T]$ for some tree T ; A is **open** iff $\sim A \doteq {}^\omega\omega \setminus A$ is closed.
- (ii) A relation $R \subseteq ({}^\omega\omega)^l$ is **closed** iff $R \doteq [T]$ for some l -tree T ; R is **open** iff $\sim R \doteq ({}^\omega\omega)^l \setminus R$ is closed.

We leave to the exercises the easy verification that this does define a topology known as the **Baire topology**: the class of open (closed) relations is closed under finite intersection (union) and arbitrary union (intersection). It may also be described as the product topology generated by the discrete topology on ω . Other standard topological notions are defined as usual; in particular, a function $F : {}^\omega\omega \rightarrow {}^\omega\omega$ is **continuous** iff for every open set A , $F^{-1}(A)$ is open.

Definition 6.11.14

- (i) Σ_1^0 is the class of all open relations;
- for all countable ordinals $\sigma \dot{>} 0$,
- (ii) $\Pi_\sigma^0 \doteq \{ \sim R : R \in \Sigma_\sigma^0 \}$;
 - (iii) $\Sigma_\sigma^0 \doteq \{ \dot{\bigcup}_{n \in \omega} R_n : \forall n (\exists \tau \dot{<} \sigma) R_n \in \Pi_\tau^0 \}$ for $\sigma \dot{>} 1$;
 - (iv) $\Delta_\sigma^0 \doteq \Sigma_\sigma^0 \dot{\cap} \Pi_\sigma^0$;
 - (v) $\text{Bo} \doteq \dot{\bigcup} \{ \Sigma_\sigma^0 \dot{\cup} \Pi_\sigma^0 : 0 \dot{<} \sigma \dot{<} \aleph_1 \}$, the class of **Borel relations**.

Of course in writing $\dot{\bigcup}_{n \in \omega} R_n$ we intend that all R_n are subsets of $({}^\omega\omega)^l$ for some common l . We summarize some basic properties of these classes, but leave the relatively easy proofs to the exercises.

Proposition 6.11.15 For all countable ordinals $\sigma \dot{>} 0$,

- (i) Σ_σ^0 , Π_σ^0 and Δ_σ^0 are closed under relational composition with continuous functions and finite union and intersection;
- (ii) Σ_σ^0 is closed under countable union; Π_σ^0 is closed under countable intersection; Δ_σ^0 is closed under complementation;
- (iii) $\Sigma_\sigma^0 \doteq \{ \sim R : R \in \Pi_\sigma^0 \}$;
 $\Pi_\sigma^0 \doteq \{ \dot{\bigcap}_{n \in \omega} R_n : \forall n (\exists \tau \dot{<} \sigma) R_n \in \Sigma_\tau^0 \}$;
- (iv) $\Sigma_\sigma^0 \dot{\cup} \Pi_\sigma^0 \dot{\subseteq} \Delta_{\sigma+1}^0$;
- (v) Δ_σ^0 is a proper subset of both Σ_σ^0 and Π_σ^0 ; $\Sigma_\sigma^0 \dot{\cup} \Pi_\sigma^0$ is a proper subset of $\Delta_{\sigma+1}^0$;
- (vi) Bo is the smallest class of relations which includes Σ_1^0 and is closed under complementation and countable union. $\square$

Note in particular that (i) implies that if A is a Borel set and F a continuous function, then also $F^{-1}(A)$ is Borel. This **Borel Hierarchy** has been studied extensively, especially in the first third of the 20th century. Together with the Projective Hierarchy defined below, it makes up some of the central subject matter of **Descriptive Set Theory**. The material of the rest of this section is a mere taste of this deep and complex area.

The most important result for us here about this hierarchy is

Borel Determinacy Theorem 6.11.16 (ZFC) $\text{AD}[\text{Bo}]$.

The proof of this is long and deep and well beyond the scope of this book, but to give a bit of the flavor, and because we will make specific use of it, we give a proof of the first step, $\text{AD}[\Pi_1^0]$. In the following, let $A_s := \{ \alpha : s \smallfrown \alpha \in A \}$, so $A \dot{=} A_{\langle \rangle}$.

Lemma 6.11.17 (ZFC) For any $A \subseteq {}^\omega\omega$,

(i) I has a winning strategy for A iff

$$\exists p [II \text{ has a winning strategy for } \sim A_{\langle p \rangle}];$$

(ii) II has a winning strategy for A iff

$$\forall p [I \text{ has a winning strategy for } \sim A_{\langle p \rangle}].$$

Proof. For (i)($\rightarrow$), for any γ ,

$$\begin{aligned} \gamma \in \text{WS}_I(A) &\leftrightarrow \forall \delta [(\gamma \# \delta) \in A] \\ &\leftrightarrow \forall \delta [(\delta \# \gamma') \in A_{\langle \gamma(\langle \rangle) \rangle}] \\ &\leftrightarrow \gamma' \in \text{WS}_{II}(\sim A_{\langle \gamma(\langle \rangle) \rangle}), \end{aligned}$$

where $\gamma'(s) := \gamma(\langle \gamma(\langle \rangle) \rangle \smallfrown s)$. For (i)($\leftarrow$),

$$\begin{aligned} \gamma \in \text{WS}_{II}(\sim A_{\langle p \rangle}) &\leftrightarrow \forall \delta [(\delta \# \gamma) \in A_{\langle p \rangle}] \\ &\leftrightarrow \forall \delta [(\gamma^p \# \delta) \in A] \\ &\leftrightarrow \gamma^p \in \text{WS}_I(A), \end{aligned}$$

where $\gamma^p(\langle \rangle) \dot{=} p$ and for all s , $\gamma^p(\langle p \rangle \smallfrown s) \dot{=} \gamma(s)$. For (ii)($\rightarrow$),

$$\begin{aligned} \delta \in \text{WS}_{II}(A) &\leftrightarrow \forall \gamma [(\gamma \# \delta) \notin A] \\ &\leftrightarrow \forall \gamma [(\delta_p \# \gamma) \notin A_{\langle p \rangle}] \\ &\leftrightarrow \forall p [\delta_p \in \text{WS}_I(\sim A_{\langle p \rangle})], \end{aligned}$$

where $\delta_p(s) := \delta(\langle p \rangle^\frown s)$. Finally, for (ii)($\leftarrow$), if $\forall p [\text{WS}_\text{I}(\sim A_{\langle p \rangle}) \neq \emptyset]$, using AC there is a function $p \mapsto \delta^p$ such that for all p , $\delta^p \in \text{WS}_\text{I}(\sim A_{\langle p \rangle})$. Then for $\delta(\langle p \rangle^\frown s) := \delta^p(s)$, easily $\delta \in \text{WS}_\text{II}(A)$. $\square$

Theorem 6.11.18 (ZFC) $\text{AD}[\Pi_1^0]$ and $\text{AD}[\Sigma_1^0]$.

Proof. By the lemma, either of these follows from the other, since

$$A \text{ is open (closed)} \leftrightarrow \forall p [A_{\langle p \rangle} \text{ is open (closed)}].$$

Suppose A is closed, so for some tree T , $A \doteq [T]$. By the lemma, for any s , if II has no winning strategy for A_s , then

$$\exists p \forall q [\text{II has no winning strategy for } A_{s \smallfrown \langle p, q \rangle}].$$

Let

$$\gamma(s) := \begin{cases} \text{least } p \forall q [\text{WS}_\text{II}(A_{s \smallfrown \langle p, q \rangle}) \doteq \emptyset]. & \text{if any;} \\ 0, & \text{otherwise.} \end{cases}$$

Now if $\text{WS}_\text{II}(A) \doteq \emptyset$, then for any strategy δ for player II, it follows by induction that for all k ,

$$\text{WS}_\text{II}(A_{\overline{\gamma \# \delta}(2k)}) \doteq \emptyset, \quad \text{whence} \quad \overline{\gamma \# \delta}(2k) \in T,$$

since for $s \notin T$, *any* strategy is a win for II in A_s . But then since T is closed under subsequence, $\forall k [\overline{\gamma \# \delta}(k) \in T]$, so $\gamma \# \delta \in A$. Thus γ is a winning strategy for player I. $\square$

The Borel Determinacy Theorem tells us that if we are to find principles of the form $\text{AD}[\mathcal{C}]$ which extend ZFC, we need to consider classes $\mathcal{C}$ larger than Bo. Analogously to Definition 4.3.4,

$$\begin{aligned} (\exists^1 R)(\alpha_0, \dots, \alpha_{l-1}) : & \leftrightarrow \exists \beta R(\alpha_0, \dots, \alpha_{l-1}, \beta); \\ (\forall^1 R)(\alpha_0, \dots, \alpha_{l-1}) : & \leftrightarrow \forall \beta R(\alpha_0, \dots, \alpha_{l-1}, \beta). \end{aligned}$$

The superscript 1 here and in the following definition indicates that we are quantifying over objects of type 1 (functions). Note that if R is an $(l+1)$ -ary relation, then $\exists^1 R$ can be viewed as the projection of R into $({}^\omega \omega)^l$.

Definition 6.11.19

- (i) $\Sigma_1^1 := \{ \dot{\exists}^1 R : R \in \Pi_1^0 \};$
 for all $n \succ 0$,
 (ii) $\Pi_n^1 := \{ \dot{\sim} R : R \in \Sigma_n^1 \};$
 (iii) $\Sigma_{n+1}^1 := \{ \dot{\exists}^1 R : R \in \Pi_n^1 \};$
 (iv) $\Delta_n^1 := \Sigma_n^1 \cap \Pi_n^1;$
 (v) $\text{Proj} := \bigcup \{ \Sigma_n^1 \cup \Pi_n^1 : n \succ 0 \},$ the class of **projective relations**.

Again there is a list of relatively simple properties of these classes which we list for reference and leave to an exercise for proof.

Proposition 6.11.20 *For all $n \succ 0$,*

- (i) Σ_n^1, Π_n^1 and Δ_n^1 are closed under relational composition with continuous functions, countable union and countable intersection;
 (ii) Σ_n^1 is closed under $\dot{\exists}^1$; Π_n^1 is closed under $\dot{\forall}^1$; Δ_n^1 is closed under complementation;
 (iii) $\Sigma_n^1 \dot{\sim} \{ \dot{\sim} R : R \in \Pi_n^1 \};$ $\Pi_{n+1}^1 \dot{\sim} \{ \dot{\forall}^1 R : R \in \Sigma_n^1 \};$
 (iv) $\Sigma_n^1 \cup \Pi_n^1 \subseteq \Delta_{n+1}^1;$
 (v) Δ_n^1 is a proper subset of both Σ_n^1 and Π_n^1 ; $\Sigma_n^1 \cup \Pi_n^1$ is a proper subset of Δ_{n+1}^1 . $\square$

Corollary 6.11.21 $\text{Bo} \subseteq \Delta_1^1.$

Proof. Immediate from Propositions 6.11.15(vi) and (i) and (ii) of the preceding proposition. $\square$

We aim next to show that in fact $\text{Bo} \dot{=} \Delta_1^1$. A more leisurely presentation of these ideas would contain many examples of sets and relations in these classes; here we content ourselves with the following, which is most important for our purposes. Recall from Definition 6.11.4 the set WO of codes for countable ordinals. For each countable ordinal σ , set

$$\text{WO}^{<\sigma} := \{ \alpha \in \text{WO} : \|\alpha\| \dot{<} \sigma \},$$

the set of codes for ordinals less than σ .

Proposition 6.11.22

- (i) $\text{WO} \in \Pi_1^1.$
 (ii) For each countable ordinal σ , $\text{WO}^{<\sigma} \in \text{Bo}.$

Proof. It is easy to see that $\mathbf{LO} := \{ \alpha : <_\alpha \text{ is a linear ordering} \}$ is a Borel set; the number quantifiers in the definition correspond to countable intersections. Then

$$\begin{aligned} \mathbf{WO} &\doteq \mathbf{LO} \dot{\cap} \{ \alpha : <_\alpha \text{ is well-founded} \} \\ &\doteq \mathbf{LO} \dot{\cap} \left\{ \alpha : \forall \beta \exists n \left([\alpha(\langle \beta(n+1), \beta(n) \rangle) \neq 1] \right) \right\} \end{aligned}$$

is Π_1^1 . We prove (ii) by ordinal induction. It is clear for $\sigma \doteq 0$, the induction steps follow from

$$\mathbf{WO}^{<\sigma+1} \doteq \bigcap_{r \in \omega} \{ \alpha : (\alpha \upharpoonright r) \in \mathbf{WO}^{<\sigma} \}$$

and if σ is a limit ordinal,

$$\mathbf{WO}^{<\sigma} \doteq \bigcup_{\tau < \sigma} \mathbf{WO}^{<\tau},$$

where $(\alpha \upharpoonright r)(\langle m, n \rangle) \doteq 1 \leftrightarrow \alpha(\langle m, n \rangle) \doteq 1 \wedge \alpha(\langle n, r \rangle) \doteq 1$, so $<_{\alpha \upharpoonright r}$ is a proper initial segment of $<_\alpha$. Note that for countable σ , the union over $\tau < \sigma$ is a countable union and thus preserves the property of being Borel. $\square$

Proposition 6.11.23 *For any set $A \in \Pi_1^1$, there exists a continuous function $\alpha \mapsto \alpha^\diamond$ such that $\forall \alpha (\alpha \in A \leftrightarrow \alpha^\diamond \in \mathbf{WO})$.*

Proof. For any $A \in \Pi_1^1$ there exists a 2-tree T such that

$$\alpha \in A \leftrightarrow \neg \exists \gamma \forall k \langle \bar{\alpha}(k), \bar{\gamma}(k) \rangle \in T \leftrightarrow [T^\alpha] \dot{\neq} \emptyset,$$

where

$$T^\alpha := \left\{ s : \langle \bar{\alpha}(\lg(s)), s \rangle \in T \right\}.$$

Let $\alpha^\diamond$ be the characteristic function of the relation

$$s <_{\alpha^\diamond} t : \leftrightarrow s, t \in T^\alpha \wedge (s <_{\text{lex}} t \vee t \subset s),$$

where

$$s <_{\text{lex}} t : \leftrightarrow \exists k < \lg(s), \lg(t) [(\forall i < k)(s)_i \doteq (t)_i \wedge (s)_k < (t)_k]$$

is the usual lexicographic ordering. Easily the function $\alpha \mapsto \alpha^\diamond$ is continuous and $<_{\alpha^\diamond}$ is always a linear ordering; we show that

$$<_{\alpha^\diamond} \text{ is not well-founded} \leftrightarrow [T^\alpha] \dot{\neq} \emptyset,$$

from which the result follows immediately. If $\gamma \in [T^\alpha]$, then $\{\bar{\gamma}(k) : k \in \dot{\omega}\}$ is a subset of the field of $<_{\alpha^\circ}$ with no least element and thus witnesses that $<_{\alpha^\circ}$ is not well-founded. Conversely, suppose that $<_{\alpha^\circ}$ is not well-founded, so using AC there exists a function δ such that $\forall k [\delta(k+1) <_{\alpha^\circ} \delta(k)]$. Let

$$\gamma(k) := \text{least } n \exists m [\bar{\beta}(k) \smallfrown \langle n \rangle \subseteq \delta(m)],$$

and prove by induction that

$$\forall k [\forall l (\delta(l) \not\prec_{\text{lex}} \bar{\gamma}(k)) \wedge \exists m (\forall l \geq m) \bar{\gamma}(k) \subseteq \delta(l)].$$

In particular, this implies $\forall k [\bar{\gamma}(k) \in T^\alpha]$ because T^α is closed under subsequence, and thus $\gamma \in [T^\alpha]$. $\square$

Lemma 6.11.24 *There exist Σ_1^1 relations $\prec_\Sigma$ and $\preceq_\Sigma$ and Π_1^1 relations $\prec_\Pi$ and $\preceq_\Pi$ such that for all α and β , if either $\alpha \in \text{WO}$ or $\beta \in \text{WO}$, then*

- (i) $\alpha \preceq_\Sigma \beta \leftrightarrow (\alpha \in \text{WO} \wedge \|\alpha\| \dot{\leq} \|\beta\|) \leftrightarrow \alpha \preceq_\Pi \beta$;
- (ii) $\alpha \prec_\Sigma \beta \leftrightarrow (\alpha \in \text{WO} \wedge \|\alpha\| \dot{<} \|\beta\|) \leftrightarrow \alpha \prec_\Pi \beta$.

Proof. Define two relations by

$$\begin{aligned} \alpha \preceq \beta &: \leftrightarrow \exists \gamma \forall m \forall n [m <_\alpha n \rightarrow \gamma(m) <_\beta \gamma(n)]; \\ \alpha \prec \beta &: \leftrightarrow \exists \gamma \exists p \forall m \forall n [m <_\alpha n \rightarrow \gamma(m) <_\beta \gamma(n) <_\beta p]. \end{aligned}$$

In words, $\alpha \preceq \beta$ iff $<_\alpha$ is isomorphic to a subordering of $<_\beta$ and $\alpha \prec \beta$ iff $<_\alpha$ is isomorphic to a subordering of a proper initial segment of $<_\beta$. These relations are both Σ_1^1 and for both $\alpha, \beta \in \text{WO}$, clearly

$$\alpha \preceq \beta \leftrightarrow \|\alpha\| \dot{\leq} \|\beta\| \quad \text{and} \quad \alpha \prec \beta \leftrightarrow \|\alpha\| \dot{<} \|\beta\|.$$

Now let

$$\alpha \preceq_\Sigma \beta : \leftrightarrow \alpha \preceq \beta \vee \beta \notin \text{WO} \quad \text{and} \quad \alpha \prec_\Sigma \beta : \leftrightarrow \alpha \prec \beta \vee \beta \notin \text{WO},$$

and

$$\alpha \prec_\Pi \beta : \leftrightarrow \neg \beta \preceq_\Sigma \alpha \quad \text{and} \quad \alpha \preceq_\Pi \beta : \leftrightarrow \neg \beta \prec_\Sigma \alpha.$$

It is straightforward to verify that these relations satisfy the given conditions; recall that for $\alpha \notin \text{WO}$, $\|\alpha\| \doteq \aleph_1$. $\square$

The relations $\preceq_\Sigma$ and $\preceq_\Pi$ are called **pre-well-orderings**. Restricted to members of WO they are reflexive, transitive, connected and well-founded; from being well-orderings they lack only antisymmetry.

Corollary 6.11.25 (*Boundedness Principle*)

- (i) $\text{WO} \notin \dot{\Sigma}_1^1$; in fact,
- (ii) for any set $B \subseteq \text{WO}$, if $B \in \Sigma_1^1$, then for some $\sigma < \aleph_1$, $B \subseteq \text{WO}^{<\sigma}$.

Proof. For (i), by Proposition 6.11.20(v) there exists a set $A \in \Pi_1^1 \setminus \Sigma_1^1$ and therefore a continuous function F such that $A \equiv F^{-1}(\text{WO})$. Then if $\text{WO} \in \Sigma_1^1$, also $A \in \Sigma_1^1$, contrary to hypothesis. For (ii), suppose that $B \in \Sigma_1^1$, that $B \subseteq \text{WO}$ and towards a contradiction that for no $\sigma < \aleph_1$, $B \subseteq \text{WO}^{<\sigma}$. Then a function α codes a well-ordering iff for some $\beta \in B$, α codes a well-ordering $\leq \|\beta\|$. It follows that

$$\text{WO} \equiv \dot{\{ \alpha : \exists \beta [\beta \in B \wedge \alpha \preceq_{\Sigma} \beta] \}}.$$

But this implies that $\text{WO} \in \Sigma_1^1$, contrary to (i). $\square$

Theorem 6.11.26 $\text{Bo} \equiv \Delta_1^1$.

Proof. By Corollary 6.11.21, it suffices to prove the inclusion $(\supseteq)$; fix $A \in \Delta_1^1$. Since $A \in \Pi_1^1$, there exists a continuous function F such that $A \equiv F^{-1}(\text{WO})$. Since $A \in \Sigma_1^1$,

$$\text{Im}(F \upharpoonright A) \equiv \dot{\{ \beta : \exists \alpha [\alpha \in A \wedge \beta \equiv F(\alpha)] \}}$$

is a Σ_1^1 subset of WO , so by (ii) of the corollary is included in some $\text{WO}^{<\sigma}$. But then, $A \equiv F^{-1}(\text{WO}^{<\sigma})$, which is Borel by Propositions 6.11.22 and 6.11.15. $\square$

Thus in terms of the projective hierarchy, the first determinacy principles that might yield new results are $\text{AD}[\Sigma_1^1]$ or $\text{AD}[\Pi_1^1]$, which by Lemma 6.11.17 are equivalent. In the remainder of this section we shall show that this is indeed the case by establishing that

$$\exists \text{MC} \rightarrow \text{AD}[\Pi_1^1] \rightarrow \aleph_1 \in \text{SI}^{\text{L}}.$$

We consider first the second of these implications, which requires some preparation.

Definition 6.11.27 A nonempty set $A \subseteq {}^\omega\omega$ is **perfect** iff

- (i) A is closed;
- (ii) $(\forall \alpha \in A) \forall k (\exists \beta \in A) [\bar{\alpha}(k) \equiv \bar{\beta}(k) \wedge \alpha \neq \beta]$.

In topological language, (ii) says that A has no isolated points. A tree is called **perfect** iff for each $s \in T$, there exist incompatible $s_0, s_1 \in T$ such that $s \subseteq s_0, s_1$; each $s \in T$ **splits** in T .

Lemma 6.11.28 For any set $A \subseteq {}^\omega\omega$,

- (i) A is perfect iff for some perfect tree T , $A \dot{=} [T]$;
- (ii) if A is perfect, then $|A| \dot{=} 2^{\aleph_0}$.

Proof. For (i), clearly if T is perfect, then so is $[T]$. Conversely, if A is perfect, let $T \dot{=} \{s : (\exists \alpha \in A) s \subseteq \alpha\}$. It is easy to check that T is a perfect tree and $A \dot{=} [T]$.

For (ii), fix a perfect set A and a perfect tree T such that $A \dot{=} [T]$. Now by the definition of perfect tree, there exists a function $f : {}^{<\omega}2 \rightarrow \text{Sq}$ such that for all $u, v \in {}^{<\omega}2$,

- (iii) $f(u) \in T$;
- (iv) $u \dot{\subset} v \rightarrow f(u) \dot{\subset} f(v)$;
- (v) $u|v \rightarrow f(u)|f(v)$

(see notational conventions at the beginning of this section). By (ii) of the definition, for each $\gamma \in {}^\omega 2$ there is a unique function α_γ such that for all k , $\bar{\alpha}_\gamma(k) \subseteq f(\bar{\gamma}(k)) \subseteq \alpha$, so by (i) of the definition, $\alpha_\gamma \in A$ and by (iii) the mapping $\gamma \mapsto \alpha_\gamma$ is injective and thus witnesses that $2^{\aleph_0} \leq |A|$. $\square$

Definition 6.11.29 For any class $\mathcal{C} \subseteq \wp({}^\omega\omega)$, $\text{Perf}(\mathcal{C})$ is the assertion that every uncountable set $A \in \mathcal{C}$ has a perfect subset. We say that $\mathcal{C}$ has the **perfect set property**.

In view of the preceding lemma, the assertion $\text{Perf}(\mathcal{C})$ is a restricted version of the Continuum Hypothesis; it implies that at least in $\mathcal{C}$ there are no sets of cardinality strictly between $\aleph_0$ and $2^{\aleph_0}$. The proof of Proposition 6.11.7 shows that AD implies $\text{Perf}(\wp({}^\omega\omega))$. Indeed, the following result that this holds for the class of closed sets was discovered by Cantor early in his study of CH and encouraged him to believe that it could be extended to all sets.

Proposition 6.11.30 $\text{Perf}(\Pi_1^0)$.

Proof. Fix a closed set A and a tree T such that $A \dot{=} [T]$, and assume that A is uncountable. It will suffice to construct a perfect tree $U \dot{\subseteq} T$, since then $B \dot{=} [U]$ is the desired perfect subset of A . For each $s \in T$, set

$$T_s \dot{=} \{s' \in T : s \sim s'\}; \quad A_s \dot{=} [T_s].$$

Clearly $T_{\langle \rangle} \dot{=} T$ and for all $s \dot{\subseteq} s'$,

$$(T_s)_{s'} \dot{=} T_{s'} \dot{\subseteq} T_s \quad \text{and} \quad A_{s'} \dot{\subseteq} A_s \dot{\subseteq} A.$$

Let $U := \{s \in T : A_s \text{ is uncountable}\}$. Clearly U is a subtree of T , and it remains only to show that U is perfect. Suppose, towards a contradiction, that some $s \in U$ has no pair of incompatible extensions in U . Since $A_t \doteq \bigcup \{A_{t \smallfrown \langle n \rangle} : n \in \omega\}$, it follows that

$$s \subseteq t \in U \rightarrow \exists! n[t \smallfrown \langle n \rangle \in U],$$

so there exists a unique function $\alpha \in [U_s]$. But then $A_s \subseteq \{\alpha\} \cup \bigcup \{A_t : t \in T \setminus U\}$, which is a countable set, contrary to hypothesis. $\square$

The following extension is sometimes described as a 2-dimensional version of this argument.

Proposition 6.11.31 $\text{Perf}(\Sigma_1^1)$.

Proof. Fix a Σ_1^1 set A and a 2-tree T such that

$$A \doteq \{\alpha : \exists \beta \forall k \langle \bar{\alpha}(k), \bar{\beta}(k) \rangle \in T\}.$$

For each $\langle s, t \rangle \in T$, set

$$\begin{aligned} T_{s,t} &\doteq \{\langle s', t' \rangle \in T : s \sim s' \wedge t \sim t'\}; \\ A_{s,t} &\doteq \{\alpha : \exists \beta \forall k \langle \bar{\alpha}(k), \bar{\beta}(k) \rangle \in T_{s,t}\}. \end{aligned}$$

Let $U := \{\langle s, t \rangle \in T : A_{s,t} \text{ is uncountable}\}$. Much as in the preceding proof, we may show that for all $\langle s, t \rangle \in U$,

$$(1) \exists s_0 \exists s_1 \exists t_0 \exists t_1 [s \subseteq s_0, s_1 \wedge t \subseteq t_0, t_1 \wedge s_0 \mid s_1 \wedge \langle s_0, t_0 \rangle, \langle s_1, t_1 \rangle \in U].$$

From (1), it is straightforward to define functions $f, g : {}^{<\omega}2 \rightarrow \dot{\Sigma}q$ such that for all $u \in {}^{<\omega}2$,

- (i) $\langle f(u), g(u) \rangle \in T$;
- (ii) $u \dot{\subset} v \rightarrow f(u) \dot{\subset} f(v) \wedge g(u) \dot{\subset} g(v)$;
- (iii) $u \mid v \rightarrow f(u) \mid f(v)$.

Now set $V := \{s : (\exists u \in {}^{<\omega}2) s \subseteq f(u)\}$. Clearly V is a perfect tree and therefore $B := [V]$ is a perfect set; we show that $B \subseteq A$. For any $\alpha \in B$, by (iii) there is a unique $\gamma \in {}^{\dot{\omega}}2$ such that for all k , $\bar{\alpha}(k) \subseteq (\bar{\gamma}(k)) \dot{\subset} \alpha$. By (ii), there is a unique function β such that for all k , $\bar{\beta}(k) \subseteq g(\bar{\gamma}(k)) \dot{\subset} \beta$, and then by (i), for all k , $\langle \bar{\alpha}(k), \bar{\beta}(k) \rangle \in T$, so $\alpha \in A$. $\square$

Proposition 6.11.32 $\text{AD}[\Pi_1^1] \rightarrow \text{Perf}(\Pi_1^1)$.

Proof. Assume $\text{AD}[\Pi_1^1]$ and fix an uncountable Π_1^1 set A . Suppose first that $A \subseteq \dot{\omega}2$. With notation as in the proof of Proposition 6.11.7, since A is not countable, there can be no winning strategy for player II in the game A^* . But A^* is easily also Π_1^1 , so under the hypothesis $\text{AD}[\Pi_1^1]$, player I must have a winning strategy γ for A^* . Thus

$$B := \{(\gamma \# \delta)^* : \delta \in \dot{\omega}2\} = \{\alpha : \exists \delta [\alpha \dot{=} (\gamma \# \delta)^*]\} \subseteq A,$$

and is uncountable. But $B \in \Sigma_1^1$, so by Proposition 6.11.31, B has a perfect subset, hence so does A .

For the general case, consider the continuous mapping $\dot{\omega}\dot{\omega} \dot{\rightarrow} \dot{\omega}2$:

$$\alpha \mapsto \alpha^+ := (\underbrace{0 \cdots 0}_\alpha 1 \underbrace{0 \cdots 0}_\alpha 1 \cdots).$$

Let $E := \{\gamma \in \dot{\omega}2 : \exists k (\forall l \geq k) \gamma(l) \dot{=} 0\}$, so $\dot{\omega}2 \dot{=} \{\alpha^+ : \alpha \in \dot{\omega}\dot{\omega}\} \dot{\cup} E$ and for $\gamma \in \dot{\omega}2 \setminus E$, there is a continuous inverse function $\gamma \mapsto \gamma^-$. Now if $A \subseteq \dot{\omega}\dot{\omega}$ is Π_1^1 and uncountable, then so is $\{\gamma \in \dot{\omega}2 \setminus E : \gamma^- \in A\}$, so by the first part this set has a perfect subset B . Then $\{\alpha : \alpha^+ \in B\}$ is an uncountable closed subset of A and thus has a perfect subset. $\square$

Definition 6.11.33 For any class $\mathcal{C} \subseteq \dot{\wp}(\dot{\omega}\dot{\omega})$, $\text{Unif}(\mathcal{C})$ is the assertion that for any relation $R \subseteq (\dot{\omega}\dot{\omega})^{l+1}$ with $R \in \mathcal{C}$, there exists $\hat{R} \subseteq R$, $\hat{R} \in \mathcal{C}$, such that

$$\forall \alpha_0 \cdots \forall \alpha_{l-1} [\exists \beta R(\alpha_0, \dots, \alpha_{l-1}, \beta) \leftrightarrow \exists! \beta \hat{R}(\alpha_0, \dots, \alpha_{l-1}, \beta)].$$

We say that $\mathcal{C}$ has the **uniformization property**.

Note that $\hat{R}$ is the graph of a function with the same domain as R so this can be thought of as a sort of definable choice principle. Note also that the Partial Selection Theorem 4.3.15 says exactly that the class of semi-recursive relations has the uniformization property (Corollary 4.3.17).

Uniformization Theorem 6.11.34 $\text{Unif}(\Pi_1^1)$.

Proof. We extend the analysis of Π_1^1 sets and relations begun in the proof of Proposition 6.11.23. Suppose that R is a (binary) Π_1^1 relation, so for some 3-tree T ,

$$(2) \quad R(\alpha, \beta) \leftrightarrow \neg \exists \gamma \forall k \langle \bar{\alpha}(k), \bar{\beta}(k), \bar{\gamma}(k) \rangle \in T.$$

Exactly as before we define

$$T^{\alpha,\beta} := \left\{ s : \langle \bar{\alpha}(\lg(s)), \bar{\beta}(\lg(s)), s \rangle \in T \right\},$$

set $(\alpha\beta)^\diamond$ to be the characteristic function of the strict linear ordering

$$s <_{(\alpha\beta)^\diamond} t : \leftrightarrow s, t \in T^{\alpha,\beta} \wedge (s <_{\text{lex}} t \vee t \subset s),$$

and conclude that

$$(3) \quad R(\alpha, \beta) \leftrightarrow \neg \exists \gamma \forall k \bar{\gamma}(k) \in T^{\alpha,\beta} \leftrightarrow [T^{\alpha,\beta}] \dot{\neq} \emptyset \leftrightarrow (\alpha\beta)^\diamond \in \text{WO}.$$

Now, since $<_{(\alpha\beta)^\diamond}$ is a countable ordering, it is a well-ordering iff it is isomorphic to a subordering of $<_{\aleph_1}$, and in this case there is a unique countable ordinal $\sigma_{\alpha,\beta}$ and function $f_{\alpha,\beta}$ which is an isomorphism of $<_{(\alpha\beta)^\diamond}$ with $<_{\sigma_{\alpha,\beta}}$. In fact, if we define for any α and β ,

$$f_{\alpha,\beta}(u) := \|(\alpha\beta)^\diamond \upharpoonright u \|,$$

the order-type of the initial segment of $<_{(\alpha\beta)^\diamond}$ determined by u , then

$$\begin{aligned} R(\alpha, \beta) &\leftrightarrow (\exists h \in \dot{\omega} \dot{\aleph}_1) \forall s \forall t [s <_{(\alpha\beta)^\diamond} t \rightarrow h(s) \dot{<} h(t)] \\ &\leftrightarrow \forall u [f_{\alpha,\beta}(u) \dot{<} \dot{\aleph}_1] \\ &\leftrightarrow f_{\alpha,\beta}(0) \dot{<} \dot{\aleph}_1. \end{aligned}$$

The second and third equivalences are based on the fact that a strict linear ordering is a well-ordering iff every proper initial segment is a well-ordering, and in the case at hand iff the proper initial segment determined by the greatest element $0 \doteq \langle \rangle$ is a well-ordering. This equivalence can be expressed in terms of a tree (in a natural extended sense) of finite sequences of countable ordinals: if

$$\begin{aligned} \mathbb{T}^{\alpha,\beta} &:= \left\{ \langle \dot{\sigma}_0, \dots, \sigma_{k-1} \rangle : k \in \dot{\omega} \wedge \right. \\ &\quad \left. (\forall s \dot{<} k)(\forall t \dot{<} k)[s <_{(\alpha\beta)^\diamond} t \rightarrow \sigma_s \dot{<} \sigma_t] \right\}, \end{aligned}$$

then

$$\begin{aligned} (4) \quad R(\alpha, \beta) &\leftrightarrow (\exists h \in \dot{\omega} \dot{\aleph}_1) \forall k (h \upharpoonright k \in \mathbb{T}^{\alpha,\beta}) \\ &\leftrightarrow [\mathbb{T}^{\alpha,\beta}] \dot{\neq} \emptyset \leftrightarrow f_{\alpha,\beta} \in [\mathbb{T}^{\alpha,\beta}]. \end{aligned}$$

When the conditions of (4) hold, $f_{\alpha,\beta}$ is the **pointwise left-most path** of $\mathbb{T}^{\alpha,\beta}$ in the sense that

$$(5) \quad R(\alpha, \beta) \rightarrow (\forall h \in \dot{\omega} \dot{\aleph}_1) \left(h \in [\mathbb{T}^{\alpha,\beta}] \rightarrow \forall s [f_{\alpha,\beta}(s) \dot{\leq} h(s)] \right).$$

This follows by induction on $f_{\alpha,\beta}(s)$, since when $f_{\alpha,\beta}(s) \dot{>} 0$, for any $h \in [\mathbb{T}^{\alpha,\beta}]$,

$$\begin{aligned} f_{\alpha,\beta}(s) &\dot{=} \bigcup \{ f_{\alpha,\beta}(t) \dot{+} 1 : t <_{(\alpha\beta)^\diamond} s \} \\ &\dot{\leq} \bigcup \{ h(t) \dot{+} 1 : t <_{(\alpha\beta)^\diamond} s \} \dot{\leq} h(s). \end{aligned}$$

Now, towards the definition of $\hat{R}$, for $v \in \dot{\mathbf{S}}\mathbf{q}$, let

$$\begin{aligned} T^{\alpha,v} &:= \left\{ s : s \dot{\leq} \mathbf{lg}(v) \wedge \langle \bar{\alpha}(\mathbf{lg}(s)), v \upharpoonright \mathbf{lg}(s), s \rangle \in T \right\}; \\ s <^{\alpha,v} t &: \leftrightarrow s, t \in T^{\alpha,v} \wedge (s <_{\text{lex}} t \vee t \subset s); \\ \mathbb{T}^\alpha &:= \left\{ \langle \sigma_0, n_0, \dots, \sigma_{k-1}, n_{k-1} \rangle : k \in \dot{\omega} \wedge \right. \\ &\quad \left. (\forall s \dot{<} k)(\forall t \dot{<} k) [s <^{\alpha, \langle n_0, \dots, n_{k-1} \rangle} t \rightarrow \sigma_s \dot{<} \sigma_t] \right\}. \end{aligned}$$

Then easily

- (6) $R(\alpha, \beta) \leftrightarrow (\exists h \in \dot{\omega}\dot{\aleph}_1) \forall k (\langle h(0), \beta(0), \dots, h(k-1), \beta(k-1) \rangle \in \mathbb{T}^\alpha);$
 (7) $\exists \beta R(\alpha, \beta) \leftrightarrow [\mathbb{T}^\alpha] \neq \emptyset.$

Any infinite path in $\mathbb{T}^\alpha$ is of the form $h \oplus \beta$ for some β such that $R(\alpha, \beta)$ and $h \in [\mathbb{T}^{\alpha,\beta}]$, where

$$(h \oplus \beta)(2k) := h(k) \quad \text{and} \quad (h \oplus \beta)(2k+1) := \beta(k).$$

In particular, when $R(\alpha, \beta)$, $f_{\alpha,\beta} \oplus \beta \in [\mathbb{T}^\alpha]$. Finally, we define

$$(8) \quad \hat{R}(\alpha, \beta) : \leftrightarrow R(\alpha, \beta) \wedge \forall \gamma [f_{\alpha,\beta} \oplus \beta \leq_{\text{lex}} f_{\alpha,\gamma} \oplus \gamma].$$

Clearly $\hat{R} \dot{\subseteq} R$ and we need to show that

- (i) $\exists \beta R(\alpha, \beta) \rightarrow \exists! \beta R(\alpha, \beta);$
 (ii) $\hat{R} \in \Pi_1^1.$

For (i), it is immediate that for each α there is at most one β such that $\hat{R}(\alpha, \beta)$. Suppose $\exists \beta R(\alpha, \beta)$. Since by (7) $[\mathbb{T}^\alpha] \neq \emptyset$, it, like any tree of finite sequences from a well-ordered set, has a **left-most path** defined by

$$g_\alpha(k) := \text{least } x \text{ } [\mathbb{T}^\alpha \text{ has an infinite path extending } (g_\alpha \upharpoonright k) \frown \langle x \rangle].$$

g_α is of the form $h_\alpha \oplus \beta_\alpha$, with $h_\alpha \in [\mathbb{T}^{\alpha,\beta_\alpha}]$ and hence $R(\alpha, \beta_\alpha)$. By (5), $f_{\alpha,\beta_\alpha} \leq_{\text{lex}} h_\alpha$ and therefore $f_{\alpha,\beta_\alpha} \oplus \beta_\alpha \leq_{\text{lex}} g_\alpha$, so by the choice of g_α ,

$f_{\alpha, \beta_\alpha} \oplus \beta_\alpha \dot{=} g_\alpha$. For each γ such that $R(\alpha, \gamma)$, $f_{\alpha, \gamma} \oplus \gamma \dot{\in} [\mathbb{T}^\alpha]$, so again by the choice of g_α , $g_\alpha \leq_{\text{lex}} f_{\alpha, \gamma} \oplus \gamma$. But when $\neg R(\alpha, \gamma)$, $g_\alpha(0) \dot{<} \aleph_1 \dot{=} f_{\alpha, \gamma}(0)$, so again $g_\alpha <_{\text{lex}} f_{\alpha, \gamma} \oplus \gamma$. Thus $\hat{R}(\alpha, \beta_\alpha)$.

Finally, to verify (ii), by the definitions of $<_{\text{lex}}$ and $f_{\alpha, \beta}$,

$$\begin{aligned} f_{\alpha, \beta} \oplus \beta <_{\text{lex}} f_{\alpha, \gamma} \oplus \gamma &\leftrightarrow \exists k \forall u [2u \dot{<} k \rightarrow \|(\alpha\beta)^\diamond \upharpoonright u\| \dot{=} \|(\alpha\gamma)^\diamond \upharpoonright u\| \\ &\quad \wedge 2u + 1 \dot{<} k \rightarrow \beta(u) \dot{=} \gamma(u) \\ &\quad \wedge k \dot{=} 2u \rightarrow \|(\alpha\beta)^\diamond \upharpoonright u\| \dot{<} \|(\alpha\gamma)^\diamond \upharpoonright u\| \\ &\quad \wedge k \dot{=} 2u + 2 \rightarrow \beta(u) \dot{<} \gamma(u)]. \end{aligned}$$

That this is a Π_1^1 relation follows easily from Lemma 6.11.24. $\square$

Proposition 6.11.35 $\text{AD}[\Pi_1^1] \rightarrow \text{Perf}(\Sigma_2^1)$.

Proof. Assume $\text{AD}[\Pi_1^1]$, and fix an uncountable Σ_2^1 set A and a Π_1^1 relation R such that $A \dot{=} \{\alpha : \exists \beta R(\alpha, \beta)\}$. By the Uniformization Theorem, there exists a Π_1^1 relation $\hat{R} \subseteq R$ such that $A \dot{=} \{\alpha : \exists! \beta \hat{R}(\alpha, \beta)\}$. Define $\alpha \oplus \beta$ by $(\alpha \oplus \beta)(2n) \dot{=} \alpha(n)$, and $(\alpha \oplus \beta)(2n + 1) \dot{=} \beta(n)$; and set $B \dot{=} \{\alpha \oplus \beta : \hat{R}(\alpha, \beta)\}$. Easily B is also Π_1^1 and the function $f(\alpha \oplus \beta) \dot{=} \alpha$ is a bijection $B \rightarrow A$, so B is uncountable. Hence by Proposition 6.11.32, B has a perfect subset C , which is in particular closed and uncountable. Let $D \dot{=} f^*C$. $D \subseteq A$ and because f is injective, D is also uncountable. Furthermore, D is Σ_1^1 , because $D \dot{=} \{\alpha : \exists \beta (\alpha \oplus \beta \dot{\in} C)\}$. Hence by Proposition 6.11.31, D has a perfect subset, which is also a perfect subset of A as desired. $\square$

The connection between $\text{Perf}(\Sigma_2^1)$ and the status of $\aleph_1$ as a cardinal in $\mathbb{L}$ is established by way of an analysis of the complexity of $\mathbb{L}$ and the well-ordering $<_{\mathbb{L}}$ restricted to $\dot{\omega}$ — that is, a classification of $\mathbb{L} \dot{\cap} \dot{\omega}$ and $<_{\mathbb{L}} \dot{\cap} (\dot{\omega}\dot{\omega})^2$ in the Projective Hierarchy. Let

$$\text{WO}^* \dot{=} \left\{ \alpha \dot{\in} \text{WO} \dot{\cap} \mathbb{L} : (\forall \delta <_{\mathbb{L}} \alpha) \|\delta\| \neq \|\alpha\| \right\}.$$

Recall from Proposition 6.5.32 that the property ‘is a well-ordering’ is Δ^{ZF} , hence $\mathbb{L}$ -absolute, so $\text{WO}^{\mathbb{L}} \dot{=} \text{WO} \dot{\cap} \mathbb{L}$ and

$$\aleph_1^{\mathbb{L}} \dot{=} \left\{ \|\alpha\| : \alpha \dot{\in} \text{WO} \dot{\cap} \mathbb{L} \right\}.$$

Hence also $\aleph_1^{\mathbb{L}} \dot{=} \left\{ \|\alpha\| : \alpha \dot{\in} \text{WO}^* \right\}$ with the additional property that each $\sigma \dot{<} \aleph_1^{\mathbb{L}}$ has exactly one code α in WO^* . We will show below

Proposition 6.11.36 $\text{WO}^* \in \Sigma_2^1$.

Then we have immediately

Theorem 6.11.37 $\text{AD}[\Pi_1^1] \rightarrow (\aleph_1^L < \aleph_1)$.

Proof. Assume $\text{AD}[\Pi_1^1]$ and suppose towards a contradiction that $\aleph_1^L \doteq \aleph_1$. Then WO^* is a Σ_2^1 set with exactly one code α for each countable ordinal. In particular, WO^* is uncountable and therefore, by Proposition 6.11.35, has a perfect subset B . On the one hand, by Lemma 6.11.28, $|B| \doteq 2^{\aleph_0}$, so B is uncountable. But B is closed, hence certainly Σ_1^1 , and a subset of WO , so by Corollary 6.11.25, for some $\sigma < \aleph_1$, $B \subseteq \text{WO}^{<\sigma}$. Since B contains at most one code for each ordinal $\tau < \sigma$, B is countable, a contradiction. $\square$

The stronger conclusion that $\aleph_1 \in \text{Sl}^L$ will use the notion of relativized constructibility $L[\gamma]$ introduced at the end of Section 6.7, but we turn first to the proof of the proposition.

Lemma 6.11.38

- (i) For any Δ^{ZF} formula ϕ , there exists an $L_\in$ -sentence χ_ϕ such that $\text{ZF} \vdash \chi_\phi$, and for any class M ,

$$\text{ZF} \vdash (M \text{ is transitive} \wedge \chi_\phi^{(M)}) \rightarrow \phi \text{ is } M\text{-absolute}.$$

- (ii) For any Δ^{ZF} set operation F , there exists an $L_\in$ -sentence χ_F such that $\text{ZF} \vdash \chi_F$, and for any class M ,

$$\begin{aligned} \text{ZF} \vdash (M \text{ is transitive} \wedge \chi_F^{(M)}) \rightarrow \\ [F \text{ is } M\text{-absolute} \wedge F \text{ is an } M\text{-set operation}]. \end{aligned}$$

Proof. This is a refinement of Theorem 6.5.30. For (i), since ϕ is Δ^{ZF} , there exist Δ_0 formulas ψ and θ such that

$$\text{ZF} \vdash \forall x [\phi(x) \leftrightarrow \exists y \psi(x, y)] \quad \text{and} \quad \text{ZF} \vdash \forall x [\phi(x) \leftrightarrow \forall z \theta(x, z)].$$

Hence there exist finitely many ZF-axioms such that if χ_ϕ denotes the conjunction of them,

$$\chi_\phi \vdash \forall x [\phi(x) \leftrightarrow \exists y \psi(x, y)] \quad \text{and} \quad \chi_\phi \vdash \forall x [\phi(x) \leftrightarrow \forall z \theta(x, z)].$$

Then, for any class M ,

$$\begin{aligned} \chi_\phi^{(M)} \vdash \forall^M x [\phi^{(M)}(x) \leftrightarrow \exists^M y \psi^{(M)}(x, y)] \quad \text{and} \\ \chi_\phi^{(M)} \vdash \forall^M x [\phi^{(M)}(x) \leftrightarrow \forall^M z \theta^{(M)}(x, z)]. \end{aligned}$$

If additionally $\mathbf{M}$ is transitive, then both ψ and θ are $\mathbf{M}$ -absolute, and under the hypothesis $\chi_\phi^{(\mathbf{M})}$, as in the proof of Theorem 6.5.30, $\forall^{\mathbf{M}}x [\phi^{(\mathbf{M})}(x) \leftrightarrow \phi(x)]$ as desired.

For (ii), it suffices to take $\chi_{\mathbf{F}}$ to be the conjunction of $\chi_{(\mathbf{F}x \dot{=} y)}$ as in (i) with enough ZF-axioms to ensure that $\chi_{\mathbf{F}} \vdash \mathbf{F}$ is a set operation. $\square$

Proposition 6.11.39 *There exists an $L_{\in}$ -sentence $\theta_{\mathbf{L}}$ such that*

- (i) $\text{ZFL} \vdash \theta_{\mathbf{L}}$;
- (ii) $\text{ZF} \vdash \forall \mathbf{m} (\mathbf{m} \text{ is transitive} \wedge \theta_{\mathbf{L}}^{(\mathbf{m})} \rightarrow$

$$\exists \sigma (\mathbf{m} \dot{=} \mathbf{L}_\sigma) \wedge <_{\mathbf{L}}^{\mathbf{m}} \dot{=} <_{\mathbf{L}} \dot{\cap} (\mathbf{m} \dot{\times} \mathbf{m}));$$

- (iii) $\text{ZFC} \vdash (\forall \tau \dot{\geq} \dot{\omega})(\exists \sigma \dot{\geq} \tau)(|\sigma| \dot{=} |\tau| \wedge \theta_{\mathbf{L}}^{(\mathbf{L}_\sigma)})$.

Proof. Let $\theta_{\mathbf{L}}$ be, in the notation of the preceding lemma,

$$\chi_{\tau \dot{\mapsto} \mathbf{L}_\tau} \wedge \chi_{<_{\mathbf{L}}} \wedge \chi_{\dot{\Sigma}} \wedge \dot{\mathbf{V}} \dot{=} \mathbf{L}.$$

Then (i) is immediate. For (ii), arguing in ZF, assume that $\mathbf{m}$ is transitive and $\theta_{\mathbf{L}}^{(\mathbf{m})}$. Then

$$\begin{aligned} \chi_{\tau \dot{\mapsto} \mathbf{L}_\tau}^{(\mathbf{m})} &\rightarrow (\forall \tau \in \mathbf{m}) \mathbf{L}_\tau^{\mathbf{m}} \dot{=} \mathbf{L}_\tau; \\ \chi_{<_{\mathbf{L}}}^{(\mathbf{m})} &\rightarrow <_{\mathbf{L}} \text{ is } \mathbf{m}\text{-absolute} \text{ — that is, } <_{\mathbf{L}}^{\mathbf{m}} \dot{=} <_{\mathbf{L}} \dot{\cap} (\mathbf{m} \dot{\times} \mathbf{m}); \\ \chi_{\dot{\Sigma}}^{(\mathbf{m})} \wedge \mathbf{m} \text{ is transitive} &\rightarrow \sigma := \dot{\mathbf{O}}_{\mathbf{r}} \dot{\cap} \mathbf{m} \text{ is a limit ordinal;} \\ (\dot{\mathbf{V}} \dot{=} \mathbf{L})^{(\mathbf{m})} &\rightarrow \mathbf{m} \dot{=} \bigcup_{\tau \in \dot{\mathbf{O}}_{\mathbf{r}}^{\mathbf{m}}} \mathbf{L}_\tau^{\mathbf{m}}. \end{aligned}$$

Hence $\mathbf{m} \dot{=} \bigcup_{\tau \in \dot{\mathbf{O}}_{\mathbf{r}} \dot{\cap} \mathbf{m}} \mathbf{L}_\tau \dot{=} \bigcup_{\tau < \sigma} \mathbf{L}_\tau \dot{=} \mathbf{L}_\sigma$. Finally, by the Strong Transitive Model Theorem 6.7.20,

$$\text{ZFC} \vdash (\forall \tau \dot{\geq} \dot{\omega}) \exists \mathbf{m} [\tau \dot{\subseteq} \mathbf{m} \wedge |\mathbf{m}| \dot{=} |\tau| \wedge \mathbf{m} \text{ is transitive} \wedge \theta_{\mathbf{L}}^{(\mathbf{m})}],$$

and thus (iii) follows by (ii). $\square$

To connect these ideas with the Projective Hierarchy, we develop a coding by functions ε of countable well-founded $L_{\in}$ -structures parallel to the coding of countable well-orderings (Definition 6.11.4).

Definition 6.11.40

- (i) For each $\varepsilon \in \dot{\omega}\dot{\omega}$, $\mathbf{m}_\varepsilon = \langle \dot{\omega}, \{ \langle p, q \rangle : \varepsilon(\langle p, q \rangle) \doteq \dot{1} \} \rangle$.
- (ii) $\mathbf{WFM} := \{ \varepsilon : \theta_{\mathbf{L}}^{(\mathbf{m}_\varepsilon)} \wedge (\text{Extensionality})^{(\mathbf{m}_\varepsilon)} \wedge \mathbf{m}_\varepsilon \text{ is well-founded} \}$.
- (iii) For $\varepsilon \in \mathbf{WFM}$, $\mathbf{n}_\varepsilon$ is the unique countable transitive set and π_ε the unique function such that $\langle \mathbf{n}_\varepsilon, \dot{\in}_{\mathbf{n}_\varepsilon} \rangle \dot{\simeq}_{\pi_\varepsilon} \mathbf{m}_\varepsilon$.

Of course, we are relying on Mostowski Collapsing 6.6.17 to guarantee the existence and uniqueness of $\mathbf{n}_\varepsilon$.

Lemma 6.11.41

- (i) $\mathbf{WFM} \in \Pi_1^1$.
- (ii) $\{ \mathbf{n}_\varepsilon : \varepsilon \in \mathbf{WFM} \} \doteq$

$$\{ m : m \text{ is a countable transitive set such that } \theta_{\mathbf{L}}^{(m)} \}$$

and this is an uncountable subset of $\{ \mathbf{L}_\sigma : \sigma < \aleph_1 \}$. In particular,

$$\mathbf{L}_{\aleph_1} \doteq \bigcup \{ \mathbf{n}_\varepsilon : \varepsilon \in \mathbf{WFM} \}.$$

Proof. Part (i) follows by a straightforward calculation. For (ii), any countable transitive structure $\langle m, \dot{\in}_m \rangle$ is isomorphic to a well-founded structure of the form $\mathbf{m}_\varepsilon$ and hence coincides with one of the form $\mathbf{n}_\varepsilon$. If $\theta_{\mathbf{L}}^{(m)}$, then $\varepsilon \in \mathbf{WFM}$, $\mathbf{n}_\varepsilon$ is $\mathbf{L}_\sigma$ for some countable ordinal σ and the set of σ which occur is unbounded in $\aleph_1$. $\square$

Theorem 6.11.42

- (i) $\mathbf{L} \dot{\cap} \dot{\omega}\dot{\omega} \in \Sigma_2^1$.
- (ii) $<_{\mathbf{L}} \dot{\cap} (\dot{\omega}\dot{\omega} \dot{\times} \dot{\omega}\dot{\omega}) \in \Sigma_2^1$.
- (iii) For any $n \geq 2$ and any relation $R \in \Sigma_n^1(\Pi_n^1)$, if

$$P(\alpha, \beta) : \leftrightarrow (\exists \delta <_{\mathbf{L}} \beta) R(\alpha, \beta, \delta) \quad \text{and}$$

$$Q(\alpha, \beta) : \leftrightarrow (\forall \delta <_{\mathbf{L}} \beta) R(\alpha, \beta, \delta),$$

then also $P, Q \in \Sigma_n^1(\Pi_n^1)$.

Proof. We note first that for each $L_\in$ -formula $\phi(x_0, \dots, x_{k-1})$,

$$\forall p_0 \dots \forall p_{k-1} \left(\{ \varepsilon : \phi(p_0, \dots, p_{k-1})^{(\mathbf{m}_\varepsilon)} \} \in \mathbf{Bo} \right).$$

The proof is a straightforward formula induction; the key point is that

$$(9) \quad \{ \varepsilon : \exists y \psi(p)^{(m_\varepsilon)} \} \doteq \bigcup_{q \in \dot{\omega}} \{ \varepsilon : \psi(p, q)^{(m_\varepsilon)} \}$$

is a countable union. Next we define relations which describe how natural numbers represent other objects in structures m_ε .

$$\begin{aligned} \text{Code}_{r,y}(\varepsilon) &: \leftrightarrow \forall q [\varepsilon(\dot{\langle} q, r \rangle) \doteq 1 \leftrightarrow (\exists z \dot{\in} y) \text{Code}_{q,z}(\varepsilon)] \\ \text{Code}_{r,\{x\}}(\varepsilon) &: \leftrightarrow \forall q [\varepsilon(\dot{\langle} q, r \rangle) \doteq 1 \leftrightarrow \text{Code}_{q,x}(\varepsilon)] \\ \text{Code}_{r,\{x,y\}}(\varepsilon) &: \leftrightarrow \forall q [\varepsilon(\dot{\langle} q, r \rangle) \doteq 1 \leftrightarrow (\text{Code}_{q,x}(\varepsilon) \vee \text{Code}_{q,y}(\varepsilon))] \\ \text{Code}_{r,\langle x,y \rangle}(\varepsilon) &: \leftrightarrow \forall q [\varepsilon(\dot{\langle} q, r \rangle) \doteq 1 \leftrightarrow \text{Code}_{q,\{x\}}(\varepsilon) \vee \text{Code}_{q,\{x,y\}}(\varepsilon)] \\ \text{Code}_r(\alpha, \varepsilon) &: \leftrightarrow \forall q [\varepsilon(\dot{\langle} q, r \rangle) \doteq 1 \leftrightarrow (\exists x \dot{\in} \dot{\omega}) \text{Code}_{q,\langle x, \alpha(x) \rangle}(\varepsilon)]. \end{aligned}$$

Again it is easy to show that for $x, y \in \dot{\omega}$ all of these relations are Borel and for $\varepsilon \in \mathbf{WFM}$ have the intended meaning — for example, $\text{Code}_{r,\{x,y\}}(\varepsilon) \leftrightarrow \pi_\varepsilon(r) \doteq \{x, y\}$. In particular,

$$(10) \quad \forall p \forall \alpha \forall \varepsilon (\varepsilon \in \mathbf{WFM} \rightarrow [\text{Code}_p(\alpha, \varepsilon) \leftrightarrow \pi_\varepsilon(p) \doteq \alpha]).$$

Now using the lemma, (i) and (ii) follow immediately from the equivalences

$$\begin{aligned} \alpha \dot{\in} \mathbf{L} &\leftrightarrow (\exists \sigma \dot{<} \dot{\aleph}_1) \alpha \dot{\in} \mathbf{L}_\sigma \\ &\leftrightarrow \exists \varepsilon [\varepsilon \in \mathbf{WFM} \wedge \alpha \dot{\in} \mathbf{n}_\varepsilon] \\ &\leftrightarrow \exists \varepsilon [\varepsilon \in \mathbf{WFM} \wedge \exists p \text{Code}_p(\alpha, \varepsilon)], \end{aligned}$$

and

$$\begin{aligned} \alpha <_{\mathbf{L}} \beta &\leftrightarrow (\exists \sigma \dot{<} \dot{\aleph}_1) [\alpha, \beta \dot{\in} \mathbf{L}_\sigma \wedge \theta_{\mathbf{L}}^{(\mathbf{L}_\sigma)} \wedge (\alpha <_{\mathbf{L}} \beta)^{(\mathbf{L}_\sigma)}] \\ &\leftrightarrow \exists \varepsilon [\varepsilon \in \mathbf{WFM} \wedge \alpha, \beta \dot{\in} \mathbf{n}_\varepsilon \wedge (\alpha <_{\mathbf{L}} \beta)^{(\mathbf{n}_\varepsilon)}] \\ &\leftrightarrow \exists \varepsilon [\varepsilon \in \mathbf{WFM} \wedge \exists p \exists q (\text{Code}_p(\alpha, \varepsilon) \wedge \\ &\quad \text{Code}_q(\beta, \varepsilon) \wedge (p <_{\mathbf{L}} q)^{(m_\varepsilon)}]. \end{aligned}$$

Suppose now that $R \in \Sigma_n^1$ for $n \geq 2$ with P and Q defined as in the

statement. Then $P, Q \in \Sigma_n^1$ because

$$\begin{aligned}
 P(\alpha, \beta) &\leftrightarrow \exists \delta [\delta <_{\mathbf{L}} \beta \wedge R(\alpha, \beta, \delta)] \\
 Q(\alpha, \beta) &\leftrightarrow \exists \varepsilon [\varepsilon \in \mathbf{WFM} \wedge \alpha, \beta \in \mathbf{n}_\varepsilon \wedge \exists \delta ((\delta <_{\mathbf{L}} \beta)^{(n_\varepsilon)} \wedge R(\alpha, \beta, \delta))] \\
 &\leftrightarrow \exists \varepsilon [\varepsilon \in \mathbf{WFM} \wedge \exists p \exists q (\mathbf{Code}_p(\alpha, \varepsilon) \wedge \mathbf{Code}_q(\beta, \varepsilon) \\
 &\quad \wedge \forall s [(s <_{\mathbf{L}} q \wedge s \in \dot{\omega} \dot{\omega})^{(m_\varepsilon)} \rightarrow \\
 &\quad \exists \delta [\mathbf{Code}_s(\delta, \varepsilon) \wedge R(\alpha, \beta, \delta)])]]]. \quad \square
 \end{aligned}$$

Proof. (of Proposition 6.11.36) By (i) of the theorem, $\mathbf{WO} \dot{\cap} \mathbf{L} \in \Sigma_2^1$, so using the relations of Lemma 6.11.24, the equivalence

$$\|\delta\| \neq \|\alpha\| \leftrightarrow \delta \prec_\Sigma \alpha \vee \alpha \prec_\Sigma \delta$$

together with (iii) of the theorem shows that $\mathbf{WO}^* \in \Sigma_2^1$. $\square$

With this we have completed the proof that $\mathbf{AD}[\mathbf{\Pi}_1^1] \rightarrow \dot{\aleph}_1^{\mathbf{L}} < \dot{\aleph}_1$, and it remains to draw also the stronger conclusion $\dot{\aleph}_1 \in \mathbf{S}^{\mathbf{L}}$ from this hypothesis. We shall make use of the notion of relative constructibility $\mathbf{L}[a]$ introduced at the end of Section 6.7. In particular, we shall be interested in $\mathbf{L}[\gamma]$ for $\gamma \in \dot{\omega} \dot{\omega}$. Since $\gamma \subseteq \dot{V}_{\dot{\omega}}$, by the corollary to Theorem 6.7.21, $\gamma \in \mathbf{L}[\gamma]$. Let $<_{\mathbf{L}[\gamma]}^1$ denote the restriction $<_{\mathbf{L}[\gamma]} \dot{\cap} (\dot{\omega} \dot{\omega} \dot{\times} \dot{\omega} \dot{\omega})$ and

$$\mathbf{WO}^*[\gamma] := \left\{ \alpha \in \mathbf{WO} : \alpha \in \mathbf{L}[\gamma] \wedge (\forall \delta <_{\mathbf{L}[\gamma]}^1 \alpha) \|\delta\| \neq \|\alpha\| \right\}.$$

Then by simple modifications of the proofs above we have

Proposition 6.11.43 *For each $\gamma \in \dot{\omega} \dot{\omega}$,*

- (i) $\mathbf{WO}^*[\gamma] \in \Sigma_2^1$;
- (ii) $\dot{\aleph}_1^{\mathbf{L}[\gamma]} \doteq \{ \|\alpha\| : \alpha \in \mathbf{WO}^*[\gamma] \}$; *in fact,*
- (iii) *for each $\sigma < \dot{\aleph}_1^{\mathbf{L}[\gamma]}$, there is exactly one $\alpha \in \mathbf{WO}^*[\gamma]$ such that $\|\alpha\| \doteq \sigma$;*
- (iv) $\mathbf{AD}[\mathbf{\Pi}_1^1] \rightarrow \forall \gamma (\dot{\aleph}_1^{\mathbf{L}[\gamma]} < \dot{\aleph}_1)$.

Theorem 6.11.44 $\mathbf{AD}[\mathbf{\Pi}_1^1] \rightarrow \dot{\aleph}_1 \in \mathbf{S}^{\mathbf{L}}$.

Proof. As in the proof of Theorem 6.11.9, it suffices to show

$$\dot{\aleph}_1 \text{ (is a limit cardinal)}^{(\mathbf{L})},$$

or equivalently,

$$(\forall \sigma \dot{<} \dot{\aleph}_1)(\exists \tau \dot{<} \dot{\aleph}_1) [\sigma \dot{<} \tau \wedge (\tau \text{ is a cardinal})^{(L)}].$$

For any $\sigma \dot{<} \dot{\aleph}_1$, there exists $\gamma \in \mathbf{WO}$ such that $\|\gamma\| \doteq \sigma$. Since $\gamma \in L[\gamma]$ and this is a class model of ZFC, it follows that $\gamma \in \mathbf{WO}^{L[\gamma]}$, so $\sigma \dot{<} \dot{\aleph}_1^{L[\gamma]}$. By the preceding proposition, $\dot{\aleph}_1^{L[\gamma]} \dot{<} \dot{\aleph}_1$, and by definition, $\dot{\aleph}_1^{L[\gamma]}$ (is a cardinal) $^{(L[\gamma])}$. Since ‘is a cardinal’ is a $\Pi^{\mathbf{ZF}}$ property, it is downward absolute, so also $\dot{\aleph}_1^{L[\gamma]}$ (is a cardinal) $^{(L)}$. $\square$

Corollary 6.11.45 $\mathbf{ZFC} \vdash \mathbf{AD}[\mathbf{\Pi}_1^1] \rightarrow \mathbf{Con}(\mathbf{ZFC})$; hence unless ZF is actually inconsistent, there is no proof formalizable in ZFC that

$$\mathbf{ZF} \text{ is consistent} \implies \mathbf{ZFC} + \mathbf{AD}[\mathbf{\Pi}_1^1] \text{ is consistent. } \square$$

Finally, to prepare for the derivation of the implication $\exists \mathbf{MC} \rightarrow \mathbf{AD}[\mathbf{\Pi}_1^1]$, we develop some additional properties of measurable cardinals.

Definition 6.11.46 A measure ultrafilter U on a measurable cardinal κ is **normal** iff for any function $f : \kappa \rightarrow \kappa$,

$$\{\sigma \dot{<} \kappa : f(\sigma) \dot{<} \sigma\} \in U \rightarrow (\exists \rho \dot{<} \kappa) \{\sigma \dot{<} \kappa : f(\sigma) \doteq \rho\} \in U.$$

Proposition 6.11.47 For any measurable cardinal κ , there exists a normal measure ultrafilter on κ .

Proof. Let κ be measurable and V any measure ultrafilter on κ . By the proof of Theorem 6.10.20, with notation as there and using V , there exists a function $h : \kappa \rightarrow \kappa$ such that $F(\hat{h}) = \kappa$. Let

$$U := \{x \subseteq \kappa : h^{-1}(x) \in V\};$$

we show that U is a normal measure ultrafilter. Properties (i)–(iii) of Definition 6.10.16 are immediate. For (iv),

$$x \notin U \leftrightarrow h^{-1}(x) \notin V \leftrightarrow h^{-1}(\kappa \setminus x) \doteq (\kappa \setminus h^{-1}(x)) \in V \leftrightarrow \kappa \setminus x \in U,$$

and for (v), for any $\sigma \dot{<} \kappa$,

$$\{\sigma\} \in U \rightarrow h^{-1}(\{\sigma\}) \in V \rightarrow \hat{h} \doteq \hat{h}_\sigma \rightarrow F(\hat{h}) \doteq \sigma,$$

contrary to hypothesis. For (vi), if $v \subseteq U$ and $|v| \dot{<} \kappa$, then for each $x \in v$, $h^{-1}(x) \in V$, so

$$h^{-1}\left(\bigcap v\right) \doteq \bigcap \{h^{-1}(x) : x \in v\} \in V \quad \text{so} \quad \bigcap v \in U.$$

Finally, to verify that U is normal, note that since $F(\hat{h}) \dot{=} \kappa$, for any $g : \kappa \dot{\rightarrow} \kappa$,

$$\begin{aligned} \{ \tau \dot{<} \kappa : g(\tau) \dot{<} h(\tau) \} \dot{\in} V &\rightarrow F(\hat{g}) \dot{<} F(\hat{h}) \\ &\rightarrow (\exists \rho \dot{<} \kappa) \hat{g} \dot{=} \hat{h}_\rho \\ &\rightarrow (\exists \rho \dot{<} \kappa) \{ \tau \dot{<} \kappa : g(\tau) \dot{=} \rho \} \dot{\in} V. \end{aligned}$$

Hence, for any $f : \kappa \dot{\rightarrow} \kappa$,

$$\begin{aligned} \{ \sigma \dot{<} \kappa : f(\sigma) \dot{<} \sigma \} \dot{\in} U &\rightarrow \{ \tau \dot{<} \kappa : f(h(\tau)) \dot{<} h(\tau) \} \dot{\in} V \\ &\rightarrow (\exists \rho \dot{<} \kappa) \{ \tau \dot{<} \kappa : f(h(\tau)) \dot{=} \rho \} \dot{\in} V \\ &\rightarrow (\exists \rho \dot{<} \kappa) \{ \sigma \dot{<} \kappa : f(\sigma) \dot{=} \rho \} \dot{\in} U. \quad \square \end{aligned}$$

Definition 6.11.48 For any cardinals $\lambda \dot{<} \kappa$, any $n \dot{\in} \dot{\omega}$ and any $X \dot{\subseteq} \kappa$,

- (i) $[X]^n := \{ a \dot{\subseteq} X : |a| \dot{=} n \}$;
- (ii) a λ -**partition** of $[\kappa]^n$ is a function $\pi : [\kappa]^n \dot{\rightarrow} \lambda$;
- (iii) a set $H \dot{\subseteq} \kappa$ is π -**homogeneous** iff for some $\sigma \dot{<} \lambda$, for all $a \dot{\in} [H]^n$, $\pi(a) \dot{=} \sigma$.

Partition Theorem 6.11.49 For any normal measure ultrafilter U on a measurable cardinal κ ,

- (i) for any $\lambda \dot{<} \kappa$ and $n \dot{\in} \dot{\omega}$, and any λ -partition π of $[\kappa]^n$, there exists a π -homogeneous set $H \dot{\in} U$ (in particular $|H| \dot{=} \kappa$);
- (ii) for any countable family $\langle \pi_k : k \dot{\in} \dot{\omega} \rangle$ of λ_k -partitions of $[\kappa]^{n_k}$, there exists a set $H \dot{\in} U$ which is simultaneously π_k -homogeneous for all $k \dot{\in} \dot{\omega}$.

Proof. For (i), we proceed by induction. For $n \dot{=} 1$, if $\pi : [\kappa]^1 \dot{\rightarrow} \lambda$, then for $\sigma \dot{<} \lambda$, let

$$z_\sigma := \{ \tau \dot{<} \kappa : \pi(\{ \tau \}) \dot{=} \sigma \}, \quad \text{so} \quad \bigcup \{ z_\sigma : \sigma \dot{<} \lambda \} \dot{=} \kappa \dot{\in} U,$$

and thus by Lemma 6.10.18 for some $\sigma \dot{<} \lambda$, $H := z_\sigma \dot{\in} U$.

Now assume the result for n as induction hypothesis and fix a λ -partition $\pi : [\kappa]^{n+1} \dot{\rightarrow} \lambda$. For each $\tau \dot{<} \kappa$, define

$$\pi_\tau : [\kappa \setminus (\tau + 1)]^n \dot{\rightarrow} \lambda \quad \text{by} \quad \pi_\tau(b) := \pi(b \dot{\cup} \{ \tau \}).$$

Since $|\kappa \setminus (\tau + 1)| \dot{=} \kappa$, it follows from the induction hypothesis that for each τ there is a π_τ -homogeneous set in U ; using AC, there exists a family $\langle H_\tau : \tau < \kappa \rangle$ such that each $H_\tau \in U$ is π_τ -homogeneous. Define

$$\pi^* : \kappa \rightarrow \lambda \quad \text{by} \quad \pi^*(\tau) := \pi_\tau(b) \quad \text{for some (any) } b \in [H_\tau]^n.$$

By the $n \dot{=} 1$ case (identifying $[\kappa]^1$ and κ), there is a π^* -homogeneous set $H^* \in U$; let ρ^* be the common value of $\pi^*(\tau)$ for $\tau \in H^*$. Finally, set

$$H := \{ \sigma \in H^* : (\forall \tau < \sigma) \sigma \in H_\tau \}.$$

First, to verify that H is π -homogeneous, fix $a \in [H]^{n+1}$. Let τ be the smallest element of a and $b := a \setminus \{\tau\}$. Then $\tau \in H^*$ and is less than all $\sigma \in b$, so $b \in [H_\tau]^n$, and thus

$$\pi(a) \dot{=} \pi(b \cup \{\tau\}) \dot{=} \pi_\tau(b) \dot{=} \pi^*(\tau) \dot{=} \rho^*.$$

Finally, towards a contradiction, suppose that $H \notin U$, so $(H^* \setminus H) \in U$. Define

$$f : \kappa \dot{\rightarrow} \kappa \quad \text{by} \quad f(\sigma) := \begin{cases} \text{least } \tau < \sigma [\sigma \notin H_\tau], & \text{if } \sigma \in H^* \setminus H; \\ 0, & \text{otherwise.} \end{cases}$$

Now $\{ \sigma : f(\sigma) < \sigma \} \in U$, so by normality of U , for some $\rho < \kappa$, $\{ \sigma : f(\sigma) \dot{=} \rho \} \in U$ and is a subset of $\kappa \setminus H_\rho$, so also $(\kappa \setminus H_\rho) \in U$, a contradiction (because $H_\rho \in U$).

Finally, for (ii), if for each $k \in \dot{\omega}$, $H_k \in U$ is π_k -homogeneous, then $\bigcap_{k \in \dot{\omega}} H_k \in U$ is simultaneously homogeneous. $\square$

Theorem 6.11.50 $\exists \text{MC} \rightarrow \text{AD}[\Pi_1^1]$.

Proof. Assume that κ is a measurable cardinal, and fix a normal measure ultrafilter U on κ . Fix also a Π_1^1 set A . As in (6) in the proof of the Uniformization Theorem 6.11.34, replacing $\aleph_1$ by κ , there exists a tree $\mathbb{T} \subseteq \dot{\omega} \dot{\times} <^\omega \kappa$ such that

$$\alpha \in A \leftrightarrow (\exists h \in {}^\omega \kappa) \forall k \langle \bar{\alpha}(k), h \restriction k \rangle \in \mathbb{T}.$$

This represents A as the projection onto the first component of a (in a natural extension of the term) *closed* subset $\mathcal{A}$ of $\dot{\omega} \dot{\times} <^\omega \kappa$. We associate with $\mathcal{A}$ an asymmetric game (also called $\mathcal{A}$) as follows. At each k -th turn, player I plays a pair $\langle m_{2k}, \sigma_k \rangle \in \dot{\omega} \dot{\times} \kappa$, while player II plays simply some

$m_{2k+1} \in \dot{\omega}$. A strategy f for player I computes this play from (essentially) the sequence of preceding plays

$$f(\langle \dot{m}_0, \dots, m_{2k-1} \rangle, \langle \dot{\sigma}_0, \dots, \sigma_{k-1} \rangle) \doteq \langle \dot{m}_{2k}, \sigma_k \rangle,$$

while a strategy g for player II computes

$$g(\langle \dot{m}_0, \dots, m_{2k} \rangle, \langle \dot{\sigma}_0, \dots, \sigma_k \rangle) \doteq m_{2k+1}.$$

It follows by a simple modification of the proof of Theorem 6.11.18 that $\mathcal{A}$ is determined, and thus to finish the proof it will suffice to define two mappings $f \mapsto \gamma_f$ and $g \mapsto \delta_g$ such that

$$(11) \quad f \in \text{WS}_I(\mathcal{A}) \rightarrow \gamma_f \in \text{WS}_I(A);$$

$$(12) \quad g \in \text{WS}_{II}(\mathcal{A}) \rightarrow \delta_g \in \text{WS}_{II}(A).$$

Towards (11), for any $f \in \text{WS}_I(\mathcal{A})$ and any $u \in \dot{\omega}$ with $\text{lg}(u) \doteq 2k$, set

$$\gamma_f(u) \doteq \text{the unique } m \text{ such that } f(u, \langle \dot{\sigma}_0, \dots, \sigma_{k-1} \rangle) \doteq \langle \dot{m}, \sigma \rangle,$$

where $\sigma_0, \dots, \sigma_{k-1}$ are the ordinal moves generated by f responding to initial segments of u . In other words, γ_f plays according to the strategy f but discards the ordinal component of each move. Then clearly for any δ , if $g_\delta(u, \langle \dot{\sigma}_0, \dots, \sigma_k \rangle) \doteq \delta(u)$ for any $\sigma_1, \dots, \sigma_k$, then for some h ,

$$\gamma_f \# \delta \doteq \alpha \rightarrow f \# g_\delta \doteq \langle \dot{\alpha}, h \rangle \in \mathcal{A} \rightarrow \alpha \in A,$$

so γ_f is a winning strategy for player I.

Now, towards (12), suppose that $g \in \text{WS}_{II}(\mathcal{A})$. We aim to define a strategy δ_g such that for any α and γ ,

$$\gamma \# \delta_g \doteq \alpha \in A \rightarrow \exists f_\gamma \exists h [f_\gamma \# g \doteq \langle \dot{\alpha}, h \rangle \in \mathcal{A}].$$

Then, since g wins for II in $\mathcal{A}$, the right-hand side is false, hence $\alpha \notin A$, so δ_g wins for II in A . The idea is to define, when $\text{lg}(u) \doteq 2k+1$,

$$\delta_g(u) \doteq g(u, \langle \dot{\sigma}_0, \dots, \sigma_k \rangle)$$

where $\sigma_0, \dots, \sigma_k$ are ordinal moves which would be generated by a suitably chosen f_γ . These are found using the Partition Theorem. For each $k \in \dot{\omega}$, define a partition π_{k+1} of $[\kappa]^{k+1}$ as follows. For $x \in [\kappa]^{k+1}$, let $\langle \tau_0^x, \dots, \tau_k^x \rangle$ be the elements of x listed in increasing order. Let S_{k+1} be the set of all

permutations of $\{0, \dots, k\}$ and for $p \in S_{k+1}$, $\vec{p}(x)$ denotes the sequence $\langle \tau_{p(0)}^x, \dots, \tau_{p(k)}^x \rangle$. Now set

$$\pi_{k+1}(x) := \{ \langle p, u, m \rangle : p \in S_{k+1} \wedge \lg(u) \doteq 2k+1 \wedge g(u, \vec{p}(x)) \doteq m \}.$$

Easily there are $2^{\aleph_0}$ -many possible values of $\pi_{k+1}(x)$, so this may be viewed as a $2^{\aleph_0}$ -partition, and of course $2^{\aleph_0} < \kappa$. Hence, by the Partition Theorem, there exists a set $H \subseteq \kappa$ of cardinality κ homogeneous for all π_{k+1} — that is, for any k and any $x, y \in [H]^{k+1}$, $\pi_{k+1}(x) \doteq \pi_{k+1}(y)$, and thus for any $p \in S_{k+1}$ and any u with $\lg(u) \doteq 2k+1$, $g(u, \vec{p}(x)) \doteq g(u, \vec{p}(y))$. In particular, let p_u be a permutation such that for all $s, t < k$,

$$s <^u t \leftrightarrow p_u(s) < p_u(t),$$

so in particular, for any $x \in [\kappa]^{k+1}$, $s \leq^u t \leftrightarrow \tau_{p_u(s)}^x < \tau_{p_u(t)}^x$. Set

$$\delta_g(u) := g(u, \vec{p}_u(x)) \quad \text{for some (any) } x \in [H]^{k+1},$$

and, towards a contradiction, suppose that $\delta_g \notin \text{WS}_\Pi(A)$, so for some γ , $\gamma \# \delta_g \doteq \alpha \in A$. Since $<^\alpha$ is a countable well-ordering, there exists a function $h : \omega \rightarrow H$ such that for all s and t , $s <^\alpha t \leftrightarrow h(s) < h(t)$. Define $f_\gamma(u, \langle \sigma_0, \dots, \sigma_{k-1} \rangle) := \langle \gamma(u), h(k) \rangle$, and let $f_\gamma \# g \doteq \langle \beta, h' \rangle$. Clearly $h \doteq h'$ and it suffices to show by induction that $\alpha \doteq \beta$. Assume as induction hypothesis that $\bar{\alpha}(2k) \doteq \bar{\beta}(2k)$. Then

$$\begin{aligned} \beta(2k) &\doteq \gamma(\bar{\alpha}(2k)) \doteq \alpha(2k) \text{ because } \langle \beta(2k), h(k) \rangle \doteq f_\gamma(\bar{\alpha}(2k), h \upharpoonright k); \\ \beta(2k+1) &\doteq g(\bar{\alpha}(2k+1), h \upharpoonright (k+1)) \\ &\doteq g(\bar{\alpha}(2k+1), \vec{p}_{\bar{\alpha}(2k+1)}(\text{Im}(h \upharpoonright (k+1)))) \\ &\doteq \delta_g(\bar{\alpha}(2k+1)) \\ &\doteq \alpha(2k+1). \quad \square \end{aligned}$$

Corollary 6.11.51 $\exists \text{MC} \rightarrow \aleph_1 \in \text{Sl}^L$. $\square$

Collateral Reading. The material of this section belongs to the area known as Descriptive Set Theory; two superb and comprehensive references for this area are [Mo80] and [Kec]. Material concerning the Borel and Projective Hierarchies is also treated in [Hin].

Exercises

Exercise 6.11.52 Prove Proposition 6.11.15.

Hint: Use techniques from the proofs of Lemma 5.1.2 and Propositions 5.1.3 and 5.1.7. For $\sigma \geq 1$ let

$$\begin{aligned} {}^0_1U^l &:= \left\{ \langle \alpha_0, \dots, \alpha_{l-1}, \gamma \rangle : \exists k (\gamma(\langle \bar{\alpha}_0(k), \dots, \bar{\alpha}_{l-1}(k) \rangle) \neq 1) \right\}; \\ {}^0_\sigma U^l &:= \left\{ \langle \alpha_0, \dots, \alpha_{l-1}, \gamma \rangle : (\exists n \in \dot{\omega}) \left(\|\langle \gamma \rangle^{2n}\| < \sigma \wedge \right. \right. \\ &\quad \left. \left. \langle \alpha_0, \dots, \alpha_{l-1}, (\gamma)^{2n+1} \rangle \notin {}^0_{\|\langle \gamma \rangle^{2n}\|} U^l \right) \right\}; \end{aligned}$$

where $(\gamma)^m(p) := \gamma(\langle m, p \rangle)$. Show that ${}^0_\sigma U^l$ is universal for the class of l -ary Σ^0_σ relations.

Exercise 6.11.53 Prove that for all $\sigma \geq 0$,

- (i) Σ^0_σ has the reduction property but not the separation property;
- (ii) Π^0_σ has the separation property but not the reduction property.

Hint: Use the techniques of the hint for Exercise 5.1.36 together with the results of the preceding exercise.

Exercise 6.11.54 Prove Proposition 6.11.20.

Hint: As in the preceding exercise use techniques from Section 5.1. Set ${}^1_0 U^l := {}^0_1 U^l$ and

$${}^1_{n+1} U^l := \left\{ \langle \alpha_0, \dots, \alpha_{l-1}, \gamma \rangle : \exists \beta \langle \alpha_0, \dots, \alpha_{l-1}, \beta, \gamma \rangle \notin {}^1_n U^l \right\}.$$

Show that ${}^1_n U^l$ is universal for the class of l -ary Σ^1_n relations.

Exercise 6.11.55 Prove that

- (i) Π^1_1 has the reduction property but not the separation property;
- (ii) Σ^1_1 has the separation property but not the reduction property.

Hint: Use the techniques of the hint for Exercise 5.1.36 together with Proposition 6.11.23, Lemma 6.11.24 and the results of the preceding exercise.

Exercise 6.11.56 Prove in $\text{ZF} + \text{AD}$ that for any family $\langle a_n : n \in \dot{\omega} \rangle$, if for all n , $a_n \subseteq {}^\omega \dot{\omega}$ and a_n is countable, then also $\bigcup \{a_n : n \in \dot{\omega}\}$ is countable.

Exercise 6.11.57 Let CH'_1 be the assertion

$$\forall u [u \subseteq \dot{\varphi} \dot{\omega} \wedge u \text{ is uncountable} \rightarrow u \approx \dot{\varphi} \dot{\omega}],$$

and recall from [Section 6.4](#) the first version of the Continuum Hypothesis CH_1 . Thus Proposition 6.11.7 asserts that CH'_1 is a theorem of $\text{ZF} + \text{AD}$. Use the result of Exercise 6.4.65 to show that CH_1 is a theorem of $\text{ZF} + \text{AD} + \text{CAC}$. (Although AD contradicts AC , it is believed to be consistent with CAC and even with ADC .)

Exercise 6.11.58 Prove in ZFC that for any Σ^1_2 relation R there exists a family $\langle R_\sigma : \sigma \prec \aleph_1 \rangle$ of Borel relations such that $R \doteq \bigcup \{ R_\sigma : \sigma \prec \aleph_1 \}$.

Hint: Let $\text{WO}_{<\sigma} := \{ \alpha : \alpha \in \text{WO}^{\sigma^{+o1}} \vee \exists p (\alpha \upharpoonright p \in \text{WO}^{<\sigma}) \}$, where

$$m <_{\alpha \upharpoonright p} n \leftrightarrow m <_\alpha n <_\alpha p.$$

Show first that

$$\text{WO} \doteq \bigcup \{ \text{WO}^{<\sigma} : \sigma \prec \aleph_1 \} \doteq \bigcap \{ \text{WO}_{<\sigma} : \sigma \prec \aleph_1 \}.$$

Use these representations to derive the stated result for R in Σ^1_1 and Π^1_1 , then extend to Σ^1_2 . Although AC is used in the most natural proof requested here (where?), there is a considerably more complicated proof of this result just in ZF .

Exercise 6.11.59 Establish in $\text{ZF} + \text{AD}$ the following choice principle. Let f be a function with domain $\aleph_1$ such that for all $\sigma \prec \tau \prec \aleph_1$, $\emptyset \neq f(\tau) \subseteq f(\sigma)$. Prove that there exists a function g such that for all $\alpha \in \text{WO}$, $g(\alpha) \in f(\|\alpha\|)$.

Hint: Consider the game $\{ \varepsilon : \varepsilon_I \in \text{WO} \wedge \varepsilon_{II} \notin f(\|\varepsilon_I\|) \}$, where $\varepsilon_I(k) := \varepsilon(2k)$ are player I's moves and $\varepsilon_{II}(k) := \varepsilon(2k+1)$ are those of player II. Use the boundedness principle 6.11.25 to show that player I cannot have a winning strategy.

Exercise 6.11.60 Prove in $\text{ZF} + \text{AD}$ the following converse to Exercise 6.11.58: for any family $\langle R_\sigma : \sigma \prec \aleph_1 \rangle$ of Σ^1_2 relations, also $R := \bigcup \{ R_\sigma : \sigma \prec \aleph_1 \}$ is Σ^1_2 .

Hint: Apply the result of the preceding exercise to the function

$$f(\sigma) := \{ \gamma : (\exists \rho \geq \sigma) \frac{1}{2} U^l_\gamma \doteq \bigcup \{ R_\tau : \tau \prec \rho \} \},$$

where $\frac{1}{2} U^l$ is the universal relation of Exercise 6.11.54.

Exercise 6.11.61 In the proof of the Uniformization Theorem 6.11.34 we established in display (4) that (simplifying from relations to sets) for any Π_1^1 set A there exists a 2-tree $\mathbb{T} \subseteq \dot{\omega} \times^{<\omega} \dot{\aleph}_1$ such that

$$\alpha \in A \leftrightarrow (\exists h \in \dot{\omega} \dot{\aleph}_1) \forall k (\langle \dot{\bar{\alpha}}(k), h \upharpoonright k \rangle \in \mathbb{T}) \leftrightarrow [\mathbb{T}^\alpha] \neq \emptyset.$$

A set A expressible in this form is called $\dot{\aleph}_1$ -**Suslin**. Note that Σ_1^1 sets are naturally called $\dot{\aleph}_0$ -Suslin.

- (i) Prove in ZFC that every Σ_2^1 set is $\dot{\aleph}_1$ -Suslin.
- (ii) Prove in ZF + AD that every $\dot{\aleph}_1$ -Suslin set is Σ_2^1 .

Hint: For (ii), for each $\sigma \prec \dot{\aleph}_1$, let

$$A_\sigma := \left\{ \dot{\alpha} : (\exists h \in \dot{\omega} \sigma) \forall k (\langle \dot{\bar{\alpha}}(k), h \upharpoonright k \rangle \in \mathbb{T}) \right\}.$$

Show that each A_σ is a Σ_1^1 set and that $A = \bigcup \{ A_\sigma : \sigma \prec \dot{\aleph}_1 \}$. Then use the result of Exercise 6.11.60.

Exercise 6.11.62 Consider games in which each player plays a countable ordinal at each turn, and let $\text{AD}_{\dot{\aleph}_1}$ denote the principle that every set $A \subseteq \dot{\omega} \dot{\aleph}_1$ is determined. Prove that $\text{ZF} \vdash \neg \text{AD}_{\dot{\aleph}_1}$.

Hint: Find a set A for which player I cannot have a winning strategy and such that the existence of a winning strategy for player II would lead to the existence of an uncountable well-ordered subset of $\dot{\omega} \dot{\omega}$.

Chapter 7

Model Theory

We have already seen a good bit of what is usually called model theory in [Chapter 3](#); this chapter continues the investigations there into various sorts of relationships between theories and their models. For the most part we will be concerned with first-order languages, but there are several occasions when the natural languages to use are infinitary ([Section 3.7](#)), so we formulate some of the definitions and results to include these.

7.1 Partial embeddings

In [Sections 3.4–3.5](#) we made frequent use of embeddings of one structure $\mathfrak{A}$ into another $\mathfrak{B}$, and in Exercise 2.4.55 we introduced the notion of a partial embedding. Here we extend the results of that exercise and consider various families of embeddings of substructures of $\mathfrak{A}$ into $\mathfrak{B}$. As above, if $\mathbf{x} = x_0, \dots, x_{k-1}$ and $\mathbf{a} = a_0, \dots, a_{k-1}$, we write $\phi(\mathbf{x})$ and $\models_{\mathfrak{A}} \phi[\mathbf{a}]$ to indicate that the free variables of ϕ are among $\mathbf{x}$ and $\models_{\mathfrak{A}} \phi[\alpha]$ for some (any) assignment such that each $\alpha(x_i) = a_i$. $\mathbf{a} \in X$ means each $a_i \in X$, and $\eta(\mathbf{a})$ is $\eta(a_0), \dots, \eta(a_{k-1})$.

Definition 7.1.1 For any language L , any class Δ of L -formulas and any L -structures $\mathfrak{A}$ and $\mathfrak{B}$, a function η is a *partial Δ -embedding* of $\mathfrak{A}$ into $\mathfrak{B}$

— in symbols, $\eta : \mathfrak{A} \hookrightarrow_{\mathbf{p}}^{\Delta} \mathfrak{B}$ — iff for some subset $X \subseteq A$, $\eta : X \rightarrow B$ is injective and respects all formulas in Δ — that is, for any $\phi(\mathbf{x}) \in \Delta$ and any $\mathbf{a} \in X$,

$$\models_{\mathfrak{A}} \phi[\mathbf{a}] \iff \models_{\mathfrak{B}} \phi[\eta(\mathbf{a})].$$

Note that in contrast with Definition 3.4.11 we have used $\iff$ in place of $\implies$ in the definition; this reflects the fact that all the classes Δ that we will be using here are closed under negation, so the two versions are equivalent. In particular, we shall be interested in this notion mainly for the classes of formulas:

$$\text{QF} := \{ \phi : \phi \text{ is a quantifier-free formula} \},$$

$$\text{Fm}_{\omega} := \{ \phi : \phi \text{ is a first-order formula} \},$$

$$\text{Fm}_{\kappa} := \{ \phi : \phi \text{ is an } L_{\kappa\omega} \text{ formula with finitely many free variables} \},$$

for κ an infinite cardinal or ∞ . Since in general the language L is fixed in each context, we simplify the notation by omitting mention of it. When not otherwise specified, we assume that Δ is one of these classes. Then extending the definitions of Section 2.3, we say

η is a **partial embedding** iff η is a partial QF-embedding;

η is a **partial elementary embedding** iff η is a partial Fm_{ω} -embedding;

η is a **partial κ -embedding** iff η is a partial Fm_{κ} -embedding.

As in Section 3.4 there are several alternative ways to characterize this notion. Recall that L^X denotes the extension of L by new constant symbols $\dot{a}$ for each $a \in X$ and $\mathfrak{A}_X$ denotes the canonical expansion of $\mathfrak{A}$ to an L^X -structure obtained by setting $\dot{a}^{\mathfrak{A}_X} = a$. If $\eta : X \rightarrow |\mathfrak{B}|$, then we denote by $\mathfrak{B}_X^{\eta}$ the expansion of $\mathfrak{B}$ such that $\dot{a}^{\mathfrak{B}_X^{\eta}} = \eta(a)$ for each $a \in X$. Then easily, generalizing Proposition 3.4.19, if $X = \text{Dom}(\eta)$,

$$\eta : \mathfrak{A} \hookrightarrow_{\mathbf{p}}^{\Delta} \mathfrak{B} \iff \mathfrak{B}_X^{\eta} \text{ is a model of } \text{Th}^{\Delta}(\mathfrak{A}_X) \iff \mathfrak{A}_X \equiv^{\Delta} \mathfrak{B}_X^{\eta}.$$

We shall frequently be considering partial Δ -embeddings which are finite functions with $\text{Dom}(\eta) = \{a_0, \dots, a_k\}$, which we also write $\text{Dom}(\eta) = \mathbf{a}$. In this case we also usually write $(\mathfrak{A}, \mathbf{a})$ in place of $\mathfrak{A}_{\{a_0, \dots, a_k\}}$ and have

$$\eta : \mathfrak{A} \hookrightarrow_{\mathbf{p}}^{\Delta} \mathfrak{B} \iff (\mathfrak{A}, \mathbf{a}) \equiv^{\Delta} (\mathfrak{B}, \eta(\mathbf{a})).$$

In particular, for the classes listed above we write

$$\mathfrak{A} \hookrightarrow_{\mathbf{p},0} \mathfrak{B} \iff (\mathfrak{A}, \mathbf{a}) \equiv_0 (\mathfrak{B}, \eta(\mathbf{a}));$$

$$\mathfrak{A} \hookrightarrow_{\mathbf{p},\omega} \mathfrak{B} \iff (\mathfrak{A}, \mathbf{a}) \equiv (\mathfrak{B}, \eta(\mathbf{a}));$$

$$\mathfrak{A} \hookrightarrow_{\mathbf{p},\kappa} \mathfrak{B} \iff (\mathfrak{A}, \mathbf{a}) \equiv_{\kappa} (\mathfrak{B}, \eta(\mathbf{a})).$$

It will be useful to express this in still another way, and we introduce a notion which will play a major role in this chapter.

Definition 7.1.2 For any class Δ of formulas, any structure $\mathfrak{A}$ and any $\mathbf{a} \in A$,

$$\text{Tp}_{\Delta}^{\mathfrak{A}}(\mathbf{a}) := \left\{ \phi(\mathbf{x}) : \phi \in \Delta \text{ and } \models_{\mathfrak{A}} \phi[\mathbf{a}] \right\},$$

the Δ -**type** of $\mathbf{a}$ in $\mathfrak{A}$. In particular,

- (i) $\text{Tp}_0^{\mathfrak{A}}(\mathbf{a}) := \text{Tp}_{\text{QF}}^{\mathfrak{A}}(\mathbf{a})$ is the **quantifier-free type** of $\mathbf{a}$ in $\mathfrak{A}$;
- (ii) $\text{Tp}^{\mathfrak{A}}(\mathbf{a}) := \text{Tp}_{\text{Fm}_{\omega}}^{\mathfrak{A}}(\mathbf{a})$ is the **elementary type** or simply **type** of $\mathbf{a}$ in $\mathfrak{A}$;
- (iii) $\text{Tp}_{\kappa}^{\mathfrak{A}}(\mathbf{a}) := \text{Tp}_{\text{Fm}_{\kappa}}^{\mathfrak{A}}(\mathbf{a})$ is the κ -**type** of $\mathbf{a}$ in $\mathfrak{A}$.

Corollary 7.1.3 For any class Δ of formulas and any structures $\mathfrak{A}$ and $\mathfrak{B}$,

$$\eta : \mathfrak{A} \hookrightarrow_{\mathfrak{p}}^{\Delta} \mathfrak{B} \iff \text{for all } \mathbf{a} \in \text{Dom}(\eta), \quad \text{Tp}_{\Delta}^{\mathfrak{A}}(\mathbf{a}) = \text{Tp}_{\Delta}^{\mathfrak{B}}(\eta(\mathbf{a}));$$

hence if η is finite and $\text{Dom}(\eta) = \mathbf{a}$,

$$\eta : \mathfrak{A} \hookrightarrow_{\mathfrak{p}}^{\Delta} \mathfrak{B} \iff \text{Tp}_{\Delta}^{\mathfrak{A}}(\mathbf{a}) = \text{Tp}_{\Delta}^{\mathfrak{B}}(\eta(\mathbf{a})).$$

Note that we have placed no restriction on the set X , the domain of a partial embedding; in particular it need not be closed under the functions $f^{\mathfrak{A}}$ of $\mathfrak{A}$ and therefore the domain of a substructure of $\mathfrak{A}$. However, there is a canonical extension to the generated substructure.

Proposition 7.1.4 If η is a partial embedding of $\mathfrak{A}$ into $\mathfrak{B}$ with domain X , then η has a unique extension to an embedding $\bar{\eta}$ of the substructure generated by X — in symbols, $\bar{\eta} : \langle X \rangle_{\mathfrak{A}} \hookrightarrow_0 \mathfrak{B}$. If η is an elementary (κ -)partial embedding, then also $\bar{\eta}$ is an elementary (κ -)embedding.

Proof. Recall from Proposition 2.3.19 that the members of the universe of $\langle X \rangle_{\mathfrak{A}}$ are exactly the elements of A of the form $t^{\mathfrak{A}}[\alpha]$ with α an assignment taking all values in X — that is, $t^{\mathfrak{A}}[\mathbf{a}]$ for $\mathbf{a} \in X$. We define the extension by

$$\bar{\eta}(t^{\mathfrak{A}}[\mathbf{a}]) := t^{\mathfrak{B}}[\eta(\mathbf{a})].$$

This is well-defined since

$$\begin{aligned} t^{\mathfrak{A}}[\mathbf{a}] = u^{\mathfrak{A}}[\mathbf{a}] &\implies \models_{\mathfrak{A}} (t \doteq u)[\mathbf{a}] \\ &\implies \models_{\mathfrak{B}} (t \doteq u)[\eta(\mathbf{a})] \implies \bar{\eta}(t^{\mathfrak{A}}[\mathbf{a}]) = \bar{\eta}(u^{\mathfrak{A}}[\mathbf{a}]). \end{aligned}$$

To see that this is an embedding, suppose, as an example, that p is a unary relation symbol. Then

$$\begin{aligned} (t^{\mathfrak{A}}[\mathbf{a}]) \in p^{\mathfrak{A}} &\iff \models_{\mathfrak{A}} p t[\mathbf{a}] \\ &\iff \models_{\mathfrak{B}} p t[\eta(\mathbf{a})] \iff \eta(t^{\mathfrak{A}}[\mathbf{a}]) \in p^{\mathfrak{B}}. \end{aligned}$$

More generally, if $\phi(x) \in \Delta$, then

$$\begin{aligned} \models_{\mathfrak{A}} \phi[t^{\mathfrak{A}}[\mathbf{a}]] &\iff \models_{\mathfrak{A}} \phi_x(t)[\mathbf{a}] \\ &\iff \models_{\mathfrak{B}} \phi_x(t)[\eta(\mathbf{a})] \iff \models_{\mathfrak{B}} \phi[t^{\mathfrak{B}}[\eta(\mathbf{a})]]. \quad \square \end{aligned}$$

We have also immediately

Proposition 7.1.5 *For any class Δ of formulas and any structures $\mathfrak{A}$ and $\mathfrak{B}$, there exists a (finite) partial Δ -embedding of $\mathfrak{A}$ into $\mathfrak{B}$ iff $\mathfrak{A} \equiv^{\Delta} \mathfrak{B}$.*

Proof. The empty function η is a partial Δ -embedding iff $\mathfrak{A} \equiv^{\Delta} \mathfrak{B}$. $\square$

The following notions were first introduced in Exercise 2.4.55.

Definition 7.1.6 For any set $\mathcal{E}$ of partial embeddings of $\mathfrak{A}$ into $\mathfrak{B}$,

- (i) $\mathcal{E}$ has the **forward extension** property iff for each $\eta \in \mathcal{E}$ and each $c \in A$, there exists $\zeta \in \mathcal{E}$ which extends η and has $c \in \text{Dom}(\zeta)$;
- (ii) $\mathcal{E}$ has the **backward extension** property iff for each $\eta \in \mathcal{E}$ and each $d \in B$, there exists $\zeta \in \mathcal{E}$ which extends η and has $d \in \text{Im}(\zeta)$;
- (iii) $\mathcal{E}$ has the **back-and-forth** property iff it has both the forward and backward extension properties.

Definition 7.1.7 Two structures $\mathfrak{A}$ and $\mathfrak{B}$ are **partially isomorphic** — in symbols, $\mathfrak{A} \simeq_p \mathfrak{B}$ — iff there exists a non-empty set of partial embeddings from $\mathfrak{A}$ into $\mathfrak{B}$ which has the back-and-forth property.

Note that because partial embeddings are injective and the definition of back-and-forth is symmetric, $\simeq_p$ is an equivalence relation.

Examples 7.1.8

- (i) If $\mathfrak{A}$ and $\mathfrak{B}$ are isomorphic via η , then the singleton set $\{\eta\}$ trivially has the back-and-forth property.
- (ii) Let $\mathfrak{A} = \langle A, <_A \rangle$ and $\mathfrak{B} = \langle B, <_B \rangle$ be dense linear orders without endpoints and set

$$\mathcal{E} = \{ \eta : \eta \text{ is a finite partial embedding from } \mathfrak{A} \text{ into } \mathfrak{B} \}.$$

Of course, in this case a partial embedding with domain X is simply an order-preserving map from X into B . Now the density of the orderings easily implies that $\mathcal{E}$ has the back-and-forth property. For example, given $\eta \in \mathcal{E}$ and $c \in A$, there exists $d \in B$ such that for all of the finitely many $a \in \text{Dom}(\eta)$, $a <_A c$ iff $\eta(a) <_B d$. Then the extension to ζ such that $\zeta(c) = d$ again belongs to $\mathcal{E}$. This is essentially a

translation of part of the proof of Cantor's Theorem 2.4.45. If $\mathfrak{A}$ and $\mathfrak{B}$ both have, say, a least element but no greatest, then we take $\mathcal{E}$ to be the set of partial embeddings which map the least element of $\mathfrak{A}$ to the least element of $\mathfrak{B}$.

Note that in Definition 7.1.7 there was no condition on the cardinality of $\mathfrak{A}$ and $\mathfrak{B}$; partially isomorphic structures may be of different sizes. However,

Theorem 7.1.9 *For any countable structures $\mathfrak{A}$ and $\mathfrak{B}$,*

$$\mathfrak{A} \simeq_p \mathfrak{B} \implies \mathfrak{A} \simeq \mathfrak{B}.$$

Proof. Let $\mathfrak{A}$ and $\mathfrak{B}$ be countable with $\mathfrak{A} \simeq_p \mathfrak{B}$, and choose enumerations $\langle a_k : k \in \omega \rangle$ and $\langle b_k : k \in \omega \rangle$ of their universes and a non-empty set $\mathcal{E}$ of partial embeddings with the back-and-forth property. Fix any $\eta_0 \in \mathcal{E}$; we construct an increasing chain $\eta_0 \subseteq \eta_1 \subseteq \dots \subseteq \eta_k \subseteq \dots$ of elements of $\mathcal{E}$. Given η_{2k} , let η_{2k+1} be any extension of η_{2k} such that $a_k \in \text{Dom}(\eta_{2k+1})$ and η_{2k+2} any extension of η_{2k+1} such that $b_k \in \text{Im}(\eta_{2k+2})$. It is straightforward to check that the union η of this chain is an isomorphism of $\mathfrak{A}$ with $\mathfrak{B}$. $\square$

Even when partially isomorphic structures are not both countable or even of the same cardinality, there is a strong relationship.

Proposition 7.1.10 *For any structures $\mathfrak{A}$ and $\mathfrak{B}$ and any set $\mathcal{E}$ of partial embeddings from $\mathfrak{A}$ into $\mathfrak{B}$, if $\mathcal{E}$ has the back-and-forth property, then each element of $\mathcal{E}$ is actually a partial elementary embedding and indeed a partial ∞ -embedding. Hence*

$$\mathfrak{A} \simeq_p \mathfrak{B} \implies \mathfrak{A} \equiv_\infty \mathfrak{B}.$$

Proof. We prove by formula induction that for all $L_{\infty\omega}$ -formulas ϕ , for all $\eta \in \mathcal{E}$ and all $\mathbf{a} \in \text{Dom}(\eta)$,

$$\models_{\mathfrak{A}} \phi[\mathbf{a}] \iff \models_{\mathfrak{B}} \phi[\eta(\mathbf{a})].$$

This holds for atomic formulas by definition. The Boolean induction steps are trivial, even for infinite disjunctions:

$$\begin{aligned} \models_{\mathfrak{A}} \bigvee \Gamma[\mathbf{a}] &\iff \text{for some } \psi \in \Gamma, \models_{\mathfrak{A}} \psi[\mathbf{a}] \\ &\iff \text{for some } \psi \in \Gamma, \models_{\mathfrak{B}} \psi[\eta(\mathbf{a})] \iff \models_{\mathfrak{B}} \bigvee \Gamma[\eta(\mathbf{a})]. \end{aligned}$$

Suppose the equivalence holds for ψ and $\models_{\mathfrak{A}} \exists y \psi[\mathbf{a}]$. Then for some $c \in A$, $\models_{\mathfrak{A}} \psi[\mathbf{a}, c]$. By the forward extension property, there is an extension $\zeta \in \mathcal{E}$ of η such that $c \in \text{Dom}(\zeta)$. Now $\mathbf{a}, c \in \text{Dom}(\zeta)$, so by the induction hypothesis,

$\models_{\mathfrak{B}} \psi[\zeta(\mathbf{a}), \zeta(c)]$, which is the same as $\models_{\mathfrak{B}} \psi[\eta(\mathbf{a}), \zeta(c)]$. Hence $\models_{\mathfrak{B}} \exists y \psi[\eta(\mathbf{a})]$ as desired.

The converse implication follows similarly using the backward extension property. $\square$

We consider now under what circumstances the converse to either of these two implications holds. For example (Theorems 2.3.11 and 2.6.8), if $\mathfrak{A}$ and $\mathfrak{B}$ are finite and elementarily equivalent, then they are isomorphic and hence partially isomorphic. More interestingly, if two dense linear orderings are elementarily equivalent, then both or neither have greatest and least elements, so the example above shows that they are partially isomorphic. On the other hand, we showed in Proposition 2.5.11 that $T_{\text{Sc},0}$, the (complete) theory of the structure $(\omega, \text{Sc}, 0)$ has many countable non-standard models — that is, models not isomorphic to $(\omega, \text{Sc}, 0)$. It follows that such models are also not partially isomorphic despite being elementarily equivalent.

Lemma 7.1.11 *For any class Δ of formulas and any structures $\mathfrak{A}$ and $\mathfrak{B}$, if $\mathfrak{A} \equiv^{\Delta} \mathfrak{B}$ and for all $\mathbf{a} \in \mathfrak{A}$ and $\mathbf{b} \in \mathfrak{B}$ such that $(\mathfrak{A}, \mathbf{a}) \equiv^{\Delta} (\mathfrak{B}, \mathbf{b})$ we have,*

$$\begin{aligned} \text{for all } c \in A, \text{ there exists } d \in B \quad & (\mathfrak{A}, \mathbf{a}, c) \equiv^{\Delta} (\mathfrak{B}, \mathbf{b}, d), \quad \text{and} \\ \text{for all } d \in B, \text{ there exists } c \in A \quad & (\mathfrak{A}, \mathbf{a}, c) \equiv^{\Delta} (\mathfrak{B}, \mathbf{b}, d), \end{aligned}$$

then $\mathfrak{A} \simeq_p \mathfrak{B}$.

Proof. Under the given hypotheses,

$$\mathcal{E} = \{ \eta : \eta \text{ is a finite partial } \Delta\text{-embedding of } \mathfrak{A} \text{ into } \mathfrak{B} \}.$$

is nonempty and has the back-and-forth property. $\square$

Definition 7.1.12 For any class Δ of formulas and any structure $\mathfrak{A}$,

- (i) for any $\mathbf{a} \in \mathfrak{A}$, $\text{Tp}_{\Delta}^{\mathfrak{A}}(\mathbf{a})$ is **principal** iff there exists a formula $\theta \in \text{Tp}_{\Delta}^{\mathfrak{A}}(\mathbf{a})$ such that for all $\phi \in \text{Tp}_{\Delta}^{\mathfrak{A}}(\mathbf{a})$, $\models_{\mathfrak{A}} \forall \mathbf{x} [\theta \rightarrow \phi]$; we say that θ is a **generator** of $\text{Tp}_{\Delta}^{\mathfrak{A}}(\mathbf{a})$;
- (ii) $\mathfrak{A}$ is **Δ -atomic** iff for all $\mathbf{a} \in \mathfrak{A}$, $\text{Tp}_{\Delta}^{\mathfrak{A}}(\mathbf{a})$ is principal;
- (iii) $\mathfrak{A}$ is **atomic** iff it is Fm_{ω} -atomic and **κ -atomic** iff it is Fm_{κ} -atomic.

We shall explain the reason for the term ‘atomic’ in the next section.

Examples 7.1.13

- (i) If $\mathfrak{A}$ is any structure such that every element a has a name $\dot{a}$ in the language (or even is denoted by some constant term), then $\mathfrak{A}$ is atomic: a generator for $\text{Tp}^{\mathfrak{A}}(\mathbf{a})$ is the formula $x_0 \doteq \dot{a}_0 \wedge \cdots \wedge x_{k-1} \doteq \dot{a}_{k-1}$. All of the standard models of arithmetic $(\omega, \text{Sc}, 0, \dots)$ have this property.
- (ii) If $\mathfrak{A} = (A, <_A)$ is a dense linear ordering without endpoints, then $\mathfrak{A}$ is atomic: a generator for $\text{Tp}^{\mathfrak{A}}(\mathbf{a})$ is the formula θ which is the conjunction of all formulas $x_i < x_j$ such that $a_i <_A a_j$ and $x_i \doteq x_j$ such that $a_i = a_j$. This follows from the fact that $\text{Th}(\mathfrak{A}) = T_{(\text{DLO})}$ and admits quantifier-elimination — see Exercise 2.4.58.

Definition 7.1.14 A class Δ of fomulas is **first-order closed** iff it is closed under the connectives and quantifiers of first-order logic.

Of course, Fm_ω and all Fm_κ are first-order closed, but QF is not.

Proposition 7.1.15 For any first-order closed class Δ of formulas, any structures $\mathfrak{A}$ and $\mathfrak{B}$ and any $\mathbf{a} \in A$ and $\mathbf{b} \in B$, if $\mathfrak{A} \equiv^\Delta \mathfrak{B}$, θ is a generator of $\text{Tp}_\Delta^{\mathfrak{A}}(\mathbf{a})$ and $\models_{\mathfrak{B}} \theta[\mathbf{b}]$, then

$$\text{Tp}_\Delta^{\mathfrak{A}}(\mathbf{a}) = \text{Tp}_\Delta^{\mathfrak{B}}(\mathbf{b}) \quad \text{and (equivalently)} \quad (\mathfrak{A}, \mathbf{a}) \equiv^\Delta (\mathfrak{B}, \mathbf{b}).$$

Proof. Under the given hypotheses, for each formula $\phi \in \Delta$,

$$\models_{\mathfrak{A}} \phi[\mathbf{a}] \implies \models_{\mathfrak{A}} \forall \mathbf{x}[\theta \rightarrow \phi] \implies \models_{\mathfrak{B}} \forall \mathbf{x}[\theta \rightarrow \phi] \implies \models_{\mathfrak{B}} \phi[\mathbf{b}],$$

since also $\forall \mathbf{x}[\theta \rightarrow \phi] \in \Delta$. But then also for each $\phi \in \Delta$,

$$\not\models_{\mathfrak{A}} \phi[\mathbf{a}] \implies \models_{\mathfrak{A}} \neg \phi[\mathbf{a}] \implies \models_{\mathfrak{B}} \neg \phi[\mathbf{b}] \implies \not\models_{\mathfrak{B}} \phi[\mathbf{b}]. \quad \square$$

We can now give a first condition under which Δ -equivalence implies partial isomorphism.

Theorem 7.1.16 For any first-order closed class Δ of formulas and any structures $\mathfrak{A}$ and $\mathfrak{B}$, if $\mathfrak{A}$ and $\mathfrak{B}$ are both Δ -atomic, then

$$\mathfrak{A} \equiv^\Delta \mathfrak{B} \implies \mathfrak{A} \simeq_p \mathfrak{B}.$$

Proof. Given $\mathfrak{A}$ and $\mathfrak{B}$ satisfying the hypothesis, we verify that the hypotheses of Lemma 7.1.11 are satisfied. Fix $\mathbf{a}$ and $\mathbf{b}$ such that $(\mathfrak{A}, \mathbf{a}) \equiv^\Delta (\mathfrak{B}, \mathbf{b})$ and c , and choose a generator θ for $\text{Tp}_\Delta^{\mathfrak{A}}(\mathbf{a}, c)$. Then $\models_{\mathfrak{A}} \exists y \theta[\mathbf{a}]$, so since $(\mathfrak{A}, \mathbf{a}) \equiv^\Delta (\mathfrak{B}, \mathbf{b})$, also $\models_{\mathfrak{B}} \exists y \theta[\mathbf{b}]$. Let d be any witness, an element of B such that $\models_{\mathfrak{B}} \theta[\mathbf{b}, d]$. Then by the preceding proposition, $(\mathfrak{A}, \mathbf{a}, c) \equiv^\Delta (\mathfrak{B}, \mathbf{b}, d)$. The verification of the backward extension property is symmetric. $\square$

Corollary 7.1.17 *For any first-order closed class Δ of formulas and any countable structures $\mathfrak{A}$ and $\mathfrak{B}$, if $\mathfrak{A}$ and $\mathfrak{B}$ are both Δ -atomic, and $\mathfrak{A} \equiv^\Delta \mathfrak{B}$, then $\mathfrak{A} \simeq \mathfrak{B}$.*

Proof. Immediate from the preceding theorem and Theorem 7.1.9. $\square$

In Section 7.3 we shall see conditions under which structures are atomic and other conditions under which $\mathfrak{A} \equiv \mathfrak{B}$ implies that $\mathfrak{A} \simeq_p \mathfrak{B}$, but we turn here to the infinitary case.

Proposition 7.1.18 *For any structure $\mathfrak{A}$ and any infinite cardinal $\kappa > \text{Card}(\mathfrak{A})$, $\mathfrak{A}$ is κ -atomic.*

Proof. Fix $\mathfrak{A}$, $\mathbf{a} \in \mathfrak{A}$ and $\kappa > \text{Card}(\mathfrak{A})$ — we construct a generator for $\text{Tp}_\kappa^\mathfrak{A}(\mathbf{a})$. For each $\mathbf{c} \in \mathfrak{A}$, set

$$\psi^{\mathbf{c}} = \begin{cases} \top, & \text{if } \text{Tp}_\kappa^\mathfrak{A}(\mathbf{a}) = \text{Tp}_\kappa^\mathfrak{A}(\mathbf{c}); \\ \text{some } \psi \in \text{Tp}_\kappa^\mathfrak{A}(\mathbf{a}) \sim \text{Tp}_\kappa^\mathfrak{A}(\mathbf{c}), & \text{otherwise.} \end{cases}$$

There are at most $\text{Card}(\mathfrak{A})$ distinct formulas $\psi^{\mathbf{c}}$ and each is an $L_{\kappa\omega}$ -formula; hence

$$\theta := \bigwedge \{ \psi^{\mathbf{c}} : \mathbf{c} \in \mathfrak{A} \}$$

is also an $L_{\kappa\omega}$ -formula. Clearly $\theta \in \text{Tp}_\kappa^\mathfrak{A}(\mathbf{a})$, and we claim that it is a generator. Fix $\phi \in \text{Tp}_\kappa^\mathfrak{A}(\mathbf{a})$; we show that $\models_{\mathfrak{A}} \forall \mathbf{x} (\neg \phi \rightarrow \neg \theta)$. Fix $\mathbf{c} \in \mathfrak{A}$ and suppose $\models_{\mathfrak{A}} \neg \phi[\mathbf{c}]$. Then ϕ witnesses that $\text{Tp}_\kappa^\mathfrak{A}(\mathbf{a}) \neq \text{Tp}_\kappa^\mathfrak{A}(\mathbf{c})$, and by the definition, $\models_{\mathfrak{A}} \neg \psi^{\mathbf{c}}[\mathbf{c}]$. Hence also $\models_{\mathfrak{A}} \neg \theta^{\mathbf{c}}[\mathbf{c}]$. Thus $\models_{\mathfrak{A}} (\neg \phi \rightarrow \neg \theta)[\mathbf{c}]$. $\square$

Karp's Theorem 7.1.19 *For any structures $\mathfrak{A}$ and $\mathfrak{B}$ and any infinite cardinal $\kappa > \max\{\text{Card}(\mathfrak{A}), \text{Card}(\mathfrak{B})\}$,*

$$\mathfrak{A} \equiv_\kappa \mathfrak{B} \implies \mathfrak{A} \simeq_p \mathfrak{B}.$$

Proof. This now follows immediately from Theorem 7.1.16. $\square$

Corollary 7.1.20 *For any structures $\mathfrak{A}$ and $\mathfrak{B}$,*

$$\mathfrak{A} \equiv_\infty \mathfrak{B} \iff \mathfrak{A} \simeq_p \mathfrak{B}.$$

If $\mathfrak{A}$ and $\mathfrak{B}$ are both countable, then

$$\mathfrak{A} \equiv_\infty \mathfrak{B} \iff \mathfrak{A} \equiv_{\omega_1} \mathfrak{B} \iff \mathfrak{A} \simeq_p \mathfrak{B} \iff \mathfrak{A} \simeq \mathfrak{B}.$$

Proof. This is an immediate consequence of Theorem 7.1.10 and Karp's Theorem. $\square$

Recall that for a finite language L and a finite L -structure $\mathfrak{A}$ there is a single L -sentence which characterizes $\mathfrak{A}$ up to isomorphism (Theorem 2.3.11). Here we generalize this to show that for any structure for any language there is an $L_{\kappa\omega}$ -sentence for sufficiently large κ which characterizes $\mathfrak{A}$ up to partial isomorphism.

Scott's Theorem 7.1.21 *For any language L , L -structure $\mathfrak{A}$, and infinite cardinal κ greater than both $\text{Card}(\mathfrak{A})$ and $\text{Card}(L)$, there exists an $L_{\kappa\omega}$ -sentence $\chi_{\mathfrak{A}}$, called a **Scott sentence** of $\mathfrak{A}$, such that for any structure $\mathfrak{B}$,*

$$\models_{\mathfrak{B}} \chi_{\mathfrak{A}} \iff \mathfrak{A} \simeq_{\mathfrak{p}} \mathfrak{B}.$$

Hence if $\mathfrak{A}$ and $\mathfrak{B}$ are both countable, then $\models_{\mathfrak{B}} \chi_{\mathfrak{A}} \iff \mathfrak{A} \simeq \mathfrak{B}$.

Proof. Given $\mathfrak{A}$ and κ greater than $\text{Card}(\mathfrak{A})$ and $\text{Card}(L)$, for each $\mathbf{a} \in \mathfrak{A}$, using Proposition 7.1.18 choose a generator $\theta^{\mathbf{a}}$ of $\text{Tp}_{\kappa}^{\mathfrak{A}}(\mathbf{a})$; in particular, let $\theta^{(\cdot)}$ be any logically valid sentence. Let $\chi_{\mathfrak{A}}$ be the conjunction of the following $L_{\kappa\omega}$ -sentences over all $\mathbf{a} \in \mathfrak{A}$ (of all finite lengths):

- (1) $\forall \mathbf{x}[\theta^{\mathbf{a}}(\mathbf{x}) \rightarrow \bigwedge \text{Tp}_0^{\mathfrak{A}}(\mathbf{a})]$;
- (2) $\forall \mathbf{x}[\theta^{\mathbf{a}}(\mathbf{x}) \rightarrow \bigwedge_{c \in A} \exists y \theta^{\mathbf{a},c}(\mathbf{x}, y)]$;
- (3) $\forall \mathbf{x}[\theta^{\mathbf{a}}(\mathbf{x}) \rightarrow \forall y \bigvee_{c \in A} \theta^{\mathbf{a},c}(\mathbf{x}, y)]$.

That $\models_{\mathfrak{A}} \chi_{\mathfrak{A}}$ is immediate from the definition of generator, since the right-hand sides of each of the implications clearly belong to $\text{Tp}_{\kappa}^{\mathfrak{A}}(\mathbf{a})$. Hence if $\mathfrak{B}$ is any structure such that $\mathfrak{A} \simeq_{\mathfrak{p}} \mathfrak{B}$, then by Theorem 7.1.10, $\mathfrak{A} \equiv_{\infty} \mathfrak{B}$, so also $\models_{\mathfrak{B}} \chi_{\mathfrak{A}}$. Conversely, suppose that $\mathfrak{B}$ is any structure such that $\models_{\mathfrak{B}} \chi_{\mathfrak{A}}$. Set

$$\mathcal{E} := \{ \eta : \eta \text{ is a finite partial embedding of } \mathfrak{A} \text{ into } \mathfrak{B} \\ \text{such that for } \mathbf{a} = \text{Dom}(\eta), \models_{\mathfrak{B}} \theta^{\mathbf{a}}[\eta(\mathbf{a})] \}.$$

$\mathcal{E}$ is nonempty, since by our choice of $\theta^{(\cdot)}$ it contains the empty partial embedding. It is then straightforward to verify that $\mathcal{E}$ has the back-and-forth property. For example, given $\eta \in \mathcal{E}$, $\mathbf{a} = \text{Dom}(\eta)$, and $c \in A$, by (2),

$$\models_{\mathfrak{B}} \exists y \theta^{\mathbf{a},c}[\eta(\mathbf{a})].$$

If $d \in B$ is any witness for this and ζ is the extension of η generated by setting $\zeta(c) = d$, then by (1), ζ is a partial embedding so $\zeta \in \mathcal{E}$. The backward extension property follows similarly from (3). The second claim is now immediate from Theorem 7.1.9. $\square$

Note that after the fact we conclude that $\mathfrak{A} \equiv_{\infty} \mathfrak{B}$ from which it follows that the members of $\mathcal{E}$ are exactly the finite partial ∞ -embeddings, but this fact is not available at the beginning of the proof.

Further refinement of these results yields additional information about both the infinitary and first-order cases.

Definition 7.1.22 For any partial embedding η of $\mathfrak{A}$ into $\mathfrak{B}$ and any ordinal σ ,

- (i) η is **0-extendable**;
- (ii) if σ is a limit ordinal, then η is **σ -extendable** iff it is τ -extendable for all $\tau < \sigma$;
- (iii) η is **$(\sigma + 1)$ -extendable** iff
 - (a) for each $c \in A$, there is some σ -extendable ζ such that ζ extends η and $c \in \text{Dom}(\zeta)$;
 - (b) for each $d \in B$, there is some σ -extendable ζ such that ζ extends η and $d \in \text{Im}(\zeta)$;
- (iv) $\mathfrak{A}$ is **partially σ -embeddable** in $\mathfrak{B}$ — in symbols, $\mathfrak{A} \simeq_{\mathfrak{p}}^{\sigma} \mathfrak{B}$ — iff there exists a σ -extendable finite partial embedding of $\mathfrak{A}$ into $\mathfrak{B}$.

Proposition 7.1.23 For any structures $\mathfrak{A}$ and $\mathfrak{B}$,

$$\mathfrak{A} \simeq_{\mathfrak{p}} \mathfrak{B} \iff \text{for all ordinals } \sigma, \mathfrak{A} \simeq_{\mathfrak{p}}^{\sigma} \mathfrak{B}.$$

Proof. For the implication $(\implies)$, a straightforward induction on σ verifies that if $\mathcal{E}$ is a set of partial embeddings with the back-and-forth property, then for all σ , every $\eta \in \mathcal{E}$ is σ -extendable. Suppose now that for all σ , $\mathfrak{A} \simeq_{\mathfrak{p}}^{\sigma} \mathfrak{B}$. For each finite partial embedding η of $\mathfrak{A}$ into $\mathfrak{B}$, set

$$\sigma_{\eta} := \begin{cases} \text{greatest } \sigma [\eta \text{ is } \sigma\text{-extendable}], & \text{if such } \sigma \text{ exists;} \\ 0, & \text{otherwise;} \end{cases}$$

and let

$$\rho := \sup^{+} \{ \sigma_{\eta} : \eta \text{ is a finite partial embedding of } \mathfrak{A} \text{ into } \mathfrak{B} \},$$

where for any set X of ordinals, $\sup^{+} X := \bigcup \{ \sigma +_o 1 : \sigma \in X \}$ is the least ordinal strictly greater than all $\sigma \in X$. Since by hypothesis, $\mathfrak{A} \simeq_{\mathfrak{p}}^{\rho} \mathfrak{B}$, there exist ρ -extendable partial embeddings, and it is clear from the definition of ρ that every ρ -extendable finite partial embedding is also σ -extendable for all $\sigma \geq \rho$, in particular for $\sigma = \rho + 1$. It follows easily that the set of ρ -extendable finite partial embeddings has the back-and-forth property and hence witnesses that $\mathfrak{A} \simeq_{\mathfrak{p}} \mathfrak{B}$. $\square$

We next define the syntactic notion that corresponds to σ -extendability of partial embeddings. To avoid tedious repetition in the following, we shall write the definitions and results for infinitary languages $L_{\kappa\omega}$ without special attention to the finitary (first-order) language L . However, L is for almost all purposes the same as $L_{\omega\omega}$, and we leave it to the reader to formulate the finitary versions.

Definition 7.1.24 For any infinite cardinal κ , the **quantifier rank** $\text{qr}(\phi)$ of an $L_{\kappa\omega}$ -formula is defined recursively by the following clauses:

- (i) if ϕ is atomic, then $\text{qr}(\phi) = 0$;
- (ii) $\text{qr}(\neg \psi) = \text{qr}(\psi)$; $\text{qr}(\psi \vee \theta) = \max\{\text{qr}(\psi), \text{qr}(\theta)\}$;
- (iii) $\text{qr}(\bigvee \Gamma) = \sup\{\text{qr}(\psi) : \psi \in \Gamma\}$;
- (iv) $\text{qr}(\exists x \psi) = \text{qr}(\psi) + 1$.

Note that first-order formulas have finite quantifier rank.

Much of the preceding development can be refined to take account of quantifier rank with only minor changes, and we simply introduce the appropriate notation and state the facts. First we set $\mathfrak{A} \equiv_{\kappa}^{\sigma} \mathfrak{B}$ iff $\mathfrak{A}$ and $\mathfrak{B}$ have the same true $L_{\kappa\omega}$ -sentences of quantifier rank at most σ . For $\mathbf{a} \in \mathfrak{A}$, the $(\kappa; \sigma)$ -**type** of $\mathbf{a}$, $\text{Tp}_{\kappa; \sigma}^{\mathfrak{A}}(\mathbf{a})$ is the set of all $L_{\kappa\omega}$ -formulas with at most the variables $\mathbf{x}$ free and of quantifier rank at most σ which are satisfied by $\mathbf{a}$. A partial embedding η is a **partial** $(\kappa; \sigma)$ -**embedding** iff it respects all $L_{\kappa\omega}$ -formulas of quantifier rank at most σ and we have

η is a partial $(\kappa; \sigma)$ -embedding

$$\begin{aligned} \iff & \text{ for all } \mathbf{a} \in \text{Dom}(\eta), \quad \text{Tp}_{\kappa; \sigma}^{\mathfrak{A}}(\mathbf{a}) = \text{Tp}_{\kappa; \sigma}^{\mathfrak{B}}(\eta(\mathbf{a})) \\ \iff & \text{ for all } \mathbf{a} \in \text{Dom}(\eta), \quad (\mathfrak{A}, \mathbf{a}) \equiv_{\kappa}^{\sigma} (\mathfrak{B}, \eta(\mathbf{a})). \end{aligned}$$

Theorem 7.1.25 For any structures $\mathfrak{A}$ and $\mathfrak{B}$, any partial embedding η from $\mathfrak{A}$ into $\mathfrak{B}$ and any ordinal σ ,

- (i) if η is σ -extendable, then η is a partial $(\infty; \sigma)$ embedding;
- (ii) if κ is greater than $\max\{\text{Card}(\mathfrak{A}), \text{Card}(\mathfrak{B})\}$ and η is a finite partial $(\kappa; \sigma)$ -embedding, then η is σ -extendable.

Proof. Part (i) is proved by a straightforward induction on σ ; we give only the successor step. Suppose that the result is true for σ and that η is a $(\sigma + 1)$ -extendable partial embedding. We prove by formula induction that η respects all formulas ϕ of quantifier rank at most $\sigma + 1$. The basis and the Boolean induction steps are trivial. If ϕ is $\exists y \psi$ and $\models_{\mathfrak{A}} \phi[\mathbf{a}]$, then for some $c \in A$, $\models_{\mathfrak{A}} \psi[\mathbf{a}, c]$. Let ζ be a σ -extendable extension of η such that $c \in \text{Dom}(\zeta)$. Since ψ has quantifier rank at most σ , by the induction hypotheses ζ respects ψ so $\models_{\mathfrak{B}} \psi[\zeta(\mathbf{a}), \zeta(c)]$. Hence $\models_{\mathfrak{B}} \phi[\eta(\mathbf{a})]$ as desired.

Now for (ii), suppose that κ is a large enough cardinal; we prove the implication by induction on σ . The basis and limit cases are again trivial, so assume as induction hypothesis that it holds for σ and that η is a finite partial $(\kappa; \sigma + 1)$ -embedding. Then for $\mathbf{a} \in \text{Dom}(\eta)$ we have $(\mathfrak{A}, \mathbf{a}) \equiv_{\kappa}^{\sigma+1} (\mathfrak{B}, \eta(\mathbf{a}))$.

To show that η is $(\sigma + 1)$ -forward extendable we need, given $c \in A$, to find a σ -extendable extension ζ with $c \in \text{Dom}(\zeta)$. It suffices to find $d \in B$ such that $(\mathfrak{A}, \mathbf{a}, c) \equiv_{\kappa}^{\sigma} (\mathfrak{B}, \eta(\mathbf{a}), d)$, since then with $\zeta(c) = d$, ζ is a $(\kappa; \sigma)$ embedding, so by the induction hypothesis is σ -extendable. Suppose towards a contradiction that there is no such $d \in B$. Then for each $d \in B$, there exists an $L_{\kappa\omega}$ -formula $\psi_d(\mathbf{x}, y)$ with $\text{qr}(\psi_d) \leq \sigma$ such that

$$\psi_d \in \text{Tp}_{\kappa; \sigma}^{\mathfrak{A}}(\mathbf{a}, c) \quad \text{but} \quad \neg \psi_d \in \text{Tp}_{\kappa; \sigma}^{\mathfrak{B}}(\eta(\mathbf{a}), d).$$

Let $\theta(\mathbf{x}, y)$ be $\bigwedge \{ \psi_d : d \in B \}$, so θ is also an $L_{\kappa\omega}$ -formula with $\text{qr}(\theta) \leq \sigma$. Then $\exists y \theta$ has quantifier rank at most $\sigma + 1$, $\models_{\mathfrak{A}} \exists y \theta[\mathbf{a}]$ but $\not\models_{\mathfrak{B}} \neg \exists y \theta[\eta(\mathbf{a})]$, contrary to the hypothesis that η is a partial $(\kappa; \sigma + 1)$ embedding. A parallel argument gives backward extendability. $\square$

Corollary 7.1.26 *For any structures $\mathfrak{A}$ and $\mathfrak{B}$, any κ greater than $\max\{\text{Card}(\mathfrak{A}), \text{Card}(\mathfrak{B})\}$ and any ordinal σ ,*

$$\mathfrak{A} \simeq_{\mathfrak{p}}^{\sigma} \mathfrak{B} \iff \mathfrak{A} \equiv_{\mathfrak{p}}^{\sigma} \mathfrak{B}. \quad \square$$

Note that by generalizing both sides over all ordinals σ we obtain a slightly different proof of Karp's Theorem. For the finitary language L we have the

Corollary 7.1.27 *For any structures $\mathfrak{A}$ and $\mathfrak{B}$,*

$$[\text{for all } n \in \omega, \mathfrak{A} \simeq_{\mathfrak{p}}^n \mathfrak{B}] \implies \mathfrak{A} \equiv \mathfrak{B}.$$

Proof. Clearly $\mathfrak{A} \equiv \mathfrak{B}$ iff for all $n \in \omega$, $\mathfrak{A} \equiv_{\mathfrak{p}}^n \mathfrak{B}$. $\square$

If the language L is finite, we shall show that the converse of this last corollary holds as well. If $\mathfrak{A}$ and $\mathfrak{B}$ are finite, then the proof of (ii) of the theorem shows that if η is a partial elementary embedding, then for all $n \in \omega$, η is n -extendable. The following proposition allows us to avoid this finiteness hypothesis in the case that the language is finite and relational; we sketch in an exercise how this may be extended to all finite languages. Let us temporarily call a first-order formula ϕ an (n, k) -**formula** iff ϕ has quantifier rank at most n and the free variables of ϕ are among $v_0, \dots, v_{k-1}$.

Proposition 7.1.28 *For any finite relational language L (no function symbols) and each $n, k \in \omega$, there exists a finite set $\Delta^{(n, k)}$ of (n, k) -formulas such that every (n, k) -formula is logically equivalent to a member of $\Delta^{(n, k)}$.*

Proof. Note first that for any finite set Δ of formulas, there is a finite set $\text{Bool}^*(\Delta) \subseteq \text{Bool}(\Delta)$ such that every member of $\text{Bool}(\Delta)$ is logically equivalent to a member of $\text{Bool}^*(\Delta)$ — for example, we may take $\text{Bool}^*(\Delta)$

to be the set of disjunctive normal forms formed from members of Δ . Now set

$$\begin{aligned}\Delta^{(0,k)} &:= \text{Bool}^*(\{\psi : \psi \text{ is atomic with variables among } v_0, \dots, v_{k-1}\}); \\ \Delta^{(n+1,k)} &:= \text{Bool}^*(\Delta^{(n,k)} \cup \{\exists v_k \psi : \psi \in \Delta^{(n,k+1)}\}).\end{aligned}$$

The assumption that L is finite and relational implies that $\Delta^{(0,k)}$ is finite and hence all $\Delta^{(n,k)}$ are finite. The remainder of the proof is a straightforward induction on n . $\square$

Fraïssé's Theorem 7.1.29 *For any finite relational language L , any L -structures $\mathfrak{A}$ and $\mathfrak{B}$, any finite partial embedding η from $\mathfrak{A}$ into $\mathfrak{B}$ and all $n \in \omega$,*

$$\eta \text{ is } n\text{-extendable} \iff \eta \text{ is an } (\omega; n)\text{-partial embedding.}$$

Hence for all $n \in \omega$, $\mathfrak{A} \simeq_{\mathfrak{p}}^n \mathfrak{B} \iff \mathfrak{A} \equiv_{\omega}^n \mathfrak{B}$, so

$$\mathfrak{A} \equiv \mathfrak{B} \iff [\text{for all } n \in \omega, \mathfrak{A} \simeq_{\mathfrak{p}}^n \mathfrak{B}].$$

Proof. The implication $(\implies)$ is immediate from Theorem 7.1.25; the stronger conclusion there that η is an $(\infty; n)$ -partial embedding is equivalent here because of the restriction to finite relational languages. We prove the implication $(\impliedby)$ by induction on n . It is obvious for $n = 0$, so assume that it holds for n and let η be a finite $(\omega; n+1)$ -partial embedding with, say, $\text{Dom}(\eta) = \mathbf{a} = a_0, \dots, a_{k-1}$. For each $c \in A$ and $\theta(v_0, \dots, v_k)$, set

$$\begin{aligned}\theta^c &:= \begin{cases} \theta, & \text{if } \models_{\mathfrak{A}} \theta[\mathbf{a}, c]; \\ \neg \theta, & \text{otherwise;} \end{cases} \\ \chi^c &:= \bigwedge \{ \theta^c : \theta \in \Delta^{(n,k+1)} \}.\end{aligned}$$

Then $\models_{\mathfrak{A}} \exists v_k \chi^c[\mathbf{a}]$ and $\text{qr}(\exists v_k \chi^c) = n+1$, so by hypothesis $\models_{\mathfrak{B}} \exists v_k \chi^c[\eta(\mathbf{a})]$. Let d be an element of B such that $\models_{\mathfrak{B}} \chi^c[\eta(\mathbf{a}), d]$, and ζ the extension of η with $\zeta(c) = d$. Now if ϕ is any $(n, k+1)$ -formula, it is logically equivalent to some $\theta \in \Delta^{(n,k+1)}$ and clearly

$$\models_{\mathfrak{A}} \phi[\mathbf{a}, c] \iff \models_{\mathfrak{B}} \phi[\zeta(\mathbf{a}, c)],$$

so ζ is an (ω, n) -embedding and hence by the induction hypothesis is n -extendable. The backward extension property is treated in a parallel way, and we conclude that η is $(n+1)$ -extendable as required. The remainder of the claim is now immediate. $\square$

The right-hand side of the second equivalence of this theorem has an interesting formulation in terms of games. With any two structures $\mathfrak{A}$ and $\mathfrak{B}$ with disjoint universes we associate the game $G(\mathfrak{A}, \mathfrak{B})$ played by two players I and II according to the following rules. I has the first move and must choose a natural number $n > 0$ and either $a_0 \in A$ or $b_0 \in B$. Then II plays an element from the other structure: $b_0 \in B$ or $a_0 \in A$, respectively. I now chooses either $a_1 \in A$ or $b_1 \in B$ and II plays either b_1 or a_1 . Play proceeds for n pairs of moves and at the end II wins iff the function $\eta(a_i) = b_i$ ($i < n$) is a partial embedding from $\mathfrak{A}$ into $\mathfrak{B}$; otherwise I wins. A **winning strategy** for II is a set of rules for responses to the moves of I that guarantee that II always wins.

Proposition 7.1.30 *For any language L and any disjoint L -structures $\mathfrak{A}$ and $\mathfrak{B}$,*

$$[\text{for all } n \in \omega, \quad \mathfrak{A} \simeq_{\mathfrak{p}}^n \mathfrak{B}] \quad \Longleftrightarrow \quad \text{II has a winning strategy for } G(\mathfrak{A}, \mathfrak{B}).$$

Proof. Suppose first that for all n , $\mathfrak{A} \simeq_{\mathfrak{p}}^n \mathfrak{B}$. Then II may guarantee a win by playing as follows. Suppose the first move of I is n and a_0 . Then II chooses b_0 such that the function $a_0 \mapsto b_0$ is an $(n-1)$ -extendable partial embedding; such a b_0 exists because the empty partial embedding, as a subset of every partial embedding, is n -extendable. If, now, I plays b_1 , then II chooses a_1 such that $(a_0, a_1) \mapsto (b_0, b_1)$ is $(n-2)$ -extendable. At the end of play, clearly II wins.

Now suppose that II has a winning strategy. Fix n and consider only plays of I which start with n as the first move. We show that for any $i \leq n$, sequences $a_0, \dots, a_{n-i-1}, b_0, \dots, b_{n-i-1}$ generated by a partial play in which II follows the winning strategy for $(n-i)$ -many moves, the function $\eta(a_j) = b_j$ ($j < n-i$) is an i -extendable partial embedding. In particular, for $i = n$, the empty partial embedding is n -extendable, so $\mathfrak{A} \simeq_{\mathfrak{p}}^n \mathfrak{B}$. For $i = 0$ this is immediate from the definition of strategy. Suppose it is true for i and sequences $a_0, \dots, a_{n-i-2}, b_0, \dots, b_{n-i-2}$ are generated by $n-(i+1)$ many moves. Since II's strategy provides an appropriate response to counter I's choice of either a_{n-i-1} or b_{n-i-1} , the function $\eta(a_j) = b_j$ ($j < n-i-1$) has the requisite extensions to an i -extendable partial embedding to ensure that it is $(i+1)$ -extendable. $\square$

Ehrenfeucht's Theorem 7.1.31 *For any finite relational language L and any disjoint L -structures $\mathfrak{A}$ and $\mathfrak{B}$,*

$$\mathfrak{A} \equiv \mathfrak{B} \quad \Longleftrightarrow \quad \text{II has a winning strategy for } G(\mathfrak{A}, \mathfrak{B}). \quad \square$$

Collateral Reading. The material of this section can also be found, in various combinations, in [Ch-Ke], [Ho93], [Kei] and [Dic]. The last of these covers many extensions of the results to larger infinitary languages $L_{\kappa\lambda}$.

Definition 7.1.32 A structure $\mathfrak{A}$ is called *rigid* iff it has no non-trivial automorphisms.

Exercises

Exercise 7.1.33 Prove that for any countable structure $\mathfrak{A}$ the following are equivalent:

- (i) $\mathfrak{A}$ has fewer than $2^{\aleph_0}$ automorphisms;
- (ii) for some finite sequence $\mathbf{a}$ of elements of $\mathfrak{A}$, $(\mathfrak{A}, \mathbf{a})$ is rigid;
- (iii) $\mathfrak{A}$ has at most $\aleph_0$ automorphisms.

Hint: For the implication (i) $\implies$ (ii), consider the set $\mathcal{E}$ of finite partial embeddings η of $\mathfrak{A}$ into itself such that if $\mathbf{a} = \text{Dom}(\eta)$, $(\mathfrak{A}, \mathbf{a}) \simeq_p (\mathfrak{A}, \eta(\mathbf{a}))$. Show that every element of $\mathcal{E}$ has two distinct extensions to members of $\mathcal{E}$.

Exercise 7.1.34 Use the result of the preceding exercise to give the following variant of the Corollary to Karp's Theorem 7.1.19: for any countable structure $\mathfrak{A}$ with fewer than $2^{\aleph_0}$ automorphisms and any structure $\mathfrak{B}$,

$$\mathfrak{A} \simeq_p \mathfrak{B} \iff \mathfrak{A} \simeq \mathfrak{B}.$$

Hint: Show that under the hypothesis there are $L_{\omega_1\omega}$ -formulas $\theta_a(x)$ for each $a \in A$ such that

$$\models_{\mathfrak{A}} \bigwedge_{a \in A} \exists! x \theta_a(x) \wedge \forall x \bigvee_{a \in A} \theta_a(x).$$

Conclude that if $\mathfrak{A} \simeq_p \mathfrak{B}$, then $\mathfrak{B}$ is countable.

Exercise 7.1.35 Show, strengthening Karp's Theorem, that for any structures $\mathfrak{A}$ and $\mathfrak{B}$ and any infinite cardinal $\kappa > \min\{\text{Card}(\mathfrak{A}), \text{Card}(\mathfrak{B})\}$,

$$\mathfrak{A} \equiv_{\kappa} \mathfrak{B} \implies \mathfrak{A} \simeq_p \mathfrak{B}.$$

Exercise 7.1.36 Call $\mathfrak{A}$ $(\kappa; \sigma)$ -*atomic* iff every $(\kappa; \sigma)$ -type of a sequence of elements of $\mathfrak{A}$ is principal — that is, implied in $\mathfrak{A}$ by a single member. Show that for any structure $\mathfrak{A}$, any ordinal σ and any infinite cardinal κ greater than $\text{Card}(\mathfrak{A})$, $\mathfrak{A}$ is $(\kappa; \sigma)$ -atomic.

Exercise 7.1.37 Give a new proof of Scott's Theorem 7.1.21 along the following lines. Fix a structure $\mathfrak{A}$ and an infinite cardinal κ greater than $\text{Card}(\mathfrak{A})$. For each $\mathbf{a} \in A$, let

$$\begin{aligned}\theta_{\mathfrak{A},0}^{\mathbf{a}}(\mathbf{x}) &= \bigwedge \text{Tp}_0^{\mathfrak{A}}(\mathbf{a}); \\ \theta_{\mathfrak{A},\sigma}^{\mathbf{a}}(\mathbf{x}) &= \bigwedge_{\tau < \sigma} \theta_{\mathfrak{A},\tau}^{\mathbf{a}}(\mathbf{x}), \quad \text{for } \sigma \text{ a limit ordinal;} \\ \theta_{\mathfrak{A},\sigma+1}^{\mathbf{a}}(\mathbf{x}) &= \theta_{\mathfrak{A},\sigma}^{\mathbf{a}} \wedge \bigwedge_{c \in A} \exists y \theta_{\mathfrak{A},\sigma}^{\mathbf{a},c}(\mathbf{x}, y) \wedge \forall y \bigvee_{c \in A} \theta_{\mathfrak{A},\sigma}^{\mathbf{a},c}(\mathbf{x}, y).\end{aligned}$$

Show that for all σ and $\mathbf{a}$,

- (i) $\text{qr}(\theta_{\mathfrak{A},\sigma}^{\mathbf{a}}) = \sigma$ and $\models_{\mathfrak{A}} \theta_{\mathfrak{A},\sigma}^{\mathbf{a}}[\mathbf{a}]$;
- (ii) for any other L -structure $\mathfrak{B}$, any finite partial embedding η from $\mathfrak{A}$ into $\mathfrak{B}$, and $\mathbf{a} = \text{Dom}(\eta)$,

$$\eta \text{ is } \sigma\text{-extendable} \iff \models_{\mathfrak{B}} \theta_{\mathfrak{A},\sigma}^{\mathbf{a}}[\eta(\mathbf{a})];$$

- (iii) in particular, $\mathfrak{A} \simeq_{\mathfrak{p}}^{\sigma} \mathfrak{B} \iff \models_{\mathfrak{B}} \theta_{\mathfrak{A},\sigma}^{()}$.

As in the proof of Proposition 7.1.23, for each finite embedding η of $\mathfrak{A}$ into itself, set

$$\sigma_{\eta} := \begin{cases} \text{greatest } \sigma \text{ } [\eta \text{ is } \sigma\text{-extendable}], & \text{if such } \sigma \text{ exists;} \\ 0, & \text{otherwise;} \end{cases}$$

and let

$$\rho(\mathfrak{A}) = \sup^+ \{ \sigma_{\eta} : \eta \text{ is a finite partial embedding of } \mathfrak{A} \text{ into itself } \}.$$

$\rho(\mathfrak{A})$ is called the **Scott rank** of $\mathfrak{A}$. Now let $\chi_{\mathfrak{A}}$ be the sentence

$$\theta_{\mathfrak{A},\rho(\mathfrak{A})}^{()} \wedge \bigwedge_{\mathbf{a} \in A} \forall \mathbf{x} [\theta_{\mathfrak{A},\rho(\mathfrak{A})}^{\mathbf{a}} \rightarrow \theta_{\mathfrak{A},\rho(\mathfrak{A})+1}^{\mathbf{a}}].$$

- (iv) Show that for any structure $\mathfrak{B}$, $\models_{\mathfrak{B}} \chi_{\mathfrak{A}} \iff \mathfrak{A} \simeq_{\mathfrak{p}} \mathfrak{B}$.

Exercise 7.1.38 Show that the sentence $\chi_{\mathfrak{A}}$ defined in the preceding exercise is a genuine invariant in the sense that

$$\mathfrak{A} \simeq_{\mathfrak{p}} \mathfrak{B} \iff \chi_{\mathfrak{A}} = \chi_{\mathfrak{B}}.$$

Exercise 7.1.39 Show along the following lines that the Ehrenfeucht and Fraïssé Theorems hold for arbitrary finite languages. Recall from Definition 2.6.17 the notion of a formula being unnested and from Proposition 2.6.18 that every formula is logically equivalent to an unnested formula. Refine this proposition to show that for each $n, k \in \omega$ there exists a finite set $\Delta_u^{(n,k)}$ of unnested (n, k) -formulas such that every unnested (n, k) -formula is logically equivalent to a member of $\Delta_u^{(n,k)}$. Now follow the earlier proof with appropriate modifications.

For the following two exercises use the definitions and results of Exercises 2.3.79–2.3.82.

Exercise 7.1.40 Assume that $\mathfrak{A}$ and $\mathfrak{B}$ are minimal structures, $\mathfrak{A} \equiv \mathfrak{B}$, $X \subseteq A$ is $\mathfrak{A}$ -independent, $Y \subseteq B$ is $\mathfrak{B}$ -independent and $\eta : X \rightarrow Y$ is a bijection. Let $\mathfrak{C} \subseteq \mathfrak{A}$ and $\mathfrak{D} \subseteq \mathfrak{B}$ be the substructures with universes $\text{acl}_{\mathfrak{A}} X$ and $\text{acl}_{\mathfrak{B}} Y$, respectively. Show that

- (i) η is a partial elementary embedding;
- (ii) η has an extension to an isomorphism ζ of $\mathfrak{C}$ and $\mathfrak{D}$.

Hint: For (i), prove by induction on k that for all $a_0, \dots, a_k$ and all formulas $\phi(x_0, \dots, x_k)$,

$$\models_{\mathfrak{A}} \phi[a_0, \dots, a_k] \iff \models_{\mathfrak{B}} \phi[\eta(a_0), \dots, \eta(a_k)].$$

For (ii), let ζ be maximal among extensions of η which are partial elementary embeddings with $\text{Dom}(\zeta) \subseteq \text{acl}_{\mathfrak{A}} X$.

Exercise 7.1.41 Prove that if $\mathfrak{A}$ and $\mathfrak{B}$ are minimal structures of the same uncountable cardinality, then

$$\mathfrak{A} \equiv \mathfrak{B} \implies \mathfrak{A} \simeq \mathfrak{B}.$$

7.2 Boolean algebras, ultrafilters and types

In [Sections 1.7](#) and [3.3](#) we presented enough of the theory of Boolean algebras to describe alternative proofs for the Countable Completeness and Compactness Theorems. In particular, the ultrafilters on a Boolean algebra played a central role. In preparation for the remainder of this chapter, we develop these notions further. Recall from Proposition 1.7.15 that for a propositional theory T , the mapping

$$U \mapsto V_U \quad \text{where} \quad V_U(\phi) := \begin{cases} \mathbf{T}, & \text{if } [\phi]_T \in U; \\ \mathbf{F}, & \text{otherwise;} \end{cases}$$

is a bijective correspondence between ultrafilters on $\mathcal{L}_T$, the Lindenbaum algebra of T , and models (truth assignments) of T . An alternative formulation which will be more suitable for the first-order case is

$$U \mapsto \Phi_U := \{ \phi : [\phi]_T \in U \},$$

which is easily seen to be a bijective correspondence between ultrafilters on $\mathcal{L}_T$ and complete propositional theories which extend T . This same definition makes perfect sense for the Lindenbaum algebra of a first-order theory T , but Φ_U is of course not a theory since it contains formulas as well as sentences. This (and much else, as we shall see below) suggests the following definitions. This material is one of the rare instances for which it is useful to refer to the fixed list of variables $v_0, \dots, v_n, \dots$ of a first-order language L .

Definition 7.2.1 For any language L , any L -theory T and any $n \in \omega$,

- (i) Fm_L^n denotes the set of all L -formulas with at most the variables $v_0, \dots, v_{n-1}$ free;
- (ii) for any $\phi \in \text{Fm}_L^n$, $[\phi]_T^n$ denotes $\{ \psi \in \text{Fm}_L^n : T \models \phi \leftrightarrow \psi \}$;
- (iii) $\mathcal{L}_T^n$ is the Boolean algebra with elements $[\phi]_T^n$ for $\phi \in \text{Fm}_L^n$ with the same operations as for $\mathcal{L}_T$.

We leave to the reader the easy verification that $\mathcal{L}_T^n$ is a Boolean algebra. Note that although each $\mathcal{L}_T^n$ is easily isomorphic to a subalgebra of $\mathcal{L}_T$, it is not literally a subalgebra, since $[\phi]_T^n$ is not the same as $[\phi]_T$.

For the following discussion we fix a language L and an L -theory T . With any filter F on $\mathcal{L}_T^n$, we associate the set

$$\Phi_F := \{ \phi \in \text{Fm}_L^n : [\phi]_T^n \in F \}.$$

Translating the defining properties of filters gives

Proposition 7.2.2 For any n , any filter F in $\mathcal{L}_T^n$ and any $\phi, \psi \in \text{Fm}_L^n$,

- (i) $\top \in \Phi_F$;
- (ii) if $\phi \in \Phi_F$ and $T \models \phi \rightarrow \psi$, then also $\psi \in \Phi_F$;
- (iii) if both $\phi, \psi \in \Phi_F$, then also $\phi \wedge \psi \in \Phi_F$;
- (iv) if F is an ultrafilter, then $\neg \phi \in \Phi_F$ iff $\phi \notin \Phi_F$. $\square$

Extending these properties, we have

Proposition 7.2.3 For any n , any filter F in $\mathcal{L}_T^n$ and any $\psi \in \mathbf{Fm}_L^n$,

- (i) $T \subseteq \Phi_F \subset \mathbf{Fm}_L^n$;
- (ii) if $\Phi_F \models \psi$, then also $\psi \in \Phi_F$;
- (iii) for any set $\Phi \subseteq \mathbf{Fm}_L^n$, if both
 - (a) $T \subseteq \Phi$ and
 - (b) for any $\psi \in \mathbf{Fm}_L^n$, if $\Phi \models \psi$ then also $\psi \in \Phi$,

then for some filter F on $\mathcal{L}_T^n$, $\Phi = \Phi_F$.

Proof. For (i), if $\phi \in T$, then $[\phi]_T^n = \mathbb{I} \in F$ and $[\mathbf{F}]_T^n = \mathbb{O} \notin F$. For (ii), if $\Phi_F \models \psi$, then by compactness, for some $\phi_0, \dots, \phi_{k-1} \in \Phi_F$, $\models \phi_0 \wedge \dots \wedge \phi_{k-1} \rightarrow \psi$. It follows by (ii) and (iii) of the preceding proposition that $\psi \in \Phi_F$. Finally, suppose that Φ satisfies the hypotheses of (iii) and set $F := \{[\phi]_T^n : \phi \in \Phi\}$. It is easy to verify that F is a filter and that $\Phi \subseteq \Phi_F$. Conversely, if $\phi \in \Phi_F$, then $[\phi]_T^n \in F$, so for some $\psi \in \Phi$, $[\phi]_T^n = [\psi]_T^n$. Then $T \models \phi \leftrightarrow \psi$, so since $T \subseteq \Phi$, $\Phi \models \phi$ and thus also $\phi \in \Phi$. $\square$

Now we have the analogue of the propositional results for the case $n = 0$:

Corollary 7.2.4 The mapping $F \mapsto \Phi_F$ is a bijective correspondence between filters on $\mathcal{L}_T^0$ and consistent theories which extend T . Φ_F is a complete theory iff F is an ultrafilter. $\square$

For $n > 0$, we need to introduce some terminology. Recall from the preceding section that for any structure $\mathfrak{A}$ and a sequence of elements $\mathbf{a} \in \mathfrak{A}$,

$$\text{Tp}^{\mathfrak{A}}(\mathbf{a}) = \left\{ \phi \in \mathbf{Fm}_L^n : \mathfrak{A} \models \phi[\mathbf{a}] \right\}.$$

Clearly $\Phi := \text{Tp}^{\mathfrak{A}}(\mathbf{a})$ satisfies the conditions of part (iii) of the preceding proposition.

Definition 7.2.5 For any $n > 0$, any $\Phi \subseteq \mathbf{Fm}_L^n$ and any L -theory T ,

- (i) Φ is an ***n-type of T*** iff for some model $\mathfrak{A}$ of T and some $\mathbf{a} = a_0, \dots, a_{n-1} \in \mathfrak{A}$, $\Phi \subseteq \text{Tp}^{\mathfrak{A}}(\mathbf{a})$;
- (ii) Φ is a ***closed n-type of T*** iff in addition $T \subseteq \Phi$ and for any $\psi \in \mathbf{Fm}_L^n$, if $\Phi \models \psi$ then also $\psi \in \Phi$;
- (iii) Φ is a ***complete n-type of T*** iff in addition for any $\phi \in \mathbf{Fm}_L^n$, $\neg \phi \in \Phi$ iff $\phi \notin \Phi$.

For $\Phi \subseteq \mathbf{Fm}_L^n$, let $\mathbf{c} = c_0, \dots, c_{n-1}$ be new constant symbols, $L(\mathbf{c})$ the extension of L by $\mathbf{c}$ and $\Phi(\mathbf{c}) := \{ \phi(\mathbf{c}) : \phi \in \Phi \}$.

Lemma 7.2.6 For any $n > 0$, any $\Phi \subseteq \text{Fm}_L^n$ and any L -theory T ,

- (i) Φ is an n -type of T iff $T \cup \Phi(\mathbf{c})$ is a consistent set of $L(\mathbf{c})$ -sentences;
- (ii) Φ is a closed n -type of T iff $\Phi(\mathbf{c})$ is a consistent $L(\mathbf{c})$ -theory which extends T ;
- (iii) Φ is a complete n -type of T iff $\Phi(\mathbf{c})$ is a complete $L(\mathbf{c})$ -theory which extends T ;
- (iv) Φ is a complete n -type of T iff $\Phi = \text{Tp}^{\mathfrak{A}}(\mathbf{a})$ for some model $\mathfrak{A}$ of T and $\mathbf{a} \in \mathfrak{A}$.

Proof. These are nearly immediate from the definitions. For example, if $\Phi(\mathbf{c})$ is a consistent $L(\mathbf{c})$ -theory which extends T , then it has a model $(\mathfrak{A}, \mathbf{a})$. Clearly $\mathfrak{A}$ is a model of T and $\Phi \subseteq \text{Tp}^{\mathfrak{A}}(\mathbf{a})$. $\square$

Proposition 7.2.7 For each $n > 0$, the mapping $F \mapsto \Phi_F$ is a bijective correspondence between filters on $\mathcal{L}_T^n$ and closed n -types of T . Φ_F is a complete n -type of T iff F is an ultrafilter.

Proof. It follows from the two propositions above that for any filter F on $\mathcal{L}_T^n$, $\Phi_F(\mathbf{c})$ is a consistent $L(\mathbf{c})$ -theory extending T , so by the lemma, ϕ is an n -type of T . That the mapping is bijective and that ultrafilters correspond to complete types also follows from the propositions. $\square$

We shall be very involved with types in [Sections 7.3–7.5](#) below, but here we consider some other features of Boolean algebras.

Definition 7.2.8 For any Boolean algebra $\mathcal{A} = (A, \oplus, \otimes, \bar{}, \mathbb{O}, \mathbb{I})$,

- (i) an element $a \in A$ is an **atom** iff $a \neq \mathbb{O}$ but there is no $b < a$ such that $b \neq \mathbb{O}$;
- (ii) $\mathcal{A}$ is **atomic** iff for every $b \in A$, if $b \neq \mathbb{O}$, then there exists $a \leq b$ such that a is an atom;
- (iii) $\mathcal{A}$ is **atomless** iff $\mathcal{A}$ has no atoms.

Examples 7.2.9

- (i) In the full power set $\wp(X)$ of a nonempty set, the atoms are the singleton sets and the algebra is atomic.
- (ii) If $[\theta]_T$ is an atom in the Lindenbaum algebra $\mathcal{L}_T$ of a consistent theory, then by (ii) of the following proposition, for all formulas ψ

$$T \models \theta \rightarrow \psi \quad \text{or} \quad T \models \theta \rightarrow \neg \psi.$$

This is a very strong requirement, since generally ψ will have variables free that are not free in θ . This happens only in the case that T is

a theory which has a one-element model and θ is the sentence which asserts that the universe has exactly one element and what is true about it. If θ can be satisfied in models of T with more than one element, then for variables x and y not occurring in θ , neither $T \models \theta \rightarrow x \doteq y$ nor $T \models \theta \rightarrow \neg x \doteq y$ can hold.

- (iii) If T is a complete theory, then $\mathcal{L}_T^0$ is the two element algebra, $\mathbb{I}$ is an atom, and the algebra is atomic. If T is not complete, then $[\theta]_T^0$ is an atom just in case $T \cup \{\theta\}$ generates a complete theory. For example, this is the case for T_{DLO} and θ the sentence $F \wedge \neg L$ which says that there is a first but no last element.
- (iv) In $\mathcal{L}_T^1$, a formula $\theta(v_0)$ is an atom iff in the language with a new constant symbol c , $T \cup \{\theta(c)\}$ generates a complete theory. For example, if T is a complete theory of arithmetic and θ is the formula $v_0 \doteq \dot{5}$.
- (v) The algebra of clopen subsets of the irrational numbers is atomless.

Proposition 7.2.10 *For any Boolean algebra $\mathcal{A}$ and any $a \in \mathcal{A}$, the following are equivalent,*

- (i) *a is an atom;*
- (ii) *for all $b \in \mathcal{A}$, $a \leq b$ or $a \leq \bar{b}$;*
- (iii) *the principal filter U_a generated by a is an ultrafilter.*

Proof. Assume first (i) and that $a \not\leq b$. then $a \otimes \bar{b}$ is not equal to $\mathbb{O}$ but is below the atom a so must equal a . Hence $a \leq \bar{b}$. That (ii) implies (iii) is immediate. Assume (iii) and that $b < a$. Since $b \notin U_a$, $\bar{b} \in U_a$ so $a \leq \bar{b}$. Thus $b \leq \bar{a}$ so $b \leq a \otimes \bar{a} = \mathbb{O}$ and $b = \mathbb{O}$ as desired. $\square$

Corollary 7.2.11 *For any Boolean algebra $\mathcal{A}$ and any ultrafilter U of $\mathcal{A}$, U is principal iff it contains an atom.*

Proof. The implication $(\implies)$ is immediate from the preceding proposition. For $(\impliedby)$, suppose that U is an ultrafilter and contains the atom a . For each $b \in U$ we must have $a \leq b$, since otherwise by (ii) $a \leq \bar{b}$, hence $\bar{b} \in U$, so $\mathbb{O} = b \otimes \bar{b} \in U$, a contradiction. $\square$

Definition 7.2.12 For any Boolean algebra $\mathcal{A}$, $S(\mathcal{A})$ denotes the set of ultrafilters of $\mathcal{A}$.

In the Stone Representation Theorem 1.7.14, we showed that any Boolean algebra $\mathcal{A}$ is isomorphic to a field of subsets of $S(\mathcal{A})$. These subsets are actually the clopen sets of a natural topology on $S(\mathcal{A})$, which we now define.

Definition 7.2.13 For any Boolean algebra $\mathcal{A}$, the **basic open sets** of $S(\mathcal{A})$ are the sets $a^* := \{U \in S(\mathcal{A}) : a \in U\}$. As usual, the **open sets** are arbitrary unions of basic open sets.

Theorem 7.2.14 *With this topology, $S(\mathcal{A})$ is a totally disconnected compact Hausdorff space such that the set of all a^* for $a \in \mathcal{A}$ is exactly the set of all clopen sets.*

Proof. Each basis set a^* is clopen, since $a^* = S(\mathcal{A}) \sim (\bar{a})^*$ and thus is also closed. Hence the space is totally disconnected. To see that it is Hausdorff, let U and V be any distinct elements of $S(\mathcal{A})$. If there is some $a \in U \sim V$, then $U \in a^*$, but $V \notin a^*$, so $V \in (\bar{a})^*$ and these neighborhoods separate U and V as required.

To see that the space is compact, let $\mathcal{C}$ be a class of closed sets such that $\bigcap \mathcal{C} = \emptyset$; we aim to show that there is a finite subclass of $\mathcal{C}$ which has empty intersection. Let

$$B := \{b \in A : \text{for some } X \in \mathcal{C}, \quad X \subseteq b^*\}.$$

We claim first that B does not have the finite intersection property (FIP; Definition 1.7.11); suppose towards a contradiction that it does. Then there exists an ultrafilter $U \supseteq B$. By hypothesis, $U \notin \bigcap \mathcal{C}$, so there is some $X \in \mathcal{C}$ such that $U \not\subseteq X$. Since X is closed, there exists a basic open set b^* disjoint from X such that $U \in b^*$. Then $X \subseteq \bar{b}^*$, so $\bar{b} \in B$, but clearly $U \not\subseteq \bar{b}^*$, so $\bar{b} \notin U$, a contradiction.

Now we may choose $b_0, \dots, b_n \in B$ with $b_0 \otimes \dots \otimes b_n = \mathbb{O}$ and for each $i \leq n$, $X_i \in \mathcal{C}$ with $X_i \subseteq b_i^*$. Now $X_0 \cap \dots \cap X_n = \emptyset$, since if any ultrafilter U belonged to this set, then for each $i \leq n$, $U \in b_i^*$ so $b_i \in U$, from which it follows that $\mathbb{O} \in U$, a contradiction.

Finally, if X is any clopen set, it is a union of some collection of basis sets b^* . Because X is closed and the space is compact, X is compact and is thus already covered by finitely many of these b^* . Thus X is of the form $b_0^* \cup \dots \cup b_k^* = (b_0 \oplus \dots \oplus b_k)^*$. $\square$

Corollary 7.2.15 *For any Boolean algebra $\mathcal{A}$, $\mathcal{A}$ is isomorphic to the algebra $\text{CLO}(S(\mathcal{A}))$ of clopen subsets of $S(\mathcal{A})$.*

Proof. Immediate from the preceding theorem and the proof of the Stone Representation Theorem. $\square$

Remark 7.2.16 In Exercises 3.3.40 and 3.3.42 we sketched the proof that for any topological space, the regular open sets form a complete Boolean algebra. It is immediate that any clopen set is regular open, so in particular $\text{CLO}(S(\mathcal{A})) \subseteq \text{RO}(S(\mathcal{A}))$ and thus $\mathcal{A} \hookrightarrow_0 \text{RO}(S(\mathcal{A}))$. In an exercise below we invite the reader to prove that $\text{RO}(S(\mathcal{A}))$ is the smallest complete Boolean algebra which includes (an isomorphic copy of) $\mathcal{A}$ — that is, for any complete Boolean algebra $\mathcal{B}$, if $\text{CLO}(S(\mathcal{A})) \subseteq \mathcal{B}$, then also $\text{RO}(S(\mathcal{A})) \subseteq \mathcal{B}$. This makes sense to call $\text{RO}(S(\mathcal{A}))$ the **completion** of $\mathcal{A}$.

Finally, we consider some facts about the cardinality of $S(\mathcal{A})$. First we have

Proposition 7.2.17 *For any Boolean algebra $\mathcal{A}$, the following are equivalent,*

- (i) $\mathcal{A}$ is finite;
- (ii) $S(\mathcal{A})$ is finite;
- (iii) every member of $S(\mathcal{A})$ is a principal ultrafilter.

Proof. It is immediate that (ii) follows from (i). Conversely, suppose that $S(\mathcal{A})$ has k elements $U_0, \dots, U_{k-1}$; we show that $\mathcal{A}$ can have at most 2^k elements. Otherwise, there must exist distinct a and b such that for all $i < k$, $a \in U_i$ iff $b \in U_i$. But since $a \neq b$, one of $\bar{a} \otimes b \neq \mathbb{O}$ or $a \otimes \bar{b} \neq \mathbb{O}$; these are parallel, so suppose the former. The set $\{\bar{a}, b\}$ has the FIP so is included in some ultrafilter U . But then $a \notin U$ while $b \in U$ so U cannot be one of $U_0, \dots, U_{k-1}$.

That (i) implies (iii) is easy: if $\mathcal{A}$ is finite, then any ultrafilter is also finite and is clearly generated by the inf of its elements. For the converse, suppose that $\mathcal{A}$ is infinite; we show that $\mathcal{A}$ has a non-principal ultrafilter. If $\mathcal{A}$ has no atoms, then by Proposition 7.2.10, $\mathcal{A}$ has no principal ultrafilters. Otherwise, let $C := \{\bar{a} : a \text{ is an atom of } \mathcal{A}\}$. Any ultrafilter $U \supseteq C$ contains no atom so by Corollary 7.2.11 cannot be principal; hence it suffices to show that C has the FIP. If it does not, there exist $\bar{a}_0, \dots, \bar{a}_n \in C$ such that $\bar{a}_0 \otimes \dots \otimes \bar{a}_n = \mathbb{O}$. It follows that $a_0 \oplus \dots \oplus a_n = \mathbb{I}$, so for any $b \in \mathcal{A}$,

$$b = (b \otimes a_0) \oplus \dots \oplus (b \otimes a_n).$$

However, since each a_i is an atom and $\mathbb{O} \leq (b \otimes a_i) \leq a_i$, this product is either $\mathbb{O}$ or a_i . Hence there can be at most 2^{n+1} such b , contrary to the hypothesis that $\mathcal{A}$ is infinite. $\square$

Now we consider conditions under which $S(\mathcal{A})$ is infinite. If $\mathcal{A}$ is infinite, say of cardinality κ , then since $S(\mathcal{A})$ is a subset of $\wp(\mathcal{A})$, the cardinality of $S(\mathcal{A})$ lies between $\aleph_0$ and 2^κ . We shall give two conditions under which the cardinality is at least $2^{\aleph_0}$, which for countable $\mathcal{A}$ is the largest possible.

Definition 7.2.18 For any Boolean algebra $\mathcal{A}$ and any set $X \subseteq \mathcal{A}$,

- (i) an element $a \in \mathcal{A}$ **splits** in X iff there exist $d, e \in X$ such that $\mathbb{O} < d, e < a$ and d and e are **disjoint** — that is, $d \otimes e = \mathbb{O}$;
- (ii) X is **perfect** iff $X \neq \emptyset$ and every element $a \in X$ splits in X ;
- (iii) X is **interval closed** iff for any $a \leq b \leq c$ in $\mathcal{A}$, if both $a, c \in X$, then also $b \in X$.

Lemma 7.2.19 *For any Boolean algebra $\mathcal{A}$ and any set $X \subseteq A$, if X is interval closed and $\mathbb{0} \notin X$, then for each $a \in X$,*

$$a \text{ splits in } X \iff \exists b \in A [a \otimes b \in X \text{ and } a \otimes \bar{b} \in X].$$

Proof. Suppose first that a splits in X and fix disjoint $d, e \in X$ such that $\mathbb{0} < d, e < a$. Then $a \otimes d = d \in X$ and $e \leq a \otimes \bar{d} \leq a$, so by interval closure also $a \otimes \bar{d} \in X$. For the converse implication, suppose that b satisfies the right-hand side and let $d := a \otimes b$ and $e := a \otimes \bar{b}$. Clearly d and e are disjoint and $\mathbb{0} \leq d, e \leq a$; we show both inequalities are proper. If $d = a$, then $a \leq b$, so $\mathbb{0} = a \otimes \bar{b} \in X$, contrary to hypothesis. If $d = \mathbb{0}$, then $a \leq \bar{b}$ and $\mathbb{0} = a \otimes b \in X$, again contrary to hypothesis. The arguments for e are entirely parallel. $\square$

Note that the definition of atom may be rephrased as follows: a is not an atom iff if $a > \mathbb{0}$ then the *interval* $(\mathbb{0}, a) := \{b : \mathbb{0} < b < a\}$ is nonempty. In fact this implies a stronger condition.

Lemma 7.2.20 *For any Boolean algebra $\mathcal{A}$ and any $a \in A$, $a > \mathbb{0}$, a is not an atom iff a splits in $(\mathbb{0}, a)$.*

Proof. The implication $(\Leftarrow)$ is immediate from the definition. Suppose that a is not an atom and fix b such that $\mathbb{0} < b < a$; we use the criterion of the preceding lemma. Clearly $a \otimes b = b \in (\mathbb{0}, a)$ and $\mathbb{0} \leq a \otimes \bar{b} \leq a$. If $a \otimes \bar{b} = \mathbb{0}$, then $a \leq b$, contrary to assumption, and if $a \otimes \bar{b} = a$, then $a \leq \bar{b}$, so $b \leq \bar{b}$ and $b = \mathbb{0}$, again contrary to assumption. Hence also $a \otimes \bar{b} \in (\mathbb{0}, a)$, so a splits in $(\mathbb{0}, a)$. $\square$

Proposition 7.2.21 *For any Boolean algebra $\mathcal{A}$, if $\mathcal{A}$ is not atomic, then there is a perfect set $X \subseteq A$.*

Proof. Under the hypothesis, let

$$X := \{a \in A : \mathbb{0} < a \text{ and there is no atom } b \text{ of } \mathcal{A} \text{ with } b \leq a\}.$$

Since $\mathcal{A}$ is not atomic, X is nonempty. For any $a \in X$, a is not an atom so by the preceding lemma, a splits in $(\mathbb{0}, a)$. But clearly $(\mathbb{0}, a) \subseteq X$, so a splits also in X . $\square$

The reader should compare the next result with Lemma 6.11.28.

Theorem 7.2.22 *For any Boolean algebra $\mathcal{A}$, if there exists a perfect set $X \subseteq A$, then $\text{Card}(S(\mathcal{A})) \geq 2^{\aleph_0}$; in fact, there exists a countable subalgebra $\mathcal{B} \subseteq \mathcal{A}$ such that $\text{Card}(S(\mathcal{B})) = 2^{\aleph_0}$.*

Proof. Under the given hypotheses, we construct first a family of elements of X indexed by finite sequences s of 0s and 1s. Let $a_\emptyset$ be any element of

a perfect set X . Given $a_s \in X$, from the definition of perfect there exist disjoint $d, e \in X$ such that $\mathbb{0} < d, e < a$; take $a_{s(0)} := d$ and $a_{s(1)} := e$.

Now, for any function $f \in {}^\omega 2$, it follows that $\{a_{f \upharpoonright n} : n \in \omega\}$ has the FIP so is contained in some ultrafilter U_f . If $f \neq g$, say $f \upharpoonright n = g \upharpoonright n$ but $f(n) \neq g(n)$, then $a_{f \upharpoonright n+1} \otimes a_{g \upharpoonright n+1} = \mathbb{0}$, and it follows that $U_f \neq U_g$. Hence the mapping $f \mapsto U_f$ is an injective mapping ${}^\omega 2 \rightarrow S(\mathcal{A})$ and the cardinality of $S(\mathcal{A})$ is at least that of ${}^\omega 2$ — that is, $2^{\aleph_0}$. The only relevant elements of $\mathcal{A}$ here are the countably many a_s , so if $\mathcal{B}$ is the subalgebra of $\mathcal{A}$ generated by these, we have also $\text{Card}(S(\mathcal{B})) \geq 2^{\aleph_0}$, but since $\mathcal{B}$ is countable, also $\text{Card}(S(\mathcal{B})) \leq 2^{\aleph_0}$, so these cardinals are equal. $\square$

Corollary 7.2.23 *For any Boolean algebra $\mathcal{A}$, if $\mathcal{A}$ is not atomic, then $\text{Card}(S(\mathcal{A})) \geq 2^{\aleph_0}$; in fact there exists a countable subalgebra $\mathcal{B} \subseteq \mathcal{A}$ such that $\text{Card}(S(\mathcal{B})) = 2^{\aleph_0}$. $\square$*

Finally, we get the same conclusion under quite a different hypothesis (cf. Proposition 6.11.30).

Cantor-Bendixson Theorem 7.2.24 *For any infinite Boolean algebra $\mathcal{A}$, if $\text{Card}(S(\mathcal{A})) > \text{Card}(\mathcal{A})$, then $\text{Card}(S(\mathcal{A})) \geq 2^{\aleph_0}$; in fact, there exists a countable subalgebra $\mathcal{B} \subseteq \mathcal{A}$ such that $\text{Card}(S(\mathcal{B})) = 2^{\aleph_0}$.*

Proof. Under the given hypothesis, let $\kappa = \text{Card}(\mathcal{A})$ and set

$$X := \{a \in \mathcal{A} : a \text{ belongs to more than } \kappa\text{-many ultrafilters}\}.$$

By the theorem it suffices to show that X is perfect. X is nonempty, since by the hypothesis there are more than κ -many ultrafilters and each one contains $\mathbb{1}$, so $\mathbb{1} \in X$. Suppose towards a contradiction that there is an $a \in X$ which does not split in X . Clearly X satisfies the hypothesis of Lemma 7.2.19, so for each $b \in \mathcal{A}$, one of $a \otimes b$ and $a \otimes \bar{b}$ does not belong to X and hence is a member of at most κ -many ultrafilters. Since a is the Boolean sum of $a \otimes b$ and $a \otimes \bar{b}$, we have for any ultrafilter U ,

$$a \in U \implies [a \otimes b \in U \text{ or } a \otimes \bar{b} \in U].$$

Thus (since κ is infinite) the other one of the pair must belong to more than κ -many ultrafilters. Hence if we set $V := \{b \in \mathcal{A} : a \otimes b \in X\}$, we have for each $b \in \mathcal{A}$ that $b \in V$ iff $\bar{b} \notin V$. But it is easy to check that V satisfies the other conditions of the definition and is itself an ultrafilter.

We conclude the proof by showing that contrary to hypothesis, a belongs to at most κ many ultrafilters U . For each $b \in V$, let

$$\mathcal{U}_b := \{U : U \text{ is an ultrafilter and } a \otimes \bar{b} \in U\}.$$

Note that for all $b \in V$, $\text{Card}(\mathcal{U}_b) \leq \kappa$. For each ultrafilter U such that $a \in U$, either

$$[\text{for all } b \in V, \quad a \otimes b \in U] \quad \text{or} \quad [\text{for some } b \in V, \quad a \otimes \bar{b} \in U].$$

In the first case, $V \subseteq U$ from which it easily follows that $U = V$. In the second case, for some $b \in V$, $U \in \mathcal{U}_b$. Hence all of the ultrafilters of which a is a member belong to the set

$$\{V\} \cup \bigcup \{\mathcal{U}_b : b \in V\}.$$

This is a union of κ many sets each of cardinality at most κ , so is itself of cardinality at most κ . This contradiction finally verifies that X is perfect and finishes the proof. $\square$

Remark 7.2.25 In the exercises we suggest another proof of this result which resembles the original one more closely.

Since we will be applying many of these results to the Lindenbaum algebras $\mathcal{L}_T^n$, it will be useful to state explicitly their interpretation in this case. As usual we write $\mathbf{v}$ for $v_0, \dots, v_{n-1}$, with n determined by context.

Definition 7.2.26 For any theory T and any formula $\phi(\mathbf{v})$, ϕ is **T -consistent** iff $T \not\models \neg \phi(\mathbf{v})$, or equivalently $T \not\models \forall \mathbf{v} \neg \phi(\mathbf{v})$.

Hence we have also

$$\phi(\mathbf{v}) \text{ is } T\text{-consistent} \iff [\phi]_T^n \neq \mathbf{0} \iff T \cup \{\exists \mathbf{v} \phi(\mathbf{v})\} \text{ is consistent,}$$

and if T is complete, then $\phi(\mathbf{v})$ is T -consistent $\iff T \models \exists \mathbf{v} \phi(\mathbf{v})$. Now, extending Proposition 7.2.7,

Proposition 7.2.27 For any theory T , $n > 0$ and $\Phi \subseteq \text{Fm}_L^n$, the following are equivalent:

- (i) Φ is an n -type of T ;
- (ii) for every finite $\Gamma \subseteq \Phi$, $\bigwedge \Gamma$ is T -consistent;
- (iii) $\{[\phi]_T^n : \phi \in \Phi\}$ has the finite intersection property in $\mathcal{L}_T^n$.

Proof. The implications (i) $\implies$ (ii) $\iff$ (iii) follow easily from the preceding remarks. Suppose (ii) holds; by Lemma 7.2.6, to deduce (i) it suffices to show that $T \cup \Phi(\mathbf{c})$ is consistent. But (ii) implies that for any finite $\Gamma \subseteq \Phi$, $T \cup \Gamma(\mathbf{c})$ is consistent, so the result follows by compactness. $\square$

An atom of $\mathcal{L}_T^n$ is an element $[\phi]_T^n$ such that ϕ is T -consistent but there is no T -consistent formula $\psi \in \text{Fm}_L^n$ such that $T \models \psi \rightarrow \phi$ but $T \not\models \phi \rightarrow \psi$. We also (imprecisely) call ϕ itself an **atom** of $\mathcal{L}_T^n$.

Now Propositions 7.2.7 and 7.2.10 give immediately

Proposition 7.2.28 *For any language L , any L -theory T , any $n > 0$ and any $\theta(\mathbf{v}) \in \mathbf{Fm}_L^n$, the following are equivalent:*

- (i) $\theta(\mathbf{v})$ is an atom of $\mathcal{L}_T^n$;
- (ii) for all $\phi(\mathbf{v}) \in \mathbf{Fm}_L^n$, $T \models \theta(\mathbf{v}) \rightarrow \phi(\mathbf{v})$ or $T \models \theta(\mathbf{v}) \rightarrow \neg \phi(\mathbf{v})$;
- (iii) $\{\phi(\mathbf{v}) : T \models \theta(\mathbf{v}) \rightarrow \phi(\mathbf{v})\}$ is a complete n -type of T . $\square$

A formula $\theta(\mathbf{v})$ satisfying these conditions is also called a **complete formula** (relative to T). Generalizing the terminology again we have

Definition 7.2.29 For any theory T , any $n > 0$ and any n -type Φ of T , Φ is **principal** iff there exists a T -consistent formula $\theta(\mathbf{v})$ such that for all $\phi \in \Phi$, $T \models \theta \rightarrow \phi$. θ is called a **generator** of Φ .

Note that in general the generator θ of a principal type Φ need not be a member of Φ , but if Φ is complete, then $\theta \in \Phi$, since otherwise $\neg\theta \in \Phi$, hence $T \models \theta \rightarrow \neg\theta$ and $T \models \neg\theta$ contrary to the assumption that θ is T -consistent. It is easy to check that under the correspondence of Proposition 7.2.7, the principal complete n -types correspond exactly to the principal ultrafilters of $\mathcal{L}_T^n$. Two useful characterizations are

Corollary 7.2.30 *For any theory T and any type Φ of T ,*

- (i) Φ is non-principal iff the $\mathcal{L}_T^n$ -inf: $\prod \{[\phi]_T^n : \phi \in \Phi\} = \mathbb{O}$ iff the $\mathcal{L}_T$ -inf: $\prod \{[\phi]_T : \phi \in \Phi\} = \mathbb{O}$;
- (ii) if T is complete, then Φ is principal iff Φ contains a complete formula.

Proof. Part (ii) is immediate by Corollary 7.2.11. For (i), note that $\prod \{[\phi]_T^n : \phi \in \Phi\} = \mathbb{O}$ says just that for any formula $\psi(\mathbf{v}) \in \mathbf{Fm}_L^n$,

$$\text{if for all } \phi \in \Phi, [\psi]_T^n \leq [\phi]_T^n, \text{ then } [\psi]_T^n = \mathbb{O},$$

which is a translation of the definition of non-principality. If this condition holds, then for any formula $\theta(\mathbf{v}, \mathbf{x})$, suppose that for all $\phi \in \Phi$, $[\theta]_T \leq [\phi]_T$ — that is, $T \models \theta \rightarrow \phi$. Then since the variables $\mathbf{x}$ are not free in ϕ , also $T \models \exists \mathbf{x} \theta \rightarrow \phi$, so $[\exists \mathbf{x} \theta]_T^n = \mathbb{O}$. Hence $T \models \neg \exists \mathbf{x} \theta$, so also $T \models \neg \exists \mathbf{v} \exists \mathbf{x} \theta$ and thus $[\theta]_T = \mathbb{O}$. $\square$

With this we can explain the terminology of Definition 7.1.12.

Corollary 7.2.31 *For any structure $\mathfrak{A}$, $\mathfrak{A}$ is atomic iff for all $n > 0$ and all n -tuples $\mathbf{a} \in A$, there exists a complete formula θ (an atom of $\mathcal{L}_{\text{Th}(\mathfrak{A})}^n$) such that $\models_{\mathfrak{A}} \theta[\mathbf{a}]$. $\square$*

Next we introduce the notation

Definition 7.2.32 For any theory T , $S^n(T)$ denotes the set of complete n -types of T .

$S^n(T)$ is the set of all types of the form $\text{Tp}^{\mathfrak{A}}(\mathbf{a})$ for $\mathfrak{A}$ a model of T and $\mathbf{a}$ an n -tuple of elements of $\mathfrak{A}$. By Proposition 7.2.7, $S^n(T)$ is in bijective correspondence with $S(\mathcal{L}_T^n)$, the set of ultrafilters on the Lindenbaum algebra of formulas with n free variables under T -equivalence.

Translating Proposition 7.2.17 we have

Corollary 7.2.33 For any theory T and each $n > 0$, the following are equivalent,

- (i) $\mathcal{L}_T^n$ is finite — that is, there exists a finite list $\psi_0, \dots, \psi_k \in \text{Fm}_L^n$ such that for every $\phi \in \text{Fm}_L^n$, $T \models \phi \leftrightarrow \psi_i$ for some $i \leq k$;
- (ii) T has only finitely many complete n -types — that is, $S^n(T)$ is finite;
- (iii) every n -type of T is principal. $\square$

Lemmas 7.2.19 and 7.2.20 give

Corollary 7.2.34 For any theory T , any $n > 0$ and any $\phi \in \text{Fm}_L^n$, if ϕ is T -consistent, then ϕ is not an atom of $\mathcal{L}_T^n$ iff there exists $\psi \in \text{Fm}_L^n$ such that both $\phi \wedge \psi$ and $\phi \wedge \neg\psi$ are T -consistent. $\square$

Finally we have the translations of the results 7.2.21–7.2.24 concerning large algebras.

Theorem 7.2.35 For any theory T and any $n > 0$, if $\mathcal{L}_T^n$ contains a perfect set — in particular if $\mathcal{L}_T^n$ is not atomic — then $\text{Card}(S^n(T)) \geq 2^{\aleph_0}$; in fact, there exists a countable subtheory T^* of T such that $\text{Card}(S^n(T^*)) = 2^{\aleph_0}$.

Proof. The first clause is immediate from Theorem 7.2.22, and we have that there is a countable subalgebra $\mathcal{B} \subseteq \mathcal{L}_T^n$ such that $\text{Card}(S(\mathcal{B})) = 2^{\aleph_0}$. For each $b \in \mathcal{B}$, choose $\phi_b \in b$ and let L^* be the smallest sublanguage of L such that all ϕ_b are L^* -formulas; clearly L^* is a countable language. Let $T^* = T \cap \text{Fm}_{L^*}$; T^* is an L^* -theory. The mapping $b \mapsto [\phi_b]_{T^*}^n$ is clearly an injection of $\mathcal{B}$ into $\mathcal{L}_{T^*}^n$. For each ultrafilter U of $\mathcal{B}$, the image of U under this mapping is a subset of $\mathcal{L}_{T^*}^n$ with the finite intersection property which is therefore included in some ultrafilter U^* of $\mathcal{L}_{T^*}^n$. It is easy to see that the mapping $U \mapsto U^*$ is an injection $S(\mathcal{B}) \rightarrow S(\mathcal{L}_{T^*}^n)$, so also $\text{Card}(S^n(T^*)) = \text{Card}(S(\mathcal{L}_{T^*}^n)) = 2^{\aleph_0}$. $\square$

Theorem 7.2.36 For any theory T and any $n > 0$, if $\text{Card}(S^n(T)) > \text{Card}(\mathcal{L}_T^n) \geq \aleph_0$, then $\text{Card}(S^n(T)) \geq 2^{\aleph_0}$; in fact, there exists a countable subtheory T^* of T such that $\text{Card}(S^n(T^*)) = 2^{\aleph_0}$.

Proof. This is immediate from the Cantor-Bendixon Theorem 7.2.24 and the method of the preceding proof. $\square$

Exercises

Exercise 7.2.37 Show that every finite Boolean algebra is atomic.

Exercise 7.2.38 Show that for any atomic Boolean algebra $\mathcal{A}$, if $b^* := \{a \leq b : a \text{ is an atom}\}$, then the mapping $b \mapsto b^*$ is an isomorphism of $\mathcal{A}$ with a field of sets.

Exercise 7.2.39 Fill in the details of the fact mentioned in Remark 7.2.16 — that is, show that for any Boolean algebra $\mathcal{A}$ and any complete Boolean algebra $\mathcal{B}$, if $\text{CLO}(S(\mathcal{A})) \subseteq \mathcal{B}$, then also $\text{RO}(S(\mathcal{A})) \subseteq \mathcal{B}$.

Exercise 7.2.40 Show that for any L -structure $\mathfrak{A}$ and any n -type Φ of $\text{Th}(\mathfrak{A})$, Φ is realized in some elementary extension of $\mathfrak{A}$.

Exercise 7.2.41 Give another proof of the Cantor-Bendixon Theorem along the following lines. Define a mapping $X \mapsto X'$ on subsets of a Boolean algebra $\mathcal{A}$ by

$$X' := \{a \in X : a \text{ splits in } X\},$$

and a transfinite sequence

$$X_0 = A; \quad X_{(\sigma)} = \bigcap_{\tau < \sigma} X_\tau; \quad X_\sigma = (X_{(\sigma)})'.$$

Prove that

- (i) for some ordinal $\bar{\sigma} < \text{Card}(\mathcal{A})^+$, $X_{\bar{\sigma}} = X_{(\bar{\sigma})}$;
- (ii) $X_{\bar{\sigma}}$ is a perfect set (the key point is that $X_{\bar{\sigma}} \neq \emptyset$).

Exercise 7.2.42 Prove Proposition 7.2.27.

Exercise 7.2.43 Show that for any ultrafilter U in a Boolean algebra $\mathcal{A}$, U is isolated in $S(\mathcal{A})$ iff U is a principal ultrafilter. Because of this, a principal type is sometimes called an *isolated type*.

Exercise 7.2.44 Let T be the theory of Exercise 2.4.52. Show that $\{\neg x \doteq c_n : n \in \omega\}$ is a non-principal type of T .

7.3 Countable models of countable theories

In this and the next two sections we shall study the models of a countable first-order theory T . We shall be interested in various special sorts of models as well as the number of (isomorphically) distinct models of T . In this section we concentrate on countable models.

Definition 7.3.1 For any theory T and any infinite cardinal κ , $\mu(T, \kappa)$ is the cardinality of the set of isomorphism classes of models of T of cardinality κ .

One of the motivations for this line of inquiry is to understand what can and cannot be expressed in a first-order language. To let the language say as much as it can, we generally want the theory to be complete. This still leaves a great deal of variation. Recall that T is called κ -categorical iff any two models of cardinality κ are isomorphic — that is, $\mu(T, \kappa) = 1$. Cantor's Theorem 2.4.45 shows that $T_{(\text{DLO})}$ is $\aleph_0$ -categorical, while the existence of non-standard models for the various theories of arithmetic entails that these theories are not $\aleph_0$ -categorical. Clearly $\mu(T_{\text{DLO}}, \aleph_0) = 4$.

By a simple counting argument we have

Proposition 7.3.2 For any countable consistent theory T and any infinite cardinal κ ,

$$1 \leq \mu(T, \kappa) \leq 2^\kappa.$$

Proof. Any structure of cardinality κ is isomorphic to one whose universe is κ . For each function or relation symbol, there are at most 2^κ ways to interpret it as a function or relation on ω , so altogether there are at most $\aleph_0 \times 2^\kappa = 2^\kappa$ many L -structures with κ as universe. $\square$

If T is a complete theory, then $\mathcal{L}_T^0$ is the 2-element algebra and $\{\mathbb{I}\}$ is the only ultrafilter. However, $\mathcal{L}_T^n$ and $S^n(T)$ are generally more complex for $n > 0$. Note that a complete n -type of T is an infinite set of formulas and has uncountably many subsets, each of which is a type. Thus the interesting measure of complexity is the number of *complete* n -types of T — that is, the size of $S^n(T)$. In general, we have

Proposition 7.3.3 For any countable complete theory T and $n > 0$,

$$1 \leq \text{Card}(S^n(T)) \leq 2^{\aleph_0}.$$

Proof. Since T is consistent, it has a model and hence at least one complete type. Since the language is countable, there are only $2^{\aleph_0}$ sets of formulas, so at most this many complete types. $\square$

Definition 7.3.4 For any model $\mathfrak{A}$ of T and any n -type Φ of T , Φ is *realized* in $\mathfrak{A}$ iff for some $\mathbf{a} = a_0, \dots, a_{n-1} \in \mathfrak{A}$, $\Phi \subseteq \text{Tp}^{\mathfrak{A}}(\mathbf{a})$; otherwise Φ is *omitted* in $\mathfrak{A}$. We also say that $\mathbf{a}$ *realizes* Φ in $\mathfrak{A}$.

By definition every type of T is realized in some model of T ; more specifically,

Proposition 7.3.5 *For any countable theory T , any $n > 0$ and any $\Phi \subseteq \text{Fm}_L^n$, Φ is an n -type of T iff Φ is realized in some countable model of T .*

Proof. Immediate from the Downward Löwenheim-Skolem Theorem. $\square$

Examples 7.3.6

- (i) Let T be $T_{(\text{DLO})}$, the theory of dense linear orderings with neither first nor last element. From the proof sketched following Corollary 2.4.47, it follows that there is only one complete 1-type, since any formula satisfied in a model by one element is satisfied in another model by any other element. Another way to see this is to use Cantor's Theorem from that section to show that for any two countable models $\mathfrak{A}$ and $\mathfrak{B}$, and any $a \in A$ and $b \in B$, there is an isomorphism $\eta : \mathfrak{A} \simeq \mathfrak{B}$ such that $\eta(a) = b$.
- (ii) There are three complete 2-types of $T_{(\text{DLO})}$, one containing each of the formulas $v_0 < v_1$, $v_0 \doteq v_1$ and $v_1 < v_0$. In general for each n there are finitely many complete n -types of $T_{(\text{DLO})}$ — that is, $\text{Card}(S^n(T_{(\text{DLO})}))$ is finite. This is in accord with Ryll-Nardzewski's Theorem below.
- (iii) Let T be $\text{Th}(\omega, \text{Sc}, 0)$; there are at least countably many complete 1-types Φ_n containing, respectively the formula $v_0 \doteq n$ for each $n \in \omega$, so $\text{Card}(S^1(T)) \geq \aleph_0$.
- (iv) Let T be $\text{Th}(\Omega)$, the complete theory of the standard model of arithmetic. Let P denote the set of primes. For any set $X \subseteq P$, let

$$\Phi_X := \{ \exists y(p \dot{\times} y \doteq v_0) : p \in X \} \cup \{ \neg \exists y(p \dot{\times} y \doteq v_0) : p \in P \setminus X \}.$$

By Proposition 7.2.27, each Φ_X is a 1-type of T . Clearly if $X \neq Y$, then Φ_X and Φ_Y are incompatible and must have only distinct extensions to complete types. Hence $\text{Card}(S^1(\text{Th}(\Omega))) = 2^{\aleph_0}$. Of course, only Φ_X for finite X are realized in Ω ; the others must be realized in non-standard models of $\text{Th}(\Omega)$.

One of the themes that will play a major role in the rest of this section and the following ones is the relationship between the number of complete types of a theory and the number of models of various cardinalities. A first simple relationship is

Proposition 7.3.7 *For any countable consistent theory T and all $n > 0$, $\text{Card}(S^n(T)) \leq \aleph_0 \times \mu(T, \aleph_0)$. Hence*

- (i) *if $\mu(T, \aleph_0) \leq \aleph_0$, then for all $n > 0$, $\text{Card}(S^n(T)) \leq \aleph_0$;*
- (ii) *if for any n , $\text{Card}(S^n(T))$ has its maximum value $2^{\aleph_0}$, then so does $\mu(T, \aleph_0)$.*

Proof. Since two distinct n types cannot be realized by the same n -tuple of elements of a structure, each countable model realizes at most countably many n types. Since by Proposition 7.3.5 every complete type is realized in some countable model, altogether there are at most $\aleph_0 \times \mu(T, \aleph_0)$ of them. Note that this number may include up to countably many complete types realized in finite models of T . Parts (i) and (ii) are now immediate. $\square$

The theory $T_{\text{Sc},0} := \text{Th}(\omega, \text{Sc}, 0)$ was analyzed in Section 2.5. It follows easily from this analysis that $\mu(T_{\text{Sc},0}, \aleph_0) = \aleph_0$ and hence from the preceding proposition that for each $n > 0$, $\text{Card}(S^n(T_{\text{Sc},0})) \leq \aleph_0$ so by Example 7.3.6(iii), $\text{Card}(S^n(T_{\text{Sc},0})) = \aleph_0$. Similarly it follows from the example (iv) above that $\mu(\text{Th}(\Omega), \aleph_0) = 2^{\aleph_0}$.

Proposition 7.3.8 *For any countable consistent theory T and all $n > 0$, if $\text{Card}(S^n(T)) < 2^{\aleph_0}$, then $\text{Card}(S^n(T)) \leq \aleph_0$.*

Proof. This is an immediate consequence of Theorem 7.2.36. $\square$

Of course, if we assume the Continuum Hypothesis, there are no cardinal numbers κ such that $\aleph_0 < \kappa < 2^{\aleph_0}$ and this proposition is trivial. Furthermore, the same conclusion would hold for $\mu(T, \aleph_0)$. However, without this assumption it is a long-standing open problem.

Vaught's Conjecture 7.3.9 *For any countable consistent theory T , if $\mu(T, \aleph_0) < 2^{\aleph_0}$, then $\mu(T, \aleph_0) \leq \aleph_0$.*

This conjecture was first announced in 1963 and has been a focal point for a major part of the research in Model Theory since then. Note that Vaught's Conjecture would follow from the converse of Proposition 7.3.7(i); however there exist theories T such that all $S^n(T)$ are countable but $\mu(T, \aleph_0) = 2^{\aleph_0}$ (see exercises). There have been many partial results; some will be mentioned in the sections below when the necessary notions have been introduced. One that can be stated here is

Theorem 7.3.10 (Morley) *For any countable consistent theory T , if $\mu(T, \aleph_0) < 2^{\aleph_0}$, then $\mu(T, \aleph_0) \leq \aleph_1$.*

Proof. Left as an exercise (with hints!). $\square$

Proposition 7.3.11 *For any complete theory T and any principal type Φ of T , Φ is realized in every model of T .*

Proof. Suppose that T is complete and Φ is a principal type of T with generator θ . Then by the discussion following Definition 7.2.26, $T \models \exists \mathbf{x}\theta$, so for any model $\mathfrak{A}$ of T , there exists $\mathbf{a} \in \mathfrak{A}$ such that $\models_{\mathfrak{A}} \theta[\mathbf{a}]$. But then for all $\phi \in \Phi$, since $T \models \theta \rightarrow \phi$, also $\models_{\mathfrak{A}} \phi[\mathbf{a}]$, so Φ is realized in $\mathfrak{A}$. $\square$

Recall from Definition 7.1.12 that (in the current terminology) a structure $\mathfrak{A}$ is atomic iff every complete type realized in $\mathfrak{A}$ is principal (with respect to $\text{Th}(\mathfrak{A})$).

Proposition 7.3.12 *For any theory T , if for all $n > 0$ every complete n -type of T is principal, then every model of T is atomic. Hence if T is also complete, then for any two models $\mathfrak{A}$ and $\mathfrak{B}$ of T , $\mathfrak{A} \simeq_p \mathfrak{B}$, and T is $\aleph_0$ -categorical.*

Proof. This is immediate from the preceding remarks and Theorems 7.1.9 and 7.1.16. $\square$

To prove a converse to this result, we need the following, which will have many other uses.

Omitting Types Theorem 7.3.13 *For any countable consistent theory T and any non-principal type Φ of T , there is a countable model of T which omits Φ . More generally, for any countable family $\langle \Phi_k : k \in \omega \rangle$ of non-principal types of T , there is a countable model of T which omits all Φ_k .*

Proof. We prove first the version for a single non-principal n -type Φ . The proof is based on an extension of the construction used in the proof of the Countable Compactness Theorem and may be done in any of the versions of that construction based on Henkin complete sets, notions of consistency, ultraproducts, or Boolean algebra. We shall adopt the Boolean algebra version and rely heavily on the material from Section 3.3. In the Lindenbaum algebra $\mathcal{L}_T$ of T , for each sequence $\mathbf{i} = i_0, \dots, i_{n-1}$ of distinct natural numbers, let $\mathbf{v}_i := v_{i_0}, \dots, v_{i_{n-1}}$ and

$$C_i := \{ [\phi(\mathbf{v}_i)]_T : \phi(\mathbf{v}) \in \Phi \}.$$

It follows easily from Corollary 7.2.30 that for each $\mathbf{i}$, $\prod C_i = \mathbb{O}$. By the Rasiowa-Sikorski Theorem 3.3.30, there is an ultrafilter U on $\mathcal{L}_T$ which is quantifier-preserving and preserves all of the infs $\prod C_i = \mathbb{O}$ for all finite sequences $\mathbf{i}$ of distinct natural numbers. Let $\mathfrak{A}$ be the structure defined in the proof of Theorem 3.3.29; recall that the universe A of this structure is $\{ \tilde{v}_i : i \in \omega \}$. Hence, if Φ is an n -type, to verify that Φ is omitted in $\mathfrak{A}$ it suffices to check that no n -tuple $(\tilde{v}_{i_0}, \dots, \tilde{v}_{i_{n-1}})$ satisfies all $\phi \in \Phi$. But since $\mathbb{O} \notin U$ and U preserves $\prod C_i = \mathbb{O}$, it follows that for some $\phi \in \Phi$, $[\phi(v_{i_0}, \dots, v_{i_{n-1}})]_T^n \notin U$, so by the proof of Theorem 3.3.29, $\models_{\mathfrak{A}} \neg \phi[(\tilde{v}_{i_0}, \dots, \tilde{v}_{i_{n-1}})]$ as desired.

The last clause of the theorem follows by the observation that even with countably many non-principal types Φ_k , the total number of infs to be preserved is still countable. $\square$

Corollary 7.3.14 *For any countable complete theory T and any type Φ of T , Φ is realized in every model of T iff Φ is principal. $\square$*

Corollary 7.3.15 *For any countable complete theory T , T is $\aleph_0$ -categorical iff every complete type of T is principal.*

Proof. The implication ($\Leftarrow$) is Proposition 7.3.12 above. Suppose that there is a type Φ of T which is not principal. Then there exist countable models $\mathfrak{A}$ and $\mathfrak{B}$ of T such that Φ is realized in $\mathfrak{A}$ and omitted in $\mathfrak{B}$. Clearly $\mathfrak{A}$ and $\mathfrak{B}$ are not isomorphic so T is not $\aleph_0$ -categorical. $\square$

Combining this with Proposition 7.2.33, we have

Ryll-Nardzewski's Theorem 7.3.16 *For any countable complete theory T , the following are equivalent.*

- (i) T is $\aleph_0$ -categorical;
- (ii) for all $n > 0$, $\mathcal{L}_T^n$ is finite;
- (iii) for all $n > 0$, $S^n(T)$ is finite;
- (iv) for all $n > 0$, every n -type of T is principal. $\square$

In particular, from this and Proposition 7.3.12 we have that if for all $n > 0$, $S^n(T)$ is finite, then T has a countable atomic model. By Corollary 7.1.17, if T is complete, such a model is unique up to isomorphism. This can be strengthened to

Proposition 7.3.17 *For any countable complete theory T , if any of the following hold*

- (i) for all $n > 0$, $S^n(T)$ is countable;
- (ii) for all $n > 0$, $\text{Card}(S^n(T)) < 2^{\aleph_0}$;
- (iii) $\mu(T, \aleph_0) < 2^{\aleph_0}$;

then T has a (unique) countable atomic model.

Proof. Suppose first that for all $n > 0$, $S^n(T)$ is countable. Then in particular there are at most countably many non-principal complete types of T , so by the Omitting Types Theorem 7.3.13 there is a countable model $\mathfrak{A}$ of T which omits them all. Clearly $\mathfrak{A}$ is atomic. Condition (ii) suffices by Proposition 7.3.8. Similarly, if $\mu(T, \aleph_0) < 2^{\aleph_0}$, then by Proposition 7.3.7, again condition (ii) is satisfied. $\square$

Finally, we have a condition which is both necessary and sufficient.

Proposition 7.3.18 *For any countable complete theory T , T has a countable atomic model iff for all $n > 0$, $\mathcal{L}_T^n$ is an atomic Boolean algebra.*

Proof. Suppose first that T has an atomic model $\mathfrak{A}$. For any non-zero element $[\psi(\mathbf{v})]_T^n$ of $\mathcal{L}_T^n$, $T \not\models \neg \psi(\mathbf{v})$, so since T is complete, $T \models \exists \mathbf{v} \psi(\mathbf{v})$. Hence, for some $\mathbf{a} \in \mathfrak{A}$, $\models_{\mathfrak{A}} \psi[\mathbf{a}]$. Since $\mathfrak{A}$ is atomic, by Corollary 7.2.31 there is an atom $[\phi]_T^n$ such that $\models_{\mathfrak{A}} \phi[\mathbf{a}]$. Since clearly it is not the case that $[\phi]_T^n \leq [\neg \psi]_T^n$, we have $[\phi]_T^n \leq [\psi]_T^n$. Hence $\mathcal{L}_T^n$ is atomic.

For the converse, suppose that for all $n > 0$ $\mathcal{L}_T^n$ is atomic and set

$$\Delta_n := \{ \neg \phi(v_0, \dots, v_{n-1}) : [\phi]_T^n \text{ is an atom of } \mathcal{L}_T^n \}.$$

Each Δ_n is a non-principal type, since if θ is any formula such that $T \models \theta \rightarrow \neg \phi$ for every ϕ such that $[\phi]_T^n$ is an atom of $\mathcal{L}_T^n$, then there can be no atom $[\phi]_T^n$ of $\mathcal{L}_T^n$ such that $[\phi]_T^n \leq [\theta]_T^n$; since $\mathcal{L}_T^n$ is atomic, it follows that $[\theta]_T^n = \mathbf{0}$. Hence by the Omitting Types Theorem 7.3.13, there exists a countable model $\mathfrak{A}$ of T which omits all of the types Δ_n for $n > 0$. For any complete type Φ realized in $\mathfrak{A}$, $\Delta_n \not\subseteq \Phi$, so for some atom $[\phi]_T^n$, $\neg \phi \notin \Phi$, whence $\phi \in \Phi$ and thus Φ is principal by Corollary 7.2.30. Thus $\mathfrak{A}$ is an atomic model. $\square$

A countable atomic model of a countable complete theory T is *small* in the sense that it realizes the smallest collection of types possible, the principal ones. We show next that it is small in another sense: it is elementarily embeddable in any model of T . We need to adapt the technique of partial embeddings of Section 7.1 — in particular Theorem 7.1.9.

Proposition 7.3.19 *For any countable L -structure $\mathfrak{A}$ and any L -structure $\mathfrak{B}$, if there exists a set $\mathcal{E}$ of partial elementary embeddings of $\mathfrak{A}$ into $\mathfrak{B}$ with the forward extension property, then $\mathfrak{A} \hookrightarrow \mathfrak{B}$.*

Proof. Left as an exercise. $\square$

As with the back-and-forth property, we most often use for $\mathcal{E}$ the set of all finite elementary partial embeddings in the following form.

Corollary 7.3.20 *For any countable L -structure $\mathfrak{A}$ and any L -structure $\mathfrak{B}$, if $\mathfrak{A} \equiv \mathfrak{B}$ and for any $\mathbf{a} \in \mathfrak{A}$ and $\mathbf{b} \in \mathfrak{B}$ such that $(\mathfrak{A}, \mathbf{a}) \equiv (\mathfrak{B}, \mathbf{b})$ we have (cf. Lemma 7.1.11)*

$$\text{for each } c \in \mathfrak{A}, \quad \text{there exists } d \in \mathfrak{B} \text{ such that } (\mathfrak{A}, \mathbf{a}, c) \equiv (\mathfrak{B}, \mathbf{b}, d),$$

then $\mathfrak{A} \hookrightarrow \mathfrak{B}$. $\square$

Recall from Definition 3.5.20 that for any complete theory T , a model $\mathfrak{A}$ of T is elementarily prime iff $\mathfrak{A} \hookrightarrow \mathfrak{B}$ for every model $\mathfrak{B}$ of T .

Theorem 7.3.21 *For any countable complete theory T and any $\mathfrak{A}$, $\mathfrak{A}$ is a countable atomic model of T iff $\mathfrak{A}$ is an elementarily prime model of T .*

Proof. Let $\mathfrak{A}$ be a countable atomic model of T and $\mathfrak{B}$ any model of T . Since T is complete, $\mathfrak{A} \equiv \mathfrak{B}$. To verify the hypothesis of the preceding corollary, suppose that $(\mathfrak{A}, \mathbf{a}) \equiv (\mathfrak{B}, \mathbf{b})$ and c is any element of $\mathfrak{A}$. Since $\mathfrak{A}$ is atomic, there exists a generator θ of $\text{Tp}^{\mathfrak{A}}(\mathbf{a}, c)$. Since $\models_{\mathfrak{A}} \theta[\mathbf{a}, c]$, also $\models_{\mathfrak{A}} \exists y \theta[\mathbf{a}]$, so $\models_{\mathfrak{B}} \exists y \theta[\mathbf{b}]$, and hence for some $d \in \mathfrak{B}$, $\models_{\mathfrak{B}} \theta[\mathbf{b}, d]$. Since θ is a generator it follows that $\text{Tp}^{\mathfrak{A}}(\mathbf{a}, c) = \text{Tp}^{\mathfrak{B}}(\mathbf{b}, d)$, and equivalently $(\mathfrak{A}, \mathbf{a}, c) \equiv (\mathfrak{B}, \mathbf{b}, d)$ as required.

For the converse, it is clear from the Löwenheim-Skolem Theorem that a prime model of a countable theory must be countable. Suppose that $\mathfrak{A}$ is not atomic, so for some $\mathbf{a} \in \mathfrak{A}$, $\text{Tp}^{\mathfrak{A}}(\mathbf{a})$ is not a principal type. By the Omitting Types Theorem, T has a model $\mathfrak{B}$ which omits this type. But if η were an elementary embedding of $\mathfrak{A}$ into $\mathfrak{B}$, then $\text{Tp}^{\mathfrak{A}}(\mathbf{a}) = \text{Tp}^{\mathfrak{B}}(\eta(\mathbf{a}))$, so this type would be realized in $\mathfrak{B}$. Hence $\mathfrak{A}$ is not an elementarily prime model. $\square$

The models of the preceding theorem have the property that all types realized in $\mathfrak{A}$ are also realized in $\mathfrak{B}$. This property alone is almost, but not quite, enough to guarantee that $\mathfrak{A} \hookrightarrow \mathfrak{B}$. The extra hypothesis needed is

Definition 7.3.22 A structure $\mathfrak{B}$ is $\aleph_0$ -**homogeneous** iff for any $\mathbf{b}, \mathbf{b}' \in \mathfrak{B}$, if $(\mathfrak{B}, \mathbf{b}) \equiv (\mathfrak{B}, \mathbf{b}')$, then

for each $d \in \mathfrak{B}$, there exists $d' \in \mathfrak{B}$ such that $(\mathfrak{B}, \mathbf{b}, d) \equiv (\mathfrak{B}, \mathbf{b}', d')$.

Proposition 7.3.23 *Every atomic structure is $\aleph_0$ -homogeneous.*

Proof. By the technique of the proof of the preceding theorem. $\square$

Theorem 7.3.24 *For any countable complete theory T and any models $\mathfrak{A}$ and $\mathfrak{B}$ of T , if $\mathfrak{A}$ is countable, every type of T realized in $\mathfrak{A}$ is also realized in $\mathfrak{B}$ and $\mathfrak{B}$ is $\aleph_0$ -homogeneous, then $\mathfrak{A} \hookrightarrow \mathfrak{B}$.*

Proof. We again use the criterion of the preceding corollary. Suppose that $(\mathfrak{A}, \mathbf{a}) \equiv (\mathfrak{B}, \mathbf{b})$ and c is any element of $\mathfrak{A}$. Since $\text{Tp}^{\mathfrak{A}}(\mathbf{a}, c)$ is realized in $\mathfrak{A}$ it is also realized in $\mathfrak{B}$, so for some $\mathbf{b}', d' \in \mathfrak{B}$ we have $(\mathfrak{A}, \mathbf{a}, c) \equiv (\mathfrak{B}, \mathbf{b}', d')$. Then by $\aleph_0$ -homogeneity, there is a $d \in \mathfrak{B}$ as required. $\square$

We turn now to characterizing models of a complete theory T which are *large* in the following two senses.

Definition 7.3.25 For any complete theory T and any model $\mathfrak{B}$ of T ,

- (i) $\mathfrak{B}$ is **weakly saturated** iff every type of T is realized in $\mathfrak{B}$;
- (ii) $\mathfrak{B}$ is **countably universal** iff for every countable model $\mathfrak{A}$ of T , $\mathfrak{A} \hookrightarrow \mathfrak{B}$.

The easy relationships between these notions are

Proposition 7.3.26 *For any countable complete theory T and any model $\mathfrak{B}$ of T ,*

- (i) *if $\mathfrak{B}$ is countably universal, then $\mathfrak{B}$ is weakly saturated;*
- (ii) *if $\mathfrak{B}$ is weakly saturated and $\aleph_0$ -homogeneous, then $\mathfrak{B}$ is countably universal.*

Proof. For (i), if $\mathfrak{B}$ is countably universal and Φ is any type of T , then Φ is realized in some countable model $\mathfrak{A}$ of T , say by $\mathbf{a}$. By hypothesis there exists $\eta : \mathfrak{A} \hookrightarrow \mathfrak{B}$. Then for each $\phi \in \Phi$, since $\models_{\mathfrak{A}} \phi[\mathbf{a}]$, also $\models_{\mathfrak{B}} \phi[\eta(\mathbf{a})]$, so $\eta(\mathbf{a})$ realizes Φ in $\mathfrak{B}$. Part (ii) is immediate from Theorem 7.3.24. $\square$

As the terminology suggests, there is a stronger notion of saturation.

Definition 7.3.27 A structure $\mathfrak{B}$ is $\aleph_0$ -**saturated** iff for all finite sequences $\mathbf{b} \in \mathfrak{B}$, $(\mathfrak{B}, \mathbf{b})$ is a weakly saturated model (of $\text{Th}(\mathfrak{B}, \mathbf{b})$).

In other words, while weak saturation requires that $\mathfrak{B}$ realize any type that can be consistently described by a set of formulas in the base language L , $\aleph_0$ -saturation extends this to all types that can be described using L enriched by any finitely many parameters from $\mathfrak{B}$. Since we will be using these types often, we introduce a streamlined notation.

Definition 7.3.28 For any structure $\mathfrak{A}$ and any Φ ,

- (i) Φ is a (complete) *n -type of $\mathfrak{A}$* iff it is a (complete) n -type of $\text{Th}(\mathfrak{A})$;
- (ii) $S^n(\mathfrak{A})$ denotes $S^n(\text{Th}(\mathfrak{A}))$, the set of complete n -types of $\mathfrak{A}$;
- (iii) $\mathfrak{A}$ is $\aleph_0$ -**categorical** iff $\text{Th}(\mathfrak{A})$ is $\aleph_0$ -categorical.

Lemma 7.3.29 *For any structure $\mathfrak{A}$ for a countable language and any $\mathbf{a} = a_0, \dots, a_{m-1} \in \mathfrak{A}$,*

- (i) *for each $n > 0$ there is a injective mapping from $S^n(\mathfrak{A}, \mathbf{a})$ into $S^{m+n}(\mathfrak{A})$ given by $\Phi \mapsto \Phi^*$, where*

$$\Phi^* := \{ \phi(\mathbf{x}, \mathbf{y}) : \phi(\dot{\mathbf{a}}, \mathbf{y}) \in \Phi \};$$

in particular, $\text{Card}(S^n(\mathfrak{A})) \leq \text{Card}(S^n(\mathfrak{A}, \mathbf{a})) \leq \text{Card}(S^{m+n}(\mathfrak{A}))$;

- (ii) $\mathfrak{A}$ is $\aleph_0$ -saturated iff $(\mathfrak{A}, \mathbf{a})$ is $\aleph_0$ -saturated;
- (iii) $\mathfrak{A}$ is $\aleph_0$ -categorical iff $(\mathfrak{A}, \mathbf{a})$ is $\aleph_0$ -categorical.

Proof. Parts (i) and (ii) are immediate from the definitions. We have by (i) that

$$\text{for all } n \in \omega [S^n(\mathfrak{A}) \text{ is finite}] \iff \text{for all } n \in \omega [S^n(\mathfrak{A}, \mathbf{a}) \text{ is finite}],$$

and (iii) follows by Theorem 7.3.16. $\square$

$\aleph_0$ -saturated models have the same uniqueness properties as atomic models (cf. Theorem 7.1.16 and its corollary).

Proposition 7.3.30 *For any complete theory T and any models $\mathfrak{A}$ and $\mathfrak{B}$ of T , if both $\mathfrak{A}$ and $\mathfrak{B}$ are $\aleph_0$ -saturated, then $\mathfrak{A} \simeq_p \mathfrak{B}$. Hence if both are countable, then $\mathfrak{A} \simeq \mathfrak{B}$.*

Proof. We use the criterion of Lemma 7.1.11. Since T is complete, $\mathfrak{A} \equiv \mathfrak{B}$, so suppose that $\mathbf{a} \in \mathfrak{A}$ and $\mathbf{b} \in \mathfrak{B}$ are such that $(\mathfrak{A}, \mathbf{a}) \equiv (\mathfrak{B}, \mathbf{b})$. For any $c \in \mathfrak{A}$, $\text{Tp}^{(\mathfrak{A}, \mathbf{a})}(c)$ is a 1-type of $(\mathfrak{A}, \mathbf{a})$, hence one of $(\mathfrak{B}, \mathbf{b})$, and because $\mathfrak{B}$ is $\aleph_0$ -saturated this type is realized by some element $d \in \mathfrak{B}$. It follows that $(\mathfrak{A}, \mathbf{a}, c) \equiv (\mathfrak{B}, \mathbf{b}, d)$ as required. The argument for the backward extension property is parallel with the roles of $\mathfrak{A}$ and $\mathfrak{B}$ reversed. The final sentence now follows from Theorem 7.1.9. $\square$

Let us temporarily call a structure **very weakly saturated** iff every 1-type of $\mathfrak{B}$ is realized in $\mathfrak{B}$.

Lemma 7.3.31 *For any structure $\mathfrak{B}$ for a countable language, if for all finite sequences $\mathbf{b} \in \mathfrak{B}$ $(\mathfrak{B}, \mathbf{b})$ is very weakly saturated, then $\mathfrak{B}$ is $\aleph_0$ -saturated.*

Proof. Under the hypothesis, we prove by induction on n that for every finite sequence $\mathbf{b} \in \mathfrak{B}$, every n -type of $(\mathfrak{B}, \mathbf{b})$ is realized in $(\mathfrak{B}, \mathbf{b})$. The basis $n = 1$ is the hypothesis, so assume this for n and let Φ be an $(n + 1)$ -type of $(\mathfrak{B}, \mathbf{b})$; without loss of generality suppose that Φ is closed under conjunction. Clearly the set of formulas

$$\Psi := \left\{ \exists v_n \phi(\dot{\mathbf{b}}, \mathbf{v}, v_n) : \phi \in \Phi \right\}$$

is an n -type of $(\mathfrak{B}, \mathbf{b})$, so by the induction hypothesis is realized, say by $\mathbf{c}$ in $(\mathfrak{B}, \mathbf{b})$. Then

$$\Theta := \left\{ \phi(\dot{\mathbf{b}}, \dot{\mathbf{c}}, v_n) : \phi \in \Phi \right\}$$

is a 1-type of $(\mathfrak{B}, \mathbf{b}, \mathbf{c})$ so by hypothesis is realized, say by d . Then $(\mathbf{c}, d)$ realizes Φ . $\square$

Theorem 7.3.32 *For any structure $\mathfrak{B}$ for a countable language, the following are equivalent:*

- (i) $\mathfrak{B}$ is $\aleph_0$ -saturated;
- (ii) $\mathfrak{B}$ is weakly saturated and $\aleph_0$ -homogeneous;
- (iii) $\mathfrak{B}$ is countably universal and $\aleph_0$ -homogeneous.

Proof. The equivalence of (ii) and (iii) is immediate from Proposition 7.3.26, and we shall show the equivalence of (i) and (ii). From (i), certainly $\mathfrak{B}$ is weakly saturated and we show that it is $\aleph_0$ -homogeneous. Suppose that for some $\mathbf{b}, \mathbf{b}' \in \mathfrak{B}$, $(\mathfrak{B}, \mathbf{b}) \equiv (\mathfrak{B}, \mathbf{b}')$. As in the proof of the preceding proposition, for any $d \in \mathfrak{B}$, $\text{Tp}^{(\mathfrak{B}, \mathbf{b})}(d)$ is a type of $(\mathfrak{B}, \mathbf{b})$, hence one

of $(\mathfrak{B}, \mathbf{b}')$ so is realized by some $d' \in \mathfrak{B}$. Then $(\mathfrak{B}, \mathbf{b}, d) \equiv (\mathfrak{B}, \mathbf{b}', d')$ as required.

Assume now that $\mathfrak{B}$ satisfies condition (ii) and that for some $\mathbf{b} \in \mathfrak{B}$, Φ is a complete 1-type of $(\mathfrak{B}, \mathbf{b})$; by the lemma, and the fact that every 1-type has an extension to a complete 1-type, it suffices to show that Φ is realized in $(\mathfrak{B}, \mathbf{b})$. In the notation of Lemma 7.3.29, Φ^* is an $(n+1)$ -type of $\mathfrak{B}$ so by weak saturation is realized in $\mathfrak{B}$, say by $(\mathbf{b}', c')$. Because Φ is complete, $(\mathfrak{B}, \mathbf{b}) \equiv (\mathfrak{B}, \mathbf{b}')$, so by homogeneity, there exists $c \in \mathfrak{B}$ such that $(\mathfrak{B}, \mathbf{b}, c) \equiv (\mathfrak{B}, \mathbf{b}', c')$. Thus also $(\mathbf{b}, c)$ realizes Φ^* in $\mathfrak{B}$, so c realizes Φ in $(\mathfrak{B}, \mathbf{b})$. $\square$

We address next the question of what sorts of $\aleph_0$ -saturated models exist.

Definition 7.3.33 For any structures $\mathfrak{A} \prec \mathfrak{B}$, $\mathfrak{B}$ is $\aleph_0$ -**saturated over** $\mathfrak{A}$ iff for all $\mathbf{a} \in \mathfrak{A}$, every type of $(\mathfrak{A}, \mathbf{a})$ is realized in $(\mathfrak{B}, \mathbf{a})$.

Proposition 7.3.34 *For any countable complete theory T and any model $\mathfrak{A}$ of T , there exists a model $\mathfrak{B}$ of T such that*

- (i) $\mathfrak{A} \prec \mathfrak{B}$;
- (ii) $\mathfrak{B}$ is $\aleph_0$ -saturated over $\mathfrak{A}$;
- (iii) if $\text{Card}(\mathfrak{A}) \leq 2^{\aleph_0}$, then also $\text{Card}(\mathfrak{B}) \leq 2^{\aleph_0}$;
- (iv) if $\mathfrak{A}$ is countable and for all $n > 0$, $S^n(T)$ is countable, then $\mathfrak{B}$ is countable.

Proof. Given $\mathfrak{A}$, for each $\mathbf{a} \in \mathfrak{A}$, each $n > 0$ and each complete n -type Φ of $(\mathfrak{A}, \mathbf{a})$, let $\mathbf{c}_\Phi = c_{0,\Phi}, \dots, c_{n-1,\Phi}$ be new constant symbols and let $\Phi(\mathbf{c}_\Phi)$ denote the set of all $\phi(\mathbf{c}_\Phi)$ for $\phi \in \Phi$. Let Γ be $\text{Th}(\mathfrak{A}_A)$ together with the union of all such $\Phi(\mathbf{c}_\Phi)$. A straightforward compactness argument verifies that Γ is consistent; hence by the First Extension Lemma 3.4.21 there exists a structure $\mathfrak{B}^+$ such that $\mathfrak{B}^+ \upharpoonright L$ is an elementary extension of $\mathfrak{A}$ and $\mathfrak{B}^+$ is a model of all $\Phi(\mathbf{c}_\Phi)$. Clearly $\mathfrak{B}$ is $\aleph_0$ -saturated over $\mathfrak{A}$.

Since for each $\mathbf{a}$ the language of $(\mathfrak{A}, \mathbf{a})$ is countable, by Proposition 7.3.3, the theory has at most $2^{\aleph_0}$ -many types, and if $\mathfrak{A}$ has cardinality at most $2^{\aleph_0}$, there are at most this many finite sequences $\mathbf{a}$ of elements of $\mathfrak{A}$. Hence altogether there are at most $2^{\aleph_0} \times 2^{\aleph_0} = 2^{\aleph_0}$ many such complete types Φ , and since we adjoin only finitely many constant symbols for each Φ , the language of Γ has cardinality at most $2^{\aleph_0}$ (including also the names for elements of A). Hence by the Löwenheim-Skolem Theorem we may choose the model $\mathfrak{B}$ to be of cardinality at most $2^{\aleph_0}$.

If $\mathfrak{A}$ is countable and for all $n > 0$, $S^n(T)$ is countable, then by Lemma 7.3.29 each $(\mathfrak{A}, \mathbf{a})$ has at most countably many types and there are at most countably many finite sequences $\mathbf{a}$, so there are countably many types Φ and

hence the language of Γ is countable, so in this case we may choose $\mathfrak{B}$ to be countable. $\square$

Theorem 7.3.35 *For any countable complete theory T ,*

- (i) *T has an $\aleph_0$ -saturated model of cardinality at most $2^{\aleph_0}$;*
- (ii) *T has a countable $\aleph_0$ -saturated model iff for all $n > 0$, $S^n(T)$ is countable.*

Proof. For (i) and the implication ($\Leftarrow$) of (ii), let $\mathfrak{A}_0$ be any countable model of T . By iterating the construction of the preceding proposition, there exists an elementary chain $\langle \mathfrak{A}_n : n \in \omega \rangle$ of models of T such that for each n , $\mathfrak{A}_{n+1}$ is $\aleph_0$ -saturated over $\mathfrak{A}_n$. Let $\mathfrak{B}$ be the union of this chain. For any $\mathbf{b} \in \mathfrak{B}$ and any type Φ of $(\mathfrak{B}, \mathbf{b})$, there is some n such that $\mathbf{b} \in \mathfrak{A}_n$. Since $\mathfrak{A}_n \prec \mathfrak{B}$, $\text{Th}(\mathfrak{A}_n, \mathbf{b}) = \text{Th}(\mathfrak{B}, \mathbf{b})$ so Φ is also a type of $(\mathfrak{A}_n, \mathbf{b})$, and thus is realized in $\mathfrak{A}_{n+1}$. But then, since $\mathfrak{A}_{n+1} \prec \mathfrak{B}$, Φ is also realized in $(\mathfrak{B}, \mathbf{b})$.

With no further hypotheses, each $\mathfrak{A}_n$ may be chosen to have cardinality at most $2^{\aleph_0}$ and hence $\mathfrak{B}$ has cardinality at most $2^{\aleph_0}$. If for each $n > 0$, $S^n(T)$ is countable, then by part (iv) of the proposition, each $\mathfrak{A}_n$ may be chosen to be countable, hence also $\mathfrak{B}$ is countable.

Finally, suppose that T does have a countable $\aleph_0$ -saturated model $\mathfrak{B}$. In particular, $\mathfrak{B}$ realizes each member of each $S^n(T)$. But a countable structure can realize at most countably many types altogether, so each $S^n(T)$ must be countable. $\square$

Corollary 7.3.36 *For any countable complete theory T , if any of the following hold*

- (i) *for all $n > 0$, $S^n(T)$ is countable;*
- (ii) *for all $n > 0$, $\text{Card}(S^n(T)) < 2^{\aleph_0}$;*
- (iii) *$\mu(T, \aleph_0) < 2^{\aleph_0}$;*

then T has a countable $\aleph_0$ -saturated model.

Proof. This follows from the preceding theorem as in the proof of Proposition 7.3.17. $\square$

Corollary 7.3.37 *For any countable complete theory T , if T has a countable $\aleph_0$ -saturated model, then T also has a countable atomic model; these models coincide iff T is $\aleph_0$ -categorical.*

Proof. The first part follows from Proposition 7.3.17. If T is $\aleph_0$ -categorical, then it has both a countable atomic and a countable $\aleph_0$ -saturated model; since it has (up to isomorphism) only one model altogether, these must coincide. Conversely, if a single countable model of T is both atomic and

even weakly saturated, then every type of T is realized in an atomic model so is principal. Then by Proposition 7.3.12, T is $\aleph_0$ -categorical. $\square$

Corollary 7.3.38 *For any countable complete theory T , T is $\aleph_0$ -categorical iff every countable model of T is $\aleph_0$ -saturated. $\square$*

We know examples of $\aleph_0$ -categorical theories ($\mu(T, \aleph_0) = 1$), and exercises below provide examples of theories T such that $\mu(T, \aleph_0) = 3, 4, \dots$. In this light the following theorem is somewhat surprising.

Theorem 7.3.39 *For any countable complete theory T , $\mu(T, \aleph_0) \neq 2$.*

Proof. Towards a contradiction, suppose that T is a theory such that $\mu(T, \aleph_0) = 2$. Then by Proposition 7.3.17 and Corollary 7.3.36, T has both a countable atomic model $\mathfrak{A}$ and a countable $\aleph_0$ -saturated model $\mathfrak{B}$. Since T is not $\aleph_0$ -categorical, by the last corollary but one above these models are not isomorphic and thus $\mathfrak{B}$ is not atomic. Hence there exists some finite sequence $\mathbf{b} \in \mathfrak{B}$ such that $\text{Tp}^{\mathfrak{B}}(\mathbf{b})$ is non-principal. By Lemma 7.3.29, $(\mathfrak{B}, \mathbf{b})$ is still $\aleph_0$ -saturated, but is not $\aleph_0$ -categorical; hence there exists a countable structure $(\mathfrak{C}, \mathbf{c})$ which is elementarily equivalent to $(\mathfrak{B}, \mathbf{b})$ but not isomorphic to $(\mathfrak{B}, \mathbf{b})$ and hence not $\aleph_0$ -saturated. Again by Lemma 7.3.29, also $\mathfrak{C}$ is not $\aleph_0$ -saturated and hence is not isomorphic to $\mathfrak{B}$. But clearly $\text{Tp}^{\mathfrak{C}}(\mathbf{c})$ is non-principal, so $\mathfrak{C}$ is also not atomic and hence is not isomorphic to $\mathfrak{A}$. Thus $\mu(T, \aleph_0) \geq 3$, contrary to assumption. $\square$

Collateral Reading. Most of the results in this section appear in [Ch-Ke; Chapter 2] and also in other model-theory texts such as [Ho93] or [Mar].

Exercises

Exercise 7.3.40 Let L be any countable language which includes a constant symbol $\dot{0}$ and a unary function symbol $\dot{S}$; as usual $\dot{n}$ denotes the term with n occurrences of $\dot{S}$ followed by $\dot{0}$. An ω -**structure** is any L -structure $\mathfrak{A}$ such that the universe of A is exactly $\{\dot{n}^{\mathfrak{A}} : n \in \omega\}$. An ω -**theory** is an L -theory T which is closed under the ω -**rule**: for any formula ϕ with at most x free,

$$\text{if for all } n \in \omega, \quad \phi_x(\dot{n}) \in T \quad \text{then} \quad \forall x \phi \in T.$$

Prove that an ω -theory is consistent iff it has an ω -model — that is, a model which is an ω -structure.

Exercise 7.3.41 Show that the Omitting Types Theorem can fail for uncountable languages.

Hint: Let L be a language with an uncountable set C of constant symbols (and possibly other non-logical symbols) and T the theory generated by the sentences $\neg c \doteq c'$ for c and c' distinct members of C .

Exercise 7.3.42 Give a proof of Proposition 7.3.19.

Exercise 7.3.43 Let L be the language with one binary relation symbol $\dot{<}$ and a denumerable sequence of constant symbols $\langle c_n : n \in \omega \rangle$; let T be the theory generated by $T_{(\text{DLO})}$ and the sentences $\{c_n \dot{<} c_{n+1} : n \in \omega\}$.

- (i) Show that $\mu(T, \aleph_0) = 3$ by verifying that every countable model of T is isomorphic to one of

$$\mathbb{Q}_2 := (\mathbb{Q}, 1, 1.9, 1.99, 1.999, \dots)$$

$$\mathbb{Q}_\pi := (\mathbb{Q}, 3, 3.1, 3.14, 3.141, \dots)$$

$$\mathbb{Q}_\infty := (\mathbb{Q}, 1, 2, 3, 4, \dots).$$

- (ii) Explain why T has both an atomic model and an $\aleph_0$ -saturated model.
- (iii) Which of $\mathbb{Q}_2$, $\mathbb{Q}_\pi$ and $\mathbb{Q}_\infty$ is the atomic model? For each type Φ of a finite sequence of elements of that model give the generator of Φ and for each of the other two models find a non-principal type that is realized.
- (iv) Which two of $\mathbb{Q}_2$, $\mathbb{Q}_\pi$ and $\mathbb{Q}_\infty$ are weakly saturated? Which one of these is also $\aleph_0$ -homogeneous and therefore $\aleph_0$ -saturated? Why is the other one not $\aleph_0$ -homogeneous?
- (v) Which of the models are countably universal?

Exercise 7.3.44 Generalize the method of the preceding exercise to show that for all $n \geq 3$ there exists a theory T such that $\mu(T, \aleph_0) = n$.

Hint: For $n = 4$ adjoin a unary relation symbol p to the language of the preceding exercise. Let T be the theory generated by $T_{(\text{DLO})}$, $\{c_n \dot{<} c_{n+1} : n \in \omega\}$, $\{pc_n : n \in \omega\}$ and the assertions that both p and its complement are dense. The four models are the analogues of $\mathbb{Q}_\pi$ and $\mathbb{Q}_\infty$ together with models having limits in or out of p .

Exercise 7.3.45 Suppose that T is a countable complete theory such that $1 < \mu(T, \aleph_0) < \omega$. Prove that T has a countable model which is weakly saturated but not $\aleph_0$ -saturated.

Exercise 7.3.46 Let $\mathfrak{A}$ be a non-standard countable model of Peano Arithmetic $\mathcal{P}$, and let L be L_Ω extended by a single constant symbol c . Let T be the L -theory generated by $\text{Th}_A(\mathfrak{A}) \cup \{\dot{a} \dot{<} c : a \in A\}$.

- (i) Show that for each $a \in A \sim \{\dot{n}^{\mathfrak{A}} : n \in \omega\}$,

$$\Phi_a := \{v \dot{<} \dot{a}\} \cup \{\neg v \dot{=} \dot{b} : b \in A\}$$

is a non-principal 1-type of T .

- (ii) Conclude that there exists a proper elementary end extension $\mathfrak{B}$ of $\mathfrak{A}$ (Remark 4.6.12).

Hint: For (i), suppose that Φ_a had a generator $\theta(v, c)$, so

$$T \models \theta(v, c) \rightarrow v \dot{<} a;$$

$$\text{for all } b \in A, \quad T \models \neg \theta(\dot{b}, a);$$

$$T \cup \{\exists v \theta(v, c)\} \text{ is consistent.}$$

Show that $\models_{\mathfrak{A}} \forall x (\exists y \dot{\geq} x) \exists v \theta(v, y)$ and use induction to strengthen this to

$$\models_{\mathfrak{A}} \exists v \forall x (\exists y \dot{\geq} x) \theta(v, y).$$

Conclude that for some $b \in A$, $T \cup \{\theta(\dot{b}, c)\}$ is consistent, a contradiction. Now for (ii) use the Omitting Types Theorem.

Exercise 7.3.47 Prove that if $\mathfrak{B}$ is a countable $\aleph_0$ -homogeneous structure and $\mathbf{b}, \mathbf{c} \in \mathfrak{B}$ are such that $(\mathfrak{B}, \mathbf{b}) \equiv (\mathfrak{B}, \mathbf{c})$, then there is an automorphism η of $\mathfrak{B}$ such that $\eta(\mathbf{b}) = \mathbf{c}$.

Exercise 7.3.48 Suppose that $\mathfrak{A}$ and $\mathfrak{B}$ are both $\aleph_0$ -homogeneous and realize exactly the same n -types for all n . Prove that $\mathfrak{A} \simeq_{\mathfrak{p}} \mathfrak{B}$.

Exercise 7.3.49 Suppose that $\langle \mathfrak{A}_\sigma : \sigma < \tau < \aleph_1 \rangle$ is a countable elementary chain of $\aleph_0$ -homogeneous structures. Show that

(i) $\mathfrak{A} := \bigcup_{\sigma < \tau} \mathfrak{A}_\sigma$ is also $\aleph_0$ -homogeneous;

(ii) if for all $\sigma < \tau$, $\mathfrak{A}_\sigma \simeq \mathfrak{A}_0$, then also $\mathfrak{A} \simeq \mathfrak{A}_0$.

Exercise 7.3.50 Show that for any countable structure $\mathfrak{A}$, there is a countable $\aleph_0$ -homogeneous structure $\mathfrak{B}$ such that $\mathfrak{A} \prec \mathfrak{B}$.

Hint: First show that $\mathfrak{A}$ has an elementary extension $\mathfrak{A}'$ such that for any $\mathbf{a}, \mathbf{b} \in \mathfrak{A}$, if $(\mathfrak{A}, \mathbf{a}) \equiv (\mathfrak{A}, \mathbf{b})$, then for each $c \in \mathfrak{A}$, there exists $d \in \mathfrak{A}'$ such that $(\mathfrak{A}, \mathbf{a}, c) \equiv (\mathfrak{A}', \mathbf{b}, d)$.

Exercise 7.3.51 Let T be the theory of algebraically closed fields of characteristic 0. Prove that $\mu(T, \aleph_0) = \aleph_0$.

Exercise 7.3.52 Give another proof of the Omitting Types Theorem 7.3.13 in the style of the original Henkin proof of the Compactness Theorem in [Section 3.1](#).

Exercise 7.3.53 Give a proof of Theorem 7.3.10 along the following lines; many ideas used here are similar to those of Exercise 7.1.37. Fix a language L , an L -theory T and a model $\mathfrak{A}$ of T . Recall from Definition 3.7.19 and Proposition 3.7.21 the notion of a fragment of an infinitary language; although the definition there is for $L_{\omega_1\omega}$, it makes sense for any $L_{\kappa\omega}$. For any fragment Φ , the Φ - n -**type** of $\mathbf{a} = a_0, \dots, a_{n-1} \in A$ is the set

$$\text{Tp}_{\Phi}^{\mathfrak{A}}(\mathbf{a}) := \left\{ \phi(\mathbf{v}) \in \Phi : \models_{\mathfrak{A}} \phi[\mathbf{a}] \right\},$$

Ψ is a Φ -**type of** T iff Ψ is a subset of some $\text{Tp}_{\Phi}^{\mathfrak{A}}(\mathbf{a})$, Ψ is complete iff for each $\phi \in \Phi$ either $\phi \in \Psi$ or $\neg \phi \in \Psi$, and we denote by $S_{\Phi}^n(T)$ the set of all complete Φ - n -types of T . Define a transfinite sequence of fragments Φ_{σ} as follows: $\Phi_0 := L$; for limit ordinals σ , $\Phi_{\sigma} := \bigcup_{\tau < \sigma} \Phi_{\tau}^{\mathfrak{A}}$; and

$$\Phi_{\sigma+1} := \text{Frag} \left(\Phi_{\sigma} \cup \left\{ \bigwedge \Psi : \Psi \in S_{\Phi_{\sigma}}^n(T), \quad n \in \omega \right\} \right).$$

For each $\mathbf{a} \in A$, let $\theta_{\sigma}^{\mathbf{a}}(\mathbf{x}) := \bigwedge \text{Tp}_{\Phi_{\sigma}}^{\mathfrak{A}}(\mathbf{a})$. It is immediate from the definitions that

- (i) $\theta_{\sigma}^{\mathbf{a}} \in \Phi_{\sigma+1}$;
- (ii) for any L -structure $\mathfrak{B}$ and any $\mathbf{b} \in B$,

$$\models_{\mathfrak{B}} \theta_{\sigma}^{\mathbf{a}}[\mathbf{b}] \iff \text{Tp}_{\Phi_{\sigma}}^{\mathfrak{A}}(\mathbf{a}) = \text{Tp}_{\Phi_{\sigma}}^{\mathfrak{B}}(\mathbf{b}).$$

For each $\mathbf{a}, \mathbf{c} \in A$, set

$$\rho^{\mathfrak{A}}(\mathbf{a}, \mathbf{c}) := \begin{cases} \text{least } \sigma & \left[\text{Tp}_{\Phi_{\sigma}}^{\mathfrak{A}}(\mathbf{a}) \neq \text{Tp}_{\Phi_{\sigma}}^{\mathfrak{A}}(\mathbf{c}) \right], & \text{if such } \sigma \text{ exists;} \\ 0, & \text{otherwise.} \end{cases}$$

Finally, let

$$\rho(\mathfrak{A}) := \sup^+ \left\{ \rho^{\mathfrak{A}}(\mathbf{a}, \mathbf{c}) : \mathbf{a}, \mathbf{c} \in A \right\} \quad \text{and} \quad \chi_{\mathfrak{A}} := \theta_{\rho(\mathfrak{A})+2}^{()}. \quad \square$$

Show that

- (iii) for each $\mathbf{a} \in A$, $\models_{\mathfrak{A}} \forall \mathbf{x} \left[\theta_{\rho(\mathfrak{A})}^{\mathbf{a}}(\mathbf{x}) \rightarrow \theta_{\rho(\mathfrak{A})+1}^{\mathbf{a}}(\mathbf{x}) \right]$;
- (iv) for any L -structure $\mathfrak{B}$, $\models_{\mathfrak{B}} \chi_{\mathfrak{A}} \iff \mathfrak{A} \simeq_{\rho} \mathfrak{B}$;
- (v) if $\mu(T, \aleph_0) < 2^{\aleph_0}$, then for all $\sigma < \aleph_1$, Φ_{σ} is a countable fragment;
- (vi) if $\mu(T, \aleph_0) < 2^{\aleph_0}$, then $\mu(T, \aleph_0) \leq \aleph_1$.

Hint: For the implication (iv)($\implies$), use the set

$$\mathcal{E} := \left\{ \eta : \eta \text{ is a finite partial embedding with domain } \mathbf{a} \right. \\ \left. \text{such that } \models_{\mathfrak{B}} \theta_{\rho(\mathfrak{A})}^{\mathbf{a}}[\eta(\mathbf{a})] \right\}.$$

For (v), Proceed by induction and prove also that each $S_{\Phi_{\sigma}}^n(T)$ is countable. For (vi), show that for each countable model $\mathfrak{A}$ of T , $\chi_{\mathfrak{A}}$ belongs to Φ_{σ} for some $\sigma < \aleph_1$; then use (iv) and (v).

Exercise 7.3.54 Let T be the theory of Exercise 2.4.61.

- (i) Use the quantifier-elimination result for T to describe all of the complete 1-types of T .
- (ii) Show that for all $n > 0$, $S^n(T)$ is countable.
- (iii) Show that $\mu(T, \aleph_0) = 2^{\aleph_0}$.

These results show that the converse of Proposition 7.3.7(i) does not hold and thus does not provide a proof of Vaught's Conjecture.

The next few exercises provide an introduction to an interesting variant of $\aleph_0$ -saturation. We assume that for each finite language involved some Gödel numbering (Section 4.4) has been fixed. Then a set of formulas, in particular a type, is recursive iff the corresponding set of Gödel numbers is recursive.

Definition 7.3.55 A structure $\mathfrak{A}$ is *recursively saturated* iff for all finite sequences $\mathbf{a} \in \mathfrak{A}$, every recursive type of $\text{Th}(\mathfrak{A}, \mathbf{a})$ is realized in $\mathfrak{A}$.

Exercise 7.3.56 Let L be a finite language. Show that for any L -structure $\mathfrak{A}$,

- (i) $\mathfrak{A}$ is $\aleph_0$ -saturated $\implies \mathfrak{A}$ is recursively saturated;
- (ii) if $\mathfrak{A}$ is countable, then there exists a countable recursively saturated structure $\mathfrak{B}$ such that $\mathfrak{A} \prec \mathfrak{B}$;
- (iii) $\mathfrak{A}$ is recursively saturated $\implies \mathfrak{A}$ is $\aleph_0$ -homogeneous.

Hint: For (iii), given $(\mathfrak{A}, \mathbf{a}) \equiv (\mathfrak{A}, \mathbf{b})$ and $c \in A$, consider the recursive type $\{ \phi(\dot{\mathbf{a}}, \dot{c}) \leftrightarrow \phi(\dot{\mathbf{b}}, x) : \phi \text{ a formula} \}$.

Definition 7.3.57 For any two structures $\mathfrak{A}$ and $\mathfrak{B}$, let $\mathfrak{A} \dot{\cup} \mathfrak{B}$ denote the structure with universe $A \cup B$, unary relations A and B and distinct copies of the relations, functions and distinguished elements of $\mathfrak{A}$ and $\mathfrak{B}$, with the functions extended in any canonical way to the full universe.

Exercise 7.3.58 Show that for any finite language L and any L -structures $\mathfrak{A}$ and $\mathfrak{B}$, if $\mathfrak{A} \equiv \mathfrak{B}$ and $\mathfrak{A} \dot{\cup} \mathfrak{B}$ is recursively saturated, then $\mathfrak{A} \simeq_p \mathfrak{B}$. Hence if both are countable, then $\mathfrak{A} \simeq \mathfrak{B}$.

Exercise 7.3.59 Use the two preceding exercises to give a new proof of the Countable Robinson Interpolation Theorem 3.3.12 (for finite languages) along the following lines. With notation as in the theorem, suppose that Γ_0 and Γ_1 have no Robinson interpolant. Show that

- (i) $\Gamma_0 \cup [\bar{\Gamma}_1 \cap \text{Sent}_L]$ is consistent; fix a countable model $\mathfrak{A}$;

- (ii) $[\text{Th}(\mathfrak{A}) \cap \text{Sent}_L] \cup \Gamma_1$ is consistent; fix a countable model $\mathfrak{B}$;
- (iii) any countable recursively saturated elementary extension of $\mathfrak{A} \dot{\cup} \mathfrak{B}$ is of the form $\mathfrak{C} \dot{\cup} \mathfrak{D}$ with $\mathfrak{C} \restriction L \simeq \mathfrak{D} \restriction L$.

Now use these facts to construct a model of $\Gamma_0 \cup \Gamma_1$.

Exercise 7.3.60 Use recursively saturated models to show that in Exercise 3.5.55 the hypothesis that p is weakly implicitly T^+ - L -definable may be replaced by the apparently weaker condition: for any L -structure $\mathfrak{A}$ and any $P, Q \subseteq |\mathfrak{A}|^k$, if both $(\mathfrak{A}, P)$ and $(\mathfrak{A}, Q)$ are models of T^+ and $(\mathfrak{A}, P) \simeq (\mathfrak{A}, Q)$, then $P = Q$. This is known as **Svenonius' Theorem**. Note that for T^+ a complete theory, this condition is equivalent to the explicit T^+ - L -definability of p .

7.4 Uncountable models of countable theories

This section and the next are largely devoted to a proof of Morley's Theorem; the reader is encouraged to read at this point the background and summary of this result at the beginning of [Section 7.5](#).

In Proposition 7.3.17 and Corollary 7.3.36 we established that if for all $n > 0$, $S^n(T)$ is countable, then T has both a countable atomic model and a countable $\aleph_0$ -saturated model. Here we shall want to construct larger models of T which are either atomic or have stronger saturation properties.

Definition 7.4.1 For any cardinal κ and any structure $\mathfrak{B}$, $\mathfrak{B}$ is **κ -saturated** iff for any set $X \subseteq B$ with $\text{Card}(X) < \kappa$, $\mathfrak{B}_X$ is weakly saturated. $\mathfrak{B}$ is **saturated** iff $\mathfrak{B}$ is $\text{Card}(\mathfrak{B})$ -saturated.

In this terminology, the conclusion of the result quoted above is that T has countable atomic and saturated models. The first new notion here is that of a structure $\mathfrak{B}$ being $\aleph_1$ -saturated iff it realizes all types of every expansion $\mathfrak{B}_X$, with X a countable subset of $\mathfrak{B}$. Note that such a structure $\mathfrak{B}$ could not itself be countable, since if it were, it would have to realize the 1-type $\{\neg x \doteq b : b \in B\}$, a contradiction — more generally with the same proof

Proposition 7.4.2 For any structure $\mathfrak{B}$ and any infinite cardinal κ , if $\mathfrak{B}$ is κ -saturated, then $\text{Card}(\mathfrak{B}) \geq \kappa$. $\square$

We shall see soon that the techniques of Corollary 7.3.36 suffice to construct for any countable theory T and any infinite cardinal κ , a κ^+ -saturated model of cardinality at most 2^κ — for example, an $\aleph_1$ -saturated model of power $2^{\aleph_0}$. Under the assumption of the Generalized Continuum Hypothesis (GCH), such a model has also cardinality κ^+ and is therefore saturated, but we would like to achieve this without assuming the GCH.

Recall that to construct a countable saturated model for a theory T , the hypothesis that $S^n(T)$ is countable for all $n > 0$ suffices because it entails that for any model $\mathfrak{B}$ of T and any $\mathbf{b} \in \mathfrak{B}$ also all $S^n(\mathfrak{B}, \mathbf{b})$ are countable. Here, however, we shall need to consider types which involve larger sets of parameters from the model for which the corresponding size limitation is not automatic.

Definition 7.4.3 For any theory T and any infinite cardinal κ , T is κ -**stable** iff for every model $\mathfrak{B}$ of T , all $X \subseteq \mathfrak{B}$ with $\text{Card}(X) \leq \kappa$, and all $n > 0$, $S^n(\mathfrak{B}_X)$ has cardinality at most κ . Traditionally we write ω -stable in place of $\aleph_0$ -stable.

Note that a type of $\mathfrak{B}_X$ is a set of L^X -formulas; if $\text{Card}(X) = \kappa$, then L^X has κ -many formulas and there are thus up to 2^κ -many types. Thus κ -stability is a real limitation. The property of κ -stability can also be expressed in a slightly different way.

Definition 7.4.4 For any theory T and any infinite cardinal κ , an extension T^+ of T by constants to a language with exactly κ -many new constant symbols is called a κ -**inessential** extension. An **inessential** extension is one that is κ -inessential for some κ .

Note that any complete κ -inessential extension of T is (essentially) of the form $\text{Th}(\mathfrak{B}_X)$ for some model $\mathfrak{B}$ of T and some set $X \subseteq B$ of cardinality κ .

Corollary 7.4.5 For any theory T and any infinite cardinal κ , T is κ -stable iff for every $n > 0$ and every complete κ -inessential extension T^+ of T , $S^n(T^+)$ has cardinality at most κ . $\square$

The construction of (partially) saturated models proceeds quite analogously to the countable case (Proposition 7.3.34).

Definition 7.4.6 For any structures $\mathfrak{A} \subseteq \mathfrak{B}$ and any infinite cardinal κ , $\mathfrak{B}$ is κ -**saturated over** $\mathfrak{A}$ iff for every set $X \subseteq A$ of cardinality less than κ , every type of $\mathfrak{A}_X$ is realized in $\mathfrak{B}$.

Proposition 7.4.7 For any countable complete theory T , any infinite cardinal κ and any model $\mathfrak{A}$ of T , there exists a model $\mathfrak{B}$ of T such that

- (i) $\mathfrak{A} \prec \mathfrak{B}$;
- (ii) $\mathfrak{B}$ is κ^+ -saturated over $\mathfrak{A}$;
- (iii) if $\text{Card}(\mathfrak{A}) \leq 2^\kappa$, then also $\text{Card}(\mathfrak{B}) \leq 2^\kappa$;
- (iv) if $\text{Card}(\mathfrak{A}) \leq \kappa$ and T is κ -stable, then also $\text{Card}(\mathfrak{B}) \leq \kappa$.

Proof. Given $\mathfrak{A}$, for each $X \subseteq A$ of power at most κ , each $n > 0$ and each complete n -type Φ of $(\mathfrak{A}_X)$, let $\mathbf{c}_\Phi = c_{0,\Phi}, \dots, c_{n-1,\Phi}$ be new constant symbols and let $\Phi(\mathbf{c}_\Phi)$ denote the set of all set of all $\phi(\mathbf{c}_\Phi)$ for $\phi \in \Phi$. Let

Γ be $\text{Th}(\mathfrak{A}_A)$ together with the union of all such $\Phi(\mathbf{c}_\Phi)$. A straightforward compactness argument verifies that Γ is consistent; hence by the First Extension Lemma 3.4.21 there exists a structure $\mathfrak{B}^+$ such that $\mathfrak{B}^+ \upharpoonright L$ is an elementary extension of $\mathfrak{A}$ and $\mathfrak{B}^+$ is a model of all $\Phi(\mathbf{c}_\Phi)$. Clearly $\mathfrak{B}$ is κ^+ -saturated over $\mathfrak{A}$.

Since for each $X \subseteq A$ of power at most κ the language of $(\mathfrak{A}_X)$ has cardinality κ , there are at most 2^κ -many types of $\mathfrak{A}_X$, and if $\mathfrak{A}$ has cardinality at most 2^κ , there are at most $(2^\kappa)^\kappa = 2^\kappa$ many such X . Hence altogether there are at most $2^\kappa \times 2^\kappa = 2^\kappa$ many such complete types Φ , and since we adjoin only finitely many constant symbols for each Φ , the language of Γ has cardinality at most 2^κ . Hence by the Löwenheim-Skolem Theorem we may choose the model $\mathfrak{B}$ to be of cardinality at most 2^κ .

If $\mathfrak{A}$ has cardinality at most κ it suffices to consider types Φ of the one structure $(\mathfrak{A}_A)$, and if T is κ -stable there are at most κ many of these. Hence we adjoin at most κ many new constant symbols, so the language of Γ has cardinality at most κ and by the downward Löwenheim-Skolem Theorem we may choose $\mathfrak{B}$ to be also of cardinality at most κ . $\square$

Theorem 7.4.8 *For any countable complete theory T and any infinite cardinal κ ,*

- (i) *T has a κ^+ -saturated model of cardinality at most 2^κ ;*
- (ii) *if T is κ -stable, then T has a saturated model of cardinality κ^+ .*

Proof. For (i) and the implication $(\Leftarrow)$ of (ii), let $\mathfrak{A}_0$ be any model of T of cardinality at most κ . By iterating the construction of the preceding proposition, there exists an elementary chain $\langle \mathfrak{A}_\sigma : \sigma < \kappa^+ \rangle$ of models of T such that for all $\sigma < \kappa^+$, $\mathfrak{A}_{\sigma+1}$ is κ^+ -saturated over $\mathfrak{A}_\sigma$. Let $\mathfrak{B}$ be the union of this chain. For any $X \subseteq B$ of cardinality at most κ and any type Φ of $(\mathfrak{B}_X)$, there is some $\sigma < \kappa^+$ such that $X \subseteq A_\sigma$. Since $\mathfrak{A}_\sigma \prec \mathfrak{B}$, $\text{Th}((\mathfrak{A}_\sigma)_X) = \text{Th}(\mathfrak{B}_X)$ so Φ is also a type of $(\mathfrak{A}_\sigma)_X$, and thus is realized in $(\mathfrak{A}_{\sigma+1})_X$. But then, since $\mathfrak{A}_{\sigma+1} \prec \mathfrak{B}$, Φ is also realized in $\mathfrak{B}_X$.

With no further hypotheses, each $\mathfrak{A}_\sigma$ may be chosen to have cardinality at most 2^κ and hence $\mathfrak{B}$ has cardinality at most $\kappa^+ \times 2^\kappa = 2^\kappa$. If T is κ -stable, then by part (iv) of the proposition, each $\mathfrak{A}_\sigma$ may be chosen to be of cardinality at most κ , hence also $\mathfrak{B}$ is of cardinality at most $\kappa^+ \times \kappa = \kappa^+$. By Proposition 7.4.2, $\mathfrak{B}$ has cardinality exactly κ^+ . $\square$

Proposition 7.4.9 *For any countable theory T , if T is ω -stable, then*

- (i) *for every inessential extension T^+ of T and every $m > 0$, $\mathcal{L}_{T^+}^m$ is atomic and contains no perfect set;*
- (ii) *for every infinite cardinal κ , T is κ -stable.*

Proof. For (i), suppose that there exists an inessential extension T^+ such that $\mathcal{L}_{T^+}^m$ contains a perfect set. By Theorem 7.2.35, there exists a countable subtheory T^* such that $S^m(T^*)$ is uncountable. By the proof of Theorem 7.2.35 we may assume that $T \subseteq T^*$ — that is, T^* is an $\aleph_0$ -inessential extension of T . Hence T is not ω -stable. That $\mathcal{L}_{T^*}^m$ is atomic now follows by Proposition 7.2.21.

For (ii), if for some κ , T is not κ -stable, then there exists a κ -inessential extension T^+ of T and $m > 0$ such that $S^m(T^+)$ has cardinality greater than κ . Since $\mathcal{L}_{T^+}^m$ has cardinality κ , the conclusion follows from Theorem 7.2.36 as in the first part. $\square$

Theorem 7.4.10 *For any countable complete ω -stable theory T , any infinite regular cardinal κ and any cardinal $\lambda \geq \kappa$, T has a κ -saturated model of cardinality λ .*

Proof. Let $\mathfrak{A}_0$ be any model of T of cardinality λ ; we construct an elementary chain $\langle \mathfrak{A}_\sigma : \sigma < \kappa \rangle$ of models of T of cardinality λ as follows. For limit σ , $\mathfrak{A}_\sigma$ is the union of $\mathfrak{A}_\tau$ for $\tau < \sigma$; since $\sigma \leq \lambda$, this preserves the cardinality λ . Since T is ω -stable, by the preceding proposition, T is also λ -stable, so under the induction hypothesis that $\text{Card}(\mathfrak{A}_\sigma) = \lambda$ we may choose, by Proposition 7.4.7, $\mathfrak{A}_{\sigma+1}$ to be an elementary extension of $\mathfrak{A}_\sigma$ of cardinality λ which is λ -saturated over $\mathfrak{A}_\sigma$.

Let $\mathfrak{B} = \bigcup_{\sigma < \kappa} \mathfrak{A}_\sigma$; again because $\kappa \leq \lambda$, $\mathfrak{B}$ has cardinality λ , and we need only show that $\mathfrak{B}$ is κ -saturated. By the regularity of κ , for any set $X \subseteq B$ of cardinality less than κ there exists $\sigma < \kappa$ such that $X \subseteq A_\sigma$. Since $\mathfrak{A}_\sigma \prec \mathfrak{B}$, every type of $\mathfrak{B}_X$ is also one of $(\mathfrak{A}_\sigma)_X$ and is thus realized in $(\mathfrak{A}_{\sigma+1})_X$. Since $\mathfrak{A}_{\sigma+1} \prec \mathfrak{B}$, the same elements realize the type in $\mathfrak{B}_X$. $\square$

Definition 7.4.11 For any structures $\mathfrak{A}$ and $\mathfrak{B}$ and any infinite cardinal κ ,

- (i) a set $\mathcal{E}$ of partial embeddings of $\mathfrak{A}$ into $\mathfrak{B}$ has the **κ -back-and-forth property** iff it has the back-and-forth property (Definition 7.1.6) and is closed under unions of subsets $\mathcal{D}$ such that $\bigcup \mathcal{D}$ is of cardinality less than κ and $\mathcal{D}$ is linearly ordered by inclusion;
- (ii) $\mathfrak{A}$ and $\mathfrak{B}$ are **κ -partially isomorphic** — in symbols, $\mathfrak{A} \simeq_{\mathfrak{p}, \kappa} \mathfrak{B}$ — iff there exists a non-empty set of partial embeddings from $\mathfrak{A}$ into $\mathfrak{B}$ which has the κ -back-and-forth property.

Lemma 7.4.12 *For any structures $\mathfrak{A}$ and $\mathfrak{B}$ and infinite cardinal κ , if a set $\mathcal{E}$ of partial embeddings of $\mathfrak{A}$ into $\mathfrak{B}$ has the κ -back-and-forth property, then so does the set*

$\{ \eta : \eta \text{ is a subfunction of some member of } \mathcal{E} \text{ of cardinality less than } \kappa \}.$

Proof. Immediate. $\square$

Theorem 7.4.13 *For any infinite cardinal κ and any structures $\mathfrak{A}$ and $\mathfrak{B}$, both of cardinality at most κ ,*

$$\mathfrak{A} \simeq_{p,\kappa} \mathfrak{B} \implies \mathfrak{A} \simeq \mathfrak{B}.$$

Proof. This is closely parallel with the proof of Theorem 7.1.9. Let $\mathfrak{A}$ and $\mathfrak{B}$ be of cardinality at most κ with $\mathfrak{A} \simeq_{p,\kappa} \mathfrak{B}$, and choose enumerations $\langle a_\sigma : \sigma < \kappa \rangle$ and $\langle b_\sigma : \sigma < \kappa \rangle$ of their universes and a non-empty set $\mathcal{E}$ of partial embeddings with the κ -back-and-forth property. By the lemma we may assume that elements of $\mathcal{E}$ are of cardinality less than κ and that $\mathcal{E}$ contains all subfunctions of its elements. Let $\eta_0 \in \mathcal{E}$ be the empty partial embedding; we construct an increasing chain $\langle \eta_\sigma : \sigma < \kappa \rangle$ of elements of $\mathcal{E}$ with the property that η_σ has cardinality $\text{Card}(\sigma)$. Given η_σ with σ even, let $\eta_{\sigma+1}$ be any minimal extension of η_σ such that $a_\sigma \in \text{Dom}(\eta_\sigma)$. If σ is odd, choose $\eta_{\sigma+1}$ such that $b_\sigma \in \text{Im}(\eta_{\sigma+1})$. For limit σ , set $\eta_\sigma = \bigcup \{ \eta_\tau : \tau < \sigma \}$, so also in this case $\eta_\sigma \in \mathcal{E}$. It is straightforward to check that the union η of this chain is an isomorphism of $\mathfrak{A}$ with $\mathfrak{B}$. $\square$

Proposition 7.4.14 *For any structures $\mathfrak{A}$ and $\mathfrak{B}$ and any infinite cardinal κ , if both $\mathfrak{A}$ and $\mathfrak{B}$ are κ -saturated, then $\mathfrak{A} \simeq_{p,\kappa} \mathfrak{B}$; hence if both are also of cardinality κ , then $\mathfrak{A} \simeq \mathfrak{B}$.*

Proof. This is closely parallel with the proof of Proposition 7.3.30. Let $\mathcal{E}$ denote the set of partial elementary embeddings from $\mathfrak{A}$ into $\mathfrak{B}$ of cardinality less than κ . For any $\eta \in \mathcal{E}$, let $X = \text{Dom}(\eta)$ and $Y = \text{Im}(\eta)$, and fix $c \in A$. $\text{Tp}^{\mathfrak{A}_X}(c)$ is a type of $\mathfrak{A}_X$ and since $\mathfrak{A}_X \equiv \mathfrak{B}_Y$, also a type of $\mathfrak{B}_Y$. Since Y is of cardinality less than κ , this type must also be realized in $\mathfrak{B}_Y$, say by d . Then the extension of η to ζ such that $\zeta(c) = d$ is a member of $\mathcal{E}$ as required. The argument for the backward extension property is parallel. $\square$

Theorem 7.4.15 *For any countable complete theory T , if T is ω -stable, then for every infinite cardinal κ ,*

$$T \text{ is } \kappa\text{-categorical} \iff \text{every model of } T \text{ of cardinality } \kappa \text{ is saturated.}$$

Proof. Suppose first that every model of T of cardinality κ is saturated. Then by the preceding proposition, any two models of T of cardinality κ are isomorphic; hence T is κ -categorical.

Suppose now that T is κ -categorical, and let $\mathfrak{A}$ be the unique (up to isomorphism) model of T of cardinality κ . If κ is a regular cardinal, then by Theorem 7.4.10 there exists a saturated model of T of cardinality κ , so $\mathfrak{A}$ is saturated. Since any model $\mathfrak{B}$ of cardinality κ is isomorphic to $\mathfrak{A}$, also $\mathfrak{B}$ is saturated. Suppose now that κ is a singular cardinal. By Theorem 7.4.10, for each cardinal $\mu < \kappa$, there exists a model of T of cardinality μ which is

μ^+ -saturated. Hence for each such μ , $\mathfrak{A}$ is μ^+ -saturated. But this implies easily that $\mathfrak{A}$ is in fact κ -saturated. $\square$

We turn now to the existence of uncountable atomic models. In the countable case, we were able to construct atomic models easily using the Omitting Types Theorem 7.3.13. This method will not suffice to construct the uncountable atomic models we need here of the form $\mathfrak{A}_X$ with uncountable X because Omitting Types does not generally hold for uncountable languages (Exercise 7.3.41).

Remark 7.4.16 Although $\mathcal{L}_T^n$ is defined in terms of formulas with free variables among the fixed list $\mathbf{v} = v_0, \dots, v_{n-1}$, an isomorphic structure is obtained using formulas with free variables in any other list $\mathbf{x} = x_0, \dots, x_{n-1}$ of distinct variables. We use this implicitly in the following two results.

Lemma 7.4.17 *For any countable complete theory T , any formulas $\theta(\mathbf{x}, \mathbf{y})$ and $\chi(\mathbf{x}, \mathbf{y}, \mathbf{z})$ with $m + n$, respectively $m + n + p$, free variables, and any inessential extension T^+ of T containing constant symbols $\dot{\mathbf{a}}$ and $\dot{\mathbf{b}}$ such that $T^+ \models \theta(\dot{\mathbf{a}}, \dot{\mathbf{b}})$, if $\theta(\mathbf{x}, \mathbf{y})$ is an atom of $\mathcal{L}_T^{m+n}$ and $\chi(\dot{\mathbf{a}}, \dot{\mathbf{b}}, \mathbf{z})$ is an atom of $\mathcal{L}_{T^+}^p$, then $\theta' := \exists \mathbf{y}[\theta(\mathbf{x}, \mathbf{y}) \wedge \chi(\mathbf{x}, \mathbf{y}, \mathbf{z})]$ is an atom of $\mathcal{L}_T^{m+p}$.*

Proof. Suppose towards a contradiction that θ' is not an atom. Since clearly θ' is T -consistent, by Corollary 7.2.34 there is a formula $\psi(\mathbf{x}, \mathbf{z})$ such that both $\theta' \wedge \psi$ and $\theta' \wedge \neg\psi$ are T -consistent. Then both $\theta \wedge \chi \wedge \psi$ and $\theta \wedge \chi \wedge \neg\psi$ are T -consistent, so both $\theta \wedge \exists \mathbf{z}[\chi \wedge \psi]$ and $\theta \wedge \exists \mathbf{z}[\chi \wedge \neg\psi]$ are T -consistent. Since θ is an atom — that is, a complete formula — it follows that

$$T \models \theta \rightarrow \exists \mathbf{z}[\chi \wedge \psi] \quad \text{and} \quad T \models \theta \rightarrow \exists \mathbf{z}[\chi \wedge \neg\psi].$$

Then, since $T^+ \models \theta(\dot{\mathbf{a}}, \dot{\mathbf{b}})$,

$$T^+ \models \exists \mathbf{z}[\chi(\dot{\mathbf{a}}, \dot{\mathbf{b}}, \mathbf{z}) \wedge \psi(\dot{\mathbf{a}}, \mathbf{z})] \quad \text{and} \quad T^+ \models \exists \mathbf{z}[\chi(\dot{\mathbf{a}}, \dot{\mathbf{b}}, \mathbf{z}) \wedge \neg\psi(\dot{\mathbf{a}}, \mathbf{z})].$$

Hence both $\chi(\dot{\mathbf{a}}, \dot{\mathbf{b}}, \mathbf{z}) \wedge \psi(\dot{\mathbf{a}}, \mathbf{z})$ and $\chi(\dot{\mathbf{a}}, \dot{\mathbf{b}}, \mathbf{z}) \wedge \neg\psi(\dot{\mathbf{a}}, \mathbf{z})$ are T^+ -consistent, which contradicts the hypothesis that $\chi(\dot{\mathbf{a}}, \dot{\mathbf{b}}, \mathbf{z})$ is an atom. $\square$

Theorem 7.4.18 *For any ω -stable countable theory T , every complete inessential extension of T has an atomic model. More generally, for any model $\mathfrak{B}$ of T , any infinite cardinal $\kappa \leq \text{Card}(\mathfrak{B})$ and any set $X \subseteq B$ of cardinality at most κ , there exists a model $\mathfrak{A} \prec \mathfrak{B}$ of cardinality κ such that $X \subseteq A$, and $\mathfrak{A}_X$ is atomic.*

Proof. Assume the hypotheses of the second assertion; without loss of generality we may assume that $\mathfrak{B}$ is of cardinality κ . We shall construct a sequence $\langle b_\sigma : \sigma < \kappa \rangle$ of elements of B such that if for all $\sigma < \kappa$ we set

$$A_\sigma := X \cup \{b_\tau : \tau < \sigma\},$$

we have

- (1) $\text{Tp}^{\mathfrak{B}_{A_\sigma}}(b_\sigma)$ is principal;
- (2) for any L^{A_σ} -formula $\phi(z)$, if $\models_{\mathfrak{B}_{A_\sigma}} \exists z \phi(z)$, then for some $\tau \geq \sigma$,
 $\models_{\mathfrak{B}_{A_\sigma}} \phi(z)[b_\tau]$.

Let Γ denote the set of all L^B -formulas $\phi(z)$ such that $\models_{\mathfrak{B}_B} \exists z \phi(z)$ and fix a well-ordering $<_\Gamma$ of Γ of order-type κ . Given $\langle b_\tau : \tau < \sigma \rangle$ we choose b_σ as follows. Let

$$\Delta_\sigma := \{ \phi \in \Gamma : \phi \text{ is an } L^{A_\sigma}\text{-formula but for all } \tau < \sigma, \quad \models_{\mathfrak{B}_{A_\sigma}} \neg \phi[b_\tau] \}.$$

If $\Delta_\sigma = \emptyset$, let ϕ_σ be $x \doteq x$; otherwise, let ϕ_σ be the $<_\Gamma$ -least element of Δ_σ . Since T is ω -stable, the Lindenbaum algebra $\mathcal{L}_{\mathfrak{B}_{A_\sigma}}^1$ is atomic (Proposition 7.4.9), so there is an atom $\chi_\sigma(z)$ below ϕ_σ — that is, $\models_{\mathfrak{B}_{A_\sigma}} \forall z [\chi_\sigma \rightarrow \phi_\sigma]$. Choose b_σ to be any element of B such that $\models_{\mathfrak{B}_{A_\sigma}} \chi_\sigma[b_\sigma]$, so also $\models_{\mathfrak{B}_{A_\sigma}} \phi_\sigma[b_\sigma]$.

Property (1) is immediate by Corollary 7.2.30, since an atom generates a principal type. Towards (2), if $\models_{\mathfrak{B}_{A_\sigma}} \exists z \phi(z)$, then for every $\tau \geq \sigma$, ϕ is an L^{A_τ} -formula, so if $\models_{\mathfrak{B}_{A_\sigma}} \neg \phi[b_\tau]$, $\phi \in \Delta_\tau$. But if ϕ is in the v -th position in $<_\Gamma$, then for some $\tau < \sigma + v$, ϕ will be the least element of Δ_τ so will be satisfied by b_τ .

Let $A := \{ b_\sigma : \sigma < \kappa \}$; clearly $\text{Card}(A) = \kappa$. We first note that $X \subseteq A$, since for any $a \in X$, we have $\models_{\mathfrak{B}_X} \exists x (a \doteq x)$, so by (2), for some $\tau \geq 0$, $b_\tau = a$. Similarly, A contains all the distinguished elements of $\mathfrak{B}$ and is closed under the functions $f^{\mathfrak{B}}$ of $\mathfrak{B}$: for any $\mathbf{a} \in A$, choose σ such that $\mathbf{a} \in A_\sigma$. Then

$$\models_{\mathfrak{B}_{A_\sigma}} \exists x (f^{\mathfrak{B}} \mathbf{a} \doteq x) \quad \text{so for some } \tau \geq \sigma, \quad f^{\mathfrak{B}}(\mathbf{a}) = b_\tau.$$

Hence A is the universe of a unique substructure $\mathfrak{A} \subseteq \mathfrak{B}$. To see that in fact $\mathfrak{A} \prec \mathfrak{B}$ we use Tarski's Criterion 2.3.33: if ϕ is an L^A -formula such that $\models_{\mathfrak{B}_A} \exists x \phi$, then by (2), for some $b \in A$, $\models_{\mathfrak{B}_A} \phi[b]$.

Finally we verify that $\mathfrak{A}_X$ is atomic. For any $\mathbf{a} := b_{\sigma_0}, \dots, b_{\sigma_m} \in A$, let the *rank* of $\mathbf{a}$ be $|\mathbf{a}| := \max\{\sigma_0, \dots, \sigma_{m-1}\}$. We proceed by induction on rank to show that for each $\mathbf{a} \in A$ there exists an L^X formula θ which is an atom of $\mathcal{L}_{\mathfrak{A}_X}^m$ and $\models_{\mathfrak{A}_X} \theta[\mathbf{a}]$. Assume as induction hypothesis that this holds for all $\mathbf{a}$ of rank $< \sigma$. A typical sequence of rank σ is of the form $\mathbf{a}, b_\sigma$ with $|\mathbf{a}| < \sigma$. The formula $\chi_\sigma(z) = \chi_\sigma(\dot{\mathbf{a}}, \dot{\mathbf{b}}, z)$ chosen in the construction may include in addition to the names $\dot{\mathbf{a}}$ also names $\dot{\mathbf{b}} = \dot{b}_{\tau_0}, \dots, \dot{b}_{\tau_n}$ for other elements of A_σ , but still $|\mathbf{a}, \mathbf{b}| < \sigma$, so by the induction hypothesis, there exists an atom

$\theta(\mathbf{x}, \mathbf{y})$ of $\mathcal{L}_{\mathfrak{A}_X}^{m+n}$ such that $\models_{\mathfrak{A}_X} \theta[\mathbf{a}, \mathbf{b}]$. By construction $\chi_\sigma(\dot{\mathbf{a}}, \dot{\mathbf{b}}, z)$ is an atom of $\mathcal{L}_{\mathfrak{A}_{A_\sigma}}^1$, so by the lemma, $\theta' := \exists \mathbf{y}[\theta(\mathbf{x}, \mathbf{y}) \wedge \chi(\mathbf{x}, \mathbf{y}, z)]$ is an atom of $\mathcal{L}_{\mathfrak{A}_X}^{m+1}$. Clearly $\models_{\mathfrak{A}_X} \theta'[\mathbf{a}, b_\sigma]$ as desired. $\square$

Exercises

Exercise 7.4.19 Modify the proof of Theorem 7.4.18 to give another proof of one direction of Proposition 7.3.18: for any countable complete theory T , if for all $n > 0$, $\mathcal{L}_T^n$ is an atomic Boolean algebra then T has a countable atomic model.

Exercise 7.4.20 For any complete theory T , any model $\mathfrak{A}$ of T and any $X \subseteq A$, $\mathfrak{A}$ is **prime over** X iff for any model $\mathfrak{B}$ of T , any partial elementary embedding $\eta : \mathfrak{A} \hookrightarrow_{\text{p}} \mathfrak{B}$ with $\text{Dom}(\eta) = X$ has an extension to an elementary embedding $\zeta : \mathfrak{A} \hookrightarrow \mathfrak{B}$. Prove that the model constructed in the proof of Theorem 7.4.18 is also prime over X .

Exercise 7.4.21 Suppose that T is a countable complete theory and κ an infinite cardinal such that for every model $\mathfrak{B}$ and every $X \subseteq \mathfrak{B}$ with $\text{Card}(X) \leq \kappa$, $S^1(\mathfrak{B}_X)$ has cardinality at most κ . Show that T is κ -stable.

Exercise 7.4.22 A linear ordering $\mathfrak{A} = (A, <_A)$ is κ -dense iff for any $X, Y \subseteq A$ such that $\text{Card}(X), \text{Card}(Y) < \kappa$ and $(\forall a \in X)(\forall b \in Y) a <_A b$ there exists $c \in A$ such that $(\forall a \in X)(\forall b \in Y) a <_A c <_A b$. Prove that if $\mathfrak{A}$ is a dense linear ordering without endpoints, then for any infinite cardinal κ , $\mathfrak{A}$ is κ -dense iff $\mathfrak{A}$ is κ -saturated.

Exercise 7.4.23 Let $\langle \mathfrak{A}_i : i \in \omega \rangle$ be a family of L -structures and $\mathcal{U}$ a non-principal ultrafilter on ω . Show that the ultraproduct $\mathfrak{A} := \prod_{\mathcal{U}} \mathfrak{A}_i$ is $\aleph_1$ -saturated.

Hint: For X a countable subset of A , choose $\hat{F}_a = a$ for each $a \in X$, set $X^i := \{F_a(i) : a \in X\} \subseteq A_i$, and use Łoś's Theorem 3.3.19 to show that for any L^X -sentence θ ,

$$\models_{\mathfrak{A}_X} \theta \iff \{i : \models_{\mathfrak{A}_{i, X^i}} \theta^i\} \in \mathcal{U},$$

where θ^i is θ with each occurrence of a constant symbol $\dot{a}$ replaced by $F_a(i)$. Now if $\Phi = \{\phi_k(x) : k \in \omega\}$ is a 1-type of $\text{Th}(\mathfrak{A}_X)$, conclude that for all k ,

$$\left\{ i : \models_{\mathfrak{A}_{i, X^i}} \exists x [\phi_0^i(x) \wedge \dots \wedge \phi_k^i(x)] \right\} \in \mathcal{U},$$

and use this to define a function G such that $\hat{G}$ realizes ϕ in $\mathfrak{A}$.

Exercise 7.4.24 Let L be the language with a unary relation symbol p_n for each $n \in \omega$, and T the L -theory generated by the axioms

$$\phi_{X,Y} : \exists x \left[\bigwedge_{m \in X} p_m x \wedge \bigwedge_{n \in Y} \neg p_n x \right]$$

for all disjoint pairs of finite sets $X, Y \subseteq \omega$.

- (i) Show that T is complete (Exercise 2.4.57).
- (ii) Describe $\mathcal{L}_T^1$ and $S^1(T)$ and draw conclusions about the countable models of T ; is T ω -stable or $\aleph_0$ -categorical?
- (iii) Show that for $\kappa \geq 2^{\aleph_0}$, T is κ -stable (such a T is called **superstable**).

Exercise 7.4.25 Let L be the language with a binary relation symbol $\dot{E}_n$ for each $n \in \omega$ and T the L -theory generated by the axioms which say

- (i) each $\dot{E}_n$ is an equivalence relation;
- (ii) each $\dot{E}_n$ has infinitely many equivalence classes;
- (iii) for all n , $\forall x_0 \dots \forall x_n \exists y \left(\dot{E}_0 x_0 y \wedge \dots \wedge \dot{E}_n x_n y \right)$.

Show that for all infinite cardinals κ , T is κ -stable iff $\kappa^{\aleph_0} = \kappa$. Hence T is **stable** (κ -stable for some cardinal κ) but not superstable.

7.5 Morley's Theorem

In 1954 Łoś observed that all known countable theories T fall into one of the following four categories and conjectured that these are the only possibilities:

- $(+, -)$ T is $\aleph_0$ -categorical but for all uncountable cardinals κ , T is not κ -categorical.
- $(-, +)$ T is not $\aleph_0$ -categorical but for all uncountable cardinals κ , T is κ -categorical.
- $(+, +)$ For all infinite cardinals κ , T is κ -categorical.
- $(-, -)$ For all infinite cardinals κ , T is not κ -categorical.

We have seen examples of each of these categories above, and we only mention them now (see also Exercise 2.5.27).

- $(+, -)$ $T_{(\text{DLO})}$, the theory of dense linear orderings without first or last element (Cantor's Theorem 2.4.45).

- ($-$, $+$) The theory T_{ACF} of algebraically closed fields of characteristic 0 or a prime p (Theorem 3.4.31); $T_{\text{Sc},0}$, the theory of the structure $(\omega, \text{Sc}, 0)$ (Corollary 2.5.12).
- ($+$, $+$) Pure identity theory, the logical validities in the language $L_=$ with no non-logical symbols (Corollary 2.5.3).
- ($-$, $-$) The theory of the standard structure Ω of arithmetic; the logical validities in a language with a single unary relation symbol p .

Morley proved the correctness of this conjecture in 1965. We give here a simplification of his proof due largely to Baldwin and Lachlan using ideas of Keisler and Marsh. It suffices to establish the two implications

$$\begin{aligned} \exists \kappa > \aleph_0 [T \text{ is } \kappa\text{-categorical}] &\implies T \text{ is } \aleph_1\text{-categorical} \\ &\implies \forall \kappa > \aleph_0 [T \text{ is } \kappa\text{-categorical}], \end{aligned}$$

which we shall refer to as the **Downward** and **Upward** versions of the theorem.

The Downward version depends on the following two notions.

Definition 7.5.1 For any structure $\mathfrak{A}$,

- (i) $\mathfrak{A}$ is **modest** iff for each countable set $X \subseteq A$, the expanded structure $\mathfrak{A}_X$ realizes at most countably many types of its theory; otherwise $\mathfrak{A}$ is **immodest**;
- (ii) a structure $\mathfrak{B}$ is **modest over** $\mathfrak{A}$ iff $\mathfrak{A} \prec \mathfrak{B}$ and for each countable set $X \subseteq A$, every type of $\mathfrak{B}_X$ realized in $\mathfrak{B}_X$ is already realized in $\mathfrak{A}_X$.

It is immediate that every model of a complete ω -stable theory T is modest, but in fact the property is more widespread.

First Modesty Theorem 7.5.2 *For any countable theory T with infinite models and any infinite cardinal κ , there exists a modest model of T of cardinality κ .*

We shall postpone the proof of this result to focus on its use. In contrast we have

Proposition 7.5.3 *For any countable theory T , if T is not ω -stable, then for each uncountable cardinal κ , there exists a model of T of cardinality κ which is not modest.*

Proof. A straightforward application of compactness. $\square$

Corollary 7.5.4 *For any countable theory T , if T is κ -categorical for some uncountable cardinal κ , then T is ω -stable.*

Proof. If T is not ω -stable, then for each uncountable κ has both modest and immodest models of cardinality κ ; these are not isomorphic, so T is not κ -categorical. $\square$

The rest of the proof of the Downward version relies on the

Second Modesty Theorem 7.5.5 *For any ω -stable countable theory T , any uncountable model $\mathfrak{A}$ of T and any cardinal $\kappa \geq \text{Card}(\mathfrak{A})$, there exists model $\mathfrak{B}$ of cardinality κ which is modest over $\mathfrak{A}$.*

Again we postpone the proof.

Downward Morley Theorem 7.5.6 *For any countable theory T , if for some uncountable cardinal κ , T is κ -categorical, then T is $\aleph_1$ -categorical.*

Proof. Assume that for some uncountable cardinal κ , T is κ -categorical. Let $\mathfrak{A}$ be any model of T of cardinality $\aleph_1$; by Theorem 7.4.15 it will suffice to show that $\mathfrak{A}$ is saturated. Fix a countable set $X \subseteq A$. Since by the corollary T is ω -stable, there exists a model $\mathfrak{B}$ of cardinality κ modest over $\mathfrak{A}$. By Theorem 7.4.15, $\mathfrak{B}$ is saturated, in particular $\aleph_1$ -saturated, so every type of $\mathfrak{B}_X$ is realized in $\mathfrak{B}_X$, hence in $\mathfrak{A}_X$. But since $\mathfrak{A}_X \equiv \mathfrak{B}_X$, these are exactly the types of $\mathfrak{A}_X$. $\square$

The proof of the Upward version will use the following notion.

Definition 7.5.7 A pair $(\mathfrak{A}, \theta(x))$ is a **two-cardinal pair** iff $\theta^{\mathfrak{A}}$ is infinite but of cardinality less than that of $\mathfrak{A}$.

Two-Cardinal Theorem 7.5.8 *For any structure $\mathfrak{A}$ and formula θ , if $(\mathfrak{A}, \theta(x))$ is a two-cardinal pair, then there exist structures $\mathfrak{B}$ and $\mathfrak{C}$ such that*

- (i) $\mathfrak{B} \prec \mathfrak{A}$ and $\mathfrak{B}$ is countable;
- (ii) $\mathfrak{B} \prec \mathfrak{C}$ and $\text{Card}(\mathfrak{C}) = \aleph_1$;
- (iii) $\theta^{\mathfrak{B}} = \theta^{\mathfrak{C}}$.

We again postpone the proof.

Upward Morley Theorem 7.5.9 *For any countable theory T , if T is $\aleph_1$ -categorical, then for all uncountable cardinals κ , T is κ -categorical.*

Proof. We assume that T is $\aleph_1$ -categorical, fix an arbitrary uncountable cardinal κ and suppose towards a contradiction that T is not κ -categorical so by Theorem 7.4.15 has a model $\mathfrak{A}$ of cardinality κ which is not saturated. Hence there exists a set $X \subseteq A$ of cardinality less than κ and a type Φ of $\mathfrak{A}_X$

which is not realized in $\mathfrak{A}_X$. By the method of the proof of Lemma 7.3.31 we may assume that Φ is a 1-type. Since X has cardinality less than κ , so does Φ and we may choose a set $I \subseteq A$ of cardinality less than κ such that there is a bijective correspondence $i \mapsto \phi_i$ of I onto Φ . Let R and S be the binary relations on A defined by

$$\begin{aligned} R(i, a) &\iff i \in I, a \in X, \text{ and } \dot{a} \text{ occurs in } \phi_i; \\ S(i, b) &\iff i \in I \text{ and } \models_{\mathfrak{A}_X} \phi_i[b]. \end{aligned}$$

Let $\mathfrak{A}'$ denote the structure $(\mathfrak{A}, I, R, S)$; clearly $(\mathfrak{A}', \dot{I}x)$ is a two-cardinal pair.

By the Two-Cardinal Theorem there exist structures

$$\mathfrak{B}' = (\mathfrak{B}, I \cap B, R \cap B \times B, S \cap B \times B) \quad \text{and} \quad \mathfrak{C}' = (\mathfrak{C}, \dot{I}^{\mathfrak{C}'}, \dot{R}^{\mathfrak{C}'}, \dot{S}^{\mathfrak{C}'})$$

such that

- (i) $\mathfrak{B}' \prec \mathfrak{A}'$ and $\mathfrak{B}$ is countable;
- (ii) $\mathfrak{B}' \prec \mathfrak{C}'$ and $\text{Card}(\mathfrak{C}) = \aleph_1$;
- (iii) $I \cap B = I \cap C$.

Let Φ_B denote the type $\{\phi_i : i \in I \cap B\}$. We aim to show that Φ_B is a type of $\mathfrak{C}_B$ which is not realized in $\mathfrak{C}_B$; it follows that $\mathfrak{C}$ is not saturated and hence by Theorem 7.4.15 that T is not $\aleph_1$ -categorical, contrary to hypothesis.

We verify first that the constant symbols which occur in the members of Φ_B are names for members of B . For $i \in I \cap B$, let $\dot{a}_0, \dots, \dot{a}_{n-1}$ be the constant symbols which occur in ϕ_i . Then

$$\models_{\mathfrak{A}'} \exists y_0 \cdots \exists y_{n-1} \left(\bigwedge_{j < k < n} \neg y_j \dot{=} y_k \wedge \bigwedge_{j < n} \dot{R}(x, y_j) \right) [i],$$

so as $\mathfrak{B}' \prec \mathfrak{A}'$, this same formula is satisfied by i also in $\mathfrak{B}'$. It follows that $a_0, \dots, a_{n-1} \in B$.

To verify that Φ_B is a type of $\mathfrak{C}_B$, we need (using Proposition 7.2.27) to show that for any finitely many $i_0, \dots, i_p \in I$, the formula $\exists y \bigwedge_{k \leq p} \phi_{i_k}(y)$ is true in $\mathfrak{C}_B$. First note that because for any $i \in I$

$$\models_{\mathfrak{A}'_B} \forall y [\phi_i(y) \leftrightarrow S(x, y)] [i],$$

the same formula holds in $\mathfrak{B}'_B$ and thus in $\mathfrak{C}'_B$ for $i \in I \cap B$. Hence since

$$\models_{\mathfrak{A}'} \forall x_0 \cdots \forall x_p \left[\bigwedge_{k \leq p} \dot{I}x_k \rightarrow \exists y \bigwedge_{k \leq p} \dot{S}(x_k, y) \right],$$

the same sentence is true in $\mathfrak{C}'$, and it follows that $\models_{\mathfrak{C}_B} \exists y \bigwedge_{k \leq p} \phi_{i_k}(y)$.

Finally, suppose towards a contradiction that an element c realizes Φ_B in $\mathfrak{C}_B$. Then for all $i \in I \cap B$, $\models_{\mathfrak{C}_B} \phi_i[c]$, so $\models_{\mathfrak{C}} \dot{S}(x, y)[i, c]$. Hence $\models_{\mathfrak{C}} \exists y \forall x [\dot{I}x \rightarrow \dot{S}(x, y)]$. But then this same sentence is true in $\mathfrak{A}'$, which implies that Φ is realized in $\mathfrak{A}$, contrary to hypothesis. $\square$

To complete the proof of Morley's Theorem, it remains to fill in the proofs of the First and Second Modesty and Two-Cardinal Theorems.

Proof. (of the Two-Cardinal Theorem 7.5.8) Let $(\mathfrak{A}, \theta)$ be a two-cardinal pair: $\theta^{\mathfrak{A}}$ is infinite of cardinality less than that of $\mathfrak{A}$. By replacing $\mathfrak{A}$ by an elementary submodel if necessary, we may assume that for some infinite cardinal λ , $\theta^{\mathfrak{A}}$ is of cardinality λ and $\mathfrak{A}$ is of cardinality λ^+ . Fix a well-ordering $\leq$ of A in type λ^+ such that $\theta^{\mathfrak{A}}$ forms an initial segment. In the following we write $\exists^{\text{cf}} y$ as an abbreviation for $\forall z (\exists y \dot{\geq} z)$ (in the language of $(\mathfrak{A}, \leq)$). Note that if $\models_{(\mathfrak{A}, \leq)} \exists^{\text{cf}} y \psi(y)$, then $\psi^{\mathfrak{A}}$ has cardinality λ^+ .

Just by cardinality considerations, for any formula $\phi(x, y)$,

$$(1) \quad \models_{(\mathfrak{A}, \leq)} \exists^{\text{cf}} y \exists x [\theta(x) \wedge \phi(x, y)] \rightarrow \exists x [\theta(x) \wedge \exists^{\text{cf}} y \phi(x, y)].$$

Let $(\mathfrak{B}, \leq)$ be any denumerable elementary submodel of $(\mathfrak{A}, \leq)$ — to minimize notational complexity we denote here and below the restrictions of the ordering $\leq$ by the same symbol. We show first that there exists a denumerable proper elementary extension $(\mathfrak{B}', \leq)$ of $(\mathfrak{B}, \leq)$ such that $\theta^{\mathfrak{B}} = \theta^{\mathfrak{B}'}$. By the First Extension Lemma 3.4.21 it suffices to show that there exists a model of the theory T^+ generated by $\text{Th}(\mathfrak{B}_B, \leq) \cup \{\dot{b} < \dot{c} : b \in B\}$ (for a new constant symbol $\dot{c}$) which omits the type $\Phi := \{\theta\} \cup \{\neg x \dot{=} \dot{b} : b \in \theta^{\mathfrak{B}}\}$, so by the Omitting Types Theorem 7.3.13 it suffices to show that Φ is non-principal over T^+ .

We establish first that for any formula χ of L^B ,

$$(2) \quad T^+ \models \chi(\dot{c}) \iff \models_{(\mathfrak{B}, \leq)} \neg \exists^{\text{cf}} y \neg \chi(y).$$

Note that the right-hand side may be expressed in the following two ways:

$$\begin{aligned} \models_{(\mathfrak{B}, \leq)} \neg \exists^{\text{cf}} y \neg \chi(y) &\iff \models_{(\mathfrak{B}, \leq)} \exists z (\forall y \dot{\geq} z) \chi(y) \\ &\iff \text{for some } b \in B, \quad \text{Th}((\mathfrak{B}, \leq)_B) \models (\forall y \dot{\geq} \dot{b}) \chi(y). \end{aligned}$$

Suppose first that this holds. Then since $T^+ \models \dot{b} < \dot{c}$, it follows that $T^+ \models \chi(\dot{c})$. Conversely, if $T^+ \models \chi(\dot{c})$, then by compactness, for some

$b_0, \dots, b_n,$

$$\begin{aligned} \text{Th}((\mathfrak{B}, \leq)_B) &\models \bigwedge_{i \leq n} \dot{b}_i < \dot{c} \rightarrow \chi(\dot{c}) \quad \text{hence} \\ \text{Th}((\mathfrak{B}, \leq)_B) &\models \forall y \left[\bigwedge_{i \leq n} \dot{b}_i < y \rightarrow \chi(y) \right], \end{aligned}$$

because $\dot{c}$ does not occur in $\text{Th}((\mathfrak{B}, \leq)_B)$. Hence if b is any element of B larger than all of $b_0, \dots, b_n$, we have $\text{Th}((\mathfrak{B}, \leq)_B) \models (\forall y \geq \dot{b})\chi(y)$ and thus $\models_{(\mathfrak{B}, \leq)} \neg \exists^{\text{cf}} y \neg \chi(y)$.

Towards a contradiction, suppose that Φ has a generator $\psi = \psi(x, \dot{c})$ (ψ may also contain some finite number of constant symbols $\dot{b}$, for $b \in \theta^{\mathfrak{B}}$). For each $b \in \theta^{\mathfrak{B}}$,

$$T^+ \models \psi(x, \dot{c}) \rightarrow \neg x \doteq \dot{b} \quad \text{hence} \quad T^+ \models \neg \psi(\dot{b}, \dot{c}),$$

so by (2),

$$\models_{(\mathfrak{B}, \leq)} \neg \exists^{\text{cf}} y \psi(\dot{b}, y).$$

Hence

$$\models_{(\mathfrak{B}, \leq)} \forall x [\theta(x) \rightarrow \neg \exists^{\text{cf}} y \psi(x, y)],$$

and by (1) and the fact that $(\mathfrak{B}, \leq) \prec (\mathfrak{A}, \leq)$,

$$\models_{(\mathfrak{B}, \leq)} \neg \exists^{\text{cf}} y \exists x [\theta(x) \wedge \psi(x, y)],$$

and by (2),

$$T^+ \models \neg \exists x [\theta(x) \wedge \psi(x, \dot{c})].$$

Since ψ is a generator of Φ and $\theta \in \Phi$, we have also $T^+ \models \psi(x, \dot{c}) \rightarrow \theta(x)$; it follows that $T^+ \models \neg \exists x \psi(x, \dot{c})$ and $\psi(x, \dot{c})$ is inconsistent with T^+ so cannot be a generator of Φ after all.

Finally, we construct the desired model $\mathfrak{C}$ by iterating this construction. Let $\mathfrak{B}_0 = \mathfrak{B}$, for limit ordinals σ , $\mathfrak{B}_\sigma = \bigcup_{\tau < \sigma} \mathfrak{B}_\tau$ and $\mathfrak{B}_{\sigma+1} = \mathfrak{B}'_\sigma$. Let $\mathfrak{C} = \mathfrak{B}_{\aleph_1}$. Since each $\mathfrak{B}_\sigma$ is denumerable but the extensions are proper at successor stages, $\mathfrak{C}$ has cardinality $\aleph_1$. Easily for all σ , $\theta^{\mathfrak{B}_\sigma} = \theta^{\mathfrak{B}}$, so $\mathfrak{C}$ is as desired. $\square$

The following notion is central to the proof of the First Modesty Theorem.

Definition 7.5.10 For any structure $\mathfrak{B}$ and $Y \subseteq B$ and any linear ordering $<_Y$ of Y , Y is a **set of indiscernibles (SOI)** for $\mathfrak{B}$ (with respect to $<_Y$) iff for any sequences $\mathbf{a} := a_0, \dots, a_n$ and $\mathbf{b} := b_0, \dots, b_n$ from Y , if $a_0 <_Y \dots <_Y a_n$ and $b_0 <_Y \dots <_Y b_n$, then

$$(\mathfrak{B}, \mathbf{a}) \equiv (\mathfrak{B}, \mathbf{b}).$$

Note that there is nothing special about considering only increasing sequences: if we write

$$\mathbf{a} \sim_Y \mathbf{b} \iff \text{for all } i < j \leq n, \quad a_i <_Y a_j \iff b_i <_Y b_j,$$

then

Corollary 7.5.11 For any structure $\mathfrak{B}$, any $Y \subseteq B$ and any linear ordering $<_Y$ of Y , if Y is an SOI for $\mathfrak{B}$, then for any sequences $\mathbf{a} \sim_Y \mathbf{b}$ from Y ,

$$(\mathfrak{B}, \mathbf{a}) \equiv (\mathfrak{B}, \mathbf{b}).$$

Proof. For any sequence $\mathbf{a}$, let π be a permutation such that $\pi(\mathbf{a})$ is the same sequence in increasing order. If $\mathbf{a} \sim_Y \mathbf{b}$, then also $\pi(\mathbf{b})$ is in increasing order. For any formula ϕ , let ϕ^π be the result of applying π to the free variables of ϕ . Then if $\mathbf{a} \sim_Y \mathbf{b}$,

$$\models_{\mathfrak{B}} \phi[\mathbf{a}] \iff \models_{\mathfrak{B}} \phi^\pi[\pi(\mathbf{a})] \iff \models_{\mathfrak{B}} \phi^\pi[\pi(\mathbf{b})] \iff \models_{\mathfrak{B}} \phi[\mathbf{b}]. \quad \square$$

Recall that the substructure $\langle Y \rangle_{\mathfrak{B}}$ of a structure $\mathfrak{B}$ generated by a set $Y \subseteq B$ is the unique substructure of $\mathfrak{B}$ whose universe is the smallest set including Y and all distinguished elements of $\mathfrak{B}$, and closed under all of the functions of $\mathfrak{B}$. Generally, $\langle Y \rangle_{\mathfrak{B}}$ is not an *elementary* substructure of $\mathfrak{B}$, but it will be if the language is sufficiently rich.

Recall from Definition 2.6.25 the notion of a Skolem complete theory T and the basic facts about it: (1) if $\mathfrak{B}$ is a model of T and $\mathfrak{A} \subseteq \mathfrak{B}$, then $\mathfrak{A} \prec \mathfrak{B}$; (2) every countable theory T has a countable Skolem complete extension T^* such that every model of T has an expansion to a model of T^* ; (3) a structure is called Skolem complete iff its theory $\text{Th}(\mathfrak{B})$ is a Skolem complete theory.

The relationship of indiscernibles and types is given by

Theorem 7.5.12 *For any Skolem complete structure $\mathfrak{B}$ for a countable language and any set $Y \subseteq B$ which is a SOI for $\mathfrak{B}$ with respect to a linear ordering $<_Y$, $\mathfrak{A} := \langle Y \rangle_{\mathfrak{B}}$ is an elementary substructure which realizes only countably many types of $\mathfrak{B}$. If $<_Y$ is a well-ordering, then $\mathfrak{A}$ is modest.*

Proof. Each element of $\mathfrak{A}$ is of the form $t^{\mathfrak{B}}[\mathbf{a}]$ for some term t and some increasing sequence $\mathbf{a} = a_0, \dots, a_{n-1} \in Y$. Since $\mathfrak{A} \subseteq \mathfrak{B}$, this element is also $t^{\mathfrak{A}}[\mathbf{a}]$. For any other increasing $\mathbf{b} = b_0, \dots, b_{n-1} \in Y$, by the definition of indiscernibility, for any formula $\phi(x)$ with one free variable,

$$\models_{\mathfrak{B}} \phi[t^{\mathfrak{B}}[\mathbf{a}]] \iff \models_{\mathfrak{B}} \phi(t)[\mathbf{a}] \iff \models_{\mathfrak{B}} \phi(t)[\mathbf{b}] \iff \models_{\mathfrak{B}} \phi[t^{\mathfrak{B}}[\mathbf{b}]].$$

Since $\mathfrak{B}$ is Skolem complete, $\mathfrak{A} \prec \mathfrak{B}$ so it follows that the same equivalences hold in $\mathfrak{A}$ and thus $\text{Tp}^{\mathfrak{A}}(t^{\mathfrak{A}}[\mathbf{a}]) = \text{Tp}^{\mathfrak{A}}(t^{\mathfrak{A}}[\mathbf{b}])$. Hence for each term t , elements of $\mathfrak{A}$ of the form $t^{\mathfrak{A}}[\mathbf{a}]$ for increasing $\mathbf{a}$ realize the same 1-type. Since there are only countably many terms, at most countably many 1-types can be realized. An analogous argument shows that only countably many n -types are realized in $\mathfrak{A}$.

Now suppose that $<_Y$ is a well-ordering and fix a countable set $X \subseteq A$; we aim to show that at most countably many types of $\mathfrak{A}_X$ are realized in $\mathfrak{A}_X$. For each $c \in X$, fix a term u and $\mathbf{d} \in Y$ such that $c = u^{\mathfrak{A}}[\mathbf{d}]$. Let Z be the countable set of all $d \in Y$ which are selected in this way. For any increasing sequences $\mathbf{a} := a_0, \dots, a_n$ and $\mathbf{b} := b_0, \dots, b_n$ from Y , set

$$\mathbf{a} \sim_{Y;Z} \mathbf{b} \iff \text{for all } i \leq n \text{ and all } d \in Z, \quad a_i <_Y d \iff b_i <_Y d.$$

For any L^X -formula $\phi = \phi(\dot{\mathbf{c}})$ with one free variable and parameters $\mathbf{c} \in X$, let $\mathbf{u}$ and $\mathbf{d} \in Z$ be the terms and elements chosen above such that $\mathbf{c} = \mathbf{u}^{\mathfrak{A}}[\mathbf{d}]$. Then for any term t and increasing sequences $\mathbf{a}$ and $\mathbf{b}$, if $\mathbf{a} \sim_{Y;Z} \mathbf{b}$, then $\mathbf{a}, \mathbf{d} \sim_Y \mathbf{b}, \mathbf{d}$, so

$$\begin{aligned} \models_{\mathfrak{A}} \phi(\dot{\mathbf{c}})[t^{\mathfrak{A}}[\mathbf{a}]] &\iff \models_{\mathfrak{A}} \phi(t, \mathbf{u})[\mathbf{a}, \mathbf{d}] \\ &\iff \models_{\mathfrak{A}} \phi(t, \mathbf{u})[\mathbf{b}, \mathbf{d}] \iff \models_{\mathfrak{A}} \phi(\dot{\mathbf{c}})[t^{\mathfrak{A}}[\mathbf{b}]]. \end{aligned}$$

Thus, for each term t , elements of $\mathfrak{A}$ of the form $t^{\mathfrak{A}}[\mathbf{a}]$ for increasing $\mathbf{a}$ realize at most as many 1-types as there are $\sim_{Y;Z}$ -equivalence classes, and it suffices to show that this number is $\aleph_0$.

For each $a \in Y$, let

$$d_a := \begin{cases} \text{the } <_Y\text{-least } d \in Z \text{ such that } a <_Y d, & \text{if any;} \\ \infty, & \text{otherwise.} \end{cases}$$

Of course, it is here that we use the hypothesis that $<_Y$ is a well-ordering. Now easily

$$\mathbf{a} \sim_{Y;Z} \mathbf{b} \iff \text{for } i \leq n, \quad d_{a_i} = d_{b_i},$$

and there are only countably many possible values for $d_{a_0}, \dots, d_{a_n}$. $\square$

To conclude the proof of the First Modesty Theorem, it remains to show that SOI exist. This will follow from a standard result of set theory (cf. Definition 6.11.48 and the following Partition Theorem). Compare also Exercise 1.4.48.

Definition 7.5.13 For any set A and any $n \in \omega$,

- (i) $[A]^n$ denotes the collection of n -element subsets of A ;
- (ii) for any $P \subseteq [A]^n$, a set $H \subseteq A$ is **homogeneous for P** iff either $[H]^n \subseteq P$ or $[H]^n \cap P = \emptyset$.

Ramsey's Theorem 7.5.14 For any infinite set A , any $n > 0$ and any $P \subseteq [A]^n$, there exists an infinite set $H \subseteq A$ homogeneous for P .

Proof. We proceed by induction on n . For $n = 1$ we take $H := P$ if P is infinite, otherwise $H := A \sim P$. Assume the result for n as induction hypothesis and fix $P \subseteq [A]^n$. Consider the assertion

$$(3) \quad (\forall^{\text{inf}} X \subseteq A) (\exists a \in X) (\exists^{\text{inf}} Y \subseteq X \sim \{a\}) (\forall Z \in [Y]^n) Z \cup \{a\} \in P.$$

Suppose first that (3) holds; we define recursively infinite sets A_k and $a_k \in A_k$. Take $A_0 := A$. Given A_k , by (3) there exist $a_k \in A_k$ and an infinite $A_{k+1} \subseteq A_k \sim \{a_k\}$ such that for all $Z \in [A_{k+1}]^n$, $Z \cup \{a_k\} \in P$. Let $H := \{a_k : k \in \omega\}$. Then for any $Z \in [H]^{n+1}$, let k be smallest such that $a_k \in Z$. Then $Z \sim \{a_k\} \in [A_{k+1}]^n$ so $Z \in P$.

If, on the other hand, (3) fails, fix an infinite set $\overline{X} \subseteq A$ such that

$$(4) \quad (\forall a \in \overline{X}) (\forall^{\text{inf}} Y \subseteq \overline{X} \sim \{a\}) (\exists Z \in [Y]^n) Z \cup \{a\} \notin P.$$

For each infinite $X \subseteq \overline{X}$ and each $a \in X$, set

$$P_{X,a} := \{Z \in [X \sim \{a\}]^n : Z \cup \{a\} \in P\}.$$

Condition (4) implies that for each infinite $X \subseteq \overline{X}$ and $a \in X$ there is no infinite $Y \subseteq (X \sim \{a\})$ such that $[Y]^n \subseteq P_{X,a}$, so by the induction hypothesis there is an infinite $Y \subseteq (X \sim \{a\})$ such that $[Y]^n \cap P_{X,a} = \emptyset$. Thus we have

$$(5) \quad (\forall^{\text{inf}} X \subseteq \overline{X}) (\forall a \in X) (\exists^{\text{inf}} Y \subseteq X \sim \{a\}) (\forall Z \in [Y]^n) Z \cup \{a\} \notin P.$$

Now if we construct the sequences A_k and a_k as before, with $A_0 = \overline{X}$ and $H := \{a_k : k \in \omega\}$, we have easily that for any $Z \in [H]^{n+1}$ that $Z \notin P$. $\square$

Corollary 7.5.15 *For any infinite set A , any $k, n > 0$ and any $P_0, \dots, P_k$ which partition $[A]^n$ — that is, $P_0 \cup \dots \cup P_k = [A]^n$ and for $i \neq j$, $P_i \cap P_j = \emptyset$ — there exists an infinite set $H \subseteq A$ and an $i \leq k$ such that $[H]^n \subseteq P_i$.*

Proof. We proceed by induction on k . The case $k = 1$ is Ramsey's Theorem; assume the result for k as induction hypothesis and fix $P_0, \dots, P_{k+1}$ which partition $[A]^n$. Let $Q_i := P_i$ for $i < k$ and $Q_k := P_k \cup P_{k+1}$. By the induction hypothesis there exists an infinite set $H \subseteq A$ and $i \leq k$ such that $[H]^n \subseteq Q_i$. If $i < k$, this H suffices; if $i = k$, apply the case $k = 1$ to H to find an infinite $H' \subseteq H$ such that either $[H']^n \subseteq P_k$ or $[H']^n \subseteq P_{k+1}$. $\square$

Ehrenfeucht-Mostowski Theorem 7.5.16 *For any theory T with infinite models and any linearly ordered set $(Y, <_Y)$, there exists a model $\mathfrak{B}$ of T such that $Y \subseteq B$ and is an SOI for $\mathfrak{B}$ with respect to $<_Y$.*

Proof. From the language L of T form an extension L^+ by adjoining constant symbols $\dot{c}$ for each $c \in Y$. Let Γ denote the following set of L^+ -sentences:

$$T \cup \left\{ \neg \dot{c} \dot{=} \dot{d} : c \neq d, c, d \in Y \right\} \\ \cup \left\{ \phi(\dot{\mathbf{c}}) \leftrightarrow \phi(\dot{\mathbf{d}}) : \phi(\mathbf{x}) \text{ a formula and } \mathbf{c} \text{ and } \mathbf{d} \text{ are increasing} \right\}.$$

It clearly suffices to show that Γ is consistent, since any model of Γ will contain an isomorphic copy of $(Y, <_Y)$ which is an SOI. Let Δ be any finite subset of Γ , $\mathfrak{A}$ any infinite model of T , and $<_A$ any linear ordering of A . Choose lists $\phi_0, \dots, \phi_{k-1}$, $\dot{c}_0, \dots, \dot{c}_{l-1}$ and $v_0, \dots, v_{n-1}$ such that the members of Δ contain equivalences and inequalities involving only the ϕ_i and $\dot{c}_j$, $c_0, \dots, c_{l-1}$ is increasing, and the free variables of all the ϕ_i are among $v_0, \dots, v_{n-1}$. Define an equivalence relation on $[A]^n$ as follows. For any $Y, Z \in [A]^n$, let $\mathbf{a}$ and $\mathbf{b}$ be the elements of Y and Z , respectively, arranged in increasing order with respect to $<_A$. Then set

$$Y \sim Z : \Longleftrightarrow \text{for all } i < k, \quad \models_{\mathfrak{A}} \phi_i[\mathbf{a}] \Longleftrightarrow \models_{\mathfrak{A}} \phi_i[\mathbf{b}].$$

Clearly there are at most 2^k many equivalence classes $P_0, \dots, P_{2^k-1}$ which partition $[A]^n$, so by the corollary there exists an infinite set H such that for some $i < 2^k$, $[H]^n \subseteq P_i$. Choose $e_0, \dots, e_{l-1} \in H$ such that for $i < j < l$, $e_i <_A e_j$. Then easily $(\mathfrak{A}, e_0, \dots, e_{l-1})$ is a model of Δ . Thus by Compactness, Γ is consistent. $\square$

Proof. (of the First Modesty Theorem) Let L be a countable language, T an L -theory which has infinite models and κ an infinite cardinal. By Proposition 2.6.27 there exists a countable Skolem complete extension T^* of

T . Fix $(Y, <_Y)$ a well-ordering of cardinality κ . By the preceding theorem, there exists a model $\mathfrak{B}^*$ of T^* such that $Y \subseteq B^*$ and is an SOI for $\mathfrak{B}^*$. By Theorem 7.5.12, $\mathfrak{A}^* := \langle Y \rangle_{\mathfrak{B}^*}$ is modest. Furthermore, $\mathfrak{A}^*$ has the same cardinality as Y , namely κ , and is a model of T^* . Hence $\mathfrak{A} := \mathfrak{A}^* \upharpoonright L$ is a modest model of T of cardinality κ . $\square$

Finally, we give the

Proof. (of the Second Modesty Theorem) Let $\mathfrak{A}$ be an uncountable model of a countable ω -stable theory T and κ a cardinal $\geq \text{Card}(\mathfrak{A}) =: \lambda$. We construct first a proper elementary extension $\mathfrak{B}$ of $\mathfrak{A}$ which is modest over $\mathfrak{A}$ but has the same cardinality λ ; the desired model is then obtained by iterating this construction κ times.

In the Lindenbaum algebra $\mathcal{L}_{\mathfrak{A}_A}^1$, let

$$Z := \{ [\phi(x)]_{\mathfrak{A}_A}^1 : \phi^{\mathfrak{A}_A} \text{ is uncountable} \}.$$

Clearly Z is nonempty — for example, $[x \dot{=} x]_{\mathfrak{A}_A}^1 \in Z$ — but by Proposition 7.4.9, Z is not a perfect set and by Lemma 7.2.19, there exists an L^A -formula $\phi(x)$ such that $\phi^{\mathfrak{A}_A}$ is uncountable, but for each $L^{\mathfrak{A}}$ -formula ψ , exactly one of $(\phi \wedge \psi)^{\mathfrak{A}_A}$ and $(\phi \wedge \neg \psi)^{\mathfrak{A}_A}$ is uncountable. Fix such a ϕ and set

$$\Phi := \{ \psi(x) : \psi \text{ is an } L^A\text{-formula and } (\phi \wedge \psi)^{\mathfrak{A}_A} \text{ is uncountable} \}.$$

We establish first that

$$(6) \quad \text{for every countable } \Phi_0 \subseteq \Phi, \quad \Phi_0 \text{ is realized in } \mathfrak{A}_A.$$

Indeed, for any such Φ_0 , let $Y := \bigcap \{ \psi^{\mathfrak{A}_A} : \psi \in \Phi_0 \}$. $\phi^{\mathfrak{A}_A} \sim Y$ is a countable union of the countable sets $(\phi \wedge \neg \psi)^{\mathfrak{A}_A}$, so is countable. Hence $\phi^{\mathfrak{A}_A} \cap Y$ is uncountable, in particular nonempty; any element of this set realizes Φ_0 .

Now by Proposition 7.3.5, Φ is a type of $\mathfrak{A}_A$ and by the choice of ϕ , Φ is a complete type. However, since for each $a \in A$ the formula $\neg x \dot{=} a$ clearly belongs to Φ , Φ is not realized in $\mathfrak{A}_A$. Let $\dot{c}$ be a new constant symbol, so $\Phi(\dot{c}) := \{ \psi(\dot{c}) : \psi \in \Phi \}$ is a complete inessential extension of $\text{Th}(\mathfrak{A}_A)$ and hence also of T . By Theorem 7.4.18, $\Phi(\dot{c})$ has an atomic model $\mathfrak{B}_{A \cup \{c\}}$ of cardinality λ with $\mathfrak{A}_A \prec \mathfrak{B}_A$. Since $c := \dot{c}^{\mathfrak{B}_A}$ realizes Φ in $\mathfrak{B}_A$, $\mathfrak{B}_A$ is a proper extension.

It remains to show that $\mathfrak{B}$ is modest over $\mathfrak{A}$. Let $X \subseteq A$ be any countable set and $\Psi(\mathbf{y})$ an n -type of $\mathfrak{B}_X$, hence also an n -type of $\mathfrak{B}_{A \cup \{c\}}$. Since $\mathfrak{B}_{A \cup \{c\}}$ is atomic, Ψ is principal; let $\theta(\mathbf{y}, \dot{c})$ be a generator (where we suppress explicit mention of constants $\dot{a}$ for $a \in A$ which occur in θ). Since $\Phi(\dot{c})$ is complete, we have

$$\Phi(\dot{c}) \models \exists \mathbf{y} \theta(\mathbf{y}, \dot{c}) \quad \text{and} \quad \text{for all } \chi \in \Psi, \quad \Phi(\dot{c}) \models \theta(\mathbf{y}, \dot{c}) \rightarrow \chi(\mathbf{y}).$$

Since X is countable, L^X is a countable language, hence Ψ is countable, and altogether there are countably many consequences of $\Phi(\dot{c})$. By compactness each requires only finitely much of $\Phi(\dot{c})$ so there exists a countable $\Phi_0 \subseteq \Phi$ such that

$$\Phi_0(\dot{c}) \models \exists \mathbf{y} \theta(\mathbf{y}, \dot{c}) \quad \text{and} \quad \text{for all } \chi \in \Psi, \quad \Phi_0(\dot{c}) \models \theta(\mathbf{y}, \dot{c}) \rightarrow \chi(\mathbf{y}).$$

But by (6), Φ_0 is realized in $\mathfrak{A}_A$ by some element $d \in A$, so we have

$$\models_{\mathfrak{A}_A} \exists \mathbf{y} \theta(\mathbf{y}, \dot{d}) \quad \text{and} \quad \text{for all } \chi \in \Psi, \quad \models_{\mathfrak{A}_A} \theta(\mathbf{y}, \dot{d}) \rightarrow \chi(\mathbf{y}).$$

It follows that Ψ is realized in $\mathfrak{A}_A$, hence in $\mathfrak{A}_X$ as desired.

Finally, to construct an elementary extension of $\mathfrak{A}$ of cardinality κ modest over $\mathfrak{A}$, let $\mathfrak{B}_0 = \mathfrak{A}$ and for each ordinal $\sigma \leq \kappa$ let $\mathfrak{B}_{\sigma+1}$ be a proper elementary extension of $\mathfrak{B}_\sigma$, of the same cardinality, and modest over $\mathfrak{B}_\sigma$. For limit ordinals σ , $\mathfrak{B}_\sigma := \bigcup_{\tau < \sigma} \mathfrak{B}_\tau$. Easily by induction each $\mathfrak{B}_\sigma$ is modest over $\mathfrak{A}$ and of cardinality $\text{Card}(\sigma)$. Hence $\mathfrak{B}_\kappa$ is the desired model. $\square$

Collateral Reading. The proof of Morley's Theorem given here is essentially the same as that in [Ch-Ke] and avoids the notion of Morley rank. For proofs which use the notion, which is essential for further work in this direction, see [Mar], [Bue], [Pi83] and [Bal]. The latter three also contain proofs of the Baldwin-Lachlan Theorem (see the last exercise below). All of these books contain a great deal of further work in stability theory, one of the central areas of current research in model theory. A still more advanced book is [Pi96]. Indiscernibles have many other applications in model theory beyond the ones explored here: see [Ho93].

Exercises

Exercise 7.5.17 Use the results of Exercises 7.3.49, 7.3.56 and 7.3.58 to give a proof along the following lines of a weak version of the Two Cardinal Theorem 7.5.8: if $(\mathfrak{A}, \theta(x))$ is a two-cardinal pair, then there exists a structure $\mathfrak{C}$ of cardinality $\aleph_1$ such that $\theta^{\mathfrak{C}}$ is countable. Given the hypotheses show that

- (i) there exists a proper elementary substructure $\mathfrak{A}_0$ of $\mathfrak{A}$ such that $\theta^{\mathfrak{A}_0} = \theta^{\mathfrak{A}}$;
- (ii) there exists a countable recursively saturated structure of the form $\mathfrak{B}_0 \dot{\cup} \mathfrak{B}_1$ elementarily equivalent to $\mathfrak{A}_0 \dot{\cup} \mathfrak{A}$; it follows that
 - (a) $\mathfrak{B}_0$ is a proper elementary substructure of $\mathfrak{B}_1$;
 - (b) $\mathfrak{B}_0 \simeq \mathfrak{B}_1$;

- (c) $\theta^{\aleph_0} = \theta^{\aleph_1}$;
- (d) $\mathfrak{B}_0$ and $\mathfrak{B}_1$ are $\aleph_0$ -homogeneous.

Because of (b), $\mathfrak{B}_1$ now has a similar proper elementary extension $\mathfrak{B}_2$; use this to construct an appropriate elementary chain $\langle \mathfrak{B}_\sigma : \sigma < \aleph_1 \rangle$ using Exercise 7.3.49 at the limit stages.

Exercise 7.5.18 Call a theory T a **two-cardinal theory** iff there exists a two-cardinal pair $(\mathfrak{A}, \theta(x))$ with $\mathfrak{A}$ a model of T . Note that by the Two-Cardinal Theorem, we may assume that $\text{Card}(\mathfrak{A}) = \aleph_1$ and $\theta^{\aleph_1}$ is denumerable. Prove that no two-cardinal theory is $\aleph_1$ categorical.

Hint: Construct a second model $\mathfrak{B}$ of T such that $\text{Card}(\mathfrak{B}) = \text{Card}(\theta^{\aleph_1}) = \aleph_1$.

Exercise 7.5.19 Improve the result of the preceding exercise to show that if T even has a finite inessential extension T^+ which is a two-cardinal theory, then T is not $\aleph_1$ categorical.

Hint: Suppose $\mathfrak{A}$ is a model of T and $\mathbf{a} \in \mathfrak{A}$ such that $((\mathfrak{A}, \mathbf{a}), \theta(\dot{\mathbf{a}}, x))$ witnesses that T^+ is two-cardinal. Construct a model $\mathfrak{B}$ of T of cardinality $\aleph_1$ such that for every $\mathbf{b} \in \mathfrak{B}$, $\theta(\dot{\mathbf{b}}, x)^{(\mathfrak{B}, \mathbf{b})}$ is either finite or uncountable.

Exercise 7.5.20 Call a theory T **Vaughtian** iff there exist models $\mathfrak{A}$ and $\mathfrak{B}$ of T and a formula $\theta(x)$ such that $\mathfrak{A} \prec \mathfrak{B}$, $\mathfrak{A} \neq \mathfrak{B}$ but $\theta^{\aleph_1} = \theta^{\aleph_2}$. Show that every two-cardinal theory is Vaughtian.

Definition 7.5.21 A theory T is **strongly minimal** iff every model of T is minimal (in the sense of Definition 2.3.78).

Exercise 7.5.22 Use the notions and results of Exercises 2.3.79–2.3.82 to show that for any countable strongly minimal theory T ,

- (i) for every uncountable cardinal κ , T is κ -categorical;
- (ii) T has at most countably many isomorphism classes of countable models — that is, $\mu(T, \aleph_0) \leq \aleph_0$.

Hint: Fix an uncountable cardinal κ and two models $\mathfrak{A}$ and $\mathfrak{B}$ of T of cardinality κ . By Exercise 2.3.82 there exist independent sets $X \subseteq A$ and $Y \subseteq B$ such that $A = \text{acl}_{\mathfrak{A}} X$, $B = \text{acl}_{\mathfrak{B}} Y$ and $|X| = \kappa = |Y|$. Let $f : X \rightarrow Y$ be any bijection. Show first by induction on n that for any formula $\phi(x_0, \dots, x_{n-1})$ and any $a_0, \dots, a_{n-1} \in X$,

$$\models_{\mathfrak{A}} \phi[a_0, \dots, a_{n-1}] \iff \models_{\mathfrak{B}} \phi[f(a_0), \dots, f(a_{n-1})].$$

Thus f is a partial elementary embedding of $\mathfrak{A}$ into $\mathfrak{B}$. Apply Zorn's Lemma to find a maximal element $\bar{g}$ of

$\{g : g \text{ is a partial elementary embedding of } \mathfrak{A} \text{ into } \mathfrak{B} \text{ which extends } f\}$.

Show along the following lines that in fact $\bar{g} : A \rightarrow B$ is a bijection and hence an isomorphism of $\mathfrak{A}$ and $\mathfrak{B}$. Suppose $a \in A \sim \text{Dom } \bar{g}$ so for some $c_0, \dots, c_{k-1} \in X$ and $n \in \omega$,

$$\models_{\mathfrak{A}} \phi[a, c_0, \dots, c_{k-1}] \wedge \exists^{=n} x \phi(x)[c_0, \dots, c_{k-1}],$$

and hence also

$$\models_{\mathfrak{B}} \exists^{=n} x \phi(x)[f(c_0), \dots, f(c_{k-1})].$$

Show that for at least one of the n -many $b \in B$ such that

$$\models_{\mathfrak{B}} \phi[b, f(c_0), \dots, f(c_{k-1})],$$

$\bar{g} \cup \{(a, b)\}$ is elementary, contrary to the maximality of $\bar{g}$. Hence $\text{Dom } \bar{g} = A$; use a parallel argument to show $\text{Im } \bar{g} = B$. Finally, prove (ii) by a similar analysis of the countable models of T .

With considerable additional effort, beyond the scope of this book, this argument can be extended to give a proof of the following strengthening of Morley's Theorem.

Baldwin-Lachlan Theorem 7.5.23 *For any countable theory T , if T is κ -categorical for some uncountable cardinal κ , then*

- (i) *for every uncountable cardinal κ , T is κ -categorical;*
- (ii) $\mu(T, \aleph_0) = 1$ or $\aleph_0$. $\square$

7.6 Abstract logics

In [Sections 3.6–3.7](#) we described logics that extend the power of first-order logic in several different ways. We observed that each of these logics lacks one or more of the basic structural properties of first-order logic, particularly those expressed by the Compactness, Löwenheim-Skolem and Enumerability Theorems. The goal of this section is to establish two results which assert that this is a necessary feature: there is no proper extension of first-order logic that satisfies the Löwenheim-Skolem Theorem and either the Compactness or Enumerability Theorems, even in weak versions.

To formulate this precisely, we need first to specify what we mean in general by a logic and what constitutes a proper extension of first-order

logic. Recall that a **signature** μ is an object which specifies for each $n > 0$ a number (possibly infinite) of n -ary relations and functions and a number of distinguished elements. Two μ -structures have the same “shape” and may be compared by various relationships such as (partial) isomorphism, elementary equivalence etc. For each signature (except the one with no relations, functions or distinguished elements) there are infinitely many first-order languages obtained by choosing a particular relation (function, constant) symbol to represent each of the relations (functions, distinguished elements) specified by the signature. Although these are all formally different, they are in an obvious sense isomorphic. In most contexts it is more convenient to work with a particular language and leave the signature implicit.

An abstract logic as defined below is not given in terms of languages or symbols, so formally we are forced to consider signatures explicitly. However, it is still more convenient (and familiar) to describe a signature by giving one possible choice of symbols for a corresponding first-order language, so in practice we shall think of a signature as an object $\mu := (\mathbf{Rs}_\mu, \mathbf{Fs}_\mu, \mathbf{Cs}_\mu, \nu_\mu)$ with $\nu_\mu : \mathbf{Rs}_\mu \cup \mathbf{Fs}_\mu \rightarrow \omega \sim \{0\}$ with the understanding that the particular choice of symbols is irrelevant. Thus, for example, a signature μ' is an **extension** of μ — in symbols, $\mu \subseteq \mu'$ — iff each $\mathbf{Rs}_\mu \subseteq \mathbf{Rs}_{\mu'}$ etc. and $\nu_{\mu'}$ is an extension of ν_μ . When $\mu \subseteq \mu'$ and $\mathfrak{B}$ is a μ' -structure, then the μ -**reduct** $\mathfrak{B} \upharpoonright \mu$ is the unique μ -structure with the same universe as $\mathfrak{B}$ which agrees with $\mathfrak{B}$ in the interpretation of the symbols of μ .

Definition 7.6.1 A **logic** $\mathcal{L}$ consists of a mapping $\mu \mapsto \text{Sent}_\mu^\mathcal{L}$ and a binary relation $\models^\mathcal{L}$ such that,

- (i) for each signature μ , $\text{Sent}_\mu^\mathcal{L}$ is a non-empty set whose elements are called the μ -**sentences** of $\mathcal{L}$; an $\mathcal{L}$ -**sentence** is a member of $\text{Sent}_\mu^\mathcal{L}$ for some μ ;
- (ii) for any ϕ and $\mathfrak{A}$, if $\models^\mathcal{L}_{\mathfrak{A}} \phi$, then for some μ , $\phi \in \text{Sent}_\mu^\mathcal{L}$ and $\mathfrak{A}$ is a μ -structure;
- (iii) for any signatures $\mu \subseteq \mu'$, $\text{Sent}_\mu^\mathcal{L} \subseteq \text{Sent}_{\mu'}^\mathcal{L}$;
- (iv) (**isomorphism property**) for any μ , any $\phi \in \text{Sent}_\mu^\mathcal{L}$ and any μ -structures $\mathfrak{A}$ and $\mathfrak{B}$, if $\models^\mathcal{L}_{\mathfrak{A}} \phi$ and $\mathfrak{A} \simeq \mathfrak{B}$, then also $\models^\mathcal{L}_{\mathfrak{B}} \phi$;
- (v) (**reduct property**) for any signatures $\mu \subseteq \mu'$, any μ' -structure $\mathfrak{A}'$ and any $\phi \in \text{Sent}_\mu^\mathcal{L}$,

$$\models^\mathcal{L}_{\mathfrak{A}'} \phi \iff \models^\mathcal{L}_{\mathfrak{A}' \upharpoonright \mu} \phi;$$

- (vi) (**Boolean property**) for any $\phi, \psi \in \text{Sent}_\mu^\mathcal{L}$, there exist $\neg \phi, \phi \vee \psi \in \text{Sent}_\mu^\mathcal{L}$ such that

$$\models^\mathcal{L}_{\mathfrak{A}} \neg \phi \iff \text{not } \models^\mathcal{L}_{\mathfrak{A}} \phi \quad \text{and} \quad \models^\mathcal{L}_{\mathfrak{A}} \phi \vee \psi \iff \models^\mathcal{L}_{\mathfrak{A}} \phi \quad \text{or} \quad \models^\mathcal{L}_{\mathfrak{A}} \psi.$$

We write also $\phi \wedge \psi$ for $\neg(\neg\phi \vee \neg\psi)$, $\phi \rightarrow \psi$ for $\neg\phi \vee \psi$ and $\phi \leftrightarrow \psi$ for $(\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi)$.

Note that $\mathcal{L}$ -sentences are not required to have any particular structure. There are no atomic formulas, connectives, variables or quantifiers; sentences are purely abstract objects. Hence there can be no proofs by induction on formulas; we write the Boolean “connectives” in raised position to emphasize this difference. On the other hand, the interpretations for $\mathcal{L}$ -sentences are the same structures that we have used throughout this book. A still more general notion of logic, which we shall not consider here, would replace these also with abstract objects subject to certain conditions. This definition includes almost all of the examples we have encountered: first-order logic $\mathcal{L}^1$, monadic second-order logic $\mathcal{L}^{2m}$, full second-order logic $\mathcal{L}^2$, weak monadic second-order logic $\mathcal{L}^{w2m}$ and the infinitary logics $\mathcal{L}_{\kappa\omega}$. Note, however, that the logic of finite structures does not fall under this definition because the class of structures is restricted.

We adapt much of the terminology of first-order logic to the general case.

Definition 7.6.2 For any logic $\mathcal{L}$, any signature μ , any $\phi \in \text{Sent}_\mu^\mathcal{L}$ and any $\Gamma \subseteq \text{Sent}_\mu^\mathcal{L}$,

- (i) $\text{Mod}_\mu^\mathcal{L}(\phi) := \left\{ \mathfrak{A} : \mathfrak{A} \text{ is a } \mu\text{-structure and } \mathfrak{A} \models_\mathcal{L} \phi \right\};$
- (ii) $\text{Mod}_\mu^\mathcal{L}(\Gamma) := \bigcap \left\{ \text{Mod}_\mu^\mathcal{L}(\phi) : \phi \in \Gamma \right\};$
- (iii) $\Gamma \models_\mathcal{L} \phi \iff \text{Mod}_\mu^\mathcal{L}(\Gamma) \subseteq \text{Mod}_\mu^\mathcal{L}(\phi);$ when $\Gamma = \emptyset$ we write simply $\models_\mathcal{L} \phi$;
- (iv) for any μ -structure $\mathfrak{A}$, $\text{Th}^\mathcal{L}(\mathfrak{A}) := \left\{ \phi \in \text{Sent}_\mu^\mathcal{L} : \mathfrak{A} \models_\mathcal{L} \phi \right\};$
- (v) for any μ -structures $\mathfrak{A}$ and $\mathfrak{B}$, $\mathfrak{A} \equiv^\mathcal{L} \mathfrak{B}$ iff $\text{Th}^\mathcal{L}(\mathfrak{A}) = \text{Th}^\mathcal{L}(\mathfrak{B})$.

Definition 7.6.3 For any logics $\mathcal{K}$ and $\mathcal{L}$,

- (i) $\mathcal{L}$ *translates* $\mathcal{K}$ *at* μ iff there exists a mapping $\phi \mapsto \phi^\mathcal{L}$ of $\text{Sent}_\mu^\mathcal{K} \rightarrow \text{Sent}_\mu^\mathcal{L}$ such that for all $\phi \in \text{Sent}_\mu^\mathcal{K}$, $\text{Mod}_\mu^\mathcal{K}(\phi) = \text{Mod}_\mu^\mathcal{L}(\phi^\mathcal{L})$; for $\Gamma \subseteq \text{Sent}_\mu^\mathcal{K}$, $\Gamma^\mathcal{L} := \{ \phi^\mathcal{L} : \phi \in \Gamma \}$;
- (ii) $\mathcal{L}$ *translates* $\mathcal{K}$ — in symbols, $\mathcal{K} \leq \mathcal{L}$ — iff $\mathcal{L}$ translates $\mathcal{K}$ at every signature;
- (iii) $\mathcal{K}$ and $\mathcal{L}$ are *equivalent logics* — in symbols, $\mathcal{K} \sim \mathcal{L}$ — iff $\mathcal{K} \leq \mathcal{L}$ and $\mathcal{L} \leq \mathcal{K}$.

Examples 7.6.4

- (i) Clearly if $\mathcal{K}$ is (in the obvious sense) a sublogic of $\mathcal{L}$, then $\mathcal{K} \leq \mathcal{L}$, and $\mathcal{L}^1$ is a sublogic of every particular logic we have considered. Full second-order logic $\mathcal{L}^2$ translates full weak second-order logic $\mathcal{L}^{w2}$ because finiteness is definable in $\mathcal{L}^2$, but for the monadic versions it is not obvious that $\mathcal{L}^{w2m} \leq \mathcal{L}^{2m}$.
- (ii) A more interesting example is that $\mathcal{L}^{w2m} \leq \mathcal{L}_{\omega_1\omega}$. We define a mapping $\phi \mapsto \phi^{\omega_1\omega}$ of weak monadic second-order sentences to $\mathcal{L}_{\omega_1\omega}$ sentences which preserves meaning in the sense of the definition. This goes recursively on sentences with the main step being

$$(\exists X[\dots X t \dots])^{\omega_1\omega} = \bigvee_{k \in \omega} (\exists x_0 \dots \exists x_{k-1} [\dots (t = x_0 \vee \dots \vee t = x_{k-1}) \dots])^{\omega_1\omega}.$$

- (iii) On the other hand, $\mathcal{L}_{\omega_1\omega} \not\leq \mathcal{L}^2$ because there are $2^{\aleph_0}$ -many $\mathcal{L}_{\omega_1\omega}$ model classes, but only countably many $\mathcal{L}^2$ model classes.

Remark 7.6.5 When $\mathcal{K} \leq \mathcal{L}$ we may without loss of generality assume that for any $\phi, \psi \in \text{Sent}_{\mu}^{\mathcal{K}}$, $(\neg \phi)^{\mathcal{L}} = \neg \phi^{\mathcal{L}}$ and $(\phi \vee \psi)^{\mathcal{L}} = \phi^{\mathcal{L}} \vee \psi^{\mathcal{L}}$, and similarly for the other connectives.

Recall from Proposition 2.3.15 that for any structure $\mathfrak{C}$ for a signature μ and any subset $A \subseteq C$, there exists a (unique) substructure $\mathfrak{A}_{\mathfrak{C}} \subseteq \mathfrak{C}$ with universe A iff A is μ -**closed** — that is, A contains all of the distinguished elements of $\mathfrak{C}$ and is closed under all of the functions of $\mathfrak{C}$. In particular, if μ is relational — that is, has no function or constant symbols — then this is true for every $A \subseteq C$.

A familiar property of most standard logics that we shall need is relational translation (Definition 2.6.31). For any signature μ , there is a natural signature μ^{rel} in which each n -ary function symbol f is replaced by a $(n+1)$ -ary relation symbol p_f . For this purpose, constant symbols are treated as 0-ary function symbols. Similarly, for each μ -structure $\mathfrak{A}$ there is a unique corresponding μ^{rel} -structure $\mathfrak{A}^{\text{rel}}$.

Definition 7.6.6 A logic $\mathcal{L}$ is a **regular logic** iff

- (i) $\mathcal{L}$ translates first-order logic: $\mathcal{L}^1 \leq \mathcal{L}$;
- (ii) $\mathcal{L}$ admits **quantifier relativization**: for any signature μ and any extension μ' including a new unary relation symbol $\dot{A}$, there is a mapping $\phi \mapsto \phi^{(\dot{A})}$ of $\text{Sent}_{\mu}^{\mathcal{L}}$ into $\text{Sent}_{\mu'}^{\mathcal{L}}$ such that for any μ -structure $\mathfrak{C}$, any μ -closed set $A \subseteq C$ and any $\phi \in \text{Sent}_{\mu}^{\mathcal{L}}$,

$$\models_{(\mathfrak{A}; A)}^{\mathcal{L}} \phi^{(\dot{A})} \iff \models_{\mathfrak{A}_{\mathfrak{C}}}^{\mathcal{L}} \phi;$$

- (iii) $\mathcal{L}$ admits **relational translation**: for each signature μ , there is a mapping $\phi \mapsto \phi^{\text{rel}}$ of $\text{Sent}_\mu^\mathcal{L}$ into $\text{Sent}_{\mu^{\text{rel}}}^\mathcal{L}$ such that for each $\phi \in \text{Sent}_\mu^\mathcal{L}$ and each μ -structure $\mathfrak{A}$,

$$\models_{\mathfrak{A}}^\mathcal{L} \phi \iff \models_{\mathfrak{A}^{\text{rel}}}^\mathcal{L} \phi^{\text{rel}}.$$

All of our example logics are regular; for $\mathcal{L}^1$ see Propositions 2.6.32 and 2.6.44. Note that although we use the term *quantifier relativization*, since the $\mathcal{L}$ -sentences of a general logic need not have any quantifiers as such, the actual mapping of (ii) may in principle be quite different from the quantifier relativization of standard logics.

We are nearing the end of the definitions, but before the first results, we need still a few more. Recall from Definition 7.1.22 the relation $\mathfrak{A} \simeq_{\mathfrak{p}}^n \mathfrak{B}$ which holds iff there exists an n -extendable partial embedding from $\mathfrak{A}$ into $\mathfrak{B}$. By Fraïssé's Theorem 7.1.29, $\mathfrak{A} \simeq_{\mathfrak{p}}^n \mathfrak{B}$ iff the two structures are equivalent with respect to first-order sentences of quantifier rank at most n .

Definition 7.6.7 For any logic $\mathcal{L}$ and any signature μ ,

- (i) $\mathcal{L}$ has the **finite dependence property at μ** iff for any $\phi \in \text{Sent}_\mu^\mathcal{L}$, there exists a finite signature $\mu_\phi \subseteq \mu$ such that for any μ -structures $\mathfrak{A}$ and $\mathfrak{B}$,

$$\mathfrak{A} \upharpoonright \mu_\phi \simeq \mathfrak{B} \upharpoonright \mu_\phi \implies \left[\models_{\mathfrak{A}}^\mathcal{L} \phi \iff \models_{\mathfrak{B}}^\mathcal{L} \phi \right];$$

- (ii) $\mathcal{L}$ has the **strong finite dependence property at μ** iff for any $\phi \in \text{Sent}_\mu^\mathcal{L}$, there exist a finite signature $\mu_\phi \subseteq \mu$ and $n \in \omega$ such that for any μ -structures $\mathfrak{A}$ and $\mathfrak{B}$,

$$\mathfrak{A} \upharpoonright \mu_\phi \simeq_{\mathfrak{p}}^n \mathfrak{B} \upharpoonright \mu_\phi \implies \left[\models_{\mathfrak{A}}^\mathcal{L} \phi \iff \models_{\mathfrak{B}}^\mathcal{L} \phi \right];$$

- (iii) $\mathcal{L}$ is a **compact logic** iff whenever $\Gamma \models^\mathcal{L} \phi$, there exists a finite subset $\Gamma_0 \subseteq \Gamma$ such that $\Gamma_0 \models^\mathcal{L} \phi$.

Examples 7.6.8 First-order logic $\mathcal{L}^1$ is compact and has the strong finite dependence property at every signature, $\mathcal{L}^2$ has the finite dependence property, but $\mathcal{L}_{\omega_1\omega}$ has none of these properties.

Proposition 7.6.9 For any regular logic $\mathcal{L}$, if $\mathcal{L}$ is compact, then $\mathcal{L}$ has the finite dependence property at every signature.

Proof. Fix a signature μ and let μ' be an extension of μ containing new unary relations symbols $\dot{A}$ and $\dot{B}$ and a new unary function symbol f . There is a natural set Γ of first-order μ' -sentences such that any μ' -structure $\mathfrak{C}$ is a model of Γ iff

- (i) $\dot{A}^{\mathfrak{C}}$ and $\dot{B}^{\mathfrak{C}}$ are μ -closed so are universes of substructures $\mathfrak{A}_{\mathfrak{C}}$ and $\mathfrak{B}_{\mathfrak{C}}$, respectively;
- (ii) $f^{\mathfrak{C}}$ is an isomorphism of $\mathfrak{A}_{\mathfrak{C}}$ onto $\mathfrak{B}_{\mathfrak{C}}$.

Since $\mathcal{L}$ translates $\mathcal{L}^1$, there is a corresponding set $\Gamma^{\mathcal{L}}$ of $\mathcal{L}$ -sentences such that $\text{Mod}_{\mu'}^{\mathcal{L}^1}(\Gamma) = \text{Mod}_{\mu'}^{\mathcal{L}}(\Gamma^{\mathcal{L}})$. For each $\mathfrak{C}$ in this set and each $\phi \in \text{Sent}_{\mu}^{\mathcal{L}}$, using the reduct, isomorphism and quantifier relativization properties,

$$\models_{\mathfrak{C}}^{\mathcal{L}} \phi^{(\dot{A})} \iff \models_{\mathfrak{A}_{\mathfrak{C}}}^{\mathcal{L}} \phi \iff \models_{\mathfrak{B}_{\mathfrak{C}}}^{\mathcal{L}} \phi \iff \models_{\mathfrak{C}}^{\mathcal{L}} \phi^{(\dot{B})}.$$

Thus, using the Boolean property of $\mathcal{L}$, we have

$$\Gamma^{\mathcal{L}} \models^{\mathcal{L}} \phi^{(\dot{A})} \leftrightarrow \phi^{(\dot{B})},$$

so by the compactness assumed for $\mathcal{L}$, there is some finite subset $\Gamma_0 \subseteq \Gamma$ such that $\Gamma_0^{\mathcal{L}} \models^{\mathcal{L}} \phi^{(\dot{A})} \leftrightarrow \phi^{(\dot{B})}$. Let μ_{ϕ} be the smallest signature such that all members of Γ_0 are first-order μ_{ϕ} -sentences; clearly μ_{ϕ} is finite and it follows that $\Gamma_0^{\mathcal{L}} \subseteq \text{Sent}_{\mu_{\phi}}^{\mathcal{L}}$. To verify that μ_{ϕ} satisfies the definition of finite dependence, suppose that $\mathfrak{A}$ and $\mathfrak{B}$ are any μ -structures such that $\eta : \mathfrak{A} \upharpoonright \mu_{\phi} \simeq \mathfrak{B} \upharpoonright \mu_{\phi}$; by the isomorphism property we may assume that $A \cap B = \emptyset$. We may construct a μ' -structure $\mathfrak{C}$ with $C = A \cup B$, $\mathfrak{A}_{\mathfrak{C}} = \mathfrak{A}$, $\mathfrak{B}_{\mathfrak{C}} = \mathfrak{B}$, and $f^{\mathfrak{C}} \upharpoonright \dot{A}^{\mathfrak{C}} = \eta$. Clearly $\mathfrak{C}$ is a model of Γ_0 , hence belongs to $\text{Mod}_{\mu'}^{\mathcal{L}}(\Gamma_0^{\mathcal{L}})$. Then $\models_{\mathfrak{C}}^{\mathcal{L}} \phi^{(\dot{A})} \leftrightarrow \phi^{(\dot{B})}$, so

$$\models_{\mathfrak{A}}^{\mathcal{L}} \phi \iff \models_{\mathfrak{C}}^{\mathcal{L}} \phi^{(\dot{A})} \iff \models_{\mathfrak{C}}^{\mathcal{L}} \phi^{(\dot{B})} \iff \models_{\mathfrak{B}}^{\mathcal{L}} \phi,$$

as desired. $\square$

Proposition 7.6.10 *For any regular logic $\mathcal{L}$ and any relational signature μ , if $\mathcal{L}$ has the strong finite dependence property at μ , then $\mathcal{L}^1$ translates $\mathcal{L}$ at μ .*

Proof. Under the hypothesis of the strong finite dependence property, we shall show that for each $\phi \in \text{Sent}_{\mu}^{\mathcal{L}}$, there exists a first-order μ -sentence ϕ^1 such that $\text{Mod}^{\mathcal{L}}(\phi) = \text{Mod}(\phi^1)$. Fix such a ϕ . By hypothesis there exist a finite signature $\mu_{\phi} \subseteq \mu$ and an $n \in \omega$ such that for any μ -structures $\mathfrak{A}$ and $\mathfrak{B}$,

$$\mathfrak{A} \upharpoonright \mu_{\phi} \simeq_{\mathfrak{p}}^n \mathfrak{B} \upharpoonright \mu_{\phi} \implies \left[\models_{\mathfrak{A}}^{\mathcal{L}} \phi \iff \models_{\mathfrak{B}}^{\mathcal{L}} \phi \right].$$

Recall from Proposition 7.1.28 that there exists a finite set $\Delta_{\mu_{\phi}}^{(n,0)}$ of first-order sentences of quantifier rank at most n such that every such sentence is logically equivalent to a member of $\Delta_{\mu_{\phi}}^{(n,0)}$. For each μ -structure $\mathfrak{A}$ set

$$\chi_{\mathfrak{A},n} := \bigwedge \left\{ \psi : \psi \in \Delta_{\mu_{\phi}}^{(n,0)} \text{ and } \models_{\mathfrak{A}} \psi \right\}.$$

Then, by Fraïssé's Theorem 7.1.29, for any μ -structure $\mathfrak{B}$,

$$\mathfrak{A} \restriction \mu_\phi \simeq_p^n \mathfrak{B} \restriction \mu_\phi \iff \mathfrak{A} \restriction \mu_\phi \equiv_\omega^n \mathfrak{B} \restriction \mu_\phi \iff \models_{\mathfrak{B}} \chi_{\mathfrak{A},n}.$$

Let

$$\phi^1 := \bigvee \left\{ \chi_{\mathfrak{A},n} : \mathfrak{A} \in \text{Mod}^{\mathcal{L}}(\phi) \right\}.$$

Since $\Delta_{\mu_\phi}^{(n,0)}$ is finite, there are at most finitely many distinct sentences $\chi_{\mathfrak{A},n}$, so ϕ^1 is a first-order sentence. Obviously $\text{Mod}^{\mathcal{L}}(\phi) \subseteq \text{Mod}(\phi^1)$. If $\mathfrak{B}$ is a model of ϕ^1 it satisfies some $\chi_{\mathfrak{A},n}$ for $\mathfrak{A} \in \text{Mod}^{\mathcal{L}}(\phi)$. Then by the last displayed equivalence, $\mathfrak{A} \restriction \mu_\phi \simeq_p^n \mathfrak{B} \restriction \mu_\phi$, so by the hypothesis $\models_{\mathfrak{B}}^{\mathcal{L}} \phi$ as desired. $\square$

Proposition 7.6.11 *For any logic $\mathcal{L}$ and any signature μ , if $\mathcal{L}^1$ translates $\mathcal{L}$ at μ^{rel} , then $\mathcal{L}^1$ translates $\mathcal{L}$ also at μ .*

Proof. By hypothesis, for each $\phi \in \text{Sent}_\mu^{\mathcal{L}}$ there exists a first-order μ^{rel} -sentence $(\phi^{\text{rel}})^1$ such that for any μ -structure $\mathfrak{A}$,

$$\models_{\mathfrak{A}}^{\mathcal{L}} \phi \iff \models_{\mathfrak{A}^{\text{rel}}}^{\mathcal{L}} \phi^{\text{rel}} \iff \models_{\mathfrak{A}^{\text{rel}}}^{\mathcal{L}^1} (\phi^{\text{rel}})^1.$$

Now let ϕ^1 be the first-order μ -sentence obtained by replacing each occurrence of an atomic formula $p_f \mathbf{x} y$ by $f \mathbf{x} \doteq y$ and $p_c y$ by $c \doteq y$. Clearly $(\phi^1)^{\text{rel}}$ is exactly $(\phi^{\text{rel}})^1$, so we have

$$\models_{\mathfrak{A}^{\text{rel}}}^{\mathcal{L}^1} (\phi^{\text{rel}})^1 \iff \models_{\mathfrak{A}}^{\mathcal{L}^1} \phi^1,$$

and thus ϕ^1 is the desired $\mathcal{L}^1$ -translation of ϕ . $\square$

Recall (Definition 3.6.13) that the Löwenheim number $\ell(\mathcal{L})$ of a logic $\mathcal{L}$ is the smallest cardinal number κ such that for any $\mathcal{L}$ -sentence ϕ , if $\text{Mod}^{\mathcal{L}}(\phi) \neq \emptyset$, then there exists $\mathfrak{A} \in \text{Mod}^{\mathcal{L}}(\phi)$ of cardinality at most κ .

Lindström's First Theorem 7.6.12 *For any regular logic $\mathcal{L}$, if $\mathcal{L}$ is compact and $\ell(\mathcal{L}) = \omega$, then $\mathcal{L}$ is equivalent to first-order logic.*

Proof. By the two preceding propositions, it will suffice to show that under the hypotheses, for every relational signature μ , $\mathcal{L}$ has the strong finite dependence property at μ . Suppose, towards a contradiction, that this is not the case — that is, for some relational signature μ and some $\phi \in \text{Sent}_\mu^{\mathcal{L}}$, for each $n \in \omega$ there exist μ -structures $\mathfrak{A}_n$ and $\mathfrak{B}_n$ such that

$$(1) \quad \mathfrak{A}_n \restriction \mu_\phi \simeq_p^n \mathfrak{B}_n \restriction \mu_\phi, \quad \models_{\mathfrak{A}_n}^{\mathcal{L}} \phi, \quad \text{but} \quad \not\models_{\mathfrak{B}_n}^{\mathcal{L}} \phi.$$

Here μ_ϕ is the finite subsignature of μ guaranteed by Proposition 7.6.9 such that for any μ -structures $\mathfrak{A}$ and $\mathfrak{B}$,

$$(2) \quad \mathfrak{A} \upharpoonright \mu_\phi \simeq \mathfrak{B} \upharpoonright \mu_\phi \quad \implies \quad \left[\models_{\mathfrak{A}}^{\mathcal{L}} \phi \iff \models_{\mathfrak{B}}^{\mathcal{L}} \phi \right].$$

We shall define a signature $\mu' \supseteq \mu$, containing a new unary relation symbol $\dot{M}$ and a sentence $\psi \in \text{Sent}_{\mu'}^{\mathcal{L}}$, such that

$$(3) \quad \text{for each } n \in \omega, \text{ exists } \mathfrak{C}_n \in \text{Mod}_{\mu'}^{\mathcal{L}}(\psi) \text{ with } \text{Card}(\dot{M}^{\mathfrak{C}_n}) = n + 1;$$

and

$$(4) \quad \text{for every } \mathfrak{C} \in \text{Mod}_{\mu'}^{\mathcal{L}}(\psi), \dot{M}^{\mathfrak{C}} \text{ is finite.}$$

From (3) and (4) we have an immediate contradiction: if Γ denotes the set consisting of ψ and the $\mathcal{L}$ -translations of the first-order sentences $\exists^{\geq n} x \dot{M}x$ for all $n > 0$, then Γ is consistent by (3) and the assumed compactness of $\mathcal{L}$, but any model of Γ would contradict (4).

Fix a choice of $\mathfrak{A}_n$ and $\mathfrak{B}_n$ which satisfy (1); by the isomorphism property we may assume that $A_n \cap B_n = \emptyset$. For each n and $i \leq n$, let $\mathcal{E}_i^n$ be

$$\{ \eta : \eta \text{ is an } i\text{-extendable partial embedding of } \mathfrak{A}_n \upharpoonright \mu_\phi \text{ into } \mathfrak{B}_n \upharpoonright \mu_\phi \},$$

so by hypothesis, $\mathcal{E}_i^n \neq \emptyset$. The signature μ' is obtained by adjoining to μ new relation symbols

$$\dot{A}_-, \quad \dot{B}_-, \quad \dot{M}_-, \quad \dot{E}_{--}, \quad \dot{J}_{---}, \quad \text{and} \quad \dot{<}.$$

The μ' -structure $\mathfrak{C}_n$ is defined as follows. The universe is

$$C_n := A_n \cup B_n \cup \{0, 1, \dots, n\} \cup \bigcup_{i \leq n} \mathcal{E}_i^n.$$

Relations corresponding to the relation symbols of μ are defined on A_n and B_n in such a way that these sets are universes of substructures of $\mathfrak{C} \upharpoonright \mu$ isomorphic to $\mathfrak{A}_n$ and $\mathfrak{B}_n$, respectively. The new symbols are interpreted as follows:

$$\begin{aligned} \dot{A}^{\mathfrak{C}_n} &:= A_n; & \dot{B}^{\mathfrak{C}_n} &:= B_n; & \dot{M}^{\mathfrak{C}_n} &:= \{0, 1, \dots, n\}; \\ \dot{E}^{\mathfrak{C}_n} &:= \{ (\eta, i) : i \leq n \text{ and } \eta \in \mathcal{E}_i^n \}; & \dot{<}^{\mathfrak{C}_n} &:= \{ (i, j) : i < j \leq n \}; \\ \dot{J}^{\mathfrak{C}_n} &:= \{ (\eta, a, b) : \text{for some } i \leq n, \eta \in \mathcal{E}_i^n \text{ and } \eta(a) = b \}. \end{aligned}$$

Let $\text{ImPr}(u, v)$ abbreviate ‘ v is an immediate predecessor of u in the ordering $\dot{<}$ ’. The sentence $\psi \in \text{Sent}_{\mu'}^{\mathcal{L}}$ is designed to describe much of the structure $\mathfrak{C}_n$; it is the conjunction of

(i) the $\mathcal{L}$ -sentence $\phi^{(\dot{A})} \wedge (\neg \phi)^{(\dot{B})}$

with the $\mathcal{L}$ -translations of the universal closures of the following first-order sentences:

- (ii) $\exists x \dot{A}x \wedge \exists x \dot{B}x$;
- (iii) ‘ $\dot{<}$ is a discrete linear ordering of $\dot{M}$ with last element’;
- (iv) $\dot{M}u \leftrightarrow \exists e \dot{E}eu$;
- (v) $\dot{J}exy \rightarrow \exists u \dot{E}eu \wedge \dot{A}x \wedge \dot{B}y$;
- (vi) $\dot{J}exy \wedge \dot{J}ex'y' \rightarrow [x \dot{=} x' \leftrightarrow y \dot{=} y']$;
- (vii) $\bigwedge_{i < k} \dot{J}ex_i y_i \rightarrow [px_0, \dots, x_{k-1} \leftrightarrow py_0, \dots, y_{k-1}]$ for each of the (finitely many) k -ary μ_ϕ -relation symbols p ;
- (viii) $\dot{E}eu \wedge \dot{A}x' \wedge \text{ImPr}(u, v) \rightarrow \exists z [\dot{E}zv \wedge \forall x \forall y (\dot{J}exy \rightarrow \dot{J}zxy) \wedge \exists y' \dot{J}zx'y']$;
- (ix) $\dot{E}eu \wedge \dot{B}y' \wedge \text{ImPr}(u, v) \rightarrow \exists z [\dot{E}zv \wedge \forall x \forall y (\dot{J}exy \rightarrow \dot{J}zxy) \wedge \exists x' \dot{J}zx'y']$.

It is straightforward to verify that $\mathfrak{C}_n \in \text{Mod}_{\mu'}^{\mathcal{L}}(\psi)$, and obviously $\text{Card}(\dot{M}^{\mathfrak{C}_n}) = n + 1$, so (3) holds. Suppose contrary to (4) that there exists $\mathfrak{C} \in \text{Mod}_{\mu'}^{\mathcal{L}}(\psi)$ with $\dot{M}^{\mathfrak{C}}$ infinite. We claim first that there exists a countable such model. Let f be a new unary function symbol. By the assumption above, the $\mathcal{L}$ -sentence

$$\psi \wedge \left['f \text{ is an injection of } \dot{M} \text{ into a proper subset of } \dot{M}' \right]^{\mathcal{L}}$$

has a model; hence since $\ell(\mathcal{L}) = \omega$, it has a countable model (which we also call) $\mathfrak{C}$. The second clause guarantees that $\dot{M}^{\mathfrak{C}}$, and hence $\mathfrak{C}$ is infinite, so $\mathfrak{C}$ is denumerable.

For each $u \in \dot{M}^{\mathfrak{C}}$ and each e such that $(e, u) \in \dot{E}^{\mathfrak{C}}$, let

$$\eta_e := \left\{ (a, b) : (e, a, b) \in \dot{J}^{\mathfrak{C}} \right\} \quad \text{and} \quad \mathcal{E}_u := \left\{ \eta_e : (e, u) \in \dot{E}^{\mathfrak{C}} \right\}.$$

Using (ii), let $\mathfrak{A}$ and $\mathfrak{B}$ denote the substructures of $\mathfrak{C}$ with universes $A := \dot{A}^{\mathfrak{C}}$ and $B := \dot{B}^{\mathfrak{C}}$, respectively. By (v)–(vii), each η_e is a partial embedding of $\mathfrak{A} \upharpoonright \mu_\phi$ into $\mathfrak{B} \upharpoonright \mu_\phi$ and by (iv) each $\mathcal{E}_u$ is nonempty. Furthermore, by (viii) and (ix), the collection of sets $\mathcal{E}_u$ have the following extension property:

$\eta \in \mathcal{E}_u, a \in A$, and v is the immediate predecessor of u

$$\implies \text{for some } \zeta \in \mathcal{E}_v [\eta \subseteq \zeta \text{ and } a \in \text{Dom}(\zeta)],$$

and

$\eta \in \mathcal{E}_u, b \in B$, and v is the immediate predecessor of u

$$\implies \text{for some } \zeta \in \mathcal{E}_v[\eta \subseteq \zeta \text{ and } b \in \text{Im}(\zeta)].$$

Using (iii), let u_0 be the last element of C in the discrete ordering $\prec^c$ and for each $n \in \omega$, let u_{n+1} be the immediate predecessor of u_n . Since $\dot{M}^c$ is infinite, all u_n exist. Set $\mathcal{E} := \bigcup_{n \in \omega} \mathcal{E}_{u_n}$. It is easy to verify that $\mathcal{E}$ has the back-and-forth property and thus witnesses that $\mathfrak{A} \upharpoonright \mu_\phi \simeq_{\mathbf{p}} \mathfrak{B} \upharpoonright \mu_\phi$. Since these are countable structures, it follows by Theorem 7.1.9 that $\mathfrak{A} \upharpoonright \mu_\phi \simeq \mathfrak{B} \upharpoonright \mu_\phi$. But by (i) and the relativization property, $\models_{\mathfrak{A}}^{\mathcal{L}} \phi$ and $\models_{\mathfrak{B}}^{\mathcal{L}} \neg \phi$, contrary to (2). Hence also (4) holds. $\square$

We head now for the second theorem of Lindström in which the compactness condition is replaced by one involving the effectiveness of the notion of logical validity. To formulate this condition in the general setting of abstract logics requires several new notions.

Definition 7.6.13 μ is a **recursive signature** iff it is finite or it is denumerable and there is an enumeration $\langle s_n : n \in \omega \rangle$ of the non-logical symbols of μ such that the function $n \mapsto \nu_\mu(s_n)$ is recursive.

Definition 7.6.14 An **effective logic** is a logic $\mathcal{L}$ with the finite dependence property together with a family of functions $\text{gn}_\mu^{\mathcal{L}} : \text{Sent}_\mu^{\mathcal{L}} \rightarrow \omega$ for each recursive signature μ such that

- (i) $\text{Im}(\text{gn}_\mu^{\mathcal{L}})$ is a recursive set;
- (ii) the mappings

$$\text{gn}_\mu^{\mathcal{L}}(\phi) \mapsto \text{gn}_\mu^{\mathcal{L}}(\neg \phi) \quad \text{and} \quad (\text{gn}_\mu^{\mathcal{L}}(\phi), \text{gn}_\mu^{\mathcal{L}}(\psi)) \mapsto \text{gn}_\mu^{\mathcal{L}}(\phi \vee \psi)$$

are recursive;

- (iii) for any recursive signature $\mu' \supseteq \mu$, the mapping $\text{gn}_\mu^{\mathcal{L}}(\phi) \mapsto \text{gn}_{\mu'}^{\mathcal{L}}(\phi)$ is recursive.

Definition 7.6.15 For any effective logics $\mathcal{K}$ and $\mathcal{L}$,

- (i) $\mathcal{L}$ **effectively translates** $\mathcal{K}$ — in symbols, $\mathcal{K} \leq_{\text{eff}} \mathcal{L}$ — iff for each recursive signature μ , there exists a mapping $\phi \mapsto \phi^{\mathcal{L}}$ of $\text{Sent}_\mu^{\mathcal{K}}$ into $\text{Sent}_\mu^{\mathcal{L}}$ such that for all $\phi \in \text{Sent}_\mu^{\mathcal{K}}$, $\text{Mod}_\mu^{\mathcal{K}}(\phi) = \text{Mod}_\mu^{\mathcal{L}}(\phi^{\mathcal{L}})$ and the mapping $\text{gn}_\mu^{\mathcal{K}}(\phi) \mapsto \text{gn}_\mu^{\mathcal{L}}(\phi^{\mathcal{L}})$ is recursive;
- (ii) $\mathcal{K}$ and $\mathcal{L}$ are **effectively equivalent logics** — in symbols, $\mathcal{K} \sim_{\text{eff}} \mathcal{L}$ — iff $\mathcal{K} \leq_{\text{eff}} \mathcal{L}$ and $\mathcal{L} \leq_{\text{eff}} \mathcal{K}$;

- (iii) $\mathcal{L}$ is **effectively regular** iff $\mathcal{L}$ is regular, $\mathcal{L}^1 \leq_{\text{eff}} \mathcal{L}$, and for each recursive signature μ the mappings $\text{gn}_\mu^\mathcal{L}(\phi) \mapsto \text{gn}_\mu^\mathcal{L}(\phi^{(A)})$ and $\text{gn}_\mu^\mathcal{L}(\phi) \mapsto \text{gn}_\mu^\mathcal{L}(\phi^{\text{rel}})$ are recursive;
- (iv) $\mathcal{L}$ has the **validity enumeration property** iff for each recursive signature μ ,

$$\left\{ \text{gn}_\mu^\mathcal{L}(\phi) : \models \phi \text{ and } \phi \in \text{Sent}_\mu^\mathcal{L} \right\}$$

is recursively enumerable.

The key result we use here is Trachtenbrot's Theorem 5.5.18, which in the current notation reads: there exists a finite signature μ_{Tr} such that the set

$$\text{FinVal}_{\mu_{\text{Tr}}} := \left\{ \chi : \chi \in \text{Sent}_{\mu_{\text{Tr}}}^{\mathcal{L}^1} \text{ and for all finite } \mu_{\text{Tr}}\text{-structures } \mathfrak{A}, \models_{\mathfrak{A}}^{\mathcal{L}^1} \chi \right\}$$

is not recursively enumerable — that is, $\left\{ \text{gn}_{\mu_{\text{Tr}}}^{\mathcal{L}^1}(\chi) : \chi \in \text{FinVal}_{\mu_{\text{Tr}}} \right\}$ is not an r.e. set of natural numbers.

Lindström's Second Theorem 7.6.16 *For any effectively regular logic $\mathcal{L}$, if $\mathcal{L}$ has the validity enumeration property and $\ell(\mathcal{L}) = \omega$, then $\mathcal{L}$ is effectively equivalent to first-order logic.*

Proof. We show first that under the hypotheses, $\mathcal{L}^1$ translates $\mathcal{L}$ at every recursive relational signature. It follows by Proposition 7.6.11 that this holds for every recursive signature, and we finish by showing that these translations are effective. Suppose toward a contradiction that there is a recursive relational signature μ such that $\mathcal{L}^1$ does not translate $\mathcal{L}$ at μ , so by Proposition 7.6.10, $\mathcal{L}$ does not have the strong finite dependence property at μ . Exactly as in the proof of Lindström's First Theorem, we may conclude that there is a signature $\mu' \supseteq \mu$ containing a new unary relation $\dot{M}$ and a sentence $\psi \in \text{Sent}_{\mu'}^\mathcal{L}$ such that

- (3) for each $n \in \omega$, there exists $\mathfrak{C}_n \in \text{Mod}_{\mu'}^\mathcal{L}(\psi)$ with $\text{Card}(\dot{M}^{\mathfrak{C}_n}) = n + 1$;

and

- (4) for every $\mathfrak{C} \in \text{Mod}_{\mu'}^\mathcal{L}(\psi)$, $\dot{M}^\mathfrak{C}$ is finite.

Note that this part of the earlier proof used the hypothesis that $\ell(\mathcal{L}) = \omega$, but depended on the compactness assumption only to conclude that $\mathcal{L}$ has the finite dependence property, which we have assumed here (as part of the definition of an effective logic).

Let μ_{Tr} be a signature disjoint from μ' as in Trachtenbrot's Theorem and μ'' the union of μ' and μ_{Tr} . By the hypothesis that $\mathcal{L}^1 \leq_{\text{eff}} \mathcal{L}$, there exists

a translation mapping $\chi \mapsto \chi^{\mathcal{L}}$ which is recursive on Gödel numbers. We claim that for all $\chi \in \text{Sent}_{\mu_{\text{Tr}}}$,

$$\chi \in \text{FinVal}_{\mu_{\text{Tr}}} \iff \models \psi \rightarrow (\chi^{\mathcal{L}})^{(\dot{M})}.$$

Indeed, as $\mathfrak{C}$ varies over $\text{Mod}_{\mu''}^{\mathcal{L}}(\psi)$, the structures $\mathfrak{M}_{\mathfrak{C}} \upharpoonright \mu_{\text{Tr}}$ with universe $\dot{M}^{\mathfrak{C}}$ vary over isomorphic copies of all finite μ_{Tr} -structures, and the claim follows by the reduct property. Now by the definition of effectively regular there exists a recursive function F such that for all $\chi \in \text{Sent}_{\mu_{\text{Tr}}}$,

$$F(\text{gn}_{\mu_{\text{Tr}}}^{\mathcal{L}^1}(\chi)) = \text{gn}_{\mu''}^{\mathcal{L}}(\psi \rightarrow (\chi^{\mathcal{L}})^{(\dot{M})}),$$

so from the validity enumeration property we conclude that

$$\left\{ \text{gn}_{\mu_{\text{Tr}}}^{\mathcal{L}^1}(\chi) : \chi \in \text{FinVal}_{\mu_{\text{Tr}}} \right\}$$

is many-one reducible to a recursively enumerable set, hence is recursively enumerable, contrary to Trachtenbrot's Theorem.

We have now shown that for each recursive signature μ there exists an $\mathcal{L}^1$ -translation $\phi \mapsto \phi^1$ of $\mathcal{L}$ at μ . Equivalently, for all $\phi \in \text{Sent}_{\mu}^{\mathcal{L}}$, $\models \phi \leftrightarrow (\phi^1)^{\mathcal{L}}$. By the validity enumeration property there exists a recursive relation R such that for all $\theta \in \text{Sent}_{\mu}^{\mathcal{L}}$,

$$\models \theta \iff \exists m R(m, \text{gn}_{\mu}^{\mathcal{L}}(\theta)).$$

Fix an enumeration $\langle \psi_n : n \in \omega \rangle$ of $\text{Sent}_{\mu}^{\mathcal{L}^{\infty}}$ such that the function $n \mapsto \text{gn}_{\mu}^{\mathcal{L}^1}(\psi_n)$ is recursive, let G be the recursive function such that for all $\phi \in \text{Sent}_{\mu}^{\mathcal{L}}$,

$$G(\text{gn}_{\mu}^{\mathcal{L}}(\phi)) = (\text{least } \langle m, n \rangle R(m, \text{gn}_{\mu}^{\mathcal{L}}(\phi \leftrightarrow (\psi_n)^{\mathcal{L}})))_1,$$

and set $\phi^* := \psi_{G(\text{gn}_{\mu}^{\mathcal{L}}(\phi))}$. Then $\phi \mapsto \phi^*$ witnesses that $\mathcal{L} \leq_{\text{eff}} \mathcal{L}^1$. $\square$

Collateral Reading. [Eb-Fl-Th] and [Ch-Ke] present proofs of Lindström's theorems.

Exercises

Exercise 7.6.17 Let $\mathcal{L}$ be a regular logic with $\ell(\mathcal{L}) = \omega$ and μ any signature. Show that for any two μ -structures $\mathfrak{A}$ and $\mathfrak{B}$, if $\mathfrak{A} \simeq_{\text{p}} \mathfrak{B}$, then $\mathfrak{A} \equiv^{\mathcal{L}} \mathfrak{B}$.

Hint: Expand μ to a signature μ' in such a way that there is a μ' -sentence which implies that any model consists of a pair of partially isomorphic structures (together with some other relations).

Chapter 8

Recursion Theory

8.1 Many-one degrees and r.e. sets

The topic to be explored in this chapter can roughly be characterized as the study of the complexity of sets of natural numbers, particularly the recursively enumerable (r.e.) sets. In this accounting, a recursive set is considered simple because there is an algorithm for deciding membership in the set. A non-recursive set is more complex, and we shall see that there are interesting ways to measure how complex a set is. One such measure considered above is the arithmetical hierarchy of [Section 5.1](#). The recursive sets are those in the class Δ_1 , the r.e. sets are the Σ_1 sets, and as the subscript n increases, sets in Σ_n or Π_n are increasingly complex in the sense that more quantifiers are required to give their definitions.

Another measure of relative complexity was introduced in Definition 4.3.23.

Definition 8.1.1 For any $A, B \subseteq \omega$, A is *many-one reducible* or *m-reducible* to B — in symbols, $A \leq_m B$ — iff there exists a recursive function f such that for all $m \in \omega$, $m \in A \iff f(m) \in B$. Note that this may also be written more compactly $A = f^{-1}(B)$.

If $A \leq_m B$, then A is at most as complex as B in the following sense: from a procedure $\mathcal{B}$ for determining membership in B , we can generate algorithmically a procedure $\mathcal{A}$ for determining membership in A by setting $\mathcal{A}(m) := \mathcal{B}(f(m))$. If B is non-recursive, then $\mathcal{B}$ is not an algorithm but makes use of some non-effective information; no additional non-effective information is needed to compute values of $\mathcal{A}$. In practice we shall think of these procedures as simply the characteristic functions of the sets involved and use the same symbol to denote the two. Thus we will write

$$A(m) = \begin{cases} 0, & \text{if } m \notin A; \\ 1, & \text{if } m \in A; \end{cases}$$

and $A \leq_m B$ iff there exists a recursive function f such that for all $m \in \omega$, $A(m) = B(f(m))$.

This relationship may also be described as follows. Faced with the problem of computing a particular value $A(m)$ — that is, answering the question whether m is a member of A or not — we may use the services of an **oracle** for B and ask one question: is $f(m)$ a member of B ? The (correct) answer we receive is also the correct answer for the original question. For example, if A is any r.e. set, then $A \leq_m \text{gn}(\mathbf{Q})$, since if $\psi(x)$ is a formula that weakly $\mathbf{Q}$ -represents A , then $m \in A \iff \text{Sb}^1(\text{gn}(\psi), m) \in \text{gn}(\mathbf{Q})$. The proofs in Examples 5.1.10 were based on similar ideas.

This formulation suggests various generalizations in which we may ask several questions of the oracle, with later questions depending on the answers to earlier ones, and interpret the answers in more complex ways. The most general version of this idea is a rough description of *Turing reducibility*, which we will study starting in Section 8.2. Several intermediate notions have also been extensively studied under such names as *truth-table reducibility*, *bounded truth-table reducibility* and *bounded Turing reducibility*, and we consider the first of these in the exercises. See also the Collateral Reading suggestions at the end of the section.

A few elementary properties of $\mathbf{m}$ -reducibility were assigned in Exercises 4.3.26 and 4.3.27, and we review them first.

Definition 8.1.2 For any $A, B \subseteq \omega$,

- (i) A is **strict many-one reducible** or **strict $\mathbf{m}$ -reducible** to B — in symbols, $A <_m B$ — iff $A \leq_m B$ but $B \not\leq_m A$;
- (ii) A and B are **many-one equivalent** or **$\mathbf{m}$ -equivalent** — in symbols, $A \equiv_m B$ — iff $A \leq_m B$ and $B \leq_m A$;
- (iii) the **many-one degree** or **$\mathbf{m}$ -degree** of A is the set

$$\text{dg}_m(A) := \{ B : A \equiv_m B \};$$

- (iv) $\mathbf{dg}_m(A) \leq \mathbf{dg}_m(B)$ iff $A \leq_m B$; $\mathbf{dg}_m(A) < \mathbf{dg}_m(B)$ iff $A <_m B$;
- (v) $\mathbf{Dg}_m = \{ \mathbf{dg}_m(A) : A \subseteq \omega \}$, the set of all m -degrees.

Proposition 8.1.3

- (i) $<_m$ is irreflexive and transitive.
- (ii) $\leq_m$ is reflexive and transitive.
- (iii) $\equiv_m$ is an equivalence relation.

Proof. Immediate from the definitions. $\square$

It follows that the ordering relations on $\mathbf{Dg}_m$ are well-defined.

Proposition 8.1.4

- (i) For any A and B , if $A \leq_m B$, then

$$\begin{aligned} B \text{ is recursive} &\implies A \text{ is recursive} \quad \text{and} \\ B \text{ is (co-)r.e.} &\implies A \text{ is (co-)r.e.} \end{aligned}$$

- (ii) $\mathbf{dg}_m(\emptyset) = \{ \emptyset \}$ and $\mathbf{dg}_m(\omega) = \{ \omega \}$ are minimal m -degrees.
- (iii) If $\mathbf{0} := \{ A : A \text{ is recursive} \wedge A \neq \emptyset, \omega \}$, then
 - (a) $\mathbf{0}$ is an m -degree;
 - (b) $\mathbf{dg}_m(\emptyset), \mathbf{dg}_m(\omega) < \mathbf{0}$;
 - (c) for any non-recursive set A , $\mathbf{0} < \mathbf{dg}_m(A)$.
- (iv) Any pair of m -degrees has a least upper bound.
- (v) There is no maximal m -degree.
- (vi) The m -degrees distinct from $\mathbf{dg}_m(\emptyset)$ and $\mathbf{dg}_m(\omega)$ are not linearly ordered.

Proof. Part (i) is Proposition 4.3.25, and (ii) is immediate from the fact that

$$A = f^{-1}(\emptyset) \implies A = \emptyset \quad \text{and} \quad A = f^{-1}(\omega) \implies A = \omega.$$

Towards (iii), we first establish two facts: for any A and B ,

- (1) if $B \in \mathbf{0}$ and $A \leq_m B$, then

$$A \text{ is recursive} \quad \text{and} \quad (B \leq_m A \implies A \neq \emptyset, \omega);$$
- (2) A is recursive and $B \neq \emptyset, \omega \implies A \leq_m B$.

(1) is immediate from (i) and (ii); for (2), fix any such B and fix $p \in B$ and $q \notin B$. Then, for any recursive set A , set

$$f_A(m) = \begin{cases} p, & \text{if } m \in A; \\ q, & \text{if } m \notin A. \end{cases}$$

Then $A = f_A^{-1}(B)$, and f_A is a recursive function so $A \leq_m B$.

Now by (1) and (2), for any A and B ,

$$A \equiv_m B \in \mathbf{0} \implies A \in \mathbf{0} \quad \text{and} \quad A, B \in \mathbf{0} \implies A \equiv_m B,$$

so $\mathbf{0}$ is the $\mathbf{m}$ -degree of any of its members. Parts (b) and (c) are also immediate from (1) and (2).

For (iv) it is easy to verify that the least upper bound of $\mathbf{dg}_m(A)$ and $\mathbf{dg}_m(B)$ is the $\mathbf{m}$ -degree of the set

$$A \oplus B := \{2m : m \in A\} \cup \{2m + 1 : m \in B\}.$$

For (v), note that since there are only denumerably many recursive functions, for any set B there are only denumerably many $A \leq_m B$. Hence there exists C such that either $B <_m C$ or B and C are incomparable. In either case the $\mathbf{m}$ -degree of $B \oplus C$ is strictly greater than $\mathbf{dg}_m(B)$, so this is not maximal. Finally, for (vi), let A be an r.e. set which is not recursive, for example $\mathbf{gn}(\mathbf{Q})$ (Corollaries 4.1.15 and 4.6.18). Then since $\overline{A}$ is not r.e. and A is not co-r.e. (Corollary 4.3.7), neither $A \leq_m \overline{A}$ nor $\overline{A} \leq_m A$. $\square$

Note that we do not claim (because it is false) that any pair of $\mathbf{m}$ -degrees has a greatest lower bound. We shall be interested here primarily in the $\mathbf{m}$ -degrees of r.e. sets.

Definition 8.1.5 For any set C ,

- (i) C is **$\mathbf{m}$ -complete** iff C is r.e. and for every r.e. set A , $A \leq_m C$;
- (ii) $\mathbf{dg}_m(C)$ is an ***r.e. m-degree*** iff it contains an r.e. set — that is, there exists an r.e. set A such that $A \equiv_m C$.

Proposition 8.1.6 *Each of the following sets is $\mathbf{m}$ -complete:*

- (i) $\mathbf{Th}^\Sigma(\Omega)$, the set of true Σ sentences of Ω (Definition 5.1.20);
- (ii) $\mathbf{PartComp}$ (Definition 5.2.17);
- (iii) the unary halting problem, $W := \{ \langle a, m \rangle : \exists n[\{a\}(m) \simeq n] \}$ (Definition 5.2.24);
- (iv) $\mathbf{gn}(T)$ for any consistent recursively axiomatizable theory T which extends $\mathbf{Q}$.

Proof. The $\mathbf{m}$ -completeness of (i) was the content of Exercise 5.1.33. That each of (ii)–(iv) is r.e. is (part of) the content of Proposition 5.2.18, Theorem 5.2.25 and the Enumerability Theorem 4.4.17. Now let A be any r.e. set. By Corollary 4.3.19, for some partial recursive function F , $A = \text{Dom}(F)$, so by Theorem 5.2.23, for some $a \in \omega$, $A = \text{Dom}\{a\}$. Then for the recursive function $G(m) = \langle a, m \rangle$, we have $A = G^{-1}(W)$. Thus W is $\mathbf{m}$ -complete.

Now let $Z(a) := \langle 3, 1, a, \langle 0, 1, 0 \rangle \rangle$ be an index for the composition of the partial function $\{a\}$ with the unary constant 0 function. Then

$$\begin{aligned} m \in A &\iff m \in \text{Dom}\{a\} \iff \{Z(a)\}(m) \simeq 0 \\ &\iff \langle Z(a), m, 0 \rangle \in \text{PartComp}, \end{aligned}$$

and with $H(m) := \langle Z(a), m, 0 \rangle$, we have $A = H^{-1}(\text{PartComp})$. Thus also PartComp is $\mathbf{m}$ -complete.

Finally, if T is any consistent recursively axiomatizable theory which extends $\mathbf{Q}$, by Theorem 4.6.30, every r.e. set A is weakly T -representable — that is, for some formula θ and all m ,

$$m \in A \iff T \vdash \theta_x(m) \iff \text{Sb}^1(\text{gn}(\theta), m) \in \text{gn}(T).$$

Since Sb^1 is recursive, it follows that $\text{gn}(T)$ is $\mathbf{m}$ -complete. This was also Exercise 4.6.47(i). $\square$

Corollary 8.1.7 $\mathbf{1} := \{C : C \text{ is } \mathbf{m}\text{-complete}\}$ is the unique largest r.e. $\mathbf{m}$ -degree. Furthermore, $\mathbf{0} < \mathbf{1}$.

Proof. Using the transitivity of $\leq_{\mathbf{m}}$ it is immediate that if C is $\mathbf{m}$ -complete, then for any r.e. set D , $C \equiv_{\mathbf{m}} D$ iff D is $\mathbf{m}$ -complete so $\mathbf{1}$ is an $\mathbf{m}$ -degree, and it is clearly the largest r.e. $\mathbf{m}$ -degree. That $\mathbf{0} < \mathbf{1}$ follows from the fact that there exist non-recursive r.e. sets, so any $\mathbf{m}$ -complete set must be non-recursive. $\square$

This may be seen as an abstract form of the First Incompleteness Theorem, and a tentative conclusion one might draw from the proposition is that seemingly every example of a non-recursive r.e. set is in fact $\mathbf{m}$ -complete. It is natural to ask if there are any r.e. sets A of intermediate complexity: non-recursive but not complete or equivalently such that $\mathbf{0} < \text{dg}_{\mathbf{m}}(A) < \mathbf{1}$. We shall soon construct many such sets.

In what follows, it will be essential to fix an enumeration of the r.e. sets. At the same time we introduce some notational changes. Since from now on in this chapter we shall never be dealing with formal languages and their formulas and variables, we shall begin to use letters x, y etc. as informal variables ranging over ω and letters ϕ, ψ etc. as variables over (partial)

functions. Letters $A, B, \dots$ are variables over sets of natural numbers, and $\overline{A}$ denotes the complement $\omega \sim A$ of A .

Recall from Definition 5.2.21 that $\langle \{a\} : a \in \omega \rangle$ is a list of the partial recursive functions via the defining equivalence

$$\{a\}(x_0, \dots, x_{k-1}) \simeq y \iff \langle a, x_0, \dots, x_{k-1}, y \rangle \in \text{PartComp};$$

we call a an **index** for $\{a\}$. Recall also that

$$\{a\}(x_0, \dots, x_{k-1}) \downarrow \iff (x_0, \dots, x_{k-1}) \in \text{Dom}(\{a\});$$

otherwise $\{a\}(x_0, \dots, x_{k-1}) \uparrow$. Formally every natural number a is an index, but by the definition of **PartComp**, unless a is of a carefully prescribed form, the corresponding function $\{a\}$ is the empty function. In particular, $\{a\} \neq \emptyset$ only if a is of the form $\langle i, k, \dots \rangle$, and in this case $\{a\}$ is a k -ary function, so

$$\{a\}(m_0, \dots, m_{k-1}) \downarrow \implies (a)_1 = k.$$

Definition 8.1.8

- (i) For all $a \in \omega$, $W_a := \{x : \{a\}(x) \downarrow\}$.
- (ii) $K := \{a : a \in W_a\}$.

Proposition 8.1.9 *A set A is r.e. iff for some $a \in \omega$, $A = W_a$.*

Proof. This is a restatement of part of Theorem 4.3.16(ii). $\square$

Since K is a diagonal set, the methods of the Diagonal Lemma 4.1.21 show easily that

Proposition 8.1.10 *K is r.e. but not recursive.*

Proof. That K is r.e. follows immediately from the fact that $K \leq_m W$: $a \in K \iff \langle a, a \rangle \in W$. Now if K were recursive, then both K and $\overline{K}$ would be r.e., so in particular for some $\bar{a}$, $\overline{K} = W_{\bar{a}}$. But then

$$\bar{a} \in K \iff \bar{a} \in W_{\bar{a}} \iff \bar{a} \in \overline{K} \iff \bar{a} \notin K,$$

a contradiction. Hence K is not recursive. $\square$

It is not immediately obvious that K is **m**-complete; to show this we need a basic technical result. Compare here also the similar result of Exercise 4.6.47(ii).

Proposition 8.1.11 *For each $m, n \in \omega$, there exists a primitive recursive function S_n^m such that for all $a, x_0, \dots, x_{m-1}$ and $y_0, \dots, y_{n-1}$,*

$$\{S_n^m(a, x_0, \dots, x_{m-1})\}(y_0, \dots, y_{n-1}) \simeq \{a\}(x_0, \dots, x_{m-1}, y_0, \dots, y_{n-1}).$$

Proof. Note that the functions S_n^m compute indices from indices — in effect they take an index for an $(m+n)$ -ary partial recursive function and for each fixed choice of the first m arguments compute an index for the value as a function of the remaining n arguments. It is a simple calculation, using the definitions at the end of [Section 5.2](#), to verify that the following definitions do the job:

$$\begin{aligned} S_n^1(a, x) &:= \langle 3, n, a, \langle 0, n, x \rangle, \langle 1, n, 0 \rangle, \dots, \langle 1, n, n-1 \rangle \rangle; \\ S_n^{m+1}(a, x_0, \dots, x_m) &:= S_n^1(S_{n+1}^m(a, x_0, \dots, x_{m-1}), x_m). \quad \square \end{aligned}$$

Note the similarity of these functions with the substitution functions Sb^k (Corollary 4.4.6) which produce the Gödel number of a formula with certain substitutions made.

Proposition 8.1.12 *K is $\mathbf{m}$ -complete.*

Proof. For any r.e. set A , let F be the partial function defined by

$$F(x, y) \simeq \begin{cases} 0, & \text{if } x \in A; \\ \uparrow, & \text{otherwise.} \end{cases}$$

The graph of F is clearly semi-recursive, so F is partial recursive (Corollary 4.3.13); let a be an index for F . Then for any $x \in \omega$,

$$x \in A \iff \forall y \{S_1^1(a, x)\}(y) \downarrow \iff S_1^1(a, x) \in K.$$

Hence $A \leq_m K$ and K is $\mathbf{m}$ -complete. $\square$

Although K still does not provide us with an example of a non-recursive r.e. set which is not $\mathbf{m}$ -complete, it does lead to a method for finding one. The key observation is that $\bar{K}$ is not r.e. in the following strong way.

Definition 8.1.13 A set P is **productive** iff there exists a partial recursive function ϕ such that for each index a , if $W_a \subseteq P$, then $\phi(a)$ is defined and is a member of $P \sim W_a$.

A productive set is in a sense effectively not r.e. since whenever $W_a \subseteq P$, $\phi(a)$ provides a witness to the fact that $P \neq W_a$.

Proposition 8.1.14 *$\bar{K}$ is productive.*

Proof. For any a , if $W_a \subseteq \bar{K}$, then a is a member of $\bar{K} \sim W_a$. $\square$

Proposition 8.1.15 *For any P and Q , if P is productive and $P \leq_m Q$, then also Q is productive.*

Proof. Suppose that P is productive with associated function ϕ and that f is a recursive function such that $P = f^{-1}(Q)$. Let b be an index for f and set $g(a) := \langle 3, 1, a, b \rangle$. Then for any a , $W_{g(a)} = f^{-1}(W_a)$, since for any x ,

$$\begin{aligned} x \in f^{-1}(W_a) &\iff f(x) \in W_a \\ &\iff \{a\}(f(x)) \downarrow \iff \{g(a)\}(x) \downarrow. \end{aligned}$$

Hence, for each a ,

$$\begin{aligned} W_a \subseteq Q &\implies W_{g(a)} \subseteq P \\ &\implies \phi(g(a)) \in P \sim W_{g(a)} \implies f(\phi(g(a))) \in Q \sim W_a. \quad \square \end{aligned}$$

Corollary 8.1.16 *For any A , if A is $\mathbf{m}$ -complete, then $\overline{A}$ is productive.*

Proof. If A is $\mathbf{m}$ -complete, then in particular $\mathbf{K} \leq_m A$, so also $\overline{\mathbf{K}} \leq_m \overline{A}$ and $\overline{A}$ is productive. $\square$

The key to constructing a non-recursive r.e. set such that $\overline{A}$ is not productive and thus A is not $\mathbf{m}$ -complete is the following property.

Proposition 8.1.17 *Every productive set has an infinite r.e. subset.*

Proof. Let P be productive with associated function ϕ . The partial function F such that

$$F(a, x) \simeq 0 \iff \{a\}(x) \downarrow \quad \text{or} \quad \phi(a) \simeq x$$

has semi-recursive graph so is partial recursive. Let b be an index for F and set $f(a) := S_1^1(b, a)$. Then for each a

$$x \in W_{f(a)} \iff x \in W_a \quad \text{or} \quad \phi(a) \simeq x.$$

Let e be an index such that $W_e = \emptyset$ and g the function defined recursively by

$$g(0) = e \quad \text{and} \quad \text{for all } n, \quad g(n+1) = f(g(n)).$$

It is straightforward to check that $\phi \circ g$ is total recursive and enumerates an infinite r.e. subset of P . $\square$

Definition 8.1.18 A set A is **simple** iff A is r.e. and $\overline{A}$ is infinite but has no infinite r.e. subset.

Corollary 8.1.19 *Any simple set is a non-recursive r.e. set which is not m-complete.*

Proof. If A is simple, then clearly $\overline{A}$ is not r.e., so A is non-recursive. But also $\overline{A}$ is not productive, so A is not m-complete. $\square$

There are two natural proofs of the existence of simple sets, one short and slightly obscure, the other considerably longer but much more useful in the development of techniques for obtaining further results. First the short proof.

Theorem 8.1.20 *There exists a simple set and hence an r.e. set which is neither recursive nor m-complete.*

Proof. Let P be the relation defined by

$$P(a, x) \iff x \in W_a \text{ and } x > 2a.$$

Clearly P is semi-recursive, so by the Partial Selection Theorem 4.3.15, there exists a partial recursive function F with domain $\{a : \exists x P(a, x)\}$ such that for all a in this domain, $P(a, F(a))$. Let A be the image of F ; we show that A is simple. As the image of a partial recursive function, A is r.e. (Corollary 4.3.19). Since for all a in $\text{Dom}(F)$, $F(a) > 2a$, at most a members of A are in each initial segment $\{0, 1, \dots, 2a\}$ and thus this segment contains at least $a + 1$ elements of $\overline{A}$; it follows that $\overline{A}$ is infinite. Finally, suppose W_a is infinite. Then W_a has elements $x > 2a$, so $F(a)$ is defined and is therefore a member of $A \cap W_a$, and $W_a \not\subseteq \overline{A}$. $\square$

Once we know there is at least one such set, and hence an r.e. m-degree $\mathbf{d}$ such that $\mathbf{0} < \mathbf{d} < \mathbf{1}$, it is natural to ask if there are others, how many, and how they are ordered. The methods needed for these results are significantly more complicated and need some preparation in the form of a new characterization of r.e. sets.

Definition 8.1.21 A *stage enumeration* of a set A is a sequence of sets $\langle A_s : s \in \omega \rangle$ such that for all s ,

- (i) A_s is finite;
- (ii) $A_s \subseteq A_{s+1}$;
- (iii) $A = \bigcup_{s \in \omega} A_s$.

A stage enumeration is *recursive* iff also

- (iv) $\{(x, s) : x \in A_s\}$ is recursive.

The sets A_s can be viewed as increasingly good finite approximations to A . The terminology “stage enumeration” arises from a method of construction (or definition) which we will use for many of the proofs in the remainder of

this chapter. We will construct a set A by a process best described as taking place in an infinite sequence of stages; $A_0 = \emptyset$, and at stage s we determine A_{s+1} by adding a finite number of elements (usually at most one) to A_s . The recursiveness of the enumeration will follow from the effectiveness of the conditions which govern these additions.

We shall soon show that a set A is r.e. iff A has a recursive stage enumeration. One direction is clear: if A has a recursive stage enumeration then A is r.e., since

$$x \in A \iff \exists s [x \in A_s].$$

Towards the converse, recall from [Section 5.2](#) that **PartComp** is inductively defined by a recursive induction system $\mathcal{X}$, and therefore there is a recursive relation $\text{Derivof}^{\mathcal{X}}$ such that

$$\begin{aligned} \{a\}(x_0, \dots, x_{k-1}) \simeq y &\iff \langle a, x_0, \dots, x_{k-1}, y \rangle \in \text{PartComp} \\ &\iff \exists u \text{ Derivof}^{\mathcal{X}}(u, \langle a, x_0, \dots, x_{k-1}, y \rangle). \end{aligned}$$

A natural extension of Remark 5.2.16 shows that such a witness u can naturally be interpreted as coding the scratchwork for a computation of the value y of the partial recursive function $\{a\}$ applied to arguments $x_0, \dots, x_{k-1}$. Hence, a natural notion of approximation is to restrict the value of this witness. Since we are concentrating here on sets, and therefore unary functions, we state the definitions only for this case, although the extensions are obvious.

Definition 8.1.22 For any a, x, y and s ,

- (i) $\{a\}_s(x) \simeq y \iff (\exists u < s) \text{Derivof}^{\mathcal{X}}(u, \langle a, x, y \rangle)$;
- (ii) $W_{a,s} = \{x : \{a\}_s(x) \downarrow\}$.

The exact form of these definitions will be largely irrelevant; the principal attributes we use are

Proposition 8.1.23 For all s, a, x and y ,

- (i) $\{a\}(x) \simeq y \iff \exists s \{a\}_s(x) \simeq y$;
- (ii) $\{a\}_s \simeq y \implies (\forall t \geq s) \{a\}_t(x) \simeq y$;
- (iii) $x \in W_a \iff \exists s [x \in W_{a,s}]$;
- (iv) $\{a\}_s(x) \simeq y$ and $x \in W_{a,s}$ are recursive relations;
- (v) $\{a\}_s(x) \simeq y \implies a, x, y < s$;
- (vi) $x \in W_{a,s} \implies a, x < s$;
- (vii) $\{a\}_s$ is a finite function; $W_{a,s}$ is a finite set;
- (viii) there is at most one x such that $x \notin W_{a,s}$ but $x \in W_{a,s+1}$.

Proof. Parts (i)–(v) are obvious from the definitions. (vi) follows from the definition of Derivof^χ and hence of χ -derivation: if a number u codes a derivation which witnesses that $\langle a, x, y \rangle \in \text{PartComp}$, then u codes a sequence whose last component is $\langle a, x, y \rangle$ and by the way that we code sequences, $a, x, y < \langle a, x, y \rangle < u < s$. (vii) follows immediately as does (viii), since the only new derivation $< s + 1$ is the one with code s . $\square$

Corollary 8.1.24 *For any set A , A is r.e. iff it has a recursive stage enumeration.*

Proof. The implication ($\Leftarrow$) was noted above. If A is r.e., then for some index a , $A = W_a$ (Proposition 8.1.9). Then $\langle W_{a,s} : s \in \omega \rangle$ is a recursive stage enumeration of A . $\square$

Note that this provides a *uniform* stage enumeration of all r.e. sets in the sense that not only is each individual relation $\{(x, s) : x \in W_{a,s}\}$ recursive, but also the relation $\{(a, x, s) : x \in W_{a,s}\}$ is recursive. This fact will be used repeatedly without further mention.

We are now ready to give the second

Proof. (of Theorem 8.1.20) Our goal is to construct an r.e. set A such that $\bar{A}$ is infinite and A has a non-empty intersection with every infinite r.e. set. It will therefore suffice to ensure that for each $a \in \omega$, A satisfies the **requirements**

$$\begin{aligned} P_a : W_a \text{ infinite} &\implies A \cap W_a \neq \emptyset \\ N_a : A \cap \{x : x \leq 2a\} &\text{ has at most } a \text{ elements.} \end{aligned}$$

Note that N_a guarantees that at least half of the numbers in each initial segment of ω belong to $\bar{A}$. The construction proceeds in infinitely many stages s to build a stage enumeration $\langle A_s : s \in \omega \rangle$ of a set A which is defined by $A := \bigcup_{s \in \omega} A_s$. At stage s we determine the set A_{s+1} as either the same as A_s or as $A_s \cup \{x_a\}$, where x_a will be chosen from $W_{a,s}$ (for an appropriate a) and will therefore witness that $A \cap W_a \neq \emptyset$ — that is, that P_a is satisfied. As we proceed, those P_a that have been satisfied in this way are called **marked**; we need to ensure that for each a such that W_a is infinite, P_a is eventually marked.

Let $R(a) := 2a$. $R(a)$ is a **restraint** to our attempts to satisfy P_a in that we shall require any potential witness x_a to be as large as $R(a)$; this device ensures the satisfaction of N_a because it ensures that $A \cap \{x : x < 2a\} \subseteq \{x_0, \dots, x_{a-1}\}$, but it also potentially interferes with our ability to satisfy P_a . We say that P_a **needs attention** at stage s iff $a < s$, P_a is not marked and it could be satisfied at this stage — that is,

$$(\exists x < s)[x \in W_{a,s} \text{ and } x \geq R(a)].$$

If no P_a needs attention at stage s , then $A_{s+1} = A_s$ and each P_a is marked at stage $s + 1$ iff it was marked at stage s . Otherwise, choose the smallest index a such that P_a needs attention at stage s and let it **get attention** by performing the following steps:

- (i) $x_a := \text{least } x < s[x \in W_{a,s} \text{ and } x \geq R(a)]$;
- (ii) $A_{s+1} := A_s \cup \{x_a\}$;
- (iii) P_a is marked at stage $s + 1$; for all $b \neq a$, P_b is marked at stage $s + 1$ iff it was marked at stage s .

This completes the construction of A and we now verify that for all a ,

- (1) P_a gets attention at most once;
- (2) if W_a is infinite, then P_a is marked at some stage;
- (3) if P_a is marked at any stage, then P_a is satisfied;
- (4) P_a is satisfied;
- (5) N_a is satisfied.

By the construction, P_a gets attention only if it is not marked and whenever P_a gets attention it is marked at that stage, remains marked at all subsequent stages and therefore never gets attention again. Therefore (1) holds, and it follows that there exists a stage s_a such that for no $b < a$ does P_b get attention at any stage $s \geq s_a$.

For (2), suppose that W_a is infinite. Then there exists $s \geq s_a$ and $x \in W_{a,s}$ such that $x \geq R(a)$. If P_a is not already marked at s , then P_a needs attention at s , and since $s \geq s_a$, a will be the smallest such index so P_a will get attention at stage s and become marked at stage $s + 1$. (3) is immediate, since P_a becomes marked only by getting attention, and this process provides a number x_a which is a common element of A and W_a . For (4), either W_a is finite and P_a is automatically satisfied, or W_a is infinite and P_a is satisfied by (2) and (3). Finally, (5) holds because the elements of $A \cap \{x : x < 2a\}$ are among those $x_0, x_1, \dots, x_{a-1}$ which are defined at some stage and enumerated into A .

Finally, the description of the construction constitutes an algorithm for determining, given x and s , whether or not $x \in A_s$. By Church's Thesis (or an explicit coding) this guarantees that the relation $x \in A_s$ is recursive, so $\langle A_s : s \in \omega \rangle$ is a recursive stage enumeration of A and hence A is r.e. $\square$

In the preceding proof, each P_a is a **positive requirement** in that its satisfaction requires the positive action of enumerating a number x_a into A at some stage. N_a is a **negative requirement** in that its satisfaction requires keeping certain numbers out of A . The conflict between these two goals is easily resolved by a notion of priority. For each a , by requiring $x_a \geq R(a)$ we are implementing the idea that the requirements $N_0, \dots, N_{a-1}$ have higher

priority than P_a : x_a is defined and enumerated into A only if it can be found large enough so that it does not conflict with any N_b for $b < a$. Since x_a is needed only in case W_a is infinite, we can always eventually satisfy P_a .

This idea of resolving conflicting requirements by a system of priorities is fundamental in most of the remaining results of this chapter. As a first step in developing this method, we give still another construction of a simple set. This time we construct simultaneously an explicit witness to the fact that A is not $\mathbf{m}$ -complete.

Theorem 8.1.25 *There exist r.e. sets A and B such that A is simple and $B \not\leq_{\mathbf{m}} A$.*

Proof. It will suffice to construct r.e. sets A and B which for each $a, b \in \omega$ satisfy the requirements

$$\begin{aligned} P_a : W_a \text{ infinite} &\implies A \cap W_a \neq \emptyset \\ Q_b : \{b\} \text{ total} &\implies \exists y \exists u [\{b\}(y) \simeq u \wedge (y \in B \iff u \notin A)] \\ N_a : A \cap \{x : x < 2a\} &\text{ has at most } a \text{ elements.} \end{aligned}$$

Note that Q_b simply asserts that $B \neq \{b\}^{-1}(A)$. Hence the cumulative import of Q_b for all b is that $B \not\leq_{\mathbf{m}} A$. We construct $A := \bigcup_{s \in \omega} A_s$ and $B := \bigcup_{s \in \omega} B_s$ simultaneously in stages. Requirements P_a and N_a are satisfied much as before. To satisfy Q_b , at stage s we attempt to find numbers y_b and u_b such that $\{b\}_s(y_b) \simeq u_b$ and $y_b \notin B_s$. If (I) $u_b \in A_s$, then $u_b \in A$ and we aim to ensure that $y_b \notin B$. If (II) $u_b \notin A_s$, we enumerate y_b into $B_{s+1} \subseteq B$ and try to ensure that $u_b \notin A$. Thus satisfaction of Q_b has both a positive and a negative component.

The important new feature here is that the negative condition in case (II) may conflict with our strategy for satisfying some requirement P_a at a later stage t : the number u_b that we desire to exclude from A may be exactly the number x_a that we want to put into A_{t+1} . This conflict is again resolved by assigning priorities: if $a \leq b$, then we assign P_a higher priority than Q_b and allow x_a to enter A thereby satisfying P_a but possibly *injuring* Q_b ; if $b < a$, then Q_b has the higher priority and we shall require that x_a be at least as large as a restraint value $R_t(a) > u_b$, which ensures that no such injury takes place. If Q_b is satisfied at some stage but later injured, we need to try at a still later stage to satisfy it again, generally with new values of y_b and u_b ; since only finitely many P_a are allowed to injure Q_b , and each one gets attention at most once, Q_b eventually has highest priority among active requirements and will be permanently satisfied. Similarly, only finitely many Q_b are allowed to restrain each P_a so it too can eventually be satisfied.

Note that there could also be a conflict in case (I) if the number y_b that we need to keep out of B were the same as some other y_c that we needed to

put into B at a later stage to satisfy Q_c . We will avoid this type of conflict by choosing witnesses y_b of the form $\langle b, z \rangle$ and therefore different for each Q -requirement. This is possible because Q_b is automatically satisfied unless $\{b\}$ is a total function, so we are free to choose the witnesses y_b , whereas the x_a are necessarily elements of W_a so may not be freely chosen.

Set

$$R_s(a) := 1 + \max\{u_b : b < a \text{ and } Q_b \text{ is marked at } s\} \cup \{2a\}.$$

As will emerge from the description of the construction below, each of the finitely many times a requirement Q_b gets attention, possibly new values of y_b and u_b are chosen, so they should technically be written $y_{b,s}$ and $u_{b,s}$ to indicate this dependence on the stage. We avoid this notional complication and think of them as the time-dependent variables in the execution of a computer program whose values are periodically updated. In contrast with the preceding proof, this restraint value varies with s , and the restriction to $b < a$ reflects the idea that P_a must respect only Q_b with $b < a$ and ensures that $R_s(a)$ does not grow without bound and prevent the satisfaction of P_a . P_a needs attention at stage s iff $a < s$, P_a is not marked at stage s and

$$(\exists x < s)[x \in W_{a,s} \text{ and } x \geq R_s(a)].$$

Q_b needs attention at stage s iff $b < s$, Q_b is not marked at stage s and

$$(\exists y < s)[\{b\}_s(y) \downarrow \wedge y \notin B_s \wedge \exists z(y = \langle b, z \rangle)].$$

If no P_a or Q_b needs attention at stage s , then $A_{s+1} = A_s$, $B_{s+1} = B_s$ and each P_a and Q_b is marked at stage $s+1$ iff it was marked at stage s . Otherwise, choose the requirement of highest priority — that is the first in the sequence

$$P_0, Q_0, P_1, Q_1, \dots, P_a, Q_a, \dots$$

which needs attention at stage s and let it **get attention** as follows. If the chosen requirement is P_a , perform the following steps:

- (i) $x_a := \text{least } x < s[x \in W_{a,s} \text{ and } x \geq R_s(a)]$;
- (ii) $A_{s+1} := A_s \cup \{x_a\}$; $B_{s+1} := B_s$;
- (iii) P_a is marked at stage $s+1$; for $b \neq a$, P_b is marked at stage $s+1$ iff it was marked at stage s ;
- (iv) for all $b < a$, Q_b is marked at stage $s+1$ iff it was marked at stage s ; for all $b \geq a$, Q_b is unmarked at stage $s+1$ and is said to have been **injured**.

If the chosen requirement is Q_b , perform the following steps:

- (i) $y_b := \text{least } y < s \left[\{b\}_s(y) \downarrow \wedge y \notin B_s \wedge \exists z(y = \langle b, z \rangle) \right];$
 $u_b := \{b\}_s(y_b);$
- (ii) if $u_b \notin A_s$, then $B_{s+1} := B_s \cup \{y_b\}$; otherwise $B_{s+1} := B_s$;
 $A_{s+1} := A_s$;
- (iii) for all a , P_a is marked at stage $s + 1$ iff it was marked at stage s ;
- (iv) Q_b is marked at stage $s + 1$; for all $a \neq b$, Q_a is marked at stage $s + 1$ iff it was marked at stage s .

This completes the construction of A and B . Note that in the construction a requirement becomes marked at the stage following the one that it gets attention. P_a once marked never later becomes unmarked whereas Q_b is injured and becomes unmarked at any later stage at which some P_a with $a \leq b$ gets attention. If Q_b is marked at stage $s + 1$ and never subsequently unmarked we say that it is **permanently marked**. We now verify that this construction ensures that for all a and b ,

- (1) P_a gets attention at most once; Q_b gets attention at most finitely often;
- (2) $\lim_{s \rightarrow \infty} R_s(a)$ exists and is finite;
- (3) if W_a is infinite, then P_a is marked at some stage;
- (4) if P_a is marked at some stage, then P_a is satisfied;
- (5) if $\{b\}$ is total, then Q_b is permanently marked at some stage;
- (6) if Q_b is permanently marked at some stage, then Q_b is satisfied;
- (7) P_a is satisfied;
- (8) Q_b is satisfied;
- (9) N_a is satisfied.

For (1), it is clear as before that each P_a gets attention at most at one stage, at that stage is marked and stays forever marked. On the other hand, Q_b will become unmarked each time a P_a with $a \leq b$ gets attention, but this will happen at most $b + 1$ many times so Q_b can get attention at most $b + 2$ many times.

For (2), note that the value of $R_s(a)$ changes at most at stages at which the marked status of some Q_b with $b < a$ changes and hence at most at stages s at which some P_b or Q_b with $b < a$ gets attention. By (1) there exists s_a such that for all $s \geq s_a$ this is not the case at stage s and therefore for all such s , $R_s(a)$ has the same (limiting) value; we denote this value by $R(a)$.

(3) and (4) are now immediate as before using the values of s_a and $R(a)$ just defined, and (7) follows. (9) is also clear as before.

Towards (5), suppose that $\{b\}$ is total. By (1), there exists $t_b \geq s_b$ such that none of P_a ($a \leq b$) or Q_a ($a < b$) gets attention at any stage $s \geq t_b$.

If Q_b is already marked at stage t_b , then it is permanently marked there. Otherwise, we claim that there exists a stage $s \geq t_b$ such that Q_b needs attention at s . Indeed, for any $y = \langle b, z \rangle \notin B_{t_b}$, there is $s \geq t_b$ such that $\{b\}_s(y) \downarrow$ (because $\{b\}$ is total). If Q_b is still not marked at s , then still $y \notin B_s$, since only attention to Q_b can put an element of the form $y = \langle b, z \rangle$ into B . Hence Q_b will need attention at s and will therefore get attention (because it now has highest priority) and become marked. The marking will be permanent, again by the definition of t_b .

Now towards (6), suppose that Q_b gets attention and becomes permanently marked at stage s . Then either

- (I) $u_b \in A_s$ and $y_b \notin B_{s+1}$ or
- (II) $u_b \notin A_s$ and $y_b \in B_{s+1}$.

In case (I), clearly $u_b \in A$ and also $y_b \notin B$, since as argued above, no action other than giving attention to Q_b can put y_b into B . In case (II), clearly $y_b \in B$, and it suffices to prove by induction that for all $t \geq s$, $u_b \notin A_t$, since then also $u_b \notin A$. This holds for $t = s$, so assume as induction hypothesis that $u_b \notin A_t$. If no P_a gets attention at stage t , then $A_{t+1} = A_t$. Otherwise, for some a , $A_{t+1} = A_t \cup \{x_a\}$. Since Q_b is not injured (because it does not become unmarked) at t , necessarily $b < a$, and thus $x_a \geq R_t(a) > u_b$, so again $u_b \notin A_{t+1}$.

Now (8) is immediate from (5) and (6). Finally note that all of the computations required at stage s — of $R_s(a)$ and x_a , or of y_b and u_b — are recursive so the construction is effective and defines recursive stage enumerations of A and B and thus both sets are r.e. $\square$

The device used in this proof to keep separate the witnesses for different Q_b will be used repeatedly and is worth introducing a special notation.

Definition 8.1.26 For any set A and any a ,

$$A^{[a]} := \{ \langle a, z \rangle : \langle a, z \rangle \in A \}$$

is called the a -th **column** of A .

The terminology comes from viewing (part of) A as a subset of the ‘plane’ $\omega \times \omega$ or an infinite matrix via the coding of ordered pairs. Note that $\omega^{[a]}$ is the set of all numbers of the form $\langle a, z \rangle$. We can now extend the methods of the preceding proof to show that the r.e. m -degrees have a very complex order structure. The first step in this direction is that the ordering is not linear.

Theorem 8.1.27 *There exist r.e. sets A and B such that $A \not\leq_m B$ and $B \not\leq_m A$.*

Proof. We follow the pattern of the preceding proof replacing the requirements P_a and N_a there by ones analogous to Q_a . Because of the symmetry, it simplifies the argument to rename the requirements as follows:

$$\begin{aligned} P_{2a} : \quad \{a\} \text{ total} &\implies \exists x \exists u [\{a\}(x) \simeq u \wedge (x \in A \iff u \notin B)]; \\ P_{2a+1} : \quad \{a\} \text{ total} &\implies \exists x \exists u [\{a\}(x) \simeq u \wedge (x \in B \iff u \notin A)]. \end{aligned}$$

Set

$$R_s(k) := 1 + \max\{u_l : l < k \text{ and } P_l \text{ is marked at } s\}.$$

As above, the value of u_l varies also with the stage s . Then P_{2a} needs attention at stage s iff $2a < s$, P_{2a} is not marked at stage s and

$$(\exists x < s) [\{a\}_s(x) \downarrow \wedge x \geq R_s(2a) \wedge x \notin A_s \wedge x \in \omega^{[a]}].$$

Similarly, P_{2a+1} needs attention at stage s iff $2a+1 < s$, P_{2a+1} is not marked at stage s and

$$(\exists x < s) [\{a\}_s(x) \downarrow \wedge x \geq R_s(2a+1) \wedge x \notin B_s \wedge x \in \omega^{[a]}].$$

Note that in contrast to the preceding proof, all requirements have both a positive and a negative component, are capable both of injuring and being injured and must obey restraints. If no P_k needs attention at stage s , then $A_{s+1} = A_s$, $B_{s+1} = B_s$ and each P_k is marked at stage $s+1$ iff it was marked at stage s . Otherwise, choose the smallest k such that P_k needs attention at s . If $k = 2a$, let P_k get attention by performing the following steps:

- (i) $x_k := \text{least } x < s [\{a\}_s(x) \downarrow \wedge x \geq R_s(k) \wedge x \notin A_s \wedge x \in \omega^{[a]}];$
 $u_k := \{a\}_s(x_k);$
- (ii) if $u_k \notin B_s$, then $A_{s+1} := A_s \cup \{x_k\}$; otherwise $A_{s+1} := A_s$;
 $B_{s+1} := B_s$;
- (iii) P_k is marked at stage $s+1$; for all $l < k$, P_l is marked at stage $s+1$ iff it was marked at stage s ; for all $l > k$, P_l is unmarked at stage $s+1$ and is said to have been injured.

If $k = 2a+1$, the same steps are performed with the roles of A and B interchanged. In this construction, each P_k becomes marked at the stage following one at which it gets attention and later becomes unmarked at any stage following one at which any P_l with $l < k$ gets attention. As before, we say that P_k is permanently marked at stage s iff it is marked for all stages $t \geq s$. We now verify that this construction ensures that for all a and k ,

- (1) P_k gets attention at most finitely often;
- (2) $\lim_{s \rightarrow \infty} R_s(k)$ exists and is finite;
- (3) if $\{a\}$ is total, then P_{2a} and P_{2a+1} are permanently marked at some stage;
- (4) if P_k is permanently marked at some stage, then P_k is satisfied;
- (5) P_k is satisfied.

We prove (1) by induction. Assume as induction hypothesis that for all $l < k$, P_l gets attention at most finitely often. It follows that there exists a stage s_k such that no such P_l gets attention at any stage $s \geq s_k$. Then P_k gets attention at at most one stage $s > s_k$, since for any such s , P_k is marked at stage $s + 1$ and (by induction) at all $t > s$ so is permanently marked. Hence also P_k gets attention at most finitely often.

The value of $R_s(k)$ changes at most at stages s at which the marked status of some P_l with $l < k$ changes and hence at most at stages at which some such P_l gets attention. Hence $R_s(k)$ has the same value for all $s \geq s_k$. We denote this value by $R(k)$.

For (3), assume that $\{a\}$ is total. If P_{2a} is already marked at stage s_{2a} , then it is permanently marked there. Otherwise, we claim that there exists a stage $s \geq s_{2a}$ such that P_{2a} needs attention at stage s . Indeed, for any $x \in \omega^{[a]}$ such that $x > R(2a)$ and $x \notin A_{s_{2a}}$, there is some $s \geq s_{2a}$ such that $\{a\}_s(x) \downarrow$. If P_{2a} is still unmarked at s , then still $x \notin A_s$ (because only attention to P_{2a} could cause such an x to enter A) so P_{2a} needs attention at s . Since $s \geq s_{2a}$, P_{2a} has highest priority and will get attention at s and be permanently marked. The argument for P_{2a+1} is parallel with the roles of A and B interchanged.

For (4), suppose that P_{2a} gets attention and becomes permanently marked at stage s . Then either

- (I) $u_{2a} \in B_s$ and $x_{2a} \notin A_{s+1}$ or
- (II) $u_{2a} \notin B_s$ and $x_{2a} \in A_{s+1}$.

In case (I), clearly $u_{2a} \in B$ and also $x_{2a} \notin A$, since only attention to P_{2a} can put a number $x_{2a} \in \omega^{[a]}$ into A . In case (II), clearly $x_{2a} \in A$ and it suffices to prove that $u_{2a} \notin B$; equivalently to prove by induction that for all $t > s$, $u_{2a} \notin B_t$. This holds for $t = s + 1$, so assume as induction hypothesis that it is true for t . If no requirement P_{2b+1} gets attention at stage t , then $B_{t+1} = B_t$. Otherwise, for some b , $B_{t+1} = B_t \cup \{x_{2b+1}\}$. Since P_{2a} is not injured at stage t , $2b + 1 > 2a$, hence $x_{2b+1} \geq R_t(2b + 1) > u_{2a}$, so again $u_{2a} \notin B_{t+1}$.

The argument for P_{2a+1} is parallel. Finally, (5) is immediate from (3) and (4). $\square$

Definition 8.1.28 For any $I \subseteq \omega$ and any family of sets $\langle A^n : n \in \omega \rangle$,

- (i) $\bigoplus \langle A^n : n \in I \rangle := \{ \langle n, x \rangle : x \in A^n \wedge n \in I \}$;
- (ii) $\langle A^n : n \in \omega \rangle$ is **m-independent** iff for all $n \in \omega$,

$$A^n \not\leq_m \bigoplus \langle A^p : p \neq n \rangle;$$

- (iii) $\langle A^n : n \in \omega \rangle$ is an **r.e. family** iff $\bigoplus \langle A^n : n \in \omega \rangle$ is r.e.

Note that if $\langle A^n : n \in \omega \rangle$ is m-independent, then the sets A^n are pairwise m-incomparable and if $\langle A^n : n \in \omega \rangle$ is an r.e. family, then also each A^n is r.e.

Theorem 8.1.29 *There exists an m-independent r.e. family.*

Proof. We view the construction as one of a single r.e. set

$$A = \bigoplus \langle A^n : n \in \omega \rangle, \text{ where } A^n = \{ x : \langle n, x \rangle \in A \}.$$

At stage s , $A_s^n := \{ x : \langle n, x \rangle \in A_s \}$. We need for A to satisfy for each a and n the requirement

$$P_{\langle a, n \rangle} : \{a\} \text{ total} \implies \exists x \exists u [\{a\}(x) \simeq u \wedge (x \in A^n \iff u \notin B^n)],$$

where $B^n := \bigoplus \langle A^p : p \neq n \rangle$. We say that $P_{\langle a, n \rangle}$ needs attention at stage s iff $\langle a, n \rangle < s$, it is not marked at s and

$$(\exists x < s) \left[\{a\}_s(x) \downarrow \wedge x \geq R_s(\langle a, n \rangle) \wedge x \notin A_s^n \wedge x \in \omega^{[\langle a, n \rangle]} \right],$$

where

$$R_s(\langle a, n \rangle) := 1 + \max\{u_l : l < \langle a, n \rangle \text{ and } P_l \text{ is marked at } s\}.$$

If no P_k needs attention at stage s , then $A_{s+1} = A_s$, $B_{s+1} = B_s$, and each P_k is marked at stage $s+1$ iff it was marked at stage s . Otherwise, choose the smallest $k = \langle a, n \rangle$ such that P_k needs attention and perform the following steps:

- (i) $x_k := \text{least } x < s [\{a\}_s(x) \downarrow \wedge x \geq R_s(k) \wedge x \notin A_s^n \wedge x \in \omega^{[k]}]$;
 $u_k := \{a\}_s(x_k)$;
- (ii) if $u_k \notin B_s^n$, then $A_{s+1} := A_s \cup \{\langle n, x_k \rangle\}$; otherwise $A_{s+1} := A_s$;
- (iii) P_k is marked at stage $s+1$; for all $l < k$, P_l is marked at stage $s+1$ iff it was marked at stage s ; for all $l > k$, P_l is unmarked at stage $s+1$ and is said to have been injured.

The proof that this construction achieves its goal is a straightforward adaptation of the preceding techniques and is left as an exercise. $\square$

Theorem 8.1.30 *Every recursive partial ordering of ω is embeddable in the r.e. $\mathbf{m}$ -degrees — that is, for any recursive partial ordering $\preceq$ of ω , there exists a family $\langle C_i : i \in \omega \rangle$ of r.e. sets such that for all i and j ,*

$$i \preceq j \iff C_i \leq_{\mathbf{m}} C_j.$$

Proof. Let $\langle A^n : n \in \omega \rangle$ be an $\mathbf{m}$ -independent r.e. family and as above $B^n = \bigoplus \langle A^p : p \neq n \rangle$. For each i , set

$$C_i := \bigoplus \langle A^n : n \preceq i \rangle := \{ \langle n, x \rangle : x \in A^n \wedge n \preceq i \}.$$

Clearly each C_i is r.e. For each i , let f_i be a recursive function such that

$$f_i(\langle n, x \rangle) = \begin{cases} \langle n, x \rangle, & \text{if } n \preceq i; \\ 0, & \text{otherwise.} \end{cases}$$

Then if $i \preceq j$, we have $C_i = f_i^{-1}(C_j)$, so $C_i \leq_{\mathbf{m}} C_j$ as desired. On the other hand, if $i \not\preceq j$, then C_j has no elements of the form $\langle i, x \rangle$, so $C_j = f_j^{-1}(B^i)$. Hence if $C_i \leq_{\mathbf{m}} C_j$, then also $A^i \leq_{\mathbf{m}} C_i \leq_{\mathbf{m}} C_j \leq_{\mathbf{m}} B^i$, so $A^i \leq_{\mathbf{m}} B^i$ contrary to the definition of independence. Thus $C_i \not\leq_{\mathbf{m}} C_j$. $\square$

Lemma 8.1.31 *There exists a recursive partial ordering of ω such that every countable partial ordering is embeddable in it.*

Proof. (sketch) Build a recursive partial ordering in stages. At each stage adjoin points which realize every set of relationships with previous elements which are consistent with the notion of partial ordering. To embed a given countable partial ordering, enumerate its universe and define the embedding in stages. At the n -th stage choose for the image of the n -th element any element of the recursive partial ordering which bears the same relationships to the preceding image points as the n -th element bears to its predecessors in the enumeration. $\square$

Theorem 8.1.32 *Every countable partial ordering is embeddable in the r.e. $\mathbf{m}$ -degrees — that is, for any partial ordering $\preceq$ of ω , there exists a family $\langle C_i : i \in \omega \rangle$ of r.e. sets such that for all i and j ,*

$$i \preceq j \iff C_i \leq_{\mathbf{m}} C_j.$$

Proof. Immediate from the preceding theorem and the lemma. $\square$

Collateral Reading. The leading contemporary references for all of the material of this chapter are [Soa], [Od89] and [Od99], although the older classic [Rog] is still useful for many purposes. As in the chapters on Set Theory and Model Theory, the topics selected for inclusion here represent a

narrow focussed excursion into a large and complex area. Most treatments of priority arguments begin with their use in the context of Turing degrees as in the following sections; the simplified versions for $\mathbf{m}$ -degrees are seldom seen. On the other hand, there is a large literature on so-called **strong reducibilities**; [Od89; Chapter III] and several other sections throughout both [Od89] and [Od99] are a comprehensive survey.

Exercises

Exercise 8.1.33 Show that every productive set has an infinite recursive subset.

Exercise 8.1.34 Complete the proof of Theorem 8.1.29.

Exercise 8.1.35 Prove that in the proof of Theorem 8.1.27, each requirement P_k gets attention at most 2^k many times.

Exercise 8.1.36 (Rice's Theorem) An **index set** is a set $A \subseteq \omega$ such that $a \in A \wedge \{a\} = \{b\} \implies b \in A$. Prove that for any index set other than $\emptyset$ and ω , either $\mathbf{K} \leq_{\mathbf{m}} A$ or $\mathbf{K} \leq_{\mathbf{m}} \overline{A}$.

Hint: Let e be an index for the empty function and consider cases $e \in A$ or $e \notin A$.

Many other reducibilities among sets have been studied. In Sections 8.2–8.6 below we concentrate on Turing reducibility; in the following series of exercises we explore the intermediate **truth-table reducibility**.

For any $B \subseteq \omega$, let V_B denote the unique truth assignment for the propositional language L of Chapter 1 such that $V_B(p_n) = \mathbf{T} \iff n \in B$. Assign Gödel numbers to L -sentences as in Section 4.4, and set

$$B^{\text{tt}} := \{ \text{gn}(\phi) : \phi \text{ is an } L\text{-sentence and } V_B(\phi) = \mathbf{T} \}.$$

Then A is **truth-table reducible** to B — in symbols, $A \leq_{\text{tt}} B$ — iff $A \leq_{\mathbf{m}} B^{\text{tt}}$.

Thus $A \leq_{\mathbf{m}} B^{\text{tt}}$ iff for some recursive function f each question $x \in A$ is determined by the truth value of a propositional combination with Gödel number $f(x)$ of questions of the form $y \in B$. We set also

$$\begin{aligned} A <_{\text{tt}} B &\iff A \leq_{\text{tt}} B \text{ but } B \not\leq_{\text{tt}} A, \\ A \equiv_{\text{tt}} B &\iff A \leq_{\text{tt}} B \text{ and } B \leq_{\text{tt}} A, \end{aligned}$$

and $\text{dg}_{\text{tt}}(A) := \{ B : A \equiv_{\text{tt}} B \}$, the **truth-table degree** of A . The truth table degrees inherit an ordering by setting $\text{dg}_{\text{tt}}(A) \leq \text{dg}_{\text{tt}}(B)$ iff $A \leq_{\text{tt}} B$. It follows from the following exercise that this is well-defined.

Exercise 8.1.37 Show that

- (i) for all A and B , $A \leq_m B \implies A \leq_{tt} B$, but not in general conversely;
- (ii) $\leq_{tt}$ is reflexive and transitive;
- (iii) $<_{tt}$ is irreflexive and transitive;
- (iv) $\equiv_{tt}$ is an equivalence relation.

Exercise 8.1.38 Show that

- (i) $\{A : A \text{ is recursive}\}$ is the least **tt**-degree;
- (ii) any pair of **tt**-degrees has a least upper bound;
- (iii) there is no maximal **tt**-degree.

Hint: For (i) note first that if B is recursive, then also B^{tt} is recursive.

An r.e. set C is called **tt-complete** iff for every r.e. set A , $A \leq_{tt} C$.

Exercise 8.1.39 Prove that there exists a simple set which is **tt**-complete.

Hint: Let A be the simple set constructed in Theorem 8.1.20. For each $x \in \omega$, set

$$E_x := \{2^{2x} + 1, \dots, 2^{2x+2}\} \quad \text{and} \quad B := A \cup \bigcup \{E_x : x \in \mathbb{K}\}.$$

Show that E_x has more than 2^{2x+1} elements and therefore $E_x \cap \overline{A} \neq \emptyset$. Show that B is simple and **tt**-complete.

Call an r.e. set **hypersimple** iff there is no recursive function h such that for all $n \in \omega$, $\overline{A}$ has at least n elements $\leq h(n)$.

Exercise 8.1.40 Show that

- (i) every hypersimple set is simple;
- (ii) the simple set of Theorem 8.1.20 is not hypersimple;
- (iii) there exists a hypersimple set.

Hint: For (iii), let f be a recursive function such that $B := \text{Im}(f)$ is a non-recursive r.e. set and set $A := \{m : (\exists n > m) f(n) < f(m)\}$. Show that if h is a recursive function such that for all n , $\overline{A}$ has at least n elements $\leq h(n)$, then for all n , $n \in B \leftrightarrow n \in \{f(0), \dots, f(h(n))\}$, so B would be recursive.

Exercise 8.1.41 Prove that if A is $\mathbf{tt}$ -complete, then A is not hypersimple. Hence there exists an r.e. set which is non-recursive and not $\mathbf{tt}$ -complete.

Hint: Fix a recursive function f such that $x \in \mathbf{K} \iff f(x) \in A^{\mathbf{tt}}$ and show that there is a recursive function g such that for all n , $\overline{A} \cap [n, g(n)) \neq \emptyset$. For each $B \subseteq [0, n)$, let $B_\infty := B \cup [n, \infty)$. B_∞ is recursive, hence so is $B_\infty^{\mathbf{tt}}$ and we can find effectively an index a_B such that

$$W_{a_B} = \{x : f(x) \notin B_\infty^{\mathbf{tt}}\}.$$

Set $g(n) := 1 +$

$$\max_{B \subseteq [0, n)} \{m : p_m \text{ occurs in the sentence with Gödel number } f(a_B)\}.$$

Show that for $B = A \cap [0, n)$, $f(a_B) \in A^{\mathbf{tt}} \iff f(a_B) \notin B_\infty^{\mathbf{tt}}$, and deduce that $A \cap [n, g(n)) \neq \emptyset$.

Exercise 8.1.42 Prove the following results known as the **Recursion Theorem**:

- (i) for any unary recursive function f there exists an index e such that $\{e\} = \{f(e)\}$, hence also $W_e = W_{f(e)}$;
- (ii) for any binary recursive function f , there exists a recursive function g such that for all x , $\{g(x)\} = \{f(g(x), x)\}$, hence also $W_{g(x)} = W_{f(g(x), x)}$.

Hint: For (i), use the function S_1^1 of Proposition 8.1.11 to define a recursive function h such that for all a and x , $\{h(a)\}(x) \simeq \{\{a\}(a)\}(x)$. Choose an index d such that $\{d\} = f \circ h$ and set $e := h(d)$.

Exercise 8.1.43 Generalize the preceding exercise to show that for any two recursive functions f and g there exist indices d and e such that $\{d\} = \{f(d, e)\}$ and $\{e\} = \{g(d, e)\}$.

Exercise 8.1.44 Show that for every productive set P ,

- (i) there exists a total recursive function h such that for all a , if $W_a \subseteq P$, then $h(a) \in P \sim W_a$;
- (ii) $\overline{\mathbf{K}} \leq_m P$ (converse of Corollary 8.1.16).

Hint: For (i), if P has the associated partial recursive function ϕ , find a recursive function g such that $W_{g(a)} = W_a$ if $\phi(a) \downarrow$ and otherwise is $\emptyset$, so for all a , either $\phi(a) \downarrow$ or $\phi(g(a)) \downarrow$. For (ii), find a recursive function f such that $W_{f(a, x)}$ is the singleton set $\{h(a)\}$ if $x \in \mathbf{K}$ and otherwise is $\emptyset$. Use the Recursion Theorem to obtain g such that for all x , $W_{g(x)} = W_{f(g(x), x)}$ and show that $x \in \overline{\mathbf{K}} \iff h(g(x)) \in P$.

Remark 8.1.45 An r.e. set A such that $\overline{A}$ is productive is called *creative*. By the preceding exercise a creative set is non-recursive and hence by Exercise 4.6.49 provides a witness to the incompleteness of any recursively axiomatizable L_Ω -theory extending Q — that is, a proof of the First Incompleteness Theorem. Thus the existence of creative sets is sometimes interpreted as verifying the essentially *creative* nature of mathematics — no fixed recursively axiomatizable theory suffices to prove all true theorems even of a very restricted form $\psi(\dot{m})$ or $\neg\psi(\dot{m})$ for $m \in \omega$.

8.2 Turing reducibility

As we sketched following Definition 8.1.1, Turing reducibility is the most general notion of computing membership information about one set from another. When $A \leq_m B$, in computing a value $A(x)$ we have access to the B -oracle only to answer a single question $y \stackrel{?}{\in} B$ and the answer has to be taken as given. When $A \leq_{tt} B$, we may get the answer to a Boolean combination of such questions. The relationship $A \leq_T B$ is characterized (intuitively) by the availability of the B -oracle at all points in the computing process with freedom to evaluate answers and use them to determine new questions. Formally, we introduce (the characteristic function of) B as a new initial function (cf. Definition 4.2.5).

Definition 8.2.1 For any set $B \subseteq \omega$, the class $\text{Rec}[B]$ of functions *recursive relative to B* or simply *B -recursive* is the smallest class of functions which contains the characteristic function of B and the initial functions $(Cs_n^k, Pr_i^k, \text{ and } Sc_i^k)$ and is closed under composition, primitive recursion and search.

Remark 8.2.2 In this and the following definitions, the relativizing function B could easily be replaced by any total function with only minor changes and very little gain in generality. Furthermore, we could base our development of relative recursiveness on any of the other characterizations of recursiveness described in Section 5.4; we explore two of these variations in the exercises.

It is immediate that all recursive functions are B -recursive for every B , and the definitions and arguments of Sections 4.2–4.3 can be repeated nearly verbatim for recursiveness relative to B . We list the most important of these and leave it to the exercises to verify details. To focus attention on the results to follow, we state most results about relations only for sets (unary relations).

- (1) A relation R is B -recursive iff its characteristic function K_R is B -recursive.

- (2) The class of B -recursive functions and relations is closed under the Boolean operations, bounded search, definition by cases, bounded quantification and course-of-values recursion.
- (3) A set A is B -r.e. iff $A = \emptyset$ or $A = \text{Im}(F)$ for some B -recursive function F iff $A = \{x : \exists y R(x, y)\}$ for some B -recursive relation R .
- (4) A is B -recursive iff both A and $\overline{A}$ are B -r.e.
- (5) The class of **B -partial recursive functions** is the smallest class of partial functions which contains the characteristic function of B and the initial functions, and is closed under composition, primitive recursion and search.
- (6) A is B -r.e. iff $A = \text{Dom}(G)$ for some B -partial recursive function G .

In a similar way, the development of [Section 5.2](#) can be extended to the relativized case. We define $\text{PartComp}[B]$ as the smallest set of natural numbers as in Definition 5.2.17 with the additional clause: for all $z \in \omega$,

$$\langle \langle 6, 1 \rangle, z, B(z) \rangle \in \text{PartComp}[B].$$

Then we define for each a ,

$$\{a\}^B(x_0, \dots, x_{k-1}) \simeq y \iff \langle a, x_0, \dots, x_{k-1}, y \rangle \in \text{PartComp}[B],$$

and prove as in Theorem 5.2.23 that

- (7) a function F is B -partial recursive iff for some $a \in \omega$, $F = \{a\}^B$.

Also with $W_a^B := \{x : \{a\}^B(x) \downarrow\}$ we have

- (8) A is B -r.e. iff for some $a \in \omega$, $A = W_a^B$.

If $\mathcal{X}[B]$ denotes the induction system which defines $\text{PartComp}[B]$, then $\text{Derivof}^{\mathcal{X}[B]}$ is a B -recursive relation and

$$\{a\}^B(x_0, \dots, x_{k-1}) \simeq y \iff \exists u \text{ Derivof}^{\mathcal{X}[B]}(u, \langle a, x_0, \dots, x_{k-1}, y \rangle),$$

so we set, parallel to Definition 8.1.22,

- (9) $\{a\}_s^B(x) \simeq y : \iff (\exists u < s) \text{ Derivof}^{\mathcal{X}[B]}(u, \langle a, x, y \rangle);$
- (10) $W_{a,s}^B := \{x : \{a\}_s^B(x) \downarrow\}.$

Then the relativized version of 8.1.23 holds.

Proposition 8.2.3 For all B, s, a, x and y ,

- (i) $\{a\}_s^B(x) \simeq y \iff \exists s \{a\}_s^B(x) \simeq y$;
- (ii) $\{a\}_s^B(x) \simeq y \implies (\forall t \geq s) \{a\}_t^B(x) \simeq y$;
- (iii) $x \in W_a^B \iff \exists s [x \in W_{a,s}^B]$;
- (iv) $\{a\}_s^B(x) \simeq y$ and $x \in W_{a,s}^B$ are B -recursive relations;
- (v) $\{a\}_s^B(x) \simeq y \implies a, x, y < s$;
- (vi) $x \in W_{a,s}^B \implies a, x < s$;
- (vii) $\{a\}_s^B$ is a finite function; $W_{a,s}^B$ is a finite set;
- (viii) there is at most one x such that $x \notin W_{a,s}^B$ but $x \in W_{a,s+1}^B$.

Proof. Left as an exercise. $\square$

Definition 8.2.4 For any sets A and B ,

- (i) A is **Turing reducible** to B — in symbols, $A \leq_T B$ — iff A is B -recursive;
- (ii) A is **strict Turing reducible** to B — in symbols, $A <_T B$ — iff $A \leq_T B$ but $B \not\leq_T A$;
- (iii) A and B are **Turing equivalent** or **T-equivalent** — in symbols, $A \equiv_T B$ — iff $A \leq_T B$ and $B \leq_T A$;
- (iv) the **Turing degree** or **T-degree** of A is the set

$$\text{dg } A := \{ B : B \equiv_T A \};$$

- (v) $\text{dg } A \leq \text{dg } B$ iff $A \leq_T B$; $\text{dg } A < \text{dg } B$ iff $A <_T B$;
- (vi) $\text{Dg}_T := \{ \text{dg } A : A \subseteq \omega \}$, the set of all T -degrees.

That $\equiv_T$ is an equivalence relation is slightly less trivial than the corresponding fact for $\equiv_m$; the only point at issue is transitivity, which is part (i) of

Lemma 8.2.5 For any sets A, B and C ,

- (i) if $A \leq_T B \leq_T C$, then $A \leq_T C$;
- (ii) if B is recursive and $A \leq_T B$, then A is recursive;
- (iii) if A is recursive, then $A \leq_T B$;
- (iv) if $A \leq_m B$, then $A \leq_T B$.

Proof. Part (i) follows if we know that

$$B \leq_T C \implies \text{Rec}[B] \subseteq \text{Rec}[C].$$

This is easy to verify by $\text{Rec}[B]$ -induction. Similarly, (ii) may be rephrased as

$$B \text{ is recursive} \implies \text{Rec}[B] \subseteq \text{Rec},$$

which is also verified by an easy induction. (iii) is a restatement of the fact that $\text{Rec} \subseteq \text{Rec}[B]$ and (iv) is immediate. $\square$

Proposition 8.2.6

- (i) $\{A : A \text{ is recursive}\}$ is the unique smallest T -degree.
- (ii) Every pair of T -degrees has a least upper bound.
- (iii) There is no maximal T -degree.

Proof. (i) is immediate from the preceding lemma. Parts (ii) and (iii) are proved essentially as are the corresponding parts of Proposition 8.1.4, and we leave the details as an exercise. $\square$

In practice, when we want to verify that some relationship $A \leq_T B$ holds, we shall give an informal argument that relies on the usual techniques of effective calculation, which we know can be translated (programmed) in the formalism of recursion, together with appeals to the B -oracle. For example, if we define a function f via a primitive recursion: $f(0, y) = g(y)$ and $f(x + 1, y) = h(f(x, y), x, y)$ from a recursive function g and a B -recursive function h , then also f is B -recursive. A particularly important case of this arises when B is $\mathbf{K}$ (or some other $\mathbf{m}$ -complete or T -complete set). Since every r.e. set D is $\mathbf{K}$ -recursive, any question about membership in D can be answered by the $\mathbf{K}$ -oracle. By extension, the same is true about any question which involves one or more existential quantifiers — for example, if P is a recursive relation, then $\exists z \exists u \exists v P(x, y, z, u, v)$ holds iff $\langle x, y \rangle$ belongs to a certain r.e. set and can therefore be determined by the $\mathbf{K}$ -oracle. Note that since also $\forall z \forall u \forall v P(x, y, z, u, v)$ holds iff $\langle x, y \rangle$ does *not* belong to another r.e. set, this question can also be answered by $\mathbf{K}$. In other words, questions about membership in Σ_1 or Π_1 relations can be answered by the $\mathbf{K}$ -oracle. This does not, of course, extend to relations in higher levels of the arithmetical hierarchy. We refer to this method as the **one-quantifier principle**.

Although the preceding proposition highlights common properties of the two reducibilities, there are also many differences. For example, since the class of B -recursive sets is closed under the Boolean operations, $A \equiv_T \bar{A}$, so

$$A \leq_T B \quad \text{and} \quad B \text{ is r.e.} \quad \text{do not imply that } A \text{ is r.e.}$$

Hence the easy proof in Proposition 8.1.4 that the $\mathbf{m}$ -degrees are not linearly ordered cannot be adapted for T -degrees.

To construct incomparable T -degrees (and much more) we need first to formulate a precise formal counterpart to the following intuitively obvious

fact: if $\{a\}^B(x) \simeq y$, then the computation uses only finitely much information about B — the B -oracle is asked only finitely many questions. To make this precise we use the method of Example 1.2.15(iv). Let

$$Y := \{ \langle \langle 6, 1 \rangle, z, B(z) \rangle : z \in \omega \},$$

and define the **use** of a computation $\{a\}^B(x)$ to be

$$u(B; a, x) \simeq 1 + \max \{ z : \langle \langle 6, 1 \rangle, z, B(z) \rangle \in \text{Supp}_Y(\langle a, x, \{a\}^B(x) \rangle) \},$$

if $\{a\}^B(z) \downarrow$ and 0 otherwise. Intuitively, $u(B; a, x)$ is the smallest number greater than all z such that the value of $B(z)$ is “used” in the computation of $\{a\}^B(x)$. We set also

$$u(B; a, x, s) := \begin{cases} u(B; a, x), & \text{if } \{a\}_s^B(x) \downarrow; \\ 0, & \text{otherwise.} \end{cases}$$

First Use Theorem 8.2.7 For all a, x, s, B and C ,

(i) if $\{a\}_s^B(x) \downarrow$ then $(\forall t \geq s)$,

$$u(B; a, x, t) = u(B; a, x, s) = u(B; a, x);$$

(ii) if $\{a\}^B(x) \downarrow$ and $B \upharpoonright u(B; a, x) = C \upharpoonright u(B; a, x)$, then

$$\{a\}^B(x) \simeq \{a\}^C(x) \quad \text{and} \quad u(B; a, x) = u(C; a, x).$$

Proof. Immediate from the definitions. $\square$

Now we can give another version of Proposition 8.2.3(iii). Let us temporarily use variables σ and τ to represent both finite sequences of 0s and 1s and their codes. The **length** of σ is denoted by $\text{lg}(\sigma)$. We write $\sigma \subset A$ to mean that σ is an initial segment of the characteristic function of A — that is, for all $x < \text{lg}(\sigma)$, $\sigma(x) = A(x)$ or equivalently, $\sigma = A \upharpoonright \text{lg}(\sigma)$.

Definition 8.2.8 For all a, x, y, s and σ ,

(i) $\{a\}_s^\sigma(x) \simeq y \iff$ for some $B \supset \sigma$

$$\{a\}_s^B(x) \simeq y \quad \text{and} \quad u(B; a, x, s) < \text{lg}(\sigma);$$

(ii) $W_{a,s}^\sigma := \{x : \exists y \{a\}_s^\sigma(x) \simeq y\}$.

Second Use Theorem 8.2.9 For all a, x, y, s and B ,

- (i) $\{a\}_s^B(x) \simeq y \iff (\exists \sigma \subset B) \{a\}_s^\sigma(x) \simeq y;$
- (ii) $\{a\}_s^B(x) \simeq y \iff \exists s(\exists \sigma \subset B) \{a\}_s^\sigma(x) \simeq y;$
- (iii) $x \in W_{a,s}^B \iff (\exists \sigma \subset B) x \in W_{a,s}^\sigma;$
- (iv) $x \in W_a^B \iff \exists s(\exists \sigma \subset B) x \in W_{a,s}^\sigma;$
- (v) $\{a\}_s^\sigma(x) \simeq y \implies \text{for all } B \supset \sigma,$
 $\{a\}_s^B(x) \simeq y \text{ and } u(B; a, x, s) < \lg(\sigma);$
- (vi) $x \in W_{a,s}^\sigma \implies \text{for all } B \supset \sigma, x \in W_{a,s}^B;$
- (vii) $\{a\}_s^\sigma(x) \simeq y$ is a recursive relation of $a, x, y, s, \sigma;$
 $x \in W_{a,s}^\sigma$ is a recursive relation of $a, x, s, \sigma.$

Proof. For (i)–(iv) it suffices to use $\sigma = B \upharpoonright u(B; a, x)$. (v) and (vi) follow from the First Use Theorem, and then (vii) holds because if $B_\sigma := \{x : \sigma(x) = 1\}$, then

$$\begin{aligned} \{a\}_s^\sigma \simeq y &\iff \{a\}_s^{B_\sigma}(x) \simeq y \text{ and } u(B_\sigma; a, x, s) < \lg(\sigma); \\ x \in W_{a,s}^\sigma &\iff (\exists y < s) \{a\}_s^\sigma(x) \simeq y. \quad \square \end{aligned}$$

At this point we can prove that the partial ordering of T -degrees is not linear.

Kleene-Post Theorem 8.2.10 There exist sets A and B such that $A \not\leq_T B$ and $B \not\leq_T A$. Furthermore, $A, B \leq_T \mathcal{K}$.

Proof. This proof should be contrasted with that of Theorem 8.1.27; it is easier in that we are not trying to make A and B be r.e. (we do that below) but harder in that we are preventing T -comparability, not just m-comparability. It will suffice to construct A and B to satisfy the following requirements for all $a \in \omega$:

$$\begin{aligned} P_{2a} : \quad \{a\}^B \text{ total} &\implies \exists x [A(x) \neq \{a\}^B(x)]; \\ P_{2a+1} : \quad \{a\}^A \text{ total} &\implies \exists x [B(x) \neq \{a\}^A(x)]. \end{aligned}$$

At each stage s we shall determine sequences σ_{s+1} and τ_{s+1} of 0s and 1s which will represent initial segments of the characteristic functions of A and B . We will ensure that as s increases the lengths of these sequences are unbounded and hence uniquely determine A and B .

At stage s , if $s = 2a$ we consider two cases. If

$$(\exists \tau \supset \tau_s) \exists t \{a\}_t^\tau(\lg(\sigma_s)) \downarrow,$$

then choose τ_{s+1} as the least such τ , and if $\{a\}_t^{\tau_{s+1}}(\lg(\sigma_s)) \simeq i$, set $\sigma_{s+1} := \sigma_s \smallfrown \langle 1 \overset{\circ}{-} i \rangle$. Otherwise, set $\sigma_{s+1} := \sigma_s \smallfrown \langle 0 \rangle$ and $\tau_{s+1} := \tau_s \smallfrown \langle 0 \rangle$. If $s = 2a + 1$, proceed similarly with the roles of σ and τ interchanged.

Now set $A := \lim_{s \rightarrow \infty} \sigma_s$ and $B := \lim_{s \rightarrow \infty} \tau_s$, and suppose that $\{a\}^B$ is total. Then at stage $s = 2a$, by the Second Use Theorem there is $\tau \subset B$ and t such that $\{a\}_t^\tau(\lg(\sigma_s)) \downarrow$. Clearly this property is preserved by extending τ so there is such a $\tau \supset \tau_s$. Hence for $x = \lg(\sigma_s)$,

$$\{a\}^B(x) \simeq \{a\}_t^{\tau_{s+1}}(x) \neq \sigma_{s+1}(x) = A(x).$$

Similarly the action at stage $s = 2a + 1$ guarantees that $B \neq \{a\}^A$.

Finally, the case distinction is determined by a condition described by two existential quantifiers preceding a recursive relation, so by the one-quantifier principle can be determined using the K-oracle. Since otherwise the construction is recursive, the functions $s \mapsto \sigma_s$ and $s \mapsto \tau_s$ are K-recursive. Hence so are A and B , since $A(x) = \sigma_s(x)$, where s is the smallest stage such that $\lg(\sigma_s) > x$, and similarly for B . $\square$

In the preceding proof, the sets A and B are not (obviously) r.e. for two reasons: (1) at each stage we determine both that certain numbers belong to these sets and that other numbers do not — that is, we are enumerating both the sets and their complements — and (2) the case distinction is not effective so these enumerations are not recursive. Of course, (2) is unavoidable given (1), since otherwise the sets *and* their complements would be r.e. and hence recursive, which is clearly impossible. The same argument as above does establish that all of A , B and their complements are K-r.e., and hence gives an alternative proof that they are K-recursive. Of course, the fact that we are in this case satisfied with building A and B to be K-recursive is what allows us to settle each P_s at stage s without the injury and backtracking of earlier (and later) proofs.

Our goal for the rest of this section is to show that the structure of the r.e. T -degrees has much the same complexity as the r.e. m -degrees. Let $\mathbf{0}$ now denote the T -degree of the recursive sets. At the other extreme (among r.e. sets) we have

Definition 8.2.11 For any set C ,

- (i) C is **Turing complete** or **T-complete** iff C is r.e. and for every r.e. set A , $A \leq_T C$;

- (ii) $\mathbf{dg}(C)$ is an **r.e. T -degree** iff it contains an r.e. set — that is, there exists an r.e. set A such that $A \equiv_T C$;
- (iii) $\mathbf{0}' := \{C : C \text{ is } T\text{-complete}\}$.

Corollary 8.2.12 $\mathbf{0}'$ is the unique greatest r.e. T -degree. $\square$

Every m -complete set is T -complete, but not necessarily conversely; by Exercise 8.1.39 some simple sets are T -complete. Hence we need a new strategy even to show that there exist non-recursive r.e. sets which are not T -complete — that is, that there is an r.e. T -degree $\mathbf{d}$ such that $\mathbf{0} < \mathbf{d} < \mathbf{0}'$.

We can construct such a set along the lines of Theorem 8.1.25 by building a simple (hence non-recursive) set A and simultaneously an r.e. set $B \not\leq_T A$. Before getting into the proof, we need to discuss a new complication. Requirements P_a and N_a designed to make A simple will be exactly as before, but to ensure that $B \not\leq_T A$, we need to arrange that for each index b , $B \neq \{b\}^A$ by satisfying a requirement of the form

$$Q_b : \quad \{b\}^A \text{ total} \quad \implies \quad \exists y \exists i \left[\{b\}^A(y) \simeq i \wedge (y \in B \iff i = 0) \right].$$

At stage s we have only partial knowledge about A — the finite subset A_s — and to satisfy Q_b we shall attempt to find numbers y_b and i_b such that $\{b\}_s^{A_s}(y_b) \simeq i_b$ and $y_b \notin B_s$. If (I) $i_b > 0$, then our goal is to ensure that $y_b \notin B$ and $\{b\}^A(y_b) \simeq i_b$. If (II) $i_b = 0$, then we put $y_b \in B_{s+1} \subseteq B$ and again try to ensure that $\{b\}^A(y_b) \simeq i_b$. As in the proof of Theorem 8.1.25, the value of $B(y_b)$ is easy to control, but it is more difficult to make sure that the value of $\{b\}^A(y_b)$ stays equal to i_b . The computation $\{b\}_s^{A_s}(y_b) \simeq i_b$ uses information about A_s which may change as the construction proceeds — that is, it may use the fact that certain $z \notin A_s$, but in the process of satisfying a requirement P_a at some stage $t > s$, we may want to put some such $z = x_a$ into $A_{t+1} \subseteq A$. The key point is that any such z is less than $u_{b,s} := u(A_s; b, y_b, s)$ and we can adopt the same priority mechanism as before: if $a \leq b$, then we accord P_a higher priority than Q_b and allow such an x_a to enter A at stage t , thereby injure Q_b , and cause it to become unmarked so that it will be given attention again at a later stage; if $b < a$, then we require that x_a be larger than a restraint value $R_t(a) \geq u_{b,s}$ so that it will not injure Q_b . It will be useful to establish some terminology.

Definition 8.2.13 For any stage enumeration $\langle A_s : s \in \omega \rangle$ of a set A and any b, y and $s \leq t$, if $u_{b,s} := u(A_s; b, y, s)$, then

- (i) $\{b\}_s^{A_s}(y) \downarrow$ **correctly to** $t \iff$

$$\{b\}_s^{A_s}(y) \downarrow \quad \text{and} \quad A_t \upharpoonright u_{b,s} = A_s \upharpoonright u_{b,s};$$

$$(ii) \{b\}_s^{A_s}(y) \downarrow \text{correctly} \iff$$

$$\{b\}_s^{A_s}(y) \downarrow \quad \text{and} \quad A \upharpoonright u_{b,s} = A_s \upharpoonright u_{b,s}.$$

By the following Correctness Lemma, if $\{b\}_s^{A_s}(y) \simeq z$ correctly, then not only is the computed value correct — $\{b\}_s^A(y) \simeq z$ — but also the information about A used to carry out the computation is correct.

Correctness Lemma 8.2.14 *For any stage enumeration $\langle A_s : s \in \omega \rangle$ of a set A and any b, y and $s \leq t$, if $u_{b,s} := u(A_s; b, y, s)$, then*

$$(i) \{b\}_s^{A_s}(y) \downarrow \text{correctly to } t \implies \text{for all } r \text{ such that } s \leq r \leq t,$$

$$\{b\}_r^{A_r}(y) \downarrow \text{correctly to } t, \quad \{b\}_r^{A_r}(y) \simeq \{b\}_s^{A_s}(y), \quad \text{and} \\ u(A_r; b, y, r) = u_{b,s};$$

$$(ii) \{b\}_s^{A_s}(y) \downarrow \text{correctly} \iff (\forall t \geq s) \{b\}_s^{A_s}(y) \downarrow \text{correctly to } t \\ \iff (\forall t \geq s) \{b\}_t^{A_t}(y) \downarrow \text{correctly};$$

$$(iii) \{b\}_s^{A_s}(y) \downarrow \text{correctly} \implies$$

$$\{b\}^A(y) \simeq \{b\}_s^{A_s}(y) \quad \text{and} \quad u(A; b, y) = u_{b,s};$$

$$(iv) \{b\}^A(y) \downarrow \implies \exists s [\{b\}_s^{A_s}(y) \downarrow \text{correctly}];$$

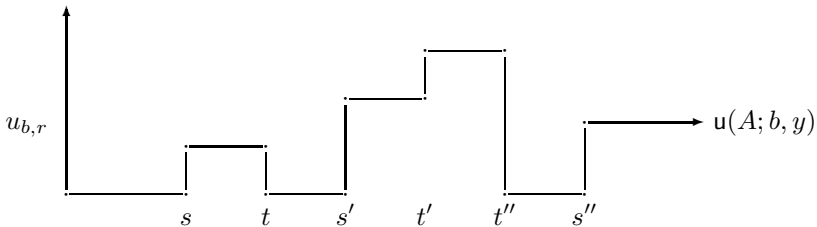
$$(v) \{b\}_s^{A_s}(y) \downarrow \text{incorrectly} \implies$$

$$(\exists t \geq s) A_s \upharpoonright u_{b,s} = A_t \upharpoonright u_{b,s} \neq A_{t+1} \upharpoonright u_{b,s};$$

we say that the computation is **destroyed** at stage $t + 1$.

Proof. Parts (i)–(iii) follow from the First Use Theorem. For (iv), it suffices to choose s large enough that $\{b\}_s^A(y) \downarrow$ and $A_s \upharpoonright u(A; b, y) = A \upharpoonright u(A; b, y)$ — that is, for all $x < u(A; b, y)$, if $x \in A$ then already $x \in A_s$. (v) is immediate. $\square$

It is helpful to consider a possible graph of $u_{b,r}$ against r :



This illustrates a pattern in which for $r < s$, $\{b\}_r^{A_r}(y) \uparrow$, so $u_{b,r} = 0$. Then $\{b\}_s^{A_s}(y) \downarrow$ correctly to t , but for some $z < u_{b,s} = u_{b,t}$, $z \in A_{t+1} \sim A_t$, the computation is destroyed at stage $t+1$, and the use $u_{b,t+1}$ drops back to 0. At stage s' a new computation is established, which again is destroyed at stage t' , but this time the new element of $A_{t'+1}$ allows the immediate establishment of a third computation which survives until stage t'' . Finally at stage s'' is established a correct computation which witnesses that $\{b\}^A(y) \downarrow$. If $\{b\}^A(y) \uparrow$, then patterns like those between s and t or between s' and t'' might repeat infinitely often or the use $u_{b,r}$ might be eventually constantly 0.

We are now ready to prove

Theorem 8.2.15 *There exist r.e. sets A and B such that A is simple and $B \not\leq_T A$; in particular A is non-recursive but not T -complete.*

Proof. As discussed above, it will suffice to construct r.e. sets A and B which for each $a, b \in \omega$ satisfy the requirements

$$\begin{aligned} P_a : \quad & W_a \text{ infinite} \implies A \cap W_a \neq \emptyset \\ Q_b : \quad & \{b\}^A \text{ total} \implies \exists y \exists i \left[\{b\}^A(y) \simeq i \wedge (y \in B \iff i = 0) \right] \\ N_a : \quad & A \cap \{x : x \leq 2a\} \text{ has at most } a \text{ elements.} \end{aligned}$$

We construct $A := \bigcup_{s \in \omega} A_s$ and $B := \bigcup_{s \in \omega} B_s$ in stages as usual and follow closely the parallel Theorem 8.1.25. Set the restraint function

$$R_s(a) := \max\{u_b : b < a \text{ and } Q_b \text{ is marked at } s\} \cup \{2a\},$$

where $u_b := u(A_s; b, y_b, s)$ and varies with s . P_a **needs attention** at stage s iff $a < s$, P_a is not marked at stage s and

$$(\exists x < s)[x \in W_{a,s} \text{ and } x \geq R_s(a)].$$

Q_b needs attention at stage s iff $b < s$, Q_b is not marked at stage s and

$$(\exists y < s)[\{b\}_s^{A_s}(y) \downarrow \wedge y \notin B_s \wedge y \in \omega^{[b]}].$$

If no P_a or Q_b needs attention at stage s , then $A_{s+1} = A_s$, $B_{s+1} = B_s$ and each P_a and Q_b is marked at stage $s+1$ iff it was marked at stage s . Otherwise, choose the first requirement in the priority list

$$P_0, Q_0, P_1, Q_1, \dots, P_a, Q_a, \dots$$

which needs attention at stage s and let it **get attention** as follows. If the chosen requirement is P_a , perform the following steps:

- (i) $x_a := \text{least } x < s[x \in W_{a,s} \text{ and } x \geq R_s(a)]$;
- (ii) $A_{s+1} := A_s \cup \{x_a\}$; $B_{s+1} := B_s$;
- (iii) P_a is marked at stage $s+1$; for $b \neq a$, P_b is marked at stage $s+1$ iff it was marked at stage s ;
- (iv) for all $b < a$, Q_b is marked at stage $s+1$ iff it was marked at stage s ; for all $b \geq a$, Q_b is unmarked at stage $s+1$ and is said to have been ***injured***.

If the chosen requirement is Q_b , perform the steps:

- (i) $y_b := \text{least } y < s[\{b\}_s^{A_s}(y) \downarrow \wedge y \notin B_s \wedge y \in \omega^{[b]}]$;
 $u_b := u(A_s; b, y_b, s)$;
 $i_b := \{b\}_s^{A_s}(y_b)$;
- (ii) if $i_b = 0$, then $B_{s+1} := B_s \cup \{y_b\}$; otherwise $B_{s+1} := B_s$;
 $A_{s+1} := A_s$;
- (iii) for all a , P_a is marked at stage $s+1$ iff it was marked at stage s ;
- (iv) Q_b is marked at stage $s+1$; for all $a \neq b$, Q_a is marked at stage $s+1$ iff it was marked at stage s .

This completes the construction of A and B . Note also that as before the values of y_b , u_b and i_b do not change as long as Q_b remains marked but are redefined each time that Q_b is re-marked. We now verify that this construction ensures that for all a and b ,

- (1) P_a gets attention at most once; Q_b gets attention at most finitely often;
- (2) $\lim_{s \rightarrow \infty} R_s(a)$ exists and is finite;
- (3) if W_a is infinite, then P_a is marked at some stage;
- (4) if P_a is marked at some stage, then P_a is satisfied;
- (5) if $\{b\}^A$ is total, then Q_b is permanently marked at some stage;
- (6) if Q_b is permanently marked at some stage, then Q_b is satisfied;
- (7) P_a is satisfied;
- (8) Q_b is satisfied;
- (9) N_a is satisfied.

(1)–(4), (7) and (9) hold as before; although the restraint $R_s(a)$ is calculated differently, it still changes only at stages s when the marked status of some Q_b with $b < a$ changes.

Towards (5), suppose that $\{b\}^A$ is total. By (1), there exists a stage t_b such that none of P_a ($a \leq b$) or Q_a ($a < b$) gets attention at any stage $s \geq t_b$. If Q_b is already marked at stage t_b , then it is permanently marked there. Otherwise, we claim that there exists a stage $s \geq t_b$ such that Q_b

needs attention at s . Indeed, by the Correctness Lemma 8.2.14 for any y , so in particular for any $y = \langle b, z \rangle \notin B_{t_b}$, there is $s \geq t_b$ such that $\{b\}_s^{A_s}(y) \downarrow$ (because $\{b\}^A$ is total). If Q_b is still not marked at s , then still $y \notin B_s$, since only attention to Q_b can put a number $y \in \omega^{[b]}$ into B . Hence Q_b will need attention at s and being of highest priority will therefore get attention and become marked. The marking will be permanent, again by the definition of t_b .

Now towards (6), suppose that Q_b gets attention and becomes permanently marked at stage s . Then with $\{b\}_s^{A_s}(y_b) \simeq i_b$, either

- (I) $i_b > 0$ and $y_b \notin B_{s+1}$ or
- (II) $i_b = 0$ and $y_b \in B_{s+1}$.

In either case, as before the status of y_b relative to B will not change after stage $s + 1$, and it suffices to prove that $\{b\}^A(y_b) \simeq i_b$. Hence, by the Correctness Lemma 8.2.14, it suffices to prove by induction that for all $t \geq s$, $A_t \upharpoonright u_b = A_s \upharpoonright u_b$ so the computation at stage s is correct. This holds for $t = s$, so assume as induction hypothesis that $A_t \upharpoonright u_b = A_s \upharpoonright u_b$. If no P_a gets attention at stage t , then $A_{t+1} = A_t$. Otherwise, for some a , $A_{t+1} = A_t \cup \{x_a\}$. Since Q_b is not injured (because it does not become unmarked) at t , necessarily $b < a$, and thus $x_a \geq R_t(a) \geq u_b$, so again $A_{t+1} \upharpoonright u_b = A_t \upharpoonright u_b$.

As before, (8) is immediate from (5) and (6). To complete the argument we must verify that each step of the construction is effective in order to conclude that A and B are r.e. The new element here is the use $u(A_s; b, y_b, s)$, which is effectively computable from the finite set A_s . $\square$

Friedberg-Mučnik Theorem 8.2.16 *There exist r.e. sets A and B such that $A \not\leq_T B$ and $B \not\leq_T A$.*

Proof. This is now a simple combination of the techniques of the preceding proof with those of the many-one version Theorem 8.1.27. We need for A and B to satisfy for each a the requirements

$$P_{2a} : \quad \{a\}^B \text{ total} \implies \exists x \exists i \left[\{a\}^B(x) \simeq i \wedge (x \in A \iff i = 0) \right];$$

$$P_{2a+1} : \quad \{a\}^A \text{ total} \implies \exists x \exists i \left[\{a\}^A(x) \simeq i \wedge (x \in B \iff i = 0) \right].$$

Of course, P_{2a} guarantees that $A \neq \{a\}^B$ and P_{2a+1} guarantees that $B \neq \{a\}^A$. Set the restraint function

$$R_s(k) := \max\{u_l : l < k \text{ and } P_l \text{ is marked at } s\}.$$

Then P_{2a} needs attention at stage s iff $2a < s$, P_{2a} is not marked at stage s and

$$(\exists x < s) [\{a\}_s^{B_s}(x) \downarrow \wedge x \geq R_s(2a) \wedge x \notin A_s \wedge x \in \omega^{[a]}].$$

Similarly, P_{2a+1} needs attention at stage s iff $2a+1 < s$, P_{2a+1} is not marked at stage s and

$$(\exists x < s)[\{a\}_s^{A_s}(x) \downarrow \wedge x \geq R_s(2a+1) \wedge x \notin B_s \wedge x \in \omega^{[a]}].$$

If no P_k needs attention at stage s , then $A_{s+1} = A_s$, $B_{s+1} = B_s$ and each P_k is marked at stage $s+1$ iff it was marked at stage s . Otherwise, choose the smallest k such that P_k needs attention at s . If $k = 2a$, let P_k get attention by performing the following steps:

- (i) $x_k := \text{least } x < s[\{a\}_s^{B_s}(x) \downarrow \wedge x \geq R_s(k) \wedge x \notin A_s \wedge x \in \omega^{[a]}];$
 $u_k := u(B_s; a, x_k, s);$
 $i_k := \{a\}_s^{B_s}(x_k);$
- (ii) if $i_k = 0$, then $A_{s+1} := A_s \cup \{x_k\}$; otherwise $A_{s+1} := A_s$;
 $B_{s+1} := B_s$;
- (iii) P_k is marked at stage $s+1$; for all $l < k$, P_l is marked at stage $s+1$ iff it was marked at stage s ; for all $l > k$, P_l is unmarked at stage $s+1$ and is said to have been ***injured***.

If $k = 2a+1$, the same steps are performed with the roles of A and B interchanged. We leave the remainder of the proof as an exercise. $\square$

Similarly, we leave it to the reader to make the necessary modifications in the results at the end of the preceding section to prove

Theorem 8.2.17 *Every countable partial ordering is embeddable in the r.e. T -degrees — that is, for any partial ordering $\preceq$ of ω , there exists a family $\langle C_i : i \in \omega \rangle$ of r.e. sets such that for all i and j ,*

$$i \preceq j \iff C_i \leq_T C_j. \quad \square$$

Collateral Reading. All of the material of [Sections 8.2–8.6](#) of this chapter (and much more) are covered in [Soa; [Chapters VII–VIII](#)] and [Od99; Chapters X–XI]. Some parts, with a quite different exposition, are in [Coo; Chapter 12]. We recommend strongly a parallel reading of some of this for a deeper understanding of the results.

Exercises

Exercise 8.2.18 Prove Proposition 8.2.3.

Exercise 8.2.19 Prove that for any sets A and B , A is B -r.e. iff $A = \{x : \exists u \langle x, B \upharpoonright u \rangle \in W\}$ for some r.e. set W , where $B \upharpoonright u$ here denotes $\langle B(0), B(1), \dots, B(u-1) \rangle$.

Exercise 8.2.20 Prove that there exists a hypersimple set which is T -complete.

Hint: Use the construction of Exercise 8.1.40 with B a T -complete set. Show that $A \equiv_T B$.

Exercise 8.2.21 Complete the proof of Proposition 8.2.6.

Exercise 8.2.22 Complete the proof of the Friedberg-Mučnik Theorem 8.2.16.

Exercise 8.2.23 Prove Theorem 8.2.17.

Exercise 8.2.24 Show that we can characterize the class of B -partial recursive functions (for any $B \subseteq \omega$) in terms of equational derivability as follows. Let $L[B]$ be as in Definition 5.4.1 with one additional distinguished function symbol g . Define the notion of $\Gamma[B]$ -theorems as before except that now there are infinitely many “logical axioms”: all equations of the form $g\dot{m} \doteq \dot{1}$ for $m \in B$ and $g\dot{m} \doteq \dot{0}$ for $m \notin B$. Prove that a partial function F is B -equationally derivable iff it is B -partial recursive.

Exercise 8.2.25 Give the appropriate definitions and prove that a partial function is B -register computable iff it is B -partial recursive.

Exercise 8.2.26 Prove that for any non-recursive set B , there exists a set A such that $A \not\leq_T B$ and $B \not\leq_T A$.

Hint: At even stages $s = 2a$ proceed as in the Kleene-Post Theorem 8.2.10 to ensure that $A \neq \{a\}^B$. At odd stages $s = 2a + 1$, if there exist $\sigma, \sigma' \supseteq \sigma_s$ such that $\{a\}^\sigma \neq \{a\}^{\sigma'}$, it is possible to choose σ_{s+1} such that $\{a\}^{\sigma_{s+1}} \not\subseteq B$. Otherwise, argue that if $\{a\}^A$ is total, then it is recursive so not equal to B .

Exercise 8.2.27 A unary function f is called **fixed-point free** (FPF) iff $\forall a [W_a \neq W_{f(a)}]$ and **diagonally non-recursive** (DNR) iff $\forall a [f(a) \neq \{a\}(a)]$. Show that

- (i) no recursive function is either FPF or DNR;
- (ii) for every FPF function f there is a DNR function g such that $g \leq_T f$;
- (iii) for every DNR function f there is a FPF function g such that $g \leq_T f$.

Hint: For (ii) use a variant of the argument for Exercise 8.1.42(i). For (iii), given a DNR function f , find a recursive function h such that for all a , if $W_a \neq \emptyset$, then for all x , $\{h(a)\}(x) \in W_a$, and choose $g \leq_T f$ such that for all a , $W_{g(a)}$ is the singleton set $\{f(h(a))\}$. (A more complex argument shows that in (ii) and (iii) $\leq_T$ can be replaced by $\equiv_T$.)

8.3 The jump operator

In the preceding section we introduced the notions of a set A being B -recursive or B -r.e. and sketched some initial steps in relativizing the theory of recursive and r.e. sets and functions to an arbitrary B . Our first goal in this section is to carry this relativization further. First, in addition to the several characterizations of [Section 8.2](#), we have

Proposition 8.3.1 *For any sets A and B , A is B -r.e. iff A has a B -recursive stage enumeration.*

Proof. First, suppose A is B -r.e. so for some index a , $A = W_a^B$. Clearly $\langle W_{a,s}^B : s \in \omega \rangle$ is a stage enumeration of A , and we show that it is B -recursive. By the Second Use Theorem 8.2.9, for all x ,

$$x \in W_{a,s}^B \iff (\exists \sigma \subseteq B) x \in W_{a,s}^\sigma,$$

and $x \in W_{a,s}^\sigma$ is a recursive relation. Furthermore, by the definition of $W_{a,s}^\sigma$,

$$x \in W_{a,s}^B \iff \exists \sigma [\text{lg}(\sigma) < s \wedge \sigma \subseteq B \wedge x \in W_{a,s}^\sigma].$$

This is really a bounded quantifier, since σ is a sequence of 0s and 1s, so has code less than the code for the sequence of length s of all 1s. Furthermore the relation $\sigma \subseteq B$ is B -recursive, so the relation $x \in W_{a,s}^B$ is also B -recursive as desired.

The converse goes just like the unrelativized version: if $\langle A_s : s \in \omega \rangle$ is a B -recursive stage enumeration of A , then $x \in A \iff \exists s [x \in A_s]$, so A is B -r.e. $\square$

Proposition 8.3.2 *For any sets B and C ,*

- (i) $B \leq_T C \implies \text{for all } A [A \text{ is } B\text{-r.e.} \implies A \text{ is } C\text{-r.e.}]$;
- (ii) $B \equiv_T C \implies \text{for all } A [A \text{ is } B\text{-r.e.} \iff A \text{ is } C\text{-r.e.}]$.

Proof. This is immediate from Lemma 8.2.5 together with any one of the characterizations of relative recursive enumerability. $\square$

Definition 8.3.3 For any sets B and C ,

- (i) C is **m-complete** $[B]$ (**T-complete** $[B]$) iff C is B -r.e. and for every B -r.e. set A , $A \leq_m C$ ($A \leq_T C$);
- (ii) $\text{dg}_m(C)$ ($\text{dg}(C)$) is a B -r.e. m -degree (T -degree) iff it contains a B -r.e. set — that is, there exists a B -r.e. set A such that $A \equiv_m C$ ($A \equiv_T C$).

Remark 8.3.4 There is another natural way that we could define these notions: if we think of relativization to B as providing access to the B -oracle at all times, then it would seem reasonable to replace $\leq_m$ and $\leq_T$ in the preceding definition by the relations

$$\begin{aligned} A \leq_m^B C &\iff \text{exists a } B\text{-recursive function } f \text{ such that } A = f^{-1}(C); \\ A \leq_T^B C &\iff A \text{ is } (B, C)\text{-recursive.} \end{aligned}$$

Here ‘ (B, C) -recursive’ has the obvious definition in terms of two oracles or equivalently as ‘ $(B \oplus C)$ -recursive’. However, the simpler definitions as given are traditional and will be used here.

Proposition 8.3.5 *For any B , each of the following sets is m -complete $[B]$ and hence also T -complete $[B]$:*

- (i) $\text{PartComp}[B]$;
- (ii) $W^B := \{ \langle a, x \rangle : \exists y[\{a\}^B(x) \simeq y] \} = \{ \langle a, x \rangle : x \in W_a^B \}$;
- (iii) $K^B := \{ a : a \in W_a^B \}$.

Proof. The proofs are almost identical to the earlier unrelativized ones (Propositions 8.1.6 and 8.1.12) and are omitted. Note in particular that the functions S_n^m have the same effect for B -computations. $\square$

Definition 8.3.6 For any set B , K^B is called the **jump** of B and is denoted B' .

The most important properties of the jump are given by

Proposition 8.3.7 *For any set B ,*

- (i) B' is B -r.e.;
- (ii) $B <_T B'$; in particular, B' is not B -recursive;
- (iii) for any A , A is B -r.e. $\iff A \leq_m B'$;
- (iv) for any C , $B \leq_T C \iff B' \leq_m C'$.

Proof. Parts (i) and (ii) are simply the relativized versions of Proposition 8.1.10 with the same proofs. The implication $(\implies)$ of (iii) is the m -completeness $[B]$ of K^B and $(\impliedby)$ follows easily from (i). For (iv) $(\implies)$, if $B \leq_T C$, then by (i), B' is B -r.e., hence by Proposition 8.3.2 also C -r.e., so by (iii), $B' \leq_m C'$. Conversely, if $B' \leq_m C'$, then since both B and $\overline{B}$ are B -recursive, hence B -r.e., by (iii) $(\implies)$ and transitivity, both $B, \overline{B} \leq_m C'$, so by (iii) $(\impliedby)$, both are C -r.e. and hence both are C -recursive; in particular, $B \leq_T C$. $\square$

It follows from (iv) that $B \equiv_T C \implies B' \equiv_T C'$ — that is, the jump operator is well-defined on T -degrees. We shall see below that the converse of this implication does not hold in general — that is, the mapping $\mathbf{b} \mapsto \mathbf{b}'$ in the following definition is not injective.

Definition 8.3.8 For any T -degree $\mathbf{b}$, $\mathbf{b}' := \text{dg}(B')$ for some (any) $B \in \mathbf{b}$.

This notation is consistent with the notation $\mathbf{0}'$ for $\text{dg}(\mathbf{K})$ from Section 8.2 once we observe that $\mathbf{K} \equiv_m \mathbf{K}^\emptyset (= \emptyset') \equiv_m \mathbf{K}^B (= B')$ for any recursive set B (exercise). The reader should also note carefully the use of $\leq_m$ in (iii) and (iv) of the proposition. In general

$$A \leq_T B' \not\Rightarrow A \text{ is } B\text{-r.e.}$$

since, for example, $\overline{B'} \leq_T B'$, but is not B -r.e. by (ii) of the proposition. The corresponding question for T -degrees is more subtle:

$\mathbf{a} \leq \mathbf{b}' \stackrel{?}{\implies} \mathbf{a} \text{ is } \mathbf{b}\text{-r.e.}$ in particular $\mathbf{a} \leq \mathbf{0}' \stackrel{?}{\implies} \mathbf{a} \text{ is r.e.},$
or equivalently,

$$A \leq_T \emptyset' \stackrel{?}{\implies} A \equiv_T D \text{ for some r.e. set } D.$$

Similarly, for (iv) it is natural to ask if $B' \leq_T C' \stackrel{?}{\implies} B \leq_T C$. We shall answer both of these questions in the negative at the end of this section, but for now we give some equivalent characterizations.

Definition 8.3.9 A set A is *(B-)recursively approximable* iff there exists a (B) -recursive family $\langle A_s : s \in \omega \rangle$ such that $\forall x [A(x) = \lim_{s \rightarrow \infty} A_s(x)]$, which we also write $A = \lim_{s \rightarrow \infty} A_s$. A *modulus* of the approximation is a function g such that $\forall x (\forall s \geq g(x)) [A(x) = A_s(x)]$.

Examples 8.3.10

- (i) A stage enumeration is a special sort of recursive approximation in which all of the sets A_s are finite, $A_0 = \emptyset$ and the sequence is increasing. As s increases, the status of a particular number x changes at most once when it enters A_s .
- (ii) Given recursive stage enumerations

$$\langle A_s^+ : s \in \omega \rangle \quad \text{and} \quad \langle A_s^- : s \in \omega \rangle$$

of r.e. sets A^+ and A^- , respectively, let $A_s = A_s^+ \sim A_s^-$. Then $\langle A_s : s \in \omega \rangle$ is a recursive approximation to $A := A^+ \sim A^-$. Such an A is called *d.r.e. (difference of r.e. sets)*. Here the status of a number x with respect to A_s may change twice if it enters A_s^+ , and hence A_s , and then at $t > s$ enters A_t^- and hence *leaves* A_t .

In a general recursive approximation a number x may enter and leave the approximating sequence any finite number of times; a modulus gives a bound for the termination of this process.

Lemma 8.3.11 *For any sets A , B and D , if A has a B -recursive approximation with a D -recursive modulus, then $A \leq_T B \oplus D$. In particular, if A has a recursive approximation with a recursive modulus, then A is recursive.*

Proof. If g is a D -recursive modulus of the B -recursive approximation $\langle A_s : s \in \omega \rangle$, then for all x , $A(x) = A_{g(x)}(x)$. $\square$

We refer to this principle often as **stage bounding**.

Limit Lemma 8.3.12 *For any set A ,*

- (i) $A \leq_T \emptyset'$ iff A is recursively approximable;
- more generally, for any set B ,
- (ii) $A \leq_T B'$ iff A is B -recursively approximable.

Proof. We prove the second version. Suppose first that $A \leq_T B'$, and fix an index a such that $A = \{a\}^{B'}$. Since B' is B -r.e., it has a B -recursive stage enumeration $\langle C_s : s \in \omega \rangle$. Set

$$A_s := \left\{ x : \{a\}_s^{C_s}(x) \simeq 1 \right\}.$$

Clearly this is a B -recursive family; if $x \in A$, then for all sufficiently large s , $\{a\}_s^{C_s}(x) \simeq 1$ so $x \in A_s$, and if $x \notin A$, then for all sufficiently large s , $\{a\}_s^{C_s}(x) \simeq 0$ so $x \notin A_s$. Hence $A(x) = \lim_{s \rightarrow \infty} A_s(x)$.

Suppose now that $\langle A_s : s \in \omega \rangle$ is a B -recursive family which approximates A , and let

$$D := \left\{ \langle s, x \rangle : \exists t [s < t \wedge A_s(x) \neq A_t(x)] \right\};$$

$$h(x) := \text{least } s [\langle s, x \rangle \notin D].$$

Then h is a D -recursive modulus, so $A \leq_T B \oplus D$. But D is clearly B -r.e., so $D \leq_T B'$ and hence also $A \leq_T B'$. $\square$

Definition 8.3.13 A set A is **of (B) -r.e. T -degree** iff $\text{dg } A$ is a (B) -r.e. T -degree — that is, $A \equiv_T D$ for some (B) -r.e. set D .

Clearly if A is of r.e. T -degree, then $A \leq_T \emptyset'$.

Modulus Lemma 8.3.14 *For any set A , A is of r.e. T -degree iff A is recursively approximable with an A -recursive modulus.*

Proof. Suppose first that A is of r.e. T -degree, say $A \equiv_T D$ with D r.e., and let d be an index such that $D = W_d$. Set $D_s := W_{d,s}$, choose an index a such that $A = \{a\}^D$, and set $A_s := \left\{ x : \{a\}_s^{D_s}(x) \simeq 1 \right\}$. By the Correctness

Lemma 8.2.14, for each x there exists s such that $\{a\}_s^{D_s}(x) \downarrow$ correctly and thus $\langle A_s : s \in \omega \rangle$ is a recursive approximation of A . Furthermore, if

$$g(x) := \text{least } s (\exists u \leq s) \left[\{a\}_s^{D_s \upharpoonright u}(x) \downarrow \wedge D_s \upharpoonright u = D \upharpoonright u \right],$$

then g is a D -recursive modulus for the approximation and, since $D \leq_T A$, also an A -recursive modulus.

Now suppose that $\langle A_s : s \in \omega \rangle$ is a recursive approximation to A and g is an A -recursive modulus. As in the preceding proof, let

$$D := \{ \langle s, x \rangle : \exists t [s < t \wedge A_s(x) \neq A_t(x)] \}.$$

$A \leq_T D$ as before, but now we have also

$$D = \{ \langle s, x \rangle : \exists t [s < t \leq g(x) \wedge A_s(x) \neq A_t(x)] \},$$

so since g is A -recursive, also $D \leq_T A$. $\square$

Definition 8.3.15 A set $A \leq_T \emptyset'$ is **low** iff $A' \equiv_T \emptyset'$.

Since $\emptyset' \leq_T A'$ for every A , a low set is one whose jump has the lowest possible value. Of course, since $A <_T A'$, a low set is not T -complete and indeed should be thought of as “nearly recursive”, since its jump is T -equivalent with that of a recursive set. As an application of the Limit Lemma, we show that a minor modification in the proofs of Theorems 8.2.15 and 8.2.16 allows us to construct sets A and B as in those theorems which are in addition low.

Proposition 8.3.16 *There exist an r.e. set A such that A is both simple (hence non-recursive) and low.*

Proof. We construct an r.e. set A which for each $a, b \in \omega$ satisfies the requirements

$$\begin{aligned} P_a : W_a \text{ infinite} &\implies A \cap W_a \neq \emptyset \\ Q_b : \forall t (\exists s \geq t) \{b\}_s^{A_s}(b) \downarrow &\implies \{b\}^A(b) \downarrow \\ N_a : A \cap \{x : x \leq 2a\} &\text{ has at most } a \text{ elements.} \end{aligned}$$

As usual, the requirements P_a and N_a guarantee that A is simple. The requirements Q_b guarantee that A' is recursively approximable, so by the Limit Lemma 8.3.12 $A' \leq_T \emptyset'$. Indeed, by the Correctness Lemma 8.2.14 and Q_b ,

$$\begin{aligned} \{b\}^A(b) \downarrow &\implies \exists t [\{b\}_t^{A_t}(b) \downarrow \text{ correctly}] \implies \exists t (\forall s \geq t) \{b\}_s^{A_s}(b) \downarrow; \\ \{b\}^A(b) \uparrow &\implies \exists t (\forall s \geq t) \{b\}_s^{A_s}(b) \uparrow; \end{aligned}$$

so

$$A' = \lim_{s \rightarrow \infty} \left\{ b : \{b\}_s^{A_s}(b) \downarrow \right\}.$$

We construct $A := \bigcup_{s \in \omega} A_s$ in stages as usual and follow closely the parallel Theorem 8.2.15. Set the restraint function

$$R_s(a) := \max\{u_b : b < a \text{ and } Q_b \text{ is marked at } s\} \cup \{2a\},$$

where $u_b := u(A_s; b, b, s)$. P_a **needs attention** at stage s iff $a < s$, P_a is not marked at stage s and

$$(\exists x < s)[x \in W_{a,s} \text{ and } x \geq R_s(a)].$$

Q_b needs attention at stage s iff $b < s$, Q_b is not marked at stage s and $\{b\}_s^{A_s}(b) \downarrow$. If no P_a or Q_b needs attention at stage s , then $A_{s+1} = A_s$ and each P_a and Q_b is marked at stage $s+1$ iff it was marked at stage s . Otherwise, choose the first requirement in the priority list

$$P_0, Q_0, P_1, Q_1, \dots, P_a, Q_a, \dots$$

which needs attention at stage s and let it **get attention** as follows. If the chosen requirement is P_a , perform the same steps as in the proof of Theorem 8.2.15:

- (i) $x_a := \text{least } x < s[x \in W_{a,s} \text{ and } x \geq R_s(a)]$;
- (ii) $A_{s+1} := A_s \cup \{x_a\}$;
- (iii) P_a is marked at stage $s+1$; for $b \neq a$, P_b is marked at stage $s+1$ iff it was marked at stage s ;
- (iv) for all $b < a$, Q_b is marked at stage $s+1$ iff it was marked at stage s ; for all $b \geq a$, Q_b is unmarked at stage $s+1$ and is said to have been **injured**.

If the chosen requirement is Q_b , perform the simpler steps:

- (i) $u_b := u(A_s; b, b, s)$;
- (ii) $A_{s+1} := A_s$;
- (iii) for all a , P_a is marked at stage $s+1$ iff it was marked at stage s ;
- (iv) Q_b is marked at stage $s+1$; for all $a \neq b$, Q_a is marked at stage $s+1$ iff it was marked at stage s .

This completes the construction of A , and we need to verify that this construction ensures that for all a and b ,

- (1) P_a gets attention at most once; Q_b gets attention at most finitely often;

- (2) $\lim_{s \rightarrow \infty} R_s(a)$ exists and is finite;
- (3) if W_a is infinite, then P_a is marked at some stage;
- (4) if P_a is marked at some stage, then P_a is satisfied;
- (5) if $\forall t (\exists s \geq t) \{b\}_s^{A_s}(b) \downarrow$, then Q_b is permanently marked at some stage;
- (6) if Q_b is permanently marked at some stage, then $\{b\}^A(b) \downarrow$;
- (7) P_a is satisfied;
- (8) Q_b is satisfied;
- (9) N_a is satisfied.

(1)–(4), (7) and (9) hold as before. For (5), suppose that $\forall t (\exists s \geq t) \{b\}_s^{A_s}(b) \downarrow$. By (1), there exists a stage t_b such that none of P_a ($a \leq b$) or Q_a ($a < b$) gets attention at any stage $s \geq t_b$. If Q_b is already marked at stage t_b , then it is permanently marked there. Otherwise, by hypothesis there exists a stage $s \geq t_b$ such that $\{b\}_s^{A_s}(b) \downarrow$ and therefore if Q_b is still not marked at s , then Q_b will need attention at s and being of highest priority will therefore get attention and become marked. The marking will be permanent again by the definition of t_b .

For (6), if Q_b gets attention and becomes permanently marked at stage s , then as before we may prove by induction that for all $t \geq s$, $A_t \upharpoonright u_b = A_s \upharpoonright u_b$, so by the Correctness Lemma 8.2.14, $\{b\}^A(b) \downarrow$.

As before, (8) is immediate from (5) and (6) and each step of the construction is effective, so the stage enumeration is recursive and A is r.e. $\square$

We leave as an exercise the application to a similar technique to prove the following improvement of the Friedberg-Mučnik Theorem 8.2.16.

Theorem 8.3.17 *There exist low r.e. sets A and B such that $A \not\leq_T B$ and $B \not\leq_T A$. $\square$*

A natural question about any function, in particular the jump operator, is to determine its image — which sets (degrees) are jumps of other sets (degrees). Since sets of the form A' have a very special structure, there is no more interesting answer to the set version of this question than simply the definition. However, for degrees, the answer is surprisingly simple: a degree $\mathbf{s}$ is of the form $\mathbf{a}'$ for some degree $\mathbf{a}$ iff it satisfies the minimal necessary condition that $\mathbf{0}' \leq \mathbf{s}$.

Friedberg Jump Theorem 8.3.18 *For every set S , if $\emptyset' \leq_T S$, then there exists a set A such that $A' \equiv_T S$.*

This result is immediate from the following more general version, since if $\emptyset' \leq_T S$, then $S \oplus \emptyset' \equiv_T S$.

Proposition 8.3.19 *For every set S , there exists a set A such that*

$$A' \equiv_T S \oplus \emptyset' \equiv_T A \oplus \emptyset'.$$

Proof. This proof is somewhat in the style of the Kleene-Post Theorem 8.2.10. Let S be given; we construct an increasing sequence of approximations $\langle \sigma_a : a \in \omega \rangle$ to (the characteristic function of) A . $\sigma_0 := \emptyset$ and recursively,

$$\begin{aligned} \sigma_a^+ &:= \sigma_a \frown (S(a)); \\ \sigma_{a+1} &:= \begin{cases} \text{least } \tau \supseteq \sigma_a^+ \left[\{a\}_{\lg(\tau)}^\tau(a) \downarrow \right], & \text{if such a } \tau \text{ exists;} \\ \sigma_a^+ & \text{otherwise.} \end{cases} \end{aligned}$$

Let $A := \lim_{a \rightarrow \infty} \sigma_a$ and set $f(a) := (\text{the code of}) \sigma_a$. Since trivially $A \oplus \emptyset' \leq_T A'$, it will suffice to establish

- (10) $A' \leq_T f$ and $S \leq_T f$;
- (11) $f \leq_T A \oplus \emptyset'$ and $f \leq_T S \oplus \emptyset'$.

For (10) we have

$$\begin{aligned} \{a\}^A(a) \downarrow &\iff \{a\}_{\lg(\sigma_{a+1})}^{\sigma_{a+1}}(a) \downarrow, \quad \text{and} \\ S(a) &= \sigma_{a+1}(\lg(\sigma_a)). \end{aligned}$$

For (11), note that the mapping $\sigma_a \mapsto \sigma_a^+$ can be computed from either S or A and the mapping $\sigma_a^+ \mapsto \sigma_{a+1}$ can be computed from $\emptyset'$ by the one-quantifier principle. $\square$

In the next section we shall establish a refinement of the Friedberg Jump Theorem to the effect that if additionally S is $\emptyset'$ -r.e., then there exists an r.e. set A such that $A' \equiv_T S$. Note in contrast that in the preceding theorem, if $\emptyset' <_T S$, then $A \not\leq_T \emptyset'$ so in particular A is *not* r.e.

In addition to establishing the Jump Theorem, this proposition provides a relatively easy answer to the following general question: if two sets A and B are related by T -computability in a certain way, what can we conclude about the relationship of A' and B' ? We know, for example, that

$$A \equiv_T B \implies A' \equiv_T B',$$

and

$$A <_T B \implies A' <_T B' \quad \text{or} \quad A' \equiv_T B'.$$

Furthermore, both possibilities of the second implication are realized, since

$$\emptyset <_T \emptyset' \quad \text{and} \quad \emptyset' <_T \emptyset'',$$

while if A is the low simple set constructed in Proposition 8.3.16, then

$$\emptyset <_T A \quad \text{and} \quad \emptyset' \equiv_T A'.$$

Note that this also implies that the jump operator is not injective on T -degrees. What remains is to ask what can happen if A and B are T -incomparable — that is, $A \not\leq_T B$ and $B \not\leq_T A$, which we temporarily denote by $A|B$.

Proposition 8.3.20 *There exist sets A, B, C, D, E and F such that,*

- (i) $A|B$ and $A' \equiv_T B'$;
- (ii) $C|D$ and $C' <_T D'$;
- (iii) $E|F$ and $E'|F'$.

Proof. For (i) let A be $\emptyset'$ and by the preceding proposition choose B such that $B' \equiv_T \emptyset'' \equiv_T B \oplus \emptyset'$. Then clearly $A' \equiv B'$, but

$$\begin{aligned} A \leq_T B &\implies B \equiv_T B \oplus \emptyset' \equiv_T B' \quad \text{and} \\ B \leq_T A &\implies \emptyset' \equiv_T B \oplus \emptyset' \equiv_T \emptyset'', \end{aligned}$$

both of which are contradictions. Thus $A|B$. For (ii) we take C to be $\emptyset'$ and choose D such that $D' \equiv_T \emptyset''' \equiv_T D \oplus \emptyset'$. Now clearly $C' <_T D'$, but as for (i), $C|D$. For (iii), it is not hard to relativize the proof of the Kleene-Post Theorem 8.2.10 to construct two sets S and T such that $\emptyset' \leq_T S, T$ and $S|T$. Then by the Friedberg Jump Theorem there exist sets E and F such that $E' \equiv_T S$ and $F' \equiv_T T$. By the above discussion, also $E|F$. $\square$

The final result of this section will answer the question posed in the discussion following Definition 8.3.8 by establishing that there exists a degree $\mathbf{a} \leq \mathbf{0}'$ which is not an r.e. degree. For any set A ,

$$A \text{ is of r.e. degree} \iff \exists a \exists b \exists c [A = \{a\}^{W_c} \quad \text{and} \quad W_c = \{b\}^A].$$

Hence it will suffice to construct a set $A \leq_T \emptyset'$ which for each a, b and c satisfies the requirement

$$P_{\langle a, b, c \rangle} : \quad A \neq \{a\}^{W_c} \quad \text{or} \quad W_c \neq \{b\}^A$$

and is therefore not of r.e. degree. Obviously such an A cannot be r.e., but we shall show that it can be d.r.e. (Examples 8.3.10) and therefore $A \leq_T \emptyset'$.

Theorem 8.3.21 *There exists a d.r.e. set A which is not of r.e. degree.*

Proof. As noted above, we need for A to satisfy for all a, b and c the requirement $P_{\langle a, b, c \rangle}$. The construction is in stages and is a priority argument; we define two recursive stage enumerations $\langle A_s^+ : s \in \omega \rangle$ and $\langle A_s^- : s \in \omega \rangle$ of r.e. sets A^+ and A^- and take $A := A^+ \sim A^-$. As in Examples 8.3.10, we set $A_s := A_s^+ \sim A_s^-$.

Before beginning the formal proof, it will be useful to sketch the strategy for satisfying a single requirement $P_{\langle a, b, c \rangle}$. The *first strategy* aims to ensure that $A \neq \{a\}^{W_c}$ much as in previous examples by finding at some stage s numbers $x \notin A_s^+ \cup A_s^-$ and i such that $\{a\}_s^{W_{c,s}}(x) \simeq i$ and arranging that

$$x \in A \iff i = 0 \quad \text{and} \quad \{a\}^{W_c}(x) \simeq i.$$

The first of these conditions is relatively easy as before: if $i = 0$, we enumerate $x \in A_{s+1}^+$ and otherwise do nothing. Appropriate choice of x will guarantee that no action taken to satisfy some other requirement will change the status of x by enumerating x into either A_{t+1}^+ or A_{t+1}^- for any $t > s$. Let $u := u(W_{c,s}; a, x, s)$. If $W_{c,s} \upharpoonright u = W_c \upharpoonright u$, then $\{a\}_s^{W_{c,s}}(x) \simeq i$ correctly so the second condition also holds and $P_{\langle a, b, c \rangle}$ is satisfied. However, we need a *second strategy* to deal with the case that $W_{c,s} \upharpoonright u \neq W_c \upharpoonright u$, a circumstance over which we have no control; in this case we aim to ensure that $W_c \neq \{b\}^A$. To prepare for this eventuality, which, of course, cannot be known at stage s , we do not give $P_{\langle a, b, c \rangle}$ attention at stage s unless it is also true that

$$(\forall y < u) \{b\}_s^{A_s}(y) \simeq W_{c,s}(y).$$

Let $v := \max\{u(A_s; b, y, s) : y < u\}$. Then the second strategy aims to ensure that $A_s \upharpoonright v = A \upharpoonright v$, since then for all $y < u$,

$$\{b\}^A(y) \simeq \{b\}_s^{A_s}(y) \simeq W_{c,s}(y),$$

and if $W_{c,s} \upharpoonright u \neq W_c \upharpoonright u$, then for at least one such y , this value differs from $W_c(y)$ and therefore $\{b\}^A \neq W_c$. When eventually $P_{\langle a, b, c \rangle}$ has highest priority, restraints will prevent action taken to satisfy any other requirement from changing the status of any $z < v$. However, at stage s we may have exercised the first strategy by enumerating $x \in A_{s+1}^+$, so if $x < v$, then already $A_{s+1} \upharpoonright v \neq A_s \upharpoonright v$. Now if the need for the second strategy becomes known at some stage $t > s$ when for some $y < u$, $y \in W_{c,t} \sim W_{c,t-1}$, we can enumerate $x \in A_{t+1}^-$, so that $A_s \upharpoonright v = A_{t+1} \upharpoonright v = A \upharpoonright v$, and the second strategy succeeds.

The formal proof essentially consists in mixing the by now standard priority mechanism with this outline. Set the restraint function

$$R_s(k) := \max\{v_l : l < k \text{ and } P_l \text{ is marked at } s\},$$

where if $l = \langle d, e, f \rangle$ and $t_l < s$ is the stage at which P_l became marked,

$$\begin{aligned} u_l &:= u(W_{f,t_l}; d, x_l, t_l) \quad \text{and} \\ v_l &:= \max\{u(A_{t_l}; e, y, t_l) : y < u_l\}. \end{aligned}$$

As usual, these quantities vary with s , and the construction will guarantee that they are defined when P_l is marked at stage s . Let

$$\begin{aligned} \text{Witness}(x, k, s) : \iff & \left[\{a\}_s^{W_{c,s}}(x) \downarrow \wedge x \geq R_s(k) \wedge x \notin A_s^+ \cup A_s^- \right. \\ & \left. \wedge x \in \omega^{[k]} \wedge (\forall y < u) \{b\}_s^{A_s}(y) \simeq W_{c,s}(y) \right], \end{aligned}$$

where $k = \langle a, b, c \rangle$ and $u := u(W_{c,s}; a, x, s)$. Then we say that P_k *needs first attention* at stage s iff $k < s$, P_k is not marked at s and $(\exists x < s) \text{Witness}(x, k, s)$. P_k *needs second attention* at stage s iff it received first attention at some stage $s_0 < s$, has been marked at all t for $s_0 < t \leq s$ and $W_{c,s} \upharpoonright u \neq W_{c,s_0} \upharpoonright u$.

If no P_k needs either first or second attention at stage s , then $A_{s+1}^\pm = A_s^\pm$ and each P_k is marked at stage $s+1$ iff it was marked at stage s . Otherwise, let $k = \langle a, b, c \rangle$ be the smallest index such that P_k needs attention. If this is first attention, we perform the following steps:

- (i) $x_k := \text{least } x < s [\text{Witness}(x, k, s)];$
 $u_k := u(W_{c,s}; a, x_k, s);$
 $v_k := \max\{u(A_s; b, y, s) : y < u_k\};$
 $i_k := \{a\}_s^{W_{c,s}}(x_k);$
- (ii) if $i_k = 0$, then $A_{s+1}^+ = A_s^+ \cup \{x_k\}$, otherwise $A_{s+1}^+ = A_s^+;$
 $A_{s+1}^- = A_s^-.$

If this is second attention, with the values of (i) already defined, we set instead

- (ii) $A_{s+1}^+ = A_s^+$ and $A_{s+1}^- = A_s^- \cup \{x_k\}.$

In either case we have

- (iii) P_k is marked at stage $s+1$; for all $l < k$, P_l is marked at stage $s+1$ iff it was marked at stage s ; for all $l > k$, P_l is unmarked at stage $s+1$ and is said to have been injured.

This completes the construction of A , and we need to verify that for all $k = \langle a, b, c \rangle$,

- (1) P_k gets attention at most finitely often;

- (2) $\lim_{s \rightarrow \infty} R_s(k)$ exists and is finite;
- (3) if both $\{a\}^{W_c}$ and $\{b\}^A$ are total, then P_k is permanently marked at some stage;
- (4) if P_k is permanently marked at stage s and $W_{c,s} \upharpoonright u_k = W_c \upharpoonright u_k$, then P_k is satisfied because $\{a\}^{W_c} \neq A$;
- (5) if P_k is permanently marked at stage s and $W_{c,s} \upharpoonright u_k \neq W_c \upharpoonright u_k$, then P_k is satisfied because $\{b\}^A \neq W_c$;
- (6) P_k is satisfied.

These steps are now a relatively routine adaptation of similar ones in earlier proofs and are left to the reader.

Exercises

Exercise 8.3.22 Prove that for any recursive set B ,

$$K \equiv_m K^\emptyset (= \emptyset') \equiv_m K^B (= B').$$

Exercise 8.3.23 State and prove a relativized form of the Modulus Lemma 8.3.14.

Exercise 8.3.24 Prove that for any sets A and B , A is B' -r.e. iff $A = \liminf_{s \rightarrow \infty} A_s := \{x : \exists s (\forall t \geq s) x \in A_s\}$ for some B -recursive family $\langle A_s : s \in \omega \rangle$.

Exercise 8.3.25 Prove Theorem 8.3.17.

Exercise 8.3.26 Complete the proof of Theorem 8.3.21.

Exercise 8.3.27 Prove that the set A constructed in Exercise 8.2.26 satisfies $A \leq_T B'$.

Exercise 8.3.28 A set $A \subseteq \omega$ is called *n -r.e.* (for $n \in \omega$) iff A has a recursive approximation $\langle A_s : s \in \omega \rangle$ such that $A_0 = \emptyset$ and for all x ,

$$\text{Card}\{s : A_s(x) \neq A_{s+1}(x)\} \leq n.$$

Show that for all A and n ,

- (i) A is 0-r.e. iff $A = \emptyset$;
- (ii) A is $(n+1)$ -r.e. iff $A = B \sim C$ for some r.e. set B and some n -r.e. set C ;
- (iii) if A is n -r.e., then $A \leq_T \emptyset'$;
- (iv) there exists an $(n+1)$ -r.e. set which is not n -r.e. (as a set, compare the next exercise).

Exercise 8.3.29 Strengthen (iv) of the preceding exercise and Theorem 8.3.21 by proving that there exists a 3-r.e. set A which is not of d.r.e. degree — that is, $A \not\equiv_T D$ for any d.r.e. set D .

Hint: Construct r.e. sets A^0 , A^1 and A^2 such that $A := A^0 \sim (A^1 \sim A^2)$ satisfies all of the requirements

$$P_{\langle a, b, c, d \rangle} : \quad A \neq \{a\}^{W_c \sim W_d} \quad \text{or} \quad W_c \sim W_d \neq \{b\}^A.$$

Exercise 8.3.30 Show that for any n and any $(n+1)$ -r.e. set A there exists an n -r.e. set $D \leq_T A$ such that A is D -r.e.

Hint: If $A = B \sim C$ as in Exercise 8.3.28(iii), let f be a recursive function such that $B = \text{Im}(f)$ and take $D := f^{-1}(C)$.

Exercise 8.3.31 Establish the following connections known as **Post's Theorem** between the ideas of this section and the Arithmetical Hierarchy (Definition 5.1.1; see also Exercise 5.1.31). Define recursively $\emptyset^{(0)} := \emptyset$ and $\emptyset^{(n+1)} := \emptyset^{(n)'}.$ Show that for all $n \in \omega$ and any set A ,

- (i) $A \leq_m \emptyset^{(n+1)} \iff A \in \Sigma_{n+1} \iff A \text{ is } \emptyset^{(n)}\text{-r.e.};$
- (ii) $A \leq_T \emptyset^{(n)} \iff A \in \Delta_{n+1}.$

Exercise 8.3.32 Prove that there exists a non-recursive r.e. set A with the following strong lowness property: every A -r.e. set C such that $A \leq_T C$ is of d.r.e. degree.

Hint: Let A be constructed as in Proposition 8.3.16 and

$$A'_s := \{b : \{b\}_s^{A_s}(b) \downarrow\} \quad \text{so} \quad \lim_{s \rightarrow \infty} A'_s = A'.$$

Since C is A -r.e., there exists a recursive function f such that $C = f^{-1}(A')$. Set $E_0 = F_0 := \emptyset$ and for all s ,

$$\begin{aligned} E_{s+1} &:= E_s \cup \{ \langle x, y \rangle : f(x) \in (A'_{s+1} \sim A'_s) \wedge (\forall z < y) \langle x, z \rangle \in F_s \}; \\ F_{s+1} &:= F_s \cup \{ \langle x, y \rangle : f(x) \in (A'_s \sim A'_{s+1}) \wedge \langle x, y \rangle \in E_s \}. \end{aligned}$$

Let $E := \bigcup_{s \in \omega} E_s$ and $F := \bigcup_{s \in \omega} F_s$. Use the fact that a requirement Q_b is injured at most $b+1$ times to show that $C \leq_T E \sim F$ and the assumption $A \leq_T C$ to show that $E \sim F \leq_T C$. (A considerably more difficult proof shows that this result cannot be improved to assert that each A as above is of r.e. degree.)

Exercise 8.3.33 Prove the *Arslanov Completeness Criterion*: for any r.e. set A ,

$$A \text{ is } T\text{-complete} \iff \text{there exists a FPF function } f \leq_T A.$$

(See Exercise 8.2.27.)

Hint: For $(\Leftarrow)$, given $f \leq_T A$, by a variant of the Modulus Lemma 8.3.14 there exists a recursive sequence $\langle f_s : s \in \omega \rangle$ and an A -recursive function m^f such that for all x and $s \geq m^f(x)$, $f_s(x) = f(x)$. Fix a recursive stage enumeration of K and set $m^K(x) \simeq \text{least } s[x \in K_s]$. Use the Recursion Theorem (Exercise 8.1.42) to show that there is a recursive function h such that for all x ,

$$W_{h(x)} = \begin{cases} W_{f_{m^K(x)}(h(x))}, & \text{if } x \in K; \\ \emptyset, & \text{otherwise.} \end{cases}$$

Prove that $m^f(h(x))$ is an A -recursive modulus for the stage enumeration of K so $K \leq_T A$.

8.4 Upper bounds

Our eventual goal for the next three sections is to show that for any pair $D <_T E$ of r.e. sets there is an r.e. set A with $D <_T A <_T E$ and further that given S (subject to minimal necessary conditions), A may be chosen such that $A' \equiv_T S$. Of course, the first part simply asserts that the partial ordering $<$ on the r.e. T -degrees is dense. To realize this goal, given $D <_T E$ and S , there are six distinct conditions which such an A must satisfy, each of which requires a separate strategy:

$$\begin{array}{lll} (1) D \leq_T A & (3) A \leq_T E & (5) S \leq_T A' \\ (2) A \not\leq_T D & (4) E \not\leq_T A & (6) A' \leq_T S. \end{array}$$

(1), (2) and (5) are *positive* conditions which assert that A is complex enough; these will be satisfied by strategies which involve enumeration of certain numbers into A . (3), (4) and (6) are *negative* conditions which assert that A is not too complex; these will be satisfied by imposing restraints on enumeration into A . With considerably more effort (cf. [Soa; Exercises VIII.4.7–10]) we could improve the density result to an extension of Theorem 8.2.17: given r.e. T -degrees $\mathbf{d} < \mathbf{e}$, every countable partial ordering is embeddable in the r.e. T -degrees between $\mathbf{d}$ and $\mathbf{e}$.

In this section we consider the special case $D = \emptyset$, so (1) is always true and to satisfy (2) it suffices to make A simple, hence non-recursive. Furthermore, we postpone consideration of conditions (5) and (6) involving the jump. Thus we focus on (3) and (4), the relationship between A and E .

The **permitting method** illustrated in the following two theorems provides a relatively simple way to impose upper bounds on the constructions of Section 8.2 and other finite injury constructions. However, it is generally not sufficient for use with the infinite injury constructions of Sections 8.5–8.6.

Theorem 8.4.1 *For any non-recursive r.e. set E , there exists a simple set $A \leq_T E$.*

Proof. Fix a non-recursive r.e. set E ; the conditions to be satisfied are

$$(2) A \not\leq_T \emptyset \qquad (3) A \leq_T E.$$

We follow the pattern of Theorem 8.1.20 with a modification to achieve (3), and we aim to ensure that for each $a \in \omega$ the following requirements are satisfied:

$$\begin{aligned} P_a : W_a \text{ infinite} &\implies A \cap W_a \neq \emptyset \\ N_a : A \cap \{x : x < 2a\} &\text{ has at most } a \text{ elements.} \end{aligned}$$

Fix a recursive stage enumeration $\langle E_s : s \in \omega \rangle$ of E . As before set $R(a) := 2a$. We say that P_a **needs attention** at stage s iff P_a is not marked at s and

$$(\exists x < s)[x \in W_{a,s} \wedge x \geq R(a) \wedge (\exists y < x) y \in E_{s+1} \sim E_s].$$

We say that the appearance of $y < x$ in E_{s+1} **permits** the enumeration of x into A_{s+1} . If no P_a needs attention at stage s , then $A_{s+1} = A_s$ and each P_a is marked at stage $s+1$ iff it was marked at stage s . Otherwise, choose the smallest index a such that P_a needs attention at stage s and let it **get attention** by performing the following steps:

- (i) $x_a := \text{least } x < s[x \in W_{a,s} \wedge x \geq R(a) \wedge (\exists y < x) y \in E_{s+1} \sim E_s]$;
- (ii) $A_{s+1} := A_s \cup \{x_a\}$;
- (iii) P_a is marked at stage $s+1$; for all $b \neq a$, P_b is marked at stage $s+1$ iff it is marked at stage s .

This completes the construction of A , and we proceed to verify that for all a ,

- (2.1) P_a gets attention at most once;
- (2.2) if W_a is infinite, then P_a is marked at some stage;
- (2.3) if P_a is marked at any stage, then P_a is satisfied;
- (2) P_a and N_a are satisfied (so A is simple and hence non-recursive);
- (3) $A \leq_T E$.

(2.1) and (2.3) follow exactly as before (in the second proof of Theorem 8.1.20), and (2) is immediate from (2.1)–(2.3). However, (2.2) requires a new proof, since we have made it more difficult for P_a to get attention. Let s_a be least such that for no $b < a$ does P_b get attention at any stage $s \geq s_a$. Suppose that W_a is infinite but P_a never becomes marked; by the definition of s_a , this implies that for no $s \geq s_a$ does P_a need attention. At all sufficiently large stages $s > s_a$ there exist $x \in W_{a,s}$ such that $x \geq R(a)$; if P_a does not need attention at s it is because there is no $y < x$ in $E_{s+1} \sim E_s$ which permits x . Since for $s < t$ we have $W_{a,s} \subseteq W_{a,t}$, it follows that x is also not permitted at any stage $t \geq s$; that is for all x and $s > s_a$,

$$x \in W_{a,s} \wedge x \geq R(a) \implies (\forall t \geq s) \neg (\exists y < x) y \in E_{t+1} \sim E_t.$$

In other words,

$$x \in W_{a,s} \wedge x \geq R(a) \implies E_s \upharpoonright x = E \upharpoonright x.$$

Hence, if we define a recursive function g by

$$g(y) = \text{least } s \ (\exists x < s) [x \in W_{a,s} \wedge x \geq R(a) \wedge y < x],$$

then for all y , $y \in E \iff y \in E_{g(y)}$, so by stage bounding E is recursive, contrary to hypothesis.

For (3), let h be the function $h(x) = \text{least } s [E_s \upharpoonright x = E \upharpoonright x]$; clearly h is E -recursive. For any $s \geq h(x)$, no $y < x$ appears in $E_{s+1} \sim E_s$, so by the construction, $x \notin A_{s+1} \sim A_s$. Hence $x \in A \iff x \in A_{h(x)}$, from which it follows, again by stage bounding, that also $A \leq_T E$. $\square$

The permitting method may be combined with the priority technique of the Friedberg-Mučnik Theorem 8.2.16 to prove

Theorem 8.4.2 *For any non-recursive r.e. set E , there exist r.e. sets $A \leq_T E$ and $B \leq_T E$ such that neither is Turing reducible to the other. In particular $\emptyset <_T A, B <_T E$.*

Proof. Left as an exercise. $\square$

However, we shall take a different and more general approach to satisfying condition (4) that $E \not\leq_T A$. Note that here, although this condition is superficially the same as the condition $B \not\leq_T A$ of Theorem 8.2.15, in fact the situation is radically different; there B was a set that we were constructing so that we could put elements into B to further our goals, whereas here E is a given set over which we have no control, so the strategy must be a negative one relying on restraints to exclude appropriate elements from A . We first introduce some functions which measure the extent to which an index b witnesses that $E \leq_T A$.

Definition 8.4.3 For any sets A and E , the *length of agreement* function is

$$\ell^{AE}(b) := \begin{cases} \infty, & \text{if } \{b\}^A = E; \\ \text{least } y \left[\{b\}^A(y) \not\simeq E(y) \right], & \text{otherwise.} \end{cases}$$

Note that $\{b\}^A(y) \not\simeq E(y)$ may be true either because $\{b\}^A(y) \uparrow$ or because $\{b\}^A(y) \simeq i \neq E(y)$. Then, $E \not\leq_T A$ iff for all b , $\ell^{AE}(b) < \infty$. The key to many of the proofs below is the use of an appropriate recursive approximation to $\ell^{AE}(b)$. The simplest one is

Definition 8.4.4 For any stage enumerations

$$\langle A_s : s \in \omega \rangle \quad \text{and} \quad \langle E_s : s \in \omega \rangle$$

of sets A and E and any b and s ,

$$\ell^{AE}(b, s) := \text{least } y \left[\{b\}_s^{A_s}(y) \not\simeq E_s(y) \right].$$

For any y , we say that $\ell^{AE}(b, s) \geq y$ **correctly** iff $\ell^{AE}(b, s) \geq y$, and for all $z < y$,

$$\{b\}_s^{A_s}(z) \downarrow \text{ correctly} \quad \text{and} \quad E_s(z) = E(z)$$

(see Definition 8.2.13).

This approximation has some natural properties, but we shall see that it is not ideal for all purposes. First we have

Correctness Lemma for Length 8.4.5 (CLL) For any recursive stage enumerations $\langle A_s : s \in \omega \rangle$ and $\langle E_s : s \in \omega \rangle$ of r.e. sets A and E and all b , s and y ,

- (i) if $y \leq \ell^{AE}(b)$, then there exists s such that $\ell^{AE}(b, s) \geq y$ correctly;
- (ii) if $\ell^{AE}(b, s) \geq y$ correctly, then $y \leq \ell^{AE}(b)$, and for all $t \geq s$, $\ell^{AE}(b, t) \geq y$ correctly;
- (iii) if $\ell^{AE}(b) \geq y$, $\ell^{AE}(b, s) \geq y$ and for all $z < y$, $\{b\}_s^{A_s}(z) \downarrow$ correctly, then $\ell^{AE}(b, s) \geq y$ correctly.

Proof. For (i), since $y \leq \ell^{AE}(b)$, for all $z < y$, $\{b\}^A(z) \downarrow$, so by the Correctness Lemma 8.2.14 there exists t_z such that $\{b\}_s^{A_s}(z) \downarrow$ correctly for all $s \geq t_z$. Now if we choose $s \geq \max\{t_z : z < y\}$ large enough so that for all $z < y$ also $E_s(z) = E(z)$, then for all $z < y$,

$$\{b\}_s^{A_s}(z) \simeq \{b\}^A(z) \simeq E(z) = E_s(z),$$

so $\ell^{AE}(b, s) \geq y$ correctly. For (ii), if $\ell^{AE}(b, s) \geq y$ correctly, then for all $t \geq s$ and $z < y$, by the Correctness Lemma,

$$\{b\}^A(z) \simeq \{b\}_t^{A_t}(z) \simeq \{b\}_s^{A_s}(z) \simeq E_s(z) = E(z).$$

Hence $y \leq \ell^{AE}(b)$. Furthermore, since $E_s \subseteq E_t \subseteq E$, also $E_t(z) = E(z)$, so also $\ell^{AE}(b, t) \geq y$, correctly by the Correctness Lemma. For (iii), under all of the stated hypotheses, for all $z < y$,

$$E_s(z) = \{b\}_s^{A_s}(z) \simeq \{b\}^A(z) \simeq E(z),$$

and it follows that $\ell^{AE}(b, s) \geq y$ correctly. $\square$

Corollary 8.4.6 *For any stage enumerations*

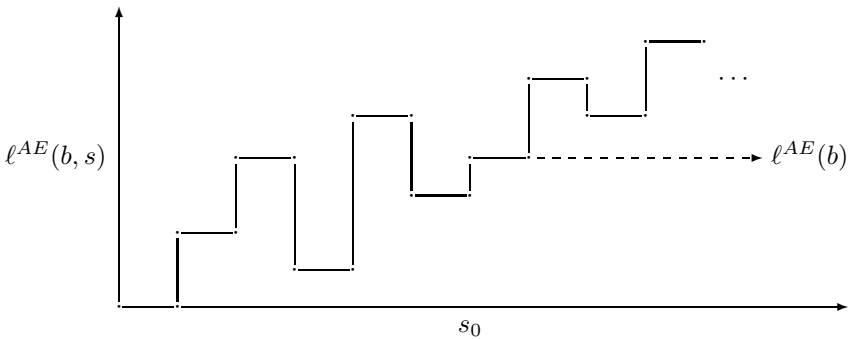
$$\langle A_s : s \in \omega \rangle \quad \text{and} \quad \langle E_s : s \in \omega \rangle$$

of sets A and E and any b ,

- (i) *if $\ell^{AE}(b) = \infty$, then $\lim_{s \rightarrow \infty} \ell^{AE}(b, s) = \infty$;*
- (ii) *if $\ell^{AE}(b) < \infty$, then for all sufficiently large s , $\ell^{AE}(b, s) \geq \ell^{AE}(b)$.*

Proof. Immediate from CLL(i) and (ii). $\square$

The following schematic graph indicates a typical pattern of behavior for $\ell^{AE}(b, s)$ when $\ell^{AE}(b) < \infty$. The value of $\ell^{AE}(b, s)$ rises and falls according as correct and incorrect information appears in the approximations A_s and E_s . At stage s_0 , correct computations and values for $y < \ell^{AE}(b)$ are in place, and for $s > s_0$, $\ell^{AE}(b, s)$ may vary above $\ell^{AE}(b)$ due to estimates which depend on information about A or E which is ultimately incorrect.



Definition 8.4.7 For any recursive stage enumerations $\langle A_s : s \in \omega \rangle$ and $\langle E_s : s \in \omega \rangle$ of r.e. sets A and E , a **correctness function** for an index b is a (partial) function h such that for all $y \leq \ell^{AE}(b)$, $h(y)$ is defined and $\ell^{AE}(b, h(y)) \geq y$ correctly.

Remark 8.4.8

- (i) It follows from CLL(iii) that there is always a partial A -recursive correctness function h : for every b

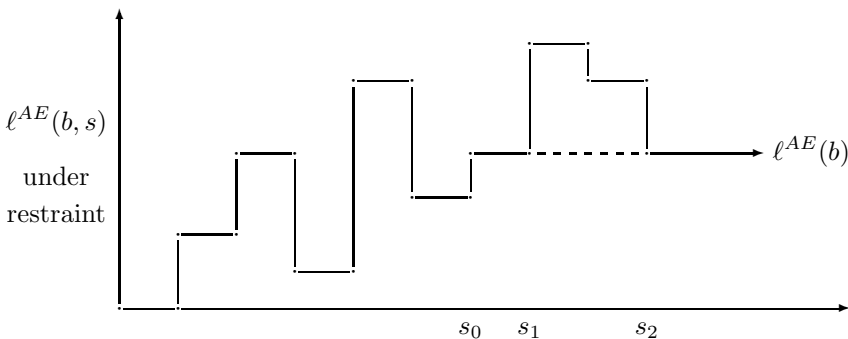
$$h(y) \simeq \text{least } s \left[\ell^{AE}(b, s) \geq y \quad \text{and} \quad (\forall z < y) \{b\}_s^{A_s}(z) \downarrow \text{ correctly} \right].$$

Furthermore, if $\{b\}^A = E$ (so $\ell^{AE}(b) = \infty$), then for any correctness function h for b , we have for all y , $E(y) = E_{h(y+1)}(y)$, and hence by stage bounding E is h -recursive. Of course, combining these two observations does not give any new information, but a key idea of the proof below is to arrange the construction so that for each b , if $\{b\}^A = E$, then there is a *recursive* correctness function and thus E is recursive. Since this will be contrary to hypothesis, it follows that no such b exist and hence $E \not\leq_T A$.

- (ii) Generally part (ii) of the corollary cannot be strengthened to say that also when $\ell^{AE}(b) < \infty$ we have

$$\lim_{s \rightarrow \infty} \ell^{AE}(b, s) = \ell^{AE}(b),$$

or even that for infinitely many s , $\ell^{AE}(b, s) = \ell^{AE}(b)$. Another key point in the proofs below is that these will be true of the set A constructed there. The restraints will ensure that if at some large enough stage s_1 we have $\ell^{AE}(b, s_1) > \ell^{AE}(b)$, then the computation $\{b\}_{s_1}^{A_{s_1}}(\ell^{AE}(b))$ will be correct, so by the definition of $\ell^{AE}(b)$ necessarily $E_{s_1}(\ell^{AE}(b))$ is incorrect and at some later stage s_2 , $\ell^{AE}(b)$ will enter E so that $\ell^{AE}(b, s)$ can never later be larger than $\ell^{AE}(b)$.



Our first and simplest application of this technique is

Theorem 8.4.9 *For any non-recursive r.e. set E , there exists a simple set A such that $E \not\leq_T A$.*

Proof. Given a non-recursive r.e. set E , we want to construct A such that

$$(2) A \not\leq_T \emptyset \quad (4) E \not\leq_T A.$$

It will suffice to ensure that for each a and b ,

$$\begin{aligned} P_a : \quad & W_a \text{ infinite} \implies A \cap W_a \neq \emptyset \\ Q_b : \quad & \{b\}^A \neq E \\ N_a : \quad & \{x : x < 2a\} \cap A \text{ has at most } a \text{ elements.} \end{aligned}$$

Fix a recursive stage enumeration $\langle E_s : s \in \omega \rangle$ of E . As usual we shall define a recursive stage enumeration $\langle A_s : s \in \omega \rangle$ and with it the values of the length of agreement function $\ell^{AE}(b, s)$ (which we now denote simply by $\ell(b, s)$). Given A_s , set the restraint functions

$$\begin{aligned} r(b, s) &:= \max\{u(A_s; b, z, s) : z \leq \ell(b, s)\}; \\ R_s(a) &:= \max\{r(b, s) : b < a\} \cup \{2a\}. \end{aligned}$$

We say as usual that P_a needs attention at stage s iff $a < s$, P_a is not marked at s and

$$(\exists x < s)[x \in W_{a,s} \text{ and } x \geq R_s(a)].$$

Note that the requirements Q_b , being treated by a purely negative strategy, never get attention and are never marked. If no P_a needs attention at stage s , then $A_{s+1} = A_s$ and each P_a is marked at stage $s+1$ iff it was marked at stage s . Otherwise, choose the least a such that P_a needs attention and give it attention by performing the following steps:

- (i) $x_a := \text{least } x < s [x \in W_{a,s} \text{ and } x \geq R_s(a)]$;
- (ii) $A_{s+1} := A_s \cup \{x_a\}$;
- (iii) P_a is marked at stage $s+1$; for $b \neq a$, P_b is marked at stage $s+1$ iff it was marked at stage s .

Although we do not explicitly describe any injury to Q_b with $a \leq b$, the values of $\ell(b, s+1)$ and $r(b, s+1)$ may differ from $\ell(b, s)$ and $r(b, s)$ because of this action as well as because of differences of E_{s+1} from E_s . This completes the construction of A .

Clearly each P_a gets attention at most once; hence there exists a stage t_b such that for no $a \leq b$ does P_a get attention at any stage $s \geq t_b$. We call such s ***b-safe*** because restraints $r(b, s)$ imposed for the sake of Q_b are respected at s .

We shall show that for each b ,

$$\lim_{s \rightarrow \infty} \ell(b, s) = \ell(b) < \infty \quad \text{and} \quad \lim_{s \rightarrow \infty} r(b, s) < \infty.$$

The first guarantees that Q_b is satisfied, and the second for all $b < a$ ensures that $R_s(a)$ is bounded and thus P_a is eventually satisfied. The rough idea is as follows. Suppose that $\{b\}^A = E$, so $\ell(b) = \infty$. Then $\ell(b, s)$ increases without bound, monotonically for $s \geq t_b$. Furthermore, if $\ell(b, s) > z$ at a stage $s > t_b$, then the computation $\{b\}_s^{A_s}(z) \simeq E_s(z)$ is correct so has value $\{b\}^A(z) \simeq E(z)$. Hence $h(z) := \text{least } s [\ell(b, s) > z]$ is a recursive correctness function and we conclude that E is recursive, contrary to hypothesis. Thus $\ell(b) < \infty$ and for $s \geq t_b$, $\ell(b, s) \leq \ell(b)$ and $r(b, s) \leq \max\{u(A_s; b, z, s) : z \leq \ell(b)\}$ and therefore reaches a finite limiting value.

To make this more precise, we first establish the following consequences of the construction. We say that $\ell(b, s) \geq y$ **very correctly** iff $\ell(b, s) \geq y$ correctly and if $\{b\}_s^{A_s}(y) \downarrow$, then also $\{b\}_s^{A_s}(y) \downarrow$ correctly. Then for all a , b , s and y ,

- (A) if s is b -safe, $y \leq \ell(b)$ and $\ell(b, s) \geq y$, then $\ell(b, s) \geq y$ very correctly;
- (B1) $\lim_{s \rightarrow \infty} \ell(b, s) = \ell(b)$;
- (B2) if $\ell(b) < \infty$ for all $b < a$, then $\lim_{s \rightarrow \infty} R_s(a) =: R(a)$ exists and is finite.

For (A), let s be a b -safe stage; we prove by induction on $y \leq \ell(b)$ that

$$\ell(b, s) \geq y \implies \ell(b, s) \geq y \text{ very correctly.}$$

Assume as induction hypothesis that this holds for y and suppose that $y+1 \leq \ell(b)$ and $\ell(b, s) \geq y+1$; hence $\ell(b, s) \geq y$ very correctly. (The basis case $y = 0$ is identical without any use of an induction hypothesis.) By CLL(iii) $\ell(b, s) \geq y+1$ correctly, and it suffices to show that if $\{b\}_s^{A_s}(y+1) \downarrow$, then it converges correctly — that is, with $u := u(A_s; b, y+1, s)$, for all $t \geq s$, $A_t \upharpoonright u = A_s \upharpoonright u$. This is immediate for $t = s$, so assume as induction hypothesis that it holds for t . By the construction, either $A_{t+1} = A_t$ or for some a , P_a gets attention at stage t and $A_{t+1} = A_t \cup \{x_a\}$. In the latter case we have

$$\begin{aligned} x_a &\geq R_t(a) \geq r(b, t) \quad \text{since } a > b \text{ because } s \text{ is } b\text{-safe} \\ &\geq u(A_t; b, y+1, t) \quad \text{because by CLL(ii), } \ell(b, t) \geq y+1 \\ &\geq u \quad \text{by the Correctness Lemma.} \end{aligned}$$

Hence, in either case $A_{t+1} \upharpoonright u = A_t \upharpoonright u = A_s \upharpoonright u$ as desired.

(B1) is immediate from the Corollary to the CLL in case $\ell(b) = \infty$. If $\ell(b) < \infty$, then by the same Corollary, for all sufficiently large s , $\ell(b, s) \geq \ell(b)$. Choose t large enough that for all $z \leq \ell(b)$, $E_t(z) = E(z)$. Suppose towards a contradiction that for some b -safe stage $s \geq t$, $\ell(b, s) \geq \ell(b) + 1$. By (A), $\ell(b, s) \geq \ell(b)$ very correctly, and since $\{b\}_s^{A_s}(\ell(b)) \downarrow$, this computation is correct. But then $\ell(b, s) \geq \ell(b) + 1$ correctly, so by CLL(ii), $\ell(b) \geq \ell(b) + 1$, a contradiction.

Now (B2) follows, since for sufficiently large s , if $\ell(b) < \infty$,

$$\begin{aligned} r(b, s) &= \max\{u(A_s; b, z, s) : z \leq \ell(b, s)\} \\ &= \max\{u(A; b, z) : z \leq \ell(b)\} =: r(b). \end{aligned}$$

Hence under the hypothesis of (B2), for sufficiently large s , $R_s(a)$ has the constant value

$$R(a) := \max\{r(b) : b < a\}.$$

Now are in a position to show

- (4) for all b , Q_b is satisfied; hence $E \not\leq_T A$;
- (2) for all a , P_a and N_a are satisfied; hence A is simple and therefore non-recursive.

For (4), suppose to the contrary that for some b , $\{b\}^A = E$. By (B1),

$$\ell(b) = \infty = \lim_{s \rightarrow \infty} \ell(b, s),$$

so the function

$$h(y) := \text{least } s \text{ [} s \text{ is } b\text{-safe and } \ell(b, s) \geq y \text{]}$$

is total. Furthermore, h is recursive, and by (A), for all y , $\ell(b, h(y)) \geq y$ correctly — that is, h is a recursive correctness function for b — and thus $E_{h(y+1)}(y) = E(y)$, so by stage bounding, E is recursive, contrary to hypothesis.

In particular, for every a , the hypothesis of (B2) is satisfied, so

$$\lim_{s \rightarrow \infty} R_s(a) =: R(a) < \infty.$$

Hence if W_a is infinite, then some element of W_a greater than $R(a)$ will eventually be enumerated into A ; thus (2) holds. $\square$

We can combine the methods of the preceding two proofs to establish the **lower density** of the r.e. degrees.

Theorem 8.4.10 *For any non-recursive r.e. set E , there exists a simple set A such that $A <_T E$; hence $\emptyset <_T A <_T E$.*

Proof. Given any non-recursive r.e. set E , we construct A to satisfy the conditions

$$\begin{array}{ll} (3) & A \leq_T E \\ (2) & A \not\leq_T \emptyset \qquad (4) & E \not\leq_T A. \end{array}$$

The requirements P_a , Q_b and N_a and the length and restraint functions are exactly as in the preceding proof. The only change in the construction is that the condition for requirement P_a needing attention becomes

$$(\exists x < s)[x \in W_{a,s} \text{ and } x \geq R_s(a) \text{ and } (\exists y < x) y \in E_{s+1} \sim E_s].$$

The proof of (4) is exactly as in the preceding proof, while the proofs of (2) and (3) are like the corresponding parts of the proof of Theorem 8.4.1. We leave the details to an exercise. $\square$

Exercises

Exercise 8.4.11 Prove that for any sets $D \leq_T E$ there exists a simple set A such that $D \leq_T A \leq_T E$. Conclude that every r.e. T -degree contains a simple set and in particular that there exist simple sets which are T -complete.

Exercise 8.4.12 Prove Theorem 8.4.2.

Exercise 8.4.13 Use this result to improve Theorem 8.2.17 to assert the embeddability of every countable partial ordering in the r.e. T -degrees below any given r.e. degree.

Exercise 8.4.14 Verify the claim made in Remark 8.4.8(ii) that there exist recursive stage enumerations $\langle A_s : s \in \omega \rangle$ and $\langle E_s : s \in \omega \rangle$ of r.e. sets A and E and an index b such that $\ell^{AE}(b) < \infty$, but for all sufficiently large s , $\ell^{AE}(b, s) > \ell^{AE}(b)$.

Exercise 8.4.15 Complete the proof of Theorem 8.4.10.

8.5 Jumps of r.e. sets

The Friedberg Jump Theorem 8.3.18 tells us that if $\emptyset' \leq_T S$, then there exists a set A such that $A' \equiv_T S$. We ask now for which S could we find such an A which is r.e.? If A is r.e., then $A \leq_T \emptyset'$, so since A' is A -r.e., it is also $\emptyset'$ -r.e. Hence the most we could expect is that for any S ,

$$\emptyset' \leq_T S \text{ and } S \text{ is } \emptyset'\text{-r.e.} \implies \text{exists an r.e. set } A [A' \equiv_T S].$$

We shall prove this below in the more general form: for any r.e. set E ,

$$\emptyset' \leq_T S \text{ and } S \text{ is } E\text{-r.e.} \implies \text{exists an r.e. set } A <_T E [A' \equiv_T S],$$

and finally in [Section 8.6](#), for any r.e. sets $D <_T E$,

$$D' \leq_T S \text{ and } S \text{ is } E\text{-r.e.} \implies$$

$$\text{exists an r.e. set } A [D <_T A <_T E \text{ and } A' \equiv_T S].$$

In preparation we need to analyze more closely the process of approximating computations $\{b\}^A(x)$ from an r.e. set A by computations $\{b\}_s^{A_s}(x)$ based on a recursive stage enumeration $\langle A_s : s \in \omega \rangle$ of A . First, by the Correctness Lemma,

$$\{b\}^A(x) \downarrow \implies \exists s \left[\{b\}_s^{A_s}(x) \downarrow \text{ correctly} \right] \implies \exists s (\forall t \geq s) \{b\}_t^{A_t}(x) \downarrow.$$

Furthermore, the approximations $\{b\}_t^{A_t}(x)$ for sufficiently large t all have the same value and use as $\{b\}^A(x)$. If $\{b\}^A(x) \uparrow$, however, the sequence of approximations can have a more complex behavior; for infinitely many s , $\{b\}_s^{A_s}(x)$ may converge incorrectly — that is, with use u such that $A_s \upharpoonright u \neq A \upharpoonright u$. Indeed, it is possible to construct an example (cf. Exercise 8.4.14) for which

$$\exists s (\forall t \geq s) \{b\}_t^{A_t}(x) \downarrow \text{ but } \{b\}^A(x) \uparrow.$$

We shall introduce a modified notion of computation which does not have this anomaly.

Definition 8.5.1 For any stage enumeration $\langle A_s : s \in \omega \rangle$ of a set A , set

$$p_s := \begin{cases} \text{least } p [p \in A_s \sim A_{s-1}], & \text{if } A_s \sim A_{s-1} \neq \emptyset; \\ \max A_s \cup \{s\}, & \text{otherwise;} \end{cases}$$

$$\widehat{\{b\}}_s^{A_s}(x) : \simeq \begin{cases} \{b\}_s^{A_s}(x), & \text{if } u(A_s; b, x, s) \leq p_s; \\ \uparrow, & \text{otherwise.} \end{cases}$$

We say that $\widehat{\{b\}}_s^{A_s}(x) \downarrow$ **correctly** iff $\widehat{\{b\}}_s^{A_s}(x) \downarrow$ and the computation is correct.

Part of the idea is that when a **hat computation** is destroyed because A changes below the use, it is undefined for at least one stage.

Lemma 8.5.2 *If $\widehat{\{b\}}_{s-1}^{A_{s-1}}(x) \downarrow$ with use $u = u(A_{s-1}; b, s-1, x)$ and $A_{s-1} \upharpoonright u \neq A_s \upharpoonright u$, then $\widehat{\{b\}}_s^{A_s}(x) \uparrow$.*

Proof. Under the assumptions clearly $p_s < u$. If $\widehat{\{b\}}_s^{A_s}(x) \downarrow$, then $u(A_s; b, x, s) \leq p_s$, so by the definition of p_s , $A_{s-1} \upharpoonright u(A_s; b, s, x) = A_s \upharpoonright u(A_s; b, s, x)$. But then $u(A_s; b, s, x)$ would also be the use of $\widehat{\{b\}}_{s-1}^{A_{s-1}}(x)$, that is, u , a contradiction. $\square$

Theorem 8.5.3 *For any stage enumeration $\langle A_s : s \in \omega \rangle$ of a set A and any b and x ,*

$$\{b\}^A(x) \downarrow \iff \exists s (\forall t \geq s) \widehat{\{b\}}_t^{A_t}(x) \downarrow.$$

Proof. If $\{b\}^A(x) \downarrow$, then by the Correctness Lemma there exists s such that $\{b\}_s^{A_s}(x) \downarrow$ correctly with use $u := u(A_s; b, s, x)$. For any $t \geq s$, $A_t \upharpoonright u = A \upharpoonright u$, so clearly $p_t \geq u$ and thus $\widehat{\{b\}}_t^{A_t}(x) \downarrow$. Conversely, if for all $t \geq s$, $\widehat{\{b\}}_t^{A_t}(x) \downarrow$, then $\{b\}_s^{A_s}(x) \downarrow$ with use $u := u(A_s; b, s, x)$, and for all $t \geq s$, $p_t \geq u$, so $A_s \upharpoonright u = A \upharpoonright u$, $\{b\}_t^{A_t}(x) \downarrow$ correctly, and thus $\{b\}^A(x) \downarrow$. $\square$

These hat computations have another extremely useful property.

Definition 8.5.4 s is a **true stage** in the stage enumeration $\langle A_s : s \in \omega \rangle$ of a set A iff $A_s \upharpoonright p_s = A \upharpoonright p_s$. T^A denotes the set of true stages.

Note that T^A depends on the stage enumeration of A ; we shall often use this and other such notations with the convention that some fixed enumeration is intended. Generally A will be an r.e. set and the stage enumeration is recursive.

Lemma 8.5.5 *For any infinite r.e. set A , T^A is infinite and $A \equiv_T \mathsf{T}^A$.*

Proof. For any s , let $q_s = \text{least } x [x \in A \sim A_s]$. Then for some $t > s$, $p_t = q_s$ and thus $t \in \mathsf{T}^A$. Clearly $\mathsf{T}^A \leq_T A$, but also $A \leq_T \mathsf{T}^A$, since if $h(x) = \text{least } s [s \in \mathsf{T}^A \wedge p_s > x]$, then h is T^A -recursive and for each x , $x \in A \iff x \in A_{h(x)}$. $\square$

Proposition 8.5.6 *For any stage enumeration $\langle A_s : s \in \omega \rangle$ of a set A and any b ,*

- (i) *if $\widehat{\{b\}}_s^{A_s}(x) \downarrow$ and $s \in \mathsf{T}^A$, then $\{b\}_s^{A_s}(x) \downarrow$ correctly;*
- (ii) *if $\{b\}_s^{A_s}(x) \downarrow$ correctly, then for all $t > s$, $\widehat{\{b\}}_t^{A_t}(x) \downarrow$ correctly.*

Proof. Under the hypotheses of (i), $u(A_s; b, x, s) \leq p_s$ and $A_s \upharpoonright p_s = A \upharpoonright p_s$. Hence also $A_s \upharpoonright u(A_s; b, x, s) = A \upharpoonright u(A_s; b, x, s)$. For (ii), note that if $\{b\}_s^{A_s}(x) \downarrow$ correctly, then for any $t > s$, $p_t \geq u(A_s; b, x, s)$. $\square$

Corollary 8.5.7 *For any stage enumeration of a set A and any b and x ,*

$$\{b\}^A(x) \downarrow \iff \exists s \left[s \in \mathbb{T}^A \wedge \widehat{\{b\}}_s^{A_s}(x) \downarrow \right]. \quad \square$$

It is also natural to define

$$\hat{u}(A_s; b, x, s) := \begin{cases} u(A_s; b, x, s), & \text{if } \widehat{\{b\}}_x^{A_x}(s) \downarrow; \\ 0, & \text{otherwise.} \end{cases}$$

Corollary 8.5.8 *For any r.e. set A and any b and x ,*

- (i) *if $\{b\}^A(x) \downarrow$, then $\lim_{s \rightarrow \infty} \hat{u}(A_s; b, x, s) = u(A; b, x)$;*
- (ii) *if $\{b\}^A(x) \uparrow$, then $\lim_{s \in \mathbb{T}^A} \hat{u}(A_s; b, x, s) = 0$; in particular, $\hat{u}(A_s; b, x, s) = 0$ for arbitrarily large values of s . $\square$*

The last property is sometimes described by writing

$$\liminf_{s \rightarrow \infty} \hat{u}(A_s; b, x, s) = 0.$$

In either case of the corollary, at stages t either before the limit is reached in (i) or between true stages in (ii), it may happen that $\widehat{\{b\}}_t^{A_t}(x) \downarrow$ with a value of $\hat{u}(A_t; b, x, t)$ which is unrelated to the limit or 0 and that this computation is later destroyed. Indeed, in case (ii), this may happen infinitely often, potentially with unbounded use, which we could describe by writing $\limsup_{s \rightarrow \infty} \hat{u}(A_s; b, x, s) = \infty$.

We turn now to the problem of constructing an r.e. set A with prescribed jump S . The first task is to build into A enough information so that $S \leq_T A'$. We say that A is **columnwise recursive** iff for all a , the a -th column $A^{[a]}$ of A is recursive. Note that a columnwise recursive set certainly need not be recursive — see the exercises.

Definition 8.5.9 A set F **effectively jump codes** a set S iff there exists an S -recursive function g such that

$$a \in S \implies F^{[a]} = \{ \langle a, y \rangle : y < g(a) \} \quad \text{and} \quad a \notin S \implies F^{[a]} = \omega^{[a]}.$$

Note that such an F is columnwise recursive; the following explains the terminology.

Proposition 8.5.10 *For any r.e. set E and any E -r.e. set S such that $E \leq_T S$, there exists an r.e. set $F \leq_T E$ which effectively jump codes S , and for any such F , $S \leq_T F'$.*

Proof. Since S is E -r.e., there exists an index e such that $S = W_e^E$. Hence, for all a and some recursive stage enumeration $\langle E_s : s \in \omega \rangle$ of E ,

$$a \in S \iff \{e\}^E(a) \downarrow \iff \exists s (\forall t \geq s) \widehat{\{e\}}_t^{E_t}(a) \downarrow.$$

Let

$$F := \left\{ \langle a, y \rangle : (\exists t \geq y) \widehat{\{e\}}_t^{E_t}(a) \uparrow \right\};$$

$$g(a) := \begin{cases} 0, & \text{if } a \notin S; \\ \text{least } s \{e\}_s^{E_s}(a) \downarrow \text{ correctly,} & \text{if } a \in S. \end{cases}$$

The displayed implications are immediate, and $F \leq_T E$ because

$$\langle a, y \rangle \in F \iff \widehat{\{e\}}_y^{E_y}(a) \uparrow \quad \text{or} \quad \widehat{\{e\}}_y^{E_y}(a) \downarrow \text{ incorrectly,}$$

and $S(a) = 1 - \lim_{y \rightarrow \infty} F(\langle a, y \rangle)$, which is F' -recursive by the Limit Lemma 8.3.12. $\square$

Definition 8.5.11 A is **thick** in F iff $A \subseteq F$ and for all a , $F^{[a]} \sim A^{[a]}$ is finite.

Corollary 8.5.12 *If F effectively jump codes S and A is thick in F , then $S \leq_T A'$.*

Proof. If A is thick in F , then for each a ,

$$\lim_{y \rightarrow \infty} A(\langle a, y \rangle) = \lim_{y \rightarrow \infty} F(\langle a, y \rangle). \quad \square$$

Thus to ensure that $S \leq_T A'$, it suffices to arrange that for all a the requirement

$$P_a : \quad F^{[a]} \sim A^{[a]} \text{ is finite,}$$

is satisfied. The strategy for ensuring $A' \leq_T S$ is a bit more complicated. By Corollary 8.5.7, for all b ,

$$b \in A' \iff \exists s \left[s \in \mathbb{T}^A \wedge \widehat{\{b\}}_s^{A_s}(b) \downarrow \right].$$

Of course, this is of no direct help, since the right-hand side is $\Sigma_1[A]$, but not in general S -recursive. By subjecting P_a to the restraint function

$$\hat{R}_s(a) = \max\{\hat{u}(A_s; b, b, s) : b < a\},$$

we shall arrange that any injuries to a computation $\widehat{\{b\}_s^{A_s}}(b)$ will be due to the enumeration into A of elements of the form $\langle a, y \rangle$ with $a \leq b$ — that is, for the sake of a requirement P_a of higher priority. Hence, if s is a stage which is *partially true* in the sense that s belongs to the set

$$\mathsf{T}_b^A := \left\{ s : A_s^{[\leq b]} \upharpoonright p_s = A^{[\leq b]} \upharpoonright p_s \right\},$$

where $A^{[\leq b]} = \bigcup_{a \leq b} A^{[a]}$, then there will be no further injuries. Thus we will have

$$b \in A' \iff \exists s \left[s \in \mathsf{T}_b^A \wedge \widehat{\{b\}_s^{A_s}}(b) \downarrow \right].$$

The result will then follow once we evaluate carefully the complexity of T_b^A .

This restraint threatens to interfere with the satisfaction of the positive requirements P_a , since as noted above, for indices b such that $\{b\}^A(b) \uparrow$, $\hat{u}(A_s; b, b, s)$ may take on arbitrarily large values, and hence the same is true of $\hat{R}_s(a)$. However, by the same mechanism, for any such b , $\{b\}_s^{A_s}(b) \uparrow$ for each $s \in \mathsf{T}_b^A$, and thus for $s \in \mathsf{T}_{<a}^A := \bigcap_{b < a} \mathsf{T}_b^A$, $\hat{R}_s(a)$ is bounded by the maximum of the uses of correct computations, which is a fixed finite value $\hat{R}(a)$. Thus there are still infinitely many opportunities to enumerate into A all elements of F greater than or equal to $\hat{R}(a)$.

Theorem 8.5.13 *For any r.e. set E and any E -r.e. set S such that $\emptyset' \leq_T S$, there exists an r.e. set $A \leq_T E$ such that $A' \equiv_T S$.*

Proof. Fix E and S as in the hypothesis; the conditions to be satisfied are

$$\begin{aligned} (3) \quad & A \leq_T E & (5) \quad & S \leq_T A' \\ (6) \quad & A' \leq_T S \end{aligned}$$

(from Section 8.4). Since $\emptyset' \leq_T S$, also $E \leq_T S$ so there exists an r.e. set $F \leq_T E$ which effectively jump codes S . Let P_a and $\hat{R}_s(a)$ be as in the sketch above. The construction is simply as follows: at stage s enumerate into A_{s+1} all numbers $\langle a, y \rangle \in F_s$ such that $\langle a, y \rangle \geq \hat{R}_s(a)$ and $\langle a, y \rangle \notin A_s$.

This construction is very different from earlier ones. For a such that $F^{[a]}$ is infinite, P_a must get attention infinitely often to ensure that all but finitely many of the elements of $F^{[a]}$ are eventually enumerated into A . This, in turn, has a potentially serious effect on the sequence of values $\hat{u}(A_s; b, b, s)$ for $b \geq a$, which contribute to the restraint values $\hat{R}_s(c)$ for $c > b$; since an enumeration of some $\langle a, y \rangle$ into $A^{[a]}$ is not required to respect $\hat{u}(A_s; b, b, s)$, it may cause an injury to the computation $\widehat{\{b\}_s^{A_s}}(b)$. If this happens infinitely often, it potentially leads to the situation discussed following Corollary 8.5.8

and to unbounded values of $\hat{R}_s(c)$, which, in turn, might prevent the satisfaction of P_c . However, at each true stage s , all values $\hat{u}(A_s; b, b, s)$ are eventually bounded by either $u(A; b, b)$ or 0 and thus also $\hat{R}_s(c)$ is bounded, say by $\hat{R}(c)$. It follows that at such stages all elements of $F^{[c]}$ greater than or equal to $\hat{R}(c)$ will eventually be enumerated into $A^{[c]}$, which is enough to satisfy P_c . Because a computation may be injured infinitely often, the argument is described as an **infinite injury** argument in contrast to the **finite injury** arguments of the preceding sections.

Note that indeed if our construction is to succeed, some computations must sustain infinitely many injuries, since otherwise

$$A_s := \left\{ b : \widehat{\{b\}}_s^{A_s}(b) \downarrow \right\}$$

would be a recursive family such that $A'(b) = \lim_{s \rightarrow \infty} A_s(b)$ and thus by the Limit Lemma $A' \equiv_T \emptyset'$, which is in general inconsistent with our goal of ensuring $S \leq_T A'$.

Before addressing directly the requirements, we establish some consequences of the construction. We show that for all a, b, s and y ,

- (A) if $s \in \mathbb{T}_b^A$ and $\widehat{\{b\}}_s^{A_s}(b) \downarrow$, then $\widehat{\{b\}}_s^{A_s}(b) \downarrow$ correctly;
- (B) $\lim_{s \in \mathbb{T}_{<a}^A} \hat{R}_s(a) =: \hat{R}(a)$ exists and is finite;
- (C) if $s \in \mathbb{T}_{<a}^A$, then for all $t \geq s$, $\hat{R}_s(a) \leq \hat{R}_t(a)$.

For (A), assume that $s \in \mathbb{T}_b^A$ and $\widehat{\{b\}}_s^{A_s}(b) \downarrow$. Let $u := \hat{u}(A_s; b, b, s)$; we prove by induction that for all $t \geq s$, $A_t \upharpoonright u = A_s \upharpoonright u$. This is true for $t = s$, so assume it for t as induction hypothesis. Any element $x \in A_{t+1} \sim A_t$ is of the form $x = \langle c, z \rangle$ with $x \geq R_t(c)$. If $c \leq a$, then

$$x \in A_{t+1}^{[\leq a]} \sim A_s^{[\leq a]} \quad \text{so} \quad x \geq p_s \geq u,$$

because $s \in \mathbb{T}_a^A$. If $c > a$, then

$$x \geq R_t(c) \geq \hat{u}(A_t; b, b, t) = u,$$

by the definition of R_t , the induction hypothesis and the Correctness Lemma. Hence, in either case $x \geq u$, so $A_{t+1} \upharpoonright u = A_t \upharpoonright u = A_s \upharpoonright u$ as desired.

For (B), it follows from (A) that for each b , for sufficiently large $s \in \mathbb{T}_b^A$, $\hat{u}(A_s; b, b, s) = u(A; b, b)$: either 0 for $b \notin A'$ or the use of the computation for $b \in A'$. Hence

$$\lim_{s \in \mathbb{T}_{<a}^A} \hat{R}_s(a) = \hat{R}(a) := \max\{u(A; b, b) : b < a\}.$$

Now (C) follows easily, since for $t \geq s$, for each $b < a$, if $s \in \mathbb{T}_b^A$, then either $\widehat{\{b\}}_s^{A_s}(b) \uparrow$ so $\hat{u}(A_s; b, b, s) = 0$, or $\widehat{\{b\}}_s^{A_s}(b) \downarrow$ so by (A) and the Correctness Lemma, $\hat{u}(A_s; b, b, s) = \hat{u}(A_t; b, b, t)$. In either case, $\hat{u}(A_s; b, b, s) \leq \hat{u}(A_t; b, b, t)$. Hence

$$\hat{R}_s(a) = \max\{\hat{u}(A_s; b, b, s) : b < a\} \leq \max\{\hat{u}(A_t; b, b, t) : b < a\} = \hat{R}_t(a).$$

Now we can establish the following:

- (5) for all a , P_a is satisfied; hence $S \leq_T A'$;
- (3) there exists a recursive function α such that for all a , $\{\alpha(a)\}^E = A^{[a]}$; hence $A \leq_T E$;
- (6) there exists an S -recursive function β such that for all a , $\{\beta(a)\} = A^{[a]}$; hence $A' \leq_T S$.

For (5), by (B), at sufficiently large stages $s \in \mathbb{T}_a^A$, $\hat{R}_s(a) = \hat{R}(a)$, so all elements of $F_s^{[a]}$ greater than or equal to $\hat{R}(a)$ which are not already in A_s will be enumerated into A_{s+1} . Since there are infinitely many such stages, $F^{[a]} \sim A^{[a]} \subseteq \{0, \dots, \hat{R}(a) - 1\}$ and P_a is satisfied.

Now P_a guarantees that $A^{[a]}$ is either finite (if $a \in S$) or cofinite (if $a \notin S$); in either case $A^{[a]}$ is recursive, and it follows immediately from the definition that also $\mathbb{T}_a^A$ is recursive. A itself is of course not recursive, so despite this there cannot be a recursive function α such that $\{\alpha(a)\} = A^{[a]}$. The assertions of (3) and (6) are two weaker versions.

For (3), assume recursively that $\alpha(b)$ is defined for $b < a$. Now, let $\alpha(a)$ be an E -index which acts as follows:

$$\{\alpha(a)\}^E(\langle a, y \rangle) \simeq \begin{cases} 0, & \text{if } \langle a, y \rangle \notin F^{[a]}; \\ A_{f(a,y)+1}(\langle a, y \rangle), & \text{otherwise;} \end{cases}$$

where for $\langle a, y \rangle \in F^{[a]}$,

$$f(a, y) := \text{least } s \left[\langle a, y \rangle \in F_s^{[a]} \quad \text{and} \quad s \in \mathbb{T}_{<a}^A \right].$$

To see that this gives the correct result, note that by (C), for all $s \geq f(a, y)$, $\hat{R}_{f(a,y)}(a) \leq \hat{R}_s(a)$. Hence if $\langle a, y \rangle$ is in A , it is already in $A_{f(a,y)+1}$.

To see that $\alpha(a)$ is recursively computable from the values $\alpha(b)$ for $b < a$, note that since $\mathbb{T}_{<a}^A$ is simply defined in terms of $A^{[b]}$ for $b < a$, we may compute an E -index for $\mathbb{T}_{<a}^A$. Then we use the fact that $F \leq_T E$. That $A \leq_T E$ now follows immediately.

The proof of the first part of (6) is similar with one new twist. From the definition of effectively jump codes, it follows easily that there exists an S -recursive function ε such that for all a , $\{\varepsilon(a)\} = F^{[a]}$. Then in place of the direct appeal to the F -oracle to decide whether or not $x = \langle a, y \rangle \in F$ (equivalently $x \in F^{[a]}$) in the definition of $\alpha(a)$, we can define $\beta(a)$ by computing $\{\varepsilon(a)\}(x)$ to answer such questions.

Finally, to prove that $A' \leq_T S$, we first note that for all b ,

$$b \in A' \iff \exists s \left[s \in \mathbb{T}_b^A \wedge \widehat{\{b\}}_s^{A_s}(b) \downarrow \right].$$

The implication ($\implies$) is immediate from Corollary 8.5.7 and ($\impliedby$) follows from (A). Now as in the proof of (3), from the values of $\beta(a)$ for $a \leq b$ it is easy to compute a value $\delta(b)$ such that $\{\delta(b)\} = \mathbb{T}_b^A$; thus also δ is S -recursive. Now let J be the relation defined by

$$J(b, d) \iff \exists s \left[\{d\}(s) \simeq 1 \wedge \widehat{\{b\}}_s^{A_s}(b) \downarrow \right].$$

By the one-quantifier principle, $J \leq_T \emptyset' \leq_T S$, and by the above, $b \in A'$ iff $J(b, \delta(b))$, so since δ is S -recursive, also $A' \leq_T S$. $\square$

The ultimate goal of this section is to strengthen the conclusion of this theorem from $A \leq_T E$ to $A <_T E$. The strategy is to combine the preceding technique with that of Theorem 8.4.9 to ensure that additionally $E \not\leq_T A$. However, this entails a further complication, and to prepare the way, we consider first the special case in which $E = \emptyset'$ (a complete r.e. set) and $S = \emptyset''$. Note that every r.e. set A satisfies $A \leq_T \emptyset'$, hence also $A' \leq_T \emptyset''$. Therefore, dually to Definition 8.3.15, we call an r.e. set **high** iff $A' \equiv_T \emptyset''$. Then the theorem below is dual to Proposition 8.3.16.

The proof will be based on a revised approximation to the length of agreement function introduced in Definition 8.4.3.

Definition 8.5.14 For any stage enumerations

$$\langle A_s : s \in \omega \rangle \quad \text{and} \quad \langle E_s : s \in \omega \rangle$$

of sets A and E and any b and s ,

$$\hat{\ell}^{AE}(b, s) := \text{least } y \left[\widehat{\{b\}}_s^{A_s}(y) \neq E_s(y) \right].$$

Furthermore, for any y , $\hat{\ell}^{AE}(b, s) \geq y$ **correctly** iff $\hat{\ell}^{AE}(b, s) \geq y$, and for all $z < y$,

$$\widehat{\{b\}}_s^{A_s}(z) \downarrow \text{ correctly} \quad \text{and} \quad E_s(z) = E(z).$$

Thus $\hat{\ell}^{AE}(b, s)$ is an alternative measure of the extent of the agreement between $\{b\}^A$ and E at stage s . Clearly $\hat{\ell}^{AE}(b, s) \leq \ell^{AE}(b, s)$. The properties of the CLL hold also for $\hat{\ell}$ in place of ℓ .

Correctness Lemma for $\widehat{\text{Length}}$ 8.5.15 (CL $\widehat{\text{L}}$) *For any stage enumerations $\langle A_s : s \in \omega \rangle$ and $\langle E_s : s \in \omega \rangle$ of sets A and E and all b, s and y ,*

- (i) *if $y \leq \ell^{AE}(b)$, then there exists s such that $\hat{\ell}^{AE}(b, s) \geq y$ correctly;*
- (ii) *if $\hat{\ell}^{AE}(b, s) \geq y$ correctly, then $y \leq \ell^{AE}(b)$ and for all $t \geq s$, $\hat{\ell}^{AE}(b, t) \geq y$ correctly;*
- (iii) *if $\ell^{AE}(b) \geq y$, $\hat{\ell}^{AE}(b, s) \geq y$, and for all $z < y$, $\widehat{\{b\}}_s^{A_s}(z) \downarrow$ correctly — in particular if $s \in \mathsf{T}^A$ — then $\hat{\ell}^{AE}(b, s) \geq y$ correctly.*

Proof. For (i), if $y \leq \ell^{AE}(b)$, then by the CLL, for some s , $\ell^{AE}(b, s) \geq y$ correctly. Then $\hat{\ell}^{AE}(b, s+1) \geq y$ correctly, since by correctness, for each $z < y$, $u(A_s; b, z, s) \leq p_{s+1}$. Parts (ii) and (iii) are immediate from the corresponding parts of CLL and Proposition 8.5.6. $\square$

Corollary 8.5.16 *For any stage enumerations*

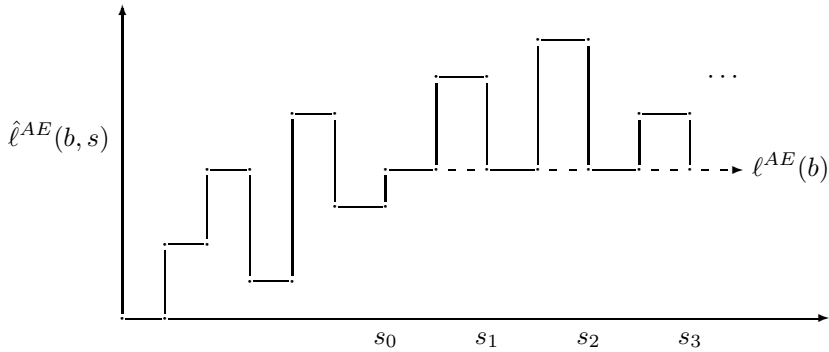
$$\langle A_s : s \in \omega \rangle \quad \text{and} \quad \langle E_s : s \in \omega \rangle$$

of sets A and E and any b ,

- (i) *if $\ell^{AE}(b) = \infty$, then $\lim_{s \rightarrow \infty} \hat{\ell}^{AE}(b, s) = \infty$;*
- (ii) *if $\ell^{AE}(b) < \infty$, then for all sufficiently large s , $\hat{\ell}^{AE}(b, s) \geq \ell^{AE}(b)$ and for all sufficiently large $s \in \mathsf{T}^A$, $\hat{\ell}^{AE}(b, s) = \ell^{AE}(b)$.*

Proof. Part (i) is immediate from CL $\widehat{\text{L}}$. For (ii), choose $\bar{t}$ large enough that for all $z \leq \ell^{AE}(b)$, $E_{\bar{t}}(z) = E(z)$ and suppose for some $s \in \mathsf{T}^A$ with $s \geq \bar{t}$ that $\hat{\ell}^{AE}(b, s) \geq \ell^{AE}(b) + 1$. Then, by Proposition 8.5.6, for all $z \leq \ell^{AE}(b)$, $\widehat{\{b\}}_s^{A_s}(z) \downarrow$ correctly, so by the choice of t , $\hat{\ell}^{AE}(b, s) \geq \ell^{AE}(b) + 1$ correctly, contradicting CL $\widehat{\text{L}}$ (ii). $\square$

We may picture the behavior of $\hat{\ell}^{AE}(b, s)$ when $\ell^{AE}(b) < \infty$ as in the following schematic graph. Once the correct computations for $y < \ell^{AE}(b)$ are established at s_0 , despite arbitrarily large values in between, at later stages $s_1, s_2, s_3, \dots \in \mathsf{T}^A$ the value of $\hat{\ell}^{AE}(b, s)$ returns to its $\liminf$ bound $\ell^{AE}(b)$.



Theorem 8.5.17 *There exists an incomplete high r.e. set — that is, an r.e. set A such that $A <_T \emptyset'$ and $A' \equiv_T \emptyset''$.*

Proof. The conditions to be satisfied here are

$$\begin{array}{ll} (3) A \leq_T \emptyset' & (5) \emptyset'' \leq_T A' \\ (4) \emptyset' \not\leq_T A & (6) A' \leq_T \emptyset''. \end{array}$$

Properties (3) and (6) are trivial (because A is r.e.). Since $\emptyset''$ is $\emptyset'$ -r.e., there exists an r.e. set F which effectively jump codes $\emptyset''$. We shall construct A thick in F while protecting computations of the form $\{a\}_s^{A_s}(x) \simeq \emptyset'(x)$ sufficiently to ensure that if $\emptyset'$ were A -recursive, then it would be recursive, a contradiction. Thus, we aim to satisfy the requirements

$$\begin{array}{ll} P_a : & F^{[a]} \sim A^{[a]} \text{ is finite} \\ N_a : & \{a\}^A \neq \emptyset'. \end{array}$$

The restraints which guarantee the satisfaction of the requirements N_a are similar to those of Theorem 8.4.9 using $\hat{\ell} := \hat{\ell}^{A\emptyset'}$ in place of ℓ :

$$\begin{array}{l} \hat{r}(b, s) := \max \left\{ \hat{u}(A_s; b, z, s) : z \leq \hat{\ell}(b, s) \right\}; \\ \hat{R}_s(a) := \max \{ \hat{r}(b, s) : b < a \}. \end{array}$$

The construction is now as for Theorem 8.5.13: at stage s enumerate into A_{s+1} all numbers $\langle a, y \rangle \in F_s$ such that $\langle a, y \rangle \geq \hat{R}_s(a)$ and $\langle a, y \rangle \notin A_s$.

As above, we first establish some consequences of the construction. We say that $\hat{\ell}(b, s) \geq y$ **very correctly** iff $\hat{\ell}(b, s) \geq y$ correctly and if $\widehat{\{b\}}_s^{A_s}(y) \downarrow$, then also $\widehat{\{b\}}_s^{A_s}(y) \downarrow$ correctly. Then for all a, b, s and y ,

- (A) if $s \in \mathbb{T}_b^A$, $y \leq \ell(b)$ and $\hat{\ell}(b, s) \geq y$, then $\hat{\ell}(b, s) \geq y$ very correctly;
 (B1) $\lim_{s \in \mathbb{T}_b^A} \hat{\ell}(b, s) = \ell(b)$;
 (B2) if for all $b < a$, $\ell(b) < \infty$, then $\lim_{s \in \mathbb{T}_{<a}^A} \hat{R}_s(a) =: R(a)$ exists and is finite.

For (A), let $s \in \mathbb{T}_b^A$; we prove by induction on $y \leq \hat{\ell}(b)$ that

$$\hat{\ell}(b, s) \geq y \implies \hat{\ell}(b, s) \geq y \text{ very correctly.}$$

Assume as induction hypothesis that this holds for y , and suppose that $y + 1 \leq \ell(b)$ and $\hat{\ell}(b, s) \geq y + 1$; hence $\hat{\ell}(b, s) \geq y$ very correctly. (The basis case $y = 0$ is identical without any use of an induction hypothesis.) By $\text{CLL}\widehat{\text{L}}(\text{iii})$ $\hat{\ell}(b, s) \geq y + 1$ correctly, and it suffices to show that $\widehat{\{b\}}_s^{A_s}(y + 1) \downarrow$ correctly — that is, with $u := \hat{u}(A_s; b, y + 1, s)$, for all $t \geq s$, $A_t \upharpoonright u = A_s \upharpoonright u$. This is immediate for $t = s$, so assume as induction hypothesis that it holds for t . By the construction, any element $x \in A_{t+1} \sim A_t$ is of the form $x = \langle c, z \rangle$ with $x \geq \hat{R}_t(c)$. If $c \leq b$, then

$$x \in A^{[\leq b]} \sim A_s^{[\leq b]} \quad \text{so} \quad x \geq p_s \geq u$$

because $s \in \mathbb{T}_b^A$. If $c > b$, then

$$\begin{aligned} x &\geq \hat{R}_t(c) \geq \hat{r}(b, t) \\ &\geq \hat{u}(A_t; b, y + 1, t) \quad \text{because by } \text{CLL}\widehat{\text{L}}(\text{ii}), \hat{\ell}(b, t) \geq y + 1 \\ &\geq u \quad \text{by the Correctness Lemma.} \end{aligned}$$

Hence, in either case $A_{t+1} \upharpoonright u = A_t \upharpoonright u = A_s \upharpoonright u$ as desired.

(B1) is immediate from the corollary to the CLL in case $\ell(b) = \infty$. If $\ell(b) < \infty$, then by the same corollary, for all sufficiently large s , $\hat{\ell}(b, s) \geq \ell(b)$. Choose $\bar{t}$ large enough that for all $z \leq \ell(b)$, $E_{\bar{t}}(z) = E(z)$. Suppose towards a contradiction that for some $s \in \mathbb{T}_b^A$ with $s \geq \bar{t}$ that $\hat{\ell}(b, s) \geq \ell(b) + 1$. By (A), $\hat{\ell}(b, s) \geq \ell(b)$ very correctly, and since $\widehat{\{b\}}_s^{A_s}(\ell(b)) \downarrow$, this computation is correct. But then by the choice of $\bar{t}$, $\hat{\ell}(b, s) \geq \ell(b) + 1$ correctly, so by $\text{CLL}\widehat{\text{L}}(\text{ii})$, $\ell(b) \geq \ell(b) + 1$, a contradiction.

Now (B2) follows, since for sufficiently large $s \in \mathbb{T}_b^A$, if $\ell(b) < \infty$, by (B1) and (A),

$$\begin{aligned} \hat{r}(b, s) &= \max \left\{ \hat{u}(A_s; b, z, s) : z \leq \hat{\ell}(b, s) \right\} \\ &= \max \{ u(A; b, z) : z \leq \ell(b) \} =: r(b). \end{aligned}$$

Hence under the hypothesis of (B2), for sufficiently large $s \in \mathbb{T}_{<a}^A$, $\hat{R}_s(a)$ has the constant value $R(a) := \max \{ r(b) : b < a \}$.

We are now in a position to prove that for all a ,

(5a) P_a is satisfied;

(4a) $A^{[a]}$ and T_a^A are recursive and N_a is satisfied.

In contrast with the analogous part of the proof of Theorem 8.4.9, we prove (5a) and (4a) simultaneously by induction on a . Assume as induction hypothesis that (5b) and (4b) are true for all $b < a$. In particular, for all $b < a$, $\ell(b)$ is finite (since N_b is satisfied), so by (B2), for all sufficiently large $s \in \mathsf{T}_{<a}^A$, $\hat{R}_s(a) = R(a)$. Hence all elements of $F_s^{[a]}$ greater than $R(a)$ will be enumerated into A_{s+1} , so $F^{[a]} \sim A^{[a]} \subseteq \{0, \dots, R(a) - 1\}$ and P_a is satisfied.

P_a guarantees that $A^{[a]}$ is either finite (if $a \in \emptyset''$) or cofinite (if $a \notin \emptyset''$), so in either case recursive. By the induction hypothesis, the same is true for all $b < a$, so $A^{[\leq a]}$ is recursive and hence easily from its definition so is T_a^A .

For (4a), suppose to the contrary that, $\{a\}^A = \emptyset'$. By (B1), $\ell(a) = \infty = \lim_{s \in \mathsf{T}_a^A} \hat{\ell}(a, s)$, so the function

$$h(y) := \text{least } s \left[s \in \mathsf{T}_a^A \text{ and } \hat{\ell}(a, s) \geq y \right]$$

is total. Furthermore, by the preceding remark, h is recursive, and by (A), for all y , $\hat{\ell}(a, h(y)) \geq y$ correctly. Thus h is a recursive correctness function for b , so $\emptyset'_{h(y+1)}(y) = \emptyset'(y)$, and by stage bounding, $\emptyset'$ is recursive, a contradiction.

From these we conclude immediately as before that (4) and (5) hold. $\square$

The final theorem of this section, stated below, is the natural combination of Theorems 8.5.13 and 8.5.17. Its proof is also a combination of the techniques of these two proofs, but this combination requires a further subtle modification of the approximating length function to guarantee the property (C) of the proof of Theorem 8.5.13 for a restraint function based on length of agreement as in Theorem 8.5.17. Note that it follows from (A) in the preceding proof that as long as $\hat{\ell}(b, s) \leq \ell(b)$, then

$$s \in \mathsf{T}_b^A \text{ and } s \leq t \implies \hat{\ell}(b, s) \leq \hat{\ell}(b, t).$$

For the analogue of (C) to hold, however, we need this monotonicity without the extra hypothesis $\hat{\ell}(b, s) \leq \ell(b)$, but this is not generally true for $\hat{\ell}$ because although $s \in \mathsf{T}_b^A$ ensures that computations $\{b\}_{s^A}^{A_s}(z)$ for $z \leq \hat{\ell}(b, s)$ are correct and hence equal to $\{b\}^A(z)$, the values $E_s(z)$ may not be correct unless also $\{b\}^A(z) \simeq E(z)$. The solution is to use a length function which is insensitive to changes in the values of the $E_s(z)$ — once an agreement on a number z is recorded at a stage t , it will be preserved at $s \geq t$ as long as the computation is still correct at s . This idea is the basis of the

Definition 8.5.18 For any stage enumerations

$$\langle A_s : s \in \omega \rangle \quad \text{and} \quad \langle E_s : s \in \omega \rangle$$

of sets A and E ,

$$\ell^{+AE}(b, s) := \max \left\{ y : \exists s' \leq s \left(y \leq \hat{\ell}^{AE}(b, s') \wedge \forall z \leq y \left[\widehat{\{b\}}_{s'}^{A_{s'}}(z) \downarrow \implies \widehat{\{b\}}_{s'}^{A_{s'}}(z) \downarrow \text{correctly to } s \right] \right) \right\}.$$

Clearly $\hat{\ell}^{AE}(b, s) \leq \ell^{+AE}(b, s)$, and the key to the success of this function is that it has most of the same properties as $\hat{\ell}^{AE}$. Note that $\ell^{+AE}(b, s) \geq y$ does *not* in general imply that $\hat{\ell}^{AE}(b, s) \geq y$ because although for all $z < y$,

$$\widehat{\{b\}}_s^{A_s}(z) \simeq \widehat{\{b\}}_{s'}^{A_{s'}}(z) \simeq E_{s'}(z),$$

for some z we may have $E_s(z) \neq E_{s'}(z)$. Hence it is not appropriate to introduce a notion: ' $\ell^{+AE}(b, s) \geq y$ correctly', and the corresponding Correctness Lemma has only one part.

Correctness Lemma for Length⁺ 8.5.19 (CLL⁺) *For any stage enumerations*

$$\langle A_s : s \in \omega \rangle \quad \text{and} \quad \langle E_s : s \in \omega \rangle$$

of sets A and E and all b, s and y , if $\ell^{AE}(b) \geq y$, $\ell^{+AE}(b, s) \geq y$, and for all $z < y$, $\widehat{\{b\}}_s^{A_s}(z) \downarrow$ correctly — in particular if $s \in \mathbb{T}^A$ — then $\hat{\ell}^{AE}(b, s) \geq y$ correctly.

Proof. Assume the hypotheses. Since $\ell^{+AE}(b, s) \geq y$, for each $z < y$ there is an $s' \leq s$ such that $z < \hat{\ell}^{AE}(b, s')$ and $\widehat{\{b\}}_{s'}^{A_{s'}}(z) \downarrow$ correctly to s . Since also $\widehat{\{b\}}_s^{A_s}(z) \downarrow$ correctly, we have $\widehat{\{b\}}_{s'}^{A_{s'}}(z) \downarrow$ correctly, so

$$E_{s'}(z) = \widehat{\{b\}}_{s'}^{A_{s'}}(z) \simeq \widehat{\{b\}}_s^{A_s}(z) \simeq \{b\}^A(z) \simeq E(z),$$

because $z < \ell^{AE}(b)$. Since $E_{s'} \subseteq E_s \subseteq E$, also $E_s(z) = \widehat{\{b\}}_s^{A_s}(z) \simeq E(z)$. It follows that $\hat{\ell}^{AE}(b, s) \geq y$ correctly. $\square$

Corollary 8.5.20 *For any stage enumerations*

$$\langle A_s : s \in \omega \rangle \quad \text{and} \quad \langle E_s : s \in \omega \rangle$$

of sets A and E and any b ,

- (i) if $\ell^{AE}(b) = \infty$, then $\lim_{s \rightarrow \infty} \ell^{+AE}(b, s) = \infty$;
- (ii) if $\ell^{AE}(b) < \infty$, then for all sufficiently large s , $\ell^{+AE}(b, s) \geq \ell^{AE}(b)$, and for all sufficiently large $s \in \mathbb{T}^A$, $\ell^{+AE}(b, s) = \ell^{+AE}(b)$, where

$$\bar{t} := \text{least } t \left(\forall z \leq \ell^{AE}(b) [E_t(z) = E(z)] \text{ and } \hat{\ell}^{AE}(b, t) \geq \ell^{AE}(b) \right)$$

$$\ell^{+AE}(b) := \max \left\{ y : \exists s' \leq \bar{t} [y \leq \hat{\ell}^{AE}(b, s') \wedge (\forall z < y) \{ \widehat{b} \}_{s'}^{A_{s'}}(z) \downarrow \text{ correctly}] \right\}.$$

Proof. Immediately from the definitions, for all $s \geq \bar{t}$, $\ell^{+AE}(b, s) \geq \ell^{+AE}(b) \geq \ell^{AE}(b)$. Suppose for some $s \in \mathbb{T}^A$ with $s \geq \bar{t}$, we have $\ell^{+AE}(b, s) \geq \ell^{+AE}(b) + 1$. Then for some $s' \leq s$, $\hat{\ell}^{AE}(b, s') \geq \ell^{+AE}(b) + 1 \geq \ell^{AE}(b) + 1$, and for all $z \leq \ell^{AE}(b)$, $\{ \widehat{b} \}_{s'}^{A_{s'}}(z) \downarrow$ correctly to s . Since $s \in \mathbb{T}^A$, by Proposition 8.5.6, $\{ \widehat{b} \}_s^{A_s}(z) \downarrow$ correctly, so also $\{ \widehat{b} \}_{s'}^{A_{s'}}(z) \downarrow$ correctly. By the definition of $\ell^{+AE}(b)$, $s' > t$, so also for all $z \leq \ell^{AE}(b)$, $E_{s'}(z) = E(z)$. Hence $\hat{\ell}^{AE}(b, s') \geq \ell^{AE}(b) + 1$ correctly, contrary to CLL(ii). Hence for all such s , $\ell^{+AE}(b, s) = \ell^{+AE}(b)$. $\square$

Sacks Jump Theorem 8.5.21 *For any non-recursive r.e. set E and any E -r.e. set S such that $\emptyset' \leq_T S$, there exists an r.e. set $A <_T E$ such that $A' \equiv_T S$.*

Proof. The conditions to be satisfied here are

$$\begin{array}{ll} (3) \ A \leq_T E & (5) \ S \leq_T A' \\ (4) \ E \not\leq_T A & (6) \ A' \leq_T S. \end{array}$$

Fix E and S as in the hypothesis and $F \leq_T E$ which effectively jump codes S . The strategy is again to make A thick in F while protecting computations of the form $\{b\}_s^{A_s}(b) \downarrow$ sufficiently to ensure that $A' \leq_T S$ and those of the form $\{a\}_s^{A_s}(x) \simeq E_s(x)$ to ensure that if E were recursive in A , then actually E would be recursive, contrary to hypothesis. In fact, by a simple device it suffices to do the second of these. Thus we aim to satisfy the requirements

$$\begin{array}{ll} P_a : & F^{[a]} \sim A^{[a]} \text{ is finite} \\ N_a : & \{a\}^A \neq E \end{array}$$

subject to the restraints

$$R_s^+(a) = \max\{r^+(b, s) : b < a\},$$

where

$$r^+(b, s) = \max\{\hat{u}(A_s; b, z, s) : z \leq \ell^+(b, s)\}.$$

(As usual we write simply ℓ^+ for ℓ^{+AE} .) The construction is now as for Theorems 8.5.13 and 8.5.17 using the new restraint function: at stage s enumerate into A_{s+1} all numbers $\langle a, y \rangle \in F_s$ such that $\langle a, y \rangle \geq R_s^+(a)$ and $\langle a, y \rangle \notin A_s$.

We shall prove first that for all a, b, s and y ,

- (A) if $s \in \mathbb{T}_b^A$, $y \leq \ell(b)$, and $\ell^+(b, s) \geq y$, then $\hat{\ell}(b, s) \geq y$ very correctly;
- (B1) $\lim_{s \in \mathbb{T}_b^A} \ell^+(b, s) =: \ell^+(b)$ exists and is finite; $\ell^+(b) \geq \ell(b)$;
- (B2) if for all $b < a$, $\ell(b) < \infty$, then $\lim_{s \in \mathbb{T}_{<a}^A} R_s^+(a) =: R^+(a)$ exists and is finite;
- (C1) if $s \in \mathbb{T}_b^A$, then for all $t \geq s$, $\ell^+(b, s) \leq \ell^+(b, t)$ and $r^+(b, s) \leq r^+(b, t)$;
- (C2) if $s \in \mathbb{T}_{<a}^A$, then for all $t \geq s$, $R_s^+(a) \leq R_t^+(a)$.

Now (A) follows from CLL^+ exactly as (A) in the proof of Theorem 8.5.17 follows from CLL . For (C1), suppose that $s \in \mathbb{T}_b^A$. Then by (A), for all $y \leq \ell^+(b, s)$, if $\widehat{\{b\}}_s^{A_s}(y) \downarrow$, then $\widehat{\{b\}}_s^{A_s}(y) \downarrow$ correctly, so for each $t \geq s$, $\widehat{\{b\}}_s^{A_s}(y) \downarrow$ correctly to t . Hence directly from the definition, $\ell^+(b, s) \leq \ell^+(b, t)$ (see the discussion preceding Definition 8.5.18). Furthermore,

$$\begin{aligned} r^+(b, s) &= \max\{\hat{u}(A_s; b, z, s) : z \leq \ell^+(b, s)\} \\ &\leq \max\{\hat{u}(A_t; b, z, t) : z \leq \ell^+(b, t)\} = r^+(b, t). \end{aligned}$$

For (C2), if $s \in \mathbb{T}_{<a}^A$, then

$$\begin{aligned} R_s^+(a) &= \max\{r^+(b, s) : b < a\} \\ &\leq \max\{r^+(b, t) : b < a\} = R_t^+(a). \end{aligned}$$

(B1) is immediate from the corollary to CLL^+ in case $\ell(b) = \infty$ and is proved using (A) similarly to the proof of the preceding corollary in the case that $\ell(b)$ is finite. Now (B2) follows, since for sufficiently large $s \in \mathbb{T}_b^A$, if $\ell(b) < \infty$, then

$$\begin{aligned} r^+(b, s) &= \max\{\hat{u}(A_s; b, z, s) : z \leq \ell^+(b, s)\} \\ &= \max\{u(A; b, z) : z \leq \ell^+(b)\} =: r^+(b). \end{aligned}$$

Hence under the hypothesis of (B2), for sufficiently large $s \in \mathbb{T}_{<a}^A$, $R_s^+(a)$ has the constant value $R^+(a) := \max\{r^+(b) : b < a\}$.

We are now in a position to prove exactly as in the proof of Theorem 8.5.17, using (B2), by induction that for all a ,

(5a) P_a is satisfied;

(4a) $A^{[a]}$ and $\mathbb{T}_a^A$ are recursive and N_a is satisfied;

and hence that (5) and (4) hold. Finally, we can show exactly as in the proof of Theorem 8.5.13, using (C2), that

(3) there exists a recursive function α such that for all a , $\{\alpha(a)\}^E = A^{[a]}$; hence $A \leq_T E$;

(6) there exists an S -recursive function β such that for all a , $\{\beta(a)\} = A^{[a]}$;

and it remains to derive $A' \leq_T S$ from (6). It is easy to describe a recursive function h such that for all b , x and X ,

$$\{h(b)\}^X(x) \simeq \begin{cases} E(0), & \text{if } \{b\}(b) \downarrow; \\ \uparrow, & \text{otherwise.} \end{cases}$$

Then

$$\begin{aligned} \{b\}^A(b) \downarrow &\iff \ell(h(b)) \geq 1 \\ &\iff \exists s [s \in \mathbb{T}_b^A \wedge \{\widehat{h(b)}\}_s^{A_s}(0) \simeq E(0)]. \end{aligned}$$

From this expression, $A' \leq_T S$ follows as at the end of the proof of Theorem 8.5.13. $\square$

Exercises

Exercise 8.5.22 Prove that for any set B there exists a columnwise recursive set A such that $A \equiv_T B$.

8.6 Lower bounds

In this section we address the problem of carrying out the constructions of Sections 8.4–8.5 above a given set D — that is, satisfying the conditions (1) $D \leq_T A$ and (2) $A \not\leq_T D$ in addition to (3)–(6). It turns out that (1) is relatively easy to handle but a significant new idea is required for (2). To introduce the methods we prove first

Theorem 8.6.1 *For any r.e. sets $D \leq_T E$ and any E-r.e. set S such that $D' \leq_T S$, there exists an r.e. set A such that $D \leq_T A \leq_T E$ and $A' \equiv_T S$.*

Proof. Given D , E and S as in the hypothesis, we aim to satisfy the conditions

$$\begin{array}{lll} (1) D \leq_T A & (3) A \leq_T E & (5) S \leq_T A' \\ & & (6) A' \leq_T S. \end{array}$$

We follow closely the proof of Theorem 8.5.13 and concentrate on the differences. Fix an r.e. set $F \leq_T E$ which jump-codes S in a slightly modified way: $F^{[0]} = \emptyset$ and

$$a \in S \implies F^{[a+1]} \text{ is finite} \quad \text{and} \quad a \notin S \implies F^{[a+1]} = \omega^{[a+1]},$$

and fix recursive stage enumerations of D and F . We shall ensure that $D \leq_T A$ by arranging that

$$P_0 : \quad A^{[0]} = \{ \langle 0, y \rangle : y \in D \},$$

and that $S \leq_T A'$ by satisfying for all a

$$P_{a+1} : \quad F^{[a+1]} \sim A^{[a+1]} \text{ is finite.}$$

The restraints are as before

$$\hat{R}_s(a) = \max\{ \hat{u}(A_s; b, b, s) : b < a \}.$$

Now we can describe the construction: at stage s enumerate into A_{s+1} all numbers of one of the forms

- (i) $\langle 0, y \rangle$ for each $y \in D_{s+1} \sim D_s$;
- (ii) $\langle a, y \rangle \in F_s$ such that $\langle a, y \rangle \geq \hat{R}_s(a)$ and $\langle a, y \rangle \notin A_s$.

Now it is immediate that

- (1) P_0 is satisfied; hence $D \leq_T A$;

Furthermore, we can show exactly as in Theorem 8.5.13 that for all a , b , s and y ,

- (A) if $s \in \mathbb{T}_b^A$ and $\widehat{\{b\}}_s^{A_s}(b) \downarrow$, then $\widehat{\{b\}}_s^{A_s}(b) \downarrow$ correctly;
- (B) $\lim_{s \in \mathbb{T}_{<a}^A} \hat{R}_s(a) := \hat{R}(a)$ exists and is finite;
- (C) if $s \in \mathbb{T}_{<a}^A$, then for all $t \geq s$, $\hat{R}_s(a) \leq \hat{R}_t(a)$;

so as in that proof

(5) for all $a > 0$, P_a is satisfied; hence $S \leq_T A'$.

It follows that for each $a > 0$, $A^{[a]}$ is either finite or cofinite, hence recursive and thus for all a , $A^{[a]}$ is D -recursive. To complete the proof we need to establish that

- (3) there exists a recursive function α such that for all a , $\{\alpha(a)\}^E = A^{[a]}$; hence $A \leq_T E$;
- (6) there exists an S -recursive function β such that for all a , $\{\beta(a)\}^D = A^{[a]}$; hence $A' \leq_T S$.

There are appropriate values $\alpha(0)$ and $\beta(0)$ because

$$A^{[0]} = \{ \langle 0, y \rangle : y \in D \} \leq_T D \quad \text{and} \quad D \leq_T E.$$

This is, of course, why we need $\{\beta(a)\}^D$ instead of simply $\{\beta(a)\}$. The recursion for the remaining values is exactly as in Theorem 8.5.13, and we omit it.

Finally, to prove that $A' \leq_T S$, we have as before that for all b ,

$$b \in A' \iff \exists s \left[s \in \mathbb{T}_b^A \wedge \widehat{\{b\}}_s^{A_s}(b) \downarrow \right].$$

From the values of $\beta(a)$ for $a \leq b$, it is easy to compute a value $\delta(b)$ such that $\{\delta(b)\}^D = \mathbb{T}_b^A$; thus also δ is S -recursive. Now let J be the relation defined by

$$J(b, d) \iff \exists s \left[\{d\}^D(s) \simeq 1 \wedge \widehat{\{b\}}_s^{A_s}(b) \downarrow \right].$$

By the one-quantifier principle relativized to D , $J \leq_T D' \leq_T S$ (by hypothesis), and by the above, $b \in A'$ iff $J(b, \delta(b))$, so since δ is S -recursive, also $A' \leq_T S$. $\square$

No new techniques beyond those of [Section 8.5](#) are needed to strengthen this to

Theorem 8.6.2 *For any r.e. sets $D <_T E$ and any E -r.e. set S such that $D' \leq_T S$, there exists an r.e. set A such that $D \leq_T A <_T E$ and $A' \equiv_T S$.*

Proof. Given D , E and S as in the hypothesis, we aim to show that

- | | | |
|------------------|----------------------|---------------------|
| (1) $D \leq_T A$ | (3) $A \leq_T E$ | (5) $S \leq_T A'$ |
| | (4) $E \not\leq_T A$ | (6) $A' \leq_T S$. |

This is proved by a straightforward adaptation of the proof of Theorem 8.5.21 as in the preceding proof. $\square$

The final step is to arrange in this result that $D <_T A$ — that is, also (2) $A \not\leq_T D$. We shall of course accomplish this by satisfying for all a the requirement

$$Q_a : \quad \{a\}^D \neq A,$$

and we make use of the length of agreement function $\ell^{DA}(a)$ and the sequence of approximations $\hat{\ell}^{DA}(a, s)$. However, these will be playing a somewhat different role from that of the functions ℓ^{AE} and $\hat{\ell}^{AE}$ because the roles of the given set (D or E) and the set being constructed A are reversed. To satisfy Q_a means to ensure that $\ell^{DA}(a) < \infty$. The strategy for accomplishing this is roughly as follows. For each x , if at stage s both $x \in E_s$ and $x < \hat{\ell}^{DA}(a, s)$, then we enumerate into A_{s+1} a **marker** $\mathbf{m}_x^a$ — some number such that the mapping $x \mapsto \mathbf{m}_x^a$ is an injective recursive function. Then for $x < \ell^{DA}(a)$,

$$x \in E \quad \Longleftrightarrow \quad \mathbf{m}_x^a \in A.$$

Now if $A = \{a\}^D$, then $\ell^{DA}(a) = \infty$, and this equivalence holds for all x , so $E \leq_m A \leq_T D$, hence $E \leq_T D$, contrary to hypothesis. We conclude that $A \neq \{a\}^D$ and Q_a is satisfied.

Unfortunately this simple mechanism is not compatible with the other goals of the construction because the restraints which enable the satisfaction of the negative conditions may not permit the enumeration of $\mathbf{m}_x^a$. To get around this difficulty we modify the strategy; instead of a single marker $\mathbf{m}_x^a$ for x , we assign an infinite sequence $\mathbf{m}_{x,t}^a$ ($t \in \omega$) of markers and at each stage s designate certain of these as **qualified**. When x appears in E_s , then all qualified markers which do not violate the restraints are enumerated into A_{s+1} . The key is that we will be able to identify one of these, which will not be restrained, via a D -recursive function g and conclude that

$$x \in E \quad \Longleftrightarrow \quad \mathbf{m}_{x,g(x)}^a \in A,$$

hence $E \leq_T A \leq_T D$ as above, contrary to hypothesis. The delicate point which qualification handles is to do this in such a way that A remains effectively columnwise D -recursive as expressed in (6) below. We first illustrate this technique in the simplest possible setting.

Theorem 8.6.3 *For any r.e. sets $D <_T E$, there exists an r.e. set A such that $D <_T A \leq_T E$ and $A' \equiv_T D'$.*

Proof. Given D and E as in the hypothesis, we aim to construct A to satisfy the following conditions

- | | | |
|----------------------|------------------|--------------------|
| (1) $D \leq_T A$ | (3) $A \leq_T E$ | (5) $D' \leq_T A'$ |
| (2) $A \not\leq_T D$ | | (6) $A' \leq_T D'$ |

Of course (5) follows immediately from (1), and as above we guarantee (1) by arranging that $A^{[0]} = \{ \langle 0, y \rangle : y \in D \}$. The restraints which guarantee (3) and (6) are exactly the same as in Theorem 8.5.13:

$$\hat{R}_s(a) := \max\{ \hat{u}(A_s; b, b, s) : b < a \}.$$

For (2), we proceed as in the sketch above. For all a, x and t , set $\mathbf{m}_{x,t}^a := \langle a+1, \langle x, t \rangle \rangle$ and say that $\mathbf{m}_{x,t}^a$ is **qualified at** s iff

$$t \leq s \quad \text{and} \quad \forall v \left[t \leq v \leq s \implies x < \hat{\ell}^{DA}(a, v) \right]$$

and **well-qualified at** s iff in addition

$$x \in E_s \quad \text{and} \quad \mathbf{m}_{x,t}^a \geq \hat{R}_s(a+1).$$

Note that once $\mathbf{m}_{x,t}^a$ is disqualified at some $s \geq t$ where $\hat{\ell}^{DA}(a, s) \leq x$, it is never qualified again.

We can now describe the construction as follows: at stage s enumerate into A_{s+1} all numbers of one of the forms

- (i) $\langle 0, x \rangle$ for $x \in D_{s+1} \sim D_s$
- (ii) $\mathbf{m}_{x,t}^a$, for $a, x, t < s$, which are well-qualified at s .

Because the restraints are the same, we have again exactly as in Theorem 8.5.13 that for all a, b, s and y ,

- (A) if $s \in \mathbb{T}_b^A$ and $\widehat{\{b\}}_s^{A_s}(b) \downarrow$, then $\widehat{\{b\}}_s^{A_s}(b) \downarrow$ correctly;
- (B) $\lim_{s \in \mathbb{T}_{<a}^A} \hat{R}_s(a) =: \hat{R}(a)$ exists and is finite;
- (C) if $s \in \mathbb{T}_{<a}^A$, then for all $t \geq s$, $\hat{R}_s(a) \leq \hat{R}_t(a)$.

Suppose, contrary to (2), that for some a , $\{a\}^D = A$, so $\ell^{DA}(a) = \infty$. Set for all x ,

$$g(x) := \text{least } t \left[t \in \mathbb{T}^D \quad \text{and} \quad \hat{\ell}^{DA}(a, t) > x \quad \text{and} \quad \mathbf{m}_{x,t}^a > \hat{R}(a+1) \right].$$

By (B) and $\widehat{\text{CL}\hat{L}}(\text{iii})$, $g(x)$ is well-defined and is a correctness function for a . It follows that for all $s \geq g(x)$, $\mathbf{m}_{x,g(x)}^a$ is qualified at s . Furthermore, $\mathbf{m}_{x,g(x)}^a$ is infinitely often (at $s \in \mathbb{T}_{<a+1}^A$) unrestrained, so eventually well-qualified and hence enumerated into A iff $x \in E$. In other words, for all x ,

$$x \in E \quad \iff \quad \mathbf{m}_{x,g(x)}^a \in A.$$

Since $\mathsf{T}^D \leq_T D$, g is D -recursive and as in the sketch above, we have $E \leq_T D$, contrary to hypothesis. Hence (2) is satisfied.

To establish (3) and (6), we shall show as in previous proofs that

- (3) there exists a recursive function α such that for all a , $\{\alpha(a)\}^E = A^{[a]}$; hence $A \leq_T E$;
- (6) there exists a D' -recursive function β such that for all a , $\{\beta(a)\}^D = A^{[a]}$; hence $A' \leq_T D'$.

For (3), let $\alpha(0)$ be any E -index for $A^{[0]} = \{\langle 0, y \rangle : y \in D\}$ and assume recursively that $\alpha(b)$ is defined for $b < a+1$. Now, let $\alpha(a+1)$ be an E -index which acts as follows:

$$\{\alpha(a+1)\}^E(\mathbf{m}_{x,t}^a) \simeq \begin{cases} 0, & \text{if } x \notin E; \\ A_{f(a,x,t)+1}(\mathbf{m}_{x,t}^a), & \text{if } x \in E; \end{cases}$$

where for $x \in E$,

$$f(a, x, t) := \text{least } s \geq t \left[x \in E_s \wedge s \in \mathsf{T}_{<a+1}^A \right].$$

To see that this gives the correct result, note that by (C), for all $s \geq f(a, x, t)$, $\hat{R}_{f(a,x,t)}(a+1) \leq \hat{R}_s(a+1)$. Hence if $\mathbf{m}_{x,t}^a$ is well-qualified at some $s \geq f(a, x, t)$, then it is already well-qualified at $f(a, x, t)$.

To see that $\alpha(a)$ is recursively computable from the values $\alpha(b)$, note that since $\mathsf{T}_{<a+1}^A$ is simply defined in terms of $A^{[b]}$ for $b < a+1$, we may compute recursively from the values $\alpha(b)$ for $b < a+1$ an E -index for $\mathsf{T}_{<a+1}^A$.

Towards (6), note first that by (2), for all a , $\ell^{DA}(a) < \infty$. Let $\beta(0)$ be any D -index for $A^{[0]}$, and assume that values $\beta(b)$ for $b < a+1$ are given. Define

$$j(a, x, t) := \begin{cases} \text{least } s \geq t \left[\hat{\ell}^{DA}(a, s) = \ell^{DA}(a) \right], & \text{if } x \geq \ell^{DA}(a); \\ \text{least } s \geq t \left[\hat{R}_s(a+1) = \hat{R}(a+1) \right], & \text{if } x < \ell^{DA}(a). \end{cases}$$

j is well-defined by (B) and the corollary to CLL $\hat{\mathbf{L}}$, 8.5.16. Now, let $\beta(a+1)$ be a D -index which acts as follows:

$$\{\beta(a+1)\}^D(\mathbf{m}_{x,t}^a) \simeq \begin{cases} A_{j(a,x,t)+1}(\mathbf{m}_{x,t}^a), & \text{if } x \geq \ell^{DA}(a) \text{ or } t \geq h(a); \\ A(\mathbf{m}_{x,t}^a), & \text{otherwise;} \end{cases}$$

where

$$h(a) := \text{least } t \left[E_t \upharpoonright \ell^{DA}(a) = E \upharpoonright \ell^{DA}(a) \wedge \right. \\ \left. \forall s \geq t \left(\ell^{DA}(a) \leq \hat{\ell}^{DA}(a, s) \wedge \hat{R}(a+1) \leq \hat{R}_s(a+1) \right) \right].$$

To see that this gives the correct result — that is, for all a and y , $\{\beta(a+1)\}^D(y) = A^{[a+1]}(y)$ — consider two cases. Of course, we need only consider y of the form $\mathbf{m}_{x,t}^a$, since no other numbers are enumerated into $A^{[a+1]}$. If $x \geq \ell^{DA}(a)$, then since for each t , $\mathbf{m}_{x,t}^a$ is permanently disqualified at stage $j(a, x, t)$, it will never be enumerated into A at any stage $s > j(a, x, t)$. Suppose now that $x < \ell^{DA}(a)$. If $t < h(a)$, the value is correct by definition. Suppose that $t \geq h(a)$ and $\mathbf{m}_{x,t}^a \in A$. Then for some $s \geq t$, $\mathbf{m}_{x,t}^a$ is well-qualified at s — that is, $x \in E_s$ and $\mathbf{m}_{x,t}^a \geq \hat{R}_s(a+1)$. But since $j(a, x, t) \geq t \geq h(a)$, $\mathbf{m}_{x,t}^a$ is qualified at $j(a, x, t)$, $x \in E_{j(a,x,t)}$ and

$$\mathbf{m}_{x,t}^a \geq \hat{R}_s(a+1) \geq \hat{R}(a+1) = \hat{R}_{j(a,x,t)}(a+1).$$

Hence $\mathbf{m}_{x,t}^a$ is well-qualified at $j(a, x, t)$, so $\mathbf{m}_{x,t}^a \in A_{j(a,x,t)+1}$.

Finally, we need to verify that $\beta(a)$ is D' -computable from the values $\beta(b)$ for $b < a$. First, since $A \leq_T \emptyset' \leq_T D'$, the value of $\ell^{DA}(a)$ can be D' -computed. Let

$$s_a := \text{least } s \left[(\forall t > s) t \in \mathbf{T}_{<a+1}^A \implies \hat{R}_t(a+1) \leq \hat{R}_s(a+1) \right].$$

This exists by (C) and as in the proof of (3), from the values $\beta(b)$ for $b < a+1$ we can recursively compute a D -index for $\mathbf{T}_{<a+1}^A$ and hence, by the one-quantifier principle, D' -compute s_a . Since by (B) and (C), $\hat{R}(a+1) = \hat{R}_{s_a}(a+1)$, it follows that we can D' -compute $\hat{R}(a+1)$. The function

$$f(a, y, z) := \text{least } t \left[E_t \upharpoonright y = E \upharpoonright y \wedge \forall s \geq t \left(y \leq \hat{\ell}^{DA}(a, s) \wedge z \leq \hat{R}_s(a+1) \right) \right]$$

is $\emptyset'$ -recursive, and $h(a) = f(a, \ell^{DA}(a), \hat{R}(a+1))$, so also $h(a)$ can be D' -computed. Finally, since $A \leq_T D'$, we can D' -compute the values of $A(\mathbf{m}_{x,t}^a)$ for the finitely many exceptional cases $x < \ell^{DA}(a)$ and $t < h(a)$ and hence an appropriate value for $\beta(a+1)$. That (6) $A' \leq_T D'$ now follows as in several previous cases. $\square$

It remains to apply the techniques we have developed in various combinations. First, we have the following, which was historically the first use of this technique.

Sacks Density Theorem 8.6.4 *For any r.e. sets $D <_T E$, there exists an r.e. set A such that $D <_T A <_T E$.*

Proof. Given r.e. sets $D <_T E$, we aim to construct an r.e. set A such that

- | | |
|----------------------|------------------------|
| (1) $D \leq_T A$ | (3) $A \leq_T E$ |
| (2) $A \not\leq_T D$ | (4) $E \not\leq_T A$. |

Thus we need to satisfy the requirements

$$\begin{aligned} Q_a : \quad & \{a\}^D \neq A \\ N_a : \quad & \{a\}^A \neq E \end{aligned}$$

subject to the restraints

$$R_s^+(a) = \max\{r^+(b, s) : b < a\},$$

where

$$r^+(b, s) = \max\{\hat{u}(A_s; b, z, s) : z \leq \ell^{+AE}(b, s)\}.$$

(As usual we write simply ℓ^+ for ℓ^{+AE} as defined in 8.5.18.) The positive requirements Q_a are, of course, designed to guarantee (2), (1) is handled as usual by arranging that $A^{[0]} = \{\langle 0, y \rangle : y \in D\}$, and the restraints are used to guarantee (3) and (4).

We now say as in the preceding, with the restraint function R^+ in place of $\hat{R}$, that for all a, x and t , $\mathbf{m}_{x,t}^a := \langle a+1, \langle x, t \rangle \rangle$ is **qualified at s** iff

$$t \leq s \quad \text{and} \quad \forall v \left[t \leq v \leq s \implies x < \hat{\ell}^{DA}(a, v) \right]$$

and **well-qualified at s** iff in addition

$$x \in E_s \quad \text{and} \quad \mathbf{m}_{x,t}^a \geq R_s^+(a+1).$$

We can now describe the construction as follows: at stage s enumerate into A_{s+1} all numbers of one of the forms

- (i) $\langle 0, y \rangle$ for $y \in D_{s+1} \sim D_s$,
- (ii) $\mathbf{m}_{x,t}^a$, for $a, x, t < s$, which are well-qualified at s .

Exactly as in the proof of Theorem 8.5.21, we have that for all a, b, s and y

- (A) if $s \in T_b^A$, $y \leq \ell(b)$ and $\ell^+(b, s) \geq y$, then $\hat{\ell}(b, s) \geq y$ correctly;
- (B1) if $\ell(b) < \infty$, then $\lim_{s \in T_b^A} \ell^+(b, s) =: \ell^+(b)$ exists and is finite; $\ell^+(b) \geq \ell(b)$;
- (B2) if for all $b < a$, $\ell(b) < \infty$, then $\lim_{s \in T_{<a}^A} R_s^+(a) =: R^+(a)$ exists and is finite;
- (C1) if $s \in T_b^A$, then for all $t \geq s$, $\ell^+(b, s) \leq \ell^+(b, t)$ and $r^+(b, s) \leq r^+(b, t)$;
- (C2) if $s \in T_{<a}^A$, then for all $t \geq s$, $R_s^+(a) \leq R_t^+(a)$.

Now (1) and (3) are proved exactly as in the preceding theorem. To complete the proof note first that immediately from the construction, $A^{[0]}$

and $\mathsf{T}_{\leq 0}^A$ are D -recursive and hence as in other proofs, N_0 is satisfied. We establish by induction that for all a ,

(2a) Q_a is satisfied;

(4a) $A^{[a+1]}$ and T_{a+1}^A are D -recursive and N_{a+1} is satisfied.

Assume as induction hypothesis that (2b) and (4b) hold for all $b < a$. Hence for all $b < a + 1$, $\ell(b) < \infty$ and thus by (B2), $\lim_{s \in \mathsf{T}_{\leq a}^A} R_s^+(a + 1) =: R^+(a + 1)$ exists and is finite. Suppose towards a contradiction that (2a) fails, so $\ell^{DA}(a) = \infty$. Set for all x ,

$$g(x) := \text{least } t \left[t \in \mathsf{T}^D \text{ and } \hat{\ell}^{DA}(a, t) > x \text{ and } \mathbf{m}_{x,t}^a \geq R^+(a + 1) \right].$$

By (B2) and CLL(iii), $g(x)$ is well-defined and is a correctness function for a . It follows that for all $s \geq g(x)$, $\mathbf{m}_{x,g(x)}^a$ is qualified at s . Furthermore, $\mathbf{m}_{x,g(x)}^a$ is infinitely often (at $s \in \mathsf{T}_{< a+1}^A$) unrestrained, so eventually well-qualified and hence enumerated into A iff $x \in E$. In other words, for all x ,

$$x \in E \iff \mathbf{m}_{x,g(x)}^a \in A.$$

Since $\mathsf{T}^D \leq_T D$, g is D -recursive, and as in the sketch above we have $E \leq_T D$, contrary to hypothesis. Hence (2a) is satisfied.

Towards (4a), we first argue that $A^{[a+1]}$ is D -recursive. This argument is essentially the argument for (6) in the proof of Theorem 8.6.3 without the complication of evaluating the complexity of computing the index $\beta(a + 1)$ — to emphasize this simplification we change the notation slightly. Explicitly, we have from (2a) that $\ell^{DA}(a) < \infty$, and we define

$$j_a(x, t) := \begin{cases} \text{least } s \geq t \left[\hat{\ell}^{DA}(a, s) = \ell^{DA}(a) \right], & \text{if } x \geq \ell^{DA}(a); \\ \text{least } s \geq t \left[R_s^+(a + 1) = R^+(a + 1) \right], & \text{if } x < \ell^{DA}(a). \end{cases}$$

j_a is well-defined by (B2) and the corollary to CLL⁺ and is recursive: the numbers $\ell^{DA}(a)$ and $R^+(a + 1)$ occur simply as fixed parameters. Now, for each a , let β_a be the D -recursive function

$$\beta_a(\mathbf{m}_{x,t}^a) := \begin{cases} A_{j_a(x,t)+1}(\mathbf{m}_{x,t}^a), & \text{if } x \geq \ell^{DA}(a) \text{ or } t \geq t_a; \\ A(\mathbf{m}_{x,t}^a), & \text{otherwise;} \end{cases}$$

where

$$t_a := \text{least } t \left[E_t \restriction \ell^{DA}(a) = E \restriction \ell^{DA}(a) \wedge \forall s \geq t \left(\ell^{DA}(a) \leq \hat{\ell}^{DA}(a, s) \wedge R^+(a + 1) \leq R_s^+(a + 1) \right) \right].$$

Again, t_a and the finitely many values of A used in the second clause are merely fixed parameters. To see that this gives the correct result — that is, for all a and y , $\beta_a(x) = A^{[a+1]}(y)$ — consider two cases. Of course, we need only consider y of the form $\mathbf{m}_{x,t}^a$, since no other numbers are ever enumerated into $A^{[a+1]}$. If $x \geq \ell^{DA}(a)$, then since for each t , $\mathbf{m}_{x,t}^a$ is permanently disqualified at stage $j_a(x, t)$, it will never be enumerated into A at any stage $s > j_a(x, t)$. Suppose now that $x < \ell^{DA}(a)$. If $t < t_a$, the value is correct by definition. Suppose that $t \geq t_a$ and $\mathbf{m}_{x,t}^a \in A$. Then for some $s \geq t$, $\mathbf{m}_{x,t}^a$ is well-qualified at s — that is, $x \in E_s$ and $\mathbf{m}_{x,t}^a \geq R_s^+(a+1)$. But since $j_a(x, t) \geq t \geq t_a$, $\mathbf{m}_{x,t}^a$ is qualified at $j_a(x, t)$, $x \in E_{j_a(x,t)}$ and

$$\mathbf{m}_{x,t}^a \geq R_s^+(a+1) \geq R^+(a+1) = R_{j_a(x,t)}^+(a+1).$$

Hence $\mathbf{m}_{x,t}^a$ is well-qualified at $j_a(x, t)$, so $\mathbf{m}_{x,t}^a \in A_{j_a(x,t)+1}$. Note that for $a > 0$, β_a is in fact a *recursive* function; the oracle D is needed only for the 0th column.

It now follows as usual that also T_{a+1}^A is D -recursive and that N_{a+1} is satisfied: otherwise using (A) and Corollary 8.5.20,

$$h(y) := \text{least } s \ [s \in \mathsf{T}_{a+1}^A \quad \text{and} \quad \ell^+(a+1, s) \geq y]$$

is a D -recursive correctness function for a , and thus $E_{h(y+1)}(y) = E(y)$, so by stage bounding, E is D -recursive, contrary to hypothesis. $\square$

Although in this version, we do not need the evaluation of the complexity of computing a D -index $\beta(a)$ for the function β_a as in both the preceding and following proofs, we could carry this out exactly as above and come to the additional conclusion that the set A constructed here also satisfies (6) $A' \leq_T D'$, hence $A' \equiv_T D'$.

Finally, we combine all of the techniques of [Sections 8.4–8.6](#) to prove the

Jump Interpolation Theorem 8.6.5 *For any r.e. sets $D <_T E$ and any E-r.e. set S such that $D' \leq_T S$, there exists an r.e. set A such that $D <_T A <_T E$ and $A' \equiv_T S$.*

Proof. Given D , E and S as in the hypothesis, we shall construct A to satisfy all of the conditions

$$\begin{array}{lll} (1) \ D \leq_T A & (3) \ A \leq_T E & (5) \ S \leq_T A' \\ (2) \ A \not\leq_T D & (4) \ E \not\leq_T A & (6) \ A' \leq_T S. \end{array}$$

The main new complexity here is that because there are two sorts of positive requirements (for conditions (2) and (5)), we need to interleave action to

satisfy them. This requires some minor adjustments in some of the earlier strategies. We now choose $F \leq_T E$ such that

$$a \in S \implies F^{[2a+1]} \text{ is finite} \quad \text{and} \quad a \notin S \implies F^{[2a+1]} = \omega^{[2a+1]}$$

and aim to satisfy for all a ,

$$\begin{aligned} P_0 : \quad & A^{[0]} = \{ \langle 0, y \rangle : y \in D \} \\ P_{2a+1} : \quad & F^{[2a+1]} \sim A^{[2a+1]} \text{ is finite} \\ P_{2a+2} : \quad & \{a\}^D \neq A \\ N_a : \quad & \{a\}^A \neq E. \end{aligned}$$

Clearly satisfaction of P_0 guarantees $D \leq_T A$, and satisfaction of all P_{2a} for $a > 0$ guarantees $S \leq_T A'$. The restraint function R_s^+ is defined exactly as in the preceding proof. We take the markers $\mathbf{m}_{x,t}^a := \langle 2a+2, \langle x, t \rangle \rangle$, and we say that $\mathbf{m}_{x,t}^a$ is qualified at s as in the previous proof and well-qualified at s iff in addition

$$x \in E_s \quad \text{and} \quad \mathbf{m}_{x,t}^a \geq R_s^+(2a+2).$$

We can now describe the construction as follows: at stage s enumerate into A_{s+1} all numbers of one of the forms

- (i) $\langle 0, y \rangle$ for $y \in D_{s+1} \sim D_s$,
- (ii) $\langle 2a+1, y \rangle \in F_s$ such that $\langle 2a+1, y \rangle > R_s^+(2a+1)$ and $\langle 2a+1, y \rangle \notin A_s$,
- (iii) $\mathbf{m}_{x,t}^a$ for $a, x, t < s$ which are well-qualified at s .

Properties (A), (B1), (B2), (C1) and (C2) again follow exactly as in Theorem 8.5.21. We establish (2)–(6) by defining a recursive function α and an S -recursive function β and proving by induction that for all k ,

- $(2+5k)$ P_k is satisfied;
- $(3k)$ $\{\alpha(k)\}^E = A^{[k]}$;
- $(4+6k)$ $\{\beta(k)\}^D = A^{[k]}$ and N_k is satisfied.

For all k , the second clause of $(4+6k)$ follows from the first as before: $A^{[k]}$ is D -recursive, hence so is $\mathbb{T}_k^A$, and if $\lim_{s \rightarrow \infty} \ell^+(k, s) = \infty$ and

$$h(y) := \text{least } s \ [s \in \mathbb{T}_k^A \text{ and } \ell^+(k, s) \geq y],$$

h is a D -recursive correctness function for k , and thus $E_{h(y+1)}(y) = E(y)$, so by stage bounding, E is D -recursive, contrary to hypothesis.

Assume as induction hypothesis that $(2+5l)$ and $(4+6l)$ hold for all $l < k$. Hence for all $l < k$, $\ell(l) < \infty$ and thus by (B2), $\lim_{s \in \mathbb{T}_{<k}^A} R_s^+(a) =: R^+(k)$ exists and is finite. If $k = 0$, $R^+(0) = 0$ and all numbers of the form $\langle 0, y \rangle$ for $y \in D$ will be enumerated into $A^{[0]}$. Hence P_0 is satisfied and we may take $\alpha(0)$ ($\beta(0)$) to be any E (D)-index for $A^{[0]}$.

Suppose that $k = 2a + 1$. Then all elements of $F^{[2a+1]}$ greater than $R^+(2a + 1)$ will eventually be enumerated into A , so P_{2a+1} is satisfied. Let $\alpha(2a + 1)$ be an E -index which acts as follows:

$$\{\alpha(2a + 1)\}^E(\langle 2a + 1, y \rangle) \simeq \begin{cases} 0, & \text{if } \langle 2a + 1, y \rangle \notin F^{[2a+1]}; \\ A_{f(a,y)+1}(\langle 2a + 1, y \rangle), & \text{otherwise;} \end{cases}$$

where for $\langle 2a + 1, y \rangle \in F^{[2a+1]}$,

$$f(a, y) := \text{least } s \left[\langle 2a + 1, y \rangle \in F_s^{[2a+1]} \quad \text{and} \quad s \in \mathbb{T}_{<2a+1}^A \right].$$

This gives the correct result because, by (C2), for all $s \geq \langle 2a + 1, y \rangle$, $R_{f(a,y)}^+(a) \leq R_s^+(a)$, so if $\langle 2a + 1, y \rangle \in A_s$, then already $\langle 2a + 1, y \rangle \in A_{f(a,y)+1}$.

Similarly, we take $\beta(2a + 1)$ to be an index for the computation described by $\alpha(2a + 1)$ above with the appeals to the E -oracle to decide membership in $F^{[2a+1]}$ replaced by use of the function $\{\varepsilon(2a + 1)\} = F^{[2a+1]}$ for an S -recursive function ε as described in the proof of Theorem 8.5.13.

Suppose now that $k = 2a + 2$. P_{2a+2} is satisfied as in Theorem 8.6.4: if $\{a\}^A = D$, then the function

$$g(x) := \text{least } t \left[t \in \mathbb{T}^D \quad \text{and} \quad \hat{\ell}^{DA}(a, t) > x \quad \text{and} \quad \mathbf{m}_{x,t}^a \geq R^+(2a + 2) \right]$$

is D -recursive and for all x ,

$$x \in E \quad \Longleftrightarrow \quad \mathbf{m}_{x,g(x)}^a \in A,$$

so $E \leq_T D$, contrary to hypothesis. We take $\alpha(2a + 2)$ to be an E -index such that

$$\{\alpha(2a + 2)\}^E(\mathbf{m}_{x,t}^a) \simeq \begin{cases} 0, & \text{if } x \notin E; \\ A_{f(a,x,t)+1}(\mathbf{m}_{x,t}^a), & \text{if } x \in E; \end{cases}$$

where for $x \in E$,

$$f(a, x, t) := \text{least } s \geq t \left[x \in E_s \quad \wedge \quad s \in \mathbb{T}_{<2a+2}^A \right].$$

This gives the correct result because by (C2) for all $s \geq f(a, x, t)$,

$$R_{f(a,x,t)}^+(a) \leq R_s^+(a),$$

so if $\mathbf{m}_{x,t}^a$ is well-qualified at some $s \geq f(a, x, t)$, then it is already well-qualified at $f(a, x, t)$. $\alpha(a)$ is recursively computable from the values $\alpha(b)$, because as before we may compute recursively from the values $\alpha(l)$ for $l < 2a + 2$ an E -index for $\mathbf{T}_{<2a+2}^A$.

To specify $\beta(2a + 2)$, we proceed as in the proof of Theorem 8.6.3, define a D -recursive function

$$j(a, x, t) := \begin{cases} \text{least } s \geq t \left[\hat{\ell}^{DA}(a, s) = \ell^{DA}(a) \right], & \text{if } x \geq \ell^{DA}(a); \\ \text{least } s \geq t \left[R_s^+(2a + 2) = R^+(2a + 2) \right], & \text{if } x < \ell^{DA}(a); \end{cases}$$

and choose $\beta(2a + 2)$ to be a D -index such that

$$\{\beta(2a + 2)\}^D(\mathbf{m}_{x,t}^a) \simeq \begin{cases} A_{j(a,x,t)+1}(\mathbf{m}_{x,t}^a), & \text{if } x \geq \ell^{DA}(a) \quad \text{or} \quad t \geq h(a); \\ A(\mathbf{m}_{x,t}^a), & \text{otherwise;} \end{cases}$$

where

$$h(a) := \text{least } t \left[E_t \upharpoonright \ell^{DA}(a) = E \upharpoonright \ell^{DA}(a) \wedge \forall s \geq t \left(\ell^{DA}(a) \leq \hat{\ell}^{DA}(a, s) \wedge R^+(2a + 2) \leq R_s^+(2a + 2) \right) \right].$$

That such an index can be D' - and hence S -computed from the values $\beta(l)$ for $l < 2a + 2$ is shown as in the corresponding part of Theorem 8.6.3. $\square$

References

Numbers in square brackets indicate pages on which the reference is cited.

- [Ac-Si-Wa] PETER ACZEL, HAROLD SIMMONS, STANLEY S. WAINER, *Proof theory*, Papers from the Leeds Summer School and Conference, Cambridge University Press, Cambridge, 1992, x+306 pp. ISBN: 0-521-41413-X.
- [Ba-Fe] J. BARWISE AND S. FEFERMAN (EDS), *Model-theoretic logics*, Perspectives in Mathematical Logic, Springer-Verlag, New York, 1985, xviii+893 pp. ISBN: 0-387-90936-2. [291]
- [Ba75] JON BARWISE, *Admissible sets and structures: An approach to definability theory*, Perspectives in Mathematical Logic, Springer-Verlag, Berlin-New York, 1975, xiii+394 pp. ISBN: 3-540-07451-1, 0-387-07451-1. [308]
- [Ba77] JON BARWISE (ED.), *Handbook of mathematical logic*, Studies in Logic and the Foundations of Mathematics 90, North-Holland Publishing Co., Amsterdam-New York-Oxford, 1977, xi+1165 pp. ISBN: 0-7204-2285-X. [30, 61, 71, 308, 325, 430, 452, 470, 523]
- [Bal] JOHN T. BALDWIN, *Fundamentals of stability theory*, Perspectives in Mathematical Logic, Springer-Verlag, Berlin, 1988, xiv+447 pp. ISBN 3-540-15298-9. [719]

- [Bo-Bu-Je] GEORGE S. BOOLOS, JOHN P. BURGESS, RICHARD C. JEFFREY, *Computability and logic*, Fourth edition, Cambridge University Press, Cambridge, 2002, xii+356 pp. ISBN: 0-521-80975-4.
- [Bo-Gr-Gu] EGON BOERGER, ERICH GRAEDEL, YURI GUREVICH, *The classical decision problem*, Reprint of the 1997 original, Universitext., Springer-Verlag, Berlin, 2001, x+482 pp. ISBN: 3-540-42324-9. [452]
- [Bue] STEVEN BUECHLER, *Essential stability theory*, Perspectives in Mathematical Logic, Springer-Verlag, Berlin, 1996, xiv+355 pp. ISBN: 3-540-61011-1. [719]
- [Bus] SAMUEL R. BUSS (ED.), *Handbook of proof theory*, Studies in Logic and the Foundations of Mathematics 137, North-Holland Publishing Co., Amsterdam, 1998, x+811 pp. ISBN 0-444-89840-9. [216]
- [Ch-Ke] C. C. CHANG, H. J. KEISLER, *Model theory*, Third edition, Studies in Logic and the Foundations of Mathematics 73, North-Holland Publishing Co., Amsterdam, 1990, xvi+650 pp. ISBN: 0-444-88054-2. [71, 136, 268, 668, 695, 719, 732]
- [Coo] S. BARRY COOPER, *Computability theory*, Chapman Hall CRC, Boca Raton, FL, 2004, x+409 pp. ISBN: 1-58488-237-9. [768]
- [Cut] NIGEL CUTLAND, *Computability: An introduction to recursive function theory*, Cambridge University Press, Cambridge-New York, 1980, x+251 pp. ISBN: 0-521-22384-9; 0-521-29465-7. [61, 340, 406, 443]
- [Dav] MARTIN DAVIS (ED.), *The undecidable: Basic papers on undecidable propositions, unsolvable problems and computable functions*, Raven Press, Hewlett, NY, 1965, ii+440 pp. [325, 443, 470]
- [De84] KEITH J. DEVLIN, *Constructibility*, Perspectives in Mathematical Logic, Springer-Verlag, Berlin, 1984, xi+425 pp. ISBN: 3-540-13258-9. [575]
- [De93] KEITH J. DEVLIN, *The joy of sets: Fundamentals of contemporary set theory*, Second edition, Undergraduate Texts in Mathematics, Springer-Verlag, New York, 1993, x+192 pp. ISBN: 0-387-94094-4.
- [Dic] M. A. DICKMANN, *Large infinitary languages: Model theory*, Studies in Logic and the Foundations of Mathematics 83, North-Holland Publishing Co., Amsterdam-Oxford; American Elsevier Publishing Co., Inc., New York, 1975, xv+464 pp. ISBN: 0-7204-2200-0, 0-444-10622-7. [298, 300, 308, 668]

- [Dr-Si] F. R. DRAKE, D. SINGH, *Intermediate set theory*, John Wiley & Sons, Ltd., Chichester, 1996, x+234 pp. ISBN: 0-471-96496-4.
- [Dra] FRANK R. DRAKE, *Set theory: An introduction to large cardinals*, Studies in Logic and the foundations of mathematics 76, North-Holland Publishing Co., Amsterdam, 1974, xii+351 pp. ISBN: 0-7204-2200-0, 0-444-10535-2. [619]
- [Eb-Fl-Th] H.-D. EBBINGHAUS, J. FLUM, W. THOMAS, *Mathematical logic*, Second edition, Undergraduate Texts in Mathematics, Springer-Verlag, New York, 1994, x+289 pp. ISBN: 0-387-94258-0. [20, 61, 96, 291, 308, 732]
- [Eb-Fl] HEINZ-DIETER EBBINGHAUS, JÖRG FLUM, *Finite model theory*, Second edition, Perspectives in Mathematical Logic, Springer-Verlag, Berlin, 1999, xiv+360 pp. ISBN: 3-540-65758-4. [292]
- [En01] HERBERT B. ENDERTON, *A mathematical introduction to logic*, Second edition, Harcourt/Academic Press, Burlington, MA, 2001, xii+317 pp. ISBN: 0-12-238452-0. [20, 30, 61, 96, 171, 291, 324]
- [En77] HERBERT B. ENDERTON, *Elements of set theory*, Academic Press [Harcourt Brace Jovanovich, Publishers], New York-London, 1977, xiv+279 pp. ISBN: 0-12-238440-7. [470, 495, 508]
- [Go-Ju] MARTIN GOLDSTERN, HAIM JUDAH, *The incompleteness phenomenon: A new course in mathematical logic*, with a foreword by Saharon Shelah, A K Peters, Ltd., Wellesley, MA, 1995, xiv+247 pp. ISBN: 1-56881-029-6.
- [Gri] EDWARD R. GRIFFOR (ED.), *Handbook of computability theory*, Studies in Logic and the Foundations of Mathematics 140, North-Holland Publishing Co., Amsterdam, 1999, xii+727 pp. ISBN 0-444-89882-4. [340, 443]
- [Hin] PETER G. HINMAN, *Recursion-theoretic hierarchies*, Perspectives in Mathematical Logic, Springer-Verlag, Berlin, 1978, xii+480 pp. ISBN: 3-540-07904-1. [340, 435, 651]
- [Ho-Ru] PAUL HOWARD AND JEAN E. RUBIN, *Consequences of the axiom of choice*, Mathematical Surveys and Monographs 59, American Mathematical Society, Providence, RI, 1998, viii+432 pp. ISBN: 0-8218-0977-6. [523]
- [Ho85] WILFRID HODGES, *Building models by games*, London Mathematical Society Student Texts 2, Cambridge University Press, Cambridge, 1985, vi+311 pp. ISBN: 0-521-26897-4.

- [Ho93] WILFRID HODGES, *Model theory*, Encyclopedia of Mathematics and its Applications 42, Cambridge University Press, Cambridge, 1993, xiv+772 pp. ISBN: 0-521-30442-3. [268, 668, 695, 719]
- [Ho97] WILFRID HODGES, *A shorter model theory*, Cambridge University Press, Cambridge, 1997, x+310 pp. ISBN: 0-521-58713-1. [268]
- [Hod] RICHARD E. HODEL, *An introduction to mathematical logic*, PWS Publishing Company, Boston, 1995, xiv+491 pp. ISBN: 0-534-94440-X. [20, 216]
- [Hu-Lo] ALBERT E. HURD, PETER A. LOEB, *An introduction to non-standard real analysis*, Pure and Applied Mathematics 118, Academic Press, Inc., Orlando, FL, 1985, xii+232 pp. ISBN: 0-12-362440-1. [268]
- [Hun] THOMAS W. HUNGERFORD, *Algebra*, Graduate Texts in Mathematics 73, Springer-Verlag, New York, 1974, xxiii+502 pp. ISBN: 0-387-90518-9, 3-540-90518-9. [246]
- [Hr-Je] KAREL HRBACEK, THOMAS JECH, *Introduction to set theory*, Third edition. Monographs and Textbooks in Pure and Applied Mathematics 220, Marcel Dekker, Inc., New York, 1999, xii+291 pp. ISBN: 0-8247-7915-0.
- [Jec] THOMAS JECH, *Set theory*, Springer Monographs in Mathematics, third millenium edition, Springer-Verlag, Berlin, 2003, xiv+769 pp. ISBN 3-540-44085-2. [470, 508, 523, 548, 575, 593, 603, 619]
- [Ju-We96] WINFRIED JUST, MARTIN WEESE, *Discovering modern set theory I: Set-theoretic tools for every mathematician*, Graduate Studies in Mathematics 8, American Mathematical Society, Providence, RI, 1996, xviii+210 pp. ISBN: 0-8218-0266-6.
- [Ju-We97] WINFRIED JUST, MARTIN WEESE, *Discovering modern set theory II: Set-theoretic tools for every mathematician*, Graduate Studies in Mathematics 18, American Mathematical Society, Providence, RI, 1997, xiv+224 pp. ISBN: 0-8218-0528-2.
- [Kan] AKIHIRO KANAMORI, *The higher infinite*, Springer Monographs in Mathematics, Second edition, Springer-Verlag, Berlin, 2003, xxii+536 pp. ISBN: 3-540-00384-3. [619]
- [Kar] CAROL R. KARP, *Languages with expressions of infinite length*, North-Holland Publishing Co., Amsterdam, 1964, xix+183 pp. [300]

- [Kay] RICHARD KAYE, *Models of Peano arithmetic*, Oxford Logic Guides 15, Oxford University Press, New York, 1991, x+292 pp. ISBN: 0-19-853213-X. [171, 388]
- [Kec] ALEXANDER S. KECHRIS, *Classical descriptive set theory*, Graduate Texts in Mathematics 156, Springer-Verlag, New York, 1995, xviii+402 pp. ISBN: 0-387-94374-9. [651]
- [Kei] H. JEROME KEISLER, *Model theory for infinitary logic*, Studies in Logic and the Foundations of Mathematics 62, North-Holland Publishing Co., Amsterdam-London, 1971, x+208 pp. ISBN: 0-7204-2258-2. [71, 308, 668]
- [Kl02] STEPHEN C. KLEENE, *Mathematical logic*, Reprint of the 1967 original, Dover Publications, Inc., Mineola, NY, 2002, xiv+398 pp. ISBN: 0-486-42533-9.
- [Kl52] STEPHEN C. KLEENE, *Introduction to metamathematics*, D. Van Nostrand Co., Inc., New York, 1952, x+550 pp. [9, 430]
- [Kop] SABINE KOPPELBERG, *Handbook of Boolean algebras Vol. 1*, Edited by J. Donald Monk and Robert Bonnet, North-Holland Publishing Co., Amsterdam, 1989, xx+312 pp. ISBN 0-444-70261-X. [71]
- [Kun] KENNETH KUNEN, *Set theory: An introduction to independence proofs*, Reprint of the 1980 original, Studies in Logic and the Foundations of Mathematics 102, North-Holland Publishing Co., Amsterdam, 1983, xvi+313 pp. ISBN: 0-444-86839-9. [548, 575, 593, 603]
- [Ler] MANUEL LERMAN, *Degrees of unsolvability: Local and global theory*, Perspectives in Mathematical Logic, Springer-Verlag, Berlin, 1983, xiii+307 pp. ISBN: 3-540-12155-2.
- [Lev] AZRIEL LEVY, *Basic set theory*, Reprint of the 1979 original, Dover Publications, Inc., Mineola, NY, 2002, xiv+398 pp. ISBN: 0-486-42079-5. [523]
- [Ma-Me-Pi] D. MARKER, M. MESSMER, A. PILLAY, *Model theory of fields*, Lecture Notes in Logic 5, Springer-Verlag, Berlin, 1996, x+154 pp. ISBN: 3-540-60741-2. [248]
- [Mal] JEROME MALITZ, *Introduction to mathematical logic: Set theory, computable functions, model theory*, Undergraduate Texts in Mathematics, Springer-Verlag, New York-Heidelberg, 1979, x+198 pp. ISBN: 0-387-90346-1. [203]
- [Mar] DAVID MARKER, *Model theory: An introduction*, Graduate Texts in Mathematics 217, Springer-Verlag, New York, 2002, viii+342 pp. ISBN: 0-387-98760-6. [248, 388, 695, 719]

- [Mat] YURI V. MATIYASEVICH, *Hilbert's tenth problem*, Foundations of Computing Series, MIT Press, Cambridge, MA, 1993, xxiv+264 pp. ISBN: 0-262-13295-8. [406]
- [Men] ELLIOTT MENDELSON, *Introduction to mathematical logic*, Fourth edition, Chapman & Hall, London, 1997, x+440 pp. ISBN: 0-412-80830-7. [20, 96, 216]
- [Mo74] YIANNIS N. MOSCHOVAKIS, *Elementary induction on abstract structures*, Studies in Logic and the Foundations of Mathematics 77, North-Holland Publishing Co., Amsterdam-London; American Elsevier Publishing Co., Inc., New York, 1974, x+218 pp. ISBN: 0-7204-2200-0, 0-444-10537-9. [30]
- [Mo80] YIANNIS N. MOSCHOVAKIS, *Descriptive set theory*, Studies in Logic and the Foundations of Mathematics 100, North-Holland Publishing Co., Amsterdam-New York, 1980, xii+637 pp. ISBN: 0-444-85305-7. [651]
- [Mo94] YIANNIS N. MOSCHOVAKIS, *Notes on set theory*, Undergraduate Texts in Mathematics, Springer-Verlag, New York, 1994, xiv+272 pp. ISBN: 0-387-94180-0. [470, 495]
- [Mon] J. DONALD MONK, *Mathematical logic*, Graduate Texts in Mathematics 37, Springer-Verlag, New York-Heidelberg, 1976, x+531 pp. ISBN: 0-387-90170-1, 3-540-90170-1. [71, 171, 308, 430, 446, 448, 452, 484]
- [Moo] GREGORY H. MOORE, *Zermelo's axiom of choice: Its origins, development, and influence*, Studies in the History of Mathematics and Physical Sciences 8, Springer-Verlag, New York, 1982, xiv+410 pp. ISBN: 0-387-90670-3. [523]
- [Na-Ne] ERNEST NAGEL, JAMES R. NEWMAN, *Gödel's proof*, Revised edition with a new foreword by Douglas R. Hofstadter, New York University Press, New York, 2001, xxvi+129 pp. ISBN: 0-8147-5816-9.
- [Ne-Sh] ANIL NERODE, RICHARD SHORE, *Logic for applications*, Second edition, Graduate Texts in Computer Science, Springer-Verlag, New York, 1997, xiv+456 pp. ISBN: 0-387-94893-7.
- [Od89] P.G. ODIFREDDI, *Classical recursion theory*, Studies in Logic and the Foundations of Mathematics 125, North-Holland Publishing Co., Amsterdam, 1989, xviii+668 pp. ISBN: 0-444-87295-7. [340, 435, 443, 752]
- [Od99] P. G. ODIFREDDI, *Classical recursion theory, Vol. II*, Studies in Logic and the Foundations of Mathematics 143, North-Holland Publishing Co., Amsterdam, 1999, xvi+949 pp. ISBN: 0-444-50205-X. [340, 752, 768]

- [Ott] MARTIN OTTO, *Bounded variable logics and counting: A study in finite models*, Lecture Notes in Logic 9, Springer-Verlag, Berlin, 1997, x+183 pp. ISBN: 3-540-62037-0. [292]
- [Pi83] ANAND PILLAY, *An introduction to stability theory*, Oxford Logic Guides 8, The Clarendon Press, Oxford University Press, New York, 1983, xii+146 pp. ISBN: 0-19-853186-9. [719]
- [Pi96] ANAND PILLAY, *Geometric stability theory*, Oxford Logic Guides 32, Oxford Science Publications, The Clarendon Press, Oxford University Press, New York, 1996, x+361 pp. ISBN: 0-19-853437-X. [719]
- [Rog] HARTLEY ROGERS, JR., *Theory of recursive functions and effective computability*, Second edition, MIT Press, Cambridge, MA, 1987, xxii+482 pp. ISBN: 0-262-68052-1. [61, 340, 752]
- [Rot] PHILIPP ROTHMALER, *Introduction to model theory*, Algebra, Logic and Applications 15, Gordon and Breach Science Publishers, Amsterdam, 2000, xvi+305 pp. ISBN: 90-5699-287-2. [268]
- [Ru-Ru] HERMAN RUBIN, JEAN E. RUBIN, *Equivalents of the axiom of choice II*, Studies in Logic and the Foundations of Mathematics 116, North-Holland Publishing Co., Amsterdam, 1985, xxviii+322 pp. ISBN: 0-444-87708-8. [523]
- [Sa72] GERALD E. SACKS, *Saturated model theory*, Mathematics Lecture Note Series, W. A. Benjamin, Inc., Reading, MA, 1972, xii+335 pp. ISBN: 0-805-38380-8, 0-805-38381-6.
- [Sa90] GERALD E. SACKS, *Higher recursion theory*, Perspectives in Mathematical Logic, Springer-Verlag, Berlin, 1990, xvi+344 pp. ISBN: 3-540-19305-7.
- [Sh01a] JOSEPH R. SHOENFIELD, *Recursion theory*, Lecture Notes in Logic 1, Association for Symbolic Logic, Urbana, IL, A K Peters, Ltd., Natick, MA, 2001, vi+84 pp. ISBN: 1-56881-149-7.
- [Sh01b] JOSEPH R. SHOENFIELD, *Mathematical logic*, Reprint of the 1973 second printing, Association for Symbolic Logic, Urbana, IL, A K Peters, Ltd., Natick, MA, 2001, viii+344 pp. ISBN: 1-56881-135-7. [20, 61, 96]
- [She] SAHARON SHELAH, *Classification theory and the number of non-isomorphic models*, Second edition, Studies in Logic and the Foundations of Mathematics 92, North-Holland Publishing Co., Amsterdam, 1990, xxxiv+705 pp. ISBN: 0-444-70260-1.
- [Sim] STEPHEN G. SIMPSON, *Subsystems of second order arithmetic*, Perspectives in Mathematical Logic, Springer-Verlag, Berlin, 1999, xiv+445 pp. ISBN: 3-540-64882-8. [292]

- [Smu] RAYMOND M. SMULLYAN, *Gödel's incompleteness theorems*, Oxford Logic Guides 19, The Clarendon Press, Oxford University Press, New York, 1992, xvi+139 pp. ISBN: 0-19-504672-2.
- [Soa] ROBERT I. SOARE, *Recursively enumerable sets and degrees*, Perspectives in Mathematical Logic, Springer-Verlag, Berlin, 1987, xviii+437 pp. ISBN: 3-540-15299-7. [752, 768, 783]
- [Tr-Sch] A. S. TROELSTRA, H. SCHWICHTENBERG, *Basic proof theory*, Second edition, Cambridge Tracts in Theoretical Computer Science 43, Cambridge University Press, Cambridge, 2000, xii+417 pp. ISBN: 0-521-77911-1. [216]
- [Tr-vD88a] A. S. TROELSTRA, D. VAN DALEN, *Constructivism in mathematics I: An introduction*, Studies in Logic and the Foundations of Mathematics 121, North-Holland Publishing Co., Amsterdam, 1988, xx+342+XIV pp. ISBN: 0-444-70266-0; 0-444-70506-6.
- [Tr-vD88b] A. S. TROELSTRA, D. VAN DALEN, *Constructivism in mathematics II: An introduction*, Studies in Logic and the Foundations of Mathematics 123, North-Holland Publishing Co., Amsterdam, 1988, i-xviii and 345–880 and I–LII pp. ISBN: 0-444-70358-6.
- [Var] MOSHE Y. VARDI ET. AL., *Finite model theory and its applications*, Springer-Verlag, New York-Heidelberg, 2005, 300 pp. ISBN: 3-540-00428-9. [292]
- [Vau] ROBERT L. VAUGHT, *Set theory: An introduction.*, Second edition, Birkhuser Boston, Inc., Boston, MA, 1995, x+167 pp. ISBN: 0-8176-3697-8. [508]
- [vHe] JEAN VAN HEIJENOORT, *From Frege to Gödel: A source book in mathematical logic, 1879–1931*, Harvard University Press, Cambridge, MA, 1967, xi+660 pp. [9, 325]

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Symbol Index

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$\Gamma \models \phi$	ϕ is a logical consequence of Γ	4

Chapter 1. Propositional Logic and Other Fundamentals

1.1 The propositional language

Symb_L	set of symbols of language L	14
$\neg$	not; negation symbol	14
$\vee$	or; disjunction symbol	14
$\wedge$	and; conjunction symbol	14
$\rightarrow$	implies; implication symbol	14
$\leftrightarrow$	iff, if and only if; bi-implication symbol	14
Expr_L	set of expressions of language L	14
$\text{Sent}_n, \text{Sent}_L$	sets of sentences of language L	15
T	truth	17
F	falsity	17
$A \sim B$	difference of two sets	18

1.2 Induction and recursion

$\mathcal{X} = (X, X_0, \mathcal{H})$	induction system	21
$\overline{\mathcal{X}}, \overline{X}$	closure set of $\mathcal{X}$	21
At	set of atomic sentences occurring	27
SubSent	set of subsentences	28
Supp _Y	Y-support	28
A	Ackermann function	31
lg(ϕ)	length of ϕ	31

1.3 Propositional semantics

$\phi_0 \wedge \cdots \wedge \phi_n$	iterated conjunction	32
$\phi_0 \vee \cdots \vee \phi_n$	iterated disjunction	32
$\models \phi$	ϕ is a tautology	32
$\phi \models \psi$	ϕ and ψ are tautologically equivalent	32
$\phi \equiv \psi$	ψ is a tautological consequence of ϕ	32
T	a canonical tautology	33
F	a canonical falsehood	33
Sent _{$\neg, \vee$}	not-or sentences	35
TA	set of truth assignments	35
$\phi' \leftarrow \phi_{\chi \mapsto \chi'}$	substitution of χ' for χ	39
$\phi \downarrow \psi$	nor or neither nor	40
$\phi \mid \psi$	nand or Sheffer stroke	40
At ⁺ (ϕ)	sentence symbols occurring positively	41
At ⁻ (ϕ)	sentence symbols occurring negatively	41

1.4 Propositional theories

Mod(Γ)	set of models of Γ	42
$\Gamma \models \psi$	ψ is a tautological consequence of Γ	42
Th(Γ)	theory generated by axioms	43
Th(V)	theory of a truth assignment	44
Th(K)	theory of a set of truth assignments	45
PCT	Propositional Compactness Theorem	47
$\Gamma \models^* \psi$	ψ is a finite tautological consequence of Γ	47
Th [*] (Γ)	set of finite tautological consequences of Γ	47
APCT	Alternative PCT	47
$\bigwedge \Gamma$	conjunction of a set of sentences	50
$\bigvee \Gamma$	disjunction of a set of sentences	50

Δ^T	sentences T -equivalent to one in Δ	52
$\text{Th}^\Delta(\Gamma)$	$\text{Th}(\Gamma) \cap \Delta$	52
$\text{Th}^\Delta(V)$	$\text{Th}(V) \cap \Delta$	52
$[A]^n$	set of n -element subsets of A	53

1.5 Decidability and effective enumerability

PET	Propositional Enumerability Theorem	60
Gr_F	graph of a function	63

1.6 Other constructions

$V_{\mathcal{U}}(\phi)$	truth assignment of an ultrafilter	69
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1.7 Topology and Boolean algebra

$\mathcal{A} = (A, \oplus, \otimes, \bar{\cdot}, \mathbb{O}, \mathbb{I})$	Boolean algebra	75
$\phi \equiv_{\Gamma} \psi$	Γ -tautologically equivalent	75
$[\phi]_{\Gamma}$	class of ϕ in a Lindenbaum algebra	75
$\mathcal{L}_{\Gamma} = (\mathcal{S}_{\Gamma}, \oplus, \otimes, \bar{\cdot}, \mathbb{O}, \mathbb{I})$	Lindenbaum algebra of Γ	76
$a \leq b$	ordering in a Boolean algebra	77

Chapter 2. First-Order Logic

2.1 Syntax and semantics of first-order languages

$\mathfrak{A} := (A, R, S, \dots)$	general structure	83
$\Omega := (\omega, <, +, \times, \text{Sc}, 0)$	standard model of arithmetic	85
Sc	successor function $m \mapsto m + 1$	85
Rs_L	set of relation symbols of L	86
Fs_L	set of function symbols of L	86
Cs_L	set of constant symbols of L	86
ν_L	rank function of L	86
$\text{Vb} = \{v_n : n \in \omega\}$	set of variables	86
$\doteq$	equality symbol	86
$\exists$	existential quantifier	86
$\forall$	universal quantifier	86
$L_{=}$	language of equality	86
$L_{<}$	language for ordering	87
$L_{\in}$	language of set theory	87
L_{Ω}	language of arithmetic	87

Te_L	set of terms of language L	88
AtFm_L	set of atomic formulas of language L	89
Fm_L	set of formulas of language L	89
$ \mathfrak{A} $	universe of structure $\mathfrak{A}$	91
$p^{\mathfrak{A}} \subseteq A^{\nu_L(p)}$	relation interpreting p	91
$f^{\mathfrak{A}} : A^{\nu_L(f)} \rightarrow A$	function interpreting f	91
$c^{\mathfrak{A}} \in A$	distinguished element interpreting c	91
$t^{\mathfrak{A}}[\alpha]$	denotation of a term under an assignment	91
$V_{\mathfrak{A},\alpha}$	first-order truth assignment	91
$\alpha_{x \mapsto a}$	assignment altered at x	92
$\models_{\mathfrak{A}} \phi[\alpha]$	α satisfies ϕ in $\mathfrak{A}$	93

2.2 Basic semantics

$\models \phi$	ϕ is logically valid	96
$\phi \models \psi$	ϕ and ψ are logically equivalent	96
$\phi \models \psi$	ψ is a logical consequence of ϕ	96
$\Gamma \models \psi$	ψ is a logical consequence of Γ	96
$\bigwedge \Gamma$	conjunction of a set of formulas	97
$\bigvee \Gamma$	disjunction of a set of formulas	97
$\models \phi$	ϕ is a tautology	97
$\phi \models \psi$	ϕ and ψ are tautologically equivalent	97
$\phi \models \psi$	ψ is a tautological consequence of ϕ	97
$\text{Fv}(\phi)$	set of free variables of ϕ	100
$\models_{\mathfrak{A}} \phi[a_0, \dots, a_{k-1}]$	formula ϕ is satisfied in $\mathfrak{A}$ by $a_0, \dots, a_{k-1}$	102
$\models_{\mathfrak{A}} \phi$	sentence ϕ is true in $\mathfrak{A}$	102
$\mathbf{T}$	canonical logically valid sentence	103
$\mathbf{F}$	canonical logically false sentence	103
$\exists \geq n$	there exist at least n elements	103
θ_{SLO}	axiom for strict linear ordering	104
θ_{DLO}	axiom for dense linear ordering	104
$\models_{\mathfrak{A}} \phi$	formula ϕ is universally valid in $\mathfrak{A}$	105
$t_x(u)$	substitution in a term	105
$\phi_x(u)$	substitution in a formula	105
$\overline{\forall \psi}$	universal closure of ψ	109
$\overline{\exists \psi}$	existential closure of ψ	109
$\overline{\forall \Gamma}$	set of universal closures of $\psi \in \Gamma$	109
$\text{Bool}(\Gamma)$	Boolean combinations of members of Γ	110
QF	set of quantifier-free formulas	110

2.3 Structures

$\mathfrak{A} \simeq \mathfrak{B}$	$\mathfrak{A}$ and $\mathfrak{B}$ are isomorphic structures	115
$\mathfrak{A} \equiv \mathfrak{B}$	elementarily equivalent structures	116
$\alpha^\eta(x) = \eta(\alpha(x))$	assignment induced by a mapping	116
$\mathfrak{A} \subseteq \mathfrak{B}$	substructure and extension	119
$\langle X \rangle_{\mathfrak{B}} \subseteq \mathfrak{B}$	substructure generated by X	120
$\eta : \mathfrak{A} \rightarrow \mathfrak{B}$	homomorphism	124
$\eta : \mathfrak{A} \hookrightarrow_0 \mathfrak{B}$	embedding	124
$\mathfrak{A} \prec \mathfrak{B}$	elementary substructure and extension	125
$\text{CLS}\downarrow$	Cbl Downward Löwenheim-Skolem	127
$\phi^{\mathfrak{A}}$	relation defined by ϕ	128
$\phi_{b_0, \dots, b_{n-1}}^{\mathfrak{A}}$	relation defined by ϕ from parameters	128
$\eta : \mathfrak{A} \hookrightarrow \mathfrak{B}$	elementary embedding	130
$\mathfrak{A} \times \mathfrak{B}$	direct product	130
$\prod_{i \in I} \mathfrak{A}_i$	infinite direct product	133
$\bigcup \langle \mathfrak{A}_i : i \in \omega \rangle$	union of a chain	134
$\text{acl}_{\mathfrak{A}} X$	algebraic closure of X	138
$\exists^{\geq m} x \phi$	at least m objects satisfy ϕ	138

2.4 Theories

$\text{Mod}(\Gamma)$	set of models of Γ	140
$\text{Th}(\Gamma)$	theory generated by axioms	140
$\text{Th}(\mathfrak{A})$	theory of a structure	141
$\text{Th}(K)$	theory of a class of structures	142
$\forall \Delta$	set of $\forall \mathbf{y} \theta$ for $\theta \in \Delta$	144
$\exists \Delta$	set of $\exists \mathbf{y} \theta$ for $\theta \in \Delta$	144
$\neg \Delta$	set of $\neg \theta$ for $\theta \in \Delta$	144
$\exists_n, \forall_n$	formulas with n quantifier alternations	144
$\theta_{[\text{DLO}]}, \theta_{[\text{DLO}]}, \dots$	axioms for dense linear ordering	154
$T_{[\text{DLO}]}, T_{[\text{DLO}]}, \dots$	theories of dense linear ordering	154
$\mathfrak{A} \simeq_{\text{p}} \mathfrak{B}$	partially isomorphic structures	158

2.5 Arithmetic

$(S1)$ and $(S2)$	successor axioms for ω	161
$I_{\phi, x}$	instance of the induction schema	163
$T_{\text{Sc}, 0}$	theory of successor and zero	164
Ω_+	natural numbers with addition	166
$(L1)$ and $(L2)$	order axioms for ω	166

$(A1)$ and $(A2)$	addition axioms for ω	166
T_+	theory of addition arithmetic	166
$(M1)$ and $(M2)$	multiplication axioms for ω	171

2.6 Changing languages

$L \subseteq L^+$	sublanguage and extension	173
$\mathfrak{A} = \mathfrak{A}^+ \upharpoonright L$	reduct of a structure to a sublanguage	174
$\text{Th}_L(\Gamma)$	L -theory generated by Γ	174
L^X	extension language with names for $a \in X$	176
$\mathfrak{A}_X$	L^X -expansion of $\mathfrak{A}$	176
$\exists! y \phi$	exists a unique y such that ϕ	177
ϕ^{Sk}	Skolem normal form of ϕ	184
T^{Sk}	Skolem extension of T	185
ϕ^{Hb}	Herbrand normal form of ϕ	185
L^{rel}	relational translation of L	185
$\phi^{(\text{M})}$	relativization of ϕ to M	189
$T^{(\text{M})}$	relativization of T to M	189

Chapter 3. Completeness and Compactness

3.1 Countable compactness

CCT	Countable Compactness Theorem	194
$\Gamma \models^* \psi$	ψ is a finite logical consequence	194
$\text{Th}^*(\Gamma)$	set of finite sentence consequences	194
ACCT	Alternative CCT	194
$L^\bullet, \phi^\bullet$	extension with new constants	200

3.2 Countable completeness

LogAx_L	set of logical axioms	205
$\text{Thm}_L(\Gamma)$	set of Γ - L -theorems	206
$\text{Th}_L^-(\Gamma)$	sentences in $\text{Thm}_L(\Gamma)$	206
$\Gamma \vdash_L \psi$	$\psi \in \text{Thm}_L(\Gamma)$	206
CCplT	Countable Completeness Theorem	210
ACCplT	Alternative Cbl. Completeness Theorem	210

3.3 Other constructions

$F \sim_{\mathcal{U}} G$	equivalent modulo an ultrafilter	225
$\prod_{i \in I} \mathfrak{A}_i / \mathcal{U} := \prod_{\mathcal{U}} \mathfrak{A}_i$	ultraproduct of a family of structures	225

$\phi \equiv_{\Gamma} \psi$	ϕ and ψ are Γ -logically equivalent	229
$[\phi]_{\Gamma}$	class of ϕ in a Lindenbaum algebra	229
$\phi \equiv_{\Gamma}^* \psi$	ϕ and ψ are finitely Γ -equivalent	229
$\sum B = c$	sup in a Boolean algebra	230
$\prod B = c$	inf in a Boolean algebra	230
$\text{Rs}^+(\phi)$	relation symbols occurring positively	234
$\text{Rs}^-(\phi)$	relation symbols occurring negatively	234
$\text{RO}(\mathcal{T})$	regular open algebra of a topology $\mathcal{T}$	236

3.4 Uncountable languages and structures

$L^{\mathfrak{A}}$	extension language with names for $a \in \mathfrak{A}$	237
$\mathbf{x} := x_0, \dots, x_{k-1}$	list of variables	241
$\phi(\mathbf{x})$	abbreviation for $\phi(x_0, \dots, x_{k-1})$	241
$\mathbf{c} := c_0, \dots, c_{k-1}$	list of constant symbols	241
$\phi(\mathbf{c})$	substitution of c_i for x_i in $\phi(\mathbf{x})$	241
$\mathbf{a} := a_0, \dots, a_{k-1} \in A$	list of elements of A	241
$\models_{\mathfrak{A}} \phi[\mathbf{a}]$	satisfaction with x_i assigned to a_i	241
$\eta(\mathbf{a})$	pointwise mapping $(\eta(a_0), \dots, \eta(a_{k-1}))$	241
$\mathfrak{A} \prec^{\Delta} \mathfrak{B}$	Δ -substructure and extension	241
$\eta : \mathfrak{A} \hookrightarrow^{\Delta} \mathfrak{B}$	Δ -embedding	241
$\mathfrak{B}_X^{\eta}$	expansion of $\mathfrak{B}$ to an L^X -structure	244
$\text{Th}^{\Delta}(\mathfrak{A}_X)$	Δ -theory of $\mathfrak{A}$ in L^X	244
$\mathfrak{A}_X \equiv^{\Delta} \mathfrak{B}_Y$	Δ -equivalent in L^X	244
L_{FL}	language of field theory	246
T_{FL}	theory of fields	246
Char_p	sentence expressing characteristic p	246
$T_{\text{FL}(p)}$	theory of fields of characteristic p	246
$T_{\text{FL}(0)}$	theory of fields of characteristic 0	246
$T_{\text{ACF}(p)}$	theory of alg. closed fields of char. p	247

3.5 Applications of compactness

Δ^T	set of formulas T -equivalent to one in Δ	252
$\eta : \mathfrak{A} \rightarrow^{\Delta} \mathfrak{B}$	Δ -homomorphism	274

3.6 Higher-order logic

$L^{2\text{m}}$	monadic second-order language	276
$\text{Vb}^1 = \{V_n : n \in \omega\}$	set of set variables	276
$\text{Vb}^0 = \{v_n : n \in \omega\}$	set of individual variables	276

Fm_L^{2m}	set of monadic second-order formulas	276
θ_{WO}	L^{2m} -axiom for well-ordering	277
θ_{Ind}	L^{2m} -induction axiom	278
θ_{ω}	L^{2m} -axiom for successor	278
θ_{Ω}	L^{2m} -axiom for arithmetic	279
θ_{CWO}	L^{2m} -axiom for countable well-orderings	279
$\theta_{(\text{CDLO})}$	L^{2m} -axiom for complete dense orderings	280
$\exists_1^1$	set of existential second-order formulas	280
$\forall_1^1$	set of universal second-order formulas	280
$t_{\kappa}(\mathcal{L})$	smallest λ s.t. $\mathcal{L}$ is (κ, λ) -compact	281
$\ell_{\kappa}(\mathcal{L})$	smallest λ s.t. $\mathcal{L}$ has (κ, λ) -LS $\downarrow$	281
$h_{\kappa}(\mathcal{L})$	smallest λ s.t. $\mathcal{L}$ has (κ, λ) -LS $\uparrow$	281
$t(\mathcal{L}) := t_{\infty}(\mathcal{L})$	Tarski number of $\mathcal{L}$	282
$\ell(\mathcal{L}) := \ell_1(\mathcal{L})$	Löwenheim number of $\mathcal{L}$	282
$h(\mathcal{L}) := h_1(\mathcal{L})$	Hanf number of $\mathcal{L}$	282
κ_{ω}	first beth fixed point	283
θ_{Fin}	$\mathcal{L}^2$ -sentence expressing finiteness	289
$\theta_{\aleph_0}$	$\mathcal{L}^2$ -sentence expressing countability	289
$\theta_{\aleph_1}$	$\mathcal{L}^2$ -sentence expressing power at most $\aleph_1$	289
A^{σ}	domain over A of type σ	290
$\mathfrak{A} \simeq_p^{2m} \mathfrak{B}$	2m-partially isomorphic structures	292

3.7 Infinitary logic

$L_{\kappa\omega}$	infinitary languages	293
$\bigvee$	infinite disjunction symbol	293
Vb^{κ}	variables of $L_{\kappa\omega}$	293
$\text{Fm}_{L_{\kappa\omega}}$	$L_{\kappa\omega}$ -formulas	293
$\bigwedge$	infinite conjunction symbol	294
$L_{\kappa\lambda}$	languages with infinitary quantification	294
$\text{SubFm}(\phi)$	set of subformulas of ϕ	295
θ_{fin}	$L_{\omega_1\omega}$ -axiom for finiteness	295
θ_{ω}	$L_{\omega_1\omega}$ -axiom for successor	295
θ_{Ω}	$L_{\omega_1\omega}$ -axiom for arithmetic	295
$\text{Thm}_{L_{\kappa\omega}}^{\kappa\omega}(\Gamma)$	set of $L_{\kappa\omega}$ -theorems from Γ	300
$\Gamma \vdash_{\kappa\omega, L} \psi$	ψ is an $L_{\kappa\omega}$ -theorem from Γ	300
$\text{Frag}(\Theta)$	smallest fragment which includes Θ	301

Chapter 4. Incompleteness and Undecidability

4.1 A first look

Q	Robinson Arithmetic	311
$(L3)$	axiom of linearity of order	311
P	Peano Arithmetic	311
DH	Definability Hypothesis	313
$\chi_0, \chi_1, \dots, \chi_m, \dots$	enumeration of L_Ω -formulas	313
$\#(\phi)$	least index of ϕ in this enumeration	313
RH	Representability Hypothesis	316
U_a	section of relation U	320
D_U	diagonal set of relation U	320
$Th(\Omega)^\#$	indices of sentences true in Ω	321
Sb	diagonalization function	321

4.2 Recursive functions and relations

Cs_n^k	constant functions	328
Pr_i^k	projection functions	328
Sc_i^k	successor functions	328
Cmp_l^k	composition functionals	328
Rec^{k+1}	recursion functionals	328
$Prim$	set of primitive recursive functions	328
Se^k	search functionals	329
Rec	set of recursive functions	329
$(\exists p < q) \dots$	bounded existential number quantifier	329
$(\forall p < q) \dots$	bounded universal number quantifier	329
$Pred$	predecessor function	331
$\circ$	truncated subtraction	331
Sg	signum function	331
K_R	characteristic function of relation R	331
least $p < n$	bounded search	332
$m n$	m divides n	334
p_n	n -th prime number	334
$<_\omega \omega$	set of finite sequences of natural numbers	334
$\langle m_0, \dots, m_{k-1} \rangle^k$	sequence code	334
$(s)_i$	i -th component of a coded sequence	334
$lg(s)$	length of sequence coded by s	335
$Sq(s)$	s is a sequence number	335

$s * t$	concatenation of two sequence codes	335
$\overline{F}$	course-of-values function	335
Rec'	alternative char. of recursive functions	336
Op	ordered pair function	337
Sq^k	sequence coding function	338
$\text{Cp}(t, i)$	i -th component of sequence t	339

4.3 Recursively enumerable sets and relations

$\exists R$	projection of relation R	342
$\text{Graph}(F)$	graph of F	346
$F(m_0, \dots, m_{k-1}) \downarrow$	F is defined on $m_0, \dots, m_{k-1}$	346
$F(m_0, \dots, m_{k-1}) \uparrow$	F is undefined on $m_0, \dots, m_{k-1}$	347
$F(\mathbf{m}) \simeq G(\mathbf{n})$	equality for partial functions	347
Cmp_l^k	composition functionals	347
Rec^{k+1}	recursion functionals	347
$\text{Se}^k(G)$	partial search functionals	347
$X \leq_m Y$	X is many-one reducible to Y	350
$X \equiv_m Y$	X and Y are many-one equivalent	350
$\text{dg}_m(X)$	many-one degree of X	351
K_R^+	positive characteristic function of R	351

4.4 Gödel numbering

$\text{gn}(t)$	Gödel number of a term	354
$\text{gn}(\phi)$	Gödel number of a formula	354
$\text{gn}(\Gamma)$	set of Gödel numbers of a set of formulas	354
Te	set of Gödel numbers of terms	355
Afo	set of Gödel numbers of atomic formulas	355
Fo	set of Gödel numbers of formulas	355
Sub	substitution function	356
Sb^k	numeral substitution functions	357
$\text{Free}(m, i)$	v_i is free in $\text{gn}^{-1}(m)$	358
$\text{Freesub}(m, i, q)$	$\text{gn}^{-1}(q)$ is freely subst. for v_i in $\text{gn}^{-1}(m)$	358
Sent	set of Gödel numbers of sentences	358
Proof_Γ	set of codes of proofs from Γ	360
Proofof_Γ	relation between proofs and last lines	360
Taut	set of Gödel numbers of tautologies	360
Eqax	set of Gödel numbers of equality axioms	360
$\exists\text{-ax}$	set of Gödel numbers of $\exists$ -axioms	360
ModPon	set of Gödel numbers of MP inferences	360

$\exists$ -rule	set of Gödel numbers of $\exists$ -rule inferences	360
$A_p(n) = A(p, n)$	Ackermann function	363

4.5 Definability in arithmetic I

4.6 Representability: First Incompleteness Theorem

$(\exists x \dot{<} t)$	bounded existential quantification	374
$(\forall x \dot{<} t)$	bounded universal quantification	374
U_T^k	universal semi-recursive relations	378
$\ulcorner \phi \urcorner$	name for ϕ	380

Chapter 5. Topics in Definability

5.1 Definability in arithmetic II

$\exists R$	projection of relation R	393
$\forall R$	co-projection of relation R	394
$\Sigma_n, \Pi_n, \Delta_n$	classes of the arithmetical hierarchy	394
$U^k := U_Q^k$	universal semi-recursive relations	396
Emp	set of indices of the empty set	396
Fin	set of indices of finite r.e. sets	396
Inf	set of indices of infinite r.e. sets	396
Rec	set of indices of recursive r.e. sets	396
${}_n U^k$	universal Σ_n relations	397
Σ formulas	no unbounded universal quantifiers	401
Π formulas	no unbounded existential quantifiers	401
$\text{Th}^\Sigma(\Omega)$	set of Σ sentences true in Ω	405

5.2 Indexing

TE	set of Gödel numbers of terms	410
THM_Γ	set of Gödel numbers of theorems from Γ	410
$\text{Deriv}^\mathcal{X}$	set of codes of $\mathcal{X}$ -derivations	411
$\text{Derivof}^\mathcal{X}$	codes of $\mathcal{X}$ -derivations and result	411
PrimInd	set of primitive recursive indices	412
$[a]$	primitive recursive function with index a	412
PrimInd^k	set of k -ary primitive recursive indices	412
U_{Prim}^k	universal primitive recursive functions	413
PrimComp	set of primitive recursive computations	415
PartComp	set of partial recursive computations	418

$\{a\}$	partial recursive function with index a	418
U^k	universal partial recursive functions	418
W^k	halting problems	419

5.3 Second Incompleteness Theorem

Con_T	sentence expressing consistency of T	422
(DC1), (DC2)	derivability conditions	423
ConCat	formal concatenation operator	424

5.4 Church's Thesis

$\text{Thm}_L(\Gamma)$	set of equations provable from Γ	435
$\mathcal{M} = (M, Q, q_I)$	Turing machine	438

5.5 Applications to other languages and theories

Val_L	set of valid sentences	449
Sat_L	set of satisfiable sentences	449
FinVal_L	set of finitely valid sentences	450
FinSat_L	set of finitely satisfiable sentences	450

Chapter 6. Set Theory

IST	Informal Set Theory	456
AST	Axiomatic Set Theory	456
ZF	Zermelo-Fraenkel Set Theory	456
FAST	Formalized Axiomatic Set Theory	456
ZFC	Zermelo-Fraenkel with Axiom of Choice	456

6.1 Zermelo-Fraenkel set theory

$L \in$	language of set theory	456
$\emptyset$	empty class	458
$\dot{V}$	universal class	458
Ru	Russell class	458
$x \dot{\cup} y$	union operation	458
$x \dot{\cap} y$	intersection operation	458
$x \setminus y$	set difference operation	458
$\dot{S}x$	successor operation	458
$\dot{\cup} x$	Union operation	458
$\dot{\cap} x$	Intersection operation	458
$\dot{\wp} x$	powerclass operation	458

$\{ \dot{z} : \phi(x, z) \}$	class $\{ \{ z : \phi(x, z) \} \}$ is a set	459
$B \dot{\cup} C$	union of two classes	461
$B \dot{\cap} C$	intersection of two classes	461
$\dot{\bigcup} B$	Union of a class	461
$\dot{\bigcap} B$	Intersection of a class	461
$F(Gx)$	composition of class operations	461
F^*	image operation under F	462
$\dot{\omega}$	class of natural numbers	462
$\{ \dot{x}, y \}$	pair set	464
$F^\cap(x_0, \dots, x_{k-1}, u)$	$F(x_0, \dots, x_{k-1}) \cap u$	464
V_σ	σ -th stage of the universe V	468

6.2 Mathematics in set theory I

$\langle \dot{x}, y \rangle$	ordered pair	472
$\langle x_0, \dots, x_k \rangle$	ordered $(k+1)$ -tuple	472
$u \dot{\times} v$	Cartesian product	473
$\dot{\text{Op}}(z)$	z is an ordered pair	473
$\text{Rel}(r)$	r is a relation	473
$\text{Dom}(r)$	domain of r	473
$\text{Im}(r)$	image of r	474
$\text{Fld}(r)$	field of r	474
$\text{Fn}(f)$	f is a function	474
$\text{Apl}(f, x)$	function application	474
$\text{Inj}(f)$	f is injective	474
$f : x \dot{\rightarrow} y$	f maps x into y	474
${}^x y$	set of functions from x into y	474
$g \circ f$	f composed with g	475
$f \dot{\leftarrow} 1$	inverse of f	475
$f \dot{\restriction} u$	restriction of f to u	475
$f^* u$	f -image of u	475
$r \dot{\simeq}_f s$	r and s are isomorphic via f	478
$\dot{k}$	representation of $k \in \omega$ as a set	478
$\dot{<}_{\dot{\omega}}$	ordering of the natural numbers	480
$\dot{+}$	addition of natural numbers	482
$\dot{\times}$	multiplication of natural numbers	482
$x \dot{\approx}_f y$	x and y are equinumerous via f	485
$\dot{<}_{\dot{\omega}} u$	set of finite sequences from u	486
$\dot{\mathbb{Z}}$	representation of integers in ZF	486
$\dot{\mathbb{Q}}$	representation of rational numbers in ZF	487

$\dot{\mathbb{R}}$	representation of real numbers in ZF	487
$\dot{\mathbb{C}}$	representation of complex numbers in ZF	487
$\dot{\mathbb{I}}(x)$	inductive closure of $\mathbb{I}$	488
$\mathbb{I}_{\dot{\omega}}(u)$	inductive operator defining $\dot{\omega}$	489
$\mathbb{I}^y(x)$	y -th iterate of $\mathbb{I}$	490
$\text{TC}(x)$	transitive closure of x	491
$\text{TC}^+(x)$	complete transitive closure of x	491
$\mathbb{I}_{\text{HF}}, \text{HF}$	hereditarily finite sets	494
$\mathbb{I}_{\text{HFO}}, \text{HFO}(x)$	hereditarily finite sets over x	494

6.3 Ordinal numbers: induction and recursion

$(\exists x \dot{\in} y)$	$\dot{\in}$ -bounded existential quantifier	497
$(\forall x \dot{\in} y)$	$\dot{\in}$ -bounded universal quantifier	497
$\dot{\text{Or}}$	class of ordinal numbers	497
$\rho, \sigma, \tau, v, \dots$	variables over ordinals	497
$\exists \sigma \phi(\sigma)$	$\exists x [x \text{ is an ordinal} \wedge \phi(x)]$	497
$\dot{<} \dot{\text{Or}}$	class ordering of the ordinals	498
$\dot{<}_u$	ordering of the ordinals in u	498
$\dot{+}_o, \dot{\times}_o$	ordinal addition and multiplication	502
$\dot{\mathbb{V}}_{\sigma}$	stages of the hierarchy of the universe	506
rank	ordinal rank	507

6.4 Cardinal numbers

AC	Axiom of Choice	510
$x \dot{\preceq} y$	x is of less or equal size than y	510
$x \dot{\prec} y$	x is of strictly smaller size than y	510
CH_1	first version of the Continuum Hypothesis	514
GCH_1	first version of the Generalized Cont. Hyp.	514
Ha	Hartogs' operation	514
$\text{CH}_2, \text{GCH}_2$	second versions of the (Gen.) Cont. Hyp.	515
$\text{ep}(x)$	equipotency class of x	515
$\dot{ } x \dot{ }$	cardinal of x	516
$\dot{\text{Ca}}$	class of cardinal numbers	517
κ, λ, μ and ν	variables over cardinals	517
κ^+	cardinal successor to κ	518
$\dot{2}^{\kappa}$	cardinal of the powerset of κ	518
$\text{CH}_3, \text{GCH}_3$	third versions of the (Gen.) Cont. Hyp.	518
$\dot{\aleph}_{\sigma}$	enumeration of the infinite cardinals	519
$\text{CH}_4, \text{GCH}_4$	fourth versions of the (Gen.) Cont. Hyp.	519

$\kappa \dot{+}_c \lambda$	cardinal addition	519
$\kappa \dot{\times}_c \lambda$	cardinal multiplication	519
$\text{cf}(\rho)$	cofinality of a limit ordinal	522
κ^λ	cardinal exponentiation	523
CAC	Countable Axiom of Choice	526
ADC	Axiom of Dependent Choice	527

6.5 Models and independence

$\phi^{(M)}$	relativization of ϕ to the class M	528
$\exists^M x, \forall^M x$	quantifiers bounded to a class M	529
F^M	relativization of a class operation	529
Δ_0	bounded-quantifier $L_\in$ -formulas	540
$\Sigma^{\text{ZF}}, \Pi^{\text{ZF}}, \Delta^{\text{ZF}}$	ZF-existential and universal $L_\in$ -formulas	540
$\Sigma^T, \Pi^T, \Delta^T$	T-existential and universal $L_\in$ -formulas	540

6.6 Mathematics in set theory II

$\dot{=}, \dot{=}, \dot{\vee}, \dot{\exists}$	logical symbols of FAST	550
vb	set of $\dot{\vee}$ -variables	550
$\ \ell\ $	cardinality of a language	550
$\text{AtTe}(\ell)$	set of ℓ -atomic terms	550
$\text{Te}(\ell)$	set of ℓ -terms	551
$\text{Sent}(\ell)$	set of ℓ -sentences	551
$\text{Th}(\gamma, \ell)$	ℓ -theory generated by γ	553
$\text{Con}(t)$	t is consistent	553
$\dot{\text{P}}, \dot{\text{ZF}} \text{ and } \dot{\text{ZFC}}$	formalized versions of P, ZF and ZFC	556

6.7 The constructible universe

$\dot{\rho}_{\text{def}}(u)$	definable powerset operation	568
L	the constructible universe	568
ZFL	theory generated by $\text{ZF} \cup \{ \dot{\vee} \dot{=} \text{L} \}$	569
$\text{rank}_{\text{L}}(x)$	L-rank of x	570
$<_{\text{L}_\sigma}$	well-ordering of L_σ	572
$<_{\text{L}}$	class well-ordering of L	572
$\text{L}[a]$	universe of sets constructible from a	575
HOD	class of hereditarily ordinal definable sets	576
OD	class of ordinal definable sets	576

6.8 Generic extensions

ZFL_m	$\text{ZFL} + m$ is a cbl. trans. class model of ZFL	578
$\mathbb{FF}(u, v)$	forcing with finite functions u into v	579
$\text{Na}^{\mathbb{P}}$	class of $\mathbb{P}$ -names	582
$\text{rank}_{\mathbb{P}}(b)$	rank of a name	583
a_G	G -interpretation of the name a	583
$m[G]$	generic extension of m by G	583
$\check{x}$	canonical name for $x \in m$	583
$\check{G}$	canonical name for generic set G	583
$\check{F}$	canonical name for class operation F	584
$p \Vdash \phi(a_0, \dots, a_{k-1})$	p forces $\phi(a_0, \dots, a_{k-1})$	586
$c \dot{\in}_p a$	strong form of $p \Vdash c \dot{\in} a$	587
$p \Vdash \phi$	formalized forcing relation	587
$\mathring{F}^m(a)$	bound for names of elements of $F^{m[G]}(a_G)$	588

6.9 Forcing

$p \Vdash \phi$	formalized forcing relation	596
$q \Vdash^* c \dot{\in} b$	temporary forcing relation	597

6.10 Large cardinals

$\text{cf}(\rho)$	cofinality of a limit ordinal	606
WI	class of uncbl. weakly inacc. cardinals	609
SI	class of uncbl. strongly inacc. cardinals	609
$\exists \text{WI}$	exists an uncbl. weakly inacc. cardinal	609
$\exists \text{SI}$	exists an uncbl. strongly inacc. cardinal	609
$\dot{V}_{\text{SI}}$	smallest strongly inaccessible rank	610
L_{WI}	smallest weakly inaccessible level of L	611
MC	class of uncountable measurable cardinals	613
$\exists \text{MC}$	exists an uncountable measurable cardinal	613
N and j	elementary embedding of the universe	614
$\phi^{(M, E)}$	relativization to class structure (M, E)	615
x_E	set represented by x in (M, E)	616

6.11 Determinacy

$\langle \overset{\circ}{m}, n \rangle$	formal ordered pair	622
$\langle \overset{\circ}{m}, n \rangle \frown \langle \overset{\circ}{p}, q, r \rangle$	formal concatenation operator	622
$\mathring{S}q$	formal set of sequence numbers	623
$\gamma \# \delta$	play resulting from two strategies	623

$WS_I(A)$	set of winning strategies for player I	623
$WS_{II}(A)$	set of winning strategies for player II	623
$AD[\mathcal{C}]$	all games in $\mathcal{C}$ are determined	623
AD	Axiom of Determinacy	623
$\varepsilon_I(k)$	sequence of plays of player I	624
$\varepsilon_{II}(k)$	sequence of plays of player II	624
WO	set of codes for well-orderings of ω	625
$\ \alpha\ $	countable ordinal coded by α	625
$[T]$	set of infinite branches of a tree	628
Bo	set of Borel relations	629
$\Pi_\sigma^0, \Sigma_\sigma^0, \Delta_\sigma^0$	classes of the Borel Hierarchy	629
A_s	extensions of s in A	630
Proj	set of projective relations	632
$\Pi_n^1, \Sigma_n^1, \Delta_n^1$	classes of the Projective Hierarchy	632
$WO_{<\sigma}$	codes for well-orderings shorter than σ	632
LO	set of codes for linear orderings	633
$\prec_\Sigma, \preceq_\Sigma, \prec_\Pi, \preceq_\Pi$	Σ_1^1 and Π_1^1 pre-well-ordering relations	634
$\text{Perf}(\mathcal{C})$	$\mathcal{C}$ has the perfect set property	636
$\text{Unif}(\mathcal{C})$	$\mathcal{C}$ has the uniformization property	638
WO^*	codes for constructibly cbl. ordinals	641
$[X]^n$	set of n -element subsets of X	648

Chapter 7. Model Theory

7.1 Partial embeddings

$\eta : \mathfrak{A} \hookrightarrow_{\mathfrak{p}}^\Delta \mathfrak{B}$	partial Δ -embedding	656
QF	set of quantifier-free formulas	656
Fm_ω	set of first-order formulas	656
Fm_κ	set of $L_{\kappa\omega}$ -formulas	656
$\text{Dom}(\eta) = \mathbf{a}$	$\text{Dom}(\eta) = \{a_0, \dots, a_k\}$	656
$(\mathfrak{A}, \mathbf{a})$	$\mathfrak{A}$ with $a_0, \dots, a_k$ distinguished	656
$\mathfrak{A} \hookrightarrow_{\mathfrak{p}, 0} \mathfrak{B}$	partial embedding	656
$\mathfrak{A} \hookrightarrow_{\mathfrak{p}, \omega} \mathfrak{B}$	partial elementary embedding	656
$\mathfrak{A} \hookrightarrow_{\mathfrak{p}, \kappa} \mathfrak{B}$	partial κ -embedding	656
$\text{Tp}_{\Delta}^{\mathfrak{A}}(\mathbf{a})$	Δ -type of $\mathbf{a}$ in $\mathfrak{A}$	657
$\text{Tp}_0^{\mathfrak{A}}(\mathbf{a})$	quantifier-free type of $\mathbf{a}$ in $\mathfrak{A}$	657
$\text{Tp}^{\mathfrak{A}}(\mathbf{a})$	(elementary) type of $\mathbf{a}$ in $\mathfrak{A}$	657
$\text{Tp}_\kappa^{\mathfrak{A}}(\mathbf{a})$	κ -type of $\mathbf{a}$ in $\mathfrak{A}$	657
$\mathfrak{A} \simeq_{\mathfrak{p}} \mathfrak{B}$	partially isomorphic	658

$\mathfrak{A} \simeq_{\mathfrak{p}}^{\sigma} \mathfrak{B}$	partially σ -embeddable	664
$\text{qr}(\phi)$	quantifier-rank of ϕ	665
$\mathfrak{A} \equiv_{\kappa}^{\sigma} \mathfrak{B}$	equivalent for $L_{\kappa\omega}$ -sentences of $\text{qr} \leq \sigma$	665
$\text{Tp}_{\kappa;\sigma}^{\mathfrak{A}}(\mathbf{a})$	$(\kappa; \sigma)$ -type of $\mathbf{a}$ in $\mathfrak{A}$	665

7.2 Boolean algebras, ultrafilters and types

Fm_L^n	L -formulas with at most $v_0, \dots, v_{n-1}$ free	672
$[\phi]_T^n$	T -equivalence class of ϕ	672
$\mathcal{L}_T^n$	Boolean algebra of n -variable formulas	672
Φ_F	set of formulas corresponding to a filter F	672
$S(\mathcal{A})$	set of ultrafilters of $\mathcal{A}$	675
a^*	basic open set of $S(\mathcal{A})$	675
$\text{CLO}(S(\mathcal{A}))$	algebra of clopen subsets of $S(\mathcal{A})$	676
$\text{RO}(S(\mathcal{A}))$	regular open algebra of $S(\mathcal{A})$	676
$S^n(T)$	set of complete n -types of T	682

7.3 Countable models of countable theories

$\mu(T, \kappa)$	number of models of T of cardinality κ	684
$S^n(\mathfrak{A})$	set of complete n -types of $\text{Th}(\mathfrak{A})$	691

7.4 Uncountable models of countable theories

$\mathfrak{A} \simeq_{\mathfrak{p}, \kappa} \mathfrak{B}$	κ -partially isomorphic	703
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7.5 Morley's Theorem

SOI	set of indiscernibles	714
$[A]^n$	set of n -element subsets of A	716

7.6 Abstract logics

$\models^{\mathcal{L}}$	truth relation for the logic $\mathcal{L}$	722
$\text{Sent}_{\mu}^{\mathcal{L}}$	set of μ -sentences of the logic $\mathcal{L}$	722
$\neg \phi, \phi \vee \psi$	“connectives” of an abstract logic	722
$\text{Mod}_{\mu}^{\mathcal{L}}(\phi)$	set of μ -models of ϕ	723
$\text{Mod}_{\mu}^{\mathcal{L}}(\Gamma)$	set of μ -models of Γ	723
$\Gamma \models^{\mathcal{L}} \phi$	ϕ is true in every model of Γ	723
$\text{Th}^{\mathcal{L}}(\mathfrak{A})$	set of $\mathcal{L}$ -sentences true in $\mathfrak{A}$	723
$\mathfrak{A} \equiv^{\mathcal{L}} \mathfrak{B}$	$\mathfrak{A}$ and $\mathfrak{B}$ have the same $\mathcal{L}$ -theories	723
$\mathcal{K} \leq \mathcal{L}$	$\mathcal{L}$ translates $\mathcal{K}$	723
$\mathcal{K} \sim \mathcal{L}$	$\mathcal{K}$ and $\mathcal{L}$ are equivalent logics	723

$\phi \mapsto \phi^{(A)}$	quantifier relativization	724
$\phi \mapsto \phi^{\text{rel}}$	relational translation	725
$\text{gn}_\mu^{\mathcal{L}} : \text{Sent}_\mu^{\mathcal{L}} \rightarrow \omega$	Gödel numbering of the logic $\mathcal{L}$	730
$\mathcal{K} \leq_{\text{eff}} \mathcal{L}$	$\mathcal{L}$ effectively translates $\mathcal{K}$	730
$\mathcal{K} \sim_{\text{eff}} \mathcal{L}$	$\mathcal{K}$ and $\mathcal{L}$ are effectively equivalent	730

Chapter 8. Recursion Theory

8.1 Many-one degrees and r.e. sets

$A \leq_m B$	many-one reducible	733
$A <_m B$	strict many-one reducible	734
$A \equiv_m B$	many-one equivalent	734
$\text{dg}_m(A)$	many-one degree	734
Dg_m	set of many-one degrees	735
$\mathbf{0}$	many-one degree of recursive sets	735
$A \oplus B$	least upper bound of m-degrees	736
$\mathbf{1}$	largest r.e. many-one degree	737
$\{a\}$	partial recursive function with index a	738
$\{a\}(x_0, \dots, x_{k-1}) \downarrow$	partial recursive function is defined	738
$\{a\}(x_0, \dots, x_{k-1}) \uparrow$	partial recursive function is undefined	738
W_a	r.e. set with index a	738
$\mathbf{K}$	diagonal many-one complete r.e. set	738
S_n^m	substitution fns for partial rec. indices	739
$\{a\}_s(x) \simeq y$	approximations to $\{a\}$	742
$W_{a,s}$	stage enumeration of W_a	742
$A^{[a]}$	a -th column of A	748
$\bigoplus \langle A^n : n \in I \rangle$	coding of a family of sets	751

8.2 Turing reducibility

$\text{PartComp}[B]$	partial computations relative to B	757
$\{a\}^B$	B -partial recursive functions	757
W_a^B	B -r.e. set with index a	757
$\{a\}_s^B(x) \simeq y$	approx. to B -partial rec. functions	757
$W_{a,s}^B$	stages of B -r.e. sets	757
$A \leq_T B$	Turing reducible	758
$A <_T B$	strict Turing reducible	758
$A \equiv_T B$	Turing equivalent	758
$\text{dg } A$	Turing degree of A	758
Dg_T	set of Turing degrees	758

$u(B; a, x)$	use of a computation	760
$u(B; a, x, s)$	use of a stage computation	760
$\{a\}_s^\sigma(x) \simeq y$	stage computation from initial segment	760
$W_{a,s}^\sigma$	stage of a set r.e. from initial segment	760
$\mathbf{0}'$	the complete r.e. Turing degree	763
FPF	set of fixed-point free functions	769
DNR	set of diagonally non-recursive functions	769

8.3 The jump operator

W^B, K^B	m-complete $[B]$ sets	771
B'	jump of a set B	771
$\mathbf{b}'$	jump of a Turing degree $\mathbf{b}$	772

8.4 Upper bounds

$\ell^{AE}(b)$	length of agreement	786
$\ell^{AE}(b, s)$	recursive approx. to length of agreement	786
CLL	Correctness Lemma for Length	786

8.5 Jumps of r.e. sets

p_s	least new element of A_s	793
$\widehat{\{b\}}_s^{A_s}(x)$	hat computation	793
T^A	set of true stages	794
$\hat{u}(A_s; b, x, s)$	hat use	795
T_b^A	set of partially true stages	797
$A^{[\leq b]}$	union of first $b + 1$ columns of A	797
$\mathsf{T}_{<a}^A$	$\bigcap_{b < a} \mathsf{T}_b^A$	797
$\hat{\ell}^{AE}(b, s)$	hat approx. to length of agreement	800
$\widehat{\text{CLL}}$	Correctness Lemma for $\widehat{\text{Length}}$	801
$\ell^{+AE}(b, s)$	modified hat approx. to length	805
CLL^+	Correctness Lemma for Length^+	805

8.6 Lower bounds

$m_{x,t}^a$	markers for coding E into A	811
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